Algebraic Geometry

— *EYNTKA* Series —

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Contents

Algebraic Geometry originated as the study of the zero-sets of polynomials, namely it was (and in many ways behind all the abstraction predominantly still is) the solution to a system of polynomials. By explicitly working with the zero sets of polynomials, we can discover a great multitude of geometric results within algebra, however as the field progressed it was discovered zero sets are in many ways insufficient to capture all the geometric data. For some quick examples:

1. it was discovered in the mid 20th century that results in number theory seem eerily similar to the study of 1 dimensional curves.
2. The galois group of a field share many of the same properties of as the fundamental group of a topological space.
3. The fact that a polynomial may have a root of high multiplicity is lost in the classical geometric picture $V(f)$, but this is often key information.
4. Number theorists do not work over algebraically closed fields as they are looking for integer (or integral) solutions to polynomial equations. However, these do not have the same nice geometric correspondence within classical algebraic geometry as the Nullstellensatz works well when the field is algebraically closed.
5. All rings can be represented as the quotient of $\mathbb{Z}[x_i, \mid i \in \mathcal{I}]$ or $k[x_i, \mid i \in \mathcal{I}]$ for some indexing set I, hence all ring (not just reduced finitely generated rings over algebraically closed fields) have some notion of being a function with evaluation
6. When studying a family of varieties (ex. $x^2 + ty^2 = 1$ as t varies) the resulting object will often have "bad" singularities which varieties are poorly equipped to work with.

Such realizations lead to the need to ground algebraic geometry with a new object that encapsulates all of these ideas. this new object is the *scheme*. As compared to varieties that requires an ambient space and working over polynomial rings over fields, schemes are *intrinsic* in that they don't require any ambient space to be defined in, and they generalize classical algebraic geometry to be able to capture the geometry of both Diophantine equations (for number theory) as well as the geometry of (quasi-projective) varieties and other more novel. Perhaps key in their success over other variations is how well they also remember "multiplicity" information of roots of polynomials. This data shall be paramount in many results we shall introduce[1].

This book shall be written with graduate or well-read undergraduates in mind. The reason is to show how inter-connected schemes are to various other fields of mathematics. For example:

1. The geometry of schemes draws many parallels to complex geometry and some results from complex analysis (ex. Hartog's lemma from Complex analysis over several variables shall have a strong appearance when discussing normal schemes).
2. A sheaf draws many similarities to the bundles that are put on manifolds; in fact a manifold can be defined equivalently by using a sheaf of differentiable functions instead of an atlas. Some results such as Poincaré duality shall be proven for schemes (see 6.5)
3. Chapter 6 shall strongly rely on the readers intuitions from algebraic topology and use modern homological algebra.

[1]We shall see for Elliptic Curves that the finite-generation of certain torsion groups shall rely on Schemes in a key way to remember multiplicity

4. Many of the properties in commutative algebra shall become interesting geometric properties in algebraic geometry.

Essentially, anything non-optional material from EYNTKA Undergraduate Algebra ([**NateClassicalAlgGeo**]), EYNTKA Differential Geometry ([7]), and EYNTKA Algebraic Topology ([6]) will be considered as assumed prerequisites. Some results from EYNTKA complex analysis ([**NateComplexAnalysis**]) will be drawn on as well given the strong connection between complex geometry and the geometry of schemes.

All essential category theory will be assumed; it can be reviewed in EYNTKA Undergraduate Algebra.

Throughout this book, we shall use the following terminology:

- Let k be a field, $\overline{k}$ the algebraic closure of k
- Let R an arbitrary ring, $M_n(R)$ the matrix ring over R, $R_\mathfrak{p}$ the localization of R by $R \setminus \mathfrak{p}$, R_f the localization of R by $\{f, f^2, ...\}$, $\text{Frac}(R)$ the field of fractions of R is R is an integral domain
- A is an arbitrary R-algebra.

see [**NateClassicalAlgGeo**].

adding new citation: [1]

The material covered in the preliminary chapter is at a break-neck pace and is only there for expository purposes for the reader to see if they have the right intuitions to cover algebraic geometry in its greater generality. The first main Part, Scheme Theory, focuses heavily on building up all the necessary theoretical framework to have a comfortable grasp of the material and is heavily inspired by the many famous texts covering the material (ex. Hartshorne, Shafarevich, Vakil, Serre, Eisenbud, and so forth).

Preliminary Chapter

In this chapter, we shall quickly review elementary concepts linking algebra and geometry without doing any proofs, that is we are going to explore some classical algebraic geometry. For all the important details, see [**NateClassicalAlgGeo**].

0.1 Recall Classical Algebraic Geometry

If A is an R-algebra, then by the universal property of free R-algebras we have if S is a set of generators for A,

$$\begin{tikzcd} R[S] \arrow[r, "\varphi"] & A \\ S \arrow[u, "\iota"] \arrow[ur, "\iota"'] & \end{tikzcd}$$

If S is finite, then $R[S] \cong R[x_1, x_2, \dots, x_n]$ by relabelling S. Since every ring is a $\mathbb{Z}$-algebra, we see that polynomial rings are central to the study of general rings. It is thus fruitful to realize that polynomial rings can naturally be interpreted as functions, that is:

$$f \in R[S],\ R^S \to R[S]^\vee \qquad f \mapsto f(s_i)_{i \in S}$$

In the finite case:

$$f \in R[x_1, x_2, \dots, x_n], \qquad R^n \to R[x_1, x_2, \dots, x_n]^\vee \qquad (a_1, a_2, \dots, a_n) \mapsto (f \mapsto f(a_1, a_2, \dots, a_n))$$

We don't want to consider R^S as having any structure besides being an affine space, hence we shall label it as $\mathbb{A}_R^S$ or in the finite case $\mathbb{A}_R^n$. If the ring is evident in context, we shall write $\mathbb{A}^n$.

0.1.1 Conventions in Number Theory In a number-theoretic context, $\mathbb{A}^n$ often implicitly means $\mathbb{A}_{\mathbb{C}}^n$ (or $\mathbb{A}_{\mathbb{Q}}^n$), with $\mathbb{A}^n(R)$ denoting the R-points. This convention arises because polynomials over $\mathbb{C}$ always have solutions, which is not guaranteed over $\mathbb{R}$ or $\mathbb{Q}$.

For now, we shall stick to the finitely generated case. Note that in $\mathbb{A}^n_R$, the point $(0,\ldots,0)$ has no special property, as we are treating this space as simply a collection of points. To emphasize this geometric interpretation, any $p \in \mathbb{A}^n_R$ will be called a *point*, and for

$$p = (r_1, r_2, \ldots, r_n)$$

each r_i is called a *coordinate* of p. Hence, we may view any $f \in R[x_1, x_2, \ldots, x_n]$ as a function:

$$f : \mathbb{A}^n_R \to R \qquad (r_1, r_2, \ldots, r_n) \mapsto f(r_1, r_2, \ldots, r_n)$$

Algebraic Sets and the Zariski Topology

The zeros of polynomials form many interesting algebraic shapes. A prototypical example is $V(x^2 + y^2 - 1) \subseteq \mathbb{A}^2_{\mathbb{R}}$, the unit circle.

Definition 0.1.2: Zero Set

For $S \subseteq R[x_1, \ldots, x_n]$, the *zero set* (or *vanishing locus*) of S is

$$\mathcal{Z}(S) = \{p \in \mathbb{A}^n_R \mid f(p) = 0, \ \forall f \in S\}$$

Definition 0.1.3: Algebraic Set

A subset $X \subseteq \mathbb{A}^n_R$ is called an *algebraic set* (or *closed set*) if there exists $S \subseteq R[x_1, \ldots, x_n]$ such that $\mathcal{Z}(S) = X$.

The collection of algebraic sets satisfies the axioms for closed sets of a topology:

1. $\mathbb{A}^n_R = \mathcal{Z}(0)$ and $\emptyset = \mathcal{Z}(1)$ are algebraic.
2. Finite unions: $\mathcal{Z}(S_1) \cup \mathcal{Z}(S_2) = \mathcal{Z}(S_1 \cdot S_2)$ where $S_1 \cdot S_2 = \{fg : f \in S_1, g \in S_2\}$.
3. Arbitrary intersections: $\bigcap_\alpha \mathcal{Z}(S_\alpha) = \mathcal{Z}\big(\bigcup_\alpha S_\alpha\big)$.

Definition 0.1.4: Zariski Topology

The *Zariski topology* on $\mathbb{A}^n_R$ is the topology whose closed sets are precisely the algebraic sets.

0.1.5 Properties of the Zariski Topology The Zariski topology is quite coarse compared to the Euclidean topology on $\mathbb{R}^n$ or $\mathbb{C}^n$:

1. It is generally *not* Hausdorff: in $\mathbb{A}^1_k$, the only proper closed subsets are finite sets of points, so any two nonempty open sets intersect.
2. Every nonempty open set is dense.
3. Despite this, the Zariski topology is well-suited to algebraic geometry because its closed sets are precisely the sets defined by polynomial equations.

The Ideal–Variety Correspondence

Conversely, given a subset $S \subseteq \mathbb{A}^n_R$, we can ask which polynomials vanish on S:

Definition 0.1.6: Ideal of a Set

For $S \subseteq \mathbb{A}^n_R$, the *ideal of* S is

$$\mathcal{I}(S) = \{f \in R[x_1, \ldots, x_n] \mid f(p) = 0,\ \forall p \in S\}$$

One readily checks that $\mathcal{I}(S)$ is always an ideal in $R[x_1, \ldots, x_n]$. Moreover, $\mathcal{I}(S)$ is always a *radical* ideal, i.e., $f^n \in \mathcal{I}(S)$ implies $f \in \mathcal{I}(S)$. This is immediate: if f^n vanishes on S, then f vanishes on S.

The operators $\mathcal{Z}$ and $\mathcal{I}$ form an order-reversing correspondence:

$$S_1 \subseteq S_2 \implies \mathcal{I}(S_1) \supseteq \mathcal{I}(S_2), \qquad I_1 \subseteq I_2 \implies \mathcal{Z}(I_1) \supseteq \mathcal{Z}(I_2)$$

However, this correspondence is not bijective in general. The precise relationship requires the notion of radical ideals and irreducibility.

Definition 0.1.7: Radical Ideal

An ideal $I \subseteq R$ is *radical* if $f^n \in I$ implies $f \in I$ for all $f \in R$ and $n \geq 1$. Equivalently, $I = \sqrt{I}$ where $\sqrt{I} = \{f \in R \mid \exists n \geq 1,\ f^n \in I\}$.

Definition 0.1.8: Irreducible Set

An algebraic set V is *irreducible* if whenever $V = V_1 \cup V_2$ with V_1, V_2 closed, then $V = V_1$ or $V = V_2$. Equivalently, V is irreducible if and only if any two nonempty open subsets of V intersect.

Definition 0.1.9: Variety (Classical)

An algebraic set $V \subseteq \mathbb{A}^n_k$ is an (affine) *variety* if V is irreducible.

The key algebraic characterization of irreducibility is:

Proposition 0.1.10: Irreducibility and Prime Ideals

An algebraic set V is irreducible if and only if $\mathcal{I}(V)$ is a prime ideal.

***Proof*:**
See [**NateClassicalAlgGeo**].

Example 0.1.11: Reducible vs. Irreducible
Consider $V(xy) \subseteq \mathbb{A}^2_k$. This is the union of the two coordinate axes:

$$V(xy) = V(x) \cup V(y)$$

The ideal (xy) is radical but not prime (since $xy \in (xy)$ but $x \notin (xy)$ and $y \notin (xy)$). Hence $V(xy)$

is an algebraic set but *not* a variety.

In contrast, $V(y - x^2) \subseteq \mathbb{A}_k^2$ (a parabola) is irreducible: the ideal $(y - x^2)$ is prime.

We can now state the Nullstellensatz precisely. From here on, we assume k is an algebraically closed field.

Theorem 0.1.12: Hilbert's Nullstellensatz

Let k be algebraically closed. Then:

1. **(Weak form)** If $I \subsetneq k[x_1, \ldots, x_n]$ is a proper ideal, then $\mathcal{Z}(I) \neq \emptyset$.
2. **(Strong form)** For any ideal $I \subseteq k[x_1, \ldots, x_n]$, we have $\mathcal{I}(\mathcal{Z}(I)) = \sqrt{I}$.
3. **(Maximal ideal form)** The maximal ideals of $k[x_1, \ldots, x_n]$ are precisely those of the form $\mathfrak{m}_p = (x_1 - a_1, \ldots, x_n - a_n)$ for points $p = (a_1, \ldots, a_n) \in \mathbb{A}_k^n$.

***Proof*:**
See [**NateClassicalAlgGeo**].

The strong form immediately gives:

Corollary 0.1.13: The Ideal–Variety Correspondence

Let k be algebraically closed. The maps $\mathcal{Z}$ and $\mathcal{I}$ induce inclusion-reversing bijections:

$$\{\text{radical ideals in } k[x_1, \ldots, x_n]\} \xleftrightarrow{\mathcal{Z}} \{\text{algebraic sets in } \mathbb{A}_k^n\}$$

Under this correspondence:

Ideals	$\longleftrightarrow$	**Algebraic Sets**
radical ideals	$\longleftrightarrow$	algebraic sets
prime ideals	$\longleftrightarrow$	varieties (irreducible)
maximal ideals	$\longleftrightarrow$	points

The Coordinate Ring

The passage from geometry to algebra is achieved through the coordinate ring:

Definition 0.1.14: Coordinate Ring

Let $V \subseteq \mathbb{A}_k^n$ be an algebraic set. The *coordinate ring* (or *ring of regular functions*) of V is

$$k[V] := k[x_1, \ldots, x_n]/\mathcal{I}(V)$$

The elements of $k[V]$ are equivalence classes of polynomials, where two polynomials are equivalent if they agree on V. Thus $k[V]$ can be interpreted as the ring of *polynomial functions* $V \to k$.

Proposition 0.1.15: Properties of the Coordinate Ring

Let $V \subseteq \mathbb{A}_k^n$ be an algebraic set with coordinate ring $k[V]$. Then:

1. $k[V]$ is a finitely generated k-algebra.
2. $k[V]$ is reduced (i.e., has no nilpotents) if and only if $\mathcal{I}(V)$ is radical.
3. $k[V]$ is an integral domain if and only if V is irreducible.
4. Points $p \in V$ correspond to maximal ideals $\mathfrak{m}_p \subseteq k[V]$.

***Proof*:**
See [**NateClassicalAlgGeo**].

Example 0.1.16: Coordinate Rings

1. $k[\mathbb{A}^n] = k[x_1, \ldots, x_n]$, since $\mathcal{I}(\mathbb{A}^n) = (0)$.
2. For the parabola $V = V(y - x^2) \subseteq \mathbb{A}^2$, we have $k[V] = k[x, y]/(y - x^2) \cong k[x]$, reflecting that $V \cong \mathbb{A}^1$.
3. For the circle $V = V(x^2 + y^2 - 1) \subseteq \mathbb{A}_{\mathbb{C}}^2$, we have $k[V] = \mathbb{C}[x, y]/(x^2 + y^2 - 1)$. This is an integral domain since $(x^2 + y^2 - 1)$ is prime over $\mathbb{C}$.
4. For the nodal cubic $V = V(y^2 - x^3 - x^2) \subseteq \mathbb{A}^2$, the coordinate ring is $k[V] = k[x, y]/(y^2 - x^3 - x^2)$. This is an integral domain (the curve is irreducible), but V is singular at the origin.

Morphisms of Affine Varieties

To form a category of varieties, we need morphisms.

Definition 0.1.17: Regular Map / Morphism

Let $V \subseteq \mathbb{A}^n$ and $W \subseteq \mathbb{A}^m$ be algebraic sets. A *regular map* (or *morphism*) $\varphi : V \to W$ is a map of the form

$$\varphi(p) = (f_1(p), f_2(p), \ldots, f_m(p))$$

where each $f_i \in k[V]$. That is, φ is given by polynomial functions in coordinates.

Proposition 0.1.18: Morphisms and Algebra Homomorphisms

A regular map $\varphi : V \to W$ induces a k-algebra homomorphism

$$\varphi^* : k[W] \to k[V], \qquad g \mapsto g \circ \varphi$$

called the *pullback*. Conversely, every k-algebra homomorphism $k[W] \to k[V]$ arises as φ^* for a unique regular map $\varphi : V \to W$.

***Proof*:**
See [**NateClassicalAlgGeo**].

Note the contravariance: a map $V \to W$ gives a map $k[W] \to k[V]$ in the opposite direction. This leads to the fundamental equivalence:

Theorem 0.1.19: Equivalence of Categories

Let k be algebraically closed. There is a contravariant equivalence of categories:

$$\text{Aff-Var}_k \underset{\text{mSpec}}{\overset{V \mapsto k[V]}{\rightleftarrows}} \left(\text{Alg}_k^{\text{f.g., red}}\right)^{\text{op}}$$

where Aff-Var_k is the category of affine varieties over k with regular maps, and $\text{Alg}_k^{\text{f.g., red}}$ is the category of finitely generated reduced k-algebras with k-algebra homomorphisms.

***Proof*:**
The functor $V \mapsto k[V]$ sends varieties to their coordinate rings and sends morphisms $\varphi : V \to W$ to pullbacks $\varphi^* : k[W] \to k[V]$. The quasi-inverse sends a finitely generated reduced k-algebra A to its maximal spectrum $\text{mSpec}(A)$, the set of maximal ideals with the Zariski topology. The Nullstellensatz guarantees that for $A = k[V]$, the points of V are in bijection with maximal ideals of A. See [**NateClassicalAlgGeo**] for details.

0.1.20 Why Reduced Algebras? The restriction to reduced algebras (no nilpotents) corresponds to the fact that classical varieties are determined by their point-sets. Modern algebraic geometry (via schemes) removes this restriction, allowing nilpotent elements to encode infinitesimal data – this is one of the key generalizations.

Local Rings and Regularity

To study the local behavior of a variety at a point, we first introduce the algebraic machinery of localization. This process allows us to systematically make elements of a ring invertible, essentially allowing us to study rings "locally" by focusing on fractions whose denominators do not vanish at the point of interest.

Definition 0.1.21: Localization of a Ring

Let R be a commutative ring. A subset $S \subseteq R$ is *multiplicatively closed* if $1 \in S$ and for all $a, b \in S$, we have $ab \in S$. The *localization* of R at S, denoted $S^{-1}R$, is the set of equivalence classes of fractions $\frac{r}{s}$ with $r \in R$ and $s \in S$, modulo the equivalence relation:

$$\frac{r}{s} \sim \frac{r'}{s'} \iff \text{there exists } t \in S \text{ such that } t(rs' - r's) = 0$$

This becomes a ring under the standard addition and multiplication of fractions.

Localization is completely characterized by the following universal property, which states that $S^{-1}R$ is the "most efficient" way to force the elements of S to become units.

Proposition 0.1.22: Universal Property of Localization

Let S be a multiplicative set of a ring R. The natural map $\iota : R \to S^{-1}R$ given by $\iota(r) = \frac{r}{1}$ is a ring homomorphism that maps elements of S to units in $S^{-1}R$. Furthermore, if A is any ring and $g : R \to A$ is a ring homomorphism such that $g(S) \subseteq A^\times$ (the group of units of A), then there exists a unique ring homomorphism $h : S^{-1}R \to A$ such that $h \circ \iota = g$.

***Proof*:**

We define $h\left(\frac{r}{s}\right) = g(r)g(s)^{-1}$. To see that h is well-defined, suppose $\frac{r}{s} = \frac{r'}{s'}$. Then there is some $t \in S$ such that $t(rs' - r's) = 0$. Applying g gives $g(t)(g(r)g(s') - g(r')g(s)) = 0$. Since $t \in S$, $g(t)$ is a unit in A, so we can multiply by $g(t)^{-1}$ to obtain $g(r)g(s') = g(r')g(s)$. Multiplying by $g(s)^{-1}g(s')^{-1}$ yields $g(r)g(s)^{-1} = g(r')g(s')^{-1}$, confirming h is well-defined.

It is straightforward to verify that h is a ring homomorphism. For uniqueness, if $h' : S^{-1}R \to A$ satisfies $h' \circ \iota = g$, then $h'\left(\frac{r}{1}\right) = g(r)$ and $h'\left(\frac{1}{s}\right) = h'\left(\frac{s}{1}\right)^{-1} = g(s)^{-1}$. Thus, $h'\left(\frac{r}{s}\right) = g(r)g(s)^{-1} = h\left(\frac{r}{s}\right)$.

An important consequence of this universal property is that localization is a functor.

Proposition 0.1.23: Functoriality of Localization

Let R, R' be rings with multiplicative sets S, S' respectively. If $\varphi : R \to R'$ is a ring homomorphism such that $\varphi(S) \subseteq S'$, then there is a unique induced ring homomorphism $\widetilde{\varphi} : S^{-1}R \to (S')^{-1}R'$ making the following diagram commute:

$$\widetilde{\varphi} \circ \iota = \iota' \circ \varphi$$

where ι and ι' are the natural localization maps.

***Proof*:**

Consider the composition $\iota' \circ \varphi : R \to R' \to (S')^{-1}R'$. For any $s \in S$, we have $\varphi(s) \in S'$. Therefore, $\iota'(\varphi(s))$ is a unit in $(S')^{-1}R'$. By the universal property of $S^{-1}R$ applied to the homomorphism $\iota' \circ \varphi$, there exists a unique ring homomorphism $\widetilde{\varphi} : S^{-1}R \to (S')^{-1}R'$ satisfying

$\widetilde{\varphi} \circ \iota = \iota' \circ \varphi$. Explicitly, this map is given by $\widetilde{\varphi}\left(\frac{r}{s}\right) = \frac{\varphi(r)}{\varphi(s)}$.

We now apply this general algebraic construction to the coordinate ring of a variety to study its local geometry.

Definition 0.1.24: Local Ring at a Point

Let V be an affine variety and $p \in V$ a point with corresponding maximal ideal $\mathfrak{m}_p \subseteq k[V]$. The set $S = k[V] \setminus \mathfrak{m}_p$ is multiplicatively closed. The *local ring of V at p* is the localization of $k[V]$ at S:

$$\mathcal{O}_{V,p} := k[V]_{\mathfrak{m}_p} = \left\{ \frac{f}{g} \;\middle|\; f, g \in k[V],\ g(p) \neq 0 \right\}$$

This is a local ring with unique maximal ideal $\mathfrak{m}_{V,p} = \mathfrak{m}_p \cdot \mathcal{O}_{V,p}$.

The local ring $\mathcal{O}_{V,p}$ captures the "germs" of regular functions near p: two functions are identified if they agree on some neighborhood of p.

Definition 0.1.25: Zariski Tangent Space

The *Zariski tangent space* to V at p is the k-vector space

$$T_pV := \left(\mathfrak{m}_{V,p}/\mathfrak{m}_{V,p}^2\right)^\vee = \operatorname{Hom}_k\left(\mathfrak{m}_{V,p}/\mathfrak{m}_{V,p}^2, k\right)$$

The *Zariski cotangent space* is $T_p^*V := \mathfrak{m}_{V,p}/\mathfrak{m}_{V,p}^2$.

0.1.26 Tangent Space via Derivations Equivalently, T_pV can be defined as the space of k-derivations $\mathcal{O}_{V,p} \to k$ at p, or as the space of "ε-points": morphisms $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2) \to V$ centered at p. This latter perspective generalizes naturally to schemes.

Definition 0.1.27: Smooth Point, Singular Point

Let V be a variety of dimension d (see below). A point $p \in V$ is:

- *smooth* (or *regular*) if $\dim_k T_pV = d$;
- *singular* if $\dim_k T_pV > d$.

A variety is *smooth* (or *nonsingular*) if every point is smooth.

Example 0.1.28: Singularities

1. The parabola $V(y - x^2)$ is smooth everywhere. At any point, $\dim T_pV = 1 = \dim V$.
2. The cuspidal cubic $V(y^2 - x^3)$ has a singularity (cusp) at the origin. One computes $\dim T_0V = 2 > 1 = \dim V$.
3. The nodal cubic $V(y^2 - x^2(x+1))$ has a node at the origin where two branches cross.

Proposition 0.1.29: Regularity and Local Rings

A point $p \in V$ is smooth if and only if $\mathcal{O}_{V,p}$ is a *regular local ring*, i.e., $\dim_k(\mathfrak{m}_{V,p}/\mathfrak{m}_{V,p}^2) = \dim \mathcal{O}_{V,p}$ (Krull dimension).

***Proof*:**
See [**NateClassicalAlgGeo**].

Rational Functions and Dimension

For an irreducible variety, we can form the field of fractions of its coordinate ring:

Definition 0.1.30: Function Field

Let V be an affine variety (irreducible). The *function field* of V is

$$k(V) := \mathrm{Frac}(k[V])$$

Elements of $k(V)$ are called *rational functions* on V. A rational function f/g is defined on the open set $\{p \in V : g(p) \neq 0\}$.

Definition 0.1.31: Dimension

The *dimension* of an affine variety V is

$$\dim V := \mathrm{tr.deg}_k\ k(V)$$

the transcendence degree of the function field over k. Equivalently, $\dim V = \dim k[V]$, the Krull dimension of the coordinate ring.

Example 0.1.32: Dimensions

1. $\dim \mathbb{A}^n = n$, since $k(\mathbb{A}^n) = k(x_1, \dots, x_n)$ has transcendence degree n.
2. Any plane curve $V(f) \subseteq \mathbb{A}^2$ with f irreducible has dimension 1.
3. A hypersurface $V(f) \subseteq \mathbb{A}^n$ with f irreducible has dimension $n-1$.

Proposition 0.1.33: Dimension and Chains of Primes

For an affine variety V, we have $\dim V = \dim k[V]$, where the Krull dimension of $k[V]$ is the supremum of lengths of chains of prime ideals

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_d$$

in $k[V]$.

***Proof*:**
See [**NateClassicalAlgGeo**].

0.2 Recall Projective Spaces

Projective space is the natural setting for algebraic geometry. Many phenomena that are awkward or incomplete in affine space become clean and uniform in projective space: intersection theory works properly (Bézout's theorem), curves have well-defined degrees, and there are no "missing points at infinity." This section reviews the essential constructions.

Definition and Homogeneous Coordinates

Definition 0.2.1: Projective Space

Let k be a field. *Projective n-space* over k is

$$\mathbb{P}^n_k := \left(k^{n+1} \setminus \{0\}\right) / \sim$$

where $(a_0, a_1, \ldots, a_n) \sim (\lambda a_0, \lambda a_1, \ldots, \lambda a_n)$ for all $\lambda \in k^\times$. The equivalence class of $(a_0, \ldots, a_n)$ is denoted $[a_0 : a_1 : \cdots : a_n]$ and called a *point* in $\mathbb{P}^n$. The a_i are *homogeneous coordinates.*

0.2.2 Projective Space as Lines Equivalently, $\mathbb{P}^n_k$ is the set of 1-dimensional linear subspaces (lines through the origin) in k^{n+1}. This is sometimes denoted $\mathbb{P}(k^{n+1})$ or $\mathrm{Proj}(k^{n+1})$.

Example 0.2.3: Low-Dimensional Projective Spaces

1. $\mathbb{P}^0$ is a single point.
2. $\mathbb{P}^1$ is the *projective line*. Points are $[a_0 : a_1]$ with $(a_0, a_1) \neq (0, 0)$. If $a_1 \neq 0$, we can normalize to $[t : 1]$ for $t = a_0/a_1 \in k$. The remaining point $[1 : 0]$ is the "point at infinity." Thus $\mathbb{P}^1 \cong \mathbb{A}^1 \sqcup \{\infty\}$.
3. $\mathbb{P}^2$ is the *projective plane*, the natural home for plane curves.

Affine Charts

Projective space is covered by affine open sets:

Definition 0.2.4: Standard Affine Charts

For each $i \in \{0, 1, \ldots, n\}$, define the *standard affine open*

$$U_i := \{[a_0 : \cdots : a_n] \in \mathbb{P}^n \mid a_i \neq 0\}$$

There is an isomorphism $\varphi_i : U_i \xrightarrow{\sim} \mathbb{A}^n$ given by

$$\varphi_i([a_0 : \cdots : a_n]) = \left(\frac{a_0}{a_i}, \ldots, \widehat{\frac{a_i}{a_i}}, \ldots, \frac{a_n}{a_i}\right)$$

where the hat denotes omission.

Proposition 0.2.5: Affine Cover of Projective Space

The standard affine opens cover $\mathbb{P}^n$:

$$\mathbb{P}^n = U_0 \cup U_1 \cup \cdots \cup U_n$$

The complement of U_i is the hyperplane $H_i = \{[a_0 : \cdots : a_n] : a_i = 0\} \cong \mathbb{P}^{n-1}$.

Example 0.2.6: The Projective Line Revisited

For $\mathbb{P}^1$, we have two charts:

- $U_0 = \{[a_0 : a_1] : a_0 \neq 0\} \cong \mathbb{A}^1$ via $[a_0 : a_1] \mapsto a_1/a_0$
- $U_1 = \{[a_0 : a_1] : a_1 \neq 0\} \cong \mathbb{A}^1$ via $[a_0 : a_1] \mapsto a_0/a_1$

On the overlap $U_0 \cap U_1$, if $t = a_1/a_0$ is the coordinate on U_0 and $s = a_0/a_1$ on U_1, then $s = 1/t$. This is the standard gluing of two copies of $\mathbb{A}^1$ to form $\mathbb{P}^1$.

Homogeneous Polynomials and Graded Rings

Polynomials do not define functions on $\mathbb{P}^n$ directly, since $f(\lambda a_0, \ldots, \lambda a_n) \neq f(a_0, \ldots, a_n)$ in general. However, the *vanishing* of homogeneous polynomials is well-defined.

Definition 0.2.7: Homogeneous Polynomial

A polynomial $f \in k[x_0, x_1, \ldots, x_n]$ is *homogeneous of degree* d if every monomial in f has total degree d. Equivalently, $f(\lambda x_0, \ldots, \lambda x_n) = \lambda^d f(x_0, \ldots, x_n)$ for all $\lambda \in k$.

Definition 0.2.8: Graded Ring

The polynomial ring $S = k[x_0, \ldots, x_n]$ has a *grading* $S = \bigoplus_{d \geq 0} S_d$, where S_d is the k-vector space of homogeneous polynomials of degree d (including 0). We have $S_d \cdot S_e \subseteq S_{d+e}$.

Definition 0.2.9: Homogeneous Ideal

An ideal $I \subseteq k[x_0, \ldots, x_n]$ is *homogeneous* if it is generated by homogeneous polynomials. Equivalently, $I = \bigoplus_{d \geq 0}(I \cap S_d)$.

Proposition 0.2.10: Characterization of Homogeneous Ideals

An ideal I is homogeneous if and only if for every $f \in I$, all homogeneous components of f are also in I.

***Proof*:**
See [**NateClassicalAlgGeo**].

Projective Varieties

Definition 0.2.11: Projective Algebraic Set

For a set T of homogeneous polynomials in $k[x_0, \ldots, x_n]$, the *projective zero set* is

$$\mathcal{Z}_+(T) := \{[a_0 : \cdots : a_n] \in \mathbb{P}^n \mid f(a_0, \ldots, a_n) = 0, \ \forall f \in T\}$$

This is well-defined since f homogeneous implies $f(\lambda \mathbf{a}) = \lambda^d f(\mathbf{a})$, so $f(\mathbf{a}) = 0 \iff f(\lambda \mathbf{a}) = 0$. A subset $V \subseteq \mathbb{P}^n$ is a *projective algebraic set* if $V = \mathcal{Z}_+(T)$ for some set T of homogeneous polynomials.

Definition 0.2.12: Homogeneous Ideal of a Set

For $V \subseteq \mathbb{P}^n$, define

$$\mathcal{I}_+(V) := \text{ideal generated by } \{f \in k[x_0, \ldots, x_n] \text{ homogeneous} \mid f(p) = 0, \ \forall p \in V\}$$

This is always a homogeneous radical ideal.

The projective algebraic sets form the closed sets of the *Zariski topology on* $\mathbb{P}^n$.

Definition 0.2.13: Projective Variety

A projective algebraic set $V \subseteq \mathbb{P}^n$ is a *projective variety* if it is irreducible in the Zariski topology.

0.2.14 The Irrelevant Ideal The ideal $\mathfrak{m}_+ = (x_0, x_1, \ldots, x_n) \subseteq k[x_0, \ldots, x_n]$ is called the *irrelevant ideal*. It satisfies $\mathcal{Z}_+(\mathfrak{m}_+) = \emptyset$, since the only common zero of all x_i in k^{n+1} is the origin, which does not correspond to a point in $\mathbb{P}^n$. This is an important subtlety in the projective Nullstellensatz.

Theorem 0.2.15: Projective Nullstellensatz

Let k be algebraically closed and let $I \subseteq k[x_0, \ldots, x_n]$ be a homogeneous ideal. Then:

1. $\mathcal{Z}_+(I) = \emptyset$ if and only if $\sqrt{I} \supseteq \mathfrak{m}_+^N$ for some $N \geq 1$ (equivalently, $\sqrt{I} = \mathfrak{m}_+$ or $\sqrt{I} = (1)$).
2. If $\mathcal{Z}_+(I) \neq \emptyset$, then $\mathcal{I}_+(\mathcal{Z}_+(I)) = \sqrt{I}$.

Hence $\mathcal{Z}_+$ and $\mathcal{I}_+$ give a bijection between:

$$\left\{\text{homogeneous radical ideals } I \text{ with } \sqrt{I} \neq \mathfrak{m}_+\right\} \longleftrightarrow \{\text{projective algebraic sets in } \mathbb{P}^n\}$$

***Proof*:**
See [**NateClassicalAlgGeo**].

Homogeneous Coordinate Ring and the Proj Construction

Definition 0.2.16: Homogeneous Coordinate Ring

For a projective variety $V = \mathcal{Z}_+(I) \subseteq \mathbb{P}^n$, the *homogeneous coordinate ring* is the graded ring

$$S(V) := k[x_0, \ldots, x_n]/I$$

with the induced grading $S(V) = \bigoplus_{d \geq 0} S(V)_d$.

0.2.17 Homogeneous Coordinate Ring is Not Intrinsic Unlike the affine case, the homogeneous coordinate ring $S(V)$ depends on the embedding $V \hookrightarrow \mathbb{P}^n$, not just on V as an abstract variety. Different embeddings of the same variety can give non-isomorphic graded rings.

Definition 0.2.18: The Proj Construction

For a graded ring $S = \bigoplus_{d \geq 0} S_d$, we define

$$\mathrm{Proj}(S) := \{\mathfrak{p} \in \mathrm{Spec}(S) \mid \mathfrak{p} \text{ is homogeneous and } \mathfrak{p} \not\supseteq S_+\}$$

where $S_+ = \bigoplus_{d > 0} S_d$ is the irrelevant ideal. This set is equipped with a natural topology and structure sheaf making it a scheme. For $S = k[x_0, \ldots, x_n]$, we recover $\mathrm{Proj}(S) = \mathbb{P}^n_k$.

Relation Between Affine and Projective

The affine charts give a way to study projective varieties locally:

Definition 0.2.19: Affine Cone

For a projective variety $V \subseteq \mathbb{P}^n$, the *affine cone* over V is

$$C(V) := \mathcal{Z}(\mathcal{I}_+(V)) \subseteq \mathbb{A}^{n+1}$$

This is an affine variety in $\mathbb{A}^{n+1}$, consisting of all lines through the origin that correspond to points of V, together with the origin itself.

Proposition 0.2.20: Affine Pieces of Projective Varieties

Let $V \subseteq \mathbb{P}^n$ be a projective variety and $U_i \cong \mathbb{A}^n$ a standard affine chart. Then $V \cap U_i$ is an affine variety in $\mathbb{A}^n$.
Explicitly, if $V = \mathcal{Z}_+(f_1, \dots, f_r)$ with each f_j homogeneous of degree d_j, then $V \cap U_i = \mathcal{Z}(f_1^{(i)}, \dots, f_r^{(i)})$ where

$$f_j^{(i)}(y_0, \dots, \widehat{y_i}, \dots, y_n) := f_j(y_0, \dots, y_{i-1}, 1, y_{i+1}, \dots, y_n)$$

is the *dehomogenization* of f_j with respect to x_i.

Definition 0.2.21: Homogenization

Conversely, given $g \in k[y_1, \dots, y_n]$ of degree d, its *homogenization* with respect to x_0 is

$$g^h(x_0, x_1, \dots, x_n) := x_0^d \cdot g\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right)$$

This is homogeneous of degree d.

Example 0.2.22: Projective Closure

The parabola $V(y - x^2) \subseteq \mathbb{A}^2$ has homogenization $yz - x^2 \in k[x, y, z]$. Its projective closure is $\overline{V} = \mathcal{Z}_+(yz - x^2) \subseteq \mathbb{P}^2$. The "new" point $[0 : 1 : 0]$ (where $z = 0$ and $x = 0$) is the point at infinity where the parabola "closes up."

Quasi-Projective Varieties

Definition 0.2.23: Quasi-Projective Variety

A *quasi-projective variety* is a locally closed subset of $\mathbb{P}^n$, i.e., an open subset of a projective variety. Equivalently, it is the intersection of an open set and a closed set in $\mathbb{P}^n$.

Proposition 0.2.24: Affine and Projective are Quasi-Projective

Both affine varieties and projective varieties are quasi-projective:

1. A projective variety $V \subseteq \mathbb{P}^n$ is quasi-projective (it is closed, hence locally closed).
2. An affine variety $V \subseteq \mathbb{A}^n \cong U_0 \subseteq \mathbb{P}^n$ is quasi-projective (it is closed in the open set U_0).

The category of quasi-projective varieties with regular maps is the classical category of "algebraic varieties" in much of the literature.

Morphisms of Projective Varieties

Defining morphisms of projective varieties requires care, since homogeneous polynomials do not define functions.

Definition 0.2.25: Morphism of Projective Varieties

Let $V \subseteq \mathbb{P}^n$ and $W \subseteq \mathbb{P}^m$ be projective varieties. A *morphism* $\varphi : V \to W$ is a map that is locally given by homogeneous polynomials of the same degree. That is, for each $p \in V$, there exist homogeneous polynomials $f_0, \dots, f_m \in k[x_0, \dots, x_n]$ of the same degree d such that:

1. $f_0, \dots, f_m$ do not simultaneously vanish on a neighborhood of p in V.
2. $\varphi(q) = [f_0(q) : \cdots : f_m(q)]$ for all q in this neighborhood.

0.2.26 Local Nature of Morphisms Different homogeneous polynomials may be needed on different open subsets of V. On overlaps, they must agree as maps to $\mathbb{P}^m$. This local definition is necessary because no single tuple of homogeneous polynomials may work globally.

Example 0.2.27: The Veronese Embedding

The *Veronese embedding* of degree d is

$$\nu_d : \mathbb{P}^n \to \mathbb{P}^N, \qquad [x_0 : \cdots : x_n] \mapsto [\cdots : x_0^{i_0} \cdots x_n^{i_n} : \cdots]$$

where the coordinates on $\mathbb{P}^N$ are indexed by monomials of degree d, so $N = \binom{n+d}{d} - 1$.

For example, $\nu_2 : \mathbb{P}^1 \to \mathbb{P}^2$ is given by $[s : t] \mapsto [s^2 : st : t^2]$. The image is the conic $\mathcal{Z}_+(xz - y^2)$, showing that $\mathbb{P}^1$ is isomorphic to this conic.

Example 0.2.28: The Segre Embedding

The *Segre embedding* is

$$\sigma : \mathbb{P}^n \times \mathbb{P}^m \to \mathbb{P}^{(n+1)(m+1)-1}, \qquad ([x_i], [y_j]) \mapsto [x_i y_j]_{0 \le i \le n,\, 0 \le j \le m}$$

For example, $\sigma : \mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^3$ is given by

$$([s : t], [u : v]) \mapsto [su : sv : tu : tv]$$

The image is the quadric surface $\mathcal{Z}_+(xw - yz) \subseteq \mathbb{P}^3$.

The Segre embedding shows that $\mathbb{P}^n \times \mathbb{P}^m$ can be given the structure of a projective variety.

Products of Projective Varieties

Proposition 0.2.29: Products of Projective Varieties

Let $V \subseteq \mathbb{P}^n$ and $W \subseteq \mathbb{P}^m$ be projective varieties. Then $V \times W$, embedded via the Segre embedding $\mathbb{P}^n \times \mathbb{P}^m \hookrightarrow \mathbb{P}^{(n+1)(m+1)-1}$, is a projective variety.

***Proof*:**
See [**NateClassicalAlgGeo**].

0.2.30 Contrast with Affine Products For affine varieties, products are straightforward: $V \times W \subseteq \mathbb{A}^n \times \mathbb{A}^m = \mathbb{A}^{n+m}$, and $k[V \times W] \cong k[V] \otimes_k k[W]$. The projective case requires the Segre embedding because $\mathbb{P}^n \times \mathbb{P}^m \ncong \mathbb{P}^{n+m}$.

Completeness and Properness

Projective varieties enjoy a crucial property not shared by affine varieties:

Definition 0.2.31: Complete Variety

A variety V is *complete* if for every variety W, the projection $\pi : V \times W \to W$ is a closed map (sends closed sets to closed sets).

Theorem 0.2.32: Projective Varieties are Complete

Every projective variety is complete.

***Proof*:**
See [**NateClassicalAlgGeo**].

0.2.33 Completeness as Compactness Completeness is the algebraic analogue of compactness. Over $\mathbb{C}$, a projective variety is compact in the classical (Euclidean) topology. Affine varieties of positive dimension are never complete: for example, the projection $\mathbb{A}^1 \times \mathbb{A}^1 \to \mathbb{A}^1$ does not send the closed hyperbola $\mathcal{Z}(xy - 1)$ to a closed set.

Proposition 0.2.34: Affine and Complete Implies Finite

If V is both affine and complete, then V is a finite set of points.

***Proof*:**
See [**NateClassicalAlgGeo**].

0.3 Noetherian Induction or Well-founded Induction

This is something that will come up in later proofs (for example, see lemma 5.1.14). This is a form of induction that generalizes the usual induction where the well-ordered set is $\mathbb{N}$. Recall that a binary relation R on S is a subset of $S \times S$. For $(a, b) \in R$, we may write $a \preceq b$ or $b \succeq a$. Then the symbol $\preceq$ is sometimes called the binary relation. A binary relation is a strict partial order if it is:

1. Irreflexive: $\forall x \in A, \neg(x < x)$
2. Transitive: $\forall x, y, z \in A, (x < y \wedge y < z) \implies x < z$
3. Asymmetric: $\forall x, y \in A, x < y \implies \neg(y < x)$

Given a set S with a strict partial order $\preceq$, a descending chain is a sequence $a_1, a_2, a_3, \ldots$ of elements in A such that $a_1 \succeq a_2 \succeq a_3 \succeq \ldots$

Definition 0.3.1: Well-founded Relation

A strict partial order $\preceq$ on a set A is well-founded if there exist no infinite descending chains in A with respect to $\preceq$.

Definition 0.3.2: Minimal Element

Given a subset S of a well-founded set A, an element $m \in S$ is minimal if there is no element $x \in S$ such that $x \preceq m$.

Theorem 0.3.3: Noetherian Induction

Let $(A, \preceq)$ be a set with a well-founded relation $\preceq$, and let P be a property defined on A. If for all $x \in A : [\forall y \in A, (y \preceq x \implies P(y))] \implies P(x)$ then $P(x)$ holds for all $x \in A$.

***Proof*:**
see cite:HERE.

The following may be an insightful alternative:

Theorem 0.3.4: Noetherian Induction - Alternative Form

Let $(A, \preceq)$ be a set with a well-founded relation $\preceq$. If $S \subseteq A$ is a subset such that for all minimal elements m of $A \backslash S$, we have $m \in S$, then $S = A$.

***Proof*:**
see cite:HERE.

The connection between these forms lies in taking S to be the set of elements where P holds, and showing that if S isn't all of A, then any minimal element of $A \backslash S$ must actually be in S, giving a contradiction.

0.4 Categorical Reminder

Category theory provides the language and organizational framework for modern algebraic geometry. Schemes are defined as certain locally ringed spaces, morphisms are defined via universal properties, and many constructions (fiber products, sheafification, etc.) are instances of general categorical machinery. This section collects the essential concepts.

Categories, Functors, and Natural Transformations

Definition 0.4.1: Category

A *category* $\mathcal{C}$ consists of:

1. A class $\mathrm{Ob}(\mathcal{C})$ of *objects*.
2. For each pair of objects X, Y, a set of *morphisms* from X to Y denoted $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ or $\mathrm{Mor}_{\mathbb{C}}(X, Y)$.
3. For each triple X, Y, Z, a *composition* map $\mathrm{Hom}(Y, Z) \times \mathrm{Hom}(X, Y) \to \mathrm{Hom}(X, Z)$.
4. For each object X, an *identity* morphism $\mathrm{id}_X \in \mathrm{Hom}(X, X)$.

These satisfy associativity of composition and the identity laws.

Example 0.4.2: Categories in Algebra and Geometry

1. **Set**: sets and functions.
2. **Grp**, **Ab**, **Ring**, **CRing**: groups, abelian groups, rings, commutative rings.
3. $\mathbf{Mod}_R$: R-modules and R-linear maps.
4. **Top**: topological spaces and continuous maps.
5. $\mathbf{Aff}_k$: affine varieties over k and regular maps.

Definition 0.4.3: Functor

A *(covariant) functor* $F : \mathcal{C} \to \mathcal{D}$ consists of:

1. A map $F : \mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{D})$.
2. For each $X, Y \in \mathcal{C}$, a map $F : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$.

These must satisfy $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$ and $F(g \circ f) = F(g) \circ F(f)$.
A *contravariant functor* $F : \mathcal{C} \to \mathcal{D}$ reverses arrows: $F : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(Y), F(X))$ with $F(g \circ f) = F(f) \circ F(g)$. Equivalently, it is a covariant functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$.

Example 0.4.4: Functors in Algebraic Geometry

1. The coordinate ring functor $V \mapsto k[V]$ is a contravariant functor $\mathbf{Aff}_k \to \mathbf{CRing}$.
2. Forgetful functors: $\mathbf{Grp} \to \mathbf{Set}$, etc.
3. We shall define the spectrum functor $A \mapsto \mathrm{Spec}\,(A)$ is a contravariant functor $\mathbf{CRing} \to \mathbf{Sch}$.
4. We shall define the global sections functor $X \mapsto \Gamma(X, \mathcal{O}_X)$ is a contravariant functor $\mathbf{Sch} \to \mathbf{CRing}$.

Definition 0.4.5: Natural Transformation

Let $F, G : \mathcal{C} \to \mathcal{D}$ be functors. A *natural transformation* $\eta : F \Rightarrow G$ is a family of morphisms $\eta_X : F(X) \to G(X)$ for each $X \in \mathcal{C}$, such that for every morphism $f : X \to Y$ in $\mathcal{C}$, the following diagram commutes:

$$\begin{tikzcd} F(X) \arrow[r, "\eta_X"] \arrow[d, "F(f)"'] & G(X) \arrow[d, "G(f)"] \\ F(Y) \arrow[r, "\eta_Y"'] & G(Y) \end{tikzcd}$$

A *natural isomorphism* is a natural transformation where each η_X is an isomorphism.

Definition 0.4.6: Equivalence of Categories

An *equivalence of categories* between $\mathcal{C}$ and $\mathcal{D}$ consists of functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ together with natural isomorphisms $G \circ F \cong \mathrm{id}_{\mathcal{C}}$ and $F \circ G \cong \mathrm{id}_{\mathcal{D}}$.
A functor F is an equivalence if and only if it is:

- *Fully faithful*: $F : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$ is a bijection for all X, Y.
- *Essentially surjective*: every object of $\mathcal{D}$ is isomorphic to $F(X)$ for some $X \in \mathcal{C}$.

Example 0.4.7: The Classical Equivalence

The equivalence $\mathbf{Aff}_k \simeq (\mathbf{Alg}_k^{\text{f.g., red}})^{\mathrm{op}}$ from section 0.1 is an equivalence of categories: the coordinate ring functor is fully faithful and essentially surjective.

Universal Properties: Limits and Colimits

Many constructions in mathematics are characterized by universal properties. Limits and colimits provide a unified framework.

Definition 0.4.8: Diagram and Cone

Let J be a small category (the "index category"). A *diagram* of shape J in $\mathcal{C}$ is a functor $D : J \to \mathcal{C}$.
A *cone* over D with vertex $X \in \mathcal{C}$ is a collection of morphisms $\pi_j : X \to D(j)$ for each $j \in J$, such that for every morphism $\alpha : j \to j'$ in J, we have $D(\alpha) \circ \pi_j = \pi_{j'}$.

Definition 0.4.9: Limit

The *limit* of a diagram $D : J \to \mathcal{C}$, denoted $\varprojlim D$ or $\lim_J D$, is a cone $(\pi_j : L \to D(j))_{j\in J}$ that is universal: for any other cone $(f_j : X \to D(j))$, there exists a unique morphism $f : X \to L$ such that $\pi_j \circ f = f_j$ for all j.

$$\begin{tikzcd} X \arrow[d, dashed, "\exists! f"'] \arrow[dr, "f_j"] & \\ L \arrow[r, "\pi_j"'] & D(j) \end{tikzcd}$$

Definition 0.4.10: Colimit

Dually, a *cocone* under D with vertex X is a collection $\iota_j : D(j) \to X$ compatible with the diagram. The *colimit* $\varinjlim D$ is the universal cocone.

Example 0.4.11: Specific Limits and Colimits

Shape J	**Limit**	**Colimit**
Discrete (no arrows)	Product $\prod_i X_i$	Coproduct $\coprod_i X_i$
$\bullet \rightrightarrows \bullet$	Equalizer	Coequalizer
$\bullet \to \bullet \leftarrow \bullet$	Pullback (fiber product)	—
$\bullet \leftarrow \bullet \to \bullet$	—	Pushout
Empty	Terminal object $*$	Initial object $\emptyset$
Directed poset	Directed (inverse) limit	Directed (direct) limit

Products and Coproducts

Definition 0.4.12: Product

The *product* of objects $\{X_i\}_{i\in I}$ is an object $\prod_i X_i$ together with projection morphisms $\pi_j : \prod_i X_i \to X_j$ such that for any object Y with morphisms $f_j : Y \to X_j$, there exists a unique $f : Y \to \prod_i X_i$ with $\pi_j \circ f = f_j$.

Definition 0.4.13: Coproduct

Dually, the *coproduct* $\coprod_i X_i$ has inclusion morphisms $\iota_j : X_j \to \coprod_i X_i$ universal for morphisms out of the X_j.

Example 0.4.14: Products and Coproducts in Various Categories

Category	Product	Coproduct
Set	Cartesian product	Disjoint union
Grp	Direct product	Free product
Ab, $\mathbf{Mod}_R$	Direct product	Direct sum
CRing	Direct product	Tensor product $A \otimes_{\mathbb{Z}} B$
Top	Product topology	Disjoint union topology
Sch	Fiber product over $\operatorname{Spec} \mathbb{Z}$	Disjoint union

Fiber products are ubiquitous in algebraic geometry: base change, intersections, and fiber constructions all use them.

Definition 0.4.15: Fiber Product / Pullback

Given morphisms $f : X \to S$ and $g : Y \to S$, the *fiber product* (or *pullback*) is an object $X \times_S Y$ with morphisms $p : X \times_S Y \to X$ and $q : X \times_S Y \to Y$ such that $f \circ p = g \circ q$, universal with this property:

$$\begin{tikzcd} X \times_S Y \arrow[r, "q"] \arrow[d, "p"'] \arrow[dr, phantom, "\square"] & Y \arrow[d, "g"] \\ X \arrow[r, "f"'] & S \end{tikzcd}$$

For any Z with morphisms to X and Y making the square commute, there is a unique morphism $Z \to X \times_S Y$.

Example 0.4.16: Fiber Products in Sets

In **Set**, the fiber product is

$$X \times_S Y = \{(x, y) \in X \times Y : f(x) = g(y)\}$$

If $S = *$ is a point, this is the ordinary Cartesian product.

Adjoint Functors

Adjunctions capture many important relationships between categories.

Definition 0.4.17: Adjoint Pair

Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ be functors. We say F is *left adjoint* to G (and G is *right adjoint* to F), written $F \dashv G$, if there is a natural bijection

$$\mathrm{Hom}_{\mathcal{D}}(F(X), Y) \cong \mathrm{Hom}_{\mathcal{C}}(X, G(Y))$$

for all $X \in \mathcal{C}$ and $Y \in \mathcal{D}$.

Proposition 0.4.18: Characterizations of Adjunctions

The following are equivalent to $F \dashv G$:

1. Natural bijection $\mathrm{Hom}_{\mathcal{D}}(F(X), Y) \cong \mathrm{Hom}_{\mathcal{C}}(X, G(Y))$.
2. Existence of a *unit* $\eta : \mathrm{id}_{\mathcal{C}} \Rightarrow G \circ F$ and *counit* $\varepsilon : F \circ G \Rightarrow \mathrm{id}_{\mathcal{D}}$ satisfying the triangle identities.
3. F preserves colimits and satisfies a solution set condition (Freyd's adjoint functor theorem).

***Proof*:**
See any standard reference on category theory, e.g., Mac Lane [**NateClassicalAlgGeo**].

Example 0.4.19: Adjunctions in Algebra and Geometry

1. **Free-Forgetful**: The free group functor $F : \mathbf{Set} \to \mathbf{Grp}$ is left adjoint to the forgetful functor $U : \mathbf{Grp} \to \mathbf{Set}$:

$$\mathrm{Hom}_{\mathbf{Grp}}(F(S), G) \cong \mathrm{Hom}_{\mathbf{Set}}(S, U(G))$$

2. **Tensor-Hom**: For R-modules, $(-) \otimes_R M \dashv \mathrm{Hom}_R(M, -)$:

$$\mathrm{Hom}_R(N \otimes_R M, P) \cong \mathrm{Hom}_R(N, \mathrm{Hom}_R(M, P))$$

3. **Inverse and Direct Image**: For a continuous map $f : X \to Y$, the inverse image functor f^{-1} on sheaves is left adjoint to the direct image f_*.

Two adjunctions we shall encounter are:

1. **Spec-Global Sections**: The functor $\mathrm{Spec} : \mathbf{CRing}^{\mathrm{op}} \to \mathbf{Sch}$ is left adjoint to $\Gamma : \mathbf{Sch} \to \mathbf{CRing}^{\mathrm{op}}$:

$$\mathrm{Hom}_{\mathbf{Sch}}(\mathrm{Spec}\,(A), X) \cong \mathrm{Hom}_{\mathbf{CRing}}(\Gamma(X, \mathcal{O}_X), A)$$

2. **Sheafification**: The sheafification functor is left adjoint to the inclusion of sheaves into presheaves.

Proposition 0.4.20: Adjoints and Limits

1. Left adjoints preserve colimits.
2. Right adjoints preserve limits.

In particular, if $F \dashv G$, then F preserves coproducts, coequalizers, and pushouts, while G preserves products, equalizers, and pullbacks.

***Proof*:**
See [**NateClassicalAlgGeo**].

Representable Functors and the Yoneda Lemma

The Yoneda perspective is fundamental to modern algebraic geometry: schemes can be understood through their "functor of points."

Definition 0.4.21: Representable Functor

A functor $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ is *representable* if there exists an object $X \in \mathcal{C}$ and a natural isomorphism

$$F \cong \mathrm{Hom}_{\mathcal{C}}(-, X) =: h_X$$

The object X is called the *representing object.* We say "X represents F" or "F is represented by X."

Theorem 0.4.22: Yoneda Lemma

Let $\mathcal{C}$ be a category, $X \in \mathcal{C}$, and $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ a functor. There is a natural bijection

$$\mathrm{Nat}(h_X, F) \cong F(X)$$

where $\mathrm{Nat}(h_X, F)$ is the set of natural transformations from h_X to F. The bijection sends $\eta : h_X \Rightarrow F$ to $\eta_X(\mathrm{id}_X) \in F(X)$.

***Proof*:**
See [**NateClassicalAlgGeo**].

Corollary 0.4.23: Yoneda Embedding

The functor $h : \mathcal{C} \to \mathbf{Fun}(\mathcal{C}^{\mathrm{op}}, \mathbf{Set})$ given by $X \mapsto h_X = \mathrm{Hom}(-, X)$ is fully faithful. In particular:

1. If $h_X \cong h_Y$, then $X \cong Y$ (representing objects are unique up to unique isomorphism).
2. $\mathrm{Hom}_{\mathcal{C}}(X, Y) \cong \mathrm{Nat}(h_X, h_Y)$.

Example 0.4.24: Moduli Problems as Functors
Many moduli problems are phrased as functors $F : \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$:

- $F(S) = \{\text{families of elliptic curves over } S\}$
- $F(S) = \{\text{closed subschemes of } \mathbb{P}^n_S \text{ flat over } S\}$ (Hilbert functor)

If F is representable by a scheme M, then M is the *moduli space*. Often F is not representable by a scheme but by an algebraic stack.

Filtered Colimits

Filtered colimits arise naturally when taking stalks of sheaves and germs of functions.

Definition 0.4.25: Filtered Category

A category J is *filtered* if:

1. J is nonempty.
2. For any $i, j \in J$, there exists $k \in J$ with morphisms $i \to k$ and $j \to k$.
3. For any parallel morphisms $f, g : i \to j$, there exists $h : j \to k$ with $h \circ f = h \circ g$.

A *directed set* (poset where every pair has an upper bound) is filtered when viewed as a category.

Definition 0.4.26: Filtered Colimit

A *filtered colimit* (or *direct limit*) is a colimit over a filtered index category.

Example 0.4.27: Localization as Filtered Colimit
For a ring A and multiplicative set S, the localization is

$$S^{-1}A = \varinjlim_{s \in S} A_s$$

where $A_s = A[1/s]$ and the colimit is over S ordered by divisibility.

Proposition 0.4.28: Filtered Colimits Commute with Finite Limits

In **Set** (and more generally in any *finitely presentable* category), filtered colimits commute with finite limits. In particular, filtered colimits are exact in module categories.

***Proof*:**
See [**NateClassicalAlgGeo**].

Abelian Categories

For homological algebra and cohomology, we need abelian categories.

Definition 0.4.29: Abelian Category

A category $\mathcal{A}$ is *abelian* if:

1. $\mathcal{A}$ has a zero object (both initial and terminal).
2. $\mathcal{A}$ has all binary products and coproducts (which coincide: biproducts).
3. Every morphism has a kernel and cokernel.
4. Every monomorphism is a kernel; every epimorphism is a cokernel.

Example 0.4.30: Abelian Categories

1. $\mathbf{Ab}$, $\mathbf{Mod}_R$ for any ring R.
2. $\mathbf{Sh}(X, \mathbf{Ab})$: sheaves of abelian groups on X.
3. $\mathbf{QCoh}(X)$: quasi-coherent sheaves on a scheme X.
4. $\mathbf{Coh}(X)$: coherent sheaves on a Noetherian scheme X.

0.4.31 Homological Algebra in Abelian Categories In an abelian category, one can define:

- Exact sequences: $A \to B \to C$ is exact at B if $\ker(B \to C) = \operatorname{im}(A \to B)$.
- Chain complexes and homology.
- Derived functors (if there are enough injectives or projectives).

Sheaf cohomology $H^i(X, \mathcal{F})$ is defined as the right derived functors of the global sections functor $\Gamma(X, -)$.

0.4.1 Summary: Categorical Tools for Schemes

Concept	Use in Algebraic Geometry
Equivalence of categories	Classical: $\mathbf{Aff}_k \simeq (\mathbf{Alg}_k^{\text{f.g.,red}})^{\text{op}}$
Fiber products	Base change, fibers, intersections
Adjunctions	$\operatorname{Spec} \dashv \Gamma$; $f^{-1} \dashv f_*$; sheafification
Representable functors	Schemes via functor of points
Yoneda lemma	Universal properties; moduli problems
Filtered colimits	Stalks, localizations
Abelian categories	Sheaf cohomology, derived categories

0.5 *Key Results from Differential Geometry

Many constructions in algebraic geometry—tangent bundles, differential forms, cohomology, metrics—have their origins in differential geometry. This section reviews the essential concepts from the smooth category that will inform and motivate their algebraic counterparts.

Throughout this section, "smooth" means C^∞ (infinitely differentiable).

Smooth Manifolds

Definition 0.5.1: Smooth Manifold

A *topological manifold* of dimension n is a Hausdorff, second-countable topological space M that is locally homeomorphic to $\mathbb{R}^n$. A *chart* is a pair (U, φ) where $U \subseteq M$ is open and $\varphi : U \xrightarrow{\sim} \varphi(U) \subseteq \mathbb{R}^n$ is a homeomorphism.
A *smooth structure* on M is a maximal atlas $\{(U_\alpha, \varphi_\alpha)\}$ such that all transition maps

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

are smooth (as maps between open subsets of $\mathbb{R}^n$).
A *smooth manifold* is a topological manifold equipped with a smooth structure.

Example 0.5.2: Examples of Smooth Manifolds

1. $\mathbb{R}^n$ with the standard atlas (single chart $\mathrm{id} : \mathbb{R}^n \to \mathbb{R}^n$).
2. Open subsets of $\mathbb{R}^n$ (or of any smooth manifold).
3. The n-sphere $S^n = \{x \in \mathbb{R}^{n+1} : |x| = 1\}$.
4. Real and complex projective spaces $\mathbb{RP}^n$, $\mathbb{CP}^n$.
5. Lie groups: $\mathrm{GL}_n(\mathbb{R})$, $\mathrm{O}(n)$, $\mathrm{U}(n)$, $\mathrm{SL}_n(\mathbb{R})$, etc.
6. Any smooth algebraic variety over $\mathbb{R}$ or $\mathbb{C}$ (with its classical topology).

Definition 0.5.3: Smooth Map

A map $f : M \to N$ between smooth manifolds is *smooth* if for every $p \in M$ and charts (U, φ) around p and (V, ψ) around $f(p)$, the composition

$$\psi \circ f \circ \varphi^{-1} : \varphi(U \cap f^{-1}(V)) \to \psi(V)$$

is smooth as a map between open subsets of Euclidean spaces.
A *diffeomorphism* is a smooth bijection with smooth inverse.

Definition 0.5.4: Tangent Space (via Derivations)

Let M be a smooth manifold and $p \in M$. A *derivation at p* is an $\mathbb{R}$-linear map $v : C^\infty(M) \to \mathbb{R}$ satisfying the Leibniz rule:

$$v(fg) = f(p) \cdot v(g) + g(p) \cdot v(f)$$

The *tangent space* T_pM is the vector space of all derivations at p.

Proposition 0.5.5: Dimension of Tangent Space

If $\dim M = n$, then $\dim T_pM = n$ for all $p \in M$. In local coordinates $(x^1, \ldots, x^n)$ centered at p, a basis for T_pM is given by the partial derivative operators:

$$\left\{ \frac{\partial}{\partial x^1}\bigg|_p, \ldots, \frac{\partial}{\partial x^n}\bigg|_p \right\}$$

Definition 0.5.6: Differential / Pushforward

Let $f : M \to N$ be smooth and $p \in M$. The *differential* (or *pushforward*) of f at p is the linear map

$$df_p = f_{*,p} : T_pM \to T_{f(p)}N, \qquad (df_p(v))(g) := v(g \circ f)$$

for $v \in T_pM$ and $g \in C^\infty(N)$.

Definition 0.5.7: Tangent Bundle

The *tangent bundle* of M is the disjoint union

$$TM := \bigsqcup_{p \in M} T_pM = \{(p, v) : p \in M,\ v \in T_pM\}$$

equipped with the natural projection $\pi : TM \to M$, $(p, v) \mapsto p$. This is a smooth manifold of dimension $2n$ and a vector bundle of rank n over M.

Definition 0.5.8: Vector Field

A *vector field* on M is a smooth section of the tangent bundle, i.e., a smooth map $X : M \to TM$ such that $\pi \circ X = \mathrm{id}_M$. Equivalently, X assigns to each $p \in M$ a tangent vector $X_p \in T_pM$, varying smoothly with p.
The space of vector fields is denoted $\mathfrak{X}(M)$ or $\Gamma(TM)$.

Proposition 0.5.9: Vector Fields as Derivations

A vector field $X \in \mathfrak{X}(M)$ defines a derivation $X : C^\infty(M) \to C^\infty(M)$ by $(Xf)(p) := X_p(f)$. This gives an isomorphism between $\mathfrak{X}(M)$ and the $C^\infty(M)$-module of derivations of $C^\infty(M)$.

Definition 0.5.10: Lie Bracket

The *Lie bracket* of vector fields $X, Y \in \mathfrak{X}(M)$ is the vector field $[X, Y]$ defined by

$$[X, Y](f) := X(Y(f)) - Y(X(f))$$

This makes $\mathfrak{X}(M)$ into a Lie algebra over $\mathbb{R}$.

Cotangent Spaces and Differential Forms

Definition 0.5.11: Cotangent Space

The *cotangent space* at $p \in M$ is the dual vector space

$$T_p^*M := (T_pM)^* = \mathrm{Hom}_{\mathbb{R}}(T_pM, \mathbb{R})$$

Elements of T_p^*M are called *cotangent vectors* or *covectors*.

Definition 0.5.12: Differential of a Function

For $f \in C^\infty(M)$ and $p \in M$, the *differential* $df_p \in T_p^*M$ is defined by

$$df_p(v) := v(f) \quad \text{for } v \in T_pM$$

In local coordinates, $df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i$, where $\{dx^1, \ldots, dx^n\}$ is the dual basis to $\{\partial/\partial x^1, \ldots, \partial/\partial x^n\}$.

Definition 0.5.13: Cotangent Bundle

The *cotangent bundle* is $T^*M := \bigsqcup_{p \in M} T_p^*M$, a vector bundle of rank n over M.

Definition 0.5.14: Differential k-Forms

A *differential k-form* (or simply *k-form*) on M is a smooth section of $\bigwedge^k T^*M$. That is, a k-form ω assigns to each $p \in M$ an alternating k-linear map

$$\omega_p : \underbrace{T_pM \times \cdots \times T_pM}_{k \text{ times}} \to \mathbb{R}$$

varying smoothly with p.
The space of k-forms is denoted $\Omega^k(M)$. We set $\Omega^0(M) = C^\infty(M)$.

Example 0.5.15: Differential Forms in Coordinates

In local coordinates $(x^1, \dots, x^n)$, every k-form can be written as

$$\omega = \sum_{i_1 < \cdots < i_k} f_{i_1 \dots i_k} \, dx^{i_1} \wedge \cdots \wedge dx^{i_k}$$

where $f_{i_1 \dots i_k} \in C^\infty(M)$ and $\wedge$ denotes the wedge product.

Definition 0.5.16: Wedge Product

The *wedge product* $\wedge : \Omega^k(M) \times \Omega^\ell(M) \to \Omega^{k+\ell}(M)$ is the unique bilinear, associative operation satisfying:

1. $df \wedge dg = -dg \wedge df$ for $f, g \in C^\infty(M)$.
2. $(f\omega) \wedge \eta = f(\omega \wedge \eta) = \omega \wedge (f\eta)$ for $f \in C^\infty(M)$.
3. $\omega \wedge \eta = (-1)^{k\ell} \eta \wedge \omega$ for $\omega \in \Omega^k$, $\eta \in \Omega^\ell$.

This makes $\Omega^*(M) := \bigoplus_{k \geq 0} \Omega^k(M)$ into a graded-commutative algebra.

Definition 0.5.17: Pullback of Forms

Let $f : M \to N$ be smooth. The *pullback* $f^* : \Omega^k(N) \to \Omega^k(M)$ is defined by

$$(f^*\omega)_p(v_1, \dots, v_k) := \omega_{f(p)}(df_p(v_1), \dots, df_p(v_k))$$

The pullback satisfies $f^*(\omega \wedge \eta) = f^*\omega \wedge f^*\eta$ and $(g \circ f)^* = f^* \circ g^*$.

Exterior Derivative and de Rham Cohomology

Definition 0.5.18: Exterior Derivative

The *exterior derivative* is the unique family of $\mathbb{R}$-linear maps $d : \Omega^k(M) \to \Omega^{k+1}(M)$ satisfying:

1. On $\Omega^0(M) = C^\infty(M)$, d is the differential: $(df)(X) = X(f)$.
2. $d \circ d = 0$.
3. $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$ for $\omega \in \Omega^k$ (graded Leibniz rule).

Example 0.5.19: Exterior Derivative in Coordinates

For $\omega = \sum_I f_I \, dx^I$ (where I is a multi-index), we have

$$d\omega = \sum_I \sum_{j=1}^n \frac{\partial f_I}{\partial x^j} dx^j \wedge dx^I$$

For instance, if $\omega = f\,dx + g\,dy$ on $\mathbb{R}^2$, then

$$d\omega = \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}\right) dx \wedge dy$$

Proposition 0.5.20: Naturality of d

The exterior derivative commutes with pullback: for $f : M \to N$ smooth,

$$f^* \circ d = d \circ f^*$$

That is, $f^*(d\omega) = d(f^*\omega)$ for all $\omega \in \Omega^*(N)$.

Definition 0.5.21: de Rham Complex

The *de Rham complex* of M is the cochain complex

$$0 \to \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \Omega^2(M) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^n(M) \to 0$$

A form ω is *closed* if $d\omega = 0$, and *exact* if $\omega = d\eta$ for some η. Since $d^2 = 0$, every exact form is closed.

Definition 0.5.22: de Rham Cohomology

The *k-th de Rham cohomology* of M is

$$H^k_{\mathrm{dR}}(M) := \frac{\ker(d : \Omega^k \to \Omega^{k+1})}{\operatorname{im}(d : \Omega^{k-1} \to \Omega^k)} = \frac{\{\text{closed } k\text{-forms}\}}{\{\text{exact } k\text{-forms}\}}$$

This is a real vector space measuring the "failure of closed forms to be exact."

Theorem 0.5.23: de Rham's Theorem

For a smooth manifold M, there is a natural isomorphism

$$H^k_{\mathrm{dR}}(M) \cong H^k(M; \mathbb{R})$$

where $H^k(M; \mathbb{R})$ is the singular cohomology of M with real coefficients. The isomorphism is given by integration of forms over cycles.

***Proof*:**
See [7]

Example 0.5.24: de Rham Cohomology Examples

1. $H^k_{\mathrm{dR}}(\mathbb{R}^n) = 0$ for $k > 0$, and $H^0_{\mathrm{dR}}(\mathbb{R}^n) = \mathbb{R}$ (Poincaré lemma).
2. $H^0_{\mathrm{dR}}(M) = \mathbb{R}^{\pi_0(M)}$, where $\pi_0(M)$ is the number of connected components.

3. $H^1_{\mathrm{dR}}(S^1) = \mathbb{R}$, generated by $[d\theta]$.
4. $H^k_{\mathrm{dR}}(S^n) = \mathbb{R}$ for $k = 0, n$, and 0 otherwise.

0.5.25 Algebraic de Rham Cohomology For a smooth algebraic variety X over a field k of characteristic zero, one can define *algebraic de Rham cohomology* using the complex of algebraic differential forms $\Omega^*_{X/k}$. Over $\mathbb{C}$, this agrees with singular cohomology by Grothendieck's algebraic de Rham theorem.

Vector Bundles

Definition 0.5.26: Vector Bundle

A *(real) vector bundle of rank* r over a smooth manifold M is a smooth manifold E together with:

1. A smooth surjection $\pi : E \to M$.
2. For each $p \in M$, the fiber $E_p := \pi^{-1}(p)$ has the structure of an r-dimensional real vector space.
3. Local triviality: each $p \in M$ has a neighborhood U with a diffeomorphism $\varphi : \pi^{-1}(U) \xrightarrow{\sim} U \times \mathbb{R}^r$ such that φ is linear on each fiber.

Example 0.5.27: Vector Bundle Examples

1. The *trivial bundle* $M \times \mathbb{R}^r \to M$.
2. The tangent bundle TM and cotangent bundle T^*M.
3. The *tautological line bundle* $\mathcal{O}(-1)$ over $\mathbb{RP}^n$ or $\mathbb{CP}^n$: the fiber over a line ℓ is ℓ itself.
4. Exterior powers $\bigwedge^k T^*M$ (whose sections are k-forms).
5. The *canonical bundle* $K_M := \bigwedge^n T^*M$ (top exterior power).

Definition 0.5.28: Section of a Vector Bundle

A *section* of $\pi : E \to M$ is a smooth map $s : M \to E$ with $\pi \circ s = \mathrm{id}_M$. The space of sections is denoted $\Gamma(E)$ or $\Gamma(M, E)$. This is a $C^\infty(M)$-module.

Definition 0.5.29: Operations on Vector Bundles

Given vector bundles E, F over M:

- *Direct sum*: $(E \oplus F)_p = E_p \oplus F_p$.
- *Tensor product*: $(E \otimes F)_p = E_p \otimes_{\mathbb{R}} F_p$.
- *Dual*: $E_p^* = (E_p)^*$.
- *Exterior powers*: $(\bigwedge^k E)_p = \bigwedge^k E_p$.
- *Pullback*: for $f : N \to M$, $(f^* E)_q = E_{f(q)}$.

Definition 0.5.30: Transition Functions

If $\{U_\alpha\}$ is an open cover of M with local trivializations $\varphi_\alpha : E|_{U_\alpha} \xrightarrow{\sim} U_\alpha \times \mathbb{R}^r$, the *transition functions* are

$$g_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathrm{GL}_r(\mathbb{R}), \qquad \varphi_\alpha \circ \varphi_\beta^{-1}(p, v) = (p, g_{\alpha\beta}(p) \cdot v)$$

These satisfy the cocycle condition: $g_{\alpha\beta} g_{\beta\gamma} = g_{\alpha\gamma}$ on triple overlaps.

0.5.31 Line Bundles and Divisors A *line bundle* is a vector bundle of rank 1. Line bundles are classified by $H^1(M; C^\infty(M, \mathbb{R}^\times))$ via transition functions. In algebraic geometry, line bundles on a variety X correspond to Cartier divisors and form the Picard group $\mathrm{Pic}(X)$.

Connections and Curvature

Definition 0.5.32: Connection on a Vector Bundle

A *connection* (or *covariant derivative*) on a vector bundle $E \to M$ is an $\mathbb{R}$-linear map

$$\nabla : \Gamma(E) \to \Gamma(T^* M \otimes E) = \Omega^1(M, E)$$

satisfying the Leibniz rule: for $f \in C^\infty(M)$ and $s \in \Gamma(E)$,

$$\nabla(fs) = df \otimes s + f \nabla s$$

For a vector field X, we write $\nabla_X s := \iota_X(\nabla s) \in \Gamma(E)$ for the covariant derivative of s in direction X.

0.5.33 Connections Exist Every vector bundle over a paracompact manifold admits a connection. Connections form an affine space over $\Omega^1(M, \mathrm{End}(E))$: if ∇ is a connection and $A \in \Omega^1(M, \mathrm{End}(E))$, then $\nabla + A$ is also a connection.

Definition 0.5.34: Curvature

The *curvature* of a connection ∇ is the $\mathrm{End}(E)$-valued 2-form $F_\nabla \in \Omega^2(M, \mathrm{End}(E))$ defined by

$$F_\nabla(X,Y) := \nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X,Y]}$$

for vector fields X, Y. A connection is *flat* if $F_\nabla = 0$.

Proposition 0.5.35: Curvature as Obstruction

A connection ∇ is flat if and only if parallel transport is path-independent (depends only on the homotopy class of the path). Flat connections on a trivial bundle correspond to representations $\pi_1(M) \to \mathrm{GL}_r(\mathbb{R})$.

Riemannian Metrics

Definition 0.5.36: Riemannian Metric

A *Riemannian metric* on M is a smooth assignment of an inner product $g_p : T_pM \times T_pM \to \mathbb{R}$ to each $p \in M$. In local coordinates, $g = \sum_{i,j} g_{ij}\, dx^i \otimes dx^j$ where $(g_{ij}(p))$ is a positive definite symmetric matrix for each p.
A *Riemannian manifold* is a pair (M, g).

Example 0.5.37: Riemannian Metrics

1. The standard metric on $\mathbb{R}^n$: $g = dx^1 \otimes dx^1 + \cdots + dx^n \otimes dx^n$.
2. The round metric on S^n (induced from $\mathbb{R}^{n+1}$).
3. The hyperbolic metric on the upper half-plane: $g = \frac{dx^2+dy^2}{y^2}$.
4. The Fubini–Study metric on $\mathbb{CP}^n$.

Proposition 0.5.38: Existence of Riemannian Metrics

Every smooth manifold admits a Riemannian metric. (This uses partitions of unity.)

Definition 0.5.39: Levi-Civita Connection

On a Riemannian manifold (M, g), there is a unique connection ∇ on TM that is:

1. *Metric-compatible*: $X(g(Y,Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$.
2. *Torsion-free*: $\nabla_X Y - \nabla_Y X = [X,Y]$.

This is called the *Levi-Civita connection*.

Definition 0.5.40: Geodesics

A *geodesic* is a curve $\gamma : I \to M$ satisfying $\nabla_{\dot\gamma}\dot\gamma = 0$. Geodesics are locally length-minimizing curves.

Integration and Orientation

Definition 0.5.41: Orientation

An *orientation* on an n-manifold M is a choice of smoothly varying orientation on each tangent space T_pM. Equivalently, it is a nowhere-vanishing n-form $\omega \in \Omega^n(M)$, or a trivialization of the canonical bundle $K_M = \bigwedge^n T^*M$.
A manifold is *orientable* if it admits an orientation, and *oriented* if one has been chosen.

Example 0.5.42: Orientability

1. $\mathbb{R}^n$, S^n, and $\mathbb{CP}^n$ are orientable.
2. $\mathbb{RP}^n$ is orientable if and only if n is odd.
3. The Möbius strip is not orientable.

Definition 0.5.43: Integration of Top Forms

Let M be an oriented n-manifold and $\omega \in \Omega^n_c(M)$ a compactly supported n-form. The *integral* $\int_M \omega$ is defined by:

1. In a single oriented chart (U, φ) with $\mathrm{supp}(\omega) \subseteq U$: write $\omega = f\, dx^1 \wedge \cdots \wedge dx^n$ and set
$$\int_M \omega := \int_{\varphi(U)} f(\varphi^{-1}(x))\, dx^1 \cdots dx^n$$
2. For general ω, use a partition of unity to reduce to the local case.

Theorem 0.5.44: Stokes' Theorem

Let M be an oriented n-manifold with boundary ∂M (given the induced orientation), and let $\omega \in \Omega^{n-1}_c(M)$. Then
$$\int_M d\omega = \int_{\partial M} \omega$$
In particular, if M is closed (compact without boundary), then $\int_M d\omega = 0$.

Proof:
See [7]

0.5.45 Classical Theorems as Special Cases Stokes' theorem generalizes:

- The fundamental theorem of calculus ($n = 1$).
- Green's theorem ($n = 2$, planar region).
- The classical Stokes theorem ($n = 3$, surface in $\mathbb{R}^3$).
- The divergence theorem ($n = 3$, solid region in $\mathbb{R}^3$).

Partitions of Unity

Definition 0.5.46: Partition of Unity

Let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of M. A *partition of unity subordinate to* $\{U_\alpha\}$ is a collection of smooth functions $\{\rho_\alpha : M \to [0, 1]\}_{\alpha \in A}$ such that:

1. $\operatorname{supp}(\rho_\alpha) \subseteq U_\alpha$ for each α.
2. The collection $\{\operatorname{supp}(\rho_\alpha)\}$ is locally finite.
3. $\sum_\alpha \rho_\alpha(p) = 1$ for all $p \in M$.

Theorem 0.5.47: Existence of Partitions of Unity

Every open cover of a smooth manifold admits a subordinate partition of unity.

***Proof*:**
See [7].

0.5.48 Partitions of Unity in Algebraic Geometry Partitions of unity are a crucial tool in differential geometry, enabling:

- Existence of Riemannian metrics.
- Existence of connections.
- Extension of local constructions to global ones.
- Proof that smooth manifolds are paracompact.

Importantly, partitions of unity do not exist in algebraic geometry. Polynomial or algebraic functions cannot be "bump functions." This is one reason why sheaf cohomology becomes essential in algebraic geometry: it replaces the role of partitions of unity in extending local to global.

0.5.1 Summary: Differential vs. Algebraic Geometry

Concept	Differential Geometry	Algebraic Geometry
Local model	$\mathbb{R}^n$ or $\mathbb{C}^n$	$\operatorname{Spec}(A)$
Functions	$C^\infty(M)$	$\mathcal{O}_X(X)$ (regular functions)
Tangent space	T_pM (derivations)	$(\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee$
Cotangent space	T_p^*M	$\mathfrak{m}_p/\mathfrak{m}_p^2$
Vector bundle	Smooth fiber bundle	Locally free sheaf
Differential forms	$\Omega^k(M)$	$\Omega^k_{X/k}$ (Kähler differentials)
Cohomology	de Rham $H^*_{\mathrm{dR}}(M)$	Sheaf cohomology $H^*(X, \mathcal{F})$
Partitions of unity	Exist	Do not exist
Gluing	Smooth	Via sheaf axioms

The absence of partitions of unity in the algebraic world makes sheaf-theoretic methods indispensable. Many constructions that are "automatic" in differential geometry (e.g., existence of metrics) require careful sheaf-cohomological arguments in algebraic geometry.

0.6 *Key Results from Complex Geometry

Complex geometry sits at the intersection of differential geometry and algebraic geometry. Over $\mathbb{C}$, algebraic varieties carry both a Zariski topology (algebraic) and a classical topology (analytic), and Serre's GAGA theorems establish connections between these perspectives. This section reviews the essential concepts from complex analytic geometry that inform and enrich algebraic geometry over $\mathbb{C}$.

Complex Manifolds

Definition 0.6.1: Complex Manifold

A *complex manifold* of dimension n is a smooth manifold M of real dimension $2n$ equipped with an atlas $\{(U_\alpha, \varphi_\alpha)\}$ where each $\varphi_\alpha : U_\alpha \xrightarrow{\sim} \varphi_\alpha(U_\alpha) \subseteq \mathbb{C}^n$ is a homeomorphism onto an open subset of $\mathbb{C}^n$, and all transition maps

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

are *holomorphic* (i.e., complex analytic).

Definition 0.6.2: Holomorphic Function

A function $f : U \to \mathbb{C}$ on an open set $U \subseteq \mathbb{C}^n$ is *holomorphic* if it satisfies the Cauchy–Riemann equations in each variable, or equivalently, if it is complex differentiable at every point. In coordinates $z_j = x_j + iy_j$, this means

$$\frac{\partial f}{\partial \bar{z}_j} = 0 \quad \text{for all } j$$

where $\frac{\partial}{\partial \bar{z}_j} = \frac{1}{2}\left(\frac{\partial}{\partial x_j} + i\frac{\partial}{\partial y_j}\right)$.

Definition 0.6.3: Holomorphic Map

A map $f : M \to N$ between complex manifolds is *holomorphic* if it is holomorphic in local coordinates. A *biholomorphism* is a holomorphic bijection with holomorphic inverse.

Example 0.6.4: Complex Manifolds

1. $\mathbb{C}^n$ with the standard complex structure.
2. Open subsets of $\mathbb{C}^n$ (or of any complex manifold).
3. Complex projective space $\mathbb{CP}^n$: covered by $n+1$ charts $U_i = \{[z_0 : \cdots : z_n] : z_i \neq 0\} \cong \mathbb{C}^n$.
4. Smooth projective varieties over $\mathbb{C}$ (with the classical topology).
5. Complex tori $\mathbb{C}^n/\Lambda$ where $\Lambda \cong \mathbb{Z}^{2n}$ is a lattice.
6. Riemann surfaces: complex manifolds of dimension 1.

0.6.5 Smooth vs. Complex vs. Algebraic We have strict inclusions of categories:

$$\{\text{smooth projective varieties}/\mathbb{C}\} \subsetneq \{\text{compact Kähler manifolds}\} \subsetneq \{\text{compact complex manifolds}\}$$

Not every complex manifold is algebraic (e.g., generic complex tori of dimension ≥ 2), and not every compact complex manifold is Kähler (e.g., most Hopf surfaces).

Almost Complex and Complex Structures

Definition 0.6.6: Almost Complex Structure

An *almost complex structure* on a smooth manifold M is a smooth bundle endomorphism $J : TM \to TM$ satisfying $J^2 = -\mathrm{id}$. This makes each tangent space T_pM into a complex vector space via $(a + bi) \cdot v := av + bJ(v)$.

Proposition 0.6.7: Complex Manifolds Have Almost Complex Structures

Every complex manifold has a natural almost complex structure: in holomorphic coordinates $(z_1, \ldots, z_n)$ with $z_j = x_j + iy_j$,

$$J\left(\frac{\partial}{\partial x_j}\right) = \frac{\partial}{\partial y_j}, \qquad J\left(\frac{\partial}{\partial y_j}\right) = -\frac{\partial}{\partial x_j}$$

The converse is not automatic:

Definition 0.6.8: Integrable Almost Complex Structure

An almost complex structure J is *integrable* if it arises from a complex manifold structure. The *Nijenhuis tensor* N_J measures the obstruction to integrability.

Theorem 0.6.9: Newlander–Nirenberg Theorem

An almost complex structure J is integrable if and only if the Nijenhuis tensor vanishes: $N_J = 0$.

***Proof*:**
see [9]

Complexified Tangent Bundle and (p, q)-Forms

Definition 0.6.10: Complexified Tangent Bundle

For a complex manifold M, the *complexified tangent bundle* is $T_{\mathbb{C}}M := TM \otimes_{\mathbb{R}} \mathbb{C}$. The almost complex structure J extends $\mathbb{C}$-linearly and has eigenvalues $\pm i$. This gives a decomposition

$$T_{\mathbb{C}}M = T^{1,0}M \oplus T^{0,1}M$$

where $T^{1,0}M$ is the $+i$ eigenspace and $T^{0,1}M$ is the $-i$ eigenspace.
In local coordinates, $T^{1,0}M$ is spanned by $\frac{\partial}{\partial z_j} := \frac{1}{2}\left(\frac{\partial}{\partial x_j} - i\frac{\partial}{\partial y_j}\right)$ and $T^{0,1}M$ by $\frac{\partial}{\partial \bar{z}_j} := \frac{1}{2}\left(\frac{\partial}{\partial x_j} + i\frac{\partial}{\partial y_j}\right)$.

Definition 0.6.11: (p,q)-Forms

The complexified cotangent bundle similarly decomposes: $T^*_{\mathbb{C}}M = (T^{1,0}M)^* \oplus (T^{0,1}M)^*$. A *differential form of type* (p,q) is a section of

$$\bigwedge^{p,q} T^*_{\mathbb{C}}M := \bigwedge^{p} (T^{1,0}M)^* \otimes \bigwedge^{q} (T^{0,1}M)^*$$

The space of (p,q)-forms is denoted $\Omega^{p,q}(M)$ or $\mathcal{A}^{p,q}(M)$.

Proposition 0.6.12: Decomposition of Forms

The space of complex k-forms decomposes as

$$\Omega^k(M,\mathbb{C}) = \bigoplus_{p+q=k} \Omega^{p,q}(M)$$

In local coordinates, a (p,q)-form can be written as

$$\omega = \sum_{|I|=p,|J|=q} f_{I,J}\, dz^I \wedge d\bar{z}^J$$

where $dz^I = dz_{i_1} \wedge \cdots \wedge dz_{i_p}$ and $d\bar{z}^J = d\bar{z}_{j_1} \wedge \cdots \wedge d\bar{z}_{j_q}$.

The Dolbeault Operators

Definition 0.6.13: Dolbeault Operators

The exterior derivative $d : \Omega^k(M,\mathbb{C}) \to \Omega^{k+1}(M,\mathbb{C})$ decomposes as $d = \partial + \bar{\partial}$ where:

$$\partial : \Omega^{p,q}(M) \to \Omega^{p+1,q}(M)$$
$$\bar{\partial} : \Omega^{p,q}(M) \to \Omega^{p,q+1}(M)$$

These satisfy $\partial^2 = 0$, $\bar{\partial}^2 = 0$, and $\partial\bar{\partial} + \bar{\partial}\partial = 0$.

Example 0.6.14: Dolbeault Operators in Coordinates

For $f \in C^\infty(M,\mathbb{C}) = \Omega^{0,0}(M)$:

$$\partial f = \sum_{j=1}^{n} \frac{\partial f}{\partial z_j} dz_j, \qquad \bar{\partial} f = \sum_{j=1}^{n} \frac{\partial f}{\partial \bar{z}_j} d\bar{z}_j$$

A function is holomorphic if and only if $\bar{\partial} f = 0$.

Definition 0.6.15: Dolbeault Complex

For each p, the *Dolbeault complex* is

$$0 \to \Omega^{p,0}(M) \xrightarrow{\bar{\partial}} \Omega^{p,1}(M) \xrightarrow{\bar{\partial}} \Omega^{p,2}(M) \xrightarrow{\bar{\partial}} \cdots \xrightarrow{\bar{\partial}} \Omega^{p,n}(M) \to 0$$

Definition 0.6.16: Dolbeault Cohomology

The *Dolbeault cohomology groups* are

$$H^{p,q}_{\bar{\partial}}(M) := \frac{\ker(\bar{\partial} : \Omega^{p,q} \to \Omega^{p,q+1})}{\operatorname{im}(\bar{\partial} : \Omega^{p,q-1} \to \Omega^{p,q})}$$

These are complex vector spaces measuring the failure of the $\bar{\partial}$-Poincaré lemma.

Theorem 0.6.17: Dolbeault's Theorem

For a complex manifold M, there is a natural isomorphism

$$H^{p,q}_{\bar{\partial}}(M) \cong H^q(M, \Omega^p_M)$$

where Ω^p_M is the sheaf of holomorphic p-forms.

***Proof*:**
See [9]

Hermitian and Kähler Metrics

Definition 0.6.18: Hermitian Metric

A *Hermitian metric* on a complex manifold M is a smoothly varying Hermitian inner product h_p on each $T^{1,0}_p M$. Equivalently, it is a Riemannian metric g such that $g(JX, JY) = g(X, Y)$ for all tangent vectors X, Y.
In local coordinates, $h = \sum_{j,k} h_{j\bar{k}}\, dz_j \otimes d\bar{z}_k$ with $(h_{j\bar{k}})$ a positive definite Hermitian matrix.

Definition 0.6.19: Fundamental Form

The *fundamental form* (or *associated* $(1,1)$*-form*) of a Hermitian metric h is

$$\omega := -\operatorname{Im}(h) = \frac{i}{2} \sum_{j,k} h_{j\bar{k}}\, dz_j \wedge d\bar{z}_k$$

This is a real $(1,1)$-form satisfying $\omega(X, JX) > 0$ for nonzero X.

Definition 0.6.20: Kähler Manifold

A Hermitian metric h is *Kähler* if its fundamental form ω is closed: $d\omega = 0$. A *Kähler manifold* is a complex manifold equipped with a Kähler metric. The form ω is then called the *Kähler form*.

Proposition 0.6.21: Equivalent Characterizations of Kähler

The following are equivalent for a Hermitian manifold (M, h):

1. $d\omega = 0$ (the Kähler condition).
2. The Levi-Civita connection ∇ of $g = \mathrm{Re}(h)$ satisfies $\nabla J = 0$.
3. Locally, $\omega = i\partial\bar{\partial}\varphi$ for some real function φ (a *Kähler potential*).
4. The holonomy of g is contained in $\mathrm{U}(n) \subset \mathrm{SO}(2n)$.

***Proof*:**
See [9].

Example 0.6.22: Kähler Manifolds

1. $\mathbb{C}^n$ with the standard metric $\omega = \frac{i}{2}\sum_j dz_j \wedge d\bar{z}_j$.
2. $\mathbb{CP}^n$ with the *Fubini–Study metric*:
$$\omega_{\mathrm{FS}} = \frac{i}{2\pi}\partial\bar{\partial}\log\left(|z_0|^2 + \cdots + |z_n|^2\right)$$
3. Any smooth projective variety over $\mathbb{C}$ (inherits Kähler structure from $\mathbb{CP}^n$).
4. Complex submanifolds of Kähler manifolds are Kähler.
5. Complex tori $\mathbb{C}^n/\Lambda$ with the flat metric.

Example 0.6.23: Non-Kähler Manifolds

1. The *Hopf surface* $(\mathbb{C}^2 \setminus \{0\})/\langle z \mapsto 2z\rangle \cong S^1 \times S^3$ is a compact complex surface with $b_2 = 0$, hence not Kähler (Kähler manifolds have $b_2 \geq 1$ by the existence of $[\omega]$).
2. Most compact complex manifolds are not Kähler.

Hodge Theory

Hodge theory provides a bridge between the topology, complex structure, and Riemannian geometry of a Kähler manifold.

Definition 0.6.24: Hodge Star and Laplacian

On a Riemannian manifold (M, g) of dimension n, the *Hodge star* $* : \Omega^k(M) \to \Omega^{n-k}(M)$ is defined by

$$\alpha \wedge *\beta = g(\alpha, \beta)\,\mathrm{vol}_g$$

The *Hodge Laplacian* is $\Delta := dd^* + d^*d$ where $d^* := (-1)^{n(k+1)+1} * d*$ is the formal adjoint of d.
A form is *harmonic* if $\Delta\omega = 0$.

Theorem 0.6.25: Hodge Theorem (Riemannian)

On a compact oriented Riemannian manifold M:

1. Every de Rham cohomology class contains a unique harmonic representative.
2. $H^k_{\mathrm{dR}}(M) \cong \mathcal{H}^k(M) := \{\omega \in \Omega^k(M) : \Delta\omega = 0\}$.

***Proof*:**
See [9]

On a Kähler manifold, there are additional Laplacians:

Definition 0.6.26: Dolbeault Laplacians

On a Hermitian manifold, define

$$\Delta_\partial := \partial\partial^* + \partial^*\partial, \qquad \Delta_{\bar\partial} := \bar\partial\bar\partial^* + \bar\partial^*\bar\partial$$

Theorem 0.6.27: Kähler Identities

On a Kähler manifold:

$$\Delta = 2\Delta_\partial = 2\Delta_{\bar\partial}$$

In particular, a form is harmonic if and only if it is ∂-harmonic if and only if it is $\bar\partial$-harmonic.

***Proof*:**
See [9]

Theorem 0.6.28: Hodge Decomposition

On a compact Kähler manifold M:

1. There is a decomposition

$$H^k(M, \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(M)$$

where $H^{p,q}(M) \cong H^{p,q}_{\bar{\partial}}(M) \cong H^q(M, \Omega^p_M)$.

2. Complex conjugation gives $\overline{H^{p,q}(M)} = H^{q,p}(M)$.
3. The decomposition is independent of the choice of Kähler metric.

***Proof*:**
See [9]

Definition 0.6.29: Hodge Numbers

The *Hodge numbers* of a compact Kähler manifold are

$$h^{p,q}(M) := \dim_{\mathbb{C}} H^{p,q}(M)$$

These satisfy:

1. $h^{p,q} = h^{q,p}$ (complex conjugation).
2. $h^{p,q} = h^{n-p,n-q}$ (Serre duality).
3. $b_k = \sum_{p+q=k} h^{p,q}$ (Betti numbers).

The Hodge numbers are often displayed in the *Hodge diamond*.

Example 0.6.30: Hodge Diamond of $\mathbb{CP}^n$

For $\mathbb{CP}^n$, the only nonzero Hodge numbers are $h^{p,p} = 1$ for $0 \leq p \leq n$:

$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 0 & & 1 & & 0 \\ & \cdots & & \cdots & \end{array}$$

This reflects $H^{2k}(\mathbb{CP}^n, \mathbb{C}) = \mathbb{C}$ and $H^{2k+1}(\mathbb{CP}^n, \mathbb{C}) = 0$.

Example 0.6.31: Hodge Diamond of a Curve
For a smooth projective curve C of genus g:

$$\begin{array}{ccc} & 1 & \\ g & & g \\ & 1 & \end{array}$$

Here $h^{1,0} = h^{0,1} = g = \dim H^0(C, \Omega^1_C) = \dim H^1(C, \mathcal{O}_C)$.

Line Bundles and Divisors

Definition 0.6.32: Divisor (Complex Manifold)

A *divisor* on a complex manifold M is a formal $\mathbb{Z}$-linear combination $D = \sum_i n_i V_i$ where each V_i is an irreducible analytic hypersurface (codimension 1) and $n_i \in \mathbb{Z}$. The group of divisors is denoted $\mathrm{Div}(M)$.

Definition 0.6.33: Line Bundle Associated to a Divisor

To a divisor D, one associates a holomorphic line bundle $\mathcal{O}(D)$ (or $\mathcal{L}(D)$) whose sections over U are meromorphic functions f with $(f) + D|_U \geq 0$, i.e., f may have poles along negative components of D and must vanish along positive components.

Proposition 0.6.34: Divisors and Line Bundles

On a compact complex manifold M:

1. Every holomorphic line bundle L is isomorphic to $\mathcal{O}(D)$ for some divisor D.
2. $\mathcal{O}(D_1) \cong \mathcal{O}(D_2)$ if and only if $D_1 - D_2 = (f)$ for some meromorphic function f (linear equivalence).
3. The *Picard group* $\mathrm{Pic}(M) = \{\text{line bundles}\}/\cong$ satisfies $\mathrm{Pic}(M) \cong \mathrm{Div}(M)/\sim$.

Definition 0.6.35: First Chern Class

The *first Chern class* of a line bundle L is a cohomology class $c_1(L) \in H^2(M, \mathbb{Z})$ (or $H^{1,1}(M) \cap H^2(M, \mathbb{Z})$ for Kähler manifolds). It can be computed as:

$$c_1(L) = \frac{i}{2\pi}[F_\nabla]$$

where F_∇ is the curvature of any connection on L.

Proposition 0.6.36: Chern Class Properties

1. $c_1(L_1 \otimes L_2) = c_1(L_1) + c_1(L_2)$.
2. $c_1(L^*) = -c_1(L)$.
3. $c_1(\mathcal{O}(D)) = [D]$ (the cohomology class of D).
4. On $\mathbb{CP}^n$: $c_1(\mathcal{O}(1)) = [\omega_{\mathrm{FS}}]$, the class of a hyperplane.

0.6.1 Summary: Complex Geometry and Algebraic Geometry

This table has more than what was covered for the purpose of reference:

Concept	**Complex Analytic**	**Algebraic**
Space	Complex manifold	Smooth variety / scheme
Functions	Holomorphic	Regular (polynomial)
Topology	Classical (Euclidean)	Zariski
Local model	$\mathbb{C}^n$	Spec$\,(A)$
Differential forms	(p,q)-forms, $\bar{\partial}$	Kähler differentials $\Omega^p_{X/k}$
Cohomology	Dolbeault $H^{p,q}_{\bar{\partial}}$	Sheaf cohomology $H^q(X, \Omega^p)$
Vector bundles	Holomorphic bundles	Locally free sheaves
Divisors	Analytic hypersurfaces	Weil/Cartier divisors
Line bundles	Pic(M)	Pic(X)
Metrics	Hermitian, Kähler	(No direct analogue)
Key theorem	Hodge decomposition	Serre duality, GAGA

0.6.37 Foreshadowing GAGA For projective varieties over $\mathbb{C}$, one can freely pass between algebraic and analytic viewpoints via GAGA:

- **Algebraic methods**: Work over any field, use scheme theory, étale cohomology.
- **Analytic methods**: Use Hodge theory, L^2-methods, Kähler geometry, transcendental techniques.

Many results (e.g., Hodge conjecture, Torelli theorems) involve both perspectives essentially.

Part I

Scheme Theory

I believe there will be more than just basic alg geo. This part will go over the core theory, Hartshorne, Vakil, Eisenbud, and so forth. Other parts will surely be specializations.

(I want to mention somewhere that for all the concepts we shall generalize from geometry, the integral shall be interesting as it will be recovers using cohomology by using the pushforward and trace maps).

1

Sheaves

Roughly speaking, Sheaf Theory is the theory of augmenting a topological space[1] that adds information to each open set (adds "local" information) which can be "glued" to get information in larger open sets, and can be "restricted" to get information on smaller sets. The prototypical example of a sheaf is the set of smooth functions on a manifold. Smooth functions on a manifold can be restricted to open subsets of a manifold which are homeomorphic to $U \subseteq \mathbb{R}^n$ and if a smooth function on each part of the manifold is defined in a compatible way (i.e. they agree on intersections), they can be glued back together to form a function on the entire manifold. As an open covering of M can always be chosen so that each U is simply connected connected, the non-trivial information is in the ability or choice of gluing of functions.

A key motivator behind sheaves is that they can hold the "geometric" information of a space. The reader may recall that a vector-space and subspaces can be characterized via their dual spaces, that is all functions from the vector space to $\mathbb{F}$. A similar result holds for Varieties with the coordinate ring uniquely characterizing a Variety. This duality was found to exist everywhere in mathematics, to name three more examples:

- A classical example that started the duality between algebra and geometry is that of $C(X)$ and X where X is a compact space and $C(X)$ all continuous function on X. By taking the stone-Cech compactification of a space X, the space can be achieved for any space (see [12, p.144] for details)
- Similarly, a radon measure's can be characterized using the duals of $C_c(X)$ where X is a locally compact space (classical result can be found in EYTNKA at [23])
- Spin manifolds can be characterized by spectral triples (see [5])

[1] more generally, a topoi, which concretely translates in our case to a category whose objects are open sets and morphisms are inclusions

- Sets with set-function can be thought of as the most "abstract" geometric object (they only have information about containment). There is an appropriate dual called the *complete atomic Boolean algebras.* The reader may be interested in knowing that Boolean algebra's characterize logical operations, which gives an interesting geometric interpretation to logic[2].

There is a long list of such examples (the reader may be curious to checkout this link). This gave rise to the idea that a geometric space can be understood by studying some subset of function on this space with certain. In algebraic cases, there are relatively few functions that are defined globally on a space, but many that are defined on local components (the prototypical example of this would be $\mathbb{C}$[3]).[4]

Why do functions capture this geometric information? One can think of functions as tools for representing relations. If our function is of the form $f : X \to \mathbb{R}$, then this is relating X to numbers, i.e. quantification, usually X being a topological space and f being continuous for some notion of closeness. If we have a continuous function of the form $f : X \to \mathbb{R}^n$ (or $\mathbb{C}^n$), with X a topological space, then we may pullback a lot of geometric results from $\mathbb{R}^n$ (or $\mathbb{C}^n$) onto X, depending on how f is defined (ex. it can be shown that f is a diffeomorphism if and only if it pulls back the differential structure of $\mathbb{R}^n$). Very often, it is not possible to define a diffeomorphism $f : X \to \mathbb{R}^n$, however we may define it on components. Functions have the property that we may restrict them or construct them by defining what happening at each point (or in the continuous case, on each open set where they agree on intersections[5]). But you may say that morphisms in a category already capture this intuition. This is indeed the case, what makes functions special is that you can construct larger morphisms from smaller morphisms with an appropriate notion of *gluing* and *restricting.* In particular, the closeness given by the notion of topological spaces and the locality properties inspired by functions shall be the two main properties of sheaves. As is the course of mathematics, it may be asked if these properties of functions can be generalized, and we may ask if we can take special categories which have these features; this is called a *site.* For an introduction to algebraic geometry, limiting to the sites defined on topological spaces is plenty sufficient and shall be the focus of this book.

A particularly important realization will be that commutative rings will in many ways behave like functions on a space, that is if R is a ring, there is a sense in which the elements of R will be the "functions" on a special space (namely, the space $\operatorname{Spec}(R)$). This shall be elaborated on in chapter 2 and was commented on in the front-matter. At this stage it is motivating to know that there is a reason why we may care about sheaves other than the sheaf of functions, namely the interpolation between commutative rings and functions.

Motivating Example from Vakil

Before exploring sheaves in generality, it is useful to have the prototypical example of a sheaf in mind in more detail:

[2] This connection is further refined using Topoi

[3] entire holomorphic are necessarily power series, while holomorphic on (not necessarily simply connected) open sets are analytic and may even have non-trivial Laurent expansion around holes

[4] see this link for more details

[5] by the gluing lemma

Example 1.0.1: Sheaf Of Differentiable Functions On Manifold

Let M be a smooth manifold. Then the we shall constuct the sheaf of differentiable function on M. For each open subset $U \subseteq M$, we shall assocaite to it the set of smooth function $C^\infty(U)$. Denote this collection by $\mathcal{O}(U)$ (this will be common notation for sheaves). Then if we have an open subset of U, $V \subseteq U$, we may restrict the function from U to function on V, namely we have a map $\text{res}_{U,V} : \mathcal{O}(U) \to \mathcal{O}(V)$ mapping $f \mapsto f|_V$. Naturally, if we have $W \hookrightarrow V \hookrightarrow U$, then the following commutative diagram.

$$\begin{tikzcd} & V \arrow[dr, "\text{res}_{V,W}"] & \\ U \arrow[ur, "\text{res}_{U,V}"] \arrow[rr, "\text{res}_{U,W}"'] & & W \end{tikzcd}$$

This collection of maps satisfy two important properties. In the following let and $U = \bigcup_i U_i$.

1. *Defining Locally*: Let's say $f_1, f_2 \in \mathcal{O}(U)$, Then if f_1 and f_2 agree on all restriction, that is $\text{res}_{U,U_i} f_1 = \text{res}_{U,U_j} f_2$, then $f_1 = f_2$
2. *Gluing*: If there exists a collection $f_i \in \mathcal{O}(U_i)$ such that f_i and f_j agree on intersections, that is $\text{res}_{U_i,U_i \cap U_j} f_i = \text{res}_{U_j,U_i \cap U_j} f_j$, then there is a $f \in \mathcal{O}(U)$ such that $\text{res}_{U,U_i} f = f_i$ for all i.

Then a sheaf will respect these properties. Note that in fact, we may define a manifold M to be $(M, \mathcal{O}_M)$ where M Is a topological space and $\mathcal{O}_M$ is a sheaf of smooth functions on M. If instead $\mathcal{O}(U)$ was the collection of continuous functions, we would get a topological manifold. Even more generally, we may let the functions be set functions, for set functions work well under restrictions and we may define set functions locally and construct set function via gluing.

Note that Diffeology can be thought of as a generalization of this result or of a special case of sheaf theory, and we shall explore it briefly at the end of this chapter.

Example 1.0.2: Sheaf On Set Functions

Take $\text{Hom}_{\mathbf{Set}}(X, Y)$ to be the collection of arbitrary set functions (or equivalently, let X, Y have the discrete topology so that all set functions are continuous). Then given an arbitrary collection of sets $\{U_i\}_{i \in I}$ of X with functions $f_i : U_I \to Y$, if the collection of functions agree on all pair-wise intersections $U_i \cap U_j$, $i, j \in I$, then there exists a unique function f on $\bigcup_{i \in I} U_i$ that restricts to all these functions. Notice the lack of essentially any structure. This shows that the heavy-lifter insuring the gluability of local data is the *associativity* axiom functions must satisfy.

So if even set functions can form a sheaf, then why did I write "nice" function earlier? As it turns out, what we shall do is abstract the notion of function! In particular, we shall look at systems where we may have the behavior of localizing and gluing "functions" on a topological space. Hence, in the following abstraction, we will consider $\mathcal{O}_M$ to be the data that stores a collection of objects that *behave like functions.*

A key property that distinguishes functions from general morphisms is the notion of evaluation (taking $f(x)$ for some element x and function f). Since we are abstracting the notion of function, we shall need a notion of *evaluation*. To motivate this generalization, let's come back to $(M, \mathcal{O}_M)$ and consider some $\mathcal{O}(U) = C^\infty(U)$ for some open $U \subseteq M$ and take $p \in U$. Recall that $\mathcal{O}(U)$ is a ring and that the *germ* at p is collection of equivalence relations:

$$[f] = \{(f, U) \mid p \in U,\ f \in \mathcal{O}(U)\}$$
$$(f, U) \sim (g, V) \ \Leftrightarrow W \subseteq U, W \subseteq V, p \in W,\ f|_W = g|_W$$

that is, $[f] = [g]$ if and only if there is an open neighborhood $W \subseteq U, V$ containing p ($p \in W$) such that $f|_W = g|_W$ (in the language of sheaf theory, $\mathrm{res}_{U,W} f = \mathrm{res}_{V,W} g$). Denote this collection $\mathcal{O}(U)_p$. The key is that the germ of $\mathcal{O}(U)$ at p is a *local ring* with maximal ideal $\mathfrak{m}_p$ of germs such that $[f](p) = 0$ (or simply $f(p) = 0$ if unambiguous). Then:

$$0 \to \mathfrak{m}_p \to \mathcal{O}(U)_p \xrightarrow{f \mapsto f(p)} \mathbb{R} \to 0$$

and all elements $\mathcal{O}(U) \setminus \mathfrak{m}_p$ are invertible. Hence, we shall see that we may thing of "evalutation" of f at some point $p \in U$ as taking first the germ $\mathcal{O}(U)_p$ and quotient it by the maximal ideal $\mathfrak{m}_p$, $\mathcal{O}(U)_p/\mathfrak{m}_p$, and then taking $\overline{f} \in \mathcal{O}(U)_p/\mathfrak{m}_p$ to be the value of f. Note that the above construction required that $\mathcal{O}(U)$ is a *local ring*. This is evidently the case when $\mathcal{O}(U) = C^\infty(U)$, however in general sheaf theory this ring will not in general be local. In fact, $\mathcal{O}(U)$ will not even need to be a ring! When each $\mathcal{O}(U)$ is a ring, then we shall say that X is a *ringed space*, and when all the germs are local rings, then we shall say that X is a *locally ringed space*.

A final word on generalization: we have been working over a topological space, however we may abstract further and work over a general category; in this way, we would define the notion of a "topology" on a category, known as a *Grothendeick topology* (a category with this structure is called a *site*), and develop sheaf theory accordingly. Though this is fruitful, this generlization shall be ommited for now to concentrate on the geoemtric properties at hand when using a topological space.

1.1 Definitions and Examples

We start by formalizing the above intuitions.

Definition 1.1.1: Presheaf Axiomatic Definition

Let X be a topological space. Then a *presheaf* over a category $\mathbf{C}$ on X, denoted $\mathscr{F}_X$ contains the following data:

1. for each open set $U \subseteq X$, there is an object c of $\mathbf{C}$ and denote it by $\mathscr{F}(U)$. If C is a category whose objects are sets with additioanl structure, then the elements of $\mathscr{F}(U)$ are called the *sections* of $\mathscr{F}$ over U. If we say that s is a section of $\mathscr{F}$, then we mean that s is a section of $\mathscr{F}$ over X.

2. For each inclusion map $U \hookrightarrow V$, there is a morphism $\text{res}_{V,U} : \mathscr{F}(V) \to \mathscr{F}(U)$ called the *restriction* morphism (or *restriction map*) such that

   (a) $\text{res}_{U,U} = \text{id}_U$

   (b) If $W \hookrightarrow V \hookrightarrow U$, then the following diagram commutes

$$\begin{tikzcd}
 & \mathscr{F}(V) \arrow[dr, "\text{res}_{V,W}"] & \\
\mathscr{F}(U) \arrow[ur, "\text{res}_{U,V}"] \arrow[rr, "\text{res}_{U,W}"'] & & \mathscr{F}(W)
\end{tikzcd}$$

The collection of all presheaves on a topological space X is denoted $\mathbf{PSh}_{\mathbf{C}}(X)$ (or $\mathbf{PSh}(X)$ if unambiguous). If U is clear from context, we shall often write $\text{res}_{U,V} f$ as $f|_V$.

The nomenclature "section" will be made clear in proposition 1.1.13. When it is no ambiguous the image of the restriction morphism for $s \in \mathscr{F}(U)$ shall be denoted:

$$\text{res}_{U,V}(s) = s|_V$$

This notation can be ambiguous as it is not explicit which $U \subseteq X$ is taken from which $s \in \mathscr{F}(U)$, however it will be unambiguous in-context.

When constructing sections of non-affine scheme's in chapter 2, it shall be important to have internalized the following consequence of the restriction maps. Let $U = U_1 \cup U_2$ and $W = U_1 \cap U_2$. Then $W \subseteq U_1 \subseteq U$ and $W \subseteq U_2 \subseteq U$. Then by definition of the restriction map, the following diagram commutes:

$$\begin{tikzcd}
 & \mathscr{F}(U_1) \arrow[dr, "\text{res}_{U_1,W}"] & \\
F(U) \arrow[ur, "\text{res}_{U,U_1}"] \arrow[rr, "\text{res}_{U,W}"] \arrow[dr, "\text{res}_{U,U_2}"'] & & F(W) \\
 & \mathscr{F}(U_2) \arrow[ur, "\text{res}_{U_2,W}"'] &
\end{tikzcd}$$

Importantly:

$$f|_W = (f|_{U_1})|_W = (f|_{U_2})|_W$$

The key take-away is that a larger section on U can be restricted to different sections on an open cover of U, but these sections map to the same restriction on intersections. In this way, presheaves preserve elements "going down". We shall see example of how "going up" fails after introducing some more basic terminology.

For practical computational purposes, it shall be important to re-interpret a presheaf in more categorical language, Notice that it is equivalent to define a presheaf as a *contravariant functor*. Namely

Definition 1.1.2: Presheaf Functorial Definition

Let X be a topological space and let $O(X)$ be the category whose objects are the open sets of X and whose morphism are $\subseteq$ (i.e. the category induced by the partial order given by $\subseteq$). Then a *presheaf* over $\mathbf{C}$ is a contravariant functor from $O(X)$ to $\mathbf{C}$, that is a functor $\mathscr{F} : O(X)^{op} \to \mathbf{C}$.

Usually, our presheaves shall be over **Set**, **Ab**, Mod_k or **CRing**[6].

An important notion for geometric objects is their behavior locally. The extreme (or limit) of the notation of locality is the behavior around a point p. This idea originated when studying germs of differential functions in Differential Geometry. These are well-defined on presheaves;

Definition 1.1.3: Stalk as Equivalence Relation

let $(X, \mathcal{O}_X)$ be a presheaf. Then the *stalks* at $p \in X$ is the collection of equivalence class:

$$[f] = \{(f, U) \mid p \in U,\ f \in \mathcal{O}(U)\}$$
$$(f, U) \sim (g, V) \qquad \Leftrightarrow \qquad W \subseteq U, W \subseteq V, p \in W,\ \text{res}_{U,W} f = \text{res}_{V,W} g$$

The stalks at p is usually denoted $\mathcal{O}_{X,p}$ or $\mathcal{O}_p$ if unambiguous in context.

If $\mathbf{C}$ is a category whose objects are augmented sets[a], then if $p \in U$ and $f \in \mathscr{F}(U)$, $[f] \in \mathscr{F}_p$ is called the *germ* of f at p.

[a] ex. **Set**, **Ring**, **Grp**, etc.

Note that in general, the notion of "evaluation" of a stalk/germ as it is usually defined on differential geometry won't make sense; though the symbol f is written the open sets need not contain functions or objects that act like function. We shall later define nice presheaves in which we may recover the notion of evaluation and treat the elements like functions. There is again a categorical way of interpreting stalks that is useful to us:

Definition 1.1.4: Stalks As Colimits

Let $\mathscr{F} \in \mathbf{PSh}_{\mathbf{C}}(X)$ be a presheaf where $\mathbf{C}$ is a category with small filtered colimits. Then a *stalk* at $p \in X$ is:

$$\mathscr{F}_p := \varinjlim \mathscr{F}(U)$$

that is $\mathscr{F}_p$ is the colimit over all $\mathscr{F}(U)$ where $p \in U$.

The adjectives "small filtered colimit" are only there to guarantee that the colimit exists. Most base category that will be worked over shall have the stronger property of bieng *cocompelte* (having all colimits) and often also being *complete* (having all limits). Categories that are complete and cocomplete include:

[6] Technically, **Ab** is $\mathbf{Mod}_{\mathbb{Z}}$, however distinguishing these two is useful as we usually mean k to be a field

1. **Set** (Category of Sets)
2. **Grp** (Category of Groups)
3. **Ab** (Category of Abelian Groups)
4. **Ring** (Category of Rings)
5. R-**Mod** (Category of modules over a ring R)
6. **Top** (Category of Topological Spaces)

Hence, this shows us that as long as the colimit exist in our category, we may define the notion of stalks for presheaves over that category without needing to reference the notion of elements. Note that it is the colimit and not the limit since we need $\mathscr{F}_p$ to "represent" the behaviour around p rather than "embed" germs.

One of the most important properties of stalks is their ability to be defined locally. First, if $U \subseteq X$ is an open set, we can define the sub-presheaf on U by letting $\mathcal{F}|_U(V) = \mathcal{F}(V \cap U)$. The following proposition will be really useful later when we need to work with stalks between larger and smaller (pre)sheaves, and mirrors the property in differential geometry (recall [7, chapter 2.1])

Proposition 1.1.5: Stalk Isomorphism

Let X be a presheaf of sets and $U \subseteq X$ an open set. Then for any $x \in U$:

$$\mathscr{F}_{X,x} \cong_{\mathbf{C}} \mathscr{F}_{U,x}$$

***Proof*:**
We shall prove this in the case where $\mathbf{C} = \mathbf{Set}$. The general case is left as an exercise.

Let us start by defining the map $\varphi : \mathscr{F}_{X,x} \to \mathscr{F}_{U,x}$. Take $[(f, V)] \in \mathcal{F}_{X,x}$ such that $x \in V$. As $x \in U$, $x \in U \cap V$. Hence, notice that:

$$(f, V) \sim (f|_{U\cap V}, U \cap V) \qquad \text{as} \qquad \operatorname{res}_{U,U\cap V} f = \operatorname{res}_{U\cap V,U\cap V} f|_{U\cap V} = f|_{U\cap V}$$

Then define the map between stalk via:

$$\varphi : (f, V) \sim (f|_{U\cap V}, U \cap V) \mapsto (f|_{U\cap V}, U \cap V)$$

This map is well-defined. Take $(f, V_1) \sim (g, V_2)$ where $V_1, V_2 \subseteq X$ so that we have to show that $(f|_{U\cap V_1}, U \cap V_1) \sim (g|_{U\cap V_2}, U \cap V_2)$. As $(f, V_1) \sim (g, V_2)$ there exists $W \subseteq V_1, V_2$ such that

$$f|_W = g|_W$$

Take $U \cap W \subseteq W \subseteq V_1, V_2$. Then by the restriction axioms we have that:

$$f|_{U\cap W} = g|_{U\cap W}$$

So that $(f|_{U\cap W}, U \cap W) \sim (g|_{U\cap W}, U \cap W)$. Then as:

$$\operatorname{res}_{U\cap V_1,U\cap W}(f|_{U\cap V_1}) = \operatorname{res}_{U\cap W,U\cap W}(f|_{U\cap W})$$

(and similarly for V_2), we get:

$$(f|_{U\cap V_1}, U\cap V_1) \sim (f|_{U\cap W}, U\cap W) \sim (g|_{U\cap W}, U\cap W) \sim (g|_{U\cap V_2}, U\cap V_2)$$

and hence the map is well-defined. This map has a natural inverse, namely

$$\varphi^{-1} : \mathscr{F}_{U,x} \to \mathscr{F}_{X,x} \qquad (f, V) \mapsto (f, V)$$

which can be shown to be well-defined and $\varphi \circ \varphi^{-1} = \varphi^{-1} \circ \varphi = \mathrm{id}$, hence these sets are bijective, which are the isomorphisms in set. To generalize to categories that are augmented sets, take advantage of the restriction maps being morphisms.

Now, presheaves allow for the notion of restricting and localizing, that is there is structure preserved "going down". But there are no condition on any structure being preserved "going up". In particular, not all presheaves are amenable to gluing local information to create more global information (think of bounded holomorphic functions on open subsets of $\mathbb{C}$). We must add this condition axiomatically:

Definition 1.1.6: Sheaf using Axioms

Let $\mathscr{F} \in \mathbf{PSh}_C(X)$ be a presheaf over a category $\mathbf{C}$ whose objects are sets possibly with additional structure. Then $\mathscr{F}$ is also a sheaf if it satisfies the following two axioms;

1. *Gluing Axiom (Existence)*: Let $U = \bigcup_i U_i$. If there exists a collection $f_i \in \mathscr{F}(U_i)$ such that $\mathrm{res}_{U_i, U_i\cap U_j} f_i = \mathrm{res}_{U_j, U_i\cap U_j} f_j$, then there is a $f \in \mathscr{F}(U)$ such that $\mathrm{res}_{U,U_i} f = f_i$ for all i.

2. *Locality Axiom (Uniqueness).* Let $U = \bigcup_i U_i$. If $f_1, f_2 \in \mathscr{F}(U)$, then if $\mathrm{res}_{U,U_i} f_1 = \mathrm{res}_{U,U_i} f_2$ for all i, then $f_1 = f_2$

3. $\mathscr{F}(\emptyset)$ is associated with the terminal object (if we're working over set, $\mathscr{F}(\emptyset)$ is a singleton).

If only the locality axiom is satisfied, then our presheaf is called a *separated presheaf.* The collection of all sheaves on a topological space X is denoted $\mathbf{Sh}_C(X)$ (or $\mathbf{Sh}(X)$ if unambiguous)

We may think of the locality axiom telling us there is *at most* one way to glue, while the gluing axiom is telling us there is *at least* one way to glue. A separated presheaf is so named as it allows sections to "separate" points (there a cannot be two sections identical on restrictions of an open covering).

1.1.7 Separation and Evaluation Later, the notion of "evaluation" shall be defined for sections where the objects of $\mathbf{C}$ behave like sets of functions. Being separated *does not* imply a function is defined by the evaluation at every point as the intuition of separated algebra for the Stone-Weierstrass Theorem would imply. This is because evaluation on locally ringed spaces will carry more data that needs to be taken into account (ex. see lemma 2.2.6).

The following is sometimes (redundantly) included in the definition of a presheaf:

Lemma 1.1.8: Sheaf and Zero Section

Let $\mathbf{C}$ be a category with the zero object 0 and $\mathscr{F}$ a sheaf (or particularly, a separated presheaf) on X. Let $\bigcup_{i\in I} U_i = X$ be an open cover and $s \in \mathscr{F}(X)$. Then if $s|_{U_i} = 0$ for all i, then $s = 0$.

***Proof*:**
exercise.

Notice that we are taking elements of each $\mathscr{F}(U)$ because $\mathscr{F}(U)$ is an object in a category whose objects are sets with possibly additional structure. This is a deviation from the generalization presented in the definition of presheaves as $\mathbf{C}$ may be more general. There is in fact a way to define a sheaf for general categories:

Proposition 1.1.9: Sheaf Categorical Interpretation

Let $\mathscr{F} \in \mathbf{PSh}(X)$ be a presheaf, and let $\cdot$ be a terminal object (in **Set**, an arbitrary 1-point set). Let $U = \bigcup_i U_i$, and consider:

$$\cdot \to \mathscr{F}(U) \to \prod_{i\in I} \mathscr{F}(U_i) \rightrightarrows \prod_{i,j\in I} \mathscr{F}(U_i \cap U_j)$$

for any indexing set I. Then if the sequence is exact at $\mathscr{F}(U)$, the locality axiom holds, and if it is exact at $\prod \mathscr{F}(U_i)$, the gluing axiom holds.

***Proof*:**
This shall be proven when $\mathbf{C} = \mathbf{Ab}$, the general result can be proven by the reader interested in practicing their category theory.

Assume $\mathscr{F}(U) \xrightarrow{\alpha} \prod \mathscr{F}(U_i)$ is injective so that

$$s \hookrightarrow (s|_{U\cap U_i})_i$$

Then by definition of injectivity any s_a, s_b mapping to this collection of restricted sections is equal, $s_a = s_b$, giving the locality.

Now, let $s_i = \mathscr{F}(U_i)$. Take the equalizer:

$$\prod_{i\in I} \mathscr{F}(U_i) \xrightarrow{\beta,\gamma} \prod_{j\in J} \mathscr{F}(U_i \cap U_j)$$

where:

$$\beta((s_i)_i) = \left(s_i|_{U_i\cap U_j}\right)_{i,j\in I} \qquad \text{and} \qquad \gamma((s_i)_i) = \left(s_j|_{U_i\cap U_j}\right)_{i,j\in I}$$

For example Let $I = \{1, 2\}$ and $U_1 \cup U_2 = U$. Let us not put in the "redundant" elements for the pairs $(1, 1), (2, 2)$, and $(2, 1)$; the last pair is not automatically excluded in the definition to avoid putting an unnecessary ordering on I, and the former two are not excluded for the simplicity of notation. As we are working with abelian groups, we can turn the equalizer into:

$$\cdot \to \mathscr{F}(U) \to \prod_{i\in I} \mathscr{F}(U_i) \xrightarrow{s_i|_{U_i\cap U_j} - s_j|_{U_i\cap U_j}} \prod_{i,j\in I} \mathscr{F}(U_i \cap U_j)$$

Thus the kernel of the equalize is when these two are equal, namely:

$$\beta(s_1, s_2) = (s_1|_{U_1 \cap U_2}) \qquad \text{and} \qquad \gamma(s_1, s_2) = (s_2|_{U_1 \cap U_2})$$

are equal:

$$(s_1|_{U_1 \cap U_2}) = \beta(s_1, s_2) = \gamma(s_1, s_2) = (s_2|_{U_1 \cap U_2})$$

Giving the gluing condition. In general, let us denote the kernel of the equalizer $\mathrm{Eq}(\beta, \gamma)$. By exactness we get that:

$$\mathrm{im}(\alpha) = \mathrm{Eq}(\beta, \gamma)$$

Thus, every element in the equalizer comes from a section in U. If α is also a monomorphism, it is unique, as we sought to show.

1.1.10 Foreshadowing Čech Cohomology

What if we extend this to have further equalizers? Sheaves that satisfy higher-order intersection-exactness conditions are a proper subset of sheaves (over non-trivial categories and topological spaces). Such sheaves can be thought of as never having any gluing obstruction, and hence have trivial cohomology. We shall return to these ideas when exploring Čech cohomology in section 6.1.1.

Example 1.1.11: Presheaves failing exactness (i.e. gluability and locality)

Let X be any topological space and define:

$$\mathcal{F}(X) = \mathbb{Z} \qquad \mathcal{F}(U) = 0 \qquad (\text{for } U \subsetneq X)$$

Then this sheaf is gluable, but not separated. If $U \subsetneq X$ is covered, then it is clear that gluing must hold. If $U = X = \cup_i U_i$, then that restrictions holds as any $z \in \mathbb{Z}$ has $z|_{U_i} = 0$ and so they certainly agree on all intersections. Hence, we see that each $z \in \mathbb{Z}$ is glued by a bunch of 0-elements!

Example 1.1.12: Presheaves And Sheaves

Most of these examples will be on sheaves of sets.

1. Verify that the sheaf of differentiable functions over M, namely $\mathcal{O}_M$, is indeed a sheaf. More generally, if X is any topological space, then the data defined by $\mathscr{F}(U) = \mathrm{Hom}_{\mathbf{C}}(U, \mathbb{R})$ where $\mathbf{C}$ is either **Top** or **Set** is a sheaf.

   Note how in a sheaf, we are "forced" to have interesting global behaviour if there are local section on non-intersecting sets. Consider $U_1 = (0, 1)$ and $U_2 = (3, 4)$. Then $U_1 \cap U_2 = 0$, so $\mathscr{F}(U_1 \cap U_2) = \{*\}$ is the terminal object (a singleton). If there is a function $f \in \mathscr{F}(U_1)$ and $g \in \mathscr{F}(U_2)$, then the restriction map

$$\mathrm{res}_{U_1,\emptyset} f = * = \mathrm{res}_{U_2,\emptyset}$$

   Meaning they *have* to agree on intersection. Thus, by the gluability axiom, there must exist a function F on $U_1 \cup U_2$ that restricts to f, g. For functions, this is obvious as it is the function defined by:

$$F(x) = \begin{cases} f(x) & x \in (0,1) \\ g(x) & x \in (3,4) \end{cases}$$

This becomes more interesting when the category in question is not obviously interpret able as functions on a space (ex. some sheaves represent logical proposition, see ref:HERE).

2. (*Restriction of Sheaves*) Let $\mathscr{F} \in \mathbf{Sh}(X)$ and $U \subseteq X$ an open subset of X. Then the *restriction* of $\mathscr{F}$ to U, denoted $\mathscr{F}|_U$ has as data for each open $V \subseteq U$ $\mathscr{F}|_U(V) = \mathscr{F}(V)$ (note hos it is important that U is open for well-definedness). We shall later see how we may define restrictions of sheaves to arbitrary sets[a]

3. (*Skyscraper Sheaf*) Let X be a topological space, $p \in X$, and S be any set. Let $\iota_p : \{p\} \to X$ be the inclusion map. Then define the sheaf of sets $(\iota_p)_* S$ on X (not on S!) to be:

$$(\iota_p)_* S(U) = \begin{cases} S & p \in U \\ \{e\} & p \notin U \end{cases}$$

This sheaf is called the *skyscraper sheaf.* The name comes from the fact that the informal picture for this sheaf looks like a skyscraper at p. For sheaves of Abelian groups, $\{e\}$ is the trivial group. More generally, $(\iota_p)_*(U)$ would be the final object if $p \notin U$.

4. (*Constant Presheaf*) Let X be a topological space and S be any set. Define $\underline{S}_{\text{pre}}(U) = S$ for all open $U \subseteq X$ Then it should be verified that $\underline{S}_{\text{pre}}$ forms a presheaf on X, called the *constant presheaf associated to* S. This set is not in general a sheaf if $|S| > 1$: take $X = U_1 \coprod U_2$ to be the disjoint union of two (open) sets, take the constant presheaf so that all restriction maps are constant maps. Chose $s_1 \in S$ and $s_2 \in S$ which are distinct: $s_1 \neq s_2$. Then by the gluing axiom, there would have to exist a $s \in S$ such that $s|_{U_1} = s_1$ and $s|_{U_2} = s_2$ and their intersection is empty and hence tautologically follow the gluing axiom, however as we have the identity map we have $s_1 = s = s_2$ however $s_1 \neq s_2$.

   This is fixed by a process called *sheafification* which will add the appropriate section equivalent to (s_1, s_2) that would restrict to both of these elements.

   What makes the sky-scrapper sheaf special is the "distinguishing" of a point making it act like a generalization of an indicator function. In the above example, one of the two sets will not have the point p.

5. (*Constant Sheaf*) Again let X be a topological space and S be any set. For every open $U \subseteq X$, let $\mathscr{F}(U)$ be the collection of all *locally constant* continuous functions from U to S, that is if $f \in \mathscr{F}(U)$, for each $p \in U$ there is an open neighborhood $V \subseteq U$ containing p, $p \in V \subseteq U$, such that $f|_V : V \to S$ is constant (on euclidean space, this is equivalent to having a constant function on each connected component). Then this sheaf is called the *constant sheaf associated to* S, and is denoted $\underline{S}$.

   Another way of describing this sheaf is if we endow S with the discrete topology and let $\mathscr{F}(U)$ be the continuous maps $U \to S$.

6. A great example of a pre-sheaf on $\mathbb{C}$ that is not a sheaf is the sheaf of bounded holomorphic functions on $\mathbb{C}$. Recall that bounded entire holomorphic functions (defined on all of $\mathbb{C}$) must be constant by Louisville's Theorem, however there do exist bounded functions on open subsets of $\mathbb{C}$. For example , on each disk of radius r, $\mathbb{D}_r^2$, the holomorphic function z^2 is

bounded but $\mathbb{C} = \cup_r \mathbb{D}_r^2$ has no bounded holomorphic function for which the restriction to any disk $\mathbb{D}_r^2$ is z^2; hence the gluing axiom fails.

Another example would be the sheaf composed of holomorphic functions admitting holomorphic square roots, as there is no global square-root function on $\mathbb{C}$ due to non-trivial monodromy.

7. (*Sheaf of continuous maps*) Let X and Y be topological spaces. Define $\mathscr{F}(U) = \mathrm{Hom}_{\mathbf{Top}}(U, Y)$. This sheaf is a generalization sheaf of differentiable functions and the constant sheaf, namely

   - The sheaf of differentiable functions has $Y = \mathbb{R}$
   - the constant sheaf has $Y = S$ with the discrete topology.

   If Y is also a topological group, then the continuous maps to Y form a sheaf of groups.

8. (*Sheaf of section of a maps*) This is an important special case of the above sheaf. Let X, Y be topological spaces and let $\mu : Y \to X$ be a continuous map. Then the sections of μ (in the sense of the existence of a map $s : X \to Y$ such that $\mu \circ s = \mathrm{id}$) form a sheaf. In particular, for each open $U \subseteq X$, $\mathscr{F}(U)$ is the set of section maps $s : U \to Y$ (continuous maps $s : U \to Y$ such that $\mu \circ s = \mathrm{id}|_U$). Vector bundles and vector fields are a good example of such a sheaf.

9. The following examples shows how different sheaves can be put on the same object to characterize different type of information. Throughout this example, let $X = \mathbb{A}_k^1$ be the (classical) affine line (see definition 0.1.9). The open subsets in the Zariski topology are all cofinite subsets. If $U \subseteq \mathbb{A}_k^1$ is open let $a_1, ..., a_n$ be the points that are omitted from U.

   (a) (Sheaf of regular function) Then letting $f_i = (x - a_i)$ and $F = \{f_1, ..., f_n\}$, define $\mathscr{F}(U) = k[x]_F$ i.e. $\mathscr{F}(U)$ is the ring generated by $k[x]$ and $1/f_i$ for each i. The reader should verify that this is indeed a pre-sheaf, and even a sheaf.

   (b) (Sheaf of invertible functions) Let X be as above, and let $\mathscr{F}(U) = (k[x]_F)^*$, that is consider only the invertible functions. Then X is again a sheaf.

   (c) (sheaf of roots of unit) Again consider X as above. Then let $\mathscr{F}(U)$ be the subgroup of $k[x]_F^*$ consisting of all roots of unity present in $\mathscr{F}(U)$. This forms a sheaf

   (d) (Sheaf of differential forms) If the reader is familiar with differential geometry, let $k = \mathbb{R}$ and let $\mathscr{F}(U)$ be the collection of all closed forms on U (or more generally, all smooth forms on U). Note that only $\mathscr{F}(\mathbb{A}_{\mathbb{R}}^1)$ are all the closed forms exact.

[a] see ref:HERE

If $\mathscr{F} \in \mathbf{PSh}(U)$ and $f \in \mathscr{F}(U)$ for some open $U \subseteq X$, then f is called a section. The following proposition motivates this idea

Proposition 1.1.13: Space of Sections of (Pre)Sheaf: étale spaces

Let $\mathscr{F}$ be a presheaf of sets. Then there exists a topological space F and a local homeomorphism $\pi : F \to X$ so that $\mathscr{F}$ can be represented as the sheaf of sections from F to X. The space F is called the *étalé space*. In particular, F is the *étalé space* of $\mathscr{F}$.

Furthermore, there is an equivalence between the category $\mathbf{Sh}(X)$ and the category of étalé spaces over X (with bundle morphisms as morphisms).

The idea is that (pre)sheaves correspond to a space F that is it is locally homeomorphic to X. Thus, geometrically sheaves could be thought of as local homeomorphisms, though the total space F may be rather ugly as shall be shown in example 1.1.14. Note that local homeomorphisms need not be surjective; this is a common misconceptions as many examples of surjective local homeomorphisms are covering spaces, but examples such as the sky-scrapper sheaf should show why this need not be the case.

***Proof*:**
Let F be the disjoint union of the stalks on F.

$$F = \coprod_{x \in X} \mathscr{F}_x$$

Then $\pi : F \to X$ is naturally a set map. Define a basis for F as follows: for each $s \in \mathscr{F}(U)$, take the collection

$$\{[s]_x \in \mathscr{F}_x \mid x \in U\}$$

These form a basis for F (note that each stalk has the discrete topology). With the setup, it is now easy to see the equivalence of categories.

The étale space may seem like it ought to be pretty similar to X being locally homeomorphic, however this can be far from the case:

Example 1.1.14: Non-Hausdorff Étale Space

Take $X = \mathbb{R}$ be the real numbers and take $\mathcal{F}$ to be the smooth functions on $\mathbb{R}$. Take the constant germ around 0 as well as the germ which can be represented by the function:

$$g(x) = \begin{cases} x\sin(1/x) & x\sin(1/x) > 0 \\ 0 & x\sin(1/x) \leq 0 \\ 0 & x < 0 \end{cases}$$

In other words, g is zero when x is negative and when the function $x\sin(1/x)$ would be negative; we want to keep when $x\sin(1/x)$ is positive. Then these two are different points in the étale space, but *cannot* be separated by open sets.

Note that if we were working with $X = \mathbb{C}$ and took analytic or holomorphic functions, then this cannot happen as we have a non-isolated 0 without all of g being zero. This turns out to be the main impingement for the étale space being Hausdorff. In general sheaves of analytic functions shall form Hausdorff étale spaces.

1.1.1 Stalks and Sheaves

As mentioned earlier, presheaves induce stalks. It was mentioned that sheaves but *not* presheaves can be characterized on the level of stalks, meaning the stalks of a sheaf shall give us all the information about the morphism a sheaf. This in fact almost characterizes sheaves from presheaves, as it in fact characterizes *separated presheaves* (recall they satisfy the locality but not gluing axiom). The following shows the value of working with separated presheaves:

Lemma 1.1.15: Sections Determined By Germs

Let $\mathscr{F}$ be a sheaf (more specifically, a separated presheaf). Then the map:

$$\mathscr{F}(U) \to \prod_{p \in U} (\mathscr{F}_p)$$

is injective

This map being injective justifies the term "separated", namely the points are separated by the sections.

***Proof*:**
Let f be the map and consider $f(s) = (s_x)_{x \in U}$ Say:

$$(s_x)_{x \in U} = (t_x)_{x \in U} \qquad \text{equiv.} \qquad [(s, V_x)]_{x \in U} = [(t, W_x)]_{x \in U}$$

Then by definition there exists a $C_x \subseteq V_x \cap W_x$ such that

$$s|_{C_x} = t|_{C_x}$$

Now as $x \in C_x$ for each $x \in U$ we have $U = \bigcup_{x \in U} C_x$ and furthermore for for any pair C_x, C_y in the open cover we have:

$$\text{res}_{C_x, C_x \cap C_y} s = \text{res}_{C_y, C_x \cap C_y} t$$

hence, by the gluing axiom $s = t$, as we sought to show.

Note that we can show the locality axiom form this argument:

Corollary 1.1.16: Global Section Over Local Sections

Let $\mathscr{F}$ be a sheaf on X and $U_i \subseteq X$ is an open cover. Then:

$$\mathscr{F}(X) \to \prod_i \mathscr{F}(U_i)$$

is injective

***Proof*:**
Map $s \mapsto \left(s|_{U_i}\right)_i$. Consider

$$\left(s|_{U_i}\right)_i = \left(t|_{U_i}\right)_i$$

then s and t agree on intersections of U_i, hence there exists a unique element in $\mathcal{F}(X)$ for both these elements, namely $s = t$

In example 1.1.11, the sheaf $\mathcal{F}(X) = \mathbb{Z}$ and $\mathcal{F}(U) = 0$ for $U \subsetneq X$ was not a separated presheaf, and certainly $\mathcal{F}(X) \to \prod_{x\in X} \mathcal{F}_x$ is *not* injective, as each $\mathcal{F}_x$ consist of a single equivalence class representing the zero element (and hence $\prod_{x\in X} \mathcal{F}_x \equiv \{[0]\}$, and $\mathcal{F}(X) = \mathbb{Z}$. This example shows that we can somehow construct a nonzero object from a bunch of zero-objects. This can be thought of as being the key-blocker, especially in the case of abelian categories where most of the time the pre-image of 0 is only 0. This motivate the following:

Definition 1.1.17: Support of Sections

Let $\mathscr{F}$ be a sheaf of an abelian group on X, and s a global section of $\mathscr{F}$. Then the *support* of s, denoted $\mathrm{Supp}(s)$, is the points $p \in X$ where s has a nonzero germ:

$$\mathrm{Supp}(s) := \{p \in X \mid s_p \neq 0 \text{ in } \mathscr{F}_p\}$$

This notion differs a bit from the usual notion of support in analysis, in particular notice that it is the germ itself that is nonzero and not that the evaluation of the germ s_p at the point p that is nonzero.

Lemma 1.1.18: Support is Closed

Let $\mathscr{F}$ be a sheaf of abelian groups on X and $s \in \mathscr{F}(X)$. Then $\mathrm{Supp}(s)$ is closed

***Proof*:**
We'll show $\mathrm{Supp}(s)^c$ is open, that is the set of all points $p \in X$ such that $s_p = 0$ is open. If $s_p = 0$ in the stalk, there exists an open subset $W_p \subseteq X$ such that $s|_{W_p} = 0$. Taking the union of arbitrary open sets is open, and hence taking the union of all W_p cover $\mathrm{Supp}(s)^c$ making $\mathrm{Supp}(s)$ closed, completing the proof.

The equivalent in analysis is the definition of the *closed support* of a function. Due to this, the compliment of the support of an element s is the 0-section. The section can also be defined on presheaves, however there are examples of presheaves where there are nonzero sections that "live nowhere" because it is 0 in every stalk[7]

Let us now take a closer look at $\prod_{p\in U}(\mathscr{F}_p)$. This set is much bigger than sections of a sheaf. We thus take the special sub-collection that associates with $\mathscr{F}(U)$ (essentially characterizing the image of $\mathscr{F}(U)$:

[7]Take $\mathscr{F}$ to be the sheaf of holomorphic functions on $\mathbb{C}$, $\mathscr{G}$ the sheaf of bounded holomorphic functions. Then take the sheaf $\mathcal{H} = \mathscr{F}/\mathscr{G}$. Then show that $\mathcal{H}_x = 0$, however $\mathcal{H}(\mathbb{C}) = \{\text{entire functions}\}/\mathbb{C}$

Definition 1.1.19: Compatible Germs

Let $\mathscr{F}$ be a sheaf on X and $U \subseteq X$ an open set. Then we say an elements $(s_p)_{p\in U} \in \prod_{p\in U} \mathscr{F}_p$ consists of *compatible germs* if there exists some representative open sets U_p for s_p:

$$U_p \subseteq U \text{ open}, \ \tilde{s}_p \in \mathscr{F}(U_p)$$

such that the germ of $\tilde{s}_p$ for all $q \in U_p$ is s_q. In other words, if $\{U_i\}$ is an open cover of U where $p \in U_i$ for all i, then s_p is the germ of f_i at p.

Hence, a compatible germ is the collection of germs that are "compatible" when taking subsets, that is *restrictions*. Using the gluing axiom, it is a good exercise to show that any choice of compatible germs for a sheaf $\mathscr{F}$ over U is the image of a section of $\mathscr{F}$ over U, that is we have successfully identified the right hand side of the equation in lemma 1.1.15. Note how important it is that we use sheaves (or at least separated presheaves) for the proof. The proof also motivate the agricultural terminology of "sheaf", for in agriculture a sheaf is a collection of "Stalks" (which is the main stem of a plant).

1.1.20 Remark When working with sheaves of sets or some structures on sets, the elements of the section can literally be thought of as functions using the above definition, namely function of the form $s : U \to \coprod_{p\in U} \mathscr{F}_p$. The generalization presented above allows to generalize the notion of "open set U" so that the topological space X is replaced special category called a *site*, see ref:HERE.

1.1.2 Sheaf from a Topological Base

When defining a differential or topological structure, one of the strategies that is taken is to define pieces of the manifold in a compatible way, or said differently define it on some basis of the manifold. This section endeavours to do this for sheaves.

Let X be a topological space with a sheaf $\mathscr{F}$. Take a basis $\{B_i\}$ for X and take the subset of $\mathscr{F}$ consisting of $\mathscr{F}(B_i)$ along with all the appropriate restriction maps. Then there is enough Stalk information to recover all of $\mathscr{F}$ (show this), and hence we my want to start with such a collection to generate a sheaf. This collection alone will generate a presheaf, we will require the basis to satisfy the locality and gluability axioms:

Definition 1.1.21: Sheaf On A Base

Let X be a topological space and $\mathcal{B}$ be a topological base. Then this collection is caleld a *sheaf on a base* if it satisfies:

1. **Base locality axiom**: If $B = \cup_i B_i$, then if $f, g \in \mathscr{F}(B)$ are such $\mathrm{res}_{B,B_i} f = \mathrm{res}_{B,B_i} g$ for all i, then $f = g$.
2. **Base Gluability**: If $B = \cup_i B_i$, and $f_i \in \mathscr{F}(B_i)$ such that f_i agrees with f_j on any basic open set contained in $B_i \cap B_j$, then there exists $f \in \mathscr{F}(B)$ such that $\mathrm{res}_{B,B_i} f = f_i$ for all i

Theorem 1.1.22: Define Sheaf From Base

Let $\{B_i\}$ be a base for the topological space X and let F be a of sets on the base. Then there is a sheaf $\mathscr{F}$ extending F with isomorphisms $\mathscr{F}(B_i) \cong F(B_i)$ agreeing with restrictions. Furthermore, this sheaf is unique up to unique isomorhpism

***Proof*:**
The sheaf $\mathscr{F}$ will be defined as the sheaf of compatible germs of F. To this end, define the stalk of a presheaf F on a base at $p \in X$ by taking:

$$F_p = \varinjlim F(B_i)$$

Define:

$$\mathscr{F}(U) =$$
$$\{(f_p \in F_p)_{p\in U} \mid \forall p \in U, \exists B \text{ with } p \in B \subseteq U,\ s \in F(B) \text{ that is compatible: } s_q = f_q\ \forall q \in B\}$$

Then the map $F(B) \to \mathscr{F}(B)$ is an isomorphism.

Lots more to cover here that is worth going over

1.1.3 Gluing Sheaves on an Open Cover

Just as we can construct a sheaf from data on a topological base, we can construct a sheaf from sheaves defined on the pieces of an open cover, provided the local sheaves are compatible on overlaps. This result is foundational: it will be used in chapter 2 to construct the structure sheaf when gluing schemes.

Theorem 1.1.23: Gluing Sheaves On An Open Cover

Let X be a topological space and $\{U_i\}_{i\in I}$ an open cover. Suppose we are given:

1. a sheaf $\mathscr{F}_i$ on each U_i,
2. isomorphisms of sheaves $\varphi_{ij} : \mathscr{F}_i|_{U_i\cap U_j} \xrightarrow{\sim} \mathscr{F}_j|_{U_i\cap U_j}$ for all i, j

satisfying the cocycle condition on triple overlaps:

$$\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik} \quad \text{on } U_i \cap U_j \cap U_k$$

Then there exists a sheaf $\mathscr{F}$ on X, unique up to unique isomorphism, together with isomorphisms $\psi_i : \mathscr{F}|_{U_i} \xrightarrow{\sim} \mathscr{F}_i$ such that $\varphi_{ij} \circ \psi_i = \psi_j$ on $U_i \cap U_j$.

***Proof*:**
For each open set $V \subseteq X$, define:

$$\mathscr{F}(V) = \left\{(s_i)_{i\in I} \;\middle|\; s_i \in \mathscr{F}_i(V \cap U_i), \quad \varphi_{ij}\left(s_i|_{V\cap U_i\cap U_j}\right) = s_j|_{V\cap U_i\cap U_j}\ \forall i, j\right\}$$

with restriction maps defined componentwise: if $W \subseteq V$ then $\mathrm{res}_{V,W}((s_i)_i) = (s_i|_{W\cap U_i})_i$.

$\mathscr{F}$ is a presheaf. The restriction of a compatible tuple is compatible (as φ_{ij} commutes with restriction, being a sheaf morphism), and the identity and composition axioms are inherited componentwise.

Locality axiom. Let $V = \bigcup_\alpha V_\alpha$ and suppose $(s_i), (t_i) \in \mathscr{F}(V)$ satisfy $(s_i)|_{V_\alpha} = (t_i)|_{V_\alpha}$ for all α. Then for each i, we have $s_i|_{V_\alpha \cap U_i} = t_i|_{V_\alpha \cap U_i}$ for all α. Since $\{V_\alpha \cap U_i\}_\alpha$ is an open cover of $V \cap U_i$ and $\mathscr{F}_i$ is a sheaf (satisfying locality), $s_i = t_i$ for all i.

Gluing axiom. Let $V = \bigcup_\alpha V_\alpha$ and suppose $(s_i^{(\alpha)})_i \in \mathscr{F}(V_\alpha)$ agree on overlaps, i.e. $s_i^{(\alpha)}|_{V_\alpha \cap V_\beta \cap U_i} = s_i^{(\beta)}|_{V_\alpha \cap V_\beta \cap U_i}$ for all α, β, i. For each fixed i, the sections $s_i^{(\alpha)} \in \mathscr{F}_i(V_\alpha \cap U_i)$ agree on overlaps $V_\alpha \cap V_\beta \cap U_i$. As $\mathscr{F}_i$ is a sheaf, there exists a unique $s_i \in \mathscr{F}_i(V \cap U_i)$ with $s_i|_{V_\alpha \cap U_i} = s_i^{(\alpha)}$ for all α.

It remains to check the compatibility condition for the glued tuple $(s_i)_i$: we need $\varphi_{ij}(s_i|_{V \cap U_i \cap U_j}) = s_j|_{V \cap U_i \cap U_j}$. Both sides are sections of $\mathscr{F}_j(V \cap U_i \cap U_j)$ that agree on each $V_\alpha \cap U_i \cap U_j$ (since $(s_i^{(\alpha)})_i \in \mathscr{F}(V_\alpha)$ satisfies the compatibility condition). By the locality axiom for $\mathscr{F}_j$, they are equal.

Hence $\mathscr{F}$ is a sheaf. The isomorphisms $\psi_i : \mathscr{F}|_{U_i} \to \mathscr{F}_i$ are given by projection onto the i-th component: $(s_j)_j \mapsto s_i$, which is an isomorphism because the compatibility conditions determine all other components on U_i via the φ_{ij}. Uniqueness of $\mathscr{F}$ follows from corollary 1.3.4 applied to the open cover.

1.2 Morphisms and Exactness

Now that we have defined (pre)sheaves, it is natural to define morphism between sheaves to show that $\mathbf{PSh}(X)$ and $\mathbf{Sh}(X)$ form a category. Sheaf morphisms shall be over the same object, and hence can be thought of as the equivalent of looking at local homeomorphisms between two étale spaces:

Definition 1.2.1: Morphism Of Sheaves

Let $\mathscr{F}, \mathscr{G} \in \mathbf{PSh}(X)$ (note that $\mathbf{Sh}(X) \subseteq \mathbf{PSh}(X)$). Then a *morphism of (pre)sheaves* (over the chosen category) $\varphi : \mathscr{F} \to \mathscr{G}$ is a natural transformation between the two functors. Concretely, it contains the data $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ such that if $V \subseteq U$ then the following diagram commutes:

$$\begin{tikzcd}
\mathscr{F}(U) \arrow[r, "\varphi(U)"] \arrow[d, "\mathrm{res}_{U,V}"'] & \mathscr{G}(U) \arrow[d, "\mathrm{res}_{U,V}"] \\
\mathscr{F}(V) \arrow[r, "\varphi(V)"'] & \mathscr{G}(V)
\end{tikzcd}$$

that is

$$\mathrm{res}_{U,V}(\varphi(U)(-)) = \varphi(V)(\mathrm{res}_{U,V}(-))$$

The collection of morphisms between two sheaves is denoted $\mathrm{Hom}_{\mathbf{Sh}(X)}(\mathscr{F}, \mathscr{G})$ or $\mathrm{Mor}_{\mathbf{Sh}(X)}(\mathscr{F}, \mathscr{G})$

To emphasize when the domain and codomain of a presheaf morphism is a sheaf, the morphism shall be called a *sheaf morphism*. A useful geometric intuition of sheaf morphisms is that they apply a transformation to each map (ex. $\exp(f)(z) = e^{f(z)}$). With the morphisms and objects defined, it is easy to show that they form a category. We shall denote the category of sheaves over a category $\mathbf{C}$ on a space X as $\mathbf{C}_X$. Thus:

$$\mathbf{Set}_X \qquad \mathbf{Ab}_X \qquad \mathbf{Ring}_X$$

are the categories of sheaves over sets, abelian groups, and rings respectively. Since the morphisms are the same, the category $\mathbf{Sh}(X)$ is a full subcategory of $\mathbf{PSh}(X)$. For presheaves, we shall denote the category with an extra superscript:

$$\mathbf{Set}_X^{\text{pre}} \qquad \mathbf{Ab}_X^{\text{pre}} \qquad \mathbf{Ring}_X^{\text{pre}}$$

Example 1.2.2: Morphism of Presheaves and Sheaves

(If the reader is familiar with Riemann surface) Let X be a Riemann surface and $\mathcal{O}_X(-P)$ be the collection of meromorphic function $f \in K^*(X)$ such that $\text{div}(f) + (-p) \geq 0$ ($P \in X$ a point). Then if $\mathcal{O}_X$ is the sheaf of holomorphic functions on X there are sheaf morphisms:

$$0 \to \mathcal{O}_K(-P) \to \mathcal{O}_X \to \kappa_P \to 0$$

where κ_P is the sky-scrapper sheaf at P, namely:

$$\kappa_P = \begin{cases} (0) & x \neq P \\ \mathbb{C} & x = P \end{cases}$$

This is even an exact sequence of sheaves, but we need to build-up to show how to properly define exactness in $\mathbf{C}_X$.

From now on, we shall be working with *abelian categories* unless stated otherwise: they all have 0 objects, they are closed under finite products and coproducts, and their kernel/cokernels behave well with epimorphisms and monomorphisms; see section 0.4 for a refresher or [**NateClassicalAlgGeo**] for more details. We shall build-up to showing that $\mathbf{C}_X$ and $\mathbf{C}_X^{\text{pre}}$ are abelian categories.

As a presheaf morphism is a natural transformation, it important to emphasize that it does not make sense to say that a sheaf morphism is injective or surjective. Instead, it can be a *monomorphism* or an *epimorphism* and have corresponding kernel, cokernels, and images. Let us find objects in $\mathbf{PSh}(X)$ and $\mathbf{Sh}(X)$ that can represent these kernel, cokernel, and image. Starting with the kernel of a presheaf morphism $\varphi : \mathscr{F} \to \mathscr{G}$ which is defined to be be object $\mathcal{K}$ with the morphism $K : \mathcal{K} \to \mathcal{F}$ such that $\varphi \circ K = 0$ universally.

Proposition 1.2.3: Kernels in Presheaf Category

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between presheaves. Then $\ker \varphi$ can be defined to be the kernel of each $\varphi(U)$:

$$\mathcal{K}(U) = \ker(\varphi(U))$$

The resulting presheaf is denoted $\ker_{\text{pre}}(\varphi)$. Furthermore, if $\mathscr{F}$ is a sheaf and $\mathscr{G}$ satisfies at least the locality axiom (in particular if it is also a sheaf), then $\mathcal{K}$ is a sheaf.

> ***Proof*:**
> chase the diagram, and use a similar separated/locality condition in the case of a sheaf as was done before.

Note that as the representation of the kernel for sheaves is the same as the characterization of presheaves, we can just write $\ker(\varphi)$ without any ambiguity. As the monomorphism condition can be checked on open sets, the following usual definition of injectivity is recovered on the level of open sets:

Definition 1.2.4: Injective Presheaf and Sheaf Morphism

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between shaves or presheaves. Then φ is *injective* if $\ker \varphi(U) = 0$ for all open $U \subseteq X$.

The image of a presheaf morphism $\varphi : \mathscr{F} \to \mathscr{G}$ is defined to be be object $\mathscr{I}$ with the monomorphism $I : \mathscr{G} \to \mathscr{I}$ and a morphism $E : \mathscr{F} \to \mathscr{I}$ such that $\varphi = E \circ I$ universally, namely:

$$\begin{tikzcd}
\mathscr{F} \arrow[rr, "\varphi"] \arrow[dr, "E"] \arrow[ddr, bend right, "E'"'] & & \mathscr{G} \\
& \mathscr{I} \arrow[ur, "I"] \arrow[d, dashed, "\exists! v"] & \\
& \mathscr{I}' \arrow[uur, bend right, "I'"'] &
\end{tikzcd}$$

The reader familiar with category may be more familiar with images in abelian categories defined as:

$$\operatorname{im}(\varphi) = \ker(\operatorname{coker}(\varphi))$$

As we are working with abelian categories, $\ker(\operatorname{coker}(\varphi)) \cong \operatorname{coker}(\ker(\varphi))$, and hence there is no need of defining the coimage.

Proposition 1.2.5: Images in Presheaf and Sheaf Category

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between presheaves. Then $\operatorname{im}_{\text{pre}}\varphi$ can be defined to be the image of each $\varphi(U)$:

$$\operatorname{im}_{\text{pre}}\varphi(U) = \operatorname{im}(\varphi(U))$$

and similarly for cokernels:

$$\operatorname{coker}_{\text{pre}}(\varphi(U)) = \frac{\mathscr{G}(U)}{\operatorname{im}(\varphi(U))}$$

The resulting $\operatorname{im}_{\text{pre}}(\varphi)$ and $\operatorname{coker}_{\text{pre}}(\varphi)$ is called the *image presheaf* and *cokernel presheaf* respectively.

This is *not* necessarily the case if $\mathscr{F}$ is a sheaf. Images and cokernel in $\mathbf{C}_X$ will require a bit more work to be characterized.

> ***Proof*:**
> chase the diagram.

Definition 1.2.6: Surjective Presheaf Morphism

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between presheaves. Then φ is *surjective* if $\operatorname{im}(\varphi(U)) = \mathscr{G}(U)$ for all open $U \subseteq X$.

As noted earlier, images of sheaves need not be sheaves:

Example 1.2.7: Sheaves: difficulty With Cokernels and Images

1. Let $X = \mathbb{C}$ with the usual topology, $\underline{\mathbb{Z}}$ be the constant sheaf on X, $\mathcal{O}_X$ the sheaf of holomorphic functions, and $\mathscr{F}$ the *presheaf* of functions admitting a holomorphic logarithm. Then there is an exact sequence of presheaves on X:

$$0 \to \underline{\mathbb{Z}} \to \mathcal{O}_X \to \mathscr{F} \to 0$$

where $\underline{\mathbb{Z}} \to \mathcal{O}_X$ is the natural inclusion on each open set, and $\mathcal{O}_X \to \mathscr{F}$ is given by $f \mapsto \exp(2\pi i f)$ (again, over each open set). This is indeed a short exact sequence, for if $U \subseteq V \subseteq \mathbb{C}$, then:

$$\begin{tikzcd}
0 \arrow[r] & \underline{\mathbb{Z}}(V) \arrow[r] \arrow[d, "\text{res}"] & \mathcal{O}_X(V) \arrow[r] \arrow[d, "\text{res}"] & \mathscr{F}(V) \arrow[r] \arrow[d, "\text{res}"] & 0 \\
0 \arrow[r] & \underline{\mathbb{Z}}(U) \arrow[r] & \mathcal{O}_X(U) \arrow[r] & \mathscr{F}(U) \arrow[r] & 0
\end{tikzcd}$$

commutes. Certainly each $\underline{\mathbb{Z}}(V) \to \mathcal{O}_X(V)$ is an inclusion, and $\mathcal{O}_X(V) \to \mathscr{F}(V)$ is surjective (by definition) with kernel being the constant functions $\mathbb{Z}$ since if f admits a logarithmic inverse, then $f = \log(g)$, so $g \mapsto \exp(2\pi i \log g) = f$.

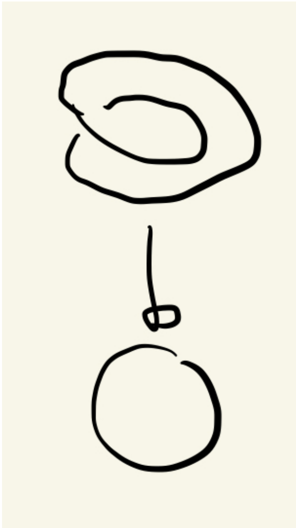
Then it should be clear that $\mathscr{F}$ is *not* a sheaf: take $U = \mathbb{C}^*$, cover it with $U_1 = \mathbb{C} \setminus \mathbb{R}_{\geq 0}$ and $U_2 = \mathbb{C} \setminus \mathbb{R}_{\leq 0}$. Then $\mathscr{F}(U)$, $\mathscr{F}(U_1)$, and $\mathscr{F}(U_2)$ each have multiple non-zero functions admitting logarithms (see [**NateComplexAnalysis**]). If $\mathscr{F}$ were a sheaf, then as z admits logarithmic inverse on U_1 and U_2 and tautologically $z|_{U_1 \cap U_2} = z|_{U_1 \cap U_2}$ there should be a function defined on U restricting to these two, that restricts to both of these, but no such function exists (consider that such a function would have to be continuous, but going around the origin introduces non-trivial monodromy).

2. Consider the following similar but subtly different example. Consider $\mathcal{O}_{\mathbb{C}}^*$ the sheaf of all non-vanishing holomorphic functions (this sheaf is strictly "bigger" than the above $\mathscr{F}$, ex. $\mathscr{F}(\mathbb{C}^*) \subsetneq \mathcal{O}_{\mathbb{C}}^*(\mathbb{C}^*)$). Consider the following short exact sequence of *sheaves*:

$$0 \to \underline{\mathbb{Z}} \xrightarrow{\iota} \mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}_{\mathbb{C}}^* \xrightarrow{!} 0$$

Notice that on $U = \mathbb{C}^*$, z does not vanish and so $z \in \mathcal{O}_{\mathbb{C}}^*(U)$. However, z does not admit a logarithmic inverse (again see [**NateComplexAnalysis**]). This would imply that $\exp(\mathbb{C}^*)$ is *not* surjective and so exp is not surjective on all open sets. And yet it is indeed the case that there exists a morphism $\xrightarrow{!}$, namely exp is an *epimorphism* on sheaves. This will be an example of an epimorphism that is *not* characterize by surjectivity on open sets, as shall be explained by defining sheafification after these examples.

3. For the reader less comfortable with complex analysis, here is a more rudimentary example: take the sheaf of continuous sections of the circle S^1. For our first sheaf of sections, let $\pi : S^1 \to S^1$ be the identity (the trivial bundle), and let $\mathcal{F}_1$ be the sheaf of sections of this bundle. The second sheaf of section will with the projection from the twisted circle, which can be visualized as:



Label this shape S^1_2. The second sheaf $\mathcal{F}_2$ is all the continuous sections of all the open subsets of the codomain for the map $\pi : S^1_2 \to S^1$ with a reasonable definition of projection map mirroring the above image (over the complex numbers, we can take $z \mapsto z^2$). Note that $\mathcal{F}_2(S^2) = 0$, i.e. it has no global sections. Then there is certainly a surjective continuous map φ satisfying the commutative diagram:

$$\begin{tikzcd} S^1_2 \arrow[rr, "\varphi"] \arrow[dr] & & S^1 \arrow[dl] \\ & S^1 & \end{tikzcd}$$

However, there is no surjective maps:

$$\mathcal{F}_2(X) \to \mathcal{F}_1(X)$$

namely since there no continuous section from S^1 to S^1_2 that would satisfy the projection condition.

In the above, the first example shows that exactness has no issue with pre-sheaves, and in the second example we claimed the short exact sequence of sheaves was still exact in the category of sheaves. Notice that the image of exp in $\mathcal{O}^*_{\mathbb{C}}$ was the presheaf $\mathscr{F}$ in the first example. What we shall show is that this image uniquely determines $\mathcal{O}^*_{\mathbb{C}}$. In the same vein in which we can from an abelian group via a abelian semi-group using groupification, or a normal subgroup by normalizing a subgroup, we can from a sheaf from a presheaf via sheafification. As is usual, this is done in a universal way that is adjoint to the forgetful functor. This sheafification shall be the solution to defining right notion of surjection for sheaves:

Theorem 1.2.8: Sheafification

Let $\mathscr{F}$ be a presheaf on X. Then there exists a sheaf $\mathscr{F}^{\mathrm{sh}}$ and a map $\mathrm{sh} : \mathscr{F} \to \mathscr{F}^{\mathrm{sh}}$ such that for any sheaf $\mathscr{G}$ and any presheaf morphism $g : \mathscr{F} \to \mathscr{G}$, there exists a unique morphism of sheaves $f : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}$ such that the following diagram commutes:

$$\begin{tikzcd}
 & \mathscr{F}^{\mathrm{sh}} \arrow[d, "{g,\ \text{sheaf mor.}}"] \\
\mathscr{F} \arrow[ur, "\mathrm{sh}"] \arrow[r, "f", "\text{presheaf mor.}"'] & \mathscr{G}
\end{tikzcd}$$

The sheafification of the presheaf $\mathscr{F}$ is often denoted $\mathscr{F}^+$.

The idea will be to remove all section that violate locality, and add sections to respect gluability. Another good take is the following post here.

***Proof*:**

Let $\mathscr{F}$ be the presheaf in the question. Define $\mathscr{F}^{\mathrm{sh}}$ as the set of compatible germs of the presheaf $\mathcal{F}$ over U, namely:

$$\mathscr{F}^{\mathrm{sh}}(U) := \{(f_p \in \mathscr{F}_p)_{p\in U} \mid \forall p \in U, \exists p \in V \subseteq U \text{ open and } s \in \mathscr{F}(V) \text{ such that } s_q = f_q, \forall q \in V\}$$

Note this resemble to the étale space. To see that $\mathscr{F}^{\mathrm{sh}}$ is a sheaf, first note that restriction is defined by restricting the domain of the tuple of germs. For the gluing axiom, if we have sections $s^{(i)} \in \mathscr{F}^{\mathrm{sh}}(U_i)$ agreeing on overlaps, we can define a global tuple s on $U = \bigcup U_i$ by setting $s_p = s_p^{(i)}$ for $p \in U_i$. This tuple belongs to $\mathscr{F}^{\mathrm{sh}}(U)$ because the condition of being "locally represented by a section of $\mathscr{F}$" is satisfied locally by the witnesses for each $s^{(i)}$.

We define the map $\mathrm{sh} : \mathscr{F} \to \mathscr{F}^{\mathrm{sh}}$ by sending a section $\sigma \in \mathscr{F}(U)$ to the tuple of its germs $((\sigma)_p)_{p\in U}$, which is clearly a presheaf morphism.

To prove the universal property, let $\mathscr{G}$ be a sheaf and $g : \mathscr{F} \to \mathscr{G}$ a presheaf morphism. We wish to construct a unique $f : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}$. Let $s \in \mathscr{F}^{\mathrm{sh}}(U)$. By definition, there exists an open cover $\{V_i\}$ of U and sections $\sigma_i \in \mathscr{F}(V_i)$ such that $s|_{V_i} = \mathrm{sh}(\sigma_i)$ (i.e., the germs of s match the germs of σ_i on V_i). We propose to define $f(s)$ by gluing the sections $g_{V_i}(\sigma_i) \in \mathscr{G}(V_i)$.

To do this, we must verify that $g(\sigma_i)$ and $g(\sigma_j)$ agree on $V_i \cap V_j$. For any point $p \in V_i \cap V_j$, we have $(\sigma_i)_p = s_p = (\sigma_j)_p$. By the definition of germs, there exists a small neighborhood $W_p \subseteq V_i \cap V_j$ such that $\sigma_i|_{W_p} = \sigma_j|_{W_p}$. Since g is a presheaf morphism, it respects restriction, so $g(\sigma_i)|_{W_p} = g(\sigma_j)|_{W_p}$. Because $\mathscr{G}$ is a sheaf (specifically, separated), and the sections $g(\sigma_i)$ and $g(\sigma_j)$ agree on a neighborhood of every point in the intersection, they must be equal on the entire intersection $V_i \cap V_j$. Thus, by the gluing axiom of $\mathscr{G}$, there exists a unique section $T \in \mathscr{G}(U)$ such that $T|_{V_i} = g(\sigma_i)$. We define $f_U(s) = T$. This map is unique because any morphism f must respect restriction, forcing it to agree with this local construction.

The “failure” of a presheaf being a sheaf can be measured as the lack of exactness of at the $\prod_i F(U_i)$ in proposition 1.1.9.

Example 1.2.9: Sheafification

To gain intuition, show the following sheafifications:

1. Show that $(\underline{S}_{\text{pre}})^{\text{sh}} = \underline{S}$, that is the presheaf of constant function sheafifies to the sheaf of constant functions.

2. Let $\mathscr{F}$ on $\mathbb{R}$ be all bounded continuous functions on each open set. Show that $\mathscr{F}^{\text{sh}}$ is *all* continuous functions on each open subset.

3. Show that $\mathcal{H}$ on $\mathbb{C}$ of non-vanishing holomorphic functions that admit a logarithm sheafify to all non-vanishing holomorphic functions.

4. Let $\mathscr{F}$ be a non-trivial presheaf where all stalks are trivial (see exercise ref:HERE). Show that $\mathscr{F}^{\text{sh}}$ is trivial.

Corollary 1.2.10: Sheafification and Stalks

Let $\mathcal{F} \in \mathbf{PSh}(X)$ be a presheaf. Then for all $p \in X$

$$\mathcal{F}_p \cong \mathcal{F}_p^{\text{sh}}$$

***Proof*:**

Use something similar given in the proof of proposition 1.1.5

Let us now put together the universal properties we've shown so far to get that the category of sheaves also have cokernels:

Proposition 1.2.11: Sheaf Morphisms Have Cokernels

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a sheaf morphism. Then it has a cokernel and it is the sheafification of the cokernel-presheaf.

***Proof*:**

Recall that the universal property of the cokernel can be seen as the colimit of the following diagram:

$$\begin{tikzcd} \mathcal{F} \arrow[r, "\varphi"] \arrow[d] & \mathcal{G} \\ 0 & \end{tikzcd}$$

As we know, the cokernel forms a presheaf $\mathcal{H}_{\text{pre}}$. By sheafification we have a unique map

$\mathrm{sh}:\mathcal{H}_{\mathrm{pre}}\to\mathcal{H}$. Now let $\mathcal{A}$ be any sheaf and consider the following diagram:

$$\begin{tikzcd}
\mathcal{F} \arrow[r, "\varphi"] \arrow[d] & \mathcal{G} \arrow[d] \arrow[rrdd, bend left] & & \\
0 \arrow[r] \arrow[rrrd] & \mathcal{H}_{\mathrm{pre}} \arrow[r, "\mathrm{sh}"] & \mathcal{H} \arrow[rd, dashed, "\exists!"] & \\
& & & \mathcal{A}
\end{tikzcd}$$

By the universal property of cokernels for presheaves, we have a unique map $\mathcal{H}_{\mathrm{pre}}\to\mathcal{A}$. By the universal property of sheafification, we will get a unique map composed from the two unique universal maps giving us $\mathcal{H}\to\mathcal{A}$, as we sought to show.

Corollary 1.2.12: Image and Sheafification

Let $\varphi\in\mathbf{Sh}(X)$ be a morphism of sheaves over an abelian category. Then the image of φ is the sheafification of the image presheaf.

***Proof*:**
Recall that in an abelian category, the image of a morphism is defined as the kernel of its cokernel:

$$\mathrm{Im}(\varphi)\cong\mathrm{Ker}(\mathcal{G}\to\mathrm{coker}(\varphi))$$

Let $\mathcal{K}_{\mathrm{pre}}$ be the presheaf cokernel of φ. We have, by definition, a short exact sequence of presheaves:

$$0\to\mathrm{Im}_{\mathrm{pre}}(\varphi)\to\mathcal{G}\to\mathcal{K}_{\mathrm{pre}}\to 0$$

Since the sheafification functor is exact (it preserves exact sequences and finite limits), applying it to the sequence above yields an exact sequence of sheaves:

$$0\to(\mathrm{Im}_{\mathrm{pre}}(\varphi))^{\mathrm{sh}}\to\mathcal{G}^{\mathrm{sh}}\to(\mathcal{K}_{\mathrm{pre}})^{\mathrm{sh}}\to 0$$

Since $\mathcal{G}$ is already a sheaf, $\mathcal{G}^{\mathrm{sh}}\cong\mathcal{G}$. By proposition 1.2.11, $(\mathcal{K}_{\mathrm{pre}})^{\mathrm{sh}}$ is the sheaf cokernel of φ. Consequently, the object $(\mathrm{Im}_{\mathrm{pre}}(\varphi))^{\mathrm{sh}}$ is the kernel of the morphism $\mathcal{G}\to\mathrm{coker}(\varphi)$, which is precisely the definition of the image sheaf.

Definition 1.2.13: Surjective Sheaf Morphism

Let $\varphi:\mathscr{F}\to\mathscr{G}$ be a morphism of sheaves and $\mathrm{im}\varphi$ the image presheaf. Then the cokernel and image of φ is the sheafification of the cokernel-presheaf and image-presheaf respectively. The morphism φ is *surjective* if

$$(\mathrm{im}_{\mathrm{pre}}\varphi)^{\mathrm{sh}}=\mathscr{G}$$

Using this, we see that the map $\mathcal{O}_{\mathbb{C}}\xrightarrow{\exp}\mathcal{O}_{\mathbb{C}}^{*}$ is *surjective* in the category of sheaves, even though it is not surjective in the category of presheaves. Hence, exactness in presheaves and sheaves is *different*. The following example using simpler categories should demonstrate why this is not so strange:

Example 1.2.14: Ab Vs TF-Ab

Let **Ab** be the category of abelian groups and TF-Ab be the category of torsion-free abelian groups. Consider the morphism $\varphi : \mathbb{Z} \to \mathbb{Z}$, $\varphi(n) = 2n$. This morphism exists in both categories. The cokernel of φ in **Ab** is $\mathbb{Z}/2\mathbb{Z}$, which is torsion. To find the cokernel in TF-AB, the "best" torsion-free group must be found so that its composition with φ is universal. It isn't hard to see this is 0, so:

$$\operatorname{coker}_{\mathbf{Ab}}(\varphi) = \mathbb{Z}/2\mathbb{Z} \neq 0 = \operatorname{coker}_{\text{TF-Ab}}(\varphi)$$

Thus, though the morphism in each category are the same, the epimorphism differ.

Thus, a theoretical understanding of epimorphisms in $\mathbf{C}_X$ is done. However, it is computationally intractable to always sheafify the image sheaf to check for surjectivity. As sheaves are created to be "local", we shall be able to shall be able to check their condition on *stalks*. Proving this shall fit in better into the broader framework on when the functor taking $\mathscr{F} \mapsto \mathscr{F}(U)$ and $\mathscr{F} \mapsto \mathscr{F}_p$ is exact.

Functors from (pre)sheaves: simplifying checking exactness

As presheaves and sheaves store all local information, we may want that there are some natural functors that preserve some exactness to extract that information. The two natural candidates for functors to would be $-(U)$ taking a (pre)sheaf to its sections over an open set U, and $(-)_p$ the functor taking a (pre)sheaf to it's stalk at p.

It is clear that $-(U) : \mathbf{C}_X \to \mathbf{C}$ is a functor for:

$$\mathscr{F} \mapsto \mathscr{F}(U) \qquad (\varphi : \mathscr{F} \to \mathscr{G}) \mapsto (\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U))$$

preserves composition's an identities. Let us show $(-)_p$ is also a functor:

Proposition 1.2.15: Morphism Of (Pre)Sheaves Induces Morphism On Stalks

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of (pre)sheaves over any category $\mathbf{C}$ having colimits (so that stalks are well-defined). Then there is an induced morphism on stalks $\varphi_p : \mathscr{F}_p \to \mathscr{G}_p$, that is, it induced a functor $\mathbf{C}_X \to \mathbf{C}$.

Proof:

This is definition pushing: For each open $U \subseteq X$ we have the morphism $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$. Define the map:

$$[(f, U)]_p \mapsto [(\varphi(U)(f), U]_p$$

For this map to be well-defined if $(f, U) \sim (f', V)$ then we need:

$$[(f', V)]_p \mapsto [(\varphi(V)(f'), V]_p = [(\varphi(U)(f), U]_p$$

Hence we need that $((\varphi(V)(f'), V) \sim ((\varphi(U)(f), U)$. As $(f, U) \sim (f', V)$ we have there exists $W \subseteq U, V$ such that

$$\operatorname{res}_{U,W} f = \operatorname{res}_{V,W} f' = g$$

Then as φ is a pre-sheaf morphism we have:

$$\begin{tikzcd} \varphi(U) : f \arrow[r, mapsto] \arrow[d, "\mathrm{res}_{U,W}"] & \varphi(U)(f) \arrow[d, "\mathrm{res}_{U,W}"] \\ \varphi(W) : \mathrm{res}_{U,W}(f) = g \arrow[r, mapsto] & \varphi(W)(g) \end{tikzcd} \qquad \begin{tikzcd} \varphi(V) : f' \arrow[r, mapsto] \arrow[d, "\mathrm{res}_{V,W}"] & \varphi(U)(f') \arrow[d, "\mathrm{res}_{V,W}"] \\ \varphi(W) : \mathrm{res}_{V,W}(f) = g \arrow[r, mapsto] & \varphi(W)(g) \end{tikzcd}$$

which implies

$$\mathrm{res}_{U,W}\varphi(U)(f) = \mathrm{res}_{V,W}\varphi(V)(f')$$

But that means they are similar, $((\varphi(V)(f'), V) \sim ((\varphi(U)(f), U)$, as we sought to show

Let us now see how these functors preserve monomorphisms/epimorphisms (i.e. injectivity and surjectivity on opens/stalks). Starting with open subsets:

Proposition 1.2.16: On Presheaf: Open Set Functor is Exact

The functor $-(U) : \mathbf{PSh}_{\mathbf{C}}(X) \to \mathbf{C}$ (where $\mathbf{C}$ is abelian) is an exact functor

On **Sh**, the functor $-(U)$ is left-exact, and hence preserves injective morphisms.

***Proof*:**
Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a monomorphism of presheaves between presheaves. By definition, for any presheaf $\mathcal{H}$ and any morphisms $f, g : \mathcal{H} \to \mathcal{F}$, the relation $\varphi \circ f = \varphi \circ g$ implies $f = g$. Since morphisms of presheaves are natural transformations defined component-wise, the equality of morphisms holds if and only if it holds for every open set U. Thus, the condition implies:

$$\varphi(U) \circ f(U) = \varphi(U) \circ g(U) \implies f(U) = g(U)$$

This is precisely the definition of a monomorphism in the target abelian category $\mathbf{C}$, giving left-exactness and preservation of injectivity. A symmetric argument holds for epimorphisms: if $\psi : \mathcal{G} \to \mathcal{Q}$ is an epimorphism, then $\alpha \circ \psi = \beta \circ \psi \implies \alpha = \beta$, which translates component-wise to $\psi(U)$ being an epimorphism and preserving surjectivity in $\mathbf{C}$. Since the functor $-(U)$ preserves both monomorphisms (kernels) and epimorphisms (cokernels), it is exact on presheaves.

Turning to the category of sheaves $\mathbf{Sh}(X)$, the functor $-(U)$ remains left-exact. This is because the inclusion $\mathbf{Sh}(X) \hookrightarrow \mathbf{PSh}(X)$ preserves limits: the sheafification function $(-)^{\mathrm{sh}} : \mathbf{PSh}(X) \to \mathbf{Sh}(X)$ is *left adjoint* to the inclusion $\iota : \mathbf{PSh}(X) \to \mathbf{Sh}(X)$, hence the inclusion is a *right adjoint* so by ref:HERE (put it in section 0.4) it preserves limits. As kernels are limits they are preserved. Consequently, if φ is a monomorphism of sheaves, it is also a monomorphism of presheaves, and as shown above, $\varphi(U)$ is a monomorphism (and thus injective) in $\mathbf{C}$. However, $-(U)$ is not right-exact on sheaves. While epimorphisms in presheaves imply component-wise surjectivity, there are not enough sheaves for component-wise surjectivity of sheaves and sheafification is required. As seen in example 1.2.7, a map can be an epimorphism of sheaves (locally surjective) without the component map $\varphi(U)$ being surjective, failing exactness on the right (ex. take the global section functor Γ which is will not perserve exactness of example 2).

1.2.17 Remark In Vakil's proof of this result, [17], he uses “”indicator sheaves” to verify these cancellations. In categorical terms, to test the value of a section $s \in \mathcal{F}(U)$, one uses the Yoneda Lemma. The indicator sheaf mentioned corresponds to the representable presheaf h_U (which is the sheafification of the constant section on subsets of U and empty elsewhere). Mapping out of h_U allows one to probe the specific algebraic properties of $\mathcal{F}(U)$ essentially element-by-element.

Proposition 1.2.18: Exactness of Presheaves On The Level of Open Sets

An sequence of presheaves

$$0 \to \mathscr{F}_1 \to \mathscr{F}_2 \to \cdots \to \mathscr{F}_n \to 0$$

is exact if and only if the sequence

$$0 \to \mathscr{F}_1(U) \to \mathscr{F}_2(U) \to \cdots \to \mathscr{F}_n(U) \to 0$$

is exact for all U.

***Proof*:**
The only if direction is induction on proposition 1.2.16. Conversely, if $0 \to \mathscr{F}_1(U) \to \mathscr{F}_2(U) \to \cdots \to \mathscr{F}_n(U) \to 0$ is exact for each open $U \subseteq X$, then as we already have morphism of presheaves we have these commute with restrictions and $\mathrm{im}\mathscr{F}_i = \ker \mathscr{F}_i$, completing the proof.

Corollary 1.2.19: Exactness of Stalk For Presheaves

Let $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{H} \to 0$ be an exact sequence of presheaves. Then if filtered colimits exists (so that stalks exist), then

$$0 \to \mathscr{F}_p \to \mathscr{G}_g \to \mathscr{H}_p \to 0$$

is exact.

***Proof*:**
Use that each functor $-(U)$ is exact for each U; so we can take their colimit.

Note that for presheaves, exactness on all stalks does not imply exactness on presheaves as example 1.2.7(2) shows.

Theorem 1.2.20: Presheaves Over Ab Category Form Ab Category

The category $\mathbf{PSh}_{\mathbf{C}}(X)$ over an abelian category C is an abelian category

***Proof*:**
The main thing that wasn't checked was how to define addition of morphism. By the above work, this can be done on each open set: if $\varphi, \psi : \mathcal{F} \to \mathcal{G}$ then $\varphi + \psi$ defined via open sets as:

$$\varphi + \psi(U) := \varphi(U) + \psi(U)$$

is a map of presheaf (it respects restriction by the commutativity of both natural transformation), and so the hom-sets form an abelian group. Then as we also have a 0-object, and we have finite products (again, something the reader could verify), we have an additive category.

To show kernels and cokernels exist, by proposition 1.2.18 it suffices to check on the level of open sets. Then we define:

$$(\ker_{\text{pre}} \varphi)(U) = \ker \varphi(U)$$

The key is to consider the following diagram where $U \hookrightarrow V$:

$$\begin{tikzcd} 0 \arrow[r] & \ker_{\text{pre}}(\varphi(V)) \arrow[r] \arrow[d, dashed, "\exists!"] & \mathcal{F}(V) \arrow[r] \arrow[d, "\text{res}_{V,U}"] & \mathcal{G}(V) \arrow[d, "\text{res}_{V,U}"] \\ 0 \arrow[r] & (\ker_{\text{pre}} \varphi)(U) \arrow[r] & \mathcal{F}(U) \arrow[r] & \mathcal{G}(U) \end{tikzcd}$$

which shows that this is indeed well-defined. The reader can further check that this does indeed satisfy the universal property of the kernel. By dualizing the argument, we can do the same for the cokernel, and hence the additive category of presheaves has kernels and cokernels, making it abelian.

To keep track of all morphisms between open sets, we box the following definition:

Definition 1.2.21: Sheaf Hom

Let $\mathscr{F}, \mathscr{G} \in \mathbf{Sh_C}(X)$. Then the *sheaf hom* is defined to be:

$$\mathcal{H}om(\mathscr{F}, \mathscr{G})(U) := \text{Mor}(\mathscr{F}_U, \mathscr{G}|_U)$$

where $\text{Mor}(\mathscr{F}|_U, \mathscr{G}|_U)$ are all morphism of (pre)sheaves. The dual of $\mathscr{F}$ is:

$$\mathscr{F}^\vee = \text{Hom}_{\text{Mod}_{\mathcal{O}_X}}(\mathscr{F}, \mathcal{O}_X)$$

Just like Hom, $\mathcal{H}om$ is a (co/contra)variant functor, depending on which position you choose.

The situation is not so clean for $(-)_p$, namely the functor on $\mathbf{Sh_C}(X)$ is not always exact for *any* abelian category; it need not be the case that filtered colimits preserve exactness, and hence that $(-)_p$ preserve exactness (think of the category of finite abelian groups which doesn't have $\varinjlim_p \mathbb{Z}/p^n\mathbb{Z}$). Grothendieck axiomatically added this condition to abelian categories (such categories are called *Grothendieck categories*). Fortunately for us, this will never be an issue as all the categories of interest to us shall have this property, namely:

1. abelian groups, more generally R-Mod
2. Sheaves of abelian groups (more generally $\mathcal{O}_X$-modules which are defined later)
3. the product of Grothendieck categories

4. In ref:HERE, it shall be important to have internal Hom's: given a small category **C** and a Grothendieck category **G**, all functors from **C** to **G** (with morphism being natural transformation) form Grothendieck categories.

Proposition 1.2.22: Sheaves and exactness For Stalk

Let $\mathbf{Sh}(X)$ be the category of sheaves over any R-Mod. Then $(-)_p : \mathbf{Sh}(X) \to R$-Mod is exact.

***Proof*:**
Recall that the stalk of a sheaf $\mathcal{F}$ at a point p is defined as the colimit over the filtered system of open neighborhoods of p:

$$\mathcal{F}_p = \varinjlim_{p \in U} \mathcal{F}(U)$$

By the general properties of abelian categories, the colimit functor is always right-exact (it preserves cokernels). The non-trivial property required is left-exactness. In the category of R-modules (unlike the category of finite abelian groups mentioned above), filtered colimits commute with finite limits. Consequently, the filtered colimit commutes with kernels.

Therefore, if $0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$ is an exact sequence of sheaves, applying the stalk functor yields a sequence of filtered colimits. Since exactness is preserved by filtered colimits in R-Mod, the resulting sequence of stalks

$$0 \to \mathcal{F}_p \to \mathcal{G}_p \to \mathcal{H}_p \to 0$$

is exact.

With the right notion of compatible stalks defined, we may show how to determine morphisms of sheaves at the level of stalks, namely a scheme morphisms is determined by stalks, and a scheme morphism that is an isomorphism on stalks induces an inverse scheme morphism:

Proposition 1.2.23: Morphisms Determined By Stalks

Let φ_1, φ_2 be morphisms from a presheaf of sets $\mathscr{F}$ to a *sheaf* of sets $\mathscr{G}$ that induce the same maps on each stalk. Then $\varphi_1 = \varphi_2$

***Proof*:**
For every open U, $\varphi_1(U) = \varphi_2(U)$. Let $s \in \mathscr{F}(U)$. Since $\mathscr{G}$ is a sheaf, the section $t = \varphi_1(U)(s)$ is uniquely determined by its germs $(t_p)_{p \in U}$.

For any $p \in U$, the germ of the image is the image of the germ:

$$(\varphi_1(s))_p = (\varphi_1)_p(s_p)$$

By assumption, $(\varphi_1)_p = (\varphi_2)_p$, so:

$$(\varphi_1)_p(s_p) = (\varphi_2)_p(s_p) = (\varphi_2(s))_p$$

Since $\varphi_1(s)$ and $\varphi_2(s)$ have identical germs at every point in U, and $\mathscr{G}$ is a sheaf (satisfying the identity axiom), the sections must be identical. Thus $\varphi_1(U)(s) = \varphi_2(U)(s)$.

The above proof can be summarized by the following diagram:

$$\begin{tikzcd} \mathscr{F}(U) \arrow[r] \arrow[d, hook] & \mathscr{G}(U) \arrow[d, hook] \\ \prod_{p\in U} \mathscr{F}_p \arrow[r] & \prod_{p\in U} \mathscr{G}_p \end{tikzcd}$$

Proposition 1.2.24: Stalks-Local Isomorphism induces Isomorphism

Let φ be a morphism of sheaves. Then φ is an isomorphism if and only if it induces an isomorphism for all stalks.

***Proof*:**
Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a sheaf morphism. If φ is an isomorphism, then certainly each φ_p is an isomorphism, hence say each φ_p is an isomorphism. Then it must be shown that each $\varphi(U)$ is an isomorphism. For injectivity, say for $s \in \mathscr{F}(U)$ $\varphi(s) = 0 \in \mathscr{G}(U)$. Then for each $p \in U$, $\varphi(s)_p = 0 \in \mathscr{G}_p$. As φ_p is injective at each p, we have $s_p = 0 \in \mathscr{F}_p$. Using the definition of stalk, we have that s is zero on some open set, namely $s|_{W_p} = 0$ for $W_p \subseteq U$. As U is covered by such neighborhoods, by the gluing axiom we have $s = 0$

For surjectivity, let $t \in \mathscr{G}(U)$. Consider $t_p \in \mathscr{G}_p$. As φ_p is surjective, we have $s_p \in \mathscr{F}_F$ such that $\varphi_p(s_p) = t_p$. Now, let $(s(p), V_p)$ be a representative of s_p on the set V_p containing the point p. Then $\varphi(s(p))$ and $t|_{V_p}$ are two elements of $\mathscr{G}(V_p)$ whose germs at p are the same. Thus, replacing V_p by a smaller neighborhood if necessary, we have $\varphi(s(p)) = t|_{V_p} \in \mathscr{G}(V_p)$.

Now, U is covered by open sets V_p where on each V_p we have $s(p) \in \mathscr{F}(V_p)$ and $\varphi(s(p)) = t|_{V_p}$. If p, q are two points, then $s(p)|_{V_p\cap V_q}$ and $s(q)|_{V_p\cap V_q}$ are two sections of $\mathscr{F}(V_p \cap V_q)$ which are both sent to $t|_{V_p\cap V_q}$ via φ. As φ was shown to be injective, they are equal. then by the gluing axiom, there exists an $s \in \mathscr{F}(U)$ such that

$$s|_{V_p} = s(p) \qquad \forall p \in U$$

Finally, to show that $\varphi(s) = t$, notice that $\varphi(s)$ and t are two section of $\mathscr{F}(U)$ where for each p $\varphi(s)|_{V_p} = t|_{V_p}$, hence they glue to create $\varphi(s) = t$, as we sought to show.

This shows that a *morphism* $\varphi : \mathscr{F} \to \mathscr{G}$ can be determined on the level of stalks. This does not imply that if two *sheaves* have isomorphic stalks, then they are isomorphic! You need to already have a sheaf homomorphism which in turn induces the isomorphic stalks. Conceptually, this is saying that being locally isomorphic does not imply it is globally isomorphic; there are plenty concrete example form differential geometry to choose from.

Example 1.2.25: Isomorphic Stalks, Not Isomorphic Sheaves

1. Let X be a Hausdorff space. Consider the constant sheaf $\underline{\mathbb{Z}}$ and the sky-scrapper sheaf $\iota_*(\mathbb{Z})$. Show that the stalks at each point is $\mathbb{Z}$, however these two sheaves are not isomorphic.

2. The following will rely on the reader's intuitions of varieties (see [**NateClassicalAlgGeo**]). Take the variety associated to the coordinate ring:

$$A = \mathbb{R}[x, y, z]/(x^2 + y^2 + z^2 - 1)$$

which is the 2-sphere in $\mathbb{R}^3$. Take the tangent bundle, i.e. A-module, of all derivations $E = \mathrm{Der}_{\mathbb{R}}(A)$. Then E is a rank 2 locally trivial module since for any open subset or distinguished open subset, $E_x \cong \mathbb{R}^2$, showing each stalk is isomorphic. However, $E \ncong A^2$.

3. Another famous example would be the trivial line bundle on S^1 vs. the möbius bundle.

As stalks define maps on some open neighbourhood, we can use this condition to find the closest condition that allows to check if a map between topological spaces is sheaf map on the level of open sets:

Proposition 1.2.26: Characterizing Stalks Inducing Sheaf Morphism

Let X be a topological space with two sheaves $\mathscr{F},\mathscr{G}$. Let $\varphi_p : \mathscr{F}_p \to \mathscr{G}_p$ be a collection of stalk morphisms for all $p \in X$. Then this gives a map of sheaves $\Phi : \mathscr{F} \to \mathscr{G}$ if and only if for all open subsets $U \subseteq X$, for all sections $s \in \mathscr{F}(U)$, there is an open cover $U = \cup_i U_i$ and sections $t_i \in \mathscr{G}(U_i)$ such that

$$\varphi_p(s_p) = (t_i)_p$$

for all $p \in U_i$ and all i.

***Proof*:**
($\Rightarrow$) Assume $\Phi : \mathscr{F} \to \mathscr{G}$ is a morphism of sheaves. Then for any open U and section $s \in \mathscr{F}(U)$, we can simply take the trivial cover $U_1 = U$. Define $t_1 = \Phi(U)(s) \in \mathscr{G}(U)$. By the definition of the induced map on stalks (proposition 1.2.15), for any $p \in U$, the map φ_p sends the germ of s to the germ of $\Phi(U)(s)$. Thus, $\varphi_p(s_p) = (t_1)_p$, satisfying the condition.

($\Leftarrow$) Conversely, assume the local condition holds. We must construct a morphism $\Phi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ for every open set U. Let $s \in \mathscr{F}(U)$. By hypothesis, there exists a cover $\{U_i\}$ and sections $t_i \in \mathscr{G}(U_i)$ such that for every $p \in U_i$, the germ of t_i at p is $\varphi_p(s_p)$.

We first show that these local sections t_i glue together. Consider the intersection $U_{ij} = U_i \cap U_j$. For any point $p \in U_{ij}$, we have:

$$(t_i)_p = \varphi_p(s_p) = (t_j)_p$$

Since $\mathscr{G}$ is a sheaf (specifically, a separated presheaf), sections are determined by their germs (lemma 1.1.15). Because t_i and t_j have identical germs at every point in U_{ij}, they must be equal as sections:

$$\mathrm{res}_{U_i,U_{ij}}(t_i) = \mathrm{res}_{U_j,U_{ij}}(t_j)$$

Since the sections agree on overlaps, the Gluing Axiom for $\mathscr{G}$ implies there exists a unique section $t \in \mathscr{G}(U)$ such that $t|_{U_i} = t_i$. We define $\Phi(U)(s) = t$.

To ensure Φ is well-defined, suppose there was another cover $\{V_j\}$ and sections $\{r_j\}$ satisfying the condition, yielding a glued section t'. For any $p \in U$, $t_p = \varphi_p(s_p)$ (by the construction on the cover U_i) and $t'_p = \varphi_p(s_p)$ (by the construction on V_j). Thus t and t' have the same germs everywhere, and since $\mathscr{G}$ is a sheaf, $t = t'$.

Finally, Φ commutes with restrictions naturally because the construction is defined pointwise via germs: for $V \subseteq U$, the germs of $\Phi(V)(s|_V)$ are $\varphi_p((s|_V)_p) = \varphi_p(s_p)$, which matches the germs of $(\Phi(U)(s))|_V$.

Summarizing the results for morphisms of sheaves:

Theorem 1.2.27: Equivalence of Mono- and Epimorphisms

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves on X (over **Set** or an abelian category).

The following conditions are equivalent:

1. φ is a monomorphism in the category of sheaves.
2. φ is injective on the level of stalks: $\varphi_p : \mathcal{F}_p \to \mathcal{G}_p$ is injective for all $p \in X$.
3. φ is injective on the level of open sets: $\varphi(U) : \mathcal{F}(U) \to \mathcal{G}(U)$ is injective for all open $U \subseteq X$.

The following conditions are equivalent:

1. φ is an epimorphism in the category of sheaves.
2. φ is surjective on the level of stalks: $\varphi_p : \mathcal{F}_p \to \mathcal{G}_p$ is surjective for all $p \in X$.

If $\varphi(U)$ is surjective for all open sets U, then φ is an epimorphism (surjectivity on sections implies surjectivity on stalks). The converse is *false*: an epimorphism of sheaves need not be surjective on global sections (as seen in the exponential sequence).

***Proof*:**
This is combining the results from earlier. In particular, apply the results just proven about stalks to the proof of proposition 1.2.16.

Theorem 1.2.28: Sheaves Over Ab Category Form Ab Category

The category $\mathbf{Sh}_{\mathbf{C}}(X)$ over an abelian category C is an abelian category

***Proof*:**
exercise.

Morphisms and Stalks

Since the stalk functor $(-)_p$ is exact by proposition 1.2.22, it preserves all finite limits and colimits. This yields the following immediate corollaries.

Corollary 1.2.29: Commuting Kernel and Stalk

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. Then the stalk of the kernel is the kernel of the stalk:

$$(\ker(\varphi))_p \cong \ker(\varphi_p)$$

***Proof*:**
Since $(-)_p$ is an exact functor, it is left-exact. Left-exact functors preserve kernels.

Corollary 1.2.30: Commuting Cokernel and Stalk

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. Then the stalk of the cokernel is the cokernel of the stalk:

$$(\mathrm{coker}(\varphi))_p \cong \mathrm{coker}(\varphi_p)$$

***Proof*:**
Since $(-)_p$ is an exact functor, it is right-exact. Right-exact functors preserve cokernels.

Due to sheafification, the surjectivity of a sheaf morphism is no longer conditional on the surjectivity of sections on all open sets. This leads to the following counter-intuitive examples.

Example 1.2.31: Surjective on Stalks $\neq$ Surjective on Global Sections

1. (if you know Complex Analysis): Let $X = \mathbb{C} \setminus \{0\}$. Let $\mathcal{O}_X$ be the sheaf of holomorphic functions and Ω^1_X be the sheaf of holomorphic differential 1-forms. Define $\varphi : \mathcal{O}_X \to \Omega^1_X$ by $f \mapsto df$.
   - *Locally:* On any simply connected disk U, every closed form is exact. Since holomorphic forms are closed, $\omega = df$ has a solution. Thus φ_p is surjective.
   - *Globally:* The form $\omega = \frac{dz}{z}$ is a global section of $\Omega^1_X(X)$. However, its primitive is $\log(z)$, which is not a function on $\mathbb{C} \setminus \{0\}$. Thus $\frac{dz}{z} \notin \mathrm{im}(\varphi(X))$.
2. Let $X = \mathbb{R}$, $\mathscr{F} = \underline{\mathbb{Z}}$ (constant sheaf), and $\mathscr{G} = i_{p*}(\mathbb{Z}) \oplus i_{q*}(\mathbb{Z})$ (skyscrapers at distinct points p, q). Define $\varphi : \mathscr{F} \to \mathscr{G}$ by restriction.
   - *Stalks:* At p, $\mathscr{F}_p = \mathbb{Z}$ and $\mathscr{G}_p = \mathbb{Z} \oplus 0 \cong \mathbb{Z}$. The map is identity (surjective). Similarly at q. Elsewhere, both are 0.
   - *Globally:* $\mathscr{F}(X) = \mathbb{Z}$ (constant functions). $\mathscr{G}(X) = \mathbb{Z} \oplus \mathbb{Z}$. The map sends $n \mapsto (n, n)$. The image is the diagonal, which is not the whole plane $\mathbb{Z} \oplus \mathbb{Z}$.

Corollary 1.2.32: Commuting Image and Stalk

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. Then the stalk of the image is the image of the stalk:

$$(\mathrm{im}(\varphi))_p \cong \mathrm{im}(\varphi_p)$$

***Proof*:**
In an abelian category, $\mathrm{im}(\varphi) = \ker(\mathrm{coker}(\varphi))$. Since $(-)_p$ commutes with both kernels and cokernels, it commutes with images.

1.2.33 Summary

- **Presheaves** form an abelian category because computation is done on the level of open sets.
- **Sheaves** form an abelian category because computation is done on the level of stalks.

Exercise 1.2.1

1. Show that $\varphi : \mathscr{F} \to \mathscr{G}$ is surjective if and only if for every $s \in \mathscr{G}(U)$, there exists an open covering $\bigcup_i U_i = U$ and elements $t_i \in \mathscr{F}(U_i)$ such that $\varphi(t_i) = s|_{U_i}$ (hint: recall φ is surjective if and only if it is surjective on stalks).

   This exercise will come back in proposition 5.1.16

2. Let $(X, \mathcal{O}_X)$ be a ringed space and $U \subseteq X$ an open subset. Show that we can define non-vanishing distinguished open subset by:

$$\begin{tikzcd} U_f \arrow[r] \arrow[d] & \mathcal{O}_X^* \arrow[d] \\ U \arrow[r, "f"] & \mathcal{O}_X \end{tikzcd}$$

   that is, $U_f = f^{-1}(\mathcal{O}_X^*) = \left\{x \in U \;\middle|\; f(x) \in \mathcal{O}_{X,x}^*\right\}$. Show these form a sheaf.

3. Define a *local ringed space* to be a ringed space $(X, \mathcal{O}_X)$ such that for all open $U \subseteq X$ and all $f \in \mathcal{O}_X(U)$, $U = U_f \cup U_{1-f}$, namely for all $x \in U$, f or $1 - f$ is invertible.

   a) Show that if X, $\operatorname{mSpec} \mathcal{O}_X(X)$ is a singleton.

   b) Let $C(X)$ be all continuous real function on X. Show that $(X, \mathbb{C}(X))$ is local.

4. Let $(X, \mathcal{O}_X)$ is a local ringed space, $U \subseteq X$ open, and $f \in \mathcal{O}_X(U)$. Then if $f_1 + \cdots + f_r = 1$, then $\bigcup U_i = U$.

5. Let $\mathscr{F}$ be a sheaf of sets, show that $\mathcal{H}om(\underline{\{p\}}, \mathscr{F}) \cong \mathscr{F}$

6. Let $\mathscr{F}$ be a sheaf of groups, show that $\mathcal{H}om(\underline{\mathbb{Z}}, \mathscr{F}) \cong \mathscr{F}$

7. It is tempting to then think that $\mathcal{H}om(\mathscr{F}, \mathscr{G})_p \cong \operatorname{Hom}(\mathscr{F}_p, \mathscr{G}_p)$. However this is *not* the case: think of a counter-example. There is a map from one of these sets to another, what is it?

1.3 Grothendieck's Functors: Pushforward, Pullback, and more

We next endeavor to move sheaves around from one topological space to another given a continuous map $f : X \to Y$. Grothendieck identified 4 key ways of doing so that come in adjoint pairs:

1. The pushforward and pullback: f_* and f^*
2. the proper direct image and proper inverse image: $f_!$ and $f^!$

These are some of the key functors in Grothendieck *6 functor formalism* where the adjoint pair Hom and $\otimes$ is missing. These functor's are more naturally introduced in a cohomological setting and will be present in chapter 6.

1.3.1 Pushforward and Pullback

The easiest of these is the pushforward. If $U \subseteq V \subseteq Y$ and $f : X \to Y$, then $f^{-1}(U) \subseteq f^{-1}(V)$. Hence, f^{-1} can be thought of as a contravariant functor preserving $\subseteq$. In fact, more is true: f^{-1} also preserves the properties in proposition 1.1.9, and hence works on sheaves as well:

Definition 1.3.1: Pushforward Of (Pre)sheaf

Let $f : X \to Y$ be a continuous map and $\mathscr{F} \in \mathbf{PSh}(X)$. Then the *pushforward* or *direct image* of $\mathscr{F}$ through f is defined on every open set $U \subseteq Y$ to be

$$f_*\mathscr{F}(U) = \mathscr{F}(f^{-1}(U))$$

and induces a (per)sheaf on Y. The pushforward defines a [covariant] functor $f_* : \mathbf{Sh}(X) \to \mathbf{Sh}(Y)$ sending:

$$f_* : \mathscr{F} \to f_*(\mathscr{F})$$

If working with a particular category, for example **Set**, we can write $f_* : \mathbf{Set}_X \to \mathbf{Set}_Y$.

The term "pushforward" comes from the fact that we are taking a sheaf on X and defining a sheaf on Y through the map $f : X \to Y$. The pullback shall do the same thing in reverse.

It is perhaps unsurprising that f^{-1} preserves pre-sheaves, however it is a little less obvious that it preserves sheaves as well; it is worth thinking this through. To me, this shows that sheaves are a very natural concept, as the usually topological function f^{-1} naturally preserves them. This also motivates the Grothendieck topology that will be introduced much later. We have already seen an example of a pushforward: the skyscraper sheaf is induced that pushing forward along the inclusion map $\iota_p : \{p\} \to X$ the constant sheaf $\underline{S}$.

Proposition 1.3.2: Pushforward Induces Map On Stalks

Let $f : X \to Y$ be a continuous map and $\mathscr{F} \in \mathbf{Sh}(X)$. Then for each $\pi(p) = q$, there exists a map

$$(f_*\mathscr{F})_q \to \mathscr{F}_p$$

between the stalks of $f_*\mathscr{F}$ and $\mathscr{F}$.

***Proof*:**
exercise. This can be verified using the equivalence class definition or the universal property. Doing it using equivalence classes, define the map

$$[s]_q \mapsto [f^{-1}(s)]_p$$

Proposition 1.3.3: Sheaf Morphisms Determined by a Basis

Let X be a topological space and $\mathcal{B}$ a basis for the topology. Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on X. Suppose that for every basis set $B \in \mathcal{B}$, we have a morphism $\varphi_B : \mathcal{F}(B) \to \mathcal{G}(B)$ such that for any inclusion of basis sets $V \subseteq U$ (where $U, V \in \mathcal{B}$), the maps commute with restriction:

$$\varphi_V(s|_V) = (\varphi_U(s))|_V$$

Then there exists a unique sheaf morphism $\Phi : \mathcal{F} \to \mathcal{G}$ extending this collection.

***Proof*:**
Let $U \subseteq X$ be an arbitrary open set. Since $\mathcal{B}$ is a basis, there exists a cover $U = \bigcup_{i \in I} B_i$ where $B_i \in \mathcal{B}$. We wish to construct a map $\Phi_U : \mathcal{F}(U) \to \mathcal{G}(U)$.

Take a section $s \in \mathcal{F}(U)$. We restrict this section to the basis elements to get $s_i = s|_{B_i} \in \mathcal{F}(B_i)$. Using our hypothesis, we can map these to the target sheaf:

$$t_i := \varphi_{B_i}(s_i) \in \mathcal{G}(B_i)$$

We now claim that the collection $\{t_i\}$ glues together to form a global section $t \in \mathcal{G}(U)$. By the sheaf axioms for $\mathcal{G}$, it suffices to show that t_i and t_j agree on the overlap $B_i \cap B_j$.

Unlike the specific case of affine schemes, the intersection $B_i \cap B_j$ might not be a single basis element. However, by the definition of a topology basis, we can cover the intersection by smaller basis elements:

$$B_i \cap B_j = \bigcup_k V_k \quad \text{where } V_k \in \mathcal{B}$$

Since $V_k \subseteq B_i$, our compatibility hypothesis tells us that restricting the mapped section is the same as mapping the restricted section:

$$t_i|_{V_k} = (\varphi_{B_i}(s|_{B_i}))|_{V_k} = \varphi_{V_k}(s|_{V_k})$$

By the exact same logic for j, we have $t_j|_{V_k} = \varphi_{V_k}(s|_{V_k})$. Thus, t_i and t_j agree on every V_k covering the intersection. Since $\mathcal{G}$ is a sheaf (specifically, using the identity axiom locally), this implies t_i and t_j agree on the entire intersection $B_i \cap B_j$.

Since the sections are compatible, there exists a unique $t \in \mathcal{G}(U)$ such that $t|_{B_i} = t_i$. We define $\Phi_U(s) = t$.

To ensure this is well-defined, one must check that choosing a different basis cover for U yields the same result. This follows from standard refinement arguments (taking the union of two covers and applying the uniqueness of gluing on the common refinement). Uniqueness of Φ is guaranteed because two sheaf morphisms agreeing on a basis must agree everywhere.

Corollary 1.3.4: Sheaf Isomorphism Determined By A Basis

Let X be a topological space and $\mathcal{B}$ a basis for the topology. Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on X. Suppose that for every basis set $B \in \mathcal{B}$, we have an isomorphism $\varphi_B : \mathcal{F}(B) \to \mathcal{G}(B)$ such that for any inclusion of basis sets $V \subseteq U$ (where $U, V \in \mathcal{B}$), the maps commute with restriction:

$$\varphi_V(s|_V) = (\varphi_U(s))|_V$$

Then there exists a unique sheaf isomorphism $\Phi : \mathcal{F} \to \mathcal{G}$ extending this collection.

***Proof*:**
As each φ_B is an isomorphism, they each have an inverse. It is easy to show that they satisfy the same restriction compatibility condition, hence by proposition 1.3.3 there exists a global inverse morphism, and hence φ is an isomorphism.

Pullback

So far, most of the discussion of sheaves where with the assumption of a topological X. In example 1.1.12, there were examples of sheaves being pushed forward (Definition 1.3.1) via a continuous map $f : X \to Y$. Namely the sheaf $\mathscr{F}$ maps to the sheaf $f_*\mathscr{F}$ on Y defined via:

$$\pi_*\mathscr{F}(U) = \mathscr{F}\left(f^{-1}(U)\right)$$

There is an adjoint, the *pullback*, called the inverse image sheaf. It is called a pullback as it takes a sheaf $\mathscr{F}$ and Y and defines a sheaf $f^*(\mathscr{F})$ on X. It is not as clear as the case for the push-forward as the image of an open set need not be open, hence the existence is more subtle (similar to the existence of cokernels for groups). The adjoint will be important not only for having more than one perspective on sheaves, but for definition morphisms of schemes.

Definition 1.3.5: Inverse Image Sheaf

Let $f : X \to Y$ be a function function. and $\mathscr{F}, \mathscr{G}$ be sheaves on X and Y respectively. Then the *inverse image sheaf* or *pullback* $f^*\mathscr{G}$ on X is the sheaf associated to the presheaf:

$$f^*_{\text{pre}}\mathscr{G}(U) = \varinjlim_{V \supseteq f(U)} \mathscr{G}(V)$$

or more concisely

$$f^*\mathscr{G} = (f^*_{\text{pre}}\mathscr{G})^{\text{sh}}$$

Giving us a map $f^* : \mathbf{Sh}(Y) \to \mathbf{Sh}(X)$.

1.3.6 On Vakil's Notation Vakil uses the f^{-1} notation intend of f^* as he wants to reserve the f^* notation for pulling back $\mathcal{O}_X$-modules:

$$f^*\mathcal{G} := f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$$

This is because we still want pullback as defined in this section to apply to $\mathcal{O}_X$-modules. We shall instead keep track of which category we are performing the operations.

The construction of the sheaf is done in the usual quotient way: the elements of $f^*_{\text{pre}}\mathscr{F}(U)$ are equivalent classes $(s, U), (t, V)$ where $(s, U) \sim (t, V)$ if there exists a $W \subseteq U \cap V$ where $f(U) \subseteq W$ and the restriction of s, t are equal on W.

The sheafification is crucial:

Example 1.3.7: Presheaf Inverse Image
Let $f : Y \to X$ where $Y = X \coprod X$ and f is the natural projection map and a sheaf $\mathscr{G}$ defined on X. Then by definition, we would get $f^*_{\text{pre}}\mathscr{G}(U) = \mathscr{G}(U)$, however the sheafification will get:

$$f^*\mathscr{G}(U) = \mathscr{G}(U) \times \mathscr{G}(U)$$

showing the two are in general not equal.

Proposition 1.3.8: push-forward Sheaf Adjoint

Let $f : X \to Y$ be a continuous functions and $\mathscr{F}, \mathscr{G}$ be sheaves on X and Y respectfully. Then there is a bijection:

$$\text{Mor}_X(f^*\mathscr{G}, \mathscr{G}) \leftrightarrow \text{Mor}_Y(\mathscr{G}, f_*\mathscr{F})$$

That is, they are adjoints

Proof:
exercise.

One advantage of the inverse image sheaf is the computation of stalk

Proposition 1.3.9: Stalks Via Inverse Image Sheaf

Let $f : X \to Y$ be a continuous function with $\mathscr{G}$ a sheaf on Y. Then:

$$(f^*\mathscr{G})_p = \mathscr{F}_{f(p)}$$

Proof:
exercise, follows more naturally via the definition.

In fact, the adjoint of this map is the *skyscraper sheaf*, namely given $f : \{p\} \to X$ where $p \in X$, then $f^*\mathscr{G} = \mathscr{G}_p$. Conversely, if we define the sheaf $\mathscr{F}(\{p\}) = S$, then $f_*\mathscr{F}(U)$ is the skyscraper sheaf.

will get back to this when it is more needed

1.3.2 Proper Direct and Inverse Image

While the standard pushforward f_* and pullback f^* are sufficient for many geometric operations, they lack certain "clean" properties regarding the supports of sections:

Example 1.3.10: Smeared Section

Consider $X = \mathbb{R} - \{0\}$ with the euclidean topology and take the inclusion $j : X \hookrightarrow \mathbb{R}$. Then let $\mathscr{F}$ on X be the $\mathbb{Z}$-constant sheaf. Then what is $j_*\mathscr{F}$ at 0? Consider that on any open neighbourhood $(-\epsilon, \epsilon)$, the pushforward is $(-\epsilon, 0) \cup (0, \epsilon)$. We can choose two arbitrary choices of constant functions. Hence:

$$j_*(\mathscr{F})_0 \cong \mathbb{Z} \oplus \mathbb{Z}$$

it is *not* 0!

For those who already encountered the Zariski topology on $\mathbb{A}^1_k$ (k infinite), consider $X = \mathbb{A}^1_k \setminus \{0\}$ and the inclusion $j : X \hookrightarrow \mathbb{A}^1_k$. Then as the open sets of $\mathbb{A}^1_k$ are cofinite, hence any open subset U containing 0 is cofinite; it is easy to show that $X \cap U$ is always *connected*. Hence, if we put the $\mathbb{Z}$-constant sheaf on X, $j_*(\mathscr{F})_0 = \mathbb{Z}$. The functor $j_!$ that will be defined soon will be characterized by $j_!(\mathscr{F})_x = 0$ if $x \notin X$.

To address this, Grothendieck introduced the "shriek" functors (denoted with an exclamation mark in the subscript, ex. $j_!$, or in Hartshorne with a tick in the subscript, ex. $j_{\prime}$), which handle supports more strictly. These functors behave differently depending on whether the map is an open immersion or a closed immersion.

Extension by Zero (Open Immersion)

Let $j : U \hookrightarrow X$ be an inclusion of an open set. The standard pushforward j_* "smears" the information from U up to the boundary in X. Often, we wish to extend a sheaf from U to X such that it is exactly zero outside of U.

Definition 1.3.11: Extension by Zero

Let $j : U \to X$ be an open immersion and $\mathscr{F}$ a sheaf on U. The *extension by zero*, denoted $j_!\mathscr{F}$, is the sheaf on X associated to the presheaf:

$$j_!(V) = \begin{cases} \mathscr{F}(V) & \text{if } V \subseteq U \\ 0 & \text{otherwise} \end{cases}$$

This functor $j_! : \mathbf{Sh}(U) \to \mathbf{Sh}(X)$ is also called the *proper direct image* or *lower shriek*.

The behavior of this functor is best understood at the level of stalks. Unlike j_*, which may have non-zero stalks on the boundary ∂U, $j_!$ is "clean":

$$(j_!\mathscr{F})_p = \begin{cases} \mathscr{F}_p & \text{if } p \in U \\ 0 & \text{if } p \notin U \end{cases}$$

Just as f^* is the left adjoint to f_*, the extension by zero provides a *left* adjoint to the restriction.

Proposition 1.3.12: Adjunction of Extension by Zero

Let $j : U \to X$ be an open immersion. Then $j_!$ is the left adjoint to the pullback (restriction) j^*. That is, for $\mathscr{F} \in \mathbf{Sh}(U)$ and $\mathscr{G} \in \mathbf{Sh}(X)$:

$$\mathrm{Mor}_X(j_!\mathscr{F}, \mathscr{G}) \cong \mathrm{Mor}_U(\mathscr{F}, j^*\mathscr{G})$$

***Proof*:**
exercise. (Hint: A map from zero is unique, so the morphisms on X are entirely determined by their behavior on the open set U).

Sections with Support (Closed Immersion)

Conversely, let $i : Z \hookrightarrow X$ be a closed immersion. The standard pullback i^* restricts sections to Z, effectively "forgetting" everything outside. The right adjoint to the pushforward i_* is the functor that finds sections supported *only* on Z.

Definition 1.3.13: Exceptional Inverse Image

Let $i : Z \to X$ be a closed immersion. The *exceptional inverse image*, denoted $i^!\mathscr{F}$ (or sometimes $\mathcal{H}^0_Z(\mathscr{F})$), is the functor $i^! : \mathbf{Sh}(X) \to \mathbf{Sh}(Z)$ defined by the subgroup of sections:

$$\Gamma(V, i^!\mathscr{F}) = \{s \in \Gamma(V, i^*\mathscr{F}) \big| \mathrm{supp}(s) \subseteq Z\}$$

It serves as the right adjoint to the pushforward i_*.

1.3.3 Hom and Tensor

here (I think this is a good place for it, to round off the 6-functor formalism.)

Summary of the Six Operations

It is common to group these functors into a "Six Operation" formalism. While defined generally for derived categories, for sheaves on topological spaces, they exhibit a specific symmetry between open and closed maps.

1. **For an Open Immersion j:** The exceptional inverse image collapses to the standard pullback ($j^! \cong j^*$), leaving us with the triple $(j_!, j^*, j_*)$.
2. **For a Closed Immersion i:** The proper direct image collapses to the standard pushforward ($i_! \cong i_*$), leaving us with the triple $(i^*, i_*, i^!)$.

These functors allow for the decomposition of a sheaf $\mathscr{F}$ on X into its parts on an open set U and the closed complement Z via the fundamental exact sequence:

$$0 \to j_!j^*\mathscr{F} \to \mathscr{F} \to i_*i^*\mathscr{F} \to 0$$

Exercise 1.3.1

1. Let $U \subseteq Y$ open and $\iota : U \hookrightarrow Y$ be the inclusion. Show that $\iota^* \mathscr{G} \cong \mathscr{G}|_U$.
2. Show that f^* is an exact functor (naturally on abelian categories, however is just as good to show it for any ${}_R\mathrm{Mod}$)
3. Let $U \subseteq X$ be an open subset of a topological space so that $V = X \setminus U$ is closed. Let $\mathscr{F}$ be a sheaf on V, and $\iota : V \hookrightarrow X$ the inclusion.
   a) Show that $(\iota_* \mathscr{F})_p \cong \mathscr{F}_p$ if $p \in V$ and 0 otherwise.
   b) Let $j_!(\mathscr{F})$ be a sheaf on X given by
   $$W \mapsto \mathscr{F}(W) \qquad W \mapsto 0$$
   where the first happens when $W \subseteq V$, and the second in all other cases. Show that $(j_! \mathscr{F})_p \cong \mathscr{F}_p$ if $p \in U$ and 0 otherwise. .
   c) conclude there is a short exact sequence $0 \to j_!(\mathscr{F}|_U) \to \mathscr{F} \to \iota_*(\mathscr{F}|_V) \to 0$.

1.4 Ringed Space and Locally Ringed Space

Having defined the morphisms in $\mathbf{Sh}(X)$, an important class of sheaves to distinguish are sheaves over rings, as one of the ultimate goals in this Part of the textbook is the generalize the Nullstellensatz to arbitrary (commutative) rings:

Theorem 1.4.1: (Commutative) Ringed Objects in Sheaf

let $\mathbf{Sh}_{\mathbf{C}}(X)$ be the category of sheaves over a category $\mathbf{C}$ that has finite products and a terminal object. Then commutative ringed object exists, and the subcategory $\mathcal{R} \subseteq \mathbf{Sh}_{\mathbf{C}}(X)$ of commutative ringed object is equivalent to the category of sheaves $\mathbf{Sh}_{\mathscr{R}(C)}(X)$ on the *commutative ringed objects* of $\mathbf{C}$.

***Proof*:**
This is a lot of diagram chasing. With some of the proofs later in this section, this result can be proved more easily on the level of stalks.

We shall almost always work in the case of $\mathbf{C} = \mathbf{Set}$ so that $\mathscr{R}(\mathbf{C}) = \mathbf{Ring}$:

Definition 1.4.2: Ringed Space And Structure Sheaf

Let $\mathbf{Sh}_{\mathbf{Ring}}(X)$ be the category of sheaves over $\mathbf{Ring}$ on X. Then given a sheaf of [commutative] rings over X, denote it $\mathcal{O}_X$, the tuple $(X, \mathcal{O}_X)$ is called a *ringed space*. The sheaf $\mathcal{O}_X$ is called the *structure sheaf*, and the sections of the structure sheaf $\mathcal{O}_X$ over an open subset U are called *[regular] functions on U*.

If it is unambiguous, the $\mathcal{O}_X$ is dropped and X is simply called the ringed space.

1.4.3 Warning Though these are called functions, they are elements of an arbitrary [commutative] ring. The reader should have regular function from [11, chapter 22.3] in mind.

Before defining morphisms formally, it is useful to understand why the sheaf morphism appears to go in the "opposite" direction. Geometrically, if we have a continuous map $f : X \to Y$ and a "function" φ on an open set $V \subseteq Y$ (i.e., $\varphi \in \mathcal{O}_Y(V)$), we naturally obtain a function on X by *pulling back* φ via composition: $\varphi \circ f$. The domain of this new function is $f^{-1}(V)$.

In the language of sheaves, this operation assigns to a section in $\mathcal{O}_Y(V)$ a section in $\mathcal{O}_X(f^{-1}(V))$. Recall that by definition of the pushforward sheaf, $(f_*\mathcal{O}_X)(V) := \mathcal{O}_X(f^{-1}(V))$. Therefore, the geometric notion of pulling back functions induces an algebraic map $\mathcal{O}_Y(V) \to (f_*\mathcal{O}_X)(V)$ for every open set V, or globally, a sheaf morphism $\mathcal{O}_Y \to f_*\mathcal{O}_X$.

Definition 1.4.4: Ringed Space Morphism

A map $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_X)$ is a *ringed space morphism* if $f : X \to Y$ is a continuous map and $f^\# : \mathcal{O}_Y \to f_*(\mathcal{O}_X)$ is a sheaf morphism.

It is easy to see that these form a category and shall often be denoted **RingedSp**.

Example 1.4.5: Ringed Space

1. The simplest and prototypical ringed space is again the sheaf of [real/complex] continuous/smooth/holomorphic functions on a manifold/variety with component-wise addition and multiplication.
2. Schemes in the next part shall form a large class of ringed spaces
3. vector fields on a manifold do not form a ringed space as they are not closed under multiplication (they are closed under the lie bracket operation). However, they shall form a $\mathcal{O}_X$-module over manifold, as shall be explored in chapter 5.

An important sheaf that shall often be used is the sheaf of units:

Definition 1.4.6: Unit Sheaf

Let $(X, \mathcal{O}_X)$ be a ringed space. Then $\mathcal{O}_X^*$ is the sheaf given by the fibered product:

$$\begin{tikzcd} \mathcal{O}_X^* \arrow[r] \arrow[d] & \{1\} \arrow[d] \\ \mathcal{O}_X \times \mathcal{O}_X \arrow[r, "\cdot"] & \mathcal{O}_X \end{tikzcd}$$

We shall almost never refer to the categories interpretations of the sheaf of rings in this Part of the book (it shall be important in ref:HERE), however ringed spaces are the central object of study in algebraic geometry.

As elements the sections in $\mathcal{O}_X(U)$ are called functions, there should be a notion of evaluation. When working over coordinate rings, evaluation of f at a point a is equivalent to modding f by the (maximal) ideal $(x-a)$. This works over coordinate rings since by the Nullstellensatz the maximal ideals correspond to points. For a general (commutative) ring, there can be many more pathological behaviours, namely it is not clear how to associate to each point $x \in X$ a maximal ideal $\mathfrak{m} \subseteq R$.

The first natural place to consider is the behaviour at the stalk, namely each point x is associated to the ring $\mathcal{O}_{X,x}$. Is there a natural choice of ideal in this ring? In this case, a natural choice would mean there is only *one* choice of ideal, in which case any $f \in \mathcal{O}_X(U)$ where $x \in U$ maps into $\mathcal{O}_{X,x}$ and can be modded by this maximal ideal. In general the answer is no:

Example 1.4.7: Stalk that is not a Local Ring

Take $X = \{p, q\}$ with the topology given by $\mathcal{T}_X = \{\emptyset, \{p\}, X\}$. Let:

$$\mathcal{O}_X(\emptyset) = \mathbf{0} \qquad \mathcal{O}_X(\{p\}) = k \qquad \mathcal{O}_X(X) = k \times k$$

where k is a field with the only non-trivial restriction map being

$$\mathcal{O}_X(X) \to \mathcal{O}_X(\{p\}) \qquad (a, b) \mapsto a$$

Then the stalk at p and q are:

$$\mathcal{O}_{X,p} = k \qquad \text{and} \qquad \mathcal{O}_{X,q} = k \times k$$

showing that the stalk at q has more than one maximal ideal (namely $k \times \{0\}$ and $\{0\} \times k$).

When working over coordinate rings of varieties, all stalks will be local rings (exercise 1.4.1(1.4.1 (1))). In fact, the same holds for (smooth, complex, topological) manifolds [7, section 2.1]. This motivate focusing on ringed spaces whose stalks are local rings:

Definition 1.4.8: Locally Ringed Space

Let X be a ringed space. Then X is a *locally ringed space* if its stalks are local rings.

Definition 1.4.9: Evaluation on Locally Ringed Space

Let $(X, \mathcal{O}_X)$ be a locally ringed space. Then as every stalk is a local ring, it has a unique maximal ideal, hence a unique field

$$\mathcal{O}_{X,x}/\mathfrak{m}_x = \kappa(x)$$

called the *residue field*. Define *evaluation* of an element $f \in \mathcal{O}_X(U)$ where $x \in U$ by defining:

$$f(x) := \overline{f} \bmod \mathfrak{m}_x$$

where $\overline{f}$ is the image of f in $\mathcal{O}_X(U) \to \mathcal{O}_{X,x}$.

In general, ringed space morphisms *do not* preserve the local ring structure of the stalks (an example is worked through at the beginning of chapter 3, see example 3.1.1). This condition must be imposed:

Definition 1.4.10: Locally Ringed Space Morphism

Let X, Y be locally ringed spaces. Then a morphism of locally ringed spaces $\left(\pi, \pi^{\#}\right) : X \to Y$ is a morphism of ringed spaces such that the induced map of stalks $\pi^{\sharp} : \mathcal{O}_{Y,q} \to \pi_* \mathcal{O}_{X,p}$ (where $\pi(p) = q$) are local ring homomorphisms (the pre-image of the maximal ideal is contained in the maximal ideal).

It can be checked that locally ringed spaces with locally ringed morphisms form a category which shall often be denoted **LocRingedSp** or **LRS**.

Example 1.4.11: Manifolds and Locally ringed sheaves

The two basic examples are varieties and manifolds. Abstracting, topological spaces with continuous function form locally ringed spaces. Complex analytic space (which generalize varieties by taking the vanishing set of holomorphic functions) also form locally ringed spaces. p-adic spaces (central to number theory and modern representation theory) form locally ringed spaces. More generally, Algebraic Spaces and Stacks (defined in Part ref:HERE) form locally ringed spaces.

Overall, locally ringed spaces are the "right" over-arching category when considering the duality between algebra and geometry.

Definition 1.4.12: Isomorphism Of Ringed Spaces

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be two ringed spaces. Then an isomorphism or ringed space is a map $\pi : X \to Y$ where:

1. π is a homeomorphism
2. There is an isomorphism of sheaves Between $\mathcal{O}_X$ and $\mathcal{O}_Y$ where two two sheaves are given via the pushforward π_* (i.e. $\mathcal{O}_Y \cong \pi_* \mathcal{O}_X$ on Y)[a]

[a] By proposition 1.2.24, it is enough to check this isomorphism on stalks

Exercise 1.4.1

1. Show that the stalks of coordinate rings at maximal ideals are all local rings

1.5 *Diffeology

(would love to add a word here)

Pontificating about Points

(I think it is worth writing here my thoughts on what really constitutes a point. Points are fundamental in geometry, you can say there are the necessary indivisible object, and Treating points as secial collections of functions *I think* makes sense because of the result on locally compact spaces withe the upsisde down pi isomorphism, and the more elementary result on compact groups)

(Also, This may veer into "functor of points" as well as "Yoneda's lemma" as well as Distributions as well as Mitchell's theorem were that any abelian category can be seen as an R-module, and as R-modules have elements, we get something reminiscent to recovering points)

2

Schemes I: Basics

Recall the following correspondence:

$$\left\{\text{Affine Varieties}\right\} \underset{\mathcal{Z}(S)}{\overset{\mathcal{I}(V)}{\rightleftarrows}} \left\{\begin{array}{c}\text{Finitely Generated,}\\ \text{Reduced rings}\\ \text{over algebraically closed fields}\end{array}\right\} \tag{2.1}$$

This setups an algebraic-geometric correspondence, where the algebraic object can be thought of as functions on the geometric object, or conversely that the algebraic objects can induce a new geometric space with a notion of "points" (or more generally local information).

finish this above!!

The question naturally arises then if this duality can be extended to general commutative rings. In particular, does there exist some topological space such that:

$$\left\{\,???\,\right\} \underset{\mathcal{Z}(S)}{\overset{\mathcal{I}(V)}{\rightleftarrows}} \left\{\text{Commutative Ring}\right\}$$

The necessity of commutativity stems from the need of prime ideals being well-defined[1]. For a more geometric motivation, smooth/holomorphic functions over a real/complex manifold $\mathbb{C}^\infty(M)/\mathcal{H}(M)$ form a commutative ring since they map into $\mathbb{R}$ or $\mathbb{C}$, and hence we copy this behaviour.

Why do this? This abstraction is not simply for abstractions-sake. It is because the dictionary between algebra and geometry that has been built up to is incredibly useful, and will fix some of the problems of algebraic varieties, for example:

[1]The theory of "non-commutative geometry" using non-commutative rings can be explored in *Spectral Theory* in a class on functional analysis (see [8])

1. There are many algebras over more general rings that have geometric interpretations. For a rather trivial but illustrative example, the coordinate ring of a line whose points are in some arbitrary (commutative) ring R (the coordinate ring being $R[x]$) currently does not have a good representation since the Nullstellensatz requires k to be an algebraically closed fields. Another example that may at first seem "un-geometric" but turns out to have many strong geometric properties are rings such as $\mathbb{Z}$, $\mathbb{Z}[i]$ (the Gaussian integers), and more generally *algebraic number rings* (see [11, chapter 22]). Mathematicians have found that these corresponds naturally to looking for integer solutions (or integral solutions when working over an algebraic ring) of a curves (see [11, chapter 22]). In particular, the element of $\mathbb{Z}[i] = \mathbb{Z}[x]/(x^2+1)$ should be functions that "evaluate" at points that would find information mod $\mathfrak{p}$ to solutions of x^2+1.

2. As we saw in front-matter, localization gives us powerful tools to explore the behavior of a variety at a point. In the theoretical framework given by the Nullstellensatz, we cannot consider such rings, for example, if we take the coordinate ring of an affine line, $k[x]$, takes the closure $k(x)$, or study $k[x]_{(x)}$, we loose finite generation. However, as we saw these certainly still have interpretations as functions on a geometric space (the former being the rational functions on a variety, the latter being the "germs"[2] around $0 \in \mathbb{A}^1_k$).

3. Varieties loose multiplicity information. Take for example two varieties over $\mathbb{C}$ defined by the equations
$$y = x^2 \qquad \text{and} \qquad y = 0$$
then if we take their intersection, we get a point $(0,0)$. The corresponding affine variety is given by taking the union of quotient of the two coordinate rings, namely:
$$\frac{k[x,y]}{(y, x^2 - y)} = \frac{k[x]}{(x^2)}$$
Usually, we would omit the x^2 term and just use $k[x]/(x)$ for the coordinate ring, however this eliminate the multiplicity information, namely that the variety $y = x^2$ "doubly intersects" the variety $y = 0$. The ring $k[x]/(x^2)$ keeps track that we have a "double-point", while $k[x]/(x)$ does not. This is a achieved by the fact that the ring has nilpotent elements.

4. (If you know differential geometry) A geometrical object that has many algebraic parallels is the vector bundle. Most vector bundles are very algebraic, however many are not projective (cannot be embedded into $\mathbb{P}^n_k$, or even $\mathbb{P}^n_R$ for some ring R). We thus need need to generalize better our notion of projective space. This shall lead to different sheaves on spectra and will be studied in chapter 5.

This build-up will introduce to us the *affine scheme*. Many familiar geometric objects shall look like affine schemes, but not all familiar geometric objects shall look like them (notably, projective space). We will need to generalize a bit more and define a *scheme* which is locally an affine scheme. This shall complete the liaison that algebraic geometry promises between algebra and geometry.

We shall then focus on how what properties of commutative rings translate to geometric properties on a scheme, and conversely when imposing desirable geometric properties how does this translate as algebraic properties. Properties such as being an integral domain, PID, UFD, Noetherian, reduced, and so forth will have geometric interpretations, and conversely notions like dimension, smoothness, open/closed immersions, and so forth shall have algebraic counter-parts.

[2] Somewhat surprisingly, germs are also a "purely algebraic" notion given by nilpotent elements and localization, and hence appear as schemes

2.1 Defining Spec

Recall that the maximal ideals of $k[x_1, x_2, \cdots, x_n]/V$ correspond to points in a variety V (theorem 0.1.12). It may then be natural to say that the collection of maximal ideals should be considered the "points" of an arbitrary ring R. These collection were called the spectrum (plural spectra). For such an association to work, there would need to be a functor translating maps between rings to maps between the collection of maximal ideals. In particular, we would want that the functor translating a ring homomorphism $\varphi : R \to S$ to the associated continuous function mapping the maximal ideal $\mathfrak{m} \subseteq S$ to a maximal ideal $\varphi^{-1}(\mathfrak{m})$ (recall that regular function corresponds contra-variantly to k-algebra homomorphisms), However, the pre-image of a maximal ideal need not be maximal breaking functoriality (ex. take $\mathbb{Z} \hookrightarrow \mathbb{Q}$. Then (0) is maximal in $\mathbb{Q}$, but certainly not maximal in $\mathbb{Z}$).

When the theory was originally being developed within spectral theory from functional analysis (see [8]), the spaces were compact Hausdorff spaces, in which case it would have sufficed to work with the maximal ideals as within this setting the pre-image of maximal ideals (of the ring of continuous functions) is usually maximal. However, as demonstrated in the example above this doesn't hold for general rings.

Fortunately, if we loosen our requirement a bit and consider prime ideals, then it is indeed the case that $\varphi^{-1}(P)$ is prime, making it the more natural choice of "points" for functorial purposes, even though it shall produce some more "complicated" points (see example 2.1.3). Working with primes also brings the added benefit of incurring a lot more structure which shall turn out to be very insightful and make the addition of primes not only necessary for the definition to work, but to explain a lot of phenomena[3].

Definition 2.1.1: Spectrum

Let R be an [arbitrary] commutative ring. Then $\operatorname{Spec} R$ is the collection of prime ideals of R. Let $\operatorname{mSpec}(R)$ represent the collection of maximal ideals of R.

We call an element of $\operatorname{Spec} R$ a *point*, and we call an element of R a *regular functions*

In the special case where R is a coordinate ring $k[x_1, ..., x_n]/I(V)$, $\operatorname{Spec} R$ corresponds to points of an affine variety as well as "extra" points associated to primes that are not maximal (ex. $(y-x^2) \subseteq k[x, y]$) along with the zero ideal which is prime in integral coordinate rings; the intuition behind having these extra points shall be elucidated throughout this section.

To avoid confusion between points of $\operatorname{Spec} R$ and ideals of R, we shall write points in square brackets, for example $[\mathfrak{p}] \in \operatorname{Spec} R$ for $\mathfrak{p} \subseteq R$. There is some merit in writing it in the same notation as the equivalence relation notation, as points on in the topology are defined up to the radical of an ideal, for example for $(x^2) \subseteq \mathbb{C}[x]$, $V(\{(x^2)\}) = [(x)]$ (the $V(-)$ function will be defined in section 2.1.1, or see [11, chapter 21]). As we shall see, once a structure sheaf is introduced, the sets $\operatorname{Spec} \mathbb{C}[x]/(x^2)$ and $\operatorname{Spec} \mathbb{C}[x]/(x)$ which each have one element shall not be have the same structure sheaf added to them. We shall see that in most cases, we can interpret the maximal ideals as points and can write an ideal of the form $[(x_1 - a_1, ..., x_n - a_n)]$ as $(a_1, ..., a_n)$ without ambiguity, however this is not always the case (we shall define non-trivial schemes that shall have no such point[4]). In example 2.1.3,

[3] For example, recall that if k was not algebraically closed, there exists regular functions that are not polynomials. Another is the distinct characterization of divisors in number theory which unifies describing the same cohomological properties of schemes

[4] In particular, there shall be no closed points

we shall see that we can write any point associated to a maximal ideal from a coordinate ring in this way.

Another advantage of working with spectra is that there is no need to reference an ambient space for the algebraic subsets, the information is "within" the space! In particular, every quasi-projective variety in [**NateClassicalAlgGeo**] was a subset of $\mathbb{P}^n_k$, while a spectra need not be a subset of $\mathbb{P}^n_k$! In fact, there shall be schemes that are not embed-able in projective space, though they are few and far between[5]. We shall later see under which condition we can embed a scheme into $\mathbb{P}^n_R$ for a ring R, and naturally this shall always be be the case for schemes associated to quasi-projective varieties (covered in section 5.3.3).

When working with coordinate rings R of a variety V, we may consider R to be the set of functions on V, since we may have a notion of "evaluation" for each element of R. We can do the same for a general commutative ring R, motivating the notation f for a general element of a commutative ring R ($f \in R$) and for calling such elements *regular functions*. Notice that if $p(x) \in k[x]$, then we may find the value of the evaluation $p(a)$ for some $a \in k$ by doing

$$\frac{p(x)}{(x-a)} = p(a) \bmod (x-a)$$

To represent the above more formally, every element $f \in k[x]$ (for some algebraically closed field k) is associated to an element of $\mathrm{Hom}_{\mathbf{Set}}\left(\mathrm{Spec}\,(k[x]), \coprod_{(x-a)\in \mathrm{Spec}\,(k[x])} k[x]/(x-a)\right)$, in particular:

$$k[x] \to \mathrm{Hom}_{\mathbf{Set}}\left(\mathrm{Spec}\,(k[x]), \coprod_{(x-a)\in \mathrm{Spec}\,(k[x])} \frac{k[x]}{(x-a)}\right)$$

$$f \mapsto ((x-a) \mapsto f \bmod (x-a))$$

Stare at the above until it makes sense; this is essentially a result of the Nullstellensatz ([**NateClassicalAlgGeo**]). Notably, we can think of a function as holding local information (indeed, the "most local" as it gives point-wise information rather than open-set local, component local, germ local, and so forth) at each point of the spectrum[6]. We extract this point-wise information by evaluating, or in the case of spectra by modding.

Recall that if we take subsets of the classical affine space, we can often define more regular functions[7], for example by taking the (classical) affine line minus the origin, $\mathbb{A}^{\setminus}_k\{0\}$, we have $f(x) = 1/x$ is a well-defined regular function in the classical sense. We shall thus want to focus on all possible functions from $\mathrm{Frac}(R)$[8] that are regular at a point $\mathfrak{p} \in \mathrm{Spec}\, R$; we saw in [11, chapter 21.6] that this collection is given by localizing at the prime: $R_{\mathfrak{p}}$ (for $R = k[x]$ with $\mathfrak{p} = (x-a)$ we had $k[x]_{(x-a)}$). This collection is called the *stalk* at the point $\mathfrak{p}$. Notice that the codomain of the hom-set can be thought of as the disjoint union of field of fractions of the germs $k[x]_{(x-a)}$ as

$$\frac{k[x]_{(x-a)}}{\mathfrak{m}_{(x-a)}} \cong \mathrm{Frac}\left(\frac{k[x]}{(x-a)}\right) \cong \mathrm{Frac}(k) = k$$

[5]We shall see in ref:HERE that all proper scheme are covered by a projective scheme, hence we shall require some "awkward blow-ups" to forcefully construct a scheme that cannot be embedded in projective space

[6]or more generally, at each point of a geometric space, like a manifold

[7]though not always, for example the punctured plane has the same algebraic functions as the plane, see example 2.2.51

[8]Technically, it is required that R be an integral domain as this corresponds to irreducible subsets. Each irreducible subsets will have its own $\mathrm{Frac}(R_i)$ of possible regular functions

which show that we recover the same notion of evaluation when working over the localization. This greater generality shall be important when we define the sheaf on $\operatorname{Spec} R$ namely since the ring $R_{\mathfrak{p}}$ shall be a *local ring* that has only a single maximal ideal $\mathfrak{m}_{\mathfrak{p}}$ making it unambiguous which ideal to mod by (as explored in section 1.4), and so we shall use it for our definition:

Definition 2.1.2: Value Of Regular Function

Let $f \in R$. Then the *value* of f at the point $\mathfrak{p} \in \operatorname{Spec}(R)$ is:

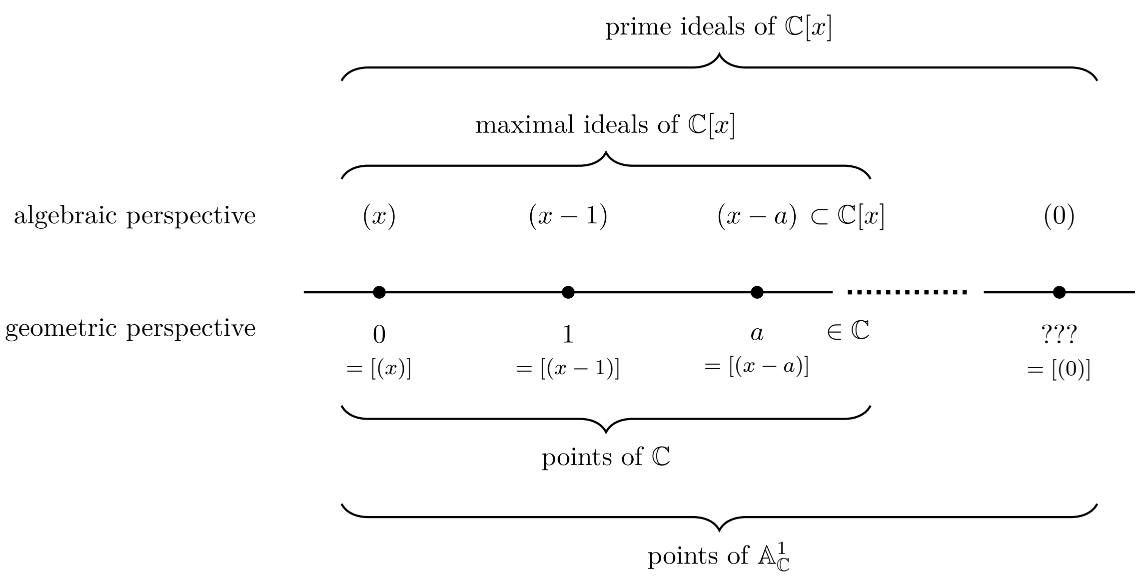
$$f(\mathfrak{p}) := \overline{f} \bmod \mathfrak{m}_{\mathfrak{p}} \in R_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}$$

Let us now give multiple examples to get some grasp on spec and the functions on Spec

Example 2.1.3: Spectra and [regular] functions

This is a long set of examples, the reader should cover as many as they need to feel comfortable with the notion of spectrum; they may return here later for further elucidation on spectra.

1. Take $\operatorname{Spec}(\mathbb{C}[x])$ which consists of all prime (in fact maximal) ideals $(x-a)$ for $a \in \mathbb{C}$ and the prime ideal (0). This spectrum is denoted $\mathbb{A}^1_{\mathbb{C}}$; more generally the spectrum of $R[x_1, \ldots, x_n]$ is denoted $\mathbb{A}^n_R$. As $\mathbb{C}[x]$ is a finitely generated reduced ring over an algebraically closed field, the maximal ideals correspond exactly to the classical points of $\mathbb{A}^1_{\mathbb{C}}$ by the Nullstellensatz, with the exception of the special point (0) called the *generic point* which we shall return to soon. To visualize this set, imagine a real line (to remember that it is a 1-dimensional object over $\mathbb{C}$):



The generic point was located "away" from the line, but that is not in fact accurate. The problem is that it is a single point that is near every other point, while still being a single point. What it means for points to be near shall be defined shortly when the Zariski topology on spectra is introduced, but for now you can imagine that this causes issues when trying to draw such a point on a Euclidean piece of paper. There are many ways to visualize it: some write it as a cloud of points in the corner, others at some edge of the visual. Personally, I find it fruitful to think of it as a "higher-dimensional" point that lives over all other points of $\mathbb{A}^1_{\mathbb{C}}$. This perspective shall be corroborated in section **??**.

Let us evaluate some regular functions $f \in \mathbb{C}[x]$. Let's say $f = x^2 + 2x + 1$ and input $[(x-3)]$. Then the answer is

$$16 \bmod (x-3) = f(3) \bmod (x-3)$$

which can be thought of as our usual answer $f(3)$! Notice that $\mathbb{C}[x]/(x-a) \cong \mathbb{C}$, and hence the codomain when evaluating at each point is the same. We shall see this is not the case soon. We can think of the fact that all the codomains of evaluation coincide as being a special symmetry for varieties over algebraically closed fields (that the evaluation information locally can all be embedded into a single global function-space).

Let us now plug in the generic point (0) into our function f (i.e. the point that isn't in our usual visualization of the complex line). We get:

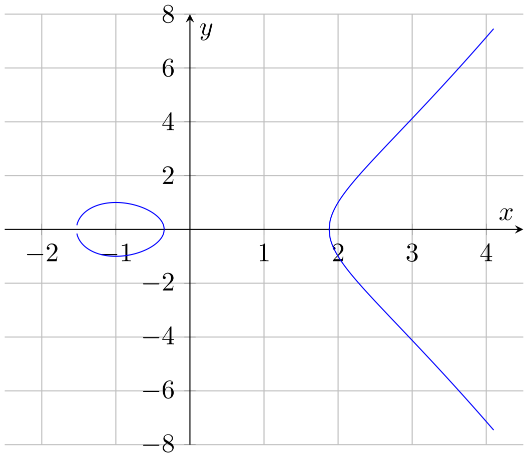
$$f \bmod 0 \in \mathbb{C}[x]$$

that is, it gives back the function itself, and is the only point for which the codomain of f is not isomorphic to $\mathbb{C}$. You can think of (0) as a "1"-[complex]-dimensional point. When we have spectra of higher dimensional (irreducible) schemes, say dimension n, then the generic point will be "n"-dimensional. We can't yet define dimension as we are still working generically with sets; this shall be rectified in section 2.6.

More generally, this entire discussion works just as well over any algebraically closed field $k = \overline{k}$, as the completeness of $\mathbb{C}$ did not factor into the above results.

Often when working with $\overline{k}$-polynomials over several variables, we shall write the maximal ideals as tuples of points, e.g. $[(x-2, y-3)] \in \mathbb{A}^2_k$ will be represented as $(2,3) \in \mathbb{A}^2_k$. We shall often then denote the generic point $[(0)]$ as η to distinguish it from the point 0 which is associated to the ideal (x).

2. Take $y^2 - x^3 + 3x - 1 \in \mathbb{C}[x, y]$ and consider Spec $\frac{\mathbb{C}[x,y]}{(y^2-x^3+3x-1)}$. This spectrum corresponds to the elliptic curve $y^2 = x^3 - 3x + 1$. The spectrum will contain all the maximal ideals of the ring $\frac{\mathbb{C}[x,y]}{(y^2-x^3+3x-1)}$ as well as a generic point η corresponding to $[(0)]$ representing the entire space, and hence can be visualized as:



More generally, all the points of an affine algebraic set correspond to the maximal ideals of a spectrum, and so a subset of the geometry of spectra includes the classical geometry of curves, surfaces, and so forth!

3. Here is a Spectra that has no corresponding algebraic set in the classical sense: $\mathrm{Spec}(\mathbb{Z}) = \{(2),(5),(7),\cdots,(p),...,(0)\}$. In other words, $\mathrm{Spec}(\mathbb{Z}) = \{(p)\big|p \text{ is prime}\} \cup \{0\}$ and $\mathrm{mSpec}(\mathbb{Z}) = \mathrm{Spec}(\mathbb{Z})\backslash\{(0)\}$.

   Let us evaluate some of the regular function $n \in \mathbb{Z}$ on the points $\mathrm{Spec}\,(\mathbb{Z})$. The regular function 15 at the point $[(7)] \in \mathrm{Spec}\,(\mathbb{Z})$ is $15 \bmod (7) = 1 \bmod (7)$. The regular function 9 evaluated at $[(3)]$ is $0 \bmod (3)$. We may even say it has a double-zero at 3.

   Hence, functions on $\mathrm{Spec}\,(\mathbb{Z})$ return some information about divisibility by primes.

4. For any field k, $\mathrm{Spec}\,k = \mathrm{mSpec}\,k = \{(0)\}$ (or $\mathrm{Spec}\,k = \{\eta\}$). For example, $\mathrm{Spec}\,\mathbb{Z}/3\mathbb{Z} = \{(0)\}$. Thus, fields are like "points" within algebraic geometry. An advantage of adding a structure sheaf will be to remember the original field k: we shall see that as schemes $\mathrm{Spec}\,K \ncong \mathrm{Spec}\,L$ if $K \ncong L$, as their sheaves will differ. Though the geometric picture as spectra is trivial, they shall still hold valuable non-trivial information.

   The evaluation of each function at the single point returns the function modulo (0), namely the function itself as an element of k:

$$3 \bmod 0 = 3 \in k$$

5. Take $\mathrm{Spec}\,\mathbb{R}[x]$. This consists of $[(0)]$, all prime (even maximal) ideals of the form $[(x-a)]$, and all prime (even maximal) ideals that can be represented by an irreducible 2nd degree polynomial $\left[\left(x^2+bx+c\right)\right]$, for example $\left[\left(x^2+1\right)\right]$. Evaluating at any of these points (besides $[(0)]$) will give a field: if evaluating points of the form $(x-a)$ the field will be isomorphic to $\mathbb{R}$, and it evaluating points of the form $\left(x^2+ax+b\right)$, the field will be isomorphic to $\mathbb{C}$. As every irreducible quadratic splits into conjugate pairs of linear terms over $\mathbb{C}$, we can think of $\mathrm{Spec}(\mathbb{R}[x])$ as being a copy of $\mathbb{C}$ folded over itself attaching the conjugate pairs (don't continue reading until this is clear)
   Let us take the function x^3-1 and evaluate it as some point. At the point $(x-2)$ we get

7 mod $(x-2)$, or in our classical interpretation, $f(2)$. At the point $\left(x^2+1\right)$, we get:

$$x^3 - 1 \equiv -x - 1 \bmod \left(x^2+1\right)$$

which we can think of as being $-i-1$. Hence, the Spectra of polynomials over fields that are not algebraically closed will behave more like algebraically closed fields as the non-linear maximal primes shall play the role of the required "additional points"! Hence, one piece of the puzzle for the "additional points" in a spectra is that they allow us to think of the spectrum as being "algebraically closed"!

6. Generalizing the above result, show that the points of $\operatorname{Spec}\mathbb{Q}[x]$ (minus the generic point 0) are in bijective correspondence with the points of $\overline{\mathbb{Q}}$, the algebraic closure of $\mathbb{Q}$, where the points of $\overline{\mathbb{Q}}$ are glue to their galois-conjugates: $\mathrm{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$.

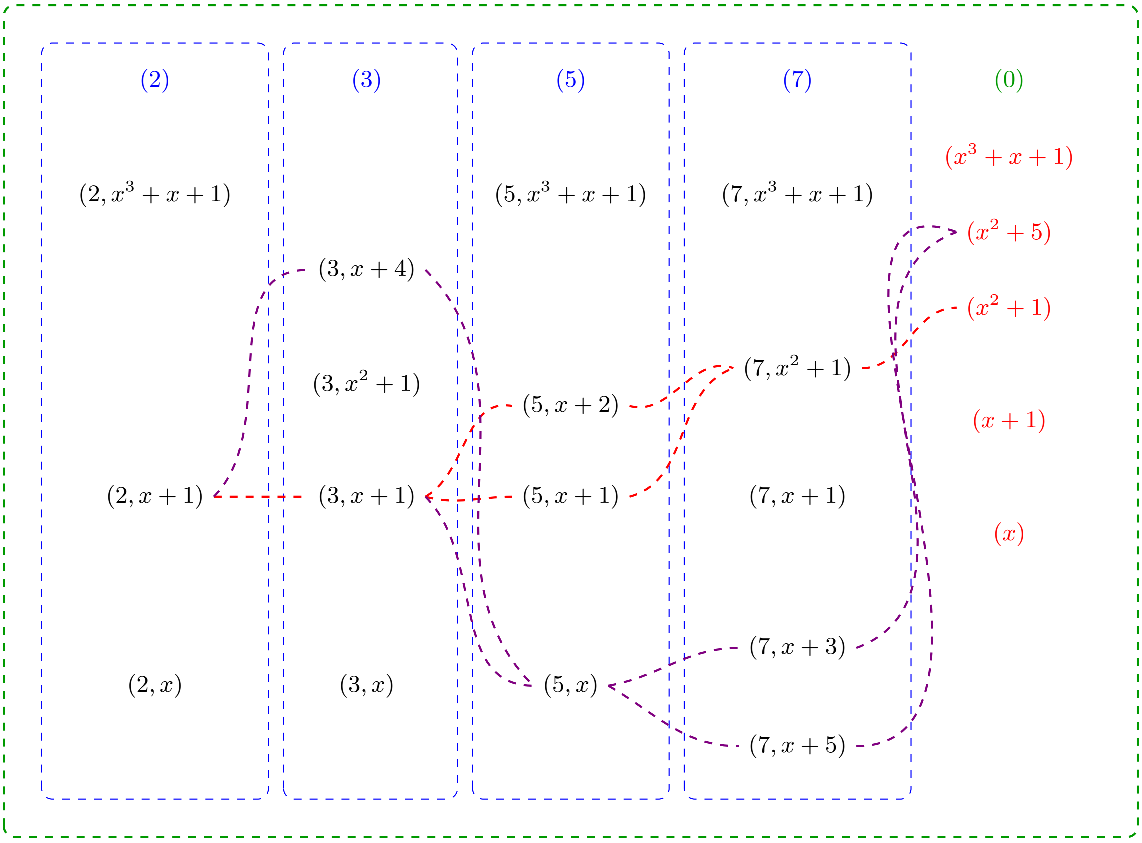
7. Consider Spec $\mathbb{Z}[x]$. Unlike $\mathbb{Q}[x]$, the ring $\mathbb{Z}[x]$ is not a PID and has Krull dimension 2, As it is not a PID, it has prime ideals of the form $(p, f(x))$ where $f(x)$ and $f(x)$ mod p is irreducible. This is the first example where the irreducible functions of a polynomial ring are not in one-to-one correspondence with the points, there are more points than there are irreducible functions[a]! The maximal ideals are of the form

$$(p, f(x))$$

while the prime ideals are of the form:

$$(p) \quad (f(x)) \quad (p, f(x))$$

There are now non-maximal prime ideals, namely the principal ideals (p) for prime p and principal ideals $(f(x))$ where $f(x)$ is irreducible over $\mathbb{Z}$ (or equivalently by Gauss's lemma over $\mathbb{Q}$). To understand this spectrum better, the following is a part of a visual of Spec $\mathbb{Z}[x]$:



Let us parse this image: the maximal points are all of the form $(p, f(x))$ and they are the black points. The prime (p) are contained in each maximal ideal $(p, f(x))$. They are like a more specialized generic point, where they are not near all the points, but near every point for which the ideal representing the point contains (p). Due to this, each column is surround by dotted blue lines that indicating the generic location of each prime $(\ p\ )$. The entire image is surround by a green dotted line which represents the generic point $[(0)]$ and how it is close to every point. Notice that there are gaps in the columns exactly where the irreducible polynomial $f(x)$ is reducible mod$p$. For example $x^2+1 \bmod 2$ is reducible with solution $x = 1$, and so $\left(2, x^2+1\right)$ is not prime, namely given $\mathbb{Z}[x]/\left(2, x^2+1\right) = (\mathbb{Z}/2\mathbb{Z})[\omega]$ we have

$$(\omega+1)(\omega+1) = \omega^2 + 2\omega + 1 = 1 + 1 = 0$$

showing that $\left(2, x^2+1\right)$ is not prime, and hence not a point.
Next, each point $(f(x))$ is also a generic point, however their "shape" is not a simple as the points (p). The simplest case is when a point $(p, g(x))$ contains $f(x)$, namely $f(x) = g(x)$. In the image, we see that $\left[\left(3, x^2+1\right)\right]$ is a point and hence $\left[\left(x^2+1\right)\right]$ is near this point. On the other hand, both $(5, x+2), (5, x+1)$ are contained in $\left(x^2+1\right)$, precisely because x^2+1 factors into a product of those two linear terms mod 5 . Hence $\left[\left(x^2+1\right)\right]$ is near $[(5, x+2)]$ and $[(5, x+1)]$. This closeness is shown via a red-line passing near these points and represents the fact that $\left[\left(x^2+1\right)\right]$ is near each of these points. Essentially, this represents the quadratic residues of x^2+1 !

The same thing has been done for the point $\left[\left(x^2+5\right)\right]$. Some extra points have been added to the image to demonstrate more clearly the branching behavior. Notice that $\left[\left(x^2+5\right)\right]$ and $\left[\left(x^2+1\right)\right]$ "intersect", for example at $[(2, x+1)]$ and at $[(13, x+5)]$. These give us some number-theoretic information about these equations!
The way I like to interpret these "fuzzy" points corresponding to the irreducible polynomials from some intuition given by complex analysis. This is not unreasonable as the spectrum in many ways mirror the behavior of algebraically closed fields and complex analysis is the analysis of functions that are locally infinite polynomials, and indeed these two fields are very closed linked, something we shall cover in ref:HERE.
For now, consider the following: if we have a "germ" around a single point of a holomorphic function (i.e. the information given by its power-series representation around an arbitrarily small neighborhood around the point), we in fact have information about the entire holomorphic function. From this perspective, it doesn't matter at which point we choose to take the power-series representation, it will return equivalent information. Thus, the information does not live at any point point in particular but "generically" at all these points. If we choose to "specialize" and specify which data we want, we shall evaluate further to a single maximal ideal.
Notice that $\mathbb{Z}[x]$ is seen on a two-dimensional grid. We shall later see that $\operatorname{Spec} \mathbb{Z}[x]$ is naturally seen as a 2 -dimensional scheme, while $\operatorname{Spec} \mathbb{Q}[x]$ shall be 1-dimensional. Furthermore, the red-lines and blue dotted lines will be seen as 1-dimensional points, the generic point will be seen as a 2 -dimensional point, while all the maximal ideals will be 0 -dimensional points.

Finally, let us consider evaluation of functions $f \in \mathbb{Z}[x]$ at points. At maximal points, we shall get values in the field $\mathbb{F}_p(\omega)$[b] where ω is a root of the irreducible polynomial. For example, take $[(7, x^3+x+1)] \in \operatorname{Spec} \mathbb{Z}[x]$ and $x^2-4 \in \mathbb{Z}[x]$. Then the value will be in $(\mathbb{Z}/7\mathbb{Z})(\omega)$ where ω is an element such that $\omega^3+\omega+1=0$. Then we would get that the value of x^2-4 is $\omega^2+3 \in \mathbb{F}_7(\omega)$.

Another example would be evaluating at the point $[(3, x^2+1)]$ the function x^4-1. Working in $\mathbb{Z}[x]/(3, x^2+1) \cong \mathbb{F}_3[x]/(x^2+1) \cong \mathbb{F}_9$, we use $x^2 \equiv -1$ to compute:

$$x^4-1 \equiv (x^2)^2-1 \equiv (-1)^2-1=1-1=0 \pmod{3,\, x^2+1}$$

In other words, $f([(3, x^2+1)])=0 \in \mathbb{F}_3(i)$. As we are evaluating at maximal ideals, we shall always get a "number" as the output; in particular, since we can always embed finite fields into an algebraic closure, the output of a function at a maximal point always lies within an algebraically closed field.

Let us now evaluate at point that corresponds to a non-maximal prime ideal, say at 3. The output of each function $f \in \mathbb{Z}[x]$ will then be:

$$\bar{f} \in (\mathbb{Z}/3\mathbb{Z})[x]=\mathbb{F}_3[x]$$

The way I like to think about these output is that we are asking for some of the information stored in f, namely it shall return enough information about f to determine the generic behavior of that function around $[(3)]$. If we wanted the generic behavior around a different generic point, for example $\left[\left(x^2+1\right)\right]$, then we would get:

$$f(i) \in \mathbb{Z}[i]$$

Namely, if we choose any irreducible polynomial, we are choosing which new element of an integral extension we are adding that we then evaluate the function f at. We may then extract further information by modding by different primes. It shall often be the case that evaluating at a generic point corresponds to reducing the information of a function down to a smaller geometric object or to modding information.

8. If you know some number theory, $\operatorname{Spec}\mathbb{Z}[\sqrt{-5}]$ is the quintessential example of a ring that has primes that are *not* principle, for example $(2, 1+\sqrt{-5})$ is prime but is provably not principal (see [11, chapter 22]). Evaluating at (2) and $(1+\sqrt{-5})$ will return 0, however the zero set of both of these function's shall be larger than $(2, 1+\sqrt{-5})$; we shall require the vanishing set to contain both of these.

9. The following is another example of a spectrum with a single element. Take $\operatorname{Spec}(k[x]/(x^2))$, the spectrum of a ring with nilpotent elements. This ring can be thought of as an extension of k by an element $\epsilon = x \bmod x^2$ that is thought to be "infinitely small". It has the interesting property that $\epsilon^2 = 0$, even though ϵ itself is nonzero. As the spectrum only contains one point[c], namely $[(\epsilon)]$, this can be thought of as a "thicker" point (as shall be explored in greater detail in section 2.2.2). The ring $k[x]/(x^2)$ is called the *dual numbers* and shall be a recurring ring to study the behaviour of nilpotent elements. This ring has the interesting property of having the element $\epsilon \in k[\epsilon]$ which always evaluates to 0 even though it is not the zero element! This gives us our first example of a function[d] that is not defined by what it does at every point, namely because the functions 0 and ϵ have the same outputs.

   More generally, $k[x]/(x^n)$ has a single element in its spectrum. As k and $k[x]/(x^n)$ have the same underlying topological space, what shall distinguish them will be the sheaf structure associated to these two spectra (they shall not be isomorphic as schemes).

10. The above ring only had one point in its spectrum. There are many spectrum's with only two points: a generic point and a non-generic point. A large class of such rings shall be *local rings*. For example $k[x]_{(x-a)}$ will have (0) and $(x-a)$ as the two points of the spectrum. Another example example of a spectrum with only two points is $k[[x]]$ which has (x) and (0) as its two points. As spectra, $k[x]_{(x)}$ and $k[[x]]$ are indistinguishable. What shall distinguish these two spectra shall be the sheaf-structure associated to these two spectra.

11. Let us take a look at the number theoretic equivalent of localization: take $\mathbb{Z}_{(p)}$, the localization of $\mathbb{Z}$ at $\mathbb{Z}\backslash(p)$. This spectrum contains two points, $[(0)]$ and $[(p)]$. In particular, $\operatorname{Spec}\mathbb{Z}_{(p)}$ is not a single point like $\operatorname{Spec}\mathbb{F}_p$, it somehow has some more "fuzz" around it's single point $[(p)]$. This can be thought of as $\operatorname{Spec}\mathbb{Z}_{(p)}$ "remembering" that there is some arbitrarily small neighborhood around the point $[(p)]$ that it must keep track of. Once we introduce a topology on spectra's, we'll see that the closure of $[(p)]$ is itself (it is a closed point), while the closure of $[(0)]$ is $\{[(0)], [(p)]\}$, motivating the idea that $[(0)]$ "remembers" that there is some more information outside of just the single point! This shall be even more clear once we introduce the notion of dimension: the point $[(p)]$ will be zero-dimensional, while $[(0)]$ will be one-dimensional!

12. Let $R = k[x] \times k[x]$. What is $\operatorname{Spec} R$? Notice that $\operatorname{Spec} R$ does not contain just one generic point, but two! Importantly, $[(0)]$ is not prime as $(1,0)\cdot(0,1) \in (0)$ but $(1,0), (0,1) \notin (0)$.

    In general, the primes of a product ring $A \times B$ are all of the form $\mathfrak{p} \times B$ or $A \times \mathfrak{q}$ where $\mathfrak{p} \in \operatorname{Spec}(A)$ and $\mathfrak{q} \in \operatorname{Spec}(B)$ (exercise). Hence $\operatorname{Spec}(A \times B) \cong \operatorname{Spec}(A) \sqcup \operatorname{Spec}(B)$: the spectrum of a product is the disjoint union of the spectra. In our case, $\operatorname{Spec}(k[x] \times k[x]) \cong \mathbb{A}^1_k \sqcup \mathbb{A}^1_k$, two disjoint copies of the affine line, each with its own generic point.

13. Take $\operatorname{Spec} \mathbb{F}_p[x] = \mathbb{A}^1_{\mathbb{F}_p}$. As it is a Euclidean domain, the points are (0) and $(f(x))$ for all irreducible polynomials over $\mathbb{F}_p$. One of the interesting facets of $\mathbb{A}^1_{\mathbb{F}_p}$ is that its closed points are far richer than the p elements of $\mathbb{F}_p$: for each $n \geq 1$, each irreducible polynomial of degree n over $\mathbb{F}_p$ gives a closed point whose residue field is $\mathbb{F}_{p^n}$. Since there are irreducible polynomials of every degree, $\mathbb{A}^1_{\mathbb{F}_p}$ has infinitely many closed points despite $\mathbb{F}_p$ being finite. This richness in points will allow us to distinguish functions on $\mathbb{F}_p$ by their evaluation: recall that a polynomial is not distinguished by evaluation at all points in $\mathbb{F}_p$ (e.g. $x^p - x$ vanishes everywhere on $\mathbb{F}_p$ but is nonzero), however a polynomial *will* be distinguished by its evaluation at all points in $\operatorname{Spec}(\mathbb{F}_p[x])$, as we shall show using proposition 2.1.4.

14. Take $\operatorname{Spec}(\overline{k}[x, y]) = \mathbb{A}^2_{\overline{k}}$ (e.g. $\overline{k} = \mathbb{C}$). Then the points are:

$$(0) \qquad (x-a, y-b) \qquad (f(x,y))$$

where $f(x,y)$ is an irreducible polynomial in $\overline{k}[x,y]$. Visually, all the primes of the form $(x-a, y-b)$ correspond to the usual points in $\mathbb{C}^2$. The points $(f(x,y))$ can be thought of as tracing out their curve in $\mathbb{C}^2$; for example $(y - x^2)$ is the point that is the curve $y = x^2$. Notice how the maximal ideals correspond to "0"-dimensional points, while the ideal $(y - x^2)$ corresponds to a "1"-dimensional point. Similarly, (0) is a "2"-dimensional point. We shall come back to this through Krull dimensions in section 2.6.

More generally, by the Nullstellensatz the maximal points of $\overline{k}[x_1, \ldots, x_n]$ are the ideals of the form:

$$(x_1 - a_1, x_2 - a_2, \ldots, x_n - a_n)$$

The prime ideals of this ring don't lend themselves to a nice classification. Each prime ideal corresponds to an irreducible variety of a certain dimension, and hence the classification of the prime ideals is equivalent to the classification of irreducible varieties, which is a rather difficult task that is carried out in special cases, and more generally leads to the theory of moduli spaces. Some context on this classification in a classical setting was covered in [**NateClassicalAlgGeo**].

15. Find the points of $\operatorname{Spec}(k[x_1, \ldots, x_n])$ where k is not algebraically closed, ex. $\mathbb{Q}$; generalize example 5[e].

[a]This sort of exploration on the correspondence between objects that "principal" (or locally principal) and how much they are not will be the focus of section 5.4

[b]Recall that $k[\omega] = k(\omega)$, or re-prove this

[c]Prove that $[(0)]$ is not a prime ideal: $\epsilon \cdot \epsilon = 0 \in (0)$ but $\epsilon \notin (0)$

[d]granted, a generalization of the notion of function

[e]If you solved it, note that $(\sqrt{2}, \sqrt{2})$ is glued to $(-\sqrt{2}, -\sqrt{2})$ but *not* to $(\sqrt{2}, -\sqrt{2})$

2.1.1 Zariski Topology

Having defined the set that represents the space, the next thing to do is give a topology to this set. This topology would have to be a generalization of the Zariski topology defined on $k[x_1, ..., x_n]$ using the definition of regular function that was just introduced:

Definition 2.1.4: Zariski Topology

Let R be a commutative ring and $\operatorname{Spec}(R)$ the collection of prime ideals of R. Then define the *closed sets* for the Zariski topology on $\operatorname{Spec}(R)$ to be:

$$V(S) = \{x \in \operatorname{Spec}(R) \mid f(x) = 0, \ \forall f \in S\} = \{[\mathfrak{p}] \in \operatorname{Spec}(R) \mid S \subseteq \mathfrak{p}\}$$

Take a moment to see how this parallels the condition that all the functions in the coordinate ring must equal zero when evaluated at $\mathfrak{p}$. the notation $f(x) = 0$ will eventually be correctly re-interpreted when we show the sheaf naturally induced by R is a *locally ringed space*, and hence evaluation is well-defined for any ring element. It is an exercise in commutative algebra to check that this is indeed a topology, namely:

Lemma 2.1.5: Zariski-closed Sets Form Topology

Let $\operatorname{Spec}(R)$ be the collection of primes and let:

$$\mathcal{T} = \{V(S) \mid S \subseteq R\}$$

Then:

1. $\mathcal{T}$ contain the empty set and $\operatorname{Spec}(R)$ (these are both the vanishing set of single elements)
2. $\mathcal{T}$ is closed under arbitrary unions and finite intersections, namely:

$$\cap_i V(I_i) = V\left(\sum_i I_i\right)$$

for arbitrary indexing set, and that

$$V(I_1) \cup V(I_2) = V(I_1 I_2)$$

Furthermore, for any radical $I \subseteq R$, $V(I) \cong \operatorname{Spec} R/I$ as as topological spaces[a].

[a]and as sheaves, which we'll see later

***Proof*:**
see [11, chapter 21.1].

Note that the vanishing set $V(-)$ defined for the spectra of rings usually have more points than their counterpart for coordinate rings in classical algebraic geometry as they will contains primes that are not maximal ideals (particularly for rings of "dimension" ≥ 2 as we have briefly discussed), and these primes will be within $V(S)$ for appropriate S.

The reader should check how this topology compares to the topology for classical algebraic sets. For example, as there exist generic points in most spectra we have that this space is no longer T_1 (it shall soon be shown that generic points are not closed). If R is Noetherian, then $\operatorname{Spec} R$ is a Noetherian topological space; see exercise 2.1.1 and section 2.1.3 for more comparisons.

Just like for varieties, we may define the inverse function $I(-)$, that is for [nonempty] $S \subseteq \operatorname{Spec}(R)$,

$$I(S) \subseteq R \qquad I(S) = \bigcap_{[\mathfrak{p}] \in S} \mathfrak{p}$$

Hence $I(S)$ is a radical ideal (recall the intersection of prime ideals is a radical ideal, think $2\mathbb{Z} \cap 3\mathbb{Z} = 6\mathbb{Z}$). Just like in classical algebraic geometry:

Proposition 2.1.6: Properties of $I(-)$ and $V(-)$

1. $I(S)$ is an ideal of R, and furthermore it is always *radical*.
2. $I(-)$ is inclusion-reversing
3. $I(A \cup B) = I(A) \cap I(B)$, and $I(A \cap B) = \sqrt{I(A) + I(B)}$.
4. $I(\overline{S}) = I(S)$
5. $V(I(S)) = \overline{S}$ and $I(V(J)) = \sqrt{J}$.
6. If S is closed, then $V(I(S)) = S$. If I is radical, then $I(V(J)) = J$.
7. $I(-)$ and $V(-)$ are an inclusion-reversing bijection between closed subsets of $\operatorname{Spec}(R)$ and radical ideals of R. This is called a *galois correspondence*.

Hence:

$$\text{closed subset} \subseteq \operatorname{Spec}(R) \qquad \Leftrightarrow \qquad \text{radical ideal of } R$$

***Proof*:**
exercise.

Thus, we get the (surjective) correspondence:

$$\{\text{Radical ideals of } R\} \underset{V(-)}{\overset{I(-)}{\rightleftarrows}} \{\text{closed sets of } \operatorname{Spec}(R)\} \tag{2.2}$$

Notice that prime ideals are radical and hence are part of this association. This correspondence does not extend to all ideals until a sheaf, the structure sheaf, is introduced (see section 2.2.2).

As we see, the Zariski topology on $\operatorname{Spec}(R)$ is a generalization of the Zariski topology on $k[x_1, ..., x_n]$. The closed sets of $\operatorname{Spec}(R)$ will contain the "traditional" points as well as the new points, for example $V(xy, yz)$ will contain all the "traditional" points where $y = 0$ or $x = z = 0$, as well as the generic point that vanish on $V(xy)$ and $V(yz)$ (i.e. $[(y)]$ vanishes on this set). Just like for affine varieties, $\operatorname{Spec}(R)$ is almost never Hausdorff. This is even clearer in $\operatorname{Spec}(R)$, for if there are primes that are not maximal, these are *not* closed points (in a Hausdorff or T_1 space, every point is closed)

Example 2.1.7: Non-closed Points
Take $\mathbb{A}^2_{\mathbb{C}}$ with $[(y^2 - x)]$ and $[(x-2, y-4)]$. The point $[(y^2 - x)]$ is not closed. Notice that:

$$[(y-4, x-2)] \in V(I([(y^2-x)])) = V(y^2 - x) \overset{\text{pr. } 2.1.6}{=} \overline{\{[(y^2-x)]\}}$$

Showing that the point is *not* closed, namely points associated to non-maximal prime ideals are not closed. However, notice the set $V(I([(y^2-x)]))$ is closed by definition. In a moment, the point $[(y^2-x)]$ will be seen as the generic point of the above closed set.

In fact, due to Hilbert's Nullstellensatz, we know that in $\mathbb{A}^n_k$ the maximal ideals correspond to the "traditional points", that is the closed points. From here on, we shall call the "traditional points" (more generally points associated to maximal ideals) the *closed points*, and the "extra" or "bonus" points that were being referred to shall be called the *non-closed points* or *generic points*.

With a topology, we can now link points like $[(x-2, y-4)]$ to points like $[(y-x^2)]$. As these are point, it doesn't make sense to say one point contains another. However we we be able to link these points using closure:

Definition 2.1.8: Specialization And Generization

Let $x, y \in X$ be two points in a topological space. Then we say that x is a *specialization* of y and y a *generization* of x if

$$x \in \overline{\{y\}}$$

Algebraically, $[\mathfrak{q}]$ is a specialization of $[\mathfrak{p}]$ if and only if $\mathfrak{p} \subseteq \mathfrak{q}$, hence $V(\mathfrak{p}) = \overline{\{[\mathfrak{p}]\}}$.

We can now define generic point more precisely:

Definition 2.1.9: Generic Point

Let X be a topological space. Then $p \in X$ is a *generic point* for a closed subset K iff

$$\overline{\{p\}} = K$$

By this definition, closed points are technically generic points associated the closet set of a maximal idea; we shall only use the term generic point for closed sets that are not given by a maximal ideal. With generic points and the Zariski topology defined, the idea that generic points of a closed set (including all of $\text{Spec}(R)$) are close to all the points in the closed subset (it is contained in all the open subsets (or all the open subsets of $V(S) \cap U$).

Let us end with upgrading proposition proposition 2.2.7

Proposition 2.1.10: Nilpotent Spectrum Homeomorphism

Let R be a ring and I an ideal of nilpotent elements. Then

$$\text{Spec}\, R/I \to \text{Spec}\, R$$

is a homeomorphism.

***Proof*:**
The map is a bijective continuous map. Show the inverse is continuous.

Non-closed points shall have their use in maps such as $k[x, y] \to k[x](y)$ as well as in intersection theory where the intersection or "creases" of schemes shall often be generic points. We shall see that there is a "local-ring" associated to both closed and non-closed points alike and their reduction behaviour is the same (though naturally being in different dimensions means that the behaviour may differ in exactitude's).

2.1.2 (Topological) Basis: Distinguished Open Sets

As the topology of $|\mathrm{Spec}\,(R)|$ is generated by the closed sets $V(S)$, and these sets are the intersection of:

$$V(S) = \bigcap_{f\in S} V(f)$$

the sets $V(f)$ generate the topology of $\mathrm{Spec}\,(R)$. We may want to call these sets "distinguished closed sets", however we shall favor their open counterpart due to their simpler behaviour when introducing sheaf theory to spectra[9]:

Definition 2.1.11: Distinguished Open Sets

The sets $D(f)$ are called the *distinguished open sets*, *basic open sets*, or *principal open sets*, and are defined to be

$$D(f) = X_f = |\mathrm{Spec}\,(R)| \setminus V(f) = \{\mathfrak{p} \in \mathrm{Spec}\,(R) \mid f \notin \mathfrak{p}\}$$

that is, it is the collection of prime ideals of R that do not contain f. Any open set U can be written as:

$$U = \mathrm{Spec}\,(R) \setminus V(S) = \mathrm{Spec}\,(R) \setminus \bigcap_{f\in S} V(f) = \bigcup_{f\in S} D(f)$$

Writing it as $D(f)$ may be helpful; Prof. Ravi Vakil calls these "doesn't-vanish sets" which may serve as a helpful mnemonic. These sets are usually not closed (especially if $\mathrm{Spec}\,(R)$ is connected, in which case unless $D(f) = \emptyset$ it cannot be), and hence they are not the vanishing set for any functions in R. However, they are still isomorphic to a spectrum:

Proposition 2.1.12: Distinguished Open Sets and Localization

Let $D(f) \subseteq \mathrm{Spec}\,(R)$ be a distinguished open set. Then:

$$D(f) \cong_{\mathbf{Top}} \mathrm{Spec}\,(R_f)$$

justifying the X_f notation. Furthermore, if $g = f_1 f_2 \cdots f_n$, then

$$\bigcap_{i=1,\cdots,n} (\mathrm{Spec}\,R)_{f_i} = (\mathrm{Spec}\,R)_g$$

or as distinguished open sets if $D(f_1), \ldots, D(f_n) \subseteq \mathrm{Spec}\,(R)$, then their intersection is the distinguished open

$$D(f_1 \cdots f_n)$$

[9]There shall be a natural unique open subscheme structure up to isomorphism, but infinitely many natural closed subscheme structures up to isomorphism due to the possibility of nilpotents.

***Proof*:**
Recall that $R_f = R[f^{-1}]$, and that $\mathfrak{p} \subseteq R[f^{-1}]$ iff "$f \notin \mathfrak{p}$ and $\mathfrak{p} \in R$". This is exactly the points of $D(f)$; it is left to check that the map is bi-continuous. The second part is left as an exercise.

Corollary 2.1.13: Distinguished Open Sets Form Topological Basis

The collection of distinguished open sets $\{D(f)\}_{f\in R}$ from a topological basis for $\operatorname{Spec} R$ with the Zariski topology

***Proof*:**
Distinguished opens cover $\operatorname{Spec} R$ by definition, and proposition 2.1.12 show they are closed under intersections.

A mistake the reader may make is to believe that each $D(f)$ can be thought of as the spectrum minus some points, but this need not be the case:

Example 2.1.14: Spectrum and Distinguished Open Sets
Take $X = \operatorname{Spec}(k[x, y])$ and take $U = X \setminus \{[(x, y)]\}$, i.e. X minus the origin (which is a closed set as it is a maximal ideal). Show that $U \ncong D(f)$ for any f (hint: notice that (x, y) is not principal). This shows an important distinction between points and functions.

Show further that U can be represented as the union of two distinguished open sets.

Another result we may want to conclude is that each open set $U \subseteq \operatorname{Spec}(R)$ corresponds to $\operatorname{Spec}(R_S)$ for an appropriate S, i.e. $U \cong_{\mathbf{Top}} \operatorname{Spec}(R_S)$. This shall be shown to be *false*; the curious reader may jump to example 2.2.51 for a counter-example.

The following result, while elementary, is used repeatedly in gluing arguments and in the proof of the Affine Communication Lemma. It says that a distinguished open inside a distinguished open is itself distinguished in the ambient spectrum.

Proposition 2.1.15: Distinguished Open of a Distinguished Open

Let $D(f) \subseteq \operatorname{Spec}(R)$ be a distinguished open, so that $D(f) \cong \operatorname{Spec}(R_f)$. Let $g' \in R_f$ and consider the distinguished open $D(g') \subseteq \operatorname{Spec}(R_f)$. Then $D(g')$, viewed as a subset of $\operatorname{Spec}(R)$, is itself a distinguished open of $\operatorname{Spec}(R)$.
More precisely, if $g' = h/f^n$ with $h \in R$ and $n \geq 0$, then:

$$\operatorname{Spec}((R_f)_{g'}) = \operatorname{Spec}(R_{fh}).$$

In particular, $D(g') = D(fh)$ as subsets of $\operatorname{Spec}(R)$.

***Proof*:**
Since f is already a unit in R_f, inverting $g' = h/f^n$ in R_f is the same as inverting h in R_f:

$$(R_f)_{h/f^n} = (R_f)_h.$$

Now $(R_f)_h$ is the localization of R at the multiplicative set generated by f and h, which is the same as localizing at fh:

$$(R_f)_h = R_{fh}.$$

(Formally: $(R_f)_h = R[f^{-1}][h^{-1}] = R[(fh)^{-1}] = R_{fh}$, since inverting fh inverts both f and h, and conversely.) The corresponding open set is $D(fh) = D(f) \cap D(h) \subseteq \operatorname{Spec}(R)$.

2.1.16 Remark The proposition says that the collection of distinguished open sets is closed under "refinement": passing to a smaller distinguished open inside a distinguished open stays within the class. This is what makes distinguished opens a good basis for defining sheaves—one never needs to leave the class when restricting.

Let us record two further useful consequences.

Corollary 2.1.17: Unit Ideal and Distinguished Cover

Elements $f_1, \ldots, f_n \in R$ generate the unit ideal $(f_1, \ldots, f_n) = R$ if and only if $\operatorname{Spec}(R) = \bigcup_{i=1}^n D(f_i)$.

***Proof*:**
A prime $\mathfrak{p}$ is not in $\bigcup D(f_i)$ if and only if $f_i \in \mathfrak{p}$ for all i, if and only if $(f_1, \ldots, f_n) \subseteq \mathfrak{p}$. Thus $\bigcup D(f_i) = \operatorname{Spec}(R)$ if and only if no prime contains all the f_i, if and only if $(f_1, \ldots, f_n) = R$.

Corollary 2.1.18: Powers Generate the Unit Ideal

If $f_1, \ldots, f_n$ generate the unit ideal in R, then for any positive integers $k_1, \ldots, k_n$, the powers $f_1^{k_1}, \ldots, f_n^{k_n}$ also generate the unit ideal.

***Proof*:**
We have $D(f_i) = D(f_i^{k_i})$ (a prime contains f_i if and only if it contains $f_i^{k_i}$), so $\bigcup D(f_i^{k_i}) = \bigcup D(f_i) = \operatorname{Spec}(R)$, which by the previous corollary means $(f_1^{k_1}, \ldots, f_n^{k_n}) = R$.

2.1.3 Topological Properties

Let us see what classical topological properties applies to $\operatorname{Spec} R$. An important consequence is that for any open cover of $\operatorname{Spec}(R)$ given by distinguished open sets, there exists a finite open sub-cover. To see this, notice that $\cup_{i\in I} D(f_i) = \operatorname{Spec}(R)$ For some indexing set I if and only if $(\{f_i\}_{i\in I}) = R$ that is the ideal generated by the set $\{f_i\}_{i\in I}$. Then 1 must be inside that ideal, and must be a finite linear combination, and the result follows by the algebraic/geometric correspondence. This shows that $\operatorname{Spec}(R)$ is always compact in the topological sense. We may wish that this implies that $\operatorname{Spec}(R)$ has the usual nice properties of compactness, but this is very much not the case (notice this implies $\mathbb{A}^n_{\mathbb{C}}$ is compact). This turns out to be a consequence of lacking Hausdroffness. For this reason, a different name is given to compactness:

Definition 2.1.19: Quasicompact

Let X be a topological space. Then if X is compact and non-Hausdorff, X is *quasicompact*

As an exercise, show that if R is Noetherian, every subset of Spec (R) is quasicompact. In section 3.5.4, we shall show how the notion of proper maps from analysis is the right way of defining the notion of compactness which recovers many of the nice results expected from compactness.

The companion property of [quasi]compactness is connectedness. Recall a set is connected if it cannot be written as a disjoint of two nonempty open sets. Disconnectedness of Spec is equivalent to a ring representable as the product of two rings, $R = R_1 \times R_2$, or if the reader remembers the brief discussion in Representation theory(see [11, chapter 23.2]), if R has idempotent elements

Proposition 2.1.20: Spectrum Connectedness And Idempotent Elements

IF $R = R_1 \times \cdots \times R_n = \prod_i^n R_i$ is a ring, then show that

$$\operatorname{Spec}\left(\prod_i^n R_i\right) = \coprod_i \operatorname{Spec}(R_i)$$

More generally, show that Spec (R) is not connected if R is isomorphic to the product of two zero-rings, that is there exists $a_1, a_2 \in R$ st $a_1^2 = a_2, a_2^2 = a_2, a_1 + a_2 = 1$.

***Proof*:**
good exercise.

Note that for the homeomorphism, the product must be finite, if the product is infinite then Spec $(\prod_i R_i)$ is larger than the right hand side. In fact, an infinite disjoint union of Spec (R_i) cannot be represented as Spec (R) for an appropriate R; it can be seen as the "simplest" example of a scheme that is not an affine scheme (see example 2.2.12(6)).

A notion similar to connectedness is *irreducibility*. Recall that a topological space is said to be *irreducible* if it is nonempty and it is not the union of two proper closed subsets. Written differently, X is irreducible if whenever $X = Y \cup Z$ with Y, Z closed in X, then $Y = X$ or $Z = X$[10]. Note that Y, Z may have nonempty intersection. In the euclidean topology, this notion is much less important (think $\mathbb{C}$), however in the Zariski topology we shall see that many interesting sets are irreducible. The intuition stems from traditional varieties where irreducibility corresponds to irreducible polynomials, and indeed this intuition when properly generalized translates over to irreducible [closed] sets in the Zariski topology of Spectra.

Let $X = Y \cup Z$ for closed sets Y, Z. Then there are radical ideals J_X, J_Y, J_Z such that $V(J_X) = V(J_Y) \cup V(J_Z) = V(J_Y J_Z) = V(J_Y \cap J_Z)$. As these are all radical ideals, $J_X = J_Y \cap J_Z$. Then if J_X is prime, it is an exercise in commutative algebra to show that either $J_Y \subseteq J_Z$ or $J_Z \subseteq J_Y$. Hence, irreducibility is in correspondence with prime ideals:

[10]The finiteness is important, for there may be irreducible sets that contain within them reducible sets of lower dimension

$$\{\text{prime ideals of } R\} \underset{V(-)}{\overset{I(-)}{\rightleftarrows}} \{\text{closed irreducible sets of } \mathrm{Spec}\,(R)\} \tag{2.3}$$

If we look at irreducible closed sets, then $V(-)$ and $I(-)$ gives a bijection between the irreducible closed subsets of $\mathrm{Spec}\,(R)$ and the prime ideals of R, in other words there is a bijection between the *points* of $\mathrm{Spec}\,(R)$ and the irreducible closed subsets of $\mathrm{Spec}\,(R)$. Since it is a bijection, every irreducible closed subset of $\mathrm{Spec}\,(R)$ has *exactly* one generic point associated to it.

As usual, we define the Noetherian property on $\mathrm{Spec}\,(R)$ to mean that the descending chain condition for closed sets is satisfied (to mirror the ascending chain of ideals in R). In the Noetherian case, all closed sets can be decomposed into irreducibles:

Proposition 2.1.21: Noetherian Decomposition

Let X be a Noetherian topological space. Then every nonempty closed subset Z can be expressed uniquely as a finite union

$$Z = Z_1 \cup \cdots \cup Z_n$$

of irreducible closed subsets, none contained in another.That is , any closed subset Z has a finite number of pieces.

***Proof*:**
see [**NateClassicalAlgGeo**].

The reader should check the equivalent result for prime ideals

2.1.4 Ring to Top Functor: Induced Spectra Maps

As spectra are given by rings, we shall want to translate over ring homomorphisms to maps between spectra:

Definition 2.1.22: Induced Spectra Map

let $Y = \mathrm{Spec}\,(S)$ and $X = \mathrm{Spec}\,(R)$ be two spectrum. Then given the ring homomorphism: $\varphi : R \to S$, the induced *spectrum map* $\varphi^a : \mathrm{Spec}\,(S) \to \mathrm{Spec}\,(R)$ is given by:

$$\varphi^a(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$$

The reader ought to verify that if $\varphi : R \to S$ is a ring homomorphism, then $\varphi^a : \mathrm{Spec}\,(S) \to \mathrm{Spec}\,(R)$ sending $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$ is functorial (recall or reprove that the pre-image of a prime ideal is prime). We now have two functions happening: φ^a given by ring homomorphisms, and regular functions given by ring elements. Often, φ^a is called a map, while the elements of R are called functions (following a similar convention from differential geometry).

Example 2.1.23: Maps between Spectra: φ^a

1. Show that the surjective map $R \to R/I$ induce an injective map $\mathrm{Spec}\,(R/I) \to \mathrm{Spec}\,(R)$. Algebraically, the surjective map adds relations. Geometrically, we are taking the points with the extra algebraic relations given by I and associating them to the points in R.

2. Show that the injective map $R \to S^{-1}R$ induce a injective map $\mathrm{Spec}\,(S^{-1}R) \to \mathrm{Spec}\,(R)$. Algebraically, the map $R \to S^{-1}R$ takes (integral) elements[a] and maps it to units.

3. Here is a concrete example generalizing the usual notion of a regular morphism as introduced in[**NateClassicalAlgGeo**]: Take $\mathbb{A}^1_{\mathbb{C}} = \mathrm{Spec}\,\mathbb{C}[x]$ which by Nullstellensatz we can think of as the traditional points $\mathbb{C}$ along with the generic point (0). Consider the ring homomorphism between these rings defined by:

$$\varphi : \mathbb{C}[y] \to \mathbb{C}[x] \qquad y \mapsto x^2$$

Note this is not the squaring map, as that is not a ring morphism, but rather the map that maps the variable y to x^2 and then extends linearly. Then the map on the sets Spec 's is given by:

$$\varphi^a : \mathrm{Spec}\,(\mathbb{C}[x]) \to \mathrm{Spec}\,(\mathbb{C}[y])$$

$$\begin{aligned}
\varphi^a\,(\mathfrak{p}) &= \varphi^{-1}\,((x-a)) \\
&= \left\{ f(y) \in \mathbb{C}[y] \;\middle|\; f(x^2) \in (x-a) \right\} \\
&= \left\{ f(y) \in \mathbb{C}[y] \;\middle|\; x-a \middle| f(x^2) \right\} \\
&= \left\{ f(y) \in \mathbb{C}[y] \;\middle|\; f(x^2) = (x-a)q(x) \right\} \\
&= (y-a^2)
\end{aligned}$$

In one line:

$$\varphi^a([x-a]) = [(y-a^2)]$$

verify that

$$(\varphi^a)^{-1}\,((y-a)) = \{(x-\sqrt{a}), (x+\sqrt{a})\}$$

Perhaps more intuitively, this is the Spec-equivalent of the more usual map $\mathbb{C} \to \mathbb{C}$ given by $z \to z^2$ which can be thought of as the double-covering of $\mathbb{C}$ with branching point 0. Note the importance of visualizing schemes using complex numbers, or more generally an algebraically closed field, even if we replaced $\mathbb{C}$ with $\mathbb{R}$; for example the image of (x^2+1) is $(y+1)$ (i.e. $-i^2 = -1$) and $(\varphi^a)^{-1}\left((y^2+1)\right) = \{(x^2+1)\}$ where the information about the 2 points is captured in the degree of the (irreducible) polynomial in the pre-image (recall they are glued together in $\mathrm{Spec}\,\mathbb{R}[x]$).

4. More generally, any collection of regular functions can produce a spectra map, a "regular map" (in particular, this mirror's regular maps between affine varieties). Let k be any field and take polynomial $f_1, ..., f_n \in k[x_1, ..., x_m]$. Then define:

$$\varphi : k[y_1, ..., y_n] \to k[x_1, ..., x_m] \qquad y_i \to f_i$$

Let A represent the domain and B the codomain. Then show that for any ideal $I \subseteq A$, $J \subseteq B$ such that $\varphi(I) \subseteq J$, φ induces a map of sets:

$$\mathrm{Spec}\,(k[x_1, ..., x_m]/J) \to \mathrm{Spec}\,(k[y_1, ..., y_n]/I)$$

More particularly, show that this maps sends the point $[(x_1 - a_1, ..., x_m - a_m)]$ (traditionally, $(a_1, ..., a_m) \in k^m$ to the point traditionally represented as:

$$(f_1(a_1, ..., a_m), ..., f_n(a_1, ..., a_m)) \in k^n$$

From now on, we shall sometimes take liberties in spectra maps between spectra resembling varieties by defining polynomial maps on the maximal ideals (this shall pose no problems with extending the definition to the prime ideals or to the later sheaf structure that shall be added by theorem 3.7.1).

5. The map $\mathbb{A}^1_k \to \mathbb{A}^1_k$ given by $a \mapsto p(a)$ can always be represented as a map to the graph:

$$\mathbb{A}^1_k \to \mathbb{A}^2_k \qquad a \mapsto (a, p(a))$$

Furthermore, by the projection map $\pi : \mathbb{A}^2_k \to \mathbb{A}^1_k$ given by $(a, b) \mapsto a$, we shall see that this maps $\mathbb{A}^1_k$ isomorphic to all of its graphs (this works even for higher dimensions). Note that this does not extend if we have more variable where more than one term is not the identity, for example:

$$\mathbb{A}^1_k \to \mathbb{A}^3_k \qquad t \mapsto (t, t^2, t^3)$$

maps to the point of $\mathbb{A}^1_k$ to the associated to the variety $\mathcal{Z}(x-y^2, x-z^3)$ which in [**NateClassicalAlgGeo**] was shown to be rational but not isomorphic. Another example that isn't even rational is a map from $\mathbb{P}^1_k$ to an elliptic curve E.

6. This example gives a way of thinking of $\mathbb{A}^n_{\mathbb{Z}} = \operatorname{Spec}(\mathbb{Z}[x_1, ..., x_n])$ as an "$\mathbb{A}^n$-bundle" on $\operatorname{Spec}(\mathbb{Z})$ and is an important example for number theory. Define the map $\pi : \mathbb{A}^n_{\mathbb{Z}} \to \operatorname{Spec}(\mathbb{Z})$ given by the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}[x_1, ..., x_n]$. Then show there is a bijection between $\pi^{-1}([(p)])$ where $(p) \neq (0)$and $\mathbb{A}^n_{\mathbb{F}_p}$! If $(p) = (0)$ then the pre-image is $\mathbb{A}^n_k$. The picture of this bundle can be seen as follows:

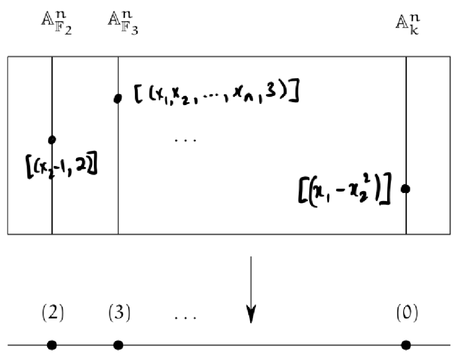


Figure 2.1: Bundle on $\operatorname{Spec}\mathbb{Z}$

7. Consider $\varphi : \mathbb{Z} \hookrightarrow k[x]$. Then $\varphi^a : \operatorname{Spec}\mathbb{C}[x] \to \operatorname{Spec}\mathbb{Z}$ is the zero map, namely $\varphi^a(\operatorname{Spec} k[x]) = \{[(0)]\}$. From this perspective, we can see that classical field theory has trivial number theory as it has no interesting fibers over $\operatorname{Spec}\mathbb{Z}$.

8. Let $\varphi : R \to S$ be an integral extension. Show that $\mathrm{Spec}(S) \to \mathrm{Spec}(R)$ is surjective (hint: Lying Over). This results also motivates the terminology "lying over" geometrically. Recall that is R or S is a field, the other must be too. Show that S being a field and the map being surjective can be used to conclude R is a field. What is the geometric argument if R was a field?

9. Let us see why it is important to work with primes instead of maximal ideals. Consider Spec $k(x)[y] \to$ Spec $k[x]$ induced by $k[x] \to k(x)[y]$ where $k[x]$ maps to itself (notably, all elements become units). Then as $k(x)[y]$ is a PID, all of its primes are maximal and are of the form $(y - f(x))$ for some $f(x) \in k(x)$. Then the pre-image of this ideal under the inclusion map must be the ideal (0), which is certainly not maximal (in fact it is the generic point). Hence the map Spec $k(x)[y] \to$ Spec $k[x]$ will map $[(y - f(x))] \mapsto \eta$, notably a 0 -dimensional point mapped to a 1-dimensional point!

10. Let R be a ring with nilpotence and $I = \sqrt{(0)}$ the nilradical. Take $\varphi : R \to R/I$ to be the usual projection. Then φ^a is a bijection; the key is to recall that the nilradical intersection of all prime ideals, and hence any prime $\overline{\mathfrak{p}} \subseteq R/I$ which is of the for $\mathfrak{p} + I$ must in fact equal itself: $\mathfrak{p} + I = \mathfrak{p}$ (as $I \subseteq \mathfrak{p}$). The example to keep in mind should be $R = \mathbb{C}[x]/(x^2)$ with $I = (x)$.

[a]Note that if an element is nilpotent, localizing it will map it to zero

Lemma 2.1.24: Induced Sepctra Map Is Continuous

Let $\varphi : R \to S$ be a ring homomorphism. Then φ^a is a continuous map

***Proof*:**
It suffices to show the pre-image of a closed set is closed, so take $V(S) \subseteq \mathrm{Spec}\,(R)$ and consider $(\varphi^a)^{-1}(V(I))$. Show that this is equal to $V(\varphi(I))$, showing it is closed.

The map φ^a given by the natural inverting function is a naturally continuous map, that is φ^a is a continuous function between the Spec of two rings with the Zariski topology. Hence Spec is a [contravariant] functor from **CRing** to **Top**. Furthermore, continuity can be shown on distinguished open sets, namely as they form a basis we can show that:

$$(\varphi^a)^{-1}\left(D([\mathfrak{p}]) = D(\varphi([\mathfrak{p}])\right)$$

Note that there will be continuous maps between the spectrum of rings that are not φ^a,

Example 2.1.25: Continuous Not Induced By Ring Homomorphism
The map from $\mathbb{A}^1_{\mathbb{C}}$ to itself that is the identity on all points except $[(x)]$ and $[(x-1)]$ (which will be represented with 0 and 1) where it swaps these two values is *not* a polynomial and hence cannot have been induced by a ring homomorphism, however it *is* continuous. If $0, 1 \notin V(S)$, then $\varphi^{-1}(V(S)) = V(S)$. Similarly if both values are inside. If 0 or 1 is in $V(S)$, say $0 \in V(S)$, then $\varphi^{-1}(V(S)) = V(S) \setminus \{0\} \cup \{1\}$ which is closed as all finite subsets of $\mathbb{A}^1_{\mathbb{C}}$ are closed.

Can you now find two ring homomorphisms that induce the same Spectra map? See exercise 2.1.1(2.1.1 (13))

Exercise 2.1.1(2.1.1 (14)) goes over some of the properties preserved by continuous maps. Most of the properties that we shall be want for schemes to have (ex. separatedness, properness, smoothness, multiplicity information, finiteness conditions) shall require more than continuity.

The correspondence is terse as Spec does not carry enough information to fully and faithfully correspond to ring homomorphism. In chapter 3 it shall be shown that scheme morphisms are the right object to carry this information (namely, they shall insure that around every point, that is each stalk, the spectra map behaves like a polynomial).

2.1.5 *Subsets of Spectra via quotient and Localization

Two common constructions that came up in a few example is the process of taking the quotient and the localization of a ring R. These form important subsets of $\operatorname{Spec}(R)$, namely by the 4th isomorphism theorem and the prime-correspondence theorem for prime ideals in localization (see [11, chapter 9.3] and [11, chapter 20.5]). By these theorems, $\operatorname{Spec}(R/I), \operatorname{Spec}(S^{-1}R) \subseteq \operatorname{Spec}(R)$. If $I = \mathfrak{p}$ and $S = R \backslash \mathfrak{p}$,the first corresponds to all prime ideals containing $\mathfrak{p}$, while the second corresponds to all prime ideals that are contained in the prime ideal $\mathfrak{p}$[11] These two sets need not union to the entire $\operatorname{Spec}(R)$ (take $\mathbb{Z}$ and 5), however they do correspond to two "opposite" ways of carving out a part of $\operatorname{Spec}(R)$. In the following visual is a very common illustration:

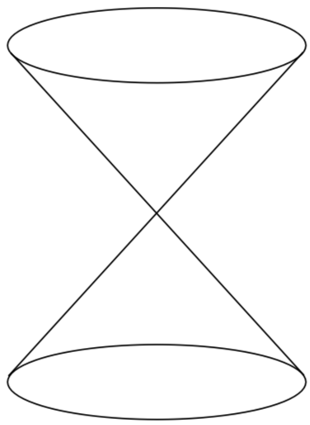


Figure 2.2: partial decomposition of $\operatorname{Spec}(R)$

Geometrically, $\operatorname{Spec}(R/I)$ can be thought of as "carving out" a part of $\operatorname{Spec}(R)$. For example, take $R = \mathbb{C}[x, y, z]$ and $\mathfrak{p} = (x^2 + y^2 - z)$. If we were working with affine complex varieties, then we would say this is the 3D-parabola in $\mathbb{C}^3$, however as we are on $\operatorname{Spec}(R/\mathfrak{p})$ this would come with some extra points (which ones?). For $\operatorname{Spec}(S^{-1}R)$, we may think it as the opposite (we shall cover the special case of $R_{\mathfrak{p}}$ afterwards). For example let's say $S = \{1, f, f^2, ...\}$ and take $S^{-1}R$. Then the primes of $\operatorname{Spec}(S^{-1}R)$ would be all primes that do not containing f, that is all points where f doesn't vanish!

[11]More generally it is all prime ideals not having trivial intersection with the multiplicative set.

As an illustration, the set $\operatorname{Spec}(\mathbb{C}[x,y]_{y-x^2})$ are the points in the affine plane, minus all the points on the parabola $y - x^2$, namely all the singletons (a, a^2) for $a \in \mathbb{C}$ (so $[(x-a, y-a^2)]$), as well as the point $[(y-x^2)]$.

Proposition 2.1.26: Closed And Open Subsets

Let R be a commutative ring, $I \subseteq R$ an ideal, $f \in R$, $\mathfrak{p} \subseteq R$ prime, and S a multiplicative set. Then:

1. $\operatorname{Spec}(R/I) \subseteq \operatorname{Spec}(R)$ is a closed subset
2. $\operatorname{Spec}(R_f)$ is an open subset
3. $\operatorname{Spec}\left(S^{-1}R\right) \subseteq \operatorname{Spec}(R)$ need not be open or closed in general. In particular $\operatorname{Spec}(R_{\mathfrak{p}}) \subseteq \operatorname{Spec}(R)$ need not be open or closed. However, it is an embedding, namely $\iota : \operatorname{Spec} S^{-1}R \hookrightarrow \operatorname{Spec} R$ is injective, continuous, and closed.

***Proof*:**

1. Think of $V(I)$ (induce the topology on $\operatorname{Spec}(R/I)$ via the inclusion map)
2. Think $\operatorname{Spec}(R) \setminus V(f)$
3. think of $\operatorname{Spec}(\mathbb{Q}) \subseteq \operatorname{Spec}(\mathbb{Z})$ and $\operatorname{Spec}(\mathbb{Z}_{\mathfrak{p}}) \to \operatorname{Spec}(\mathbb{Z})$ induced by $\mathbb{Z} \to \mathbb{Z}_{\mathfrak{p}}$ (this shows that $\operatorname{Spec}(R_{\mathfrak{p}})$ need not be open.

For the case of $S = R - \mathfrak{p}$, we can think of $\operatorname{Spec}(R_{\mathfrak{p}})$ as keeping near the $\mathfrak{p}$, namely since the prime ideals away from $\mathfrak{p}$ will be turned into units, the primes intersecting $R \setminus \mathfrak{p}$ will be eliminated. For example, if $R = \mathbb{C}[x,y]$ and $\mathfrak{p} = (x,y)$, then we keep all the points corresponding to being "near" the origin, that is the point $[(x,y)]$, i.e. $(0,0)$, and the 1-dimensional points $[(f(x,y))]$ where $f(0,0) = 0$ (i.e. where $f \equiv 0 \bmod (x,y)$) that is irreducible curves that pass through the origin. In the special case where $R = k[x]$, then the localization will always only have two points, usually called the classical point and the generic point. These shall eventually be seen as geometric interpretations of DVR.

Using these geometric results, they can lend intuition towards algebraic manipulation:

Example 2.1.27: Illustration Of Quotient And Localization

Using the above intuitions, we can quickly come up with some intuition isomorphisms. Take $R = \mathbb{C}[x,y]/(xy)$. Visually, this is the union of the y and x-axis. If we localize this at x, then by our above intuition this should eliminated all points where the "function" x vanishes. The function x vanishes at the y-axis, and so we must be left with the x-axis minus the origin. This happens to be $\operatorname{Spec} \mathbb{C}[x]_x$. In fact, these two rings are isomorphic!

$$\left(\frac{\mathbb{C}[x,y]}{(xy)}\right)_x \cong \mathbb{C}[x]_x$$

which the reader should check.

Exercise 2.1.1

1. Show that $D(f) \cap D(g) = D(fg)$.
2. $D(f) \subseteq D(g)$ iff $f^n \in (g)$ iff g is invertible in R_f. How would you prove this via "geometric explanation"? For an algebraic explanation, change to $V(f) \supseteq V(g)$ and take $I(-)$ to inverse and conclude.
3. $D(f) = \emptyset$ iff f is in the Nilradical.
4. Show that any connected component is the union of irreducible components (think of the $+$ shape, i.e. $\text{Spec}\,(\mathbb{C}[x, y]/(xy))$).
5. Show that $\mathbb{A}^1_{\mathbb{C}}$ is irreducible.
6. Show that in an irreducible topological space, any nonempty open set is dense.
7. If X is a topological and Z an irreducible subspace, then $\overline{Z}$ is irreducible as well
8. Show that an irreducible space is connected.
9. If A is a integral domain, $\text{Spec}\,(A)$ is irreducible.
10. Show that $\text{Spec}\,(\mathbb{C}[x, y]/(xy))$ is connected but reducible.
11. Show that there is a natural bijection irreducible components (maximal irreducible subsets) of $\text{Spec}\,(R)$ and the *minimal* prime ideals of R.
12. these next two exercises show us how the theory of classical varieties persists with Schemes
    a) Let A be a finitely generated k-algebra. Then show that the closed points of $\text{Spec}\,(A)$ are dense; show that if $f \in A$ and $D(f)$ is nonempty, then $D(f)$ contains a closed point of $\text{Spec}\,(A)$[12]
    b) Let $A = \overline{k}[x_1, ..., x_n]/I$ where the Nilradical is trivial, and let $X = \text{Spec}\,(A) \subseteq \mathbb{A}^n_{\overline{k}}$. Show that [regular] functions on X are determined by their valued on the closed points. Hint: if f, g are different [regular] functions on X, then $f - g$ is nowhere zero on an open subset of X. Use the above result.
13. Find two ring homomorphism that induce the same spectra map.
14. Let $f : X \to Y$ be a continuous map.
    a) Show that irreducible sets map to irreducible sets to irreducible sets.
    b) Show that if X is Noetherian, then the image of f is Noetherian
    c) Show that the dimension of X is greater than or equal to the dimension of the image of Y (also see 2.6).

2.2 Schemes

The spectrum of a ring with the Zariski Topology does not hold all data provided by a commutative ring, for example nilpotence is forgotten. The *structure sheaf* will be added to $\text{Spec}\, R$ in order to add this extra information (namely, we shall have the collection of regular function defined on each

[12]Hint: A_f is also a finitely generated k-algebra use generalized Nullstellensatz. Show the closed points of the Spec of a finitely generated k-algebra are those for which the residue field is a finite extension of k

open set as part of the data we are tracking). The resulting object is called the *affine scheme*.

After defining the affine scheme, we shall require one more crucial abstraction that links it to differential geometry: we shall want our object to the glued-together open sets. In differential geometry, these would all be open subsets of euclidean space, usually simply connected and all isomorphic. A key distinguishing feature in algebraic geometry is that the sets that can be glued can be very different, namely $\operatorname{Spec} R$ for all commutative rings R will be the sets from which we can select, and as the reader has already seen these sets can have very different topologies.

2.2.1 The Structure Sheaf for Affine Schemes

When working with geometric objects (varieties, real and complex manifolds, etc), we study the local behaviour of the space, namely by studying some given information on open sets or germs/stalks. In the theory of differential manifold, the information at each local was the "linear" equation (recall that one way of defining the tangent bundle is by taking all the derivations of smooth functions, which can be shown to "preserve" the linear part of each smooth function). For an affine scheme, we are interested in the functions functions are *regular* on open sets. By section 2.1.2, it is enough to define the regular functions on distinguished open sets. For those motivated by analytic parallels, Complex manifold have a sheaf of meromorphic functions which are defined on the open subsets of the complex manifold, and we usually have holomorphic functions on the entire manifold.

To motivate the following sheaf, recall that on $\mathbb{C}^n$, if we take open subsets there are many more holomorphic functions that could be defined on that open subset, namely that rational functions are well-defined when the roots of the denominator (which are not cancel by the numerator) are outside the open set. For example, on $\mathbb{C}^2 \setminus \{x\text{-axis} \cup y\text{-axis}\}$, the function

$$\frac{x^2+4x+y+1}{x^5y^4}$$

is well-defined. Moving this to Spec, in $\mathbb{A}^2_{\mathbb{C}}$, we ought to have the above function to be well-defined on $D(xy)$. Furthermore, we want the collection of global sections, $\mathcal{O}_X$, to be R, mirroring the "global section" on $\mathbb{C}^2$ being the (non-rational) holomorphic functions in two variables.

As we saw in theorem 1.1.22, it suffice to define a sheaf on a base of $\operatorname{Spec}(R)$, namely on the distinguished open sets. Thus, we may define on $D(f) \subseteq \operatorname{Spec}(R)$ the ring that is the localization of R at the multiplicative subset of all function that do not vanish on $V(f)$ (note how this is subtly different than functions that do not vanish on f, what if f is nilpotent?). This can be represented as:

$$\mathcal{O}_X(D(f)) = \mathcal{O}_X(X_f) \cong R_f$$

First off, this is a presheaf: if $D(f) \supseteq D(g)$, by exercise 2.1.1 there is a minimal $n \in \mathbb{N}_{>0}$ such that $g^n \in (f)$ so $rf = g^n$. Then as $D(g^n) = D(g)$, we may define the restriction map:

$$\operatorname{res}_{X_f, X_g} : R_f \to R_g \qquad \frac{s}{f^m} \mapsto \frac{r^m s}{g^{nm}}$$

which shows we have defined a presheaf on a base. We next must show that it satisfies the sheaf on base axioms.

Proposition 2.2.1: Distinguished Opens form Sheaf-Basis for $\operatorname{Spec} R$

Let $X = \operatorname{Spec} R$, and suppose that $D(f)$ is covered by distinguished open sets $D(f_a) \subseteq D(f)$. Then:

1. If $g, h \in R_f$ are equal in R_{f_a} when passing through the restriction map for every a, then $g = h$.
2. If for each a, there exists a $g_a \in R_{f_a}$ such that for each pair a, b, the images of g_a and g_b in $R_{f_a f_b}$ are equal, then there is an element $g \in R_f$ whose image in R_{f_a} is g_a.

In other words, $\mathcal{O}_X$ defines a $\mathcal{B}$-sheaf on the distinguished base, and by theorem 1.1.22, $\mathcal{O}_X$ extends to a sheaf on X.

***Proof*:**
Let $\operatorname{Spec}(R) = \bigcup_{i \in I} D(f_i)$, which is equivalent to saying the f_i generate the unit ideal: $(f_i)_{i \in I} = R$. We show the locality and gluability axioms, or in categorical terms the exactness of:

$$0 \longrightarrow R \longrightarrow \prod_{i \in I} R_{f_i} \rightrightarrows \prod_{i \neq j \in I} R_{f_i f_j}$$

Starting with locality. As $s_1 = s_2$ is equivalent to $s_1 - s_2 = 0$, it suffices to show the result in the case where $s = 0$. By quasicompactness we may pass to a finite subcover:

$$\operatorname{Spec}(R) = \bigcup_{i=1}^{n} D(f_i) \qquad (f_1, \ldots, f_n) = R$$

Suppose that for some $s \in R$, $\operatorname{res}_{\operatorname{Spec}(R), D(f_i)} s = 0$ for all i. We must show $s = 0$. By definition of R_{f_i}:

$$s|_{D(f_i)} = 0 \qquad \text{if and only if} \qquad f_i^{m_i} s = 0 \quad \text{for some } m_i \geq 0$$

Taking $m = \max_i(m_i)$, we have $f_i^m s = 0$ for all i. Now, as $D(f_i) = D(f_i^m)$, the cover $\bigcup D(f_i^m) = \operatorname{Spec}(R)$ still holds, so $(f_1^m, \ldots, f_n^m) = R$. Thus, for appropriate choices $r_i \in R$ with $\sum_{i=1}^n r_i f_i^m = 1$, we get:

$$s = 1 \cdot s = \left(\sum_i r_i f_i^m \right) s = \sum_i r_i (f_i^m s) = 0$$

For general $D(f)$, repeat the above argument replacing R with R_f (quasi-compactness of $D(f)$ ensures the cover can still be taken to be finite, and the same algebraic manipulations apply in R_f).

We next show gluability[a]. Starting with the case $\operatorname{Spec}(R) = \bigcup_i D(f_i)$, suppose there are elements in each R_{f_i} that agree on overlaps $R_{f_i f_j}$ (note that intersections of distinguished open sets are distinguished open sets). As there may be infinitely many sets on which there are overlaps, it cannot be assumed that I is finite. However, the finite case is a good start, say $I = \{1, \ldots, n\}$. Then we have elements $\frac{a_i}{f_i^{l_i}} \in R_{f_i}$ agreeing on the overlaps $R_{f_i f_j}$, meaning:

$$(f_i^{l_i} f_j^{l_j})^{m_{ij}} (f_j^{l_j} a_i - f_i^{l_i} a_j) = 0$$

Simplifying notation by writing $g_i = f_i^{l_i}$:

$$(g_i g_j)^{m_{ij}}(g_j a_i - g_i a_j) = 0$$

As there are finitely many m_{ij}, take $m = \max_{i,j}(m_{ij})$:

$$(g_i g_j)^{m}(g_j a_i - g_i a_j) = 0$$

Next, convert to a different set of distinguished opens: take $b_i = a_i g_i^m$ and $h_i = g_i^{m+1}$ so that $D(h_i) = D(g_i)$. Then on each $D(h_i)$, we have the section b_i/h_i, and the overlap condition simplifies to:

$$h_j b_i = h_i b_j$$

Noting that $\bigcup_i D(h_i) = \mathrm{Spec}\,(R)$ gives $1 = \sum_{i=1}^n r_i h_i$ for appropriate $r_i \in R$, define:

$$r = \sum_i r_i b_i$$

Then r is the element of R that restricts to b_j/h_j on each $D(h_j)$:

$$r h_j = \sum_i r_i b_i h_j = \sum_i r_i h_i b_j = \left(\sum_i r_i h_i\right) b_j = b_j$$

showing gluability in the finite case. For the infinite case, we use the previously shown base-locality axiom. By quasi-compactness, we can choose a finite subcover $D(f_1), \ldots, D(f_n)$ and construct r as above. For any $z \in I \setminus \{1, \ldots, n\}$, repeat the argument on $\{1, \ldots, n, z\}$ to get $r' \in R$ which restricts to a_i for $i \in \{1, \ldots, n, z\}$. By base locality, $r' = r$. Hence r restricts to $a_z/f_z^{l_z}$ for all z.

To finish off the proof, we must handle the case of an arbitrary $D(f)$ rather than all of $\mathrm{Spec}\,(R)$. This follows by applying the same argument with R replaced by R_f throughout.

[a]Vakil pointed out that Serre called this a "partition of unity argument," which is the local-to-global principle physically manifested in differential geometry – a rather good perspective on what is about to happen.

It may be tempting to say that $\mathcal{O}_{\mathrm{Spec}\,(R)}(U)$ for any open subset U is a localization of R at some multiplicatively closed subset S, however this is not true:

Example 2.2.2: Subtle Local Section Of Affine Scheme

Let $R = k[x, y, z, w]/(xw - yz)$; $\mathrm{Spec}\, R$ is called the *quadric cone*. Note that R is an integral domain as $(xw - yz)$ is prime[a]. Let $U = D(y) \cup D(w)$. Then a section $s \in \mathcal{O}(U)$ is determined by a choice of $f \in R_y$ and $g \in R_w$ that agree on R_{yw}, namely $\mathcal{O}(U)$ is determined by the left exact sequence:

$$0 \to \mathcal{O}(U) \hookrightarrow R_y \oplus R_w \xrightarrow{(f,g) \mapsto f-g} R_{yw}$$

The kernel of the difference map is $\mathcal{O}(U)$. Take $(x/y,\, z/w)$ and consider:

$$\frac{x}{y} = \frac{z}{w} \qquad \text{if and only if} \qquad xw - zy = 0$$

where there is no need for a factor of $(yw)^n$ in front, as there are no zero-divisors. In R, this is exactly the defining relation of the quotient, hence this equality holds. Therefore there is an element $t \in \mathcal{O}(U)$ such that:

$$t \mapsto \left(\frac{x}{y}, \frac{z}{w}\right)$$

Next, for any $r \in R$, certainly $r \in R_y, R_w$ and $r - r = 0$, so $r \in \mathcal{O}(U)$. Is it possible that $t = r$ for some $r \in R$? As the map $\mathcal{O}(U) \hookrightarrow R_y \oplus R_w$ is injective, if $t = r$ then since $r \mapsto (r, r)$ we would need $r = x/y = z/w$, but y and w are not invertible in R, and hence $t \notin R$. Thus the ring $R[t] \neq R$. Certainly $R[t] \subseteq \mathcal{O}(U)$ and any R-polynomial $f(t) \in R[t]$ maps to:

$$f(t) \mapsto \left(f\left(\frac{x}{y}\right), f\left(\frac{z}{w}\right)\right)$$

Let us show that $yt - x = 0$ (equivalently $wt - z = 0$). By definition of a sheaf, this can be checked on the open cover $\{D(y), D(w)\}$. By construction (recall that restriction maps are ring homomorphisms and that $r \mapsto r$ under these maps):

$$(yt - x)|_{D(y)} = y \cdot \frac{x}{y} - x = x - x = 0$$

and

$$(yt - x)|_{D(w)} = y \cdot \frac{z}{w} - x = \frac{yz - xw}{w} = \frac{0}{w} = 0$$

giving $yt - x = 0$. Thus:

$$R[t] \cong \frac{R[T]}{(yT - x)} \cong R\left[\frac{x}{y}\right]$$

Note that y is *not* a unit in $R[x/y]$. Let us show $\mathcal{O}(U) \cong R[t]$. If $s \in \mathcal{O}(U)$,

$$s|_{D(y)} = \frac{a}{y^n} \qquad \text{and} \qquad s|_{D(w)} = \frac{b}{w^m}$$

for some $a, b \in R$ and $n, m \in \mathbb{N}$. They must agree on $D(yw)$:

$$\left.\frac{a}{y^n}\right|_{D(yw)} = \frac{aw^n}{(yw)^n} = \frac{by^m}{(yw)^m} = \left.\frac{b}{w^m}\right|_{D(yw)}$$

Since R is an integral domain, this simplifies to $aw^m = by^n$ in R.

To show that $s \in R[t]$, we use the key relations $x = ty$ and $z = tw$ (which hold in $R[t]$ since $yt = x$ and $wt = z$). In R_y, the element a/y^n can be rewritten by substituting $x = ty$ and $z = tw$: every monomial in a involving x or z can be expressed using t, y, and w. More precisely, any element of R_y can be written as a polynomial in t with coefficients in $k[y, y^{-1}, w]$ (after eliminating x and z using the relations). Similarly, any element of R_w is a polynomial in t with coefficients in $k[y, w, w^{-1}]$. The overlap condition $aw^m = by^n$ ensures that the two expressions, when restricted to R_{yw}, give the same polynomial in t. By the locality axiom (already proved), this polynomial is unique, and it lies in $R[t]$ (no negative powers of y or w are needed, as one can verify by comparing degrees). Hence $s \in R[t]$, and so $\mathcal{O}_X(U) \cong R[x/y]$.

Now, $R[x/y]$ is *not* a localization of R. Indeed, $R[x/y] \ncong R$ (since $x/y \notin R$), yet $R[x/y]$ has the same units as R (namely k^*): no element becomes invertible. Since any nontrivial localization R_S contains units not present in R, and isomorphisms must preserve units, no localization of R is isomorphic to $R[x/y]$.

More generally, given an arbitrary open subset U, we can cover it by affine opens $\{U_i\}_{i\in I}$ and cover the intersections $U_i \cap U_j$ by affine opens $\{U_{ijk}\}_{k\in K_{ij}}$. Then the sheaf condition gives that $\mathcal{O}_X(U)$ is the equalizer of:

$$\prod_{i\in I} \mathcal{O}_X(U_i) \rightrightarrows \prod_{i,j\in I,\, k\in K_{ij}} \mathcal{O}_X(U_{ijk})$$

The general idea is that $\mathcal{O}_X(U)$ may contain rational functions f/g where g is nonzero on all of U. In practice, we shall often only require the local sections on distinguished open sets or on affine open sets, and many computations reduce to these two cases.

Note that even the following weaker hope fails: given $U \subseteq \operatorname{Spec} R$, is $\mathcal{O}_X(U)$ isomorphic to the localization of R at the multiplicative set S of all elements that do not vanish at any point of U? Consider $U = D(x) \cup D(y) \subseteq \mathbb{A}_k^2 = \operatorname{Spec} k[x,y]$. We will show in example 2.2.51 that U is not affine. However, $\mathcal{O}_{\mathbb{A}_k^2}(D(x) \cup D(y)) \cong k[x,y]$ (a Hartogs-type phenomenon: regular functions extend across a codimension-2 locus). This is just $k[x,y]$ itself, yet $D(x) \cup D(y) \ncong \mathbb{A}_k^2 = \operatorname{Spec} k[x,y]$.

This shows that the relationship between local geometry and algebra is more nuanced than the affine case suggests.

[a] As $k[x,y,z,w]$ is a UFD, a principal ideal (p) where p is irreducible is prime. It is tedious but rudimentary to show that $xw - yz$ is irreducible.

Keeping track of the structure sheaf $\mathcal{O}_{\operatorname{Spec} R}$ along with $\operatorname{Spec}(R)$ gives us our first definition we need:

Definition 2.2.3: Affine Scheme

Let $X = \operatorname{Spec}(R)$ be the spectrum of the ring R with the Zariski topology, and let $\mathcal{O}_{\operatorname{Spec}(R)}$ be the structure sheaf. Then $(\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ is called an *affine scheme*.

In some textbooks, an affine scheme is defined as a locally ringed space (recall definition 1.4.2) $(X, \mathcal{O}_X)$ that is an *affine scheme* if it is locally-ringed-space isomorphic to $(\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ for appropriate ring R.

With the sheaf defined, we have now "justified" thinking of elements on an arbitrary ring R as functions, as

$$\mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec} R) = \mathcal{O}_{\operatorname{Spec}(R)}(D(1)) = R_1 = R$$

Even more is true, due to the section's being constructed through localization of R, they may be ratio's of functions, and we give elements of section a particular name:

Definition 2.2.4: Regular Functions

Let $X = \operatorname{Spec} R$ be an and $U \subseteq X$ an open subset. Then the set $\mathcal{O}_X(U)$ is called the set of *regular functions* on U.

Note that like complex manifolds (and unlike general smooth manifolds), there are in general relatively few functions in the global section of an as compared to the global section (in the global section of an affine scheme, all rational functions are regular, that is they are R). On the other hand, $\mathcal{O}_X(U)$ may have many interesting rational functions. In fact, in many cases all open subsets of X are dense meaning the local behavior can give a lot of fruitful information on the entire scheme (as shall be the case for abstract varieties, see ref:HERE)

Example 2.2.5: Affine Schemes

All spectra can be equipped with a structure sheaf by definition and so they are all affine schemes. For less obvious examples:

1. If $k \ncong k'$ are distinct fields fields, then as schemes $\operatorname{Spec}(k) \ncong \operatorname{Spec}(k')$ as their sheaves are different. Since there is only one point $[(0)]$ an all such Spectra, the functions (elements of the sheaf) will all return themselves, which the reader can think of as there only being constant functions, where each sheaf have different constant functions.

2. Let $R = k[x]_{(x)}$. Then $\operatorname{Spec} R$ contains two points, $[(x)]$ and $[(0)]$ and has 3 open sets, namely

$$\emptyset \quad \{[(0)]\} \quad \{[(0)], [(x)]\} = \operatorname{Spec} R$$

Note that U and $\emptyset$ are distinguished open sets (since $\{(0)\} = X_x$). The corresponding structure sheaf is:

$$\mathcal{O}_X(X) = R = k[x]_{(x)}$$
$$\mathcal{O}_X(U) = k(x)$$

The restriction map is the natural inclusion. This schemes is an example of a *local scheme*.

3. Let $R = k[[x]]$, the completion of $k[x]$ at x. The only points of Spec R are again $[(0)]$ and $[(x)]$, and so it has the same open sets. The corresponding structure sheaf is:

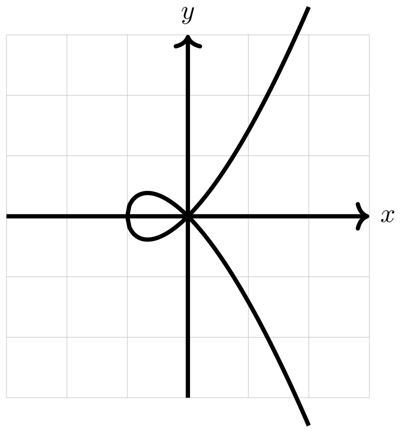
$$\mathcal{O}_X(X) = R = k[[x]]$$
$$\mathcal{O}_X(U) = \operatorname{Frac}(k[[x]])$$

Thus, we see that there can be schemes with very similar sheaves and identical spectra, but are distinct.

This scheme is considered to be *even more local* than localization. Consider $\mathbb{A}^2_k$. Then the sequence of maps:

$$k[x,y] \hookrightarrow k[x,y]_{(x,y)} \hookrightarrow k[[x,y]]$$

induces a sequence of inclusion spectra maps $Y \to X \to \mathbb{A}^2_k$, showing how the completion at the point (x,y) is "smaller" than the localization at that point. To demonstrate this, assume k does not have characteristic 2 or 3 and consider the affine scheme given by $y^2 = x^2 + x^3$:



Note that $y^2 - x^2 - x^3$ is irreducible, and localizing at (x, y) keeps the corresponding $(\frac{k[x,y]}{(y^2-x^2-x^3)})_{(x,y)}$ with only two points. However, in

$$\frac{k[[x,y]]}{(y^2 - x^2 - x^3)}$$

we have the access to power series expansion around $(0, 0)$, and we have that the power series:

$$x + \frac{1}{2}x^2 - \frac{1}{8}x^3 + \cdots$$

belongs to this ring, and notably this power series is the square-root of $x^2 + x^3$. Thus, we may write:

$$y^2 - x^3 - x^2 = (y - \sqrt{x^3 + x^2})(y + \sqrt{x^3 + x^2})$$

and so we get that $\frac{k[[x,y]]}{(y^2-x^3-x^2)}$ is *not* an integral domain, and in particular we shall see that $\operatorname{Spec} \frac{k[[x,y]]}{(y^2-x^3-x^2)}$ has *two* irreducible components corresponding to the two branches at $(0, 0)$.

4. The reader may recall that most power series do not satisfy an algebraic equation. For this reason, the above ring is often considered too big, and instead we take an intermediary ring H which contains all power series that satisfy algebraic equations over $\operatorname{Frac}(R)$. This is called the *Henselization* of $\operatorname{Spec} R$ at a point $\mathfrak{p}$.

5. If X, Y are then the product of X and Y are:

$$\operatorname{Spec}(R) \times \operatorname{Spec}(S) = \operatorname{Spec}(R \otimes_{\mathbb{Z}} S) = \operatorname{Spec}(R \otimes S)$$

where the tensor product with no base ring by default assume it is over the initial ring. The generalization of this concept shall bee shown in section 3.2.

2.2.2 Ideal Correspondence in Affine Schemes

So far, the dictionary between certain types of ideals of R and the geometric equivalent in $\operatorname{Spec}(R)$ has been establish, namely through $V(-)$ and $I(-)$:

maximal ideal of R	$\Leftrightarrow$	closed points of $\operatorname{Spec}(R)$
minimal prime ideals of R	$\Leftrightarrow$	irreducible components of $\operatorname{Spec}(R)$
prime ideal of R	$\Leftrightarrow$	irreducible closed subset of $\operatorname{Spec}(R)$
radical ideals of R	$\Leftrightarrow$	closed subsets of $\operatorname{Spec}(R)$

What of a general ideal $J \subseteq R$? In section 2.1.1, this information was "lost" as $I(V(J)) = \sqrt{J}$. However, a sheaf would capture this information!

Let's start with the simplest case of $R = \mathbb{C}[x]/(x^2)$ with $X = \operatorname{Spec}(R)$ that was mentioned in example 2.1.3(6). If we consider the set $\operatorname{Spec}(R)$, then this is just the origin, i.e. $[(x)]$. Namely, by proposition 2.1.10[13],

$$\operatorname{Spec}(\mathbb{C}[x]/(x^2)) \cong_{\mathbf{Top}} \operatorname{Spec}(\mathbb{C}[x]/(x)) \cong_{\mathbf{Top}} \{\overline{(x)}\} \equiv \{0\}$$

[13] Note that this is a topological isomorphism, these are not scheme isomorphic as we shall see in section 3.1.1!

Note that (0) is no longer prime since $x^2 \in (0)$ but $x \notin (0)$, and thus $[(0)] \notin \operatorname{Spec}(\mathbb{C}[x]/(x^2))$. To show that the $\left(\operatorname{Spec}\left(\mathbb{C}[x]/\left(x^2\right)\right), \mathcal{O}_{\operatorname{Spec}(\mathbb{C}[x]/(x^2))}\right)$ is different from $\left(\operatorname{Spec}(\mathbb{C}[x]/(x)), \mathcal{O}_{\operatorname{Spec}\,(\mathbb{C}[x]/(x)))}\right)$, let us first label:

$$(X, \mathcal{O}_X) = \left(\operatorname{Spec}\frac{\mathbb{C}[x]}{(x^2)}, \mathcal{O}_{\operatorname{Spec}\mathbb{C}[x]/(x^2)}\right)$$

Consider the image of a polynomial $f(x)$ (i.e. an element in the global section) in $\mathbb{C}[x]/\left(x^2\right)$. For example if $\overline{f(x)} = \overline{(x-a)^2} \in \mathbb{C}[x]/\left(x^2\right)$, then:

$$\bar{f} = \overline{(x-a)^2} = \overline{x^2 - 2ax + a^2} = \overline{-2ax + a^2} \in \mathbb{C}[x]/\left(x^2\right) = \mathcal{O}_X(X)$$

Evaluating at the single point $[(x)] = \epsilon$ we get

$$\bar{f}(\epsilon) = a^2$$

showing that evaluating at ϵ acts like plugging in zero. as x is a function, we get:

$$x(\epsilon) = 0$$

even though x is *not* the zero-function. In fact, one very important algebraic fact that is "obvious" for traditional functions but is not necessarily the case for ring with nilpotents is that a function that is zero everywhere will have to be zero:

Lemma 2.2.6: Nilpotence and Nonzero "Zero" Functions

Let R be a ring with nilpotence (i.e. a nonreduced ring). Then $\mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R))$ contains a function that evaluate to 0 everywhere but is not zero.

***Proof*:**
Let $f \in R$ be a nilpotence element so that $f \neq 0$ but $f^n = 0$. Then f is contained in each prime ideal, for $ff^{n-1} = f^n = 0 \in \mathfrak{p}$, and so either $f, f^{n-1} \in \mathfrak{p}$, which we can then reduce to having $f \in \mathfrak{p}$. Hence it is in the Nilradical, and so as a function

$$f(\mathfrak{p}) = 0$$

However $f \neq 0$, as we sought to show.

For completeness, we register this result whose proof was outlined in example 2.1.23(8). As it turns out, the failure of a regular function to be determined by its evaluation at points can completely be put at the feet of nilpotents:

Proposition 2.2.7: Spectrum And Nilpotent Elements

Let R be a ring and I an ideal of nilpotent elements. Then

$$\operatorname{Spec} R/I \to \operatorname{Spec} R$$

is a homeomorphism. As a consequence, two regular functions (elements of R) have the same value if they differ by a nilpotent element.

Thus, as topological spaces, nilpotents have no effect. Their presence is felt within the sheaf we put on spectra (see section 2.2.1)

> ***Proof*:**
> Hint: recall that the nilradical is equal to the intersection of all prime elements (see [11, chapter 22]).

Hence if the nilradical is zero, $\sqrt{(0)} = (0)$, functions (in global sections of affine schemes) are determined by their points[14]! In the new language we developed, this say that a function vanishes at every point of an affine scheme if and only if it is nilpotent! Note how this contrasts to the case of (classical) varieties where we required k to be algebraically closed for the same result, showing the greater generality achieved by having the extra prime points!

How to then visualize the $\operatorname{Spec}(R)$ along with its structure sheaf? It is usually done by picturing a "cloud" or "fuzz" around the point:

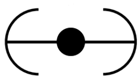


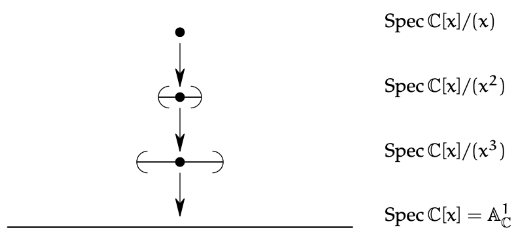
Figure 2.3: Picture of $(X, \mathcal{O}_X)$, not just $X = \operatorname{Spec}(R)$

This can be thought of as the "first degree germ" information around the point (recall that in $\mathbb{C}[\epsilon]$, $f(a+\epsilon) = f(a) + f'(a)\epsilon$, showing that the output of f at a plus the "fudge" of ϵ gives $f(a)$ plus an infinitesimal amount in the direction $f'(a)$. The sequence of restrictions $\mathbb{C}[x] \to \mathbb{C}[x]/(x^2) \to \mathbb{C}[x]/(x)$ should be thought of as inching closer to $f(0)$:

$$\begin{tikzcd}
\mathbb{C}[x] \arrow[r, two heads] & \mathbb{C}[x]/(x^2) \arrow[r, two heads] & \mathbb{C}[x]/(x) \\
f(x) \arrow[rr, mapsto] & & f(0),
\end{tikzcd}$$

The reason the point is visualized to have the fuzz is that it is the point that creates this "effect" when a function evaluates at this point, in other words remembering the functions behaviour changes the geometric interpretation of the point. If we take, $\mathbb{C}[x]/(x^3)$, then then the spectrum will again be a single point, and when finding an element from the global section, the value, 1st derivative, and 2nd derivative will be remembered. This can either be thought of a larger fuzz or a fuzz that is "bending" with more degree's of precision. We can have the following chain of inclusions:

[14] Proposition 2.4.5 shall push this result to its maximal generalization on quasicompact schemes



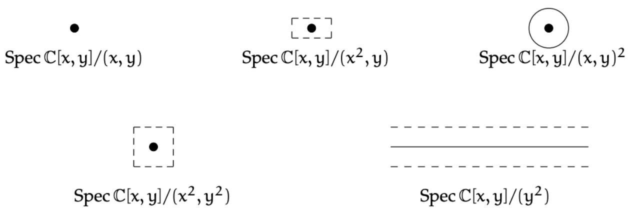
Going to higher dimensions and considering:

$$\operatorname{Spec}\left(\frac{\mathbb{C}[x,y]}{(x,y)^2}\right)$$

this can be thought of as remembering the derivative in all directions, and can be represented by a circle.

Example 2.2.8: Geometric Interpretation of Nilpotence

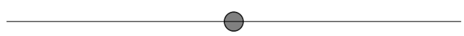
1. The reader should stare at this image to make sure it makes sense:



2. consider,

$$\operatorname{Spec}\left(\frac{\mathbb{C}[x,y]}{(y^2,xy)}\right)$$

Then this can be thought of as the intersection of a thickened x-axis and the "+" created by xy, that is this can be thought of as the x-axis as well as the first-order differential information around the origin.



Once we introduce stalks in section 2.2.5, these points will be more detectable. Notice that $[(x)]$ is *no longer* a generic point since $y^2 = 0 \in \overline{(x)}$ but $y \notin \overline{(x)}$, and hence the ideal is not prime, thus no longer a point. On the other hand, the function y when evaluating at the points $[\overline{(x-a,y-b)}] = [\overline{(x-a,y)}]$ is 0 at each point

3. Consider

$$\operatorname{Spec}\left(\frac{\mathbb{C}[x,y]}{(y^2)}\right)$$

By proposition 2.1.10, this is homeomorphic to $\mathbb{A}^1_{\mathbb{C}}$. As this is a rather simple example, this can be seen concretely: if $\mathfrak{p} \subseteq \mathbb{C}[x,y]/(y^2)$ is prime and contains y^2, then by primality $y \in \mathfrak{p}$, and hence the prime ideals correspond to those of $\mathbb{C}[x,y]/(y^2,y) \cong \mathbb{C}[x]$, i.e. those of $\mathbb{A}^1_{\mathbb{C}}$.

Now, in $\mathbb{C}[x,y]/(y^2)$, there cannot be y-terms of degree higher than 1: if $F(x,y) \in \mathbb{C}[x,y]/(y^2)$, then:

$$F(x,y) = f(x) + g(x)\,y$$

This can be thought of as $F(x,y)$ "remembering" some first-order information in the y-direction, as though it were the restriction of a function on $\mathbb{A}^2_{\mathbb{C}}$ to an infinitesimal neighborhood of the x-axis. In this decomposition, $f(x)$ is the value on the x-axis and $g(x)$ plays the role of $\partial F/\partial y|_{y=0}$ – a tangent direction at each point. Algebraic operations respect this: for instance,

$$(f_1 + g_1 y)(f_2 + g_2 y) = f_1 f_2 + (f_1 g_2 + f_2 g_1)\,y$$

which mirrors the product rule for derivatives.

Another behaviour that may seem strange is the addition of new units! Consider $1 + xy$. Since y is nilpotent ($y^2 = 0$), the element xy is also nilpotent: $(xy)^2 = x^2y^2 = 0$. We can verify directly that $1 + xy$ is a unit:

$$(1 + xy)(1 - xy) = 1 - (xy)^2 = 1 - 0 = 1$$

More generally, an element $f(x) + g(x)\,y$ is a unit in $\mathbb{C}[x,y]/(y^2)$ if and only if $f(x)$ is a unit in $\mathbb{C}[x]$, i.e. $f(x) \in \mathbb{C}^*$ is a nonzero constant. This is because the nilpotent part $g(x)\,y$ can always be "absorbed" by the geometric series trick $(1+n)^{-1} = 1 - n$ when $n^2 = 0$, but the non-nilpotent part $f(x)$ must already be invertible.

Geometrically, this means the unit group of $\mathbb{C}[x,y]/(y^2)$ is *much larger* than that of $\mathbb{C}[x]$. For $\mathbb{C}[x]$, the only units are the nonzero constants $\mathbb{C}^*$. For the "fuzzy line" $\mathbb{C}[x,y]/(y^2)$, every element $a + g(x)\,y$ with $a \in \mathbb{C}^*$ is a unit – regardless of the choice of $g(x)$. The unit group is thus $\mathbb{C}^* \times \mathbb{C}[x]$, an infinite-dimensional group. All of these units evaluate to the constant a at every point (since y maps to 0 in every residue field), yet they are genuinely distinct elements of the ring, distinguished by their "tangent data" $g(x)$. In this sense, $1+g(x)\,y$ is an invertible function that is infinitesimally close to 1: it equals 1 on the underlying topological space but carries a nontrivial first-order perturbation. This is the multiplicative counterpart of the earlier observation that y is a nonzero function that evaluates to 0 everywhere – here, $1 + g(x)\,y$ is a non-constant unit that evaluates to the *same* constant everywhere.

Observe that $D(x)$ is the "punctured fuzzy line": it removes the point $[(x)] \in \operatorname{Spec}(\mathbb{C}[x,y]/(y^2))$ but retains the nilpotent thickening at every remaining point. On the other hand, $D(y) = \emptyset$, since every prime ideal contains y (as $y^2 = 0 \in \mathfrak{p}$ implies $y \in \mathfrak{p}$ by primality). Hence the underlying topological space $|\operatorname{Spec} R|$ does not detect the nilpotent "germ" information carried by y – but the structure sheaf does, as y is a nonzero section of $\mathcal{O}_X$ that vanishes at every point.

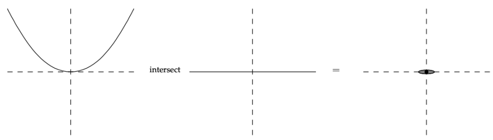
Intersection Theory and Family of Schemes

We shall cover intersection theory extensively in Part II. The intersection theory of curves as covered in [11, chapter 21.6] gives insightful ways in which the discussion above has practical applications. Recall that the multiplicity information of a curve is defined as an equivalence class of elements of $k[x, y]$ up to scaling. This was to distinguish f and f^2 in $k[x, y]$, something varieties cannot do in classical algebraic geometry. Schemes naturally capture multiplicity information.

For example, consider the following:

$$\begin{aligned} \operatorname{Spec}\left(\mathbb{C}[x, y]/(y - x^2)\right) \cap \operatorname{Spec}\left(\mathbb{C}[x, y]/(y)\right) &= \operatorname{Spec}\left(\mathbb{C}[x, y]/(y - x^2, y)\right) \\ &= \operatorname{Spec}\left(\mathbb{C}[x, y]/(y, x^2)\right) \end{aligned}$$

That is, intersecting the parabola $y = x^2$ with the x-axis produces a fuzzy point that remembers the multiplicity:



Let us push this idea further and identify the local subscheme associated to a curve passing through the origin with a given tangent direction. Let $R = k[\epsilon]/(\epsilon^2)$ be the dual numbers, and consider a surjection:

$$\varphi : k[x, y] \to R$$

that sends the origin to the unique point of $\operatorname{Spec} R$, i.e. $\varphi(x), \varphi(y) \in (\epsilon)$. The most general such map sends $x \mapsto b\epsilon$ and $y \mapsto -a\epsilon$ for some $a, b \in k$ not both zero (different choices of (a, b) up to scaling give different tangent directions). As $\epsilon^2 = 0$, the map φ vanishes on $(x, y)^2 = (x^2, xy, y^2)$, and hence φ factors through:

$$\overline{\varphi} : \frac{k[x, y]}{(x^2, xy, y^2)} \to R$$

As R is 2-dimensional while $k[x, y]/(x^2, xy, y^2)$ is 3-dimensional (with basis $\{1, x, y\}$), the map $\overline{\varphi}$ has a 1-dimensional kernel. One can check that this kernel is generated by $ax + by$ (the linear form vanishing in the chosen tangent direction), and hence we get the isomorphism:

$$X_{a,b} = \operatorname{Spec} \frac{k[x, y]}{(x^2, xy, y^2, ax + by)} \cong \operatorname{Spec} R$$

How can we interpret this scheme?

- Algebraically, we can interpret $X_{a,b} \subseteq \mathbb{A}^2_k$ as the subscheme associated to the ideal containing all functions $f \in k[x, y]$ that vanish at the origin *and* have partial derivatives satisfying:

$$b \left.\frac{\partial f}{\partial x}\right|_0 - a \left.\frac{\partial f}{\partial y}\right|_0 = 0$$

  This says that the gradient of f at the origin is proportional to (a, b), or equivalently, $f = c(ax + by) + g(x, y)$ where $g(x, y) \in (x, y)^2$ consists of higher-order terms.

- Geometrically, $X_{a,b}$ is the first-order neighborhood of the origin along the line $ax+by=0$. It arises as the composite
$$\operatorname{Spec}\frac{k[t]}{(t^2)} \hookrightarrow \mathbb{A}^1_k \xrightarrow{t\mapsto(bt,-at)} \mathbb{A}^2_k$$
which embeds a "fuzzy point" into $\mathbb{A}^2$ along the tangent direction $(b,-a)$.

We interpret $X_{a,b}$ as containing the point $(0,0)$ together with the infinitesimally near information in the direction specified by the line $ax+by=0$ (the "tangent-line" information!). Such a scheme arises naturally when intersecting two curves non-transversally (the example with the parabola and the line is the simplest such case).

Another place where such schemes naturally arise is when looking at the *limit of families of schemes*. To understand this, take the case of two points $(0,0),(a,b)\in\mathbb{A}^2_k$. These two points together form a scheme:
$$X_1=\operatorname{Spec}R=\operatorname{Spec}\frac{k[x,y]}{(x,y)\cap(x-a,\,y-b)}=\operatorname{Spec}\frac{k[x,y]}{(x^2-ax,\,xy-bx,\,xy-ay,\,y^2-by)}$$
It should not be hard to see (e.g. via the Chinese Remainder Theorem) that $R\cong k\times k$.

Now, suppose (a,b) moves towards $(0,0)$ along a polynomial curve. In particular, say there are two polynomial functions $a(t),b(t)$ such that $(a(0),b(0))=(0,0)$. Let $X_t=\{(0,0),\,(a(t),b(t))\}$. Then what scheme would we expect to get when $t\to 0$? The advantage of using schemes for intersection theory, as outlined above, suggests that this limit should remember what direction the points came from – and indeed it does. Even more than this, the limit shall still be a length-2 scheme: two points that have "collided" but retained a memory of their approach direction.

Let us elucidate the details. Take
$$I_t=(x,y)\cap(x-a(t),\,y-b(t))=(x^2-a(t)x,\;xy-b(t)x,\;xy-a(t)y,\;y^2-b(t)y)$$
As $t\to 0$, the ideal I_0 must contain x^2, xy, and y^2. Furthermore, I_t always contains a linear form:
$$a(t)y-b(t)x=(xy-b(t)x)-(xy-a(t)y)$$
Writing $a(t)=a_1t+a_2t^2+\cdots$ and $b(t)=b_1t+b_2t^2+\cdots$ (since $a(0)=b(0)=0$), we can divide by t to obtain an element of I_t for $t\neq 0$:
$$\frac{a(t)y-b(t)x}{t}=a_1y-b_1x+t\cdot(a_2y-b_2x+\cdots)$$
Taking $t\to 0$, the limit ideal contains:
$$(x^2,\,xy,\,y^2,\,a_1y-b_1x)\subseteq I_0$$
These are in fact equal. To see this, note that the quotient $k[x,y]/I_t$ is a k-vector space of dimension 2 for each $t\neq 0$ (since I_t cuts out two distinct points). By a continuity argument (in the sense of flat limits), the limiting quotient $k[x,y]/I_0$ also has dimension 2[15]. Since the quotient by the left-hand side (x^2,xy,y^2,a_1y-b_1x) already has dimension 2 (with basis $\{1,b_1x\}$ if $b_1\neq 0$, or $\{1,a_1y\}$ if $a_1\neq 0$), the inclusion must be an equality. Labelling $\alpha=a_1$, $\beta=b_1$ and $X_0=X_{\alpha,\beta}$, we may write:
$$\lim_{t\to 0}X_t=X_{\alpha,\beta}$$

[15]This argument is more subtle without the finite-dimensionality hypothesis, which shall be addressed when we cover this in fuller generality in section 6.8.

In higher dimensions, things get more complicated, and when our schemes are no longer local even more so. We shall see in section 6.8 that what we have been examining is a *flat family* of schemes. Flatness provides the correct conditions for a parameterized family of schemes to have well-behaved limits, generalizing the "continuity" argument used above.

Exercise 2.2.1

1. Show that for any polynomial $f \in k[x]$, if $a + b\epsilon \in \mathbb{C}(\epsilon)$, then
$$f(a + b\epsilon) = f(a) + bf'(a)\epsilon$$

2. let k be algebraically closed, R be a local k-algebra of dimension 2 with maximal ideal $\mathfrak{m}$. Show that $R \cong k[x]/(x^2)$ (hint: use some dimension argument and Nakayama's lemma to conclude $\mathfrak{m}^2 = 0$).
3. Let k be algebraically closed. Find two non-isomorphic local k-algebras of dimension 3. Geometrically, we can think of this as representing the extra flexibility in directions or local schemes can capture.
4. Let k be algebraically closed and $\operatorname{Spec} R = \operatorname{Spec} k[x_1, ..., x_n]/I \subseteq \mathbb{A}^n_k$ be a local subscheme of dimension 0 and degree 3 (R has dimension 3 over k) supported at the origin. Then $\operatorname{Spec} R$ is either isomorphic to $\operatorname{Spec} k[x]/(x^3)$ or $\operatorname{Spec} k[x,y]/(x^2, xy, y^2)$, and these two schemes are not isomorphic to each other.
5. Let R be a ring that is not a field. Show that the restriction map on the structure sheaf is almost never surjective. Hence, there are more local functions than global function.
6. Let R be a reduced ring (contains no nilpotent elements). Show that the restriction maps are injective, that is each function on a larger open set can only restrict to one function. Give a counter-example for a ring with nilpotent elements.

2.2.3 Schemes

The last step in the definition of schemes is to observe that many geometric objects are not the spectrum of a ring, but are *locally* the spectrum of a ring. The prototypical examples are projective spaces, their closed subsets, and sequences of blow ups—none of which are in general affine.

Definition 2.2.9: Scheme

A *scheme* is a locally ringed space $(X, \mathcal{O}_X)$ that is locally affine: there exists an open cover $X = \bigcup_i U_i$ such that for each i, there is a ring R_i and an isomorphism of locally ringed spaces
$$(U_i,\ \mathcal{O}_X|_{U_i}) \cong_{\mathbf{LRS}} (\operatorname{Spec}(R_i),\ \mathcal{O}_{\operatorname{Spec}(R_i)}).$$
Such a cover is called an *affine open cover* of X.

The requirement that $(X, \mathcal{O}_X)$ be a locally ringed space—that is, that each stalk $\mathcal{O}_{X,x}$ be a local ring—is essential. Without it, the "evaluation" of a section at a point (via the residue field $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$) would not be well-defined, and the connection to geometry would be lost. We saw in section 1.4 that locally ringed spaces are the correct generalization of the spaces that arise in both differential and algebraic geometry. Since the stalk of $\mathcal{O}_X$ at a point $x \in X$ is a local ring (by the locally ringed space condition), we denote it $\mathcal{O}_{X,x}$, its maximal ideal $\mathfrak{m}_x$, and its residue field $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$.

The topology of affine schemes is generated by distinguished open sets. Lemma 2.2.29 shall show that given $\operatorname{Spec} R, \operatorname{Spec} S \subseteq X$ where their intersection is non-empty, the intersection can be covered by distinguished open sets that are distinguished on both $\operatorname{Spec} R$ and $\operatorname{Spec} S$. Hence the topology of schemes is also generated by distinguished open sets. See section 2.8 for a deeper discussion about the purely topological properties of schemes.

The way to interpret the agreement on intersections is that locally a scheme respects the algebraic relations imposed by a ring R_i, and the compatibility on $U_i \cap U_j$ (when nonempty) corresponds to the algebraic relations between R_i and R_j being consistent. A detailed example is presented in example 2.2.55.

2.2.10 Lack of $V(-)$ and $I(-)$? The topology of an affine scheme is characterized by the ideals of its global ring. General schemes are no longer tethered to the global sections (as shall be momentarily demonstrated). However, there is a generalization of $V(-)$ and $I(-)$ that works with the (quasi-coherent) *ideal sheaves* of a scheme X; the details are covered in section 5.1.4. From this perspective, the structure sheaf $\mathcal{O}_X$ can be thought of as generalizing the notion of a ring to a "local" object, remembering information that no single ring can capture.

Definition 2.2.11: Morphism of Schemes

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be schemes. Then a *morphism* of schemes is a morphism of locally ringed spaces $(\varphi, \varphi^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$.

A *isomorphism of schemes* is an isomorphism of locally ringed spaces: a homeomorphism $f : X \to Y$ together with a sheaf isomorphism $f^\# : \mathcal{O}_Y \xrightarrow{\sim} f_*\mathcal{O}_X$.

Not every morphism of ringed spaces preserves local rings (see example 3.1.1). The full exploration of the nuances shall be covered in section 3.1.1.

Example 2.2.12: Schemes

1. All affine schemes are schemes, covered by a single open set.
2. The key non-affine scheme is projective space $\mathbb{P}^n_k$. We shall construct the general projective scheme $\mathbb{P}^n_R$ in section 2.3; for now, the reader can verify that the usual projective space as defined in [**NateClassicalAlgGeo**] is a scheme by exhibiting the standard affine charts $U_i = \{x_i \neq 0\} \cong \mathbb{A}^n_k$. Any closed subvariety of $\mathbb{P}^n_k$ (with the appropriate structure sheaf) is also a scheme, typically not affine.
3. The following is the smallest non-affine scheme. Take $X = \{p, q_1, q_2\}$ with the topology generated by $X_1 = \{p, q_1\}$ and $X_2 = \{p, q_2\}$ as a sub-base. Define the sheaf of rings by
$$\mathcal{O}(X) = \mathcal{O}(X_1) = \mathcal{O}(X_2) = K[x]_{(x)}, \qquad \mathcal{O}(\{p\}) = K(x),$$
with restriction maps $\mathcal{O}(X) \to \mathcal{O}(X_i)$ the identity and $\mathcal{O}(X_i) \to \mathcal{O}(\{p\})$ the natural localization. This presheaf is a sheaf, and $(X, \mathcal{O})$ is a scheme but *not* an affine scheme: the

ring $K[x]_{(x)}$ has two points in its spectrum, but X has three. Geometrically, this is the "germ of a doubled point"—the generic point of $\mathbb{A}^1$ has been split into two.

4. By proposition 2.1.20, a finite disjoint union of schemes is a scheme. An arbitrary (infinite) disjoint union of affine schemes is a scheme but not an affine scheme, as it fails to be quasi-compact.

5. All affine schemes are "separated" (a property to be defined in section 3.5.1). There exist non-separated schemes; the simplest is the "line with doubled origin," constructed by gluing two copies of $\mathbb{A}^1_k$ along $\mathbb{A}^1_k \setminus \{0\}$.

6. If the reader has already encountered blow-ups, the schemes associated to a blow-up are typically not affine.

7. The open subset $D(x) \cup D(y) \subseteq \mathbb{A}^2_k$ is a scheme that is quasi-compact, separated, and a subset of an affine scheme, yet is not itself affine. The details are in Example 2.2.51.

8. Let $X = \{*\}$ be a singleton and $\mathcal{O}_X(X) = R$ for a commutative ring R. Is $(X, \mathcal{O}_X)$ a scheme? First, it must be affine so $X = \operatorname{Spec}(R)$. Next, $\operatorname{Spec} R$ must have a single prime ideal. This must be a 0-dimensional is a local ring. For example:

$$R \in \{k, \mathbb{Z}/p^n\mathbb{Z}, k[x]/(x^m)\}$$

Note there are local rings with more than one point, for example $k[[x]]$.

It is not always immediate whether a scheme is affine. For example, $\mathbb{A}^1_{\mathbb{C}} \setminus \{[(x)]\}$ may seem non-affine since it is not an affine variety in the classical sense, but it is isomorphic to $\operatorname{Spec}(\mathbb{C}[x, x^{-1}])$. On the other hand, $\mathbb{A}^2_{\mathbb{C}} \setminus \{[(x, y)]\}$ is genuinely not affine (Example 2.2.51). We shall establish tools for detecting affineness shortly.

Given a scheme X, we write $|X|$ for the underlying topological space. The elements of $\mathcal{O}_X(U)$ are called *sections* over U, and sections of $\mathcal{O}_X(X)$ are called *global sections*. These are variously denoted

$$\mathcal{O}_X(U), \qquad \Gamma(U, \mathcal{O}_X), \qquad H^0(U, \mathcal{O}_X),$$

depending on context: the first is convenient when only the structure sheaf is present; the second will be useful when multiple sheaves on X are in play; and the third hints at the fact that the global section functor $\Gamma(X, -)$ is left-exact but generally not exact, with its right-derived functors $H^i(X, -)$ forming the cohomology theory developed in chapter 6.

Recovering the ring from an affine scheme

A natural question arises: if a scheme X happens to be affine, can we recover the ring R from the locally ringed space $(X, \mathcal{O}_X)$? The answer is yes, and the ring is simply the global sections.

Proposition 2.2.13: Distinguished Open Sets and Schemes

Let R be a ring and $f \in R$. Then:

$$(D(f),\ \mathcal{O}_{\mathrm{Spec}\,(R)}|_{D(f)}) \ \cong\ (\mathrm{Spec}\,(R_f),\ \mathcal{O}_{\mathrm{Spec}\,(R_f)}).$$

In particular, taking $f = 1$: if $(X, \mathcal{O}_X)$ is an affine scheme, meaning $(X, \mathcal{O}_X) \cong (\mathrm{Spec}\,(R), \mathcal{O}_{\mathrm{Spec}\,(R)})$ for some ring R, then R is recovered as the global sections:

$$\mathcal{O}_X(X) \ \cong\ R.$$

***Proof*:**
The first statement is the scheme-theoretic upgrade of Proposition 2.1.12: the homeomorphism $D(f) \cong \mathrm{Spec}\,(R_f)$ extends to an isomorphism of locally ringed spaces because the structure sheaf on $D(f)$ is defined by the same localization data as $\mathcal{O}_{\mathrm{Spec}\,(R_f)}$.

For the recovery of R: since X is affine, $X = D(1) = \mathrm{Spec}\,(R)$, and

$$\mathcal{O}_X(X) = \mathcal{O}_X(D(1)) = R_1 \cong R,$$

where the middle equality is the definition of the structure sheaf on $\mathrm{Spec}\,(R)$.

The motivation behind this proposition is that for affine schemes, the global sections completely determine the scheme. This is emphatically *not* the case for general schemes: the global sections of $\mathbb{P}^n_k$ are just k (for $n \geq 1$), which carries no information about the geometry of projective space. The structure sheaf, not the global sections, is the fundamental invariant.

Proposition 2.2.14: Characterization of Affine Schemes

Let $(X, \mathcal{O}_X)$ be a locally ringed space with $R = \mathcal{O}_X(X)$. Then $(X, \mathcal{O}_X) \cong (\mathrm{Spec}\,(R), \mathcal{O}_{\mathrm{Spec}\,(R)})$ if and only if:

1. For every $f \in R$, $\mathcal{O}_X(U_f) \cong R_f$, where $U_f = \{x \in X : f \notin \mathfrak{m}_x\}$.
2. The natural map $X \to |\mathrm{Spec}\,(R)|$ (sending x to the prime $\mathfrak{m}_x \cap R$) is a homeomorphism.

***Proof*:**
Exercise. (The content is that conditions (1) and (2) together give an isomorphism of locally ringed spaces, not merely of topological spaces or of ringed spaces.)

Theorem 2.2.15: Ring Homomorphisms and Affine Scheme Isomorphisms

Let R and S be rings. An isomorphism of locally ringed spaces $\pi : \mathrm{Spec}\,(R) \to \mathrm{Spec}\,(S)$ induces an isomorphism $\pi^\# : S \xrightarrow{\sim} R$ on global sections. Conversely, an isomorphism $\varphi : S \xrightarrow{\sim} R$ induces an isomorphism $\mathrm{Spec}\,(\varphi) : \mathrm{Spec}\,(R) \xrightarrow{\sim} \mathrm{Spec}\,(S)$, and these two constructions are inverse to each other.

***Proof*:**
Corollary of theorem 3.1.4.

Given a scheme X, the collection of distinguished open sets from any affine cover forms a basis for the topology on X. This is important because the intersection of two affine opens need not itself be affine (see section 3.5.1), but distinguished opens within each affine chart suffice.

Proposition 2.2.16: Distinguished Open Sets Form a Basis

Let X be a scheme and $\{U_i = \operatorname{Spec}(R_i)\}$ an affine open cover. Then the collection of all distinguished open sets $D(f) \subseteq U_i$, as i ranges over the index set and f ranges over R_i, forms a basis for the topology on X.

***Proof*:**
Every open subset of X restricts to an open subset of each U_i, which is a union of distinguished opens in U_i (since distinguished opens form a basis for the Zariski topology on $\operatorname{Spec}(R_i)$). The union over all i covers the original open set.

Finiteness conditions and base ring

In algebra, every ring is a $\mathbb{Z}$-algebra, and additional structure arises by specifying a base ring. The scheme-theoretic counterpart is the following.

Definition 2.2.17: R-scheme and Finite Type Schemes

An *R-scheme* (or a *scheme over R*) is a scheme X together with a morphism $X \to \operatorname{Spec}(R)$, called the *structure morphism*. Equivalently, the sections of $\mathcal{O}_X$ are R-algebras and all restriction maps are R-algebra homomorphisms.
An R-scheme X is *locally of finite type over R* if it can be covered by affine opens $\operatorname{Spec}(S_i)$ where each S_i is a finitely generated R-algebra. If X is also quasi-compact, then X is *of finite type over R*.

In number theory, integral extensions S/R (where S is a finitely generated R-module) play a central role. The scheme-theoretic analogue is:

Definition 2.2.18: Finite R-Schemes

An R-scheme X is *locally finite over R* if it can be covered by affine opens $\operatorname{Spec}(S_i)$ where each S_i is a finitely generated R-module. If X is quasi-compact, then X is *finite over R*.

If $R = k$ is a field (or more generally an Artinian ring), then a finite R-scheme is a finite collection of "fat points"—spectra of Artinian local k-algebras.

2.2.19 Terminology Warning Do not confuse *finite type* with *finite*. A scheme is locally of finite type if its coordinate rings are finitely generated *R-algebras* (finitely many generators as a ring); it is locally finite if its coordinate rings are finitely generated *R-modules* (finitely many generators as a module). The latter is a much stronger condition: $k[x]$ is a finitely generated k-algebra but not a finitely generated k-module.

Example 2.2.20: Finite Type R-schemes

1. Affine n-space $\mathbb{A}^n_k = \operatorname{Spec}(k[x_1, \ldots, x_n])$ is of finite type over k.
2. For polynomials $f_1, \ldots, f_m \in R[x_1, \ldots, x_n]$, the scheme $\operatorname{Spec}(R[x_1, \ldots, x_n]/(f_1, \ldots, f_m))$ is of finite type over R. Most classical intuitions from algebraic sets concern $\overline{k}$-schemes of finite type over $\overline{k}$ (abstract varieties shall require further properties; see section 3.7).
3. If R is the localization of a polynomial ring over k at a prime, then $\operatorname{Spec}(R)$ is generally not of finite type over k. Intuitively, local rings live in a "different dimension" from the global picture.
4. If the fibers of $X \to \operatorname{Spec}(R)$ are infinite-dimensional, then X is not of finite type over R.

2.2.4 Comparing Schemes and Real/Complex Manifolds

Before proceeding to further explore properties of schemes, it is worth taking a moment to catalogue how schemes and real smooth manifolds have some fundamentally different geometric properties:

1. The connected scheme given by $V(x) \cup V(yz) \subseteq \mathbb{A}^3_k$ has two irreducible components of different dimensions (as we shall more rigorously show in section 2.6). A connected manifold (smooth or even topological) is of a single, well-defined dimension at every point.
2. An abstract manifold cannot self-intersect[16], while schemes (even affine schemes, even varieties) can: consider $V(y^2 - x^3 - x^2) \subseteq \mathbb{A}^2_k$, which has a node at the origin.
3. The patches of a manifold are very regular, being diffeomorphic to $\mathbb{R}^n$; in this sense manifold patches all carry "linear smooth" information, and there is essentially no local diversity from one patch to another. The affine patches of a scheme, by contrast, may be "pathological"—for example, the local rings can be non-reduced or non-Noetherian (in the latter case there may even be uncountably many connected components). This reflects how affine schemes represent "local algebraic information," which is far richer than the "local smooth linear information" that real manifolds encode. Some of the conditions and properties we shall introduce later (e.g. Noetherianity, regularity) will restore some order.
4. In Manifold Theory, the notion of irreducible component is trivial, since in any Hausdorff space the only irreducible subsets are singletons. From an analytic perspective, this existence of non-trivial irreducible components in schemes can be thought of as a consequence of the failure of the Zariski topology to separate points by functions; part of this phenomenon is the

[16] An *immersion* of a manifold may of course have self-intersections in the target, but the source manifold itself is not singular at such points.

existence of "large" (i.e. non-closed) points for schemes. More generally, the correspondence between irreducible closed sets and non-closed points (in at least Noetherian schemes) shows that schemes encode much more "global" information about codimension subsets than real smooth manifolds do.

5. The topology of $\mathbb{R}^n$ (and hence of $\mathbb{C}^n$ as a real manifold) is the product topology: the sets

$$\prod_{i=1}^{n}(a_i, b_i), \qquad a_i, b_i \in \mathbb{R},\ a_i < b_i,$$

form a topological basis for $\mathbb{R}^n$. This is far from the case for $\mathbb{A}^n_k$: the Zariski topology on $\mathbb{A}^n_k$ is *not* the product of the Zariski topologies on $\mathbb{A}^1_k$ (indeed, the product would be the cofinite topology, whose closed sets are always finite unions of points, whereas $\mathbb{A}^2_k$ has closed curves). However, there will still be a meaningful way to construct the scheme $\mathbb{A}^n_k$ from $\mathbb{A}^1_k$ via the *fibered product* (see section 3.2).

6. If a scheme X is irreducible, all its nonempty open sets are dense. Many of the schemes we shall work with will be unions of irreducible subschemes, where for each irreducible subscheme every nonempty affine open subset is dense in that subscheme. Dense open subsets are far from typical for manifolds (e.g. a disjoint union of two manifolds has open subsets contained in a single component, which are not dense).

7. More generally, the behaviour of schemes matches more closely with the behaviour of complex analytic geometry than with smooth real geometry. We shall draw such analogies often: for example, *Hartogs's Extension Lemma* ([**NateComplexAnalysis**]) will have parallels with normal schemes, the existence of certain holomorphic/meromorphic functions will tie into divisor theory, compact complex Riemann surfaces will correspond to projective curves, and so forth.

8. (embedding properties) Any smooth real manifold M can be embedded into $\mathbb{R}^N$ for appropriate N (Whitney Embedding Theorem). For classical affine (resp. projective) varieties of dimension n, they essentially by definition embed into $\mathbb{A}^N_k$ (resp. $\mathbb{P}^N_k$) for some N. For complex manifolds, it is *not* the case that they all embed in $\mathbb{C}^N$[17], and in particular no compact complex manifold of positive dimension embeds into $\mathbb{C}^N$. This is because complex manifolds possess much more rigidity than real manifolds [9][18]. Schemes exhibit this same type of rigidity. The only types of schemes that have a universal embedding theorem into one of the classical spaces $\mathbb{A}^N_k$ or $\mathbb{P}^N_k$ are curves (see **??**) which embed into $\mathbb{P}^3$. the category of all surfaces already fail to have such a universal embedding theorem (by example 7.3.8).

   By the Kodaira Embedding Theorem [9], a compact Kähler manifold with a positive line bundle can be embedded into projective space. Proper schemes (the algebraic analogue of compact complex manifolds) with an *ample line bundle* (the algebraic analogue of a positive line bundle) similarly embed into projective space. In fact, this means that compact Kähler manifolds with positive line bundle are (as complex analytic spaces) isomorphic to a smooth proper scheme with an ample bundle. This is the closest to a general embedding theorem that is widely used, and will be proven in section 5.3.3.

[17] If a compact complex manifold does embed holomorphically, it must reduce to a point, since any global holomorphic function on a compact complex manifold is constant. Non-compact complex manifolds that embed holomorphically in $\mathbb{C}^N$ are called *Stein manifolds* and share many properties of quasi-projective varieties.

[18] For instance, the complex torus $\mathbb{C}/\Lambda$ depends on the lattice Λ up to biholomorphism, giving a one-complex-parameter family of non-isomorphic structures.

For a closed immersion $X \hookrightarrow \mathbb{A}^n_R$ so that $X \cong V \subseteq \mathbb{A}^n_R$ where V is a closed subscheme, X is topologically a sub-variety, though it may have a more complicated sheaf structure (for more details see section 3.7).

For an open embedding $X \hookrightarrow \mathbb{A}^n_R$ so that $X \cong U \subseteq \mathbb{A}^n_R$ where U is open, X is a *quasi-affine subscheme*, which by definition is an open subscheme of affine space. Their characterization is more subtle: the map $X \to \operatorname{Spec}(\mathcal{O}_X(X))$ must be an open embedding and, as will be discussed in section 5.1.1, all quasi-coherent sheaves of $\mathcal{O}_X$-modules must be generated by global sections. Verifying that every $\mathcal{O}_X$-module is generated by global sections is difficult; there is a simpler characterization given by the Goodman–Hartshorne Theorem [13], but it is still a nontrivial condition, illustrating the greater complexities that arise when working with non-proper schemes.

More generally, much later in ref:HERE, there shall be other interesting ambient spaces into which certain types of schemes may be embedded (e.g. toric varieties, Grassmannians, flag varieties, etc.), each serving different purposes depending on the moduli problem under study.

9. Schemes can have "doubled points": recall that the line with two origins is a standard counterexample for manifolds, as it fails to be Hausdorff at the origin. In scheme theory the analogous phenomenon is permitted; indeed, many moduli spaces that are schemes are naturally non-separated schemes (see section 3.3.6).

10. A scheme remembers multiplicity. For example, on a classical point there is little room for interesting functions, but scheme-theoretic functions (i.e. regular functions on $\operatorname{Spec} k[x]/(x^2)$) can encode multiplicity and infinitesimal information. A smooth manifold $(M, \mathcal{A})$ with its maximal atlas does not retain such data.

11. There can be multiple structure sheaves on the same underlying topological space, defining genuinely different schemes (think of the different fields that can serve as the structure sheaf over a single point). In contrast, manifolds of dimension ≤ 3 have unique differential structures up to diffeomorphism (by work of Radó in dimension 2 and Moise in dimension 3); in fact, the categories of piecewise-linear, topological, and smooth manifolds coincide in these dimensions, showing how "linear" manifold theory is in low dimensions. In section 3.2 we shall introduce the process of *base change*, by which one can change the structure sheaf without changing the underlying topological space.

12. Though we have not defined an atlas for schemes, Theorem 2.2.54 will show that schemes can be constructed by gluing affine patches, just as a manifold is built from its atlas, see section 2.2.7. (want to say why not as useful here).

13. The natural additional structure on a smooth manifold is a *tangent bundle*, whose transition functions take the form $D(\mathbf{y} \circ \mathbf{x}^{-1})$; that is, they are not just smooth maps between open subsets but must respect a $\mathrm{GL}_n(\mathbb{R})$-action given by the Jacobian. We shall define the algebraic analogue for schemes (the sheaf of Kähler differentials and the tangent sheaf) in chapter 5, section 5.5, though more care is required since schemes are not "locally linear" everywhere in the way smooth manifolds are. Much of the intuition from Poincaré duality (see [6, section 4.3]) will inform the generalization to Serre duality.

14. Schemes can better capture different levels of *local* behaviour. For example, a germ of a smooth manifold at a point is not itself a smooth manifold, but the analogous object for a scheme—the spectrum of a local ring—is again a scheme, called a *local scheme*. This is because commutative

rings faithfully encode this local information, and (affine) schemes are the geometric realization of commutative rings.

15. Scheme functions are more "rigid": on a smooth manifold, partitions of unity exist thanks to the flexibility of the Euclidean topology, but no exact analogue holds for schemes. This rigidity, however, also adds structure and sometimes simplifies definitions (e.g. closed subschemes, introduced in section 3.1.3, are a rather "natural" concept compared to the work that goes into verifying that closed submanifolds are locally diffeomorphic to $\mathbb{R}^k \subseteq \mathbb{R}^n$).

    As smooth functions on each $U \subseteq M$ form a sheaf, sheaves that admit a "partition of unity" are called *fine sheaves*. A weaker notion of *soft sheaf* will be introduced in section 6.1; this is the closest scheme-theoretic analogue of the partition-of-unity property.

16. Schemes unify "arithmetic" and "geometry" in a way that manifold theory is not equipped for. As examples, there are no manifold analogues of $\operatorname{Spec}\mathbb{Z}$, $\operatorname{Spec}\mathcal{O}_K$ (where $\mathcal{O}_K$ is a ring of integers of a number field), or $\operatorname{Spec}\mathbb{Z}[t]$. In this way, scheme theory gives geometric meaning to objects that are not tractable in differential geometry. A striking consequence is that different points of a single scheme can have residue fields of different characteristics (e.g. on Spec $\mathbb{Z}$, the residue field at the prime (p) is $\mathbb{F}_p$ while the residue field at the generic point is $\mathbb{Q}$)—a phenomenon with no manifold analogue. To work in a more "purely geometric" setting, one often restricts to $\overline{k}$-schemes where $\overline{k}$ denotes the algebraic closure of a field of characteristic 0. In fact, when defining Z-valued points in section 3.3.2 for a scheme Z, if $Z = \operatorname{Spec}\overline{k}$ then such points are called *geometric points*. Working over a non-algebraically closed field often signals that the scheme carries arithmetic information.

17. Schemes have non-closed ("generic") points. Such points are essential: for instance, the ring map $k[x, y] \to k(x)[y]$ (localizing away from the irreducible polynomials in x) induces a morphism Spec $k(x)[y] \to$ Spec $k[x, y]$ whose image consists of points lying over the generic point of Spec $k[x]$. We shall later see that the image of such a morphism is *dense*.

18. Though the above list highlights many differences, many of the usual geometric tools remain applicable in the study of schemes. For example, $\mathbb{C}[x, y]/(x^2 - y^3)$ is not a normal ring. This can be hard to verify algebraically, but viewing Spec $\mathbb{C}[x, y]/(x^2 - y^3)$ as a curve in $\mathbb{C}^2$ (in the Euclidean topology), the fact that it has a cusp at the origin—and hence is singular there, at a codimension-0 point of the curve—is precisely why it fails to be normal. (Normality for varieties is equivalent to regularity in codimension 1, also known as Serre's condition R_1 together with S_2.) This type of geometric reasoning can be powerful: one can study the smoothness (or more specifically the normality) of a variety to determine the integral closure of its coordinate ring.

Though this is a long list consisting mostly of differences, very often in practice (especially in the setting of k-schemes, or when studying curves, surfaces, moduli spaces, etc.) we work with schemes that are proper (the correct generalization of compact manifolds) and have many methods to "resolve" singularities (e.g. eliminate nodes, cusps, etc.) and make the resulting scheme regular[19]. In fact, as we shall see in section 6.9, a result known as *GAGA* (Géométrie Algébrique et Géométrie Analytique) shows that proper schemes over $\mathbb{C}$ with appropriate coherent sheaves are equivalent, as categories, to compact complex analytic spaces with their coherent analytic sheaves. Naturally, arithmetic schemes will not be part of this manifold-scheme correspondence.

[19]Though as we shall see, there is no universal resolution of singularities in all characteristics and dimensions. Resolution has been completely proved in characteristic 0 (Hironaka) and in low dimensions, but in positive characteristic the general case remains open. Moreover, the singularity information itself is sometimes important and should not always be discarded.

2.2.5 Evaluation and Stalks on Schemes

In differential geometry, the stalks of a manifold can be used define the evaluation of a function. This same trick is endeavored for scheme:

Proposition 2.2.21: Stalks of Schemes

Let $\operatorname{Spec}(R)$ be a scheme. Then the stalk at $\mathcal{O}_{\operatorname{Spec}(R)}$ at the point $[\mathfrak{p}]$ is isomorphic to the local ring $R_{\mathfrak{p}}$.

$$\mathcal{O}_{\operatorname{Spec}(R),\mathfrak{p}} \cong R_{\mathfrak{p}}$$

Furthermore, for any $[\mathfrak{p}] \in X$, as $[\mathfrak{p}] \in \operatorname{Spec}(R_i)$ for appropriate R_i, we have

$$\mathcal{O}_{X,[\mathfrak{p}]} \cong \mathcal{O}_{\operatorname{Spec}(R_i),[\mathfrak{p}]}$$

***Proof*:**
For the first part, prove that $R_{\mathfrak{p}}$ is the colimit of R_f where $f \notin \mathfrak{p}$. The second part comes from proposition 1.1.5.

Definition 2.2.22: Local Scheme

Let X be a scheme. Then X is a *local scheme* if it has a unique closed points.

The scheme $(\operatorname{Spec} \mathcal{O}_{\operatorname{Spec} R,\mathfrak{p}}, \mathcal{O}_{\operatorname{Spec} R,\mathfrak{p}})$ is called a *local scheme* at the point $\mathfrak{p}$.

The reader should show that the local scheme at the point $\mathfrak{p}$ is the intersection of all open subsets containing $\mathfrak{p}$. Hence, local schemes are the germs of schemes. Note that as there are "points within points", namely as there are non-closed points in schemes, then there is also stalks that are subsets of stalks, in particular if $\mathfrak{p}$ and $\mathfrak{m}$ is a prime and maximal ideal respectively where $\mathfrak{p} \subsetneq \mathfrak{m} \in \operatorname{Spec} R$ so that $[\mathfrak{m}] \in \overline{[\mathfrak{p}]}$, then $\mathcal{O}_{X,\mathfrak{p}} \cong R_{\mathfrak{p}}$ is a localization of $\mathcal{O}_{X,\mathfrak{m}} \cong R_{\mathfrak{m}}$. Due to this, notice that the local schemes in higher dimensions (section 2.6) can have local schemes contained within them. For example, the scheme $\operatorname{Spec} k[x,y]_{(x,y)}$ is the local scheme at the origin, it contain a single closed and a non-closed point for every irreducible curve in $\mathbb{A}^2_k$. Notice too how the local schemes are still schemes, as compared to the collection of germs at a point along with the point is not considered to be a manifold.[20]

The main idea is that the single maximal ideal represents that there is one point at which we can evaluate for each germ. As all local rings have a maximal ideal, this motivates remembering the following notion form commutative algebra:

Definition 2.2.23: Residue Field

Let R be a commutative ring and $\mathfrak{p} \subseteq R$ a prime ideal. Then the ring of fractions of the integral domain domain $R/\mathfrak{p}$ is called the *residue field*, i.e. $\operatorname{Frac}(R/\mathfrak{p})$ is the residue field of $\mathfrak{p}$ (with respect to R). This field is often denoted $\kappa(\mathfrak{p})$.

[20] In fact, germs not being manifolds can be seen as a lacking property in the category of manifolds as this shows manifolds are not closed under filtered colimits. Synthetic differential geometry is an attempt to rectify this.

The use of kappa in $\kappa(\mathfrak{p})$ is due to the fact that there can be many different residue fields at different points. For example in $\operatorname{Spec}\mathbb{R}[x]$, all residue fields are isomorphic to either $\mathbb{R}$ or $\mathbb{C}$ (depending if the localization is being taken at a non-maximal or maximal prime ideal).

Using residue fields, we can generalize the evaluation of a regular function (i.e. the evaluation of any element in the sections of $\mathcal{O}_X$). Recall that:

$$R_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}} \cong \operatorname{Frac}(R/\mathfrak{p})$$

as localization and taking the quotient commutes (see [11, section 20.5]). This makes it much easier to evaluate a function if we can determine R. Then as the valuation of a function is usually in $R/\mathfrak{p}$, as rational functions are now permitted on (proper) open subsets of X, the evaluation will be in $\operatorname{Frac}(R/\mathfrak{p})$. For a general point $x \in X$ in a scheme, we may diminish to the to define evaluation.

Definition 2.2.24: Evaluation of Function [on a Scheme]

Let X be a scheme, $U \subseteq X$ an open subset, and $f \in \mathcal{O}_X(U)$ a section. Then the value of f at the point $[\mathfrak{p}] \in U$ is the element:

$$f(\mathfrak{p}) := \bar{f} \in \mathcal{O}_{X,\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}} \cong \operatorname{Frac}(R/\mathfrak{p}) = \kappa(\mathfrak{p})$$

where $\operatorname{Spec}(R)$ is some affine local neighborhood of $\mathfrak{p}$.

As the quotient of any ring by a prime ideal is an integral domain, by taking its field of fraction we are forcing the codomain of each function to be a disjoint union of fields:

$$f : U \to \coprod_{\mathfrak{p}} \mathcal{O}_{U,\mathfrak{p}} \quad (\text{ and in affine case }) \quad f : U \to \coprod_{\mathfrak{p}\in U} \operatorname{Frac}(R/\mathfrak{p})$$

If $\mathfrak{p}$ is maximal, then $\operatorname{Frac}(R/\mathfrak{p}) = R/\mathfrak{m}$. For certain rings, it suffices to evaluate sections only over maximal points (namely Jacobson Schemes, see corollary 3.7.6). In the case of affine Jacobson schemes, we can consider:

$$f : U \to \coprod_{\mathfrak{p}\in U} R/\mathfrak{m}$$

Varieties will be an example of Jacobson schemes. If $R/\mathfrak{m}_i \cong R/\mathfrak{m}_j$ for all maximal ideals (which shall be shown to be the case for varieties), then we can simply write:

$$f : U \to R/\mathfrak{m} \cong k$$

which recovers the notion of evaluation from classical algebraic geometry!

2.2.25 Foreshadowing: What Type of Function? As f is defined above, it is a set function between a relatively nice space U and a bad codomain. Can this function be characterized, or represented, by a known nicer function? In fact, it shall be representable with relatively simple scheme morphisms, recovering how in classical algebraic geometry sections are regular functions; details in proposition 3.3.8.

Example 2.2.26: Evaluation On Affine Schemes
The key idea is that the stalk $R_\mathfrak{p}$ gives all function's that will not vanish at $\mathfrak{p}$, and hence can divide by the function at that point, and then taking the quotient by the maximal ideal evaluate the function at that point, equivalently considering the value of the rational function in $\mathrm{Frac}(R/\mathfrak{p})$:

1. Take $X = \mathbb{C}[x]$. Then each $f/g \in \mathcal{O}_X(U)$ is a rational polynomial where $g(\mathfrak{p}) \neq 0$ for all $\mathfrak{p} \in U$. For example if $U = D(x)$, then $h = \frac{x+1}{x} \in \mathcal{O}_X(D(x))$ and:

$$h(3) = \frac{3+1}{3} = \frac{4}{3} \in \mathbb{C} = \frac{\mathbb{C}[x]}{(x-3)} \cong \mathcal{O}_{X,3}/(x-3)$$
$$h(-1) = \frac{-1+1}{-1} = 0 \in \mathbb{C} = \frac{\mathbb{C}[x]}{(x+1)} \cong \mathcal{O}_{X,-1}(x+1)$$

   As the points of $\mathbb{C}[x]$ correspond to the variety, we see that the evaluation will always be in $\mathbb{C}$, and there is no need to take the field of fractions as $\mathrm{Frac}(\mathbb{C}) = \mathbb{C}$.
   Another important example, take (want the vanishing at 0 , but something feels off)

2. Take $\mathbb{A}^2_k$ for a field k where char $\neq 2$. Then in the open set U away from the y-axis and the curve $y^2 - x^5$ has in it's local section $\mathcal{O}_{\mathbb{A}^2_k}(U)$ the rational function $\left(x^2+y^2\right)/x\left(y^2-x^5\right) \in \mathcal{O}_{\mathbb{A}^2_k}(U)$. It's value at the point $(2,4)$ (i.e. $[(x-2, y-4)]$) give a values in $\mathrm{Frac}(R/(x-2, y-4)) = \mathrm{Frac}(k[x,y]/(x-2,y-4)) \cong R_{(x-2,y-4)}/\overline{(x-2,y-4)}$. Then:

$$\frac{x^2+y^2}{x\left(y^2-x^5\right)} \bmod (x-2, y-4) = \frac{2^2+4^2}{2\left(4^2-2^5\right)} \in \mathbb{C} = \frac{\mathbb{C}[x,y]}{(x-2,y-4)} \cong \mathcal{O}_{\mathbb{A}^2_k,(2,4)}$$

   The value at the point $[(y)]$, which is the generic point of the x axis, is

$$\frac{x^2+y^2}{x\left(y^2-x^5\right)} \bmod (y) = \frac{x^2}{-x^6} = -1\frac{1}{x^4} \in \mathbb{C}(x) \cong \mathrm{Frac}(\mathbb{C}[x,y]/(y)) \cong \mathcal{O}_{\mathbb{A}^2_k,[(y)]}$$

3. Take $X = \mathrm{Spec}(\mathbb{Z})$ and take the open set $U = \mathrm{Spec}(\mathbb{Z})\backslash\{(2),(7)\}$. This open set has many rational function, but let's take $27/4$. Then evaluating this function at different points of U we get:

$$(5) : 2/(-1) \equiv -2 \equiv 3 \bmod 5$$
$$(0) : 27/4$$
$$(3) : 0/1 = 0 \bmod 3$$

Notice that at 3, the function became 0. A (rational) function whose output is 0 is said to *vanish* at that point. If $\mathfrak{m}_\mathfrak{p}$ represents the maximal ideal of $\mathcal{O}_{X,\mathfrak{p}}$, then functions on an open subset U (i.e. section in $\mathscr{F}(U)$) have values at each point of U whose value lies in $\kappa(\mathfrak{p})$. A function *vanishes* at $\mathfrak{p}$ if its value at $\mathfrak{p}$ is 0. This idea is not well-defined on ringed spaces in general! There is furthermore the following properties mirroring manifold theory:

Proposition 2.2.27: Properties of Functions on Locally Ringed Spaces

Let f be function on a locally ringed space X. Then:

1. The subset of X where f vanishes is closed
2. if f vanishes nowhere, then f is invertible

Note that the property "if f vanishes everywhere, then $f = 0$" is not one of the properties. This will be addressed in proposition 2.4 .10 .

***Proof*:**

1. the subset of X where the germ of f is invertible is open
2. If $f \not\equiv 0 \bmod \mathfrak{p}$, then in particular $f \not\equiv 0 \bmod \mathfrak{m}$ for all maximal ideal, meaning f is contained in no maximal ideal. But then f must be a unit. If f where not a unit, then there would exist an $\mathfrak{m}$ that f belongs to, namely the maximal ideal containing f. But only units are not contained in any maximal ideal (as the ring is non-trivial)

Note how this is is *not* true on a variety defined in the classical sense:

Example 2.2.28: On Variety: Vanishes Nowhere, Not Invertible

Take

$$\overline{x - 100} \in \frac{\mathbb{R}[x, y]}{(x^2 + y^2 - 1)}$$

The reader should check that this function vanishes nowhere on $V(x^2 + y^2 - 1)$ as a variety, but it is *not* in invertible (i.e it is not a unit). However, $\overline{x - 100}$ *does* vanish on $\operatorname{Spec} \mathbb{R}[x, y]/(x^2 + y^2 - 1)$; can the reader find what prime ideal it vanishes on?

This ties in with the intuition that the spectrum of $\mathbb{R}[x, y]/(x^2 + y^2 - 1)$ would be some "gluing" of $\mathbb{C}[x, y]/(x^2 + y^2 - 1)$ via galois conjugate.

Finally, a moment should be taken to interpret evaluating at generic points. Let us consider $X = \mathbb{A}_k^1$ and let η be the generic point. Then looking at the fiber at η, we see that:

$$\mathcal{O}_{\mathbb{A}_k^1, \eta} = (k[x])_{(0)} = k(x)$$

that is, everything is inverted. The earlier intuition about $R_{\mathfrak{p}}$ is that we only evaluate functions *near* the point $\mathfrak{p}$ and every other $\mathfrak{q} \in \operatorname{Spec} R$ can be inverted, so this means that η is near *no* points as we can invert every point! This contrasts with the intuition that $\overline{\eta} = \mathbb{A}_k^1$ which seems to suggest that it is *close* to every point. This is where the nomenclature *generic* can be of use: it doesn't "see" what global sections capture, and can be thought of as "ignoring" any constraints given by points. This also explains the evaluation at the generic point, it returns the function itself motivating how it is not "particularly" extracting any information.

2.2.6 Independence of Affine Covering

A scheme X can be covered by affine open subsets in many different ways. Unlike the situation for smooth manifolds in low dimensions—where Moise's theorem [20] guarantees a unique smooth structure on topological manifolds of dimension ≤ 3—schemes admit no such rigidity. Even in dimension zero, the topological space underlying a field is always a single closed point, yet different fields give non-isomorphic affine schemes. It is therefore essential to know when a property of a scheme can be checked on *any* affine covering, rather than depending on a particular choice.

The tool for this is the *Affine Communication Lemma*. It identifies a class of properties—those that are stable under localization and can be recovered from a generating cover—for which checking on a single affine cover suffices. Such properties are called *affine-local*, and they include most of the "geometric" properties of schemes (reducedness, Noetherianness, integrality, normality, etc.). Properties that depend on "arithmetic" or global information—such as the number of connected components—are typically not affine-local.

The key preliminary observation is that the intersection of two affine opens, while generally not affine, can always be covered by open sets that are simultaneously distinguished in both.

Lemma 2.2.29: Build-up Lemma

Let $\operatorname{Spec}(R)$ and $\operatorname{Spec}(S)$ be affine open subschemes of a scheme X. Then $\operatorname{Spec}(R) \cap \operatorname{Spec}(S)$ can be covered by open sets that are simultaneously distinguished open in $\operatorname{Spec}(R)$ and distinguished open in $\operatorname{Spec}(S)$.

***Proof*:**
Let $[\mathfrak{p}] \in \operatorname{Spec}(R) \cap \operatorname{Spec}(S)$. Since $\operatorname{Spec}(R) \cap \operatorname{Spec}(S)$ is open in $\operatorname{Spec}(R)$, and the distinguished open sets form a basis for the Zariski topology (section 2.1.2), there exists $f \in R$ with $[\mathfrak{p}] \in \operatorname{Spec}(R_f) \subseteq \operatorname{Spec}(R) \cap \operatorname{Spec}(S)$. Now, as

$$\operatorname{Spec}(R_f) \subseteq \operatorname{Spec}(R) \cap \operatorname{Spec}(S) \subseteq \operatorname{Spec}(S)$$

$\operatorname{Spec}(R_f)$ is an open subset of $\operatorname{Spec}(S)$. Thus, by construction of the topology on $\operatorname{Spec}(S)$ using distinguished opens (given by S), there exists $g \in S$ with

$$[\mathfrak{p}] \in \operatorname{Spec}(S_g) \subseteq \operatorname{Spec}(R_f) \cap \operatorname{Spec}(S) \subseteq \operatorname{Spec}(R) \cap \operatorname{Spec}(S).$$

In particular, $\operatorname{Spec}(S_g) \subseteq \operatorname{Spec}(R_f)$.

Now, $g \in \mathcal{O}_X(\operatorname{Spec}(S))$ restricts to an element $g' \in \mathcal{O}_X(\operatorname{Spec}(R_f)) = R_f$, and the locus in $\operatorname{Spec}(R_f)$ where g is nonvanishing is exactly $\operatorname{Spec}(S_g)$, so:

$$\operatorname{Spec}(S_g) = \operatorname{Spec}((R_f)_{g'}).$$

By definition of R_f, write $g' = h/f^n$ for some $n \geq 0$ and $h \in R$. By Proposition 2.1.15, a distinguished open of a distinguished open is distinguished:

$$(R_f)_{g'} = (R_f)_{h/f^n} = R_{fh}$$

where the last equality holds since f is already invertible in R_f. Hence:

$$\operatorname{Spec}(S_g) = \operatorname{Spec}(R_{fh}) = D(fh) \subseteq \operatorname{Spec}(R).$$

This is a distinguished open in $\operatorname{Spec}(R)$, and it is simultaneously equal to $\operatorname{Spec}(S_g) = D(g) \subseteq \operatorname{Spec}(S)$, a distinguished open in $\operatorname{Spec}(S)$. Since $[\mathfrak{p}]$ was arbitrary, these sets cover $\operatorname{Spec}(R) \cap \operatorname{Spec}(S)$.

With this in hand, we can state and prove the main tool.

Theorem 2.2.30: Affine Communication Lemma

Let X be a scheme and let Q be a property enjoyed by some affine open subschemes of X. Suppose Q satisfies:

1. *(Localization)* If $\operatorname{Spec}(A) \subseteq X$ satisfies Q and $f \in A$, then $\operatorname{Spec}(A_f)$ satisfies Q.
2. *(Gluing)* If $f_1, \ldots, f_n \in A$ generate the unit ideal $(f_1, \ldots, f_n) = A$ and each $\operatorname{Spec}(A_{f_i})$ satisfies Q, then $\operatorname{Spec}(A)$ satisfies Q.

If there exists an affine open cover $X = \bigcup_i \operatorname{Spec}(R^{(i)})$ such that each $\operatorname{Spec}(R^{(i)})$ satisfies Q, then *every* affine open subscheme of X satisfies Q.

***Proof*:**
Let $\operatorname{Spec}(A) \subseteq X$ be an arbitrary affine open subscheme. We must show $\operatorname{Spec}(A)$ satisfies Q. For each i, the intersection $\operatorname{Spec}(A) \cap \operatorname{Spec}(R^{(i)})$ is an open subset of both $\operatorname{Spec}(A)$ and $\operatorname{Spec}(R^{(i)})$. By the Build-up Lemma (lemma 2.2.29), this intersection can be covered by open sets that are simultaneously distinguished in $\operatorname{Spec}(A)$ and distinguished in $\operatorname{Spec}(R^{(i)})$.

Since $\operatorname{Spec}(R^{(i)})$ satisfies Q, condition (1) implies that each of these distinguished opens (viewed as localizations of $R^{(i)}$) also satisfies Q. But these same opens are also distinguished opens of $\operatorname{Spec}(A)$, say of the form $\operatorname{Spec}(A_{g_j})$ for various $g_j \in A$.

Now, $\operatorname{Spec}(A)$ is quasi-compact, and the union $\bigcup_i (\operatorname{Spec}(A) \cap \operatorname{Spec}(R^{(i)})) = \operatorname{Spec}(A)$ covers it. Extracting a finite subcover from among the $\operatorname{Spec}(A_{g_j})$, we obtain elements $g_1, \ldots, g_m \in A$ such that:

(a) each $\operatorname{Spec}(A_{g_j})$ satisfies Q (inherited from the $R^{(i)}$ via localization), and

(b) $\operatorname{Spec}(A) = \bigcup_{j=1}^m \operatorname{Spec}(A_{g_j})$, which means $g_1, \ldots, g_m$ generate the unit ideal in A.

By condition (2), $\operatorname{Spec}(A)$ satisfies Q.

The power of the lemma is that once the two conditions (localization and gluing) are verified for a property Q—which is typically a purely algebraic statement about rings—the conclusion is that Q holds for *every* affine open, not just those in the chosen cover. This means the property can be attributed to the scheme X itself, independent of the presentation.

The Affine Communication Lemma gives a uniform method for proving that properties of schemes are well-defined. We collect some important instances;

Example 2.2.31: Affine-local Properties

1. *Reducedness*: Let Q be the property "A is a reduced ring" (i.e., A has no nonzero nilpotents). We verify the two conditions:

   (a) *Localization:* If A is reduced, then A_f is reduced. Indeed, if $(a/f^n)^m = 0$ in A_f, then $f^k a^m = 0$ in A for some k, hence $(f^j a)^m = 0$ for j large enough, so $f^j a = 0$, hence $a/f^n = 0$ in A_f.

(b) *Gluing:* If $(f_1, \dots, f_n) = A$ and each A_{f_i} is reduced, then A is reduced. Indeed, if $a^m = 0$ in A, then $a^m = 0$ in each A_{f_i}, so $a = 0$ in each A_{f_i} (as A_{f_i} is reduced), meaning $f_i^{k_i} a = 0$ in A for some k_i. Since the f_i generate the unit ideal, so do the $f_i^{k_i}$ (see section 2.1.2), hence $a = 0$.

By the Affine Communication Lemma, a scheme X is reduced if and only if some (equivalently, every) affine open cover consists of reduced rings. This shall be the foundation for definition given in section 2.4.1.

2. *Noetherian*: Let Q be the property "A is a Noetherian ring."

(a) *Localization:* If A is Noetherian, then A_f is Noetherian (a standard result in commutative algebra: every ideal of A_f is an extension of an ideal of A, and extensions of finitely generated ideals are finitely generated).

(b) *Gluing:* Suppose $(f_1, \dots, f_n) = A$ and each A_{f_i} is Noetherian. Let $I \subseteq A$ be an ideal. Each extension $I \cdot A_{f_i}$ is finitely generated, say by elements $a_{i,1}/f_i^{k_1}, \dots, a_{i,r_i}/f_i^{k_{r_i}}$. Clearing denominators, we obtain finitely many elements of I whose images generate $I \cdot A_{f_i}$ for each i. Since the f_i generate the unit ideal, these elements generate I itself.

A scheme is *locally Noetherian* if it admits an affine cover by spectra of Noetherian rings. By the Affine Communication Lemma, this is equivalent to every affine open being the spectrum of a Noetherian ring.

3. *Normality*: Let Q be the property "A is a normal ring" (integrally closed in its total ring of fractions). This can be checked via the Affine Communication Lemma: if A is normal then A_f is normal, and normality can be recovered from a generating cover. Thus, a scheme is *normal* if and only if every affine open is the spectrum of a normal ring.

Local Properties and the Affine Communication Lemma

As more properties of schemes are defined, we must be precise about what it means for a scheme to "have" a property and how to check for it. There at least two natural levels at which a property can be detected, and they form a hierarchy.

Definition 2.2.32: Property of Schemes

A *property* P of schemes is a class of schemes closed under isomorphism: if X satisfies P and $Y \cong X$, then Y satisfies P.

Definition 2.2.33: Affine-local and Stalk-local Properties

Let P be a property of schemes.

1. P is *affine-local* if for any scheme X and any affine open cover $X = \bigcup U_i$, the scheme X satisfies P if and only if every U_i satisfies P.

2. P is *Zariski-local* if for any scheme X and any (Zariski) open cover $X = \bigcup U_i$, the scheme X satisfies P if and only if every U_i satisfies P.

3. P is *stalk-local* if for any scheme X, X satisfies P if and only if for every point $x \in X$, the affine scheme $\operatorname{Spec} \mathcal{O}_{X,x}$ satisfies P.

The Affine Communication Lemma provides a practical test for affine-locality: it suffices to verify the localization and gluing conditions for the corresponding ring property. As sheaf maps can always be verified stalk-locally, stalk-locality is another natural condition to consider. We shall see that morphism properties are not always verified affine-locally; those that are give rise to *affine morphisms* (see definition 3.4.5).

Note that $\operatorname{Spec} \mathcal{O}_{X,x}$ *need not* be open in X, hence $\{\operatorname{Spec} \mathcal{O}_{X,x}\}_{x \in X}$ need not be an affine open cover. For example $X = \mathbb{A}^1_{\mathbb{C}}$ and $\mathcal{O}_{X,2} \cong \operatorname{Spec} k[x]_{(x-2)}$ so that $\operatorname{Spec} \mathcal{O}_{X,2}$ has only 2 points but any open subscheme of X has cofinitely many points plus the generic point. Thus, a stalk-local is not the condition that it suffices to check the specific "stalk-cover".

Zariski-local properties are another natural choice, but as any scheme has a topological basis of distinguished open sets, these are identical to affine-locals:

Proposition 2.2.34: Affine-local Iff Zariski-local

Let P be a property of schemes. Then P is affine-local if and only if it is Zariski-local.

***Proof*:**
exercise.

It is natural to ask whether stalk-locality is a stronger or weaker condition than affine-locality. The answer is that it is strictly stronger.

Proposition 2.2.35: Stalk-local implies Affine-local

If a property of a scheme is stalk-local, then it is affine-local.

***Proof*:**
Let P be stalk-local and let $X = \bigcup U_i$ be an affine open cover.

($\Rightarrow$) Suppose X satisfies P. For any $x \in U_i$, the stalk $\mathcal{O}_{U_i,x} \cong \mathcal{O}_{X,x}$ by proposition 1.1.5, and $\operatorname{Spec} \mathcal{O}_{X,x}$ satisfies property P since X satisfies P. As this holds for all $x \in U_i$, the scheme U_i satisfies P by stalk-locality.

($\Leftarrow$) Suppose every U_i satisfies P. For any $x \in X$, choose some U_i containing x. Then $\mathcal{O}_{X,x} \cong \mathcal{O}_{U_i,x}$ satisfies the local-ring property since U_i satisfies P. As x was arbitrary, X satisfies P.

Common examples are:

1. "X is reduced" is stalk-local with respect to "$\mathcal{O}_{X,x}$ has no nonzero nilpotents"
2. "X is regular" is stalk-local with respect to "$\mathcal{O}_{X,x}$ is a regular local ring"
3. "X is normal" is stalk-local with respect to "$\mathcal{O}_{X,x}$ is an integrally closed domain."

The converse is false: there exist affine-local properties that cannot be detected at the level of stalks. The issue is that stalks capture "infinitely close" behavior but lose information about finiteness conditions that involve the global structure of a neighborhood.

Example 2.2.36: Affine-local Does Not Imply Stalk-local

Consider the property "locally Noetherian." As shown in example 2.2.31, this is affine-local. We show it is not stalk-local.

Let $R = \prod_{i=1}^{\infty} \mathbb{F}_2$, the countably infinite product of copies of $\mathbb{F}_2$. The ring R is not Noetherian: the ideals $I_n = \{(a_1, a_2, \ldots) : a_1 = \cdots = a_n = 0\}$ form a strictly ascending chain. Thus $\mathrm{Spec}\,(R)$ is not locally Noetherian.

However, every stalk of $\mathcal{O}_{\mathrm{Spec}\,(R)}$ is Noetherian. For any prime ideal $\mathfrak{p} \subseteq R$, the localization $R_{\mathfrak{p}}$ is isomorphic to $\mathbb{F}_2$. To see this, recall that R is a Boolean ring ($a^2 = a$ for all a), so every prime ideal is maximal with residue field $\mathbb{F}_2$. Since R is zero-dimensional and Boolean, the localization at any prime is a field.

If "locally Noetherian" were stalk-local, the fact that all stalks are Noetherian fields would force R to be Noetherian—a contradiction. Thus stalk-locality is strictly stronger than affine-locality.

The upshot is a hierarchy of "locality" conditions for scheme properties:

$$\begin{tikzcd}
 & & \text{closed under open restrictions} \\
\text{stalk-local} \arrow[r, Rightarrow] & \text{affine-local} \arrow[ur, Rightarrow] \arrow[dr, Rightarrow] & \\
 & & \text{closed under gluing over open covers}
\end{tikzcd}$$

The first implication is strict, as the example above shows. In practice, most "algebraic" properties (reduced, normal, regular, Cohen–Macaulay) are stalk-local, while "finiteness" properties (Noetherian, finite type) are affine-local but not stalk-local.

2.2.37 Advanced Foreshadowing: Other Covers Notice how the collection of maps $\operatorname{Spec}\mathcal{O}_{X,x} \to X$ for all $x \in X$ can still be seen as a cover which is neither open nor closed. Such a cover still has important properties; in this case it fpqc (fidèlement plate et quasi-compacte). We shall not address such covers in the textbook, however the idea of replacing open sets with covering maps with "desirable" properties will be a key intuition in defining the *étale topology* using *étale coverings* in section ref:HERE.
Readers interested in Descent Theory [4] may be interested to know that the above hierarchy is often extended to the following:

$$\text{fpqc-local} \implies \text{fppf-local} \implies \text{étale-local} \implies \text{Zariski-local}$$

where stalk-local becomes a part of fpqc-local.

2.2.7 *Affine Atlases

By definition, a scheme is a locally ringed space that admits some affine open cover. The structure sheaf $\mathcal{O}_X$ is part of the data: it is given, and the affine cover witnesses that it is "locally affine." But there is an equivalent formulation—closer in spirit to the definition of a smooth manifold—in which the structure sheaf is not given a priori, but is *recovered* from an atlas. This perspective clarifies the sense in which a scheme is "independent of coordinates."

Recall that a smooth manifold of dimension n can be defined in two equivalent ways:

1. A topological space M equipped with a sheaf C^∞_M of smooth functions such that (M, C^∞_M) is locally isomorphic to $(\mathbb{R}^n, C^\infty_{\mathbb{R}^n})$.

2. A topological space M equipped with a *maximal smooth atlas*: a maximal collection of charts (U_i, φ_i) where $\varphi_i : U_i \xrightarrow{\sim} V_i \subseteq \mathbb{R}^n$ is a homeomorphism, and the transition maps $\varphi_j \circ \varphi_i^{-1}$ are smooth (see section 0.5)

The two definitions are equivalent because the sheaf C^∞_M can be recovered from the atlas: a function $f : U \to \mathbb{R}$ is smooth if and only if $f \circ \varphi_i^{-1}$ is smooth for every chart (U_i, φ_i) with $U_i \subseteq U$. Conversely, the atlas is recoverable from the sheaf: a chart is precisely a local isomorphism of the sheaf with the standard model.

In the manifold setting, most textbooks adopt definition (2) because the transition maps are concrete and checkable. In scheme theory, definition (1) has historically been preferred because the structure sheaf interacts well with the algebraic machinery of localization, sheaf cohomology, and base change. Nevertheless, the atlas formulation works equally well and is illuminating.

Affine atlases for schemes

Definition 2.2.38: Affine Chart

Let X be a topological space. An *affine chart* on X is a pair (U, φ) where $U \subseteq X$ is open and $\varphi : U \xrightarrow{\sim} |\operatorname{Spec}(R)|$ is a homeomorphism for some commutative ring R. The ring R is called the *coordinate ring* of the chart.

Two charts (U_i, φ_i) with coordinate ring R_i and (U_j, φ_j) with coordinate ring R_j must be compatible on their overlap. In the manifold case, compatibility means the transition map is smooth. For schemes, it means the transition map is "algebraic" in the following sense.

Definition 2.2.39: Compatibility of Charts

Two affine charts (U_i, φ_i) and (U_j, φ_j) on a topological space X are *compatible* if for every point $x \in U_i \cap U_j$, there exist elements $f \in R_i$ and $g \in R_j$ such that:

1. $x \in \varphi_i^{-1}(D(f)) \cap \varphi_j^{-1}(D(g))$,
2. $\varphi_i^{-1}(D(f)) = \varphi_j^{-1}(D(g))$ as subsets of X, and
3. The composite homeomorphism

$$D(f) \xrightarrow{\varphi_i^{-1}} \varphi_i^{-1}(D(f)) \;=\; \varphi_j^{-1}(D(g)) \xrightarrow{\varphi_j} D(g)$$

is induced by a ring isomorphism $(R_j)_g \xrightarrow{\sim} (R_i)_f$.

Condition (3) is the key algebraic content: the transition "map" between two charts must not merely be a homeomorphism of topological spaces, but must arise from an isomorphism of the corresponding localized rings. This is the condition that fails for arbitrary homeomorphisms between spectra—and it is exactly what the structure sheaf encodes in the locally ringed space definition.

Definition 2.2.40: Affine Atlas

An *affine atlas* on a topological space X is a collection $\mathcal{A} = \{(U_i, \varphi_i)\}_{i \in I}$ of pairwise compatible affine charts such that $X = \bigcup_{i \in I} U_i$.

Definition 2.2.41: Maximal Affine Atlas

An affine atlas $\mathcal{A}$ is *maximal* if it contains every affine chart on X that is compatible with all charts already in $\mathcal{A}$. That is, if (V, ψ) is an affine chart compatible with every $(U_i, \varphi_i) \in \mathcal{A}$, then $(V, \psi) \in \mathcal{A}$.

Definition 2.2.42: Scheme via Atlas

A *scheme* is a pair $(X, \mathcal{A})$ where X is a topological space and $\mathcal{A}$ is a maximal affine atlas on X.

This definition makes no reference to sheaves, locally ringed spaces, or stalks. The algebraic data is encoded entirely in the transition isomorphisms between charts: the atlas definition is equivalent to definition 2.2.9. The equivalence rests on two constructions: recovering the structure sheaf from an atlas, and recovering the maximal atlas from a locally ringed space.

Proposition 2.2.43: From Atlas to Structure Sheaf

Let $(X, \mathcal{A})$ be a scheme in the sense of definition 2.2.42. Then there exists a unique sheaf of rings $\mathcal{O}_X$ on X such that $(X, \mathcal{O}_X)$ is a scheme in the sense of definition 2.2.9, and for every chart $(U, \varphi) \in \mathcal{A}$ with coordinate ring R:

$$(U,\ \mathcal{O}_X|_U)\ \cong\ (\operatorname{Spec}(R),\ \mathcal{O}_{\operatorname{Spec}(R)}).$$

***Proof*:**
Define $\mathcal{O}_X$ on the basis of opens $\{U : (U, \varphi) \in \mathcal{A} \text{ for some } \varphi\}$ by setting $\mathcal{O}_X(U) = R$ whenever $\varphi : U \xrightarrow{\sim} \operatorname{Spec}(R)$. For a distinguished open $D(f) \subseteq \operatorname{Spec}(R)$, the chart $(\varphi^{-1}(D(f)),\ \varphi|_{D(f)})$ is in $\mathcal{A}$ (by maximality) with coordinate ring R_f, so we set $\mathcal{O}_X(\varphi^{-1}(D(f))) = R_f$. The restriction maps are the natural localization maps $R \to R_f$.

The compatibility condition on charts ensures that this is well-defined on overlaps: if two charts (U_i, φ_i) and (U_j, φ_j) both contain a point x, the transition isomorphism $(R_j)_g \cong (R_i)_f$ identifies the sections. By the sheaf axiom for the structure sheaf of $\operatorname{Spec}(R)$, this basis assignment extends uniquely to a sheaf on all of X.

The stalks of $\mathcal{O}_X$ are local rings: at a point $x \in U_i$, the stalk $\mathcal{O}_{X,x}$ is the localization $(R_i)_{\mathfrak{p}}$ where $\mathfrak{p} = \varphi_i(x)$, which is local. Hence $(X, \mathcal{O}_X)$ is a locally ringed space, and each chart gives an affine open, so it is a scheme.

Proposition 2.2.44: From Locally Ringed Space to Maximal Atlas

Let $(X, \mathcal{O}_X)$ be a scheme in the sense of definition 2.2.9. Then the collection

$$\mathcal{A}_{\max} = \left\{ (U, \varphi) : \begin{array}{l} U \subseteq X \text{ is open, } \varphi : (U, \mathcal{O}_X|_U) \xrightarrow{\sim} (\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)}) \\ \text{is an isomorphism of locally ringed spaces, for some } R \end{array} \right\}$$

is a maximal affine atlas on X. The atlas $\mathcal{A}_{\max}$ depends only on $(X, \mathcal{O}_X)$, not on any particular affine open cover.

***Proof*:**
The charts in $\mathcal{A}_{\max}$ are pairwise compatible: if (U_i, φ_i) and (U_j, φ_j) are in $\mathcal{A}_{\max}$, then on $U_i \cap U_j$ the isomorphisms φ_i and φ_j are both isomorphisms of locally ringed spaces, and the transition map is an isomorphism of the appropriate localizations (this is the content of the Build-up Lemma, lemma 2.2.29). The atlas covers X because the definition of a scheme guarantees the existence of at least one affine cover. It is maximal by construction: every affine open is included.

These two constructions are inverse to each other: starting from an atlas, building the sheaf, and taking the maximal atlas of the resulting scheme recovers the original atlas; starting from a scheme, taking $\mathcal{A}_{\max}$, and building the sheaf from it recovers the original structure sheaf.

Just as for manifolds, the practical consequence is that one need not specify a maximal atlas. Any atlas suffices.

Proposition 2.2.45: Uniqueness of Maximal Atlas

Let $\mathcal{A}$ be any affine atlas on a topological space X. Then there exists a unique maximal affine atlas $\mathcal{A}_{\max}$ containing $\mathcal{A}$. Explicitly,

$$\mathcal{A}_{\max} = \{(V,\psi) : (V,\psi) \text{ is compatible with every } (U,\varphi) \in \mathcal{A}\}.$$

***Proof*:**
One must verify that any two charts in $\mathcal{A}_{\max}$ are compatible with each other (not just with the charts in $\mathcal{A}$). Let (V_1,ψ_1) and (V_2,ψ_2) be in $\mathcal{A}_{\max}$, and let $x \in V_1 \cap V_2$. Choose a chart $(U,\varphi) \in \mathcal{A}$ with $x \in U$. By compatibility of (V_1,ψ_1) with (U,φ), there exist distinguished opens around x where the transition is an isomorphism of localizations. By compatibility of (V_2,ψ_2) with (U,φ), the same holds. Composing (and using that distinguished opens of distinguished opens are distinguished, by proposition 2.1.15), we obtain a transition isomorphism between localizations of V_1 and V_2.

Uniqueness is immediate: any maximal atlas containing $\mathcal{A}$ must contain every compatible chart, hence must equal $\mathcal{A}_{\max}$.

The upshot is that defining a scheme requires only specifying a topological space together with *any* affine atlas—the maximal atlas is then uniquely determined. The Affine Communication Lemma (theorem 2.2.30) is the tool that ensures properties of the scheme can be checked on the given atlas rather than on the full maximal atlas: its localization and gluing conditions propagate a property from any covering atlas to all affine opens.

The atlas viewpoint also gives an explicit construction of schemes from purely algebraic data, analogous to constructing a manifold by specifying charts and transition maps. Suppose we are given:

1. Rings $R_1, \dots, R_n$.
2. For each pair (i,j), elements $f_{ij} \in R_i$ specifying open subsets $U_{ij} = D(f_{ij}) \subseteq \operatorname{Spec}(R_i)$.
3. Ring isomorphisms $\varphi_{ij} : (R_i)_{f_{ij}} \xrightarrow{\sim} (R_j)_{f_{ji}}$.

Subject to the *cocycle condition*: for each triple (i,j,k), the composition $\varphi_{jk} \circ \varphi_{ij}$ agrees with φ_{ik} on the appropriate localizations.

These data determine a unique scheme X (up to unique isomorphism) by the gluing construction of section 2.2.9: the underlying set is the disjoint union $\coprod_i |\operatorname{Spec}(R_i)|$ modulo the identifications given by the φ_{ij}, the topology is the quotient topology, and the affine atlas consists of the images of the $\operatorname{Spec}(R_i)$. Conversely, every scheme with a given finite affine cover arises in this way: the rings are the coordinate rings of the charts, and the transition isomorphisms are read off from the structure sheaf on overlaps.

2.2.8 Subschemes: Open

We seek the natural notions of subschemes induced by the parent scheme. As a scheme has a topology, a natural starting point is to look for open or closed subsets with a sheaf naturally induced by the parent scheme[21]. To define the topology of open and closed subschemes is easy. The difficult part is to insure the subscheme has a sheaf structure that is in a natural way compatible with the parent scheme. This shall require a deeper discussion on monomorphic scheme morphisms as the sheaf structure of subschemes is "non-trivial", and hence this discussion is mostly delayed till section 3.1.3. There is one type of subscheme that shall be used in this chapter and requires no use of scheme morphism since it is directly compatible with sheaves: *open subschemes*.

Proposition 2.2.46: Open Subscheme

Let X be a scheme and $U \subseteq X$ an open subset. Then $(U, \mathcal{O}_X|_U)$ is a scheme.

***Proof*:**
U is certainly a subspace and a subsheaf with local rings as stalks. To show it is locally affine, for any $x \in U$ choose affine open $\operatorname{Spec} R \cong V \subseteq X$ where $x \in U \cap V$. As $D(f)$ forms a (topological) basis for $\operatorname{Spec} R$, there exists an f_0 such that $x \in D(f_0) \subseteq U \cap V \subseteq U$, showing that U is covered in affine, completing the proof.

Definition 2.2.47: Open Subscheme

Let X be a scheme. Then $(U, \mathcal{O}_X|_U)$ is called an *open subscheme*. If U is also an affine scheme, then U is often said to be an *affine open subscheme* or *affine open subset*.

If a scheme is isomorphic to an open subscheme, we shall say that the isomorphism map (onto the image) is an open embedding:

Definition 2.2.48: Open Embedding

A map of schemes $\pi : X \to Y$ is an open embedding if:

$$(X, \mathcal{O}_X) \xrightarrow{\sim} (U, \mathcal{O}_Y|_U) \hookrightarrow (Y, \mathcal{O}_Y)$$

2.2.49 Remark about Hartshorne's Definition of Open Subscheme Note that in Hartshorne, an open subscheme was technically not defined as $(U, \mathcal{O}_X|_U)$ but as $(U, [\mathcal{O}_X|_U])$ where $[\mathcal{O}_X|_U]$ is the equivalence class of all isomorphic sheaves. This is in fact not correct: isomorphic sheaves shall give technically different open subschemes (which are of course isomorphic). Hence, Hartshorne claims that an open subscheme has a unique subscheme structure, not unique up to isomorphism.

Corollary 2.2.50: Subspace and Zariski Topology On Open Subscheme

Let $U = \operatorname{Spec}(R) \subseteq X$ be an affine open subset and take $(U, \mathcal{O}_X|_U)$. Then the Zariski topology on U is the subspace topology on U with respect to X

[21] Categorically, a natural starting point is to look for subobjects, but surprisingly these are not so useful in **Sch**; see section 3.1.3

***Proof*:**
exercise.

The prototypical example of an open subscheme would be the open sets $D_+(x_i) \subseteq \mathbb{P}^n_k$ of projective space or $D(f) \subseteq \operatorname{Spec} R$ of affine space. Note that an open sub-scheme of an affine sub-scheme need not be affine:

Example 2.2.51: Open Sub-Scheme need not be Affine
Let $R = k[x, y]$, $X = \mathbb{A}^2_k = \operatorname{Spec}(k[x, y])$, and consider $U = \mathbb{A}^2_k \setminus \{(x, y)\}$ which may be considered the plane without the origin. Before seeing why this is a scheme, try to show that U is indeed an open subset (it is the compliment of intersection of certain closed subsets, equivalently the union of certain open sets, there is more than one way of creating it).

Now, there does not exist an R such that $\operatorname{Spec}(R) \cong U$. It may be tempting to say that there ought to be a distinguished open set of $\mathbb{A}^2_k$ that produces U, but the reader should think through why this is not the case (hint: (x, y) is not principle). However, U can certainly be described as the union of two distinguished open sets, namely:

$$U = D(x) \cup D(y)$$

What is the global section on the open sub-scheme U? This would be all functions in $D(x) \cup D(y)$ that agree on the intersection, $D(x) \cap D(y) = D(xy) = R_{xy} = k[x, y]_{xy}$. Computing the local sections, as $D(x)$ is all points in $\mathbb{A}^2_k$ that do not vanish in the function x, and $D(y)$ is all the points that do not vanish in the function y, we get:

$$\mathcal{O}_U(D(x)) = R_x = k\left[x, y, \frac{1}{x}\right] \qquad \mathcal{O}_U(D(y)) = R_y = k\left[x, y, \frac{1}{y}\right]$$

Now consider the functions that agree on the intersection: $R_x \cap R_y \subseteq R_{xy}$. These are all rationals functions with only powers of x in the denominator, and only powers of y in the denominator (as it must be in the intersection). But this leaves only the elements of R: $R_x \cap R_y = R$ (where R is identified with it's canonical embedding in R_{xy}). Thus:

$$\mathcal{O}_{\mathbb{A}^2_k}(U) = \mathcal{O}_U(U) = k[x, y]$$

in other words, the global section of U is identical to the global section of $\mathbb{A}^2_k$. Now, if U was affine, then by proposition 2.2.13, we would have that the is determined by the global ring, that is:

$$U \cong \mathbb{A}^2_k$$

that is, this is a scheme isomorphism, and hence also a homeomorphic map between prime ideals that must respect $V(-)$ and $I(-)$. But in U we have:

$$V(x) \cap V(y) = \emptyset$$

while this is certainly not the case in $\mathbb{A}^2_k$ (as it gives $[(x - 0, y - 0)]$). Hence U is *not* an affine scheme!

The main idea is that $D(f)$ misses a codimension 1 subset if R is Noetherian integral domain (see exercise 5.3.1 (1)), where Noetherian is to avoid infinite pathologies and integral domain is to insure there is only one irreducible component (see section 2.4). More generally, given R is Noetherian and an integral domain, $\operatorname{Spec} R_S$ for some $S \subseteq R$ not containing units misses a (pure) codimension 1 subset (as it is open). The reason that higher codimensional sets "don't matter" when considering global sections very reminiscent of complex geometry, namely this is an algebraic version of *Hartog's Lemma* (see [**NateComplexAnalysis**]).

It is possible to have an affine open subscheme of an affine scheme, and hence by dualizing we may look for algebraic conditions on the ring homomorphisms. We state the result for those familiar with flatness (see [**NateClassicalAlgGeo**]) but leave the proof till flatness is introduced in ref:HERE:

Theorem 2.2.52: Characterizing Affine Open Immersions

$\mathrm{Spec}\,(S) \to \mathrm{Spec}\,(R)$ is an open immersion if and only $R \to S$ is flat, of finite presentation, and an epimorphism.

***Proof*:**
see https://mathoverflow.net/questions/20782/ring-theoretic-characterization-of-open-affines.

2.2.53 For the expert: Cover Condition The reader well-versed in descent theory may have noticed this is almost the condition of an fppf covering, with faithful (i.e. surjective) the epimorphism condition. This motivates how fppf covers are the local open covers and appropriately generalize Zariski covers, see ref:HERE.

2.2.9 Gluing (Coproduct) of Schemes

In Differential Geometry, one of the most common way of constructing a manifold, and indeed what is at the heart of geometry, is the process of gluing pieces together with some amount of consistency (see [7, chapter 1]). More generally, gluing manifolds, and gluing schemes, is a special case of a pushout (or fibered coproduct) $X \cup_Z Y$:

$$\begin{tikzcd}
& Z \arrow[dl] \arrow[dr] & \\
X \arrow[dr] \arrow[ddr, dashed, bend right] & & Y \arrow[dl] \arrow[ddl, dashed, bend left] \\
& X \cup_Z Y \arrow[d, dashed, "\exists!"] & \\
& Q &
\end{tikzcd}$$

The fibered product $X \times_Z Y$ can be thought of as taking a product of spaces $X \times Y$ and restricting to where the maps agree (i.e., carving out a subspace), while the fibered coproduct can be thought of as taking the disjoint union of spaces and doing some gluing. The fibered product shall be more nuanced as defining $X \times Y$ does not sense and hence is left for section 3.2.

Just like for manifolds, we can't glue schemes any way we want and get another scheme, we must be a bit more careful. Example 3.2.7 shall give an example of "bad" gluing.

The key idea for coproducts of schemes is to ensure that the gluing preserves the local affine structure (note that for a manifold it also has to preserve Hausdorffness). Hence, it is marginally easier to construct schemes via gluing:

Theorem 2.2.54: Coproduct (Pushout) of Schemes

Let $\{X_i\}_{i\in I}$ be a collection of schemes. Then if there exists:

1. A collection of open subschemes, not necessarily an open cover, $X_{ij} \subseteq X_i$ (where $X_{ii} = X_i$)
2. a collection of isomorphisms $f_{ij} : X_{ij} \to X_{ji}$ (where f_{ii} would be the identity) where

$$f_{ij}(X_{ij} \cap X_{ik}) \subseteq X_{ji} \cap X_{jk}$$

such that the isomorphisms agree on triple intersection[a]:

$$f_{ik}|_{X_{ij}\cap X_{ik}} = f_{jk}|_{X_{ji}\cap X_{jk}} \circ f_{ij}|_{X_{ij}\cap X_{ik}}$$

then there is a unique scheme X (up to unique isomorphism) with open subsets isomorphic to X_i respecting the above gluing conditions.

[a] this is called the *cocycle condition*

This is a translation of the techniques from differential geometry (this condition may look more familiar if written as $\mathbf{y} \circ \mathbf{x}^{-1} : \mathbf{x}(U \cap V) \to \mathbf{y}(U \cap V)$). The cocycle condition can be visualized like so:

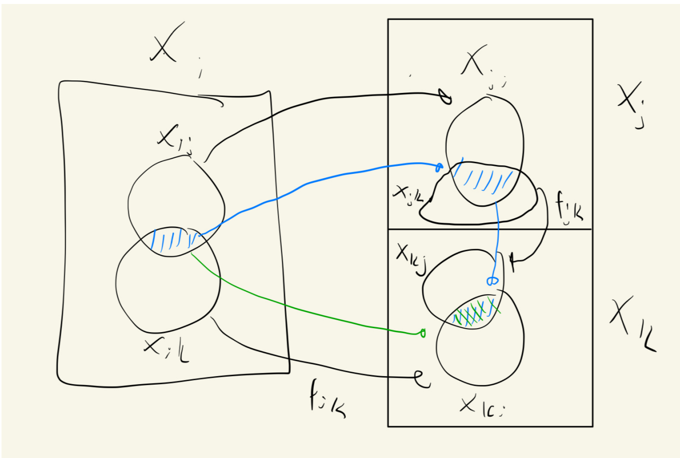


Figure 2.4: Green map must match blue map

Note the cocycle condition forces:

1. when $i = j = k$ that $\varphi_{ii} = \mathrm{id}$
2. when $i = k$ that $\varphi_{ij} = \varphi_{jk}^{-1}$

***Proof*:**
Define X as the quotient of the disjoint union of the schemes X_i:

$$X = \frac{\coprod_{i \in I} X_i}{(X_{ij} \ni x \sim f_{ij}(x) \in X_{ji})}$$

where the equivalence relation is defined by identifying points via the gluing maps: $x \sim y$ if $x \in X_{ij}$, $y \in X_{ji}$, and $f_{ij}(x) = y$. The cocycle condition ensures that this relation is transitive (and symmetry follows from $f_{ji} = f_{ij}^{-1}$, which is a consequence of the cocycle condition with $k = i$), so this is a valid equivalence relation. We endow X with the quotient topology relative to the natural projection map $\pi : \coprod X_i \to X$.

Let $\iota_i : X_i \to X$ denote the restriction of π to the component X_i, and let $U_i = \iota_i(X_i)$ be its image in X. Because the X_{ij} are open sets, one can verify that ι_i is a homeomorphism onto the open set U_i. The sets U_i form an open cover of X, where each U_i is a topological copy of X_i.

With the topological space established, we define the structure sheaf $\mathcal{O}_X$. For any open set $V \subseteq X$, we define a section $s \in \mathcal{O}_X(V)$ to be a collection of sections $\{s_i\}_{i \in I}$ where $s_i \in \mathcal{O}_{X_i}(\iota_i^{-1}(V))$, satisfying the compatibility condition on overlaps:

$$s_i\big|_{\iota_i^{-1}(V) \cap X_{ij}} = f_{ij}^{\#}\left(s_j\big|_{\iota_j^{-1}(V) \cap X_{ji}}\right)$$

This defines a valid sheaf of rings by theorem 1.1.23, applied to the open cover $\{U_i\}$ with sheaves $(\iota_i)_* \mathcal{O}_{X_i}$ and transition isomorphisms induced by $f_{ij}^{\#}$.

Finally, we must verify that the ringed space $(X, \mathcal{O}_X)$ is indeed a scheme. It suffices to check this locally. Consider the open set U_i equipped with the restricted sheaf $\mathcal{O}_X|_{U_i}$. By our construction of the global sections, the sheaf $\mathcal{O}_X|_{U_i}$ is canonically isomorphic to the sheaf $\mathcal{O}_{X_i}$ via the map ι_i. Thus, we have an isomorphism of ringed spaces $(U_i, \mathcal{O}_X|_{U_i}) \cong (X_i, \mathcal{O}_{X_i})$. Since each X_i is a scheme by hypothesis, each open set U_i is a scheme. As X is covered by open subsets that are schemes (and hence covered by affine schemes), X is a scheme as we sought to show.

What would the resulting global sections of a glued scheme look like? Consider the case of two schemes X_1, X_2 glued along $X_{12} \subseteq X_1$ and $X_{21} \subseteq X_2$ via an isomorphism $f_{12} : X_{12} \xrightarrow{\sim} X_{21}$, with inclusion maps $\iota_i : X_i \to X$. Then:

$$\mathcal{O}_X(U) =$$
$$\left\{ (s_1, s_2) \;\middle|\; s_1 \in \mathcal{O}_{X_1}(\iota_1^{-1}(U)),\ s_2 \in \mathcal{O}_{X_2}(\iota_2^{-1}(U)) \text{ such that } s_1|_{\iota_1^{-1}(U) \cap X_{12}} = f_{12}^{\#}\Big(s_2|_{\iota_2^{-1}(U) \cap X_{21}}\Big) \right\}$$

This formula is common enough to be given a name: we call it the *gluing formula*.

For three schemes X_1, X_2, X_3, the formula extends naturally: a global section is a triple $(s_1, s_2, s_3) \in \mathcal{O}_{X_1}(X_1) \times \mathcal{O}_{X_2}(X_2) \times \mathcal{O}_{X_3}(X_3)$ satisfying the pairwise compatibility conditions on all overlaps simultaneously:

$$s_i\big|_{X_{ij}} = f_{ij}^{\#}\Big(s_j\big|_{X_{ji}}\Big) \qquad \text{for all } i, j \in \{1, 2, 3\}$$

Note that one does not glue two schemes first and then attach the third; the cocycle condition ensures that all pairwise constraints are mutually consistent, so all three (or finitely many) pieces are handled at once. The pattern for n schemes is identical: global sections are n-tuples satisfying $\binom{n}{2}$ compatibility conditions.

When thinking about global sections of glued schemes, it is helpful to view the gluing maps as imposing additional *relations*: the global sections of the glued scheme are the elements of the product $\prod_i \mathcal{O}_{X_i}(X_i)$ that are compatible under all the transition isomorphisms. The more gluing constraints there are, the smaller the ring of global sections becomes.

Example 2.2.55: Gluing [affine] Schemes

Let $X = \operatorname{Spec}(k[x])$ and $Y = \operatorname{Spec}(k[y])$. Take $D(x) \subseteq X$ and $D(y) \subseteq Y$, which is the collection of all prime ideals that do not contain x and y respectively. As shown in proposition ref:HERE, these subsets are isomorphic to:

$$D(x) \cong \operatorname{Spec}\left(k\left[x, \frac{1}{x}\right]\right) \qquad D(y) \cong \operatorname{Spec}\left(k\left[y, \frac{1}{y}\right]\right)$$

As a diagram, we have:

$$\begin{tikzcd} \operatorname{Spec}(k[x]) & & \operatorname{Spec}(k[y]) \\ & \operatorname{Spec}(k[x,1/x]) \arrow[ul] \arrow[ur] & \end{tikzcd} \quad \text{and} \quad \begin{tikzcd} k[x] \arrow[dr] & & k[y] \arrow[dl] \\ & k[x,1/x] & \end{tikzcd}$$

or it may be more helpful to see this as:

$$\begin{tikzcd} \operatorname{Spec} k[x] & & \operatorname{Spec} k[y] \\ & \operatorname{Spec} k[x,1/x] \cong \operatorname{Spec} k[y,1/y] \arrow[ul] \arrow[ur] & \end{tikzcd}$$

and

$$\begin{tikzcd} k[x] \arrow[dr] & & k[y] \arrow[dl] \\ & k[x,1/x] \cong k[y,1/y] & \end{tikzcd}$$

where the cocycle condition must be satisfied by the isomorphisms. Notice how these are similar to restriction conditions: we are "manually" figuring out how the restrictions should work so that by uniqueness of gluing local sections on a sheaf we can get the global sections. These two open subsets can be glued in at least two different ways to produce two non-affine schemes that will be familiar to the reader:

1. (Line with double origin). Define the isomorphism $D(x) \cong D(y)$ via the ring isomorphism $\varphi^{\#} : k[y, 1/y] \to k[x, 1/x]$ given by $y \mapsto x$. This is a scheme by theorem 2.2.54. Label this scheme S_1. We claim S_1 is *not* affine, even though its global sections coincide with those of $\mathbb{A}^1_k$.

   By the gluing formula, a global section of $\mathcal{O}_{S_1}$ is a pair $(f(x), g(y)) \in k[x] \times k[y]$ satisfying the compatibility condition:

   $$f(x) = \varphi^{\#}(g(y)) = g(x) \qquad \text{in } k[x, 1/x]$$

Since $k[x] \hookrightarrow k[x, 1/x]$ is injective, this simply says $f(x) = g(x)$ as polynomials (after identifying the variable names). In particular, any polynomial satisfies this condition, and so:

$$\mathcal{O}_{S_1}(S_1) \cong k[x]$$

By proposition 2.2.13, an affine scheme is determined by its global sections, so if S_1 were affine it would be isomorphic to $\operatorname{Spec}(k[x]) = \mathbb{A}^1_k$. But this is impossible: S_1 has two distinct closed points (the two origins) that are not separated by any open set, while $\mathbb{A}^1_k$ does not. Hence S_1 is not affine. The same construction works in higher dimensions (e.g. $\mathbb{A}^n_k$ with double origin).

In differential geometry, such spaces are examples of non-Hausdorff manifolds, however all non-trivial schemes already fail to be Hausdorff. The relevant condition for schemes is *separatedness*: spaces with double-origins fail to be separated (as a hint: show that the intersection of the two copies of $\mathbb{A}^1_k$ inside S_1 is $D(x)$, which is not affine), while $\mathbb{A}^n_k$ is separated.

2. (Projective space). Define the isomorphism $D(x) \cong D(y)$ via the ring isomorphism $\varphi^{\#}$: $k[y, 1/y] \to k[x, 1/x]$ defined by $y \mapsto 1/x$ (equivalently, $1/y \mapsto x$). The reader will recognize the resulting scheme as $\mathbb{P}^1_k$. When $k \in \{\mathbb{R}, \mathbb{C}\}$, this is isomorphic (as a manifold) to a circle and a sphere respectively, while in dimension n, $\mathbb{P}^n_k$ cannot be embedded in $\mathbb{A}^n_k$. For schemes, one must account for the generic point (0) and the non-closed points given by irreducible polynomials (recall that linear factors that are Galois conjugates are identified).

   Let us show that $\mathbb{P}^1_k$ is not affine by computing $\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1)$. A global section is a pair $(f(x), g(y)) \in k[x] \times k[y]$ satisfying:

$$f(x) = \varphi^{\#}(g(y)) = g(1/x) \qquad \text{in } k[x, 1/x]$$

   Write $f(x) = \sum_{n=0}^{N} a_n x^n$ and $g(y) = \sum_{m=0}^{M} b_m y^m$, so that:

$$g(1/x) = \sum_{m=0}^{M} b_m\, x^{-m}$$

   In $k[x, 1/x]$, the Laurent monomials $\{x^n\}_{n\in\mathbb{Z}}$ are linearly independent over k. The left-hand side $f(x)$ involves only non-negative powers of x, while the right-hand side $g(1/x)$ involves only non-positive powers. For these to be equal, both must be constant:

$$a_n = 0 \text{ for } n \geq 1, \qquad b_m = 0 \text{ for } m \geq 1, \qquad a_0 = b_0$$

   Hence f and g are both the same constant, giving:

$$\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1) = k$$

   By proposition 2.2.13, if $\mathbb{P}^1$ were affine then $\mathbb{P}^1 \cong \operatorname{Spec}(\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1)) = \operatorname{Spec}(k)$, which is a single point. But $\mathbb{P}^1_k$ has more than one point, so $\mathbb{P}^1_k$ is not affine[a].

[a] Those familiar with complex analysis may recall that the only holomorphic functions on $\mathbb{P}^1_{\mathbb{C}}$ (the Riemann sphere) are constants, which is a manifestation of the same phenomenon.

2.2.10 *Gluing Affine Schemes and Fiber Products of Rings

Let us now specialize to the case where the pieces being glued are affine. Recall that the functors $\operatorname{Spec}(-)$ and $\Gamma(-)$ are contravariant and form an adjunction that exchanges limits and colimits. Hence, the cofibered product (pushout) of schemes dualizes to the fiber product (pullback) of rings. If $X_1 = \operatorname{Spec}(A)$, $X_2 = \operatorname{Spec}(B)$, and the overlap maps factor through a ring C, then the gluing formula for global sections becomes:

$$\mathcal{O}_X(X) = A \times_C B := \{(a, b) \in A \times B \mid \text{images of } a \text{ and } b \text{ agree in } C\}$$

This is dual to the familiar fact that the fibered product of affine schemes corresponds to the tensor product of rings:

$$\operatorname{Spec} A \times_{\operatorname{Spec} C} \operatorname{Spec} B \cong \operatorname{Spec}(A \otimes_C B)$$

In both cases, the contravariance of Spec exchanges limits and colimits, and hence exchanges tensor products (colimits in **CRing**) with fiber products (limits in **CRing**). More generally, for an arbitrary number of affine pieces, the global sections are the limit of the diagram of rings formed by all the overlap maps.

It is natural to ask: since **CRing** is both complete and cocomplete, does $\operatorname{Spec}(A \times_C B)$ give the pushout of $\operatorname{Spec}(A)$ and $\operatorname{Spec}(B)$ over $\operatorname{Spec}(C)$ in the category of schemes? The answer is *no* in general. The subtlety is that the universal property must be tested against different classes of objects depending on the ambient category:

- *In **Aff***: the object $\operatorname{Spec}(A \times_C B)$ satisfies the universal property of the pushout with respect to maps into affine schemes, and this pushout always exists since **CRing** is complete (so fiber products of rings always exist).
- *In **Sch***: the pushout must satisfy the universal property for maps into *all* schemes. A morphism from an affine scheme to a non-affine target Q is not determined by a ring homomorphism out of $\Gamma(Q, \mathcal{O}_Q)$ (since Q is not affine, the adjunction $\operatorname{Spec} \dashv \Gamma$ does not apply), so the affine pushout may fail the universal property.

In categorical language, the inclusion **Aff** $\hookrightarrow$ **Sch** does not preserve pushouts. The pushout computed in **Sch** can be genuinely non-affine: the line with doubled origin and the projective line from example 2.2.55 are both pushouts of affine schemes in **Sch** that are not affine. In each case, $\operatorname{Spec}(A \times_C B)$ exists as an affine scheme, but it is *not* isomorphic to the glued scheme X – it is only the "best affine approximation" to the pushout.

The preceding discussion raises a natural question: if $\operatorname{Spec}(A \times_C B)$ is not the pushout in **Sch**, what *is* it? The answer is given by the adjunction between global sections and Spec.

Proposition 2.2.56: $\Gamma \dashv \operatorname{Spec}$ Adjunction

The global sections functor $\Gamma : \mathbf{Sch} \to \mathbf{CRing}^{op}$ is left adjoint to $\operatorname{Spec} : \mathbf{CRing}^{op} \to \mathbf{Sch}$:

$$\operatorname{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec}(R)) \cong \operatorname{Hom}_{\mathbf{CRing}}(R, \Gamma(X, \mathcal{O}_X))$$

for any scheme X and any ring R.

***Proof*:**
See proposition 3.1.12

Since left adjoints preserve colimits and right adjoints preserve limits, this single adjunction encodes both dualities between schemes and rings:

- Γ (left adjoint) sends pushouts in **Sch** to colimits in $\mathbf{CRing}^{op}$, i.e. limits (fiber products) in **CRing**:
$$\Gamma(X_1 \cup_Z X_2) \cong \Gamma(X_1) \times_{\Gamma(Z)} \Gamma(X_2)$$
This is why the gluing formula computes global sections as a fiber product of rings – it is a formal consequence of the adjunction.
- Spec (right adjoint) sends limits in $\mathbf{CRing}^{op}$ (i.e. colimits in **CRing**, such as tensor products) to limits (fiber products) in **Sch**:
$$\operatorname{Spec}(A \otimes_C B) \cong \operatorname{Spec}(A) \times_{\operatorname{Spec}(C)} \operatorname{Spec}(B)$$
- Spec , being a right adjoint, does *not* preserve colimits in general. The fiber product $A \times_C B$ is a limit in **CRing**, hence a colimit in $\mathbf{CRing}^{op}$, and there is no reason for Spec to preserve it.

Definition 2.2.57: Affinization

Let X be a scheme. The *affinization* of X is the affine scheme $X^{\mathbf{Aff}} := \operatorname{Spec}(\Gamma(X, \mathcal{O}_X))$, together with the canonical morphism

$$\eta_X : X \to \operatorname{Spec}(\Gamma(X, \mathcal{O}_X))$$

given by the unit of the $\Gamma \dashv \operatorname{Spec}$ adjunction.

By the universal property of adjunctions, η_X is the universal map from X to an affine scheme: any morphism $X \to \operatorname{Spec}(R)$ factors uniquely through η_X. For the pushout $X = X_1 \cup_Z X_2$ of affine schemes, since $\Gamma(X) \cong A \times_C B$, this gives:

$$X_1 \cup_Z X_2 \xrightarrow{\eta} \operatorname{Spec}(A \times_C B)$$

Hence $\operatorname{Spec}(A \times_C B)$ is not the pushout in **Sch**, but rather the *best affine approximation* to the pushout – the closest affine scheme that the glued scheme maps to.

Example 2.2.58: Affinization of Glued Schemes
Returning to the two gluings from example 2.2.55:

1. *Line with doubled origin.* Here $A \times_C B \cong k[x]$, so the affinization is $\mathbb{A}^1_k$. The map $\eta : S_1 \to \mathbb{A}^1_k$ identifies the two origins to a single point, collapsing exactly the non-affine pathology.
2. *Projective line.* Here $A \times_C B \cong k$, so the affinization is $\operatorname{Spec}(k)$, a single point. The map $\eta : \mathbb{P}^1_k \to \operatorname{Spec}(k)$ collapses the entire projective line to a point, reflecting the fact that $\mathbb{P}^1_k$ has only constant global functions.

In both cases, the affinization map η "kills" precisely the geometric information that cannot be captured by global sections.

Exercise 2.2.2

1. Let X be the line with two origins, $Y = \mathbb{A}^1_k$ the affine line, and $\pi : X \to Y$ the natural projection. Compute $f_*(\mathcal{O}_X)_0$, the talk of the pushforward at 0.
2. Let X be a scheme. Show that the affine open subsets form a base.
3. A subset Z of a scheme X is closed if and only if there exists an open cover of X by affine subschemes, $\{U_i = \operatorname{Spec}(A_i)\}_{i \in I}$, such that for every $i \in I$, the intersection $Z \cap U_i$ is a closed subset of the affine scheme U_i. In other words, a closed subset of X is characterized as being locally closed on an affine open covering.

2.3 Projective Schemes

Recall projective space from [7]. In [**NateClassicalAlgGeo**] a few key properties of projective space $\mathbb{P}^n_{\mathbb{C}}$ were proved, namely:

1. Projective space was constructed by gluing together affine spaces. The cocycle condition is straightforward to verify over any ring (the transition maps are rational functions that do not require algebraic closure), though understanding the closed points is simpler when $k = \overline{k}$. We shall construct Proj in a more "natural" way without any choice of coordinates.
2. There shall be a homogeneous coordinate ring S from which we can construct the scheme $\operatorname{Proj} S$. Just like in the classical setting, S shall not be the ring of regular functions on $\operatorname{Proj} S$: the global sections of $\operatorname{Proj} S$ shall be constant, and $\operatorname{Proj} S$ shall behave like the "compact" sets in algebraic geometry (namely the proper sets, as mentioned in [**NateClassicalAlgGeo**]).

   We shall show that the algebraic object S will be associated to a different sheaf over $\operatorname{Proj} S$ that is not the structure sheaf, see section 5.3.
3. The properties of Bézout's theorem shall turn out to be the special case of the theory of *divisors* on projective space. See section **??** for more details.

Classically, n-dimensional projective space $\mathbb{RP}^n$ or $\mathbb{CP}^n$ is defined on $\mathbb{R}^{n+1} \setminus \{0\}$ or $\mathbb{C}^{n+1} \setminus \{0\}$ by taking an equivalence class:

$$(x_0, \dots, x_n) \sim (y_0, \dots, y_n) \qquad \Leftrightarrow \qquad (x_0, \dots, x_n) = (\lambda y_0, \dots, \lambda y_n) \qquad \text{for some } \lambda \neq 0$$

This produces an n-dimensional smooth manifold with the usual patches defined as:

$$U_i = D(x_i) = \{[x_0 : \cdots : x_n] \in \mathbb{P}^n \mid x_i \neq 0\}$$

where, using the notation $x_{k,j} = x_k/x_j$ for the affine coordinates on U_j, the transition maps were defined as (see [7, chapter 1]):

$$\varphi_i \circ \varphi_j^{-1}(x_{0,j}, \dots, \widehat{x_{j,j}}, \dots, x_{n,j}) = \left(\frac{x_{0,j}}{x_{i,j}}, \dots, \frac{1}{x_{i,j}}, \dots, \frac{\widehat{x_{i,j}}}{x_{i,j}}, \dots, \frac{x_{n,j}}{x_{i,j}} \right)$$

In [**NateClassicalAlgGeo**], we saw we can do the same for any field k, namely since inverses exist. Let us do the same but with a general ring R and the graded polynomial ring $S = R[x_0, x_1, \dots, x_n]$

with the standard grading ($\deg x_i = 1$, R in degree 0), and $S_+ = \bigoplus_{d>0} S_d$ being the irrelevant ideal. Though S is naturally $\mathbb{N}$-graded, we shall work in the setting of $\mathbb{Z}$-graded rings since localizations introduce negative degrees[22].

As a set, define:

Definition 2.3.1: Projective Space – as a Set

$$\operatorname{Proj} S = \{[\mathfrak{p}] \mid \mathfrak{p} \subseteq S \text{ is a homogeneous prime with } S_+ \not\subseteq \mathfrak{p}\}$$

The condition $S_+ \not\subseteq \mathfrak{p}$ excludes the "irrelevant" prime consisting of all positive-degree elements, which would correspond to the non-existent point $[0 : \cdots : 0]$ in projective space. With the set defined, $\operatorname{Proj} S$ has the Zariski topology induced by homogeneous elements of positive degree:

$$V_+(T) = \{[\mathfrak{p}] \in \operatorname{Proj} S \mid T \subseteq \mathfrak{p}\}$$

The basis for the topology can be given by the positive-degree homogeneous elements:

$$D_+(f) = \{[\mathfrak{p}] \in \operatorname{Proj} S \mid f \notin \mathfrak{p}\}$$

If $f = x_i$, we would like this to correspond to the usual affine space $\mathbb{A}^n_R$. In the classical case, we got this isomorphism by doing:

$$\begin{aligned}
\{[x_0 : \cdots : x_n] \mid x_i \neq 0\} &= \left\{ \left[\frac{x_0}{x_i} : \cdots : 1 : \cdots : \frac{x_n}{x_i}\right] \;\middle|\; x_i \neq 0 \right\} \\
&\cong k\left[\frac{x_0}{x_i}, \ldots, \frac{x_n}{x_i}\right] \\
&= k[x_{0,i}, \ldots, x_{n,i}] && \text{relabeling} \\
&= \mathbb{A}^n_k
\end{aligned}$$

Notice how these are all degree 0 in the grading of S. These sets are not quite the same as $\operatorname{Spec} S_f$; however it turns out that taking $D_+(f)$ and working with the degree-0 part of the localization gives us exactly the desired result:

Lemma 2.3.2: Characterizing Basic Open Sets For Projective Space

Let $f \in S$ be a homogeneous element of positive degree, and define $S_{(f)} := (S_f)_0$ to be the degree-0 part of the localization. Then:

$$D_+(f) \cong \operatorname{Spec}\left(S_{(f)}\right)$$

***Proof*:**
We construct explicit inverse maps. Define the forward map $\Phi : D_+(f) \to \operatorname{Spec}\left(S_{(f)}\right)$ by:

$$\Phi(\mathfrak{p}) = \mathfrak{p} \cdot S_f \cap S_{(f)}$$

Since $f \notin \mathfrak{p}$, the extension $\mathfrak{p} \cdot S_f$ is a proper homogeneous prime of S_f, and its intersection with

[22] More general gradings are sometimes useful, however we shall not need this either in this book or for a while

the subring $S_{(f)}$ is a prime ideal of $S_{(f)}$.

For the inverse, let $e = \deg f$ and define $\Psi : \operatorname{Spec}(S_{(f)}) \to D_+(f)$ by:

$$\Psi(\mathfrak{q}) = \bigoplus_{d\geq 0} \left\{ a \in S_d \;\middle|\; \frac{a^e}{f^d} \in \mathfrak{q} \right\}$$

Note that a^e has degree de and f^d has degree de, so $a^e/f^d \in S_{(f)}$ and the condition makes sense.

$f \notin \Psi(\mathfrak{q})$: If $f \in \Psi(\mathfrak{q})$, then $f^e/f^e = 1 \in \mathfrak{q}$, contradicting $\mathfrak{q}$ being proper.

Closure under addition: Let $a, b \in S_d$ with $a^e/f^d, b^e/f^d \in \mathfrak{q}$. Consider a binomial term $a^j b^{e-j}$ with $0 \leq j \leq e$. Observe:

$$\left(\frac{a^j b^{e-j}}{f^d}\right)^e = \left(\frac{a^e}{f^d}\right)^j \cdot \left(\frac{b^e}{f^d}\right)^{e-j} \in \mathfrak{q}$$

Since $\mathfrak{q}$ is prime (hence radical: $x^n \in \mathfrak{q} \Rightarrow x \in \mathfrak{q}$), we get $a^j b^{e-j}/f^d \in \mathfrak{q}$ for each j. Summing over all binomial terms gives $(a+b)^e/f^d \in \mathfrak{q}$.

Closure under S-multiplication: If $a \in \Psi(\mathfrak{q})_d$ and $c \in S_k$, then $(ca)^e/f^{d+k} = (c^e/f^k)(a^e/f^d)$. The second factor lies in $\mathfrak{q}$ and the first in $S_{(f)}$, so the product lies in $\mathfrak{q}$.

Primality: If $a \in S_d, b \in S_k$ with $ab \in \Psi(\mathfrak{q})$, then $(ab)^e/f^{d+k} = (a^e/f^d)(b^e/f^k) \in \mathfrak{q}$. Since $\mathfrak{q}$ is prime, either $a^e/f^d \in \mathfrak{q}$ or $b^e/f^k \in \mathfrak{q}$, giving $a \in \Psi(\mathfrak{q})$ or $b \in \Psi(\mathfrak{q})$.

One can verify that Φ and Ψ are inverse to each other: the key step is that $a \in \mathfrak{p}_d$ implies $a^e/f^d \in \mathfrak{p}S_f \cap S_{(f)}$, and conversely if $a^e/f^d \in \mathfrak{p}S_f$ then $a \in \mathfrak{p}$ by primality (since $f \notin \mathfrak{p}$). For the homeomorphism, the preimage of a basic open $D(g^e/f^{\deg g}) \subseteq \operatorname{Spec}(S_{(f)})$ is $D_+(fg) \cap D_+(f) = D_+(fg)$, so both sides have matching bases.

Let us apply this to $f = x_i$:

$$\begin{aligned} D_+(x_i) &= \{[\mathfrak{p}] \in \operatorname{Proj} S \mid x_i \notin \mathfrak{p}\} \\ &\cong \operatorname{Spec}\left(S_{(x_i)}\right) \\ &= \operatorname{Spec}\left(\text{degree-0 elements of } R[x_0, \dots, x_n, 1/x_i]\right) \\ &\overset{!}{=} \operatorname{Spec} R\left[\frac{x_0}{x_i}, \dots, \frac{x_n}{x_i}\right] \\ &\cong \operatorname{Spec} \frac{R[x_{0,i}, \dots, x_{n,i}]}{(x_{i,i} - 1)} \\ &\cong \mathbb{A}^n_R \end{aligned}$$

where $\overset{!}{=}$ comes from the fact that $1/x_i$ is now a unit of degree -1, hence each x_j/x_i is of degree 0, and these elements generate $S_{(x_i)}$ over R. Relabeling x_j/x_i as $x_{j,i}$ and "omitting" $x_{i,i}$ (which equals one, equivalently we quotient by $(x_{i,i} - 1)$, a form of dehomogenizing), we see that it is isomorphic to the usual $\mathbb{A}^n_R$, and hence we recover the usual affine cover! Note the importance of taking the degree-0 component, for

$$\operatorname{Spec} R[x_0, \dots, x_n, 1/x_i] \cong \mathbb{A}^{n+1}_R \setminus \{x_i = 0\}$$

Let us now construct the structure sheaf on $\operatorname{Proj} S$. We shall see that $\operatorname{Proj} S$ is not affine (we hope not, as the classical projective variety is not affine!), and so it is not determined by its global sections in the way that affine schemes are. Nonetheless, the construction below endows $\operatorname{Proj} S$ with a natural scheme structure. As we just saw, each $D_+(f)$ is affine and hence we certainly have an affine cover. Let us ensure that the gluing satisfies the cocycle condition from theorem 2.2.54, namely that all intersections glue appropriately to form a scheme.

Theorem 2.3.3: Structure Sheaf On Proj

Take the Zariski topology on $\operatorname{Proj} S$ given by the basis $D_+(f) \cong \operatorname{Spec}(S_{(f)})$. Then $\operatorname{Proj} S$ is a scheme.

***Proof*:**
Let $f, g \in S$ be homogeneous elements of positive degree and take $D_+(f), D_+(g)$ with non-trivial overlap. It is not difficult to show that $D_+(f) \cap D_+(g) = D_+(fg)$. We verify the cocycle condition from theorem 2.2.54. Consider:

$$D(g^{\deg f}/f^{\deg g}) \subseteq D_+(f) \qquad \text{and} \qquad D(f^{\deg g}/g^{\deg f}) \subseteq D_+(g)$$

We shall show that:

$$S_{(f)}\left[\frac{1}{g^{\deg f}/f^{\deg g}}\right] \cong S_{(fg)} \cong S_{(g)}\left[\frac{1}{f^{\deg g}/g^{\deg f}}\right]$$

This will induce the desired isomorphism on spectra. There is a natural map:

$$\varphi : S_{(f)} \to S_{(fg)}$$

mapping $h/f^k \mapsto hg^k/(fg)^k$. Taking $U = \left\{(g^{\deg f}/f^{\deg g})^k \;\middle|\; k \in \mathbb{N}\right\}$, notice that $\varphi(U)$ consists of units in $S_{(fg)}$, thus there is a natural map:

$$\widetilde{\varphi} : S_{(f)}\left[\frac{1}{g^{\deg f}/f^{\deg g}}\right] \to S_{(fg)}$$

It is straightforward to verify that this map is in fact an isomorphism (injectivity is easy, and matching degrees gives surjectivity). The same argument works in the other direction, giving the desired result. Thus, we have established the desired ring isomorphism, which induces the required isomorphism of schemes, completing the proof.

Definition 2.3.4: Projective Scheme

Let $S_\bullet$ be a $\mathbb{Z}$-graded ring with $S_0 = R$. Then $\operatorname{Proj} S_\bullet$ is called the projective scheme associated to $S_\bullet$. If $S = R[x_0, \ldots, x_n]$ with the standard grading, then the above construction defines the classical projective n-scheme or prototypical projective scheme over R, and is denoted $\mathbb{P}^n_R$ or $\mathbb{P}^n$ if unambiguous.

We can naturally recover the notion of quasiprojective as open subschemes of projective schemes:

Definition 2.3.5: Quasiprojective Scheme

A *quasiprojective R-scheme* is an open subscheme of a projective R-scheme.

Notice that the scalars can be defined for any ring, which generalizes the result from classical algebraic geometry. If $R = k$ is an algebraically closed field, the closed points of $\mathbb{P}^n_k$ will behave like the traditional points of projective space defined in classical algebraic geometry.

Proposition 2.3.6: Global Sections of $\operatorname{Proj} S_\bullet$

Let $S_\bullet$ be an $\mathbb{N}$-graded ring generated by S_1 as an S_0-algebra (e.g. $S = R[x_0, \ldots, x_n]$ with the standard grading), and let $X = \operatorname{Proj} S_\bullet$. Then:

$$\mathcal{O}_X(X) = S_0$$

Hence $\mathbb{P}^n_R$ is not an affine scheme (the case $n = 1$ was given in example 2.2.55).

***Proof*:**
exercise

Recall from [**NateClassicalAlgGeo**] that homogeneous polynomials of $k[x_0, \ldots, x_n]$ cut out closed sets from $\mathbb{P}^n_k$. This is because if $a \in k^{n+1}$ is a root of a homogeneous polynomial, so are all scalar multiples of a. In modern algebraic geometry, the global sections $\mathcal{O}_X(X)$ are regarded as the "functions" on a scheme X. However, $\mathcal{O}_{\mathbb{P}^n_R}(\mathbb{P}^n_R) = R$ by proposition 2.3.6, so the graded ring S that *generates* $\operatorname{Proj} S$ does not manifest as the structure sheaf.

How, then, should we interpret S? The answer is that each graded piece S_d gives rise to a sheaf $\mathcal{O}(d)$ on $\mathbb{P}^n_R$ called a *twisting sheaf* (or *Serre twist*). These are examples of *line bundles* (invertible sheaves), and the structure sheaf $\mathcal{O}_{\mathbb{P}^n_R}$ is the special case $\mathcal{O}(0)$. We shall show in proposition 5.3.10 that the graded ring S can be recovered from these sheaves:

$$S_d \cong \Gamma(\mathbb{P}^n_R, \mathcal{O}(d))$$

This is significant because line bundles and their global sections control the geometry of morphisms *from* projective space: a morphism $\mathbb{P}^n_R \to \mathbb{P}^m_R$ is determined by a choice of $m + 1$ global sections of a line bundle on $\mathbb{P}^n_R$ (with no common vanishing), a correspondence that will be made precise in section 5.3.3. In this way, the graded ring S – despite not being the ring of global functions – encodes the "embedding data" of projective space.

The reader ought to remember that if $f(x_{0,i}, \ldots, x_{n,i})$ is a homogeneous polynomial of degree d on $D(x_i)$, then when converting coordinates to $D(x_j)$ we get:

$$f(x_{0,i}, \ldots, x_{n,i}) = x_{j,i}^{\deg f} f(x_{0,j}, \ldots, x_{n,j})$$

Example 2.3.7: Projective Scheme Homogeneous Polynomial Cut

There are many examples in [**NateClassicalAlgGeo**], and hence we shall just put a few as a refresher.

1. Take $C = V(x^2 + y^2 - z^2) \subseteq \mathbb{P}^2_k$ for an algebraically closed field k. Then if $z = 0$, we would get $[(0, 0, 0)] \in C$, however this point doesn't exist in $\mathbb{P}^n_k$. Hence consider $z \neq 0$. Then

for each solution $[(a, b, c)] = [(\lambda a, \lambda b, \lambda c)] \in C$. In k^3, the solutions form a cone of points. Loosely visualizing $\mathbb{P}_k^2$ as a sphere (we should glue the antipodal points), we see that this resembles the image:

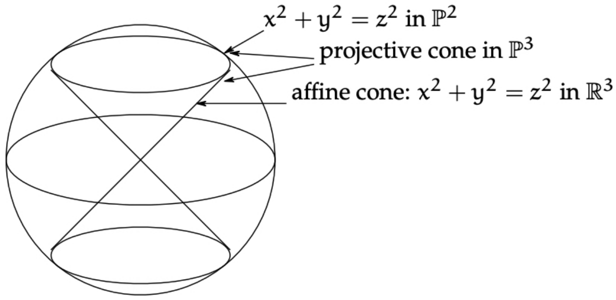


Figure 2.5: Visualization of $V(x^2 + y^2 - z^2)$

Such vanishing sets are called *vanishing cones*.

2. $\mathbb{P}_R^0 = \operatorname{Proj} R[x_0] \cong \operatorname{Spec} R$, hence all 0-dimensional projective schemes are affine. Note then that $\mathbb{P}_k^0 = \{\overline{0}\}$ since $\operatorname{Spec} k$ is a singleton; or this can be verified directly by noticing that the only non-irrelevant homogeneous prime ideal of $k[x]$ is (0), for all other homogeneous prime ideals are of the form (x^n) and hence contain S_+.

3. For an example that is not a hypersurface, take $V(wz - xy, x^2 - wy, y^2 - xz)$. This is called the *twisted cubic curve*. This ideal is prime, and the variety is isomorphic to $\mathbb{P}_k^1$ (hint: consider $k[w, x, y, z] \to k[s, t]$, $(w, x, y, z) \mapsto (s^3, s^2t, st^2, t^3)$).

4. When $R = k$ is a field, the definition of $\mathbb{P}_k^n$ as a scheme matches the classical definition of projective space as a quotient of $k^{n+1} \setminus \{0\}$ by $k^\times$. The points in the scheme correspond exactly to lines through the origin (plus the generic point). However, when R is not a field, $\mathbb{P}_R^n$ behaves like a "compact" object in a very powerful way. Consider the case where R is a Discrete Valuation Ring (DVR), for example $R = k[[t]]$ or $\mathbb{Z}_{(p)}$, with fraction field K. Consider a point $P \in \mathbb{P}_K^n$ (i.e., a point in projective space over the fraction field). We can write this point in coordinates $[f_0 : \cdots : f_n]$ where $f_i \in K$. We might ask: does this point "extend" to a point in $\mathbb{P}_R^n$? In other words, if we have a curve defined for $t \neq 0$, does it have a unique limit as $t \to 0$?

   The answer is yes. Since we are in projective space, we can scale coordinates by any non-zero element.

   (a) Let $v(f_i)$ be the valuation of each coordinate.
   (b) Multiply the entire tuple by t^{-m}, where $m = \min_i(v(f_i))$.
   (c) Now all coordinates are in R, and at least one coordinate is a unit (valuation 0).
   (d) We can now safely set $t = 0$ to find the unique limit point in the special fiber $\mathbb{P}_k^n \subset \mathbb{P}_R^n$.

   For example, if $R = k[[t]]$ and $P = [1/t^2 : 1/t] \in \mathbb{P}_K^1$, we scale by t^2 to get $[1 : t]$. Setting $t = 0$ gives the limit point $[1 : 0]$. This property – that any map from the punctured curve $\operatorname{Spec} K$ extends uniquely to the whole curve $\operatorname{Spec} R$ – is called the *Valuative Criterion for Properness*. It essentially states that projective schemes are "compact" (curves don't run off to infinity without hitting a limit). This implies that $\mathbb{P}^n_R$ is connected in a way affine schemes are not (an affine line with a missing origin cannot be patched!). See section 3.5.4 for more details.

5. In an affine scheme, two polynomials over a field cut out the same vanishing set if they differ by a scalar. This is not the case for homogeneous polynomials cutting out parts of $\mathbb{P}^n$: consider $x^2y = 0$ and $xy^2 = 0$. The problem is that these two are homogenizations of different polynomials that produce the same set. However, as we know from section 2.2.2, schemes remember multiplicity. This behaviour shall later be properly taken into account by saying that two polynomials shall cut out different *closed subschemes* (rather than the same *closed subset*). With this addendum, it shall once again be the case that two polynomials cut out the same closed subscheme if and only if they differ by a scalar multiple, as closed subschemes shall essentially correspond to polynomial cut-outs (as sets) that have sheaves remembering multiplicity information.

Proposition 2.3.8: Irreducibility of classical projective scheme

Let R be a domain. Then the scheme $\mathbb{P}^n_R$ is irreducible and is of finite type over R.

As irreducible spaces are connected, $\mathbb{P}^n_k$ is connected.

***Proof*:**
Since R is a domain, $S = R[x_0, \dots, x_n]$ is a domain, so the zero ideal (0) is a homogeneous prime not containing S_+. Hence $(0) \in \operatorname{Proj} S$ is a point. For each i, $x_i \notin (0)$, so $(0) \in D_+(x_i)$ for every i. Since $\{D_+(x_i)\}_{i=0}^n$ covers $\operatorname{Proj} S$, the generic point (0) lies in every basic open of the standard cover. As any nonempty open set contains some $D_+(x_i)$, it contains (0), so $\{(0)\}$ is dense and $\mathbb{P}^n_R$ is irreducible. Finite type over R follows from each chart $D_+(x_i) \cong \mathbb{A}^n_R = \operatorname{Spec} R[x_1, \dots, x_n]$ being of finite type over R, with finitely many charts.

To conclude this section, let us formally define the affine and projective cone:

Definition 2.3.9: Affine Cone

Let $S_\bullet$ be a graded ring. Then the affine cone of $\operatorname{Proj} S_\bullet$ is $\operatorname{Spec}(S_\bullet)$.

Note that $\operatorname{Spec}(S_\bullet)$ depends on $S_\bullet$, not just $\operatorname{Proj}(S_\bullet)$. Think of the affine cone as eliminating the points at infinity and adding the origin. In fact, given a field k, there is a natural map $\operatorname{Spec} S_\bullet \setminus V(S_+) \to \operatorname{Proj} S_\bullet$ (more generally, if $S_\bullet$ is a finitely generated graded ring over a base ring R, there is a natural morphism).

Definition 2.3.10: Projective Cone

Take $\operatorname{Spec} S_\bullet$. Then the projective cone or projective completion is $\operatorname{Proj}(S_\bullet[t])$ for some indeterminate t of degree 1.

Take for example $\operatorname{Proj}(k[x, y, z]/(z^2 - x^2 - y^2))$. Then the projective cone is $\operatorname{Proj}(k[x, y, z, t]/(z^2 - x^2 - y^2))$. The vanishing locus $V_+(t)$ returns the original projective scheme, and taking the complement (the distinguished open set $D_+(t)$) gives $\operatorname{Spec}(S_\bullet)$. This construction can be thought of as the opposite of the affine cone, where we attach the points at infinity of a conic curve.

The Non-Uniqueness of Graded Rings

Before moving on, we must highlight a fundamental difference between the Spec and Proj constructions. The spectrum functor provides a perfect anti-equivalence between commutative rings and affine schemes; if $\operatorname{Spec} A \cong \operatorname{Spec} B$ as schemes, then $A \cong B$ as rings.

The Proj construction completely fails this property. Two wildly different graded rings can produce the exact same projective scheme. For example, let $S_\bullet = k[x, y]$ and let $T_\bullet = k[x^2, xy, y^2] \cong k[u, v, w]/(uw - v^2)$. As standard rings, these are not isomorphic: $S_\bullet$ is a Unique Factorization Domain, while $T_\bullet$ is not (since $uw = v^2$ represents two distinct factorizations into irreducibles). Geometrically, their affine cones are drastically different: $\operatorname{Spec} S_\bullet$ is the smooth plane $\mathbb{A}^2_k$, whereas $\operatorname{Spec} T_\bullet$ is a quadric cone which is singular at the origin.

However, because Proj effectively removes the origin (by restricting to homogeneous prime ideals not containing S_+) and takes a quotient by the grading, these differences vanish: $\operatorname{Proj} S_\bullet \cong \operatorname{Proj} T_\bullet \cong \mathbb{P}^1_k$ (proven in theorem 3.1.35).

In general, the Proj construction only determines a graded ring "up to truncation at finite degrees". If two graded rings $S_\bullet$ and $T_\bullet$ share the exact same graded pieces for all degrees $d \geq N$ for some large integer N, they will yield isomorphic projective schemes. We will formalize exactly how to recover the "canonical" coordinate ring of a projective scheme using twisted sheaves in section 5.3.

Proj-like Functors for other schemes?

here

2.4 Properties of Schemes

In this section, we shall cover various different properties a scheme may have. In the process, we shall see if these properties can be checked affine-locally or stalk-locally Many of the topological properties inquired about affine schemes can be asked about for general schemes. For example, general schemes need not be quasicompact, while all affine schemes are quasicompact. Properties that have been explored already are:

1. (dis)connected: In the affine case, this corresponds to the existence of non-trivial idempotent elements, for if $e \in R$ is such an element then $R \cong eR \times (1 - e)R$, and so the spectrum is a disjoint union. For the general case of schemes, it is usually given by the disjoint union of known schemes.

2. (ir)reducible: In the affine case, this corresponds to the ring having *no zero divisors*, i.e. an integral domain. Proposition 2.4.19 shall characterize irreducible schemes[23].

   Recall that an equivalent conditions to irreducibility is that every non-empty open subset is dense (as the reader should verify).

3. quasicompact: Essentially all schemes we shall work with will be quasicompact (with the "trivial" exception being the infinite disjoint union of schemes). Most of the time, non-quasicompact schemes will come from some infinite gluing.

[23]with the reducedness condition insuring no "fuzz" in the sheaf

These are global topological properties, hence they cannot be checked affine- or stalk-locally (as the reader should verify with some counter-examples).

Finiteness Conditions

The most common types of rings in commutative algebra are Noetherian rings. The geometric equivalent is the following:

Definition 2.4.1: Noetherian Scheme

Let X be a scheme. If X can be covered by affine open sets where each ring R_i of $\operatorname{Spec}(R_i)$ is Noetherian, then X is said to be a *locally Noetherian Scheme*. If X is also quasicompact, then X is said to be a *Noetherian Scheme*.

All coordinate rings are Noetherian, and hence we may think of varieties as examples of Noetherian schemes[24]. We can think of the Noetherian condition as insuring that the open affine don't have "difficult" infinite behavior. Most examples which involve non-Noetherian rings involve infinitely many variables (ex. $k[x_1, x_2, ...]$, $k[x, y, y/x, y/x^2, ...]$ and so forth). In part I; examples of non-Noetherian topological spaces we shall encounter will be the fibered product of Noetherian rings (see ref:HERE), and the special product topology put on the ring of adèles. In general, we shall see that there are many results that do not require finite generation, and hence it is worth keeping track of it when this condition is necessary.

One important consequence of being Noetherian is the existence of finitely many irreducible components:

Lemma 2.4.2: Finitely Many Components

Let X be a Noetherian scheme. Then X has finitely many irreducible components.

***Proof*:**
As X is Noetherian, it is locally Noetherian, so let it be covered be Noetherian affine open schemes. As it is quasicompact, we may take a finite subcover so $X = \bigcup_i^n \operatorname{Spec} R_i$. As each R_i is Noetherian, it can have only finitely many irreducible components (check this), and hence their union can have only finitely many irreducible components, completing the proof.

This is not true if the ring is not Noetherian: the spectrum of the ring

$$\varinjlim_{n\to\infty} \frac{k[x_i \,|\, 1 \le i \le n]}{\prod_i^n x_i}$$

has infinitely many components, but it is a quasicompact scheme (as it is affine).

Let us next turn to an important property of quasicompact schemes. When working with affine schemes, we always have that there exists closed points corresponding to the maximal ideals of the ring. When working over general schemes, there may be no closed points[25]. Fortunately, all quasicompact scheme have closed points, and even better:

[24] Once we introduce theorem 3.7.1 this will be made precise
[25] see https://math.stanford.edu/~vakil/files/schwede03.pdf

Proposition 2.4.3: Quasicompact Schemes and Closed Points

Let X be a quasicompact scheme. Then X contains closed points. Furthermore, the closure of any non-closed point contains closed points.

***Proof*:**
First, if X is affine, it certainly contains closed points as $\mathcal{O}_X(X) = R$ contain at least one maximal ideal. Furthermore, the closure of any non-closed point would contain maximal ideal as any prime ideal is contained in a maximal ideal.

Now, let X be some quasicompact scheme. Let $X = \bigcup_i^n U_i$ be a finite covering of affine open subsets. Take any closed point $\mathfrak{p}_1 \in U_1$. If it is closed in X we're done. If not, take its closure $\overline{\mathfrak{p}_1}$ in X. As $\mathfrak{p}_1$ is not closed in X, its closure contains more than $\mathfrak{p}_1$, call this point $\mathfrak{p}_2$, and without loss of generality say it is in U_2. If $\mathfrak{p}_2$ is closed in X, we are done. If not, take $\overline{\mathfrak{p}_2}$ in X. Its closure must contain a point $\mathfrak{p}_3$ that is not in U_2 or U_1 (as $\mathfrak{p}_3$ would be in the closure of $\mathfrak{p}_1$ or $\mathfrak{p}_2$ in U_1 and U_2 respectively). Continue this process until a point is closed, or if we get to $\mathfrak{p}_n \in U_n$, then the closure of this point can't contain any other point in $U_1, \ldots, U_n$, making $\mathfrak{p}_n$ closed and completing the proof.
Note that $\mathfrak{p}_n$ is in the closure of $\mathfrak{p}_1$ in X : by a topological argument it is easy to show that:

$$\overline{\mathfrak{p}_1} \supseteq \overline{\mathfrak{p}_2} \supseteq \cdots \supseteq \overline{\mathfrak{p}_n} = \mathfrak{p}_n$$

and hence the closure of any non-closed point contains a closed point.

Note that this does not mean that the closed points are dense, that is it does not imply that the closure of the collection of closed point is the entire space. Such schemes are called Jacobson schemes and are related to Jacobson rings and Jacobson radicals (see [**NateClassicalAlgGeo**]). Once we introduce (abstract) varieties, we shall see that the closed points are dense for varieties. The following gives a good set of (affine) schemes whose closed points are dense:

Lemma 2.4.4: Dense Closed Points

Let A be a finitely generated k-algebra. Then the closed points of $\operatorname{Spec} A$ are dense

***Proof*:**
It suffices show that for each $f \in A$ where $D(f) \neq \emptyset$ (i.e. f is not nilpotent, see exercise ref:HERE), $D(f)$ $\operatorname{Spec}(A_f) \cong D(f)$, and A_f certainly contains maximal ideals, there is a maximal ideal $\mathfrak{p}_f \subseteq A_f$ which corresponds to a (at least) prime ideal $\mathfrak{p} \subseteq A$ not containing f. If we show that $A/\mathfrak{p}$ is a field, we're done. By Zariski's lemma, F is a finitely generated k-algebra if it is a finitely generated k-module, hence it suffice to show that $A/\mathfrak{p}$ is a finitely generated k-vector space. As $(A/\mathfrak{p})_f \cong A_f/\mathfrak{p}_f$, if we show that $A_f/\mathfrak{p}_f$ is a finite dimensional k-vector space, we can lift to a finite basis for $A/\mathfrak{p}$.

Now, as $A_f = A[1/f]$, it is a finitely generated k-algebra and $\mathfrak{p}_f$ is maximal, $A_f/\mathfrak{p}_f$ is a finite dimensional k-vector space by Zariski. But then we get the desired conclusion

We can also show it in the following more "straightforward" with a more subtle application of Nullstellensatz. For the sake of contradiction, suppose $D(f)$ has no maximal ideals. Then for each maximal ideal $\mathfrak{m} \subseteq A$, $f \in \mathfrak{m}$. Thus:

$$f \in \bigcap_{\mathfrak{m} \subseteq A} \mathfrak{m} \stackrel{!}{=} \sqrt{(0)}$$

where $\stackrel{!}{=}$ comes as a consequence of the Nullstellensatz (key here is the finitely generation as a k-algebra). But then f is nilpotent, and so $D(f) = \emptyset$. Hence, any nonempty set contains a closed point, completing the proof.

Recall in proposition 2.3.18 it was mentioned that invertibility can be determined if a function is non-zero at every point, however a function being zero at all points requires more care to study. Part of this is because this is true for quasicompact schemes

Proposition 2.4.5: Quasicompact Schemes and Vanishing Function

Let X be a quasicompact scheme. Then if $f \in \mathcal{O}_X(X)$ vanishes at all points in X, then there exists an n such that $f^n = 0$.

***Proof*:**
Say $f([\mathfrak{p}]) = 0$ so that $f \in \mathfrak{p}$ for all $\mathfrak{p}$. Then on $U_i \cong \operatorname{Spec} R_i \subseteq X, f \in \sqrt{R_i}$, implying $f^{n_i} = 0$ for some n. Cover X with these open sets, and by Quasicompactness we only require finitely many such sets, where in each set we have an f_i and n_i such that $f_i^{n_i} = 0$. As X is a scheme, on intersection $U_i \cap U_j$ we have $g_{i,j} = f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ and as ring homomorphisms must map 0 to 0, $g_{i,j}^{\max n_i, n_j} = 0$. Take $n = \max_i n_i$ so that for each intersection $U_i \cap U_j$:

$$f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j} = g_{i,j} \quad g_{i,j}^n = 0$$

Now, as these agree on intersections and cover X, by definition there must exist a function $f \in \mathcal{O}_X(X)$ such that $f|_{U_i} = f_i$. Then as $\operatorname{res}_{X,U_i}(f^n) = \operatorname{res}_{X,U_i}(f)^n = f_i^n f_i^{n_i n'} = 0 f^{n'} = 0$, we have:

$$(f^n)|_{U_i} = \left(f|_{U_i}\right)^n = 0$$

which implies $(f^n)|_{U_i} = 0 \in R_i$. Then by guability, it must be that $f^n = 0$, completing the proof.

This is not necessarily true for a general scheme, namely since we can have nilpotence of every degree:

Example 2.4.6: Non-vanishing Function On Scheme
Take

$$X = \coprod_{n \geq 1} \operatorname{Spec}\left(k[x]/\left(x^n\right)\right)$$

Then the global function ϵ is zero every-where, namely there is only one point to evaluate to, but essentially by definition there cannot exist an n such that $\epsilon^n = 0$.

Lemma 2.4.7: Projective Scheme is Noetherian

Let $X = \mathbb{P}^n_{\mathbb{Z}}$ or $\mathbb{P}^n_k$. Then X is Noetherian.

***Proof*:**
exercise.

To conclude this section, we shall define one more type of scheme here due to the fact that all affine and projective schemes have this property, however it shall not be of great use to us until chapter ref:HERE. The following establishes some finiteness conditions on the intersection of important open subsets:

Definition 2.4.8: Quasi-separated

Let X be a topological space. Then X is *quasiseparated* if the intersection of any two quasicompact open subsets is quasicompact.

Proposition 2.4.9: Characterizing Quasi-separated Schemes

Let X be scheme. Then X is quasi-separated if and only if the intersection of any two affine open subsets is a finite union of affine open subsets

***Proof*:**
This is very similar to a compactness argument and is left as an exercise

Proposition 2.4.10: Affine Scheme and Quasiseparatedned

Let $X = \operatorname{Spec} R$ be an affine scheme. Then X is quasiseparated.

***Proof*:**
The intersection of affine open subsets is affine (example ref:HERE shows that $U \cap V \cong \operatorname{Spec}(S \otimes_R S')$ for $U, V \subseteq \operatorname{Spec} R$ and $U \cong \operatorname{Spec} S$, $V \cong \operatorname{Spec} S'$. Then by quasi-compactness this can be covered by finitely many distinguished open subsets.

From this, we can reduce to the case of affine opens and the rest follows quickly.

A large family of schemes that are quasi-separated and quasicompact are projective R-schemes, and as we shall see many schemes we shall be working with can be embedded into a projective R-scheme, and hence these two finiteness properties will be very common. There do exist non quasi-separated schemes:

Example 2.4.11: Non-quasiseparated Scheme
Let $X = \operatorname{Spec}(k[x_1, x_2, \ldots])$ and let $U = X \backslash [\mathfrak{m}]$ where $\mathfrak{m}$ is the maximal ideal $(x_1, x_2, \ldots)$. Take two copies of X and glue along U to create a double-origin scheme, namely the infinite version of the double-origin scheme seen in example 2.15. Show this is not quasiseparated.

Lemma 2.4.12: Projective Scheme Quasicompact And Quasiseperated

Let X be a projective R-scheme. Then X is quasicompact and quasiseparated

Note that R need not be Noetherian, hence any projective $\mathbb{P}^n_R$ scheme is quasicompact.

***Proof*:**
hint: affine communication lemma.

Exercise 2.4.1

1. Let X be a scheme. Show that X is a locally Noetherian scheme if and only if every open affine subset corresponds to a Noetherian ring.
2. Show that $\prod_i^\infty \operatorname{Spec} R_i$ properly embeds in $\operatorname{Spec}\left(\prod_i^\infty R_i\right)$. More precisely, the right hand side is (significantly) larger than the left hand side.

2.4.1 Reducedness and Integrality

The next important property that general rings have that coordinate rings for varieties don't have is the presence of zero-divisors and nilpotence. A ring with zero-divisors can be decomposed into a product to two rings and corresponds to a function vanishing on separate irreducible components (there must be more than 1), while a ring with nilpotence have effect on the sheaf of a ring as discussed in section 2.2.2. A ring with no nilpotent elements is said to be *reduced* (note that it may have zero-divisors[26]). A scheme that has this property of being reduced for every ring of sections is given a name:

Definition 2.4.13: Reduced Scheme

Let X be a scheme. Then X is a *reduced scheme* if $\mathcal{O}_X(U)$ is reduced for every open set U of X

Naturally, on a reduced scheme by proposition 2.4.5 a global section that evaluates to zero at every point will be the zero-section.

Proposition 2.4.14: Reducedness Is Stalk-local

Let X be a scheme. Then X is reduced if and only if non of the stalks have nonzero nilpotents. As a consequence, if $f, g \in \mathcal{O}_X(U)$ are two functions and agree on all points, then $f = g$

***Proof*:**
if X is a reduced scheme, then the localization will preserve reducedness, and hence each stalk is reduced. For the converse, we can show the contrapositive and assume there exists a $f \in \mathcal{O}_X(U)$ such that $f^n = 0$. Note that f cannot be a unit (as the reader could verify). Then some affine open subset $\operatorname{Spec} R \subseteq U$ we have $f|_U^n = f^n|_U = 0|_U$, showing that f is still nilpotent in $\mathcal{O}_X(\operatorname{Spec} R)$. Then it is a non-unit element in $\operatorname{Spec} R$, and hence belongs in some maximal ideal

[26] if D is the set of all zero-divisors of a reduced ring, then it is the union of all minimal ideals

$f \in \mathfrak{m}$. Then localizing at $[\mathfrak{m}]$, it is easy to see that f remains to be a nilpotent element, showing that not all stalks are reduced.

For the next assertion, this will be a simpler version of proposition 2.4.5 as we do not need to keep track of any exponents, and hence the Quasicompactness condition is not needed! Assume $f, g \in \mathcal{O}_X(U)$ such that $f(x) = g(x)$ for all $x \in U$. We want to show that $f = g$. Take $h = f - g$. Now, by definition $h(x) = f(x) - g(x) = 0$, so h must be contained in every prime ideal on every affine open subset of U. As $\mathcal{O}_X(U)$ is reduced, $\sqrt{(0)} = (0)$ on each affine open, and so $h|_{\operatorname{Spec} R} = 0$ for each element in the affine open covering, which by the gluging gives us $h = 0$, completing the proof.

Lemma 2.4.15: Reducedness For Affine And Projective Schemes

Let R be a reduced ring. Then $\operatorname{Spec}(R)$ is reduced. Hence, $\mathbb{A}^n$ and $\mathbb{P}^n$ are reduced.

***Proof*:**
exercise.

Proposition 2.4.16: Reducedness On Quasicompact Schemes

Let X be a quasicompact scheme. Then X is reduced if and only if it is reduced at closed points

***Proof*:**
Assume it is reduced at all closed points. As X is quasi-compact, the closure of each non-closed point contains a closed point; say $Y \in X$ is a non-closed and $x \in X$ is a closed point contained in its closure. Then $\mathcal{O}_{X,Y}$ is the localization of $\mathcal{O}_{X,x}$. As localization preserves reducedness, $\mathcal{O}_{X,Y}$ must be reduced, completing the proof.

The above showed that reducedness for quasicompact schemes is a "closed-point" local condition. It is not an open set local condition. Furthermore, it is not enough to check the global sections:

Example 2.4.17: Reducedness Edge Case

1. The scheme given by:
$$X = \operatorname{Spec} \frac{\mathbb{C}[x, y_1, y_2, \ldots]}{(y_1^2, y_2^2, \ldots, (x-1)y-1, (x-2)y_2, \ldots)}$$
is nonreduced at the closed points corresponding to the positive integers. The complement of this set is not Zariski open.
The key is that if X is also locally Noetherian, then reducedness is a locally open condition.
2. Another mistake that is easy to make: let X be scheme where the global sections is reduced. This doesn't imply X is reduced (take the scheme cut out by $x^2 = 0$ in $\mathbb{P}^2_k$; note that the global section will be k, but X will be non-reduced).

One of the most important type of reduced ring is an integral domain. If a scheme is an integral domain for each section, it is given a name:

Definition 2.4.18: Integral Scheme

Let X be a scheme. Then X is *integral* if it is nonempty and $\mathcal{O}_X(U)$ is an integral domain for every nonempty open subset $U \subseteq X$.

Recall that the product of integral domain is not an integral domain, or geometrically, the disjoint union of integral scheme need not be integral. This global behavior cannot be captured through stalks, as the reader should verify, and hence integrality is not stalk-local. Integral schemes can be thought of as schemes that are in "one piece" and have "no fuzz":

Proposition 2.4.19: Integral, then Irreducible and Reduced

Let X be a scheme. Then if X is an integral scheme, it is irreducible and reduced

It is in fact an if and only if, which we shall show in proposition 2.4.25.

***Proof*:**
If X is integral, it is certainly reduced as each ring of local sections is integral (and hence reduced). To show it is irreducible, assume $X = X_1 \cup X_2$, X_1, X_2 closed. For the sake of contradiction, let's say these sets don't contain each other. Then there exists open subsets $U \subseteq X_1, V \subseteq X_2$ such that $U \cap V = \emptyset$. Indeed, given $X_1 \not\subseteq X_2$, then here exists a point $p \in X_1 \setminus X_2$. Let

$$U = (X \setminus X_2) \cap X_1$$

Then U is open in X_1, and hence by construction in X. Similarly, define $V = (X \setminus X_1) \cap X_2$. Then $U \cap V = \emptyset$.

Then, by exercise ref:HERE, we have by an almost direct consequence of the sheaf axioms that:

$$\mathcal{O}_X(U \coprod V) \cong \mathcal{O}_X(U) \times \mathcal{O}_X(V)$$

But then $\mathcal{O}_X(U \coprod V)$ certainly contains zero-divisors, and hence cannot be integral.

An important property that integral domains is the ability to localize to obtain a fractional field. Localizing at an integral schemes generic point shall give us a "global" function field in which all local sections can embed, similarity to varieties! We first prove that an integral scheme has a unique generic point. Note the proof can be done if we replace integrality with irreducibility, that is we don't need every ring of local sections to be integral, see exercise ref:HERE.

Lemma 2.4.20: Irreducible Schemes and Generic Points

Let X be an integral scheme. Then for every irreducible closed subset $Z \subseteq X$, there exists a unique generic point η as $\overline{\eta} = Z$. In particular, as X is irreducible, then there exists a point η such that $\overline{\eta} = X$.

***Proof*:**
Let $Z \subseteq X$ be an irreducible subscheme. Then $Z = \bigcup_i U_i$ is an affine open cover. As $U_i = \operatorname{Spec} R_i$ is open and Z is irreducible, $\overline{U_i} = Z$ (exercise in topology). Now, as X is integral, $[(0_i)]$ is the generic point of U_i as (0_i) is prime. Let $\eta_i = [(0_i)]$. Then:

$$\overline{\eta_i} = Z$$

This shows a generic point exists, let's shows the generic point is unique. Say η_1, η_2 are generic points of Z so that $\overline{\eta_1} = \overline{\eta_2} = Z$. Take any affine open $U_i \subseteq Z$. Then we must have $\eta_1, \eta_2 \in U_i$. If they weren't, then $\eta_i \in Z \setminus U_i$ would be a closed and hence there exists a smaller closed set containing η_i which would imply $\overline{\eta_i} = Z \setminus U_i$, a contradiction. Thus, we get $\eta_1, \eta_2 \in U_i$. Hence, in the subspace topology

$$\overline{\eta_1} = \overline{\eta_2} = U_i$$

Showing that these are also the generic points of U_i. But as U_i is affine and integral, there can only exist a single generic point, hence $\eta_1 = \eta_2$.

In the special case where X is itself irreducible, we can conclude it has a unique generic point η, as we sought to show.

This is a purely topological fact, and hence can also be proven using only irreducibility, see exercise ref:HERE.

The existence was given by reducing to an affine open and "using" its generic point. Using this, we can quickly calculate the stalk at the generic point of an integral scheme. We give this stalk a name:

Definition 2.4.21: Rational Field

Let X be an integral scheme. Then the function field is the collection

$$\mathcal{O}_{X,\eta} = R_{(0)} = \operatorname{Frac}(R)$$

for any R given by $\operatorname{Spec}(R) \subseteq X$, and it is often denoted $\kappa(X)$. If X is a k-scheme, then it is denoted $k(X)$.

Thus, integral schemes have a similar property to that of classical affine and projective varieties, namely that their sections over different open sets can be considered as subsets of a single ring! Gluing is easy as we can see that two elements are glued if and only if they are the same element of $\kappa(X)$.

Proposition 2.4.22: Integral Scheme: Restriction Maps and Function Field

Let X be a scheme that is irreducible and reduced. If $U \subseteq V$ where $V \neq \emptyset$, then

$$\operatorname{res}_{V,U} : \mathcal{O}_X(V) \hookrightarrow \mathcal{O}_X(U)$$

is an inclusion.

***Proof*:**
As X is reduced, its sections are all determined by their value on the points, and hence can be treated as function (see example 2.4.23 on why this proof fails otherwise). Take $U \subseteq V \subseteq X$, $f, g \in \mathcal{O}_X(U)$ and assume $f|_U = g|_U$. We want to show that $f|_V = g|_V$. Define:

$$Z = \{x \in V \mid f(x) = g(x)\}$$

Z is closed in V as it is the zero set of $f - g$. By definition $U \subseteq Z$, $Z \subseteq V$, and U is open in V. As X is integral, the open subset V is irreducible. But then U is *dense* in V. As Z is closed in V and contains U, it must be that $Z = V$, but then

$$f|_V = g|_V$$

as we sought to show

Note that we require *both* the irreducibility and the reducedness, having *only* irreducibility is not enough

Example 2.4.23: Non-injective Non-reduced Scheme
Let $R = \frac{k[x,y]}{(xy,y^2)}$ so that $\operatorname{Spec} R$ is the x axis there is a bit of fluff in direction of the y-axis at the point 0. Then given $D(x) \subseteq \operatorname{Spec} R$, the map $\text{res} : \mathcal{O}_{\operatorname{Spec} R}(\operatorname{Spec} R) \to \mathcal{O}_{\operatorname{Spec} R}(D(x))$ maps $y \mapsto 0$ as

$$y \mapsto \frac{y}{1} = \frac{xy}{x} = \frac{0}{x} = 0$$

Corollary 2.4.24: Integral Scheme: Generic Point

Let X be an integral scheme. Then there exists a generic point η such that given any choice of $\operatorname{Spec}(R) \subseteq X$, there is always an inclusion:

$$\mathcal{O}_X(U) \hookrightarrow \mathcal{O}_{X,\eta} \cong \kappa(R)$$

***Proof*:**
By definition:

$$\mathcal{O}_{X,\eta} = \varinjlim_{\eta \in U} \mathcal{O}_X(U)$$

and as we saw lemma 2.4.20 the generic point is in each open set. By proposition 2.4.22, each map in the commutative diagram are injective, hence the map $\mathcal{O}_X(U)$ is injective into $\mathcal{O}_{X,\eta}$.

Proposition 2.4.25: Irreducible and Reduced, then Integral

Let X be a scheme. If X is irreducible and reduced, it is integral

***Proof*:**
Let X be irreducible and reduced. Then any open subscheme is irreducible and reduced (the irreducibility is proved topologically, and the reducedness can be deduced by the stalks). In

particular, any affine open subscheme $\operatorname{Spec} R$ must be irreducible and reduced. Certainly R must then be reduced, hence to show that $\operatorname{Spec} R$ is integral consider $xy \in R$ such that $xy = 0$. Then

$$\operatorname{Spec} R = V(0) = V(xy) = V(x) \cup V(y)$$

But that contradicts irreducedness. Hence, all ring of sections of affine open subsets are integral. For general open $U \subseteq X$, we must have an affine open $\operatorname{Spec} R \subseteq U$, and hence a restriction $\mathcal{O}_X(U) \to \mathcal{O}_X(\operatorname{Spec} R)$. By proposition 2.4.22, this map is injective, and hence $\mathcal{O}_X(U)$ is a subring of an integral domain, making it an integral domain. Hence all rings of local sections are integral domains, completing the proof.

If we have the added condition of being Noetherian, then we only need to check that it is connected and that all its stalks are integral domains:

Proposition 2.4.26: Characterizing Integrality in Noetherian Schemes

Let X be a Noetherian Scheme. Then X is integral if and only if X is nonempty, connected, and all stalks $\mathcal{O}_{X,\mathfrak{p}}$ are integral domains.

Intuitively, the stalk at the intersection of irreducible components will "remember" the generic points of the respective irreducible component.

***Proof*:**
By proposition 2.4.19 and proposition 2.4.25, X is integral if and only if it is reduced and irreducible. Then X is certainly non-empty, connected, and the stalks are integral (the localization of integral domains is integral).

Conversely, assume X is nonempty, connected, with integral stalks. First of all, as the stalks are integral, they are reduced. We must show that X is irreducible. As X is Noetherian, by lemma 2.4.2 X can be decomposed into finitely many irreducible components:

$$X = X_1 \cup X_2 \cup \cdots \cup X_n$$

As X is connected, for each i, j we must have $X_i \cap X_j \neq \emptyset$. Take an affine open subset $\operatorname{Spec} R \subseteq X$ and:

$$\operatorname{Spec} R = \bigcup_i^n (X_i \cap \operatorname{Spec} R)$$

Then as each X_i is irreducible, it is a topological argument to see that the sets $(X_i \cap \operatorname{Spec} R)$ form an irreducible decomposition of $\operatorname{Spec} R$. Hence, they must each have a generic point that corresponds to a minimal prime ideal. Hence, if the X_i are distinct, R_i must contains at least two distinct minimal prime ideals. Then if $[\mathfrak{p}] \in X_i \cap X_j$, $\mathcal{O}_{X,[\mathfrak{p}]} = (R_i)_{\mathfrak{p}}$ contains two distinct minimal prime ideals. But integral domains could only contain a single minimal prime ideal (namely the one associated to the generic point $[(0)]$), and so this stalk cannot be integral. Thus, it must be that $X_i = X_j$ for all i, j, showing that X is irreducible and completing the proof.

Exercise 2.4.2

1. Show that every point $x \in X$ in a topological space must be contains in an irreducible component (you will use Zorn's Lemma).

2. Show that connected components (maximal connected subset) are closed. Show that the connected components of Spec $(\prod_i^{\infty} \mathbb{F}_2)$ are not open.
3. Show that $D(f)$ is empty if and only if f is nilpotent. Show that if X is an integral scheme, then $D(f)$ is always nonempty and contains the generic point.
4. If $f \in R$ is nilpotent, then R_f is the zero-ring.
5. Let X be an irreducible scheme. Show that there is a unique generic point η such that $\overline{\eta} = X$ (note that lack of reducedness). More generally, show that if $Z \subseteq X$ is a (nonempty) irreducible subset, there is a unique generic point η such that $\overline{\eta} = Z$
6. Let X be a reduced scheme and $Y \hookrightarrow X$ a closed immersion where $\iota(Y) = X$. Then ι is an isomorphism.

2.4.2 Normality and Factoriality

Recall that a normal ring R has the property that it an integral domain that is is integrally closed in it's field of fractions, $\overline{R}^{\mathrm{Frac}(R)} = R$. Due to their importance in number theory, we shall take the time to develop the theory of Schemes for which all the rings of sections are normal. These shall correspond to schemes that have some nicer behaviour for their singularities. It can be thought of as a condition weaker than smoothness (smoothness shall be called regular in section REF:HERE) as normality shall still allow for singularities, however these singularities shall be more tame. In particular, the reader can intuitively have the following geometric picture in mind for subschemes of Normal Schemes:

1. In codimension 1, all singularities are resolved, meaning it is "smooth" in codim 1
2. In codimension ≥ 2, being normal means being nice enough so that rational functions defined on the compliment of the singularities can be extended to the singularity (i.e. Algebraic Hartogs's lemma applies)

Definition 2.4.27: Normal Scheme

Let X be a scheme. Then X is said to be *normal* if every stalk $\mathcal{O}_{X,\mathfrak{p}}$ is normal domains.

Proposition 2.4.28: Normal Scheme Stalk Local

Let X be a scheme. Then X is a normal scheme if and only if $\mathcal{O}_{X,\mathfrak{p}}$ is normal for each $\mathfrak{p} \in X$

***Proof*:**
Recall (or reprove) that normality is preserved when localizing (hint: recall the proof that a UFD is normal)

Note that reducedness was stalk-local, and so it is immediate that normal schemes are reduced. As being integrally closed is well-behaved under localization (recall the proof that a UFD is normal), for an if R is normal then Spec (R) is a normal scheme. If each stalk is a UFD, then it is immediately a Normal scheme. Let's give a name to such a scheme

Definition 2.4.29: Factorial Scheme

Let X be a scheme. Then X is said to be *normal* if every stalk $\mathcal{O}_{X,\mathfrak{p}}$ is a UFD.

If R is a UFD, then $\operatorname{Spec} R$ is factorial (by the above reasoning), however the converse need be true! Namely, we may have factorial schemes where the stalks are all UFD's but the global sections do not form a UFD. A quintessential example will be the spectra of elliptic curves.

Example 2.4.30: Normal Schemes

1. The schemes $\mathbb{A}^n_k$, $\mathbb{P}^n_k$ and $\operatorname{Spec}\mathbb{Z}$ are normal. If R is normal, by definition $\operatorname{Spec} R$ is normal.
2. Starting on p.164 section 5.4.8, lots and lots of good examples

Exercise 2.4.3

1. Find the structure for the following rings:
   a) $\operatorname{Spec}\mathbb{C}[x]/(x^3 - x^2)$
   b) $\operatorname{Spec}\mathbb{C}[x]/(x^2 - d)$
   c) $\operatorname{Spec}\mathbb{C}[x]/(x^2 + 1)$
2. Let X be a quasicompact scheme. Show that the closure of each point contains a closed point (this is not true for non quasicompact schemes in general)
3. Let X be a quasicompact scheme. Suppose f vanishes at all point of X. Then there exists some N st $f^n = 0$. Show that this fails when X is not quasicompact (hint: take infinite disjoint unions of $\operatorname{Spec}(R_n)$ where $R_n := k[x]/(x^n)$)
4. Let X be a quasicompact scheme. Then it suffices to cheek reducedness at closed points (show that any nonreduced point has a nonreduced closed point in its closure)
5. Show that reducedness is not in general an open condition by considering:
$$X = \operatorname{Spec}\left(\frac{\mathbb{C}[x, y_1, y_2, ...,}{(y_1^2, y_2^2, y_3^3, ..., (x-1)y_1, (x-2)y_2, (x-3)y_3), ...}\right)$$
   and $Y = \operatorname{Spec}[x]$. Show the nonreduced points of X are the closed points corresponding to the positive integers. The compliment of this set is *not* Zariski open.
6. Let X be an integral scheme. Show that if $\emptyset \neq V \subseteq U$, then the map $\operatorname{res}_{U,V} : \mathcal{O}_X(U) \to \mathcal{O}_X(V)$ is an inclusion. very important, see p.156 under exercise
7. Let X be an integral scheme, γ the generic point, and $\operatorname{Spec}(R) \cong U \subseteq X$ is any nonempty affine open subset of X. Then $\mathcal{O}_X(U) \to \mathcal{O}_{X,\gamma} = K(A)$ is an inclusion. Conclude that gluing is easy: every f_i in U_i for an open cover for U glue together if they are same element in $\operatorname{Frac}(X)$ (recall that $\operatorname{Frac}(X)$ is defined to be $\operatorname{Frac}(R)$ where R is given by $\mathcal{O}_X(\operatorname{Spec}(R))$
8. Show that integrality is not stalk local via the example $\operatorname{Spec}(A)\coprod\operatorname{Spec}(B) = \operatorname{Spec}(A \times B)$.
9. Show that reducedness is an affine local property.

10. Let X be a locally Noetherian Scheme. Then X is quasiseparated.
11. Let X be a Noetherian Scheme. Then it has a finite number of connected components, each a finite union of irreducible components.
12. Show that the ring of sections of a Noetherian Scheme need not be Noetherian.
13. Let X be a scheme of locally finite type over $\mathbb{Z}$ or k. Then X is locally Noetherian.
14. A scheme that is of finite type over any Noetherian ring is Neotherian.
15. Let X be a quasiprojective R-scheme. Then if R is Noetherian, X is a Noetherian scheme, and hence has a finite number of irreducible components.
16. Let X be a locally Noetherian scheme. Then the reduced locus is open. (Vakil p.156)
17. Let $U \subseteq X$ be an open subscheme of a projective R-scheme X. Show that U is locally of finite type R
18. Find a counter-example showing that normal schemes are not in general integral schemes (hint: think of the disjoint union of two normal scheme. For more hint, Vakil p. 162).
19. Let X be a Noetherian scheme. Then X is normal if and only if if is a finite disjoint union of integral normal schemes.

2.5 Associated Points

Associated points capture the "component structure" of a scheme, including both the obvious irreducible components and more subtle "embedded" components arising from nilpotent structure. For a reduced scheme, the associated points are simply the generic points of the irreducible components. For non-reduced schemes, there can be additional associated points that "remember" how components come together or where nilpotent thickening occurs.

The algebraic foundation is the theory of associated primes of modules, which we briefly recall before giving the scheme-theoretic definition.

2.5.1 Associated Primes of Modules

Definition 2.5.1: Associated Prime

Let R be a ring and M an R-module. A prime ideal $\mathfrak{p} \in \operatorname{Spec} R$ is an *associated prime* of M if $\mathfrak{p} = \operatorname{Ann}(m)$ for some $m \in M$. The set of associated primes of M is denoted $\operatorname{Ass}_R(M)$ or simply $\operatorname{Ass}(M)$.

Equivalently, $\mathfrak{p} \in \operatorname{Ass}(M)$ if and only if $R/\mathfrak{p}$ embeds into M as an R-submodule (via $1 \mapsto m$). The condition $\mathfrak{p} = \operatorname{Ann}(m)$ is quite stringent: it is not enough for $\mathfrak{p}$ to annihilate some element; it must be the *full* annihilator.

Example 2.5.2: Associated Primes

1. Let $R = k[x]$ and $M = R$. The only associated prime is (0): if $\mathrm{Ann}(f) = \mathfrak{p}$ for some $f \neq 0$, then $\mathfrak{p} \cdot f = 0$, but $k[x]$ is a domain so $\mathfrak{p} = (0)$.
2. Let $R = k[x, y]$ and $M = R/(xy)$. The associated primes are (x) and (y): we have $\mathrm{Ann}(\bar{x}) = (y)$ and $\mathrm{Ann}(\bar{y}) = (x)$ in the quotient.
3. Let $R = k[x, y]$ and $M = R/(y^2, xy)$. The associated primes are (y) and (x, y): we have $\mathrm{Ann}(\bar{x}) = (y)$, and $\mathrm{Ann}(\bar{y}) = (x, y)$. The prime (x, y) is an *embedded* associated prime (see below).
4. Let $R = \mathbb{Z}$ and $M = \mathbb{Z}/12\mathbb{Z}$. Then $\mathrm{Ass}(M) = \{(2), (3)\}$: we have $\mathrm{Ann}(\bar{4}) = (3)$ and $\mathrm{Ann}(\bar{3}) = (4) \supsetneq (2)$ ... but $\mathrm{Ann}(\bar{6}) = (2)$. Note (2) and (3) are the primes appearing in the primary decomposition $12 = 4 \cdot 3 = 2^2 \cdot 3$.

The key properties of associated primes in the Noetherian setting are:

Proposition 2.5.3: Properties of Associated Primes

Let R be a Noetherian ring and M a finitely generated R-module. Then:

1. $\mathrm{Ass}(M) \neq \emptyset$ if and only if $M \neq 0$.
2. $\mathrm{Ass}(M)$ is finite.
3. the set of zero divisors on M is given by:

$$\bigcup_{\mathfrak{p} \in \mathrm{Ass}(M)} \mathfrak{p} = \{r \in R \mid rm = 0 \text{ for some } m \neq 0\}$$

,

4. The minimal elements of $\mathrm{Ass}(M)$ are exactly the minimal primes of $\mathrm{Supp}(M)$ (equivalently, the minimal primes of $\mathrm{Ann}(M)$).

***Proof*:**
See [11, chapter 20] for the full proof. We sketch the key ideas:

1. If $M \neq 0$, choose $m \neq 0$. The set of ideals $\{\mathrm{Ann}(m') : m' \in Rm,\, m' \neq 0\}$ is nonempty and has a maximal element $\mathrm{Ann}(m_0)$ by the Noetherian condition. One checks this maximal annihilator is prime.
2. Primary decomposition: $0 = \bigcap_{i=1}^n Q_i$ in M gives $\mathrm{Ass}(M) \subseteq \{\sqrt{Q_1}, \ldots, \sqrt{Q_n}\}$. Equality holds for the minimal primes; the others may or may not be associated.
3. If $r \in \mathfrak{p} \in \mathrm{Ass}(M)$, write $\mathfrak{p} = \mathrm{Ann}(m)$; then $rm = 0$ with $m \neq 0$. Conversely, if $rm = 0$ for $m \neq 0$, then $r \in \mathrm{Ann}(m) \subseteq \mathfrak{p}$ for some associated prime $\mathfrak{p}$ containing $\mathrm{Ann}(m)$.
4. Follows from the relationship between associated primes and primary decomposition.

Property (3) is particularly important: it says that in a Noetherian ring, the zero divisors are precisely the elements lying in some associated prime. This will translate to the geometric statement that a function is a zero divisor if and only if it vanishes at an associated point.

Definition 2.5.4: Minimal and Embedded Primes

Let R be a Noetherian ring and M a finitely generated R-module. An associated prime $\mathfrak{p} \in \mathrm{Ass}(M)$ is:

- *minimal* (or *isolated*) if it is minimal among associated primes, i.e., there is no $\mathfrak{q} \in \mathrm{Ass}(M)$ with $\mathfrak{q} \subsetneq \mathfrak{p}$;
- *embedded* if it is not minimal, i.e., there exists $\mathfrak{q} \in \mathrm{Ass}(M)$ with $\mathfrak{q} \subsetneq \mathfrak{p}$.

Minimal associated primes correspond to the irreducible components of $\mathrm{Supp}(M)$. Embedded associated primes correspond to "hidden" structure not visible in the support as a set, but detected by the module structure. Geometrically, embedded points arise from nilpotent thickening or non-transverse intersections.

2.5.2 Associated Points of Schemes

Definition 2.5.5: Associated Point of a Scheme

Let X be a scheme. A point $x \in X$ is an *associated point* of X if, for some (equivalently, every) affine open $U = \operatorname{Spec} R$ containing x with x corresponding to $\mathfrak{p} \in \operatorname{Spec} R$, we have:

$$\mathfrak{p} \in \mathrm{Ass}_A(A)$$

The set of associated points of X is denoted $\mathrm{Ass}(X)$.

The "some/every" equivalence follows from the fact that associated primes behave well under localization and thus by the affine communication lemma: if $\mathfrak{p} \in \mathrm{Ass}(R)$ and $S \subseteq A$ is a multiplicative set with $S \cap \mathfrak{p} = \emptyset$, then $S^{-1}\mathfrak{p} \in \mathrm{Ass}(S^{-1}R)$.

Proposition 2.5.6: Characterization of Associated Points

Let X be a locally Noetherian scheme. A point $x \in X$ is an associated point if and only if the maximal ideal $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$ is an associated prime of $\mathcal{O}_{X,x}$ (as a module over itself). Equivalently, $x \in \mathrm{Ass}(X)$ if and only if there exists $f \in \mathcal{O}_{X,x}$ with $\mathrm{Ann}(f) = \mathfrak{m}_x$.

***Proof*:**
Take an affine open $U = \operatorname{Spec} A$ containing x, with x corresponding to $\mathfrak{p}$. Then $\mathcal{O}_{X,x} = A_{\mathfrak{p}}$ and $\mathfrak{m}_x = \mathfrak{p}A_{\mathfrak{p}}$. We have $\mathfrak{p} \in \mathrm{Ass}_A(A)$ if and only if $\mathfrak{p}A_{\mathfrak{p}} \in \mathrm{Ass}_{A_{\mathfrak{p}}}(A_{\mathfrak{p}})$ by the localization property of associated primes.

Proposition 2.5.7: Associated Points of Locally Noetherian Schemes

Let X be a locally Noetherian scheme. Then:

1. $\mathrm{Ass}(X)$ is a discrete subset of X: every associated point has an open neighborhood containing no other associated points.
2. If X is quasi-compact (e.g., Noetherian), then $\mathrm{Ass}(X)$ is finite.
3. If X is reduced, then $\mathrm{Ass}(X)$ equals the set of generic points of the irreducible components of X.
4. The minimal associated points are exactly the generic points of the irreducible components of X.
5. X has embedded points if and only if X is not reduced "at" those points (more precisely: X has no embedded points if and only if X is reduced).

***Proof*:**

1. Each associated point x lies in an affine open $\operatorname{Spec} A$ where it corresponds to $\mathfrak{p} \in \mathrm{Ass}(A)$. Shrinking to $\operatorname{Spec} A_{\mathfrak{p}}$, the only associated prime is $\mathfrak{p}A_{\mathfrak{p}}$ by the localization property.
2. Cover X by finitely many affine opens $\operatorname{Spec} A_i$; each has finitely many associated primes, and associated points of X lying in $\operatorname{Spec} A_i$ correspond to associated primes of A_i.
3. If X is reduced, then for any affine open $\operatorname{Spec} A$, we have A reduced (no nilpotents). A reduced Noetherian ring has $\mathrm{Ass}(A) = \mathrm{Min}(A)$ (the minimal primes), and these correspond to the generic points of irreducible components.
4. The minimal primes of A are both the minimal elements of $\mathrm{Ass}(A)$ and the generic points of the irreducible components of $\operatorname{Spec} A$.
5. If X is reduced, (3) shows there are no embedded points. Conversely, if X has no embedded points and $\mathfrak{p}$ is a minimal prime, then $A_{\mathfrak{p}}$ has unique associated prime $\mathfrak{p}A_{\mathfrak{p}}$, which forces $A_{\mathfrak{p}}$ to be a field (it's a local ring with no zero divisors of depth 0), so A is generically reduced. A bit more work shows A is reduced.

Example 2.5.8: Associated Points of Integral Schemes

1. Let R be an integral domain. Then the generic point (0) is the unique associated point of $\operatorname{Spec} R$. Indeed, $\mathrm{Ass}(R) = \{(0)\}$ since R has no zero divisors: if $\mathrm{Ann}(r) = \mathfrak{p}$ for some $r \neq 0$, then $\mathfrak{p} \cdot r = 0$ forces $\mathfrak{p} = (0)$.

   More generally, if X is an integral scheme, then $\mathrm{Ass}(X) = \{\eta\}$ where η is the generic point.
2. Let $X = \operatorname{Spec}(k[x,y]/(xy))$. This is two lines (the x-axis and y-axis) meeting at the origin. The associated primes of $k[x,y]/(xy)$ are (x) and (y):

$$\mathrm{Ann}(\bar{x}) = (y), \qquad \mathrm{Ann}(\bar{y}) = (x)$$

These are both minimal, corresponding to the generic points of the two irreducible components. The origin (x, y) is *not* an associated point: one can check that $\mathrm{Ann}(\bar{f}) \neq (x, y)$ for any $\bar{f} \in k[x, y]/(xy)$.

Geometrically: the two lines meet transversally at the origin, and transverse intersections do not create embedded points.

3. Let $X = \mathrm{Spec}\,(k[x, y]/(y^2, xy))$. This scheme has the same underlying topological space as $\mathrm{Spec}\,(k[x, y]/(y)) = \mathbb{A}^1_k$ (the x-axis), but with nilpotent "fuzz" at the origin.

   The associated primes are (y) and (x, y):

   - $\mathrm{Ann}(\bar{x}) = (y)$: if $f \cdot \bar{x} = 0$ in $k[x, y]/(y^2, xy)$, then $fx \in (y^2, xy)$, forcing $f \in (y)$.
   - $\mathrm{Ann}(\bar{y}) = (x, y)$: we have $x \cdot \bar{y} = \overline{xy} = 0$ and $y \cdot \bar{y} = \overline{y^2} = 0$, so $(x, y) \subseteq \mathrm{Ann}(\bar{y})$. Maximality is clear since $\bar{y} \neq 0$.

   Thus $\mathrm{Ass}(X) = \{[(y)], [(x, y)]\}$: the generic point of the x-axis (minimal) and the origin (embedded). The embedded point at the origin "remembers" that X is non-reduced there, even though the underlying set is just a line.

4. Continuing the previous example, let $X = \mathrm{Spec}\,(k[x, y]/(y^2, xy))$ and consider the function $\bar{y} \in \mathcal{O}_X(X)$.

   Where does $\bar{y}$ vanish? At a point $\mathfrak{p} \in X$, the function $\bar{y}$ vanishes if and only if its image in the residue field $\kappa(\mathfrak{p})$ is zero, which happens if and only if $y \in \mathfrak{p}$. Thus $\bar{y}$ vanishes at every point of X (since every prime of $k[x, y]/(y^2, xy)$ contains y).

   What is $\mathrm{Supp}(\bar{y})$*?* The support of $\bar{y}$ as a section is the set of points where $\bar{y}$ is not zero in the stalk:
   $$\mathrm{Supp}(\bar{y}) = \{\mathfrak{p} \in X \mid \bar{y} \neq 0 \in \mathcal{O}_{X,\mathfrak{p}}\}$$
   At the generic point $[(y)]$: the stalk is $(k[x, y]/(y^2, xy))_{(y)} = k(x)[y]/(y^2)$ (localize away from (y), but y remains nilpotent). Here $\bar{y} \neq 0$.

   At the origin $[(x, y)]$: the stalk is $(k[x, y]/(y^2, xy))_{(x,y)}$, and $\bar{y} \neq 0$ there as well.

   At any other closed point $[(x - a, y)]$ for $a \neq 0$: the stalk is $k[x, y]_{(x-a,y)}/(y^2, xy)$. Since $x - a$ is a unit in this localization, and $xy = 0$ implies $y = (x - a)^{-1} \cdot 0 = 0$ after multiplying by the unit... actually, let's be more careful. We have $xy = 0$, and $x = (x - a) + a$, so $((x - a) + a)y = 0$, giving $(x - a)y = -ay$. Since $x - a$ is a unit, $y = -a^{-1}(x - a)y$... hmm, this gives $y = -a^{-1}(x - a)y$, so $y(1 + a^{-1}(x - a)) = 0$. Since $a \neq 0$ and $x - a$ is in the maximal ideal, $1 + a^{-1}(x - a)$ is a unit, so $y = 0$ in this stalk.

   Thus $\mathrm{Supp}(\bar{y}) = \{[(y)], [(x, y)]\}$, which is exactly $\mathrm{Ass}(X)$. The function $\bar{y}$ is a *zero divisor* (since $(x) \cdot \bar{y} = 0$), and its support is concentrated at the associated points.

Proposition 2.5.9: Zero Divisors and Associated Points

Let X be a locally Noetherian scheme and $f \in \mathcal{O}_X(X)$ a global section. Then f is a zero divisor if and only if f vanishes at some associated point of X. Equivalently:

$$f \text{ is a non-zero-divisor} \quad \Longleftrightarrow \quad f|_\xi \neq 0 \text{ for all } \xi \in \mathrm{Ass}(X)$$

***Proof*:**
The question is local, so we may assume $X = \operatorname{Spec} A$ for a Noetherian ring A. Then f is a zero divisor in A if and only if $f \in \bigcup_{\mathfrak{p} \in \operatorname{Ass}(A)} \mathfrak{p}$ by proposition 2.5.3(3), which happens if and only if $f \in \mathfrak{p}$ for some associated prime $\mathfrak{p}$, i.e., f vanishes at the corresponding associated point.

This proposition is extremely useful: to check whether a function is a non-zero-divisor, it suffices to check non-vanishing at the finitely many associated points, rather than at every point.

Example 2.5.10: Double Line
Let $X = \operatorname{Spec}(k[x,y]/(y^2))$, the "double line" or x-axis with nilpotent thickening. The only prime ideal of $k[x,y]/(y^2)$ containing (y^2) is (y) (since (y) is radical and $(y^2) \subseteq (y)$, and any prime containing y^2 must contain y).

Is (y) an associated prime? We need $\operatorname{Ann}(f) = (y)$ for some f. Take $f = \bar{y}$: then $g \cdot \bar{y} = 0$ iff $gy \in (y^2)$ iff $g \in (y)$. So $\operatorname{Ann}(\bar{y}) = (y)$.

Thus $\operatorname{Ass}(X) = \{[(y)]\}$: just the generic point. There is no embedded point! Even though X is non-reduced, the nilpotent thickening is "uniform" along the entire line, not concentrated at any particular point.

Compare this to $k[x,y]/(y^2, xy)$, where the xy relation creates an embedded point at the origin by making y "more nilpotent" there.

Example 2.5.11: Associated Points of $\operatorname{Spec} \mathbb{Z}$
Let $X = \operatorname{Spec} \mathbb{Z}$. Since $\mathbb{Z}$ is an integral domain, the unique associated point is the generic point (0). The closed points (p) for primes p are *not* associated.

Now let $M = \mathbb{Z}/n\mathbb{Z}$ for some $n > 1$, and consider $\widetilde{M}$ as a quasi-coherent sheaf on $\operatorname{Spec} \mathbb{Z}$. The associated primes of M are the primes dividing n. For $n = 12 = 2^2 \cdot 3$, we have $\operatorname{Ass}(M) = \{(2), (3)\}$, corresponding to the points where the "support" of the module is concentrated.

Associated points play a key role in many constructions:

Proposition 2.5.12: Sections Determined by Associated Points

Let X be a locally Noetherian scheme and $\mathscr{F}$ a quasi-coherent sheaf on X. A section $s \in \mathscr{F}(X)$ is zero if and only if $s_\xi = 0$ in the stalk $\mathscr{F}_\xi$ for every associated point $\xi \in \operatorname{Ass}(X)$.

***Proof*:**
This reduces to the affine case: if $X = \operatorname{Spec} A$ and $\mathscr{F} = \widetilde{M}$ for an A-module M, then $s \in M$ is zero iff $s = 0$ in $M_\mathfrak{p}$ for all $\mathfrak{p} \in \operatorname{Ass}(M)$. This follows from the fact that the natural map $M \to \prod_{\mathfrak{p} \in \operatorname{Ass}(M)} M_\mathfrak{p}$ is injective for finitely generated modules over Noetherian rings.

2.5.13 Associated Points and Depth For a locally Noetherian scheme X, the associated points are exactly the points where the depth of the local ring is zero:

$$\mathrm{Ass}(X) = \{x \in X \mid \mathrm{depth}(\mathcal{O}_{X,x}) = 0\}$$

Recall that the depth of a local ring $(R, \mathfrak{m})$ is the length of a maximal regular sequence in $\mathfrak{m}$. A point x has depth 0 if and only if $\mathfrak{m}_x$ consists entirely of zero divisors (plus nilpotents), which happens precisely when $\mathfrak{m}_x \in \mathrm{Ass}(\mathcal{O}_{X,x})$.
This perspective connects associated points to Cohen–Macaulay and Gorenstein conditions: a Noetherian scheme is Cohen–Macaulay if and only if at every point, the depth equals the dimension, which imposes strong restrictions on the associated points.

Exercise 2.5.1

1. Let $A = k[x, y, z]/(xy, xz)$. Find $\mathrm{Ass}(A)$ and describe the associated points of $\operatorname{Spec} A$ geometrically. Which are minimal and which are embedded?
2. Let $A = k[x, y]/(x^2, xy)$. Show that $\mathrm{Ass}(A) = \{(x), (x, y)\}$. Interpret this geometrically: what is the underlying space, and where is the embedded point?
3. Prove that a Noetherian scheme X is reduced if and only if it has no embedded associated points.
4. Let X be a Noetherian scheme and $U \subseteq X$ an open subset. Show that $\mathrm{Ass}(U) = \mathrm{Ass}(X) \cap U$. (Hint: use the localization property of associated primes.)
5. Let $f : X \to Y$ be a closed immersion of Noetherian schemes. Show that $\mathrm{Ass}(X) \subseteq f^{-1}(\mathrm{Ass}(Y))$ does NOT hold in general. Give a counterexample. (Hint: consider $\operatorname{Spec}(k) \hookrightarrow \operatorname{Spec}(k[x]/(x^2))$.)
6. Let X be an integral scheme with generic point η. Show that a rational function $f \in k(X)$ is regular (i.e., lies in $\mathcal{O}_X(U)$ for some open U) if and only if it is regular at η. (This is a version of the "associated points control regularity" principle for integral schemes.)
7. (*Primary decomposition and associated points*) Let $I \subseteq R$ be an ideal in a Noetherian ring with primary decomposition $I = Q_1 \cap \cdots \cap Q_n$. Show that $\mathrm{Ass}(R/I) = \{\sqrt{Q_1}, \ldots, \sqrt{Q_n}\}$, and identify which are minimal vs. embedded.

2.6 Dimension

The reader may want to review [11, chapter 20] for an in-depth look at the algebraic counterpart. Recall that the *Krull dimension* of a commutative ring R is the supremum of the lengths of proper chains of prime ideals:

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

with indexing starting at 0. Geometrically, for $\operatorname{Spec} R$ and topological spaces more generally (including schemes), the *dimension* is the supremum of lengths of proper chains of closed irreducible subsets with indexing starting at 0. Since the closed irreducible subsets of $\operatorname{Spec} R$ are exactly the sets $V(\mathfrak{p})$

for $\mathfrak{p}$ prime, we have $\dim \operatorname{Spec} R = \operatorname{kdim} R$. Furthermore:

$$\dim \operatorname{Spec} R = \dim \operatorname{Spec} (R/\sqrt{0})$$

since $\operatorname{Spec} R$ and $\operatorname{Spec} (R/\sqrt{0})$ are homeomorphic (quotienting by the nilradical does not change the prime ideals).

If the space is not irreducible, then the dimension per component can vary (for example $\mathbb{A}^1 \cup \mathbb{A}^2 \subseteq \mathbb{A}^3$, interpreted as the x-axis union the yz-plane). Hence we usually work with irreducible schemes. If necessary, we say a scheme has *pure dimension* n (or is *equidimensional*) if all of its irreducible components have the same dimension n. Dimension respects inclusions: if $X \subseteq Y$ then $\dim X \leq \dim Y$.

Dimension	Name
0	point (or collection of points)
1	curve
2	surface
$n \geq 3$	n-fold

Lemma 2.6.1: Dimension of Open Subset

Let $U \subseteq X$ be an open subset of a topological space. Then there is a bijection between irreducible closed subsets of U and the irreducible closed subsets of X that meet U, given by $Z \mapsto \overline{Z}$ (closure in X) with inverse $W \mapsto W \cap U$.

***Proof*:**
If $Z \subseteq U$ is irreducible and closed in U, then $\overline{Z}$ is irreducible and closed in X with $\overline{Z} \cap U = Z$ (since Z is closed in U, it equals the intersection of its closure with U). Conversely, if $W \subseteq X$ is irreducible closed with $W \cap U \neq \emptyset$, then $W \cap U$ is irreducible (an open subset of an irreducible set is irreducible) and closed in U, with $\overline{W \cap U} = W$ (since $W \cap U$ is dense in W). These operations are mutually inverse.

Proposition 2.6.2: Dimension of Scheme

Let X be a scheme. Then $\dim X = n$ if and only if X admits an open cover by affine open subsets of dimension at most n, where equality is achieved by at least one affine open.

***Proof*:**
If $\dim X = n$, then each open subset in an affine cover has dimension $\leq n$ (since irreducible closed subsets of an open set correspond to those of X meeting it, by lemma 2.6.1).

Conversely, suppose every affine open in some cover has dimension $\leq n$ and at least one has dimension exactly n. We must show $\dim X = n$. The dimension is at least n since some affine open has dimension n, and a chain in U_i lifts to a chain in X via lemma 2.6.1. For the reverse inequality, suppose $\dim X > n$. Then there exists a chain of irreducible closed subsets in X:

$$Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_{n+1}$$

Since Z_0 is nonempty, it meets some affine open U_i in the cover. Then each Z_j also meets U_i

(since $Z_0 \subseteq Z_j$ and $Z_0 \cap U_i \neq \emptyset$). By lemma 2.6.1, the chain:

$$Z_0 \cap U_i \subsetneq Z_1 \cap U_i \subsetneq \cdots \subsetneq Z_{n+1} \cap U_i$$

consists of distinct irreducible closed subsets of U_i (since the closure map is a bijection). This gives $\dim U_i \geq n+1 > n$, contradicting our assumption.

Lemma 2.6.3: Integral Extensions and Dimension

Let $f : \operatorname{Spec} R \to \operatorname{Spec} S$ correspond to an integral extension $S \hookrightarrow R$. Then:

$$\dim \operatorname{Spec} R = \dim \operatorname{Spec} S$$

***Proof*:**
The going-up theorem gives $\dim \operatorname{Spec} R \geq \dim \operatorname{Spec} S$ (any chain in S lifts to a chain in R), and the incomparability theorem gives $\dim \operatorname{Spec} R \leq \dim \operatorname{Spec} S$ (primes lying over the same prime cannot be comparable). See [11, chapter 17] for the details.

2.6.4 A word on Noetherian Schemes It is *not* the case that a Noetherian scheme must be finite-dimensional: Nagata constructed a Noetherian ring of infinite Krull dimension (see Vakil, chapter 11).
However, Noetherian *local* rings are always finite-dimensional (by [11, corollary 20.42]), and so points of locally Noetherian schemes always have finite codimension.

2.6.1 Codimension

We are often concerned with subspaces that are a few dimensions lower than the ambient scheme X. Since schemes can be unions of irreducible components of different dimension, we define codimension only for irreducible subschemes:

Definition 2.6.5: Codimension

Let X be a topological space and $Y \subseteq X$ an irreducible closed subset. Then:

$$\begin{aligned} &\operatorname{codim}_X Y \\ &= \sup\{n : \text{there exists a chain } Y = Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n \text{ of irreducible closed subsets of } X\} \end{aligned}$$

For a scheme X and a point $x \in X$, the codimension of $\overline{\{x\}}$ in X equals the Krull dimension of the local ring $\mathcal{O}_{X,x}$.

For the affine case: if $\mathfrak{p} \in \operatorname{Spec} R$, then $\operatorname{codim}_{\operatorname{Spec} R} V(\mathfrak{p})$ equals the *height* of $\mathfrak{p}$, i.e. the supremum of lengths of chains:

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_h = \mathfrak{p}$$

and this equals $\operatorname{kdim} R_{\mathfrak{p}}$. For example, in $\operatorname{Spec} \mathbb{Z}$, the generic point $\eta = [(0)]$ has codimension 0, while any closed point $[(p)]$ has codimension 1.

If Y has codimension 0 in X, then Y must be an irreducible component of X.

Definition 2.6.6: Hypersurface

Let $Y \subseteq X$ be an irreducible closed subscheme. Then Y is called a *hypersurface* if $\operatorname{codim}_X Y = 1$.

Lemma 2.6.7: Sum of Dimension and Codimension

Let X be a topological space and $Y \subseteq X$ an irreducible closed subset. Then:

$$\operatorname{codim}_X Y + \dim Y \leq \dim X$$

***Proof*:**
A chain of length c descending from Y and a chain of length d ascending from Y can be concatenated to give a chain of length $c + d$ in X. Taking suprema gives $\operatorname{codim}_X Y + \dim Y \leq \dim X$.

Equality holds for irreducible varieties of finite type over a field, but *not* in general:

Example 2.6.8: Codimension Pathology

Consider the scheme $X = \operatorname{Spec}(k[x]_{(x)}[t])$, which can be thought of as a surface with a line attached at one point. Then $\dim X = 2$, but the closed point $p = (x, t)$ has dimension 0 and codimension 1 (not 2!), since the maximal chain from p is:

$$(0) \subsetneq (x) \subsetneq (x, t)$$

but (x) already has codimension 1 and dimension 1 (the "line" component), while the "surface" component (0) has dimension 2. Note too that this scheme has open subsets of *different dimension* – a phenomenon that does *not* occur for open subsets of n-manifolds.

The pathology above is excluded by the following class of rings:

Definition 2.6.9: Catenary Ring

A commutative ring R is *catenary* if for every pair of nested prime ideals $\mathfrak{q} \subseteq \mathfrak{p}$, all maximal chains of prime ideals between $\mathfrak{q}$ and $\mathfrak{p}$ have the same length.

Many rings are catenary: localizations of finitely generated $\mathbb{Z}$-algebras, complete Noetherian local rings, Dedekind domains, Cohen–Macaulay rings, and in particular all localizations of finitely generated k-algebras (see exercise 2.6.1).

2.6.2 Transcendence Dimension

Working with Krull dimension directly can be difficult. For finitely generated algebras over a field, we have a more computable invariant:

Definition 2.6.10: Transcendence Dimension

Let A be an integral domain that is a k-algebra. Then the *transcendence dimension* of A over k is:

$$\operatorname{trdeg}_k A := \operatorname{trdeg}_k \operatorname{Frac}(A)$$

i.e. the transcendence degree of the fraction field of A over k.

Theorem 2.6.11: Transcendence and Krull Dimension

Let A be a finitely generated integral domain over a field k. Then:

$$\dim \operatorname{Spec} A = \operatorname{trdeg}_k \operatorname{Frac}(A)$$

Thus, if X is an irreducible k-variety with function field $k(X)$, then $\dim X = \operatorname{trdeg}_k k(X)$.

***Proof*:**
See [11, chapter 20] for the algebraic proof. The key steps are: Noether normalization gives a finite (hence integral) extension $k[y_1, \ldots, y_d] \hookrightarrow A$ with $d = \operatorname{trdeg}_k \operatorname{Frac}(A)$; by lemma 2.6.3, $\dim \operatorname{Spec} A = \dim \mathbb{A}^d_k = d$.

For catenary rings, dimension and codimension add up:

Theorem 2.6.12: Dimension Formula for Finite Type k-Schemes

Let X be an irreducible scheme locally of finite type over a field k, $Y \subseteq X$ an irreducible closed subset, and η the generic point of Y. Then:

$$\operatorname{codim}_X Y + \dim Y = \dim X$$

equivalently:

$$\dim \mathcal{O}_{X,\eta} + \dim Y = \dim X$$

***Proof*:**
The question is local, so we may reduce to $X = \operatorname{Spec} A$ for a finitely generated k-domain A. Let $\mathfrak{p}$ be the prime corresponding to η. Then $\operatorname{codim}_X Y = \operatorname{kdim} A_{\mathfrak{p}}$ and $\dim Y = \operatorname{kdim} A/\mathfrak{p}$, so we must show $\operatorname{kdim} A_{\mathfrak{p}} + \operatorname{kdim} A/\mathfrak{p} = \operatorname{kdim} A$. This follows from the fact that finitely generated k-algebras are catenary and equidimensional (see [11, chapter 20] or Vakil, chapter 11 for a guided exercise).

2.6.3 Krull's Height Theorem and Algebraic Hartogs' Lemma

We now turn to two fundamental results about codimension-1 phenomena. Krull's theorem gives an upper bound on codimension in terms of the number of defining equations, while the algebraic Hartogs' lemma characterizes regular functions on normal schemes in terms of codimension-1 behavior. Both will be used extensively in the theory of divisors (section 5.4).

Krull's Height Theorem

The following theorem, proved in [11, corollary 20.47], is one of the most important tools for computing codimension:

Theorem 2.6.13: Krull's Height Theorem (Hauptidealsatz)

Let R be a Noetherian ring. If $\mathfrak{a} = (f_1, \ldots, f_n)$ is a proper ideal generated by n elements, then every minimal prime $\mathfrak{p}$ over $\mathfrak{a}$ has:

$$\operatorname{height}(\mathfrak{p}) \leq n$$

In particular, if $f \in R$ is neither a zero-divisor nor a unit, then every minimal prime over (f) has height exactly 1.

***Proof*:**
See [11, section 20.5]. The proof uses the dimension theorem for Noetherian local rings: localizing at $\mathfrak{p}$, the ideal $\mathfrak{a}_\mathfrak{p}$ is $\mathfrak{p}_\mathfrak{p}$-primary and generated by n elements, so $\operatorname{kdim} R_\mathfrak{p} \leq n$ by the characterization of dimension as the minimal number of generators of a parameter ideal.

Geometrically, Krull's theorem says: in a Noetherian scheme, the vanishing locus of n functions has all irreducible components of codimension $\leq n$. In particular:

Corollary 2.6.14: Principal Ideal Theorem – Geometric Form

Let X be a locally Noetherian scheme and $f \in \mathcal{O}_X(X)$ a regular function that is not a zero-divisor on any irreducible component. Then every irreducible component of $V(f)$ has codimension exactly 1 in X.

Corollary 2.6.15: Codimension Bound for Complete Intersections

Let X be a locally Noetherian scheme and $Z = V(f_1, \ldots, f_n) \subseteq X$. Then every irreducible component of Z has codimension $\leq n$. If equality holds for every component, Z is called a *(set-theoretic) complete intersection*.

Example 2.6.16: Krull's Theorem in Practice

1. In $\mathbb{A}^3_k = \operatorname{Spec} k[x, y, z]$, the ideal (xy, xz) defines the vanishing locus $V(x) \cup V(y, z)$, which is the yz-plane $\cup$ the x-axis. The yz-plane has codimension 1, while the x-axis has codimension 2. Krull's theorem guarantees codimension ≤ 2, and both bounds are achieved.

2. The ideal $(x^2 + y^2 + z^2, x + y + z) \subseteq k[x, y, z]$ defines a curve in $\mathbb{A}^3_k$. By Krull's theorem, each component has codimension ≤ 2. In this case, the curve has codimension exactly 2, so it is a complete intersection.

3. The twisted cubic curve $C \subseteq \mathbb{A}^3_k$ (the image of $t \mapsto (t, t^2, t^3)$) has codimension 2 but ideal $I(C) = (y - x^2, z - x^3)$, which requires 2 generators. However, in $\mathbb{P}^3_k$ the twisted cubic requires 3 generators of its homogeneous ideal (it is *not* a complete intersection).

The converse direction of Krull's theorem also holds and is equally useful:

Proposition 2.6.17: Converse to Krull's Theorem

Let R be a Noetherian ring and $\mathfrak{p}$ a prime of height n. Then there exist $f_1, \ldots, f_n \in \mathfrak{p}$ such that $\mathfrak{p}$ is a minimal prime over $(f_1, \ldots, f_n)$.

***Proof*:**
See [11, proposition 20.48]. One constructs the f_i inductively using prime avoidance: at each step, choose f_i in $\mathfrak{p}$ but outside all minimal primes of $(f_1, \ldots, f_{i-1})$ of height $< i$.

Algebraic Hartogs' Lemma

In complex analysis, Hartogs' extension theorem states that a holomorphic function defined on the complement of a codimension-≥ 2 set extends uniquely to the ambient space. Amazingly, there is an algebraic analogue for normal schemes:

Theorem 2.6.18: Algebraic Hartogs' Lemma

Let A be a Noetherian normal domain. Then:

$$A = \bigcap_{\substack{\mathfrak{p} \in \operatorname{Spec} A \\ \operatorname{height}(\mathfrak{p})=1}} A_{\mathfrak{p}}$$

where the intersection takes place inside $\operatorname{Frac}(A)$.

In geometric language: a rational function on a Noetherian normal scheme with no poles along any codimension-1 subvariety is in fact a regular function. Equivalently, regular functions on a normal scheme are determined by their behavior at codimension-1 points – you cannot have a function that is "only irregular in codimension ≥ 2."

***Proof*:**
The inclusion $A \subseteq \bigcap_{\operatorname{ht}(\mathfrak{p})=1} A_{\mathfrak{p}}$ is clear (any element of A lies in every localization). For the reverse, suppose $x \in \operatorname{Frac}(A)$ lies in $A_{\mathfrak{p}}$ for every height-1 prime $\mathfrak{p}$. Let $I = \{a \in A \mid ax \in A\}$ be the ideal of "denominators" of x. If $x \notin A$, then $I \neq A$. Let $\mathfrak{q}$ be a minimal prime over I.

Claim: $\operatorname{height}(\mathfrak{q}) \leq 1$.

If $\operatorname{height}(\mathfrak{q}) = 0$, this is fine. If $\operatorname{height}(\mathfrak{q}) \geq 2$, then $x \in A_{\mathfrak{p}}$ for all height-1 primes $\mathfrak{p} \subseteq \mathfrak{q}$, and since $A_{\mathfrak{q}}$ is a normal local domain, $A_{\mathfrak{q}} = \bigcap_{\operatorname{ht}(\mathfrak{p})=1,\, \mathfrak{p} \subseteq \mathfrak{q}} (A_{\mathfrak{q}})_{\mathfrak{p}A_{\mathfrak{q}}}$ (applying the result in the local case, where it holds by the characterization of normal domains as S_2 and R_1). This gives $x \in A_{\mathfrak{q}}$, contradicting $I \subseteq \mathfrak{q}$. So $\operatorname{height}(\mathfrak{q}) = 1$.

But then $x \in A_{\mathfrak{q}}$ by hypothesis, so there exist $a \in A$, $s \notin \mathfrak{q}$ with $x = a/s$, giving $sx = a \in A$ and hence $s \in I$. But $I \subseteq \mathfrak{q}$ and $s \notin \mathfrak{q}$, a contradiction. Therefore $I = A$ and $x \in A$.

Example 2.6.19: Algebraic Hartogs in Practice

1. Consider $\mathbb{A}^2_k = \operatorname{Spec} k[x,y]$ and the open set $U = D(x) \cup D(y) = \mathbb{A}^2 \setminus \{(0,0)\}$. We showed in example 2.2.2 that $\mathcal{O}_{\mathbb{A}^2}(U) = k[x,y]$. This is now immediate from algebraic Hartogs: the complement of U is $\{(0,0)\}$, which has codimension 2, and $k[x,y]$ is a normal domain. Hence any regular function on U extends to all of $\mathbb{A}^2$.
2. The same argument fails for $\mathbb{A}^1$: removing a point from $\mathbb{A}^1$ removes a codimension-1 subset, and indeed $\mathcal{O}_{\mathbb{A}^1}(D(x)) = k[x, 1/x] \neq k[x]$.
3. Let $A = k[x,y,z]/(x^2+y^2-z^2)$, the quadric cone, which is a normal domain (exercise). The singular point at the origin has codimension 2, so algebraic Hartogs implies that regular functions on the smooth locus extend to all of $\operatorname{Spec} A$.
4. The algebraic Hartogs' lemma *fails* without normality. Consider the nodal cubic $A = k[x,y]/(y^2 - x^2(x+1))$, which is not normal at the origin. The element y/x in the fraction field is regular at all codimension-1 points away from the node, yet $y/x \notin A$.

The connection between these two results is fascinating: Krull's theorem says that the zero locus of a single function has codimension ≤ 1, while Hartogs' lemma says that on normal schemes, a rational function is determined by its poles, which also occur in codimension 1. Both point toward the centrality of codimension-1 phenomena, which will be formalized in the theory of *Weil and Cartier divisors* in section 5.4.

Exercise 2.6.1

1. Show that a Noetherian scheme of dimension 0 has finitely many points.
2. Let $f : X \to Y$ be an integral morphism. Show that $\dim f^{-1}(y) = 0$ for all $y \in Y$ (i.e., the fibers are zero-dimensional).
3. If $\nu : \widetilde{X} \to X$ is the normalization of a scheme, show that $\dim \widetilde{X} = \dim X$.
4. If K/k is an algebraic field extension and X is a k-scheme of finite type, show that $\dim X = \dim X_K$ where $X_K = X \times_k K$ is the base change.
5. Generalize the above result: if K/k is an arbitrary field extension and X is an irreducible k-scheme of finite type, show that $\dim X_K = \dim X$.
6. Show the following rings are catenary: localizations of finitely generated $\mathbb{Z}$-algebras, complete Noetherian local rings, Dedekind domains.
7. Let X be an integral scheme locally of finite type over a field k, and let $U \subseteq X$ be a nonempty open subset. Show that $\dim U = \dim X$.
8. (*Hartogs' lemma for schemes*) Let X be a normal Noetherian integral scheme, and let $U \subseteq X$ be an open subset whose complement has codimension ≥ 2. Show that the restriction map $\mathcal{O}_X(X) \to \mathcal{O}_X(U)$ is an isomorphism. (Hint: reduce to the affine case and apply theorem 2.6.18.)
9. Show that the quadric cone $\operatorname{Spec} k[x,y,z]/(x^2+y^2-z^2)$ is a normal domain ($\operatorname{char} k \neq 2$). (Hint: show it is R_1 and S_2, or use the Jacobian criterion to show regularity in codimension 1

and then invoke Serre's criterion.)

2.7 Regular and Smooth Schemes

The notion of smoothness is central to both differential and algebraic geometry. In differential geometry, a manifold is smooth if it locally looks like $\mathbb{R}^n$. In algebraic geometry, the analogous condition is that the local ring at each point behaves like a polynomial ring – this is the notion of *regularity*. Over perfect fields (in particular, algebraically closed fields of any characteristic), regularity and smoothness coincide; over imperfect fields, smoothness is the stronger and more geometrically natural condition.

The algebraic foundations for this section – Hilbert functions, the dimension theorem, regular local rings, and their characterization via homological methods – were developed in [11, sections 20.5–20.7]. We recall the key definitions and build on them.

Definition 2.7.1: Tangent Space

Let X be a scheme and $x \in X$ a point. Then the *(Zariski) cotangent space* at x is the $\kappa(x)$-vector space:

$$\mathfrak{m}_x/\mathfrak{m}_x^2$$

where $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$ is the maximal ideal of the stalk at x, and $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$ is the residue field. The *(Zariski) tangent space* is its dual:

$$T_xX := (\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee = \mathrm{Hom}_{\kappa(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2,\, \kappa(x))$$

Definition 2.7.2: Regular Local Ring

Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field $k = R/\mathfrak{m}$. Then R is *regular* if:

$$\mathrm{kdim} R = \dim_k(\mathfrak{m}/\mathfrak{m}^2)$$

By [11, corollary 20.44], we always have $\mathrm{kdim} R \leq \dim_k(\mathfrak{m}/\mathfrak{m}^2)$, so regularity asks for equality.

Definition 2.7.3: Regular Scheme

A scheme X is *regular* if $\mathcal{O}_{X,x}$ is a regular local ring for every point $x \in X$. A scheme is *singular* at a point x if $\mathcal{O}_{X,x}$ is not regular.

Example 2.7.4: Regular and Singular Schemes

1. $\mathbb{A}^2_k$ *is regular.* For a closed point $\mathfrak{m} = (x-a, y-b)$, first observe that $\kappa(\mathfrak{m}) = k$ (since k is algebraically closed). The cotangent space $\mathfrak{m}/\mathfrak{m}^2$ is spanned by the images of $x-a$ and $y-b$, so $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 2$. The local ring $k[x,y]_\mathfrak{m}$ has a maximal chain $0 \subsetneq (x-a) \subsetneq (x-a, y-b)$, giving $\mathrm{kdim} = 2$.

   Since we are working over schemes, we also have higher-dimensional points. Consider

$[\mathfrak{p}] = [(x^2 - y)] \in \mathbb{A}^2_k$. The local ring $k[x,y]_{\mathfrak{p}}$ has dimension 1 (one can show the chain $0 \subsetneq (x^2 - y)$ is maximal), and $\mathfrak{m}_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}^2$ is a 1-dimensional $\kappa(\mathfrak{p})$-vector space. Hence this point is regular.

An important observation: $[(x^3 - y^2)] \in \mathbb{A}^2_k$ is *also* a regular point of $\mathbb{A}^2_k$, even though $x^3 - y^2$ defines a singular curve! This is because regularity at a generic point of $\mathbb{A}^2_k$ concerns the local ring of the *ambient* scheme, not the subscheme $V(x^3 - y^2)$. The point $[(x^3 - y^2)]$ is "generically regular" in $\mathbb{A}^2_k$ – further motivating the name *generic* point.

2. *The cuspidal cubic is singular.* Let $X = \operatorname{Spec} k[x,y]/(x^3 - y^2)$. Then X is not regular at the origin: the local ring $\mathcal{O}_{X,(x,y)}$ has kdim $= 1$ (since X is an irreducible curve), but the cotangent space $\mathfrak{m}/\mathfrak{m}^2$ at the origin has $\dim_k = 2$, since neither $\overline{x}$ nor $\overline{y}$ is redundant modulo $\mathfrak{m}^2$ (the relation $x^3 = y^2$ lies in $\mathfrak{m}^3$, not $\mathfrak{m}^2$). On the other hand, X is regular at every other closed point (exercise).

3. *Normal does not imply regular.* If X is a normal scheme, then its stalks are integrally closed, and all codimension-1 points are regular (by [11, proposition 20.53]). However, normality does *not* imply regularity in dimension ≥ 2. For example, $\operatorname{Spec} k[x,y,z]/(x^2 + y^2 - z^2)$ is normal but not regular at the origin (exercise: show $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 3$ while kdim $= 2$).

   However, for a 1-dimensional connected Noetherian scheme, normality *does* imply regularity: the codimension-1 points are exactly the closed points, and the generic point is always regular. In fact, a 1-dimensional normal local domain is a DVR by [11, corollary 20.55], which is a regular local ring of dimension 1.

The singular locus of a scheme is typically "small." For varieties over algebraically closed fields, this was shown in [11, proposition 20.57]: the set of singular points forms a proper closed subset.

For schemes of finite type over a field, regularity can be checked via partial derivatives, generalizing the classical criterion from [11, proposition 20.56]:

Proposition 2.7.5: Jacobian Criterion for Schemes

Let k be a field and $X = \operatorname{Spec}(k[x_1, \ldots, x_n]/(f_1, \ldots, f_m))$ a closed subscheme of $\mathbb{A}^n_k$. Let $x \in X$ be a closed point with residue field $\kappa(x)$. Then X is regular at x if and only if:

$$\operatorname{rank}\left(\frac{\partial f_i}{\partial x_j}(x)\right) = n - \dim \mathcal{O}_{X,x}$$

where the partial derivatives are evaluated in $\kappa(x)$.

***Proof*:**
This is the scheme-theoretic translation of [11, proposition 20.56]. The key computation is that the cotangent space satisfies:

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \cong \frac{\mathfrak{m}_{\mathbb{A}^n,x}}{(\mathfrak{m}_{\mathbb{A}^n,x}^2 + (f_1, \ldots, f_m))}$$

and so $\dim_{\kappa(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2) = n - \operatorname{rank}(J_x)$. Regularity is the condition that this equals $\dim \mathcal{O}_{X,x}$, which is equivalent to $\operatorname{rank}(J_x) = n - \dim \mathcal{O}_{X,x}$.

Example 2.7.6: Jacobian Criterion in Practice

1. The cuspidal cubic $X = V(y^2 - x^3) \subseteq \mathbb{A}^2_k$. The Jacobian is $(-3x^2, 2y)$. At the origin, this is $(0,0)$, which has rank $0 < 2 - 1 = 1$. At any other closed point (a, b) with $b \neq 0$, the rank is $1 = n - \dim X$. Hence X is singular only at the origin.
2. The nodal cubic $X = V(y^2 - x^2(x+1)) \subseteq \mathbb{A}^2_k$. The Jacobian is $(-3x^2 - 2x, 2y)$. At the origin, this is $(0,0)$, so the origin is again singular. At any other point on X where $y \neq 0$, the rank is 1.
3. The quadric surface $X = V(x^2 + y^2 - z^2) \subseteq \mathbb{A}^3_k$ ($\operatorname{char} k \neq 2$). The Jacobian is $(2x, 2y, -2z)$, which has rank 0 only at the origin. Hence X is regular away from the origin.

Smooth Morphisms and Smooth Schemes

Regularity is a property of a scheme in isolation, while *smoothness* is a relative notion: it concerns a morphism $X \to S$ and asks whether the fibers are "non-singular" in a way that varies well over the base. The two notions coincide over perfect fields.

Definition 2.7.7: Smooth Morphism (Preliminary)

A morphism $f : X \to S$ of schemes locally of finite type is *smooth of relative dimension n* at a point $x \in X$ if there exists an open neighborhood U of x and a factorization:

$$\begin{tikzcd} U \arrow[r, hook, "\text{open}"] \arrow[dr, "f|_U"'] & \operatorname{Spec} \frac{\mathcal{O}_S(V)[x_1,\ldots,x_{n+r}]}{(f_1,\ldots,f_r)} \arrow[d] \\ & V \end{tikzcd}$$

where V is an open neighborhood of $f(x)$ in S, and the Jacobian matrix $(\partial f_i / \partial x_j)$ has rank r at x.
The morphism f is *smooth* if it is smooth at every point. It is *étale* if it is smooth of relative dimension 0.

Definition 2.7.8: Smooth Scheme Over a Field

Let k be a field. A k-scheme X is *smooth over k* (or *k-smooth*) if the structure morphism $X \to \operatorname{Spec} k$ is smooth.

The relationship between smoothness and regularity is as follows:

Proposition 2.7.9: Smooth Implies Regular

Let k be a field and X a k-scheme locally of finite type. If X is smooth over k, then X is regular.

***Proof*:**
Smoothness over k implies that the cotangent space $\mathfrak{m}_x/\mathfrak{m}_x^2$ has dimension equal to the local dimension of X at x for every point x, which is exactly regularity. The key point is that smoothness gives the Jacobian rank condition over every residue field, not just over k itself.

The converse holds over perfect fields:

Theorem 2.7.10: Regularity and Smoothness Over Perfect Fields

Let k be a perfect field (e.g., any field of characteristic 0, or any finite field) and X a k-scheme locally of finite type. Then X is smooth over k if and only if X is regular.

***Proof*:**
The forward direction is proposition 2.7.9. For the converse, regularity gives the Jacobian rank condition at all k-rational points. The perfection of k ensures that this extends to all points: if k is perfect, then every finitely generated field extension $\kappa(x)/k$ is separably generated, and the cotangent space computation over $\kappa(x)$ agrees with the one over k. The details require the theory of Kähler differentials, which will be developed in section 5.5.

Example 2.7.11: Regularity $\neq$ Smoothness Over Imperfect Fields
Let $k = \mathbb{F}_p(t)$ (an imperfect field) and consider $X = \operatorname{Spec} k[x]/(x^p - t)$. Then X is a single closed point with residue field $k(\alpha)$ where $\alpha^p = t$. The local ring is $k(\alpha)$, which is a field and hence a regular local ring of dimension 0.

However, X is *not* smooth over k: the polynomial $f(x) = x^p - t$ has $f'(x) = px^{p-1} = 0$, so the Jacobian criterion fails. Geometrically, the extension $k(\alpha)/k$ is purely inseparable, and the fiber over $\operatorname{Spec} k$ is "infinitesimally thickened" from the perspective of k. This is the prototypical example of a regular but non-smooth scheme.

The Singular Locus

Definition 2.7.12: Singular Locus

Let X be a locally Noetherian scheme. The *singular locus* of X is the set:

$$\operatorname{Sing}(X) = \{x \in X \mid \mathcal{O}_{X,x} \text{ is not a regular local ring}\}$$

Proposition 2.7.13: Singular Locus is Closed

Let X be a scheme locally of finite type over a field k. Then $\operatorname{Sing}(X)$ is a closed subset of X.

***Proof*:**
The question is local, so we may assume $X = \operatorname{Spec}(k[x_1, \dots, x_n]/(f_1, \dots, f_m))$. By the Jacobian criterion, the singular locus is cut out by the vanishing of all $(n - \dim X) \times (n - \dim X)$ minors of the Jacobian matrix, which defines a closed set.

For irreducible varieties, the singular locus is not just closed but *proper*:

Theorem 2.7.14: Generic Smoothness

Let X be an irreducible variety over an algebraically closed field k. Then $\operatorname{Sing}(X)$ is a proper closed subset of X. In particular, the smooth locus is a dense open subset.

***Proof*:**
See [11, proposition 20.57]. The key idea is that if X is a hypersurface $V(f)$ and $\operatorname{Sing}(X) = X$, then all partial derivatives of f vanish identically on X. In characteristic 0, this forces f to be constant. In characteristic $p > 0$, it forces $f = g^p$ for some polynomial g (using algebraic closure to take p-th roots), contradicting the irreducibility of f. The general case reduces to the hypersurface case via birational equivalence.

Over non-algebraically-closed fields, the correct generalization replaces "regular" with "smooth":

Corollary 2.7.15: Generic Smoothness Over Perfect Fields

Let X be a geometrically integral[a] k-scheme of finite type over a perfect field k. Then the smooth locus of $X \to \operatorname{Spec} k$ is a dense open subset.

[a] geometrically integral means change the base ring k to $\overline{k}$; see ref:HERE

2.7.1 Properties of Regular Schemes

We collect some important structural consequences of regularity:

Proposition 2.7.16: Regular Local Rings are Domains

Every regular local ring is an integral domain.

***Proof*:**
By [11, lemma 20.54], if $G_{\mathfrak{m}}(R) \cong k[t_1, \ldots, t_d]$ is a domain and $\bigcap_n \mathfrak{m}^n = 0$ (which holds by the Krull intersection theorem for Noetherian local rings), then R is a domain.

Proposition 2.7.17: Regular Local Rings are Normal

Every regular local ring is integrally closed (normal).

***Proof*:**
See [11, corollary 20.56]. This follows from the fact that the associated graded ring of a regular local ring is a polynomial ring (hence a UFD), and one can lift the integral closure property from the associated graded ring.

Proposition 2.7.18: Regular Local Rings are UFDs

Every regular local ring is a unique factorization domain.

***Proof*:**
This is a theorem of Auslander–Buchsbaum, using the characterization of regular local rings via finite global dimension (Serre's theorem, [11, theorem 20.52]). The key step is showing that every height-1 prime in a regular local ring is principal, and then invoking the Nagata criterion for UFDs.

The chain of implications for Noetherian local domains is:

$$\text{regular} \implies \text{UFD} \implies \text{normal} \implies \text{regular in codimension 1}$$

None of these implications reverse in general (except in dimension 1, where regular, normal, and DVR are equivalent by [11, corollary 20.55]).

Proposition 2.7.19: Regular Scheme and Irreducible Components

A regular scheme is a disjoint union of its irreducible components (each of which is integral).

***Proof*:**
Since each stalk is a domain, every point lies on a unique irreducible component. Hence distinct components cannot meet, and the scheme decomposes as a disjoint union. Each component is integral (irreducible and reduced) since regularity implies the stalks are domains.

2.7.2 Dimension and Regularity for Schemes Over $\mathbb{Z}$

An important feature of the scheme-theoretic framework is that the notion of regularity applies uniformly to schemes over any base, including $\operatorname{Spec}\mathbb{Z}$. This allows number-theoretic objects to be treated geometrically:

Example 2.7.20: Regularity of Arithmetic Schemes

1. $\operatorname{Spec}\mathbb{Z}$ is a regular scheme of dimension 1: for each prime (p), the local ring $\mathbb{Z}_{(p)}$ is a DVR (hence a regular local ring of dimension 1), and the generic point has local ring $\mathbb{Q}$, a field.
2. $\operatorname{Spec}\mathbb{Z}[i]$ is a regular scheme. At each maximal ideal $\mathfrak{p}$, the local ring $\mathbb{Z}[i]_{\mathfrak{p}}$ is a DVR (since $\mathbb{Z}[i]$ is a Dedekind domain).
3. $\operatorname{Spec}\mathbb{Z}[\sqrt{-5}]$ is *not* regular at the prime $\mathfrak{p} = (2, 1 + \sqrt{-5})$: the local ring $\mathbb{Z}[\sqrt{-5}]_{\mathfrak{p}}$ is a 1-dimensional local domain that is not a DVR (since $\mathfrak{p}$ is not principal). See [11, chapter 22] for the failure of unique factorization in this ring. The normalization (integral closure in the fraction field) produces the ring of integers $\mathbb{Z}\left[\frac{1+\sqrt{-5}}{2}\right]$, which *is* a Dedekind domain and hence regular.
4. More generally, if K is a number field and $\mathcal{O}_K$ its ring of integers, then $\operatorname{Spec}\mathcal{O}_K$ is a regular scheme if and only if $\mathcal{O}_K$ is a principal ideal domain (equivalently, the class number of

K is 1). This follows from the equivalence of regularity and DVR in dimension 1 (by [11, corollary 20.55]).

Exercise 2.7.1

1. Show that $\mathbb{A}_k^n$ is a regular scheme for any field k.
2. Let $X = \operatorname{Spec} k[x,y]/(y^2 - x^3 + x)$. Show that X is a regular scheme (assuming $\operatorname{char} k \neq 2$).
3. Show that the generic point of any integral scheme is a regular point.
4. Let $X = \operatorname{Spec} k[x,y,z,w]/(xw - yz)$ be the quadric cone (from example 2.2.2). Find $\operatorname{Sing}(X)$ and show it consists of a single point.
5. Let k be a perfect field and $f \in k[x_1, \ldots, x_n]$ irreducible. Show that $V(f)$ is smooth over k if and only if f and its partial derivatives have no common zero.
6. Show that the product of regular schemes is regular (hint: use that regularity is detected at stalks, and compute the stalks of a product).
7. Let R be a regular local ring and $x \in \mathfrak{m} \setminus \mathfrak{m}^2$. Show that $R/(x)$ is again a regular local ring (of dimension one less).

2.8 *How Much Does the Topology Know?

Throughout this chapter, we have constructed schemes as locally ringed spaces: a topological space X together with a structure sheaf $\mathcal{O}_X$. The topological space carries the Zariski topology, which for affine schemes $\operatorname{Spec}(R)$ is entirely determined by the lattice of prime ideals of R. A natural question arises: how much of the scheme-theoretic structure is already visible at the level of topology?

One might initially expect a clean dichotomy: that the topological spaces arising from affine schemes form a special class, and that the topology of a non-affine scheme would look fundamentally "different"—perhaps failing some property that all affine topologies share. This turns out to be false, and understanding *why* it is false sharpens our appreciation for the role of the structure sheaf.

Spectral spaces

To state the result precisely, we need to isolate the topological properties that characterize the spaces $|\operatorname{Spec}(R)|$.

Definition 2.8.1: Spectral Space

A topological space X is a *spectral space* (or *coherent space*) if it satisfies the following four conditions:

1. X is quasi-compact.
2. X is sober: every irreducible closed subset has a unique generic point.
3. The quasi-compact open subsets of X are closed under finite intersection (i.e. X is quasi-separated, see definition 3.5.6) and form a basis for the topology.
4. Every non-empty irreducible closed subset of X has a generic point.

Condition (2) already implies (4), so the definition can be stated with just (1)–(3), but we include (4) for emphasis as it is the condition most often violated by "pathological" topological spaces. Note that condition (3) asks for two things at once: the quasi-compact opens form a basis (so there are enough of them), and a finite intersection of quasi-compact opens is again quasi-compact (a condition that can fail even for well-behaved spaces—it fails, for instance, for the Zariski topology on an infinite product of affine lines, though not for any Noetherian space).

The reader should recognize that every affine scheme satisfies these conditions. In $\operatorname{Spec}(R)$: the space is quasi-compact (section 2.1.3); it is sober by construction (generic points of irreducible closed sets correspond to prime ideals); and the distinguished open sets $D(f)$, which are quasi-compact, form a basis that is closed under finite intersection ($D(f) \cap D(g) = D(fg)$).

The remarkable fact, proved by Hochster in 1969, is that these conditions are not just necessary but *sufficient*.

Theorem 2.8.2: Hochster's Theorem

A topological space X is homeomorphic to $\operatorname{Spec}(R)$ for some commutative ring R if and only if X is a spectral space.

***Proof*:**
see [19]

The "only if" direction is the easy half: we have just verified that every $\operatorname{Spec}(R)$ is spectral. The "if" direction is: given any spectral space X, Hochster explicitly constructs a commutative ring R such that $|\operatorname{Spec}(R)| \cong X$. The construction uses the lattice of quasi-compact open subsets of X to build R as a quotient of a polynomial ring, with generators corresponding to the quasi-compact opens and relations encoding their intersection and covering behavior.

We do not reproduce Hochster's construction here, but the theorem has an immediate and perhaps surprising consequence.

Non-affine schemes with affine topology

Consider projective space $\mathbb{P}^n_k$ over a field k, with $n \geq 1$. This is emphatically not an affine scheme: its ring of global sections is just k, and $\operatorname{Spec}(k)$ is a single point, while $\mathbb{P}^n_k$ has many points. Yet the underlying topological space $|\mathbb{P}^n_k|$ is:

1. quasi-compact (it is covered by finitely many affine charts $U_i \cong \mathbb{A}^n$),
2. sober (it is a scheme, and every scheme is sober),
3. equipped with a basis of quasi-compact opens closed under finite intersection (the distinguished opens in each affine chart, together with their finite intersections, which are again quasi-compact by proposition 2.1.15).

Hence $|\mathbb{P}^n_k|$ is a spectral space, and by Hochster's theorem, there exists a commutative ring R with $|\mathbb{P}^n_k| \cong |\mathrm{Spec}\,(R)|$ as topological spaces. The topology of projective space is indistinguishable from that of an affine scheme.

This is not an isolated phenomenon. Any quasi-compact and quasi-separated scheme has an underlying spectral space, and hence is homeomorphic to $\mathrm{Spec}\,(R)$ for some ring R. This includes virtually all schemes encountered in practice: all Noetherian schemes, all schemes of finite type over a field or over $\mathbb{Z}$, all separated schemes.

Proposition 2.8.3: Topology of Quasi-Compact Quasi-Separated Schemes

Let X be a quasi-compact and quasi-separated scheme. Then $|X|$ is a spectral space, and hence there exists a commutative ring R such that $|X| \cong |\mathrm{Spec}\,(R)|$ as topological spaces.

***Proof*:**
Quasi-compactness of X gives condition (1). Every scheme is sober (a point $x \in X$ has closure $\overline{\{x\}}$ which is irreducible, and if Z is irreducible closed then its generic point is the unique point η with $\overline{\{\eta\}} = Z$; this follows from the corresponding fact for affine schemes, which glues). For condition (3): the quasi-compact opens of X include all affine opens and their finite unions, which form a basis. The quasi-separated hypothesis says exactly that the intersection of two quasi-compact opens is quasi-compact, giving closure under finite intersection.

To make this concrete, here are some examples.

Example 2.8.4: Non-Affine Schemes with Affine Topology

1. *Projective space.* As discussed above, $|\mathbb{P}^n_k|$ is spectral. Over $k = \overline{k}$ algebraically closed, the closed points of $\mathbb{P}^1_k$ are in bijection with $\overline{k} \cup \{\infty\}$, and the topology is the cofinite topology on this set together with a generic point whose closure is the whole space. This is homeomorphic to $|\mathrm{Spec}\,(k[x])|$ (which also has the cofinite topology on its closed points, plus a generic point). In particular, $|\mathbb{P}^1_k| \cong |\mathbb{A}^1_k|$ as topological spaces, even though $\mathbb{P}^1_k$ and $\mathbb{A}^1_k$ are not isomorphic as schemes.
2. *The line with doubled origin.* The non-separated scheme obtained by gluing two copies of $\mathbb{A}^1_k$ along $\mathbb{A}^1_k \setminus \{0\}$ is quasi-compact and quasi-separated, so its topology is spectral. This scheme is not affine (it is not even separated), yet topologically it is homeomorphic to $\mathrm{Spec}\,(R)$ for some ring R.
3. *The punctured plane.* The scheme $U = \mathbb{A}^2_k \setminus \{(x, y)\}$ from Example 2.2.51 is quasi-compact and quasi-separated (it is separated, being an open subscheme of an affine scheme), hence

spectral. It is not affine, yet its topology is that of some $\operatorname{Spec}(R)$.

The constructible topology and Stone spaces

Hochster's theorem tells us that a topological space is the spectrum of a ring if and only if it is spectral. But the definition of a spectral space involves a somewhat asymmetric collection of conditions. There is a cleaner perspective that explains *what structure a spectral space actually carries*, and it comes from a classical duality in topology and logic.

Given a spectral space X, define the *constructible topology* (also called the *patch topology*) on the underlying set of X by declaring both the quasi-compact open subsets *and their complements* to be open. Equivalently, the constructible topology is the coarsest topology in which every quasi-compact open of X is clopen (both open and closed).

Proposition 2.8.5: The Constructible Topology is a Stone Space

Let X be a spectral space. Then X equipped with the constructible topology is a *Stone space*: a topological space that is compact, Hausdorff, and totally disconnected.

***Proof*:**
(Sketch) The constructible topology is Hausdorff because any two distinct points of a spectral space can be separated by a quasi-compact open (one point lies in it, the other does not), which is clopen in the constructible topology. It is compact because the constructible topology is finer than the original spectral topology (which is already quasi-compact), and one can verify that any cover by constructible opens has a finite subcover using the quasi-compactness of the spectral topology together with the finiteness of intersections. Total disconnectedness follows because the clopen sets (which include all quasi-compact opens and their complements) separate points.

We denote the set X equipped with the constructible topology by X^{cons}. The identity map $X^{\mathrm{cons}} \to X$ is continuous (the constructible topology is finer), but not a homeomorphism: the constructible topology is strictly finer than the spectral topology whenever X has a non-closed point (i.e., whenever X is not discrete).

The name "constructible topology" is not a coincidence. A subset $C \subseteq X$ of a topological space is *constructible* if it belongs to the Boolean algebra generated by the open sets—equivalently, if C is a finite union of locally closed subsets ($U \cap Z$ where U is open and Z is closed). In a spectral space, the constructible subsets have a clean characterization:

Proposition 2.8.6: Constructible Sets are Clopens

Let X be a spectral space. A subset $C \subseteq X$ is constructible (in the classical sense: a finite Boolean combination of quasi-compact opens) if and only if C is clopen in the constructible topology.

This connects directly to Chevalley's theorem (**??**): if $f : X \to Y$ is a morphism of finite type between Noetherian schemes, then the image of a constructible set is constructible. In the language of the constructible topology, this says that f induces a continuous map $X^{\mathrm{cons}} \to Y^{\mathrm{cons}}$ between the associated Stone spaces, and continuous maps between Stone spaces preserve clopens. From this perspective, Chevalley's theorem is a statement about the *continuity of scheme morphisms with respect to the constructible topology*—a viewpoint that is both conceptually cleaner and technically powerful.

Stone duality and the lattice of opens

The appearance of Stone spaces is not accidental; it reflects a duality from mathematical logic and lattice theory.

Stone duality (1936) establishes an anti-equivalence of categories between Boolean algebras and Stone spaces: every Boolean algebra B is the algebra of clopen subsets of a unique Stone space $S(B)$, and conversely, every Stone space arises in this way. Restricting to distributive lattices (which need not be Boolean), one obtains *spectral spaces*: a spectral space is precisely a Stone space together with a distinguished sublattice of its Boolean algebra of clopens.

More precisely, let X^{cons} be a Stone space. A sublattice $L \subseteq \mathrm{Clopen}(X^{\mathrm{cons}})$ that forms a basis for some topology on the underlying set, and is closed under finite intersection and finite union, determines a spectral topology on X: the topology generated by L. Different choices of L give different spectral topologies on the same underlying Stone space.

This gives a clean factorization of the data in a spectral space:

$$\text{Spectral space} \;=\; \text{Stone space} \;+\; \text{Choice of lattice.}$$

The Stone space is the "combinatorial skeleton"—it knows which subsets are constructible (i.e., definable by first-order formulas in model-theoretic language). The lattice records which constructible sets are "open," i.e., which direction the specialization order runs.

Example 2.8.7: Different Spectral Structures on the Same Stone Space

Consider the Cantor set $C \subseteq [0, 1]$, which is a Stone space (compact, Hausdorff, totally disconnected). Every countable Boolean algebra has C as its Stone space. In particular, the spaces $|\mathrm{Spec}\,(\mathbb{Z})|$ and $|\mathrm{Spec}\,(k[x])|$ (for k algebraically closed) both have the same constructible topology—they are both countable spectral spaces whose Stone space is the Cantor set—but they carry different spectral topologies (different lattices of opens) corresponding to the different lattice structures of their prime ideals.

More concretely, both spaces consist of a generic point η together with countably many closed points, and the topology in both cases has $\{\eta\}$ dense and the closed points discrete in the complement. But the "arithmetic" of which opens contain which closed points differs: in $\mathrm{Spec}\,(\mathbb{Z})$, the closed point (p) lies in $D(q)$ for $q \neq p$, while in $\mathrm{Spec}\,(k[x])$, the closed point $(x - a)$ lies in $D(x - b)$ for $b \neq a$. These are different lattices on the same Stone space.

What does the topology actually store?

The data of a scheme $(X, \mathcal{O}_X)$ decomposes into three layers:

1. *The Stone space* X^{cons}: the combinatorial skeleton. It knows the set of points, the constructible subsets, and (via Stone duality) the Boolean algebra they generate. This is the coarsest invariant—it is shared by many different schemes.

2. *The spectral topology*: a choice of sublattice L of the clopens of X^{cons}, determining which constructible sets are "open." This encodes the specialization order (which points are limits of which others) and hence the notion of "generic" and "closed" points. Different spectral topologies on the same Stone space give genuinely different spaces (e.g., $\operatorname{Spec}(\mathbb{Z})$ vs. $\operatorname{Spec}(k[x])$), but by Hochster's theorem, both are realizable as $\operatorname{Spec}(R)$ for some ring.

3. *The structure sheaf* $\mathcal{O}_X$: the algebraic data. It determines which continuous maps are "algebraic," what the global sections and cohomology are, and whether the scheme is affine, projective, smooth, etc. This is the finest invariant, and it is where the geometry lives.

The passage from (3) to (2) forgets the algebra; the passage from (2) to (1) forgets the specialization order. Neither passage is reversible: many non-isomorphic schemes share the same topology (as we saw with $\mathbb{P}^1_k$ and $\mathbb{A}^1_k$), and many non-homeomorphic spectral spaces share the same Stone space (as with $\operatorname{Spec}(\mathbb{Z})$ and $\operatorname{Spec}(k[x])$).

The conclusion is that the Zariski topology, while important for defining the notion of "local" and for the combinatorics of specialization and generization, carries far less information than one might expect. The structure sheaf is the essential datum: it determines which continuous maps are "algebraic," which open subsets are "distinguished," and what "functions" look like. Two schemes can share the same topological space—or even the same constructible topology—while having completely different algebraic geometry.

3

Schemes II: Morphisms and Constructions

Scheme morphisms are inherently more *algebraic* than *smooth* in nature: they are in some ways more stringent than holomorphic maps (only "polynomial-like" functions are allowed), but they also permit singularities. In this chapter, we shall properly explore morphisms of schemes, classify many important types, and interpret them geometrically. In the process, we will be able to "recover" the correct analogues of Hausdorffness and compactness for schemes in the form of *separatedness* and *properness.* The condition of being proper will turn out to be closely related to being projective: by Chow's lemma, every proper scheme over a Noetherian base admits a surjective birational morphism from a projective scheme, showing that projective schemes are the natural "compact schemes" to consider. The chapter concludes by defining an abstract variety and showing that $\mathbf{Var}_k$ is a fully faithful subcategory of $\mathbf{Sch}_k$.

Algebraically, smooth and holomorphic morphisms share an important feature with scheme morphisms: a continuous map $f : M \to N$ between smooth manifolds is a *smooth map* if and only if $f^*(C^\infty(N)) \subseteq C^\infty(M)$ where

$$f^* : C^\infty(N) \to C^\infty(M) \qquad f^*(\varphi) = \varphi \circ f$$

and in particular if f is a homeomorphism, it is a diffeomorphism if and only if f^* is a ring isomorphism. In essence, f "preserves" and is "compatible with" all the relations on N given by smooth functions. As schemes are defined by their algebraic relations, this pullback approach generalizes naturally[1].

With this in mind, we shall require the following properties from scheme morphisms:

1. A scheme morphism $f : X \to Y$ must preserve the structure sheaf (which encodes all algebraic

[1] In the smooth case, the "if and only if" makes this an equivalent definition of smooth map; since there is only one type of "affine patch" (open subsets of $\mathbb{R}^n$), one can then recover the usual definition. Proposition 3.1.10 establishes the analogous result for schemes.

information). Concretely, for every open set $V \subseteq Y$, there should be a ring homomorphism $f^\#(V) : \mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}(V))$ that pulls back local sections on Y to local sections on X. The intuition is that we treat sections as genuine (scheme) functions, and we require their composition with f to remain a scheme function. Thus, a scheme morphism must be a *morphism of ringed spaces* (a continuous map together with a compatible sheaf map).

This additional structure is genuinely necessary: as we saw in example 2.1.25, not all continuous maps between spectra are induced by ring homomorphisms, and there is no natural way to recover a ring homomorphism from a mere continuous map. A simple instance is that $\operatorname{Spec} k$ and $\operatorname{Spec} k'$ for two non-isomorphic fields k, k' have homeomorphic underlying spaces (both are single points) yet are not isomorphic as schemes.

2. The pullback of functions must be compatible with evaluation: if $s \in \mathcal{O}_Y(V)$ vanishes at $f(x) \in Y$, then $f^\#(s)$ must vanish at $x \in X$. This means the pullback must respect the local ring structure at each point, making f a morphism of *locally* ringed spaces.

3. Ring homomorphisms between global sections and scheme morphisms between affine schemes must be dual to each other (generalizing the fact that a map between k-varieties is regular if and only if the contravariant map between coordinate rings is a k-algebra homomorphism).

This final condition means we want an equivalence of categories:

$$\mathbf{Sch}^{\mathbf{Aff}} \underset{\operatorname{Spec}(-)}{\overset{\Gamma(-)}{\rightleftarrows}} \mathbf{CRing}^{\mathrm{op}}$$

In section ref:HERE, this equivalence is generalized from affine schemes to all schemes by replacing $\mathbf{CRing}^{\mathrm{op}}$ with the appropriate category of sheaves of rings on a site (the functor-of-points perspective).

For those familiar with differential geometry [7], let us take a moment to compare and contrast smooth maps with scheme morphisms.

In differential geometry, continuous maps between manifolds can be arbitrarily well approximated by smooth maps (as a consequence of *Whitney's Approximation Theorem*). Hence, smooth maps can be thought of as "smoothing out" continuous maps – the smooth and continuous worlds are not far apart. This is *not at all* the case in the Zariski topology. Consider, for example, any permutation $\sigma : \overline{k} \to \overline{k}$ that preserves finite subsets (where $\overline{k}$ is an algebraically closed field). Such a σ induces a homeomorphism of $\mathbb{A}^1_{\overline{k}}$ in the Zariski topology – since the closed sets are exactly the finite sets and the whole space, any bijection preserving finiteness is continuous. Yet a "wild" permutation (one not given by a polynomial) will not be a scheme morphism[2]. Thus, there is no hope of approximating arbitrary continuous maps by regular maps: scheme morphisms are fundamentally more rigid than smooth maps.

In fact, regular maps behave much more like *holomorphic* maps. Both are determined by their local algebraic/analytic expressions, both satisfy analogues of the identity theorem (a regular map vanishing on a dense open must vanish everywhere), and both have fibers of controlled dimension. This similarity becomes an exact correspondence for the right categories: the GAGA theorems of Serre establish an equivalence between algebraic and analytic coherent sheaves on projective varieties over $\mathbb{C}$ (see section 6.9). Hence, when building geometric intuition for scheme morphisms via [7]

[2] Over $\mathbb{C}$, any discontinuous field automorphism $\sigma \in \operatorname{Aut}(\mathbb{C}/\mathbb{Q})$ gives such an example; these exist in abundance (assuming the axiom of choice) but none are $\mathbb{C}$-algebra maps.

and [**NateComplexAnalysis**], it is holomorphic maps, not smooth maps, that provide the better mental model. A key difference is that scheme morphisms freely allow singularities and have unique algebraic-geometric features absent from the complex-analytic or smooth settings (for example, the Frobenius morphism from example 3.1.11, or the behavior of morphisms between schemes over $\operatorname{Spec}\mathbb{Z}$).

3.1 Elementary Concepts

Some of the discussion here shall overlap with the content in section 1.4; this was done as the self-completeness of the material in this section is a priority.

Recall (definition 1.4.4) that a morphism of ringed spaces $\pi : X \to Y$ is a continuous map of topological spaces together with a map $\pi^{\#} : \mathcal{O}_Y \to \pi_*\mathcal{O}_X$ that replicates composing any function in $\mathcal{O}_Y$ with π[3]. However, this notion is not restrictive enough for scheme theory for two reasons:

(i) Not all *continuous* maps between spectra are induced by ring homomorphisms (see Example 2.1.25), so specifying a sheaf map is genuinely additional data beyond the continuous map. Again, to abuse the same example, $\operatorname{Spec}(k)$ and $\operatorname{Spec}(k')$ for non-isomorphic fields k, k' are homeomorphic (both are single points) but carry different structure sheaves.

(ii) Even among ringed space morphisms, the global sections functor $\Gamma(-)$ is *not injective* on morphism sets: distinct ringed space morphisms can induce the same ring homomorphism on global sections. The following example demonstrates this.

Example 3.1.1: Two Ringed Space Morphisms for One Ring Homomorphism
Take $R = \mathbb{Z}_{(p)}$ and $S = \mathbb{Q}$ so that $\operatorname{Spec}(R) = \{[(p)], [(0)]\}$ and $\operatorname{Spec}(S) = \{[(0)]\}$. Let us drop the parentheses and square brackets to declutter notation. The open sets of $\operatorname{Spec}(R)$ are:

$$\emptyset, \quad \{0\}, \quad \{0, p\} = \operatorname{Spec}(R)$$

The stalks of $\operatorname{Spec}(R)$ are $\mathbb{Z}_{(p)}$ at p and $\mathbb{Q}$ at 0. The stalk of $\operatorname{Spec}(S)$ is $\mathbb{Q}$ at 0.

There is only one ring homomorphism $\psi : R \to S$, namely the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$.

In the category **Sch**, this should correspond to a unique map $\iota^a : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ defined by $\psi^{-1}((0)) = (0)$. However, in the category **RingedSp** we can define two morphisms between these affine schemes:

1. *Map 1 (The "correct" map):* $\pi_1 : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ defined by $0 \mapsto 0$. The sheaf map is determined by the stalk map at the unique point $0 \in \operatorname{Spec}(S)$:

$$(\pi_1)^{\#}_0 : \mathcal{O}_{R,0} \to \mathcal{O}_{S,0} \qquad \Longrightarrow \qquad \mathbb{Q} \xrightarrow{\text{id}} \mathbb{Q}$$

This is a local ring homomorphism (both sides are fields, with maximal ideal (0) mapping to (0)). On global sections, it induces the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$.

2. *Map 2 (The "phantom" map):* $\pi_2 : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ defined by $0 \mapsto p$. Since $\{p\}$ is closed, this is a continuous map. The sheaf map is determined by the stalk map at

[3] Or equivalently by adjointness, the map $\pi^{-1}(\mathcal{O}_Y) \to \mathcal{O}_X$.

$0 \in \operatorname{Spec}(S)$:

$$(\pi_2)_0^\# : \mathcal{O}_{R,p} \to \mathcal{O}_{S,0} \qquad \Longrightarrow \qquad \mathbb{Z}_{(p)} \xrightarrow{\text{inclusion}} \mathbb{Q}$$

This is a valid ring homomorphism, so π_2 is a valid morphism of ringed spaces. On global sections, this *also* induces the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$.

Thus, we have found two distinct ringed space morphisms $\pi_1 \neq \pi_2$ that induce the *same* ring homomorphism on global sections. We reject π_2 because the stalk map $\mathbb{Z}_{(p)} \to \mathbb{Q}$ is *not* a local ring homomorphism: the maximal ideal $(p) \subseteq \mathbb{Z}_{(p)}$ maps to the unit ideal in $\mathbb{Q}$ (since p is invertible in $\mathbb{Q}$), violating the condition $\varphi(\mathfrak{m}_R) \subseteq \mathfrak{m}_S$. The definition of locally ringed space morphism correctly filters out this non-geometric map.

Thus, if $\mathbf{RingedSp}^{\mathbf{Aff}}$ denotes the category whose objects are $(\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ and whose morphisms are ringed space morphisms, the functor:

$$\Gamma(-) : \mathbf{RingedSp}^{\mathbf{Aff}} \to \mathbf{CRing}^{\mathrm{op}}$$

is surjective on hom-sets (every ring homomorphism $R \to S$ induces at least one ringed space morphism via the standard contraction-of-primes construction), but not injective (as the example above demonstrates).

The key realization is that ring homomorphisms induce local structure that must be preserved. By proposition 2.2.21, the stalks of schemes are all *local rings*. A map between local rings need not preserve maximal ideals (for instance, the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$ sends the maximal ideal (p) to the unit ideal). We thus single out maps that do preserve this structure:

Definition 3.1.2: Local Ring Homomorphism

Let $\varphi : (R, \mathfrak{m}_R) \to (S, \mathfrak{m}_S)$ be a ring homomorphism between local rings with maximal ideals $\mathfrak{m}_R$ and $\mathfrak{m}_S$ respectively. Then φ is a *local ring homomorphism* if $\varphi(\mathfrak{m}_R) \subseteq \mathfrak{m}_S$, or equivalently $\varphi^{-1}(\mathfrak{m}_S) = \mathfrak{m}_R$.

A ringed space morphism that is a local ring homomorphism at each stalk is given a name:

Definition 3.1.3: Locally Ringed Space Morphism

Let X, Y be locally ringed spaces. A *morphism of locally ringed spaces* $(\pi, \pi^\#) : X \to Y$ is a morphism of ringed spaces such that the induced stalk map $\pi_p^\# : \mathcal{O}_{Y,\pi(p)} \to \mathcal{O}_{X,p}$ is a local ring homomorphism for every $p \in X$.

Observe that ring homomorphisms *automatically* induce local maps on localizations: if $\varphi : R \to S$ is a ring homomorphism and $\mathfrak{p} \subseteq S$ is prime, then the induced map $\varphi_\mathfrak{p} : R_{\varphi^{-1}(\mathfrak{p})} \to S_\mathfrak{p}$ given by $r/t \mapsto \varphi(r)/\varphi(t)$ satisfies $\varphi_\mathfrak{p}(\mathfrak{m}_R) \subseteq \mathfrak{m}_S$. Indeed, if $r/t \in \mathfrak{m}_R$ then $r \in \varphi^{-1}(\mathfrak{p})$, so $\varphi(r) \in \mathfrak{p}$, giving $\varphi(r)/\varphi(t) \in \mathfrak{p}S_\mathfrak{p} = \mathfrak{m}_S$. Equivalently, $\mathfrak{m}_R = \varphi_\mathfrak{p}^{-1}(\mathfrak{m}_S)$[4]. Example 3.1.1 shows that this locality condition need *not* hold for arbitrary ringed space morphisms, even between locally ringed spaces. The local ring condition forces the morphism to "respect" not just the lattice structure of opens but also the evaluation of functions at each point.

[4]Note this does not imply $\varphi(\mathfrak{m}_R) = \mathfrak{m}_S$: the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Z}_{(p)}[1/q]$ for primes $p \neq q$ is a local ring homomorphism where $\varphi(\mathfrak{m}_R) \subsetneq \mathfrak{m}_S$.

With this added structure, we recover a bijection:

Theorem 3.1.4: Ring Homomorphisms and Locally Ringed Space Morphisms

Let R, S be commutative rings. Then:

1. Every ring homomorphism $\varphi : R \to S$ induces a natural morphism of locally ringed spaces:
$$(f, f^{\#}) : (\operatorname{Spec}(S), \mathcal{O}_{\operatorname{Spec}(S)}) \to (\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$$
2. Every morphism of locally ringed spaces $(\operatorname{Spec}(S), \mathcal{O}_{\operatorname{Spec}(S)}) \to (\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ induces a ring homomorphism $\varphi : R \to S$ (via global sections).
3. These operations are inverse to each other. In particular, the locally ringed space morphism is uniquely determined by the ring homomorphism on global sections:
$$\operatorname{Hom}_{\mathbf{LRS}}(\operatorname{Spec}(S), \operatorname{Spec}(R)) \cong \operatorname{Hom}_{\mathbf{CRing}}(R, S)$$

***Proof*:**
(1) Ring homomorphism ⇒ locally ringed space morphism. Let $\varphi : R \to S$ be a ring homomorphism. Define the continuous map $f : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ by the standard contraction $f(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$.

We construct the sheaf morphism $f^{\#} : \mathcal{O}_{\operatorname{Spec}(R)} \to f_*\mathcal{O}_{\operatorname{Spec}(S)}$ on the basis of distinguished open sets, using proposition 1.3.3. For each $g \in R$, we must define a map

$$f^{\#}(D(g)) : \mathcal{O}_{\operatorname{Spec}(R)}(D(g)) \longrightarrow (f_*\mathcal{O}_{\operatorname{Spec}(S)})(D(g)) = \mathcal{O}_{\operatorname{Spec}(S)}(f^{-1}(D(g))).$$

The preimage of a distinguished open under f is again distinguished:

$$\begin{aligned} f^{-1}(D(g)) &= \left\{[\mathfrak{p}] \in \operatorname{Spec}(S) \;\middle|\; \varphi^{-1}(\mathfrak{p}) \in D(g)\right\} \\ &= \left\{[\mathfrak{p}] \in \operatorname{Spec}(S) \;\middle|\; g \notin \varphi^{-1}(\mathfrak{p})\right\} \\ &= \{[\mathfrak{p}] \in \operatorname{Spec}(S) \mid \varphi(g) \notin \mathfrak{p}\} \\ &= D(\varphi(g)) \end{aligned}$$

This is a special property of the affine setting: the pullback of a "basic" geometric object is again basic. Using the identifications $\mathcal{O}_{\operatorname{Spec}(R)}(D(g)) \cong R_g$ and $\mathcal{O}_{\operatorname{Spec}(S)}(D(\varphi(g))) \cong S_{\varphi(g)}$, the global map φ induces a map on localizations. Since $\varphi(g)$ is a unit in $S_{\varphi(g)}$, the universal property of localization provides a unique ring homomorphism:

$$\Phi_g : R_g \longrightarrow S_{\varphi(g)}, \qquad \frac{a}{g^n} \longmapsto \frac{\varphi(a)}{\varphi(g)^n}.$$

These maps are compatible with restriction: if $D(h) \subseteq D(g)$ (so $h \in \sqrt{(g)}$), the diagram

$$\begin{tikzcd} R_g \arrow[r, "\Phi_g"] \arrow[d, "\mathrm{loc}"] & S_{\varphi(g)} \arrow[d, "\mathrm{loc}"] \\ R_h \arrow[r, "\Phi_h"] & S_{\varphi(h)} \end{tikzcd}$$

commutes because both paths send a/g^n to $\varphi(a)/\varphi(g)^n$ viewed in $S_{\varphi(h)}$. By proposition 1.3.3, the maps Φ_g glue to a unique sheaf morphism $f^\#$.

At each stalk, the induced map $f^\#_{\mathfrak{p}} : R_{\varphi^{-1}(\mathfrak{p})} \to S_{\mathfrak{p}}$ is the localization of φ at $\mathfrak{p}$, which is a local ring homomorphism as shown above. Hence $(f, f^\#)$ is a morphism of locally ringed spaces.

(2) Locally ringed space morphism ⇒ ring homomorphism. Given a morphism of locally ringed spaces $(f, f^\#) : \mathrm{Spec}\,(S) \to \mathrm{Spec}\,(R)$, evaluate the sheaf map on global sections:

$$f^\#(\mathrm{Spec}\,(R)) : \mathcal{O}_{\mathrm{Spec}\,(R)}(\mathrm{Spec}\,(R)) \to \mathcal{O}_{\mathrm{Spec}\,(S)}(f^{-1}(\mathrm{Spec}\,(R)))$$

Since $\mathcal{O}_{\mathrm{Spec}\,(R)}(\mathrm{Spec}\,(R)) \cong R$ and $\mathcal{O}_{\mathrm{Spec}\,(S)}(\mathrm{Spec}\,(S)) \cong S$, this yields a ring homomorphism $\varphi : R \to S$.

(3) These operations are inverse. Starting with $\varphi : R \to S$, constructing $(f, f^\#)$, and taking global sections recovers φ by construction.

Conversely, starting with a locally ringed space morphism $(f, f^\#)$ and letting $\varphi = f^\#(\mathrm{Spec}\,(R)) : R \to S$, and g be the morphism defined in step (1) given by $g(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$, we must show that $(f, f^\#)$ equals the morphism $(g, g^\#)$ constructed in step (1).

The continuous maps agree: Let $\mathfrak{p} \in \mathrm{Spec}\,(S)$. Consider the commutative diagram relating global sections to stalks:

$$\begin{tikzcd} R \arrow[r, "\varphi"] \arrow[d, "\mathrm{loc}"] & S \arrow[d, "\mathrm{loc}"] \\ R_{f(\mathfrak{p})} \arrow[r, "f^\#_{\mathfrak{p}}"] & S_{\mathfrak{p}} \end{tikzcd}$$

We compute $\varphi^{-1}(\mathfrak{p}) \subseteq R$ by tracing the preimage of $\mathfrak{m}_{S_{\mathfrak{p}}} \subseteq S_{\mathfrak{p}}$ along two paths:

- *Algebraic path (right then down):* The localization $S \to S_{\mathfrak{p}}$ pulls $\mathfrak{m}_{S_{\mathfrak{p}}}$ back to $\mathfrak{p}$. Then φ pulls $\mathfrak{p}$ back to $\varphi^{-1}(\mathfrak{p})$.
- *Geometric path (down then right):* Since $(f, f^\#)$ is a morphism of *locally* ringed spaces, $f^\#_{\mathfrak{p}}$ is a local ring homomorphism, so $(f^\#_{\mathfrak{p}})^{-1}(\mathfrak{m}_{S_{\mathfrak{p}}}) = \mathfrak{m}_{R_{f(\mathfrak{p})}}$. The localization $R \to R_{f(\mathfrak{p})}$ pulls this maximal ideal back to $f(\mathfrak{p})$.

By commutativity, both paths yield the same ideal: $f(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p}) = g(\mathfrak{p})$. Hence $f = g$ as continuous maps.

The sheaf maps agree: Since $f = g$, both $(f, f^\#)$ and $(g, g^\#)$ are sheaf morphisms over the same continuous map. To show $f^\# = g^\#$, it suffices to check equality on stalks ($\mathcal{O}_{\mathrm{Spec}\,(R)}$ is a sheaf, so morphisms out of it are determined by stalks). At each $\mathfrak{p} \in \mathrm{Spec}\,(S)$, both stalk maps are local ring homomorphisms $R_{\varphi^{-1}(\mathfrak{p})} \to S_{\mathfrak{p}}$ making the above diagram commute. Since the localization map $R \to R_{\varphi^{-1}(\mathfrak{p})}$ is an epimorphism in the category of rings (every element of $R_{\varphi^{-1}(\mathfrak{p})}$ is a fraction r/s with $r, s \in R$), any ring homomorphism out of $R_{\varphi^{-1}(\mathfrak{p})}$ that agrees with φ when precomposed with the localization map must be the unique localization $\varphi_{\mathfrak{p}}$. Hence $f^\#_{\mathfrak{p}} = g^\#_{\mathfrak{p}}$ for all $\mathfrak{p}$, giving $f^\# = g^\#$.

Affine schemes form the correct category to be the opposite category of commutative rings. We can rephrase what we just proved as:

Corollary 3.1.5: Equivalence of Rings and Affine Schemes

The category of affine schemes with locally ringed space morphisms is equivalent to the opposite category of commutative rings:

$$\mathbf{Sch}^{\mathbf{Aff}} \underset{\operatorname{Spec}(-)}{\overset{\Gamma(-)}{\rightleftarrows}} \mathbf{CRing}^{\mathrm{op}}$$

where Spec and Γ are inverse equivalences. More generally, for any locally ringed space X and any ring R, this equivalence extends to an adjunction on all of **LRS**:

$$\operatorname{Hom}_{\mathbf{CRing}}(R, \Gamma(X)) \cong \operatorname{Hom}_{\mathbf{LRS}}(X, \operatorname{Spec}(R))$$

where $\Gamma(-)$ is the left adjoint and $\operatorname{Spec}(-)$ is the right adjoint. On the subcategory of affine schemes, this adjunction restricts to an equivalence because both the unit $(X \to \operatorname{Spec}\Gamma(X))$ and the counit $(\Gamma(\operatorname{Spec}(R)) \to R)$ are isomorphisms.

Having shown that locally ringed morphisms are the right choice, let us take a moment to understand better how evaluation at a point is "preserved" by locally ringed space morphisms. Given a morphism of locally ringed spaces $\pi : X \to Y$, a point $q \in Y$ and a point $p \in X$ with $\pi(p) = q$, how do we interpret the local ring homomorphism between stalks? Say there is a section $g \in \mathcal{O}_Y(V)$ on an open subset $V \subseteq Y$ that vanishes at q, i.e., $g(q) = 0$ (equivalently, $g \in \mathfrak{m}_q$ in the stalk $\mathcal{O}_{Y,q}$). We want the pullback $\pi^{\#}(g)$ to vanish at p:

$$g(\pi(p)) = 0 \quad \Rightarrow \quad (\pi^{\#} g)(p) = 0.$$

The local ring condition ensures exactly this: since $\pi_p^{\#} : \mathcal{O}_{Y,q} \to \mathcal{O}_{X,p}$ satisfies $\pi_p^{\#}(\mathfrak{m}_q) \subseteq \mathfrak{m}_p$, the vanishing $g \in \mathfrak{m}_q$ implies $\pi^{\#}(g) \in \mathfrak{m}_p$, which means $\pi^{\#}(g)$ vanishes at p.

$$\begin{array}{ccc}
\mathcal{O}_{Y,q} & \xrightarrow{\pi_p^{\#}} & \mathcal{O}_{X,p} \\[2em]
g \in \mathfrak{m}_q & \longmapsto & \pi^{\#}(g) \in \mathfrak{m}_p \\[4em]
Y \supseteq V & & X \supseteq \pi^{-1}(V) \\
\in & & \in \\
q = \pi(p) & \xleftarrow[\pi]{} & p
\end{array}$$

The top row encodes the algebraic condition (the stalk map sends the maximal ideal into the maximal ideal); the bottom row encodes the geometric picture (the point p maps to q under π). The locally ringed space condition is the requirement that these two rows are compatible: vanishing is preserved under pullback. π is not injective, for each $p \in \pi^{-1}(q)$, there would be a local map.

This is "obvious" when g and π are our usual functions since $g(\pi(p)) = (g \circ \pi)(p)$. However, the phantom map from Example 3.1.1 shows that this can fail for general ringed space morphisms.

Consider for the last time:

$$\pi_2 : \mathrm{Spec}\,(\mathbb{Q}) \to \mathrm{Spec}\,(\mathbb{Z}_{(p)}), \qquad \pi_2([(0)]) = [(p)].$$

The function $p \in \mathcal{O}_Y(Y) = \mathbb{Z}_{(p)}$ vanishes at the closed point $[(p)]$: its value in the residue field $\kappa([(p)]) = \mathbb{Z}_{(p)}/(p) \cong \mathbb{F}_p$ is $p \bmod p = 0$. However, the pullback $\pi_2^{\#}(p)$ is $p \in \mathbb{Q}$ (via the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$), and the value of p at the unique point $[(0)] \in \mathrm{Spec}\,(\mathbb{Q})$ is $p \bmod (0) = p \neq 0$ in $\kappa([(0)]) = \mathbb{Q}$. The vanishing was not preserved.

3.1.6 Remark: the generic point of $\mathrm{Spec}\,(\mathbb{Z}_{(p)})$ The reader may find it surprising that $\{[(0)]\}$ is an open subset of $\mathrm{Spec}\,(\mathbb{Z}_{(p)})$: it is the distinguished open $D(p)$, the complement of the closed point $V(p) = \{[(p)]\}$. This is *not* typical behaviour for generic points. In $\mathrm{Spec}\,(\mathbb{Z})$ or $\mathrm{Spec}\,(k[x, y])$, the generic point is *not* open—its closure is the entire space. The peculiarity here is that $\mathbb{Z}_{(p)}$ is a local domain with exactly two prime ideals, so the space has only two points, and removing a closed point from a two-point space leaves an open point. This makes $\mathrm{Spec}\,(\mathbb{Z}_{(p)})$ the simplest non-trivial example for testing morphism conditions, precisely because the topology is so small that pathological maps are easy to construct explicitly.

3.1.7 Sections as Functions: Three Perspectives The reader may be troubled by a philosophical point: in differential geometry, a section $f \in C^\infty(U)$ *is* a function $f : U \to \mathbb{R}$, and evaluation at a point x is simply $f(x)$. For schemes, a section $f \in \mathcal{O}_X(U)$ is an element of a ring, and evaluation at x is the composite

$$\mathcal{O}_X(U) \longrightarrow \mathcal{O}_{X,x} \longrightarrow \kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x,$$

which is an additional construction. Is the datum of evaluation "missing" from the definition? The answer is no—it is encoded in the locally ringed space condition, but implicitly. There are three equivalent ways to see this.

(a) Hartshorne's approach. Define a section over U to be a function $f : U \to \coprod_{x\in U} \mathcal{O}_{X,x}$ such that $f(x) \in \mathcal{O}_{X,x}$ for each x, and such that f is "locally a quotient of ring elements." This makes sections into genuine functions and evaluation into literal function evaluation. The cost is that the codomain $\coprod \mathcal{O}_{X,x}$ varies from point to point and the definition requires a local coherence condition.

(b) The Modern approach. Keep sections as abstract ring elements. The locally ringed space axiom guarantees that each stalk $\mathcal{O}_{X,x}$ is local with a unique maximal ideal $\mathfrak{m}_x$, hence a unique residue field $\kappa(x)$, hence a unique evaluation map. The datum of evaluation is not "added"—it is a *consequence* of the locally ringed space structure. If the stalks were not local (i.e., if there were multiple maximal ideals), there would be multiple residue fields and no canonical notion of "the value of f at x." The LRS axiom is precisely the condition that evaluation is well-defined.

(c) The functor-of-points approach. By the representability of the global section functor (Proposition 3.3.8), a section $f \in \mathcal{O}_X(U)$ corresponds to a morphism $U \to \mathbb{A}^1_k$. In this picture, sections *are* morphisms, and evaluation at a point x is the composite $\mathrm{Spec}\,(\kappa(x)) \to U \to \mathbb{A}^1_k$, which is genuinely a function value.

All three perspectives carry exactly the same information. Approach (a) is most concrete; (b) is most algebraically efficient; (c) is most categorically natural. This text primarily uses (b), with (c) developed in section 3.3.

3.1.1 Scheme Morphisms

Inspired by the algebraic-geometric properties of locally ringed space morphisms, we define morphisms between schemes as locally ringed space morphisms:

Definition 3.1.8: Morphism of Schemes

Let X, Y be schemes. A morphism of locally ringed spaces $\pi : X \to Y$ is called a *morphism of schemes*. Schemes together with scheme morphisms form the category **Sch**.

One of the most important ways to control scheme morphisms is to reduce them to an affine cover. The following result is slightly more general but implies the desired result:

Lemma 3.1.9: Gluing Morphisms of Ringed Spaces

Let X, Y be ringed spaces and $X = \bigcup_i U_i$ an open cover. Suppose there exist ringed space morphisms $f_i : U_i \to Y$ that agree on overlaps:

$$f_i\big|_{U_i \cap U_j} = f_j\big|_{U_i \cap U_j}$$

Then there is a unique morphism of ringed spaces $f : X \to Y$ such that $f\big|_{U_i} = f_i$ for all i. Furthermore, if each f_i is a morphism of locally ringed spaces, then so is f.

***Proof*:**
Continuity. By the topological gluing lemma, the maps f_i glue to a unique continuous map $f : X \to Y$ with $f|_{U_i} = f_i$.

Sheaf morphism. We must construct $f^\# : \mathcal{O}_Y \to f_*\mathcal{O}_X$. For any open $V \subseteq Y$, we need a ring homomorphism $f^\#(V) : \mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}(V))$.

The open set $f^{-1}(V) \subseteq X$ is covered by $\{f^{-1}(V) \cap U_i\}_i$. Since $f|_{U_i} = f_i$, we have $f_i^{-1}(V) = f^{-1}(V) \cap U_i$. The sheaf morphism $f_i^\# : \mathcal{O}_Y \to (f_i)_*\mathcal{O}_{U_i}$ gives, for each i, a ring homomorphism:

$$f_i^\#(V) : \mathcal{O}_Y(V) \longrightarrow \mathcal{O}_{U_i}(f^{-1}(V) \cap U_i)$$

On overlaps $f^{-1}(V) \cap U_i \cap U_j$, the agreement condition $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ implies $f_i^\#|_{U_i \cap U_j} = f_j^\#|_{U_i \cap U_j}$, so:

$$f_i^\#(V)(s)\big|_{f^{-1}(V) \cap U_i \cap U_j} = f_j^\#(V)(s)\big|_{f^{-1}(V) \cap U_i \cap U_j} \qquad \forall s \in \mathcal{O}_Y(V)$$

The sets $\{f^{-1}(V) \cap U_i\}_i$ cover $f^{-1}(V)$, and the local images $f_i^\#(V)(s)$ are sections of $\mathcal{O}_X$ that agree on overlaps. By the gluability axiom of $\mathcal{O}_X$, there exists a unique section $t \in \mathcal{O}_X(f^{-1}(V))$ restricting to $f_i^\#(V)(s)$ on each $f^{-1}(V) \cap U_i$. Define $f^\#(V)(s) := t$. This is a ring homomorphism since each $f_i^\#(V)$ is, and it commutes with restriction by construction. Uniqueness of $f^\#$ follows from the locality axiom of $\mathcal{O}_X$.

Locally ringed space condition. If each f_i is an LRS morphism, then for any $p \in X$, p lies in some U_i and the stalk map $(f^\#)_p = (f_i^\#)_p$ is a local ring homomorphism by assumption. Hence f is an LRS morphism.

By lemma 3.1.9, we can control morphisms of schemes by looking at ring homomorphisms. We can even ensure the codomain is an affine scheme:

Proposition 3.1.10: Scheme Morphisms are Locally Affine

A morphism $\pi : X \to Y$ between schemes is a morphism of schemes if and only if it is *locally affine*: every point $x \in X$ has affine open neighborhoods $U = \operatorname{Spec} R \ni x$ and $V = \operatorname{Spec} S \ni \pi(x)$ with $\pi(U) \subseteq V$, such that $\pi|_U : U \to V$ is induced by a ring homomorphism $\varphi : S \to R$.

***Proof*:**
($\Rightarrow$) Scheme morphisms are locally affine. Let $\pi : X \to Y$ be a morphism of schemes and $x \in X$. Since Y is a scheme, there exists an affine open $V = \operatorname{Spec} S$ containing $\pi(x)$. By continuity, $\pi^{-1}(V)$ is an open neighborhood of x in X; since X is a scheme, it can be covered by affine opens, so there exists an affine open $U = \operatorname{Spec} R \ni x$ with $U \subseteq \pi^{-1}(V)$, hence $\pi(U) \subseteq V$.

The restricted map $\pi|_U : \operatorname{Spec} R \to \operatorname{Spec} S$ is a morphism of locally ringed spaces between affine schemes. By theorem 3.1.4, it is uniquely induced by the ring homomorphism on global sections:

$$\varphi := \Gamma(\pi|_U) : S \cong \mathcal{O}_Y(V) \xrightarrow{\pi|_U^\#(V)} \mathcal{O}_X(U) \cong R$$

($\Leftarrow$) Locally affine morphisms are scheme morphisms. Suppose $\pi : X \to Y$ is an LRS morphism and every $x \in X$ admits affine neighborhoods as above. Let $\{U_i = \operatorname{Spec} R_i\}$ be an open cover of X with corresponding affine opens $\{V_i = \operatorname{Spec} S_i\}$ in Y such that $\pi(U_i) \subseteq V_i$ and $\pi|_{U_i}$ is induced by a ring homomorphism $\varphi_i : S_i \to R_i$.

Each $\pi|_{U_i}$ is an LRS morphism (by theorem 3.1.4), so the stalk map at every $p \in U_i$ is a local ring homomorphism. Since $\{U_i\}$ covers X, the stalk map at every $p \in X$ is local. Hence π is an LRS morphism between schemes, i.e., a morphism of schemes.

Example 3.1.11: Morphisms Of Schemes

1. Recall the ring map $\varphi : \mathbb{C}[y] \to \mathbb{C}[x]$ given by $y \mapsto x^2$. Geometrically, this corresponds to the map $f : \mathbb{A}^1_x \to \mathbb{A}^1_y$ sending $a \mapsto a^2$. Let us verify the sheaf map on a specific open set. Consider the section $s = 3/(y-4) \in \mathcal{O}_{\mathbb{A}^1_y}(D(y-4))$, defined on the open set $V = D(y-4) = \mathbb{A}^1 \setminus \{4\}$. The preimage of this set is:

$$f^{-1}(V) = \left\{a \in \mathbb{C} \;\middle|\; a^2 \neq 4\right\} = \mathbb{A}^1 \setminus \{-2, 2\}$$

   The induced map on sections $f^\#_V : \mathcal{O}_{\mathbb{A}^1_y}(V) \to \mathcal{O}_{\mathbb{A}^1_x}(f^{-1}(V))$ sends s to:

$$f^\#_V(s) = \frac{3}{x^2 - 4}$$

   which is indeed a regular function on $f^{-1}(V) = D(x-2) \cap D(x+2)$.

2. Let $U \subseteq Y$ be an open subset of a scheme Y. There is a natural morphism of ringed spaces:

$$\iota : (U, \mathcal{O}_Y|_U) \to (Y, \mathcal{O}_Y)$$

This is the open embedding as seen in definition 2.2.48. It is important to note the behavior of the stalks. For any point $p \in U$, the stalk map $\iota_p^\# : \mathcal{O}_{Y,\iota(p)} \to \mathcal{O}_{U,p}$ is an *isomorphism* (specifically, the identity map). This distinguishes open embeddings from other types of immersions.

3. *Tautological Projection (Affine Cone):* Consider the projection from the punctured affine space to projective space:

$$\pi : \mathbb{A}_k^{n+1} \setminus \{0\} \to \mathbb{P}_k^n \qquad (x_0, \dots, x_n) \mapsto [x_0 : \cdots : x_n]$$

This is a morphism of schemes. To see the sheaf map, we check it on the standard affine cover of $\mathbb{P}^n$. Let $U_i = D_+(x_i) \cong \mathbb{A}_k^n$ be the standard open charts of projective space. The preimage is the distinguished open set $\pi^{-1}(U_i) = D(x_i) \subseteq \mathbb{A}^{n+1} \setminus \{0\}$. The rings of sections are:

$$\mathcal{O}_{\mathbb{P}^n}(U_i) \cong k\left[\frac{x_0}{x_i}, \dots, \frac{x_n}{x_i}\right] \qquad \text{and} \qquad \mathcal{O}_{\mathbb{A}^{n+1}\setminus\{0\}}(D(x_i)) \cong k[x_0, \dots, x_n]_{x_i}$$

The sheaf map $\pi^\#(U_i)$ is the natural inclusion of the degree-0 subring into the full localization:

$$k\left[\frac{x_0}{x_i}, \dots, \frac{x_n}{x_i}\right] \hookrightarrow k[x_0, \dots, x_n]_{x_i}$$

Since these inclusions commute with further localization (restriction), they glue to form a valid sheaf morphism. Furthermore, the topological map π restricted to $D(x_i) \to U_i$ is exactly the map of spectra induced by this ring homomorphism (given by pulling back prime ideals $\mathfrak{p} \mapsto \mathfrak{p} \cap \mathcal{O}_{\mathbb{P}^n}(U_i)$), confirming this matched pair forms a valid morphism of schemes.

4. *Structure Map of Proj:* Let $S_\bullet$ be a finitely generated graded R-algebra (where $S_0 = R$). There is a natural structure morphism:

$$\pi : \operatorname{Proj} S_\bullet \to \operatorname{Spec}(R)$$

This is induced locally. On the basic open set $D_+(f) \subseteq \operatorname{Proj} S_\bullet$ (for homogeneous f), the ring of sections is $S_{(f)}$. The map $R \to S_\bullet$ induces a map $R \to S_{(f)}$ via $r \mapsto r/1$. These local maps glue to give the global morphism π.

5. Let $\varphi : S_\bullet \to R_\bullet$ be a *surjective* homomorphism of graded rings. This induces a *closed embedding* of schemes in the opposite direction:

$$f : \operatorname{Proj} R_\bullet \hookrightarrow \operatorname{Proj} S_\bullet$$

For example, the projection $k[x, y, z] \to k[x, y, z]/(x^2 + y^2 - z^2)$ induces the inclusion of the conic into $\mathbb{P}^2$.

6. Consider the projection $\pi : \operatorname{Spec}(\mathbb{C}[x, y]) \to \operatorname{Spec}(\mathbb{C}[x])$ induced by the inclusion $\mathbb{C}[x] \hookrightarrow \mathbb{C}[x, y]$.

   - *Closed Points:* A point $(x - 3, y - a)$ maps to the contraction $(x - 3)$. Geometrically, $(3, a) \mapsto 3$.
   - *Generic Points:* Consider the generic point of the parabola, $\eta = [(y - x^2)]$. The image is the contraction $(y - x^2) \cap \mathbb{C}[x]$. Since no nonzero polynomial in $\mathbb{C}[x]$ alone lies in $(y - x^2)$, the intersection is (0). Thus, the parabola maps to the generic point of the affine line!

7. Let $p \in X$ be a point of a scheme. We can form a scheme $\operatorname{Spec}(\mathcal{O}_{X,p})$. There is a canonical morphism $\operatorname{Spec}(\mathcal{O}_{X,p}) \to X$ induced by the localization maps $\mathcal{O}_X(U) \to \mathcal{O}_{X,p}$ for open neighborhoods U of p. For $X = \mathbb{A}^1_k$ and $p = (x)$, this is the map $\operatorname{Spec}(k[x]_{(x)}) \to \operatorname{Spec}(k[x])$.

   The scheme $\operatorname{Spec}(\mathcal{O}_{X,p})$ can be thought of as the "intersection of all open neighborhoods of p," but it is *not* an open embedding: $\operatorname{Spec}(\mathcal{O}_{X,p})$ has only the points of X that specialize to p (i.e., the prime ideals contained in $\mathfrak{m}_p$), which typically includes the generic point but not other closed points. The key property is that the stalk map at p is an isomorphism $\mathcal{O}_{X,p} \xrightarrow{\sim} \mathcal{O}_{\operatorname{Spec}(\mathcal{O}_{X,p}),\, \mathfrak{m}_p}$, since localizing a local ring at its maximal ideal has no effect.

8. *The Frobenius Morphism (Non-isomorphism bijection):* Let $X = \operatorname{Spec}(\mathbb{F}_p[t])$. Consider the ring homomorphism $\varphi : \mathbb{F}_p[t] \to \mathbb{F}_p[t]$ given by $t \mapsto t^p$. This induces a scheme morphism $F : X \to X$.

   - *Topologically:* The map on points is $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$. For a closed point $(t-a)$ where $a \in \overline{\mathbb{F}_p}$, we compute:

     $$\varphi^{-1}((t-a)) = \{f \in \mathbb{F}_p[t] \mid f(t^p) \in (t-a)\} = \{f \mid f(a^p) = 0\} = (t - a^p)$$

     Since $a^p = a$ in $\mathbb{F}_p$ (by Fermat's little theorem), we get $\varphi^{-1}((t-a)) = (t-a)$. Similarly, $\varphi^{-1}((0)) = (0)$. Thus, F is the *identity* on the underlying topological space.
   - *Sheaf-theoretically:* The map on sections is $t \mapsto t^p$. This is *not* an isomorphism: it is not surjective, as t is not in the image.

   Thus, F is a homeomorphism that is not an isomorphism of schemes. This distinction is invisible to the Zariski topology but detected by the structure sheaf.

   To give some insight on how we can extract information from this geometric perspective: once we define differentials in section 5.5, we shall see that $d(t^p) = pt^{p-1}\,dt = 0$, and hence the tangent map between tangent spaces will be the *zero map* at each point. We can think of this as "infinitesimal shrinking" at each point, while globally it is the identity!

If the codomain of a scheme morphism is an affine scheme, then we can completely determine the behaviour of the scheme morphisms by looking at the ring morphism between the global sections:

Proposition 3.1.12: Adjunction of Affine Schemes

Let $\operatorname{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec}(R))$ be the collection of morphism of scheme from a scheme X to an $\operatorname{Spec}(R)$. Then it is in natural bijection to the ring homomorphisms $\operatorname{Hom}_{\mathrm{Ring}}(R, \mathcal{O}_X(X))$:

$$\operatorname{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec}(R)) \cong \operatorname{Hom}_{\mathbf{CRing}}(R, \mathcal{O}_X(X))$$

In particular, The map $\theta : X \to \operatorname{Spec}(\Gamma(X, \mathcal{O}_X))$ is an injection. Furthermore, this is an isomorphism if and only if X is an affine scheme.

Note that the ring $\mathcal{O}_X(X)$ can be very complex, even in nice cases where X is a finite-type k-scheme, for example it need not be finitely generated.

***Proof*:**
If $X = \text{Spec}(S)$, then theorem 3.1.4 gives the bijection. Next, map $\pi : X \to \text{Spec}(R)$ certainly induces a ring homomorphism $\varphi : R \to \mathcal{O}_X(X)$, we shall show that a ring homomorphism $\varphi : R \to \mathcal{O}_X(X)$ induces a unique scheme map $\pi : X \to \text{Spec}(R)$ and that this map is the inverse of the above map. By definition of a scheme:

$$\pi : \bigcup_i \text{Spec}\left(S_i\right) \to \text{Spec}(R)$$

where we get a restriction map $\pi_i : \text{Spec}\left(S_i\right) \to \text{Spec}(R)$ along with sheaf morphisms, for which the global section map is $\varphi_i : R \to \mathcal{O}_X\left(\text{Spec}\left(S_i\right)\right)$. As $\text{Spec}\, S_i \subseteq X$ is an open subset, there is a natural restriction map $\mathcal{O}_X(X) \to \mathcal{O}_X\left(\text{Spec}\, S_i\right)$. We thus get a composition:

$$R \to \mathcal{O}_X(X) \to \mathcal{O}_X\left(\text{Spec}\, S_i\right)$$

The map $R \to \mathcal{O}_X\left(\text{Spec}\, S_i\right)$ is uniquely determined by the ring homomorphism $S_i \to R$, and as X is covered by the $\text{Spec}\, S_i$, by lemma 3.1 .6 there is a unique way to glue the maps of $\mathcal{O}_X\left(\text{Spec}\, S_i\right)$ to $\mathcal{O}_X(X)$, completing the proof.

The converse is not true:

Example 3.1.13: Maps From Affine To Projective
Take $\text{Hom}_{\text{Sch}}\left(\text{Spec}\, \mathbb{Q}[x], \mathbb{P}^1_{\mathbb{Q}}\right)$ and $\text{Hom}_{\mathbf{Ring}}(\mathbb{Q}, \mathbb{Q}[x])$ (recall that $\mathcal{O}_X\left(\mathbb{P}^1_{\mathbb{Q}}\right) \cong \mathbb{Q})$). Then

$$\text{Hom}_{\text{Ring}}\left(\mathbb{Q}, \mathbb{Q}[x]\right) = \{\text{id}\}$$

namely since since the units of $\mathbb{Q}$ must map to the units of $\mathbb{Q}[x]$, which are all contains in $\mathbb{Q}$, and as it must be that $1 \mapsto 1$, we are forced to have the identity. However, there are multiple maps from $\text{Spec}\, \mathbb{Q}[x]$ to $\mathbb{P}^1_k$; simply map $\text{Spec}\, \mathbb{Q}[x]$ into the open subset $U_0 \cong k[y] \subseteq \mathbb{P}^1_k$ by taking $x \mapsto p(y)$ for some polynomial $p(y) \in k[y]$. More generally, we can let choose a field k[a]

The intuition is that $\mathcal{O}_X(X)$ in general contains less information than X as the global sections on a general scheme need to restrict to local sections on affine open subsets, and hence there may be in general less of them.

Show that the canonical morphism $X \to \text{Spec}\left(\mathcal{O}_X(X)\right)$ is an isomorphism if and only if X is a affine.

[a] Note that an [injective] ring homomorphism $k \to k$ need not be surjective, consider $F(x) \to F(x)$ given by $x \mapsto x^2$, hence the argument requires a bit more work

Using proposition 3.1.12, we can get a few more interesting results:

Example 3.1.14: Morphisms to Affine and Projective Space

1. *Maps to Affine Space are Tuples of Functions:* Let X be *any* scheme (not necessarily affine). What is a morphism $X \to \mathbb{A}^n_k$? Using the Adjunction property (proposition 3.1.12), we have:

$$\text{Hom}_{\mathbf{Sch}}(X, \mathbb{A}^n_k) \cong \text{Hom}_{\mathbf{Sch}}(X, \text{Spec}\, k[t_1, \ldots, t_n])$$
$$\cong \text{Hom}_{\mathbf{Ring}}(k[t_1, \ldots, t_n], \mathcal{O}_X(X))$$

A ring homomorphism from a polynomial ring is uniquely determined by where it sends the variables t_i. Let f_i be the image of t_i in $\Gamma(X, \mathcal{O}_X)$. Thus, a morphism to affine space is exactly equivalent to choosing n global functions $(f_1, \ldots, f_n)$ on X. This generalizes the classical fact that a map to $\mathbb{R}^n$ is given by n coordinate functions.

Notice how this also shows that there is no non-trivial map $\mathbb{P}^n_k \to \mathbb{A}^m_k$ (exercise 3.1.2(3.1.2 (4))).

s

2. *Maps to Projective Space (The need for Line Bundles):* One might hope to define a map $X \to \mathbb{P}^n_k$ similarly by choosing $n+1$ global functions $f_0, \ldots, f_n$ to serve as homogeneous coordinates $[f_0, \ldots, f_n]$. This naturally fails when using the structure sheaf since global functions on projective space are just constants, which would make such maps trivial (as illustrated in example 3.1.13). Instead, the coordinate functions x_i on $\mathbb{P}^n$ are sections of a twisted sheaf, not the structure sheaf. Consequently, to define a scheme morphism to projective space, one requires a *Line Bundle* (an invertible sheaf $\mathcal{L}$) on X and $n+1$ global sections $s_i \in \Gamma(X, \mathcal{L})$ that do not vanish simultaneously. This distinction – that maps to affine space use functions, while maps to projective space use bundles – is the primary motivation for the study of *Invertible Sheaves* in section 5.3.3.

Our last result in this section is a special case where gluing affine schemes *does* produce an affine scheme:

Corollary 3.1.15: Characterizing Affine Schemes Locally

Let X be a scheme. Then X is an affine scheme if and only if there is a finite subset of elements $f_1, \ldots, f_n \in \mathcal{O}_X(X)$ such that X_f is affine and $(f_1, \ldots, f_n) = \mathcal{O}_X(X)$

***Proof*:**
If X is affine, take $f_1 = 1$. Conversely, assume there exist $f_1, \ldots, f_n \in \mathcal{O}_X(X)$ such that each X_{f_i} is affine and $(f_1, \ldots, f_n) = \mathcal{O}_X(X)$. This ideal condition implies there exist $a_i \in \mathcal{O}_X(X)$ such that $\sum_{i=1}^n a_i f_i = 1$. Let $R = \mathcal{O}_X(X)$. We claim that $X \cong \operatorname{Spec} R$.

By proposition 3.1.12, the identity homomorphism on R induces a canonical morphism of schemes:

$$\theta : X \longrightarrow \operatorname{Spec} R$$

To show θ is an isomorphism, it suffices to verify it locally on an open cover of the target $\operatorname{Spec} R$. The condition $\sum_{i=1}^n a_i f_i = 1$ ensures that the distinguished open sets $D(f_1), \ldots, D(f_n)$ form an open cover of $\operatorname{Spec} R$.

For each i, the preimage of $D(f_i)$ under θ is precisely the locus of points in X where the global section f_i does not vanish, which is by definition X_{f_i}. Since the $D(f_i)$ cover $\operatorname{Spec} R$, their preimages X_{f_i} necessarily cover X.

Consider the restriction of θ to one of these patches:

$$\theta|_{X_{f_i}} : X_{f_i} \longrightarrow D(f_i) \cong \operatorname{Spec}(R_{f_i})$$

By our hypothesis, X_{f_i} is an affine scheme. Because the global sections of a scheme on a distinguished open set defined by a global function f_i are given by the localization of the global sections ring, we have

$$\mathcal{O}_X(X_{f_i}) \cong R_{f_i}$$

Thus, by the Adjunction property (corollary 3.1.5), the morphism $\theta|_{X_{f_i}}$ corresponds exactly to the identity ring isomorphism $R_{f_i} \xrightarrow{\sim} R_{f_i}$. Hence, each restriction $\theta|_{X_{f_i}}$ is an isomorphism of schemes.

Since θ is a morphism of schemes that restricts to an isomorphism on the sets of an open cover $\{X_{f_i}\}$ mapping to an open cover $\{D(f_i)\}$, we can apply the gluing lemma for morphisms (lemma 3.1.9). The local inverses glue uniquely to form a global inverse, proving that θ is an isomorphism and $X \cong \operatorname{Spec} R$.

3.1.2 S-Schemes

Recall that all abelian rings can be seen as $\mathbb{Z}$-algebras, and that $\mathbb{Z}$ is initial in the category of commutative rings. From this perspective, all ring homomorphism $\varphi : R \to S$ are in bijective correspondence with morphisms that make the following diagram commute:

$$\begin{tikzcd} R \arrow[rr] & & S \\ & \mathbb{Z} \arrow[ul] \arrow[ur] & \end{tikzcd}$$

Then we may generalize the notion of abelian groups to R-algebras by replacing $\mathbb{Z}$ with a ring R. We may further augment the restriction morphisms to R-algebra homomorphisms as we are working with commutative rings; this was done in definition definition 2.2.17. Now, this structure was given by a scheme morphism $X \to \operatorname{Spec} R$, and an R-scheme morphism is a scheme morphism between R-schemes $X \to Y$ where the following diagram commutes:

$$\begin{tikzcd} X \arrow[rr] \arrow[dr] & & Y \arrow[dl] \\ & \operatorname{Spec}(R) & \end{tikzcd}$$

This is the commutative diagram of fiber-preserving morphism, the same requirement that is imposed of fiber-bundle morphisms. And indeed, R-schemes behave very much like bundles over $\operatorname{Spec} R$ (as the examples bellow will demonstrates), and when the morphism $X \to \operatorname{Spec} R$ is nice enough (ex. it is étale, see ref:HERE), then it shall even be locally trivial.

Thus, it is natural to consider the base space to be an arbitrary scheme:

Definition 3.1.16: S-Scheme

Let $\mathbf{Sch}_S$ be the category who objects are scheme morphisms:

$$X \to S$$

and whose morphisms are commutative diagrams:

$$\begin{tikzcd} X \arrow[rr] \arrow[dr] & & Y \arrow[dl] \\ & S & \end{tikzcd}$$

As compositions, $f \circ \pi_X = \pi_Y$. The objects of this category or called *S-scheme.* The morphism of schemes $X \to S$ is called the *structure morphism.*

To see what extra structure is added, let us start the case of $S = \operatorname{Spec}(R)$. Then each open subset of X has the structure of an R-algebra, and these generalize definition 2.2.17. If $S = \operatorname{Spec}(R)$, then we would often say that the S-scheme (i.e. $\operatorname{Spec}(R)$-scheme) is an R-scheme

Proposition 3.1.17: Final Object Of Scheme

In **Sch**, the scheme $\operatorname{Spec}\mathbb{Z}$ is the final object. This makes the category of schemes isomorphic to the category of $\mathbb{Z}$-schemes.

If k is a field, $\operatorname{Spec} k$ is the final object in the category of k-schemes.

***Proof*:**
Recall that $\mathbb{Z}$ is initial in Ring, and thus it is easy to show it is initial in Comm-Ring, then as Schemes (resp. k-schemes) are the co-categories to rings (resp. k-algebras), the result is immediate.

Working over S-schemes shall restrict to much “simpler” cases than over general schemes. For example, if studying complex analysis and complex varieties, we would expect Spec($\mathbb{C}$) to have trivial automorphisms, and we may want to interpret $\operatorname{Spec}\left(\mathbb{R}[x]/\left(x^2+1\right)\right)$ without too much “gluing” done by the automorphism ring, namely we want $\mathbb{C}$ be be trivial.

Example 3.1.18: The Category of S-schemes
An *S-scheme* is a scheme X equipped with a morphism $\pi : X \to S$, called the structure morphism. A morphism of S-schemes $f : X \to Y$ is a morphism of schemes such that the triangle with the base S commutes .

1. *Schemes over k (“Geometry” Schemes)*: If $S = \operatorname{Spec}(k)$ for a field k, an S-scheme closely resembles an algebraic set (over k). By proposition 3.1.12, the structure morphism $X \to \operatorname{Spec}(k)$ is equivalent to the data of a k-algebra homomorphism $k \to \Gamma(X, \mathcal{O}_X)$. This fixes the field of scalars, ensuring that the functions on X are k-algebras, not just rings.

   When working with polynomials, we are often interested in k-linear homomorphisms $k[x] \to k[x]$ where k is fixed (e.g., coordinate changes). If we only looked at ring homomorphisms, include field automorphisms (like complex conjugation or the Frobenius). We enforce linearity by requiring the algebra diagram to commute:

$$\begin{tikzcd} k[x] \arrow[rr, "\varphi"] & & k[x] \\ & k \arrow[ul, "\iota"] \arrow[ur, "\iota"'] & \end{tikzcd}$$

In the language of schemes, this corresponds to a morphism of S-schemes (where $S = \mathrm{Spec}(k)$):

$$\begin{tikzcd} \mathrm{Spec}(k[x]) \arrow[rr, "f"] \arrow[dr, "\pi_X"'] & & \mathrm{Spec}(k[x]) \arrow[dl, "\pi_Y"] \\ & \mathrm{Spec}(k) & \end{tikzcd}$$

The condition $f \circ \pi_X = \pi_Y$ ensures that the morphism preserves the underlying field structure.

2. *Projective Schemes:* Projective schemes over a field, $\mathbb{P}^n_k = \mathrm{Proj}(k[x_0, \ldots, x_n])$, are naturally k-schemes. The structure map $\mathbb{P}^n_k \to \mathrm{Spec}(k)$ is induced by the inclusion of the degree 0 scalars $k \hookrightarrow k[x_0, \ldots, x_n]$.

3. *Field Extensions as Coverings:* Let L/K be a finite field extension. Then $\mathrm{Spec}(L)$ and $\mathrm{Spec}(K)$ are both single points topologically. However, the inclusion $K \hookrightarrow L$ induces a morphism of schemes $\pi : \mathrm{Spec}(L) \to \mathrm{Spec}(K)$. We don't want to include morphisms permuting K, hence we limit to k-scheme morphisms While topologically trivial, this morphism encodes the arithmetic complexity of the extension.

3.1.19 Foreshadowing Étale Fundamental Group Later, we shall see that $\mathrm{Spec}(L)$ acts as a "covering space" for $\mathrm{Spec}(K)$. This analogy can be made more precise K, L be perfect fields and let L/K be a finite extension. Then $L = k[\alpha]$ for a root of the appropriate irreducible polynomial $m_{\alpha,L/K}$. Then taking advantage of the fact that the category of K-field extensions is *isomorphic* (not just equivalent) to the category of rings of integers with base ring $\mathcal{O}_K$ (exercise if the reader forgot) The morphism $\mathrm{Spec}\,(L) \to \mathrm{Spec}\,(K)$ is associated to a (dense) morphism $\mathrm{Spec}\,(\mathcal{O}_L) \to \mathrm{Spec}\,(\mathcal{O}_K)$ (as $\mathcal{O}_K \to \mathcal{O}_L$ is injective). In section 3.5.1, we shall see morphism between affine schemes are uniquely determined by their value on dense sets. As $\mathcal{O}_L$ and $\mathcal{O}_K$ are both dimension 1 rings, these are *covering of curves*. This covering hints at the generalization of covering spaces from algebraic topology, leading to the *Étale Fundamental Group*, which generalizes the Galois group $\mathrm{Gal}(L/K)$ to schemes, allowing us to study loops and homotopy in an arithmetic setting.

Lemma 3.1.20: Stalks of k-schemes

Let X be a k-scheme and $x \in X$ so that $x = [\mathfrak{p}]$ for some prime $\mathfrak{p} \subseteq R_i = k\left[u_i\right]_i$. Then

$$\mathcal{O}_{X,\mathfrak{p}} \cong \left(k\left[u_i\right]_i\right)_{\mathfrak{p}}$$

In particular, we have that the quotient by $\mathfrak{m}_x$ is a k-algebra:

$$\kappa(x) = k(x)$$

***Proof*:**
As X is a k-scheme, each $\mathcal{O}_X(U)$ is a k-algebra, and the colimit of k-algebra's is a k-algebra (they are closed under products and equalizers, see [**NateClassicalAlgGeo**]). The rest comes from proposition 1.1.5.

Note that the residue may be other fields, ex. $\mathbb{R}(x) = \mathbb{C}$ (take $\operatorname{Spec} \mathbb{R}[t]$ where and choose a degree > 1 point).

One of the most important first impositions for many algebraic manipulations are finiteness conditions:

Definition 3.1.21: Locally Finite Type And Finite Type

Let X, Y be S-schemes. Then a morphism $f : X \to Y$ is locally of finite type if there exists an affine open cover $V_j = \operatorname{Spec}\left(S_j\right)$ of Y where $f^{-1}\left(V_i\right)$ can be covered by affine open subsets $U_{ij} = \operatorname{Spec}\left(A_{ij}\right)$ where A_{ij} is a finitely generated S_i-algebra, that is

$$f^{\#}\left(V_i\right) : S_j = \mathcal{O}_Y\left(V_j\right) \to \mathcal{O}_X\left(f^{-1}\left(V_i\right)\right) \xrightarrow{\text{res}} \mathcal{O}_Y\left(U_{ij}\right) = R_{ij}$$

induces a finitely generated S_j-algebra on each R_{ij}. If furthermore each $f^{-1}\left(V_j\right)$ can be covered by finite number of affine subsets, f is said to be of finite type.

An S-scheme X is said to be of locally finite type (resp. finite type) over S if $f : X \to S$ is of locally finite type (resp. finite type).

3.1.3 Subschemes: Closed, Locally Closed, and More

Open subschemes (section 2.2.8) are the simplest type of subscheme: given an open subset $U \subseteq X$, the restriction $(U, \mathcal{O}_X|_U)$ carries a unique scheme structure compatible with X. The situation for closed subsets is fundamentally different: a closed subset can carry *multiple* scheme structures, distinguished by the amount of "infinitesimal data" they retain. Beyond the open and closed cases, there are subschemes that are locally closed (closed inside an open subset), and there are natural "local" subobjects that do not correspond to any topological subset at all.

The reader may remark that sub-objects usually correspond to monomorphisms. While the definition is formally definable the resulting schemes for **Sch**

Closed subschemes and closed immersions

Given a surjection $R \twoheadrightarrow R/I$, by proposition 2.1.26 the image $\operatorname{Spec}(R/I) \subseteq \operatorname{Spec}(R)$ is a closed subset, and the inclusion is a morphism of locally ringed spaces. For a general scheme, this condition generalizes to requiring that the sheaf map be surjective.

The multiplicity issue deserves emphasis. Consider the closed point $\{0\} \subseteq \mathbb{A}^1_k$. As a topological subspace, $\{0\}$ is just a point. But there are infinitely many scheme structures on this point compatible with the ambient scheme: $\operatorname{Spec}(k[x]/(x))$, $\operatorname{Spec}(k[x]/(x^2))$, $\operatorname{Spec}(k[x]/(x^3))$, and so on. These differ in how much nilpotent (infinitesimal) structure they retain. The underlying reason is elementary: $0^n = 0$ for all n, so the vanishing locus of x and x^n are identical as subsets, but $k[x]/(x^n)$ remembers the first $n-1$ derivatives. In contrast, open subschemes have a *unique* compatible scheme structure: the restriction of $\mathcal{O}_X$ to the open set.

Definition 3.1.22: Closed Immersion and Closed Subscheme

Let $f : Y \to X$ be a morphism of schemes. Then f is a *closed immersion* if:

1. f induces a homeomorphism from $|Y|$ onto a closed subset of $|X|$, and
2. the sheaf map $f^\# : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is surjective (equivalently, surjective on stalks).

A *closed subscheme* of X is a closed subset $Z \subseteq |X|$ together with a scheme structure on Z such that the inclusion $\iota : Z \hookrightarrow X$ is a closed immersion.

The distinction between a closed immersion and a closed subscheme is that a closed immersion $f : Y \to X$ is a morphism (which might come with an isomorphism $Y \cong Z$ for some closed subscheme Z), while a closed subscheme is always given with the literal inclusion as the map. In practice, the two are used interchangeably.

When define closed subsets of affine schemes, we used $V(-)$. This construction can also be used to construct closed subschemes of an affine scheme, but it shall forget the sheaf-theorem information (ex. nilpotence). This shall be recovered in definition 5.1.36 by inputting *ideal shaves* in $V(-)$.

Note that the complement of a closed subscheme is a unique open subscheme (with its canonical structure), while the complement of an open subscheme admits *multiple* closed subscheme structures—usually infinitely many, corresponding to different choices of nilpotent thickening. This asymmetry between open and closed is a recurring theme. There is a natural choice of closd subscheme if nilpotent information is removed; this is a *reduced subscheme*, see exercise ref:HERE.

Example 3.1.23: Closed Subschemes

1. *A point with its reduced structure.* The inclusion $\{0\} \hookrightarrow \mathbb{A}^1_k$ corresponds to the surjection $k[x] \twoheadrightarrow k[x]/(x) \cong k$. The image is closed, the sheaf map is surjective (and remains so on all localizations), so this is a closed immersion.
2. *A fat point.* The inclusion $\operatorname{Spec}(k[x]/(x^2)) \hookrightarrow \mathbb{A}^1_k$ corresponds to $k[x] \twoheadrightarrow k[x]/(x^2)$. The image is the same closed point $\{0\}$, but the scheme structure is different: the coordinate ring $k[x]/(x^2) \cong k[\epsilon]/(\epsilon^2)$ remembers one infinitesimal direction. More generally, $\operatorname{Spec}(k[x]/(x^n)) \hookrightarrow \mathbb{A}^1_k$ is a closed immersion for every $n \geq 1$, giving countably many closed subscheme structures on the single point $\{0\}$.

   This example also shows that the structure sheaf of a closed subscheme is *not* a subsheaf of

$\mathcal{O}_X$: the quotient $k[x]/(x^2)$ is not a subring of $k[x]$.

3. *The general affine case.* For any ideal $I \subseteq R$, the surjection $R \twoheadrightarrow R/I$ induces a closed immersion $\operatorname{Spec}(R/I) \hookrightarrow \operatorname{Spec}(R)$ with image $V(I)$.

4. *Projective subspaces.* The inclusion $\mathbb{P}^1_k \hookrightarrow \mathbb{P}^2_k$ given by $[s:t] \mapsto [s:t:0]$ is a closed immersion. On the standard affine chart $U_0 = \{x_0 \neq 0\}$, it corresponds to the surjection $k[y_1, y_2] \twoheadrightarrow k[y_1, y_2]/(y_2) \cong k[y_1]$.

It is worth noting that neither of the two conditions in the definition is redundant. The surjectivity of the sheaf map is not enough. The open immersion $D(x) \cup D(y) \hookrightarrow \mathbb{A}^2_k$ has a surjective sheaf map (in fact an isomorphism on each open), but the image is open, not closed. More exotic examples arise from non-separated schemes.

Similarly, topological closedness is not enough. The normalization of the cuspidal cubic provides an example. Let $Y = \operatorname{Spec}(k[x,y]/(y^2 - x^3))$ and consider the map $\nu : \mathbb{A}^1_k \to Y$ given by $t \mapsto (t^2, t^3)$, corresponding to the ring map $k[x,y]/(y^2 - x^3) \to k[t]$ with $x \mapsto t^2$, $y \mapsto t^3$. This map is a homeomorphism onto its image (which is all of $|Y|$) and $|Y|$ is closed in itself, but the ring map is *not* surjective: the element $t \in k[t]$ is not in the image (even though $y/x = t^3/t^2 = t$ is a rational function on Y, it is not regular). The failure of surjectivity reflects the singularity of Y at the origin—or algebraically, the failure of $k[t^2, t^3]$ to be integrally closed in $k[t]$.

Proposition 3.1.24: Properties of Closed Immersions

1. A morphism $f : Y \to X$ is a closed immersion if and only if for every affine open $\operatorname{Spec}(S) \subseteq X$, the preimage $f^{-1}(\operatorname{Spec}(S)) = \operatorname{Spec}(R)$ is affine and the induced map $S \to R$ is surjective.

2. Every surjective ring homomorphism $S \twoheadrightarrow R$ induces a closed immersion $\operatorname{Spec}(R) \hookrightarrow \operatorname{Spec}(S)$.

3. The composition of two closed immersions is a closed immersion.

***Proof*:**

1. One direction is immediate: if f is a closed immersion, then on each affine open $\operatorname{Spec}(S)$ the preimage is the closed subscheme $\operatorname{Spec}(S/I)$ for the appropriate ideal I, and $S \to S/I$ is surjective. The converse uses that surjectivity of a sheaf map can be checked on an affine cover of the codomain: if $f^\#|_{\operatorname{Spec}(S)}$ is surjective for each member of an affine cover, then $f^\#$ is surjective. A proof using ideal sheaves is given in section 5.1.4.

2. The surjection $S \twoheadrightarrow R = S/I$ induces a homeomorphism $\operatorname{Spec}(R) \cong V(I) \subseteq \operatorname{Spec}(S)$ onto a closed subset, and the sheaf map $\mathcal{O}_{\operatorname{Spec}(S)} \to f_*\mathcal{O}_{\operatorname{Spec}(R)}$ is surjective on each distinguished open (it corresponds to $S_g \twoheadrightarrow (S/I)_{\overline{g}}$, which is surjective).

3. Let $Z \xrightarrow{g} Y \xrightarrow{f} X$ be closed immersions. The composition $f \circ g$ is a homeomorphism onto a closed subset (a closed subset of a closed subset is closed). On each affine open $\operatorname{Spec}(S) \subseteq X$, by part (1), $f^{-1}(\operatorname{Spec}(S)) = \operatorname{Spec}(S/I)$ and $g^{-1}(\operatorname{Spec}(S/I)) = \operatorname{Spec}(S/J)$ for ideals $I \subseteq J \subseteq S$. The composite corresponds to $S \twoheadrightarrow S/J$, which is surjective.

As closed subschemes are determined by ideals on each open subset, it is natural to ask whether they can be described globally by an "ideal subsheaf" of $\mathcal{O}_X$. This is indeed the case, and the theory is developed in section 5.1.4 once we have the language of quasi-coherent sheaves.

Locally closed subschemes: immersions

If $X = \operatorname{Spec} R$ is an affine scheme and $U \subseteq X$ is an open subset, for any $S \subseteq \mathcal{O}_X(U)$, $V(S)$ is usually *not* a closed subset but a *locally closed subset*. These are the natural local solutions to regular functions, and thus deserve some attention:

Definition 3.1.25: Locally Closed Immersion

A morphism $f : Y \to X$ is a *locally closed immersion* (or simply an *immersion*) if f factors as

$$Y \xrightarrow{i} U \xhookrightarrow{\text{open}} X$$

where $U \subseteq X$ is an open subscheme and $i : Y \hookrightarrow U$ is a closed immersion. Equivalently, f induces a homeomorphism from $|Y|$ onto a locally closed subset of $|X|$ (i.e., the intersection $C \cap U$ of a closed set C and an open set U), with $f^\#$ surjective onto the structure sheaf of Y. A *locally closed subscheme* of X is a locally closed subset with the scheme structure making the inclusion an immersion.

Proposition 3.1.26: Open and Closed Immersions are Immersions

Every open immersion and every closed immersion is a locally closed immersion.

***Proof*:**

An open immersion $U \hookrightarrow X$ factors as $U \xrightarrow{\text{id}} U \hookrightarrow X$, where id is (trivially) a closed immersion. A closed immersion $Y \hookrightarrow X$ factors as $Y \hookrightarrow X \xrightarrow{\text{id}} X$, taking $U = X$.

Closedness of a subset $C \subseteq X$ can be verified on an open cover: C is closed if and only if $C \cap U_i$ is closed in U_i for every member of an open cover $\{U_i\}$ of X. Local closedness is a strictly weaker condition: C must be the intersection of a closed set with a *single* open set, not merely locally closed in each chart.

Locally closed subsets appear naturally in at least two important contexts. First, images of morphisms of varieties are typically locally closed (or finite unions thereof); this is the content of Chevalley's theorem (**??**). Second, the diagonal morphism $\Delta : X \to X \times_S X$ is always a locally closed immersion, and the condition that Δ be a *closed* immersion is precisely the definition of separatedness (section 3.5.1).

Local schemes: beyond topological subsets

The open, closed, and locally closed subschemes above all correspond to topological subsets of X equipped with a scheme structure. There is a fourth type of "subobject" that arises naturally but does *not* correspond to any topological subset: the *local scheme* at a point.

Let X be a scheme and $x \in X$ a point. The stalk $\mathcal{O}_{X,x}$ is a local ring, and we can form the scheme $\operatorname{Spec}(\mathcal{O}_{X,x})$. There is a canonical morphism

$$\iota_x : \operatorname{Spec}(\mathcal{O}_{X,x}) \longrightarrow X$$

defined as follows. Choose any affine open $\operatorname{Spec}(R) \subseteq X$ containing x, with x corresponding to a prime $\mathfrak{p} \subseteq R$. The localization map $R \to R_{\mathfrak{p}} = \mathcal{O}_{X,x}$ induces $\operatorname{Spec}(\mathcal{O}_{X,x}) \to \operatorname{Spec}(R) \hookrightarrow X$. This morphism is independent of the choice of affine open (the stalk is a colimit over all open neighborhoods, so any two choices yield the same map).

In what sense is $\operatorname{Spec}(\mathcal{O}_{X,x})$ a "subobject" of X? The morphism ι_x is *not* an immersion in any topological sense:

Proposition 3.1.27: The Local Scheme is Not a Topological Subscheme

Let X be a scheme of finite type over a field k and let $x \in X$ be a non-isolated point. Then $\iota_x : \operatorname{Spec}(\mathcal{O}_{X,x}) \to X$ is not an open immersion, not a closed immersion, and not a locally closed immersion. In particular, the image of ι_x is not a locally closed subset of $|X|$.

***Proof*:**
The image of ι_x consists of those primes $\mathfrak{q} \subseteq R$ with $\mathfrak{q} \subseteq \mathfrak{p}$ (these are exactly the primes that survive the localization $R \to R_{\mathfrak{p}}$). This is the set of *generizations* of x: the points $y \in X$ with $x \in \overline{\{y\}}$. In particular, the image contains the generic point of every irreducible component passing through x.

This set is neither open nor closed in general. It is not closed because it contains the generic point η of X (if X is irreducible), and $\overline{\{\eta\}} = X \neq \operatorname{im}(\iota_x)$. It is not open because removing the image from X does not yield a closed set (the closure of $\{x\}$ is not contained in $X \setminus \operatorname{im}(\iota_x)$). It is not locally closed because a locally closed set in a Noetherian space is a finite union of locally closed irreducible sets, and the set of generizations of x does not have this form when x has positive codimension.

Example 3.1.28: Local Schemes

1. *The generic point of* $\mathbb{A}^1_k$. Let $X = \mathbb{A}^1_k = \operatorname{Spec}(k[x])$ and let $\eta = [(0)]$ be the generic point. Then $\mathcal{O}_{X,\eta} = k(x)$, the function field, and $\operatorname{Spec}(\mathcal{O}_{X,\eta}) = \operatorname{Spec}(k(x))$ is a single point. The morphism $\iota_\eta : \operatorname{Spec}(k(x)) \to \mathbb{A}^1_k$ sends the unique point to η. This is the inclusion of the generic point, which is neither open nor closed in $\mathbb{A}^1_k$ (its closure is all of $\mathbb{A}^1_k$, but it is not itself closed).

   Actually, for $\mathbb{A}^1_k$ the generic point *is* a locally closed subset (it is the intersection of the open set X with the closure $\overline{\{\eta\}} = X$, so it is a dense locally closed subset). But the map ι_η is still not a locally closed immersion, because $\operatorname{Spec}(k(x))$ and $\{\eta\}$ (with its induced scheme structure as a subspace of $\mathbb{A}^1_k$) are not isomorphic as locally ringed spaces: the stalk of $\mathcal{O}_{\mathbb{A}^1}$ at η is $k(x)$, but the induced structure sheaf on $\{\eta\}$ as a subspace is $\mathcal{O}_{\mathbb{A}^1}(\mathbb{A}^1) = k[x]$, not $k(x)$.

2. *A closed point of* $\mathbb{A}^2_k$. Let $X = \mathbb{A}^2_k$ and $x = [(x,y)]$, the origin. Then $\mathcal{O}_{X,x} = k[x,y]_{(x,y)}$, and $\operatorname{Spec}(\mathcal{O}_{X,x})$ has two points: the closed point (the maximal ideal (x,y)) and the generic point (the zero ideal). The image of ι_x is $\{[(x,y)], [(0)]\}$—the origin together with the generic

point of $\mathbb{A}^2$. This is not open, not closed, and not locally closed in $\mathbb{A}^2_k$.

3. *A point on a curve in* $\mathbb{A}^2_k$. Let $C = V(y - x^2) \subseteq \mathbb{A}^2_k$ and let $x = [(x, y)]$ be the origin of C. Then $\mathcal{O}_{C,x} = (k[x, y]/(y - x^2))_{(x,y)} \cong k[x]_{(x)}$, which has two primes. The image of ι_x consists of the origin and the generic point of C—but not the generic point of $\mathbb{A}^2$ (which does not lie on C).

Despite not being a topological subscheme, the local scheme Spec $(\mathcal{O}_{X,x})$ is arguably the most important "subobject" associated to a point. It captures the local algebra of X at x—the ring $\mathcal{O}_{X,x}$ encodes the germs of regular functions near x—and many properties of X at x (regularity, normality, Cohen–Macaulay-ness) are properties of this single local ring. The morphism ι_x is flat (since localization is an exact functor), which makes it well-behaved from a homological perspective even though it is topologically ill-behaved.

We can place this in context with the other types of subobjects:

Type	*Topological condition*	*Sheaf condition*	*Uniqueness*
Open subscheme	image is open	$f^\#$ is an isomorphism	unique
Closed subscheme	image is closed	$f^\#$ is surjective	not unique
Locally closed	image is loc. closed	$f^\#$ surjective onto image	not unique
Local scheme at x	image = generizations of x	$f^\#$ is localization	unique

The first three are topological subobjects: the image is a subset with a specific topological property, and the scheme structure is constrained by the sheaf map. The fourth is algebraic rather than topological: it is determined by the localization at a prime, and the image has no clean topological description. Yet it is the one that controls the local geometry.

3.1.29 Immersions as finite-type monomorphisms There is a clean characterization that explains the role of locally closed immersions. Among monomorphisms of schemes, the immersions are (essentially) the ones that are *locally of finite type*: they impose only finitely many algebraic conditions locally. All three of our topological types—open immersions, closed immersions, and locally closed immersions—are locally of finite type. The local scheme map Spec $(\mathcal{O}_{X,x}) \to X$ is a monomorphism but is *not* locally of finite type (the local ring $\mathcal{O}_{X,x}$ is generally not a finitely generated algebra over the coordinate ring of any affine chart).
This gives a clean heuristic: schemes are locally affine, and the sub-objects of an affine scheme Spec (R) that are "of finite type" are exactly the closed subschemes Spec (R/I). The locally closed immersions are what you get by globalizing this: locally on X, a finite-type sub-object is a closed subscheme of an affine chart, which is a locally closed subscheme of X.

Formal local schemes: the analytic germ

One can zoom in even further than the local scheme by taking the completion. Let $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$ be the maximal ideal. We can form the $\mathfrak{m}_x$-adic completion of the local ring, $\widehat{\mathcal{O}}_{X,x}$, and consider the affine scheme Spec $(\widehat{\mathcal{O}}_{X,x})$.

The natural map $\mathcal{O}_{X,x} \to \widehat{\mathcal{O}}_{X,x}$ induces a canonical morphism

$$\widehat{\iota}_x : \operatorname{Spec}(\widehat{\mathcal{O}}_{X,x}) \longrightarrow \operatorname{Spec}(\mathcal{O}_{X,x}) \longrightarrow X.$$

For Noetherian schemes, the completion map $\mathcal{O}_{X,x} \to \widehat{\mathcal{O}}_{X,x}$ is faithfully flat. Consequently, the induced map on spectra is surjective, meaning the image of $\widehat{\iota}_x$ in X is exactly the same as the image of the local scheme: the set of generizations of x.

If the image is the same, how is this a "smaller" neighborhood? While the local scheme captures arbitrarily small *Zariski* neighborhoods, the completion captures arbitrarily small *formal* (or analytic) neighborhoods. The Zariski topology is famously coarse, and the completion detects geometric features that the local scheme cannot.

Example 3.1.30: Formal Branches of a Node

Let $X = \operatorname{Spec}(k[x,y]/(y^2 - x^2(x+1)))$ be a nodal cubic curve over an algebraically closed field of characteristic $\neq 2$, and let $p = [(x,y)]$ be the node at the origin.

The local ring $\mathcal{O}_{X,p} = (k[x,y]/(y^2 - x^2(x+1)))_{(x,y)}$ is an integral domain; in the Zariski topology, any open neighborhood of the origin is irreducible, so the local scheme $\operatorname{Spec}(\mathcal{O}_{X,p})$ is an integral scheme.

However, analytically, the curve consists of two branches crossing at the origin. If we pass to the completion $\widehat{\mathcal{O}}_{X,p} \cong k[\![x,y]\!]/(y^2 - x^2(x+1))$, the element $x+1$ has a square root (by Hensel's Lemma, or the binomial series). Thus, the defining equation factors:

$$y^2 - x^2(x+1) = (y - x\sqrt{x+1})(y + x\sqrt{x+1}).$$

The complete local ring is not a domain! The formal local scheme $\operatorname{Spec}(\widehat{\mathcal{O}}_{X,p})$ has two irreducible components, perfectly capturing the two analytic branches of the curve passing through the node.

Geometrically, passing to the formal local scheme feels like taking a smaller, more precise subspace. Categorically, however, this marks a hard boundary: *the morphism $\widehat{\iota}_x$ is not a monomorphism.* For a map of affine schemes to be a monomorphism, the underlying ring map must be an epimorphism (meaning the multiplication map $S \otimes_R S \to S$ is an isomorphism). However, there is a fundamental algebraic fact: *any faithfully flat epimorphism of rings is an isomorphism.* Because the completion of a Noetherian local ring is faithfully flat, if it were an epimorphism, the local ring would already have to be complete!

Because $\widehat{\iota}_x$ is not a monomorphism, the formal local scheme does not belong in our hierarchy of subobjects (immersions and local schemes). It is not "injective on points" functorially, and its diagonal map is very far from being an isomorphism. Instead of strictly restricting the space of functions, completion conceptually "thickens" the local ring by adding infinite limits (Taylor series). This breaks categorical injectivity, but it is exactly the mechanism required to reveal finer analytic structure. Geometrically, we can think of the fact that the two branches of a node our now two separate points as the analytic scheme "adding" more points, and thus failing the required injectivity.

In modern algebraic geometry, one often prefers to treat the completion as a *formal scheme*, denoted $\operatorname{Spf}(\widehat{\mathcal{O}}_{X,x})$. As a formal scheme, its underlying topological space is just the single point $\{x\}$, but its sheaf of topological rings remembers the "infinitesimal thickenings" of the point. See section 5.6 for more details.

Quick Aside: Dominant morphisms

Closed immersions of affine schemes correspond to surjective ring maps $R \twoheadrightarrow R/I$. The opposite extreme—injective ring maps—does not yield a surjective scheme morphism, but it yields the next best thing.

Definition 3.1.31: Dominant Morphism

A scheme morphism $f : X \to Y$ is *dominant* if the image of f is dense in Y, i.e., $\overline{f(X)} = Y$.

Proposition 3.1.32: Injective Ring Maps and Dominant Morphisms

Let $\varphi : R \hookrightarrow S$ be an injective ring homomorphism. Then the corresponding scheme morphism $f : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ is dominant.

***Proof*:**
The image $f(\operatorname{Spec}(S))$ is the set $\{\varphi^{-1}(\mathfrak{q}) : \mathfrak{q} \in \operatorname{Spec}(S)\}$. Its closure in $\operatorname{Spec}(R)$ is $V(I)$ where $I = \bigcap_{\mathfrak{q}\in\operatorname{Spec}(S)} \varphi^{-1}(\mathfrak{q}) = \varphi^{-1}\left(\bigcap_{\mathfrak{q}\in\operatorname{Spec}(S)} \mathfrak{q}\right) = \varphi^{-1}(\operatorname{Nil}(S))$, the preimage of the nilradical of S. Since φ is injective, $\varphi^{-1}(\operatorname{Nil}(S)) \subseteq \operatorname{Nil}(R)$. Thus $V(I) \supseteq V(\operatorname{Nil}(R)) = \operatorname{Spec}(R)$, giving $\overline{f(\operatorname{Spec}(S))} = \operatorname{Spec}(R)$.

Note that such a map is *not* necessarily injective, and in fact often fails to be injective (ex. on curves it fails at the non-branching points). Thus, we do not consider such maps subschemes; dominant maps are recorded here as it is likely the reader would be curious to know how to think of injective ring homomorphisms at this early stage. A fuller treatment of dominant morphisms and rational maps is given in **??**. In particular, we will see that for morphisms of finite type, dominance can be characterized by the density of the image, and Chevalley's theorem (theorem 5.2.7) gives a precise description of the image (it is a constructible set).

Exercise 3.1.1

1. Let X be a scheme and $Z \subseteq |X|$ a closed subset. Define an ideal sheaf $\mathcal{I}$ on X by

$$\mathcal{I}(U) = \{f \in \mathcal{O}_X(U) \mid f(x) = 0 \text{ for all } x \in Z \cap U\}$$

   where $f(x) = 0$ means the image of f in $\kappa(x)$ vanishes. Suppose $\operatorname{Spec}(R) \subseteq X$ is an affine open subset such that $Z \cap \operatorname{Spec}(R) = V(I)$ for some ideal $I \subseteq R$. Show that the ideal of R corresponding to $\mathcal{I}(\operatorname{Spec}(R))$ is $\sqrt{I}$.

2. Use the previous result to conclude that the closed subscheme Z_{red} defined by $\mathcal{I}$ is a scheme whose underlying topological space is exactly Z, and whose structure sheaf has no nilpotents. Argue why Z_{red} is the unique closed subscheme structure on Z with this property.

3. Consider the closed subset $Z = \{0\}$ in the affine line $\mathbb{A}^1_k = \operatorname{Spec}(k[x])$. Describe the reduced subscheme Z_{red} and compare it to the closed subschemes defined by the ideals (x^n) for $n \geq 2$.

4. The *reduction* of a scheme X, denoted X_{red}, is the reduced closed subscheme supported on the entire underlying space $|X|$. Show that the inclusion $X_{\text{red}} \hookrightarrow X$ satisfies the following

universal property: for any reduced scheme Y and any morphism $f : Y \to X$, f factors uniquely through X_{red}.

3.1.4 More on Projective Schemes

In section 2.3, we defined the prototypical projective scheme $\mathbb{P}^n_R = \operatorname{Proj} R[x_0, \dots, x_n]$. To do geometry, we need to understand its closed subschemes.

As we will prove rigorously in the next chapter using ideal sheaves (specifically corollary 5.3.15), closed subschemes of a projective scheme are governed by homogeneous ideals. For now, we take this as our primary geometric construction: given a graded ring $S_\bullet$ and a homogeneous ideal $I \subseteq S_\bullet$, the canonical surjective graded ring homomorphism $S_\bullet \twoheadrightarrow S_\bullet/I$ induces a closed immersion:

$$\operatorname{Proj}(S_\bullet/I) \hookrightarrow \operatorname{Proj} S_\bullet$$

Conceptually, this is the exact generalization of the affine case. If $S_\bullet$ is a finitely generated graded R-algebra generated in degree 1, we can choose $n+1$ degree 1 generators to form a surjective graded map $R[x_0, \dots, x_n] \twoheadrightarrow S_\bullet$. This implies $S_\bullet \cong R[x_0, \dots, x_n]/I$, meaning $\operatorname{Proj} S_\bullet$ naturally embeds as a closed subscheme of $\mathbb{P}^n_R$. Choosing these generators is geometrically equivalent to choosing projective coordinates.

We can classify the simplest closed subschemes of $\mathbb{P}^n_k$ by the generators of their ideals:

Definition 3.1.33: Projective Hypersurface and Linear Spaces

A *hypersurface* of $\mathbb{P}^n_k$ is a closed subscheme $\operatorname{Proj} k[x_0, \dots, x_n]/(f)$ given by a single homogeneous polynomial f. The *degree* of the hypersurface is the degree of f.

A hypersurface of degree 1 is called a *hyperplane*. A hypersurface of degree $2, 3, 4, 5 \dots$ is called a *quadric*, *cubic*, *quartic*, *quintic*, and so forth.

If $n = 2$ (so we are in $\mathbb{P}^2_k$), a hypersurface is called a *curve*. A hypersurface of degree $1, 2, 3$ is called a *line*, *conic curve* (or *conic*), and *cubic curve* respectively.

A *linear space* is a closed subscheme cut out by a collection of degree 1 homogeneous polynomials (an intersection of hyperplanes). Any injective linear transformation $V \to W$ of vector spaces induces a closed immersion $\mathbb{P}(V) \hookrightarrow \mathbb{P}(W)$ whose image is a linear space.

Recall that $\mathbb{P}^n_k$ has only constant global functions, so these subschemes are cut out by regular functions defined only on open affine patches. Let us look at some classic examples.

Example 3.1.34: Closed Projective Subschemes

1. *Twisted Cubic*: Take the homogeneous ideal $I = (xy - zw, x^2 - wy, y^2 - xz)$ and look at the closed subscheme $\operatorname{Proj} k[w, x, y, z]/I \subseteq \mathbb{P}^3_k$. This shape is called the *twisted cubic*. It is a *curve*, and it is isomorphic to $\mathbb{P}^1_k$ via the map induced by the graded ring homomorphism:
$$k[w, x, y, z] \to k[s, t] \qquad (w, x, y, z) \mapsto (s^3, s^2t, st^2, t^3)$$
To see this is an isomorphism of schemes, one can take the standard affine open sets and show it gives closed immersions on the charts. For example, on the chart $D_+(w)$ where

$w \neq 0$ and $s \neq 0$, the map of degree 0 localizations is:

$$k\left[\frac{x}{w}, \frac{y}{w}, \frac{z}{w}\right] \to k\left[\frac{t}{s}\right] \qquad \frac{x}{w} \mapsto \frac{t}{s}, \ \frac{y}{w} \mapsto \frac{t^2}{s^2}, \ \frac{z}{w} \mapsto \frac{t^3}{s^3}$$

2. *Lines on a Quadric Surface*: The following shows a way to parameterize certain projective varieties. Take the quadric surface $V_+(wz - xy) \subseteq \mathbb{P}^3_k$ (with projective coordinates w, x, y, z). For any two elements $\alpha, \beta \in k$ that are not both zero, notice that as $[c, d] \in \mathbb{P}^1_k$ varies, $[\alpha c, \beta c, \alpha d, \beta d]$ forms a line completely contained on the surface. Show there is one other parameterization of this surface via lines, and show it must be the case that these are the only two families (hint: one parameterization contains $V_+(w, x)$ and the other $V_+(w, y)$).

Another important construction arises from taking a different grading of the polynomial ring, denoted $S_{d\bullet}$. This allows us to "linearize" high-degree equations. First, we must establish that altering the grading in this specific way does not change the underlying abstract scheme.

Theorem 3.1.35: Isomorphism of Veronese Subrings

Let $S_\bullet = R[x_0, \ldots, x_n]$ be the standard graded polynomial ring. For any integer $d > 0$, let $S_{d\bullet} = \bigoplus_{m=0}^{\infty} S_{md}$ be the subring consisting of degrees that are multiples of d. Then there is a canonical isomorphism of schemes:

$$\operatorname{Proj} S_{d\bullet} \cong \operatorname{Proj} S_\bullet$$

***Proof*:**
We construct this isomorphism by comparing the standard affine open covers. By lemma 2.3.2, the scheme $\operatorname{Proj} S_\bullet$ is covered by the distinguished open sets $D_+(x_i) \cong \operatorname{Spec}((S_{x_i})_0)$ for $i = 0, \ldots, n$.

For the subring $S_{d\bullet}$, the elements x_i^d have degree d and therefore reside in $S_{d\bullet}$. The corresponding distinguished open sets $D_+(x_i^d)$ cover $\operatorname{Proj} S_{d\bullet}$, and applying lemma 2.3.2 again, each is isomorphic to $\operatorname{Spec}(((S_{d\bullet})_{x_i^d})_0)$.

We claim the rings $(S_{x_i})_0$ and $((S_{d\bullet})_{x_i^d})_0$ are identically the same. An arbitrary element in $(S_{x_i})_0$ is of the form a/x_i^k, where $a \in S_k$ is a homogeneous polynomial of degree k. We can rewrite this fraction by multiplying the numerator and denominator by x_i^{kd-k}:

$$\frac{a}{x_i^k} = \frac{a x_i^{kd-k}}{x_i^{kd}} = \frac{a x_i^{kd-k}}{(x_i^d)^k}$$

The new numerator $a x_i^{kd-k}$ has degree $k + (kd - k) = kd$, meaning it is an element of S_{kd}, which is a valid component of $S_{d\bullet}$. The denominator is a power of x_i^d. Therefore, this fraction is naturally an element of $((S_{d\bullet})_{x_i^d})_0$. Since this containment goes both ways, the rings are equal. These local isomorphisms canonically glue together to yield the global isomorphism $\operatorname{Proj} S_{d\bullet} \cong \operatorname{Proj} S_\bullet$.

Definition 3.1.36: Veronese Embedding

Let $S_\bullet = k[x_0, \dots, x_n]$ and let $S_{d\bullet}$ be the d-tuple Veronese subring. By theorem 3.1.35, $\operatorname{Proj} S_{d\bullet} \cong \mathbb{P}^n_k$. We can view $S_{d\bullet}$ as being generated in "degree 1" by the monomials of degree d from the original ring. There are $N+1 = \binom{n+d}{d}$ such monomials. Mapping a new polynomial ring $k[y_0, \dots, y_N] \twoheadrightarrow S_{d\bullet}$ by sending the variables y_i to these monomials induces a closed immersion:

$$\mathbb{P}^n_k \cong \operatorname{Proj} S_{d\bullet} \hookrightarrow \mathbb{P}^N_k$$

This map is called the *d-tuple Veronese Embedding.*

The geometric utility of the Veronese embedding is that a hypersurface of degree d in $\mathbb{P}^n_k$ gets mapped exactly to a hyperplane intersection (degree 1) in $\mathbb{P}^N_k$.

3.1.37 Foreshadowing: Sheaves and Morphisms In the next chapter, we will show that morphisms to projective space are governed by invertible sheaves. The Veronese embedding is precisely the morphism $X \to \mathbb{P}^N_k$ generated by the global sections of the twisting sheaf $\mathcal{O}_X(d)$.

As noted earlier, choosing a set of degree 1 generators is equivalent to choosing an embedding into a specific ambient projective space $\mathbb{P}^N_R$. Therefore, it is necessary to distinguish the intrinsic geometry of the abstract scheme from the extrinsic geometry of its embedding.

Consider the scheme $X = \mathbb{P}^1_k$. We can embed it in several ways:

1. *Degree 1*: Use $S_\bullet = k[x, y]$. It is generated in degree 1 by 2 elements. This gives the identity embedding $\mathbb{P}^1_k \hookrightarrow \mathbb{P}^1_k$ as a straight line.
2. *Degree 2*: Consider the 2-tuple Veronese subring $S_{2\bullet} = k[x^2, xy, y^2]$. As an abstract scheme, $\operatorname{Proj} S_{2\bullet} \cong \mathbb{P}^1_k$. To write $S_{2\bullet}$ as a ring generated in degree 1, we require 3 generators: let $u = x^2$, $v = xy$, $w = y^2$. They satisfy $uw = v^2$, yielding an embedding $\mathbb{P}^1_k \hookrightarrow \mathbb{P}^2_k$. The image is a conic curve.
3. *Degree 3*: The 3-tuple Veronese subring $S_{3\bullet} = k[x^3, x^2y, xy^2, y^3]$ requires 4 generators, yielding an embedding $\mathbb{P}^1_k \hookrightarrow \mathbb{P}^3_k$. The image is the twisted cubic.

The line in $\mathbb{P}^1_k$, the conic in $\mathbb{P}^2_k$, and the twisted cubic in $\mathbb{P}^3_k$ are isomorphic as abstract schemes. Extrinsically, however, they are embedded differently, have different geometric degrees, and are cut out by different homogeneous ideals.

One might ask why we do not always restrict to degree 1 generators. Certain equations become unnecessarily complicated if forced into a standard degree 1 embedding, but simplify if we assign different *weights* (degrees) to the variables. For example, embedding certain curves or moduli spaces into standard $\mathbb{P}^N$ via Veronese embeddings increases the ambient dimension N significantly and requires many equations to cut out. This motivates the following construction.

Definition 3.1.38: Weighted Projective Space

Let $S_\bullet = k[a_1, \dots, a_n]$ be a graded ring where each variable a_i is assigned a specific degree $d_i \in \mathbb{Z}_{>0}$. Then $\operatorname{Proj} k[a_1, \dots, a_n]$ is called a *weighted projective space* and is denoted $\mathbb{P}(d_1, \dots, d_n)$.

Geometrically, over a field, assigning different weights corresponds to taking the quotient of $\mathbb{A}_k^n \setminus \{0\}$ by the action of the multiplicative group $\mathbb{G}_m$ (i.e. $(\mathbb{A}_k^*, \cdot)$) where the scaling acts as $\lambda \cdot (x_1, \ldots, x_n) = (\lambda^{d_1} x_1, \ldots, \lambda^{d_n} x_n)$.

Example 3.1.39: Weighted Projective Space Examples

1. *Show that $\mathbb{P}(m, n) \cong \mathbb{P}_k^1$ for any positive integers m, n.*

   Let $S_\bullet = k[x, y]$ with $\deg(x) = m$ and $\deg(y) = n$. The scheme is covered by the two open sets $D_+(x)$ and $D_+(y)$. For $D_+(x) \cong \operatorname{Spec}((S_x)_0)$, the degree 0 elements are generated by $u = y^m/x^n$. Thus, $(S_x)_0 \cong k[u]$, which is isomorphic to $\mathbb{A}_k^1$. Similarly, for $D_+(y) \cong \operatorname{Spec}((S_y)_0)$, the degree 0 elements are generated by $v = x^n/y^m$. Thus, $(S_y)_0 \cong k[v]$, which is also isomorphic to $\mathbb{A}_k^1$. On the intersection $D_+(xy)$, the variable v is exactly u^{-1}. This translates to gluing two copies of $\mathbb{A}_k^1$ along $\mathbb{A}_k^1 \setminus \{0\}$ via the map $u \mapsto u^{-1}$, which is the precise definition of $\mathbb{P}_k^1$.

2. *Show that $\mathbb{P}(1, 1, 2) \cong \operatorname{Proj} k[u, v, w, z]/(uw - v^2)$.*

   Let $S_\bullet = k[x, y, t]$ with $\deg(x) = 1, \deg(y) = 1$, and $\deg(t) = 2$. By theorem 3.1.35, $\operatorname{Proj} S_\bullet \cong \operatorname{Proj} S_{2\bullet}$. The subring $S_{2\bullet}$ is generated by all degree 2 monomials from $S_\bullet$. These monomials are exactly x^2, xy, y^2, and t. Let us map a standard polynomial ring into $S_{2\bullet}$ by setting $u = x^2$, $v = xy$, $w = y^2$, and $z = t$. The only algebraic relation between these generators is $uw = x^2y^2 = (xy)^2 = v^2$. The generator z is algebraically independent of the others. Therefore, $S_{2\bullet} \cong k[u, v, w, z]/(uw - v^2)$. Since the spaces are isomorphic, it follows that $\mathbb{P}(1, 1, 2)$ can be realized as the quadric cone $\operatorname{Proj} k[u, v, w, z]/(uw - v^2)$ in standard $\mathbb{P}_k^3$.

3. *Smooth Compactifications of Hyperelliptic Curves.*

   Weighted projective spaces are highly practical for compactifying curves without introducing artificial singularities. Consider the affine hyperelliptic curve $y^2 = x^5 + 1$. If we attempt to compactify this by homogenizing it into standard $\mathbb{P}_k^2$ (where all variables have degree 1), we obtain the curve $y^2z^3 = x^5 + z^5$. Calculating the Jacobian reveals that this curve has a singularity at the point at infinity $[0 : 1 : 0]$.

   However, if we embed the curve into the weighted projective space $\mathbb{P}(1, 1, 3)$ with coordinates (x, z, y) having degrees $1, 1$, and 3 respectively, we can form a homogeneous equation of total degree 6:
   $$y^2 = x^5z + z^6$$
   In this weighted space, the curve remains smooth at infinity.

3.1.5 The Category of Schemes

Having defined schemes, morphisms, subschemes, and projective schemes, we have formally constructed the category of schemes, denoted **Sch** (or $\mathbf{Sch}_S$ for S-schemes). Before moving on, it is valuable to look at the categorical properties of **Sch** as a whole.

How well-behaved is this category? Does it have limits and colimits? The short answer is that the local affine structure is well-behaved, but the global structure is much more complex.

The Affine Subcategory

Let $\mathbf{Sch}^{\mathbf{Aff}}$ be the full subcategory of affine schemes. By the definition of the spectrum functor, we have an anti-equivalence of categories:

$$\mathbf{Sch}^{\mathbf{Aff}} \simeq \mathbf{CRing}^{op}$$

Because the category of commutative rings is both complete (has all limits) and cocomplete (has all colimits), the category $\mathbf{Sch}^{\mathbf{Aff}}$ is also completely well-behaved: it has all limits and colimits. As discussed in section 2.2.10, operations here simply mirror the algebraic operations in rings (for example, the categorical coproduct of affine schemes corresponds to the direct product of their rings).

For instance, the initial object in **CRing** is $\mathbb{Z}$, which means the terminal object in $\mathbf{Sch}^{\mathbf{Aff}}$ is $\operatorname{Spec}\mathbb{Z}$. In fact, $\operatorname{Spec}\mathbb{Z}$ is the terminal object in the entirety of **Sch**. To prove this quickly, let X be any scheme, and let X be covered by affine open sets $\operatorname{Spec} A_i$. A morphism $X \to \operatorname{Spec}\mathbb{Z}$ restricts to local morphisms $\operatorname{Spec} A_i \to \operatorname{Spec}\mathbb{Z}$, which correspond exactly to ring homomorphisms $\mathbb{Z} \to A_i$. Since $\mathbb{Z}$ is the initial object in the category of rings, there exists exactly one such ring homomorphism for each A_i. Because these local morphisms are completely unique, they trivially agree on all overlaps and glue together to form a unique global morphism $X \to \operatorname{Spec}\mathbb{Z}$.

The Projective Subcategory

On the opposite end of the spectrum is the subcategory of projective schemes over a fixed base ring. While affine schemes are determined entirely by their global functions, connected projective schemes have only constant global functions.

Categorically, the projective subcategory is highly rigid and lacks the completeness of $\mathbf{Sch}^{\mathbf{Aff}}$. It does not admit infinite limits, and finite colimits (such as general pushouts) often fail to produce a scheme that remains projective. However, what the projective subcategory lacks in categorical completeness, it makes up for in geometric properness. Because projective schemes satisfy the valuative criterion for properness (as we will see later), curves do not escape to infinity without a limit. This makes the projective subcategory the ideal setting for intersection theory and global counting problems, whereas the affine category remains the setting for local algebraic calculations.

Subobjects and Immersions

In standard category theory, a "subobject" is defined as an equivalence class of monomorphisms. If we apply this strict categorical definition to **Sch**, we encounter non-geometric behavior. A morphism of schemes $X \to Y$ being a categorical monomorphism simply means it is left-cancellative.

To see why this is geometrically problematic, consider the affine subcategory $\mathbf{Sch}^{\mathbf{Aff}}$. Because of the anti-equivalence to **CRing**, a monomorphism of affine schemes corresponds exactly to an *epimorphism* of commutative rings. Ring epimorphisms certainly include surjective maps $A \to A/I$ (which yield closed immersions) and principal localizations $A \to A_f$ (which yield basic open immersions). However, ring epimorphisms also include total localizations like $\mathbb{Z} \to \mathbb{Q}$. Therefore, the map $\operatorname{Spec}\mathbb{Q} \to \operatorname{Spec}\mathbb{Z}$ is a strict categorical subobject in $\mathbf{Sch}^{\mathbf{Aff}}$, even though geometrically $\operatorname{Spec}\mathbb{Q}$ does not resemble a "piece" of $\operatorname{Spec}\mathbb{Z}$ (it is neither an open nor a closed subset).

For this reason, algebraic geometry rarely focuses on bare categorical monomorphisms. Instead, we restrict our attention to specific geometric embeddings: *open immersions* and *closed immersions* (and their composition, locally closed immersions). These immersions ensure that the subscheme maps homeomorphically onto a topological subspace of the target and that the structure sheaves behave compatibly.

In $\mathbf{Sch}^{\mathbf{Aff}}$, these geometric subobjects are strictly generated by ideal quotients and localizations. In the projective subcategory, the geometric subobjects are almost exclusively closed immersions; because projective schemes are proper (compact), any open immersion that is also a projective scheme must actually be a closed set (a union of connected components). Thus, geometric "subschemes" form a much narrower and better-behaved class of morphisms than general categorical subobjects.

Limits in Sch

When we glue affine schemes together to form general schemes, we lose some of the categorical completeness found in $\mathbf{Sch}^{\mathbf{Aff}}$.

1. *Finite limits exist:* As proved above, the category **Sch** contains a terminal object $\operatorname{Spec}\mathbb{Z}$. Furthermore, as we will see in section 3.2, the category of schemes also admits pullbacks (fibered products). A standard result in category theory dictates that any category with a terminal object and pullbacks automatically has all finite limits (including equalizers and finite products).
2. *Infinite limits generally do not exist:* Unlike affine schemes, general schemes are not closed under infinite products. For example, the infinite product $\prod_{i=1}^{\infty} \mathbb{P}^1_k$ does not exist in the category of schemes. The intuitive reason for this failure is that the Zariski topology behaves poorly under infinite Cartesian products, making it impossible to find a valid affine open cover for the resulting space (for a formal proof involving the failure of the representability of its functor of points, see ref:HERE).
3. *Cofiltered limits exist:* The category of schemes does have cofiltered limits (often called inverse limits), provided the transition maps are affine. This is easily justified: because the spectrum functor is contravariant, taking the inverse limit of these affine schemes translates algebraically to taking the direct limit (filtered colimit) of their corresponding commutative rings, which is a well-behaved algebraic operation. A classic example is the scheme of p-adic integers, which is the limit of affine schemes: $\operatorname{Spec}\mathbb{Z}_p = \varprojlim \operatorname{Spec}(\mathbb{Z}/p^n\mathbb{Z})$.

Colimits in Sch

1. *Arbitrary coproducts exist:* One of the simplest and most useful colimits in **Sch** is the coproduct. Any arbitrary collection of schemes $\{X_i\}$ has a coproduct given by their disjoint union $\coprod X_i$. As we established in section 2.2.9, this holds true for both finite and infinite collections.
2. *Finite colimits generally do not exist:* Outside of disjoint unions, the category of schemes does not have general pushouts or coequalizers. A standard counter-example involves trying to glue a scheme to itself along a generic point. For instance, the pushout of $\operatorname{Spec}(K) \rightrightarrows \mathbb{A}^1_K$ (where we attempt to glue the generic point to something else) fails to be a scheme. We will see this explicitly in example 3.2.7.
3. *Pushouts of open immersions exist:* There is one important exception to the failure of pushouts. If $f : X \to Z$ and $g : Y \to Z$ are both open immersions, their pushout always exists. This is exactly the categorical formulation of gluing schemes together, which we covered in section 2.2.9.

4. *Filtered colimits exist:* The category does admit filtered colimits (often called direct limits). A geometric example is infinite-dimensional projective space, $\mathbb{P}^\infty_k = \varinjlim \mathbb{P}^n_k$, which exists as a scheme (specifically $\operatorname{Proj} k[x_0, x_1, \dots]$), though it loses the property of being locally Noetherian.

Schemes as Colimits of Affine Schemes

We have seen that the category **Sch** contains **Aff** $=$ **CRing**$^{\mathrm{op}}$ as a full subcategory, and that general schemes are obtained by "gluing" affine pieces. We can now make the gluing process categorically precise: every scheme is a colimit of affine schemes, and the colimit has a particularly clean form.

The coequalizer picture. Let X be a scheme admitting a finite affine open cover $X = U_1 \cup \cdots \cup U_n$. Form the disjoint union

$$U = U_1 \sqcup \cdots \sqcup U_n,$$

which is itself an affine scheme: if $U_i = \operatorname{Spec}(A_i)$, then $U = \operatorname{Spec}(A_1 \times \cdots \times A_n)$. The natural map $\pi : U \to X$ (the inclusion of each U_i into X) is surjective but not injective: a point lying in $U_i \cap U_j$ appears twice in U, once in the U_i-copy and once in the U_j-copy.

To recover X from U, we must identify these redundant copies. Define the "overlap scheme"

$$R = U \times_X U = \coprod_{i,j} (U_i \cap U_j).$$

By the Build-up Lemma (lemma 2.2.29), each intersection $U_i \cap U_j$ can be covered by open sets that are simultaneously distinguished in U_i and U_j. If X is separated, the intersections $U_i \cap U_j$ are themselves affine (section 3.5.1), and R is affine as well; in general, R can be covered by affines, and the construction still works by refining the cover.

The scheme R comes equipped with two natural morphisms to U: the projections

$$s, t : R \rightrightarrows U$$

where s includes $U_i \cap U_j$ into its U_i-copy and t includes it into its U_j-copy. The scheme X is then recovered as the *coequalizer* of s and t in the category of locally ringed spaces:

$$R \underset{t}{\overset{s}{\rightrightarrows}} U \xrightarrow{\pi} X.$$

That is, X is the universal locally ringed space receiving a map from U that identifies $s(r)$ with $t(r)$ for every point $r \in R$.

Let us form an equivalence relation between the glued points categorically. We shall look at the *groupoid object* in the category of schemes (or, when everything is affine, in **Aff**). The additional structure maps are:

1. *Identity:* $e : U \to R$, the diagonal, sending a point of U_i to the "trivial overlap" of U_i with itself.
2. *Composition:* $c : R \times_{s,U,t} R \to R$, expressing transitivity: if a point of $U_i \cap U_j$ and a point of $U_j \cap U_k$ agree on U_j, they compose to a point of $U_i \cap U_k$.
3. *Inverse:* $\iota : R \to R$, the swap: a point of $U_i \cap U_j$ is also a point of $U_j \cap U_i$.

These satisfy the usual groupoid axioms (associativity, unit, inverse), which encode the fact that the relation "x and y represent the same point of X" is an equivalence relation. On the ring side, dualizing: if $U = \mathrm{Spec}\,(A)$ and $R = \mathrm{Spec}\,(B)$, the two ring homomorphisms $s^\#, t^\# : A \rightrightarrows B$ together with the identity, composition, and inverse maps make (A, B) into a *cogroupoid object* in **CRing**.

Conversely, one can *start* with affine groupoid data and produce a scheme. Given rings A and B with homomorphisms $s^\#, t^\# : A \rightrightarrows B$ and maps $e^\# : B \to A$, $c^\# : B \to B \otimes_A B$, $\iota^\# : B \to B$ satisfying the cogroupoid axioms, together with the condition that s and t are open immersions (so that the "gluing maps" identify open subsets), the coequalizer of $\mathrm{Spec}\,(B) \rightrightarrows \mathrm{Spec}\,(A)$ in the category of locally ringed spaces is a scheme.

This gives a precise sense in which the category of schemes is built from the category of affine schemes: **Sch** is obtained from **Aff** by freely adjoining coequalizers of affine groupoid presentations whose structure maps are open immersions. Every scheme admits such a presentation (by choosing an affine cover), and the result is independent of the choice (by the Affine Communication Lemma and the uniqueness of colimits).

The reader should keep this in mind when going through the functorial perspective developed in section 3.3.7, where schemes are characterized as Zariski sheaves with affine open covers. The two viewpoints are dual: the functorial approach defines schemes as certain *limits* (sheaves are defined by equalizer conditions), while the groupoid approach constructs schemes as certain *colimits* (coequalizers of equivalence relations). The equivalence of these two perspectives is a manifestation of a general principle: in algebraic geometry, limit-based and colimit-based descriptions of the same object coexist, and switching between them is sometimes the key to a proof.

Exercise 3.1.2

1. Show that the map $\mathrm{Spec}\, k(x) \to \mathrm{Spec}\, k[y]_{(y)}$ sending the unique (generic) point η to the closed point of the codomain is a valid morphism of ringed spaces, but *not* a morphism of locally ringed spaces. (This highlights why the local ring condition is necessary for scheme morphisms).
2. Let $p \in X$ be a point in a scheme. Show there is a canonical morphism of schemes $\mathrm{Spec}\,(\mathcal{O}_{X,p}) \to X$.
3. Using the above, define a canonical morphism $\mathrm{Spec}(\kappa(p)) \to X$. This is the formal scheme-theoretic morphism corresponding to the inclusion of a point.
4. Show there is no non-trivial map $\mathbb{P}^n_k \to \mathbb{A}^m_k$.

3.2 (Fiber) Product and Fibers

https://stacks.math.columbia.edu/tag/01JW

One of the most common tools in constructing new algebraic and geometric objects are limits and colimits. Given a category that is complete and cocomplete (having all limits and colimits), then:

1. limits are a combination of products and equalizers (i.e. cutting)
2. colimits are a combination of coproducts and coequalizers (i.e. gluing)

The category $\mathbf{Sch}_S$ is closed under coproducts (disjoint union), but not necessarily colimits as we require non-trivial gluing conditions (recall theorem 2.2.54). However, it *is* closed under products and equalizes (equalizer of an affine scheme R usually becomes a closed subscheme). To do this, we must show it is closed under products. Starting with the affine case, by corollary 3.1.5 it is equivalence for R-algebra to be closed under coproducts, and indeed the coproduct of A, B is $A \otimes_R B$ (see [**NateClassicalAlgGeo**]).

Theorem 3.2.1: Fibered Product Of Schemes

Let X, Y be schemes over S. Then then the fibered product of X and Y over S exists, that is the limit of the diagram exists:

$$\begin{tikzcd} X \arrow[dr] & & Y \arrow[dl] \\ & S & \end{tikzcd}$$

The fibered product is denoted $X \times_S Y$

More commonly, we would say that the fibered product satisfies the universal property that if $Z \to X$ and $Z \to Y$ commutes in the diagram, then there exists a unique morphism $Z \to X \times_S Y$ such that

$$\begin{tikzcd} & Z \arrow[d] \arrow[ddl, bend right] \arrow[ddr, bend left] & \\ & X \times_S Y \arrow[dl] \arrow[dr] & \\ X \arrow[dr] & & Y \arrow[dl] \\ & S & \end{tikzcd}$$

commutes. If S is the final object, or more concisely it is empty or "doesn't matter", then we have the usual product. If S is not specified, the product shall be defined as $X \times Y := X \times_{\mathrm{Spec}\,(\mathbb{Z})} Y$

***Proof*:**
We shall construct the products of s, and then glue them together.

Let $X = \mathrm{Spec}\,(A)$, $Y = \mathrm{Spec}\,(B)$, and $S = \mathrm{Spec}\,(R)$ be affine. Then A, B are R-algebras, and the product of X and Y over S is:

$$\mathrm{Spec}\,(A \otimes_R B)$$

This comes form the universal property of tensor products along with the property of adjoints. Indeed, if $Z \to \mathrm{Spec}\,(A \otimes_R B)$ is a scheme morphism, then this gives a ring homomorphism $A \otimes_R B$ into $\mathcal{O}_Z(Z)$. By the universal property of tensor products, a morphism of $A \otimes_R B$ into is the same as a morphism from A and B to that ring. Then by proposition 3.1.12, this gives a morphism of Z into $\mathrm{Spec}\,(A \otimes_R B)$, and this morphism will satisfy the universal property of fibered products given by the universal property of tensor products.

Let us now construct the more general case via gluing. Let X, Y be schemes. To do this, we shall reduce to a covering of $X \times_S Y$. For any open affine $U \subseteq X$, if the product existed then $\pi_1^{-1}(U) \cong U \times_S Y$ (check this[a]). Now if $\{X_i\}$ is an open covering for x, then I claim that if

$X_i \times_S X$ exists, so does $X \times_S Y$. Indeed, for each $U_{ij} = \pi_1^{-1}(X_i) \subseteq X_i \times_S Y$, by the uniqueness of the product there are unique isomorphisms: $\varphi_{ij} : U_{ij} \to U_{ji}$ for each i, j for compatible projections. Furthermore, these isomorphisms are compatible with the gluing condition (as you should check if you are not convinced!). As a remind, the condition was:

$$\varphi_{ik}|_{X_{ij} \cap X_{ik}} = \varphi_{jk} \circ \varphi_{ij}|_{X_{ij} \cap X_{ik}}$$

Thus, we get that we may glue them together by lemma 3.1.9 we have that this glued scheme is isomorphic to $X \times_S Y$, and through this isomorphism we can check that the universal property is satisfied.

By interchanging X and Y, we can now conclude from the proof on affine schemes that if X and Y are any schemes, and S is an affine, then $X \times_S Y$ exists. To show that S can be an arbitrary scheme, let $\{S_i\}$ be an open affine cover of S. Let $\alpha : X \to S$ and $\beta : Y \to S$ be two scheme morphisms, and let $X_i = \alpha^{-1}(S_i)$, $Y_i = \beta^{-1}(S_i)$. Then $X_i \times_{S_i} Y_i$ is well-defined.

Now, what's key is that this scheme is the same as the scheme $X_i \times_S Y$. To see this, we must invoke the uniqueness of the universal property, and indeed if $f : Z \to X_i$ and $g : Z \to Y$ are scheme morphisms over S, by the properties of the commutative diagram we must have that the image of v must land inside Y_i. But by universality they must be equal. Thus $X_i \times_S Y$ is well-defined for each i, and so by the above argument it glues to create $X \times_S Y$, as we sought to show.

[a]If $f : Z \to U$, compose with the inclusion map and invoke the appropriate universal properties

Example 3.2.2: Fibered Products

1. Show that $\mathbb{A}_k^n \times_k \mathbb{A}_k^m \cong \mathbb{A}_k^{n+m}$, which is exactly what we would hope from a product. Notably, prove $k[x] \otimes_k k[y] \cong k[x, y]$.

2. Similarly, if k is a characteristic 0 field show that $\mathbb{Z}[x] \otimes_{\mathbb{Z}} k[y] \cong k[x, y]$. Hence $\mathbb{A}_{\mathbb{Z}}^1 \times \mathbb{A}_k^1 \cong \mathbb{A}_k^2$.

3. Let $X = \operatorname{Spec} \mathbb{C}[x, y]/(x^2 - y)$ and $Y = \mathbb{C}[z, w]/(z^3 - w^4)$. Then:

$$X \times_{\mathbb{C}} Y \cong \operatorname{Spec} \frac{\mathbb{C}[x, y, z, w]}{(x^2 - y, z^3 - w^3)}$$

   The reader should stare at this until they understand why it is called a fibered product.

4. There is new behaviour when considering schemes being tensored over non-algebraically closed fields. Take $\operatorname{Spec} \mathbb{C} \times_{\operatorname{Spec} \mathbb{R}} \operatorname{Spec} \mathbb{C}$. As these are affine, this is $\operatorname{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C})$. As $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$ (exercise),

$$\operatorname{Spec} \mathbb{C} \times_{\operatorname{Spec} \mathbb{R}} \operatorname{Spec} \mathbb{C} \cong \operatorname{Spec}(\mathbb{C}) \amalg \operatorname{Spec}(\mathbb{C})$$

   Notice here how the choice of sheaf completely changed the answer, notably $\operatorname{Spec} \mathbb{C}$ and $\operatorname{Spec} \mathbb{R}$ are schemes over a single point, hence the fibered product diagram when only considering points is an identity map on points:

$$\begin{tikzcd} \bullet \arrow[dr] & & \bullet \arrow[dl] \\ & \bullet & \end{tikzcd}$$

and it is easy to see that in **Top** this results is a single point. However, the resulting scheme is *two* points. Thus, the fibered product here somehow "unravelled" the fact that $\operatorname{Spec}\mathbb{R}$ should be thought of as "gluing" conjugates together (think of the intuition for $\operatorname{Spec}\mathbb{R}[x]$ as a $\operatorname{Spec}\mathbb{C}[x]/\mathrm{Gal}_{\mathbb{R}}(\mathbb{C})$, the real line then don't glue to another points, but their residue field is $\mathbb{R}$ instead of $\mathbb{C}$, and hence it codimension 2 with respect to $\mathbb{C}$).

More generally show that if L/K is a finite galois extension, then

$$L \otimes_K L \cong \prod_{|\mathrm{Aut}(L/K)|} K$$

is isomorphic to $L^{|G|}$($|G|$ copies of L), hint: this is the linear independence of characters proof in geometric form. Hence, over fields fibered products capture some idea properties of galois groups.

5. Recall that the intersection of rings is a ring. This is a special case of the fibered product in the category of commutative rings, namely that if $S \to R$ and $S' \to R$ exists, then:

$$\begin{tikzcd} & S \cap S' \arrow[dl] \arrow[dr] & \\ S \arrow[dr] & & S' \arrow[dl] \\ & R & \end{tikzcd}$$

Let us see how this translates to schemes and affine schemes. Recall that the product and coproduct of affine schemes is an affine scheme:

$$\operatorname{Spec} S \times \operatorname{Spec} S' = \operatorname{Spec} S \otimes_{\mathbb{Z}} S'$$
$$\operatorname{Spec} S \amalg \operatorname{Spec} S' = \operatorname{Spec} S \times S'$$

If $U, U' \subseteq \operatorname{Spec} R$ are open affine, $U = \operatorname{Spec} S$ and $U' = \operatorname{Spec} S'$, Then as $\operatorname{Spec}(-)$ is contravariant we need that the intersection $(\operatorname{Spec}(S)) \cap (\operatorname{Spec}(S'))$ becomes the "union" of rings. Naturally, the union of rings need not be a ring, and as both S, S' contain units then $\langle S, S' \rangle = R$. Instead, the natural dual is:

$$\operatorname{Spec} S \cap \operatorname{Spec} S' = \operatorname{Spec} S \otimes_R S'$$

As we saw, the union of affine need not be affine (see example 2.2.55).

One of the most important uses of the fibered is to change the base scheme:

Definition 3.2.3: Base Change (Scheme-theoretic Pullback)

Let X be an S-scheme, and $T \to S$ a scheme morphism. Then the *base change* of X to T is the T-scheme:

$$X_T = T \times_S X$$

If $f : X \to Y$ is an S-scheme morphism, and $T \to S$ a scheme morphism, then the *base change* of f is the map $f' : X_T \to Y_T$ where $f' = \mathrm{id}_T \times_{\mathrm{id}_S} f$.

Example 3.2.4: Base Change

1. For any R-scheme X and morphism $\operatorname{Spec} S \to \operatorname{Spec} R$ (i.e. ring homomorphism $R \to S$), we get the S-scheme:
$$X' = S \times_R X$$
The category of R-schemes maps to the category of S schemes via base change.
2. If $X = R[x_1, ..., x_m]/I$, is an R-scheme and $S \to R$ is a ring homomorphism, then:
$$X_S = \frac{S[x_1, ..., x_m]}{IS}$$
that is, the set of constant change.
3. As the map $S \to R$ need not be surjective, we can get something like $X = \operatorname{Spec} \mathbb{C}$ as a $\mathbb{C}$-scheme and base change $\mathbb{R} \hookrightarrow \mathbb{C}$.
4.

As the tensor product is a very versatile operation, the fibered product is as well. Let us demonstrate by translating some important results:

Proposition 3.2.5: Base Change Properties

The following properties hold for fibered products:

1. Let $\iota : U \to Z$ be an open immersion and $\varphi : Y \to Z$ be any scheme morphism. Then $U \times_Z Y$ is isomorphic to the open subscheme of Y given by the preimage of U:
$$U \times_Z Y \cong \left(\varphi^{-1}(U), \mathcal{O}_Y|_{\varphi^{-1}(U)}\right)$$
2. *Stability of Closed Immersions:* If $X \to Z$ is a closed immersion, then for any map $Y \to Z$, the base change $X \times_Z Y \to Y$ is a closed immersion.
3. *Stability of Open Immersions:* If $X \to Z$ is an open immersion, then for any map $Y \to Z$, the base change $X \times_Z Y \to Y$ is an open immersion.

***Proof*:**
Reduction to the Affine Case (for 2 and 3): The properties of being a closed immersion or an open immersion are *local on the target.* That is, a morphism $f : W \to Y$ is a closed (resp. open) immersion if and only if there exists an open cover $\{V_i\}$ of Y such that each restriction $f^{-1}(V_i) \to V_i$ is a closed (resp. open) immersion.

Let $X \to Z$ be the immersion we are base changing. Let $\{Z_\alpha = \operatorname{Spec}(R_\alpha)\}$ be an affine open cover of Z. Since $X \to Z$ is an immersion, the restriction $X_\alpha = X \times_Z Z_\alpha \to Z_\alpha$ corresponds to a map of affine schemes (either a quotient or a localization). Now, let $\{Y_{ij} = \operatorname{Spec}(S_{ij})\}$ be an affine open cover of Y such that each Y_{ij} maps into some Z_α. Since fibered products commute with open restriction, the base change restricted to Y_{ij} is:

$$(X \times_Z Y) \times_Y Y_{ij} \cong X \times_Z Y_{ij} \cong (X \times_Z Z_\alpha) \times_{Z_\alpha} Y_{ij}$$

This reduces the problem to showing that the base change of an affine immersion by an affine morphism is an immersion. If we prove this for rings, the global property follows by gluing.

1. The fibered product is the limit in the category of schemes. Maps $W \to U \times_Z Y$ correspond to pairs of maps $W \to U$ and $W \to Y$ that agree in Z. Since $U \hookrightarrow Z$ is an inclusion, a map to U is just a map to Z whose image lies in U. Thus, a map to the product is a map $W \to Y$ such that the composition $W \to Y \xrightarrow{\varphi} Z$ factors through U. This is exactly the definition of a map into the open subscheme $\varphi^{-1}(U) \subseteq Y$.

2. *Affine Case:* Let $Z = \operatorname{Spec}(R)$, $X = \operatorname{Spec}(R/I)$, and $Y = \operatorname{Spec}(S)$. The closed immersion corresponds to the surjection $\pi : R \twoheadrightarrow R/I$. We compute the base change by tensoring with S over R. Since the tensor product is *right-exact*, applying $S \otimes_R -$ to the surjection preserves surjectivity:
$$S \cong S \otimes_R R \xrightarrow{1 \otimes \pi} S \otimes_R (R/I) \longrightarrow 0$$
Thus, the ring map corresponding to the base change $X \times_Z Y \to Y$ is surjective. Since surjective ring maps correspond to closed immersions of affine schemes, and we have established the property is local, the general base change is a closed immersion.

3. *Affine Case:* Let $Z = \operatorname{Spec}(R)$ and let $X \to Z$ correspond to a localization $R \to R_f$ for some $f \in R$ (a basic open immersion). Let $Y = \operatorname{Spec}(S)$ with map $\varphi : R \to S$. We compute the base change by tensoring:
$$S \otimes_R R_f = S \otimes_R R[f^{-1}] \cong S[\varphi(f)^{-1}] = S_{\varphi(f)}$$
The resulting map $S \to S_{\varphi(f)}$ is a localization. Since any open immersion is locally isomorphic to a basic open immersion (localization), and we have established the property is local, the general base change is an open immersion.

This motivates an important definition generalizing affine and projective schemes (idk if this should be here)

Definition 3.2.6: Affine Schemes over a Scheme

Let X be a scheme. Then:

$$\mathbb{A}^n_X = X \times_{\mathbb{Z}} \mathbb{Z}[x_1, \ldots, x_n]$$

Example 3.2.7: Pushout Need Not Exist

Consider two copies of $\mathbb{A}^1_{\mathbb{C}}$. Consider:

$$\begin{tikzcd} & \operatorname{Spec}\mathbb{C} \arrow[dl] \arrow[dr] & \\ \mathbb{A}^1_{\mathbb{C}} & & \mathbb{A}^1_{\mathbb{C}} \end{tikzcd}$$

where $[(0)] \mapsto [(0)]$ in both cases. This is like the double-origin example, except every point is a double origin. This shall be shown to not be a scheme as it is "too non-separated".
We cannot fix this by limiting to the category of affine schemes. Consider for example:

$$\begin{tikzcd} & \operatorname{Spec} k[t]_t \arrow[dl] \arrow[dr] & \\ \operatorname{Spec} k[t] & & \operatorname{Spec} k[t] \end{tikzcd}$$

where $\operatorname{Spec} k[t]_t = \mathbb{A}^1_k \backslash \{0\}$, and the maps are the maps given by $t \mapsto t$ and $t \mapsto 1/t$. Then we saw in example 2.15 that the colimit of this diagram is projective space, $\mathbb{P}^1_k$. Hence affine schemes are not in general closed under pushout. Note that the corresponding fibered product does exist in CRing, namely it will have to be:

$$\begin{tikzcd} & k \arrow[dl, dashed] \arrow[dr, dashed] & \\ k[t] \arrow[dr] & & k[t] \arrow[dl] \\ & k[t]_t & \end{tikzcd}$$

However, applying Spec to this will not return the pushout, namely the pair of adjoint functors (Spec, Γ) does not preserve pushouts.

Note however that colimits do exist in the category of locally ringed spaces. See this link (which link??) for more information.

(words here)

Definition 3.2.8: Geometrically Integral

Let S be a k-scheme. Then S is *geometrically integral* if it is integral over the algebraic closure$\overline{k}$.

The term "geometrically PROPERTY" is a common term to specify that a PROPERTY occurs for the k-scheme when taking the algebraic closure of k.

(want to include this somewhere: *Tensor Product Interpretation (Base Change):* The map constructed above is canonically isomorphic to the base change of the ring R_r to S. Using the tensor product:

$$S \otimes_R R_r \cong S \otimes_R (R[x]/\langle xr - 1\rangle) \cong S[x]/\langle x\varphi(r) - 1\rangle \cong S_{\varphi(r)}$$

This interpretation is powerful because it links the *geometric pullback* of sheaves to the *algebraic tensor product.*)

3.2.1 Fibers and family of Schemes

This gives a huge class of new object's that we may define. Let us start with defining fibers:

Definition 3.2.9: Fibers

Let $f : X \to Y$ be a morphism of scheme. Let $y \in Y$, and $k(y)$ be the residue field of y. Let $\operatorname{Spec}(k(y)) \to Y$ be the natural morphism[a]. Then the *fiber* of the morphism f over the point y is the scheme

$$X_y = X \times_Y \operatorname{Spec}(\kappa(y))$$

In the case where $X = \operatorname{Spec}(R), Y = \operatorname{Spec}(S)$ are affine, then for a point $\mathfrak{p} \in Y$:

$$f^{-1}(\mathfrak{p}) = \operatorname{Spec} R \times_S \operatorname{Spec} \kappa(\mathfrak{p}) = \operatorname{Spec}\left(R \otimes_S S_\mathfrak{p}/\mathfrak{p}S_\mathfrak{p}\right)$$

[a]exercise ref:HERE

Proposition 3.2.10: Topological Space of Fibers

Let $f : X \to Y$ be a morphism of schemes, and let $y \in Y$. The natural map $|X_y| \to f^{-1}(y)$ is a homeomorphism.

Furthermore, $X_y \to X$ is a *closed immersion* identified with the ideal $f^{\#}(\mathfrak{m}_y)$.

***Proof*:**
The question is affine local on source and target (proving bijectivity can be reduced to compatible open covers), so let's assume $X = \operatorname{Spec}(R)$ and $Y = \operatorname{Spec}(S)$ are affine schemes. Then $f : \operatorname{Spec} R \to \operatorname{Spec} S$ corresponds to a ring homomorphism $f^{\#} : S \to R$. Let y correspond to the prime ideal $\mathfrak{p}$ in S. Then:

$$X_y = X \times_Y \operatorname{Spec}(\kappa(y)) = \operatorname{Spec}(R \otimes_S S_\mathfrak{p}/\mathfrak{p}S_\mathfrak{p})$$

We shall take advantage of example **??**(1) where points are associated to maps to $\kappa(\mathfrak{q})$. Suppose $(f^{\#})^{-1}(\mathfrak{q}) = \mathfrak{p}$ where $\mathfrak{q} \in \operatorname{Spec}(R)$ (i.e. $f(\mathfrak{q}) = \mathfrak{p}$). Then Let g represent the composition $S \to R \to R_\mathfrak{q}$. Then $g : S \to R_\mathfrak{q}$ factors uniquely through $\overline{g} : S_\mathfrak{p} \to R_\mathfrak{q}$. We can extend this map by tensoring with $R \otimes_S (-)$ so that:

$$\Phi : R \otimes_S S_\mathfrak{p} \to R_\mathfrak{q} \qquad (r, s/t) = r\overline{g}(s/t)$$

Note that any element in $\mathfrak{p}S_\mathfrak{p}$ maps to $\mathfrak{q}R_\mathfrak{q}$, hence have have a unique map:

$$R \otimes_S S_\mathfrak{p}/\mathfrak{p}S_\mathfrak{p} \twoheadrightarrow R_\mathfrak{q}/\mathfrak{q}R_\mathfrak{q} = \kappa(\mathfrak{q})$$

As this map is unique, the correspondence between $k(\mathfrak{q})$ and the points of the fiber is unique, and certainly it surjective. Thus, for each point $\operatorname{Spec}(R_\mathfrak{q}/\mathfrak{q}R_\mathfrak{q})$, it is associated to a unique point in $f^{-1}(\mathfrak{p})$.

Finally, it is clear that $f^{-1}(\{y\})$ satisfies the (topological) pullback

$$\begin{tikzcd} f^{-1}(y) \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec}(k(y)) \arrow[r] & Y \end{tikzcd}$$

Taking:

$$\begin{tikzcd} X_y \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec}(k(y)) \arrow[r] & Y \end{tikzcd}$$

The maps are all continuous and follow from properties of fiber product, and the algebra above shows it is a bijection. It is easy to see it is a homeomorphism. For the closed

In some ways, fibers of schemes are more "regular" than fibers of sets, as the extra information of schemes will provide some more possible pre-images:

Example 3.2.11: Fibers

1. Take the familiar projection of the parabola onto the x axis induced by the ring map $\mathbb{Q}[x] \to \mathbb{Q}[y]$ mapping $x \mapsto y^2$. We can think of this as the map $\mathbb{Q}[x] \to \mathbb{Q}[y^2]$ where the codomain is isomorphic to:
$$\mathbb{Q}[y^2] \cong \frac{\mathbb{Q}[x,y]}{x-y^2}$$
This is sometimes called the parameterization of the map. Then the pre-image of 1 is:
$$\begin{aligned}\operatorname{Spec}\mathbb{Q}[x,y]/\left(y^2-x\right)\otimes_{\mathbb{Q}[x]}\mathbb{Q}[x]/(x-1) &\cong \operatorname{Spec}\mathbb{Q}[x,y]/\left(y^2-x, x-1\right)\\ &\cong \operatorname{Spec}\left(\mathbb{Q}[y]/\left(y^2-1\right)\right)\\ &\cong \operatorname{Spec}(\mathbb{Q}[y]/(y-1)(y+1))\\ &\overset{\text{C.R.T}}{\cong} \operatorname{Spec}(\mathbb{Q}[y]/(y-1)\times\mathbb{Q}[y]/(y+1))\\ &\cong \operatorname{Spec}(\mathbb{Q}[y]/(y-1))\amalg\operatorname{Spec}(\mathbb{Q}[y]/(y+1))\end{aligned}$$
That is, it is the points ± 1. The pre-image of 0 is:
$$\operatorname{Spec}\mathbb{Q}[x,y]/(y^2-x,x) \cong \operatorname{Spec}\mathbb{Q}[y]/(y^2)$$
Notice that the pre-image remembers multiplicity information!
As the spectrum of a polynomial ring over rationals is the quotient by the galois action on

the polynomial ring over $\bar{k}$, the pre-image of -1 exists! It is:

$$\operatorname{Spec}\mathbb{Q}[x,y]/(y^2-x,x+1)\cong\operatorname{Spec}\mathbb{Q}[y]/(y^2+1)\cong\operatorname{Spec}\mathbb{Q}[i]=\operatorname{Spec}\mathbb{Q}(i)$$

This can be thought of as "size 2" point over the base field. The pre-image of the generic point is a reduced point, and can be thought of as "size 2" over the residue field:

$$\operatorname{Spec}\mathbb{Q}[x,y]/(y^2-x)\otimes_{\mathbb{Q}[x]}\mathbb{Q}(x)\cong\operatorname{Spec}\mathbb{Q}[y]\otimes_{\mathbb{Q}[y]}\mathbb{Q}(y^2)$$

Try working through the pre-image of other points that are not commonly associated to $\mathbb{Q}$. Notice that the pre-image is the two points, or you get nonreduced behaviour that is "of degree 2" or you get a field extension of degree 2.

2. Let $X=\operatorname{Spec}(\mathbb{Z}[i])$ (the Gaussian integers) and $Y=\operatorname{Spec}(\mathbb{Z})$. The inclusion $\mathbb{Z}\hookrightarrow\mathbb{Z}[i]$ induces a morphism $\pi:X\to Y$. Let us analyze the fibers of this map (the pre-images of points).
   - *Ramification:* Consider the point $(2)\in\operatorname{Spec}(\mathbb{Z})$. The fiber is $\operatorname{Spec}(\mathbb{Z}[i]\otimes_{\mathbb{Z}}\mathbb{Z}/(2))\cong\operatorname{Spec}(\mathbb{F}_2[x]/(x^2+1))=\operatorname{Spec}(\mathbb{F}_2[x]/(x+1)^2)$. Topologically, there is one point, but the ring has a nilpotent element. The prime 2 "ramifies."
   - *Splitting:* Consider $(5)\in\operatorname{Spec}(\mathbb{Z})$. In $\mathbb{Z}[i]$, $5=(2+i)(2-i)$. The fiber corresponds to $\mathbb{F}_5[x]/(x^2+1)$. Since x^2+1 has roots $2,3$ in $\mathbb{F}_5$, this ring is $\mathbb{F}_5\times\mathbb{F}_5$. Geometrically, the point (5) pulls back to *two* distinct points.
   - *Inert:* Consider $(3)\in\operatorname{Spec}(\mathbb{Z})$. x^2+1 is irreducible in $\mathbb{F}_3$. The fiber is $\operatorname{Spec}(\mathbb{F}_9)$, a single point with a larger residue field.

   This example demonstrates that scheme morphisms provide the correct geometric framework for Algebraic Number Theory.

3. As $X\to\operatorname{Spec}(\mathbb{Z})$ is always well-defined and unique, there is always fibers over $\operatorname{Spec}(\mathbb{Z})$ and hence they are usually given a name: The fiber over (0) is often labeled $X_{\mathbb{Q}}$ (over $\mathbb{Q}$), and the fibers for all the primes are X_p (over $\mathbb{F}_p$). A common terminology is that X_p *arises by reduction* mod p of the scheme X.

4. One way of characterizing surjectivity $\operatorname{Spec}(B)\to\operatorname{Spec}(A)$ ring theoretically is to demand that for each $\mathfrak{p}\in A$, $B\otimes_{\mathbb{Z}}\kappa(\mathfrak{p})\neq 0$ (i.e. the preimage is nonempty). In section ref:HERE.

Proposition 3.2.12: Pre-image of Generic (Spreading out)

Let $f:X\to Y$ be a morphism of finite type and Y Noetherian. Then if $f^{-1}(\eta)=\emptyset$ for a generic point $\eta\in Y$, there exists an open subset $U\subseteq Y$ containing η such that $f^{-1}(U)=\emptyset$.

***Proof*:**
Since the problem is local on Y around η, we may assume $Y=\operatorname{Spec}(A)$ is affine. Since Y has a generic point η, Y is irreducible. As Y is Noetherian, we can treat A as an integral domain (modulo nilpotents, which do not affect the topology or emptiness of schemes). Let $K=\operatorname{Frac}(A)$ be the residue field of η.

Since f is of finite type, X is quasi-compact. We can cover X by a finite number of affine open subsets $X = \bigcup_{i=1}^n \operatorname{Spec}(B_i)$, where each B_i is a finitely generated A-algebra.

The hypothesis $f^{-1}(\eta) = \emptyset$ means the fiber product $X \times_Y \operatorname{Spec}(K)$ is the empty scheme. For each affine patch $\operatorname{Spec}(B_i)$, this corresponds to the ring:

$$B_i \otimes_A K = 0$$

Note that $B_i \otimes_A K$ is isomorphic to the localization of B_i by the multiplicative set $S = A \setminus \{0\}$. If a localization is the zero ring, then $1 = 0$ in that ring. By the definition of localization, this implies there exists an element $s_i \in S$ (i.e., a non-zero element $s_i \in A$) such that:

$$s_i \cdot 1_{B_i} = 0 \quad \text{in } B_i$$

Now, consider the basic open set $U_i = D(s_i) \subseteq Y$. Since $s_i \neq 0$, $\eta \in U_i$. The preimage of this open set in the chart $\operatorname{Spec}(B_i)$ is $\operatorname{Spec}((B_i)_{s_i})$. However, since s_i annihilates the unity in B_i, the localized ring is zero:

$$1 = \frac{s_i}{s_i} = \frac{1}{s_i} \cdot (s_i \cdot 1_{B_i}) = 0$$

Thus, $\operatorname{Spec}((B_i)_{s_i}) = \emptyset$.

To handle the entire scheme X, let $s = s_1 s_2 \cdots s_n$. Since A is a domain, $s \neq 0$, so $U = D(s)$ is a non-empty open set containing η. Furthermore, $U \subseteq U_i$ for all i. Consequently, the preimage $f^{-1}(U)$ is the union of the preimages restricted to each chart. Since s is a multiple of s_i, s acts as zero on every B_i, making the localization $(B_i)_s = 0$ for all i. Therefore, $f^{-1}(U) = \emptyset$.

On p. 260 of Vakil he gives more on general fibers, generic fibers, and generically finite morphisms don't know where to put these examples yet

Example 3.2.13: Parameterization of Curves using Schemes

Take

$$X = \operatorname{Spec} \frac{k[x, y, t]}{(ty - x^2)}$$
$$Y = \operatorname{Spec} k[t]$$

Let $f : X \to Y$ be given by the natural ring homomorphism $k[t] \to \frac{k[x,y,t]}{(ty-x^2)}$. Note both these schemes are integral schemes of finite type over k.

Now, identify the closed points of Y with k. For $a \neq 0 \in k$, we get the fibers correspond to the parabola's $ay = x^2$ or $y = x^2/a$. All of these are irreducible, reduced curves. At $a = 0$, we get $x^2 = 0$, which is a reduced scheme.

More generally, S-schemes can be geometrically interpreted as schemes paramaterized by the points of S whose maps are fiber-bundle morphisms. Hence, the scheme $\operatorname{Spec} k[t]$ *paramaterized* the curve.

A quick note must be done about general products and why it may be preferable to take the fibered product over a field:

Example 3.2.14: Strange Fibered Products

1. Show that $\operatorname{Spec}(\mathbb{Z}/m\mathbb{Z}) \times_{\operatorname{Spec}(\mathbb{Z})} \operatorname{Spec}(\mathbb{Z}/n\mathbb{Z}) = \emptyset$.
2. Another product of taking the "absolute" fibered product is that if $X = \operatorname{Spec}\mathbb{Z}[x]$ and $Y = \operatorname{Spec}\mathbb{Z}[y]$ then:

$$\dim X \times Y = \dim(X) + \dim(Y) - \dim(\operatorname{Spec} Z) = \dim X + \dim Y - 1$$

If we restrict to k-schemes for a field k, these two pathologies do not occur.

Put an appropriate definition here for the following

Hence, given a surjective map $Y \to X$, Y can be regarded as a collection of schemes paramaterized by Y. Conversely, we may want to study a scheme X_0, and see if there are some deformations we may want to do. Then a common way of defining a *family of algebraic deformation* is by a scheme morphisms $f : X \to Y$ where one of the fibers of f is isomorphic to X_0.

Some more here on representable Functors

Exercise 3.2.1

1. Let $\varphi : X \to Y$ be a morphism of S-scheme of finite type. Show that if $S \to S'$ is any base extension, $\varphi' : X' \to Y'$ is an S'-scheme morphism preserves finite type.
2. Let $\varphi : X \to Y$ be a scheme morphism of integral schemes. Show the fibers needn't be irreducible or reduced.
3. Let $\mathfrak{p} \in X$ be a point in a scheme. Show there is a canonical map

$$\operatorname{Spec}(\mathcal{O}_{X,\mathfrak{p}}) \to X$$

4. Using the above, define a canonical morphism $\operatorname{Spec}(\kappa(\mathfrak{p})) \to X$.
5. Perhaps put here some base-change exercises relating the tensor product exercises you had in [**NateClassicalAlgGeo**] and now give them their geometric counter-part.
6. Show if $U, V \subseteq X$ are open subschemes then $U \times_X V \cong U \cap V$.
7. Let $\varphi : B \to A$ be a ring morphism, $S \subseteq B$ a multiplicative subset, so $\varphi(S)$ is a multiplicative subset of A. Show that $\varphi(S)^{-1}A \cong A \otimes_B (S^{-1}B)$. What does this imply for the relation between localization and the fibered product?
8. Let X be a k-scheme. Show that if L/k is a field extension, $X \times_{\operatorname{Spec}(k)} \operatorname{Spec}(L)$ is an L-scheme. Show that this in fact gives a functor (define the appropriate maps, and verify).

3.2.2 *Monomorphisms in Sch

Using fiber products, we can better characterize subschemes categorically. It is natural to ask: what are the *categorical* sub-objects of a scheme? In any category, the correct notion of "sub-object" is a monomorphism: a morphism $f : Y \to X$ such that for any two morphisms $g, h : Z \rightrightarrows Y$, the equality $f \circ g = f \circ h$ implies $g = h$. In the category of sets, monomorphisms are injective functions. In the category of groups, they are injective homomorphisms. In the category of schemes, the answer is more interesting.

Proposition 3.2.15: Monomorphisms of Schemes

A morphism $f : Y \to X$ of schemes is a monomorphism if and only if the diagonal morphism $\Delta : Y \to Y \times_X Y$ is an isomorphism.

***Proof*:**
By definition, f is a monomorphism if and only if for every scheme Z, the map $f_* : \mathrm{Hom}(Z, Y) \to \mathrm{Hom}(Z, X)$ is injective. By the universal property of the fiber product, $\mathrm{Hom}(Z, Y \times_X Y) \cong \mathrm{Hom}(Z, Y) \times_{\mathrm{Hom}(Z,X)} \mathrm{Hom}(Z, Y)$, so f_* is injective for all Z if and only if the two projections $Y \times_X Y \rightrightarrows Y$ are equal, which holds if and only if Δ is an isomorphism.

For affine schemes, the characterization is purely algebraic. A morphism $\mathrm{Spec}\,(S) \to \mathrm{Spec}\,(R)$ is a monomorphism if and only if the corresponding ring map $\varphi : R \to S$ is an *epimorphism of commutative rings*: a map such that for any two ring maps $\alpha, \beta : S \rightrightarrows T$, the equality $\alpha \circ \varphi = \beta \circ \varphi$ implies $\alpha = \beta$. Equivalently, φ is an epimorphism if and only if the multiplication map $S \otimes_R S \to S$ is an isomorphism.

Note that monomorphissm of schemes are at least always injective. It is easy to see this reduces to affine patches. THen for a ring map $f : R \to S$ to be an epimorphism in the category of commutative rings, it must satisfy a strict tensor condition: the multiplication map $S \otimes_R S \to S$ must be an isomorphism. Geometrically, this injective on points, and it doesn't ramify or fold back on itself; see exercise ref:HERE.

The crucial observation is that epimorphisms of rings are *much more general* than surjections. The three main classes are:

Example 3.2.16: Epimorphisms in CRing

1. *Surjections* $R \twoheadrightarrow R/I$. These are epimorphisms because any map out of R/I is determined by its values on the images of elements of R. The corresponding monomorphisms of affine schemes are the *closed immersions*.

2. *Localizations* $R \to S^{-1}R$. These are epimorphisms because any map out of $S^{-1}R$ is determined by its values on elements of R (the images of elements of S must be invertible, which leaves no freedom). In general these maps are neither open nor closed (recall proposition 2.1.26). This includes both $R \to R_f$ (giving *open immersions* to distinguished opens) and $R \to R_\mathfrak{p}$ (giving the *local scheme* morphisms $\mathrm{Spec}\,(\mathcal{O}_{X,x}) \to X$).

   The most famous such example is $\mathbb{Z} \to \mathbb{Q}$ which is the localization away from the prime (0).

3. *Gabriel Localizations.* Most localization is done at either a single element or the compliment of a prime ideal. However, we can localize ideals. More generally, we can localize on *filters of ideals* (Gabriel localization) corresponding to an arbitrary intersection of open sets. Naturally, open sets are not closed under arbitrary intersection! These can result in pathological "fractal" sets that are still categorically a subscheme.

All four types of subobjects from the summary table are monomorphisms of schemes, but the converse is false: there exist monomorphisms that do not fall into any of the four classes. For instance, the map $\mathrm{Spec}\,(\mathbb{Q}) \to \mathrm{Spec}\,(\mathbb{Z})$ is a monomorphism (the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is an epimorphism of rings—it is the localization at the multiplicative set $\mathbb{Z} \setminus \{0\}$), but the image is the single point $\{[(0)]\}$, which is the generic point of $\mathrm{Spec}\,(\mathbb{Z})$. This is neither open, nor closed, nor locally closed: it is a local scheme map.

3.3 The Functor of Points

Throughout the development of scheme theory, we have built our geometric objects from locally ringed spaces: a topological space equipped with a structure sheaf, glued from affine pieces. This is a powerful framework, but it carries a certain asymmetry: the points of a scheme and the morphisms between schemes are defined by quite different-looking data. The goal of this section is to develop a perspective—the *functor of points*—in which schemes are understood entirely through their morphisms. The result is not merely a rephrasing; it reveals structure that the locally ringed space definition obscures, and it provides the language in which most modern moduli problems are formulated.

The idea has a simple origin. Recall from section 2.2.5 that a global section $f \in \Gamma(X, \mathcal{O}_X)$ of a scheme X assigns to each point $x \in X$ an element $f(x)$ in the residue field $\kappa(x)$, but that these residue fields vary from point to point. A global section is therefore *not* a function in the classical sense. Let us trace how this observation leads, step by step, to a functorial description of schemes.

3.3.1 From Evaluation to Morphisms

When the base field is algebraically closed, the situation is familiar. Over an algebraically closed field $k = \overline{k}$, a polynomial $f \in \overline{k}[x_1, \ldots, x_n]$ determines a genuine set-theoretic function:

$$\overline{k}[x_1, \ldots, x_n] \longrightarrow \mathrm{Hom}_{\mathbf{Set}}(\overline{k}^n, \overline{k}), \qquad f \longmapsto \big((a_1, \ldots, a_n) \mapsto f(a_1, \ldots, a_n)\big). \tag{3.1}$$

Here $\overline{k}^n$ can be identified with $\mathrm{mSpec}\,(\overline{k}[x_1, \ldots, x_n])$ via the Nullstellensatz, and each evaluation $f(a_1, \ldots, a_n)$ lands in $\overline{k} \cong \kappa(\mathfrak{m}_{(a_1,\ldots,a_n)})$. The map in equation (3.1) is injective precisely because an algebraically closed field has "enough points" to separate polynomials.

Over a general field k, this breaks down: the maximal ideals of $k[x_1, \ldots, x_n]$ no longer correspond to k-rational tuples. As discussed in example 2.1.3, $\mathrm{mSpec}\,(k[x_1, \ldots, x_n])$ contains orbits of geometric points under the absolute Galois group. One can partially recover the situation by mapping not into k, but into the affine line $\mathbb{A}^1_k$, which itself carries these additional closed points. Concretely, a polynomial $f \in k[x_1, \ldots, x_n]$ induces a morphism of k-algebras

$$k[t] \longrightarrow k[x_1, \ldots, x_n], \qquad t \longmapsto f,$$

which, by the equivalence of affine schemes with rings, corresponds to a scheme morphism $\mathbb{A}^n_k \to \mathbb{A}^1_k$. Evaluating at a closed point $\mathfrak{m}$ of $\mathbb{A}^n_k$ yields the image of f in $\kappa(\mathfrak{m})$, and this image determines a closed point of $\mathbb{A}^1_k$. Thus, we arrive at a map

$$k[x_1, \ldots, x_n] \longrightarrow \mathrm{Hom}_{\mathbf{Sch}_k}(\mathrm{mSpec}\, k[x_1, ..., x_n],\, \mathrm{mSpec}\, k[t]), \qquad f \longmapsto \text{the morphism induced by } t \mapsto f. \tag{3.2}$$

This is already an improvement: it works over any field, and the target is a set of *scheme morphisms* rather than set-theoretic functions. However, it still only detects the behavior of f at closed points. Recall that such schemes are called *Jacobson schemes* (see definition 3.7.4). For general schemes this is insufficient. For instance, on $\mathrm{Spec}\,(k[[x]])$ or $\mathrm{Spec}\,(\mathbb{Z}_{(p)})$, a function is not determined by its values at closed points alone.

The natural fix is to allow *all* prime ideals as test points. For any k-algebra A, a scheme morphism $\mathrm{Spec}\,(A) \to \mathbb{A}^1_k$ is the same data as a k-algebra map $k[t] \to A$, which since this is a k-algebra homomorphism is the same as choosing an element of A for t to map to. This gives:

$$\Gamma(\mathrm{Spec}\,(A), \mathcal{O}_{\mathrm{Spec}\,(A)}) = A \;\cong\; \mathrm{Hom}_{\mathbf{Sch}_k}(\mathrm{Spec}\,(A),\, \mathbb{A}^1_k). \tag{3.3}$$

The left-hand side is the ring of global sections; the right-hand side is the set of scheme morphisms to the affine line. Each element $f \in A$ corresponds to the unique morphism $\pi_f : \mathrm{Spec}\,(A) \to \mathbb{A}^1_k$ for which $\pi_f^{\#}(t) = f$. The "value" of f at a prime $\mathfrak{p}$ is recovered by following $t \mapsto f \mapsto f \bmod \mathfrak{p} \in \kappa(\mathfrak{p})$, which is precisely the scheme-theoretic notion of evaluation from section 2.2.5.

Two observations are now in order. First, the construction (3.3) is not specific to affine schemes: it will extend to arbitrary k-schemes X once we establish the representability of the global section functor (Proposition 3.3.8 below). Second, and more importantly, there is nothing special about $\mathbb{A}^1_k$ as a target apart from the fact that it *represents* the functor of global sections. One can ask: what happens if we replace $\mathbb{A}^1_k$ by another scheme T?

The set $\mathrm{Hom}_{\mathbf{Sch}_k}(X, T)$ makes perfect sense for any T, and different choices of T probe different aspects of X. Taking $T = \mathbb{A}^1_k$ recovers global sections; taking $T = \mathbb{G}_m$ recovers invertible global sections; taking $T = \mathbb{P}^n_k$ recovers line bundles generated by $n+1$ global sections (cf. section 5.3.3). The test object T acts as a "measuring device" that detects a particular piece of the geometry of X. The remarkable fact—a consequence of the Yoneda Lemma applied to schemes—is that if we keep track of $\mathrm{Hom}(-, X)$ for *all* possible test objects simultaneously, we recover X entirely.

3.3.2 Z-valued Points

Let us make the above discussion concrete. Consider the scheme

$$X = \mathrm{Spec}\ \frac{\mathbb{R}[x, y]}{(x^2 + y^2 - 1)},$$

whose underlying topological space consists of $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$-conjugate pairs of maximal ideals together with a generic point. The closed points do not individually look like points on a circle—many have residue field $\mathbb{C}$, not $\mathbb{R}$—and so the "usual" picture of the unit circle is not immediately visible.

If we want to recover that picture, we should look for $\mathbb{R}$-valued points: ring homomorphisms

$$\mathrm{ev}_{a,b} : \frac{\mathbb{R}[x, y]}{(x^2 + y^2 - 1)} \longrightarrow \mathbb{R}$$

sending $x \mapsto a$ and $y \mapsto b$. The quotient relation forces $a^2 + b^2 = 1$, so the collection of all such homomorphisms is in bijection with

$$X(\mathbb{R}) \;\simeq\; \left\{(a, b) \in \mathbb{R}^2 \;\middle|\; a^2 + b^2 = 1\right\},$$

which is exactly the unit circle. Equivalently, each such homomorphism factors through the evaluation map $\mathbb{R}[x, y] \to \mathbb{R}$ as a map that kills the ideal $(x^2 + y^2 - 1)$. Via the contravariant equivalence of affine schemes and rings, these ring homomorphisms correspond to scheme morphisms $\operatorname{Spec}(\mathbb{R}) \to X$. The sheaf morphism on global sections is:

$$\mathrm{ev}_{a,b}^{\#}(X) : \frac{\mathbb{R}[x,y]}{(x^2+y^2-1)} \longrightarrow \mathbb{R}.$$

Now suppose we change the target ring. Over $\mathbb{Z}$, the equation $a^2+b^2=1$ has only the four solutions $(\pm 1, 0)$ and $(0, \pm 1)$. Over $\mathbb{Q}$, a Pythagorean-triple parametrization yields infinitely many solutions such as $\left(\frac{3}{5}, \frac{4}{5}\right)$. Over $\mathbb{C}$, the circle becomes a rational curve, and the solution set is much larger still. Each choice of "coefficient ring" yields a different collection of evaluation maps, probing different arithmetic and geometric features of the same scheme.

There is one subtlety to address regarding the base. For the evaluation map $\mathrm{ev}_{a,b}$ to exist as a map of $\mathbb{R}$-algebras, the target ring must receive a structure map from $\mathbb{R}$. There is no nontrivial ring homomorphism $\mathbb{R} \to \mathbb{Q}$, so asking for $\mathbb{Q}$-valued points of the $\mathbb{R}$-scheme X is not well-posed. To allow all these evaluations simultaneously, it is natural to work over the universal base $\mathbb{Z}$. Set $X_{\mathbb{Z}} = \operatorname{Spec} \mathbb{Z}[x,y]/(x^2+y^2-1)$. Then for any commutative ring R, the R-valued points of $X_{\mathbb{Z}}$ are the ring homomorphisms $\mathbb{Z}[x,y]/(x^2+y^2-1) \to R$, or equivalently the solutions of $a^2+b^2=1$ in R. If $A \to B$ is a ring homomorphism and X_A, X_B denote the base changes, then $X_A(B) = X_B(B)$: the B-valued points see only the base-changed scheme.

It is worth pausing to understand the $\mathbb{Z}$-points of $X_{\mathbb{Z}}$ scheme-theoretically.

A $\mathbb{Z}$-point of $X_{\mathbb{Z}}$ is a morphism $\operatorname{Spec}(\mathbb{Z}) \to X_{\mathbb{Z}}$, which is a section of the structure map $X_{\mathbb{Z}} \to \operatorname{Spec}(\mathbb{Z})$. Geometrically, one can visualize $X_{\mathbb{Z}}$ as an arithmetic surface fibered over $\operatorname{Spec}(\mathbb{Z})$, with each fiber X_p being the circle $x^2+y^2=1$ reduced modulo the prime p. A $\mathbb{Z}$-point is then a "horizontal" section of this fibration: it passes through every fiber simultaneously.

These $\mathbb{Z}$-points are *not* closed points of $X_{\mathbb{Z}}$. Consider the point $P=(1,0)$, corresponding to the homomorphism $\varphi : \mathbb{Z}[x,y]/(x^2+y^2-1) \to \mathbb{Z}$ defined by $x \mapsto 1$, $y \mapsto 0$. The kernel is the prime ideal $\mathfrak{p} = (x-1, y)$, and the quotient $\mathbb{Z}[x,y]/\mathfrak{p} \cong \mathbb{Z}$. Since $\mathbb{Z}$ is not a field, $\mathfrak{p}$ is prime but not maximal, and the image of P is a closed subscheme isomorphic to $\operatorname{Spec}(\mathbb{Z})$—it has dimension 1 and residue field $\mathbb{Q}$ at its generic point. In contrast, a closed point of $X_{\mathbb{Z}}$ has dimension 0 and a finite residue field $\mathbb{F}_q$.

The four $\mathbb{Z}$-points interact with the fibers in instructive ways. Modulo 2, since $1 \equiv -1$ in $\mathbb{F}_2$, the points $(1,0)$ and $(-1,0)$ collapse to the same point of X_2, as do $(0,1)$ and $(0,-1)$. Modulo an odd prime $p \geq 3$, all four points remain distinct in $X_p(\mathbb{F}_p)$. These four "horizontal sections" are the only integer solutions on the circle; any other rational point, such as $\left(\frac{3}{5}, \frac{4}{5}\right)$, gives a $\mathbb{Q}$-point but not a $\mathbb{Z}$-point.

More generally, there is no reason to restrict the "value scheme" to be affine. One might wish to evaluate points on a torus with values in a curve, or to study morphisms between projective schemes. The evaluation interpretation as a global ring homomorphism is then lost, but this is precisely what the sheaf was built to handle: evaluation can be performed locally. Geometrically, a morphism $Z \to X$ between arbitrary schemes is the datum of compatible local evaluations. This motivates the following definition.

Definition 3.3.1: Z-valued Points

Let X be a scheme. The *Z-valued points* of X are the morphisms $Z \to X$ in the category of schemes. The set of all such morphisms is denoted

$$X(Z) := \mathrm{Hom}_{\mathbf{Sch}}(Z, X).$$

When $Z = \mathrm{Spec}\,(R)$ is affine, this is written $X(R)$.

The reader should note a potential source of confusion: a "Z-valued point" is the *morphism* $Z \to X$ itself, not any particular point in Z or in the image of Z. This terminology is justified by the fact that a morphism of schemes is not determined by the underlying continuous map of topological spaces—the sheaf map $\varphi^{\#}$ carries additional information. The full morphism is the "point".

the fibered product of two schemes need not be the schemes of these two fibered products. In a concrete sense, this is because we must be more specific with what types of points we are dealing with. If we specify the points, we in fact recover the usual "sets"[5].

Proposition 3.3.2: Fibered Product With Z-valued Points

Let X, Y be schemes and take Z-valued points on each scheme i.e. morphisms in $\mathrm{Hom}(Z, X)$ and $\mathrm{Hom}(Z, Y)$. Then show that the fibered product will be:

$$\mathrm{Hom}(W, X \otimes_W Y) = \mathrm{Hom}(Z, X) \times_{\mathrm{Hom}(Z,W)} \mathrm{Hom}(Z, Y)$$

***Proof*:**
exercise.

3.3.3 The Yoneda Embedding for Schemes

Having defined Z-valued points, we can now ask: how much of a scheme X is captured by the totality of its valued points? The answer, given by the Yoneda Lemma, is: *everything*.

Proposition 3.3.3: Scheme Morphisms from Valued Points

Let X and Y be schemes. A morphism of schemes $\varphi : X \to Y$ is equivalent to a natural transformation $\eta : h_X \to h_Y$ between the associated functors of points. That is, giving a collection of maps of sets

$$\eta_Z : X(Z) \longrightarrow Y(Z)$$

for every scheme Z, natural in Z, uniquely determines a scheme morphism $\varphi : X \to Y$, and conversely.

[5] The topological product can also be thought of as being over the right valued points

***Proof*:**
Every scheme X defines a contravariant functor $h_X : \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ by

$$h_X(Z) = \mathrm{Hom}_{\mathbf{Sch}}(Z, X) = X(Z).$$

A morphism $\varphi : X \to Y$ induces a natural transformation $\varphi_* : h_X \to h_Y$ by post-composition: for each Z and each $f \in X(Z)$, set $(\varphi_*)_Z(f) = \varphi \circ f$. Naturality follows from the associativity of composition.

Conversely, given a natural transformation $\eta : h_X \to h_Y$, we evaluate at $Z = X$. The set $X(X) = \mathrm{Hom}(X, X)$ contains the identity id_X, and we define

$$\varphi := \eta_X(\mathrm{id}_X) \in Y(X) = \mathrm{Hom}(X, Y).$$

By the Yoneda Lemma (section 0.4),

$$\mathrm{Nat}(h_X, h_Y) \;\cong\; h_Y(X) = \mathrm{Hom}(X, Y),$$

and this correspondence is bijective. Thus, the assignment $X \mapsto h_X$ defines a fully faithful embedding $\mathbf{Sch} \hookrightarrow \mathrm{Fun}(\mathbf{Sch}^{\mathrm{op}}, \mathbf{Set})$: the category of schemes embeds into the category of presheaves on $\mathbf{Sch}$.

The proof is short, but the content is not trivial. It says that a scheme is *completely determined* by how all other schemes map into it. No information is lost by passing from the locally ringed space X to the functor h_X.

A useful analogy organizes this perspective. In linear algebra, a vector $v \in V$ is determined by the collection of values $\ell(v)$ for all linear functionals $\ell \in V^*$: the dual space "separates points." Choosing a basis for V^* gives coordinates on V. The Yoneda embedding plays the same role for schemes. The functor h_X is the "dual description" of X: the scheme is determined by the collection of sets $X(Z)$ for all test schemes Z. Each choice of test scheme is analogous to choosing a basis vector in the dual—it extracts one piece of information about X.

Different test objects give genuinely different perspectives on the same scheme:

Test object Z	$X(Z)$	*What it detects*
$\mathrm{Spec}\,(\bar{k})$	geometric points	the classical variety
$\mathrm{Spec}\,(k)$	rational points	arithmetic information
$\mathrm{Spec}\,(k[\epsilon]/(\epsilon^2))$	tangent vectors	infinitesimal geometry (order 1)
$\mathrm{Spec}\,(k[t]/(t^n))$	$(n-1)$-jets	higher infinitesimal geometry
$\mathrm{Spec}\,(k[[t]])$	formal arcs	formal local behavior
$\mathrm{Spec}\,(K),\ K = k(C)$	rational curves	birational geometry
$\mathrm{Spec}\,(\mathcal{O}_{X,x})^\dagger$	the germ at x	local scheme structure
$\mathrm{Spec}\,(\hat{\mathcal{O}}_{X,x})^\dagger$	formal neighborhood	formal geometry at x

† The last two entries differ in character from the others: they depend on the scheme X itself, rather than being universal test objects. Their role is not to "probe" X from outside, but to isolate the local structure at a point $x \in X$.

The local ring $\mathcal{O}_{X,x}$ is the stalk of the structure sheaf, and $\operatorname{Spec}(\mathcal{O}_{X,x})$ is the "germ" of X at x: the limit of all open neighborhoods of x, retaining only the information visible in an arbitrarily small neighborhood. There is a canonical morphism $\operatorname{Spec}(\mathcal{O}_{X,x}) \to X$ (induced by the localization maps $\mathcal{O}_X(U) \to \mathcal{O}_{X,x}$), and any morphism $f : Y \to X$ sending a point $y \in Y$ to x factors through $\operatorname{Spec}(\mathcal{O}_{X,x})$ at the level of germs:

$$\operatorname{Spec}(\mathcal{O}_{Y,y}) \longrightarrow \operatorname{Spec}(\mathcal{O}_{X,x}) \longrightarrow X.$$

In this sense, the local ring "sees" every morphism that passes through x. This is why properties of morphisms (flatness, smoothness, étaleness) can often be checked at the level of local rings.

The completion $\hat{\mathcal{O}}_{X,x}$ captures even finer information: the formal power series behavior at x, analogous to the Taylor expansion of a smooth function. The formal arc entry $\operatorname{Spec}(k[[t]])$ in the table is in fact the special case $\operatorname{Spec}(\hat{\mathcal{O}}_{\mathbb{A}^1,0})$—the formal neighborhood of the origin in the affine line. For a general point x on a smooth n-dimensional variety, the completion is $\hat{\mathcal{O}}_{X,x} \cong k[[t_1, \ldots, t_n]]$, and the formal neighborhood is a "formal polydisk" of dimension n.

The passage from one test object to another is analogous to a change of basis. A morphism $Z' \to Z$ induces, by precomposition, a map $X(Z) \to X(Z')$ that relates the two perspectives. For instance, the closed immersion $\operatorname{Spec}(k) \hookrightarrow \operatorname{Spec}(k[\epsilon]/(\epsilon^2))$, dual to the quotient map $k[\epsilon]/(\epsilon^2) \twoheadrightarrow k$, sends a tangent vector to its basepoint. This is the "forgetful" map from infinitesimal data to point data. The inclusions among the infinitesimal test objects form a tower:

$$\operatorname{Spec}(k) \hookrightarrow \operatorname{Spec}(k[\epsilon]/(\epsilon^2)) \hookrightarrow \operatorname{Spec}(k[t]/(t^3)) \hookrightarrow \cdots \hookrightarrow \operatorname{Spec}(k[[t]])$$

where each step remembers one more order of infinitesimal information, and the formal arc $\operatorname{Spec}(k[[t]])$ is the limit, retaining all orders simultaneously.

3.3.4 Key Examples

The definition of Z-valued points subsumes many classical constructions. We now develop the most important instances in detail.

Example 3.3.4: Geometric Points

Let X be a scheme locally of finite type over a field k, and let $\overline{k}$ be an algebraic closure. A $\overline{k}$-valued point is a k-morphism

$$\varphi : \operatorname{Spec}(\overline{k}) \longrightarrow X.$$

Since $\operatorname{Spec}(\overline{k})$ has a unique point η, the morphism φ is determined by the image $y = \varphi(\eta) \in X$ together with the stalk map $\varphi_y^\# : \mathcal{O}_{X,y} \to \overline{k}$. Passing to the residue field, we obtain a k-embedding $\kappa(y) \hookrightarrow \overline{k}$.

We claim that y must be a closed point. Indeed, if y were not closed, then $\kappa(y)$ would contain an element transcendental over k; say t. In particular, as $\varphi_y(t) \in \overline{k}$ must be algebraic, there exists a k-polynomial p such that $\varphi_y(p(t)) = p(\varphi_y(t)) = 0$. As $\varphi_y(0) = 0$ and t is transcendental, $p(t) \neq 0$, demonstrating the failure of injectivity—a contradiction, since field homomorphisms are injective. Conversely, if y is closed, then $\kappa(y)/k$ is a finite extension (by Zariski's Lemma, since X is locally of finite type), and any finite extension of k embeds into $\overline{k}$. Thus, y must be closed, and every closed point arises in this way.

However, there may be *more* $\overline{k}$-valued points than closed points. For example, the $\mathbb{R}$-scheme $\operatorname{Spec}(\mathbb{R}[x]/(x^2+1)) \cong \operatorname{Spec}(\mathbb{C})$ has a single closed point, but two $\mathbb{C}$-valued points: these correspond to the two $\mathbb{R}$-algebra homomorphisms $\mathbb{C} \to \mathbb{C}$ (namely the identity and complex conjugation). In general, the number of $\overline{k}$-valued points lying over a closed point y equals the number of k-embeddings $\kappa(y) \hookrightarrow \overline{k}$, which is $[\kappa(y) : k]$ when $\kappa(y)/k$ is separable.

To obtain a bijection, one can either base-change to $\overline{k}$ (replacing X by $X_{\overline{k}} = X \times_k \operatorname{Spec}(\overline{k})$, which identifies all the Galois conjugates; see section 3.2), or work directly over an algebraically closed field. In the latter case, $\kappa(y) = \overline{k}$ for every closed point, and the correspondence

$$\left\{\operatorname{Spec}(\overline{k}) \to X\right\} \xrightarrow{\sim} \{y \in X : y \text{ is closed}\}$$

is bijective. This recovers the classical variety associated to X: the "geometric points" are exactly the closed points over an algebraically closed field.

Example 3.3.5: Generic Points and Rational Maps

Let C be a curve over $\mathbb{C}$ with function field $K = \mathbb{C}(C)$. A K-valued point of a $\mathbb{C}$-scheme X is a $\mathbb{C}$-morphism $\operatorname{Spec}(K) \to X$. Since $\operatorname{Spec}(K)$ is the generic point of C, such a morphism amounts to a rational map $C \dashrightarrow X$: it is defined on some dense open subset of C, but may not extend to all of C.

More concretely, taking $K = \mathbb{C}(t)$ (the function field of $\mathbb{A}^1_{\mathbb{C}}$), any $\mathbb{C}(t)$-valued point fits into the commutative diagram:

$$\begin{tikzcd} \operatorname{Spec}\mathbb{C}(t) \arrow[rr] \arrow[dr] & & X \arrow[dl] \\ & \operatorname{Spec}\mathbb{C} & \end{tikzcd}$$

This is the algebro-geometric analogue of a "generic curve" in X: rather than picking out a single closed point, it describes a one-parameter family of points parametrized by the generic point of the affine line. Whether such a rational map extends to a regular morphism $\mathbb{A}^1_{\mathbb{C}} \to X$ (or $C \to X$ in general) is precisely the content of the valuative criteria discussed in section 3.5.1 and section 3.5.4.

Example 3.3.6: Tangent Vectors and the Dual Numbers

Let $D = k[\epsilon]/(\epsilon^2)$ be the ring of dual numbers. The scheme $\operatorname{Spec}(D)$ is a "fat point": it has a unique closed point $\overline{o}$ (corresponding to the maximal ideal (ϵ)), but its structure sheaf remembers one infinitesimal direction. The canonical surjection $D \twoheadrightarrow k$ corresponds to a closed immersion $\iota : \operatorname{Spec}(k) \hookrightarrow \operatorname{Spec}(D)$, and we write $o = \iota^{-1}(\overline{o})$.

Let X be a k-scheme of finite type and let $x \in X$ be a closed point (so $\kappa(x) = k$). We classify all D-valued points of X based at x, namely all morphisms $\varphi : \operatorname{Spec}(D) \to X$ such that $\varphi \circ \iota$ sends o to x. Choose an affine open neighborhood $x \in \operatorname{Spec}(R) \subseteq X$; then $\operatorname{Hom}_x(\operatorname{Spec}(D), X) \cong \operatorname{Hom}_x(\operatorname{Spec}(D), \operatorname{Spec}(R))$, where the subscript x indicates that the closed point maps to x. By the equivalence of affine schemes and rings, these are in bijection with k-algebra homomorphisms $f : R \to D$ satisfying $f(\mathfrak{m}_x) \subseteq (\epsilon)$.

Now R decomposes as a k-vector space as $R = k \oplus \mathfrak{m}_x$. Indeed, as x is closed,

$$0 \to \mathfrak{m}_x \hookrightarrow R \twoheadrightarrow R/\mathfrak{m}_x \cong k \to 0$$

and as k is free this splits. Then f is the identity on k, so f is determined by its restriction to $\mathfrak{m}_x$. Since $\epsilon^2 = 0$ in D, we have $f(\mathfrak{m}_x^2) = 0$, and so f descends to a k-linear functional on $\mathfrak{m}_x/\mathfrak{m}_x^2$. Conversely, any linear functional $\ell : \mathfrak{m}_x/\mathfrak{m}_x^2 \to k$, extended by zero on $\mathfrak{m}_x^2$ and by the identity on k, determines a k-algebra map $R \to D$ sending $\mathfrak{m}_x$ into (ϵ). We conclude:

$$\mathrm{Hom}_x(\mathrm{Spec}\,(D),\, X) \;\cong\; (\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee = \Theta_{X,x},$$

the Zariski tangent space at x. This is a satisfying realization: the tangent space, originally defined via the maximal ideal and its square (cf. section 2.7), is recovered purely by probing X with the fat point $\mathrm{Spec}\,(D)$.

Moreover, if $f : X \to Y$ is a morphism of k-schemes, then post-composition with f sends a D-valued point based at x to one based at $f(x)$:

$$\mathrm{d}_x f :\; \Theta_{X,x} \longrightarrow \Theta_{Y,f(x)}, \qquad v \longmapsto f \circ v.$$

This is the differential of f, now defined without any reference to local rings or Kähler differentials—it is simply the functoriality of D-valued points.

Example 3.3.7: Group Schemes via Valued Points

By Proposition 3.3.3, a group scheme G over k can equivalently be described as a scheme such that for every k-scheme Z, the set $G(Z)$ carries a group structure, and all maps $G(Z) \to G(Z')$ induced by morphisms $Z' \to Z$ are group homomorphisms. In other words, G is a "group object" in the functor-of-points sense: the functor $h_G : \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ factors through the forgetful functor $\mathbf{Grp} \to \mathbf{Set}$.

For a concrete instance, consider the multiplicative group scheme $\mathbb{G}_m = \mathrm{Spec}\,(k[t, t^{-1}])$. For any k-algebra A:

$$\mathbb{G}_m(A) = \mathrm{Hom}_{k\text{-}\mathbf{Alg}}(k[t, t^{-1}],\, A) \;\cong\; A^\times.$$

A k-algebra homomorphism out of $k[t, t^{-1}]$ is determined by the image of t, and invertibility of t in the source forces the image to be a unit in A. Since the units of any ring form a group under multiplication, and ring homomorphisms preserve units, $\mathbb{G}_m(A)$ is a group for every A, functorially in A.

Similarly, the additive group scheme $\mathbb{G}_a = \mathrm{Spec}\,(k[t]) = \mathbb{A}_k^1$ satisfies $\mathbb{G}_a(A) = A$ with its additive group structure; and the general linear group $\mathrm{GL}_{n,k} = \mathrm{Spec}\,(k[x_{ij}, \det^{-1}])$ satisfies $\mathrm{GL}_{n,k}(A) \cong \mathrm{GL}_n(A)$ for any k-algebra A. These examples demonstrate a recurring theme: when working over a non-algebraically-closed base, or with non-reduced schemes, the functor of points often gives a cleaner description than the underlying topological space. One *defines* the algebraic group by specifying what it does on all test rings, and the scheme structure is a consequence.

3.3.5 Representable Functors in Algebraic Geometry

The examples of the previous subsection probe a fixed scheme X by varying the test object Z. We now reverse the perspective: fix the type of geometric data we wish to extract, and ask whether there exists a "universal" scheme that classifies it. Formally, given a contravariant functor $\mathcal{F} : \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$, we say $\mathcal{F}$ is *representable* if there exists a scheme M and a natural isomorphism $\mathcal{F} \cong h_M$. The scheme M is then called a *fine moduli space* for the problem encoded by $\mathcal{F}$, and by the Yoneda Lemma it is unique up to unique isomorphism.

We have already encountered the simplest instance of this idea.

Proposition 3.3.8: Global Sections and the Affine Line

Let X be a k-scheme. There is a natural bijection

$$\Gamma(X, \mathcal{O}_X) \;\cong\; \mathrm{Hom}_{\mathbf{Sch}_k}(X,\, \mathbb{A}^1_k).$$

Concretely, a global section $f \in \Gamma(X, \mathcal{O}_X)$ corresponds to the unique morphism $\pi_f : X \to \mathbb{A}^1_k$ such that $\pi_f^\#(t) = f$, where t is the coordinate on $\mathbb{A}^1_k = \mathrm{Spec}\,(k[t])$. In categorical language, the global section functor $\Gamma(-, \mathcal{O})$ is representable by $\mathbb{A}^1_k$.

***Proof*:**
By the adjunction between Spec and Γ (proposition 3.1.12):

$$\mathrm{Hom}_{\mathbf{Sch}_k}(X,\, \mathrm{Spec}\; k[t]) \;\cong\; \mathrm{Hom}_{k\text{-}\mathbf{Alg}}(k[t],\, \Gamma(X, \mathcal{O}_X)).$$

The polynomial ring $k[t]$ is the free k-algebra on one generator: a k-algebra homomorphism $k[t] \to R$ is uniquely determined by the image of t. Thus:

$$\mathrm{Hom}_{k\text{-}\mathbf{Alg}}(k[t],\, \Gamma(X, \mathcal{O}_X)) \;\cong\; \Gamma(X, \mathcal{O}_X).$$

Let us verify that the bijection matches our intuition of evaluation. Let $x \in X$ and let $\pi_f : X \to \mathbb{A}^1_k$ be the morphism corresponding to $f \in \Gamma(X, \mathcal{O}_X)$. The definition of a scheme morphism gives a commutative diagram of local rings:

$$\begin{tikzcd}
k[t] \arrow[r, "\psi"] \arrow[d] & \Gamma(X, \mathcal{O}_X) \arrow[r] & \mathcal{O}_{X,x} \arrow[d, "\text{residue}"] \\
\mathcal{O}_{\mathbb{A}^1, \pi_f(x)} \arrow[rr] & & \kappa(x)
\end{tikzcd}$$

The composition across the top sends $t \mapsto f \mapsto f(x) \in \kappa(x)$, and the image point $\pi_f(x)$ is the prime ideal $\mathfrak{q} = \{P(t) \in k[t] : P(f(x)) = 0 \text{ in } \kappa(x)\}$. This is exactly the scheme-theoretic "value" of f at x.

3.3.9 Remark: no field hypothesis is needed The above proof uses only that $k[t]$ is free on one generator; it did not use that k is a field. For any base scheme S, the relative affine line $\mathbb{A}^1_S$ represents the global section functor on $\mathbf{Sch}_S$. The same remark applies to several of the examples that follow.

The representability of $\Gamma(-, \mathcal{O})$ by $\mathbb{A}^1_k$ is the first instance of a broad pattern. Let us now survey the most important representable functors in algebraic geometry. In each case, a piece of geometric data attached to a scheme X is captured by morphisms from X into a single "classifying" scheme.

The multiplicative group. Replacing the additive structure of $\mathbb{A}^1_k$ with the multiplicative structure of $\mathbb{G}_m = \mathrm{Spec}\,(k[t, t^{-1}])$ gives:

$$\mathrm{Hom}_{\mathbf{Sch}_k}(X,\, \mathbb{G}_m) \;\cong\; \Gamma(X, \mathcal{O}_X)^\times.$$

The proof follows the same pattern: a k-algebra map $k[t, t^{-1}] \to R$ is determined by the image of t, and invertibility of t in the source forces the image to be a unit. The $\mathbb{G}_m$-valued points of X are precisely the invertible global sections.

Projective space and line bundles. The representability of projective space is quite interesting. Define the functor

$$\mathcal{F}(X) = \{(\mathcal{L},\, s_0, \ldots, s_n) \mid \mathcal{L} \in \mathrm{Pic}\,(X),\; s_i \in \Gamma(X, \mathcal{L}),\; \text{the } s_i \text{ generate } \mathcal{L}\} \,/\!\cong$$

where two tuples $(\mathcal{L}, s_0, \ldots, s_n)$ and $(\mathcal{L}', s'_0, \ldots, s'_n)$ are identified if there is an isomorphism $\mathcal{L} \xrightarrow{\sim} \mathcal{L}'$ carrying s_i to s'_i for each i. Then there is a natural bijection

$$\mathrm{Hom}_{\mathbf{Sch}_k}(X,\, \mathbb{P}^n_k) \;\cong\; \mathcal{F}(X).$$

A morphism to $\mathbb{P}^n$ is the same data as a line bundle equipped with $n+1$ globally generating sections; these sections serve as "homogeneous coordinates" for the map. This is the content of the theory developed in section 5.3.3. When $n = 1$, a map $X \to \mathbb{P}^1$ corresponds to a line bundle with two generating sections—the algebro-geometric analogue of a meromorphic function.

Grassmannians. The Grassmannian $\mathrm{Gr}(r, n)$ generalizes projective space from lines to r-dimensional subspaces. It represents the functor

$$\mathcal{F}(X) = \left\{\mathcal{O}_X^{\oplus n} \twoheadrightarrow \mathcal{E} \;\middle|\; \mathcal{E} \text{ is a locally free sheaf of rank } r\right\} /\!\cong.$$

A morphism $X \to \mathrm{Gr}(r, n)$ classifies a rank-r quotient bundle of the trivial bundle $\mathcal{O}_X^{\oplus n}$, or equivalently, a sub-bundle of rank $n - r$. When $r = 1$, this recovers the description of projective space above. The Grassmannian is itself a smooth projective variety; its functor-of-points description gives a uniform construction that works over any base ring, without the need to choose coordinates or Plücker embeddings.

The general linear group. The group scheme $\mathrm{GL}_{n,k} = \mathrm{Spec}\,(k[x_{11}, \ldots, x_{nn},\, \det^{-1}])$ represents the functor

$$\mathrm{Hom}_{\mathbf{Sch}_k}(\mathrm{Spec}\,(A),\, \mathrm{GL}_{n,k}) \;\cong\; \mathrm{GL}_n(A)$$

for any k-algebra A. A map into $\mathrm{GL}_{n,k}$ is the same as giving an invertible $n \times n$ matrix with entries in A. Combined with the group scheme structure from Example 3.3.7, this gives a definition of the general linear group that works uniformly over all base rings, including rings with nilpotents or mixed characteristic.

Hilbert schemes. Perhaps the most striking application of representability is given by the Hilbert scheme, which we develop in the next subsection.

The common thread in each of the above examples is the Yoneda philosophy: a geometric invariant of X—global sections, line bundles, quotient bundles, invertible matrices—is captured by morphisms from X into a single classifying scheme. The Yoneda Lemma ensures that the classifying scheme, when it exists, is unique up to unique isomorphism.

3.3.6 Hilbert Schemes

The representable functors considered so far classify data that is "linear" in nature: sections of a sheaf, quotients of a free module, invertible matrices. The Hilbert scheme is of a fundamentally different character: it parametrizes *subschemes* of a given projective scheme. That such a moduli problem admits a scheme as its solution is one of the most important results in the functorial approach to algebraic geometry, and it provides a rich source of examples and applications.

Flatness as the continuity condition

Before stating the result, we must address a foundational question: what does it mean for a collection of subschemes to "vary continuously" over a parameter space? Consider a family of closed subschemes $\{Z_s\}_{s\in S}$ of a fixed projective scheme $Y \subseteq \mathbb{P}^n_k$, parametrized by a scheme S. Algebraically, this family is encoded by a closed subscheme $Z \subseteq Y \times_k S$ such that each fiber $Z_s = Z \times_S \{s\}$ is the corresponding member of the family.

Not every such Z should be considered a "nice" family. For instance, one could construct a family where a curve degenerates to a union of a curve and an isolated point, causing the Hilbert polynomial to jump. The condition that prevents such pathologies is *flatness*: we require that Z be flat over S. Flatness ensures that the Hilbert polynomial $P_{Z_s}(m) = \chi(Z_s, \mathcal{O}_{Z_s}(m))$ is constant as s varies over connected components of S. This is the correct algebraic analogue of "continuous variation": the members of the family all share the same discrete numerical invariants.

The reader who has not yet encountered flatness in detail should think of it as a homological condition: a morphism $\pi : Z \to S$ is flat if the fibers Z_s do not "jump" in any cohomological sense. A thorough treatment will be given in section 6.8.

The Hilbert functor and Grothendieck's theorem

Fix a projective scheme $Y \subseteq \mathbb{P}^n_k$ and a numerical polynomial $p \in \mathbb{Q}[t]$ (recall that the Hilbert polynomial of any coherent sheaf on projective space is a polynomial with rational coefficients). The *Hilbert functor* is the contravariant functor $\mathcal{H}ilb^p_Y : \mathbf{Sch}^{\mathrm{op}}_k \to \mathbf{Set}$ defined by

$$\mathcal{H}ilb^p_Y(S) = \left\{ Z \subseteq Y \times_k S \;\middle|\; \begin{array}{l} Z \text{ is a closed subscheme, flat over } S, \\ \text{with Hilbert polynomial } p \text{ on each fiber} \end{array} \right\}.$$

For a morphism $f : S' \to S$, the induced map $\mathcal{H}ilb^p_Y(S) \to \mathcal{H}ilb^p_Y(S')$ sends a family $Z \subseteq Y \times S$ to its pullback $Z \times_S S' \subseteq Y \times S'$.

Theorem 3.3.10: Grothendieck's Existence Theorem

Let $Y \subseteq \mathbb{P}^n_k$ be a projective scheme over a Noetherian ring k, and let $p \in \mathbb{Q}[t]$ be a numerical polynomial. Then the Hilbert functor $\mathcal{H}ilb^p_Y$ is representable by a projective scheme Hilb^p_Y over k:

$$\mathrm{Hom}_{\mathbf{Sch}_k}(S, \mathrm{Hilb}^p_Y) \;\cong\; \mathcal{H}ilb^p_Y(S)$$

naturally in S.

The proof is beyond our scope (it relies on a careful analysis of Castelnuovo–Mumford regularity and the Grassmannian embedding), but the statement is worth absorbing. It says that the totality of all subschemes of Y with a given Hilbert polynomial is itself a geometric object—a projective scheme—on which one can do geometry. There is a *universal family*: a closed subscheme $\mathcal{Z} \subseteq Y \times \mathrm{Hilb}^p_Y$, flat over Hilb^p_Y, such that every flat family $Z \subseteq Y \times S$ with Hilbert polynomial p is the pullback of $\mathcal{Z}$ along a unique morphism $S \to \mathrm{Hilb}^p_Y$.

Example: the Hilbert scheme of points

The simplest interesting case arises when p is the constant polynomial $p(m) = n$ for a positive integer n. Then a closed subscheme $Z \subseteq Y$ with Hilbert polynomial $p(m) = n$ is a zero-dimensional subscheme of length n: it is supported on at most n points of Y, possibly with multiplicity (i.e., with nilpotent structure). The corresponding Hilbert scheme is denoted $\mathrm{Hilb}^n(Y)$ and is called the *Hilbert scheme of n points* on Y.

When Y is a smooth surface, $\mathrm{Hilb}^n(Y)$ turns out to be a smooth variety of dimension $2n$. This is a nontrivial theorem (due to Fogarty) and stands in contrast to the naive expectation: the "symmetric product" $\mathrm{Sym}^n(Y) = Y^n/S_n$, which parametrizes unordered n-tuples of points, is singular along the diagonals where points collide. The Hilbert scheme resolves these singularities by remembering the infinitesimal directions in which colliding points approach each other.

To see this concretely, consider $Y = \mathbb{A}^2_k$ and $n = 2$. A length-2 subscheme $Z \subseteq \mathbb{A}^2$ is one of two types:

1. Two distinct points $\{p, q\}$, corresponding to the ideal $I_p \cap I_q \subseteq k[x, y]$ where I_p and I_q are the maximal ideals of p and q. The quotient $k[x, y]/I$ has dimension 2 as a k-vector space.

2. A single point p with a tangent direction $v \in T_p\mathbb{A}^2$. If $p = (0, 0)$ and v points in the x-direction, the corresponding ideal is (y, x^2), and $k[x, y]/(y, x^2) \cong k[x]/(x^2)$ has dimension 2. The extra nilpotent direction records "how the two points came together."

Thus $\mathrm{Hilb}^2(\mathbb{A}^2)$ parametrizes pairs of distinct points together with a "blow-up" of the diagonal: at each point of the diagonal, the fiber is a $\mathbb{P}^1$ recording the tangent direction of approach. This is precisely the blow-up of $\mathrm{Sym}^2(\mathbb{A}^2)$ along its singular locus.

Tangent spaces to the Hilbert scheme

The functor-of-points perspective makes the computation of tangent spaces to moduli spaces almost mechanical. By Example 3.3.6, a tangent vector to Hilb^p_Y at a point $[Z]$ (corresponding to a subscheme $Z \subseteq Y$) is a morphism $\mathrm{Spec}\,(k[\epsilon]/(\epsilon^2)) \to \mathrm{Hilb}^p_Y$ sending the closed point to $[Z]$. By the universal property, this is the same as a flat family of subschemes of Y over $\mathrm{Spec}\,(k[\epsilon]/(\epsilon^2))$ whose special fiber is Z. One can show that such a family corresponds to a first-order deformation of Z inside Y, and that:

$$\Theta_{\mathrm{Hilb}^p_Y,\,[Z]} \;\cong\; \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{I}_Z,\, \mathcal{O}_Z)$$

where $\mathcal{I}_Z$ is the ideal sheaf of Z in Y. This identification—tangent vectors to the Hilbert scheme are the same as homomorphisms from the ideal sheaf to the structure sheaf of the subscheme—is a prototype for the general principle that tangent spaces to moduli spaces are computed by Hom or Ext groups.

3.3.7 Schemes as Functors

The reader may have noticed an apparent circularity in the functor-of-points perspective. We define the functor $h_X : \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ using the category $\mathbf{Sch}$ of schemes, but if the functor of points is to serve as a *definition* of schemes, we seem to need the category $\mathbf{Sch}$ already in hand.

The resolution, due to Grothendieck and developed systematically by Demazure and Gabriel, proceeds in stages and is worth spelling out.

Step 1: Start with rings. The category **CRing** of commutative rings is purely algebraic. We define the category of *affine schemes* as $\mathbf{Aff} = \mathbf{CRing}^{\mathrm{op}}$, the formal opposite category. No geometry has been invoked.

Step 2: Embed into functors. Any functor $F : \mathbf{CRing} \to \mathbf{Set}$ (equivalently, a presheaf on **Aff**) can be thought of as a "generalized space." An affine scheme $\mathrm{Spec}\,(A)$ defines such a functor by $h_{\mathrm{Spec}\,(A)}(R) = \mathrm{Hom}_{\mathbf{CRing}}(A, R)$. By the Yoneda Lemma, the assignment $\mathrm{Spec}\,(A) \mapsto h_{\mathrm{Spec}\,(A)}$ is a fully faithful embedding $\mathbf{Aff} \hookrightarrow \mathrm{Fun}(\mathbf{CRing}, \mathbf{Set})$.

Step 3: Impose a sheaf condition. Not every functor $\mathbf{CRing} \to \mathbf{Set}$ deserves to be called geometric. We require that F be a *sheaf* for the Zariski topology. Concretely, if $f_1, \dots, f_n$ generate the unit ideal in a ring A (so that $\mathrm{Spec}\,(A) = \bigcup_i D(f_i)$ is a cover by distinguished open sets), then the diagram

$$F(A) \longrightarrow \prod_i F(A_{f_i}) \rightrightarrows \prod_{i,j} F(A_{f_i f_j})$$

should be an equalizer. This is the local-to-global property that geometric objects possess: compatible data on an open cover should glue uniquely. A sheaf satisfying this property is called a *Zariski sheaf.*

Step 4: Require an open cover by affines. A *scheme* is a Zariski sheaf F that admits an open cover by representable subfunctors. That is, there exist rings A_i and natural transformations $h_{\mathrm{Spec}\,(A_i)} \hookrightarrow F$ that are "open immersions" (in the sense defined below) and that jointly cover F.

This definition recovers the usual category of schemes without circularity: we begin with rings, pass to representable functors (affine schemes), impose a sheaf condition (gluing), and characterize schemes as those sheaves admitting an affine open cover. The result is a full subcategory of $\mathrm{Fun}(\mathbf{CRing}, \mathbf{Set})$ defined by intrinsic conditions.

The practical payoff is significant. To define a scheme, it suffices to specify a functor $\mathcal{F} : \mathbf{CRing} \to \mathbf{Set}$ and verify the Zariski sheaf condition and the existence of an affine open cover. This is often considerably easier than constructing a locally ringed space by hand, especially for moduli problems: the Hilbert scheme was originally constructed by Grothendieck precisely by first defining the functor $\mathcal{H}ilb_Y^p$ and then proving it satisfies the above conditions. The same strategy applies to Picard schemes, moduli of curves, and many other classifying spaces that arise in algebraic geometry.

Open subfunctors

Step 4 above relies on a notion of "open immersion" for functors, which we now make precise. Let $F : \mathbf{CRing} \to \mathbf{Set}$ be a Zariski sheaf. A subfunctor $G \subseteq F$ (meaning $G(A) \subseteq F(A)$ for every ring A, compatibly with restriction maps) is called an *open subfunctor* if the following condition holds: for every ring A and every element $\xi \in F(A)$, there exist elements $f_1, \dots, f_n \in A$ generating the unit ideal such that, for each i, the image of ξ in $F(A_{f_i})$ lies in $G(A_{f_i})$.

Informally, G is open in F if membership in G can be "checked locally." To see why this is the right definition, consider the representable case. If $F = h_{\mathrm{Spec}\,(B)}$ and G corresponds to an open subscheme $U \subseteq \mathrm{Spec}\,(B)$, then a morphism $\xi : \mathrm{Spec}\,(A) \to \mathrm{Spec}\,(B)$ factors through U if and only if its image lands in U. The preimage $\xi^{-1}(U)$ is an open subset of $\mathrm{Spec}\,(A)$, and the condition that $f_1, \dots, f_n$ generate the unit ideal with $\xi|_{D(f_i)}$ landing in U says exactly that $\xi^{-1}(U)$ covers $\mathrm{Spec}\,(A)$—i.e., ξ factors through U.

A collection of open subfunctors $\{G_i \subseteq F\}$ is said to *cover* F if for every field K, the sets $G_i(K)$ jointly cover $F(K)$. The condition in Step 4 can now be restated: a Zariski sheaf F is a scheme if it admits a cover by open subfunctors, each of which is representable by an affine scheme.

Tangent spaces to functors

One of the practical advantages of the functorial viewpoint is that many geometric constructions can be performed directly on functors, without first knowing that the functor is representable. We have already seen in Example 3.3.6 that the tangent space to a scheme X at a k-rational point x is recovered by probing with the dual numbers $D = k[\epsilon]/(\epsilon^2)$. This construction generalizes immediately to arbitrary functors.

Definition 3.3.11: Tangent Space of a Functor

Let $F : \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ be a functor and let $\xi_0 \in F(\mathrm{Spec}\,(k))$ be a k-rational point of F. The *tangent space* of F at ξ_0 is

$$T_{\xi_0}F := \left\{\xi \in F(\mathrm{Spec}\,(k[\epsilon]/(\epsilon^2))) \;\middle|\; \xi \text{ maps to } \xi_0 \text{ under } F(\mathrm{Spec}\,(k) \hookrightarrow \mathrm{Spec}\,(D))\right\}.$$

When $F = h_X$ for a scheme X and ξ_0 corresponds to a closed point $x \in X$ with $\kappa(x) = k$, this recovers the Zariski tangent space $\Theta_{X,x}$ by Example 3.3.6. The key point is that the definition makes sense even when F is not representable.

For the tangent space to carry a k-vector space structure, one needs the functor F to satisfy a mild condition: $T_{\xi_0}F$ should be compatible with the k-module structure on square-zero extensions of k. This is automatic for representable functors, and holds for most functors arising in practice (including the Hilbert functor and the Picard functor).

As an application, we computed the tangent space of the Hilbert functor at the end of section 3.3.6: for a subscheme $Z \subseteq Y$ with ideal sheaf $\mathcal{I}_Z$, the tangent space $T_{[Z]}\,\mathrm{Hilb}_Y^p \cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{I}_Z, \mathcal{O}_Z)$. This computation was carried out purely at the level of the functor $\mathcal{H}ilb_Y^p$; the fact that Hilb_Y^p is actually a scheme (Theorem 3.3.10) was not needed for the computation itself, only for the conclusion that this tangent space has geometric meaning.

3.3.8 Moduli Problems and the Limits of Representability

The Hilbert scheme demonstrates the power of the functorial approach: define a moduli functor, prove it is representable, and obtain a geometric parameter space for free. A natural question is how far this method extends. Can every reasonable classification problem in algebraic geometry be solved by a representing scheme?

The answer is no, and understanding *why* it fails is as instructive as the successes.

The moduli problem for elliptic curves

An *elliptic curve* over a field k is a smooth projective curve of genus 1 equipped with a distinguished rational point (the reader unfamiliar with this notion may consult section 7.1.7 and return here afterward). Two elliptic curves are isomorphic if and only if they have the same j-invariant, an element of k.

We would like a “moduli space of elliptic curves”: a scheme M whose points correspond to isomorphism classes of elliptic curves, and such that families of elliptic curves over a base S correspond to morphisms

$S \to M$. The functor to be represented is:

$$\mathcal{M}_{1,1}(S) = \left\{ (E \to S,\, \sigma : S \to E) \;\middle|\; \begin{array}{l} E \to S \text{ is a smooth proper morphism} \\ \text{with genus-1 geometric fibers,} \\ \sigma \text{ is a section} \end{array} \right\} /\cong_S$$

where $\cong_S$ denotes isomorphism of families over S. If this functor were representable by a scheme M, there would exist a *universal family*: an elliptic curve $\mathcal{E} \to M$ such that every family $E \to S$ is the pullback of $\mathcal{E}$ along a unique morphism $S \to M$.

Over an algebraically closed field, the isomorphism classes of elliptic curves are classified by the j-invariant, so the underlying set of M should be $\mathbb{A}^1$. Indeed, there is a scheme—the j-line $\mathbb{A}^1_k$—that serves as a *coarse moduli space*: its closed points are in bijection with isomorphism classes, and for any family $E \to S$ there exists a unique morphism $S \to \mathbb{A}^1$ whose fiber over a point s has j-invariant equal to the j-invariant of E_s. But the j-line is *not* a fine moduli space. The functor $\mathcal{M}_{1,1}$ is not representable.

The failure is easy to spot: every elliptic curve E over a field of characteristic $\neq 2$ has a nontrivial automorphism, namely it has at least the inversion map $\iota : E \to E$ sending $P \mapsto -P$ (with respect to the group law). For the curves with $j = 0$ and $j = 1728$, the automorphism group is even larger ($\mathbb{Z}/6\mathbb{Z}$ and $\mathbb{Z}/4\mathbb{Z}$, respectively). These automorphisms are the obstruction to representability.

To see why concretely, suppose a fine moduli space M existed with universal family $\mathcal{E} \to M$. Consider the trivial family $E \times S \to S$ for a fixed elliptic curve E with nontrivial automorphism α. The identity map $\mathrm{id} : E \times S \to E \times S$ and the automorphism $\alpha \times \mathrm{id}_S : E \times S \to E \times S$ are *different* isomorphisms of the family, but they give rise to the *same* classifying morphism $S \to M$ (since they classify the same family up to isomorphism). This violates the requirement that the classifying morphism be unique.

More precisely, the issue is that the functor $\mathcal{M}_{1,1}$ does not satisfy the sheaf condition in sufficient generality: it fails descent for étale covers. A family that is locally trivial in the étale topology need not be globally trivial, and the automorphisms of the fibers obstruct the gluing.

Coarse moduli spaces

The j-line is an example of a *coarse moduli space*: a scheme M equipped with a natural transformation $\mathcal{M}_{1,1} \to h_M$ that is universal among maps to representable functors, and that induces a bijection $\mathcal{M}_{1,1}(\mathrm{Spec}\,(\bar{k})) \xrightarrow{\sim} M(\bar{k})$ on geometric points. A coarse moduli space classifies objects up to isomorphism at the level of points, but does not carry a universal family and does not faithfully track families over general bases.

Beyond schemes: stacks

The correct solution is to enlarge the category of geometric objects. Rather than forcing $\mathcal{M}_{1,1}(S)$ to be a *set* (of isomorphism classes), one retains it as a *groupoid*: the category whose objects are families $E \to S$ and whose morphisms are isomorphisms of families. The resulting structure is an *algebraic stack*, often denoted $\mathcal{M}_{1,1}$ as well. The stack remembers the automorphisms of each object, and this additional data resolves the descent obstruction.

An intermediate notion is that of an *algebraic space*. These arise when the moduli problem has no nontrivial automorphisms but the functor still fails to be representable by a scheme—for instance, because the "correct" topology for gluing is the étale topology rather than the Zariski topology. Algebraic spaces are Zariski-locally schemes but may not be globally so; they are to schemes what manifolds are to open subsets of $\mathbb{R}^n$.

The hierarchy, in increasing generality, is:

$$\text{Schemes} \subset \text{Algebraic Spaces} \subset \text{Algebraic Stacks} \subset \text{Higher Stacks},$$

where each step relaxes a condition on the moduli functor. Schemes require representability by a Zariski sheaf with an affine cover. Algebraic spaces replace the Zariski topology with the étale topology. Algebraic stacks allow the functor to take values in groupoids rather than sets. The development of this hierarchy is beyond our scope, but the reader should be aware that the functor-of-points perspective developed in this section is the gateway to this broader landscape: every step in the hierarchy is formulated in terms of functors and their properties.

3.3.9 *The Fourier–Mukai Transform

The functor-of-points philosophy associates to a scheme X the collection of all morphisms into X. One can push this idea further by asking: what information is carried not by morphisms, but by *sheaves* on products? This line of thought leads to the Fourier–Mukai transform, which can be understood as a "categorified" version of the change-of-basis perspective developed above. This subsection is intended as a brief orientation for the interested reader; the full theory belongs to the realm of derived categories and will not be developed in detail here.

From graphs to correspondences

Given a morphism $f : X \to Y$, its *graph* $\Gamma_f = \{(x, f(x)) \mid x \in X\}$ is a closed subscheme of $X \times Y$. The morphism f is entirely recoverable from Γ_f: project to X (an isomorphism) and then to Y.

A natural generalization is to drop the requirement that the projection to X be an isomorphism. Any closed subscheme $Z \subseteq X \times Y$—or, more generally, any sheaf $\mathcal{P}$ on $X \times Y$—defines a "correspondence" between X and Y. For a point $x \in X$, the fiber $Z_x = Z \cap (\{x\} \times Y)$ is a subscheme of Y, and as x varies, one obtains a family of subschemes of Y parametrized by X. This is precisely the kind of data that Hilbert schemes classify.

A word on derived categories

To formulate the Fourier–Mukai transform precisely, one works not with the abelian category $\mathbf{Coh}(X)$ of coherent sheaves, but with its *bounded derived category* $D^b(X)$. The motivation is that many natural operations on sheaves—tensor products, pushforwards, global sections—are not exact, and their "corrections" (the higher derived functors Tor, $R^i f_*$, H^i) carry essential geometric information. The derived category is a framework in which a sheaf is replaced by a *complex* of sheaves, and these higher derived functors become part of a single, coherent operation.

An object of $D^b(X)$ is a bounded complex

$$\cdots \to \mathcal{F}^{i-1} \to \mathcal{F}^i \to \mathcal{F}^{i+1} \to \cdots$$

of coherent sheaves, considered up to *quasi-isomorphism*: two complexes are identified if they have isomorphic cohomology sheaves. A single sheaf $\mathcal{F}$ is regarded as a complex concentrated in degree zero. The derived category retains the information of the abelian category (via the cohomology functor H^i) but also remembers the "process" of resolution, which is essential for the functoriality that follows.

The reader familiar with the construction of Ext groups via projective or injective resolutions will recognize the underlying idea: in the derived category, one works with the resolutions themselves rather than discarding them after taking cohomology.

The transform

Let X and Y be smooth projective varieties over a field k, and let $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$ denote the projection morphisms. Given a "kernel" $\mathcal{P} \in D^b(X \times Y)$, the *Fourier–Mukai transform* with kernel $\mathcal{P}$ is the functor

$$\Phi_{\mathcal{P}} : D^b(X) \longrightarrow D^b(Y), \qquad \mathcal{F} \longmapsto R\pi_{Y,*}(\mathcal{P} \otimes^L \pi_X^* \mathcal{F}),$$

where π_X^* is the (exact) pullback, $\otimes^L$ is the derived tensor product, and $R\pi_{Y,*}$ is the derived pushforward. In words: pull the sheaf $\mathcal{F}$ back from X to the product, "multiply" by the kernel $\mathcal{P}$, and "integrate" (push forward) along Y.

The analogy with the classical Fourier transform on $\mathbb{R}^n$ is worth making explicit. Recall that the classical transform sends a function f on $\mathbb{R}^n$ to a function $\hat{f}$ on the dual space $(\mathbb{R}^n)^*$ via

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x)\, e^{2\pi i \langle x, \xi \rangle}\, dx.$$

The exponential $e^{2\pi i \langle x, \xi \rangle}$ is a "kernel" on the product $\mathbb{R}^n \times (\mathbb{R}^n)^*$; the multiplication by f is the analogue of tensoring with $\pi_X^* \mathcal{F}$; and the integration over $\mathbb{R}^n$ is the analogue of the derived pushforward $R\pi_{Y,*}$. The Fourier–Mukai transform is, in this precise sense, the algebraic geometer's Fourier transform, with coherent sheaves playing the role of functions and the derived pushforward replacing integration.

From delta-points to character-points

The classical Fourier transform admits a particularly illuminating description in terms of two distinguished "bases" for the space of functions on a locally compact abelian group G. The first basis consists of the *delta functions* δ_x for $x \in G$: any function can be written as $f = \int_G f(x)\, \delta_x\, dx$, expressing f in terms of where it is "supported." The second basis consists of the *characters* χ_ξ for $\xi \in \hat{G}$ (the Pontryagin dual group): the Fourier transform rewrites f as $\hat{f} = \int_G f(x)\, \chi_\xi(x)\, dx$, expressing f in terms of its "frequency content." The Fourier transform is, at its core, a change of basis from delta-points to character-points:

$$\widehat{\delta_x} = \chi_x, \qquad \widehat{\chi_\xi} = \delta_\xi.$$

The Fourier–Mukai transform makes this change of basis precise in algebraic geometry, and the natural setting is that of *abelian varieties*. Let A be an abelian variety of dimension g over k (a smooth projective group variety—the higher-dimensional generalization of an elliptic curve). Its *dual abelian variety* $\hat{A} = \mathrm{Pic}^0(A)$ parametrizes the degree-zero line bundles on A: a point $\xi \in \hat{A}$ corresponds to a line bundle L_ξ on A that is algebraically equivalent to zero. The dual $\hat{A}$ is the algebro-geometric analogue of the Pontryagin dual $\hat{G}$.

On the product $A \times \hat{A}$ there is a canonical line bundle $\mathcal{P}$, the *Poincaré bundle*, characterized by the universal property: for each $\xi \in \hat{A}$, the restriction $\mathcal{P}|_{A\times\{\xi\}} \cong L_\xi$. The Poincaré bundle is the algebro-geometric analogue of the exponential kernel $e^{2\pi i\langle x,\xi\rangle}$: it "pairs" a point of A with a character from $\hat{A}$.

The Fourier–Mukai transform $\Phi_{\mathcal{P}} : D^b(A) \to D^b(\hat{A})$ with kernel $\mathcal{P}$ now enacts the change of basis between delta-points and character-points. The two fundamental computations are:

Deltas map to characters. Let $a \in A$ be a closed point and let $k(a)$ denote the skyscraper sheaf at a (the sheaf supported at the single point a with stalk k). The skyscraper $k(a)$ is the algebro-geometric δ-function: it is "concentrated at a" and detects nothing elsewhere. Applying the transform:

$$\Phi_{\mathcal{P}}(k(a)) = R\pi_{\hat{A},*}(\mathcal{P} \otimes^L \pi_A^* k(a)) = R\pi_{\hat{A},*}(\mathcal{P}|_{\{a\}\times\hat{A}}) = \mathcal{P}|_{\{a\}\times\hat{A}}.$$

The restriction $\mathcal{P}|_{\{a\}\times\hat{A}}$ is a line bundle on $\hat{A}$. Under the double-duality identification $A \cong \hat{\hat{A}}$, this is precisely the "character" on $\hat{A}$ associated to the point a—just as the classical Fourier transform sends δ_x to the character χ_x.

Characters map to deltas. Let $\xi \in \hat{A}$ and let $L_\xi = \mathcal{P}|_{A\times\{\xi\}}$ be the corresponding degree-zero line bundle on A. The line bundle L_ξ is the algebro-geometric "character": tensoring with L_ξ defines a "twist" of sheaves on A, analogous to multiplication by χ_ξ. One computes:

$$\Phi_{\mathcal{P}}(L_\xi) \cong k(\xi)[-g]$$

where $k(\xi)$ is the skyscraper sheaf at the point $\xi \in \hat{A}$ and $[-g]$ denotes a cohomological shift by $g = \dim A$. Up to this shift (which accounts for the difference between underived and derived pushforward on a g-dimensional variety), the character L_ξ is sent to the delta-function $k(\xi)$ on the dual—exactly as $\widehat{\chi_\xi} = \delta_\xi$ in the classical setting.

The cohomological shift $[-g]$ is the price of working in the derived category. It appears because the higher direct images $R^i\pi_{\hat{A},*}$ of L_ξ vanish for $i \neq g$ (a consequence of the fact that L_ξ has no global sections for $\xi \neq 0$, and its cohomology is concentrated in degree g). The shift is harmless: it is an automorphism of $D^b(\hat{A})$, and Mukai's theorem accounts for it by asserting that the *composition*

$$D^b(A) \xrightarrow{\Phi_{\mathcal{P}}} D^b(\hat{A}) \xrightarrow{\Phi_{\mathcal{P}^\vee}} D^b(A)$$

is isomorphic to the shift functor $(-1)_A^*[-g]$, where $(-1)_A$ is the inversion on A. This is the precise analogue of the Fourier inversion formula $\hat{\hat{f}}(x) = f(-x)$.

Mukai's theorem and Orlov representability

The computations above are instances of a fundamental result due to Mukai: the Fourier–Mukai transform $\Phi_{\mathcal{P}} : D^b(A) \to D^b(\hat{A})$ is an *equivalence of categories*. This is a strong statement: it says that the derived category of coherent sheaves on an abelian variety is equivalent to that of its dual. Two geometrically distinct varieties—A and $\hat{A}$ need not be isomorphic—are "the same" from the perspective of their sheaf theory.

More broadly, Orlov's representability theorem asserts that for smooth projective varieties, every equivalence $D^b(X) \xrightarrow{\sim} D^b(Y)$ is isomorphic to a Fourier–Mukai transform $\Phi_{\mathcal{P}}$ for some kernel $\mathcal{P} \in D^b(X \times Y)$. This is the categorified Yoneda Lemma: just as a natural transformation $h_X \to h_Y$

is determined by a single element of $\mathrm{Hom}(X, Y)$, a sufficiently nice functor $D^b(X) \to D^b(Y)$ is determined by a single object of $D^b(X \times Y)$.

The analogy with the change-of-basis perspective from section 3.3.3 is clean. The Yoneda embedding probes X by mapping test objects *into* X; the Fourier–Mukai transform probes X by its category of sheaves. The kernel $\mathcal{P}$ on $X \times Y$ mediates between these two descriptions, playing the role of a "change-of-basis matrix" between two categorical coordinate systems. On abelian varieties, this change of basis is literally the passage from delta-points to character-points—the oldest and most fundamental duality in harmonic analysis, now transplanted into algebraic geometry.

3.4 Finiteness and Affine Conditions

As schemes are rather general notions, so to scheme morphisms can be very general. The following notions are guiding principles presented by Vakil on what a good type of morphism should have. This terminology will be very useful in labeling many of the following lemmas an propositions. Let's say that C is a collection of morphisms between schemes. Then:

1. it is *local on target* if
   (a) $\pi : X \to Y$ is in the class, then for any open subset $V \subseteq Y$, the restricted homomorphism $\pi^{-1}(V) \to V$ is in the class
   (b) $\pi : X \to Y$ is a morphism, $\cup_i V_i = Y$ is an open cover where each $\pi^{-1}(V_i) \to V_i$ is in this class, then π is in this class

   Usually we consider affine opens, but it need not be the case.
2. is *local on source* if there is
   (a) $\pi : X \to Y$ is in the class, then for any open subset $U \subseteq X$, the restricted homomorphism $U \to \pi(U)$ is in the class
   (b) $\pi : X \to Y$ is a morphism, $\cup_i U_i = X$ is an open cover where each $U_i \to \pi(U_i)$ is in this class, then π is in this class
3. it is closed under composition
4. it is closed under fibered products. As a special case, is closed under change of basis from an S-scheme to a T-scheme

Finiteness Conditions

Definition 3.4.1: Quasicompact Morphism

Let $\pi : X \to Y$ be a morphism of scheme. Then π is *quasicompact* if every open affine subset $U \subseteq Y$ has quasicompact pre-image, that is $\pi^{-1}(U)$ is quasicompact.

Naturally, Quasicompactness is is closed under composition, and is local on target.

Example 3.4.2: Quasicompact Morphism

We shall in this section show that other types of morphism shall be quasicompact (finite, embeddings, proper, etc). At the moment, we shall give concrete examples:

1. Any scheme morphism of affine schemes is quasicompact, this relies on the use of the distinguished open base: Let $f : \operatorname{Spec} S \to \operatorname{Spec} R$ and $U \subseteq \operatorname{Spec} R$ be quasi-compact. Then $U = \bigcup_i^n D(g_i)$ for appropriate distinguished open sets. Then since $f^{-1}(D(g_i)) = D(f^\#(g_i))$, we have that
$$f^{-1}(U) = \bigcup_i^n D(f^\#(g_i))$$
showing the pre-image is quasi-compact.

2. Let $X = \coprod_i^\infty \operatorname{Spec} k$ and $Y = \operatorname{Spec} k$. Then the natural projection $\pi : X \to Y$ is certainly not quasicompact, the pre-image of $\operatorname{Spec} k$ not being quasicompact.

3. Let X be Noetherian schemes and $f : X \to Y$ a scheme morphism. Then f is quasicompact. This follows from the following topological property: If X is a Noetherian topological space, every subset is Noetherian.

   Hence, any non-quasicompact map *must* have X as a non-Noetherian scheme, and so in practice we shall not often see non-quasicompact schemes.

4. Let $X = \operatorname{Spec} k[x]$ and $\tilde{X}$ be the blow up at every point. Then the natural map $\varphi : \tilde{X} \to X$ is not quasicompact.

For future reference, the following definition is put down (see ref:HERE for its application):

Definition 3.4.3: Quasiseparated Morphism

Let $\pi : X \to Y$ be a morphism of scheme. Then π is *quasiseparated* if every open affine subset $U \subseteq Y$ has quasicompact pre-image.

Example 3.4.4: Quasiseparated Morphism

We shall see examples of such morphisms later.

The next important type of morphism that will always have affine open pre-images:

Definition 3.4.5: Affine Morphism

Let $\pi : X \to Y$ be a morphism of schemes. Then π is *affine* if for every affine open subset $U \subseteq Y$, $\pi^{-1}(U)$ is an affine scheme (interpreted as an open subscheme of X)

Affine morphisms can be thought of as being the closest types of morphisms between schemes that "behave" like morphisms of affine schemes, namely since the pre-image of every affine is affine we can just cover in affine sets and study those without needing to "think" about the gluing structure. Proposition 3.4.10 shall show we only need to check a morphism is affine on a single affine cover, which is not immediately obvious as it is not *a priori* clear that we cannot "gerrymander" a partition to force affine pre-images on the open subsets of the cover.

3.4.6 Advanced Intuition This is for the reader already familiar with all this material. Let $f : X \to Y$ is a morphism. Then f is an affine morphisms if and only if there exists a quasi-coherent sheaf of $\mathcal{O}_Y$-algebras $\mathcal{A}$ such that $X \cong \operatorname{Spec}_Y(\mathcal{A})$ where $\operatorname{Spec}_Y(\mathcal{A})$ is the relative spectrum with respect to S (this is a generalization of X as an S-algebra)

We shall give examples of affine morphisms after we prove some nice results, one important such example shall be that when the domain and codomain are affine, the morphism must be affine. Let us start by showing that if the domain/codomain are not affine, the morphism may not be affine:

Example 3.4.7: Non Affine Morphisms

The prototypical example of a non affine morphism is $\mathbb{P}^1_k \to \operatorname{Spec} k$ (i.e. the structure morphism). Then the pre-image is *not* an (as shown in proposition 2.3.6). Example of a non-affine scheme where the codomain is not affine is the embedding $\mathbb{A}^n_k \to \mathbb{P}^n_k$ for $\mathbb{A}^n_k$ embeds in a choice of U_i ($n = 1$ it is enough to just do $n = 1$ to see why).

Certainly affine morphisms are stable under composition, and are stable under base change. This condition would be nice to be affine-local on target. It takes some build-up:

Lemma 3.4.8: Affine Morphisms and Qcqs

Let $\pi : X \to Y$ be an affine morphism. Then it is quasicompact and quasiseparated.

***Proof*:**

The key is that *affine schemes* are quasi-separated (and quasicompact). By the definition of an affine morphism, for every affine open subset $V \subseteq Y$, the preimage $U = \pi^{-1}(V)$ is an affine scheme, say $U \cong \operatorname{Spec} A$.

1. *Quasicompactness:* Every affine scheme is quasicompact. Since the preimage of every affine open $V \subseteq Y$ is quasicompact, the morphism π is quasicompact.

2. *Quasiseparatedness:* A morphism is quasiseparated if the preimage of every affine open in Y is a quasiseparated scheme. Since $U = \operatorname{Spec} A$ is an affine scheme, it is separated (the diagonal $\Delta : U \to U \times_{\mathbb{Z}} U$ is a closed immersion), and every separated scheme is quasiseparated. Therefore, π is quasiseparated.

Lemma 3.4.9: Qcqs Lemma

Let X be a quasicompact and quasiseparated scheme and $s \in \mathcal{O}_X(X)$. Then

$$\mathcal{O}_X(X)_s \cong \mathcal{O}_X(X_s)$$

This theorem can be thought of as the generalization of the result where $X = \operatorname{Spec}(R)$, we we already know that:

$$\mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R))_s \cong, \mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R)_s) = \mathcal{O}_{\operatorname{Spec}(R)}(D(s))$$

***Proof*:**
As X is quasicompact, cover it with finitely many affine open sets $U_i = \operatorname{Spec} R_i$. take $U_{ij} = U_i \cap U_j$. Then the following is exact

$$0 \to \mathcal{O}_X(X) \to \prod_i R_i \to \prod_{i,j} \mathcal{O}_X(U_{ij})$$

As X is quasiseparated, each U_{ij} is covered by finitely many affine open sets $U_{ijk} = \operatorname{Spec}(R_{ij})$ so that the following is exact:

$$0 \to \mathcal{O}_X(X) \to \prod_i R_i \to \prod_{(i,j),k} R_{ijk}$$

Now, localization (being an exact functor) gives:

$$0 \to \mathcal{O}_X(X)_s \to \left(\prod_i R_i\right)_s \to \left(\prod_{(i,j),k} R_{ijk}\right)_s$$

which is still exact. As localization commutes with finite products (importance on finiteness), we get for appropriates values:

$$0 \to \mathcal{O}_X(X)_s \to \prod_i (R_i)_{s_i} \to \prod_{i,j,k} (R_{ijk})_{s_{ijk}}$$

which again is still exact, where the s_i and s_{ijk} is is the restriction to the rings R_i and R_{ijk}. Now, the scheme X_s can be covered by the affine open sets $\operatorname{Spec}(R_i)_{s_i}$ and $\operatorname{Spec}(R_i)_{s_i} \cap \operatorname{Spec}(R_{ijk})_{s_{ijk}}$ giving almost the same exact sequence:

$$0 \to \mathcal{O}_X(X_s) \to \prod_i (R_i)_{s_i} \to \prod_{i,j,k} (R_{ijk})_{s_{ijk}}$$

Hence, the kernel of the map

$$\mathcal{O}_X(X)_s \to \mathcal{O}_X(X_s)$$

is the same, and we get an isomorphism.

Proposition 3.4.10: Affine Morphisms Are Affine-local On Target

Let $\pi : X \to Y$ be a scheme morphism. Then π is an affine morphism if and only if there is a affine open cover $\cup_i U_i = Y$ such that $\pi^{-1}(U_i)$ is affine.

***Proof*:**
We shall want to apply the affine communication lemma. For that, we must show the conditions are satisfied. First, suppose $\pi : X \to Y$ is affine over $\operatorname{Spec}(S)$ so that $\pi^{-1}(\operatorname{Spec}(S)) = \operatorname{Spec}(R)$. Then notice that for any $s \in S$:

$$\pi^{-1}(\operatorname{Spec}(S_s)) = \operatorname{Spec}(R)_{\pi^\# s}$$

showing the first condition. For the next, suppose that $\pi : X \to \operatorname{Spec}(S)$ and $(s_1, ..., s_n)$ with

$X_{\pi^\# s_i} = \text{Spec}\,(R_i)$. We'll show that X is affine as well. To that end, let $R = \mathcal{O}_X(X)$. Then $X \to \text{Spec}\,(S)$ factors through $\alpha : X \to \text{Spec}\,(R)$ (given by the isomorphism $R \to \mathcal{O}_X(X)$), giving us:

$$\begin{tikzcd} \bigcup_i X_{\pi^\# s_i} \arrow[rr, "\alpha"] \arrow[dr, "\pi"'] & & \text{Spec}\,(R) \arrow[dl, "\beta"] \\ & \bigcup_i D(s_i) = \text{Spec}\,(S) & \end{tikzcd}$$

Then, as we are working with scheme morphisms, the pre-image of a distinguished open set is a distinguished open set and so

$$\beta^{-1}(D(s_i)) = D(\beta^\#(s_i)) \cong \text{Spec}\, R_{\beta^\# s_i}$$

and $\pi^{-1}(D(s_i)) = \text{Spec}\,(R_i)$. Now, as X is quasicompact and quasiseparated, by the Qcqs lemma we have the isomorphism:

$$A_{\beta^\# s_i} = \mathcal{O}_X(X)_{\beta^\# s_i} = \mathcal{O}_X(X_{\beta^\# s_i}) = R_i$$

Hence, α induces an isomorphism $\text{Spec}\,(R_i) \cong \text{Spec}\,(R_{\beta^\# s_i})$, and so α is an isomorphism over each $\text{Spec}\,(R_{\beta^\# s_i})$, which covers $\text{Spec}\,(R)$, and hence α is an isomorphism, and so $X \cong \text{Spec}\,(R)$, completing the proof.

From this, we can conclude that if the domain/codomain are affine, the morphisms must be affine

Corollary 3.4.11: Morphism Between Affine is Affine

Let $f : \text{Spec}\, R \to \text{Spec}\, S$ be a morphism of schemes. Then f is an affine morphism

***Proof*:**
By proposition 3.4.10, it suffices to prove it on an affine open cover of the codomain. Then naturally, take an open cover by distinguished open sets, $D(g_i)$ whose pre-image $D(f^\#(g_i))$ which is an affine scheme.

Corollary 3.4.12: Hypersurfaces and Affine Morphissm

Let $X \subseteq \text{Spec}\,(R)$ be a closed subset which is locally cut out by one equation (Z is covered by open subsets which are each given by the quotient of one equation). Then the complement of Z, Y, is affine

Note that in the global case, $Y = \text{Spec}\,(R_f)$, but Y need not be of this form for the local case.

***Proof*:**
exercise.

Example 3.4.13: Affine Morphisms

1. Take $\mathbb{A}^2_k \to \mathbb{P}^1_k$ given by $(a, b) \mapsto [a : b]$ which was shown to be a scheme morphism in example 3.1.11 (uhm, this is obviously not a morphism, where does the origin map too...

Should probably take the blow-up). This is an affine morphism: take the usual open cover U_0, U_1. Then:

$$f^{-1}(U_0) \cong \operatorname{Spec} k[x,y]_x \qquad \text{and} \qquad f^{-1}(U_1) \cong \operatorname{Spec} k[x,y]_y$$

hence by proposition 3.4.10 it is affine.

2. More generally let $\iota : X \to Y$ be either an open or closed immersion. Show that ι is an affine morphism.

The next two types of morphisms are affine morphisms where the pre-image has some extra properties. The next one can be thought of as the generalization of finite integral extensions (or more simply a finite extension or finite ring homomorphism):

Definition 3.4.14: Finite Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *finite* if for every affine open set $\operatorname{Spec}(S) \subseteq Y$, $\pi^{-1}(\operatorname{Spec}(S))$ is the spectrum of a S-algebra that is finitely generated as a S-module, i.e. it is a finite S-algebra.

Note how being a finitely generated S-module is a rather stronger condition than being a finitely generated S-algebra: it forces each element to be integral and "globally bounded" (since R is a finitely generated S-module, there can't be elements of arbitrary order). The reader should see that the composition of two finite morphisms is finite (see exercise ref:HERE)

Lemma 3.4.15: Algebra Build-up For Local On Target

Let $R \to S$ be a ring map, and $(f_1, ..., f_n) = R$ for appropriate $f_i \in R$. Then:

1. If $R_{f_i} \to S_{f_i}$ is integral, $R \to S$ is integral
2. If $R_{f_i} \to S_{f_i}$ is finite, $R \to S$ is finite

***Proof*:**

1. Take $s \in S$, and consider the ideal

$$(p \in R[x] \,|\, p(s) = 0) \subseteq R[x]$$

Then take the subset $J \subseteq R$ of the leading coefficients of the elements in I. This is in fact an ideal, as the reader could verify or see [11, section 20.5]. By clearing denominators, we get that:

$$f_i^{n_i} \in J$$

Hence, as shown in [11, section 9.6], $1 \in J$

2. The above shows that $R \to S$ is integral, for the finiteness note that we may choose a maximal N_i as the degree's of the generators are globally bounded.

Proposition 3.4.16: Finite Morphism Affine Local On Target

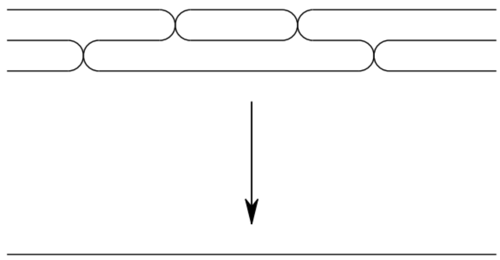
Let $f : X \to Y$ be a scheme morphism. Then it is finite if there is an affine cover $\bigcup_i \operatorname{Spec}(S_i) = Y$ such that $f^{-1}(\operatorname{Spec}(S_i))$ is the spectrum of a finite S_i-algebra.

***Proof*:**

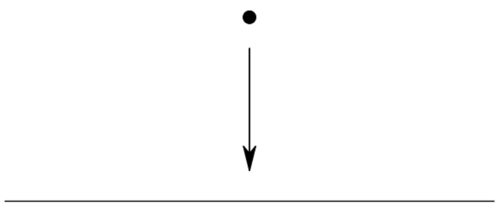
If f is finite, the result is *a fortiori* true, for the converse take $\operatorname{Spec} S_i \subseteq Y$, cover it in distinguished open sets, and apply corollary 3.1.15 and lemma 3.4.15.

Example 3.4.17: Finite Morphism

1. If L/K is a finite extension, $\operatorname{Spec}(K) \to \operatorname{Spec}(L)$ is a finite morphism.
2. Take the map $\operatorname{Spec} k[t] \to \operatorname{Spec} k[u]$ given by $u \mapsto p(t)$ for some degree n polynomial $p(t) \in k[t]$. Then this map is finite: notice that $k[t]$ is generated as a $k[u]$-module by $1, t, ..., t^{n-1}$. This can be visualized as:



3. Let $I \subseteq R$ be an ideal and take the natural schemes and map $\operatorname{Spec}(R/I) \to \operatorname{Spec}(R)$. Then this is a finite morphism, as R/I is generated as an R-module by $1 \in R/I$. This can be visualized as:



0

4. More generally, all closed immersions are finite morphisms.

5. Let $f \in R[x]$ be a polynomial whose leading coefficient is *not* a unit. Show that

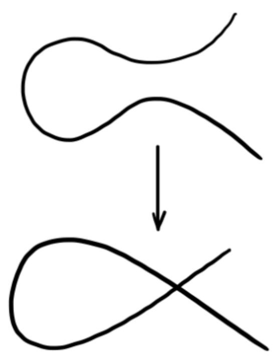
$$\operatorname{Spec} \frac{R[x]}{(f)} \to \operatorname{Spec} R$$

is *not* a finite morphism. Show that it *is* a finite morphism if the leading coefficient is a unit, making this an iff.

6. Take $\operatorname{Spec} k[t] \to \operatorname{Spec} \frac{k[x,y]}{(y^2-x^2-x^3)}$ corresponding to the ring map given by:

$$x \mapsto t^2 - 1 \qquad y \to t^3 - t$$

Namely, this shows this curve is rational. This is a finite morphism, since $k[t]$ is generated as a $\frac{k[x,y]}{(y^2-x^2-x^3)}$-module by 1 and t. This can be visualized as:



As the reader may have noticed, there is only one point that is 2-to-1. The reader should check that this will induce an isomorphism between $D(t^2-1)$ and $D(x)$, i.e. if the node is removed there is an isomorphism. This shall come up many times as an example of normalizing a curve.

7. Let $X \to \operatorname{Spec} k$ be a scheme morphism. Show that if the morphism if finite, X is the finite union of points with the discrete topology, each points having as residue field a finite extension of k. This can be visualized as:

Notice that all the fibers were finite. This is no happenstance of the chosen examples:

Proposition 3.4.18: Finite Morphism is Finite-to-one

Let $\pi : X \to Y$ be a finite morphism. Then it's fibers are finite

***Proof*:**
Since the property of the fiber is local on the target, we may assume $Y = \operatorname{Spec} B$ is affine. By the definition of a finite morphism, X is affine, say $X = \operatorname{Spec} A$, and the induced ring map $B \to A$ makes A a *finite* B-module (finitely generated as a module).

Let $y \in Y$ be a point corresponding to the prime ideal $\mathfrak{p} \subseteq B$. The fiber over y, denoted X_y, is given by the spectrum of the fiber ring:

$$A_y = A \otimes_B \kappa(y)$$

where $\kappa(y)$ is the residue field at $\mathfrak{p}$. Since A is generated by finitely many elements as a B-module, the tensor product A_y is generated by the images of those elements as a vector space over $\kappa(y)$. Thus, A_y is a finite-dimensional $\kappa(y)$-algebra.

Any finite-dimensional algebra over a field is an Artinian ring. A basic property of Artinian rings is that they have finitely many prime ideals (in fact, finitely many points, all of which are closed). Therefore, $\operatorname{Spec}(A_y)$ consists of finitely many points.

The converse is not the case

Example 3.4.19: Finite Fibers But Not Finite Morphisms

The open embedding $\mathbb{A}^2_k \setminus \{(0,0)\} \hookrightarrow \mathbb{A}^2_k$ has finite fibered (they are one-to-one), but it is not affine (as $\mathbb{A}^2_k \setminus \{(0,0)\}$ is not affine).

It also possible to have finite fibers and affine, but not finite: Take $\mathbb{A}^1_{\mathbb{C}} \setminus \{0\} \hookrightarrow \mathbb{A}^1_{\mathbb{C}}$ (intuitively, localizing is "changing perspectives" to the infinitesimal).

It looks important to add 7.3.I to get a grasp on finite morphisms to affine spectrums relation to projective scheme

Definition 3.4.20: Integral Morphism

Let $\pi : X \to Y$ be a scheme morphism. The nit is an *integral morphism* if π is affine, and for every affine open subset $\operatorname{Spec}(S) \subseteq Y$, where $\pi^{-1}(\operatorname{Spec}(S)) = \operatorname{Spec}(R)$, the induced map $S \to R$ is an integral extension.

As integral extensions must be finitely generated over the source ring, an finite morphism is a integral morphism, however an integral morphism need not be finite (think $\operatorname{Spec}\overline{\mathbb{Q}} \to \operatorname{Spec}\mathbb{Q}$). It is further easy to verify that the composition of integral maps is integral.

Proposition 3.4.21: Integral Morphisms Affine-local On Target

Let $\pi : X \to Y$ be an scheme morphism. Then π is an integral morphism if and only if there is a affine open cover $\cup_i U_i = Y$ such that we get integral extension for each pre-image.

***Proof*:**
exercise.

Proposition 3.4.22: Integral Morphism Are Closed Maps

Let $\pi : X \to Y$ be a Integral morphism. Then it is a closed map: it maps closed subsets to closed subsets

As a consequence, finite morphisms are closed

***Proof*:**
idea: reduce first to ring map., and consider the induced map. Now take $I \subseteq R$ and $J = (\pi^{\#})^{-1}(I)$ and consider the integral extension $S/J \subseteq A/I$. Apply the Lying Over theorem.

Definition 3.4.23: Locally Finite Type Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *locally of finite type* if for every affine open subset $\operatorname{Spec}(S) \subseteq Y$, and every affine open subset $\operatorname{Spec}(R) \subseteq \pi^{-1}(\operatorname{Spec}(R))$, the induced morphism $S \to R$ makes R a finitely generated S-algebra. Equivalently, $\pi^{-1}(\operatorname{Spec}(S))$ can be covered by affine open subsets $\operatorname{Spec}(R_i)$ where each R_i is a finitely generated S-algebra.

Definition 3.4.24: Finite Type Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *of finite type* if it is locally of finite type and quasicompact.

Proposition 3.4.25: Locally Finite Type Morphism Affine-local On Target

Let $\pi : X \to Y$ be an scheme morphism. Then π is locally of finite type (resp. of finite type) if there is a cover of Y by affine open sets $\operatorname{Spec}(S_i)$ such that $\pi^{-1}(\operatorname{Spec}(S_i)$ is locally finite type over S_i.

***Proof*:**
exercise.

Example 3.4.26: Finite Type Morphisms

1. All Schemes that are finite type over k are example of finite type scheme morphisms $X \to \operatorname{Spec} k$
2. The map $\mathbb{P}^n_R \to \operatorname{Spec} R$ is of finite type ($\mathbb{P}^n_R$ is covered by $n+1$ open sets)

Proposition 3.4.27: Integral and Finite Type iff Finite

1. finite morphisms are of finite type
2. A morphism is finite if it is integral and of finite type

***Proof*:**
exercise.

Definition 3.4.28: Quasifinite Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *quasifinite* if it is of finite type, and for each $\mathfrak{q} \in Y$, $\pi^{-1}(\mathfrak{q})$ is a finite set.

By proposition proposition 3.4.18 and proposition 3.4.27(1), finite morphisms are quasifinite. The converse is not true (think of $\mathbb{A}^2_k \setminus \{(0,0)\} \to \mathbb{A}^2_k$. The finite type condition is necessary, there exists morphisms with finite fibers that are not quasifinite

Example 3.4.29: Finite Fibers, But Not Quasifinite
The morphism $\operatorname{Spec}(\mathbb{C}(t)) \to \operatorname{Spec}(\mathbb{C})$ has finite fibers, but is not quasifinite. Similarly to $\operatorname{Spec}(\overline{\mathbb{Q}}) \to \operatorname{Spec}(\mathbb{Q})$.

Quasifinite morphisms shall return when discussing Zariski Main Theorem (see ref:HERE).

Vakil promises exercise 9.3.H shows why this would naturally arise

Exercise 3.4.1

1. Let $\pi : X \to Y$ be an open embedding. Then if Y is locally Noetherian, X is locally Noetherian. If Y is Noetherian, X is Noetherian.
2. Let $\pi : X \to Y$ be an open embedding. Show that if Y is quasicompact, X need not be (there is an affine counter-example)
3. Show that the composition of two finite morphism is a finite morphism.
4. Show that being an open embedding is not local on the source.

5. Let B/A be an integral extension. Then show that any ring homomorphism $B \to C$ induces an integral extensions $C \otimes_B A/C$. Conclude that integrality is preserved by base change (see scalar extensions in [**NateClassicalAlgGeo**] if you are unsure what it means to change base).

6. Show that every open embedding is locally of finite type. Hence, every quasicompact open embedding is of finite type.
7. Every open embedding into a locally Noetherian scheme is of finite type
8. The composition of two morphisms locally of finite type is locally of finite type.
9. Let $\pi : X \to Y$ be locally of finite type and Y locally Noetherian. Then X is locally Noetherian. IF $\pi : X \to Y$ is of finite type and Y is Noetherian, then X is Noetherian,
10. Show that quasifinite morphisms to $\operatorname{Spec}(k)$ are finite)
11. Let $k = \overline{\mathbb{F}_p}$. Show that the morphism $F : \mathbb{A}^n_k \to \mathbb{A}^n_k$ of k-schemes corresponding to the bijection $x_i \mapsto x_i^p$ is a bijection, and no power of F is the identity (as sets)
12. (Generalizing the above) Let X be an $\mathbb{F}_p$-scheme. Define an endomorphism $F : X \to X$ such that for each affine open subset $\operatorname{Spec}(R) \subseteq R<$ F corresponds to the map $R \to R$ given by $f \mapsto f^p$. The morphism F is called the *absolute Frobenius morphism.*

3.5 Separated and Proper Schemes

The last two major types of schemes we must cover are *separated schemes* that recover the notion of Hausdroffness for schemes, and *proper schemes* which recover a notion of compactness for schemes. As is the theme in this section, these two properties can be defined purely in terms of scheme morphisms.

From an algebraic-geometric perspective, these conditions allow for better behaviour of closed sets, whether the intersection of two closed sets is closed or whether the image of closed sets is closed is governed by separatedness and properness respectively. Both separatedness and properness can be defined in terms of morphisms, and this is in fact the more practical approach. As closed sets are locally closed set of affine scheme (i.e. vanishing sets of global sections of $\operatorname{Spec} R_i$), the preservation of closedness of these sets under various kinds of mappings is of key interest in scheme theory.

3.5.1 Separated Morphisms

Recall that if $f : X \to Y$ is a continuous map and $K \subseteq X$ is compact, then $f(K)$ is compact. In euclidean topology, this implies closed and bounded maps to closed and bounded, importantly this is a special case of when closed sets maps to closed sets. As affine schemes are quasi-compact (recall we use the term quasi-compact instead of compact due to the differing behaviour without Hausdorffness), if $K \subseteq X$ is closed it is quasi-compact, and hence its image is quasi-compact if $f : X \to Y$ is a scheme morphism (which is continuous). However, such a map *need not be closed*, namely it need not map closed sets to closed sets (recall example ref:HERE). In particular a quasicompact set need not be closed (see [14, section 3.2]).

Another example that shows the failure of separatedness is the following: Take X to be the line with double origin (recall example 2.2.55). Then the two natural inclusions $\iota_1, \iota_2 : \mathbb{A}^1_k \to X$ do not intersect on a closet set!

To recover the good properties of Hausdroffness, the notion of *separatedness* is defined. The idea is that we want to avoid cases where there can be "double points" or even more points that are intuitively arbitrarily close to each other; such points cannot be distinguished, or *separated* by (regular) functions.

In the topological setting, we saw X is Hausdorff if and only if $\Delta \subseteq X \times X$ is closed. In the case of scheme every "diagonal" (when working with fibered products) is a locally closed immersion:

Proposition 3.5.1: Diagonal Locally closed immersion

Let $f : X \to S$ be a morphism of schemes so that X is an S-scheme. Then the diagonal morphism $\delta : X \to X \times_S X$ is a locally closed immersion.

***Proof*:**

We will describe a union of open subsets of $X \times_S X$ covering the image of X, such that the image of X is a closed immersion in this union.

Say S is covered with affine open sets V_i and X is covered with affine open sets U_{ij}, with $f : U_{ij} \to V_i$. Take $W = \bigcup_{i,j} U_{ij} \times_{V_i} U_{ij}$. This is an affine open subscheme of the product $X \times_S X$ (recall this is how the product was constructed). Then the diagonal is covered by these affine open subsets $U_{ij} \times_{V_i} U_{ij}$. Any point $p \in X$ lies in some U_{ij}; then $\delta(p) \in U_{ij} \times_{V_i} U_{ij}$. Note that $\delta^{-1}(U_{ij} \times_{V_i} U_{ij}) = U_{ij}$ since certainly $U_{ij} \subset \delta^{-1}(U_{ij} \times_{V_i} U_{ij})$, and as $\pi_1 \circ \delta = \mathrm{id}_X$ (as π_1 is the first projection), $\delta^{-1}(U_{ij} \times_{V_i} U_{ij}) \subset U_{ij}$. Finally, to check that $U_{ij} \to U_{ij} \times_{V_i} U_{ij}$ is a closed immersion, say $V_i = \operatorname{Spec} B$ and $U_{ij} = \operatorname{Spec} A$. Then this corresponds to the natural ring map $A \otimes_B A \to A$ $(a_1 \otimes a_2 \mapsto a_1 a_2)$, which is certainly surjective, completing the proof.

The key is that these open subsets need not cover $X \times_S X$, hence this does not imply the embedding is closed.

In our case, a subset is closed if it is locally closed in an affine covering, this means that the diagonal can be expressed (locally) as vanishing sets. Furthermore, the product is not well-defined for schemes, as it must always be fibered over some other scheme. This leads to the following:

Definition 3.5.2: Separated Scheme

Let X be a scheme. Then X is *separated* if the map $\delta : X \to X \times_{\mathbb{Z}} X$ is a closed immersion.

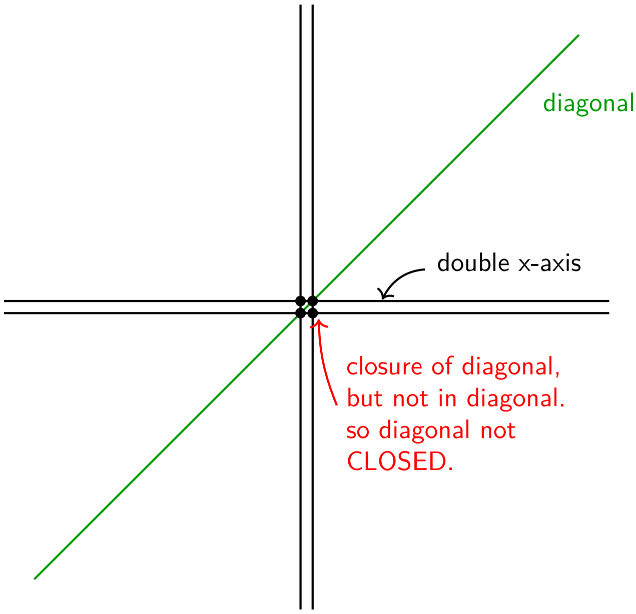
More generally, a map $X \to S$ between schemes is *separated* if the map $\delta : X \to X \times_S X$ is a closed immersion. Denote the image of δ as $\Delta(X)$ or Δ_X.

As a historical footnote, the term scheme used to mean separated scheme back before the 1960s, and the term pre-scheme was used for the modern term scheme.

Note that the first condition is equivalent to saying that the morphism $X \to \operatorname{Spec}(\mathbb{Z})$ is separated. The fact that separatedness is given by the purely topological condition of $\Delta(X)$ being closed shows its closeness to the original topological condition of Hausdorffness.

Example 3.5.3: Separated Morphisms

1. Take the line with two origins over k, label it X. Then when considering $X \times_{\operatorname{Spec} k} X$, we can think to it as in image bellow. The double origin will correspond to 4 points, but the diagonal will only have 2 points. Taking the closure of the diagonal will include all 4 points (show that any open set containing either $(0_1, 0_2)$ or $(0_2, 0_1)$ must intersect $\Delta(X)$, and hence is in the closure; see [14, section 1.5]), showing that it is *not* separated.



Note that $X \to X \times_k X$ is still a locally closed immersion. To show that it is not closed, we can take an open cover that covers *all* of $X \times_k X$ and show that for the affine open W_i containing $P_1 = (0_1, 0_2)$, $\Delta(X) \cap W_i$ is not closed. Indeed, As $P_1 \notin \Delta(X)$, $P_1 \notin \Delta(X) \cap W_i$. It is easy to see that P_1 is in the closure of this set, $P_1 \in \overline{\Delta(X) \cap W_i}$, we see that $\Delta(X) \cap W_i$ is *not* closed.

Note that if we take $X \to X$ to be the identity map, then this *is* a separated morphism, since:
$$X \cong X \times_X X$$
showing the relativeness of separatedness when picking different base schemes.

2. Let $X = \operatorname{Spec}(A) \to \operatorname{Spec}(B) = Y$ be any morphism of schemes between affine schemes. Then this is *always* a separated morphism. Unwind the definitions: when converting into rings we get the map:
$$A \otimes_B A \to A$$
which is surjective, and hence we get:
$$A \cong A \otimes_B A/\mathfrak{a}$$
that is, we are taking the quotient by an ideal, which we know will give us a closed subscheme of $X \times_Y X$ isomorphic to X.

3. More generally, if $X \to Y$ is an affine morphism, it is separated (follows from proposition 3.5.1).

4. The following scheme is a famous example of non-separatedness appearing naturally. Consider $\mathbb{C}^* \curvearrowright \mathbb{A}^2_{\mathbb{C}}$ via $t \cdot (x, y) = (tx, t^{-1}x)$; this is usually sated as "the algebraic torus acts on the affine plane with weights $(1, -1)$". Then taking $(\mathbb{A}^2_{\mathbb{C}} \setminus \{0\})/\mathbb{C}^*$ where the origin is removed as the action is non-free, the resulting space is a non-separated scheme.

   In many cases, non-separated scheme appear naturally in moduli theory.

An important example worth singling out is that of projective scheme:

Lemma 3.5.4: Projective Scheme is Separated

The scheme $\mathbb{P}^1_{\mathbb{Z}}$ is separated.

***Proof*:**
Let $X = \mathbb{P}^1_{\mathbb{Z}}$ and take a standard affine cover $U_0 = \operatorname{Spec} \mathbb{Z}[t]$ and $U_1 = \operatorname{Spec} \mathbb{Z}[u]$, glued via $u = 1/t$. To prove separatedness, we must show $\Delta(X)$ is closed in $X \times_{\mathbb{Z}} X$. It suffices to show the intersection of $\Delta(X)$ with the members of an open cover of the *product* is closed. The product is covered by the four affine patches $U_{ij} = U_i \times U_j$.

We analyze the corresponding ring maps (dual maps) $\Delta^\#$. Recall that a morphism of affine schemes is a closed immersion if and only if its ring dual map is surjective. If surjective, the image corresponds to $V(\ker(\Delta^\#))$.

- *Case 1: The standard charts* (e.g., $W_{00} = U_0 \times U_0$).
  Algebraically, $W_{00} \cong \operatorname{Spec}(\mathbb{Z}[t] \otimes_{\mathbb{Z}} \mathbb{Z}[t]) \cong \operatorname{Spec} \mathbb{Z}[t_1, t_2]$. The diagonal restricted to this chart maps U_0 into $U_0 \times U_0$. The corresponding ring dual map $\Delta^\#$ is:
  $$\varphi : \mathbb{Z}[t_1, t_2] \to \mathbb{Z}[t], \quad \text{defined by } t_1 \mapsto t,\ t_2 \mapsto t.$$
  This map is clearly surjective. By the First Isomorphism Theorem, $\mathbb{Z}[t] \cong \mathbb{Z}[t_1, t_2]/\ker(\varphi)$. The kernel is the ideal generated by $(t_1 - t_2)$. Consequently, the diagonal corresponds to the closed subscheme defined by this ideal, $V(t_1 - t_2)$, representing the line $y = x$.

- *Case 2: The mixed charts* (e.g., $W_{01} = U_0 \times U_1$).
  Algebraically, the target is $W_{01} \cong \operatorname{Spec} \mathbb{Z}[t] \times \operatorname{Spec} \mathbb{Z}[u] \cong \operatorname{Spec} \mathbb{Z}[t, u]$. The source (the part of the diagonal mapping into this chart) is the intersection $U_0 \cap U_1$. This intersection is the locus where $t \neq 0$, which is affine: $\operatorname{Spec} \mathbb{Z}[t, t^{-1}]$. The ring dual map $\Delta^\#$ maps the coordinates of the product to their values on the intersection (determined by the gluing $u = 1/t$):
  $$\psi : \mathbb{Z}[t, u] \to \mathbb{Z}[t, t^{-1}].$$
  Defined by $t \mapsto t$ and $u \mapsto t^{-1}$. Since t and t^{-1} are in the image, ψ is surjective. This implies the geometric map is a closed immersion. The algebraic geometry dictionary tells us the image is defined by the kernel of ψ:
  $$\operatorname{Spec} \mathbb{Z}[t, t^{-1}] \cong \operatorname{Spec}\left(\mathbb{Z}[t, u]/\ker(\psi)\right).$$
  Since $t(t^{-1}) - 1 = 0$ in the codomain, the element $(tu - 1)$ is in the kernel. In fact, $\ker(\psi) = (tu - 1)$. Thus, the image of the diagonal in this chart is the closed subset $V(tu - 1) \subseteq \mathbb{A}^2_{\mathbb{Z}}$, which describes the hyperbola $u = 1/t$.

Since the image is closed in every chart of an open cover of the target $X \times_{\mathbb{Z}} X$, the diagonal map is a closed immersion.

The details for the case of $\mathbb{P}^n_R$ are a bit laborious, so until I feel comfortable with them I am putting this corollary here as a refresher for my future self:

Corollary 3.5.5: Projective Space is Separated

For any ring R, the projective space $\mathbb{P}^n_R$ is separated.

***Proof*:**
Let $X = \mathbb{P}^n_R$. We use the standard affine open cover $\{U_i\}_{i=0}^n$, where $U_i = \{[x_0 : \cdots : x_n] \big| x_i \neq 0\}$. The affine ring for U_i is $R[y_0^{(i)}, \ldots, y_n^{(i)}]$ subject to $y_i^{(i)} = 1$, where the coordinates represent $y_k^{(i)} = x_k/x_i$.

To show the diagonal $\Delta : X \to X \times_R X$ is a closed immersion, we check that its image is closed in the open affine cover of the product given by $\{U_i \times U_j\}_{0 \leq i,j \leq n}$.

Consider a specific chart $W_{ij} = U_i \times U_j$.

- **Algebraic Setup:** The ring corresponding to W_{ij} is the tensor product of the rings for U_i and U_j:
$$A_{ij} = R[y_0^{(i)}, \ldots, y_n^{(i)}] \otimes_R R[z_0^{(j)}, \ldots, z_n^{(j)}]$$
where variables y come from the first factor (U_i) and z from the second (U_j), with $y_i^{(i)} = 1$ and $z_j^{(j)} = 1$.

- **The Diagonal Map:** A point lies in the image of the diagonal in this chart if it is in $U_i \cap U_j$ in both factors. The map on rings $\varphi : A_{ij} \to \mathcal{O}_X(U_i \cap U_j)$ is induced by the coordinate change formulas:
$$z_k^{(j)} = \frac{x_k}{x_j} = \frac{x_k/x_i}{x_j/x_i} = \frac{y_k^{(i)}}{y_j^{(i)}}$$
Thus, the map is defined by $y_k^{(i)} \mapsto y_k^{(i)}$ and $z_k^{(j)} \mapsto y_k^{(i)} \cdot (y_j^{(i)})^{-1}$.

- **The Closed Relations:** We can find the generators of the kernel by clearing denominators. The relations are:
$$z_k^{(j)} y_j^{(i)} - y_k^{(i)} = 0 \quad \text{for all } k = 0, \ldots, n.$$
Note specifically that when $k = i$, since $y_i^{(i)} = 1$, we get the relation $z_i^{(j)} y_j^{(i)} - 1 = 0$. This is the generalized "hyperbola" ($x_{i/j} \cdot x_{j/i} = 1$).

  The ideal generated by these polynomials $J = \langle z_k^{(j)} y_j^{(i)} - y_k^{(i)} \rangle_k$ defines a closed subset of W_{ij}. Since φ is surjective onto the localization of the polynomial ring (image is $U_i \cap U_j$), this closed set is exactly the image of the diagonal.

As the image is closed in every member of an open cover of the product, $\mathbb{P}^n_R$ is separated.

The next concept will not be too useful to us for awhile, and shall come back up when we see algebraic stacks and algebraic spaces, but is useful to define:

Definition 3.5.6: Quasi-separated

Let X be a scheme. Then X is quasi-separated if the image of $X \to X \times X$ is quasi-compact. More generally, a map $X \to S$ between schemes is quasi-separated if the image map of $X \to X \times {}_S X$ is quasi-compact.

Proposition 3.5.7: Separated Implies Quasi-separated

Let X be a separated scheme. Then X is quasi-separated.

***Proof*:**
By definition, X is separated if the diagonal morphism $\Delta : X \to X \times X$ is a closed immersion. By definition, X is quasi-separated if Δ is a quasi-compact morphism. Thus, it suffices to prove that any closed immersion is quasi-compact.

Let $f : Z \to Y$ be a closed immersion. To prove f is quasi-compact, we check the condition on an affine open cover. Let $V \subseteq Y$ be an affine open subset, say $V = \operatorname{Spec}(B)$. Because f is a closed immersion, the restriction $f|_V : f^{-1}(V) \to V$ corresponds to a surjective ring homomorphism $B \to A$ (specifically $A \cong B/I$ for the ideal I cutting out Z). Therefore, the preimage $f^{-1}(V)$ is isomorphic to $\operatorname{Spec}(A)$. Since every affine scheme is quasi-compact, $f^{-1}(V)$ is quasi-compact.

As the preimage of any affine open is quasi-compact, the map f is quasi-compact. Applying this to Δ, we conclude X is quasi-separated.

Separatedness gives us nicer behaviour on closed sets. The following results outline some important examples:

Proposition 3.5.8: Intersection of Separated affine schemes

Let $X \to S$ be a separated scheme morphism where S is affine. Then if $U, V \subseteq X$ are open affine subsets, their intersection

$$U \cap V$$

is an affine open subset.

***Proof*:**
Let $S = \operatorname{Spec} R$, $U = \operatorname{Spec} A$, and $V = \operatorname{Spec} B$.

1. **The product of affines is affine.** Recall that the product $X \times_S X$ is constructed by gluing affine charts. Specifically, the open subset $U \times_S V \subseteq X \times_S X$ is isomorphic to the spectrum of the tensor product:

$$U \times_S V \cong \operatorname{Spec}(A \otimes_R B).$$

   Thus, $U \times_S V$ is an affine open subscheme of the product.

2. **The intersection is a preimage.** Consider the diagonal morphism $\Delta : X \to X \times_S X$.

We observe that the intersection of the open sets is exactly the pullback of their product by the diagonal:

$$\Delta^{-1}(U \times_S V) = \{x \in X \big| \Delta(x) \in U \times_S V\} = \{x \in X \big| x \in U \text{ and } x \in V\} = U \cap V.$$

3. **Separatedness implies affine.** By the definition of separatedness, Δ is a closed immersion. This means that Δ induces a closed immersion of $U \cap V$ into the open subset $U \times_S V$:

$$U \cap V \hookrightarrow U \times_S V.$$

A closed immersion into an affine scheme corresponds to a surjective ring map. Since $U \times_S V$ is affine ($\operatorname{Spec}(A \otimes_R B)$), $U \cap V$ must be isomorphic to $\operatorname{Spec}((A \otimes_R B)/I)$ for some ideal I. Since the spectrum of a quotient ring is an affine scheme, $U \cap V$ is affine.

For a counter-example, taking affine *plane* with double origin we get that taking two affine lines each containing a different origin will intersect to produce the plane without the origin, which in example 2.2.51 is not affine.

Proposition 3.5.9: Graphs Are Closed

Let X be a separated S-schemes and Y any S-scheme. Then the graph of the S-morphism $f : X \to Y$:

$$\Gamma_f = \operatorname{im}((\operatorname{id}_Y, f)Y \to Y \times_S X)$$

is closed

***Proof*:**

1. Show the graph is constructed via the pullback of the diagonal $\Delta_X \subseteq X \times_S X$ along the map $id_Y \times g : Y \times_S X \to X \times_S X$.

2. pullbacks preserve closed immersions (proposition 3.2.5(2)).

For the case where it is not separated as usual let X be the line with two origins (not separated). Let Y be the affine line $\mathbb{A}^1$. Consider the morphism $g : Y \setminus \{0\} \to X$ which is the natural inclusion into the "un-doubled" part of X. What is the graph of this morphism? It's a subset of $(Y \setminus \{0\}) \times_k X$. The key is that this map cannot be extended to the origin of Y in a unique way. Indeed, $0 \in Y$ could map to either $0_1 \in X$ or $0_2 \in X$. This ambiguity means the graph of any potential extension is *not* closed; the point $(0, 0_1)$ and $(0, 0_2)$ are "limit points" of the graph but there is no natural choice of which one is in the graph. The graph is not well-behaved at the boundary. This intuition will in fact be a central way of interpreting separatedness, as shall be explored in the next section.

Proposition 3.5.10: Morphism Determined By Dense Open Subset

Let X be a reduced S-scheme, Z any scheme, and $U \subseteq Z$ a dense open subset. Then if two morphisms $f, g : Z \to X$ agree on U, $f|_U = g|_U$, then they agree everywhere:

$$f|_U = g|_U \qquad \Rightarrow \qquad f = g$$

***Proof*:**
Consider the set of points in Z where f and g agree. The key is this can be described as the preimage of the diagonal of X, namely, take $(g,h) : Z \to X \times_S X$, and:

$$E = (g,h)^{-1}(\Delta_X)$$

Then, we have a closed subset E that contains U (as every point in E are points f, g agree on), and as U is open and dense, it must be that $E = Z$, completing the proof.

3.5.2 Valuation Criterion for Separatedness

We shall next characterize separated schemes and morphisms in using valuation rings. This will be analogous to the fact that in a Hausdorff space, the limit of a sequence has at most 1 point. From this result, think of a continuous map $(0,1) \to X$. If we want to extend this to a map $[0,1] \to X$, if X is Hausdorff, there is only one possible extension. If X wasn't, say it had two origins, then there can be more than one map that can extend (the new 0 point will have two possible places to map to).

Let us now extend this idea to a map $X \to Y$, and we have an embedding $(0,1) \to X$ an identity map $(0,1) \to [0,1]$ and an embedding $(0,1] \to Y$ where we are only adding one point. Then there should only be one lift. Note that this lift is not dependent on X and Y. Let's say Y was the double-origin scheme and X was the double y-axis scheme. Then the embedding $(0,1)$ can be extended to two way into Y, but that will correspond to a unique lift.

With this intuition in mind, let us now look at what is the equivalent of the open interval for schemes. This would be a Discrete valuation ring (DVR) R. It has only two points. We can think of as $\operatorname{Spec}(\operatorname{Frac}(R))$ as the open interval and $\operatorname{Spec}(R)$ as adding the closed 0 (notice that the new point in $\operatorname{Spec}(R)$ is also a closed point, while the other point is "open" and is the generic point that can be thought of was closed to everywhere else). Then we have the following situation:

$$\begin{tikzcd}
\operatorname{Spec}(\operatorname{Frac}(R)) \arrow[r] \arrow[d] & Y \arrow[d] \\
\operatorname{Spec}(R) \arrow[r] \arrow[ur, dashed] & X
\end{tikzcd}$$

With the hope that having at most one lift corresponding to the map being separable. This is *sometimes* the case. To prove this, we require two technical lemmas regarding valuation rings and specialization.

Lemma 3.5.11: Lifts from Valuation Rings

Let R be a valuation ring of a field K. Let $T = \operatorname{Spec} R$ and $U = \operatorname{Spec} K$.

1. A morphism $\operatorname{Spec} K \to X$ is equivalent to giving a point $x_1 \in X$ and an inclusion of fields $\kappa(x_1) \subseteq K$.
2. A morphism $\operatorname{Spec} R \to X$ is equivalent to
   - giving two points $x_0, x_1 \in X$, with x_0 a specialization of x_1 (i.e., $x_0 \in \overline{\{x_1\}}$), and
   - an inclusion $\kappa(x_1) \subseteq K$, such that R dominates the local ring $\mathcal{O}_{x_0,Z}$ where $Z = \overline{\{x_1\}}$ with its reduced induced structure.

***Proof*:**

1. U is a single point scheme equipped with the structure sheaf K. To give a morphism $U \to X$ is to give a continuous map (picking a point $x_1 \in X$) and a morphism of sheaves $\mathcal{O}_{X,x_1} \to K$. This is a local homomorphism from a local ring to a field, which is equivalent to an inclusion of the residue field $\kappa(x_1) \subseteq K$.

2. Let $t_0 = \mathfrak{m}_R$ be the closed point of T and $t_1 = (0)$ be the generic point. Given a morphism $T \to X$, let x_0, x_1 be the images of t_0, t_1. Since the closure of t_1 contains t_0, by continuity the closure of x_1 must contain x_0, so x_0 is a specialization of x_1. Since T is reduced, the morphism factors through the reduced induced scheme structure on the closure $Z = \overline{\{x_1\}}$. Algebraically, $\kappa(x_1)$ is the function field of Z. The morphism gives a local homomorphism $\mathcal{O}_{x_0,Z} \to R$. Since the map on function fields is compatible with the inclusion $\kappa(x_1) \subseteq K$, this exactly means that the valuation ring R dominates the local ring $\mathcal{O}_{x_0,Z}$.

   Conversely, given the data of x_0, x_1 and the domination $\mathcal{O}_{x_0,Z} \to R$, we construct the map. The map of topological spaces is determined by the points. The map of sheaves is determined by the ring map $\mathcal{O}_{x_0,Z} \to R$ (since T is local, mapping to the global sections of T is sufficient), which composes with the natural map $\operatorname{Spec} \mathcal{O} \to X$ to give the desired morphism.

To make the conditions of lemma 3.5.11 concrete, let us look at a geometric example where T is a curve germ and X is the affine plane.

Example 3.5.12: Visualizing Valuation Lifts

1. Let $X = \mathbb{A}^2_k = \operatorname{Spec} k[x, y]$. Let $R = k[t]_{(t)}$ be the discrete valuation ring at the origin of the affine line. Then $K = k(t)$. Let $T = \operatorname{Spec} R = \{t_0, t_1\}$ where $t_0 = (t)$ is the closed point and $t_1 = (0)$ is the generic point.

   Consider the morphism $T \to X$ defined by the ring homomorphism:

   $$\varphi^\# : k[x, y] \to k[t]_{(t)}, \quad x \mapsto t, \quad y \mapsto t^2$$

   Geometrically, this represents a curve parameterized by t approaching the origin along the parabola $y = x^2$.

   (a) *Where the generic point maps ($U \to X$):* The generic point t_1 corresponds to the prime ideal $(0) \subset R$. Its image $x_1 \in X$ is the prime ideal $\mathfrak{p} = (\varphi^\#)^{-1}((0))$. Since $y - x^2 \mapsto t^2 - t^2 = 0$, the ideal $(y - x^2)$ is in the kernel. In fact, the kernel is exactly $(y - x^2)$. Thus, x_1 is the *generic point of the parabola* $V(y - x^2)$. The inclusion of fields is the map on function fields:

   $$\kappa(x_1) = \operatorname{Frac}\left(\frac{k[x, y]}{(y - x^2)}\right) \xrightarrow{\cong} k(t) = K$$

   (b) *Where the closed point maps:* The closed point t_0 corresponds to the maximal ideal $(t) \subset R$. Its image $x_0 \in X$ is $(\varphi^\#)^{-1}((t)) = (x, y)$, which is the origin.

   (c) *Specialization:* We see clearly that x_0 (the origin) is in the closure of x_1 (the whole parabola). This satisfies the condition $x_0 \in \overline{\{x_1\}}$.

(d) *Domination:* Let $Z = \overline{\{x_1\}}$ be the parabola. The local ring $\mathcal{O}_{x_0,Z}$ is the ring of functions on the parabola regular at the origin, which is isomorphic to $k[t]_{(t)}$. The map induced by $\varphi^\#$ is the identity $k[t]_{(t)} \to k[t]_{(t)}$, which is a local homomorphism (maps maximal ideal to maximal ideal), satisfying the domination condition.

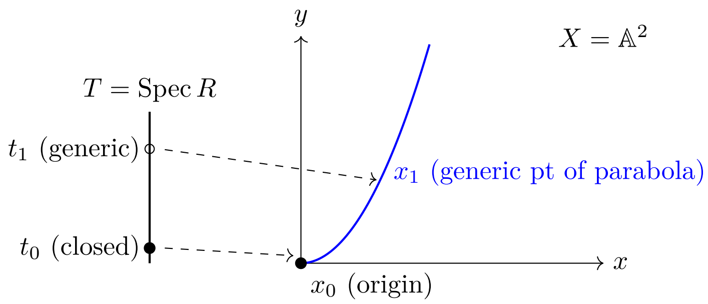


Figure 3.1: The generic point of the DVR maps to the generic point of the curve, while the closed point of the DVR maps to the limit point on the curve.

2. Let us look at a case where the generic point of T maps to the *generic point* of $X = \mathbb{A}^2_k$, rather than a curve inside it. This requires constructing a valuation on the function field $k(x,y)$.

   Let $K = k(x,y)$. We define a valuation $v : K^\times \to \mathbb{Z}$ by the "order of vanishing along the y-axis". Any rational function $f \in k(x,y)$ can be written as $f = x^n \frac{g}{h}$ where x does not divide g or h. We define $v(f) = n$.

   Let R be the valuation ring for v:

$$R = \{f \in k(x,y) \big| v(f) \geq 0\} = k(x,y)_{(x)}$$

   This is the local ring of the affine plane localized at the prime ideal (x) (the generic point of the y-axis).

   Let us analyze the map $T = \operatorname{Spec} R \to X = \operatorname{Spec} k[x,y]$ induced by the natural inclusion $k[x,y] \subset R$.

   (a) *Generic Point:* The generic point t_1 of T (ideal (0) in R) maps to the ideal (0) in $k[x,y]$. Thus, $t_1 \mapsto \eta$, the generic point of the plane.

   (b) *Closed Point:* The closed point t_0 of T (maximal ideal xR) maps to the ideal $(x) \subset k[x,y]$. This is the *generic point of the line* $x = 0$.

   (c) *Visualization:*

$$\begin{tikzcd} U = \operatorname{Spec} k(x,y) \arrow[r] \arrow[d] & \eta \text{ (generic pt of } \mathbb{A}^2) \arrow[d, dashed, "specializes"] \\ T = \operatorname{Spec} k(x,y)_{(x)} \arrow[r] & \xi \text{ (generic pt of } y\text{-axis)} \end{tikzcd}$$

   Even though t_0 is a "closed point" in the topological space T, it maps to a non-closed point in X.

(d) *Domination:* The stalk of X at the image point ξ is $\mathcal{O}_{X,\xi} = k[x,y]_{(x)}$. In this specific case, the valuation ring R is *isomorphic* to the stalk. This happens because we approached a codimension 1 point. If we approached the origin (codimension 2), R would be a further localization/quotient of the stalk.

Lemma 3.5.13: Closure and Specialization

Let $f : X \to Y$ be a quasicompact morphism of schemes (e.g., if X, Y are Noetherian). Then the subset $f(X) \subseteq Y$ is closed if and only if it is stable under specialization.

***Proof*:**
One direction is topological: closed sets are always stable under specialization (if a point is in the set, its closure is in the set). For the converse, assume $f(X)$ is stable under specialization. We want to show $\overline{f(X)} \subseteq f(X)$. We may assume X and Y are reduced, and replace Y with the reduced induced structure on $\overline{f(X)}$ so we want to show f is surjective onto Y. Let $y \in Y$. We can replace Y with an affine neighborhood of y, so assume $Y = \operatorname{Spec} B$. Since f is quasicompact, X is a finite union of affine opens $X_i = \operatorname{Spec} A_i$. Since y is in the closure of the image, $y \in \overline{f(X_i)}$ for some i. Let $Y_i = \overline{f(X_i)}$ with the reduced structure. Then $X_i \to Y_i$ corresponds to an injective ring map $B \to A_i$ (injective because the map is dominant). The point y corresponds to a prime $\mathfrak{p} \subseteq B$. Let $\mathfrak{p}' \subseteq \mathfrak{p}$ be a *minimal* prime ideal of B contained in $\mathfrak{p}$, such a prime exists by Zorn's lemma. This prime $\mathfrak{p}'$ corresponds to a point y' which specializes to y. I claim $y' \in f(X_i)$. Consider the localization $B_{\mathfrak{p}'}$. This is a field since $\mathfrak{p}'$ is minimal in a reduced ring. The map $B_{\mathfrak{p}'} \to A_i \otimes_B B_{\mathfrak{p}'}$ is an inclusion of a field into a non-zero ring as $B \to A_i$ is injective. Thus, the target ring is not the zero ring, so it has a prime ideal $\mathfrak{q}'$. This prime contracts to $\mathfrak{p}'$. Thus, y' is in the image. Since $f(X)$ is stable under specialization and $y' \rightsquigarrow y$, $y \in f(X)$ as we sought to show.

Theorem 3.5.14: Valuation Criterion for Separatedness

Let $f : X \to Y$ be a morphism of schemes, and assume that X is Noetherian. Then f is separated if and only if for any field K, and for any valuation ring R with quotient field K, $T = \operatorname{Spec} R, U = \operatorname{Spec} K$, and the morphism $i : U \to T$ is the morphism induced by the inclusion $R \subseteq K$, given a morphism of T to Y, and given a morphism of U to X, the following diagram commutes:

$$\begin{tikzcd} U \arrow[r] \arrow[d, "\iota"] & X \arrow[d, "f"] \\ T \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd}$$

and there is *at most one* morphism of T to X making the whole diagram commutative.

***Proof*:**
First, suppose f is separated. Let $h, h' : T \to X$ be two morphisms making the diagram commute. We obtain a morphism $L = (h, h') : T \to X \times_Y X$. Since the restrictions of h and h' to U are the same (given by the diagram), the generic point t_1 of T maps to the diagonal $\Delta(X) \subseteq X \times_Y X$. Since f is separated, the diagonal $\Delta(X)$ is a closed immersion, so $\Delta(X)$ is a closed set. The image of T is the closure of the image of the generic point (as $T = \operatorname{Spec} R$ has a generic point

whose closure includes the closed point). Since the generic point lands in the closed set $\Delta(X)$, the entire image $L(T)$ must lie in $\Delta(X)$. Therefore, h and h' coincide everywhere on T, so $h = h'$. The equality of the sheaf maps follows from lemma 3.5.11 as the inclusions into K are identical.

Conversely, suppose the valuation condition holds. To show f is separated, we must show that the diagonal $\Delta : X \to X \times_Y X$ is a closed immersion. Since Δ is a locally closed immersion, it suffices to show the image $\Delta(X)$ is a closed subset of $X \times_Y X$. Since X is Noetherian, the product $X \times_Y X$ is Noetherian (or at least the morphism Δ is quasicompact). By lemma 3.5.13, it suffices to show that $\Delta(X)$ is stable under specialization.

Let $\xi_1 \in \Delta(X)$ and let $\xi_1 \rightsquigarrow \xi_0$ be a specialization in $X \times_Y X$. We must show $\xi_0 \in \Delta(X)$. Let $K = \kappa(\xi_1)$ and let $\mathcal{O}$ be the local ring of ξ_0 on the subscheme $\overline{\{\xi_1\}}$ (with reduced induced structure). By domination properties of valuation rings, there exists a valuation ring R of K which dominates $\mathcal{O}$. Let $T = \operatorname{Spec} R$ and $U = \operatorname{Spec} K$. By lemma 3.5.11, this data induces a morphism $T \to X \times_Y X$ mapping the generic point to ξ_1 and the closed point to ξ_0. Composing this with the two projections $p_1, p_2 : X \times_Y X \to X$, we get two morphisms $h, h' : T \to X$. Since $\xi_1 \in \Delta(X)$, we have $p_1(\xi_1) = p_2(\xi_1)$, so h and h' agree on the generic point U. Furthermore, they both map to Y identically (as they factor through $X \times_Y X$). By the hypothesis of the theorem, there is at most one such lift, so $h = h'$. Consequently, the map $T \to X \times_Y X$ must factor through the diagonal, implying $\xi_0 = L(t_0) \in \Delta(X)$. Thus $\Delta(X)$ is stable under specialization and is closed.

Example 3.5.15: Valuation Criterion on $\mathbb{A}^1$

Let us demonstrate the mechanics of the valuation criterion by proving that the affine line $X = \mathbb{A}^1_k = \operatorname{Spec} k[t]$ is separated over $Y = \operatorname{Spec} k$.

Suppose we are given a valuation ring R with fraction field K, and the standard diagram:

$$\begin{tikzcd}
U = \operatorname{Spec} K \arrow[r, "\alpha"] \arrow[d, "\iota"'] & \operatorname{Spec} k[t] \arrow[d] \\
T = \operatorname{Spec} R \arrow[r] \arrow[ur, dashed, "\exists ? \tilde{\alpha}"] & \operatorname{Spec} k
\end{tikzcd}$$

We translate this geometric data into algebraic data (reversing arrows):

1. The morphism $\alpha : \operatorname{Spec} K \to \operatorname{Spec} k[t]$ corresponds to a k-algebra homomorphism $\alpha^\# : k[t] \to K$. This homomorphism is completely determined by the image of the generator t. Let $\alpha^\#(t) = \lambda \in K$.

2. The inclusion $\iota : \operatorname{Spec} K \to \operatorname{Spec} R$ corresponds to the inclusion of the valuation ring into its fraction field $R \hookrightarrow K$.

3. A lift $\tilde{\alpha} : \operatorname{Spec} R \to \operatorname{Spec} k[t]$ corresponds to a homomorphism $\tilde{\alpha}^\# : k[t] \to R$. Let $\tilde{\alpha}^\#(t) = r \in R$.

For the diagram to commute, we require that the composition $k[t] \xrightarrow{\tilde{\alpha}^\#} R \hookrightarrow K$ equals the original map $\alpha^\#$. Evaluating at t, this means we must have $r = \lambda$ inside K.

Now we analyze the possibilities based on the element $\lambda \in K$:

- *Case 1:* $\lambda \in R$. Then we must define $\tilde{\alpha}^{\#}(t) = \lambda$. Since the map is determined on generators, this is the *unique* homomorphism making the diagram commute.
- *Case 2:* $\lambda \notin R$. Then there is no element $r \in R$ such that $r = \lambda$. Thus, no lift exists.

In either case, the number of lifts is *at most one* (it is either 1 or 0). Therefore, $\mathbb{A}^1_k$ is separated.

Contrast this with the "Line with two origins" (X). If we map $\operatorname{Spec} K$ to the generic point of X, and R is a DVR, we essentially have a map sending t to 0 in the residue field. In the algebraic formulation, the gluing data allows for two distinct maps $T \to X$ (one to each origin) that become identical when restricted to the generic point U. This violates the uniqueness condition.

Corollary 3.5.16: Properties of Separated Morphisms

Assume that all schemes are Noetherian in the following statements.

1. Open and closed immersions are separated.
2. A composition of two separated morphisms is separated.
3. Separated morphisms are stable under base extension.
4. If $f : X \to Y$ and $f' : X' \to Y'$ are separated morphisms of schemes over a base scheme S, then the product morphism $f \times f' : X \times_S X' \to Y \times_S Y'$ is also separated.
5. If $f : X \to Y$ and $g : Y \to Z$ are two morphisms and if $g \circ f$ is separated, then f is separated.
6. A morphism $f : X \to Y$ is separated if and only if Y can be covered by open subsets V_i such that $f^{-1}(V_i) \to V_i$ is separated for each i.

***Proof*:**
These statements all follow immediately from the condition of the theorem (Valuation Criterion, theorem 3.5.14). We follow Hartshorne [16] and give the proof of (3) to illustrate the method.

Let $f : X \to Y$ be a separated morphism, let $Y' \to Y$ be any morphism, and let $X' = X \times_Y Y'$ be obtained by base extension. We must show that the induced map $f' : X' \to Y'$ is separated. So suppose we are given a valuation ring R with field of fractions K, let $T = \operatorname{Spec} R$ and $U = \operatorname{Spec} K$. Suppose we are given morphisms of T to Y' and U to X' as in the theorem, and two morphisms of T to X' making the diagram

$$\begin{tikzcd} U \arrow[r] \arrow[d, "i"'] & X' \arrow[r] \arrow[d, "f'"] & X \arrow[d, "f"] \\ T \arrow[r] \arrow[ur, shift left] \arrow[ur, shift right] & Y' \arrow[r] & Y \end{tikzcd}$$

commutative (where the double arrows represent the two potential lifts $T \to X'$). Composing with the map $X' \to X$, we obtain two morphisms of T to X. Since the outer diagram commutes with respect to f (which is separated), and the maps agree on the dense open set U, these two morphisms $T \to X$ must be the same by theorem 3.5.14.

We also have two morphisms $T \to X' \to Y'$. Since the diagram commutes, both of these must equal the given map $T \to Y'$. Thus, we have two maps $T \to X'$ such that their compositions to X are equal and their compositions to Y' are equal. But X' is the fibered product of X and Y' over Y, so by the universal property of the fibered product, a map to X' is uniquely determined by its projections to X and Y'. Therefore, the two maps of T to X' are the same. Hence f' is separated.

3.5.17 Note on Noetherian Condition You have probably noticed that in order to apply the theorem, it is not necessary to assume that all the schemes mentioned in the corollary are Noetherian. In fact, even in the theorem itself, you can get by with assuming something less than X Noetherian (see EGA I, new ed., 5.5.4).
What Hartshorne says is that if a Noetherian hypothesis will make statements and proofs substantially simpler, then he takes the hypothesis, even though it may not be necessary. This is reasonable as most cases where the valuation citation is useful is when working with stacks which will satisfy these conditions.

Rational maps to Separated Schemes

here

Exercise 3.5.1

1. add exercise 4.5 of Hartshorne: important for Weil divisors to show characterization of irred. closed subscheme of (separated noethean integral) reg. in codim 1 scheme
2. Show that $X \cong \Delta(X)$. Show that in general, $\Delta(X)$ is a locally closed immersion.
3. Show that a scheme X over S is separated if and only if it is separated over $\mathbb{Z}$.
4. If $\varphi : A \to B$ is a ring homomorphism, then the corresponding morphism $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is dominant if and only if $\ker \varphi$ is contained in the nilradical of A.

3.5.3 Universally Closed Morphisms

We now build up to the notion of properness. Proper schemes and morphisms capture the idea of compactness and "compact-preserving" maps. In topology, a map is often called *proper* if the preimage of any compact set is compact. In algebraic geometry, because the Zariski topology is so coarse (and affine schemes are almost always quasicompact), this definition is not strong enough.

Instead, we look for a property that ensures the *image* of a closed set is closed, and remains closed after any base change.

Definition 3.5.18: Universally Closed

Let $f : X \to Y$ be a morphism of schemes. Then f is *universally closed* if it is a closed map (images of closed sets are closed), and for any morphism $Y' \to Y$, the base change $f' : X \times_Y Y' \to Y'$ is also a closed map.

The intuition is that we want to avoid "missing points" at the boundary. Recall the example of $\mathbb{A}^1_k \to \operatorname{Spec} k$. This is closed, but base changing to $\mathbb{A}^1_k$ (projection of the plane to the line) maps the hyperbola $xy = 1$ to the open set $\mathbb{A}^1 \setminus \{0\}$. The "missing point" at infinity prevents the map from being universally closed.

Just as Separatedness was characterized by the *uniqueness* of lifts in the valuation criterion, Universally Closed morphisms are characterized by the *existence* of lifts.

Theorem 3.5.19: Valuation Criterion for Universally Closed Morphisms

Let $f : X \to Y$ be a quasicompact morphism of schemes. Then f is universally closed if and only if for any valuation ring R with fraction field K, and any diagram:

$$\begin{tikzcd} U = \operatorname{Spec} K \arrow[r] \arrow[d, "\iota"'] & X \arrow[d, "f"] \\ T = \operatorname{Spec} R \arrow[r] \arrow[ur, dashed, "\exists"] & Y \end{tikzcd}$$

there exists **at least one** morphism $T \to X$ making the diagram commute.

From the perspective of functors of points, the valuative criterion is naturally a statement about extending $\operatorname{Spec}(K)$-valued points to $\operatorname{Spec}(R)$-valued points.

***Proof*:**
($\Rightarrow$) Existence implies Universally Closed: Suppose the lifting property holds. We want to show f is universally closed. Let $Y' \to Y$ be any base change. We must show $f' : X' \to Y'$ is closed. Since f is quasicompact, f' is quasicompact. By lemma 3.5.13, it suffices to show that the image $f'(X')$ is stable under specialization. Let $z_1 \in X'$ and let $y_1 = f'(z_1)$. Suppose $y_1 \rightsquigarrow y_0$ is a specialization in Y'. We need to find a $z_0 \in X'$ such that $f'(z_0) = y_0$.

Let $K = \kappa(z_1)$. We have a map $\operatorname{Spec} K \to X'$. Let $\mathcal{O}$ be the local ring of y_0 in the closure $\overline{\{y_1\}}$ (with reduced structure). By domination, there exists a valuation ring R of K dominating $\mathcal{O}$. This gives us maps $U \to X'$ and $T \to Y'$. Composing with the projection $X' \to X$, we get maps $U \to X$ and $T \to Y$ fitting the hypothesis diagram. By assumption, there exists a lift $T \to X$. By the universal property of the fibered product X', the pair of maps $T \to X$ and $T \to Y'$ induce a map $T \to X'$. Let z_0 be the image of the closed point of T in X'. Since the map commutes, $f'(z_0)$ must be the image of the closed point of T in Y', which is y_0. Thus y_0 is in the image.

($\Leftarrow$) Universally Closed implies Existence: Suppose f is universally closed. Consider the diagram with $U \to X$ and $T \to Y$. We base change by $T \to Y$ to get $X_T = X \times_Y T$. The map $U \to X$ and $U \to T$ induce a map $U \to X_T$. Let ξ_1 be the image of the unique point of U in X_T. Let $Z = \overline{\{\xi_1\}}$ with reduced structure. Since f is universally closed, the projection $X_T \to T$ is closed. Thus the image of Z is closed in T. The generic point of T is in the image (it is the image of ξ_1). Since $T = \operatorname{Spec} R$, the closure of the generic point is all of T. Thus, the image of Z is all of T. Therefore, there exists a point $\xi_0 \in Z$ that maps to the closed point $t_0 \in T$.

This gives us a local homomorphism of local rings $R \to \mathcal{O}_{\xi_0, Z}$. Since R is a valuation ring of $K = \kappa(\xi_1)$, and R dominates the local ring (by construction of the map $T \to Y$), R must actually be the local ring (or dominate it via isomorphism). By lemma 3.5.11, this domination data provides a morphism $T \to X_T$ sending $t_0 \to \xi_0$ and $t_1 \to \xi_1$. Composing $T \to X_T \to X$

gives the desired lift.

3.5.4 Proper Morphisms

We now combine the concepts. We want a scheme (or morphism) that behaves like a compact Hausdorff space.

- Hausdorff iff Separated iff Uniqueness of limits.
- Compact iff Universally Closed + Finite Type iff Existence of limits.

Definition 3.5.20: Proper Morphism

A morphism $f : X \to Y$ is called *proper* if it satisfies three conditions:

1. Separated (Hausdorff-like).
2. Universally Closed (Image of closed is closed).
3. Finite Type (Finite covering / geometrically bounded).

A scheme X is said to be *proper* if the morphism $X \to \operatorname{Spec}(\mathbb{Z})$ is proper, and an R-scheme is proper if the map $X \to \operatorname{Spec}(R)$ is proper.

Proposition 3.5.21: Properties of Proper Morphisms

Assume all schemes are Noetherian.

1. A closed immersion is proper.
2. A composition of proper morphisms is proper.
3. Proper morphisms are stable under base change.
4. Products of proper morphisms are proper.
5. If $f : X \to Y$ and $g : Y \to Z$ are morphisms, g is separated, and $g \circ f$ is proper, then f is proper.
6. Properness is local on the base (target).

***Proof*:**

1. A closed immersion is separated (diagonal is iso), finite type (surjection of rings), and universally closed (closed sets in a subspace are closed in the ambient space).
2. Let $f : X \to Y$ and $g : Y \to Z$ be proper. Separatedness and finite type are preserved under composition. For universally closed: Let $Z' \to Z$. Base change $X \to Z$ by Z' is the composition of base changing g then f. Since pullback preserves closed maps, the composition is closed.
3. Finite type and separatedness are stable. Universal closedness is stable by definition.

4. Follows formally from Base Change and Composition.

5. This is best proved using the Valuation Criterion (theorem 3.5.25). Let R be a valuation ring with field K. Suppose we have a diagram:

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r] & Y \end{tikzcd}$$

We want a unique lift. Compose with $g : Y \to Z$ to get a map $\operatorname{Spec} K \to X$ and $\operatorname{Spec} R \to Z$. Since $g \circ f$ is proper, there exists a unique lift $\operatorname{Spec} R \to X$. This lift makes the big diagram commute. Does it make the triangle with Y commute? We have two maps $\operatorname{Spec} R \to Y$: one given, and one via the lift through X. Both map the generic point to the same place, and both compose with g to give the same map to Z. Since g is *separated*, by the valuation criterion for separatedness, these two maps must be equal. Thus the lift works for f.

The following two propositions give a large class of useful proper morphisms, and consequence proper schemes over Y:

Proposition 3.5.22: Finite Morphisms are Proper

Let $f : X \to Y$ be a finite morphism. Then f is proper.

***Proof*:**
A finite morphism is affine, hence separated. It is finite type (actually finite presentation if Noetherian). We need only show it is universally closed. Since properness is local on the target, assume $Y = \operatorname{Spec} R$. Then $X = \operatorname{Spec} S$ where S is a finite R-module. Let $Y' \to Y$ be a base change. The base change of a finite morphism is finite ($S \otimes_R R'$ is a finite R'-module). Thus, it suffices to show that any finite morphism is a closed map. Let $\varphi : A \to B$ be a finite ring map. Let $V(I) \subseteq \operatorname{Spec} B$ be a closed set. The image is $V(\varphi^{-1}(I))$ inside $\operatorname{Spec} A$. This follows from the *Going Up Theorem* [**NateClassicalAlgGeo**] (or the Cohen-Seidenberg Theorem). Since B is integral over A, the map $\operatorname{Spec} B \to \operatorname{Spec} A$ is closed.

If the focus is put on morphisms between affine schemes (or slightly more generally affine morphisms), then properness reduces to finiteness:

Proposition 3.5.23: Affine Proper Morphisms are Finite

Let $f : X \to Y$ be an affine morphism. Then f is proper if and only if f is finite.

***Proof*:**
($\Leftarrow$) This follows immediately from proposition 3.5.22. A finite morphism is always affine and proper.

($\Rightarrow$) Suppose f is affine and proper. We must show it is finite. Since finiteness is local on the base, we may assume $Y = \operatorname{Spec} A$ is affine. Because f is affine, $X = \operatorname{Spec} B$ can also be assumed to be affine, and so f corresponds to a ring homomorphism $\varphi : A \to B$.

Since f is proper, it is of *finite type*. This means B is a finitely generated A-algebra.

To show finiteness, we must show that φ is *integral*. We use the Valuative Criterion for Universally Closed Morphisms (theorem 3.5.19). Let K be the fraction field of B (or any field containing B) and let $R \subset K$ be any valuation ring containing A (more precisely, containing $\varphi(A)$). Consider the diagram:

$$\begin{tikzcd}
\operatorname{Spec} K \arrow[r] \arrow[d] & \operatorname{Spec} B \arrow[d] \\
\operatorname{Spec} R \arrow[r] & \operatorname{Spec} A
\end{tikzcd}$$

Since f is universally closed, a lift $\operatorname{Spec} R \to \operatorname{Spec} B$ exists. Algebraically, this means the map $B \to K$ factors through R, i.e., $B \subseteq R$.

Thus, B is contained in every valuation ring of K that contains A. A standard theorem in commutative algebra states that the integral closure of A in K is the intersection of all valuation rings of K containing A, see lemma 5.4.8 for a slightly stronger proof. Therefore, B is integral over A.

We now have that B is a finitely generated A-algebra that is integral over A. A standard result in commutative algebra states that such a ring is a finitely generated A-module (see [11, chapter 20.5]). Since B is a finite A-module, f is a finite morphism, completing the proof.

Example 3.5.24: Non-Proper Morphisms

1. $\mathbb{A}^1_k \to \operatorname{Spec} k$ is not proper. It is not universally closed (base change to $\mathbb{A}^1$ maps the hyperbola to an open set).
2. $\mathbb{A}^1_k \setminus \{0\} \to \mathbb{A}^1_k$ (open immersion). This is separated and finite type, but not closed (image is open).
3. The "Line with two origins" is not proper because it is not separated.

Theorem 3.5.25: Valuation Criterion for Properness

Let $f : X \to Y$ be a morphism of finite type, with X Noetherian. Then f is proper if and only if for every valuation ring R with fraction field K, and every diagram:

$$\begin{tikzcd}
U \arrow[r] \arrow[d, "\iota"'] & X \arrow[d, "f"] \\
T \arrow[r] \arrow[ur, dashed] & Y
\end{tikzcd}$$

there exists a *unique* lifting $T \to X$ making the diagram commute.

***Proof*:**
See derivation in previous section.

Projective Schemes and Consequences

Theorem 3.5.26: Projective Morphisms are Proper

A projective morphism between Noetherian schemes is proper.

***Proof*:**
(See previous section for proof via valuation criterion).

One of the most powerful consequences of properness is the behavior of global functions. For compact complex manifolds, Liouville's theorem states that holomorphic functions on a compact connected manifold are constant. We have the algebraic analogue:

Theorem 3.5.27: Global Sections of Proper Schemes

Let k be a field. Let X be a proper k-scheme that is reduced and connected. Then:

$$\Gamma(X, \mathcal{O}_X) = k$$

(i.e., the only global regular functions are constants).

***Proof*:**
Let $f \in \Gamma(X, \mathcal{O}_X)$ be a global section. By proposition 3.3.8, we get a scheme morphism (that we shall also denote as f)

$$f : X \to \mathbb{A}^1_k = \operatorname{Spec} k[t]$$

Consider the open immersion $\iota : \mathbb{A}^1_k \hookrightarrow \mathbb{P}^1_k$. Compose them to get $\varphi : X \to \mathbb{P}^1_k$. Since X is proper over k and $\mathbb{P}^1_k$ is separated over k, the image $\varphi(X)$ must be a *closed* subset of $\mathbb{P}^1_k$. However, f maps into the affine line, so the image $\varphi(X)$ does not contain the point at infinity, ∞. Thus, $\varphi(X)$ is a closed subset of $\mathbb{P}^1_k$ contained entirely in $\mathbb{A}^1_k$. The closed subsets of $\mathbb{P}^1_k$ are finite sets of points or the whole space. Since it misses ∞, it must be a finite set of points. Since X is connected, the image must be a single point. Since X is reduced, the map to a single point corresponds to a map to $\operatorname{Spec} k$. Thus, f maps every point to the same value $\lambda \in k$.

While Projective implies Proper, the converse is not true (though examples are pathological, like Hironaka's example). However, Chow's Lemma states that proper schemes are "birationally" projective.

Theorem 3.5.28: Chow's Lemma

Let $f : X \to S$ be a proper morphism of finite type over a Noetherian base S. Then there exists a scheme X' and a morphism $\pi : X' \to X$ such that:

1. π is a surjective projective morphism.
2. The composite $X' \to S$ is a projective morphism.
3. There exists a dense open $U \subseteq X$ such that $\pi^{-1}(U) \to U$ is an isomorphism.

***Proof*:**
(Sketch). The proof involves taking a finite cover of X by quasi-projective schemes U_i. We form the disjoint union $U = \coprod U_i$ and a map $U \to X$. We then take the closure of the graph of this map inside $X \times_S \mathbb{P}^N$ (for sufficiently large N). The resulting scheme X' dominates X and sits inside a projective space over S, making it projective. The dense open isomorphism comes from the fact that we started with a cover. See [16, Hartshorne II.4.10] for the complete derivation.

Exercise 3.5.2

1. Show that two copies of $\operatorname{Spec}(k[x_1, x_2, ...])$ minus the origin, glued at the origin, is not quasi-separated.
2. Prove that the intersection of two proper subschemes is proper.

3.6 Rational Maps

When working with [quasiprojective] varieties, we had the weaker notion of a rational function that was defined almost everywhere. A rational function was more flexible, allowing for some important classification results, the definition of Kodaira dimension, study singularities, more invariances by looking at the birational automorphisms, and some “local to global” principles. The following brings these concepts over to schemes

Definition 3.6.1: Rational Map

Let X, Y be schemes. Then a *rational map* $\pi : X \dashrightarrow Y$, is an equivalence relation:

$$(\alpha U \to Y) \sim (\beta : V \to Y) \qquad \text{if and only if} \qquad Z \subseteq U \cap V \text{ s.t. } \alpha|_Z = \beta|_Z$$

where U, V, Z are all dense open subsets[a].

If X, Y are S-schemes, then rational maps of S-schemes are defined similarly.

If the image of one of the representatives is dense, then the rational map is said to be *dominant* or *dominating*.

[a] When Y is separated, we will let $Z = U \cap V$

Note that a rational map is is an equivalence relation, not necessarily defined on a single domain.

Definition 3.6.2: Birational Map

A rational map $\pi : X \dashrightarrow Y$ is *birational* if it is dominant, and it has a dominant rational inverse. Two schemes X, Y are *birationally equivalent* if there is a birational map $\pi : X \dashrightarrow Y$.

Example 3.6.3: Rational Map

1. A scheme morphisms always gives a rational map, similarily to how all regular functions are special rational functions
2. The projection $\mathbb{P}^n_R \dashrightarrow \mathbb{P}^{n-1}_R$ projecting down a dimension is a rational map (recall the same reasoning for rational functions in [11, chapter 22.3]

Lemma 3.6.4: Dominating Rational Map on Irreducible Schemes

Let $\pi : X \dashrightarrow Y$ be a rational map between irreducible schemes. Then π is dominating iff it sends the generic point of X to the generic point of Y.

***Proof*:**
recall the properties of generic points in irreducible schemes.

We may naturally want to have a result that is similar to the correspondence of rational functions and field extensions as given in [**NateClassicalAlgGeo**]. The full correspondence requires irreducible varieties (integral finite type k-schemes), however we may at least always induce a map of function fields:

Proposition 3.6.5: Rational Maps of Integral Schemes induces Function Field Map

The collection of integrals schemes with dominant maps induces morphisms of function fields in the opposite direction.

***Proof*:**
First show that the form a category (importantly, composition works). Then think of how to create this correspondence.

Corollary 3.6.6: Birational Map on Irreducible Schemes

Let $\pi : X \dashrightarrow Y$ be a birational map between irreducible schemes. Then the induced map of function fields is an isomorphism.

For the converse failing

Example 3.6.7: Function Field Not Inducing Rational Map
Take Spec $(k[x])$ and Spec $(k(X)$ which have the same function field $k(x)$. Then certainly there is no corresponding rational map

$$\operatorname{Spec} k[x] \dashrightarrow \operatorname{Spec} k(x)$$

of k-schemes, as such a morphism would correspond to a morphism from $U \subseteq \operatorname{Spec} k[x]$ (ex. $U = \operatorname{Spec} k[x, 1/f(x)]$) to $\operatorname{Spec} k(x)$, but not map of rings $k(X) \to k[x, 1/f(x)]$ satisfies the right properties for any $f(x)$.

Proposition 3.6.8: Reduced Schemes and Birational Maps

Let X, Y be reduced schemes. Then X and nY are birational if and only if there exists a dense open subscheme $U \subseteq X$ and $V \subseteq Y$ such that

$$U \cong V$$

***Proof*:**
Generalize the proof from [**NateClassicalAlgGeo**] (If stuck, look at Vakil p. 190)

Vakil looks here at rational maps of irreducible varieties that look like translating the results over from varieties, so I will omit for now

I like his translation of the Pythagorean problem into a scheme problem

non-rational complex curve, cool presentation

3.7 Abstract Varieties

Let us use the tools we've developed to our usual notion of variety. We shall see that defining varieties abstractly (other common adverbs being inherently, integrally, intrinsically) shall give us many of the same benefits as defining schemes as spectrum. Most results about quasi-projectile varieties from [**NateClassicalAlgGeo**] readily translate to the abstract setting with a few exceptions that we shall go over.

There is a natural way in which we can embed (quasi-projective) varieties into the category $\mathbf{Sch}_{\bar{k}}$:

Theorem 3.7.1: Varieties and Schemes: Fully Faithful Functor

Let $\bar{k}$ be an algebraically closed field. Then there exists a fully faithful functor $\mathbf{Var}_{\bar{k}} \to \mathbf{Sch}_{\bar{k}}$ where $\mathbf{Var}_{\overline{k}}$ is the usual category of varieties (over $\overline{k}$) as defined in [**NateClassicalAlgGeo**].

Recall that a fully faithful means the hom-sets are in bijection, and it implies injectivity on objects (but not necessarily surjective, for example $\mathrm{Ab} \to \mathrm{Grp}$ is a fully faithful functor). We shall also stick with Hartshorne's notation for the functor, which is the letter t, it shall come up a few times later in the book.

***Proof*:**
For any (quasi-projective) variety, we shall first find the "points" that shall be the points of the scheme by working with the irreducible closed subsets. We shall then find the structure sheaf to put on it by finding all the regular functions, which shall be enough to define the functor.

First off, for any topological space X, and let $t(X)$ be the set of (nonempty) irreducible closed subsets of X. If Y is a closed subset of X, then $t(Y) \subseteq t(X), t\left(Y_1 \cup Y_2\right) = t\left(Y_1\right) \cup t\left(Y_2\right)$, and $t\left(\bigcap Y_i\right) = \bigcap t\left(Y_i\right)$. Hence, we can define a topology $t(X)$ by taking as closed sets the subsets of the form $t(Y)$, where Y is a closed subset of X. If $f : X_1 \to X_2$ is a continuous map, then we obtain a map $t(f) : t\left(X_1\right) \to t\left(X_2\right)$ by sending an irreducible closed subset to the closure of its image. Thus t is a functor on topological spaces. Furthermore, define a continuous map

$\alpha : X \to t(X)$ by $\alpha(P) = \{P\}^-$. Note that α induces a bijection between the set of open subsets of X and the set of open subsets of $t(X)$.

Next, let k be an algebraically closed field, V be a variety over k, and let $\mathcal{O}_V$ be its sheaf of regular functions. I claim that $(t(V), \alpha_* (\mathcal{O}_V))$ is a scheme over k. Since any variety can be covered by open affine sub-varieties, it will be sufficient to show that if V is affine, then $(t(V), \alpha_* (\mathcal{O}_V))$ is a scheme. Hence, let V be an affine variety with affine coordinate ring A. Define a morphism of locally ringed spaces

$$\beta : (V, \mathcal{O}_V) \to X = \operatorname{Spec} A$$

as follows: for each point $P \in V$, let $\beta(P) = \mathfrak{m}_P$, the ideal of A consisting of all regular functions which vanish at P. Then β is a bijection of V onto the set of closed points of X. It is easy to see that β is a homeomorphism onto its image. Now for any open set $U \subseteq X$, we will define a homomorphism of rings $\mathcal{O}_X(U) \to \beta_* (\mathcal{O}_V)(U) = \mathcal{O}_V\left(\beta^{-1}U\right)$. Given a section $s \in \mathcal{O}_X(U)$, and given a point $P \in \beta^{-1}(U)$, we define $s(P)$ by taking the image of s in the stalk $\mathcal{O}_{X,\beta(P)}$, which is isomorphic to the local ring $A_{\mathfrak{m}_P}$, and then passing to the quotient ring $A_{\mathfrak{m}_P}/\mathfrak{m}_P$ which is isomorphic to the field k. Thus s gives a function from $\beta^{-1}(U)$ to k. It is easy to see that this is a regular function, and that this map gives an isomorphism $\mathcal{O}_X(U) \cong \mathcal{O}_V\left(\beta^{-1}U\right)$. Finally, since the prime ideals of A are in 1-1 correspondence with the irreducible closed subsets of V, these remarks show that $(X, \mathcal{O}_X)$ is isomorphic to $(t(V), \alpha_* (\mathcal{O}_V))$, so the latter is indeed an affine scheme, completing the proof.

We have defined upon enough properties to characterize the image of this embedding:

Definition 3.7.2: k-Variety

Let X be a k-scheme. Then if X is reduced, irreducible, separated, and of finite type over k, then X is said to be a *variety over* k or a *k-variety*. If X is affine/(quasi)projective, then it is an affine/(quasi)projective variety over k respectively.

It is important to note that in [**NateClassicalAlgGeo**], a variety was an affine algebraic set that was irreducible, which motivates adding irreducible in the definition. However some authors drop the condition and "variety" would be "algebraic set". Note too that we have defined k-varieties and not varieties in general. Furthermore, some authors require the field $k = \overline{k}$ be algebraically closed as part of the definition to more closely reflect the geometric results of the Nullstellensatz, and also to avoid the following problem:

Example 3.7.3: Variety Not Closed Under Base Change

Take the $\mathbb{R}$-variety $\mathbb{C} \cong \mathbb{R}[x]/(x^2+1) = V$ which is a single point, hence irreducible. Then $C \times_{\mathbb{R}} V = \operatorname{Spec} \mathbb{C}[x](x^2+1)$ which now has two points, hence it is not irreducible and so *not* a variety as we have defined it.

While the abstract definition of a scheme allows for elegant categorical constructions, it introduces points that are not closed (the generic points corresponding to non-maximal prime ideals). Notice in the proof of theorem 3.7.1 that the map β is a bijection of the classical variety V onto *only* the closed points of $X = \operatorname{Spec} A$. One might naturally worry: do we lose algebraic information by only evaluating functions at the closed points?

This is where the concept of a *Jacobson scheme* is useful, serving as the exact geometric justification for why our classical intuition of evaluating polynomials at points perfectly captures the algebraic structure. Indeed, it is fair to say that Jacobson schemes are the "most general" setting in which the Nullstellensatz holds.

Definition 3.7.4: Jacobson Scheme

A topological space X is *Jacobson* if its closed points are dense in every closed subset. A scheme X is a *Jacobson scheme* if its Zariski topology is Jacobson.

Recall ([11, commutative-algebra]) that a *Jacobson ring* is a ring where the intersection of every maximal ideal of R containing $I \subseteq R$ is equal to the radical $\sqrt{I}$.

Proposition 3.7.5: Characterizing Jacobson Affine Schemes

An affine scheme $X = \operatorname{Spec} R$ is Jacobson if and only if R is *Jacobson*.

***Proof*:**
Let $X = \operatorname{Spec} R$. Any closed subset of X is of the form $V(I)$ for some ideal $I \subseteq R$. The closed points of X are exactly the maximal ideals $\mathrm{mSpec}(R)$. Therefore, the closed points residing inside $V(I)$ are exactly the maximal ideals containing I.

The closure of these points inside X is given by taking the vanishing locus of their intersection:

$$\overline{V(I) \cap \mathrm{mSpec}(R)} = V\left(\bigcap_{\substack{\mathfrak{m}\in\mathrm{mSpec}(R)\\ I\subseteq\mathfrak{m}}} \mathfrak{m}\right)$$

If R is a Jacobson ring, this intersection of maximal ideals is exactly $\sqrt{I}$. Thus, the closure becomes $V(\sqrt{I})$, which is equal to $V(I)$. This shows the closed points in $V(I)$ are dense in $V(I)$, making X a Jacobson space.

Conversely, if X is a Jacobson space, the closed points are dense in $V(I)$, so $V\left(\bigcap_{\mathfrak{m}\supset I}\mathfrak{m}\right) = V(I)$. Taking the radical of the defining ideals gives $\bigcap_{\mathfrak{m}\supset I}\mathfrak{m} = \sqrt{I}$, proving R is a Jacobson ring.

Let us formally prove that our geometric intuition of "density" translates exactly to this algebraic property of the nilradical, which ensures we don't lose information.

Corollary 3.7.6: Density of Closed Points and the Nilradical

Let R be a commutative ring and $X = \operatorname{Spec} R$. If the closed points of X are dense in X, then the intersection of all maximal ideals in R is exactly the nilradical $\mathrm{Nil}(R)$.

***Proof*:**
This is a direct application of the logic from proposition 3.7.5 using the zero ideal $I = (0)$.

The entire space X is the closed subset $V(0)$. The condition that the closed points are dense in

X means that the closure of the maximal ideals is the whole space:

$$V\left(\bigcap_{\mathfrak{m}\in\mathrm{mSpec}(R)}\mathfrak{m}\right)=V(0)$$

Taking the radical of the defining ideals on both sides yields:

$$\bigcap_{\mathfrak{m}\in\mathrm{mSpec}(R)}\mathfrak{m}=\sqrt{(0)}=\mathrm{Nil}(R)$$

(Note that the left side is its own radical because an intersection of prime ideals is already a radical ideal).

Because the nilradical is zero, the geometry of the maximal ideals is dense enough to completely determine the algebra. This classically justifies viewing polynomials as functions to the base field k. Furthermore, as we saw early in the chapter in section 3.3 (Functor of Points), this classical intuition is perfectly mirrored in scheme theory, where a global section $f\in\Gamma(X,\mathcal{O}_X)$ is canonically equivalent to a scheme morphism $X\to\mathbb{A}^1_k$.

Example 3.7.7: Non-Jacobson Schemes

A scheme that is not Jacobson must have a prime ideal that is not the intersection of maximal ideals. This represents a higher dimensional point that is "not covered" by closed points. Schemes likes $\operatorname{Spec}k[[x]]$ and $\operatorname{Spec}\mathbb{Z}_{(p)}$. The scheme $\operatorname{Spec}k[x]/(x^n)$ *is* Jacobson, having only (x) as its prime and maximal ideal.

Note that:

$$k[[x]]\cong\varprojlim_n\frac{k[x]}{(x^n)}$$

giving an example of how different the geometry and the algebra can behave under limits. We shall address this in section 5.6.

Having reconciled the abstract algebraic structure with our geometric intuition of points, let us look at the standard ways to construct these varieties.

Proposition 3.7.8: Affine and Projective Varieties

1. $\mathbb{A}^n_k/I$ is an affine k-variety if and only if $I\subseteq k[x_1,...,x_n]$ is a radical ideal
2. Let $I\subseteq k[x_1,...,x_n]$ be a radical graded ideal. Then $\operatorname{Proj}k[x_0,...,x_n]/I$ is a projective k-variety [a]

[a] I need not be radical, ex. $I=(x_0^2,x_0x_1,...,x_0x_n)$

***Proof*:**
exercise.

An important class of varieties are *proper varieties*. For historical reasons, namely the 19th century Italian school, the following is a common name for proper varieties;

Definition 3.7.9: Complete Variety

Let X be a k-variety. Then it is said to be *complete* if it is proper.

The adjective "complete" for a proper variety comes from the fact that we are in some way thinking of a complete variety as an extension into projective space. It can be shown (see theorem 3.5.28[6]) that if C is complete, then there exists a morphism $C \to \mathbb{P}^n$ to a projective scheme that is isomorphic to an open dense subset. There do exist complete non-projective varieties, but they are hard to come by, partly due to how bad the singularity must be[7].

Before concluding this section, it is crucial to see how finite type schemes (and thus varieties) behave under categorical products, as standard geometric products naturally map to fibered products in the category of schemes.

Proposition 3.7.10: Fibered Products of Finite Type Schemes

Let S be a k-scheme of finite type. If X and Y are k-schemes of finite type over S, then their fibered product $X \times_S Y$ is a finite type k-scheme. Consequently, the product $X \times_k Y$ of two k-varieties over an algebraically closed field is a k-variety.

***Proof*:**
Because $X \to S$ and $Y \to S$ are morphisms of finite type, their base changes are also of finite type. Specifically, the projection $X \times_S Y \to Y$ is a base change of $X \to S$, so it is of finite type. Composing this with the finite type structure morphism $Y \to \operatorname{Spec} k$, we find that $X \times_S Y \to \operatorname{Spec} k$ is of finite type.

In the specific case where $S = \operatorname{Spec} k$ and X, Y are k-varieties over an algebraically closed field k, we know $X \times_k Y$ is of finite type. Furthermore, varieties are separated by definition, and the product of separated schemes is separated. Since k is algebraically closed, the tensor product of two integral k-algebras over k is again an integral domain. Because this property holds locally on affine charts, $X \times_k Y$ is reduced and irreducible. Having satisfied all conditions, $X \times_k Y$ is a k-variety.

3.8 *The Category of Affine Schemes as Ringed Spaces

Having established that scheme morphisms must be morphisms of *locally* ringed spaces to properly reflect geometric evaluation, it is natural to ask: just how wildly does the equivalence $\mathbf{Sch}^{\mathbf{Aff}} \cong \mathbf{CRing}^{\mathrm{op}}$ fail if we drop the "locally" condition? What does the category of affine schemes look like if we restrict our morphisms to ringed space morphisms?

As we saw with the "phantom map" in example 3.1.1, multiple distinct ringed space morphisms can induce the exact same ring homomorphism on global sections. The topological map f becomes somehow decoupled from the algebraic pullback $f^\#$. Fortunately, this decoupling is not entire; it is classified by the topology of the Zariski spectrum, specifically:

[6]see Borcherds scheme video abstract and projective varieties

[7]again see the same Borcherds video for an overview. The example he gave also allows for producing algebraic spaces that are not schemes

Proposition 3.8.1: Classification of Ringed Space Morphisms

Let $X = \operatorname{Spec}(S)$ and $Y = \operatorname{Spec}(R)$ be affine schemes. A pair consisting of a continuous map $f : X \to Y$ and a ring homomorphism $\varphi : R \to S$ defines a valid morphism of ringed spaces $(f, f^\#) : X \to Y$ (where $f^\#$ on global sections is φ) if and only if:

$$\varphi^{-1}(\mathfrak{p}) \subseteq f(\mathfrak{p}) \qquad \text{for all } \mathfrak{p} \in \operatorname{Spec}(S).$$

Furthermore, when this condition holds, the sheaf map $f^\#$ is uniquely determined.

***Proof*:**
Let $g : X \to Y$ be the canonical "true" continuous map defined by $g(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$.

For $(f, f^\#)$ to be a valid morphism of ringed spaces, the global ring homomorphism $\varphi : R \to S$ must induce a well-defined local map on the stalks for every $\mathfrak{p} \in X$. Specifically, the composite map $R \to S \to S_\mathfrak{p}$ must factor uniquely through the localization of R at $f(\mathfrak{p})$:

$$\begin{tikzcd}
R \arrow[r, "\varphi"] \arrow[d, "\text{loc}"] & S \arrow[d, "\text{loc}"] \\
R_{f(\mathfrak{p})} \arrow[r, dashed, "\exists!\, f^\#_\mathfrak{p}"] & S_\mathfrak{p}
\end{tikzcd}$$

By the universal property of localization, this factorization exists if and only if every element in the denominator of $R_{f(\mathfrak{p})}$ maps to a unit in $S_\mathfrak{p}$.

- The denominators of $R_{f(\mathfrak{p})}$ are exactly the elements of the set $R \setminus f(\mathfrak{p})$.
- The units of $S_\mathfrak{p}$ are exactly the fractions with numerators in S and denominators in $S \setminus \mathfrak{p}$, meaning the image of the element in S must lie in $S \setminus \mathfrak{p}$.

Thus, we require:

$$\varphi(R \setminus f(\mathfrak{p})) \subseteq S \setminus \mathfrak{p}$$

Taking the set-theoretic complement of this condition in R, we see that an element $r \in R$ avoids the left-hand side if and only if it avoids the right-hand side. This gives:

$$\varphi^{-1}(\mathfrak{p}) \subseteq f(\mathfrak{p})$$

Since localization maps are epimorphisms in the category of commutative rings, if this factorization exists, it is unique, uniquely determining the sheaf map $f^\#$ on all stalks, and hence on all of X.

Recall that in the Zariski topology, the containment of prime ideals $\mathfrak{q} \subseteq \mathfrak{q}'$ means exactly that $\mathfrak{q}'$ is a specialization of $\mathfrak{q}$ (i.e., $\mathfrak{q}'$ lies in the closure of the point $\mathfrak{q}$, so $\mathfrak{q}' \in \overline{\{\mathfrak{q}\}}$). The proposition reveals a geometric reality about the category of ringed spaces: the algebraic condition $\varphi^{-1}(\mathfrak{p}) \subseteq f(\mathfrak{p})$ translates to:

$$f(\mathfrak{p}) \in \overline{\{g(\mathfrak{p})\}}$$

A ringed space morphism allows the topological map f to take a point $\mathfrak{p}$ and map it to *any specialization* (any point in the closure) of the "correct" algebraic target $g(\mathfrak{p})$, provided the resulting assignment f remains globally continuous.

Example 3.8.2: Revisiting the Phantom Map

Consider again $R = \mathbb{Z}_{(p)}$, $S = \mathbb{Q}$, and φ the standard inclusion. We have $\operatorname{Spec}(\mathbb{Q}) = \{[(0)]\}$ and $\operatorname{Spec}(\mathbb{Z}_{(p)}) = \{[(0)], [(p)]\}$.

The “true” algebraic target for the unique point in $\operatorname{Spec}(\mathbb{Q})$ is $g([(0)]) = \varphi^{-1}((0)) = [(0)]$.

What are the specializations of $[(0)]$ in $\operatorname{Spec}(\mathbb{Z}_{(p)})$? The closure of the generic point $[(0)]$ is the entire space, since the prime ideal (0) is contained in the maximal ideal (p). Therefore, $[(p)]$ is a valid specialization of $[(0)]$.

By proposition 3.8.1, mapping $f([(0)]) = [(p)]$ satisfies the condition $(0) \subseteq (p)$. Thus, it forms a valid ringed space morphism, giving us the phantom map π_2.

4

Schemes II.5: Group Schemes

This chapter can comfortably skipped if the reader is not interested in studying algebraic groups or abelian varieties in the near future. The chapter is placed here since it is a natural follow up to all the constructions we have presented in the previous chapter, and (I think) does not require much in the way of quasi-coherent or coherent sheaves, and similar for sheaf cohomology.

In classical algebraic geometry, an *algebraic group* is a variety that is simultaneously a group, with the multiplication and inversion maps being morphisms of varieties. The theory of algebraic groups over algebraically closed fields is rich and well-developed, encompassing the classification of reductive groups, the structure theory of Borel and torus subgroups, and the representation theory that underpins much of modern mathematics.

Over a general base scheme, however, the correct notion is that of a *group scheme*: a scheme G equipped with morphisms encoding multiplication, identity, and inversion, subject to the usual group axioms expressed diagrammatically. This generalization is essential for arithmetic applications (where the base may be $\mathrm{Spec}\,(\mathbb{Z})$ or $\mathrm{Spec}\,(\mathbb{Z}_p)$), for the study of families of groups (where the group structure may degenerate or change character from fiber to fiber), and for capturing phenomena invisible over algebraically closed fields—such as infinitesimal group schemes in characteristic p.

This section develops the foundations of group schemes. The reader should be comfortable with fiber products (section 3.2) throughout.

4.1 Definition and First Examples

Recall that in any category $\mathcal{C}$ with finite products and a terminal object e, a *group object* is an object G equipped with morphisms

$$\mu : G \times G \to G, \qquad \epsilon : e \to G, \qquad \iota : G \to G$$

satisfying the usual group axioms—associativity, identity, and inverse—expressed as commutative diagrams. In the category **Set**, a group object is an ordinary group. In the category of smooth manifolds, a group object is a Lie group. In the category of schemes, a group object is a group scheme.

Definition 4.1.1: Group Scheme

Let S be a scheme. A *group scheme over S* is an S-scheme G together with S-morphisms

$$\mu : G \times_S G \longrightarrow G, \qquad \epsilon : S \longrightarrow G, \qquad \iota : G \longrightarrow G$$

called *multiplication*, *identity*, and *inversion*, such that the following diagrams commute:
Associativity:

$$\begin{tikzcd}
G \times_S G \times_S G \arrow[r, "\mu\times\mathrm{id}"] \arrow[d, "\mathrm{id}\times\mu"'] & G \times_S G \arrow[d, "\mu"] \\
G \times_S G \arrow[r, "\mu"'] & G
\end{tikzcd}$$

Identity:

$$\begin{tikzcd}
S \times_S G \arrow[r, "\epsilon\times\mathrm{id}"] \arrow[dr, "\sim"'] & G \times_S G \arrow[d, "\mu"] & G \times_S S \arrow[l, "\mathrm{id}\times\epsilon"'] \arrow[dl, "\sim"] \\
& G &
\end{tikzcd}$$

Inverse:

$$\begin{tikzcd}
G \arrow[r, "(\iota, \mathrm{id})"] \arrow[d, "(\mathrm{id}, \iota)"'] & G \arrow[d, "\mu"] & G \\
G \times_S G \arrow[r, "\mu"'] & G & S \arrow[l, "\epsilon"'] \arrow[u, "\epsilon\circ\pi"'] \arrow[llu, bend right, "\epsilon\circ\pi"']
\end{tikzcd}$$

where $\pi : G \to S$ is the structure morphism.
A group scheme is *commutative* (or *abelian*) if additionally $\mu \circ \sigma = \mu$, where $\sigma : G \times_S G \to G \times_S G$ is the swap morphism.

The identity morphism $\epsilon : S \to G$ is a section of the structure map $\pi : G \to S$. Its image is a closed subscheme of G when G is separated over S, but in general it need not be a closed point or even a closed subscheme—it is an S-valued point of G in the sense of section 3.3.2. Over a field k, however, $\epsilon : \mathrm{Spec}\,(k) \to G$ picks out a k-rational point, which is closed if G is of finite type over k.

Proposition 4.1.2: Group Schemes over a Field are Separated

Let G be a group scheme of finite type over a field k. Then G is separated over k.

We shall assume a proposition from proven later, the intuition that translated directly from topology is that separatedness is the equivalent of Hausdorffness, and a space is Hausdorff is the diagonal is closed.

***Proof*:**
We must show the diagonal $\Delta : G \to G \times_k G$ is a closed immersion (section 3.5.1). Consider the morphism $\varphi : G \times_k G \to G$ defined by $\varphi = \mu \circ (\mathrm{id} \times \iota)$, i.e., $(g, h) \mapsto g \cdot h^{-1}$. The diagonal $\Delta(G)$ is the fiber $\varphi^{-1}(\epsilon(\mathrm{Spec}\,(k)))$. Since ϵ is a section of a morphism of finite type (and k is a field), the image of ϵ is a closed point of G. The fiber of a morphism over a closed point is a closed subscheme of the source. Hence $\Delta(G)$ is closed.

Let us now examine the fundamental examples.

Example 4.1.3: The Additive Group Scheme
The *additive group scheme* $\mathbb{G}_{a,S} = \mathbb{A}^1_S$ has underlying scheme $\mathrm{Spec}\,_S(\mathcal{O}_S[t])$, with multiplication given by the ring map

$$\mathcal{O}_S[t] \longrightarrow \mathcal{O}_S[t] \otimes_{\mathcal{O}_S} \mathcal{O}_S[t], \qquad t \longmapsto t \otimes 1 + 1 \otimes t,$$

identity $t \mapsto 0$, and inversion $t \mapsto -t$. These encode, respectively, addition, zero, and negation: for any S-scheme T,

$$\mathbb{G}_a(T) = \Gamma(T, \mathcal{O}_T)$$

with its additive group structure. Over a field k, this is simply the additive group of the coordinate ring.

Example 4.1.4: The Multiplicative Group Scheme
The *multiplicative group scheme* $\mathbb{G}_{m,S} = \mathrm{Spec}\,_S(\mathcal{O}_S[t, t^{-1}])$ has multiplication

$$t \longmapsto t \otimes t,$$

identity $t \mapsto 1$, and inversion $t \mapsto t^{-1}$. For any S-scheme T:

$$\mathbb{G}_m(T) = \Gamma(T, \mathcal{O}_T)^\times,$$

the group of invertible global sections under multiplication.

Example 4.1.5: The General Linear Group Scheme
For a positive integer n, the *general linear group scheme* is

$$\mathrm{GL}_{n,S} = \mathrm{Spec}\,_S\big(\mathcal{O}_S[x_{11}, x_{12}, \dots, x_{nn}, d]/(d \cdot \det(x_{ij}) - 1)\big).$$

The variable d is the formal inverse of $\det(x_{ij})$, so the underlying scheme parametrizes $n \times n$ matrices with invertible determinant. The multiplication map is given by matrix multiplication: the ring map sends x_{ij} to $\sum_k x_{ik} \otimes x_{kj}$, and the identity sends $x_{ij} \mapsto \delta_{ij}$. For any S-scheme T:

$$\mathrm{GL}_n(T) \cong \mathrm{GL}_n(\Gamma(T, \mathcal{O}_T)),$$

the group of invertible $n \times n$ matrices with entries in $\Gamma(T, \mathcal{O}_T)$. When $n = 1$, we recover $\mathrm{GL}_1 \cong \mathbb{G}_m$.

Example 4.1.6: Roots of Unity

For a positive integer n, the *group scheme of n-th roots of unity* is

$$\mu_n = \operatorname{Spec}(\mathbb{Z}[t]/(t^n - 1)).$$

The group structure is inherited from $\mathbb{G}_m$: the comultiplication is $t \mapsto t \otimes t$ (the same as for $\mathbb{G}_m$, but now $t^n = 1$). For any ring A:

$$\mu_n(A) = \{a \in A \mid a^n = 1\} \subseteq A^\times.$$

The behavior of μ_n depends dramatically on the characteristic of the base. Over a field k with $\operatorname{char}(k) \nmid n$, the polynomial $t^n - 1$ has n distinct roots in $\overline{k}$, and $\mu_{n,k}$ is a reduced scheme of degree n over k—an étale group scheme. Over $\overline{k}$, it is isomorphic (as a scheme) to the disjoint union of n copies of $\operatorname{Spec}(\overline{k})$, and $\mu_n(\overline{k}) \cong \mathbb{Z}/n\mathbb{Z}$.

In characteristic $p > 0$ with $p \big| n$, the situation changes. Taking $n = p$, we have $t^p - 1 = (t-1)^p$ in $\mathbb{F}_p[t]$, so

$$\mu_{p,\mathbb{F}_p} = \operatorname{Spec}(\mathbb{F}_p[t]/((t-1)^p)) = \operatorname{Spec}(\mathbb{F}_p[s]/(s^p))$$

where $s = t - 1$. This is a *non-reduced* scheme: a single fat point at $t = 1$ with a p-dimensional tangent space. The group $\mu_p(\mathbb{F}_p) = \{1\}$ is trivial, yet the scheme μ_p is non-trivial—it carries nilpotent structure that remembers the "ghost" of the p-th roots of unity that have collapsed to 1.

Example 4.1.7: Constant Group Schemes

Let Γ be a finite (abstract) group. The *constant group scheme* associated to Γ over a base S is the scheme

$$\Gamma_S = \coprod_{\gamma \in \Gamma} S,$$

a disjoint union of copies of S indexed by elements of Γ. The group structure is defined by declaring that the component indexed by γ times the component indexed by γ' lands in the component indexed by $\gamma\gamma'$. For any connected S-scheme T:

$$\Gamma_S(T) = \Gamma.$$

The constant group scheme $(\mathbb{Z}/n\mathbb{Z})_k$ over a field k is always étale (ref:HERE) and reduced, regardless of the characteristic. It should be carefully distinguished from μ_n: over $\mathbb{F}_p$, the scheme $(\mathbb{Z}/p\mathbb{Z})_{\mathbb{F}_p}$ consists of p disjoint points, while $\mu_{p,\mathbb{F}_p}$ is a single fat point.

Example 4.1.8: Elliptic Curves as Group Schemes

An elliptic curve E over a field k (see section 7.1.7) is a smooth projective curve of genus 1 equipped with a k-rational point $O \in E(k)$. The chord-and-tangent construction endows E with a group scheme structure: the multiplication is the addition law on points, $\epsilon = O$, and ι is the map $P \mapsto -P$. Unlike the previous examples, E is *not* affine—it is a projective group scheme. Commutative projective group schemes (of finite type over a field) are called *abelian varieties*; elliptic curves are abelian varieties of dimension 1.

4.1.1 The Hopf Algebra Perspective

The examples above were all defined by specifying ring maps encoding multiplication, identity, and inversion. For affine group schemes, this algebraic data has a clean formalization: the coordinate ring carries the structure of a *Hopf algebra*. This perspective is particularly useful for explicit computations and for connecting group schemes to representation theory.

Let $G = \operatorname{Spec}(A)$ be an affine group scheme over a field k. The group operations on G dualize to k-algebra maps on A:

$$\begin{aligned} \mu : G \times_k G \to G \quad &\rightsquigarrow \quad \Delta : A \to A \otimes_k A \qquad &&\text{(comultiplication)}\\ \epsilon : \operatorname{Spec}(k) \to G \quad &\rightsquigarrow \quad \varepsilon : A \to k &&\text{(counit)}\\ \iota : G \to G \quad &\rightsquigarrow \quad S : A \to A &&\text{(antipode)} \end{aligned}$$

The group axioms translate directly:

Definition 4.1.9: Hopf Algebra

A *commutative Hopf algebra* over k is a commutative k-algebra A equipped with k-algebra homomorphisms $\Delta : A \to A \otimes_k A$ (comultiplication) and $\varepsilon : A \to k$ (counit), and a k-algebra map $S : A \to A$ (antipode), satisfying:

1. *Coassociativity:* $(\Delta \otimes \mathrm{id}) \circ \Delta = (\mathrm{id} \otimes \Delta) \circ \Delta$.
2. *Counit:* $(\varepsilon \otimes \mathrm{id}) \circ \Delta = \mathrm{id} = (\mathrm{id} \otimes \varepsilon) \circ \Delta$.
3. *Antipode:* $\mu_A \circ (S \otimes \mathrm{id}) \circ \Delta = \varepsilon \cdot 1_A = \mu_A \circ (\mathrm{id} \otimes S) \circ \Delta$,
   where $\mu_A : A \otimes_k A \to A$ is the multiplication of A.

The coassociativity and counit axioms say that (A, Δ, ε) is a *coalgebra*; the antipode axiom says that S "undoes" the comultiplication in the same way that inversion undoes multiplication in a group. The key constraint is that Δ and ε must be *algebra* homomorphisms—this encodes the fact that multiplication $\mu : G \times G \to G$ is a morphism of schemes, not merely a set-theoretic map.

Proposition 4.1.10: Affine Group Schemes and Hopf Algebras

The functor $G \mapsto \mathcal{O}(G)$ is an anti-equivalence of categories:

$$\{\text{Affine group schemes over } k\}^{\mathrm{op}} \xrightarrow{\sim} \{\text{Commutative Hopf algebras over } k\}.$$

***Proof*:**
This follows from the anti-equivalence between affine k-schemes and commutative k-algebras. An affine group scheme is a group object in the category of affine k-schemes; dualizing, its coordinate ring becomes a cogroup object in k-**Alg**. The axioms of a cogroup object in a category of commutative algebras are exactly the axioms of a commutative Hopf algebra.

Let us revisit the earlier examples through this lens.

Example 4.1.11: Hopf Algebras

1. *The additive group.* The Hopf algebra of $\mathbb{G}_a = \operatorname{Spec}(k[t])$ is $k[t]$ with $\Delta(t) = t \otimes 1 + 1 \otimes t$, $\varepsilon(t) = 0$, and $S(t) = -t$. In the language of coalgebras, t is a *primitive* element: $\Delta(t) = t \otimes 1 + 1 \otimes t$.

2. *The multiplicative group.* The Hopf algebra of $\mathbb{G}_m = \operatorname{Spec}(k[t, t^{-1}])$ is $k[t, t^{-1}]$ with $\Delta(t) = t \otimes t$, $\varepsilon(t) = 1$, and $S(t) = t^{-1}$. Here t is a *group-like* element: $\Delta(t) = t \otimes t$.

3. *The general linear group.* The Hopf algebra of GL_n is $k[x_{ij}, \det^{-1}]$ with $\Delta(x_{ij}) = \sum_k x_{ik} \otimes x_{kj}$ (matrix multiplication), $\varepsilon(x_{ij}) = \delta_{ij}$ (the identity matrix), and $S(x_{ij})$ given by the (i,j)-cofactor of (x_{ij}) divided by det—this is the Cramer's rule formula for the entries of the inverse matrix.

4. *Roots of unity.* The Hopf algebra of μ_n is $k[t]/(t^n - 1)$ with the same comultiplication as $\mathbb{G}_m$. The quotient by the ideal $(t^n - 1)$ is compatible with the Hopf structure because $\Delta(t^n - 1) = t^n \otimes t^n - 1 \otimes 1 = (t^n - 1) \otimes t^n + 1 \otimes (t^n - 1)$, which lies in the ideal generated by $t^n - 1$ in $A \otimes A$. In Hopf algebra language, $(t^n - 1)$ is a *Hopf ideal*.

The Hopf algebra viewpoint makes certain constructions transparent. For instance, a *homomorphism* of group schemes $\varphi : G \to H$ corresponds to a homomorphism of Hopf algebras $\varphi^* : \mathcal{O}(H) \to \mathcal{O}(G)$ (respecting Δ, ε, and S). The kernel of φ is the group scheme whose Hopf algebra is $\mathcal{O}(G)/(\ker \varepsilon_H \cdot \mathcal{O}(G))$; we will formalize this in section 4.2.

4.1.12 Remark: non-commutative Hopf algebras If we drop the requirement that A be commutative, we obtain the notion of a *non-commutative Hopf algebra*, which corresponds to a "quantum group" rather than a group scheme. The theory of quantum groups, developed by Drinfeld and Jimbo, is a rich subject in its own right but lies outside our scope. In this text, all Hopf algebras are commutative (corresponding to the fact that the coordinate ring of an affine scheme is always commutative), though the group schemes themselves need not be.

4.1.2 Finite and Infinitesimal Group Schemes

Over an algebraically closed field of characteristic zero, the theory of affine algebraic groups is largely controlled by Lie theory: connected algebraic groups are smooth, their structure is determined by their Lie algebras, and finite group schemes are just finite groups viewed as zero-dimensional varieties. In characteristic $p > 0$, the situation is fundamentally richer: there exist group schemes that are "purely infinitesimal"—non-reduced, supported at a single point, yet carrying nontrivial group structure encoded in the nilpotent elements of their coordinate ring.

Definition 4.1.13: Finite Group Scheme

A group scheme G over a field k is *finite* if it is of the form $G = \operatorname{Spec}(A)$ where A is a finite-dimensional k-algebra. The *order* of G is $\dim_k A$.

A finite group scheme of order n has at most n geometric points, but may have fewer if A has nilpotents. The extreme cases are:

Étale group schemes. A finite group scheme is *étale* if its coordinate ring A is a product of separable field extensions of k. Over an algebraically closed field, an étale group scheme of order n is simply a disjoint union of n copies of $\mathrm{Spec}\,(k)$, and the group structure is that of an ordinary finite group. The constant group scheme $(\mathbb{Z}/n\mathbb{Z})_k$ is the prototypical example.

Infinitesimal (or connected) group schemes. A finite group scheme $G = \mathrm{Spec}\,(A)$ is *infinitesimal* if A is a local ring, i.e., G is supported at a single point. The group structure is then entirely encoded in the nilpotent elements of A. These exist only in positive characteristic (or over non-reduced bases).

We have already seen the simplest examples of both types.

Example 4.1.14: μ_p in Characteristic p

As computed in Example 4.1.6, over $\mathbb{F}_p$ the group scheme $\mu_p = \mathrm{Spec}\,(\mathbb{F}_p[t]/(t^p-1)) = \mathrm{Spec}\,(\mathbb{F}_p[s]/(s^p))$ (where $s = t-1$) is infinitesimal of order p. Its Hopf algebra structure (expressed in terms of s) is:

$$\Delta(s) = s\otimes 1 + 1\otimes s + s\otimes s, \qquad \varepsilon(s) = 0, \qquad S(s) = \sum_{i=1}^{p-1}(-1)^i s^i.$$

The comultiplication is neither purely primitive ($s\otimes 1 + 1\otimes s$) nor purely group-like ($s\otimes s$)—it is a mixture, reflecting the fact that μ_p is a "twisted" infinitesimal group.

Example 4.1.15: The Frobenius Kernel α_p

The group scheme $\alpha_p = \mathrm{Spec}\,(k[t]/(t^p))$, defined over a field k of characteristic $p>0$, is the kernel of the Frobenius endomorphism on $\mathbb{G}_a$. Its Hopf algebra is $k[t]/(t^p)$ with

$$\Delta(t) = t\otimes 1 + 1\otimes t, \qquad \varepsilon(t) = 0, \qquad S(t) = -t.$$

Note that t is primitive, just as in $\mathbb{G}_a$, but the relation $t^p = 0$ (rather than no relation) makes α_p a finite group scheme of order p. For any k-algebra A:

$$\alpha_p(A) = \{a\in A \mid a^p = 0\},$$

which is a subgroup of $(A,+)$. Over a perfect field, α_p has only the trivial k-point, yet it is a non-trivial group scheme: for instance, $\alpha_p(k[\epsilon]/(\epsilon^p)) = k\cdot\epsilon$ is a p-element group.

The schemes α_p and μ_p are both infinitesimal group schemes of order p, but they are *not* isomorphic: α_p is connected and has no non-trivial characters (homomorphisms to $\mathbb{G}_m$), while μ_p is the Cartier dual of $\mathbb{Z}/p\mathbb{Z}$ and detects p-torsion in the multiplicative group.

The contrast between α_p and μ_p illustrates a general structural result. Over a perfect field k of characteristic $p>0$, every finite group scheme G admits a canonical filtration

$$0 \;\subseteq\; G^0 \;\subseteq\; G$$

where G^0 is the *connected component of the identity* (the largest connected subgroup scheme) and G/G^0 is étale. This is the *connected-étale sequence*:

$$1 \longrightarrow G^0 \longrightarrow G \longrightarrow G^{\text{ét}} \longrightarrow 1,$$

and over a perfect field this sequence splits (non-canonically), so $G \cong G^0\times G^{\text{ét}}$. The connected part G^0 is an infinitesimal group scheme (its coordinate ring is local), and the étale part $G^{\text{ét}}$ is determined by the action of $\mathrm{Gal}(\overline{k}/k)$ on the geometric points $G(\overline{k})$.

This decomposition has no analogue in characteristic zero, where every finite group scheme over a field is étale. It is a manifestation of the richer arithmetic of positive characteristic, and plays a fundamental role in the theory of abelian varieties (where the p-torsion $A[p]$ of an abelian variety splits into connected and étale parts whose relative sizes determine the p-rank).

The Frobenius and Verschiebung

The characteristic-p phenomena above are organized by two fundamental endomorphisms. Let k be a perfect field of characteristic p, and let G be a group scheme over k.

The *absolute Frobenius* $F_{\text{abs}} : G \to G$ is the identity on the underlying topological space and acts as $a \mapsto a^p$ on the structure sheaf. This is a morphism of schemes but *not* a morphism of k-schemes (it does not commute with the structure map unless $k = \mathbb{F}_p$). To obtain a k-morphism, one passes to the *relative Frobenius* $F : G \to G^{(p)}$, where $G^{(p)}$ is the base change of G along the Frobenius $\operatorname{Spec}(k) \to \operatorname{Spec}(k)$. The relative Frobenius is a homomorphism of group schemes.

For a commutative group scheme, there is a dual morphism $V : G^{(p)} \to G$ called the *Verschiebung* (German for “shift”). It satisfies $V \circ F = [p]_G$ and $F \circ V = [p]_{G^{(p)}}$, where $[p]$ denotes multiplication by p in the group law. The kernels $\ker(F)$ and $\ker(V)$ are finite group schemes that capture, respectively, the “infinitesimal” and “étale” parts of the p-torsion.

For example, $\ker(F : \mathbb{G}_a \to \mathbb{G}_a^{(p)}) = \alpha_p$ (since F acts as $t \mapsto t^p$ on $k[t]$, and $\ker = \operatorname{Spec}(k[t]/(t^p))$), while $\ker(F : \mathbb{G}_m \to \mathbb{G}_m^{(p)}) = \mu_p$.

4.2 Homomorphisms, Kernels, and Exact Sequences

Definition 4.2.1: Homomorphism of Group Schemes

A *homomorphism* of S-group schemes $\varphi : G \to H$ is a morphism of S-schemes that is compatible with the group structures:

$$\varphi \circ \mu_G = \mu_H \circ (\varphi \times \varphi)$$

, $\varphi \circ \epsilon_G = \epsilon_H$, and $\varphi \circ \iota_G = \iota_H \circ \varphi$.

In the functor-of-points language, φ is a homomorphism if and only if for every S-scheme T, the induced map $\varphi(T) : G(T) \to H(T)$ is a group homomorphism.

Definition 4.2.2: Kernel of a Homomorphism

Let $\varphi : G \to H$ be a homomorphism of S-group schemes. The *kernel* of φ is the fiber product

$$\ker(\varphi) = G \times_H S$$

where the two maps to H are $\varphi : G \to H$ and $\epsilon_H : S \to H$.

By the universal property of fiber products, for any S-scheme T:

$$\ker(\varphi)(T) = \{g \in G(T) \mid \varphi(T)(g) = e_H \in H(T)\},$$

which is exactly the kernel of the group homomorphism $\varphi(T) : G(T) \to H(T)$. The kernel $\ker(\varphi)$ is a closed subgroup scheme of G (being a fiber of a morphism over a section).

Example 4.2.3: Classical Exact Sequences

1. *The n-th power map on $\mathbb{G}_m$.* The homomorphism $[n] : \mathbb{G}_m \to \mathbb{G}_m$ defined by $t \mapsto t^n$ has kernel μ_n. This gives a short exact sequence of group schemes:

$$1 \longrightarrow \mu_n \longrightarrow \mathbb{G}_m \xrightarrow{(\ )^n} \mathbb{G}_m \longrightarrow 1.$$

   This is the *Kummer sequence*. Over a field k with $\mathrm{char}(k) \nmid n$, passing to $\overline{k}$-points recovers the familiar exact sequence $1 \to \mu_n(\overline{k}) \to \overline{k}^\times \xrightarrow{n} \overline{k}^\times \to 1$.

2. *The determinant.* The determinant $\det : \mathrm{GL}_n \to \mathbb{G}_m$ is a homomorphism of group schemes with kernel SL_n:

$$1 \longrightarrow \mathrm{SL}_n \longrightarrow \mathrm{GL}_n \xrightarrow{\det} \mathbb{G}_m \longrightarrow 1.$$

3. *The Frobenius on $\mathbb{G}_a$ in characteristic p.* The Frobenius $F : \mathbb{G}_a \to \mathbb{G}_a$ defined by $t \mapsto t^p$ has kernel α_p:

$$0 \longrightarrow \alpha_p \longrightarrow \mathbb{G}_a \xrightarrow{F} \mathbb{G}_a \longrightarrow 0.$$

   This is the *Artin–Schreier sequence* (in its scheme-theoretic form).

A word of caution on exactness is in order. A sequence $1 \to N \xrightarrow{i} G \xrightarrow{\varphi} H \to 1$ of group schemes is said to be *exact* if: i identifies N with $\ker(\varphi)$ as a closed subgroup scheme of G, and φ is faithfully flat. The surjectivity of φ is *not* the naive statement that $\varphi(T) : G(T) \to H(T)$ is surjective for all T; this is often false even when the sequence is exact. For instance, in the Kummer sequence over $\mathbb{Q}$, the map $\mathbb{Q}^\times \xrightarrow{n} \mathbb{Q}^\times$ is not surjective (not every rational number is an n-th power), but the map of schemes $\mathbb{G}_m \xrightarrow{n} \mathbb{G}_m$ is faithfully flat. The correct notion of surjectivity for group schemes is faithfully flat descent, not surjectivity on points.

4.3 Actions of Group Schemes on Schemes

Definition 4.3.1: Action of a Group Scheme

Let G be an S-group scheme and X an S-scheme. An *action* of G on X is an S-morphism

$$\sigma : G \times_S X \longrightarrow X$$

satisfying the usual axioms: the diagram

$$\begin{tikzcd} G \times_S G \times_S X \arrow[r, "\mathrm{id}\times\sigma"] \arrow[d, "\mu\times\mathrm{id}"'] & G \times_S X \arrow[d, "\sigma"] \\ G \times_S X \arrow[r, "\sigma"'] & X \end{tikzcd}$$

commutes (associativity), and $\sigma \circ (\epsilon \times \mathrm{id}_X) = \mathrm{id}_X$ (the identity acts trivially).

In the functor-of-points language, this means: for every S-scheme T, the group $G(T)$ acts on the set $X(T)$, and these actions are compatible with base change.

Example 4.3.2: Standard Actions

1. *Translation.* Any group scheme G acts on itself by left multiplication: $\sigma(g, h) = g \cdot h$. This is the *left regular action.* Each element $g \in G(T)$ defines an automorphism $\lambda_g : G_T \to G_T$ called *left translation by g.*

2. *Conjugation.* G acts on itself by conjugation: $\sigma(g, h) = ghg^{-1}$. This is the *adjoint action* of G on itself. Its derivative at the identity gives the adjoint representation of $\mathfrak{g}$ (the Lie algebra of G) on itself; see section 4.5.

3. *Linear representations.* An action of G on $\mathbb{A}^n_S$ that is $\mathcal{O}_S$-linear corresponds to a homomorphism $G \to \mathrm{GL}_{n,S}$. This is the scheme-theoretic notion of a *linear representation.*

4. *$\mathbb{G}_m$-actions and gradings.* An action of $\mathbb{G}_m$ on an affine scheme $\operatorname{Spec}(A)$ corresponds to a $\mathbb{Z}$-grading on A: $A = \bigoplus_{d\in\mathbb{Z}} A_d$. The $\mathbb{G}_m$-equivariant morphisms are the graded ring homomorphisms. This is the algebraic origin of the weight decompositions ubiquitous in representation theory.

Definition 4.3.3: Orbits and Stabilizers

Let G act on X and let $x \in X(T)$ be a T-valued point. The *orbit map* is

$$\sigma_x : G_T \longrightarrow X_T, \qquad g \longmapsto g \cdot x.$$

The *stabilizer* (or *isotropy group*) of x is the fiber product

$$G_x = G_T \times_{X_T} T$$

where the two maps to X_T are $\sigma_x : G_T \to X_T$ and $x : T \to X_T$. For any T-scheme T':

$$G_x(T') = \{g \in G(T') \mid g \cdot x = x \in X(T')\}.$$

The stabilizer is a closed subgroup scheme of G_T.

4.4 Quotients by Group Actions

Having defined actions, we now ask: given an action $\sigma : G \times_S X \to X$, does the "quotient" X/G exist as a scheme? This question is considerably more subtle than its set-theoretic counterpart, and its resolution leads to geometric invariant theory (GIT).

The categorical quotient

The most basic notion is that of a *categorical quotient*. We want a scheme Y and a G-invariant morphism $\pi : X \to Y$ (meaning $\pi \circ \sigma = \pi \circ \mathrm{pr}_2$, where $\mathrm{pr}_2 : G \times X \to X$ is the second projection) that is universal: any G-invariant morphism $X \to Z$ factors uniquely through π.

Definition 4.4.1: Categorical and Geometric Quotients

Let G act on X over S. A *categorical quotient* is an S-scheme Y with a G-invariant morphism $\pi : X \to Y$ such that:

1. (Universal property) For every G-invariant morphism $f : X \to Z$, there exists a unique morphism $\overline{f} : Y \to Z$ with $f = \overline{f} \circ \pi$.
2. (Coinvariant ring, in the affine case) If $X = \mathrm{Spec}\,(A)$ and G acts on A by ring automorphisms, then $Y = \mathrm{Spec}\,(A^G)$ where A^G denotes the subring of G-invariants.

A categorical quotient $\pi : X \to Y$ is a *geometric quotient* if additionally:

(3) π is surjective,

(4) the fibers of π are exactly the G-orbits, and

(5) π is submersive (a subset $U \subseteq Y$ is open if and only if $\pi^{-1}(U)$ is open in X).

A geometric quotient, when it exists, is the best possible outcome: the scheme Y faithfully represents the orbit space. However, geometric quotients frequently fail to exist—the simplest example being the action of $\mathbb{G}_m$ on $\mathbb{A}^2$ by scalar multiplication, where the origin is a "bad" orbit (it is the only closed orbit of dimension less than 1).

The GIT approach

Geometric invariant theory, developed by Mumford, provides a systematic method for constructing quotients by reductive group actions on projective or quasi-projective varieties. The central idea is to work not with the full quotient, but with a *good quotient* obtained by restricting to "semistable" points.

Let G be a reductive group acting on a projective variety $X \subseteq \mathbb{P}^n$ via a linearization (an action that lifts to the homogeneous coordinate ring). The *semistable locus* X^{ss} consists of those points $x \in X$ for which there exists a G-invariant section $f \in \Gamma(X, \mathcal{O}_X(d))^G$ (for some $d > 0$) with $f(x) \neq 0$. The *stable locus* $X^s \subseteq X^{ss}$ consists of those semistable points with finite stabilizer and closed orbit in X^{ss}.

The fundamental theorem of GIT asserts that the categorical quotient $X^{ss} \to X^{ss}//G$ exists as a projective variety, and its restriction to the stable locus $X^s \to X^s/G$ is a geometric quotient. The notation "// " (double slash) is reserved for the GIT quotient to distinguish it from the naive set-theoretic quotient.

A thorough treatment of GIT lies outside the scope of this text, but we record the most important special that we use in theorem 7.1.38 to bound the number of automorphisms of most curves:

Proposition 4.4.2: Quotient by a Finite Group

Let G be a finite group acting on an affine scheme $X = \operatorname{Spec}(A)$ by ring automorphisms. Then the categorical quotient $X/G = \operatorname{Spec}(A^G)$ exists, and the morphism $\pi : X \to X/G$ is finite and surjective. If $|G|$ is invertible in A, then π is a geometric quotient and $X \to X/G$ is étale over the locus of points with trivial stabilizer.

***Proof*:**
The ring of invariants A^G is a subring of A, and A is integral over A^G by the theorem on integral extensions (each $a \in A$ satisfies the monic polynomial $\prod_{g \in G}(T - g \cdot a) \in A^G[T]$). Hence $\pi : \operatorname{Spec}(A) \to \operatorname{Spec}(A^G)$ is finite and surjective. The universal property of $\operatorname{Spec}(A^G)$ as a categorical quotient follows from the fact that G-invariant maps $A \to B$ factor uniquely through A^G. When $|G|$ is invertible, the Reynolds operator $R = \frac{1}{|G|}\sum_{g \in G} g$ is a projection $A \to A^G$, and the étaleness statement follows from the Jacobian criterion applied to the map π away from the ramification locus.

4.5 The Lie Algebra of a Group Scheme

In the theory of Lie groups, the tangent space at the identity carries the structure of a Lie algebra, and this infinitesimal invariant determines the local structure of the group. The same construction works for group schemes, using the Zariski tangent space from section 2.7. (In the following section, we will see that this tangent space admits a clean reformulation via dual-number valued points; see Example 3.3.6.)

Let G be a group scheme of finite type over a field k, with identity element $e \in G(k)$. The Zariski tangent space at e is:

$$T_eG = (\mathfrak{m}_e/\mathfrak{m}_e^2)^\vee$$

where $\mathfrak{m}_e$ is the maximal ideal of $\mathcal{O}_{G,e}$.

Definition 4.5.1: Lie Algebra of a Group Scheme

The *Lie algebra* of G is the tangent space $\mathfrak{g} = T_eG$, equipped with a k-linear bracket $[\ ,\ ] : \mathfrak{g} \otimes_k \mathfrak{g} \to \mathfrak{g}$ defined via the adjoint action (specified below). We write $\mathfrak{g} = \mathrm{Lie}(G)$.

To define the bracket, observe that the group structure on G induces a group structure on the set of dual-number points:

$$G(k[\epsilon]/(\epsilon^2))$$

is a group via the multiplication μ. The kernel of the natural map $G(k[\epsilon]/(\epsilon^2)) \to G(k)$ (induced by $\epsilon \mapsto 0$) is exactly $\mathfrak{g}$, and it sits in a split short exact sequence of groups:

$$0 \longrightarrow \mathfrak{g} \longrightarrow G(k[\epsilon]/(\epsilon^2)) \longrightarrow G(k) \longrightarrow 1.$$

In this sequence, $\mathfrak{g}$ is an *abelian* subgroup (the product of two tangent vectors $v, w \in \mathfrak{g}$ under the group law of $G(k[\epsilon]/(\epsilon^2))$ is $v + w$, since the quadratic terms in ϵ vanish). However, the *commutator* of elements in $G(k[\epsilon_1, \epsilon_2]/(\epsilon_1^2, \epsilon_2^2, \epsilon_1\epsilon_2))$ is well-defined and lands in $\mathfrak{g}$; this commutator defines the Lie bracket $[v, w]$.

More concretely, the adjoint action $\mathrm{ad} : G \to \mathrm{Aut}(\mathfrak{g})$ is defined by conjugation: for $g \in G(T)$ and $v \in \mathfrak{g} \otimes_k \mathcal{O}(T)$, set $\mathrm{ad}(g)(v) = gvg^{-1}$ (computed in $G(T[\epsilon]/(\epsilon^2))$). Differentiating ad at the identity gives a map $\mathfrak{g} \to \mathrm{End}_k(\mathfrak{g})$, and the Lie bracket is $[v, w] = (\mathrm{d}_e\mathrm{ad})(v)(w)$.

Proposition 4.5.2: Lie Algebra as Primitive Elements

Let $G = \mathrm{Spec}\,(A)$ be an affine group scheme over k with Hopf algebra (A, Δ, ε) and augmentation ideal $I = \ker(\varepsilon)$. Then:

$$\mathfrak{g} = (I/I^2)^\vee$$

as a k-vector space, and the Lie bracket corresponds to the commutator in the convolution algebra $\mathrm{Hom}_k(A, k)$. Equivalently, $\mathfrak{g}$ is the space of *primitive elements* of the dual coalgebra: those $\delta \in A^\vee$ satisfying $\delta(ab) = \varepsilon(a)\delta(b) + \delta(a)\varepsilon(b)$ for all $a, b \in A$ (i.e., δ is a derivation $A \to k$ at ε).

***Proof*:**
By the Hopf algebra structure, the augmentation ideal $I = \ker(\varepsilon)$ is the maximal ideal at the identity, so $(I/I^2)^\vee$ is the Zariski tangent space at e as expected. A k-linear functional $\delta : A \to k$ vanishing on $k \cdot 1$ corresponds to an element of $I^\vee$; it descends to $(I/I^2)^\vee$ if and only if $\delta(I^2) = 0$, which is equivalent to the derivation (Leibniz) rule $\delta(ab) = \varepsilon(a)\delta(b) + \delta(a)\varepsilon(b)$.

Example 4.5.3: Lie Algebras of Classical Group Schemes

1. $\mathbb{G}_a$*:* The augmentation ideal of $k[t]$ (with $\varepsilon(t) = 0$) is (t), and $(t)/(t^2) \cong k$. Thus $\mathrm{Lie}(\mathbb{G}_a) = k$ with the zero bracket—the Lie algebra is one-dimensional and abelian.

2. $\mathbb{G}_m$: The augmentation ideal of $k[t, t^{-1}]$ (with $\varepsilon(t) = 1$) is $(t-1)$. Setting $s = t-1$, we have $(s)/(s^2) \cong k$, so $\mathrm{Lie}(\mathbb{G}_m) = k$ with the zero bracket—again one-dimensional and abelian.

3. GL_n: The augmentation ideal of $k[x_{ij}, \det^{-1}]$ (with $\varepsilon(x_{ij}) = \delta_{ij}$) is generated by the elements $e_{ij} := x_{ij} - \delta_{ij}$. The quotient I/I^2 is freely generated by the images of the e_{ij}, so $\mathrm{Lie}(\mathrm{GL}_n) \cong M_n(k)$, the space of all $n \times n$ matrices. The Lie bracket is the matrix commutator $[A, B] = AB - BA$, which recovers the classical Lie algebra $\mathfrak{gl}_n$.

4. SL_n: The additional relation $\det(x_{ij}) = 1$ imposes the linear condition $\mathrm{tr}(e_{ij}) = 0$ at the level of I/I^2. Hence $\mathrm{Lie}(\mathrm{SL}_n) = \{A \in M_n(k) : \mathrm{tr}(A) = 0\} = \mathfrak{sl}_n$.

5. μ_n *over a field with* $\mathrm{char}(k) \nmid n$: The augmentation ideal of $k[t]/(t^n - 1)$ (with $\varepsilon(t) = 1$) is $I = (t-1)$. Writing $t^n - 1 = (t-1)(t^{n-1} + \cdots + 1)$, and noting that $t^{n-1} + \cdots + 1$ evaluates to $n \neq 0$ at $t = 1$ (hence is coprime to $t-1$), we find that $(t-1)^2$ and $t^n - 1$ together generate $(t-1)$, so $I/I^2 = 0$. Thus $\mathrm{Lie}(\mu_n) = 0$—the Lie algebra is trivial. This reflects the fact that μ_n is étale: it has no infinitesimal structure.

6. μ_p *in characteristic* p: Now $t^p - 1 = (t-1)^p$, so the coordinate ring is $k[s]/(s^p)$ with $s = t-1$, and $I/I^2 = (s)/(s^2) \cong k$. Hence $\mathrm{Lie}(\mu_p) = k$ is one-dimensional. The infinitesimal group scheme μ_p carries tangent information at its unique point, unlike its étale counterpart.

The contrast between the last two examples is worth emphasizing: μ_n for $\mathrm{char}(k) \nmid n$ is étale with trivial Lie algebra, while μ_p in characteristic p is infinitesimal with a one-dimensional Lie algebra. The Lie algebra "sees" the infinitesimal structure of the group scheme, and in characteristic p this structure is genuinely richer.

This observation foreshadows a general theme: in characteristic zero, the Lie algebra functor $G \mapsto \mathrm{Lie}(G)$ is an equivalence between connected algebraic groups and their Lie algebras (this is the scheme-theoretic incarnation of Lie's third theorem). In characteristic p, the Lie algebra loses information—it cannot distinguish between α_p and $\mathbb{G}_a$, for instance, since both have Lie $= k$. The correct replacement is the theory of *p-Lie algebras* (or *restricted Lie algebras*), which equips $\mathfrak{g}$ with an additional "p-th power" operation $x \mapsto x^{[p]}$ that captures the Frobenius structure. This theory, while important for the classification of algebraic groups in positive characteristic, lies beyond our present scope.

4.6 Cartier Duality

For finite abelian groups (or in functional analysis, a locally compact abelian group like $\mathbb{R}$ with the euclidean topology), Pontryagin duality states that a group A is isomorphic to its double dual $\mathrm{Hom}(\mathrm{Hom}(A, S^1), S^1)$. In the realm of finite commutative group schemes over a field k, there is a perfect geometric analogue where the circle group S^1 is replaced by the multiplicative group scheme $\mathbb{G}_m$.

Definition 4.6.1: Cartier Dual

Let G be a finite commutative group scheme over a field k. The *Cartier dual* of G, denoted G^D, is the group scheme representing the functor of homomorphisms from G to $\mathbb{G}_m$. That is, for any k-algebra R:

$$G^D(R) = \mathrm{Hom}_{\text{gp-sch}/R}(G_R, \mathbb{G}_{m,R}).$$

At first glance, it is not obvious that this functor is representable by a finite commutative group scheme. However, the Hopf algebra perspective makes the construction transparent and purely algebraic.

Let $G = \mathrm{Spec}\,(A)$. Since G is finite, A is a finite-dimensional k-vector space. We can form the dual vector space $A^\vee = \mathrm{Hom}_k(A, k)$. Because G is a commutative group scheme, its Hopf algebra A is both commutative and cocommutative (meaning $\sigma \circ \Delta = \Delta$, where σ swaps the tensor factors in $A \otimes_k A$).

Dualizing the structural maps of A perfectly swaps the algebra and coalgebra structures:

- The multiplication $A \otimes_k A \to A$ dualizes to a comultiplication $A^\vee \to A^\vee \otimes_k A^\vee$.
- The comultiplication $\Delta : A \to A \otimes_k A$ dualizes to a multiplication $A^\vee \otimes_k A^\vee \to A^\vee$.
- The unit $k \to A$ dualizes to a counit $A^\vee \to k$.
- The counit $\varepsilon : A \to k$ dualizes to a unit $k \to A^\vee$.
- The antipode $S : A \to A$ dualizes to an antipode $S^\vee : A^\vee \to A^\vee$.

Since A is commutative and cocommutative, $A^\vee$ is also a commutative and cocommutative Hopf algebra. Thus, it defines a new finite commutative affine group scheme.

Proposition 4.6.2: Cartier Duality Theorem

Let $G = \mathrm{Spec}\,(A)$ be a finite commutative group scheme over k. Then the Cartier dual G^D is represented by the affine group scheme $\mathrm{Spec}\,(A^\vee)$. Furthermore, the natural evaluation map gives an isomorphism:

$$G \xrightarrow{\sim} (G^D)^D.$$

Example 4.6.3: Cartier Duals of Core Examples

1. *Roots of unity and constant groups.* The most classical pairing is between the constant group scheme $(\mathbb{Z}/n\mathbb{Z})_k$ and the group of roots of unity μ_n. A homomorphism from $\mathbb{Z}/n\mathbb{Z}$ to $\mathbb{G}_m$ is completely determined by where it sends the generator $1 \in \mathbb{Z}/n\mathbb{Z}$, and this image must be an n-th root of unity. Hence, for any k-algebra R:

$$\mathrm{Hom}((\mathbb{Z}/n\mathbb{Z})_R, \mathbb{G}_{m,R}) \cong \mu_n(R).$$

   Thus, $((\mathbb{Z}/n\mathbb{Z})_k)^D \cong \mu_n$. By the duality theorem, we also have $\mu_n^D \cong (\mathbb{Z}/n\mathbb{Z})_k$.

2. *The Frobenius kernel α_p.* Let k be a perfect field of characteristic $p > 0$, and recall $\alpha_p = \mathrm{Spec}\,(k[t]/(t^p))$ with $\Delta(t) = t \otimes 1 + 1 \otimes t$. To find a homomorphism from α_p to

$\mathbb{G}_m = \operatorname{Spec}(k[u, u^{-1}])$, we must specify a group-like element in $k[t]/(t^p)$. The truncated exponential series:

$$E(t) = \sum_{i=0}^{p-1} \frac{t^i}{i!}$$

is well-defined in characteristic p (since we stop before dividing by $p!$) and satisfies $E(t+s) = E(t)E(s)$ in $k[t,s]/(t^p, s^p)$. This gives a Hopf algebra map $u \mapsto E(t)$, which defines an isomorphism $\alpha_p^D \cong \alpha_p$. Thus, unlike μ_p, the infinitesimal group scheme α_p is self-dual.

5

Schemes III: Sheaves of Modules — Projectives, Divisors, Differentials

So far, we have worked exclusively with the structure sheaf $\mathcal{O}_X$ of a scheme X. But just as the geometry of a smooth manifold M is enriched by studying vector bundles, principal bundles[1], and their sections, the geometry of a scheme is enriched by studying additional sheaves that carry an $\mathcal{O}_X$-module structure. Every vector bundle on M gives rise to a sheaf of sections that is a module over the ring of smooth functions $C^\infty(M)$; in the same way, every "bundle-like" object on a scheme gives rise to a sheaf of $\mathcal{O}_X$-modules.

The most important such sheaves are those that are *locally* isomorphic to the sheaf associated to a module: on each affine open $U_i = \operatorname{Spec} R_i$, the sheaf restricts to $\widetilde{M_i}$ for some R_i-module M_i. These are the *quasi-coherent sheaves*, and when the modules M_i satisfy a finite presentation condition they are called *coherent sheaves*[2].

Quasi-coherent sheaves can be understood as a natural extension of vector bundles. *Locally free sheaves* (the algebraic incarnation of vector bundles) are not closed under kernels and cokernels — the category of vector bundles is not abelian. The category $\mathbf{QCoh}(X)$ is obtained by adding precisely those kernels and cokernels; it is the smallest abelian subcategory of $\mathcal{O}_X$-modules containing the locally free sheaves, just as the category of all R-modules is the abelian completion of the category of free R-modules. This perspective is developed in section 5.1.3.

By working with quasi-coherent sheaves, we gain access to a wealth of new geometric objects: ideal

[1] more generally, gerbes and n-gerbes on manifolds, see ref:HERE

[2] The term "coherent" originates with Serre [3], who required sheaves to "cohere" (stick together) in a controlled way. The prefix "quasi-" signals that finite generation is dropped, at the cost of losing some finiteness results — for instance, quasi-coherent sheaves are not in general preserved under pushforward.

sheaves (encoding closed subschemes with scheme structure), twisted sheaves on projective space (from which one can recover the graded ring $S_\bullet$ of $\operatorname{Proj} S_\bullet$), skyscraper-like sheaves remembering multiplicities (generalizing example 1.1.12), and sheaves of differentials (from which we recover differential forms, tangent bundles, and smoothness criteria). The study of these objects occupies the present chapter and sets the stage for sheaf cohomology in chapter 6.

There shall be sheaves on schemes that are *not* bundle-like (not quasi-coherent), such as the sheaf given by taking all units at each local section. Therefore, we start with actions of abelian sheave on ringed spaces.

5.1 Definition and Basic Properties

We begin by sheafifying the notion of a module over a ring. If $(X, \mathcal{O}_X)$ is a ringed space, an $\mathcal{O}_X$-module is a sheaf of abelian groups on which $\mathcal{O}_X$ acts section-wise, compatibly with restriction.

Definition 5.1.1: $\mathcal{O}_X$-Module

Let $(X, \mathcal{O}_X)$ be a ringed space. An *$\mathcal{O}_X$-module* is a sheaf $\mathscr{F} \in \mathbf{Sh}_{\mathbf{Ab}}(X)$ equipped with the following data:

1. For each open $U \subseteq X$, the abelian group $\mathscr{F}(U)$ carries the structure of an $\mathcal{O}_X(U)$-module.
2. For each inclusion $U \subseteq V$ of open sets, the restriction map $\operatorname{res}_{V,U} : \mathscr{F}(V) \to \mathscr{F}(U)$ is compatible with the module structure:

$$\begin{tikzcd} \mathcal{O}_X(V) \times \mathscr{F}(V) \arrow[r, "\curvearrowright"] \arrow[d, "\operatorname{res}_{V,U} \times \operatorname{res}_{V,U}"'] & \mathscr{F}(V) \arrow[d, "\operatorname{res}_{V,U}"] \\ \mathcal{O}_X(U) \times \mathscr{F}(U) \arrow[r, "\curvearrowright"] & \mathscr{F}(U) \end{tikzcd}$$

A *morphism of $\mathcal{O}_X$-modules* $\varphi : \mathscr{F} \to \mathscr{G}$ is a morphism of sheaves of abelian groups such that for each open $U \subseteq X$, the map $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ is an $\mathcal{O}_X(U)$-module homomorphism. The collection of all such morphisms is denoted $\operatorname{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$, or simply $\operatorname{Hom}_X(\mathscr{F}, \mathscr{G})$ when the structure sheaf is understood.

The reader should think of $\mathcal{O}_X$-modules as the scheme-theoretic analogue of (sections of) fibre bundles: the structure sheaf plays the role of smooth functions, and the module structure encodes how sections transform under multiplication by functions.

Given a morphism of ringed spaces $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$, there are natural operations that transport $\mathcal{O}_X$-modules to $\mathcal{O}_Y$-modules and vice versa:

Definition 5.1.2: $\mathcal{O}_X$-module Pushforward and Pullback

Let $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces.

1. *(Direct image.)* If $\mathscr{F}$ is an $\mathcal{O}_X$-module, then the pushforward sheaf $f_*\mathscr{F}$ is naturally an $f_*\mathcal{O}_X$-module on Y. Via the sheaf map $f^\# : \mathcal{O}_Y \to f_*\mathcal{O}_X$, this acquires the structure of an $\mathcal{O}_Y$-module: for $V \subseteq Y$ open and $g \in \mathcal{O}_Y(V)$, $s \in f_*\mathscr{F}(V) = \mathscr{F}(f^{-1}(V))$, the action is $g \cdot s := f^\#(V)(g) \cdot s$.

2. *(Inverse image.)* If $\mathscr{G}$ is a $\mathcal{O}_Y$-module, then the sheaf $f^{-1}\mathscr{G}$ is an $f^{-1}\mathcal{O}_Y$-module on X. The *pullback* (or *inverse image $\mathcal{O}_X$-module*) is

$$f^*\mathscr{G} \;:=\; f^{-1}\mathscr{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X.$$

   The tensor product re-bases the module structure from $\mathcal{O}_Y$ to $\mathcal{O}_X$.

These operations form an adjoint pair: there is a natural isomorphism $\mathrm{Hom}_{\mathcal{O}_X}(f^*\mathscr{G}, \mathscr{F}) \cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathscr{G}, f_*\mathscr{F})$.

Example 5.1.3: General $\mathcal{O}_X$-Module Constructions

Fix a ringed space $(X, \mathcal{O}_X)$.

1. *The structure sheaf.* The sheaf $\mathcal{O}_X$ is an $\mathcal{O}_X$-module with the action given by ring multiplication. Since each restriction map is a ring homomorphism, compatibility with the module structure is automatic.

2. *Abelian groups as modules.* Just as every abelian group is a $\mathbb{Z}$-module, every sheaf of abelian groups $\mathscr{G}$ on X is a module over the constant sheaf $\underline{\mathbb{Z}}$. The category of $\mathcal{O}_X$-modules therefore generalizes $\mathbf{Sh_{Ab}}(X)$.

3. *Kernels, images, cokernels.* If $\varphi : \mathscr{F} \to \mathscr{G}$ is a morphism of $\mathcal{O}_X$-modules, then $\ker\varphi$, $\mathrm{im}\varphi$, and $\mathrm{coker}\varphi$ are all $\mathcal{O}_X$-modules (by the same arguments as for sheaves of abelian groups, since the module structure is inherited section-wise). The category of $\mathcal{O}_X$-modules is abelian, with products and coproducts computed section-wise.

4. *Sheaf Hom.* For $\mathcal{O}_X$-modules $\mathscr{F}, \mathscr{G}$, the assignment

$$U \longmapsto \mathrm{Hom}_{\mathcal{O}_X|_U}(\mathscr{F}|_U, \mathscr{G}|_U)$$

   defines a sheaf of $\mathcal{O}_X$-modules, the *sheaf Hom*, denoted $\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$.

5. *Tensor product.* The *tensor product* $\mathscr{F} \otimes_{\mathcal{O}_X} \mathscr{G}$ is the sheafification of the presheaf $U \mapsto \mathscr{F}(U) \otimes_{\mathcal{O}_X(U)} \mathscr{G}(U)$. Sheafification is essential here: the naive presheaf need not satisfy the gluing axiom, and the resulting sections can differ dramatically from the section-wise tensor product. See example 5.3.18 for a striking instance of this phenomenon.

The following result is a useful way

Proposition 5.1.4: Evaluation via Skyscraper Sheaves

Let X be a scheme and $x \in X$ a point. Let $\mathcal{O}_x$ denote the skyscraper sheaf at x with stalk $\kappa(x)$ (formally, the pushforward of $\mathcal{O}_{\text{Spec}\,(\kappa(x))}$ along the inclusion $i : \text{Spec}\,(\kappa(x)) \hookrightarrow X$). Then there is a natural isomorphism:

$$\text{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{O}_x) \;\cong\; \kappa(x).$$

***Proof*:**
For any sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$, morphisms from the structure sheaf $\mathcal{O}_X$ to $\mathcal{F}$ are naturally identified with the global sections of $\mathcal{F}$. Explicitly, a morphism $\varphi : \mathcal{O}_X \to \mathcal{F}$ is uniquely determined by the global section $\varphi(X)(1) \in \Gamma(X, \mathcal{F})$, giving an isomorphism:

$$\text{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{F}) \;\cong\; \Gamma(X, \mathcal{F})$$

Applying this general fact to the skyscraper sheaf $\mathcal{F} = \mathcal{O}_x$, we obtain:

$$\text{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{O}_x) \;\cong\; \Gamma(X, \mathcal{O}_x)$$

By the definition of the skyscraper sheaf, its global sections evaluate exactly to the stalk at the point x, which is the residue field $\kappa(x)$. Therefore:

$$\text{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{O}_x) \;\cong\; \kappa(x)$$

This proposition demonstrates that taking the global section of a function $f \in \Gamma(X, \mathcal{O}_X) \cong \text{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{O}_X)$ and composing it with the natural projection $\mathcal{O}_X \to \mathcal{O}_x$ precisely yields the evaluation $f(x) \in \kappa(x)$.

5.1.1 Quasi-Coherent and Coherent Sheaves

A natural hope is that, for an affine scheme $X = \text{Spec}\, R$, the category of $\mathcal{O}_X$-modules is equivalent to R-Mod. This turns out to be false: the category of $\mathcal{O}_X$-modules is too large.

Example 5.1.5: Too Many $\mathcal{O}_X$-modules Over Affine Schemes

For any such equivalence to work, the sheaf $\widetilde{M}$ associated to an R-module M would need to satisfy $\widetilde{M}_\mathfrak{p} \cong M_\mathfrak{p}$ at each prime $\mathfrak{p}$. We show that not every $\mathcal{O}_X$-module has this form.

Let $X = \text{Spec}\, \mathbb{Z}_{(2)}$, which has two points: the closed point (2) and the generic point $\eta = (0)$. The nontrivial open sets are $\{(2), \eta\} = X$ and $\{\eta\}$. The structure sheaf satisfies $\mathcal{O}_X(X) = \mathbb{Z}_{(2)}$ and $\mathcal{O}_X(\{\eta\}) = \mathbb{Q}$.

To specify an $\mathcal{O}_X$-module $\mathscr{F}$, we must give a $\mathbb{Z}_{(2)}$-module $\mathscr{F}(X)$, a $\mathbb{Q}$-module $\mathscr{F}(\{\eta\})$, and a restriction (module) homomorphism $\mathscr{F}(X) \to \mathscr{F}(\{\eta\})$. Consider the choice

$$\mathscr{F}(X) = \mathbb{Z}_{(2)}, \qquad \mathscr{F}(\{\eta\}) = 0, \qquad \mathbb{Z}_{(2)} \twoheadrightarrow 0.$$

This defines a valid $\mathcal{O}_X$-module. However, if $\mathscr{F} = \widetilde{M}$ for some $\mathbb{Z}_{(2)}$-module M, then $M = \Gamma(X, \widetilde{M}) = \mathbb{Z}_{(2)}$ and $\widetilde{M}(\{\eta\}) = M_{(0)} = M[1/2] = \mathbb{Q} \neq 0$. This contradicts $\mathscr{F}(\{\eta\}) = 0$, so $\mathscr{F}$ does not arise from any module.

The problem is that a general $\mathcal{O}_X$-module need not preserve the localization structure in compatible with the affine structure. The remedy is to restrict attention to $\mathcal{O}_X$-modules that are locally of the form $\widetilde{M}$. We first define the "affine building blocks."

Definition 5.1.6: Sheaf Associated to M

Let R be a ring and M an R-module. The *sheaf associated to M* on $\operatorname{Spec} R$ is the $\mathcal{O}_{\operatorname{Spec} R}$-module $\widetilde{M}$ defined on the basis of distinguished open sets by

$$\widetilde{M}(D(f)) \;=\; M_f \qquad \text{(the localization of } M \text{ at } f\text{)},$$

with the natural restriction maps induced by further localization. One checks (by the same argument as for the structure sheaf) that this defines a sheaf on $\operatorname{Spec} R$.

Since we now have multiple sheaves on a scheme, we adopt the standard notation $\Gamma(X, \mathscr{F})$ for the global sections of a sheaf $\mathscr{F}$ on X. In particular, $\Gamma(\operatorname{Spec} R, \widetilde{M}) \cong M$ and $\Gamma(X, \mathcal{O}_X) = \mathcal{O}_X(X)$.

Definition 5.1.7: Quasi-coherent Sheaf

Let $(X, \mathcal{O}_X)$ be a scheme. An $\mathcal{O}_X$-module $\mathscr{F}$ is *quasi-coherent* if X can be covered by affine open subsets $U_i = \operatorname{Spec} R_i$ such that for each i there exists an R_i-module M_i with

$$\mathscr{F}\big|_{U_i} \;\cong\; \widetilde{M}_i.$$

The full subcategory of quasi-coherent sheaves on X is denoted $\mathbf{QCoh}(X)$.

For the finitely generated analogue, a subtlety arises: the kernel of a map between finitely generated modules need not be finitely generated. For instance, the surjection $k[x_1, x_2, \ldots] \to k[x_1, x_2, \ldots]$ sending each x_i to 0 has kernel $(x_1, x_2, \ldots)$, which is not finitely generated. (Images and cokernels of maps between finitely generated modules *are* finitely generated, so this is specifically a kernel problem.) The correct notion must therefore be stronger than finite generation:

Definition 5.1.8: Coherent Module

Let R be a ring and M an R-module. Then M is *coherent* if it is finitely generated and for every $n \geq 1$, every R-module homomorphism $R^n \to M$ has finitely generated kernel.
A ring R is called *coherent* if it is coherent as a module over itself.

Coherence is strictly stronger than finite presentation: the latter only requires *some* surjection $R^n \twoheadrightarrow M$ to have finitely generated kernel, whereas coherence demands this for *all* maps $R^n \to M$. Over a Noetherian ring, coherence, finite presentation, and finite generation are all equivalent.

Definition 5.1.9: Coherent Sheaf

Let $(X, \mathcal{O}_X)$ be a scheme. An $\mathcal{O}_X$-module $\mathscr{F}$ is *coherent* if X can be covered by affine open subsets $U_i = \operatorname{Spec} R_i$ such that for each i there exists a coherent R_i-module M_i with $\mathscr{F}|_{U_i} \cong \widetilde{M}_i$.

5.1.10 Distinction from Hartshorne Hartshorne defines "coherent sheaf" to mean "sheaf associated to a finitely generated module," but then immediately restricts to Noetherian schemes where the distinction vanishes. Our definition agrees with the one used by EGA and most modern references: coherence is an intrinsic condition on the sheaf (finitely generated with finitely generated kernels of all maps from free sheaves), which coincides with Hartshorne's definition precisely when X is locally Noetherian.

A key feature of quasi-coherence is that it is *affine-local*: checking it on one affine cover suffices.

Proposition 5.1.11: Quasi-coherence is Affine-local

Let $(X, \mathcal{O}_X)$ be a scheme and $\mathscr{F}$ an $\mathcal{O}_X$-module. Then $\mathscr{F}$ is quasi-coherent if and only if for *every* affine open subset $U = \operatorname{Spec} R \subseteq X$, there is an isomorphism $\mathscr{F}|_U \cong \widetilde{M}$ for some R-module M.
If X is locally Noetherian, then $\mathscr{F}$ is coherent if and only if this holds with each M finitely generated (equivalently, coherent).

***Proof*:**
The "if" direction is immediate. For the "only if" direction, since the property is local and every affine scheme is covered by distinguished opens of any given affine cover, it suffices to prove the claim when $X = \operatorname{Spec} R$ is itself affine.

Let $M = \Gamma(\operatorname{Spec} R, \mathscr{F})$. We must show $\mathscr{F} \cong \widetilde{M}$. By hypothesis, there is an affine cover $\{D(g_i)\}$ of X with $\mathscr{F}|_{D(g_i)} \cong \widetilde{M_i}$ for some R_{g_i}-module M_i. Taking global sections over $D(g_i)$ gives $\mathscr{F}(D(g_i)) \cong M_i$. But $\mathscr{F}(D(g_i))$ is also obtained from M by localization at g_i: the restriction map $M = \mathscr{F}(X) \to \mathscr{F}(D(g_i))$ factors through M_{g_i}, and the quasi-coherence assumption on overlaps $D(g_i g_j)$ ensures this identification is consistent. Hence $M_i \cong M_{g_i}$, and by corollary 1.3.4 the natural map $\widetilde{M} \to \mathscr{F}$ is an isomorphism on a basis, hence an isomorphism of sheaves.

For the coherent case, if X is locally Noetherian then each R_{g_i} is Noetherian, so M_{g_i} is finitely generated if and only if it is coherent. The result follows by the same argument with the additional finite generation condition.

The affine-local property immediately yields the scheme-theoretic analogue of corollary 3.1.5:

Corollary 5.1.12: Equivalence of Categories: Module Version

If $X = \operatorname{Spec} R$, the functor $M \mapsto \widetilde{M}$ is an equivalence of categories:

$$\mathbf{QCoh}(\operatorname{Spec} R) \underset{\widetilde{(-)}}{\overset{\Gamma}{\rightleftarrows}} R\text{-Mod}$$

with inverse $\mathscr{F} \mapsto \Gamma(\operatorname{Spec} R, \mathscr{F})$. If R is Noetherian, this restricts to an equivalence between $\mathbf{Coh}(\operatorname{Spec} R)$ and finitely generated R-modules.
More generally, for any $\mathcal{O}_X$-module $\mathscr{F}$ on X and any R-module M, there is a natural adjunction isomorphism:

$$\operatorname{Hom}_{\mathcal{O}_X}(\widetilde{M}, \mathscr{F}) \cong \operatorname{Hom}_R(M, \Gamma(X, \mathscr{F})).$$

***Proof*:**
By proposition 5.1.11, every quasi-coherent sheaf on $\operatorname{Spec} R$ is of the form $\widetilde{M}$. Since $\Gamma(\operatorname{Spec} R, \widetilde{M}) \cong M$ and the tilde construction is functorial and exact, the two functors are inverse equivalences. The adjunction follows from the universal property of localization: a map $\widetilde{M} \to \mathscr{F}$ is determined by its value on global sections $M \to \Gamma(X, \mathscr{F})$, and conversely any such module map extends uniquely to a sheaf map by localization.

Example 5.1.13: Quasi-Coherent Sheaves

1. *The structure sheaf.* The structure sheaf $\mathcal{O}_X$ is quasi-coherent (even coherent): on each affine open $\operatorname{Spec} R$, it is $\widetilde{R}$. At each stalk, the $\mathcal{O}_{X,\mathfrak{p}}$-module $\mathcal{O}_{X,\mathfrak{p}} \cong R_{\mathfrak{p}}$ is free of rank one.

2. *Sheaves supported on a subvariety.* Let $X = \operatorname{Spec} k[x, y]$ and let $f \in k[x, y]$ be irreducible. The sheaf $\widetilde{k[x, y]/(f)}$ has stalks

$$\widetilde{k[x, y]/(f)}_{\mathfrak{p}} = \begin{cases} (k[x, y]/(f))_{\mathfrak{p}} = k[x, y]_{\mathfrak{p}}/(f)_{\mathfrak{p}} & \text{if } f \in \mathfrak{p}, \\ 0 & \text{if } f \notin \mathfrak{p}. \end{cases}$$

Geometrically, this is a "line bundle on $V(f)$, extended by zero to all of X." More generally, if $Y = V(\mathfrak{a}) \subseteq X$ is a closed subscheme defined by an ideal $\mathfrak{a} \subseteq R$, then the pushforward $\iota_* \mathcal{O}_Y$ is a quasi-coherent $\mathcal{O}_X$-module isomorphic to $\widetilde{R/\mathfrak{a}}$.

3. *Torsion modules over* $\operatorname{Spec} \mathbb{Z}$. Let $X = \operatorname{Spec} \mathbb{Z}$ and $M = \mathbb{Z}/12\mathbb{Z}$. The stalks of $\widetilde{M}$ are:

$$\widetilde{M}_{(2)} \cong \mathbb{Z}/4\mathbb{Z}, \qquad \widetilde{M}_{(3)} \cong \mathbb{Z}/3\mathbb{Z}, \qquad \widetilde{M}_{(p)} = 0 \text{ for } p \neq 2, 3, \qquad \widetilde{M}_{(0)} = 0.$$

The sheaf is supported on the two closed points (2) and (3), with "fibre dimensions" reflecting the prime factorization $12 = 2^2 \cdot 3$. More generally, if M is any torsion R-module, then $\widetilde{M}_{(0)} = 0$: torsion modules are not "spread" across the generic point.

4. *Eigenvalues and Jordan blocks.* Let $\alpha \in \operatorname{End}_{\mathbb{C}}(V)$ be an endomorphism of an n-dimensional $\mathbb{C}$-vector space, and view V as a $\mathbb{C}[x]$-module via $p(x) \cdot v = p(\alpha)(v)$. The associated sheaf $\widetilde{V}$ on $\mathbb{A}^1_{\mathbb{C}} = \operatorname{Spec} \mathbb{C}[x]$ is a quasi-coherent sheaf supported on the eigenvalues of α.

At a closed point λ, the stalk $\widetilde{V}_{(\lambda)} = V_{(x-\lambda)}$ is nonzero precisely when λ is an eigenvalue: if $\alpha - \lambda I$ is invertible, every element of V can be divided by $(x-\lambda)$, so the localization vanishes. If λ is an eigenvalue, the generalized eigenspace decomposition gives

$$\widetilde{V}_{(\lambda)} \cong \bigoplus_{j=1}^{r} \frac{\mathbb{C}[x]}{(x-\lambda)^{m_j}},$$

where $m_1, \ldots, m_r$ are the sizes of the Jordan blocks for λ. The sheaf $\widetilde{V}$ thus encodes the full Jordan normal form of α: the support is the spectrum of eigenvalues, and the stalk structure records the Jordan block sizes. One can make this even cleaner by descending to $\operatorname{Spec}(\mathbb{C}[x]/m_\alpha(x))$, where m_α is the minimal polynomial.

5. *Extension by zero is not quasi-coherent.* Let $U \subseteq X$ be an open subscheme of an integral scheme with inclusion $\iota : U \hookrightarrow X$. The sheaf $\iota_!\mathcal{O}_U$ (extension by zero) is an $\mathcal{O}_X$-module but is typically *not* quasi-coherent. Indeed, if $V = \operatorname{Spec} R$ is an affine open not contained in U but with $U \cap V \neq \emptyset$, then $\iota_!\mathcal{O}_U|_V$ has no nonzero global sections (since elements must vanish outside U) but is not the zero sheaf on V, and hence cannot equal $\widetilde{M}$ for any R-module M.

6. *Restriction vs. pushforward.* If Y is a closed subscheme of X, the restriction $\mathcal{O}_X|_Y$ is typically *not* an $\mathcal{O}_Y$-module. For example, let $X = \operatorname{Spec} k[x,y]$ and $Y = \operatorname{Spec} k[x,y]/(y^2)$. In $\mathcal{O}_Y$, the function y is nilpotent ($y^2 = 0$), but in $\mathcal{O}_X|_Y$ there is no such nilpotence — restriction treats Y as a topological subspace, not as a scheme. The correct construction is the pushforward $\iota_*\mathcal{O}_Y$, which *is* a quasi-coherent $\mathcal{O}_X$-module (isomorphic to $\widetilde{R/\mathfrak{a}}$ where $\mathfrak{a}$ is the ideal of Y).

7. *The Hedgehog Theorem, algebraically.* Take $R = \mathbb{R}[x,y,z]/(x^2+y^2+z^2-1)$ and let M be the submodule of R^3 consisting of triples (X,Y,Z) with $xX + yY + zZ = 0$: the module of "tangent vectors to the sphere." On each affine open, $\widetilde{M}$ is locally free of rank 2 (isomorphic to $\mathcal{O}_X^2$), but it is not globally free — its complexification and analytification would contradict the Hairy Ball Theorem.

8. *Free and locally free sheaves.* The simplest quasi-coherent sheaves are the *free* sheaves $\mathcal{O}_X^{\oplus I}$ and the *locally free* sheaves, which restrict to free sheaves on an open cover. Locally free sheaves correspond to vector bundles (theorem 5.1.27). The category of locally free sheaves is not abelian: the multiplication-by-t map $\mathcal{O}_{\mathbb{A}^1_{\mathbb{C}}} \xrightarrow{\cdot t} \mathcal{O}_{\mathbb{A}^1_{\mathbb{C}}}$ has kernel the skyscraper sheaf at the origin, which is not locally free. Quasi-coherent sheaves arise as the "abelian completion" — obtained by adjoining all kernels and cokernels of maps between locally free sheaves.

9. *Quillen–Suslin.* On affine space $\mathbb{A}^n_k = \operatorname{Spec} k[x_1, \ldots, x_n]$, every locally free sheaf is in fact free. This is the geometric content of the Quillen–Suslin theorem (formerly Serre's conjecture): every projective module over a polynomial ring is free. Nontrivial quasi-coherent sheaves on affine space therefore arise only as kernels and cokernels of maps between free sheaves.

10. *Projective and flat modules.* Let $X = \operatorname{Spec} R$ and M a projective R-module. Since projective modules over local rings are free, each stalk $\widetilde{M}_{\mathfrak{p}} \cong M_{\mathfrak{p}}$ is a free $R_{\mathfrak{p}}$-module: projective modules give the closest algebraic analogue of nontrivial vector bundles. If instead M is merely flat, the localization $M_{\mathfrak{p}}$ remains flat at each point. Geometrically, flatness is the condition that fibres vary "continuously" (cf. section 6.8.1).

11. *Number-theoretic line bundles.* Let $R = \mathbb{Z}[\sqrt{-5}]$, which is not a UFD: $2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$. The ideal $J = (2, 1+\sqrt{-5})$ is not principal, so the sheaf $\widetilde{J}$ is a line bundle on $\operatorname{Spec} R$

that is not globally free. On the distinguished opens $D(2)$ and $D(3)$, the ideal J becomes principal:

$$J \cdot R[1/2] = R[1/2] \qquad \text{(since 2 is a unit)},$$
$$J \cdot R[1/3] = (1+\sqrt{-5}) \cdot R[1/3] \qquad \text{(since } 2 = (1+\sqrt{-5})(1-\sqrt{-5})/3\text{)}.$$

Hence $\widetilde{J}$ is locally free of rank 1 but not globally free — the failure of unique factorization is detected by the nontriviality of this line bundle. More generally, the Picard group $\operatorname{Pic}(\operatorname{Spec}\mathcal{O}_K)$ of a number ring is the ideal class group of K (see section 5.4).

12. *Line bundles on* $\mathbb{P}^n_k$. There is a natural line bundle on $\mathbb{P}^n_k$ whose fibre over a point p is the one-dimensional subspace of $\mathbb{A}^{n+1}_k$ that p represents. We shall see in section 5.3 that every line bundle on $\mathbb{P}^n_k$ is isomorphic to a tensor power of this bundle or its dual.

13. *Skyscraper at the generic point.* Let X be an integral Noetherian scheme with function field $K(X)$. The constant sheaf $\underline{K(X)}$ is a quasi-coherent $\mathcal{O}_X$-module (on each affine open $\operatorname{Spec} R$, it corresponds to the R-module $\operatorname{Frac}(R)$), but it is not coherent unless X is a point.

5.1.2 Properties of Quasi-Coherent Sheaves

The following collects the basic algebraic properties of the tilde construction.

Proposition 5.1.14: Properties of Sheaf Module

Let R be a ring, $X = \operatorname{Spec} R$, M an R-module, and $\varphi : R \to S$ a ring homomorphism with $f = \varphi^a : \operatorname{Spec} S \to \operatorname{Spec} R$. Then:

1. $\widetilde{M}_{\mathfrak{p}} \cong M_{\mathfrak{p}}$ for every $\mathfrak{p} \in \operatorname{Spec} R$.
2. $\Gamma(\operatorname{Spec} R, \widetilde{M}) \cong M$.
3. The functor $M \mapsto \widetilde{M}$ is exact and fully faithful from R-Mod to $\mathcal{O}_{\operatorname{Spec} R}$-Mod.
4. $\widetilde{M \otimes_R N} \cong \widetilde{M} \otimes_{\mathcal{O}_{\operatorname{Spec} R}} \widetilde{N}$ for any R-module N.
5. $\widetilde{\bigoplus_{i\in I} M_i} \cong \bigoplus_{i\in I} \widetilde{M_i}$ for any family $\{M_i\}_{i\in I}$ of R-modules.
6. If N is an S-module, then $f_*(\widetilde{N}) \cong \widetilde{({}_R N)}$, where ${}_R N$ denotes N viewed as an R-module via φ.
7. $f^*(\widetilde{M}) \cong \widetilde{M \otimes_R S}$.

***Proof*:**
Parts (1)–(2) follow from the definition: stalks are computed as colimits of localizations M_f over distinguished opens containing $\mathfrak{p}$, which yields $M_{\mathfrak{p}}$; global sections are $M_1 = M$. Part (3): full faithfulness follows from (2) and corollary 5.1.12; exactness follows from the exactness of localization. Part (4): the isomorphism $(M \otimes_R N)_f \cong M_f \otimes_{R_f} N_f$ on each distinguished open

gives the result on a basis, hence globally. Part (5) is similar using $(\bigoplus M_i)_f \cong \bigoplus(M_i)_f$. Parts (6)–(7) follow from the canonical identifications of pushforward and pullback with restriction and extension of scalars.

It is important for exactness in (6) that the source is an affine scheme. For non-affine schemes, f_* fails to be exact — this failure is precisely what sheaf cohomology measures (chapter 6).

The next result shows that sections of quasi-coherent sheaves on affine schemes behave like elements of modules: they can be "cleared of denominators" and extended from distinguished opens.

Lemma 5.1.15: Quasi-coherence and Extending From $D(f)$

Let $X = \operatorname{Spec} R$, $f \in R$, and $\mathscr{F}$ a quasi-coherent sheaf on X. Then:

1. If $s \in \Gamma(X, \mathscr{F})$ satisfies $s|_{D(f)} = 0$, then $f^n s = 0$ for some $n > 0$.
2. If $t \in \mathscr{F}(D(f))$, then $f^n t$ extends to a global section in $\Gamma(X, \mathscr{F})$ for some $n > 0$.

Part (2) should be thought of as "clearing denominators": the section $t \in \mathscr{F}(D(f))$ may involve f in the denominator, and multiplying by a sufficiently high power of f eliminates the denominator to produce a global section. Part (1) is the quasi-coherent analogue of analytic continuation, tempered by the possibility of nilpotent elements: a section vanishing on a distinguished open must be "almost zero" globally.

***Proof*:**

By proposition 5.1.11, $\mathscr{F} \cong \widetilde{M}$ for $M = \Gamma(X, \mathscr{F})$.

(1). The condition $s|_{D(f)} = 0$ means $s/1 = 0$ in M_f. By the definition of localization, $f^n s = 0$ in M for some $n > 0$.

(2). The section $t \in \mathscr{F}(D(f)) = M_f$ can be written as $t = m/f^n$ for some $m \in M$ and $n > 0$. Then $f^n t = m/1 \in M = \Gamma(X, \mathscr{F})$, which is the desired global section.

The global section functor on an affine scheme is exact when restricted to quasi-coherent sheaves — a crucial property that distinguishes affine schemes:

Proposition 5.1.16: Exactness of Global Sections for Quasi-Coherent Sheaves

Let $X = \operatorname{Spec} R$ be an affine scheme and $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ an exact sequence of $\mathcal{O}_X$-modules with $\mathscr{F}'$ quasi-coherent. Then the sequence of global sections

$$0 \longrightarrow \Gamma(X, \mathscr{F}') \longrightarrow \Gamma(X, \mathscr{F}) \longrightarrow \Gamma(X, \mathscr{F}'') \longrightarrow 0$$

is exact.

***Proof*:**

Left-exactness of Γ holds for all sheaves (proposition 1.2.16), so it suffices to prove surjectivity on the right.

Let $s \in \Gamma(X, \mathscr{F}'')$. Since $\mathscr{F} \to \mathscr{F}''$ is surjective as a sheaf map, for each $x \in X$ there exists a distinguished open $D(g_x) \ni x$ such that $s|_{D(g_x)}$ lifts to some $\hat{s}_x \in \mathscr{F}(D(g_x))$. By quasicompactness, finitely many such opens $D(g_1), \dots, D(g_r)$ cover X. Choose lifts $t_i \in \mathscr{F}(D(g_i))$ of $s|_{D(g_i)}$.

On overlaps $D(g_i g_j)$, the sections t_i and t_j both lift s, so their difference $t_i - t_j$ lies in $\mathscr{F}'(D(g_i g_j))$. Since $\mathscr{F}'$ is quasi-coherent, lemma 5.1.15(1) gives an integer n_{ij} with $g_i^{n_{ij}}(t_i - t_j) = 0$ on $D(g_j)$ (after appropriate further localization). Taking n larger than all n_{ij} and applying lemma 5.1.15(2) to replace each t_i by $g_i^n t_i$ (which now lifts $g_i^n s$), one obtains local sections that agree on overlaps and hence glue to a global section $\bar{t} \in \Gamma(X, \mathscr{F})$ lifting $g_i^n s$ for each i.

Finally, since $D(g_1), \dots, D(g_r)$ cover X, we have $(g_1^n, \dots, g_r^n) = R$ (by corollary 2.1.18). Write $1 = \sum_i a_i g_i^n$ and let $\bar{t}_i$ be the global lift of $g_i^n s$ constructed above. Then $t := \sum_i a_i \bar{t}_i \in \Gamma(X, \mathscr{F})$ maps to $\sum_i a_i g_i^n s = s$, proving surjectivity.

This exactness characterizes affine schemes: in chapter 6, we shall see that $H^i(X, \mathscr{F}) = 0$ for all $i > 0$ and all quasi-coherent $\mathscr{F}$ if and only if X is affine (theorem 6.2.9).

The following collects closure properties of quasi-coherent and coherent sheaves:

Proposition 5.1.17: Closure Properties of Quasi-Coherent Sheaves

Let X be a scheme.

1. The kernel, cokernel, and image of any morphism of quasi-coherent sheaves are quasi-coherent. Any extension of quasi-coherent sheaves by quasi-coherent sheaves is quasi-coherent. In particular, $\mathbf{QCoh}(X)$ is an abelian category.

2. If X is locally Noetherian, the same statements hold with "quasi-coherent" replaced by "coherent."

Let $f : X \to Y$ be a morphism of schemes. Then:

3. If $\mathscr{G}$ is a quasi-coherent $\mathscr{O}_Y$-module, then $f^*\mathscr{G}$ is a quasi-coherent $\mathscr{O}_X$-module. If additionally X and Y are locally Noetherian and $\mathscr{G}$ is coherent, then $f^*\mathscr{G}$ is coherent.

4. If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module and either X is Noetherian or f is quasi-compact and quasi-separated, then $f_*\mathscr{F}$ is a quasi-coherent $\mathscr{O}_Y$-module.

***Proof*:**
All claims are local on X, so we may assume $X = \operatorname{Spec} R$ is affine.

(1) Kernels, cokernels, images. Under the equivalence $\mathbf{QCoh}(\operatorname{Spec} R) \simeq R\text{-Mod}$, kernels, cokernels, and images of $\widetilde{M} \to \widetilde{N}$ correspond to $\widetilde{\ker}$, $\widetilde{\operatorname{coker}}$, and $\widetilde{\operatorname{im}}$ respectively, by exactness of the tilde functor.

Extensions. Let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be an exact sequence with $\mathscr{F}'$ and $\mathscr{F}''$ quasi-coherent. Set $M' = \Gamma(X, \mathscr{F}')$, $M = \Gamma(X, \mathscr{F})$, and $M'' = \Gamma(X, \mathscr{F}'')$. By proposition 5.1.16 (applied with $\mathscr{F}'$ quasi-coherent), the sequence $0 \to M' \to M \to M'' \to 0$ is exact. Applying the exact functor

$\widetilde{(-)}$ gives a commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & \widetilde{M'} \arrow[r] \arrow[d, "\cong"] & \widetilde{M} \arrow[r] \arrow[d] & \widetilde{M''} \arrow[r] \arrow[d, "\cong"] & 0 \\ 0 \arrow[r] & \mathscr{F}' \arrow[r] & \mathscr{F} \arrow[r] & \mathscr{F}'' \arrow[r] & 0 \end{tikzcd}$$

where the outer vertical maps are isomorphisms (since $\mathscr{F}'$ and $\mathscr{F}''$ are quasi-coherent). By the five lemma, the middle map $\widetilde{M} \to \mathscr{F}$ is an isomorphism, so $\mathscr{F}$ is quasi-coherent.

(2) follows by adding finite generation hypotheses throughout.

(3) On affine opens, pullback corresponds to extension of scalars: $f^*(\widetilde{M}) \cong \widetilde{M \otimes_R S}$ by proposition 5.1.14(7).

(4) The key observation is that quasi-compactness and quasi-separatedness ensure $f^{-1}(V)$ is covered by finitely many affine opens whose pairwise intersections are again finitely covered by affine opens. This yields an exact sequence

$$0 \longrightarrow f_*\mathscr{F} \longrightarrow \bigoplus_i f_*(\mathscr{F}|_{U_i}) \longrightarrow \bigoplus_{i,j,k} f_*(\mathscr{F}|_{U_{ijk}})$$

where the second and third terms are quasi-coherent (since each U_i and U_{ijk} is affine), and $f_*\mathscr{F}$ is their kernel, hence quasi-coherent by (1).

Without the quasi-compactness or quasi-separatedness hypotheses, the pushforward can fail to preserve quasi-coherence:

Example 5.1.18: Pushforward Not Quasi-Coherent

Let $X = \coprod_{i\in\mathbb{N}} \operatorname{Spec}\mathbb{Z}$, $Y = \operatorname{Spec}\mathbb{Z}$, and $f : X \to Y$ the identity on each component. Take $\mathscr{F} = \mathcal{O}_X$. If $f_*\mathcal{O}_X$ were quasi-coherent, then the natural map

$$\Gamma(Y, f_*\mathcal{O}_X) \otimes_{\mathbb{Z}} \mathbb{Z}[\tfrac{1}{2}] \longrightarrow (f_*\mathcal{O}_X)(D(2))$$

would be an isomorphism. The left side is $(\prod_{i\in\mathbb{N}} \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Z}[\frac{1}{2}]$: every element has the form $(a_i/2^n)_{i\in\mathbb{N}}$ for a *fixed* exponent n. The right side is $\prod_{i\in\mathbb{N}} \mathbb{Z}[\frac{1}{2}]$, which contains elements $(b_i/2^{n_i})$ where n_i grows without bound. Such elements have no preimage, so the map is not surjective.

Even between Noetherian schemes, the pushforward of a coherent sheaf need not be coherent:

Example 5.1.19: Pushforward of Noetherian Not Coherent

The structure morphism $f : \operatorname{Spec} k[x] \to \operatorname{Spec} k$ gives $f_*\mathcal{O}_{\operatorname{Spec} k[x]} = k[x]$, which is not a finitely generated k-module. The issue is that f is not a finite morphism. Sufficient conditions for f_* to preserve coherence include f being finite, projective, or proper (see theorem 6.6.6).

For later use, we record the pushforward result from the proof of proposition 5.1.17(4) as a separate statement:

Proposition 5.1.20: Pushforward is Quasi-Coherent for Qcqs Morphisms

Let $f : X \to Y$ be a quasi-compact quasi-separated morphism of schemes. If $\mathscr{F}$ is a quasi-coherent $\mathcal{O}_X$-module, then $f_*\mathscr{F}$ is a quasi-coherent $\mathcal{O}_Y$-module.

***Proof*:**
It suffices to show that for each affine open $\operatorname{Spec} A \subseteq Y$ and each $g \in A$, the natural map

$$\Gamma(f^{-1}(\operatorname{Spec} A), \mathscr{F}) \otimes_A A_g \xrightarrow{\cong} \Gamma(f^{-1}(D(g)), \mathscr{F})$$

is an isomorphism. This can be verified using the exact sequence from the proof of proposition 5.1.17(4): the quasi-compact quasi-separated hypothesis ensures that $f^{-1}(\operatorname{Spec} A)$ admits a finite affine cover with affine pairwise intersections, reducing the claim to the affine case where it follows from the exactness of localization.

5.1.3 Locally Free Sheaves and Vector Bundles

An important class of quasi-coherent sheaves — historically the first studied — arises from vector bundles. In this subsection, we establish the equivalence between vector bundles and locally free sheaves, and explain how $\mathbf{QCoh}(X)$ can be viewed as the abelian completion of the category of vector bundles on X.

Recall from section 0.5 that a vector bundle is locally trivial: over each point x, there is an open neighborhood U and an isomorphism $\pi^{-1}(U) \cong U \times \mathbb{A}^r_k$ respecting projection and linear on fibres. The algebraic translation of "locally trivial" is "locally free."

Definition 5.1.21: Vector Bundle

Let X be a scheme over a field k. A *vector bundle of rank r* on X is a scheme E together with a morphism $\pi : E \to X$ and an open covering $\{U_i\}$ of X with isomorphisms

$$\varphi_i : \pi^{-1}(U_i) \xrightarrow{\sim} U_i \times \mathbb{A}^r_k$$

such that for every pair i, j, the composite $\varphi_j \circ \varphi_i^{-1}|_{U_i \cap U_j}$ is of the form $(\mathrm{id}, \varphi_{ij})$ where $\varphi_{ij} \in \mathrm{GL}_r(\mathcal{O}_X(U_i \cap U_j))$. A *section* of E over $U \subseteq X$ is a morphism $s : U \to E$ with $\pi \circ s = \mathrm{id}_U$.

The transition matrices φ_{ij} satisfy the cocycle conditions $\varphi_{ii} = I$ and $\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$ on triple overlaps. These are precisely the data needed to glue a sheaf from local pieces.

Definition 5.1.22: Free and Locally Free $\mathcal{O}_X$-module

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathscr{F}$ an $\mathcal{O}_X$-module. Then $\mathscr{F}$ is *free of rank r* if $\mathscr{F} \cong \mathcal{O}_X^r$. It is *locally free of rank r* if there exists an open cover $\{U_i\}$ of X with $\mathscr{F}|_{U_i} \cong \mathcal{O}_{U_i}^r$ for each i. If X is connected, the rank is globally constant.

To identify which quasi-coherent sheaves are locally free, we need to know that "freeness of the stalk" propagates to a neighborhood. In differential geometry, this propagation is immediate: a smooth map of constant rank at a point has the same rank nearby, because the rank of a matrix is detected by the nonvanishing of minors, which is an open condition. The algebraic analogue is provided by Nakayama's lemma. We now formulate a sheaf-theoretic version that will serve as our main tool.

First, some notation. For a quasi-coherent sheaf $\mathscr{F}$ on a scheme X and a point $x \in X$ with residue field $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$, the *fibre* of $\mathscr{F}$ at x is the $\kappa(x)$-vector space

$$\mathscr{F}(x) \;:=\; \mathscr{F}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x) \;=\; \mathscr{F}_x \,/\, \mathfrak{m}_x\mathscr{F}_x.$$

This is the "value" of the sheaf at the point: a vector bundle of rank r has fibres that are r-dimensional $\kappa(x)$-vector spaces, while a general coherent sheaf may have fibres of varying dimension.

Lemma 5.1.23: Geometric Nakayama Lemma

Let X be a scheme, $\mathscr{F}$ a quasi-coherent $\mathcal{O}_X$-module of finite type (i.e., locally finitely generated), and $x \in X$.

1. *(Vanishing.)* If $\mathscr{F}(x) = 0$, then there exists an open neighborhood $U \ni x$ with $\mathscr{F}|_U = 0$.
2. *(Generators lift.)* If sections $s_1, \dots, s_n \in \mathscr{F}(U)$ are defined on a neighborhood $U \ni x$ and their images $\overline{s}_1, \dots, \overline{s}_n$ generate the fibre $\mathscr{F}(x)$, then after shrinking U the induced map $\mathcal{O}_U^n \to \mathscr{F}|_U$ is surjective.
3. *(Semicontinuity of fibre dimension.)* The function

$$x \;\longmapsto\; \dim_{\kappa(x)} \mathscr{F}(x)$$

is upper semicontinuous: for each $n \geq 0$, the set $\{x \in X : \dim \mathscr{F}(x) \leq n\}$ is open.

***Proof*:**

All three statements are local, so we may assume $X = \operatorname{Spec} R$ is affine and $\mathscr{F} = \widetilde{M}$ for a finitely generated R-module $M = Rm_1 + \cdots + Rm_k$. Let $\mathfrak{p} \in \operatorname{Spec} R$ correspond to x.

(1) Vanishing. The hypothesis $\mathscr{F}(x) = 0$ translates to $M_\mathfrak{p}/\mathfrak{p}M_\mathfrak{p} = 0$, i.e., $M_\mathfrak{p} = \mathfrak{p}M_\mathfrak{p}$. By the classical Nakayama lemma (applied to the finitely generated $R_\mathfrak{p}$-module $M_\mathfrak{p}$), we conclude $M_\mathfrak{p} = 0$.

Since M is finitely generated and each generator m_i satisfies $m_i/1 = 0$ in $M_\mathfrak{p}$, there exists $f_i \notin \mathfrak{p}$ with $f_i m_i = 0$ in M. Setting $f = f_1 \cdots f_k$ gives $f \notin \mathfrak{p}$ and $M_f = 0$, so $\mathscr{F}|_{D(f)} = \widetilde{M_f} = 0$.

(2) Generators lift. After shrinking U to a distinguished open and clearing denominators (using lemma 5.1.15), we may assume the s_i arise from elements $m_1, \dots, m_n \in M$. Define $\varphi : R^n \to M$ by sending the ith basis vector to m_i, and let $C = \operatorname{coker}(\varphi) = M/(Rm_1 + \cdots + Rm_n)$.

The hypothesis that $\overline{s}_1, \dots, \overline{s}_n$ generate $\mathscr{F}(x)$ translates to $C_\mathfrak{p}/\mathfrak{p}C_\mathfrak{p} = 0$, i.e., $C(x) = 0$. Since C is a quotient of the finitely generated module M, it is itself finitely generated. Applying part (1) to $\widetilde{C}$ yields $g \notin \mathfrak{p}$ with $C_g = 0$, meaning φ is surjective after localizing at g. Hence the map $\mathcal{O}_{D(g)}^n \to \mathscr{F}|_{D(g)}$ is surjective.

(3) Semicontinuity. Suppose $\dim_{\kappa(x)} \mathscr{F}(x) \leq n$. Choose n elements $s_1, \dots, s_n$ in $\mathscr{F}(U)$ (for some neighborhood $U \ni x$) whose images generate $\mathscr{F}(x)$. By (2), after shrinking U, the map $\mathcal{O}_U^n \to \mathscr{F}|_U$ is surjective. For any $y \in U$, applying $- \otimes_{\mathcal{O}_{X,y}} \kappa(y)$ to this surjection gives a

surjection $\kappa(y)^n \twoheadrightarrow \mathscr{F}(y)$, so $\dim_{\kappa(y)} \mathscr{F}(y) \le n$. Hence $\{x \in X : \dim \mathscr{F}(x) \le n\}$ is open.

The classical Nakayama lemma is a statement about a single local ring; the geometric upgrade converts it into a *neighborhood* statement by exploiting finite generation. The mechanism is simple but powerful: finitely many generators can be simultaneously controlled by inverting finitely many elements, i.e., by passing to a single distinguished open.

Part (3) should be compared with the situation for smooth vector bundles, which have *constant* fibre dimension. For coherent sheaves on schemes, constancy of fibre dimension is a strong condition — it is closely related to local freeness, as the next proposition makes precise.

Proposition 5.1.24: Locally Free if and only if Stalks are Free

*Let X be a scheme and $\mathscr{F}$ a quasi-coherent sheaf of finite presentation (*i.e., locally the cokernel of a map between finitely generated free modules*). Then $\mathscr{F}$ is locally free of rank r if and only if $\mathscr{F}_x$ is a free $\mathcal{O}_{X,x}$-module of rank r for every $x \in X$.*
In particular, this applies to all coherent sheaves on locally Noetherian schemes, since over a Noetherian ring every finitely generated module is finitely presented.

The finite presentation hypothesis enters in a precise way: the geometric Nakayama lemma (lemma 5.1.23) requires finite type (= finitely generated) to show that generators of the fibre extend to generators of the sheaf nearby. This gives the surjectivity step. For the injectivity step, we must also apply the vanishing part of the lemma to the *kernel*, which requires the kernel to be finitely generated — and this is precisely what finite presentation guarantees.

***Proof*:**
($\Rightarrow$). If $\mathscr{F}|_U \cong \mathcal{O}_U^r$ for an open $U \ni x$, then $\mathscr{F}_x \cong \mathcal{O}_{X,x}^r$, since localization preserves free modules.

($\Leftarrow$). Suppose $\mathscr{F}_x \cong \mathcal{O}_{X,x}^r$ for all $x \in X$. Fix a point x and choose an affine neighborhood $\operatorname{Spec} R \ni x$ with $\mathscr{F}|_{\operatorname{Spec} R} \cong \widetilde{M}$, where M is a finitely presented R-module. Let $\mathfrak{p} \in \operatorname{Spec} R$ correspond to x, so that $M_{\mathfrak{p}} \cong R_{\mathfrak{p}}^r$ by hypothesis.

Step 1: Construct generators. Choose a basis $e_1, \dots, e_r$ of the free $R_{\mathfrak{p}}$-module $M_{\mathfrak{p}}$. Write $e_i = m_i/g_i$ with $m_i \in M$ and $g_i \notin \mathfrak{p}$. Replacing $\operatorname{Spec} R$ by the distinguished open $D(g_1 \cdots g_r)$ (still an affine neighborhood of x on which $\mathscr{F}$ corresponds to a finitely presented module), we may assume $e_i = m_i \in M$.

The images $\overline{m}_1, \dots, \overline{m}_r$ in the fibre $\mathscr{F}(x) = M_{\mathfrak{p}}/\mathfrak{p}M_{\mathfrak{p}} \cong \kappa(\mathfrak{p})^r$ form a basis.

Step 2: Surjectivity via geometric Nakayama. Define the map

$$\varphi : R^r \longrightarrow M, \qquad (a_1, \dots, a_r) \longmapsto a_1 m_1 + \cdots + a_r m_r.$$

Since $\overline{m}_1, \dots, \overline{m}_r$ generate $\mathscr{F}(x)$, lemma 5.1.23(2) gives a distinguished open $D(f) \ni x$ on which $\widetilde{\varphi} : \mathcal{O}_{D(f)}^r \to \mathscr{F}|_{D(f)}$ is surjective. Replacing $\operatorname{Spec} R$ by $D(f)$, we may assume φ is surjective.

Step 3: Injectivity via geometric Nakayama. Let $K = \ker(\varphi)$, so that $\widetilde{K}$ is a quasi-coherent sheaf on $\operatorname{Spec} R$ and we have the short exact sequence

$$0 \longrightarrow \widetilde{K} \longrightarrow \mathcal{O}_{\operatorname{Spec} R}^r \xrightarrow{\widetilde{\varphi}} \widetilde{M} \longrightarrow 0.$$

Since M is finitely presented, K is a finitely generated R-module[a]. At the stalk over x,

$$K_{\mathfrak{p}} \;=\; \ker(\varphi_{\mathfrak{p}} : R_{\mathfrak{p}}^r \to M_{\mathfrak{p}}).$$

But $\varphi_{\mathfrak{p}}$ sends $e_i \mapsto m_i$, which is a basis of $M_{\mathfrak{p}}$, so $\varphi_{\mathfrak{p}}$ is an isomorphism and $K_{\mathfrak{p}} = 0$. In particular, $\widetilde{K}(x) = K_{\mathfrak{p}}/\mathfrak{p}K_{\mathfrak{p}} = 0$. Since K is finitely generated, lemma 5.1.23(1) provides a distinguished open $D(g) \ni x$ with $K_g = 0$.

Conclusion. On the distinguished open $D(fg)$, the map φ is both surjective (by Step 2) and injective (by Step 3), hence an isomorphism $\mathcal{O}_{D(fg)}^r \xrightarrow{\sim} \mathscr{F}|_{D(fg)}$. Since x was arbitrary, $\mathscr{F}$ is locally free of rank r.

[a] If $R^s \xrightarrow{\alpha} R^t \to M \to 0$ is a finite presentation and $\varphi : R^r \twoheadrightarrow M$ is any surjection from a finitely generated free module, then Schanuel's lemma gives $K \oplus R^t \cong \ker(\alpha) \oplus R^r$; since $\ker(\alpha)$ is finitely generated by hypothesis, so is $K \oplus R^t$, hence K. Equivalently, being finitely presented is equivalent to *every* surjection from a finitely generated free module having finitely generated kernel.

The correspondence between vector bundles and locally free sheaves generalizes the classical fact that a vector bundle is determined by its transition functions. The key input from the scheme-theoretic side is the *relative spectrum* construction, which we develop in full generality in section 5.1.5 below. For the present theorem, we use it as follows: given a locally free sheaf $\mathcal{E}$ of rank r, the $\mathcal{O}_X$-algebra $\mathrm{Sym}(\mathcal{E}^\vee)$ is locally isomorphic to a polynomial ring in r variables, and $\underline{\mathrm{Spec}}_X(\mathrm{Sym}(\mathcal{E}^\vee))$ is the corresponding "total space" scheme over X.

Lemma 5.1.25: Coherent Sheaf Locally Free Criterion

Let $\mathcal{E}$ be a coherent sheaf on a locally Noetherian scheme X. Then:

1. A subsheaf of a locally free sheaf is locally free.
2. If $\mathcal{E}$ and $\mathcal{E}'$ are locally free of rank 1, then every nonzero morphism $\mathcal{E} \to \mathcal{E}'$ is injective.

***Proof*:**
Both statements are local and reduce to the corresponding facts about modules over local rings: a torsion-free module over a DVR is free, and a nonzero map between rank-1 free modules over a domain is injective.

The category of locally free sheaves is not abelian: quotients may fail to be locally free due to torsion. The following lemma shows how to "correct" this:

Lemma 5.1.26: Natural Quotient of Locally Free Sheaf

Let X be a Noetherian integral scheme, $\mathcal{E}$ a locally free sheaf of rank r, and $\mathcal{E}' \subseteq \mathcal{E}$ a locally free subsheaf of rank $r' \leq r$. Then there exists a locally free subsheaf $\overline{\mathcal{E}}' \subseteq \mathcal{E}$ of rank r' containing $\mathcal{E}'$ such that $\mathcal{E}/\overline{\mathcal{E}}'$ is locally free of rank $r - r'$.

***Proof*:**
Let $M = \mathcal{E}/\mathcal{E}'$ and $T \subseteq M$ its torsion subsheaf. Define $\overline{\mathcal{E}}'$ to be the kernel of the composition $\mathcal{E} \twoheadrightarrow M \twoheadrightarrow M/T$. The quotient $\mathcal{E}/\overline{\mathcal{E}}' \cong M/T$ is torsion-free and coherent, hence locally free by lemma 5.1.25(1). By the snake lemma applied to the diagram

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{E}' \arrow[r] \arrow[d, hook] & \mathcal{E} \arrow[r] \arrow[d, equal] & M \arrow[r] \arrow[d, two heads] & 0 \\ 0 \arrow[r] & \overline{\mathcal{E}}' \arrow[r] & \mathcal{E} \arrow[r] & M/T \arrow[r] & 0 \end{tikzcd}$$

we get $\overline{\mathcal{E}}'/\mathcal{E}' \cong T$, which is torsion, so $\overline{\mathcal{E}}'$ and $\mathcal{E}'$ have the same rank.

Theorem 5.1.27: Vector Bundle / Locally Free Sheaf Correspondence

Let X be a scheme over a field k. There is an equivalence of categories

$$\left\{\begin{array}{c}\text{vector bundles}\\ \text{on } X\end{array}\right\} \underset{\underline{\mathrm{Spec}}_X \mathrm{Sym}(\mathscr{F}^\vee) \leftarrow \mathscr{F}}{\overset{E \mapsto \mathcal{E}_E}{\rightleftarrows}} \left\{\begin{array}{c}\text{locally free}\\ \text{sheaves on } X\end{array}\right\}$$

where morphisms on the left are bundle maps (linear on fibres) and on the right are $\mathcal{O}_X$-module homomorphisms.

***Proof*:**
From bundles to sheaves. Let $\pi : E \to X$ be a vector bundle of rank r. Define the *sheaf of sections* $\mathcal{E}_E$ by $\mathcal{E}_E(U) = \{s : U \to E \,|\, \pi \circ s = \mathrm{id}_U\}$, with $\mathcal{O}_X$-action $(f \cdot s)_p = f(p) \cdot s_p$ and addition defined fibre-wise. On a trivializing open U_i with $\pi^{-1}(U_i) \cong U_i \times \mathbb{A}^r_k$, every section decomposes uniquely as $s = f_1 e_1 + \cdots + f_r e_r$ where e_j are the coordinate sections and $f_j \in \mathcal{O}_X(U_i)$. The map $s \mapsto (f_1, \ldots, f_r)$ gives an $\mathcal{O}_X(U_i)$-module isomorphism $\mathcal{E}_E(U_i) \cong \mathcal{O}_X(U_i)^r$, so $\mathcal{E}_E$ is locally free of rank r.

From sheaves to bundles. Let $\mathcal{E}$ be a locally free sheaf of rank r. The total space is constructed via the *relative spectrum* (section 5.1.5):

$$E \;:=\; \underline{\mathrm{Spec}}_X(\mathrm{Sym}(\mathcal{E}^\vee)),$$

where $\mathcal{E}^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{O}_X)$ is the dual sheaf and $\mathrm{Sym}(\mathcal{E}^\vee)$ is the symmetric algebra. On a trivializing open U_i where $\mathcal{E}|_{U_i} \cong \mathcal{O}^r_{U_i}$, we have $\mathrm{Sym}(\mathcal{E}^\vee)|_{U_i} \cong \mathcal{O}_X(U_i)[t_1, \ldots, t_r]$, and hence

$$\pi^{-1}(U_i) \;=\; \mathrm{Spec}\,(\mathcal{O}_X(U_i)[t_1, \ldots, t_r]) \;\cong\; U_i \times \mathbb{A}^r_k.$$

The transition functions of $\mathcal{E}$ give GL_r-valued cocycles on overlaps, furnishing E with the structure of a vector bundle. As a set,

$$E \;=\; \left\{(p, v) \;\middle|\; p \in X,\; v \in \mathcal{E}_p \otimes_{\mathcal{O}_{X,p}} \kappa(p)\right\},$$

which the reader should compare with the classical construction for curves where $E = \{(p,t) : p \in X,\; t \in \mathcal{E}_p/\mathfrak{m}_p\mathcal{E}_p\}$.

Functoriality. A bundle map $\varphi : E \to E'$ induces a sheaf morphism on sections by composition: $s \mapsto \varphi \circ s$. Conversely, a sheaf morphism $\mathcal{E} \to \mathcal{E}'$ induces a morphism of symmetric algebras (contravariantly on the duals), hence a bundle map via relative Spec. These operations are inverse to each other.

Definition 5.1.28: Line Bundle

A locally free $\mathcal{O}_X$-module of rank 1 is called a *line bundle* or *invertible sheaf*.

The term "invertible" is justified by the existence of a tensor-product inverse: if $\mathcal{L}$ is a line bundle, then $\mathcal{L}^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$ satisfies $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^\vee \cong \mathcal{O}_X$. The group of isomorphism classes of line bundles under tensor product is the *Picard group* $\operatorname{Pic}(X)$, studied in depth in section 5.4.

5.1.29 K-Theory The monoid of isomorphism classes of locally free sheaves under direct sum can be group-completed to form the *Grothendieck group* $K_0(X)$ (see [10]). The tensor product makes $K_0(X)$ into a ring. Line bundles correspond to the units of this ring under the tensor product, and the inclusion $\operatorname{Pic}(X) \hookrightarrow K_0(X)^\times$ connects divisor theory to K-theory. For affine space, the Quillen–Suslin theorem ($K_0(\mathbb{A}^n_k) \cong \mathbb{Z}$) reflects the triviality of all vector bundles.

5.1.4 Ideal Sheaves and Closed Subschemes

Having developed the general theory of quasi-coherent sheaves, we return to closed subschemes. Recall from definition 3.1.22 that a closed immersion $\iota : Y \hookrightarrow X$ is a morphism for which $\iota^\# : \mathcal{O}_X \to \iota_*\mathcal{O}_Y$ is surjective. The kernel of this surjection is an $\mathcal{O}_X$-submodule of $\mathcal{O}_X$ — an *ideal sheaf* — and it encodes the scheme structure of Y inside X.

Definition 5.1.30: Ideal Sheaf

Let $(X, \mathcal{O}_X)$ be a scheme. An *ideal sheaf* on X is an $\mathcal{O}_X$-submodule $\mathcal{I} \subseteq \mathcal{O}_X$, i.e., a sheaf of ideals: for each open $U \subseteq X$, $\mathcal{I}(U)$ is an ideal of $\mathcal{O}_X(U)$, compatibly with restriction.

Lemma 5.1.31: Kernel and Ideal Sheaves

Let $\iota : Y \hookrightarrow X$ be a closed immersion. Then the kernel

$$\mathcal{I}_{Y/X} \;:=\; \ker\bigl(\iota^\# : \mathcal{O}_X \to \iota_*\mathcal{O}_Y\bigr)$$

is an ideal sheaf on X, and there is a short exact sequence of $\mathcal{O}_X$-modules:

$$0 \longrightarrow \mathcal{I}_{Y/X} \longrightarrow \mathcal{O}_X \longrightarrow \iota_*\mathcal{O}_Y \longrightarrow 0.$$

***Proof*:**
As $\iota^\#$ is a surjective morphism of $\mathcal{O}_X$-modules, its kernel is an $\mathcal{O}_X$-submodule of $\mathcal{O}_X$, hence an ideal sheaf. The short exact sequence follows immediately.

Not every ideal sheaf arises from a closed immersion. The obstruction is quasi-coherence:

Example 5.1.32: Not All Ideal Sheaves Correspond To Closed Immersions

Let $X = \operatorname{Spec} k[x]_{(x)}$, which has two points: the closed point $\mathfrak{m} = (x)$ and the generic point $\eta = (0)$. Define an ideal sheaf $\mathcal{I}$ by

$$\mathcal{I}(X) = 0 \subseteq \mathcal{O}_X(X) = k[x]_{(x)}, \qquad \mathcal{I}(\{\eta\}) = k(x) = \mathcal{O}_X(\{\eta\}).$$

This is a valid $\mathcal{O}_X$-submodule, but it is not quasi-coherent: if $\mathcal{I} = \widetilde{I}$ for some ideal $I \subseteq k[x]_{(x)}$, then $\mathcal{I}(X) = I = 0$ forces $\mathcal{I}(\{\eta\}) = I_{(0)} = 0$, contradicting $\mathcal{I}(\{\eta\}) = k(x)$. By proposition 5.1.33 below, $\mathcal{I}$ cannot define a closed subscheme.

Proposition 5.1.33: Ideal Sheaves and Closed Immersions

Let X be a scheme and $\mathcal{I} \subseteq \mathcal{O}_X$ an ideal sheaf. Then $\mathcal{I}$ defines a closed immersion $Y \hookrightarrow X$ (with $\mathcal{O}_Y = \mathcal{O}_X/\mathcal{I}$) if and only if for every affine open $\operatorname{Spec} S \subseteq X$ and every $f \in S$, the restriction map induces an isomorphism $\mathcal{I}(\operatorname{Spec} S)_f \cong \mathcal{I}(D(f))$. This condition holds precisely when $\mathcal{I}$ is quasi-coherent.

***Proof*:**
($\Rightarrow$). If $\mathcal{I} = \mathcal{I}_{Y/X}$ for a closed immersion, then on each affine open $\operatorname{Spec} S$, the ideal sheaf restricts to $\widetilde{I}$ for $I = \mathcal{I}(\operatorname{Spec} S) \subseteq S$. The localization property $\widetilde{I}(D(f)) = I_f$ gives the required isomorphism.

($\Leftarrow$). The localization condition ensures that the quotient sheaves $\mathcal{O}_X(\operatorname{Spec} S)/\mathcal{I}(\operatorname{Spec} S)$ glue to form a well-defined quotient $\mathcal{O}_X/\mathcal{I}$, which defines the structure sheaf of the closed subscheme Y. The details are an application of the *qcqs lemma* (lemma 3.4.9).

We can now state the precise correspondence:

Proposition 5.1.34: Quasi-coherent Ideal Sheaves

Let X be a scheme. The assignment $Y \mapsto \mathcal{I}_{Y/X}$ defines a bijection

$$\{\text{closed subschemes of } X\} \xleftrightarrow{1:1} \{\text{quasi-coherent ideal sheaves on } X\}.$$

If X is locally Noetherian, then $\mathcal{I}_{Y/X}$ is coherent.

***Proof*:**
Given a closed immersion $\iota : Y \hookrightarrow X$, the ideal sheaf $\mathcal{I}_{Y/X}$ is quasi-coherent by proposition 5.1.33. Conversely, any quasi-coherent ideal sheaf $\mathcal{I}$ satisfies the localization condition of proposition 5.1.33 and hence defines a closed subscheme. Coherence in the locally Noetherian case follows because ideals of Noetherian rings are finitely generated.

Corollary 5.1.35: Ideal Sheaves On Affine Schemes

For an affine scheme $X = \operatorname{Spec} R$, there is a bijection between ideals $\mathfrak{a} \subseteq R$ and closed subschemes of X, given by $\mathfrak{a} \mapsto \operatorname{Spec}(R/\mathfrak{a})$.

With this correspondence in hand, we can define scheme-theoretic analogues of vanishing sets:

Definition 5.1.36: Vanishing Scheme

Let X be a scheme and $s \in \mathcal{O}_X(X)$ a global section. The *vanishing scheme* $V(s)$ is the closed subscheme corresponding to the quasi-coherent ideal sheaf generated by s: on each affine open $\operatorname{Spec} R \subseteq X$, it is defined by $\operatorname{Spec}(R/(s|_{\operatorname{Spec} R}))$. More generally, for a collection $S \subseteq \mathcal{O}_X(X)$, the vanishing scheme $V(S)$ corresponds to the ideal sheaf generated by S.

The advantage of vanishing *schemes* over vanishing *sets* is that they remember scheme-theoretic information such as intersection multiplicities.

Proposition 5.1.37: Properties of Vanishing Schemes

Let X be a scheme and $S_1, \ldots, S_n, \{S_i\}_{i \in \mathcal{I}}$ collections of global sections. Then:

1. $V(S_1) \cup \cdots \cup V(S_n) = V(S_1 \cap \cdots \cap S_n)$ (finite unions correspond to ideal intersections).
2. $\bigcap_{i \in \mathcal{I}} V(S_i) = V(\langle S_i \big| i \in \mathcal{I} \rangle)$ (intersections correspond to ideal sums).

***Proof*:**
The underlying topological statement follows from the corresponding properties of $V(-)$ and $I(-)$ (proposition 2.1.6). The scheme-theoretic content is that finite unions use ideal *intersections* (not products): we need $V(S) \cap V(S) = V(S)$, which holds for $S \cap S = S$ but would fail for $S \cdot S = S^2$. The key algebraic input is the inclusion $IJ \subseteq I \cap J \subseteq \sqrt{IJ}$.

Example 5.1.38: Vanishing Scheme Intersection

1. Consider $V(y - x^2) \cap V(y) \subseteq \mathbb{A}^2_k$. As a set, this is the single point $(0, 0)$. As a scheme:

$$V(y - x^2) \cap V(y) \;=\; V(y - x^2,\, y) \;=\; \operatorname{Spec} \frac{k[x, y]}{(y - x^2,\, y)} \;\cong\; \operatorname{Spec} \frac{k[x]}{(x^2)}.$$

   The intersection point has *multiplicity* 2, reflecting the tangency between the parabola and the x-axis. This information is invisible to the underlying set but recorded by the scheme structure.

2. The intersection $V(y^2 - x^2) \cap V(y)$ gives an example where distributivity of unions and intersections fails at the scheme level. Since $y^2 - x^2 = (y - x)(y + x)$, we have $V(y^2 - x^2) = V(y - x) \cup V(y + x)$ (two lines through the origin). The scheme-theoretic computation shows

$$(V(y - x) \cup V(y + x)) \cap V(y) \;\neq\; (V(y - x) \cap V(y)) \cup (V(y + x) \cap V(y)),$$

   because the left side is $\operatorname{Spec} k[x]/(x^2)$ (a double point) while the right side is $\operatorname{Spec} k[x]/(x) \sqcup \operatorname{Spec} k[x]/(x) = \operatorname{Spec} k$ (a reduced point).

5.1.5 Relative Spectrum

In the proof of theorem 5.1.27, we constructed the total space of a vector bundle as $\underline{\mathrm{Spec}}_X(\mathrm{Sym}(\mathcal{E}^\vee))$. This is an instance of a general construction that associates a scheme over X to any quasi-coherent $\mathcal{O}_X$-algebra. Relative Spec is the scheme-theoretic analogue of the total space of a fibre bundle: it globalizes Spec from a single ring to a sheaf of rings over a base.

Definition 5.1.39: $\mathcal{O}_X$-algebra

Let $(X, \mathcal{O}_X)$ be a scheme. A *quasi-coherent $\mathcal{O}_X$-algebra* is a sheaf of commutative rings $\mathcal{A}$ on X that is simultaneously a quasi-coherent $\mathcal{O}_X$-module, with the ring multiplication $\mathcal{A}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{A}(U) \to \mathcal{A}(U)$ being $\mathcal{O}_X(U)$-bilinear for each open $U \subseteq X$.
Equivalently, $\mathcal{A}$ is a ring object in the category $\mathbf{QCoh}(X)$. On an affine open $\mathrm{Spec}\, R \subseteq X$, a quasi-coherent $\mathcal{O}_X$-algebra restricts to $\widetilde{A}$ for some R-algebra A.

Example 5.1.40

1. The structure sheaf $\mathcal{O}_X$ is a quasi-coherent $\mathcal{O}_X$-algebra.
2. If $\mathcal{E}$ is a locally free sheaf of rank r, then $\mathrm{Sym}(\mathcal{E}) = \bigoplus_{n \geq 0} \mathrm{Sym}^n(\mathcal{E})$ is a quasi-coherent $\mathcal{O}_X$-algebra. On a trivializing open where $\mathcal{E} \cong \mathcal{O}_X^r$, we have $\mathrm{Sym}(\mathcal{E}) \cong \mathcal{O}_X[t_1, \ldots, t_r]$.
3. If $f : Y \to X$ is an affine morphism, then $f_*\mathcal{O}_Y$ is a quasi-coherent $\mathcal{O}_X$-algebra.

The construction of relative Spec proceeds by gluing:

Theorem 5.1.41: Relative Spectrum

Let X be a scheme and $\mathcal{A}$ a quasi-coherent $\mathcal{O}_X$-algebra. There exists a scheme $\underline{\mathrm{Spec}}_X(\mathcal{A})$ together with an affine morphism $\pi : \underline{\mathrm{Spec}}_X(\mathcal{A}) \to X$ and an isomorphism of $\mathcal{O}_X$-algebras $\pi_*\mathcal{O}_{\underline{\mathrm{Spec}}_X(\mathcal{A})} \cong \mathcal{A}$, characterized by the following properties:

1. For each affine open $U = \mathrm{Spec}\, R \subseteq X$ with $\mathcal{A}|_U \cong \widetilde{A}$, there is a canonical isomorphism $\pi^{-1}(U) \cong \mathrm{Spec}\, A$.
2. As a set, $\underline{\mathrm{Spec}}_X(\mathcal{A}) = \{(x, \mathfrak{q}) : x \in X,\ \mathfrak{q} \in \mathrm{Spec}\,(\mathcal{A}_x)\}$.
3. The fibre over a point $x \in X$ with residue field $\kappa(x)$ is
$$\pi^{-1}(x) \;\cong\; \mathrm{Spec}\,(\mathcal{A}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x)).$$

Furthermore, the construction gives an equivalence of categories:

$$\{\text{quasi-coherent } \mathcal{O}_X\text{-algebras}\} \;\simeq\; \{\text{affine morphisms } Y \to X\}^{\mathrm{op}}$$

with inverse $(\pi : Y \to X) \mapsto \pi_*\mathcal{O}_Y$.

***Proof*:**
Construction. Cover X by affine opens $U_i = \operatorname{Spec} R_i$ and write $\mathcal{A}|_{U_i} \cong \widetilde{A_i}$ for R_i-algebras A_i. On overlaps $U_i \cap U_j$, the quasi-coherence of $\mathcal{A}$ gives canonical isomorphisms between the restrictions of $\widetilde{A_i}$ and $\widetilde{A_j}$, hence isomorphisms between the corresponding spectra on the overlap. These isomorphisms satisfy the cocycle condition (being induced by identity maps of the sheaf $\mathcal{A}$), so by theorem 2.2.54 the schemes $\operatorname{Spec} A_i$ glue to a scheme $\underline{\operatorname{Spec}}_X(\mathcal{A})$ with a natural projection π to X.

Affine morphism. By construction, $\pi^{-1}(U_i) = \operatorname{Spec} A_i$ is affine for each i, so π is an affine morphism.

Equivalence. Given an affine morphism $\pi : Y \to X$, the pushforward $\pi_*\mathcal{O}_Y$ is a quasi-coherent $\mathcal{O}_X$-algebra (by proposition 5.1.20, since affine morphisms are quasi-compact and separated). On each affine open $\operatorname{Spec} R \subseteq X$, we have $\pi^{-1}(\operatorname{Spec} R) = \operatorname{Spec} A$ and $\pi_*\mathcal{O}_Y|_{\operatorname{Spec} R} \cong \widetilde{A}$. The two operations are inverse by the equivalence $\mathbf{QCoh}(\operatorname{Spec} R) \simeq R\text{-Mod}$ applied to algebras.

Description of points and fibres. The set-theoretic description follows from the gluing construction: a point of $\underline{\operatorname{Spec}}_X(\mathcal{A})$ over $x \in U_i$ is a prime ideal of $(A_i)_\mathfrak{p}$ where $\mathfrak{p}$ corresponds to x. The fibre computation follows from $\operatorname{Spec}(A_i \otimes_{R_i} \kappa(\mathfrak{p})) \cong \operatorname{Spec}(A_i/\mathfrak{p}A_i)_{\text{red}}$ after base change to the residue field.

The relative spectrum provides a unified perspective on several constructions we have already seen:

Example 5.1.42: Relative Spectrum

1. *Closed subschemes.* If $\mathcal{I} \subseteq \mathcal{O}_X$ is a quasi-coherent ideal sheaf, then $\underline{\operatorname{Spec}}_X(\mathcal{O}_X/\mathcal{I})$ is the closed subscheme defined by $\mathcal{I}$.
2. *Vector bundles.* If $\mathcal{E}$ is locally free of rank r, then $\underline{\operatorname{Spec}}_X(\operatorname{Sym}(\mathcal{E}^\vee))$ is the total space of the corresponding vector bundle, with fibres $\mathbb{A}^r$.
3. *Affine morphisms.* Every affine morphism $Y \to X$ arises as $\underline{\operatorname{Spec}}_X(\pi_*\mathcal{O}_Y) \to X$ for a unique quasi-coherent $\mathcal{O}_X$-algebra.
4. *Base change.* For any morphism $f : X' \to X$,
$$\underline{\operatorname{Spec}}_{X'}(f^*\mathcal{A}) \;\cong\; \underline{\operatorname{Spec}}_X(\mathcal{A}) \times_X X'.$$

The relative spectrum thus plays the same role for quasi-coherent algebras that Spec plays for rings: it is the universal "total space" construction, and it provides the bridge between the algebraic data (a sheaf of algebras) and the geometric data (a scheme over X).

5.2 Images of Scheme Morphisms

In the theory of sets, the image of a function is unambiguous: if $f : X \to Y$, then $f(X) = \{f(x) | x \in X\} \subseteq Y$, and it is automatically a subset. In algebraic geometry, things are not so clean. The image of a morphism of schemes need not be open, need not be closed, and need not even be locally closed—so it cannot, in general, carry a scheme structure coming from Y. This section develops two complementary responses to this problem:

1. *Chevalley's theorem*: the image is always *constructible*—a finite union of locally closed sets. This is the best purely topological statement one can make, and it has surprisingly powerful consequences for the geometry of fibers.
2. *The scheme-theoretic image*: when we insist on a scheme structure, we replace the set-theoretic image with the smallest closed subscheme through which the morphism factors. This is the "correct" notion of image in the category of schemes.

Both ideas require some commutative algebra—in particular, Grothendieck's Generic Freeness lemma, which says that finitely generated modules over a domain become free after inverting a single element. This is the key algebraic engine behind Chevalley's theorem.

5.2.1 Chevalley's Theorem

The reader who has worked primarily with affine morphisms may expect the image of a scheme morphism to be something reasonable—an open set, a closed set, or at worst a locally closed set. The following example shows that none of these hold in general.

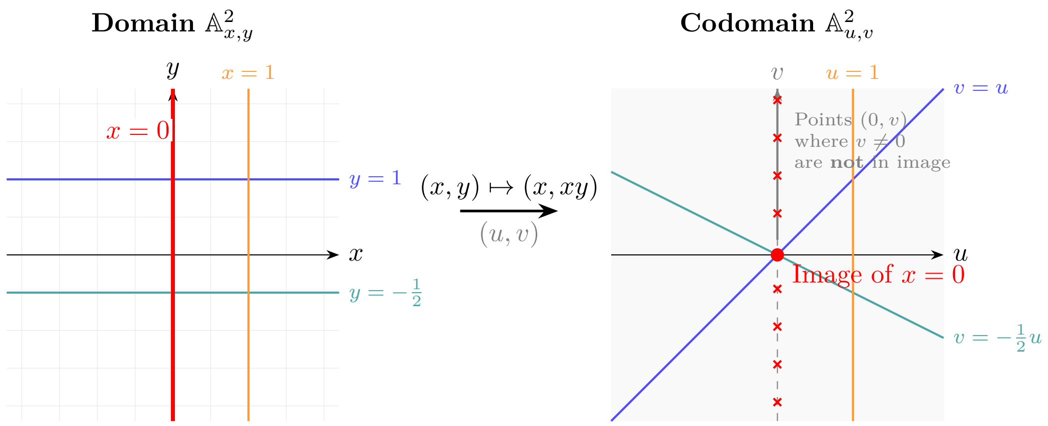
Example 5.2.1: Image That is Not Locally Closed

Let $f : \mathbb{A}^2_k \to \mathbb{A}^2_k$ be the morphism given by $(x, y) \mapsto (x, xy)$, corresponding to the ring map $k[u, v] \to k[x, y]$ with $u \mapsto x$, $v \mapsto xy$. Its image is:

$$\operatorname{im}(f) = \big(\mathbb{A}^2_k \setminus \{(u, v) \mid u = 0\}\big) \cup \{(0, 0)\}$$

In words: the entire plane with the v-axis removed, plus the origin put back in.

To see this, note that if $u = x \neq 0$, then $v = xy$ can be any element of k (by choosing $y = v/x$), so every point (u, v) with $u \neq 0$ is in the image. On the other hand, if $u = x = 0$, then $v = xy = 0$ regardless of y, so the only point on the v-axis that lies in the image is the origin.



This set is *not* locally closed. Recall that a locally closed subset is the intersection of an open set and a closed set. If $\operatorname{im}(f)$ were locally closed, its closure would be $\mathbb{A}^2_k$ (since $\operatorname{im}(f)$ is dense), and so $\operatorname{im}(f)$ would have to be open. But the complement $\mathbb{A}^2_k \setminus \operatorname{im}(f) = \{(0, v) \big| v \neq 0\}$ is not closed (its closure is the entire v-axis), so $\operatorname{im}(f)$ is not open either.

Because $\operatorname{im}(f)$ is not locally closed, it does not arise as the underlying set of any subscheme of $\mathbb{A}^2_k$.

However, the image in this example is not completely wild: it is the union of the open set $D(u)$ and the closed point $\{(0,0)\}$—a union of two locally closed sets. Chevalley's theorem says this is always the case.

Constructible Sets

To state Chevalley's theorem, we need the notion of a constructible set. The idea is simple: constructible sets are exactly the subsets you can build from open sets using complements and finite intersections. Equivalently (and more geometrically), they are finite unions of locally closed sets.

Definition 5.2.2: Constructible Set

Let X be a Noetherian topological space. A subset $S \subseteq X$ is *constructible* if it belongs to the smallest family of subsets containing all open sets and closed under:

1. complements
2. finite intersections

The first condition alone would give all open and all closed sets. Adding finite intersections then adds all locally closed sets (an open intersected with a closed set), and then all finite unions of locally closed sets (by De Morgan: a union of two constructible sets $S_1 \cup S_2 = (S_1^c \cap S_2^c)^c$ is constructible). In fact, one can show that these are *all* constructible sets:

Lemma 5.2.3: Constructible Sets as Locally Closed Sets

Let X be a Noetherian topological space. Then a subset $S \subseteq X$ is constructible if and only if it is a finite union of locally closed subsets. Consequently:

1. Finite unions and finite intersections of constructible sets are constructible.
2. If $f : X \to Y$ is continuous, then the preimage of a constructible set is constructible.

***Proof*:**
Every locally closed set is constructible (it is the intersection of an open and a closed set, both of which are in the family). Conversely, we must show that the collection of finite unions of locally closed sets is closed under complements and finite intersections.

Finite intersections. The intersection of two locally closed sets $(U_1 \cap Z_1) \cap (U_2 \cap Z_2) = (U_1 \cap U_2) \cap (Z_1 \cap Z_2)$ is locally closed. By distributing, the intersection of two finite unions of locally closed sets is again a finite union of locally closed sets.

Complements. It suffices to show the complement of a single locally closed set $U \cap Z$ is a finite union of locally closed sets. We have:

$$X \setminus (U \cap Z) = (X \setminus U) \cup (X \setminus Z)$$

The first is closed (hence locally closed) and the second is open (hence locally closed), so their union is a finite union of locally closed sets.

The two consequences are immediate: (1) follows from the definition, and (2) follows because continuous preimages preserve open sets, closed sets, intersections, and complements.

If the reader finds this reminiscent of the definition of measurable sets in measure theory, the analogy is apt: constructible sets form the "Boolean algebra" generated by open sets, just as measurable sets form the σ-algebra generated by open sets. The key difference is the finiteness constraint—no countable unions allowed.

For completeness, we record the definition given in EGA for the non-Noetherian case. A *retrocompact* open set $U \subseteq X$ is one where the inclusion $U \hookrightarrow X$ is quasi-compact. In the general definition, one replaces "open" by "retrocompact open" in definition 5.2.2. This ensures that constructible sets are described by "finite amounts of data" even when X is not Noetherian. For schemes of finite type over a Noetherian base—which includes essentially all schemes we care about—retrocompact open sets are the same as quasi-compact open sets, and the two definitions agree. We shall not need this generality.

5.2.4 Assumption For This Section We assume all topological spaces and schemes are Noetherian for the rest of this section, unless otherwise stated.

Generic Freeness

The algebraic engine behind Chevalley's theorem is Grothendieck's Generic Freeness lemma, which says that finitely generated modules over a domain become free after inverting a single element. Geometrically, this means that a coherent sheaf on an integral scheme is "generically" free—it is a vector bundle over a dense open subset.

Theorem 5.2.5: Generic Freeness (Grothendieck)

Let A be a Noetherian domain, S a finitely generated A-algebra, and M a finitely generated S-module. Then there exists a nonzero element $f \in A$ such that M_f is a free A_f-module.

Before giving the proof, let us unpack the geometric content. Taking $S = B$ and $M = B$ for a finitely generated A-algebra B: the theorem says that after shrinking $\operatorname{Spec} A$ to a distinguished open subset $D(f)$, the A_f-module B_f becomes free. In particular, $A_f \to B_f$ is faithfully flat (a nonzero free module is faithfully flat), and so the induced map $\operatorname{Spec} B_f \to \operatorname{Spec} A_f$ is *surjective*. This is the key input for Chevalley's theorem: a dominant finite type morphism from an integral scheme is surjective over a dense open.

***Proof*:**
We induct on the number of A-algebra generators of S. Write $S = A[t_1, \ldots, t_n]$ (up to taking a quotient, which only makes M a module over a ring with fewer generators; explicitly, if $S = A[t_1, \ldots, t_n]/I$ then we view M as an $A[t_1, \ldots, t_n]$-module annihilated by I, and the result for $A[t_1, \ldots, t_n]$ implies the result for S).

Base case: $n = 0$, so $S = A$. Then M is a finitely generated module over the Noetherian domain A. Let $K = \operatorname{Frac}(A)$, and let $r = \dim_K(M \otimes_A K)$. Choose elements $m_1, \ldots, m_r \in M$ whose images form a K-basis of $M \otimes_A K$, and consider the A-linear map:

$$\varphi : A^r \longrightarrow M, \qquad e_i \longmapsto m_i.$$

After tensoring with K, this becomes an isomorphism. Both $\ker(\varphi)$ and $\operatorname{coker}(\varphi)$ are finitely generated A-modules that vanish after tensoring with K. For each generator of $\ker(\varphi)$ and coker(φ), there is a nonzero element of A annihilating it (since the generator maps to zero in the localization at (0)). Let f be the product of all such elements. Then $\ker(\varphi)_f = 0$ and coker$(\varphi)_f = 0$, so $\varphi_f : A_f^r \xrightarrow{\sim} M_f$ is an isomorphism. Hence M_f is free of rank r.

Inductive step: $S = S_0[t]$ *where* $S_0 = A[t_1, \dots, t_{n-1}]$. Let M be a finitely generated $S_0[t]$-module. We split M into its t-torsion and t-torsion-free parts.

Step 1: The t-torsion part. Let $T = \{m \in M : t^k m = 0 \text{ for some } k \geq 1\}$. Since M is Noetherian, T is finitely generated, and there exists $N \geq 1$ with $t^N T = 0$. Hence T is annihilated by t^N, so T is a module over $S_0[t]/(t^N)$. Now, $S_0[t]/(t^N)$ is a finitely generated S_0-module (with S_0-basis $1, t, t^2, \dots, t^{N-1}$). Since T is finitely generated over $S_0[t]/(t^N)$, it is finitely generated over S_0. By the inductive hypothesis (applied with S_0 in place of S), there exists $f_1 \in A \setminus \{0\}$ such that T_{f_1} is a free A_{f_1}-module.

Step 2: The torsion-free part. The quotient $\overline{M} = M/T$ is a finitely generated $S_0[t]$-module on which t acts injectively. Let $K = \mathrm{Frac}(A)$. Then $\overline{M}_K = \overline{M} \otimes_A K$ is a finitely generated module over $K[t_1, \dots, t_{n-1}, t]$, and since t acts injectively, $\overline{M}_K$ is a finitely generated *torsion-free* module over $K[t]$ (torsion-free with respect to t, viewing $\overline{M}_K$ as a $K[t]$-module via the other generators). Since $K[t]$ is a PID, torsion-free and finitely generated implies free:

$$\overline{M}_K \cong K[t]^s$$

for some $s \geq 0$[a]. Choose $\overline{m}_1, \dots, \overline{m}_s \in \overline{M}$ whose images form a $K[t]$-basis of $\overline{M}_K$. The map $S_0[t]^s \to \overline{M}$ sending $e_i \mapsto \overline{m}_i$ becomes an isomorphism after tensoring with K. As in the base case, the kernel and cokernel are finitely generated and K-torsion, so there exists $f_2 \in A \setminus \{0\}$ with:

$$\overline{M}_{f_2} \cong A_{f_2}[t_1, \dots, t_{n-1}, t]^s$$

The right side is a free A_{f_2}-module (of countably infinite rank when $s > 0$, with A_{f_2}-basis given by all monomials times the standard basis vectors).

Step 3: Reassembly. Taking $f = f_1 f_2$, we have that T_f is free over A_f (from Step 1) and $\overline{M}_f$ is free over A_f (from Step 2). The short exact sequence $0 \to T_f \to M_f \to \overline{M}_f \to 0$ splits, since $\overline{M}_f$ is free (hence projective). Therefore $M_f \cong T_f \oplus \overline{M}_f$ is free over A_f.

[a] Strictly speaking, $\overline{M}_K$ is a module over $K[t_1, \dots, t_{n-1}][t]$, and we are using that it is torsion-free over $K[t_1, \dots, t_{n-1}][t]$, which is a PID when we view it as a localization of the polynomial ring. The key point is that after localizing A to K, the t-torsion-free part becomes free over the resulting ring, which is a PID in the variable t.

The geometric consequence we need is immediate:

Corollary 5.2.6: Dominant Morphisms Hit a Dense Open

Let $f : X \to Y$ be a dominant morphism of finite type between Noetherian integral schemes. Then $f(X)$ contains a dense open subset of Y.

***Proof*:**
We may assume $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$ are affine. Since f is dominant, the ring map $A \to B$ is injective (proposition 3.1.32), and B is a finitely generated A-algebra. By theorem 5.2.5 (with $S = B$ and $M = B$), there exists $a \in A \setminus \{0\}$ with B_a a free A_a-module. Since $B \neq 0$, we have $B_a \neq 0$ (localizing a nonzero module at a nonzero element of a domain gives a nonzero

module). A nonzero free module is faithfully flat, so $A_a \to B_a$ is faithfully flat, and the induced map $\operatorname{Spec} B_a \to \operatorname{Spec} A_a$ is surjective. Thus $f(X) \supseteq D(a)$, a dense open subset of Y.

Chevalley's Theorem

We can now state and prove the main result. The proof is an application of Noetherian induction (theorem 0.3.3): the corollary above gives us control over a dense open subset, and the inductive hypothesis handles the closed complement.

Theorem 5.2.7: Chevalley's Theorem

Let $f : X \to Y$ be a morphism of finite type of Noetherian schemes. Then the image of every constructible subset of X is a constructible subset of Y. In particular, $f(X)$ is constructible.

***Proof*:**
Since a constructible set is a finite union of locally closed sets, and a finite union of constructible sets is constructible, it suffices to show that the image of a locally closed set is constructible. A locally closed subset of a Noetherian scheme is again a Noetherian scheme (with its induced reduced structure), and the restriction of a finite type morphism to a locally closed subscheme is still of finite type. Thus it suffices to show that $f(X)$ is constructible.

Step 1: Reduction to the affine case. The question is local on Y: if $Y = \bigcup_j V_j$ is an open cover, then $f(X)$ is constructible in Y if and only if $f(X) \cap V_j$ is constructible in V_j for each j. So we may assume $Y = \operatorname{Spec} A$ is affine. Since f is of finite type, X is quasi-compact, so we can cover X by finitely many affine opens $X = U_1 \cup \cdots \cup U_m$. Then $f(X) = f(U_1) \cup \cdots \cup f(U_m)$, and a finite union of constructible sets is constructible. So we may assume $X = \operatorname{Spec} B$ is also affine, with B a finitely generated A-algebra.

Step 2: Noetherian induction on Y. We prove by Noetherian induction on Y: the statement holds for Y, assuming it holds for every proper closed subset $Z \subsetneq Y$ (equipped with any scheme structure making it a Noetherian scheme). This is valid because Y is a Noetherian topological space (theorem 0.3.3).

Step 3: Decompose the base. Decompose $Y = \operatorname{Spec} A$ into its (finitely many) irreducible components $Y_1, \ldots, Y_r$, where $Y_i = V(\mathfrak{p}_i)$ for the minimal primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ of A (proposition 2.1.21). The image $f(X)$ is the union of the images $f(f^{-1}(Y_i)) \to Y_i$ for $i = 1, \ldots, r$. Each $f^{-1}(Y_i) \to Y_i$ is a morphism of finite type (base change preserves finite type). If $r > 1$, each Y_i is a proper closed subset of Y, and by the inductive hypothesis the image of $f^{-1}(Y_i) \to Y_i$ is constructible in Y_i. A constructible subset of a closed subspace is constructible in the ambient space (it is still a finite union of locally closed sets), so $f(X)$ is constructible. If $r = 1$, then Y is irreducible.

Step 4: The irreducible case. We may now assume $Y = \operatorname{Spec} A$ is irreducible. Replacing A by $A/\operatorname{Nil}(A)$ does not change the underlying topological space (proposition 2.1.10), so we may further assume A is a domain (Y is integral). Similarly, replacing B by $B/\ker(A \to B) \cdot B$ does not change the image of the map on spectra. After this, the morphism $\operatorname{Spec} B \to \operatorname{Spec} A$ factors as $\operatorname{Spec} B \to \operatorname{Spec}(A/J) \hookrightarrow \operatorname{Spec} A$ for some ideal J. If $J \neq 0$, then $V(J) \subsetneq Y$ and we are done by the inductive hypothesis. If $J = 0$, then $A \hookrightarrow B$ and the morphism is dominant.

Step 5: The dominant case. We have A a Noetherian domain, B a finitely generated A-algebra with $A \hookrightarrow B$ injective, and $f : \operatorname{Spec} B \to \operatorname{Spec} A$ dominant. By corollary 5.2.6, there exists

$a \in A \setminus \{0\}$ with $D(a) \subseteq f(X)$.

Consider the closed complement $Z = V(a) \subsetneq \operatorname{Spec} A$. Since $a \neq 0$ and A is a domain, Z is a proper closed subset. The preimage $f^{-1}(Z) = \operatorname{Spec}(B/aB)$ maps to $Z = \operatorname{Spec}(A/aA)$ by a morphism of finite type. By the Noetherian inductive hypothesis, the image of $f^{-1}(Z) \to Z$ is constructible in Z, hence constructible in Y. Therefore:

$$f(X) = D(a) \cup \big(\text{constructible subset of } V(a)\big)$$

is a union of two constructible sets, hence constructible.

The proof reveals more than the statement: it shows that the image of a dominant finite type morphism *contains* a dense open subset of the target. Let us record this and a few other consequences.

Corollary 5.2.8: Image Contains a Dense Open of its Closure

Let $f : X \to Y$ be a morphism of finite type of Noetherian schemes. Then $f(X)$ contains a dense open subset of $\overline{f(X)}$.

***Proof*:**
By theorem 5.2.7, $f(X)$ is constructible, hence a finite union of locally closed sets $f(X) = \bigsqcup_{i=1}^{n}(U_i \cap Z_i)$, where U_i is open and Z_i is closed. We may assume this decomposition is irredundant. Let $\overline{f(X)}$ be the closure of $f(X)$. Since $f(X)$ is dense in $\overline{f(X)}$, at least one of the locally closed pieces, say $U_1 \cap Z_1$, must be dense in some irreducible component of $\overline{f(X)}$. Then $U_1 \cap Z_1$ contains a dense open subset of that component. Applying this argument to each irreducible component gives a dense open subset of $\overline{f(X)}$ contained in $f(X)$.

Alternatively, one can argue directly: replace Y by $\overline{f(X)}$ with its reduced induced structure. Then $f : X \to \overline{f(X)}$ is dominant, and the result follows from Step 5 of the proof of theorem 5.2.7 (applied to each irreducible component).

Corollary 5.2.9: Open and Closed Morphisms via Constructibility

Let $f : X \to Y$ be a morphism of finite type of Noetherian schemes.

1. If f is an open map, then $f(X)$ is open. If f is a closed map (e.g., if f is proper by proposition 3.5.21), then $f(X)$ is closed.
2. More generally, $f(X)$ is open (resp. closed) if and only if it is stable under generization (resp. specialization).

***Proof*:**
Part (1) is immediate. For (2), a constructible set that is stable under generization is open (in a Noetherian space, this follows from the fact that the closure of a constructible set is determined by its specializations). The closed case is dual.

5.2.2 Application: Fiber Dimension

One of the most important applications of Chevalley's theorem is to the study of how fibers vary in a family. Given a morphism $f : X \to Y$, the *fiber* over a point $y \in Y$ is the scheme $X_y = X \times_Y \operatorname{Spec} \kappa(y)$ (definition 3.2.9). The dimension of this fiber can jump as y varies, but Chevalley's theorem constrains *how* it can jump.

Definition 5.2.10: Fiber Dimension

Let $f : X \to Y$ be a morphism of schemes and $x \in X$ a point with $y = f(x)$. The *fiber dimension of f at x* is:

$$\dim_x(f) = \dim_x(X_y)$$

where $X_y = f^{-1}(y)$ is the scheme-theoretic fiber over y, and $\dim_x$ denotes the local dimension at x (the maximum of the dimensions of the irreducible components of X_y containing x).

Theorem 5.2.11: Upper Semicontinuity of Fiber Dimension

Let $f : X \to Y$ be a morphism of finite type of Noetherian schemes. Then for each integer $n \geq 0$, the set:

$$\{x \in X : \dim_x(f) \geq n\}$$

is closed in X. In other words, the function $x \mapsto \dim_x(f)$ is upper semicontinuous.

***Proof*:**
The question is local, so we may assume $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$ are affine with B a finitely generated A-algebra. For a point $x \in X$ corresponding to a prime $\mathfrak{q} \subseteq B$, with $y = f(x)$ corresponding to $\mathfrak{p} = \mathfrak{q} \cap A$, the fiber dimension at x is:

$$\dim_x(f) = \dim(B \otimes_A \kappa(\mathfrak{p}))_{\overline{\mathfrak{q}}}$$

where $\overline{\mathfrak{q}}$ is the image of $\mathfrak{q}$ in $B \otimes_A \kappa(\mathfrak{p})$. We use the following algebraic fact: for a finitely generated $\kappa(\mathfrak{p})$-algebra C and a prime $\mathfrak{q} \subseteq C$, we have $\dim C_{\mathfrak{q}} = \operatorname{trdeg}_{\kappa(\mathfrak{p})} \kappa(\mathfrak{q}) + \dim(C_{\mathfrak{q}}/\mathfrak{q}C_{\mathfrak{q}})$ by theorem 2.6.12. In particular, the fiber dimension is controlled by transcendence degrees.

We now use Chevalley's theorem. Consider the morphism $g : X \to X \times_Y X$ given by the diagonal, and its "relative" version. More directly: for each r, consider the closed subset $Z_r \subseteq X$ defined as follows. By working locally, we can express the fiber dimension condition using the rank of the Jacobian matrix of the defining equations. Alternatively, we argue as follows.

Consider the morphism $X \xrightarrow{f} Y$ and, for a given n, the subset $S_n = \{x \in X : \dim_x(f) \geq n\}$. To show S_n is closed, it suffices to show that for every irreducible closed subset $Z \subseteq X$ with $Z \cap S_n$ dense in Z, we have $Z \subseteq S_n$. This follows from a dimension-theoretic argument: the function $x \mapsto \dim_x(f)$ is constructible on X (as a consequence of Chevalley's theorem applied to appropriate projections), and a constructible function that achieves its minimum on a dense subset of an irreducible set achieves that minimum everywhere. The upper semicontinuity then follows from the fact that on each fiber, the local dimension is upper semicontinuous (theorem 2.6.12 combined with theorem 2.6.13).

A more self-contained argument proceeds by Noetherian induction on Y and uses the Generic Freeness lemma to control fibers over a dense open, much as in the proof of Chevalley's theorem.

We leave the details to the reader as a guided exercise (cf. [24, Exercise 11.4.A]).

The upshot is that fibers can "jump up" in dimension on closed subsets, but not on open ones. Intuitively, the dimension of fibers can only increase as you specialize.

Example 5.2.12: Fiber Dimension in Practice

1. *Blowing up.* Let $f : \mathrm{Bl}_0 \mathbb{A}^2_k \to \mathbb{A}^2_k$ be the blow-up of the plane at the origin (see definition 7.2.26). The generic fiber (over any point $y \neq 0$) is a single point, so $\dim(f^{-1}(y)) = 0$. But the exceptional fiber $f^{-1}(0) \cong \mathbb{P}^1_k$ has dimension 1. The fiber dimension jumps from 0 to 1 at the origin—consistent with semicontinuity, since $\{0\}$ is closed.

2. *Projection of a surface.* Let $X = V(xz-y) \subseteq \mathbb{A}^3_k$ and $f : X \to \mathbb{A}^1_k$ the projection $(x, y, z) \mapsto x$. For $x = a \neq 0$, the fiber is $V(az-y) \cong \mathbb{A}^1_k$ (a line). For $x = 0$, the fiber is $V(y) \subseteq \mathbb{A}^2_k$, which is a copy of $\mathbb{A}^1_k$ parametrized by z. So the fiber dimension is 1 everywhere—no jumping. This is consistent with f being flat (the ring map $k[x] \to k[x, y, z]/(xz-y) \cong k[x, z]$ makes $k[x, z]$ a free $k[x]$-module).

3. *The $(x, y) \mapsto (x, xy)$ example revisited.* For the morphism $f : \mathbb{A}^2_k \to \mathbb{A}^2_k$ from example 5.2.1, the fiber over a point (a, b) with $a \neq 0$ is a single point $(a, b/a)$, so $\dim = 0$. The fiber over $(0, 0)$ is $V(x, xy) = V(x) = \mathbb{A}^1_k$ (the entire y-axis), so $\dim = 1$. The fiber over $(0, b)$ with $b \neq 0$ is $V(x, xy-b) = V(x, b) = \emptyset$. The locus where $\dim_x(f) \geq 1$ is $\{x = 0\} \subseteq \mathbb{A}^2_k$, which is indeed closed.

The examples above suggest a link between flatness and the constancy of fiber dimension. We know that flat morphisms between nice schemes are equidimensional (the fiber dimension does not jump). Remarkably, under the right algebraic hypotheses, the converse is true. This is the celebrated "Miracle Flatness" theorem.

Theorem 5.2.13: Miracle Flatness Theorem

Let $f : X \to Y$ be a morphism of finite type between locally Noetherian schemes. Suppose that:

1. Y is a regular scheme (e.g., a smooth variety),
2. X is Cohen–Macaulay.

Then f is flat if and only if all non-empty fibers X_y have the exact same "expected" dimension, namely $\dim X - \dim Y$. (In other words, f is flat $\iff$ f is equidimensional).

***Proof*:**
The question is entirely local, so we may assume $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$ are affine, and we can localize at a point $x \in X$ and its image $y = f(x) \in Y$. This reduces the problem to a local homomorphism of local Noetherian rings $\varphi : (A, \mathfrak{m}) \to (B, \mathfrak{n})$.

By hypothesis, A is a regular local ring, B is a Cohen–Macaulay local ring, and the fiber dimension condition translates to the dimension formula:

$$\dim B = \dim A + \dim(B/\mathfrak{m}B).$$

We want to show that B is a flat A-module.

Since A is regular, its maximal ideal $\mathfrak{m}$ is generated by a regular sequence of parameters $a_1, \dots, a_d$, where $d = \dim A$.

Because B is Cohen–Macaulay, it is universally catenary and the dimension formula tells us that quotienting B by the ideal $\mathfrak{m}B = (a_1, \dots, a_d)B$ drops the dimension of B by exactly d. In a Cohen–Macaulay ring, any sequence of elements that drops the dimension by its length must be a regular sequence. Therefore, the images $\varphi(a_1), \dots, \varphi(a_d)$ form a regular sequence in B.

We now invoke the *Local Criterion for Flatness* (see [**NateClassicalAlgGeo**]): if $(A, \mathfrak{m}) \to (B, \mathfrak{n})$ is a local homomorphism, and an ideal $I \subseteq A$ is generated by a regular sequence that maps to a regular sequence in B, then B is flat over A. Applying this to $I = \mathfrak{m}$ completes the proof.

5.2.14 Remark: Why is it a "Miracle"? Checking flatness algebraically can be incredibly difficult, as it requires computing Tor groups or checking exactness for all modules. Checking that fibers have constant dimension is a purely topological/geometric task. The "miracle" is that geometry implies algebra here.
The hypotheses are absolutely necessary to prevent algebraic pathologies from hiding behind the geometry:

- If X is not Cohen–Macaulay, it might have embedded points (fuzz) that don't affect the topological dimension of the fibers but destroy algebraic flatness.
- If Y is not regular, the base might have singularities that force any flat family over it to also be highly singular, breaking the dimension count.

The interplay between fiber dimension and cohomology runs deep. In section 6.8.3 we shall prove a more refined semicontinuity result: for a projective morphism $f : X \to Y$ and a coherent sheaf $\mathscr{F}$ flat over Y, the function $y \mapsto h^i(X_y, \mathscr{F}_y)$ is upper semicontinuous. The Euler characteristic $y \mapsto \chi(X_y, \mathscr{F}_y)$ is locally constant. This is a far-reaching generalization whose proof uses a completely different technique (reducing to linear algebra on a complex of free modules), but the philosophical point is the same: cohomological invariants of fibers can only jump up on closed subsets.

5.2.3 The Scheme-Theoretic Image

Chevalley's theorem tells us that the set-theoretic image is constructible, but it says nothing about scheme structure. For many purposes—particularly when studying families or intersections—we need an image that lives in the category of schemes, not just the category of topological spaces. The set-theoretic image rarely has a natural scheme structure (as example 5.2.1 illustrates), so we do the next best thing: we take the smallest *closed* subscheme containing the image.

The idea is natural from the sheaf-theoretic perspective developed in this chapter. A regular function g on Y should "vanish on the image" if and only if g pulls back to zero on X, i.e., $\pi^{\#}(g) = 0$ in $\mathcal{O}_X$. The collection of all such g forms an ideal sheaf, and the corresponding closed subscheme is the scheme-theoretic image.

Definition 5.2.15: Scheme-Theoretic Image

Let $\pi : X \to Y$ be a morphism of schemes. We say the *image of π lies in a closed subscheme* $\iota : Z \hookrightarrow Y$ if the composition:

$$\mathcal{I}_{Z/Y} \longrightarrow \mathcal{O}_Y \xrightarrow{\pi^\#} \pi_*\mathcal{O}_X$$

is zero, where $\mathcal{I}_{Z/Y} = \ker(\mathcal{O}_Y \to \iota_*\mathcal{O}_Z)$ is the ideal sheaf of Z (definition 5.1.30). Equivalently, π factors through Z: every function on Y that vanishes on Z pulls back to zero on X.
The *scheme-theoretic image* of π is the closed subscheme of Y defined by the ideal sheaf:

$$\mathcal{I} = \ker\big(\mathcal{O}_Y \xrightarrow{\pi^\#} \pi_*\mathcal{O}_X\big)$$

That is, it is the smallest closed subscheme of Y through which π factors.

The scheme-theoretic image is well-defined: if the image lies in closed subschemes Z_1 and Z_2, it lies in their scheme-theoretic intersection $Z_1 \cap Z_2$ (the closed subscheme defined by $\mathcal{I}_{Z_1} + \mathcal{I}_{Z_2}$, see proposition 5.1.37). Hence the intersection of all closed subschemes containing the image exists and is the scheme-theoretic image.

Note that the scheme-theoretic image depends on the *scheme structures* of both X and Y, not just on the set-theoretic image. In particular, a non-reduced source can produce a non-reduced image, even when the set-theoretic image is a single point.

There is an important subtlety: the ideal sheaf $\mathcal{I} = \ker(\mathcal{O}_Y \to \pi_*\mathcal{O}_X)$ need not be quasi-coherent in general (the pushforward $\pi_*\mathcal{O}_X$ is not always quasi-coherent). However, by proposition 5.1.20, if π is quasi-compact and quasi-separated, then $\pi_*\mathcal{O}_X$ is quasi-coherent, and hence so is $\mathcal{I}$. In this case, the scheme-theoretic image is a honest closed subscheme (proposition 5.1.34).

The relationship between the scheme-theoretic image and the set-theoretic image is controlled by the quasi-compactness of π:

Proposition 5.2.16: Set-Theoretic Image and Scheme-Theoretic Image

Let $\pi : X \to Y$ be a quasi-compact morphism of schemes, and let Z be the scheme-theoretic image. Then the underlying set of Z is the closure of the set-theoretic image: $|Z| = \overline{\pi(X)}$.

***Proof*:**
Since π is quasi-compact, $\pi_*\mathcal{O}_X$ is quasi-coherent (proposition 5.1.20), and the question is local on Y. So assume $Y = \operatorname{Spec} A$ and $\pi_*\mathcal{O}_X(Y) = B$ (some A-algebra). The ideal sheaf $\mathcal{I}$ corresponds to $I = \ker(A \to B)$, and the scheme-theoretic image is $\operatorname{Spec}(A/I)$.

We must show $V(I) = \overline{\pi(X)}$. A prime $\mathfrak{p} \in \operatorname{Spec} A$ is in $V(I)$ if and only if $I \subseteq \mathfrak{p}$, i.e., every element of A that maps to zero in B lies in $\mathfrak{p}$. On the other hand, $\mathfrak{p} \in \overline{\pi(X)}$ if and only if $\mathfrak{p}$ is in the closure of the set-theoretic image.

If $\mathfrak{p} \in \pi(X)$, then $\mathfrak{p} = \pi(\mathfrak{q})$ for some $\mathfrak{q} \in X$, and the composition $A \to B \to \kappa(\mathfrak{q})$ sends I to zero, so $I \subseteq \mathfrak{p}$. Hence $\pi(X) \subseteq V(I)$. Since $V(I)$ is closed, $\overline{\pi(X)} \subseteq V(I)$.

For the reverse inclusion, suppose $\mathfrak{p} \supseteq I$. We must show $\mathfrak{p} \in \overline{\pi(X)}$, i.e., every open set containing $\mathfrak{p}$ meets $\pi(X)$. Let $D(a)$ be a basic open neighborhood of $\mathfrak{p}$, so $a \notin \mathfrak{p}$ and in particular $a \notin I$.

Then the image of a in B is nonzero, which means $\pi^{-1}(D(a)) \neq \emptyset$ (the function a does not vanish on X). Hence $D(a)$ meets $\pi(X)$.

Without the quasi-compactness hypothesis, the scheme-theoretic image can be strictly larger than the closure of the set-theoretic image, as the pathological example in the following illustrates:

Example 5.2.17: Scheme-Theoretic Image

1. *Non-reduced source.* The morphism $\pi : \operatorname{Spec}(k[\epsilon]/(\epsilon^2)) \to \mathbb{A}^1_k = \operatorname{Spec} k[x]$ given by $x \mapsto \epsilon$ has scheme-theoretic image $V(x^2) = \operatorname{Spec}(k[x]/(x^2))$: a polynomial $g(x)$ pulls back to zero precisely when $g(\epsilon) = 0$ in $k[\epsilon]/(\epsilon^2)$, which forces $g \in (x^2)$. The set-theoretic image is just the origin, but the scheme-theoretic image is a "fat point" of length 2, remembering the tangent direction.

   Compare this with the discussion of nilpotent thickening in example 2.2.8: the scheme-theoretic image captures the first-order infinitesimal data of the source.

2. *Reduced image from non-reduced source.* The morphism $\operatorname{Spec}(k[\epsilon]/(\epsilon^2)) \to \mathbb{A}^1_k$ given by $x \mapsto 0$ has scheme-theoretic image $V(x) = \operatorname{Spec}(k)$: now $g(x)$ pulls back to $g(0)$, which vanishes precisely when $g \in (x)$. Although the source is non-reduced, the image is reduced. The difference from (1) is that the tangent direction is "crushed" by the map—the non-reduced structure of the source is not reflected in the image.

3. *Dense image = scheme-theoretic closure.* The open immersion $j : \mathbb{A}^1_k \setminus \{0\} = \operatorname{Spec} k[t, t^{-1}] \hookrightarrow \mathbb{A}^1_k = \operatorname{Spec} k[u]$ given by $u \mapsto t$ has scheme-theoretic image all of $\mathbb{A}^1_k$: a polynomial $g(u)$ that becomes zero in $k[t, t^{-1}]$ must be the zero polynomial. This is consistent with proposition 5.2.16: the set-theoretic image is $\mathbb{A}^1_k \setminus \{0\}$, whose closure is $\mathbb{A}^1_k$. In this sense, the scheme-theoretic image is the *scheme-theoretic closure* of the image.

4. *Pathological example (non-quasi-compact source).* Consider the morphism:

$$\pi : \coprod_{n \geq 1} \operatorname{Spec}\left(\frac{k[\epsilon_n]}{(\epsilon_n^n)}\right) \longrightarrow \mathbb{A}^1_k = \operatorname{Spec} k[x]$$

   given by $x \mapsto \epsilon_n$ on the nth component. A polynomial $g(x)$ pulls back to zero on the nth component precisely when g vanishes to order $\geq n$ at the origin. Since this must hold for *all* n, the only such g is 0. The scheme-theoretic image is therefore all of $\mathbb{A}^1_k$, even though the set-theoretic image is just the origin.

   This pathology arises because the source is not quasi-compact. For quasi-compact morphisms, the scheme-theoretic image has underlying set equal to the closure of the set-theoretic image (proposition 5.2.16).

The scheme-theoretic image has a clean universal property: it is the smallest closed subscheme $Z \hookrightarrow Y$ such that π factors through Z. This makes it functorial and well-behaved in diagram chases. For instance, if $\pi = \iota \circ g$ where $\iota : Z \hookrightarrow Y$ is a closed immersion, then the scheme-theoretic image of π is contained in Z (as a closed subscheme). This is the algebraic incarnation of the geometric intuition that "the image of a composition lies in the image of the outer map."

We close with a remark connecting the two threads of this section:

5.2.18 Set-Theoretic vs. Scheme-Theoretic: When Do They Agree? For a morphism $\pi : X \to Y$ of finite type between Noetherian schemes, the set-theoretic image $\pi(X)$ is constructible (Chevalley), and the scheme-theoretic image Z has underlying set $\overline{\pi(X)}$ (the closure). These agree—i.e., $|\text{scheme-theoretic image}| = \pi(X)$—precisely when the image is already closed. This happens, for instance, when π is proper (definition 3.5.20), or more generally when π is universally closed.
Even when the underlying sets agree, the scheme-theoretic image may carry non-reduced structure (as in example (1) above). The scheme-theoretic image is the "right" notion of image whenever one cares about scheme structure—for instance, when computing intersections, studying flatness, or constructing moduli spaces.

5.3 Q.C. Sheaves on Projective Schemes

On an affine scheme $\operatorname{Spec} R$, the tilde construction $M \mapsto \widetilde{M}$ establishes an equivalence between R-modules and quasi-coherent sheaves (corollary 5.1.12). For a projective scheme $X = \operatorname{Proj} S_\bullet$, the analogous construction starts with *graded* $S_\bullet$-modules and produces quasi-coherent sheaves on X. The main result of this section is that every quasi-coherent sheaf on $\operatorname{Proj} S_\bullet$ arises in this way (proposition 5.3.14), and that — when $S_\bullet$ is a polynomial ring — the graded ring itself is recovered as a direct sum of global sections of certain natural line bundles, the *twisting sheaves* (proposition 5.3.10).

5.3.1 The Tilde Construction on Proj

Recall from theorem 2.3.3 that the structure sheaf of $X = \operatorname{Proj} S_\bullet$ is defined on distinguished opens by $\mathcal{O}_X(D_+(f)) = (S_f)_0$: we localize at a homogeneous element f and then take the degree-zero part. The same recipe generalizes from the ring $S_\bullet$ to any graded module over it.

Definition 5.3.1: Sheaf Associated to M [on Projective Scheme]

Let $S_\bullet$ be a graded ring and $M_\bullet$ a graded $S_\bullet$-module. The *sheaf associated to* $M_\bullet$ on $X = \operatorname{Proj} S_\bullet$, denoted $\widetilde{M}$, is the $\mathcal{O}_X$-module defined on the basis of projective distinguished opens by

$$\widetilde{M}(D_+(f)) \;=\; \left(M[f^{-1}]\right)_0,$$

where $f \in S_\bullet$ is a nonzero homogeneous element and $(-)_0$ denotes the degree-zero component. The restriction maps are induced by further localization.

Let us unpack this. The localization $M[f^{-1}]$ inverts f in the graded sense: a typical element is m/f^k where $m \in M$ is homogeneous. This element has degree $\deg m - k \deg f$. Taking the degree-zero part selects those fractions m/f^k where $\deg m = k \deg f$. When $M = S_\bullet$ this recovers the structure sheaf $(S_f)_0 = \mathcal{O}_X(D_+(f))$; for a general module, the degree-zero localization $(M_f)_0$ is a module over the ring $(S_f)_0$.

Concretely, if $S_\bullet = R[x_0, \ldots, x_r]$ is the standard polynomial ring graded by degree, then $D_+(x_i) \cong \operatorname{Spec}(S_{x_i})_0$ and

$$\widetilde{M}(D_+(x_i)) \;=\; (M_{x_i})_0,$$

which is a module over the affine coordinate ring $(S_{x_i})_0 \cong R[x_0/x_i, \ldots, \widehat{x_i/x_i}, \ldots, x_r/x_i]$.

Example 5.3.2: The Tilde Construction on $\mathbb{P}^1$

Let $S_\bullet = k[x, y]$ with the standard grading, so that $X = \mathbb{P}^1_k$. Take $M = S_\bullet/(x^2 - y^2)$, graded by the quotient grading, so that M has one generator in degree 0 and the relation $x^2 = y^2$ in degree 2.

On the chart $D_+(y) \cong \operatorname{Spec} k[t]$ (where $t = x/y$), the module of sections is

$$(M_y)_0 \;=\; \frac{k[x/y]}{((x/y)^2 - 1)} \;=\; \frac{k[t]}{(t^2 - 1)},$$

which is a $k[t]$-module. Similarly, on $D_+(x) \cong \operatorname{Spec} k[s]$ (where $s = y/x$):

$$(M_x)_0 \;=\; \frac{k[s]}{(1 - s^2)}.$$

Thus $\widetilde{M}$ restricts on each affine chart to the sheaf associated to a familiar quotient ring — the coordinate ring of two points $\{t = \pm 1\}$ on $\mathbb{A}^1$. The gluing of these local pieces recovers the sheaf on all of $\mathbb{P}^1$.

Proposition 5.3.3: Sheaf on Projective Scheme is Quasi-Coherent

Let $S_\bullet$ be a graded ring, $M_\bullet$ a graded $S_\bullet$-module, and $X = \operatorname{Proj} S_\bullet$. Then:

1. The stalk of $\widetilde{M}$ at a homogeneous prime $\mathfrak{p} \in X$ is $\widetilde{M}_\mathfrak{p} \cong M_{(\mathfrak{p})}$, the homogeneous localization of M at $\mathfrak{p}$.
2. $\widetilde{M}$ is a quasi-coherent $\mathcal{O}_X$-module. If $S_\bullet$ is Noetherian and $M_\bullet$ is finitely generated, then $\widetilde{M}$ is coherent.

***Proof*:**

On each basic open $D_+(f) \cong \operatorname{Spec}(S_f)_0$, the sheaf $\widetilde{M}$ restricts to the sheaf associated to the $(S_f)_0$-module $(M_f)_0$, which is quasi-coherent by the affine theory. The stalk computation follows from the same colimit argument as for the structure sheaf (theorem 2.3.3). For coherence, if $M_\bullet$ is finitely generated over a Noetherian ring $S_\bullet$, then each $(M_f)_0$ is finitely generated over the Noetherian ring $(S_f)_0$.

The structure sheaf $\mathcal{O}_X = \widetilde{S_\bullet}$ captures the degree-zero part of $S_\bullet$. To access sections of other degrees, we shift the grading:

Definition 5.3.4: Twisting Sheaf

Let $S_\bullet = \bigoplus_n S_n$ be a graded ring and $X = \operatorname{Proj} S_\bullet$. For each $n \in \mathbb{Z}$, define the *nth twisting sheaf* to be

$$\mathcal{O}_X(n) \;:=\; \widetilde{S(n)},$$

where $S(n)$ is the graded module with $S(n)_d = S_{n+d}$. The sheaf $\mathcal{O}_X(1)$ is called *the twisting sheaf* (or *Serre's twisting sheaf*) of X.

Note that $S(0) = S_\bullet$, so $\mathcal{O}_X(0) = \mathcal{O}_X$. Let us work out what the sections of $\mathcal{O}_X(n)$ look like concretely. On a distinguished open $D_+(f)$ for a homogeneous element f of degree e, the definition gives

$$\mathcal{O}_X(n)(D_+(f)) \;=\; \big(S(n)[f^{-1}]\big)_0 \;=\; \big(S[f^{-1}]\big)_n \;=\; (S_f)_n,$$

the elements of degree n in the localized ring S_f. For $n \neq 0$, this is no longer a ring, but it is a module over $\mathcal{O}_X(D_+(f)) = (S_f)_0$.

Let us see what this means explicitly on the standard projective space. Take $S_\bullet = k[x_0, \ldots, x_r]$ with the usual grading by total degree. On the chart $D_+(x_i) \cong \operatorname{Spec} k[x_0/x_i, \ldots, x_r/x_i]$, the ring $S_{x_i} = k[x_0, \ldots, x_r, x_i^{-1}]$ contains elements of every integer degree. An element of $(S_{x_i})_n$ is a sum of monomials in $x_0, \ldots, x_r, x_i^{-1}$ of total degree n. Since x_i is invertible, every such monomial can be written as x_i^n times a degree-zero monomial. More precisely, the map

$$(S_{x_i})_0 \xrightarrow[\cdot\, x_i^n]{\sim} (S_{x_i})_n, \qquad g \longmapsto x_i^n \cdot g, \tag{5.1}$$

is an isomorphism of $(S_{x_i})_0$-modules: multiplication by the unit x_i^n shifts the degree from 0 to n. This proves the following:

Proposition 5.3.5: $\mathcal{O}_X(n)$ is an Invertible Sheaf

Let $S_\bullet$ be a graded ring generated by S_1 as an S_0-algebra, and $X = \operatorname{Proj} S_\bullet$. Then $\mathcal{O}_X(n)$ is an invertible sheaf (line bundle) for each $n \in \mathbb{Z}$.

***Proof*:**

For each $f \in S_1$, multiplication by f^n defines an isomorphism $(S_f)_0 \xrightarrow{\sim} (S_f)_n$ of $(S_f)_0$-modules (as in (5.1)), giving $\mathcal{O}_X(n)|_{D_+(f)} \cong \mathcal{O}_{D_+(f)}$. Since $S_\bullet$ is generated by S_1, the opens $\{D_+(f) : f \in S_1\}$ cover X, so $\mathcal{O}_X(n)$ is locally free of rank 1.

Although $\mathcal{O}_X(n)$ is locally isomorphic to $\mathcal{O}_X$ on each chart, the sheaves $\mathcal{O}_X(n)$ for different n are *not* globally isomorphic — the local trivializations glue together differently, producing genuinely distinct line bundles. We shall now prove this by computing global sections. Before treating the general case, let us work through $\mathbb{P}^1$ in full detail.

Let $S_\bullet = k[x, y]$ and $X = \mathbb{P}^1_k = \operatorname{Proj} S_\bullet$, covered by the two charts

$$D_+(y) \cong \operatorname{Spec} k[t], \quad t = x/y,$$
$$D_+(x) \cong \operatorname{Spec} k[s], \quad s = y/x.$$

On the overlap $D_+(x) \cap D_+(y) \cong \operatorname{Spec} k[t, t^{-1}]$, we have $s = 1/t$.

A section of $\mathcal{O}(n)$ over $D_+(y)$ is an element of $(S_y)_n$: a degree-n element of $k[x, y, y^{-1}]$. Any such element can be written uniquely as $y^n \cdot p(x/y) = y^n \cdot p(t)$ for a polynomial $p \in k[t]$. Similarly, a section over $D_+(x)$ is $x^n \cdot q(y/x) = x^n \cdot q(s)$ for a polynomial $q \in k[s]$. A global section must give a pair $(p(t), q(s))$ that *agree on the overlap.* On $D_+(x) \cap D_+(y)$, both expressions represent the same element of $(S_{xy})_n$:

$$y^n \cdot p(t) \;=\; x^n \cdot q(s) \;=\; (ty)^n \cdot q(1/t) \;=\; y^n \cdot t^n\, q(1/t).$$

Cancelling y^n (which is a unit on the overlap), the gluing condition is

$$p(t) \;=\; t^n \cdot q(1/t). \tag{5.2}$$

Let us now examine this condition for several values of n.

Case $n = 0$ (the structure sheaf). The condition becomes $p(t) = q(1/t)$. Writing $p(t) = \sum_{i\geq 0} a_i t^i$ and $q(s) = \sum_{j\geq 0} b_j s^j$, the right side is $\sum_j b_j t^{-j}$. For this to equal a polynomial in t (no negative powers), we need $b_j = 0$ for $j \geq 1$, so $q(s) = b_0$ is constant. Then $p(t) = b_0$ is also constant. Hence $\Gamma(\mathbb{P}^1, \mathcal{O}) = k$, recovering the familiar fact that the only global regular functions on $\mathbb{P}^1$ are constants (proposition 2.3.6).

Case $n = 1$ (the twisting sheaf). Now $p(t) = t \cdot q(1/t)$. Write $q(s) = b_0 + b_1 s + b_2 s^2 + \cdots$. Then

$$t \cdot q(1/t) \;=\; t\left(b_0 + \frac{b_1}{t} + \frac{b_2}{t^2} + \cdots\right) \;=\; b_0 t + b_1 + \frac{b_2}{t} + \cdots$$

For $p(t)$ to be a polynomial (no negative powers of t), we need $b_2 = b_3 = \cdots = 0$, giving $q(s) = b_0 + b_1 s$. Then $p(t) = b_1 + b_0 t$. The global section is thus parametrized by the pair $(b_0, b_1) \in k^2$, or equivalently by the degree-1 polynomial $b_0 x + b_1 y \in S_1$. Hence $\Gamma(\mathbb{P}^1, \mathcal{O}(1)) \cong S_1 \cong k^2$.

Case $n = 2$. By the same analysis with $p(t) = t^2 \cdot q(1/t)$, one finds $q(s) = b_0 + b_1 s + b_2 s^2$ and $p(t) = b_2 + b_1 t + b_0 t^2$. The global sections are parametrized by $(b_0, b_1, b_2) \in k^3$, corresponding to degree-2 polynomials $b_0 x^2 + b_1 xy + b_2 y^2 \in S_2$. So $\Gamma(\mathbb{P}^1, \mathcal{O}(2)) \cong S_2 \cong k^3$.

Case $n = -1$ (negative twist). Now $p(t) = t^{-1} \cdot q(1/t)$. Write $q(s) = b_0 + b_1 s + \cdots$. Then

$$t^{-1} \cdot q(1/t) \;=\; \frac{1}{t}\left(b_0 + \frac{b_1}{t} + \cdots\right) \;=\; \frac{b_0}{t} + \frac{b_1}{t^2} + \cdots$$

For this to be a polynomial in t, *every* coefficient must vanish: $b_0 = b_1 = \cdots = 0$. Hence $q = 0$ and $p = 0$: the only global section of $\mathcal{O}(-1)$ is zero. Geometrically, the transition function t^{-1} introduces a pole that no polynomial can cancel, so no nonzero section can be defined on both charts simultaneously.

General $n < 0$. The same argument applies: $p(t) = t^n q(1/t)$ for $n < 0$ forces all coefficients of q to vanish, so $\Gamma(\mathbb{P}^1, \mathcal{O}(n)) = 0$ for every $n < 0$. Note that these sheaves often be distinguished by having different local sections, where poles of different degrees can occur.

General $n \geq 0$. The gluing condition $p(t) = t^n q(1/t)$ is satisfied by polynomials q of degree $\leq n$, giving an $(n+1)$-dimensional space. Hence $\Gamma(\mathbb{P}^1, \mathcal{O}(n)) \cong S_n \cong k^{n+1}$ for $n \geq 0$.

The pattern generalizes to $\mathbb{P}^r_k$ with $S_\bullet = k[x_0, \ldots, x_r]$. The same argument — covering by the $r+1$ standard charts and imposing the gluing conditions — yields $\Gamma(\mathbb{P}^r_k, \mathcal{O}(n)) \cong S_n$ for $n \geq 0$, the vector space of homogeneous polynomials of degree n in $r+1$ variables, which has dimension $\binom{n+r}{r}$. For $n < 0$ and $r \geq 1$, the only global section is zero (the transition functions introduce poles that cannot be compensated on every chart).

The exceptional case is $\mathbb{P}^0_k = \operatorname{Proj} k[x_0] \cong \operatorname{Spec} k$: since there is only one chart, there are no nontrivial gluing conditions, and $\Gamma(\mathbb{P}^0_k, \mathcal{O}(n)) \cong k$ for all n.

Collecting these results:

$\dim_k \Gamma(\mathbb{P}^r_k, \mathcal{O}(n))$	$\mathbb{P}^0$	$\mathbb{P}^1$	$\mathbb{P}^2$	$\mathbb{P}^3$	$\mathbb{P}^4$
$n = -2$	1	0	0	0	0
$n = -1$	1	0	0	0	0
$n = 0$	1	1	1	1	1
$n = 1$	1	2	3	4	5
$n = 2$	1	3	6	10	15
general $n \geq 0$, $r \geq 1$	—	$\binom{n+1}{1}$	$\binom{n+2}{2}$	$\binom{n+3}{3}$	$\binom{n+4}{4}$

Since the spaces of global sections have different dimensions for different n (when $r \geq 1$), the line bundles $\mathcal{O}_X(n)$ are pairwise non-isomorphic. These dimensions will be computed systematically using sheaf cohomology in theorem 6.3.1, where we shall also determine $H^i(\mathbb{P}^r, \mathcal{O}(n))$ for $i > 0$: the vanishing of H^0 for negative twists is only the beginning of the story.

5.3.6 Why "Twisting"? The name comes from the transition functions. On the overlap $D_+(x) \cap D_+(y)$ of $\mathbb{P}^1_k$, the local trivializations (5.1) identify sections of $\mathcal{O}(n)$ with elements of the structure sheaf via multiplication by y^n (on the y-chart) and x^n (on the x-chart). The transition function relating these two trivializations is

$$\varphi_{xy} = \frac{x^n}{y^n} = t^n = s^{-n},$$

which is precisely the factor appearing in the gluing condition (5.2).

For $n = 0$, the transition function is 1: the bundle is trivial. For $n = 1$, the transition function is t: traversing the base $\mathbb{P}^1$ and returning to the starting chart, the fibre is multiplied by $t \cdot (1/t) = 1$ only after passing through *both* charts. Over $\mathbb{R}$, the total space of $\mathcal{O}(1)$ on $\mathbb{P}^1_{\mathbb{R}}$ is a Möbius-like band: the fibre must be "twisted" once to close up. For general n, the transition involves $|n|$ such twists (with opposite orientation for $n < 0$).

More precisely, the transition functions define an element of the Čech cohomology group $\check{H}^1(\mathbb{P}^1, \mathcal{O}_X^\times) \cong \operatorname{Pic}(\mathbb{P}^1) \cong \mathbb{Z}$, and the integer n is exactly the degree of the line bundle. This connection between transition functions, line bundles, and Čech cohomology will be developed fully in section 5.4 and section 6.1.1.

Definition 5.3.7: n-Twisted Sheaf of $\mathscr{F}$

For any $\mathcal{O}_X$-module $\mathscr{F}$ on $X = \operatorname{Proj} S_\bullet$, the *$n$-twisted sheaf* is

$$\mathscr{F}(n) \;:=\; \mathscr{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n).$$

Twisting by $\mathcal{O}_X(n)$ is a fundamental operation: it shifts the "degree of homogeneity" of sections. Since $\mathcal{O}_X(n)$ is locally free of rank 1, tensoring with it preserves quasi-coherence, coherence, and local freeness.

Proposition 5.3.8: Properties of Twisted Sheaves

Let $S_\bullet$ be a graded ring generated by S_1 as an S_0-algebra, and $X = \operatorname{Proj} S_\bullet$. Then:

1. $\mathcal{O}_X(n) \otimes_{\mathcal{O}_X} \mathcal{O}_X(m) \cong \mathcal{O}_X(n+m)$ for all $n, m \in \mathbb{Z}$. In particular, $\mathcal{O}_X(n)$ and $\mathcal{O}_X(-n)$ are tensor-product inverses, confirming that each $\mathcal{O}_X(n)$ is invertible.

2. For any graded $S_\bullet$-module $M_\bullet$, $\widetilde{M}(n) \cong \widetilde{M(n)}$.

3. Let $T_\bullet$ be another graded ring generated by T_1 over T_0, let $\varphi : S_\bullet \to T_\bullet$ be a graded homomorphism, and let $U \subseteq \operatorname{Proj} T_\bullet$ be the open locus where $f : U \to X$ is induced by φ. Then

$$f^*(\mathcal{O}_X(n)) \;\cong\; \mathcal{O}_{\operatorname{Proj} T}(n)\big|_U.$$

***Proof*:**
(1). On each $D_+(f)$ for $f \in S_1$, both sides restrict to $(S_f)_{n+m}$: the left side via the isomorphism $(S_f)_n \otimes_{(S_f)_0} (S_f)_m \xrightarrow{\sim} (S_f)_{n+m}$ given by multiplication in S_f, and the right side by definition. These local isomorphisms are compatible with restriction (further localization commutes with multiplication), hence glue to a global isomorphism by corollary 1.3.4.

(2). On $D_+(f)$, the left side gives $(M_f)_0 \otimes_{(S_f)_0} (S_f)_n \cong (M_f)_n$, and the right gives $(M(n)_f)_0 = (M_f)_n$.

(3). Reduce to the affine case: on $D_+(g) \subseteq \operatorname{Proj} T_\bullet$ mapping into $D_+(\varphi(g)) \subseteq X$, pullback corresponds to extension of scalars, and the grading shift commutes with this operation.

Having seen that $\Gamma(\mathbb{P}^r_k, \mathcal{O}(n)) \cong S_n$, it is natural to reassemble all degrees into a single graded object and ask whether we recover the entire graded ring $S_\bullet$.

Definition 5.3.9: Graded Module Associated to $\mathscr{F}$

Let $S_\bullet$ be a graded ring, $X = \operatorname{Proj} S_\bullet$, and $\mathscr{F}$ an $\mathcal{O}_X$-module. The *graded $S_\bullet$-module associated to $\mathscr{F}$* is

$$\Gamma_*(\mathscr{F}) := \bigoplus_{n\in\mathbb{Z}} \Gamma(X, \mathscr{F}(n)),$$

with the $S_\bullet$-module structure induced by the multiplication maps $S_d \otimes \Gamma(X, \mathscr{F}(n)) \to \Gamma(X, \mathscr{F}(n+d))$ coming from proposition 5.3.8(1)–(2).

For polynomial rings, this recovers $S_\bullet$ exactly:

Proposition 5.3.10: Recovering the Graded Ring

Let R be a ring, $S_\bullet = R[x_0, \dots, x_r]$, and $X = \operatorname{Proj} S_\bullet$. Then $\Gamma_*(\mathcal{O}_X) \cong S_\bullet$.

***Proof*:**
A global section $t \in \Gamma(X, \mathcal{O}_X(n))$ is a family of sections $t_i \in \mathcal{O}_X(n)(D_+(x_i)) = (S_{x_i})_n$ that agree on overlaps. The variables x_i are not zero divisors in $S_\bullet$, so the localization maps $S_\bullet \hookrightarrow S_{x_i} \hookrightarrow S_{x_0\cdots x_r}$ are all injective, and these rings are subrings of the common localization $S' = S_{x_0\cdots x_r}$. The agreement condition on overlaps means t lies in $\bigcap_{i=0}^r (S_{x_i})_n$ inside S'.

It remains to show $\bigcap_i S_{x_i} = S_\bullet$ inside S'. Every homogeneous element of S' can be written uniquely as $x_0^{a_0} \cdots x_r^{a_r} \cdot g(x_0, \dots, x_r)$ with $a_j \in \mathbb{Z}$ and g not divisible by any x_i. Such an element lies in S_{x_i} if and only if $a_j \geq 0$ for all $j \neq i$. Belonging to *every* S_{x_i} requires $a_j \geq 0$ for all j (for each j, choose any $i \neq j$), which means the element lies in $S_\bullet$. Summing over degrees gives $\Gamma_*(\mathcal{O}_X) \cong S_\bullet$.

The reader should note two important features of this proof: the injectivity of the localization maps (which uses the fact that x_i are not zero divisors) and the combinatorial intersection argument. For a general graded ring $S_\bullet$ where the generators may be zero divisors, the conclusion can fail.

More fundamentally, the Γ_*–tilde correspondence is *not* an equivalence in either direction:

- The map $M_\bullet \mapsto \Gamma_*(\widetilde{M})$ need not recover M. For instance, if $M_0 = k$ and $M_n = 0$ for $n \neq 0$, then $\widetilde{M} = 0$ (there are no nonzero elements to localize in positive degrees), so $\Gamma_*(\widetilde{M}) = 0 \neq M$. The graded module M is concentrated in a single degree that is "invisible" to Proj.
- The map $M_\bullet \mapsto \widetilde{M}$ is not injective: the above example shows two different modules (M and 0) giving the same sheaf.

However, the other direction works perfectly: starting from a quasi-coherent sheaf and applying Γ_*, one can always recover the sheaf by applying the tilde construction. This is the key structural result of this section.

The proof requires a generalization of lemma 5.1.15 from distinguished opens of affine schemes to the projective setting. The role of the element $f \in R$ (which defines the distinguished open $D(f)$ in the affine case) is played by a global section $f \in \Gamma(X, \mathcal{L})$ of an invertible sheaf $\mathcal{L}$. The open set $X_f := \{x \in X : f_x \notin \mathfrak{m}_x \mathcal{L}_x\}$ is the locus where f "does not vanish" (cf. the evaluation of sections in definition 2.2.24); when $\mathcal{L} = \mathcal{O}_X$ this recovers the ordinary distinguished open, and when $\mathcal{L} = \mathcal{O}_X(1)$ on $\operatorname{Proj} S_\bullet$ this recovers $D_+(f)$.

Lemma 5.3.11: Extending Sections for Line Bundles

Let X be a scheme, $\mathcal{L}$ an invertible sheaf on X, $f \in \Gamma(X, \mathcal{L})$, and X_f the open set of points $x \in X$ where $f_x \notin \mathfrak{m}_x \mathcal{L}_x$. Let $\mathscr{F}$ be a quasi-coherent sheaf on X.

1. Suppose X is quasi-compact. If $s \in \Gamma(X, \mathscr{F})$ satisfies $s|_{X_f} = 0$, then $f^n s = 0$ in $\Gamma(X, \mathscr{F} \otimes \mathcal{L}^{\otimes n})$ for some $n > 0$.
2. Suppose additionally that X has a finite cover by affine opens $\{U_i\}$ with $\mathcal{L}|_{U_i}$ free and $U_i \cap U_j$ quasi-compact for all i, j. Then for any $t \in \Gamma(X_f, \mathscr{F})$, there exists $n > 0$ such that $f^n t \in \Gamma(X_f, \mathscr{F} \otimes \mathcal{L}^{\otimes n})$ extends to a global section of $\mathscr{F} \otimes \mathcal{L}^{\otimes n}$.

The reader should think of this lemma as the projective analogue of lemma 5.1.15: part (1) says "sections that vanish on X_f are killed by a power of f," and part (2) says "sections defined only on X_f can be extended globally after multiplying by a power of f." The new feature is that f is a section of a line bundle rather than a global function, so "multiplying by f^n" lands in the twisted sheaf $\mathscr{F} \otimes \mathcal{L}^{\otimes n}$ rather than in $\mathscr{F}$ itself. In the special case $\mathcal{L} = \mathcal{O}_X$, the section f is an ordinary global function, the open X_f is the distinguished open $D(f)$, and we recover lemma 5.1.15 exactly.

Before giving the proof, let us verify the hypotheses in the case we care about most.

Example 5.3.12: Hypotheses for Proj

Let $S_\bullet$ be a graded ring finitely generated by S_1 over S_0, $X = \operatorname{Proj} S_\bullet$, $\mathcal{L} = \mathcal{O}_X(1)$, and $f \in S_1$ a degree-1 element viewed as a section $f \in \Gamma(X, \mathcal{O}_X(1))$.

The open set X_f. A point $x \in X$ (corresponding to a homogeneous prime $\mathfrak{p}$) lies in X_f precisely when f_x generates the free $\mathcal{O}_{X,x}$-module $\mathcal{L}_x = \mathcal{O}_X(1)_x$, i.e., when f_x is a unit times the local generator. On the chart $D_+(g)$ for $g \in S_1$, the generator of $\mathcal{O}_X(1)|_{D_+(g)}$ is g (via (5.1)), and f_x is a unit times g precisely when $f/g \notin \mathfrak{p}$, i.e., when $\mathfrak{p} \in D_+(f)$. Hence $X_f = D_+(f)$.

The finite cover. Since $S_\bullet$ is generated by S_1, choose generators $f_0, \ldots, f_r \in S_1$; then $U_i = D_+(f_i)$ is affine (theorem 2.3.3), $\mathcal{L}|_{U_i}$ is free (by proposition 5.3.5), and $U_i \cap U_j = D_+(f_i f_j)$ is affine (hence quasi-compact).

Thus the hypotheses of both parts of lemma 5.3.11 are satisfied for any $f \in S_1$ on $\operatorname{Proj} S_\bullet$.

***Proof of lemma 5.3.11*:**

The strategy for both parts is to reduce to the affine case by trivializing $\mathcal{L}$ on a finite cover, applying lemma 5.1.15, and then patching the results.

(1) Annihilation. Cover X by finitely many affine opens $U_\alpha = \operatorname{Spec} A_\alpha$ (possible since X is quasi-compact) on which $\mathcal{L}$ is free; choose trivializations $\psi_\alpha : \mathcal{L}|_{U_\alpha} \xrightarrow{\sim} \mathcal{O}_{U_\alpha}$. Set $g_\alpha = \psi_\alpha(f|_{U_\alpha}) \in A_\alpha$, so that $X_f \cap U_\alpha = D(g_\alpha)$:

$$\mathcal{L}(U_\alpha) \xrightarrow[\sim]{\psi_\alpha} \mathcal{O}_X(U_\alpha) = A_\alpha$$

$$f|_{U_\alpha} \longmapsto g_\alpha$$

Write $\mathscr{F}|_{U_\alpha} \cong \widetilde{M_\alpha}$ for an A_α-module M_α. The section $s|_{U_\alpha}$ corresponds to an element $s_\alpha \in M_\alpha$, and the hypothesis $s|_{X_f} = 0$ gives $s_\alpha|_{D(g_\alpha)} = 0$.

By the affine annihilation lemma (lemma 5.1.15(1)), $g_\alpha^{n_\alpha} s_\alpha = 0$ in M_α for some $n_\alpha > 0$. Using the trivialization $\psi_\alpha^{\otimes n} : \mathcal{L}^{\otimes n}|_{U_\alpha} \xrightarrow{\sim} \mathcal{O}_{U_\alpha}$, the element $g_\alpha^{n_\alpha} s_\alpha$ corresponds to $(f^{n_\alpha} s)|_{U_\alpha}$ intrinsically (independent of the choice of ψ_α). Taking $n = \max_\alpha n_\alpha$ gives $f^n s|_{U_\alpha} = 0$ for every α. Since the U_α cover X, the locality axiom gives $f^n s = 0$ globally.

(2) Extension. This is more involved: we must first extend the section locally, then correct the local extensions so that they glue.

Step 1: Local extensions. On each U_i of the given finite cover, the trivialization ψ_i converts $f|_{U_i}$ to $g_i \in A_i$ and gives $X_f \cap U_i = D(g_i)$. The section $t|_{D(g_i)} \in \mathscr{F}(D(g_i)) = (M_i)_{g_i}$. By the affine extension lemma (lemma 5.1.15(2)), there exists $n_i > 0$ such that $g_i^{n_i} \cdot t|_{D(g_i)}$ extends to an element $\hat{t}_i \in M_i = \mathscr{F}(U_i)$. Intrinsically, this means $f^{n_i} t|_{X_f \cap U_i}$ extends to a section of $\mathscr{F} \otimes \mathcal{L}^{\otimes n_i}$ over U_i. Taking $N = \max_i n_i$ (and multiplying by further powers of f where needed), we may assume each $\hat{t}_i$ is a section of $\mathscr{F} \otimes \mathcal{L}^{\otimes N}$ over U_i that restricts to $f^N t$ on $X_f \cap U_i$.

Step 2: Correcting overlaps. On $U_i \cap U_j$, both $\hat{t}_i$ and $\hat{t}_j$ restrict to $f^N t$ on $X_f \cap U_i \cap U_j$, so their difference $\delta_{ij} = \hat{t}_i - \hat{t}_j$ vanishes on $X_f \cap U_i \cap U_j$. Since $U_i \cap U_j$ is quasi-compact by hypothesis, part (1) applied to δ_{ij} on $U_i \cap U_j$ gives $m_{ij} > 0$ with $f^{m_{ij}} \delta_{ij} = 0$ on $U_i \cap U_j$. Set $m = \max_{i,j} m_{ij}$ (a finite maximum since the cover is finite).

Then the sections $f^m \hat{t}_i \in \Gamma(U_i, \mathscr{F} \otimes \mathcal{L}^{\otimes(N+m)})$ satisfy $f^m \hat{t}_i = f^m \hat{t}_j$ on every overlap $U_i \cap U_j$. By the gluability axiom, they glue to a global section $\bar{t} \in \Gamma(X, \mathscr{F} \otimes \mathcal{L}^{\otimes(N+m)})$. On X_f, this restricts to $f^{N+m} t$, so $\bar{t}$ is the desired extension with $n = N + m$.

5.3.13 The Mechanism of Extension The reader may find it helpful to compare the affine and projective extension results side by side:

Affine (lemma 5.1.15)	**Projective** (lemma 5.3.11)
$f \in R$ (ring element)	$f \in \Gamma(X, \mathcal{L})$ (section of line bundle)
Open: $D(f) \subseteq \operatorname{Spec} R$	Open: $X_f \subseteq X$
$t \in M_f$ (localization)	$t \in \mathscr{F}(X_f)$ (sections on X_f)
$f^n t \in M$ (cleared denominator)	$f^n t \in \Gamma(X, \mathscr{F} \otimes \mathcal{L}^{\otimes n})$ (twisted extension)

In the affine case, multiplying by f^n "clears the denominator" and lands back in M. In the projective case, the same multiplication lands in a *different* module — the global sections of the nth twist — because f is a section of $\mathcal{L}$, not of $\mathcal{O}_X$. The price we pay for working on a non-affine scheme is that extension requires passing to a twist; the benefit is that this works for *any* quasi-compact scheme with a suitable line bundle.

We now prove the key structural result: the tilde functor applied to $\Gamma_*(\mathscr{F})$ recovers $\mathscr{F}$.

Proposition 5.3.14: Partial Equivalence for Graded Rings

Let $S_\bullet$ be a graded ring, finitely generated by S_1 as an S_0-algebra. Let $X = \operatorname{Proj} S_\bullet$ and $\mathscr{F}$ a quasi-coherent sheaf on X. Then there is a natural isomorphism

$$\beta : \widetilde{\Gamma_*(\mathscr{F})} \xrightarrow{\sim} \mathscr{F}.$$

In particular, every quasi-coherent sheaf on $\operatorname{Proj} S_\bullet$ arises from a graded $S_\bullet$-module.

The hypothesis that $S_\bullet$ is generated by S_1 is essential: it ensures that X is covered by the affine opens $D_+(f)$ for $f \in S_1$, on which the twisting sheaf $\mathcal{O}_X(1)$ becomes free.

***Proof*:**
Step 1: Constructing β. We define β on the basis of distinguished opens $D_+(f)$ for $f \in S_+$ homogeneous. A section of $\widetilde{\Gamma_*(\mathscr{F})}$ over $D_+(f)$ is an element of the homogeneous localization $\Gamma_*(\mathscr{F})_{(f)}$: a fraction m/f^d where $m \in \Gamma(X, \mathscr{F}(d))$ is a global section of the dth twist and $\deg f \cdot d$ matches $\deg m$ in the graded module.

Now f^{-d} can be viewed as a section of $\mathcal{O}_X(-d)$ over $D_+(f)$: since $\mathcal{O}_X(d)|_{D_+(f)}$ is trivial, the element f^d is a generator, and f^{-d} is its inverse in the localized sections. Tensoring, the element $m \otimes f^{-d}$ is a section of $\mathscr{F}(d) \otimes \mathcal{O}_X(-d) \cong \mathscr{F}$ over $D_+(f)$. Define

$$\beta_{D_+(f)}\left(\frac{m}{f^d}\right) := m \otimes f^{-d} \in \mathscr{F}(D_+(f)).$$

Let us verify this is well-defined. If $m/f^d = m'/f^{d'}$ in $\Gamma_*(\mathscr{F})_{(f)}$, then $f^N(f^{d'}m - f^d m') = 0$ for some N, i.e., the sections $f^{d'}m$ and $f^d m'$ agree in $\mathscr{F}(d + d' + N \deg f)$ after multiplying by f^N. Restricting to $D_+(f)$ and dividing by $f^{d+d'+N}$ (which is a unit there), we get $m \otimes f^{-d} = m' \otimes f^{-d'}$ in $\mathscr{F}(D_+(f))$. Compatibility with restriction is immediate, so β is a well-defined morphism of sheaves.

Step 2: β is an isomorphism. It suffices to show that $\beta_{D_+(f)}$ is bijective for each $f \in S_1$. By example 5.3.12, the hypotheses of lemma 5.3.11 are satisfied with $\mathcal{L} = \mathcal{O}_X(1)$ and $X_f = D_+(f)$.

Surjectivity. Let $t \in \mathscr{F}(D_+(f))$. By lemma 5.3.11(2), there exists $n > 0$ such that $f^n t$ extends to a global section $s \in \Gamma(X, \mathscr{F} \otimes \mathcal{O}_X(n)) = \Gamma(X, \mathscr{F}(n))$. Then $s/f^n \in \Gamma_*(\mathscr{F})_{(f)}$ maps under β to $s \otimes f^{-n} = t$ on $D_+(f)$.

Injectivity. Suppose $m/f^d \in \Gamma_*(\mathscr{F})_{(f)}$ maps to zero, i.e., $m|_{D_+(f)} = 0$ in $\mathscr{F}(d)(D_+(f))$. By lemma 5.3.11(1) applied to the global section m of $\mathscr{F}(d)$ (which vanishes on $X_f = D_+(f)$), there exists $n > 0$ with $f^n m = 0$ in $\Gamma(X, \mathscr{F}(d+n))$. Hence $m/f^d = 0$ in $\Gamma_*(\mathscr{F})_{(f)}$.

Let us pause to summarize the relationship between graded modules and quasi-coherent sheaves on Proj. The two functors

$$\{\text{graded } S_\bullet\text{-modules}\} \underset{\Gamma_*(-)}{\overset{\widetilde{(-)}}{\rightleftarrows}} \mathbf{QCoh}(\operatorname{Proj} S_\bullet)$$

satisfy $\widetilde{\Gamma_*(\mathscr{F})} \cong \mathscr{F}$ for every quasi-coherent $\mathscr{F}$ (proposition 5.3.14), but $\Gamma_*(\widetilde{M})$ need not be isomorphic to M. In the language of category theory, Γ_* is a *right adjoint* that is faithful but not full, and the tilde functor is a *left adjoint* that is essentially surjective. The failure of the unit $M \to \Gamma_*(\widetilde{M})$ to be an isomorphism is measured by the *saturation* of M: two graded modules give the same sheaf if and only if they agree in sufficiently large degrees. This is analogous to the fact that Proj itself depends only on the "tail" of the graded ring (definition 2.3.1).

5.3.2 Characterizing Projective Subschemes

Combining the partial equivalence with the theory of ideal sheaves from section 5.1.4, we obtain concrete descriptions of closed subschemes of projective space and of projective schemes themselves.

Corollary 5.3.15: Characterizing Projective Subschemes

Let R be a ring. Then:

1. Every closed subscheme $Y \subseteq \mathbb{P}^r_R$ is determined by a homogeneous ideal $I \subseteq S_\bullet = R[x_0, \dots, x_r]$: namely $Y \cong \operatorname{Proj}(S_\bullet/I)$.
2. A scheme Y over $\operatorname{Spec} R$ is projective if and only if $Y \cong \operatorname{Proj} S_\bullet$ for some graded ring $S_\bullet$ with $S_0 = R$ that is finitely generated by S_1 as an S_0-algebra.

***Proof*:**
(1). Let $\mathcal{I}_Y \subseteq \mathcal{O}_X$ be the ideal sheaf of Y in $X = \mathbb{P}^r_R$. Since the twisting functor is exact (tensor with a locally free sheaf preserves exact sequences) and Γ is left exact, the inclusion $\mathcal{I}_Y \hookrightarrow \mathcal{O}_X$ gives an inclusion of graded modules

$$\Gamma_*(\mathcal{I}_Y) \hookrightarrow \Gamma_*(\mathcal{O}_X) \cong S_\bullet,$$

where the isomorphism is proposition 5.3.10. Set $I = \Gamma_*(\mathcal{I}_Y)$, a homogeneous ideal of $S_\bullet$.

By proposition 5.3.14, $\mathcal{I}_Y \cong \widetilde{I}$, so Y is the closed subscheme of X defined by I, i.e., $Y \cong \operatorname{Proj}(S_\bullet/I)$. Moreover, I is the *largest* (or *saturated*) homogeneous ideal defining Y: by construction, it contains every homogeneous polynomial that vanishes on Y in every degree.

(2). By definition, Y is projective over $\operatorname{Spec} R$ if it admits a closed immersion into $\mathbb{P}^r_R$ for some r. By (1), any such Y is $\operatorname{Proj}(S_\bullet/I)$ for a homogeneous ideal $I \subseteq S_+$. Setting $S'_\bullet = S_\bullet/I$, we have $S'_0 = R$ (since $I \subseteq S_+$) and $S'_\bullet$ is finitely generated by S'_1 over S'_0.

Conversely, if $S_\bullet$ is finitely generated by $S_1 = R \cdot f_0 + \cdots + R \cdot f_r$ over $S_0 = R$, the surjection $R[x_0, \dots, x_r] \twoheadrightarrow S_\bullet$ (sending $x_i \mapsto f_i$) induces a closed immersion $\operatorname{Proj} S_\bullet \hookrightarrow \mathbb{P}^r_R$.

Corollary 5.3.16: Characterizing Hypersurfaces in Projective Space

Let R be a UFD. Then every irreducible hypersurface $X \subseteq \mathbb{P}^n_R$ corresponds to a principal homogeneous ideal $(f) \subseteq R[x_0, \dots, x_n]$.

***Proof*:**
By corollary 5.3.15, the hypersurface X corresponds to a homogeneous prime ideal $\mathfrak{p} \subseteq S_\bullet = R[x_0, \dots, x_n]$ of height 1 (since X has codimension 1 in $\mathbb{P}^n_R$). Since R is a UFD, $S_\bullet$ is also a UFD (see [**NateClassicalAlgGeo**]), and in a UFD every height-1 prime ideal is principal (exercise 5.3.1(5.3.1 (1))). Hence $\mathfrak{p} = (f)$ for some irreducible homogeneous polynomial f.

Definition 5.3.17: Degree of Projective Hypersurface

Let R be a UFD and $X \subseteq \mathbb{P}^n_R$ an irreducible hypersurface corresponding to $(f) \subseteq R[x_0, \dots, x_n]$ via corollary 5.3.16. The *degree* of X is $\deg X := \deg f$.

This is well-defined because f is determined up to a unit in R, which does not affect the degree. When $R = k$ is an algebraically closed field, the degree of a hypersurface $V(f) \subseteq \mathbb{P}^n_k$ counts (with multiplicity) the number of intersection points with a generic line, in accordance with Bézout's theorem ([**NateClassicalAlgGeo**]).

We conclude this section with the promised example showing that the tensor product of $\mathcal{O}_X$-modules cannot, in general, be computed section-wise.

Example 5.3.18: Tensor Products Require Sheafification
Let $X = \mathbb{P}^1_k$ for a field k. Consider $\mathscr{F} = \mathcal{O}_X(-1)$ and $\mathscr{G} = \mathcal{O}_X(1)$. By proposition 5.3.8(1),

$$\mathscr{F} \otimes_{\mathcal{O}_X} \mathscr{G} \;=\; \mathcal{O}_X(-1) \otimes \mathcal{O}_X(1) \;\cong\; \mathcal{O}_X(0) \;=\; \mathcal{O}_X,$$

so the global sections of the tensor product sheaf satisfy

$$(\mathscr{F} \otimes \mathscr{G})(\mathbb{P}^1_k) \;=\; \Gamma(\mathbb{P}^1_k, \mathcal{O}_X) \;\cong\; k.$$

On the other hand, the section-wise tensor product on global sections gives

$$\mathscr{F}(\mathbb{P}^1_k) \otimes_k \mathscr{G}(\mathbb{P}^1_k) \;=\; \underbrace{\Gamma(\mathbb{P}^1_k, \mathcal{O}(-1))}_{=0} \otimes_k \underbrace{\Gamma(\mathbb{P}^1_k, \mathcal{O}(1))}_{\cong k^2} \;=\; 0.$$

Hence $(\mathscr{F} \otimes \mathscr{G})(X) \cong k \ncong 0 = \mathscr{F}(X) \otimes \mathscr{G}(X)$.

The mechanism behind this is illuminating. Consider the presheaf $\mathbb{CP} : U \mapsto \mathscr{F}(U) \otimes_{\mathcal{O}_X(U)} \mathscr{G}(U)$. On each chart $U_i = D_+(x_i)$, we have $\mathscr{F}(U_i) \cong \mathscr{G}(U_i) \cong \mathcal{O}_X(U_i)$ (both are free of rank 1), so $\mathbb{CP}(U_i) \cong \mathcal{O}_X(U_i)$. But on global sections, $\mathbb{CP}(X) = 0 \otimes k^2 = 0$. The presheaf $\mathbb{CP}$ has "plenty of local sections" but "no global sections" — it violates the gluing axiom. Sheafification repairs this: the sheaf $\mathscr{F} \otimes \mathscr{G}$ is constructed by gluing the local sections of $\mathbb{CP}$, and the global sections of the sheafification can be strictly larger than those of the presheaf.

This phenomenon is purely projective: on affine schemes, the presheaf tensor product is already a sheaf, so $\widetilde{M} \otimes \widetilde{N} \;\cong\; \widetilde{M \otimes N}$ and global sections commute with tensor products (proposition 5.1.14(4)).

Exercise 5.3.1

1. *(Krull's Principal Ideal Theorem.)* Let R be a Noetherian ring and $(x) \subsetneq R$ a proper principal

ideal. Show that every minimal prime $\mathfrak{p}$ over (x) has height at most 1. *(Hint: localize at $\mathfrak{p}$ to reduce to a local ring with exactly one minimal prime over its principal ideal. Use the Artin–Rees lemma or Nakayama's lemma to show the chain of primes below $\mathfrak{p}$ has length at most 1.)*

Another example I came across is very interesting:

Example 5.3.19: Free Sheaves Need Not Be Projective Objects
Really cool! here

5.3.3 Maps to Projective Space

In example 3.1.14, we observed a fundamental asymmetry: a morphism $X \to \mathbb{A}^n_k$ is determined by n global functions $f_1, \ldots, f_n \in \mathcal{O}_X(X)$, but a morphism $X \to \mathbb{P}^n_k$ cannot be described by global functions alone — the homogeneous coordinates x_i on $\mathbb{P}^n$ are sections of the twisting sheaf $\mathcal{O}(1)$, not of the structure sheaf. We commented that defining a map to projective space requires a *line bundle* $\mathcal{L}$ on X together with global sections $s_0, \ldots, s_n \in \Gamma(X, \mathcal{L})$ that do not vanish simultaneously.

This subsection makes that observation precise. The main result (theorem 5.3.29) establishes a bijection between morphisms $X \to \mathbb{P}^n$ and equivalence classes of line bundles with generating global sections. Along the way, we develop two important tools: the notion of a *very ample sheaf* (which characterizes projective schemes), and a fundamental finite generation theorem for coherent sheaves on projective schemes (theorem 5.3.25).

Maps *from* projective space are better understood using cohomological methods (see chapter 6); theorem 3.5.28 shows that all proper schemes admit a surjective birational morphism from a projective scheme, making projective schemes the natural "compact" objects.

Twisting Sheaves on Relative Projective Space

The twisting sheaves of **??** were defined for $\operatorname{Proj} S_\bullet$ over the base $\operatorname{Spec} S_0$. For a general base scheme X, the projective space $\mathbb{P}^n_X$ is defined by base change, and the twisting sheaf is obtained by pulling back from the universal case:

Definition 5.3.20: Twisting Sheaf on $\mathbb{P}^n_X$

Let X be a scheme. The *twisting sheaf* $\mathcal{O}(1)$ on $\mathbb{P}^n_X = X \times_{\mathbb{Z}} \mathbb{P}^n_{\mathbb{Z}}$ is the pullback

$$\mathcal{O}_{\mathbb{P}^n_X}(1) \;:=\; g^*\big(\mathcal{O}_{\mathbb{P}^n_{\mathbb{Z}}}(1)\big),$$

where $g : \mathbb{P}^n_X \to \mathbb{P}^n_{\mathbb{Z}}$ is the projection. More generally, $\mathcal{O}_{\mathbb{P}^n_X}(d) := \mathcal{O}_{\mathbb{P}^n_X}(1)^{\otimes d}$ for $d \in \mathbb{Z}$.

By proposition 5.3.8(3), if $X = \operatorname{Spec} R$ and $\mathbb{P}^n_R = \operatorname{Proj} R[x_0, \ldots, x_n]$, this recovers the twisting sheaf $\widetilde{S(1)}$ defined earlier. The base-change definition ensures that all properties established over $\mathbb{Z}$ (invertibility, non-isomorphism for different twists, etc.) propagate to $\mathbb{P}^n_X$ for any base X.

Definition 5.3.21: Very Ample Sheaf

Let X be a scheme over a base scheme Y, and $\mathcal{L}$ a line bundle on X. Then $\mathcal{L}$ is *very ample relative to* Y if there exists a closed immersion $\iota : X \hookrightarrow \mathbb{P}^n_Y$ (for some n) such that

$$\iota^*\big(\mathcal{O}_{\mathbb{P}^n_Y}(1)\big) \;\cong\; \mathcal{L}.$$

The condition says that $\mathcal{L}$ is the restriction of $\mathcal{O}(1)$ under an embedding into projective space. Very ample sheaves characterize projectivity:

Lemma 5.3.22: Characterizing Projective Schemes via Very Ample Sheaves

Let Y be a Noetherian scheme. A Y-scheme X is projective over Y if and only if X is proper over Y and possesses a very ample invertible sheaf relative to Y.

***Proof*:**
($\Rightarrow$). If X is projective over Y, then by definition there exists a closed immersion $\iota : X \hookrightarrow \mathbb{P}^n_Y$. The sheaf $\mathcal{L} = \iota^*(\mathcal{O}(1))$ is very ample, and X is proper by theorem 3.5.26.

($\Leftarrow$). Suppose X is proper over Y and $\mathcal{L}$ is very ample, so $\mathcal{L} \cong \iota^*(\mathcal{O}(1))$ for some locally closed immersion $\iota : X \to \mathbb{P}^n_Y$. Since X is proper over Y and $\mathbb{P}^n_Y$ is separated over Y, the morphism ι is proper (proposition 3.5.21). A proper locally closed immersion has closed image (proposition 3.4.22), so ι is in fact a closed immersion. Hence X is projective.

Global Generation

We need a notion expressing when a sheaf has "enough global sections" to control its local behaviour.

Definition 5.3.23: Generated by Global Sections

Let X be a scheme and $\mathscr{F}$ an $\mathcal{O}_X$-module. Then $\mathscr{F}$ is *generated by global sections* if there exists a family of global sections $\{s_i\}_{i\in I} \subseteq \Gamma(X, \mathscr{F})$ whose images in the stalk $\mathscr{F}_x$ generate $\mathscr{F}_x$ as an $\mathcal{O}_{X,x}$-module for every $x \in X$.
Equivalently, $\mathscr{F}$ is generated by global sections if and only if it is a quotient of a free sheaf: the surjection $\bigoplus_{i\in I} \mathcal{O}_X \twoheadrightarrow \mathscr{F}$ sending the ith generator to s_i is surjective on stalks, hence surjective as a sheaf morphism.

Example 5.3.24: Generated by Global Sections

1. *Affine schemes.* Let $X = \operatorname{Spec} R$ and $\mathscr{F} = \widetilde{M}$. Choose any set of generators $\{m_i\}$ for M as an R-module. The corresponding global sections generate $\mathscr{F}$, since generation is checked on stalks (localizations), and generators of M remain generators after localization. Thus *every* quasi-coherent sheaf on an affine scheme is generated by global sections.
2. *The twisting sheaf.* Let $X = \operatorname{Proj} S_\bullet$ where $S_\bullet$ is generated by S_1 over S_0. The degree-1 elements $f_0, \ldots, f_r \in S_1$ define global sections of $\mathcal{O}_X(1)$ (they lie in $\Gamma(X, \mathcal{O}_X(1)) \supseteq S_1$). On $D_+(f_i)$, the section f_i generates the free module $\mathcal{O}_X(1)|_{D_+(f_i)} \cong \mathcal{O}_{D_+(f_i)}$. Since the $D_+(f_i)$

cover X, the sections $f_0, \dots, f_r$ generate $\mathcal{O}_X(1)$.

3. *Negative twists.* The sheaf $\mathcal{O}_{\mathbb{P}^r_k}(-1)$ is *not* generated by global sections for $r \geq 1$: it has no nonzero global sections at all, as we computed in **??**.

The next theorem is one of the most important results about coherent sheaves on projective schemes: after twisting by a sufficiently large power of a very ample sheaf, any coherent sheaf becomes generated by *finitely many* global sections. This should be compared with the affine case, where global generation is automatic but says nothing about projective schemes.

Theorem 5.3.25: Global Section Generation of Twisted Sheaves

Let X be a projective scheme over a Noetherian ring R, let $\mathcal{O}_X(1)$ be a very ample invertible sheaf on X, and let $\mathscr{F}$ be a coherent $\mathcal{O}_X$-module. Then there exists an integer n_0 such that for all $n \geq n_0$, the twisted sheaf $\mathscr{F}(n)$ is generated by a finite number of global sections.

***Proof*:**
Reduction to $X = \mathbb{P}^r_R$. Let $\iota : X \hookrightarrow \mathbb{P}^r_R$ be a closed immersion with $\iota^*(\mathcal{O}(1)) = \mathcal{O}_X(1)$. The pushforward $\iota_*\mathscr{F}$ is coherent on $\mathbb{P}^r_R$ (closed immersions are finite, hence preserve coherence), and $\iota_*(\mathscr{F}(n)) \cong (\iota_*\mathscr{F})(n)$ by proposition 5.3.8(3). Moreover, $\mathscr{F}(n)$ is generated by global sections if and only if $\iota_*(\mathscr{F}(n))$ is (their global sections agree, and generation is checked on the support). We may therefore assume $X = \mathbb{P}^r_R = \operatorname{Proj} R[x_0, \dots, x_r]$.

Local generators. Cover X by the standard affine charts $U_i = D_+(x_i)$ for $i = 0, \dots, r$. Since $\mathscr{F}$ is coherent, on each U_i there is a finitely generated $(S_{x_i})_0$-module M_i with $\mathscr{F}|_{U_i} \cong \widetilde{M_i}$. Choose finite sets of generators $\{s_{i,1}, \dots, s_{i,k_i}\} \subseteq M_i$.

Extending to global sections. Each $s_{i,j}$ is a section of $\mathscr{F}$ over the affine open $U_i = D_+(x_i)$. By lemma 5.3.11(2) (with $\mathcal{L} = \mathcal{O}_X(1)$ and $f = x_i$), there exists $n_{i,j} > 0$ such that $x_i^{n_{i,j}} s_{i,j}$ extends to a global section $t_{i,j} \in \Gamma(X, \mathscr{F}(n_{i,j}))$. Taking $n_0 = \max_{i,j} n_{i,j}$ (and multiplying by further powers of x_i where needed), we may assume all extensions live in $\Gamma(X, \mathscr{F}(n_0))$.

Generation. The twisted sheaf $\mathscr{F}(n_0)$ restricts on U_i to a module M_i', and the trivialization $\mathcal{O}_X(n_0)|_{U_i} \cong \mathcal{O}_{U_i}$ (via multiplication by $x_i^{n_0}$) gives an isomorphism $M_i \xrightarrow{\sim} M_i'$ sending $s_{i,j}$ to $x_i^{n_0} s_{i,j}$. Since $\{s_{i,j}\}_j$ generates M_i, the sections $\{x_i^{n_0} s_{i,j}\}_j$ generate M_i'. Hence the global sections $\{t_{i,j}\}_{i,j}$ generate $\mathscr{F}(n_0)$ on every chart U_i, and therefore everywhere. There are finitely many such sections (at most $\sum_{i=0}^r k_i$).

For $n > n_0$, the sheaf $\mathscr{F}(n) = \mathscr{F}(n_0)(n - n_0)$ is generated by products of the $t_{i,j}$ with degree-$(n - n_0)$ monomials in the x_i, which are again finitely many.

This result has two immediate and important consequences.

Corollary 5.3.26: Coherent Sheaves as Quotients of Twisted Free Sheaves

Let X be a projective scheme over a Noetherian ring R and $\mathscr{F}$ a coherent sheaf on X. Then $\mathscr{F}$ can be written as a quotient of a sheaf of the form $\bigoplus_{i=1}^m \mathcal{O}_X(n_i)$ for finitely many integers n_i.

> ***Proof***:
> By theorem 5.3.25, there exist $n \geq 0$ and a surjection $\mathcal{O}_X^m \twoheadrightarrow \mathscr{F}(n)$. Tensoring with $\mathcal{O}_X(-n)$ gives a surjection $\mathcal{O}_X(-n)^m \twoheadrightarrow \mathscr{F}$.

This should be viewed as a projective analogue of the fact that every finitely generated module over a ring is a quotient of a free module. On projective space, free modules are replaced by direct sums of line bundles. Iterating, one obtains a *free resolution* of $\mathscr{F}$ by direct sums of twisted structure sheaves — the projective analogue of a free resolution of a module. The Hilbert Syzygy Theorem asserts that such a resolution can always be chosen to have length at most $r + 1$ on $\mathbb{P}^r_R$; see section 6.4.

Finite Generation of Global Sections

On affine schemes, coherence of $\widetilde{M}$ corresponds to finite generation of M as an R-module. In particular, $\Gamma(\operatorname{Spec} R, \widetilde{M})$ is finitely generated. On projective schemes, global sections are *not* the same as the underlying module, so finite generation requires a separate argument. The following theorem provides it:

Theorem 5.3.27: Finite Generation of Coherent Sheaves over Projective Schemes

Let k be a field, R a finitely generated k-algebra, X a projective R-scheme, and $\mathscr{F}$ a coherent $\mathcal{O}_X$-module. Then $\Gamma(X, \mathscr{F})$ is a finitely generated R-module.
In particular, if $R = k$, then $\Gamma(X, \mathscr{F})$ is a finite-dimensional k-vector space.

The contrast with the affine case is stark: $\Gamma(\mathbb{A}^n_k, \mathcal{O}) = k[x_1, \ldots, x_n]$ is infinite-dimensional over k, reflecting the fact that $\mathbb{A}^n_k$ is not projective. The finiteness of global sections is a manifestation of the "compactness" of projective schemes.

> ***Proof***:
> By corollary 5.3.15, we may write $X = \operatorname{Proj} S_\bullet$ where $S_0 = R$ and $S_\bullet$ is finitely generated by S_1 as an S_0-algebra. Let $M_\bullet = \Gamma_*(\mathscr{F})$. By proposition 5.3.14, $\widetilde{M} \cong \mathscr{F}$. The proof proceeds in three steps: we first replace $M_\bullet$ by a finitely generated submodule, then reduce to a special case via filtration, and finally prove that special case using integral dependence.
>
> *Step 1: Replacing $M_\bullet$ by a finitely generated submodule.* By theorem 5.3.25, there exists n_0 such that $\mathscr{F}(n)$ is generated by finitely many global sections for all $n \geq n_0$. Let $M'_\bullet \subseteq M_\bullet$ be the graded submodule generated by these finitely many sections (for a single n_0). Then $M'_\bullet$ is a finitely generated $S_\bullet$-module, the inclusion $M' \hookrightarrow M$ induces $\widetilde{M'} \hookrightarrow \widetilde{M} = \mathscr{F}$, and twisting by n_0 gives $\widetilde{M'}(n_0) \hookrightarrow \mathscr{F}(n_0)$. Since the generators of M' in degree n_0 are exactly the global sections generating $\mathscr{F}(n_0)$, this inclusion is an isomorphism in degree n_0, hence an isomorphism of sheaves (a map of quasi-coherent sheaves that is an isomorphism after twisting is an isomorphism). Twisting back gives $\widetilde{M'} \cong \mathscr{F}$.
>
> We have shown: every coherent sheaf on $\operatorname{Proj} S_\bullet$ is the tilde of a finitely generated graded $S_\bullet$-module. It remains to show that if $M_\bullet$ is finitely generated, then $\Gamma(X, \widetilde{M}) = (M_\bullet)_0$ ... but this is *not* true in general; $\Gamma(X, \widetilde{M})$ can differ from M_0. Instead, we must show $\Gamma(X, \widetilde{M})$ is finitely generated over R directly.
>
> *Step 2: Reduction via filtration.* Since $S_\bullet$ is a Noetherian ring and $M_\bullet$ is a finitely generated graded

$S_\bullet$-module, there exists a finite filtration of graded submodules (see [**NateClassicalAlgGeo**])

$$0 = M^{(0)} \subseteq M^{(1)} \subseteq \cdots \subseteq M^{(r)} = M_\bullet$$

such that each successive quotient $M^{(i)}/M^{(i-1)}$ is isomorphic to $(S_\bullet/\mathfrak{p}_i)(n_i)$ for some homogeneous prime $\mathfrak{p}_i$ and integer n_i. Applying the tilde construction and taking global sections, the short exact sequences $0 \to \widetilde{M^{(i-1)}} \to \widetilde{M^{(i)}} \to \widetilde{M^{(i)}/M^{(i-1)}} \to 0$ give left-exact sequences

$$0 \longrightarrow \Gamma(X, \widetilde{M^{(i-1)}}) \longrightarrow \Gamma(X, \widetilde{M^{(i)}}) \longrightarrow \Gamma(X, \widetilde{M^{(i)}/M^{(i-1)}}).$$

By induction, it suffices to show that each $\Gamma(X,\, \widetilde{S_\bullet/\mathfrak{p}}(n))$ is finitely generated over R.

Step 3: The integral domain case. We are thus reduced to proving the following: let $S_\bullet$ be a graded integral domain, finitely generated by $S_1 = Rx_0 + \cdots + Rx_r$ over $S_0 = R$ (a finitely generated k-algebra), and $X = \operatorname{Proj} S_\bullet$. Then $\Gamma(X, \mathcal{O}_X(n))$ is a finitely generated R-module for each $n \in \mathbb{Z}$.

Since $S_\bullet$ is a domain, multiplication by any x_i gives an injection $S(n) \hookrightarrow S(n+1)$, hence an injection $\Gamma(X, \mathcal{O}(n)) \hookrightarrow \Gamma(X, \mathcal{O}(n+1))$. It therefore suffices to prove finite generation for all $n \geq 0$.

Set $S'_\bullet = \bigoplus_{n\geq 0} \Gamma(X, \mathcal{O}_X(n)) = \Gamma_*(\mathcal{O}_X)$. This is a graded ring containing $S_\bullet$ (via the natural map $S_n \to \Gamma(X, \mathcal{O}(n))$) and contained in $\bigcap_i S_{x_i}$ (each global section restricts to a section on each $D_+(x_i)$).

$S'_\bullet$ is integral over $S_\bullet$. Let $s' \in S'_d$ for some $d \geq 0$. Since $s' \in S_{x_i}$ for each i, there exists N (independent of i, by taking a maximum) such that $x_i^N s' \in S_\bullet$. Since the monomials in $x_0, \ldots, x_r$ of degree N generate S_N, we may enlarge N so that $y \cdot s' \in S_\bullet$ for all $y \in S_{\geq N} := \bigoplus_{e \geq N} S_e$.

By induction, $y \cdot (s')^q \in S_{\geq N}$ for all $q \geq 1$ and all $y \in S_{\geq N}$. Taking $y = x_0^N$ gives $(s')^q \in x_0^{-N} \cdot S_\bullet$ for all $q \geq 1$. The module $x_0^{-N} \cdot S_\bullet$ is a finitely generated $S_\bullet$-submodule of the quotient field of $S_\bullet$. By the standard criterion for integral dependence (see [**NateClassicalAlgGeo**]), s' is integral over $S_\bullet$.

Conclusion. Since $S_\bullet$ is a finitely generated k-algebra (hence a Nagata ring), its integral closure in the quotient field is a finitely generated $S_\bullet$-module ([**NateClassicalAlgGeo**]). Since $S'_\bullet$ is contained in this integral closure, it is a finitely generated $S_\bullet$-module as well[a]. In particular, each graded piece $S'_n = \Gamma(X, \mathcal{O}_X(n))$ is a finitely generated S_0-module, i.e., a finitely generated R-module.

[a]More precisely, $S'_\bullet$ is a graded submodule of a finitely generated module over the Noetherian ring $S_\bullet$, hence itself finitely generated.

Corollary 5.3.28: Pushforward of Coherent Sheaf by Projective Morphism

Let $f : X \to Y$ be a projective morphism of schemes of finite type over a field k. Let $\mathscr{F}$ be a coherent sheaf on X. Then $f_*\mathscr{F}$ is coherent on Y.

Recall that this fails for non-projective morphisms: example 5.1.19 showed that the pushforward of $\mathcal{O}_{\mathbb{A}^1}$ along $\mathbb{A}^1 \to \operatorname{Spec} k$ is $k[x]$, which is not finitely generated over k.

***Proof*:**
The question is local on Y, so we may assume $Y = \operatorname{Spec} R$ where R is a finitely generated k-algebra. Since f is projective, X is a projective R-scheme. By proposition 5.1.20, $f_*\mathscr{F}$ is quasi-coherent, so $f_*\mathscr{F} \cong \widetilde{N}$ for $N = \Gamma(Y, f_*\mathscr{F}) = \Gamma(X, \mathscr{F})$. By theorem 5.3.27, N is a finitely generated R-module, so $f_*\mathscr{F}$ is coherent.

5.3.4 Characterizing Morphisms to Projective Space

With the notion of global generation in hand (definition 5.3.23), we can now make precise the correspondence between morphisms to projective space and line bundles with generating sections that was previewed in the introduction to this subsection.

Recall that for a line bundle $\mathcal{L}$ (locally free of rank 1), being generated by sections $s_0, \ldots, s_n$ has a particularly clean geometric interpretation: at each point $x \in X$, the fibre $\mathcal{L}(x)$ is a one-dimensional $\kappa(x)$-vector space, and the sections span this fibre if and only if at least one s_i is nonzero at x. Equivalently, the *base locus* $\{x \in X : s_0(x) = \cdots = s_n(x) = 0\}$ is empty. This is exactly the condition ensuring that the "ratio map" $x \mapsto [s_0(x) : \cdots : s_n(x)]$ is well-defined at every point.

Before stating the theorem, let us trace through the construction on a concrete case. Consider the Veronese map $\mathbb{P}^1_k \to \mathbb{P}^2_k$, classically given by $[s : t] \mapsto [s^2 : st : t^2]$. In our language:

- The line bundle is $\mathcal{L} = \mathcal{O}_{\mathbb{P}^1}(2)$ on $\mathbb{P}^1_k$.
- The global sections are the three degree-2 monomials: $s_0 = x_0^2$, $s_1 = x_0x_1$, $s_2 = x_1^2$ in $\Gamma(\mathbb{P}^1_k, \mathcal{O}(2)) \cong k[x_0, x_1]_2$.
- At each point of $\mathbb{P}^1_k$, at least one of x_0^2, x_0x_1, x_1^2 is nonzero, so these sections generate $\mathcal{O}(2)$.
- On the open set $X_0 = D_+(x_0)$ where $s_0 = x_0^2$ does not vanish, the ratios are $s_1/s_0 = x_1/x_0$ and $s_2/s_0 = (x_1/x_0)^2$ — these are the "affine coordinates" of the map, sending $D_+(x_0)$ into the affine chart $U_0 \subseteq \mathbb{P}^2_k$ by $t \mapsto (t, t^2)$.

The image is the conic $V(y_0y_2 - y_1^2) \subseteq \mathbb{P}^2_k$, and the map is a closed immersion. The general theorem below formalizes this pattern.

Theorem 5.3.29: Determining Maps To Projective Space

Let R be a ring and X an R-scheme. Then:

1. *(From morphisms to line bundles.)* If $\varphi : X \to \mathbb{P}^n_R$ is an R-morphism, then $\mathcal{L} := \varphi^*(\mathcal{O}(1))$ is an invertible sheaf on X, and the pullback sections $s_i := \varphi^*(x_i) \in \Gamma(X, \mathcal{L})$ for $i = 0, \dots, n$ generate $\mathcal{L}$.

2. *(From line bundles to morphisms.)* If $\mathcal{L}$ is an invertible sheaf on X and $s_0, \dots, s_n \in \Gamma(X, \mathcal{L})$ are global sections that generate $\mathcal{L}$, then there exists a unique R-morphism $\varphi : X \to \mathbb{P}^n_R$ such that $\mathcal{L} \cong \varphi^*(\mathcal{O}(1))$ and $s_i = \varphi^*(x_i)$ under this isomorphism.

In particular, there is a natural bijection:

$$\mathrm{Hom}_R(X, \mathbb{P}^n_R) \xleftrightarrow{1:1} \left\{ (\mathcal{L}, s_0, \dots, s_n) \;\middle|\; \begin{array}{c} \mathcal{L} \text{ invertible on } X, \\ s_i \in \Gamma(X, \mathcal{L}) \text{ generating } \mathcal{L} \end{array} \right\} \Big/ \cong$$

where $(\mathcal{L}, s_0, \dots, s_n) \cong (\mathcal{L}', s_0', \dots, s_n')$ if there is an isomorphism $\alpha : \mathcal{L} \xrightarrow{\sim} \mathcal{L}'$ with $\alpha(s_i) = s_i'$ for all i.

***Proof*:**
Part (1): morphisms yield generating sections. Since pullback preserves locally free sheaves and their rank (proposition 5.3.8(3)), $\mathcal{L} = \varphi^*(\mathcal{O}(1))$ is an invertible sheaf on X. The homogeneous coordinates $x_0, \dots, x_n$ generate $\mathcal{O}(1)$ on $\mathbb{P}^n_R$, and pullback preserves surjections, so the sections $s_i = \varphi^*(x_i)$ generate $\mathcal{L}$.

Part (2): generating sections yield a morphism. This is the substantial direction. We construct φ by defining it on an open cover and gluing.

Step 1: The open cover. For each $i = 0, \dots, n$, define

$$X_i \;:=\; \{x \in X \mid (s_i)_x \notin \mathfrak{m}_x \mathcal{L}_x\},$$

the locus where s_i does not vanish. Since $\mathcal{L}$ is locally free of rank 1, the section s_i generates $\mathcal{L}_x$ as an $\mathcal{O}_{X,x}$-module at precisely the points where it maps to a nonzero element of the fibre. The generating hypothesis means that for each $x \in X$, some s_i does not vanish at x, so $X = X_0 \cup \cdots \cup X_n$.

Each X_i is open: on any affine open $\operatorname{Spec} A \subseteq X$ where $\mathcal{L}|_{\operatorname{Spec} A} \cong \mathcal{O}_{\operatorname{Spec} A}$, the section s_i corresponds to an element $a_i \in A$, and $X_i \cap \operatorname{Spec} A = D(a_i)$.

Step 2: Trivializing $\mathcal{L}$ on each X_i. On X_i, the section s_i generates $\mathcal{L}|_{X_i}$: for each $x \in X_i$, the element $(s_i)_x$ is a generator of the free rank-1 module $\mathcal{L}_x$. Therefore, the map

$$\psi_i : \mathcal{O}_{X_i} \xrightarrow{\sim} \mathcal{L}|_{X_i}, \qquad 1 \longmapsto s_i|_{X_i},$$

is an isomorphism of $\mathcal{O}_{X_i}$-modules. Intuitively, s_i provides a "local coordinate" for the line bundle on X_i.

Step 3: Constructing regular functions s_j/s_i. For each j, the section $s_j|_{X_i}$ is an element of

$\mathcal{L}(X_i)$. Via the trivialization ψ_i, it corresponds to a unique regular function:

$$\frac{s_j}{s_i} \; := \; \psi_i^{-1}(s_j|_{X_i}) \; \in \; \mathcal{O}_X(X_i).$$

At each stalk, this is the unique element $f \in \mathcal{O}_{X,x}$ satisfying $(s_j)_x = f \cdot (s_i)_x$ in $\mathcal{L}_x$. Such f exists and is unique because $\mathcal{L}_x$ is free of rank 1 with generator $(s_i)_x$.

In particular, $s_i/s_i = 1$, and on $X_i \cap X_j$ the ratios satisfy the multiplicative relation

$$\frac{s_k}{s_i} \; = \; \frac{s_k}{s_j} \cdot \frac{s_j}{s_i} \qquad \text{on } X_i \cap X_j, \tag{5.3}$$

since $(s_k)_x = (s_k/s_j)_x \cdot (s_j)_x = (s_k/s_j)_x \cdot (s_j/s_i)_x \cdot (s_i)_x$ and generators are unique.

Step 4: Local morphisms. The ith standard affine chart of $\mathbb{P}^n_R$ is $U_i = D_+(x_i) \cong \operatorname{Spec} R[y_0, \dots, \hat{y}_i, \dots, y_n]$, where $y_j = x_j/x_i$. Define a ring homomorphism

$$\varphi_i^\# : \; R[y_0, \dots, \hat{y}_i, \dots, y_n] \; \longrightarrow \; \mathcal{O}_X(X_i), \qquad y_j \; \longmapsto \; \frac{s_j}{s_i}.$$

This is well-defined: each s_j/s_i is a regular function on X_i by Step 3, and the map extends R-linearly (since X is an R-scheme). By theorem 3.1.4, this determines a morphism of schemes $\varphi_i : X_i \to U_i$.

Step 5: Gluing. We must show that $\varphi_i|_{X_i \cap X_j} = \varphi_j|_{X_i \cap X_j}$ as morphisms into $\mathbb{P}^n_R$. The transition functions on $\mathbb{P}^n_R$ between charts U_i and U_j are given by $y_k^{(j)} = y_k^{(i)}/y_j^{(i)}$, where $y_k^{(i)} = x_k/x_i$. Under φ_i^*, we have $y_k^{(i)} \mapsto s_k/s_i$. Hence

$$\varphi_i^*\left(\frac{y_k^{(i)}}{y_j^{(i)}}\right) \; = \; \frac{s_k/s_i}{s_j/s_i} \; = \; \frac{s_k}{s_j} \; = \; \varphi_j^*(y_k^{(j)}),$$

using (5.3). Since the local morphisms agree on overlaps, they glue by lemma 3.1.9 to a unique morphism $\varphi : X \to \mathbb{P}^n_R$.

Step 6: Identifying $\mathcal{L} \cong \varphi^*(\mathcal{O}(1))$. On U_i, the sheaf $\mathcal{O}(1)$ is trivial with generator x_i (recall from proposition 5.3.5 that multiplication by x_i trivializes $\mathcal{O}(1)$ on $D_+(x_i)$). The pullback $\varphi^*(\mathcal{O}(1))|_{X_i}$ is therefore trivial with generator $\varphi^*(x_i) = s_i$. But $\mathcal{L}|_{X_i}$ is also trivial with generator s_i (by ψ_i). These identifications are compatible on overlaps: the transition function for $\varphi^*(\mathcal{O}(1))$ between X_i and X_j is $\varphi^*(x_j/x_i) = s_j/s_i$, which is the same as the transition function for $\mathcal{L}$ (since $\psi_j^{-1} \circ \psi_i$ sends $1 \mapsto s_i/s_j$, i.e., the transition function of $\mathcal{L}$ between the ψ_i and ψ_j trivializations is s_i/s_j). Hence the local isomorphisms glue to a global isomorphism $\mathcal{L} \cong \varphi^*(\mathcal{O}(1))$ sending s_i to $\varphi^*(x_i)$.

Step 7: Uniqueness. Any R-morphism $\varphi' : X \to \mathbb{P}^n_R$ with $(\varphi')^*(\mathcal{O}(1)) \cong \mathcal{L}$ and $(\varphi')^*(x_i) = s_i$ must restrict on X_i to the morphism $X_i \to U_i$ induced by the ring map $y_j \mapsto s_j/s_i$ (since $(\varphi')^*(x_j/x_i) = s_j/s_i$ is forced). By lemma 3.1.9, φ' is uniquely determined by its restrictions to the cover $\{X_i\}$, so $\varphi' = \varphi$.

5.3.30 Comparison with the Affine Case The theorem establishes a clean parallel:

Target $\mathbb{A}^n_R$	**Target** $\mathbb{P}^n_R$
$\mathrm{Hom}_R(X, \mathbb{A}^n_R) \cong \mathcal{O}_X(X)^n$	$\mathrm{Hom}_R(X, \mathbb{P}^n_R) \cong \{(\mathcal{L}, s_0, \dots, s_n)\}/\cong$
n global functions	line bundle + $(n+1)$ generating sections
always well-defined	need base locus to be empty

In the affine case, the structure sheaf $\mathcal{O}_X$ plays the role of the line bundle, and it is always generated by a single section (1), so no additional choice is needed. In the projective case, the choice of line bundle $\mathcal{L}$ is genuine extra data — different line bundles yield morphisms into *different* projective spaces (with different embeddings). The study of which line bundles on X are generated by global sections, and how the resulting morphisms to projective space behave, is a central theme of algebraic geometry; it is developed further in section 5.4.

Example 5.3.31: Maps Into Projective Space

1. *Linear automorphisms of* $\mathbb{P}^n_k$. Let $X = \mathbb{P}^n_k$, $\mathcal{L} = \mathcal{O}(1)$, and choose generating sections $s_i = \sum_{j=0}^n a_{ij}\, x_j$ for an invertible matrix $(a_{ij}) \in \mathrm{GL}_{n+1}(k)$. The resulting morphism $\varphi : \mathbb{P}^n_k \to \mathbb{P}^n_k$ sends $[x_0 : \cdots : x_n] \mapsto [s_0 : \cdots : s_n]$, which is a linear change of coordinates — an automorphism of $\mathbb{P}^n_k$. Conversely, any automorphism of $\mathbb{P}^n_k$ arising from $\mathcal{O}(1)$ with linear sections must be of this form. Rescaling all sections by a common scalar $\lambda \in k^*$ does not change the morphism (since $[\lambda s_0 : \cdots : \lambda s_n] = [s_0 : \cdots : s_n]$), so we obtain $\mathrm{PGL}_{n+1}(k) \subseteq \mathrm{Aut}(\mathbb{P}^n_k)$. In fact, equality holds: every automorphism of $\mathbb{P}^n_k$ is linear (this follows from $\mathrm{Pic}\,(\mathbb{P}^n_k) \cong \mathbb{Z}$, proved in corollary 5.4.34).

2. *The Veronese embedding.* Let $X = \mathbb{P}^n_k$, $\mathcal{L} = \mathcal{O}(d)$ for $d \geq 1$, and take the sections to be all $N + 1 = \binom{n+d}{d}$ monomials of degree d. These generate $\mathcal{O}(d)$, so the theorem gives a morphism $\nu_d : \mathbb{P}^n_k \to \mathbb{P}^N_k$. For $n = 1$, $d = 2$:
$$\nu_2 : \mathbb{P}^1_k \longrightarrow \mathbb{P}^2_k, \qquad [s : t] \longmapsto [s^2 : st : t^2].$$
The image is the conic $V(y_0 y_2 - y_1^2)$, and ν_2 is a closed immersion. More generally, ν_d is always a closed immersion (see definition 3.1.36); this is the *d-uple Veronese embedding*. A key consequence: every degree-d hypersurface in $\mathbb{P}^n$ becomes a *hyperplane section* of the Veronese image, reducing questions about higher-degree geometry to linear geometry in a larger projective space.

3. *Maps from affine schemes.* Let $X = \operatorname{Spec} A$ be affine. Every line bundle on an affine scheme is trivial: $\mathcal{L} \cong \mathcal{O}_X$. A trivialization $\psi : \mathcal{O}_X \xrightarrow{\sim} \mathcal{L}$ sends the generating sections s_i to elements $a_i = \psi^{-1}(s_i) \in A$. The generating condition $X = X_0 \cup \cdots \cup X_n$ translates to $D(a_0) \cup \cdots \cup D(a_n) = \operatorname{Spec} A$, i.e., $(a_0, \dots, a_n) = A$. The morphism $\varphi : \operatorname{Spec} A \to \mathbb{P}^n_R$ is then given on $D(a_i)$ by the ring map $x_j/x_i \mapsto a_j/a_i \in A_{a_i}$.

   For a concrete case take $A = k[s,t]$ with $a_0 = s$, $a_1 = t$, $a_2 = 1$. Since $(s, t, 1) = A$, this defines a morphism $\mathbb{A}^2_k \to \mathbb{P}^2_k$. On $D(s)$: $y_1 \mapsto t/s$, $y_2 \mapsto 1/s$; on $D(1) = \operatorname{Spec} A$: $y_0 \mapsto s$, $y_1 \mapsto t$. This is the standard open embedding $\mathbb{A}^2_k \hookrightarrow \mathbb{P}^2_k$ sending $(s,t) \mapsto [s : t : 1]$.

4. *Projections.* Let $X = \mathbb{P}^n_k$ and choose a linear subspace $\Lambda \cong \mathbb{P}^{n-m-1}_k$ (the "centre of projection"). After a change of coordinates, $\Lambda = V(x_0, \dots, x_m)$. The sections $x_0, \dots, x_m \in$

$\Gamma(\mathbb{P}^n_k, \mathcal{O}(1))$ generate $\mathcal{O}(1)$ at every point *outside* Λ, but they all vanish on Λ. Thus, the theorem applies to $U = \mathbb{P}^n_k \setminus \Lambda$, giving a morphism $\pi : U \to \mathbb{P}^m_k$. This is the classical *projection from* Λ: the fibre $\pi^{-1}([a])$ is the $(n-m)$-plane spanned by Λ and the point $[a_0 : \cdots : a_m : 0 : \cdots : a_m : 0 : \cdots : 0]$ and Λ; geometrically, π sends a point $P \notin \Lambda$ to the unique point where the linear span $\langle P, \Lambda \rangle$ meets a complementary $\mathbb{P}^m$.

Note that π is *not* defined on Λ itself: the base locus of the sections $x_0, \ldots, x_m$ is exactly Λ. This illustrates the general principle that a collection of sections that fails to generate $\mathcal{L}$ everywhere still defines a morphism on the complement of the base locus — a *rational map* $X \dashrightarrow \mathbb{P}^m$ in the sense of section 3.6.

5. *The Segre embedding.* Let $X = \mathbb{P}^m_k \times_k \mathbb{P}^n_k$ with the line bundle $\mathcal{L} = \mathcal{O}(1,1) := \mathrm{pr}_1^*\mathcal{O}(1) \otimes \mathrm{pr}_2^*\mathcal{O}(1)$. The global sections $\{x_i \otimes y_j\}_{0 \le i \le m,\, 0 \le j \le n}$ generate $\mathcal{L}$ (at any point (P, Q), some x_i is nonzero at P and some y_j is nonzero at Q, so $x_i \otimes y_j$ is nonzero). The resulting morphism

$$\sigma : \mathbb{P}^m_k \times \mathbb{P}^n_k \longrightarrow \mathbb{P}^{(m+1)(n+1)-1}_k$$

is the *Segre embedding.* . It is a closed immersion, and its image is cut out by the 2×2 minors of the matrix (z_{ij}). The Segre embedding shows, in particular, that the product of projective schemes is again projective.

5.3.32 Base-Point-Free Linear Systems The data $(\mathcal{L}, s_0, \ldots, s_n)$ appearing in theorem 5.3.29 is closely related to the classical notion of a *linear system* on X. The vector subspace $V = \mathrm{span}_k\{s_0, \ldots, s_n\} \subseteq \Gamma(X, \mathcal{L})$ determines a linear system $|V|$, and the generating condition (empty base locus) says precisely that this linear system is *base-point free.* The morphism φ depends only on $\mathcal{L}$ and the subspace V (up to the choice of basis, which amounts to a linear automorphism of the target $\mathbb{P}^n$). Choosing a different basis of V gives a morphism that differs from φ by an element of PGL_{n+1}.
When $V = \Gamma(X, \mathcal{L})$ is the *complete* linear system and $\mathcal{L}$ is very ample, the resulting morphism is a closed immersion — this is essentially the definition of very ampleness (definition 5.3.21). Choosing a proper subspace $V \subsetneq \Gamma(X, \mathcal{L})$ gives a morphism to a *smaller* projective space, obtained by composing the complete embedding with a projection (as in example 5.3.31(4)).

5.4 Divisors: Geometric Representation of Line Bundles

Having established the correspondence between line bundles and maps to projective space (section 5.3.3), we now turn to the *geometry* of line bundles themselves. The central idea is simple: just as a regular function $f \in \mathcal{O}_X(X)$ determines a vanishing scheme $V(f)$ of codimension 1, a section of a line bundle determines a "generalized hypersurface" called a *divisor.* Since regular functions are sections of the trivial line bundle $\mathcal{O}_X$, divisors are a genuine generalization of zero sets.

Divisors measure how far the geometry of a scheme is from being controlled by global functions alone. There are two natural approaches to formalizing this idea:

1. *Weil divisors* are formal integer combinations of codimension-1 subvarieties — the "geometric" approach. They record *where* a section vanishes or has poles, and with what multiplicity.
2. *Cartier divisors* encode the same data "algebraically," as collections of local rational functions

with prescribed vanishing behaviour, glued together by invertible transition functions.

On sufficiently nice schemes (integral, Noetherian, locally factorial), these two notions coincide. Both give rise to an abelian group — the *divisor class group* — that classifies line bundles up to isomorphism, and this group is isomorphic to the Picard group $\operatorname{Pic}(X)$ under mild hypotheses. If the reader has seen the divisor theory for Dedekind domains in [11, chapter 22.7], the present section generalizes that theory to arbitrary schemes and connects it to the geometry of line bundles[3].

The intuition for divisors is best developed by starting with examples from complex analysis (see [**NateComplexAnalysis**]), where the two notions of divisor coincide and the key ideas can be seen concretely.

Divisors on $\mathbb{C}$. Let $X = \mathbb{C}$ and consider the free abelian group generated by the points of X: each element is a finite formal sum $D = \sum_i n_i[p_i]$ with $n_i \in \mathbb{Z}$ and $p_i \in \mathbb{C}$. Every such divisor determines a rational (meromorphic) function

$$f_D(z) \;=\; \prod_i (z - p_i)^{n_i},$$

which has a zero of order n_i at p_i when $n_i > 0$ and a pole of order $|n_i|$ when $n_i < 0$. Conversely, every nonzero rational function on $\mathbb{C}$ (i.e., every nonzero element of $\mathbb{C}(z)$) has finitely many zeros and poles, and hence determines a divisor. Such divisors are called *principal.* Since every divisor on $\mathbb{C}$ is principal, the quotient group is trivial:

$$\frac{\{\text{divisors on } \mathbb{C}\}}{\{\text{principal divisors}\}} \;=\; 0.$$

Divisors on $\mathbb{P}^1$. Now let $X = \mathbb{P}^1_{\mathbb{C}}$, the Riemann sphere. The divisor group is again the free abelian group on the points of X, but the principal divisors are more constrained: a meromorphic function on a compact Riemann surface must satisfy $\sum_{p \in X} \operatorname{ord}_p(f) = 0$ (the number of zeros equals the number of poles, counted with multiplicity)[4]. The *degree* map $D = \sum n_i[p_i] \mapsto \sum n_i$ therefore vanishes on principal divisors, and one can show it induces an isomorphism

$$\frac{\{\text{divisors on } \mathbb{P}^1\}}{\{\text{principal divisors}\}} \;\cong\; \mathbb{Z}.$$

The generator is the class of any single point $[p]$: two divisors are linearly equivalent if and only if they have the same total degree. Under the correspondence between divisors and line bundles, the divisor class of degree n corresponds to the twisting sheaf $\mathcal{O}_{\mathbb{P}^1}(n)$.

Divisors on an elliptic curve. Let $X = \mathbb{C}/\Lambda$ be an elliptic curve, where $\Lambda \subseteq \mathbb{C}$ is a lattice. As before, a meromorphic function on X has equal numbers of zeros and poles (counted with multiplicity), so principal divisors have degree zero. Hence the degree map again gives a surjection onto $\mathbb{Z}$, and the divisor class group fits into a short exact sequence

$$0 \longrightarrow \operatorname{Pic}^0(X) \longrightarrow \operatorname{Pic}(X) \xrightarrow{\deg} \mathbb{Z} \longrightarrow 0.$$

The degree-zero part $\operatorname{Pic}^0(X)$ is *nontrivial*: by Abel's theorem, a degree-zero divisor $D = \sum n_i[p_i]$ is principal if and only if $\sum n_i p_i = 0$ in the group law of X. This gives an isomorphism $\operatorname{Pic}^0(X) \cong X$

[3]This connection is foundational for the *Riemann–Roch theorem* (theorem 7.1.18), one of the most important tools for understanding the geometry of curves and, more broadly, for classifying projective varieties.

[4]For instance, z^2 viewed on $\mathbb{P}^1$ has a double zero at 0 and a double pole at ∞, summing to zero.

(as a group), recovering the classical fact that an elliptic curve is its own Jacobian. In particular, $\mathrm{Pic}(X)$ is *not* simply $\mathbb{Z}$ — the divisor class group detects genuinely new geometric information beyond the degree.

The progression from $\mathbb{C}$ (trivial divisor class group) to $\mathbb{P}^1$ (divisor class group $\cong \mathbb{Z}$) to elliptic curves (divisor class group $\cong \mathbb{Z} \oplus X$) illustrates a general principle: the more geometrically interesting a scheme is, the richer its divisor theory becomes. The remainder of this section develops the algebraic theory in full generality.

5.4.1 Weil Divisors

Weil divisors generalize the discussion of divisors on Riemann surfaces to higher-dimensional schemes. The key requirement is a well-defined notion of "order of vanishing" at each codimension-1 point. On a nonsingular curve, every local ring is a DVR, so the valuation provides this order directly. In higher dimensions, the codimension-1 points play the role of "prime divisors," and we need the local rings at these points to be DVRs — equivalently, regular of dimension 1.

Definition 5.4.1: Regular in Codimension 1

A scheme X is *regular in codimension* 1 if every local ring $\mathcal{O}_{X,\mathfrak{p}}$ of dimension 1 (i.e., at each codimension-1 point $\mathfrak{p}$) is a regular local ring.

Example 5.4.2: Regular in Codimension 1

1. Every regular scheme is regular in codimension 1, since all local rings are regular. In particular, every nonsingular curve satisfies this condition.
2. Every normal Noetherian scheme is regular in codimension 1. Indeed, if $\mathcal{O}_{X,\mathfrak{p}}$ is a one-dimensional integrally closed local domain, then it is a DVR (see [**NateClassicalAlgGeo**]). This is the most common hypothesis encountered in practice.
3. The scheme $X = \operatorname{Spec} k[x, y]/(xy)$ (the union of two coordinate axes) is *not* regular in codimension 1: the local ring at the origin is not a domain, hence not regular. However, each irreducible component separately is regular.

When $\mathcal{O}_{X,\mathfrak{p}}$ is a DVR with valuation $\nu_{\mathfrak{p}}$, the integer $\nu_{\mathfrak{p}}(f)$ for a nonzero rational function f measures the order of vanishing (if positive) or the order of the pole (if negative) of f along the codimension-1 subscheme $\overline{\{\mathfrak{p}\}}$. This is the only datum we need to define Weil divisors.

5.4.3 Remark: Beyond the Regular Case If $\mathcal{O}_{X,\mathfrak{p}}$ is not regular, one can still define an "order of vanishing" using the *length* $\ell_{\mathcal{O}_{X,\mathfrak{p}}}(\mathcal{O}_{X,\mathfrak{p}}/(f))$ in place of the valuation. The key property $\operatorname{ord}(fg) = \operatorname{ord}(f) + \operatorname{ord}(g)$ is preserved, as one sees from the short exact sequence

$$0 \longrightarrow \mathcal{O}_{X,\mathfrak{p}}/(g) \xrightarrow{\cdot f} \mathcal{O}_{X,\mathfrak{p}}/(fg) \longrightarrow \mathcal{O}_{X,\mathfrak{p}}/(f) \longrightarrow 0,$$

where the left map is multiplication by f (which is injective when f is a non-zero-divisor) and additivity of length gives the result. This approach works on any Noetherian scheme, but the valuation-based definition is cleaner when available, and suffices for the applications in this text.

For narrative clarity, we impose the following standing hypotheses on the ambient scheme. The conditions are chosen to ensure that: the scheme has a function field to draw rational functions from (integral), restriction maps are injective so that rational functions are unambiguous (separated), the topology is well-behaved (Noetherian), and the order of vanishing is well-defined (regular in codimension 1).

Definition 5.4.4: Weil Divisor

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1. A *prime divisor* on X is a closed integral subscheme of codimension 1. A *Weil divisor* is an element of the free abelian group generated by the prime divisors:

$$\operatorname{WDiv} X := \bigoplus_{\substack{Y \subseteq X \\ \operatorname{codim}(Y)=1}} \mathbb{Z} \cdot [Y].$$

A Weil divisor $D = \sum_i n_i [Y_i]$ is called *effective* if $n_i \geq 0$ for all i, written $D \geq 0$.

Let $Y \subseteq X$ be a prime divisor with generic point η_Y. Since Y has codimension 1 and X is regular in codimension 1, the local ring $\mathcal{O}_{X,\eta_Y}$ is a one-dimensional regular local ring, hence a DVR. Denote its valuation by ν_Y. Since X is separated, the prime divisor Y is uniquely determined by its generic point (cf. exercise **??**), and hence by its valuation ring $\mathcal{O}_{X,\eta_Y} \subseteq K(X)$.

Now let $K(X)$ denote the function field of X and let $f \in K(X)^*$ be a nonzero rational function. The integer $\nu_Y(f)$ records the behaviour of f along Y:

- $\nu_Y(f) > 0$: f vanishes along Y to order $\nu_Y(f)$ (a *zero* of order $\nu_Y(f)$).
- $\nu_Y(f) < 0$: f has a *pole* of order $|\nu_Y(f)|$ along Y.
- $\nu_Y(f) = 0$: f is regular and nonvanishing at the generic point of Y.

For the formal sum $\sum_Y \nu_Y(f)[Y]$ to be a well-defined Weil divisor, we need it to be a *finite* sum. This is guaranteed by the following lemma:

Lemma 5.4.5: Almost All Valuations Vanish

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1, and let $f \in K(X)^*$. Then $\nu_Y(f) = 0$ for all but finitely many prime divisors Y.

***Proof*:**
Since f is a rational function on an integral scheme, it is regular on some nonempty open subset; choose an affine open $U = \operatorname{Spec} R \subseteq X$ on which f is regular. The complement $C = X \setminus U$ is a proper closed subset. Since X is Noetherian, C is a finite union of irreducible closed subsets, each of codimension ≥ 1. In particular, C contains at most finitely many prime divisors of X.

It remains to show that only finitely many prime divisors of U have $\nu_Y(f) \neq 0$. On U, the function f is regular (i.e., $f \in R \subseteq K(X)$), so $\nu_Y(f) \geq 0$ for every prime divisor Y meeting U. Moreover, $\nu_Y(f) > 0$ if and only if f vanishes along $Y \cap U$, meaning $Y \cap U$ is contained in the closed subset $V(f) \subseteq U$. Since $f \neq 0$, this closed subset is proper, and a Noetherian space has only finitely many irreducible components of any given codimension. Hence only finitely many prime divisors Y of U satisfy $\nu_Y(f) > 0$.

Combining, $\nu_Y(f) \neq 0$ for at most finitely many prime divisors (those contained in C together with those in $V(f) \cap U$).

This ensures that the following formal sum is a well-defined Weil divisor:

Definition 5.4.6: Principal Divisor

Let X be as above and $f \in K(X)^*$ a nonzero rational function. The *principal divisor* (or *divisor of f*) is

$$\operatorname{div}(f) \;:=\; \sum_Y \nu_Y(f)\,[Y] \;\in\; \operatorname{WDiv} X,$$

where the sum runs over all prime divisors Y of X. The subgroup of principal divisors is

$$\operatorname{PDiv}(X) \;:=\; \{\operatorname{div}(f) \mid f \in K(X)^*\} \;\leq\; \operatorname{WDiv} X.$$

The assignment $f \mapsto \operatorname{div}(f)$ is a group homomorphism $K(X)^* \to \operatorname{WDiv} X$, since $\nu_Y(fg) = \nu_Y(f) + \nu_Y(g)$ for every prime divisor Y.

Note that $\operatorname{Div}(f/g) = \operatorname{Div} f - \operatorname{Div} g$ by the properties of discrete valuations, and hence this is even a group homomorphism from the multiplicative group $(\kappa(X))^*$ to the additive group $\operatorname{Div} X$!

Two Weil divisors D, D' are called *linearly equivalent*, written $D \sim D'$, if their difference is principal: $D - D' = \operatorname{div}(f)$ for some $f \in K(X)^*$. This is an equivalence relation (since principal divisors form a subgroup), and the quotient measures the "non-trivial" divisors on X.

Definition 5.4.7: Divisor Class Group

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1. The *divisor class group* (or *Weil divisor class group*) of X is

$$\operatorname{Cl} X \;:=\; \frac{\operatorname{WDiv} X}{\operatorname{PDiv}(X)}.$$

Two divisors representing the same class in $\operatorname{Cl} X$ are said to be *linearly equivalent*.

Before computing examples, we establish a geometric characterization of unique factorization domains in terms of divisor class groups as these characterize trivial divisor class groups of affine schemes.

The underlying algebraic input is the classical fact that a Noetherian domain is a UFD if and only if every height-1 prime ideal is principal (cf. corollary 5.3.16). The following lemma provides the geometric bridge.

Lemma 5.4.8: Normal Domain: Intersection of Height-One Localizations

Let R be a Noetherian normal domain (i.e., an integrally closed Noetherian domain). Then

$$R = \bigcap_{\substack{\mathfrak{p} \in \operatorname{Spec} R \\ \operatorname{ht}(\mathfrak{p})=1}} R_{\mathfrak{p}},$$

where the intersection is taken inside $\operatorname{Frac}(R)$.

***Proof*:**
The inclusion $R \subseteq \bigcap_{\operatorname{ht}(\mathfrak{p})=1} R_{\mathfrak{p}}$ is immediate. For the reverse, suppose $x \in \operatorname{Frac}(R) \setminus R$. Since R is normal, x is not integral over R. We claim that $x \notin R_{\mathfrak{p}}$ for some height-1 prime $\mathfrak{p}$.

Write $x = a/b$ with $a, b \in R$ and $b \neq 0$. Since $x \notin R$, we have $a \notin (b)$. Consider the quotient ideal

$$(b : a) = \{r \in R \mid ra \in (b)\}$$

this is a proper ideal of R (since $1 \notin (b : a)$). Choose a minimal prime $\mathfrak{p}$ over (b). By Krull's principal ideal theorem (theorem 2.6.13), $\operatorname{ht}(\mathfrak{p}) = 1$ (since (b) is principal and R is a domain). Now $R_{\mathfrak{p}}$ is a one-dimensional normal local domain, hence a DVR. In $R_{\mathfrak{p}}$, the element b generates an $\mathfrak{m}_{\mathfrak{p}}$-primary ideal, so $\nu_{\mathfrak{p}}(b) \geq 1$. But $a \notin bR_{\mathfrak{p}}$ (since $\mathfrak{p}$ was chosen to contain $(b : a)$), which means $\nu_{\mathfrak{p}}(a) < \nu_{\mathfrak{p}}(b)$, and hence $\nu_{\mathfrak{p}}(x) = \nu_{\mathfrak{p}}(a) - \nu_{\mathfrak{p}}(b) < 0$. In particular, $x \notin R_{\mathfrak{p}}$.

This lemma says that on a normal scheme, a rational function that is regular at every codimension-1 point is in fact a regular function. Put geometrically: regular functions on a normal scheme are determined by their behaviour along prime divisors. This is the algebraic analogue of the Hartogs extension theorem in complex analysis (compare theorem 2.6.18), and it is the key input for the following characterization.

Proposition 5.4.9: Characterizing UFDs Geometrically

Let R be a Noetherian domain. Then R is a UFD if and only if $\operatorname{Spec} R$ is normal and $\operatorname{Cl} \operatorname{Spec} R = 0$.

***Proof*:**
$(\Rightarrow)$ A UFD is integrally closed (see [**NateClassicalAlgGeo**]), so $\operatorname{Spec} R$ is normal. To show $\operatorname{Cl} \operatorname{Spec} R = 0$, it suffices to show that every prime divisor is principal. Let Y be a prime divisor with generic point η_Y corresponding to a height-1 prime $\mathfrak{p} \subseteq R$. Since R is a UFD, $\mathfrak{p}$ is principal: $\mathfrak{p} = (p)$ for some irreducible element $p \in R$. Then $\operatorname{div}(p) = [Y]$ (since $\nu_Y(p) = 1$ and $\nu_{Y'}(p) = 0$ for every other prime divisor Y', as p lies in no other height-1 prime of a UFD). Hence $[Y] = 0$ in $\operatorname{Cl} \operatorname{Spec} R$.

$(\Leftarrow)$ It suffices to show that every height-1 prime $\mathfrak{p}$ of R is principal (this characterizes Noetherian UFDs; cf. corollary 5.3.16). Let Y be the prime divisor corresponding to $\mathfrak{p}$. Since $\operatorname{Cl} \operatorname{Spec} R = 0$, the divisor $[Y]$ is principal: there exists $f \in K(X)^*$ with $\operatorname{div}(f) = [Y]$. This means:

- $\nu_Y(f) = 1$, so $f \in R_{\mathfrak{p}}$ and f generates $\mathfrak{p}R_{\mathfrak{p}}$;
- $\nu_{Y'}(f) = 0$ for every other prime divisor Y' (corresponding to every other height-1 prime $\mathfrak{q} \neq \mathfrak{p}$), so $f \in R_{\mathfrak{q}}^*$.

In particular, $\nu_{Y'}(f) \geq 0$ for all height-1 primes, so $f \in \bigcap_{\mathrm{ht}(\mathfrak{q})=1} R_{\mathfrak{q}} = R$ by lemma 5.4.8. Similarly, $\nu_{Y'}(f^{-1}) \geq 0$ for all $\mathfrak{q} \neq \mathfrak{p}$, so there is no obstruction from other primes. We claim $\mathfrak{p} = (f)$. Take any $g \in \mathfrak{p}$; then $\nu_Y(g) \geq 1$ and $\nu_{Y'}(g) \geq 0$ for all other prime divisors Y'. Hence

$$\nu_{Y'}(g/f) \ \geq \ 0$$

for *every* height-1 prime, and so $g/f \in R$ by lemma 5.4.8. This gives $g \in fR = (f)$, so $\mathfrak{p} \subseteq (f)$. Since $f \in \mathfrak{p}$ (as $\nu_Y(f) = 1 > 0$) and $\mathfrak{p}$ is prime, we conclude $\mathfrak{p} = (f)$.

Example 5.4.10: Divisor Class Groups

1. *Affine space.* The class group $\operatorname{Cl} \mathbb{A}^n_k = 0$, since $k[x_1, \ldots, x_n]$ is a UFD and proposition 5.4.9 applies. More generally, $\operatorname{Cl} \operatorname{Spec} R = 0$ for any UFD R.
2. *Dedekind domains.* Let R be a Dedekind domain (e.g., the ring of integers $\mathcal{O}_K$ of a number field K). The scheme $X = \operatorname{Spec} R$ is a normal Noetherian integral scheme of dimension 1, so it satisfies our standing hypotheses. The prime divisors are exactly the closed points of X, corresponding to the nonzero prime ideals of R. Every Weil divisor is thus a formal sum of nonzero primes — in other words, a fractional ideal written additively. Under this identification, the principal divisors correspond to principal fractional ideals, and the divisor class group $\operatorname{Cl} X$ is precisely the *ideal class group* $\operatorname{Cl}(R)$ from algebraic number theory [**NateClassicalAlgGeo**] . In particular:
   - $\operatorname{Cl} \operatorname{Spec} \mathbb{Z} = 0$, since $\mathbb{Z}$ is a PID (equivalently, a UFD).
   - $\operatorname{Cl} \operatorname{Spec} \mathbb{Z}[\sqrt{-5}] \cong \mathbb{Z}/2\mathbb{Z}$, the classical example of a non-trivial ideal class group.

   This recovers the fundamental theorem of algebraic number theory: R is a UFD if and only if $\operatorname{Cl}(R) = 0$.

We now compute the divisor class group of projective space — the fundamental example for the rest of the text. Recall that every prime divisor of $\mathbb{P}^n_k$ is a hypersurface $V_+(f)$ for some irreducible homogeneous polynomial $f \in k[x_0, \ldots, x_n]$ (unique up to scalar), and its *degree* is $\deg f$.

Proposition 5.4.11: Characterizing Divisors of Projective Space

Let k be a field, $X = \mathbb{P}^n_k$, and $H = V_+(x_0)$ the standard hyperplane. For a Weil divisor $D = \sum_i n_i[Y_i]$, define $\deg D = \sum_i n_i \deg Y_i$. Then:

1. For every $f \in K(X)^*$, one has $\deg(\operatorname{div}(f)) = 0$.
2. If D is a divisor of degree d, then $[D] = d[H]$ in $\operatorname{Cl}\mathbb{P}^n_k$.
3. The degree map induces an isomorphism $\deg\colon \operatorname{Cl}\mathbb{P}^n_k \xrightarrow{\sim} \mathbb{Z}$.

***Proof*:**
Part (1). A rational function on $\mathbb{P}^n_k$ is a ratio f/g of homogeneous polynomials of the same degree d. Factor $f = \prod_i f_i^{a_i}$ and $g = \prod_j g_j^{b_j}$ into irreducibles in the UFD $k[x_0, \ldots, x_n]$. Then

$$\operatorname{div}(f/g) \;=\; \sum_i a_i\,[V_+(f_i)] \;-\; \sum_j b_j\,[V_+(g_j)],$$

so $\deg(\operatorname{div}(f/g)) = \sum_i a_i \deg f_i - \sum_j b_j \deg g_j = \deg f - \deg g = 0$.
Part (2). Let $D = \sum_i n_i[Y_i]$ with $\deg D = d$. Each prime divisor Y_i is cut out by an irreducible homogeneous polynomial f_i of degree $\deg Y_i$. The expression

$$r \;=\; \frac{\prod_i f_i^{n_i}}{x_0^d}$$

is a well-defined element of $K(X)^*$: after collecting positive and negative exponents, the numerator and denominator are homogeneous of the same total degree (since $\sum_i n_i \deg f_i = d$). Computing its divisor:

$$\operatorname{div}(r) \;=\; \sum_i n_i\,[Y_i] \;-\; d\,[H] \;=\; D - d[H].$$

Hence $[D] = d[H]$ in $\operatorname{Cl}\mathbb{P}^n_k$.
Part (3). By (1), the degree map descends to a well-defined homomorphism $\deg\colon \operatorname{Cl}\mathbb{P}^n_k \to \mathbb{Z}$. It is surjective (since $\deg[H] = 1$) and injective (since any D of degree 0 satisfies $[D] = 0 \cdot [H] = 0$ by (2)).

Under the correspondence between line bundles and Cartier divisors (proposition 5.4.31), the class $n[H] \in \operatorname{Cl}\mathbb{P}^n_k$ corresponds to the twisting sheaf $\mathcal{O}_{\mathbb{P}^n_k}(n)$. This recovers the classification of line bundles on projective space from corollary 5.4.34.

The next result is the primary computational tool for divisor class groups. It describes how $\operatorname{Cl} X$ changes when we pass to an open subset.

Proposition 5.4.12: Divisor Class Group and Closed Subsets

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1, and let $Z \subsetneq X$ be a proper closed subset with complement $U = X \setminus Z$.

1. There is a surjective group homomorphism $\operatorname{Cl} X \twoheadrightarrow \operatorname{Cl} U$ induced by restriction: $\sum n_i[Y_i] \mapsto \sum n_i[Y_i \cap U]$ (discarding components that do not meet U).
2. If $\operatorname{codim}(Z, X) \geq 2$, then this map is an isomorphism.
3. If Z is irreducible of codimension 1 (hence a prime divisor), there is an exact sequence

$$\mathbb{Z} \xrightarrow{1 \mapsto [Z]} \operatorname{Cl} X \longrightarrow \operatorname{Cl} U \longrightarrow 0.$$

Point (2) explains why the class group is insensitive to removing closed subsets of "small" codimension. It is the divisor-theoretic shadow of the algebraic Hartogs lemma (theorem 2.6.18): regular functions extend across codimension-2 subsets, and correspondingly, the divisor theory "doesn't see" such subsets.

***Proof*:**
Part (1). Since U is a dense open subset of the integral scheme X, we have $K(U) = K(X)$. Every prime divisor of U extends uniquely to a prime divisor of X (take the closure in X), so the restriction map $\operatorname{WDiv} X \to \operatorname{WDiv} U$ is surjective on generators and hence surjective. Moreover, a principal divisor $\operatorname{div}(f)$ on X restricts to $\operatorname{div}_U(f)$ on U (the valuations at prime divisors meeting U are unchanged), so the map descends to a surjection $\operatorname{Cl} X \twoheadrightarrow \operatorname{Cl} U$.

Part (2). If $\operatorname{codim}(Z, X) \geq 2$, then Z contains no codimension-1 points of X. Hence every prime divisor of X meets U, and the restriction $\operatorname{WDiv} X \to \operatorname{WDiv} U$ is an isomorphism (not merely a surjection). For injectivity on class groups: if $D \in \operatorname{WDiv} X$ becomes principal on U, say $D|_U = \operatorname{div}_U(f)$, then $D - \operatorname{div}_X(f)$ is a Weil divisor on X supported on Z. Since Z contains no prime divisors, this forces $D - \operatorname{div}_X(f) = 0$, so D is principal. This is the divisor-theoretic counterpart of the algebraic Hartogs lemma (theorem 2.6.18): regular functions extend across codimension-2 subsets, and correspondingly the divisor theory "does not see" such subsets.

Part (3). The kernel of $\operatorname{Cl} X \to \operatorname{Cl} U$ consists of classes representable by divisors supported entirely on Z. Since Z is irreducible of codimension 1, every such divisor is an integer multiple of $[Z]$, so the kernel is exactly the image of the map $\mathbb{Z} \to \operatorname{Cl} X$, $1 \mapsto [Z]$.

By strategically removing closed subsets and applying the exact sequence in part (3), one can often reduce the computation of $\operatorname{Cl} X$ to an affine (and hence often factorial) situation. Part (2) tells us that removing "small" closed subsets does not change the class group, which frequently allows us to simplify X without losing divisor-theoretic information.

Example 5.4.13: More Divisor Class Groups

1. *Punctured affine space.* Let $X = \mathbb{A}^n_k \setminus \{0\}$ for $n \geq 2$. Since $\operatorname{Cl} \mathbb{A}^n_k = 0$ and $\{0\}$ has codimension $n \geq 2$, the surjection $\operatorname{Cl} \mathbb{A}^n_k \twoheadrightarrow \operatorname{Cl} X$ is an isomorphism (no prime divisors are lost). Hence $\operatorname{Cl} X = 0$.
2. *Complements of hypersurfaces in $\mathbb{P}^2$.* Let $Y \subseteq \mathbb{P}^2_k$ be an irreducible curve of degree d. Then

Y is a prime divisor, and proposition 5.4.12(3) gives

$$\mathbb{Z} \xrightarrow{1\mapsto[Y]} \operatorname{Cl}\mathbb{P}^2_k \longrightarrow \operatorname{Cl}(\mathbb{P}^2_k \setminus Y) \longrightarrow 0.$$

Since $[Y] = d[H]$ in $\operatorname{Cl}\mathbb{P}^2_k \cong \mathbb{Z}$ by **??**, the image of $\mathbb{Z}$ is $d\mathbb{Z} \subseteq \mathbb{Z}$, and hence

$$\operatorname{Cl}(\mathbb{P}^2_k \setminus Y) \;\cong\; \mathbb{Z}/d\mathbb{Z}.$$

For instance, the complement of a conic ($d = 2$) has class group $\mathbb{Z}/2\mathbb{Z}$, while the complement of a line ($d = 1$) has trivial class group — geometrically, every codimension-1 subvariety of $\mathbb{P}^2 \setminus L$ is already a principal divisor. The d-torsion can be understood via Bézout's theorem ([**NateClassicalAlgGeo**]): any line L meets Y in exactly d points (counted with multiplicity), so $d[L \cap (\mathbb{P}^2 \setminus Y)]$ is principal on the complement, but $[L \cap (\mathbb{P}^2 \setminus Y)]$ itself need not be.

3. *The quadric cone.* Let $\operatorname{char} k \neq 2$ and consider the quadric cone $X = \operatorname{Spec} R$ where $R = k[x,y,z]/(xy - z^2)$. The ring R is a normal domain[a], so X is normal, Noetherian, integral, and regular in codimension 1. However, R is *not* a UFD: the equation $xy = z^2$ exhibits a failure of unique factorization. By proposition 5.4.9, we expect $\operatorname{Cl} X \neq 0$.

   Let $Y = V(y,z) \subseteq X$ be the "ruling" of the cone (see fig. 5.1). Since $\mathfrak{p} = (y,z)$ is a height-1 prime of R, the subscheme Y is a prime divisor. We claim that $\operatorname{div}(y) = 2[Y]$. Indeed, in the local ring $\mathcal{O}_{X,\eta_Y} = R_{\mathfrak{p}}$, the relation $xy = z^2$ and the fact that $x \notin \mathfrak{p}$ (so x is a unit in $R_{\mathfrak{p}}$) give $y = z^2x^{-1} \in (z)^2$. Since z generates the maximal ideal of the DVR $R_{\mathfrak{p}}$, the uniformizer is z and $\nu_Y(y) = 2$.

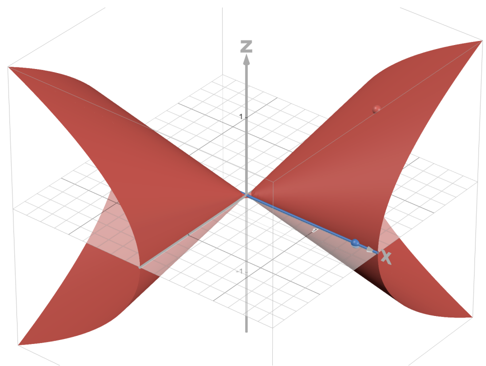


Figure 5.1: A ruling on the quadratic cone

   Now apply the exact sequence to $Z = Y$:

$$\mathbb{Z} \xrightarrow{1\mapsto[Y]} \operatorname{Cl} X \longrightarrow \operatorname{Cl}(X \setminus Y) \longrightarrow 0.$$

   The complement is $X \setminus Y = \operatorname{Spec} R_y$ (inverting y). In R_y, the relation $xy = z^2$ allows us to eliminate $x = z^2y^{-1}$, yielding $R_y \cong k[y, y^{-1}, z]$. This is a localization of the polynomial ring $k[y,z]$ (hence a UFD), so $\operatorname{Cl}(X \setminus Y) = 0$ by proposition 5.4.9. The exact sequence collapses to

$$\mathbb{Z} \longrightarrow \operatorname{Cl} X \longrightarrow 0,$$

so $\mathrm{Cl}\,X$ is cyclic, generated by $[Y]$. Since $\mathrm{div}(y) = 2[Y]$, we have $2[Y] = 0$ in $\mathrm{Cl}\,X$, giving $\mathrm{Cl}\,X \cong \mathbb{Z}/2\mathbb{Z}$ or $\mathrm{Cl}\,X = 0$.

To rule out the latter, we show that $\mathfrak{p} = (y, z)$ is not principal. Let $\mathfrak{m} = (x, y, z)$ be the maximal ideal at the vertex of the cone. The images $\overline{y}$ and $\overline{z}$ are linearly independent in the cotangent space $\mathfrak{m}/\mathfrak{m}^2 \cong k^3$ (the relation $xy - z^2 \in \mathfrak{m}^2$ imposes no constraint at the level of $\mathfrak{m}/\mathfrak{m}^2$). Hence $\mathfrak{p}$ requires at least two generators and cannot be principal. We conclude

$$\mathrm{Cl}\,X \;\cong\; \mathbb{Z}/2\mathbb{Z}.$$

Geometrically, a single ruling cannot be cut out by one equation, but the divisor $2[Y] = \mathrm{div}(y)$ *can*: the function y vanishes along Y with multiplicity 2.

4. *The smooth quadric surface.* Let $Q = V_+(xw - yz) \subseteq \mathbb{P}^3_k$. Under the Segre embedding (example 0.2.28), $Q \cong \mathbb{P}^1_k \times \mathbb{P}^1_k$, and Q carries two families of lines (rulings): "horizontal" lines $L_p = \{p\} \times \mathbb{P}^1$ and "vertical" lines $M_q = \mathbb{P}^1 \times \{q\}$ for $p, q \in \mathbb{P}^1$. Fix one line L_0 from the first family. Then $Q \setminus L_0 \cong \mathbb{A}^1 \times \mathbb{P}^1$ (removing one point from the first factor).

   Claim: $\mathrm{Cl}\,(\mathbb{A}^1 \times \mathbb{P}^1) \cong \mathbb{Z}$, generated by a vertical line. Removing a vertical section $M_0' = \mathbb{A}^1 \times \{q_0\}$ gives $\mathbb{A}^1 \times (\mathbb{P}^1 \setminus \{q_0\}) \cong \mathbb{A}^2$, which has $\mathrm{Cl} = 0$. The exact sequence

$$\mathbb{Z} \xrightarrow{1 \mapsto [M_0']} \mathrm{Cl}\,(\mathbb{A}^1 \times \mathbb{P}^1) \longrightarrow 0$$

   shows $\mathrm{Cl}\,(\mathbb{A}^1 \times \mathbb{P}^1)$ is cyclic, generated by $[M_0']$. To see that $[M_0']$ has infinite order, restrict to a fiber: the inclusion $\iota\colon \{s_0\} \times \mathbb{P}^1 \hookrightarrow \mathbb{A}^1 \times \mathbb{P}^1$ induces $\iota^*\colon \mathrm{Cl}\,(\mathbb{A}^1 \times \mathbb{P}^1) \to \mathrm{Cl}\,(\mathbb{P}^1) \cong \mathbb{Z}$ sending $[M_0'] \mapsto [\{q_0\}]$, which is a generator. Since ι^* is surjective, $[M_0']$ cannot be torsion.

   Returning to Q, the exact sequence for removing L_0 is

$$\mathbb{Z} \xrightarrow{1 \mapsto [L_0]} \mathrm{Cl}\,Q \longrightarrow \mathrm{Cl}\,(\mathbb{A}^1 \times \mathbb{P}^1) \cong \mathbb{Z} \longrightarrow 0.$$

   A vertical line M_0 on Q restricts to a generator of $\mathrm{Cl}\,(\mathbb{A}^1 \times \mathbb{P}^1)$, so the map $\mathbb{Z} \to \mathrm{Cl}\,Q$, $1 \mapsto [M_0]$ provides a splitting. Hence $\mathrm{Cl}\,Q \cong \mathbb{Z}[L_0] \oplus \mathbb{Z}[M_0]$, provided $[L_0]$ has infinite order. This follows by symmetry: swapping the two $\mathbb{P}^1$ factors exchanges horizontal and vertical rulings, so the same argument with M_0 removed shows $[L_0]$ has infinite order. Therefore

$$\mathrm{Cl}\,Q \;\cong\; \mathbb{Z} \oplus \mathbb{Z},$$

   generated by the two families of rulings. A hyperplane section of Q has class $[L_0] + [M_0]$[b]. In contrast to $\mathrm{Cl}\,\mathbb{P}^2_k \cong \mathbb{Z}$, the "degree" of a curve on Q is a *pair* of integers (a, b) recording how it meets each family of rulings — a first glimpse of how the Picard group can carry genuinely higher-rank information.

[a] A standard exercise: if $\mathrm{char}\,k \neq 2$ and $f \in k[x_1, \dots, x_n]$ is square-free, then $k[x_1, \dots, x_n, z]/(z^2 - f)$ is integrally closed.

[b] This can be checked directly: the hyperplane $V_+(x)$ meets Q in $V_+(x, yz) = V_+(x, y) \cup V_+(x, z)$, which is a horizontal ruling plus a vertical ruling.

The exact sequence in proposition 5.4.12 relates the class groups of a scheme and its open subsets. The following result provides a complementary tool, showing that the class group is insensitive to "thickening" in the affine-line direction.

Proposition 5.4.14: Class Group Stably Isomorphic

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1. Then $X \times \mathbb{A}^1$ also satisfies all of these conditions, and the projection $\pi\colon X \times \mathbb{A}^1 \to X$ induces an isomorphism

$$\pi^*\colon \operatorname{Cl} X \xrightarrow{\sim} \operatorname{Cl}(X \times \mathbb{A}^1).$$

***Proof*:**

The scheme $X \times \mathbb{A}^1$ is Noetherian (locally $\operatorname{Spec} R[t]$, which is Noetherian by Hilbert's basis theorem), integral (with function field $K(X)(t)$), and separated (since separatedness is preserved under base change). It remains to verify regularity in codimension 1. The codimension-1 points of $X \times \mathbb{A}^1$ fall into two types:

- *Type 1:* the generic point of $Y \times \mathbb{A}^1$, where Y is a prime divisor of X. The local ring is $\mathcal{O}_{X,\eta_Y}[t]$ localized at $\mathfrak{m}_{\eta_Y} \cdot \mathcal{O}_{X,\eta_Y}[t]$. Since $\mathcal{O}_{X,\eta_Y}$ is a DVR (say with uniformizer u), this localization is a DVR with the same uniformizer u.

- *Type 2:* a point ξ whose image under π is the generic point of X. The local ring is $K(X)[t]$ localized at an irreducible polynomial, hence a DVR (since $K(X)[t]$ is a PID).

Thus $X \times \mathbb{A}^1$ is regular in codimension 1.

Definition of π^*. For a Weil divisor $D = \sum n_i[Y_i]$ on X, set $\pi^*(D) = \sum n_i[Y_i \times \mathbb{A}^1]$. Each $Y_i \times \mathbb{A}^1$ is a type 1 prime divisor of $X \times \mathbb{A}^1$. If $f \in K(X)^*$, then $\pi^*(\operatorname{div}_X(f)) = \operatorname{div}_{X\times\mathbb{A}^1}(f)$ (viewing f as an element of $K(X) \subseteq K(X)(t) = K(X \times \mathbb{A}^1)$), so π^* descends to a well-defined homomorphism $\pi^*\colon \operatorname{Cl} X \to \operatorname{Cl}(X \times \mathbb{A}^1)$.

Injectivity. Suppose $D \in \operatorname{WDiv} X$ and $\pi^*(D) = \operatorname{div}(h)$ for some $h \in K(X)(t)^*$. Since $\pi^*(D)$ involves only type 1 prime divisors, h must have $\nu_Z(h) = 0$ for every type 2 prime divisor Z. Write $h = g_1/g_2$ with $g_1, g_2 \in K(X)[t]$ coprime. Any irreducible factor of g_i of positive degree in t would contribute a nonzero valuation at a type 2 prime divisor, contradicting $\nu_Z(h) = 0$. Hence $g_1, g_2 \in K(X)^*$, so $h = g_1/g_2 \in K(X)^*$. It follows that $D = \operatorname{div}_X(h)$ is principal, and π^* is injective.

Surjectivity. It suffices to show that every type 2 prime divisor on $X \times \mathbb{A}^1$ is linearly equivalent to a $\mathbb{Z}$-linear combination of type 1 prime divisors (which lie in the image of π^*). Let $Z \subseteq X \times \mathbb{A}^1$ be a type 2 prime divisor. Localizing at the generic point of X gives a prime divisor of $\operatorname{Spec} K(X)[t]$. Since $K(X)[t]$ is a PID, this corresponds to a principal ideal (f) for some irreducible $f \in K(X)[t]$. Viewed as an element of $K(X)(t)^*$, the divisor $\operatorname{div}_{X\times\mathbb{A}^1}(f)$ involves Z with coefficient 1, plus possibly type 1 prime divisors arising from the denominators of the coefficients of f. Crucially, f cannot vanish along any *other* type 2 prime divisor $Z' \neq Z$: such a Z' would correspond to a different irreducible in $K(X)[t]$, and f is irreducible. Hence

$$\operatorname{div}(f) \;=\; [Z] + (\text{type 1 divisors}),$$

showing that $[Z]$ is linearly equivalent to a combination of type 1 divisors. Since type 1 divisors are in the image of π^*, the map π^* is surjective.

This result is the scheme-theoretic upgrade of the intuition that "adding a free variable doesn't change divisor theory." As immediate applications: $\mathrm{Cl}\,(\mathbb{A}_k^n) = 0$ for all n (by induction from $\mathrm{Cl}\,(\operatorname{Spec} k) = 0$), and $\mathrm{Cl}\,(X \times \mathbb{A}^n) \cong \mathrm{Cl}\,X$ for any X satisfying our hypotheses.

We now record several further applications that illustrate the interplay between the class group, unique factorization, and geometric structure.

Example 5.4.15: Applications of Stable Isomorphism

1. *Punctured affine space.* Let $X = \mathbb{A}_k^n \setminus \{0\}$ for $n \geq 2$. The complement $\{0\}$ has codimension $n \geq 2$ in $\mathbb{A}_k^n$, so proposition 5.4.12(2) gives $\mathrm{Cl}\,X \cong \mathrm{Cl}\,\mathbb{A}_k^n = 0$.

2. *Products with affine space.* The class group of the smooth quadric surface $Q \cong \mathbb{P}^1 \times \mathbb{P}^1$ is $\mathbb{Z} \oplus \mathbb{Z}$ (example 5.4.13). By proposition 5.4.14, $\mathrm{Cl}\,(\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{A}^n) \cong \mathbb{Z} \oplus \mathbb{Z}$ for all $n \geq 0$.

3. *Normality is necessary.* Let $C = \operatorname{Spec} k[x, y]/(y^2 - x^3)$ be the cuspidal cubic. The local ring at the origin is not integrally closed (y/x satisfies $(y/x)^2 = x$ but does not lie in the local ring), so C is not normal. The class group formalism as developed here does not directly apply; one must pass to the normalization $\widetilde{C} \cong \mathbb{A}_k^1$ (via $t \mapsto (t^2, t^3)$), which has $\mathrm{Cl} = 0$. The failure of normality at a single point introduces pathologies that the Cartier divisor formalism (definition 5.4.17) handles more gracefully.

5.4.2 Cartier Divisors

Weil divisors provide a geometric language for codimension-1 phenomena, but their definition requires strong hypotheses on the ambient scheme: integrality (for the function field), regularity in codimension 1 (for the valuations), and the Noetherian condition (for finiteness). On a general scheme — possibly reducible, non-reduced, or with non-regular local rings — none of these tools are available.

Cartier divisors provide an alternative that works on any scheme. The idea is to abandon the "global" perspective of Weil divisors (formal sums of codimension-1 subvarieties) in favour of a "local" one: a Cartier divisor is the data of rational functions on an open cover that are "locally principal," glued together by units on overlaps. Since principal divisors carry no interesting divisor-class information, the gluing data is precisely what the class group detects.

On sufficiently nice schemes (integral, Noetherian, locally factorial), the two theories agree and the distinction is merely a matter of perspective. The advantage of Cartier divisors, beyond their greater generality, is their direct connection to line bundles: we will see in proposition 5.4.31 that Cartier divisor classes are in natural bijection with isomorphism classes of invertible sheaves.

To define Cartier divisors on schemes that may not be integral, we need a replacement for the function field. The correct substitute is the *sheaf of total quotient rings*, which inverts all non-zero-divisors rather than all nonzero elements.

Definition 5.4.16: Sheaf of Total Quotient Rings

Let X be a scheme. For each affine open $U = \operatorname{Spec} R \subseteq X$, let $S_U \subseteq \Gamma(U, \mathcal{O}_X)$ denote the set of non-zero-divisors, and form the total quotient ring $S_U^{-1}\Gamma(U, \mathcal{O}_X)$. This assignment defines a presheaf on the affine opens of X; its sheafification is the *sheaf of total quotient rings*, denoted $\mathscr{K}_X$.
Write $\mathscr{K}_X^*$ for the subsheaf of abelian groups whose sections over an open U are the invertible elements of $\mathscr{K}_X(U)$, and $\mathcal{O}_X^*$ for the subsheaf of units of $\mathcal{O}_X$. The inclusion $\mathcal{O}_X^* \hookrightarrow \mathscr{K}_X^*$ makes $\mathcal{O}_X^*$ a subsheaf of abelian groups, and we may form the quotient sheaf $\mathscr{K}_X^*/\mathcal{O}_X^*$.

When X is integral, every nonzero element of an integral domain is a non-zero-divisor, so the total quotient ring of $\Gamma(U, \mathcal{O}_X)$ is simply its fraction field — which is $K(X)$ for every nonempty open U. Thus $\mathscr{K}_X$ is the constant sheaf associated to $K(X)$, and we recover the usual notion of rational function. In the non-integral case, $\mathscr{K}_X$ is genuinely more subtle: its sections are "rational functions" that may have poles but are not allowed to divide by zero-divisors.

Definition 5.4.17: Cartier Divisor

Let X be a scheme. A *Cartier divisor* on X is a global section of the quotient sheaf $\mathscr{K}_X^*/\mathcal{O}_X^*$:

$$\operatorname{CDiv} X \;:=\; \Gamma(X,\ \mathscr{K}_X^*/\mathcal{O}_X^*).$$

This is an abelian group under the multiplication induced from $\mathscr{K}_X^*$. A Cartier divisor is *principal* if it lies in the image of the natural map

$$\Gamma(X, \mathscr{K}_X^*) \longrightarrow \Gamma(X, \mathscr{K}_X^*/\mathcal{O}_X^*) = \operatorname{CDiv} X.$$

Two Cartier divisors are *linearly equivalent* if they differ by a principal Cartier divisor.

Unwinding the definition of global sections of a quotient sheaf, a Cartier divisor can be represented by the following data: an open cover $\{U_i\}$ of X together with sections $f_i \in \Gamma(U_i, \mathscr{K}_X^*)$ such that on every overlap

$$\frac{f_i}{f_j} \;\in\; \Gamma(U_i \cap U_j,\ \mathcal{O}_X^*).$$

Two such collections $\{(U_i, f_i)\}$ and $\{(V_j, g_j)\}$ represent the same Cartier divisor if and only if, after passing to a common refinement, the ratios f_i/g_j are units wherever defined. The condition $f_i/f_j \in \mathcal{O}_X^*$ says that f_i and f_j agree up to an invertible regular function on the overlap — and since units contribute neither zeros nor poles, the "divisorial data" (zeros and poles of the f_i) glue to a well-defined global object.

A Cartier divisor is principal precisely when the cover can be taken to be $\{X\}$ itself: it is represented by a single global section $f \in \Gamma(X, \mathscr{K}_X^*)$.

5.4.18 Warning The group $\operatorname{CDiv} X = \Gamma(X, \mathscr{K}_X^*/\mathcal{O}_X^*)$ is *not* the same as the quotient $\Gamma(X, \mathscr{K}_X^*)/\Gamma(X, \mathcal{O}_X^*)$ in general. The former involves sections of the *quotient sheaf*, which may be represented by locally defined data that cannot be globalized. The discrepancy is measured by $H^1(X, \mathcal{O}_X^*)$, which we will identify with the Picard group in **??**.

In practice, the most frequently encountered Cartier divisors are *effective* — those with no poles:

Definition 5.4.19: Effective Cartier Divisor

A Cartier divisor $D \in \operatorname{CDiv} X$ is *effective* if it can be represented by data $\{(U_i, f_i)\}$ with each $f_i \in \Gamma(U_i, \mathcal{O}_X)$ (not merely in $\mathscr{K}_X^*$). Since $f_i \in \mathscr{K}_X^*$, each f_i is a non-zero-divisor in $\mathcal{O}_X(U_i)$. An effective Cartier divisor determines a closed subscheme $D \hookrightarrow X$ whose ideal sheaf $\mathcal{I}_D \subseteq \mathcal{O}_X$ is locally generated by f_i on each U_i. This is a closed subscheme that is *locally principal*: on each U_i, it is cut out by a single equation.

The effective Cartier divisors are precisely the closed subschemes whose ideal sheaf is an invertible $\mathcal{O}_X$-module (i.e., locally free of rank 1). This characterization — which requires no reference to $\mathscr{K}_X$ at all — is the starting point adopted by the Stacks project[5], and it has the advantage of working without any integrality or Noetherian hypotheses. We will return to this perspective in proposition 5.4.35.

When X satisfies the standing hypotheses of the previous subsection (Noetherian, integral, separated, regular in codimension 1), Cartier divisors can be compared directly with Weil divisors.

Proposition 5.4.20: Cartier Divisors to Weil Divisors

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1. There is a group homomorphism

$$\operatorname{CDiv} X \longrightarrow \operatorname{WDiv} X$$

that sends principal Cartier divisors to principal Weil divisors.

***Proof*:**
Since X is integral, $\mathscr{K}_X$ is the constant sheaf $K(X)$, so a Cartier divisor is represented by data $\{(U_i, f_i)\}$ with $f_i \in K(X)^*$ and $f_i/f_j \in \mathcal{O}_X^*(U_i \cap U_j)$. For each prime divisor Y of X, choose an index i with $\eta_Y \in U_i$ (such an i exists since the U_i cover X) and define the coefficient of $[Y]$ to be $\nu_Y(f_i)$. This is independent of the choice: if $\eta_Y \in U_j$ as well, then f_i/f_j is a unit near η_Y, so $\nu_Y(f_i/f_j) = 0$ and $\nu_Y(f_i) = \nu_Y(f_j)$.

The resulting sum $\sum_Y \nu_Y(f_i)[Y]$ is finite: since X is quasi-compact (it is Noetherian), we may take the cover $\{U_i\}$ to be finite, and on each U_i the rational function f_i has nonzero valuation at only finitely many prime divisors (by the same argument as lemma 5.4.5).

If $\{(U_i, f_i)\}$ and $\{(U_i, f_i')\}$ represent Cartier divisors D and D' (after passing to a common refinement), then $D + D'$ is represented by $\{(U_i, f_i f_i')\}$, and $\nu_Y(f_i f_i') = \nu_Y(f_i) + \nu_Y(f_i')$, so the map is a homomorphism. A principal Cartier divisor is represented by a single $f \in K(X)^*$, which maps to $\operatorname{div}(f) \in \operatorname{WDiv} X$.

The key question is: when is this map an isomorphism? It is surjective precisely when every Weil divisor is *locally principal* — that is, when each prime divisor is locally cut out by a single equation. The natural hypothesis guaranteeing this is that every local ring is a UFD (so every height-1 prime is principal).

[5] `https://stacks.math.columbia.edu`, Tag 01WR.

Proposition 5.4.21: Weil and Cartier Divisors Under Locally Factorial Hypotheses

Let X be an integral, separated, Noetherian scheme that is locally factorial (i.e., $\mathcal{O}_{X,x}$ is a UFD for all $x \in X$; cf. definition 2.4.29). Then the homomorphism of proposition 5.4.20 is an isomorphism

$$\operatorname{CDiv} X \xrightarrow{\sim} \operatorname{WDiv} X,$$

and principal Cartier divisors correspond to principal Weil divisors. In particular, there is an induced isomorphism of class groups.

***Proof*:**
Since every UFD is integrally closed, X is normal, and hence regular in codimension 1 (example 5.4.2). Because X is integral, $\mathscr{K}_X$ is the constant sheaf $K(X)$.

Injectivity. Suppose a Cartier divisor D, represented by $\{(U_i, f_i)\}$, maps to the zero Weil divisor. Then $\nu_Y(f_i) = 0$ for every prime divisor Y meeting U_i. For each affine open $V = \operatorname{Spec} R \subseteq U_i$, the element $f_i \in K(X)^*$ satisfies $\nu_{\mathfrak{p}}(f_i) = 0$ for every height-1 prime $\mathfrak{p}$ of R. Since R is a UFD (hence normal), lemma 5.4.8 gives $f_i \in R^*$, so f_i is a unit on V. As this holds for every affine open in every U_i, we have $f_i \in \mathcal{O}_X^*(U_i)$ for all i, meaning $D = 0$ in $\operatorname{CDiv} X$.

Surjectivity. Let $D = \sum n_j[Y_j]$ be a Weil divisor. For each point $x \in X$, the prime divisors through x correspond to height-1 primes of $\mathcal{O}_{X,x}$. Since $\mathcal{O}_{X,x}$ is a UFD, each such prime is principal, so $\operatorname{Cl}(\operatorname{Spec} \mathcal{O}_{X,x}) = 0$ by proposition 5.4.9. Hence the restriction of D to $\operatorname{Spec} \mathcal{O}_{X,x}$ is principal: there exists $f_x \in K(X)^*$ with

$$\nu_Y(f_x) = \begin{cases} n_j & \text{if } Y = Y_j \text{ passes through } x, \\ 0 & \text{for all other prime divisors through } x. \end{cases}$$

Now, D and $\operatorname{div}(f_x)$ can differ only at prime divisors not passing through x. Both D and $\operatorname{div}(f_x)$ involve finitely many prime divisors (the latter by lemma 5.4.5), and those not passing through x form a closed subset $Z_x \subseteq X$ with $x \notin Z_x$. Set $U_x = X \setminus Z_x$; then $D|_{U_x} = \operatorname{div}(f_x)|_{U_x}$.

It remains to verify that the collection $\{(U_x, f_x)\}_{x \in X}$ defines a Cartier divisor, i.e., that $f_x/f_{x'} \in \mathcal{O}_X^*(U_x \cap U_{x'})$ for all x, x'. On $U_x \cap U_{x'}$, both f_x and $f_{x'}$ represent D, so $\operatorname{div}(f_x/f_{x'}) = 0$ on $U_x \cap U_{x'}$. For any affine open $V = \operatorname{Spec} R \subseteq U_x \cap U_{x'}$, the ring R is a normal domain (each stalk is a UFD, hence integrally closed). By lemma 5.4.8, $f_x/f_{x'} \in \bigcap_{\operatorname{ht}(\mathfrak{p})=1} R_{\mathfrak{p}} = R$, and the same argument applied to $f_{x'}/f_x$ gives the inverse. Hence $f_x/f_{x'} \in R^* = \mathcal{O}_X^*(V)$, and since this holds on every affine open in $U_x \cap U_{x'}$, we conclude $f_x/f_{x'} \in \mathcal{O}_X^*(U_x \cap U_{x'})$.

These two constructions are mutually inverse (one verifies directly that composing in either order yields the identity), and both send principal divisors to principal divisors.

The locally factorial hypothesis is satisfied by all regular schemes (since regular local rings are UFDs by the Auslander–Buchsbaum theorem; cf. proposition 2.7.18) and by all smooth varieties over a field. In particular, the Weil and Cartier theories agree on any nonsingular variety — the setting where divisors are most commonly used. On a normal but non-factorial scheme (such as the quadric cone of example 5.4.13), the map $\operatorname{CDiv} X \to \operatorname{WDiv} X$ is a proper inclusion: the ruling $[Y]$ is a Weil divisor that is *not* locally principal at the vertex, hence has no Cartier counterpart.

When X is not locally factorial, the map $\operatorname{CDiv} X \to \operatorname{WDiv} X$ identifies Cartier divisors with a distinguished subgroup of Weil divisors:

Definition 5.4.22: Locally Principal Weil Divisor

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1. A Weil divisor $D \in \operatorname{WDiv} X$ is *locally principal* if for every $x \in X$, there exists an open neighbourhood U of x and $f \in K(X)^*$ such that $D|_U = \operatorname{div}(f)|_U$.

By construction, the image of $\operatorname{CDiv} X \to \operatorname{WDiv} X$ consists precisely of the locally principal Weil divisors. We therefore have a chain of inclusions

$$\operatorname{PDiv}(X) \subseteq \{\text{locally principal Weil divisors}\} \subseteq \operatorname{WDiv} X,$$

and the locally factorial hypothesis of proposition 5.4.21 collapses the second inclusion to an equality.

Definition 5.4.23: Cartier Divisor Class Group

Let X be a scheme. The *Cartier divisor class group* of X is

$$\operatorname{CaCl} X := \frac{\operatorname{CDiv} X}{\{\text{principal Cartier divisors}\}}.$$

When X is integral, separated, Noetherian, and locally factorial, proposition 5.4.21 gives an isomorphism $\operatorname{CaCl} X \cong \operatorname{Cl} X$.

Example 5.4.24: Cartier Divisors

1. *Projective space.* On $\mathbb{P}^n_k$, every local ring is a localization of a polynomial ring, hence a UFD. By proposition 5.4.21, every Weil divisor is Cartier, and $\operatorname{CaCl} \mathbb{P}^n_k \cong \operatorname{Cl} \mathbb{P}^n_k \cong \mathbb{Z}$. A hyperplane $H = V_+(x_0)$ is represented by the Cartier data $\{(U_i, x_0/x_i)\}_{i=0}^n$: on the standard affine chart $U_i = D_+(x_i) \cong \mathbb{A}^n$, the hyperplane is cut out by the single function $x_0/x_i \in \mathcal{O}_X(U_i)$.

2. *Smooth curves.* Every nonsingular curve is locally factorial (one-dimensional regular local rings are DVRs, hence UFDs). Thus Weil and Cartier divisors coincide, and a divisor is simply a formal sum of closed points — the classical notion of a divisor on a curve.

3. *Normal, non-factorial schemes.* On the quadric cone $X = \operatorname{Spec} k[x, y, z]/(xy - z^2)$ (example 5.4.13), the ruling $Y = V(y, z)$ is a Weil divisor that is *not* locally principal at the vertex (x, y, z): the prime ideal (y, z) is not principal in the local ring at that point. However, $2[Y] = \operatorname{div}(y)$ *is* Cartier. In this case, $\operatorname{CDiv} X \subsetneq \operatorname{WDiv} X$, and $\operatorname{CaCl} X = 0 \neq \mathbb{Z}/2\mathbb{Z} = \operatorname{Cl} X$.[a]

4. *Non-reduced schemes.* Let $X = \operatorname{Spec} k[x]/(x^2)$, the "double point." Since x is a zero-divisor, the element $x \in \mathcal{O}_X(X)$ does not define an effective Cartier divisor. The total quotient ring is $k[x]/(x^2)$ itself (no non-zero-divisors to invert beyond units), so $\mathscr{K}_X = \mathcal{O}_X$ and $\operatorname{CDiv} X = 0$. This illustrates the limitations of the Cartier formalism on highly non-reduced schemes.

[a] The Cartier class group is trivial because the only Cartier divisors on X are principal: any locally principal divisor is an even multiple of $[Y]$, hence already principal.

Example 5.4.25: Weil $\neq$ Cartier on Singular Curves

Example. Consider the nodal cubic $X = V(y^2 - x^3 - x^2) \subseteq \mathbb{P}^2_k$ (with char$k \neq 2$), which has a node at the origin $P = (0,0)$. The local ring $\mathcal{O}_{X,P} \cong k[x,y]_{(x,y)}/(y^2 - x^2 - x^3)$ is *not* a UFD: in the completion $\widehat{\mathcal{O}}_{X,P} \cong k[\![u,v]\!]/(uv)$, the element $y^2 - x^2 = x^3$ factors into two branches that cannot be separated in $\mathcal{O}_{X,P}$ itself[a]. The Weil divisor $1 \cdot P$ (a single closed point, considered as a codimension-1 subvariety) is *not* Cartier at P: its ideal sheaf $\mathcal{I}_P$ is not locally principal near P, because the maximal ideal $\mathfrak{m}_P \subseteq \mathcal{O}_{X,P}$ is not principal (it requires two generators). Hence $\mathcal{I}_P$ is a coherent ideal sheaf, but *not* an invertible sheaf.

More concretely, on the normalization $\nu\colon \widetilde{X} \cong \mathbb{P}^1 \to X$, the preimage $\nu^{-1}(P)$ consists of two points $\{P_1, P_2\}$ (the two branches), and the Weil divisor $P_1 + P_2$ on $\mathbb{P}^1$ is Cartier—but it is the pullback of P only as a *cycle*, not as a Cartier divisor on X.

The upshot:

[a] See example 3.1.30 for the formal branch structure of a node.

5.4.3 The Picard Group

The discussion of divisors on projective schemes has made it clear that there is a strong connection to the twisting sheaves $\mathcal{O}(d)$. We now make this connection precise: Cartier divisors and line bundles are two sides of the same coin, and the passage between them is the central bridge linking the geometry of divisors to the algebra of sheaves.

The reader familiar with algebraic number theory will recognize a close parallel. In a Dedekind domain R, fractional ideals are finitely generated R-submodules of Frac(R), and they form a group under multiplication. The ideal class group $\mathrm{Cl}(R)$ — fractional ideals modulo principal ones — measures the failure of unique factorization (cf. example 5.4.10). The Picard group is precisely the scheme-theoretic generalization of this construction: invertible sheaves ("line bundles") play the role of fractional ideals, the tensor product replaces ideal multiplication, and the Picard group classifies them up to isomorphism.

Proposition 5.4.26: Invertible Sheaves Form a Group

Let X be a ringed space and let $\mathcal{L}, \mathcal{L}'$ be invertible sheaves on X. Then:

1. The tensor product $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}'$ is invertible.
2. The *dual* $\mathcal{L}^\vee := \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$ is invertible, and $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^\vee \cong \mathcal{O}_X$.
3. The tensor product is associative and commutative (up to natural isomorphism) with unit $\mathcal{O}_X$.

Hence the set of isomorphism classes of invertible sheaves on X forms an abelian group under $\otimes$.

The tensor product of line bundles operates stalkwise: $(\mathcal{L} \otimes \mathcal{L}')_x \cong \mathcal{L}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{L}'_x$. Since each stalk is a free module of rank 1, this tensor product "multiplies the values" rather than increasing the rank — analogous to multiplying scalars rather than wedging differential forms. This is why the tensor product of two line bundles remains a line bundle.

***Proof*:**
For (1), choose trivialisations $\mathcal{L}|_U \cong \mathcal{O}_U$ and $\mathcal{L}'|_V \cong \mathcal{O}_V$. On $U \cap V$, $(\mathcal{L} \otimes \mathcal{L}')|_{U\cap V} \cong \mathcal{O}_{U\cap V} \otimes \mathcal{O}_{U\cap V} \cong \mathcal{O}_{U\cap V}$, so $\mathcal{L} \otimes \mathcal{L}'$ is locally free of rank 1.

For (2), on a trivialising open where $\mathcal{L}|_U \cong \mathcal{O}_U$, the dual restricts as $\mathcal{L}^\vee|_U \cong \mathcal{H}om_{\mathcal{O}_U}(\mathcal{O}_U, \mathcal{O}_U) \cong \mathcal{O}_U$, so $\mathcal{L}^\vee$ is invertible. The evaluation map

$$\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^\vee \longrightarrow \mathcal{O}_X, \qquad s \otimes \varphi \longmapsto \varphi(s)$$

is an isomorphism on each trivialising open (where it is the multiplication $R \otimes_R R \to R$), hence a global isomorphism.

Part (3) follows from the corresponding properties of the tensor product of modules.

Definition 5.4.27: Picard Group

The *Picard group* of a ringed space X is the abelian group

$$\operatorname{Pic} X \ := \ \frac{\{\text{invertible sheaves on } X\}}{\text{isomorphism}}$$

with group operation $[\mathcal{L}] \cdot [\mathcal{L}'] = [\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}']$, identity element $[\mathcal{O}_X]$, and inverse $[\mathcal{L}]^{-1} = [\mathcal{L}^\vee]$.

We shall see in theorem 5.4.32 that for integral schemes, $\operatorname{Pic} X \cong H^1(X, \mathcal{O}_X^*)$, giving a cohomological interpretation.

Proposition 5.4.28: Functoriality of the Picard Group

Let $f\colon X \to Y$ be a morphism of ringed spaces. The pullback of $\mathcal{O}_Y$-modules induces a well-defined group homomorphism

$$f^*\colon \operatorname{Pic} Y \longrightarrow \operatorname{Pic} X, \qquad [\mathcal{L}] \longmapsto [f^*\mathcal{L}].$$

Consequently, Pic defines a contravariant functor from the category of ringed spaces to the category of abelian groups.

***Proof*:**
Recall that the pullback of an $\mathcal{O}_Y$-module $\mathcal{L}$ is given by $f^*\mathcal{L} = f^{-1}\mathcal{L} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$. We must first verify that $f^*\mathcal{L}$ is an invertible sheaf on X whenever $\mathcal{L}$ is an invertible sheaf on Y.

Choose an open cover $\{V_i\}$ of Y that trivialises $\mathcal{L}$, meaning $\mathcal{L}|_{V_i} \cong \mathcal{O}_{V_i}$. For each i, let $U_i = f^{-1}(V_i)$, which forms an open cover of X. Because the pullback operation commutes with restriction to open subsets, we have

$$(f^*\mathcal{L})|_{U_i} \ \cong \ (f|_{U_i})^*(\mathcal{L}|_{V_i}) \ \cong \ (f|_{U_i})^*\mathcal{O}_{V_i} \ \cong \ \mathcal{O}_{U_i}.$$

Thus, $f^*\mathcal{L}$ is locally free of rank 1, making it an invertible sheaf on X. This ensures the assignment $[\mathcal{L}] \mapsto [f^*\mathcal{L}]$ is well-defined.

To see that f^* is a group homomorphism, we rely on the fact that the pullback functor for modules commutes with the tensor product. For any two invertible sheaves $\mathcal{L}, \mathcal{L}'$ on Y, there is a canonical isomorphism

$$f^*(\mathcal{L} \otimes_{\mathcal{O}_Y} \mathcal{L}') \;\cong\; f^*\mathcal{L} \otimes_{\mathcal{O}_X} f^*\mathcal{L}'.$$

Therefore, $f^*([\mathcal{L}] \cdot [\mathcal{L}']) = f^*([\mathcal{L}]) \cdot f^*([\mathcal{L}'])$.

Finally, the canonical isomorphisms $(g \circ f)^* \cong f^* \circ g^*$ and $(\mathrm{id}_Y)^* \cong \mathrm{id}_{\operatorname{Pic} Y}$ follow directly from the general categorical properties of module pullbacks, confirming that Pic is a contravariant functor.

Definition 5.4.29: Line Bundle Associated to a Cartier Divisor

Let X be a scheme and $D \in \operatorname{CDiv} X$ a Cartier divisor represented by data $\{(U_i, f_i)\}$ with $f_i \in \Gamma(U_i, \mathscr{K}_X^*)$ and $f_i/f_j \in \mathcal{O}_X^*(U_i \cap U_j)$. The *invertible sheaf associated to* D is the sub-$\mathcal{O}_X$-module $\mathcal{O}_X(D) \subseteq \mathscr{K}_X$ defined by

$$\mathcal{O}_X(D)|_{U_i} \;:=\; f_i^{-1} \cdot \mathcal{O}_X|_{U_i} \;\subseteq\; \mathscr{K}_X|_{U_i}.$$

On the overlap $U_i \cap U_j$, the generators satisfy $f_i^{-1} = (f_j/f_i) \cdot f_j^{-1}$ with $f_j/f_i \in \mathcal{O}_X^*$, so the local descriptions glue. The trivialisation $\mathcal{O}_X(D)|_{U_i} \xrightarrow{\sim} \mathcal{O}_X|_{U_i}$ is given by $g \mapsto g f_i$.[a]

[a]Note that X need not be integral for this construction; the sheaf $\mathscr{K}_X$ of total quotient rings replaces the function field.

When X is integral, one can describe the sections of $\mathcal{O}_X(D)$ more concretely. Since $\mathscr{K}_X$ is the constant sheaf $K(X)$, the sections over an open U are

$$\mathcal{O}_X(D)(U) \;=\; \{g \in K(X)^* \mid \operatorname{div}(g) + D \geq 0 \text{ on } U\} \;\cup\; \{0\},$$

where $\operatorname{div}(g) + D \geq 0$ means the Weil divisor is effective on U. Intuitively, $\mathcal{O}_X(D)(U)$ consists of all rational functions whose poles are "bounded by D": the divisor D prescribes how many poles are allowed along each prime divisor, and g must compensate. Historically, this is precisely how divisors first arose on Riemann surfaces — as a bookkeeping device for the poles and zeros of meromorphic functions — and the space $\mathcal{O}_X(D)(U)$ is the classical *Riemann–Roch space* $L(D)$.

Example 5.4.30: Number-Theoretic Line Bundle

Let $R = \mathbb{Z}[\sqrt{-5}]$ and $X = \operatorname{Spec} R$. Since R is a Dedekind domain, X is a normal Noetherian integral scheme of dimension 1, and every fractional ideal of R defines an invertible sheaf via the tilde construction (definition 5.1.6).

Consider the prime ideal $\mathfrak{p} = (2, 1 + \sqrt{-5})$, which lies over 2 and satisfies $\mathfrak{p}^2 = (2)$ (cf. example 5.4.10). As a Weil divisor, $[V(\mathfrak{p})]$ corresponds to the single closed point $\mathfrak{p} \in X$. This Weil divisor is Cartier (since every local ring of a Dedekind domain is a DVR, hence a UFD): at the point $\mathfrak{p}$ itself, the uniformizer $1 + \sqrt{-5}$ generates the ideal locally, while at every other closed point $\mathfrak{q} \neq \mathfrak{p}$, the ideal $\mathfrak{p}$ is the full local ring and is generated by 1.

The invertible sheaf $\widetilde{\mathfrak{p}}$ associated to this fractional ideal is *not* isomorphic to $\mathcal{O}_X$, since $\mathfrak{p}$ is not principal. Its class $[\widetilde{\mathfrak{p}}] \in \operatorname{Pic} X \cong \operatorname{Cl}(R) \cong \mathbb{Z}/2\mathbb{Z}$ is the nontrivial element, and $\widetilde{\mathfrak{p}}^{\otimes 2} \cong \widetilde{\mathfrak{p}^2} = \widetilde{(2)} \cong \mathcal{O}_X$ (the last isomorphism since (2) is principal as a fractional ideal). This recovers the ideal class group of $\mathbb{Z}[\sqrt{-5}]$ as the Picard group.

Proposition 5.4.31: Correspondence Between Line Bundles and Cartier Divisors

Let X be a scheme. The assignment $D \mapsto \mathcal{O}_X(D)$ defines a group homomorphism $\operatorname{CDiv} X \to \operatorname{Pic} X$ satisfying:

1. $\mathcal{O}_X(D + D') \cong \mathcal{O}_X(D) \otimes_{\mathcal{O}_X} \mathcal{O}_X(D')$ for all $D, D' \in \operatorname{CDiv} X$.
2. $\mathcal{O}_X(-D) \cong \mathcal{O}_X(D)^\vee$ for all $D \in \operatorname{CDiv} X$.
3. $\mathcal{O}_X(D) \cong \mathcal{O}_X$ if and only if D is a principal Cartier divisor.

In particular, the homomorphism descends to an injection $\operatorname{CaCl} X \hookrightarrow \operatorname{Pic} X$.

Part (1) pairs well with the stalkwise intuition for the tensor product: if D prescribes poles and zeros along certain prime divisors and D' prescribes its own, then $\mathcal{O}_X(D) \otimes \mathcal{O}_X(D')$ "combines" the two prescriptions, gaining the zeros and poles of both.

***Proof*:**
Part (1). If $D = \{(U_i, f_i)\}$ and $D' = \{(U_i, g_i)\}$ (after passing to a common refinement), then $D + D' = \{(U_i, f_i g_i)\}$. On U_i, the multiplication map in $\mathscr{K}_X$ gives

$$\mathcal{O}_X(D)|_{U_i} \otimes \mathcal{O}_X(D')|_{U_i} \;=\; f_i^{-1}\mathcal{O}_{U_i} \otimes g_i^{-1}\mathcal{O}_{U_i} \;\xrightarrow{\sim}\; (f_i g_i)^{-1}\mathcal{O}_{U_i} \;=\; \mathcal{O}_X(D + D')|_{U_i}.$$

These local isomorphisms are compatible on overlaps (the transition functions multiply), yielding a global isomorphism.

Part (2). The dual $\mathcal{O}_X(D)^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_X(D), \mathcal{O}_X)$ is locally generated by f_i (the functional sending $f_i^{-1} \mapsto 1$), which is $\mathcal{O}_X(-D)|_{U_i}$.

Part (3). If $D = \operatorname{div}(f)$ is principal (represented by a single global $f \in \Gamma(X, \mathscr{K}_X^*)$), then $\mathcal{O}_X(D) = f^{-1}\mathcal{O}_X \cong \mathcal{O}_X$. Conversely, if $\varphi\colon \mathcal{O}_X(D) \xrightarrow{\sim} \mathcal{O}_X$, then φ sends the local generator f_i^{-1} to some $h_i \in \mathcal{O}_X^*(U_i)$. The element $f = f_i h_i^{-1} \in \mathscr{K}_X^*$ is independent of i (on $U_i \cap U_j$, the transition functions force $f_i h_i^{-1} = f_j h_j^{-1}$), and $D = \operatorname{div}(f)$.

For a general scheme X, the injection $\operatorname{CaCl} X \hookrightarrow \operatorname{Pic} X$ may be strict: there can exist invertible sheaves that do not embed as sub-$\mathcal{O}_X$-modules of $\mathscr{K}_X$. However, the map is surjective — and hence an isomorphism — whenever X is integral:

Theorem 5.4.32: Isomorphism of Cartier Class Group and Picard Group

Let X be an integral scheme. The map $D \mapsto [\mathcal{O}_X(D)]$ induces an isomorphism of abelian groups

$$\operatorname{CaCl} X \;\xrightarrow{\sim}\; \operatorname{Pic} X.$$

***Proof*:**
Injectivity follows from proposition 5.4.31(3). For surjectivity, let $\mathcal{L}$ be an invertible sheaf. Since X is integral, the constant sheaf $\mathscr{K}_X = K(X)$ satisfies $\mathcal{L} \otimes_{\mathcal{O}_X} \mathscr{K}_X \cong \mathscr{K}_X$ (both sides are locally free $\mathscr{K}_X$-modules of rank 1, and $\mathscr{K}_X$ has a unique such module up to isomorphism). The natural map $\mathcal{L} \hookrightarrow \mathcal{L} \otimes \mathscr{K}_X$ (sending $s \mapsto s \otimes 1$) is injective since X is integral. Composing with any isomorphism $\mathcal{L} \otimes \mathscr{K}_X \xrightarrow{\sim} \mathscr{K}_X$ embeds $\mathcal{L}$ as a sub-$\mathcal{O}_X$-module of $\mathscr{K}_X$. On a trivialising open U_i,

this embedding sends the local generator of $\mathcal{L}|_{U_i}$ to some $f_i^{-1} \in K(X)^*$, and the collection $\{(U_i, f_i)\}$ defines a Cartier divisor D with $\mathcal{L} \cong \mathcal{O}_X(D)$. A different choice of isomorphism $\mathcal{L} \otimes \mathscr{K}_X \cong \mathscr{K}_X$ changes each f_i by a common factor $h \in K(X)^*$, replacing D by $D + \mathrm{div}(h)$.

Cohomological perspective. Consider the short exact sequence of abelian sheaves

$$1 \longrightarrow \mathcal{O}_X^* \longrightarrow \mathscr{K}_X^* \longrightarrow \mathscr{K}_X^*/\mathcal{O}_X^* \longrightarrow 1.$$

Since X is integral, $\mathscr{K}_X^* = K(X)^*$ is a constant sheaf on the irreducible space X, hence flasque. The long exact sequence in cohomology gives

$$\Gamma(X, \mathscr{K}_X^*) \longrightarrow \underbrace{\Gamma(X, \mathscr{K}_X^*/\mathcal{O}_X^*)}_{\mathrm{CDiv}\,X} \xrightarrow{\delta} \underbrace{H^1(X, \mathcal{O}_X^*)}_{\mathrm{Pic}\,X} \longrightarrow \underbrace{H^1(X, \mathscr{K}_X^*)}_{=0}$$

where the vanishing follows from flasqueness (corollary 6.1.7). The connecting homomorphism δ sends a Cartier divisor $\{(U_i, f_i)\}$ to the Čech 1-cocycle $\{f_i/f_j\} \in \check{H}^1(\{U_i\}, \mathcal{O}_X^*)$, which classifies the line bundle $\mathcal{O}_X(D)$. Its cokernel is zero and its kernel is exactly the image of $\Gamma(X, \mathscr{K}_X^*)$ — the principal divisors. Hence $\mathrm{CaCl}\,X \cong \mathrm{Pic}\,X$.

The cohomological proof explains the warning following definition 5.4.17: the discrepancy between global sections of the quotient sheaf $\mathscr{K}_X^*/\mathcal{O}_X^*$ and the literal quotient of global sections is precisely the connecting homomorphism δ, and the group $H^1(X, \mathcal{O}_X^*)$ classifies line bundles.

Corollary 5.4.33: Divisor Class Group and Picard Group

Let X be an integral, separated, Noetherian, locally factorial scheme. Then there is a chain of isomorphisms

$$\mathrm{Cl}\,X \xrightarrow[\textit{proposition } 5.4.21]{\sim} \mathrm{CaCl}\,X \xrightarrow[\textit{theorem } 5.4.32]{\sim} \mathrm{Pic}\,X.$$

Corollary 5.4.34: Classifying Line Bundles on Projective Space

Let k be a field. Every invertible sheaf on $\mathbb{P}_k^n$ is isomorphic to $\mathcal{O}_{\mathbb{P}_k^n}(d)$ for a unique $d \in \mathbb{Z}$. Hence

$$\mathrm{Pic}\,\mathbb{P}_k^n \;\cong\; \mathbb{Z},$$

with generator $[\mathcal{O}_{\mathbb{P}_k^n}(1)]$ corresponding to the hyperplane class.

***Proof*:**
The scheme $\mathbb{P}_k^n$ is integral, separated, Noetherian, and regular (hence locally factorial by proposition 2.7.18). By corollary 5.4.33 and proposition 5.4.11, $\mathrm{Pic}\,\mathbb{P}_k^n \cong \mathrm{Cl}\,\mathbb{P}_k^n \cong \mathbb{Z}$, generated by the hyperplane class. Since the hyperplane divisor $[H]$ corresponds to $\mathcal{O}_{\mathbb{P}_k^n}(1)$ under the construction of definition 5.4.29, the class $d[H]$ corresponds to $\mathcal{O}_{\mathbb{P}_k^n}(d)$, confirming the earlier computation of proposition 5.3.5.

We close this subsection by recording the connection between effective Cartier divisors and line bundles equipped with sections — the algebro-geometric counterpart of the classical fact that the zero locus of a holomorphic section of a holomorphic line bundle is a divisor (proposition 0.6.34).

Proposition 5.4.35: Effective Cartier Divisors and Line Bundles

Let X be an integral scheme and D an effective Cartier divisor with associated closed subscheme $Y \hookrightarrow X$ (definition 5.4.19).

1. The ideal sheaf of Y satisfies $\mathcal{I}_Y \cong \mathcal{O}_X(-D)$.
2. There is a natural bijection

$$\{\text{ effective Cartier divisors on } X\ \} \longleftrightarrow \left\{ \begin{array}{c} \text{pairs } (\mathcal{L}, s) \text{ with } \mathcal{L} \text{ invertible and} \\ 0 \neq s \in \Gamma(X, \mathcal{L}), \\ \text{up to isomorphism} \end{array} \right\}$$

Under this bijection, D corresponds to $(\mathcal{O}_X(D), s_D)$, where s_D is the "canonical section" — the element $1 \in K(X)$ viewed as a global section of $\mathcal{O}_X(D)$.

***Proof*:**
Part (1). Let D be represented by $\{(U_i, f_i)\}$ with $f_i \in \mathcal{O}_X(U_i)$ non-zero-divisors. Then $\mathcal{O}_X(-D)|_{U_i} = f_i \cdot \mathcal{O}_{U_i}$, which is precisely the ideal sheaf of Y on U_i (by definition 5.4.19, $\mathcal{I}_Y$ is locally generated by f_i). Since both sides glue by the same transition functions, we get $\mathcal{I}_Y \cong \mathcal{O}_X(-D)$ as subsheaves of $\mathcal{O}_X$.

Part (2). Given an effective $D = \{(U_i, f_i)\}$, the element $1 \in K(X)$ lies in $\mathcal{O}_X(D)(U_i) = f_i^{-1}\mathcal{O}_{U_i}$ (since $1 \cdot f_i = f_i \in \mathcal{O}_X(U_i)$), so $s_D := 1$ is a well-defined nonzero global section. Under the trivialisation $\mathcal{O}_X(D)|_{U_i} \cong \mathcal{O}_{U_i}$, the section s_D corresponds to f_i, so s_D vanishes precisely along Y.

Conversely, given $(\mathcal{L}, s)$ with $s \neq 0$, choose trivialisations $\psi_i \colon \mathcal{L}|_{U_i} \xrightarrow{\sim} \mathcal{O}_{U_i}$ and set $f_i = \psi_i(s|_{U_i})$. Since $s \neq 0$ and X is integral, each f_i is a non-zero-divisor. On overlaps, $f_i/f_j = \psi_i \circ \psi_j^{-1}(1) \in \mathcal{O}_X^*(U_i \cap U_j)$, so $\{(U_i, f_i)\}$ is an effective Cartier divisor. The two constructions are mutually inverse.

Part (1) provides a useful dictionary: the ideal sheaf of an effective Cartier divisor is the "negative" line bundle $\mathcal{O}_X(-D)$, and the short exact sequence of $\mathcal{O}_X$-modules

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_Y \longrightarrow 0$$

will be a fundamental tool in cohomological computations (section 6.3). Part (2) establishes that studying line bundles with sections is *equivalent* to studying effective divisors — a perspective that will be central to the theory of linear systems (remark 5.3.32) and the Riemann–Roch theorem (theorem 7.1.18).

5.5 Differentials

The reader arrives at this section already fluent in its geometry. On a smooth manifold M, the cotangent space T_p^*M (definition 0.5.11) and the differential forms assembled from it (definition 0.5.14) carry the infinitesimal geometry of M, and the differential of a function $df_p \in T_p^*M$ (definition 0.5.12) records its first-order behavior at p. The point of contact with algebra is a description of these objects that uses no analysis at all. Writing $\mathfrak{m}_p$ for the ideal of functions vanishing at p,

$$T_pM \;\cong\; (\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee, \qquad T_p^*M \;\cong\; \mathfrak{m}_p/\mathfrak{m}_p^2$$

([7]), and under the second identification the covector df_p is the class of $f - f(p)$ modulo $\mathfrak{m}_p^2$. No limit and no smooth structure enter the right-hand sides: only an ideal, its square, and a quotient. This is verbatim the Zariski cotangent space introduced earlier for varieties (definition 0.1.25), and that coincidence is why differentials belong to algebraic geometry at all.

The same description settles which of the two objects is primitive. The assignment $f \mapsto df = [\,f - f(p)\,]$ produces a covector from a function by a single quotient; the tangent vector, an element of the dual, is what one obtains *afterward*, by dualizing. In the smooth category the distinction is immaterial, since both spaces are finite-dimensional and either determines the other. But the construction that survives with no appeal to limits, and so transports to an arbitrary ring, is the cotangent one. Functoriality points the same way: functions pull back along a morphism $f\colon X \to Y$, so the cotangent construction, covariant in the ring of functions, transforms along with them, while tangent vectors push forward, in the opposite direction. Algebraic geometry therefore takes the cotangent side as basic. The *sheaf of differentials* $\Omega_{X/Y}$ (definition 5.5.14) is the object it builds first; the universal derivation $d\colon b \mapsto db$ is the Leibniz and chain rules promoted to a definition; and the tangent sheaf (definition 5.5.25) appears only afterward, as its dual.

The payoff is an exact algebraic stand-in for the cotangent bundle. Once $\Omega_{X/Y}$ is in hand, its fiber at a point is canonically $\mathfrak{m}_p/\mathfrak{m}_p^2 = T_p^*$ (proposition 5.5.10): the sheaf of differentials is the cotangent bundle glued together globally, with no coordinate charts and no partitions of unity. Being the cotangent object and not its dual is what lets it see singularities. The dimension of $\mathfrak{m}_p/\mathfrak{m}_p^2$ jumps above the dimension of X exactly at a singular point (definition 0.1.27), the sheaf $\Omega_{X/Y}$ fails to be locally free there (theorem 5.5.12), and this failure is the algebraic test for smoothness (proposition 5.5.22); the tangent sheaf, a dual that is well behaved only in the smooth case, is the wrong place to look for it. Further downstream, the top exterior power of $\Omega_{X/Y}$ is the canonical sheaf standing behind Serre duality (theorem 6.5.9) and Riemann–Roch (theorem 7.1.18), and $\Omega_{X/Y}$ itself governs deformation theory. These are the section's eventual destinations; the road to them begins in commutative algebra.

We begin with that algebra: the module of Kähler differentials $\Omega_{B/A}$ attached to a ring map $A \to B$, before globalizing to schemes. The reader who knows forms on manifolds will recognize the parallel at once, since the Leibniz rule and the chain rule are built directly into the definition.

5.5.1 Kähler Differentials

Definition 5.5.1: A-Derivation

Let A be a commutative ring, B an A-algebra, and M a B-module. An A-*derivation* of B into M is a map $d\colon B \to M$ satisfying:

1. *(Additivity)* $d(b+b') = d(b) + d(b')$,
2. *(Leibniz rule)* $d(bb') = b\,d(b') + b'\,d(b)$,
3. *(Vanishing on A)* $d(a) = 0$ for all $a \in A$ (treating A as mapped into B)

The set of all A-derivations $B \to M$ is denoted $\mathrm{Der}_A(B, M)$ and carries a natural B-module structure via $(b \cdot d)(b') = b\,d(b')$.

Condition (3) is the algebraic analogue of "the derivative of a constant is zero": the elements of A, viewed as "constants" in the A-algebra B, have trivial differential. Note that (3) is automatic when A is a prime ring or prime field (generated by a single element) since then $d(1) = d(1 \cdot 1) = 2\,d(1)$ forces $d(1) = 0$. Condition (3) forces the derivation to be A-linear since for $a \in A$:

$$d(ab) = d(a)b + ad(b) = 0 + ad(b) = ad(b)$$

Definition 5.5.2: Module of Relative Differential Forms

Let A be a commutative ring and B an A-algebra. A *module of (relative) Kähler differentials* for B over A is a B-module $\Omega_{B/A}$ equipped with an A-derivation $d\colon B \to \Omega_{B/A}$ satisfying the following universal property: for every B-module M and every A-derivation $d'\colon B \to M$, there exists a unique B-module homomorphism $\varphi\colon \Omega_{B/A} \to M$ such that $d' = \varphi \circ d$:

$$\begin{tikzcd} B \arrow[r, "d"] \arrow[dr, "d'"'] & \Omega_{B/A} \arrow[d, dashed, "\exists!\,\varphi"] \\ & M \end{tikzcd}$$

Equivalently, d induces a B-module isomorphism

$$\mathrm{Hom}_B(\Omega_{B/A}, M) \xrightarrow{\sim} \mathrm{Der}_A(B, M), \qquad \varphi \longmapsto \varphi \circ d.$$

The universal property determines $(\Omega_{B/A}, d)$ up to unique isomorphism, as with any object defined by a universal property. There are two general constructions to $\Omega_{B/A}$: an explicit generators-and-relations presentation, and a more geometric one via the diagonal ideal.

Construction 1: generators and relations. Let F be the free B-module on symbols $\{db : b \in B\}$, and let R be the submodule generated by the relations

$$d(b+b') - db - db', \qquad d(bb') - b\,db' - b'\,db, \qquad da \quad (a \in A).$$

Set $\Omega_{B/A} = F/R$ and define $d\colon B \to \Omega_{B/A}$ by $b \mapsto db$. By construction, d is an A-derivation, and the universal property is immediate: any A-derivation $d'\colon B \to M$ extends uniquely to a B-module map $F \to M$ (sending $db \mapsto d'(b)$), which kills R and hence factors through $\Omega_{B/A}$.

Construction 2: the diagonal ideal. The following construction has a geometric interpretation that globalizes directly to schemes.

Proposition 5.5.3: Differentials via the Diagonal

Let B be an A-algebra. Let $\mu\colon B \otimes_A B \to B$ be the multiplication map $b \otimes b' \mapsto bb'$, and let $I = \ker\mu$. View $B \otimes_A B$ as a B-module via left multiplication ($b \cdot (b_1 \otimes b_2) = bb_1 \otimes b_2$). Then:

1. I/I^2 inherits the structure of a B-module.
2. The map $d\colon B \to I/I^2$ defined by $d(b) = 1 \otimes b - b \otimes 1 \pmod{I^2}$ is an A-derivation.
3. The pair $(I/I^2,\, d)$ satisfies the universal property of $\Omega_{B/A}$.

***Proof*:**

1. A priori, I is a $B \otimes_A B$-module, hence a B-module via either the left factor ($b \mapsto b \otimes 1$) or the right factor ($b \mapsto 1 \otimes b$). These two B-module structures agree on I/I^2: for $b \in B$ and $x \in I$, $(b \otimes 1 - 1 \otimes b) \cdot x \in I \cdot I = I^2$ (since $b \otimes 1 - 1 \otimes b \in I$), so $(b \otimes 1) \cdot x \equiv (1 \otimes b) \cdot x \pmod{I^2}$.

2. The element $1 \otimes b - b \otimes 1$ lies in I (since $\mu(1 \otimes b - b \otimes 1) = b - b = 0$), so d is well-defined. Additivity is immediate. For the Leibniz rule, compute in $B \otimes_A B$:

$$\begin{aligned} 1 \otimes bb' - bb' \otimes 1 &= (1 \otimes b - b \otimes 1)(1 \otimes b') + b(1 \otimes b' - b' \otimes 1) \\ &\equiv b'\,(1 \otimes b - b \otimes 1) + b\,(1 \otimes b' - b' \otimes 1) \pmod{I^2}, \end{aligned}$$

   where the second step uses the equality of left and right B-actions modulo I^2. Hence $d(bb') = b'\,d(b) + b\,d(b')$. For $a \in A$, $d(a) = 1 \otimes a - a \otimes 1 = 0$ since the A-algebra structure map identifies $a \otimes 1 = 1 \otimes a$ in $B \otimes_A B$.

3. Let $d'\colon B \to M$ be any A-derivation. Define a ring homomorphism $\psi\colon B \otimes_A B \to B \oplus M$ by $\psi(b_1 \otimes b_2) = (b_1b_2,\; b_1\,d'(b_2))$, where $B \oplus M$ has the square-zero extension ring structure: $(b, m)(b', m') = (bb',\; bm' + b'm)$. One verifies that ψ is a well-defined ring homomorphism (the key being that d' is an A-derivation).

   Since ψ restricts to μ on the first component, $\psi(I) \subseteq 0 \oplus M = M$. Moreover, $\psi(I^2) \subseteq M^2 = 0$ (products in M vanish in the square-zero extension). Hence ψ descends to a B-module map $\varphi\colon I/I^2 \to M$ with $\varphi(d(b)) = \varphi(1 \otimes b - b \otimes 1) = (0,\, d'(b))$, i.e., $d' = \varphi \circ d$.

   For uniqueness, observe that I is generated as a $B \otimes_A B$-module by elements of the form $1 \otimes b - b \otimes 1$ (since $b_1 \otimes b_2 = b_1b_2 \otimes 1 - b_1(b_2 \otimes 1 - 1 \otimes b_2)$, so $b_1 \otimes b_2 - b_1b_2 \otimes 1 \in I$ is a B-multiple of an element $d(b_2)$). Hence the images $d(b)$ generate I/I^2 as a B-module, forcing φ to be unique.

The geometric content of Construction 2 is the following: if $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$, then $B \otimes_A B$ is the coordinate ring of the fiber product $X \times_Y X$, and the multiplication map $\mu\colon B \otimes_A B \to B$ corresponds to the diagonal morphism $\Delta\colon X \to X \times_Y X$. The ideal $I = \ker\mu$ is the ideal sheaf of the diagonal, and I/I^2 is the *conormal sheaf* of the diagonal embedding. Thus the module of differentials is the algebraic incarnation of "functions vanishing on the diagonal, modulo those vanishing to second order", precisely the data captured by first-order Taylor expansions. This interpretation globalizes immediately to the sheaf of differentials on a morphism of schemes (cf. definition 5.5.14).

Note that we can define the higher order equivalently (these are known as a jet bundle):

$$\mathcal{P}^n_{B/A} = \frac{B \otimes_A B}{I^{n+1}}$$

but we don't focus on these in this section.

Example 5.5.4: Relative Differentials

1. *Polynomial rings.* Let $B = A[x_1, \dots, x_n]$. The Leibniz rule and the chain rule force $\Omega_{B/A}$ to be generated by $dx_1, \dots, dx_n$. Conversely, the partial derivatives $\partial/\partial x_i \colon B \to B$ are A-derivations, and the universal property provides B-linear maps $\Omega_{B/A} \to B$ sending $dx_i \mapsto 1$ and $dx_j \mapsto 0$ for $j \neq i$. Hence $dx_1, \dots, dx_n$ are linearly independent, and
$$\Omega_{B/A} \;\cong\; \bigoplus_{i=1}^{n} B\,dx_i,$$
a free B-module of rank n. The universal derivation is $d(f) = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\,dx_i$, exactly as in multivariable calculus.

2. *Quotient rings.* Let $B = A[x_1, \dots, x_n]/J$ where $J = (f_1, \dots, f_r)$. Using the two exact sequences below (propositions 5.5.5 and 5.5.6), one obtains
$$\Omega_{B/A} \;\cong\; \frac{\bigoplus_{i=1}^{n} B\,dx_i}{\sum_{j=1}^{r} B\,df_j},$$
where $df_j = \sum_i \frac{\partial f_j}{\partial x_i}\,dx_i$. This is the algebraic analogue of computing the cotangent space of a submanifold by imposing the constraint equations.

   For instance, if $B = k[x,y]/(y^2 - x^3)$ (the cuspidal cubic), then $\Omega_{B/k} = B\,dx \oplus B\,dy \,/\, (2y\,dy - 3x^2\,dx)$. At the origin $(x,y) = (0,0)$, the relation $2y\,dy - 3x^2\,dx$ vanishes identically, so $\Omega_{B/k} \otimes_B k \cong k^2$, two-dimensional rather than the expected one. This detects the singularity (cf. example 2.7.4).

3. *Separable field extensions.* Let K/k be a finite field extension. Then $\Omega_{K/k} = 0$ if and only if K/k is separable. To see this, first consider a simple extension $K = k(\alpha) \cong k[x]/(f)$, where f is the minimal polynomial of α. By the quotient ring computation above, $\Omega_{K/k} \cong K\,dx/(f'(\alpha)\,dx)$. If K/k is separable, f has no multiple roots, meaning $f'(\alpha) \neq 0$. Since K is a field, $f'(\alpha)$ is a unit, forcing $dx = 0$ and hence $\Omega_{K/k} = 0$. Conversely, if K/k is inseparable (which only occurs if $\operatorname{char} k = p > 0$), there exists some $\alpha \in K$ whose minimal polynomial takes the form $f(x) = g(x^p)$. Its formal derivative vanishes identically ($f'(x) = 0$), so the relation $f'(\alpha)\,dx = 0$ imposes no constraint on dx. Thus $\Omega_{k(\alpha)/k} \neq 0$, and hence $\Omega_{K/k} \neq 0$.

   Geometrically, a separable field extension represents a discrete covering of an isolated points, meaning there is no continuous space to move and therefore a trivial cotangent space ($\Omega_{K/k} = 0$). In contrast, an inseparable extension contains "infinitesimal fuzz" where points have collided in positive characteristic, spawning a hidden, non-trivial differential direction even in a zero-dimensional space.

4. *Localization.* If $S \subseteq B$ is a multiplicative set, then $\Omega_{S^{-1}B/A} \cong S^{-1}\Omega_{B/A}$. This follows from the universal property: an A-derivation $S^{-1}B \to M$ is determined by its restriction to B, using the quotient rule $d(b/s) = (s\,db - b\,ds)/s^2$.

The following two exact sequences are the workhorses for computing differentials. They are the algebraic counterparts of the cotangent exact sequence and the conormal exact sequence in differential geometry:

Proposition 5.5.5: First Exact Sequence (Base Change, or Fibrations)

Let $A \to B \to C$ be ring homomorphisms. Then there is a natural exact sequence of C-modules

$$\Omega_{B/A} \otimes_B C \longrightarrow \Omega_{C/A} \longrightarrow \Omega_{C/B} \longrightarrow 0.$$

The first map sends $db \otimes 1 \mapsto d(\varphi(b))$ (where $\varphi\colon B \to C$ is the structure map), and the second map sends $dc \mapsto dc$.

***Proof*:**
We verify exactness using the universal property. For any C-module M, apply $\mathrm{Hom}_C(-, M)$ to obtain

$$0 \longrightarrow \mathrm{Der}_B(C, M) \longrightarrow \mathrm{Der}_A(C, M) \longrightarrow \mathrm{Der}_A(B, M).$$

The first map is injective: a B-derivation is in particular an A-derivation. The second map sends an A-derivation $d\colon C \to M$ to its restriction $d|_B$; its kernel consists of A-derivations that vanish on B, which are exactly the B-derivations. This is a left-exact sequence of C-modules, and by the isomorphism $\mathrm{Hom}_C(\Omega_{C/A}, M) \cong \mathrm{Der}_A(C, M)$ (and its analogues), the claim follows by the Yoneda lemma.

Note that $A \to B \to C$ is has no exactness conditions set on it: geometrically this has many interpretations (this can be a fibration, it can be closed immersions, etc). This generality is partly to build good tools to construct differentials, and partly because singularities often break exactness (as demonstrated in examples in chapter 3). Geometrically, if you take the total differential of a space ($\Omega_{C/A}$) and quotient out the differentials coming from the base space ($\Omega_{B/A}$), you are left exactly with the differentials of the fibers ($\Omega_{C/B}$). In terms of tangent vectors (the dual picture), the proposition says the tangent vectors of a fiber are exactly the tangent vectors of the total space that project to zero on the base.

Proposition 5.5.6: Second Exact Sequence (Conormal)

Let B be an A-algebra, $J \subseteq B$ an ideal, and $C = B/J$. There is a natural exact sequence of C-modules

$$J/J^2 \xrightarrow{\delta} \Omega_{B/A} \otimes_B C \longrightarrow \Omega_{C/A} \longrightarrow 0,$$

where $\delta(\overline{f}) = df \otimes 1$ for $f \in J$.

***Proof*:**
Again, apply $\mathrm{Hom}_C(-, M)$ for a C-module M. The resulting sequence is

$$0 \longrightarrow \mathrm{Der}_A(C, M) \longrightarrow \mathrm{Der}_A(B, M) \xrightarrow{\alpha} \mathrm{Hom}_C(J/J^2, M),$$

where α sends an A-derivation $d\colon B \to M$ to the C-linear map $J/J^2 \to M$ given by $\overline{f} \mapsto d(f)$. (This is well-defined because for $f, g \in J$, $d(fg) = f\,d(g) + g\,d(f) = 0$ since elements of J act trivially on the C-module M, meaning $d(J^2) = 0$. Furthermore, this map is indeed C-linear: for any $b \in B$ and $f \in J$, the Leibniz rule yields $d(bf) = b\,d(f) + f\,d(b) = b\,d(f)$, again because f annihilates M.)

The kernel of α consists of A-derivations vanishing on J, which factor through $C = B/J$. Left-exactness and the Yoneda lemma give the claim.

Geometrically, this tells you how to link the intrinsic cotangent space of the subspace $\Omega_{C/A}$ to the ambient cotangent space $\Omega_{B/A} \otimes_B C$. In other word, to find the differentials of a subspace ($\Omega_{C/A}$), start with the differentials of the ambient space ($\Omega_{B/A}$) and kill the gradients of the constraint equations ($\delta(J/J^2)$).

Corollary 5.5.7: Finite Generation of Differentials

If B is a finitely generated A-algebra, then $\Omega_{B/A}$ is a finitely generated B-module.

***Proof*:**
Write $B = A[x_1, \ldots, x_n]/J$. By proposition 5.5.6, $\Omega_{B/A}$ is a quotient of $\Omega_{A[x_1,\ldots,x_n]/A} \otimes B \cong B^n$, hence finitely generated.

The next two results relate differentials to base change and to field extensions, the latter being crucial for detecting smoothness.

Proposition 5.5.8: Base Change for Differentials

Let $A \to B$ and $A \to A'$ be ring homomorphisms, and set $B' = B \otimes_A A'$. Then

$$\Omega_{B'/A'} \;\cong\; \Omega_{B/A} \otimes_B B'.$$

***Proof*:**
For any B'-module M,

$$\mathrm{Hom}_{B'}(\Omega_{B'/A'}, M) \;\cong\; \mathrm{Der}_{A'}(B', M) \;\cong\; \mathrm{Der}_A(B, M) \;\cong\; \mathrm{Hom}_B(\Omega_{B/A}, M) \;\cong\; \mathrm{Hom}_{B'}(\Omega_{B/A} \otimes_B B', M),$$

where the middle isomorphism uses the universal property of the tensor product $B' = B \otimes_A A'$: an A-derivation $B \to M$ extends uniquely to an A'-derivation $B' \to M$. The result follows by the Yoneda lemma.

Proposition 5.5.9: Differentials and Field Extensions

Let k be a field and K/k a finitely generated[a] field extension. Then $\Omega_{K/k}$ is a K-vector space and:

$$\dim_K \Omega_{K/k} \geq \mathrm{trdeg}(K/k)$$

with equality if and only if K is separably generated over k (i.e., K is a separable algebraic extension of a purely transcendental extension of k).

[a] be sure not to confuse with finite! example 5.5.4 was a special case with finite extensions

5.5.2 Differentials and Local Rings

The module of differentials detects regularity of local rings, providing the algebraic foundation for the Jacobian criterion.

Proposition 5.5.10: Differentials and Local Rings

Let $(B, \mathfrak{m})$ be a Noetherian local ring that is a localization of a finitely generated k-algebra (where k is a field), with residue field $\kappa = B/\mathfrak{m}$. Assume the residue field κ is a separable extension of k (e.g., k is perfect or algebraically closed). Then there is a natural isomorphism of κ-vector spaces

$$\Omega_{B/k} \otimes_B \kappa \;\cong\; \mathfrak{m}/\mathfrak{m}^2.$$

***Proof*:**
Apply the second exact sequence (proposition 5.5.6) to the surjection $B \twoheadrightarrow \kappa$ with kernel $\mathfrak{m}$:

$$\mathfrak{m}/\mathfrak{m}^2 \xrightarrow{\delta} \Omega_{B/k} \otimes_B \kappa \longrightarrow \Omega_{\kappa/k} \longrightarrow 0.$$

If κ/k is separable (e.g., if k is perfect), then $\Omega_{\kappa/k} = 0$, and δ is surjective. Since δ sends $\overline{f} \mapsto df \otimes 1$ and both sides have the same dimension (namely $\dim_\kappa \mathfrak{m}/\mathfrak{m}^2$), the map δ is an isomorphism.

Lemma 5.5.11: Characterizing Freeness via Residue and Quotient Fields

Let $(R, \mathfrak{m})$ be a Noetherian local domain with fraction field K, and let M be a finitely generated R-module. Then M is free of rank r if and only if

$$\dim_{R/\mathfrak{m}}(M \otimes_R R/\mathfrak{m}) \;=\; r \;=\; \dim_K(M \otimes_R K).$$

***Proof*:**
By Nakayama's lemma, M is generated by r elements if and only if $\dim_{R/\mathfrak{m}}(M/\mathfrak{m}M) = r$. If additionally $\dim_K(M \otimes_R K) = r$, then a minimal generating set of M is K-linearly independent, hence R-linearly independent (since R is a domain), so $M \cong R^r$.

Theorem 5.5.12: Characterizing Regular Local Rings via Differentials

Let B be a localization of a finitely generated k-algebra at a maximal ideal, where k is a perfect field. Set $n = \dim B$. Then B is a regular local ring if and only if $\Omega_{B/k}$ is a free B-module of rank n.

***Proof*:**
By the Nullstellensatz, the residue field κ is a finite extension of k. Since k is perfect, κ/k is separable, so $\Omega_{\kappa/k} = 0$. Thus, by proposition 5.5.10, $\dim_\kappa(\Omega_{B/k} \otimes_B \kappa) = \dim_\kappa(\mathfrak{m}/\mathfrak{m}^2)$.

Let $K = \mathrm{Frac}(B)$; then $\Omega_{B/k} \otimes_B K \cong \Omega_{K/k}$. Because B is localized at a maximal ideal, we have $\mathrm{trdeg}(K/k) = \dim B = n$, so $\Omega_{K/k}$ has dimension n by proposition 5.5.9 (using that k is perfect).

If B is regular, then $\dim_\kappa(\mathfrak{m}/\mathfrak{m}^2) = n = \dim B$, so both the fibre and the generic fibre of $\Omega_{B/k}$ have the same dimension n. By lemma 5.5.11, $\Omega_{B/k}$ is free of rank n.

Conversely, if $\Omega_{B/k}$ is free of rank n, then $\dim_\kappa(\mathfrak{m}/\mathfrak{m}^2) = n = \dim B$, which is the definition of regularity.

We shall see by box 5.5.21 that the result for maximal points is sufficient.

5.5.13 Non-Closed Points and Arbitrary Primes If B is localized at a non-maximal prime P, it represents the local ring of a positive-dimensional subvariety (i.e., its generic point). The residue field $\kappa(P)$ is then a transcendental function field over k, meaning its differentials no longer vanish ($\Omega_{\kappa(P)/k} \neq 0$). Consequently, the exact sequence from proposition 5.5.10 does not yield a simple isomorphism.
If B is a regular local ring at such a prime, $\Omega_{B/k}$ is indeed still a free module, but its rank is $\dim B + \operatorname{trdeg}_k \kappa(P)$. Geometrically, $\dim B$ captures the codimension (the normal directions transverse to the subvariety), while the transcendence degree captures the tangent directions along the subvariety itself. Restricting to maximal ideals sets the transcendence degree to zero, isolating the local geometry of a single point.

5.5.3 The Sheaf of Differentials

The algebraic constructions of the previous subsection globalize to schemes via the diagonal ideal, following the geometric interpretation of proposition 5.5.3. Let $f\colon X \to Y$ be a morphism of schemes and $\Delta\colon X \to X \times_Y X$ the diagonal morphism. By proposition 3.5.1, Δ is a locally closed immersion: the image $\Delta(X)$ is isomorphic to a closed subscheme of an open subset $W \subseteq X \times_Y X$. (When f is separated, Δ is a closed immersion into all of $X \times_Y X$, and we may take $W = X \times_Y X$.)

Definition 5.5.14: Sheaf of Differentials

Let $f\colon X \to Y$ be a morphism of schemes and let $\mathscr{I}$ be the ideal sheaf of $\Delta(X)$ in the open subset $W \subseteq X \times_Y X$ described above. The *sheaf of relative differentials* (or *cotangent sheaf*) of X over Y is the quasi-coherent $\mathcal{O}_X$-module

$$\Omega_{X/Y} \;:=\; \Delta^*(\mathscr{I}/\mathscr{I}^2),$$

The *universal derivation* $d\colon \mathcal{O}_X \to \Omega_{X/Y}$ is defined locally by $d(b) = 1 \otimes b - b \otimes 1 \pmod{\mathscr{I}^2}$, exactly as in proposition 5.5.3.

The sheaf $\mathscr{I}/\mathscr{I}^2$ is naturally an $\mathcal{O}_{\Delta(X)}$-module, and the isomorphism $X \cong \Delta(X)$ gives it the structure of an $\mathcal{O}_X$-module. Since $\mathscr{I}$ is a quasi-coherent ideal sheaf (proposition 5.1.34), the quotient $\mathscr{I}/\mathscr{I}^2$ and its pullback are quasi-coherent (proposition 5.1.17). When Y is Noetherian and f is of finite type, $\Omega_{X/Y}$ is coherent.

To see the local affine structure, let $V = \operatorname{Spec} A \subseteq Y$ and $U = \operatorname{Spec} B \subseteq X$ be affine opens with $f(U) \subseteq V$. Then $U \times_V U = \operatorname{Spec}(B \otimes_A B) \subseteq X \times_Y X$, and the diagonal $\Delta(U)$ is the closed subscheme defined by the kernel $I = \ker(B \otimes_A B \xrightarrow{\mu} B)$. Thus $\mathscr{I}/\mathscr{I}^2$ restricted to $U \times_V U$ is $\widetilde{I/I^2}$, and we recover

$$\Omega_{X/Y}|_U \;\cong\; \widetilde{\Omega_{B/A}}.$$

Hence the sheaf of differentials can equivalently be constructed by gluing the modules $\widetilde{\Omega_{B/A}}$ over an affine cover. Similarly, the derivations $d\colon B \to \Omega_{B/A}$ glue to the global derivation $d\colon \mathcal{O}_X \to \Omega_{X/Y}$.

The three fundamental properties of Kähler differentials (base change, the cotangent sequence, and the conormal sequence) globalize immediately.

Proposition 5.5.15: Sheaf of Differentials: Base Change

Let $f\colon X \to Y$ be a morphism, $g\colon Y' \to Y$ another morphism, and $f'\colon X' = X \times_Y Y' \to Y'$ the base change of f. Let $g'\colon X' \to X$ denote the projection. Then

$$\Omega_{X'/Y'} \;\cong\; (g')^* \, \Omega_{X/Y}.$$

***Proof*:**
On affine patches, this is proposition 5.5.8. The local isomorphisms are compatible on overlaps (by the uniqueness in the universal property of differentials), so they glue to a global isomorphism.

Proposition 5.5.16: Cotangent Exact Sequence

Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be morphisms of schemes. There is a natural exact sequence of $\mathcal{O}_X$-modules

$$f^*\Omega_{Y/Z} \longrightarrow \Omega_{X/Z} \longrightarrow \Omega_{X/Y} \longrightarrow 0.$$

***Proof*:**
Follows from proposition 5.5.5 on affine patches and gluing.

Proposition 5.5.17: Conormal Exact Sequence

Let $f\colon X \to Y$ be a morphism and $\iota\colon Z \hookrightarrow X$ a closed immersion with ideal sheaf $\mathscr{I}_Z \subseteq \mathcal{O}_X$. There is a natural exact sequence of $\mathcal{O}_Z$-modules

$$\mathscr{I}_Z/\mathscr{I}_Z^2 \xrightarrow{\delta} \iota^*\Omega_{X/Y} \longrightarrow \Omega_{Z/Y} \longrightarrow 0,$$

where $\delta(\overline{f}) = df \otimes 1$ for local sections f of $\mathscr{I}_Z$. The sheaf $\mathscr{I}_Z/\mathscr{I}_Z^2$ is the *conormal sheaf* of Z in X.

***Proof*:**
Follows from proposition 5.5.6 on affine patches and gluing.

The next result is specific to projective space and will be indispensable for cohomological computations throughout the rest of the book.

Theorem 5.5.18: The Euler Sequence

Let A be a ring, $Y = \operatorname{Spec} A$, and $X = \mathbb{P}^n_A$. There is an exact sequence of sheaves on X:

$$0 \longrightarrow \Omega_{X/Y} \longrightarrow \mathcal{O}_X(-1)^{\oplus(n+1)} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

In particular, $\Omega_{X/Y}$ is locally free of rank n.

Recall that $\mathcal{O}_X(-1) \cong \mathcal{O}_X(1)^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_X(1), \mathcal{O}_X)$ (proposition 5.3.5), so sections of $\mathcal{O}_X(-1)$ are naturally linear functionals on $\mathcal{O}_X(1)$. The map $\mathcal{O}_X(-1)^{n+1} \to \mathcal{O}_X$ is an evaluation map: in homogeneous coordinates, it sends an $(n+1)$-tuple of linear functionals to their pairing with the tautological coordinates $(x_0, \dots, x_n)$. The kernel of this evaluation (the "relations among the coordinates") is precisely the sheaf of differentials.

It is also instructive to duality this sequence to see that the tangent sheaf (defined rigorously soon) is the quotient of the free sheaf by a action of $\mathcal{O}_X$, just like how projective space is classically constructed:

$$O \to \mathcal{O}_X \to \mathcal{O}_X(1)^{\oplus(n+1)} \to \mathcal{T}_X \to O$$

***Proof*:**

Let $S = A[x_0, \dots, x_n]$ be the homogeneous coordinate ring of X, and let $E = S(-1)^{n+1}$ be the free graded S-module with basis $e_0, \dots, e_n$ in degree 1. Define a degree-0 homomorphism of graded S-modules $\varphi\colon E \to S$ by $e_i \mapsto x_i$, and let $M = \ker \varphi$. The exact sequence of graded modules

$$0 \longrightarrow M \longrightarrow E \xrightarrow{\varphi} S$$

gives rise to an exact sequence of sheaves on X:

$$0 \longrightarrow \widetilde{M} \longrightarrow \mathcal{O}_X(-1)^{n+1} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Note that φ is not surjective as a map of graded modules (it misses degree 0), but it is surjective in all degrees ≥ 1 and hence surjective on all stalks of the associated sheaves.

It remains to show that $\widetilde{M} \cong \Omega_{X/Y}$. We construct compatible isomorphisms on each standard affine chart and verify that they glue.

Step 1: local structure of M. On the chart $U_i = D_+(x_i) \cong \operatorname{Spec} A[x_0/x_i, \dots, x_n/x_i]$, localizing at x_i makes $\varphi_{x_i}\colon E_{x_i} \to S_{x_i}$ a surjection of free S_{x_i}-modules. The kernel M_{x_i} is free of rank n, generated by $\{x_i e_j - x_j e_i \mid j \neq i\}$. Passing to the associated sheaf (and extracting degree-0 elements), $\widetilde{M}|_{U_i}$ is free over $\mathcal{O}_{U_i}$ with basis

$$\left\{ \frac{1}{x_i}\, e_j - \frac{x_j}{x_i^2}\, e_i \right\}_{j \neq i}$$

(the factor $1/x_i$ ensures these have degree 0 in M_{x_i}).

Step 2: defining the isomorphism φ_i. On U_i, the sheaf $\Omega_{X/Y}|_{U_i}$ is the free $\mathcal{O}_{U_i}$-module with basis $\{d(x_j/x_i)\}_{j \neq i}$ (example 5.5.4). Define $\varphi_i\colon \Omega_{X/Y}|_{U_i} \to \widetilde{M}|_{U_i}$ by

$$\varphi_i\left(d\left(\frac{x_j}{x_i}\right)\right) = \frac{1}{x_i^2}(x_i\, e_j - x_j\, e_i).$$

Since both sides are free of rank n and φ_i sends a basis to a basis, this is an isomorphism.

Step 3: gluing. On the overlap $U_i \cap U_j$, the chain rule gives

$$d\left(\frac{x_k}{x_i}\right) = \frac{x_k}{x_j}\, d\left(\frac{x_j}{x_i}\right) + \frac{x_j}{x_i}\, d\left(\frac{x_k}{x_j}\right).$$

Applying φ_i to the left-hand side and φ_j to the right-hand side, both yield $\frac{1}{x_i x_j}(x_j\, e_k - x_k\, e_j)$. Hence $\varphi_i|_{U_i \cap U_j} = \varphi_j|_{U_i \cap U_j}$, and the local isomorphisms glue to a global isomorphism $\Omega_{X/Y} \cong \widetilde{M}$.

5.5.19 Relation To Canonical Sheaf Taking the top exterior power of the Euler sequence (the determinant bundle) and applying the formula:

$$\bigwedge^{\text{top}} B \;\cong\; (\bigwedge^{\text{top}} A) \otimes (\bigwedge^{\text{top}} C)$$

yields the important formula:

$$\omega_{\mathbb{P}^n_A/A} \;:=\; \bigwedge^n \Omega_{\mathbb{P}^n_A/A} \;\cong\; \mathcal{O}_{\mathbb{P}^n_A}(-n-1),$$

which will be central to Serre duality (section 6.5).

5.5.4 Smoothness and the Cotangent Sheaf

We now focus on varieties over an algebraically closed field, where the cotangent sheaf provides a clean characterization of nonsingularity.

Definition 5.5.20: Abstract Nonsingular Variety

Let X be an abstract variety over an algebraically closed field k. Then X is *nonsingular* (or *smooth over* k) if every local ring $\mathcal{O}_{X,x}$ is a regular local ring.

5.5.21 Non-Closed Points? The definition requires regularity at *all* points, not just closed ones. However, since the local ring at a non-closed point $\mathfrak{p}$ is a localization of a local ring at a closed point (namely, $\mathcal{O}_{X,\mathfrak{p}} \cong (\mathcal{O}_{X,x})_{\mathfrak{p}}$ for a suitable closed point $x \in \overline{\{\mathfrak{p}\}}$), and localization preserves regularity (see [**NateClassicalAlgGeo**]), regularity at all closed points already implies regularity everywhere. Thus the two conditions are equivalent.

Proposition 5.5.22: Nonsingular Varieties and the Cotangent Sheaf

Let X be an irreducible separated scheme of finite type over an algebraically closed field k, and let $n = \dim X$. Then $\Omega_{X/k}$ is locally free of rank n if and only if X is a nonsingular variety.

***Proof*:**
The question is local, so we may work at a closed point $x \in X$ with local ring $B = \mathcal{O}_{X,x}$ and residue field $\kappa(x) = k$ (since k is algebraically closed). The stalk $(\Omega_{X/k})_x = \Omega_{B/k}$, and by proposition 5.1.24, $\Omega_{X/k}$ is locally free of rank n at x if and only if $\Omega_{B/k}$ is a free B-module of rank n. By theorem 5.5.12, this holds if and only if B is a regular local ring of dimension n. Combining over all closed points and invoking box 5.5.21 for the non-closed points gives the result.

This characterization yields a short proof that the smooth locus is open and dense:

Corollary 5.5.23: Nonsingular Points are Open and Dense

Let X be a variety over an algebraically closed field k. Then the set of nonsingular points of X is a nonempty open subset.

***Proof*:**
Let $n = \dim X$ and let $K = K(X)$ be the function field. Since k is algebraically closed, the extension K/k is separably generated, so $\dim_K \Omega_{K/k} = \operatorname{trdeg}(K/k) = n$ by proposition 5.5.9. Now, $\Omega_{K/k}$ is the stalk of $\Omega_{X/k}$ at the generic point η of X. Since $\Omega_{X/k}$ is coherent and its generic stalk is a free K-module of rank n, proposition 5.1.24 implies that $\Omega_{X/k}$ is locally free of rank n on some nonempty open subset U containing η that is not what that proposition says actually! I thought I proved spreading out there but didn't! I actually never touched on spreading out in this book; must figure that out.. By proposition 5.5.22, the points of U are nonsingular.

The conormal exact sequence imposes strong constraints on nonsingular closed subvarieties. The following theorem refines proposition 5.5.17 when both ambient and subvariety are nonsingular.

Theorem 5.5.24: Nonsingular Closed Subschemes and Differentials

Let X be a nonsingular variety over an algebraically closed field k, and let $Y \subseteq X$ be an irreducible closed subscheme defined by the ideal sheaf $\mathscr{I}_Y$. Then Y is nonsingular if and only if the following two conditions hold:

1. $\Omega_{Y/k}$ is locally free.
2. The conormal sequence of proposition 5.5.17 is exact on the left:

$$0 \longrightarrow \mathscr{I}_Y/\mathscr{I}_Y^2 \longrightarrow \Omega_{X/k} \otimes \mathcal{O}_Y \longrightarrow \Omega_{Y/k} \longrightarrow 0.$$

When Y is nonsingular, $\mathscr{I}_Y$ is locally generated by $r = \operatorname{codim}_X(Y)$ elements, and the conormal sheaf $\mathscr{I}_Y/\mathscr{I}_Y^2$ is locally free of rank r on Y.

***Proof*:**
(⇐) Assume (1) and (2) hold, and let $q = \operatorname{rank}\Omega_{Y/k}$. Since $\Omega_{X/k}$ is locally free of rank $n = \dim X$ (proposition 5.5.22), the short exact sequence in (2) forces $\mathscr{I}_Y/\mathscr{I}_Y^2$ to be locally free of rank $n - q$. By Nakayama's lemma, $\mathscr{I}_Y$ can be locally generated by $n - q$ elements,

so $\dim Y \geq n-(n-q)=q$ (see [11, chapter 21.5]). On the other hand, at any closed point $y \in Y$, proposition 5.5.10 gives $q=\dim_k(\mathfrak{m}_y/\mathfrak{m}_y^2) \geq \dim \mathcal{O}_{Y,y} = \dim Y$. Hence $q = \dim Y$, so $\dim_k(\mathfrak{m}_y/\mathfrak{m}_y^2) = \dim Y$ and Y is nonsingular. Moreover, $n-q = \operatorname{codim}_X(Y)$.

($\Rightarrow$) Assume Y is nonsingular of dimension $q=\dim Y$. Then (1) is immediate from proposition 5.5.22. It remains to show that the conormal map $\delta\colon \mathscr{I}_Y/\mathscr{I}_Y^2 \to \Omega_{X/k} \otimes \mathcal{O}_Y$ is injective. Let $r = n-q = \operatorname{codim}_X(Y)$.

At a closed point $y \in Y$, the cokernel of δ is $\Omega_{Y/k}$, which is locally free of rank q. Hence $\ker(\Omega_{X/k} \otimes \mathcal{O}_Y \to \Omega_{Y/k})$ is locally free of rank r near y. Choose local sections $f_1, \ldots, f_r \in \mathscr{I}_Y$ in a neighbourhood of y such that $df_1, \ldots, df_r$ generate this kernel, and let $\mathscr{I}'$ be the ideal sheaf generated by $f_1, \ldots, f_r$ with associated closed subscheme Y'.

By construction, the images $\delta(\overline{f_1}), \ldots, \delta(\overline{f_r})$ span a free subsheaf of rank r in $\Omega_{X/k} \otimes \mathcal{O}_{Y'}$ near y. Applying the conormal sequence (proposition 5.5.17) to Y':

$$\mathscr{I}'/(\mathscr{I}')^2 \xrightarrow{\delta'} \Omega_{X/k} \otimes \mathcal{O}_{Y'} \longrightarrow \Omega_{Y'/k} \longrightarrow 0,$$

the map δ' is injective (its image is free of rank r), so by the argument of the forward direction, Y' is nonsingular of dimension q near y. Since $Y \subseteq Y'$ and both are integral of the same dimension, we must have $Y = Y'$ and $\mathscr{I}_Y = \mathscr{I}'$ locally near y. Hence δ is injective near every closed point, giving the global left-exactness.

5.5.5 Tangent Sheaf, Canonical Sheaf, and the Adjunction Formula

Definition 5.5.25: Tangent Sheaf and Canonical Sheaf

Let X be a nonsingular variety of dimension n over a field k. The *tangent sheaf* is

$$\mathcal{T}_X \ := \ \mathcal{H}om_{\mathcal{O}_X}(\Omega_{X/k}, \mathcal{O}_X),$$

a locally free sheaf of rank n. The *canonical sheaf* (or *dualizing sheaf*) is the invertible sheaf

$$\omega_X \ := \ \bigwedge\nolimits^n \Omega_{X/k}.$$

For projective space, the Euler sequence (theorem 5.5.18) gives $\omega_{\mathbb{P}^n_k} \cong \mathcal{O}_{\mathbb{P}^n_k}(-n-1)$. Since this sheaf has no nonzero global sections (the degree is negative), the geometric genus of projective space is zero, a first indication that $\mathbb{P}^n$ carries no "interesting" top-degree differential forms.

In [6], we established a connection between the topology of a space and its analytical structure: the dimension of the space of globally defined holomorphic 1-forms on a compact Riemann surface of genus g is exactly g. This result is historically significant since it provides one of the first explicit algebraic characterizations of a geometric and topological invariant. Rather than simply counting the "number of holes" in a surface, this equivalence demonstrates that the macroscopic shape of the manifold dictates and constrains the algebraic structure of the differential forms that can exist upon it. In modern algebraic geometry, this perspective becomes the starting point: the geometric genus of a curve is formally defined exactly by this property, as the dimension of the space of global regular 1-forms. The abstraction allows for defining the genus of varieties over arbitrary fields: by replacing complex-analytic holomorphic forms with the purely algebraic machinery of Kähler differentials, we can formally measure the "topology" of spaces:

Definition 5.5.26: Geometric Genus

Let X be a nonsingular projective variety of dimension n over a field k. The *geometric genus* of X is

$$p_g(X) := \dim_k \Gamma(X, \omega_X) = \dim_k H^0\big(X, \textstyle\bigwedge^n \Omega_{X/k}\big).$$

5.5.27 Foreshadowing: Arithmetic and Geometric Genus In [11, chapter 21.5.2], we introduced the Hilbert function and the *arithmetic genus* p_a. Serre duality (section 6.5) will show that $p_g = p_a$ for nonsingular projective *curves*, but these two invariants can differ in higher dimensions. The arithmetic genus is defined in terms of the Hilbert polynomial (definition 6.7.6), while the geometric genus measures global sections of the canonical sheaf, a more intrinsic quantity.

The following fundamental result shows that the geometric genus is a birational invariant: it cannot change under birational maps. This makes it a useful tool for showing that varieties are *not* rational.

Theorem 5.5.28: Birational Invariance of Geometric Genus

Let X and X' be nonsingular projective varieties over a field k. If X and X' are birational, then $p_g(X) = p_g(X')$.

***Proof*:**
Let $F\colon X \dashrightarrow X'$ be a birational map, and let $V \subseteq X$ be the largest open subset on which F is represented by a morphism $f\colon V \to X'$.

Step 1: the complement $X \setminus V$ has codimension ≥ 2. Let $P \in X$ be a point of codimension 1. Since X is nonsingular, $\mathcal{O}_{X,P}$ is a DVR. The birational map gives a morphism from the generic point η of X to X', i.e., a map $\operatorname{Spec} K(X) \to X'$. Since X' is projective (hence proper over k), the valuative criterion of properness (theorem 3.5.25) provides a unique extension $\operatorname{Spec} \mathcal{O}_{X,P} \to X'$, which extends to a morphism on some open neighbourhood of P. Hence $P \in V$, and every codimension-1 point lies in V.

Step 2: pullback of canonical forms. The cotangent sequence (proposition 5.5.16) gives $f^*\Omega_{X'/k} \to \Omega_{V/k}$. Since X and X' are birational, both cotangent sheaves are locally free of the same rank $n = \dim X$, and f is an isomorphism on a dense open $U \subseteq V$. Taking n-th exterior powers gives $f^*\omega_{X'} \to \omega_V$, which restricts to an isomorphism on U. This induces a map on global sections:

$$f^*\colon \Gamma(X', \omega_{X'}) \longrightarrow \Gamma(V, \omega_V).$$

A nonzero global section of an invertible sheaf on an integral scheme cannot vanish on a dense open subset (U is dense in V), so f^* is injective.

Step 3: extending sections across codimension ≥ 2. We claim that the restriction map $\Gamma(X, \omega_X) \to \Gamma(V, \omega_V)$ is an isomorphism. It suffices to check this on affine opens: for any affine open $W = \operatorname{Spec} R \subseteq X$ with $\omega_X|_W \cong \mathcal{O}_W$, the map $\Gamma(W, \mathcal{O}_W) \to \Gamma(W \cap V, \mathcal{O}_{W\cap V})$ is bijective. Since X is nonsingular, R is a normal domain, and $W \setminus (W \cap V) = W \cap (X \setminus V)$ has codimension ≥ 2 in W by Step 1. The bijectivity now follows from lemma 5.4.8: a regular function on $W \cap V$ that is bounded at all codimension-1 points extends uniquely to W.

Conclusion. Combining Steps 2 and 3, we have an injection $\Gamma(X', \omega_{X'}) \hookrightarrow \Gamma(X, \omega_X)$, so

$p_g(X') \leq p_g(X)$. By symmetry (applying the same argument to the inverse birational map F^{-1}), we obtain $p_g(X) \leq p_g(X')$. Hence $p_g(X) = p_g(X')$.

The geometric genus is therefore a birational invariant: any variety with $p_g > 0$ cannot be rational (i.e., birational to $\mathbb{P}^n$), since $p_g(\mathbb{P}^n) = 0$. For example, a smooth hypersurface of degree $d \geq n+2$ in $\mathbb{P}^n$ has $p_g > 0$ and is therefore not rational. The converse fails in general: the geometric genus alone does not determine the birational equivalence class.

We close this section with the normal and conormal sheaves and the adjunction formula, which computes the canonical sheaf of a subvariety from that of the ambient variety.

Definition 5.5.29: Normal and Conormal Sheaf

Let $Y \subseteq X$ be a nonsingular closed subvariety of a nonsingular variety X over k, with ideal sheaf $\mathscr{I}_Y$. The *conormal sheaf* of Y in X is $\mathscr{I}_Y/\mathscr{I}_Y^2$, a locally free $\mathcal{O}_Y$-module of rank $r = \operatorname{codim}_X(Y)$ (theorem 5.5.24). The *normal sheaf* is its dual:

$$\mathcal{N}_{Y/X} \;:=\; \mathcal{H}om_{\mathcal{O}_Y}(\mathscr{I}_Y/\mathscr{I}_Y^2,\, \mathcal{O}_Y),$$

also locally free of rank r.

Dualizing the exact sequence of theorem 5.5.24 gives the familiar normal bundle sequence from differential geometry (definition 0.5.7):

$$0 \longrightarrow \mathcal{T}_Y \longrightarrow \mathcal{T}_X \otimes \mathcal{O}_Y \longrightarrow \mathcal{N}_{Y/X} \longrightarrow 0.$$

This is the algebraic incarnation of the splitting of the tangent bundle of X along Y into tangential and normal directions.

Proposition 5.5.30: Adjunction Formula

Let $Y \subseteq X$ be a nonsingular closed subvariety of codimension r in a nonsingular variety X over k. Then

$$\omega_Y \;\cong\; \omega_X \otimes {\textstyle\bigwedge^r} \mathcal{N}_{Y/X} \;\otimes\; \mathcal{O}_Y.$$

In the important special case $r = 1$ (i.e., Y is a divisor), let $\mathcal{L} = \mathcal{O}_X(Y)$ be the invertible sheaf associated to Y (definition 5.4.29). Then

$$\omega_Y \;\cong\; (\omega_X \otimes \mathcal{L})\big|_Y.$$

***Proof*:**
The short exact sequence from theorem 5.5.24,

$$0 \longrightarrow \mathscr{I}_Y/\mathscr{I}_Y^2 \longrightarrow \Omega_{X/k} \otimes \mathcal{O}_Y \longrightarrow \Omega_{Y/k} \longrightarrow 0,$$

involves locally free sheaves of ranks r, $n = \dim X$, and $q = \dim Y$ respectively, with $n = q + r$. Taking top exterior powers of a short exact sequence of locally free sheaves gives

$${\textstyle\bigwedge^n}(\Omega_{X/k} \otimes \mathcal{O}_Y) \;\cong\; {\textstyle\bigwedge^r}(\mathscr{I}_Y/\mathscr{I}_Y^2) \;\otimes\; {\textstyle\bigwedge^q}\, \Omega_{Y/k}.$$

The left-hand side is $\omega_X|_Y$, and the right-hand side is ${\textstyle\bigwedge^r}(\mathscr{I}_Y/\mathscr{I}_Y^2) \otimes \omega_Y$. Tensoring both sides with $\bigwedge^r \mathcal{N}_{Y/X} = (\bigwedge^r(\mathscr{I}_Y/\mathscr{I}_Y^2))^\vee$ yields

$$\omega_Y \;\cong\; \omega_X|_Y \otimes \bigwedge^r \mathcal{N}_{Y/X}.$$

When $r = 1$, the conormal sheaf is $\mathscr{I}_Y/\mathscr{I}_Y^2 \cong \mathcal{L}^{-1}|_Y$ by proposition 5.4.35 (since $\mathscr{I}_Y \cong \mathcal{O}_X(-Y) = \mathcal{L}^{-1}$). Its dual is $\mathcal{N}_{Y/X} \cong \mathcal{L}|_Y$, so

$$\omega_Y \;\cong\; \omega_X|_Y \otimes \mathcal{L}|_Y \;=\; (\omega_X \otimes \mathcal{L})\big|_Y.$$

Example 5.5.31: Canonical Sheaves via the Adjunction Formula

1. *Hypersurfaces in $\mathbb{P}^n$.* Let $Y = V_+(f) \subseteq \mathbb{P}^n_k$ be a nonsingular hypersurface of degree d. Then $\mathcal{L} = \mathcal{O}_{\mathbb{P}^n}(d)$ and $\omega_{\mathbb{P}^n} = \mathcal{O}_{\mathbb{P}^n}(-n-1)$, so the adjunction formula gives

$$\omega_Y \;\cong\; \mathcal{O}_{\mathbb{P}^n}(-n-1+d)\big|_Y \;=\; \mathcal{O}_Y(d-n-1).$$

For a nonsingular plane curve of degree d in $\mathbb{P}^2$, $\omega_Y \cong \mathcal{O}_Y(d-3)$, and the geometric genus is $p_g = \dim_k H^0(Y, \mathcal{O}_Y(d-3)) = \binom{d-1}{2}$. This recovers the classical genus formula $g = (d-1)(d-2)/2$ for plane curves.

2. *Complete intersections.* More generally, if $Y = V_+(f_1) \cap \cdots \cap V_+(f_r)$ is a nonsingular complete intersection of hypersurfaces of degrees $d_1, \ldots, d_r$ in $\mathbb{P}^n_k$, iterating the adjunction formula gives

$$\omega_Y \;\cong\; \mathcal{O}_Y(d_1 + \cdots + d_r - n - 1).$$

In particular, Y is a *Fano variety* (i.e., $\omega_Y^\vee$ is ample) precisely when $\sum d_i < n+1$.

3. *Non-rational varieties in every dimension.* A nonsingular hypersurface Y of degree $d \geq n+2$ in $\mathbb{P}^n_k$ satisfies $\omega_Y \cong \mathcal{O}_Y(d-n-1)$ with $d-n-1 \geq 1$, so $p_g(Y) \geq 1$. By theorem 5.5.28, Y is not rational. Since such hypersurfaces exist in every dimension $n-1 \geq 1$, this produces non-rational varieties in all dimensions, a consequence of the interplay between differentials, the adjunction formula, and birational invariance.

5.5.6 *Invariant Differentials on Group Schemes

We saw in example 6.5.8 that the canonical sheaf (and more generally, the cotangent sheaf $\Omega_{X/k}$) carries geometric information about a variety X. However, when the scheme in question possesses a group structure, the situation becomes especially rigid. Algebraic group schemes are "everywhere locally the same," and this global symmetry forces their sheaves of differentials to be uniform across the entire space.

Let G be a group scheme over a field k (for instance, an affine algebraic group or an abelian variety). For any T-valued point $g \in G(T)$, the group operation defines a *left translation* morphism:

$$\ell_g \colon G_T \longrightarrow G_T, \qquad x \longmapsto g \cdot x.$$

Because G is a group, ℓ_g is an isomorphism of schemes (with inverse $\ell_{g^{-1}}$). This allows us to "transport" local geometric data from the identity element $e \in G(k)$ to any other point.

Definition 5.5.32: Left-Invariant Differentials

Let G be a group scheme over k, and let $\mu\colon G \times_k G \to G$ be the multiplication morphism. A global differential form $\omega \in \Gamma(G, \Omega_{G/k})$ is *left-invariant* if

$$\mu^*\omega \;=\; 1 \otimes \omega \quad \text{in} \quad \Gamma(G \times_k G,\, \Omega_{(G\times_k G)/k}).$$

(In the functor of points perspective, this means $\ell_g^*\omega = \omega$ for all points g.)

Let $\mathfrak{g} = T_eG$ be the Lie algebra of G, which is the Zariski tangent space at the identity. Its dual vector space, $\mathfrak{g}^\vee = (\mathfrak{m}_e/\mathfrak{m}_e^2)$, is the cotangent space at e. Any left-invariant differential form is completely determined by its value at the identity, giving a natural k-linear map from the space of left-invariant differentials to $\mathfrak{g}^\vee$. In fact, this map is an isomorphism: every cotangent vector at the identity extends uniquely to a global left-invariant differential form via translation.

This process of spreading out the tangent space at the identity to the entire group scheme yields a profound structural result.

Theorem 5.5.33: Group Schemes are Parallelizable

Let G be a group scheme of finite type over a field k. Then the cotangent sheaf $\Omega_{G/k}$ is globally free. Specifically, evaluation at the identity induces an isomorphism of $\mathcal{O}_G$-modules:

$$\Omega_{G/k} \;\cong\; \mathcal{O}_G \otimes_k \mathfrak{g}^\vee.$$

***Proof*:**
The formal proof relies on the multiplication map $\mu\colon G \times_k G \to G$ and the first projection $p_1\colon G\times_k G \to G$. Consider the map $\Phi\colon G\times_k G \to G\times_k G$ defined by $(g,h) \mapsto (g, g\cdot h)$. This is an isomorphism of schemes over G (relative to the first projection), with inverse $(g,x) \mapsto (g, g^{-1}\cdot x)$.

The isomorphism Φ transforms the second projection $p_2\colon G \times_k G \to G$ into the multiplication map μ. Therefore, it must induce an isomorphism on the pullbacks of the cotangent sheaf:

$$\Phi^* p_2^*\Omega_{G/k} \;\cong\; \mu^*\Omega_{G/k}.$$

Restricting this isomorphism to the slice $G \times \{e\}$ (i.e., pulling back along $\mathrm{id}_G \times e\colon G \to G \times_k G$) trivializes the differential components from the base, leaving an isomorphism between $\Omega_{G/k}$ and the pullback of the fiber at e, which is precisely $\mathcal{O}_G \otimes_k \mathfrak{g}^\vee$.

Corollary 5.5.34: Triviality of the Canonical Sheaf for Groups

If G is a nonsingular group scheme over k of dimension n, then its tangent sheaf $\mathcal{T}_G$ and its canonical sheaf ω_G are both trivial:

$$\mathcal{T}_G \;\cong\; \mathcal{O}_G^{\oplus n}, \qquad \text{and} \qquad \omega_G \;\cong\; \mathcal{O}_G.$$

This corollary provides a massive constraint on which varieties can be endowed with a group structure. For example, among all projective curves, only those of geometric genus $p_g = 1$ (elliptic curves) have a trivial canonical sheaf. Thus, it is geometrically impossible to put a group scheme structure on the projective line $\mathbb{P}^1_k$ (where $\omega \cong \mathcal{O}(-2)$) or on a nonsingular curve of genus $g \geq 2$. Similarly, an abelian variety of dimension g must have $\omega_A \cong \mathcal{O}_A$, which immediately distinguishes it from most other projective varieties of the same dimension and restricts its intersection theory.

5.6 *Formal Schemes and Local Geometry

As we have seen, one feature which clearly distinguishes the theory of schemes from the older classical theory of varieties is the possibility of having nilpotent elements in the structure sheaf. In particular, if $Y \subseteq X$ is a closed subvariety defined by a sheaf of ideals $\mathcal{I}$, then for any $n \geq 1$ we can consider the closed subscheme Y_n defined by the n-th power $\mathcal{I}^n$. For $n \geq 2$, this is a scheme with nilpotent elements. It carries information about Y together with the infinitesimal properties of the embedding of Y in X.

We wish to construct an object which carries information about all the infinitesimal neighborhoods Y_n of Y simultaneously. Such an object would be "thicker" than any individual Y_n, but contained inside any actual open neighborhood of Y in X. We might naturally call this the formal neighborhood of Y in X.

To see why a new geometric construction is necessary to capture this, recall that the algebraic construction of the formal power series ring $k[[x]]$ is given by the limit of the truncated polynomial rings $k[x]/(x^n)$. We shall now explore the geometric consequences of this algebraic operation.

Topologically, the scheme $\operatorname{Spec}(k[x]/(x^n))$ consists of exactly one point, as the ring is Artinian and possesses a unique prime ideal (x). Geometrically, one views this as a "fat point" – a point equipped with infinitesimal nilpotents that remember $n-1$ orders of derivatives. Since $k[[x]]$ is the algebraic limit of these rings as $n \to \infty$, one might intuitively expect the geometric limit of a sequence of one-point spaces to remain a one-point space. However, passing to the infinite limit crosses a structural threshold: the limit of infinite nilpotents coalesces into a genuine coordinate variable, making $k[[x]]$ a domain of Krull dimension 1. Consequently, the affine scheme $\operatorname{Spec}(k[[x]])$ possesses *two* points. One should visualize $\operatorname{Spec}(k[[x]])$ as the "germ" of a curve, an unimaginably small neighborhood that has lost all global properties but is just large enough to have a center and an infinitesimal edge.

Formal Completions

The fact that the Spec functor transforms a limit of one-point spaces into a two-point space demonstrates that Spec does not commute nicely with infinite limits. To rectify this topological mismatch, Grothendieck introduced formal schemes. We shall follow Hartshorne and define this globally as a completion along a closed subscheme.

Definition 5.6.1: Formal Completion

Let X be a Noetherian scheme, and let Y be a closed subscheme defined by a sheaf of ideals $\mathcal{I}$. Then we define the *formal completion* of X along Y, denoted $(\hat{X}, \mathcal{O}_{\hat{X}})$, to be the following ringed space: we take the topological space Y, and on it the sheaf of rings $\mathcal{O}_{\hat{X}} = \varprojlim \mathcal{O}_X/\mathcal{I}^n$. Here we consider each $\mathcal{O}_X/\mathcal{I}^n$ as a sheaf of rings on Y, and make them into an inverse system in the natural way.

The reader should notice that the structure sheaf $\mathcal{O}_{\hat{X}}$ actually depends only on the closed subset Y, and not on the particular scheme structure on Y. For if $\mathcal{J}$ is another sheaf of ideals defining a closed subscheme structure on Y, then since X is a Noetherian scheme, there are integers m, n such that $\mathcal{I} \supseteq \mathcal{J}^m$ and $\mathcal{J} \supseteq \mathcal{I}^n$. Thus the inverse systems $(\mathcal{O}_X/\mathcal{I}^n)$ and $(\mathcal{O}_X/\mathcal{J}^m)$ are cofinal with each other, and hence have the same inverse limit.

Furthermore, one sees easily that the stalks of the sheaf $\mathcal{O}_{\hat{X}}$ are local rings, so in fact $(\hat{X}, \mathcal{O}_{\hat{X}})$ is a locally ringed space. If $U = \operatorname{Spec} A$ is an open affine subset of X, and if $I \subseteq A$ is the ideal $\Gamma(U, \mathcal{I})$, then evaluating the sections reveals that $\Gamma(\hat{X} \cap U, \mathcal{O}_{\hat{X}}) = \hat{A}$, the I-adic completion of A. Thus the process of completing X along Y is strictly analogous to the I-adic completion of a ring. However, one should note that the local rings of $\hat{X}$ are in general *not* complete, and their dimension (which is equal to $\dim X$) is *not* equal to the dimension of the underlying topological space Y.

We shall call a formal scheme obtained by completing a single Noetherian scheme X along a closed subscheme Y *algebraizable*.[6]

Example 5.6.2: Formal Completions

1. Take X to be a Noetherian scheme, and let $Y = X$. Then clearly $\hat{X} = X$. Thus the category of Noetherian formal schemes naturally includes all Noetherian schemes.

2. Take X to be a Noetherian scheme, and let $Y = \{P\}$ be a closed point. Then $\hat{X}$ is a one point space $\{P\}$ with the completion $\hat{\mathcal{O}}_P$ of the local ring at P as its structure sheaf. An $\hat{\mathcal{O}}_P$-module M, considered as a sheaf on $\hat{X}$, is coherent if and only if M is a finitely generated module. This directly resolves our earlier algebraic paradox: $\hat{X}$ is topologically a single point, capturing the geometric intuition of a true limit of fat points without accidentally spawning a generic point.

5.6.3 Comparing Local Geometries To summarize the distinctions between these deeply related objects for a point P on a curve:

1. $\operatorname{Spec}(k[x]/(x^n))$ is a single point representing a "fat point" of finite thickness.

2. $\hat{X}$ (the formal completion at P) is a single point representing the true geometric limit of fat points (infinite formal thickness).

3. $\operatorname{Spec}(k[[x]])$ is a two-point space representing a "formal disk" or germ (a center point plus a surrounding generic neighborhood).

Universal Smoothness and Singularities

The utility of formal schemes shines when studying the local properties of varieties. In classical algebraic geometry, curves can exhibit wildly different global shapes, such as a line $y = x$ or a parabola $y = x^2$. However, if one selects a smooth point on any such curve over a field k and takes the formal completion of the local ring at that point, the resulting ring is always isomorphic to $k[[t]]$, where t is a local parameter.

[6] It is apparently quite difficult to find non-algebraizable formal schemes, maybe add an exercise based on Hartshorne p.194 later?

Because their local algebra is identical, their formal geometry is identical. Thus, $\operatorname{Spec}(k[[x]])$ does not merely represent *a* curve; it represents the fundamental atomic structure of *any* smooth curve at a smooth point. Under an infinite-power formal microscope, every smooth curve is indistinguishable from a straight line.

This universal behavior fails precisely when the geometry ceases to be smooth.

Example 5.6.4: Singularities and Formal Completions
If a curve possesses a singularity, its formal completion at that point acts as a perfect algebraic memory of the singularity, remaining distinct from $k[[x]]$.

1. *The Cusp:* Consider the curve cut out by $y^2 = x^3$, which has a sharp cusp at the origin. The formal completion of the local ring at the origin is

$$\frac{k[[x, y]]}{(y^2 - x^3)}$$

   This ring is not isomorphic to $k[[t]]$. Even infinitesimally, the formal scheme remembers the sharp turn.

2. *The Node:* Consider a curve with a self-intersection at the origin, locally modeled by $xy = 0$. Its formal completion at the crossing is

$$\frac{k[[x, y]]}{(xy)}$$

   This represents two distinct microscopic branches crossing each other. Topologically, the formal completion is still a single point, but the underlying sheaf of topological rings completely captures the crossing branches.

6

Cohomology of Schemes

Let X be a scheme with a sheaf $\mathscr{F}$. A natural question is: can we quantify the obstruction to gluing local sections into global sections? On an affine scheme, there is no obstruction: the global section functor is exact for quasi-coherent sheaves, and all local data glues. But on a projective variety, or on a scheme built by gluing affine pieces in a non-trivial way, the functor $\Gamma(X, -)$ fails to be exact, and its failure is measured by the *sheaf cohomology groups* $H^i(X, \mathscr{F}) = R^i\Gamma(X, \mathscr{F})$, the right derived functors of the global section functor. These groups are the central objects of study in this chapter.

The sheaves to which this machinery applies include the structure sheaf $\mathcal{O}_X$, the twisted sheaves $\mathcal{O}_X(n)$, the sheaves of differentials $\Omega_{X/S}$ and their exterior powers, and the pushforwards of sheaves along closed immersions. Each choice of sheaf extracts different geometric information: the cohomology of $\mathcal{O}_X$ governs the arithmetic genus, the cohomology of line bundles governs linear systems and the Riemann–Roch theorem, and the cohomology of the canonical sheaf ω_X encodes duality phenomena that constrain the topology of the variety.

Though the derived-functor definition is theoretically clean, it is computationally demanding: it requires an injective resolution, and injective sheaves are typically enormous and non-explicit. For practical computations, we introduce *Čech cohomology*, which replaces injective resolutions with the combinatorial data of an open cover. For quasi-coherent sheaves on separated Noetherian schemes with an affine cover, Čech cohomology and derived-functor cohomology coincide, and this is our main computational tool. Using it, we compute the cohomology of projective space $\mathbb{P}^n_R$ with coefficients in the twisted sheaves $\mathcal{O}(d)$ — the single most important explicit computation in the subject.

With these foundations, the chapter develops in four main directions.

Vanishing and finiteness. Two structural results govern the cohomology of schemes: the cohomology of quasi-coherent sheaves vanishes on affine schemes (section 6.2), and the cohomology of coherent sheaves on projective schemes is finite-dimensional (section 6.6). The first confirms the philosophy that cohomology measures gluing; the second ensures that the invariants we extract are "finitely complicated." Together, they justify defining the Euler characteristic $\chi(X, \mathscr{F})$, the Hilbert polynomial, and the arithmetic genus (section 6.7).

Duality. Serre duality (section 6.5) is the algebraic analogue of Poincaré duality: for a smooth projective variety X of dimension n over a field k, the cohomology groups satisfy $H^i(X, \mathscr{F}) \cong H^{n-i}(X, \mathscr{F}^\vee \otimes \omega_X)^*$ for locally free $\mathscr{F}$. The proof passes through the Ext groups and the local-to-global spectral sequence (section 6.4), which provide the natural framework for the duality pairing. As a consequence, we obtain the Riemann–Roch theorem for curves and the classical genus formulas.

Families and variation. The higher direct image sheaves $R^i f_* \mathscr{F}$ (section 6.6) relativize cohomology over a base scheme, packaging the cohomology of fibers into a single sheaf. For flat families over a Noetherian base, the semicontinuity theorem (section 6.8) shows that the fiber cohomology $h^i(X_y, \mathscr{F}_y)$ is upper semicontinuous in y, and Grauert's theorem guarantees that constancy of h^i implies local freeness of $R^i f_* \mathscr{F}$ and compatibility with base change. These results are the technical heart of moduli theory.

GAGA. For projective varieties over $\mathbb{C}$, Serre's GAGA theorems (section 6.9) identify the algebraic cohomology developed in this chapter with the analytic cohomology of the complex geometry: the analytification functor induces an equivalence $\mathbf{Coh}(X) \simeq \mathbf{Coh}(X^{\mathrm{an}})$ and isomorphisms $H^i(X, \mathscr{F}) \cong H^i(X^{\mathrm{an}}, \mathscr{F}^{\mathrm{an}})$. Through GAGA, the Hodge decomposition, the algebraic de Rham theorem, and Chow's theorem become consequences of the algebraic theory, closing the loop between the algebraic and analytic perspectives on cohomology.

Prerequisites. The reader should be comfortable with the theory of sheaves and schemes (chapters 1–3), quasi-coherent and coherent sheaves (chapter 5), and the basic properties of projective schemes and differentials (sections 2.3 and 5.5). For the GAGA section, familiarity with the complex geometry review (sections 0.6 through 6.9 of chapter 1) is assumed. The reader may also want to review [11, Part VII] for the derived-functor approach to (co)homology theory.

6.1 Definition and Basic Properties

Recall that a category $\mathbf{C}$ is *pre-additive* (or enriched over $\mathbf{Ab}$) if for every $A, B \in \mathrm{Obj}(\mathbf{C})$, the set $\mathrm{Hom}_{\mathbf{C}}(A, B)$ carries the structure of an abelian group, and composition of morphisms is bilinear:

$$f \circ (g + h) = (f \circ g) + (f \circ h) \qquad \text{and} \qquad (f + g) \circ h = (f \circ h) + (g \circ h).$$

An *additive category* is a pre-additive category that admits all finite products and coproducts, and in which these coincide (this common construction is sometimes called the *biproduct*).

An abelian category is an additive category with the "right" conditions to do cohomology[1]. Concretely (cf. definition 0.4.29), a category $\mathbf{A}$ is *abelian* if:

1. for every $A, B \in \mathrm{Obj}(\mathbf{A})$, the set $\mathrm{Hom}_{\mathbf{A}}(A, B)$ is an abelian group and composition is bilinear;
2. finite products and coproducts exist and coincide;
3. every morphism $f \in \mathrm{Hom}_{\mathbf{A}}(A, B)$ has a kernel and a cokernel;
4. every monomorphism is the kernel of its cokernel, and every epimorphism is the cokernel of its kernel;

[1]Technically, one can do cohomology on exact categories that are not abelian, such as the category of vector bundles, but it is more natural to see this as a special generalization of the abelian setting.

5. defining $\mathrm{im}(f) = \ker(\mathrm{coker}(f))$, the first isomorphism theorem gives

$$\mathrm{im}(f) = \ker(\mathrm{coker}(f)) = \mathrm{coker}(\ker(f)),$$

so every morphism factors as an epimorphism followed by a monomorphism.

The prototypical abelian category is R-**Mod**; by the Freyd–Mitchell embedding theorem, any small abelian category $\mathbf{A}$ fully-faithfully embeds into R-**Mod** via an exact functor for an appropriate (not necessarily commutative) ring R. This is useful theoretically, but for our purposes it is more practical to think of the abelian category of sheaves $\mathbf{Sh}_{\mathbf{Ab}}(X)$ as its own creature. The "sheafification" of R-**Mod** gives $\mathcal{O}_X$-**Mod**. The important full subcategories $\mathbf{QCoh}(X)$ and $\mathbf{Coh}(X)$ are both abelian (by theorem 1.2.28), and $\mathbf{QCoh}(X)$ has enough injectives (as we shall see shortly). Though not critical at the moment, it should be noted that $\mathbf{QCoh}(X)$ is not closed under the internal Hom sheaf $\mathcal{H}om$, while $\mathbf{Coh}(X)$ is.

A functor between abelian categories $F : \mathbf{A} \to \mathbf{B}$ is *additive* if for every $X, Y \in \mathrm{Obj}(\mathbf{A})$, the induced map

$$\mathrm{Hom}_{\mathbf{A}}(X, Y) \longrightarrow \mathrm{Hom}_{\mathbf{B}}(FX, FY)$$

is a group homomorphism. An additive functor F is *left exact* if it preserves kernels, equivalently if a short exact sequence $0 \to X \to Y \to Z \to 0$ yields an exact sequence

$$0 \longrightarrow FX \longrightarrow FY \longrightarrow FZ,$$

and *right exact* if it preserves cokernels, equivalently if the same short exact sequence yields

$$FX \longrightarrow FY \longrightarrow FZ \longrightarrow 0.$$

If F is both left and right exact, it is called *exact*. The most common (and arguably the most important) left-exact functors are the covariant $\mathrm{Hom}(X, -)$ and the contravariant $\mathrm{Hom}(-, Y)$. The most important right-exact functor is $- \otimes Y$[2].

Given a left-exact functor $F : \mathbf{A} \to \mathbf{B}$, we can form its *right derived functors* $R^iF : \mathbf{A} \to \mathbf{B}$ for $i \geq 0$. Abstractly, one passes from $\mathbf{A}$ to its derived category $D(\mathbf{A})$ — constructed by taking the category of cochain complexes $\mathrm{Kom}(\mathbf{A})$, passing to the homotopy category $K(\mathbf{A})$, and then localizing at quasi-isomorphisms — and the total derived functor $\mathbf{R}F : D(\mathbf{A}) \to D(\mathbf{B})$ captures the right information to "fix" exactness. The derived category does not lend itself well to direct computation, but if $\mathbf{A}$ has enough injectives, then injective resolutions are universal up to homotopy, and the right derived functors R^iF can be computed as follows: given an object $A \in \mathbf{A}$, choose an injective resolution $0 \to A \to I^0 \to I^1 \to \cdots$, apply F, and take cohomology:

$$R^iF(A) = h^i(F(I^\bullet)).$$

This is independent of the choice of injective resolution (up to canonical isomorphism). The details of all of these constructions can be found in [11, chapter 27]. A "classical" approach using δ-functors is briefly introduced at the end of that same chapter.

Recall that if R is a ring, then every R-module embeds into an injective module (see [11, chapter 16.3]). From this, we shall be able to conclude that the module categories over ringed spaces have enough injectives:

[2]The functor $X \otimes -$ is also right exact and is left adjoint to $\mathrm{Hom}(X, -)$; this adjunction is the tensor–Hom adjunction.

Proposition 6.1.1: Ringed Space Has Enough Injectives

Let $(X, \mathcal{O}_X)$ be a ringed space. Then the category $\mathcal{O}_X$-**Mod** of sheaves of $\mathcal{O}_X$-modules has enough injectives.

***Proof*:**
Let $\mathscr{F}$ be a sheaf of $\mathcal{O}_X$-modules. For each $x \in X$, the stalk $\mathscr{F}_x$ is an $\mathcal{O}_{X,x}$-module, so there exists an injective $\mathcal{O}_{X,x}$-module I_x together with an embedding $\mathscr{F}_x \hookrightarrow I_x$. For each point $x \in X$, let $j_x : \{x\} \hookrightarrow X$ denote the inclusion of the one-point space $\{x\}$, and equip $\{x\}$ with the $\mathcal{O}_{X,x}$-module I_x (viewed as a sheaf on a point). Define:

$$\mathscr{I} = \prod_{x \in X} (j_x)_*(I_x).$$

We claim that $\mathscr{I}$ is an injective $\mathcal{O}_X$-module and that $\mathscr{F}$ embeds into it.

The embedding. For each $x \in X$, the stalk of $(j_x)_*(I_x)$ at x is I_x, and the embedding $\mathscr{F}_x \hookrightarrow I_x$ extends (by the adjunction below) to a sheaf map $\mathscr{F} \to (j_x)_*(I_x)$. Taking the product over all x yields a morphism $\mathscr{F} \to \mathscr{I}$ that is injective on every stalk, hence injective as a morphism of sheaves (by theorem 1.2.27).

Injectivity of $\mathscr{I}$. For any $\mathcal{O}_X$-module $\mathscr{G}$, we have:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{G}, \mathscr{I}) \cong \prod_{x \in X} \mathrm{Hom}_{\mathcal{O}_X}\big(\mathscr{G}, (j_x)_*(I_x)\big) \cong \prod_{x \in X} \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathscr{G}_x, I_x).$$

The first isomorphism is the universal property of products. The second uses the fact that for the inclusion $j_x : \{x\} \hookrightarrow X$, the stalk functor $\mathscr{G} \mapsto \mathscr{G}_x$ is left adjoint to $(j_x)_*$[a].

Now, the functor $\mathrm{Hom}_{\mathcal{O}_X}(-, \mathscr{I})$ is the product of functors $\mathrm{Hom}_{\mathcal{O}_{X,x}}((-)_x, I_x)$. Each factor is the composition of the stalk functor $\mathscr{G} \mapsto \mathscr{G}_x$, which is exact (by **??**), with the functor $\mathrm{Hom}_{\mathcal{O}_{X,x}}(-, I_x)$, which is exact because I_x is injective. A product of exact functors is exact, so $\mathrm{Hom}_{\mathcal{O}_X}(-, \mathscr{I})$ is exact, making $\mathscr{I}$ injective.

[a] More precisely, a morphism $\mathscr{G} \to (j_x)_*(I_x)$ is determined by its value on stalks at x, since $(j_x)_*(I_x)$ is supported at $\overline{\{x\}}$ and its stalk at x is I_x. The formal adjunction is $\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{G}, (j_x)_* M) \cong \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathscr{G}_x, M)$ for any $\mathcal{O}_{X,x}$-module M.

Corollary 6.1.2: Sheaves of Abelian Groups Has Enough Injectives

Let X be a topological space. Then the category of sheaves of abelian groups on X has enough injectives.

***Proof*:**
View X as the ringed space $(X, \underline{\mathbb{Z}})$, where $\underline{\mathbb{Z}}$ is the constant sheaf. Sheaves of abelian groups on X are precisely $\underline{\mathbb{Z}}$-modules, so the result follows from proposition 6.1.1.

Definition 6.1.3: Global Section Functor and Sheaf Cohomology

Let $(X, \mathcal{O}_X)$ be a ringed space. The *global section functor*

$$\Gamma(X, -) : \mathcal{O}_X\text{-}\mathbf{Mod} \longrightarrow \mathbf{Ab}$$

is left exact (by proposition 1.2.16) but not right exact in general (see Example 1.2.7). Since $\mathcal{O}_X$-**Mod** has enough injectives, we may form the right derived functors. The i-th right derived functor

$$H^i(X, -) = R^i\Gamma(X, -)$$

is called the i-th *sheaf cohomology functor*. For a sheaf $\mathscr{F}$, the abelian group $H^i(X, \mathscr{F})$ is the i-th *cohomology group* of $\mathscr{F}$. By general properties of derived functors, $H^0(X, \mathscr{F}) = \Gamma(X, \mathscr{F})$ and a short exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ of sheaves induces a long exact sequence:

$$0 \to H^0(X, \mathscr{F}') \to H^0(X, \mathscr{F}) \to H^0(X, \mathscr{F}'') \xrightarrow{\delta} H^1(X, \mathscr{F}') \to H^1(X, \mathscr{F}) \to \cdots$$

To compute these cohomology groups, we need resolutions. Recall that it suffices to resolve by *acyclic* objects — those $\mathscr{G}$ with $H^i(X, \mathscr{G}) = 0$ for $i > 0$ — rather than by injective objects (see [11, chapter 27.4]). We shall show that the following class of sheaves is acyclic:

Definition 6.1.4: Flasque Sheaf

Let X be a topological space. A sheaf $\mathscr{F}$ on X is called *flasque* (or *flabby*) if for every inclusion of open sets $U \subseteq V$, the restriction map $\mathrm{res}_{V,U} : \mathscr{F}(V) \to \mathscr{F}(U)$ is surjective.

Proposition 6.1.5: Injective Sheaf Modules Are Flasque

Let $(X, \mathcal{O}_X)$ be a ringed space. Then any injective $\mathcal{O}_X$-module is flasque.

***Proof*:**
Let $\mathscr{I}$ be an injective $\mathcal{O}_X$-module, and let $U \subseteq V$ be open subsets of X. For any open set $W \subseteq X$, let $j_W : W \hookrightarrow X$ denote the inclusion and define $\mathcal{O}_W = (j_W)_!(\mathcal{O}_X|_W)$, the extension by zero of $\mathcal{O}_X|_W$ to all of X[a]. The inclusion $U \subseteq V$ induces a natural monomorphism of $\mathcal{O}_X$-modules $0 \to \mathcal{O}_U \to \mathcal{O}_V$. Since $\mathscr{I}$ is injective, the functor $\mathrm{Hom}_{\mathcal{O}_X}(-, \mathscr{I})$ is exact, giving a surjection:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_V, \mathscr{I}) \longrightarrow \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_U, \mathscr{I}) \longrightarrow 0.$$

But $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_W, \mathscr{I}) \cong \mathscr{I}(W)$ for any open W, since a morphism $\mathcal{O}_W \to \mathscr{I}$ is determined by the image of the identity section $1 \in \mathcal{O}_X(W)$. Hence the restriction $\mathscr{I}(V) \to \mathscr{I}(U)$ is surjective, so $\mathscr{I}$ is flasque.

[a] Recall from definition 1.3.11 that $(j_W)_!$ extends a sheaf by zero outside W. In particular, $(j_W)_!(\mathcal{O}_X|_W)(V') = \{s \in \mathcal{O}_X(V' \cap W) \big| \operatorname{supp}(s) \text{ is closed in } V'\}$ for open $V' \subseteq X$.

Before proving that flasque sheaves are acyclic, we need one key property: the global section functor is *exact* on short exact sequences of flasque sheaves. Let us record this precisely.

Lemma 6.1.6: Exactness of Global Sections on Flasque Sequences

Let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be a short exact sequence of sheaves on a topological space X. If $\mathscr{F}'$ is flasque, then the sequence

$$0 \longrightarrow \Gamma(X, \mathscr{F}') \longrightarrow \Gamma(X, \mathscr{F}) \longrightarrow \Gamma(X, \mathscr{F}'') \longrightarrow 0$$

is exact. Furthermore, if both $\mathscr{F}'$ and $\mathscr{F}$ are flasque, then $\mathscr{F}''$ is also flasque.

***Proof*:**
Left exactness of $\Gamma(X, -)$ gives exactness at $\Gamma(X, \mathscr{F}')$ and $\Gamma(X, \mathscr{F})$. We must show surjectivity on the right. Let $s'' \in \Gamma(X, \mathscr{F}'')$. Consider the collection of pairs (U, s) where $U \subseteq X$ is open and $s \in \mathscr{F}(U)$ maps to $s''|_U$ under the surjection $\mathscr{F} \to \mathscr{F}''$. This collection is non-empty (by surjectivity on stalks, we can find such pairs locally) and is partially ordered by extension. By Zorn's lemma, there is a maximal such pair (U_0, s_0).

We claim $U_0 = X$. Suppose not; pick $x \in X \setminus U_0$. By surjectivity on stalks, there is an open $V \ni x$ and $t \in \mathscr{F}(V)$ mapping to $s''|_V$. On $U_0 \cap V$, the difference $s_0|_{U_0 \cap V} - t|_{U_0 \cap V}$ maps to zero in $\mathscr{F}''(U_0 \cap V)$, so it lies in $\mathscr{F}'(U_0 \cap V)$. Since $\mathscr{F}'$ is flasque, this section extends to some $\sigma \in \mathscr{F}'(V)$. Replacing t by $t + \sigma$, we may assume s_0 and t agree on $U_0 \cap V$, and they glue to a section on $U_0 \cup V$, contradicting maximality. Hence $U_0 = X$.

For the second claim, suppose $\mathscr{F}'$ and $\mathscr{F}$ are both flasque. Let $U \subseteq V$ be open sets and $s'' \in \mathscr{F}''(V)$. By the surjectivity just proved (applied to V), there exists $s \in \mathscr{F}(V)$ mapping to s''. Since $\mathscr{F}$ is flasque, the restriction $\mathscr{F}(V) \to \mathscr{F}(U)$ is surjective, so the image of $s|_U$ in $\mathscr{F}''(U)$ equals $s''|_U$. But we already have the section s defined on all of V, and any element of $\mathscr{F}''(U)$ can be lifted to $\mathscr{F}(V)$ and then restricted. More precisely, given any $t'' \in \mathscr{F}''(U)$, apply the surjectivity result to U to get a lift $t \in \mathscr{F}(U)$, then use the flasqueness of $\mathscr{F}$ to extend t to all of V, giving a preimage in $\mathscr{F}''(V)$. Hence $\mathscr{F}''$ is flasque.

Corollary 6.1.7: Trivial Cohomology For Flasque Sheaves

Let $\mathscr{F}$ be a flasque sheaf on a topological space X. Then

$$H^i(X, \mathscr{F}) = 0 \qquad \text{for all } i > 0.$$

***Proof*:**
Since $\mathcal{O}_X$-**Mod** has enough injectives, we can embed $\mathscr{F}$ into an injective sheaf $\mathscr{I}$ and form the short exact sequence:

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{I} \longrightarrow \mathscr{G} \longrightarrow 0$$

where $\mathscr{G} = \mathscr{I}/\mathscr{F}$. By proposition 6.1.5, the sheaf $\mathscr{I}$ is flasque. Since both $\mathscr{F}$ and $\mathscr{I}$ are flasque, the quotient $\mathscr{G}$ is also flasque by the lemma above.

Since $\mathscr{F}$ is flasque, the global section functor preserves the exactness of this sequence (again by the lemma above):

$$0 \longrightarrow \Gamma(X, \mathscr{F}) \longrightarrow \Gamma(X, \mathscr{I}) \longrightarrow \Gamma(X, \mathscr{G}) \longrightarrow 0.$$

Comparing with the long exact sequence in cohomology and using $H^0 = \Gamma$, we see that the connecting homomorphism $\delta : H^0(X, \mathscr{G}) \to H^1(X, \mathscr{F})$ is zero (since the map $\Gamma(X, \mathscr{I}) \to \Gamma(X, \mathscr{G})$

is already surjective). Hence $H^1(X, \mathscr{F}) = 0$.

For $i \geq 2$, the long exact sequence gives isomorphisms $H^i(X, \mathscr{F}) \cong H^{i-1}(X, \mathscr{G})$ because $\mathscr{I}$ is injective and therefore $H^j(X, \mathscr{I}) = 0$ for all $j > 0$. Since $\mathscr{G}$ is flasque, we may repeat the argument with $\mathscr{G}$ in place of $\mathscr{F}$, obtaining $H^{i-1}(X, \mathscr{G}) = 0$ by induction on i. Hence $H^i(X, \mathscr{F}) = 0$ for all $i > 0$.

Flasque sheaves are therefore acyclic for $\Gamma(X, -)$, and we may compute cohomology using flasque resolutions rather than injective ones. Let us see some examples:

Example 6.1.8: Flasque Sheaves

1. It was just shown that all injective $\mathcal{O}_X$-modules are flasque. The converse fails: flasque sheaves need not be injective.

2. *Constant sheaves on irreducible spaces.* Let X be an irreducible topological space and A an abelian group. Then the constant sheaf $\underline{A}$ is flasque. Indeed, for any non-empty open $U \subseteq V$, both U and V are connected (every open subset of an irreducible space is connected), so $\underline{A}(V) = A = \underline{A}(U)$ and the restriction map is the identity. This need not hold on reducible spaces: on $X = \{0, 1\}$ with the discrete topology, the constant sheaf $\underline{\mathbb{Z}}$ satisfies $\underline{\mathbb{Z}}(X) = \mathbb{Z} \times \mathbb{Z}$ and $\underline{\mathbb{Z}}(\{0\}) = \mathbb{Z}$, but the restriction is projection onto the first factor, which is surjective. However, if X is, say, the union of two disjoint open intervals in $\mathbb{R}$, then a section $(1, 0) \in \underline{\mathbb{Z}}(X)$ cannot be extended to a section over a connected open set containing both components while remaining locally constant. The key point is that irreducibility forces all opens to overlap, making extension trivial.

3. *The Godement resolution.* Let $\mathscr{F}$ be any sheaf of abelian groups on X. We can always construct a canonical flasque resolution called the *Godement resolution*. For each $x \in X$, let $\iota_x : \{x\} \hookrightarrow X$ denote the inclusion. Define:

$$C^0(\mathscr{F}) = \prod_{x \in X} (\iota_x)_*(\mathscr{F}_x),$$

so that $\Gamma(U, C^0(\mathscr{F})) = \prod_{x \in U} \mathscr{F}_x$ for every open $U \subseteq X$. This sheaf is flasque because a section over U is simply a choice of element in each stalk $\mathscr{F}_x$ for $x \in U$, with no compatibility condition — so restriction just forgets the stalks outside U, which is always surjective.

Note, however, that $C^0(\mathscr{F})$ need *not* be injective. For example, if $\mathscr{F} = \underline{\mathbb{Z}}$, then each stalk of $C^0(\mathscr{F})$ is $\mathbb{Z}$, which is not a divisible group, so $C^0(\mathscr{F})$ is not injective in the category of sheaves of abelian groups (where injective stalks must be divisible).

The natural map $\epsilon : \mathscr{F} \to C^0(\mathscr{F})$, sending a section $s \in \mathscr{F}(U)$ to the tuple of its germs $(s_x)_{x \in U}$, is injective by the locality axiom for sheaves. Let $\mathscr{Q}^0 = C^0(\mathscr{F})/\mathrm{im}(\epsilon)$. Applying the same construction to $\mathscr{Q}^0$ yields $C^1(\mathscr{F}) = C^0(\mathscr{Q}^0)$, and iterating:

$$0 \longrightarrow \mathscr{F} \xrightarrow{\epsilon} C^0(\mathscr{F}) \longrightarrow C^1(\mathscr{F}) \longrightarrow C^2(\mathscr{F}) \longrightarrow \cdots$$

This is a flasque resolution of $\mathscr{F}$, and one can compute $H^i(X, \mathscr{F})$ as the cohomology of the complex $\Gamma(X, C^\bullet(\mathscr{F}))$. The Godement resolution is completely canonical (no choices involved), but the resulting sheaves are typically enormous and *not* quasi-coherent.

4. *The divisor sheaf.* Let X be an irreducible normal Noetherian scheme with function field K. The sheaf of Weil divisors is:

$$\mathscr{D}iv \cong \bigoplus_{x \in X^{(1)}} (\iota_x)_*(\underline{\mathbb{Z}})$$

where $X^{(1)}$ denotes the set of codimension-1 points of X. The exact sequence of sheaves:

$$1 \longrightarrow \mathcal{O}_X^* \longrightarrow \mathcal{K}^* \longrightarrow \mathscr{D}iv \longrightarrow 0$$

relates the invertible regular functions, the non-zero rational functions (here $\mathcal{K}^*$ is the constant sheaf associated to K^*, hence flasque by item (2) above), and the divisor sheaf. Since $\mathscr{D}iv$ is a direct sum of pushforwards of flasque sheaves, it is flasque. The long exact sequence in cohomology, together with the acyclicity of $\mathcal{K}^*$ and $\mathscr{D}iv$, gives:

$$\Gamma(X, \mathcal{K}^*) \longrightarrow \Gamma(X, \mathscr{D}iv) \longrightarrow H^1(X, \mathcal{O}_X^*) \longrightarrow 0$$

and hence an isomorphism:

$$H^1(X, \mathcal{O}_X^*) \cong \frac{\mathrm{WDiv}\,(X)}{\mathrm{PDiv}(X)} = \mathrm{Cl}\, X,$$

recovering the divisor class group from section 5.4 as cohomological data. This illustrates a recurring theme: the H^1 of a sheaf of units is a "class group" measuring the failure of local-to-global extension for invertible objects.

Acyclic does not imply flasque. Though not central to our development, it is worth recording the main class of acyclic-but-not-flasque sheaves, as it explains why sheaf cohomology in the smooth category is far simpler than in the algebraic category:

Example 6.1.9: Acyclic, But Not Injective Or Flasque

Let M be a smooth manifold and consider the sheaf C_M^∞ of smooth real-valued functions. This sheaf is *not* flasque: for instance, the function $1/x$ is a smooth section of $C_{\mathbb{R}}^\infty$ over the open set $(0, 1)$, but it cannot be extended to any smooth function on $(-1, 1)$. Nor is C_M^∞ injective in general.

However, C_M^∞ is acyclic. The key reason is the existence of *partitions of unity* subordinate to any open cover (see theorem 0.5.47): given a surjection of sheaves $\mathscr{F} \to \mathscr{G} \to 0$ and a global section $s \in \Gamma(X, \mathscr{G})$, we can lift s locally and then glue the local lifts using a partition of unity. More precisely, any sheaf of C_M^∞-modules admits partitions of unity, and such sheaves (called *fine sheaves*) are acyclic for Γ.

This is exactly the mechanism that makes de Rham cohomology computable on manifolds. The sheaves Ω_M^p of smooth differential p-forms are fine (being C_M^∞-modules), hence acyclic, and the de Rham complex $0 \to \underline{\mathbb{R}} \to \Omega^0 \to \Omega^1 \to \cdots$ is an acyclic resolution of the constant sheaf $\underline{\mathbb{R}}$. As recalled in theorem 0.5.23, this gives the isomorphism $H^i_{\mathrm{dR}}(M) \cong H^i(M, \underline{\mathbb{R}})$.

In algebraic geometry, no such luxury exists: the structure sheaf $\mathcal{O}_X$ of a scheme does not admit partitions of unity (the Zariski topology is too coarse, and algebraic functions are too rigid). This is why we must work harder — using flasque resolutions, Čech cohomology, or injective resolutions — to compute sheaf cohomology in the algebraic setting.

Let us link this to the module structure on cohomology. If $(X, \mathcal{O}_X)$ is a ringed space, then $\Gamma(X, \mathcal{O}_X)$ is a ring, and for any $\mathcal{O}_X$-module $\mathscr{F}$, the global sections $\Gamma(X, \mathscr{F})$ carry a natural $\Gamma(X, \mathcal{O}_X)$-module structure. This structure propagates to cohomology:

Corollary 6.1.10: Section and Hom Functor

Let $(X, \mathcal{O}_X)$ be a ringed space and let $R = \Gamma(X, \mathcal{O}_X)$. Then for any $\mathcal{O}_X$-module $\mathscr{F}$, the cohomology groups $H^i(X, \mathscr{F})$ carry a natural R-module structure. In particular:

1. If X is a scheme over a ring A (i.e., equipped with a structure morphism $X \to \operatorname{Spec} A$), then each $H^i(X, \mathscr{F})$ is naturally an A-module.
2. If X is a scheme over a field k, then each $H^i(X, \mathscr{F})$ is a k-vector space.

Moreover, for any $\mathcal{O}_X$-module $\mathscr{F}$ and any open subset $U \subseteq X$, there is a natural isomorphism:

$$H^i(X, \mathscr{F}) \cong R^i\Gamma(X, \mathscr{F})$$

and the connecting homomorphisms in the long exact sequence are R-module maps.

***Proof*:**
The global section functor $\Gamma(X, -)$ lands in R-Mod (not just **Ab**), and the category $\mathcal{O}_X$-**Mod** has enough injectives. The right derived functors of $\Gamma(X, -)$ computed in **Ab** and in R-Mod coincide, since the forgetful functor R-Mod $\to$ **Ab** is exact and preserves injectives. Hence the cohomology groups inherit the R-module structure. For a scheme over A, the structure morphism $X \to \operatorname{Spec} A$ gives a ring homomorphism $A \to \Gamma(X, \mathcal{O}_X) = R$, making every R-module an A-module by restriction of scalars.

6.1.1 Computation: Čech Cohomology

The derived-functor definition of sheaf cohomology is clean and canonical, but it is difficult to use in practice because injective (or even flasque) resolutions are typically non-explicit. Čech cohomology provides a more "hands-on" approach: it computes the obstruction to extending sections from an open cover $\mathcal{U}$ of a topological space X to larger open sets, using only the sections of the sheaf over finite intersections of the cover. When X is a Noetherian separated scheme, $\mathscr{F}$ is quasi-coherent, and the cover is affine, the Čech groups coincide with the derived-functor groups. This will be our main computational tool.

The following construction will be familiar to those who have seen group cohomology or simplicial cohomology in algebraic topology (see [6, chapter 4]).

Let X be a topological space, $\mathcal{U} = (U_i)_{i \in I}$ an open covering of X, and fix a well-ordering on the index set I[3]. For any finite set of indices $i_0 < \cdots < i_p$ in I, write:

$$U_{i_0,\ldots,i_p} = U_{i_0} \cap \cdots \cap U_{i_p}.$$

Let $\mathscr{F}$ be a sheaf of abelian groups on X. Define the *Čech cochain groups*:

$$C^p(\mathcal{U}, \mathscr{F}) = \prod_{i_0 < \cdots < i_p} \mathscr{F}(U_{i_0,\ldots,i_p}).$$

[3] Alternatively, one can work with the "alternating" convention: allow all tuples of indices, require that the cochain α is alternating (i.e. $\alpha_{\sigma(i_0),\ldots,\sigma(i_p)} = \operatorname{sgn}(\sigma) \cdot \alpha_{i_0,\ldots,i_p}$) and vanishes when any two indices coincide. The two approaches yield isomorphic cohomology groups.

An element $\alpha \in C^p(\mathcal{U}, \mathscr{F})$ is a collection of sections $\alpha_{i_0,\dots,i_p} \in \mathscr{F}(U_{i_0,\dots,i_p})$, one for each ordered $(p+1)$-tuple. The *coboundary map* $d : C^p(\mathcal{U}, \mathscr{F}) \to C^{p+1}(\mathcal{U}, \mathscr{F})$ is defined by:

$$(d\alpha)_{i_0,\dots,i_{p+1}} = \sum_{k=0}^{p+1} (-1)^k \, \alpha_{i_0,\dots,\widehat{i_k},\dots,i_{p+1}}\big|_{U_{i_0,\dots,i_{p+1}}}$$

where $\widehat{i_k}$ means the index i_k is omitted. A standard computation shows $d^2 = 0$, so $(C^\bullet(\mathcal{U}, \mathscr{F}), d)$ is a cochain complex.

Definition 6.1.11: Čech Cohomology

Let X be a topological space, $\mathcal{U}$ an open covering of X, and $\mathscr{F}$ a sheaf of abelian groups on X. The *Čech cohomology groups* of $\mathscr{F}$ with respect to $\mathcal{U}$ are:

$$\check{H}^p(\mathcal{U}, \mathscr{F}) = h^p\big(C^\bullet(\mathcal{U}, \mathscr{F})\big) = \frac{\ker\big(d : C^p \to C^{p+1}\big)}{\operatorname{im}\big(d : C^{p-1} \to C^p\big)}.$$

An important warning: $\check{H}^p(\mathcal{U}, -)$ is *not* in general a δ-functor. That is, a short exact sequence of sheaves need not induce a long exact sequence in Čech cohomology:

Example 6.1.12: Čech Cohomology is Not a δ-Functor

Suppose $\mathcal{U}$ consists of the single open set X itself. Then every intersection $U_{i_0,\dots,i_p}$ equals X, and there is only one index, so $C^p(\mathcal{U}, \mathscr{F}) = 0$ for $p > 0$ while $C^0(\mathcal{U}, \mathscr{F}) = \mathscr{F}(X) = \Gamma(X, \mathscr{F})$. Hence $\check{H}^0(\mathcal{U}, \mathscr{F}) = \Gamma(X, \mathscr{F})$ and $\check{H}^p(\mathcal{U}, \mathscr{F}) = 0$ for all $p > 0$.

Now let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be a short exact sequence of sheaves. If $\check{H}^*(\mathcal{U}, -)$ were a δ-functor, it would produce a long exact sequence, which in this case would read:

$$0 \longrightarrow \Gamma(X, \mathscr{F}') \longrightarrow \Gamma(X, \mathscr{F}) \longrightarrow \Gamma(X, \mathscr{F}'') \longrightarrow 0.$$

But $\Gamma(X, -)$ is not right exact in general — this is the whole point of sheaf cohomology! — so this need not be exact on the right. Hence $\check{H}^*$ cannot be a δ-functor for this covering.

This failure is not a defect of the theory but a feature of the choice of cover. As we shall see, choosing an appropriate cover (an affine cover for quasi-coherent sheaves on a separated scheme) restores the connection to derived-functor cohomology. Before proving this, let us see Čech cohomology in action:

Example 6.1.13: Čech Cohomology

1. *Differentials on $\mathbb{P}^1_k$.* Let $X = \mathbb{P}^1_k$, let $\mathscr{F} = \Omega_{X/k}$ be the sheaf of differentials, and let $\mathcal{U} = \{U, V\}$ be the standard affine cover with $U = D_+(x_0) \cong \operatorname{Spec} k[t]$ and $V = D_+(x_1) \cong \operatorname{Spec} k[u]$, where $t = x_1/x_0$ and $u = x_0/x_1 = 1/t$. The Čech complex has two terms:

$$C^0 = \Gamma(U, \Omega) \times \Gamma(V, \Omega), \qquad C^1 = \Gamma(U \cap V, \Omega).$$

The sections of the sheaf of differentials are:

$$\Gamma(U, \Omega) = k[t]\, dt, \qquad \Gamma(V, \Omega) = k[u]\, du, \qquad \Gamma(U \cap V, \Omega) = k[t, t^{-1}]\, dt.$$

On the overlap, the coordinate change gives $u = 1/t$, so $du = -t^{-2}\, dt$. The coboundary map

$d : C^0 \to C^1$ sends $(f(t)\,dt,\ g(u)\,du)$ to:

$$g(u)\,du\big|_{U\cap V} - f(t)\,dt\big|_{U\cap V} \;=\; \big(-t^{-2}g(1/t) - f(t)\big)\,dt.$$

Computing $\check{H}^0$: A cocycle $(f(t)\,dt,\ g(u)\,du) \in \ker d$ satisfies $f(t) = -t^{-2}g(1/t)$. The left side is a polynomial in t, while the right side is a polynomial in t^{-1} with no constant term (the t^{-2} factor shifts degrees). These can only agree if both are zero. Hence $\check{H}^0(\mathcal{U}, \Omega) = 0$, confirming that $\mathbb{P}^1_k$ has no nonzero global differential forms.

Computing $\check{H}^1$: The image of d consists of all Laurent polynomials of the form $\big(-t^{-2}g(1/t) - f(t)\big)\,dt$ where $f \in k[t]$ and $g \in k[u]$. Writing a general element of $k[t, t^{-1}]\,dt$ as $\sum_{n\in\mathbb{Z}} a_n t^n\,dt$, the image of d is spanned by all $t^n\,dt$ with $n \neq -1$: the terms $t^n\,dt$ for $n \geq 0$ come from $f(t)$, and the terms $t^n\,dt$ for $n \leq -2$ come from $g(u)$. The single "missing" basis element is $t^{-1}\,dt$. Therefore:

$$\check{H}^1(\mathcal{U}, \Omega) \cong k,$$

generated by the class of $t^{-1}\,dt$. This one-dimensional space reflects the genus of $\mathbb{P}^1$ being zero (recall that for a smooth projective curve of genus g, one has $\dim_k H^1(X, \Omega) = g$; here we are computing $H^1(X, \Omega)$ rather than $H^0(X, \Omega)$, which we just showed vanishes). We will revisit this when we discuss Serre duality.

2. *The circle.* Let $X = S^1$ with the Euclidean topology, and consider the constant sheaf $\underline{\mathbb{Z}}$. Take the cover $\mathcal{U} = \{U, V\}$ where U and V are two overlapping open arcs whose intersection $U \cap V = W_1 \coprod W_2$ consists of two disjoint small open intervals. Since U, V, W_1, and W_2 are all connected, the constant sheaf gives:

$$C^0 = \underline{\mathbb{Z}}(U) \times \underline{\mathbb{Z}}(V) = \mathbb{Z} \times \mathbb{Z}, \qquad C^1 = \underline{\mathbb{Z}}(U \cap V) = \underline{\mathbb{Z}}(W_1) \times \underline{\mathbb{Z}}(W_2) = \mathbb{Z} \times \mathbb{Z}.$$

The coboundary $d : C^0 \to C^1$ sends (a, b) to $(b-a,\ b-a)$, since on each connected component of $U \cap V$ the sections restrict to constants. Thus:

$$\ker d = \{(a, b) : b - a = 0\} \cong \mathbb{Z} \qquad \text{and} \qquad \mathrm{im} d = \{(c, c) : c \in \mathbb{Z}\} \cong \mathbb{Z}.$$

The cokernel $(\mathbb{Z} \times \mathbb{Z})/\{(c, c)\} \cong \mathbb{Z}$ (via $(m, n) \mapsto m - n$). Hence:

$$\check{H}^0(\mathcal{U}, \underline{\mathbb{Z}}) = \mathbb{Z}, \qquad \check{H}^1(\mathcal{U}, \underline{\mathbb{Z}}) = \mathbb{Z},$$

which agrees with the singular cohomology $H^*(S^1; \mathbb{Z})$. The class in $\check{H}^1$ is generated by the cocycle $(1, 0) \in \mathbb{Z} \times \mathbb{Z}$: the two components of $U \cap V$ see different "transition data," reflecting the non-trivial loop in S^1.

Let us now build towards the theorem that Čech cohomology computes derived-functor cohomology under appropriate hypotheses. The strategy is to show that the Čech complex arises from a resolution of the sheaf, and that this resolution can be compared to any injective resolution. The key condition is that each open set in the cover must be "cohomologically trivial" for the sheaf in question, so that the main obstruction comes from their combinations.

Lemma 6.1.14: 0th Čech Cohomology

Let X be a topological space, $\mathcal{U}$ an open covering, and $\mathscr{F}$ a sheaf of abelian groups on X. Then:

$$\check{H}^0(\mathcal{U}, \mathscr{F}) = \Gamma(X, \mathscr{F}).$$

***Proof*:**

By definition, $\check{H}^0(\mathcal{U}, \mathscr{F}) = \ker(d : C^0(\mathcal{U}, \mathscr{F}) \to C^1(\mathcal{U}, \mathscr{F}))$. An element $\alpha \in C^0$ is a collection $\{\alpha_i \in \mathscr{F}(U_i)\}_{i \in I}$. The coboundary condition $d\alpha = 0$ says that for each $i < j$,

$$(d\alpha)_{ij} = \alpha_j\big|_{U_i \cap U_j} - \alpha_i\big|_{U_i \cap U_j} = 0,$$

i.e. the sections α_i and α_j agree on the overlap $U_i \cap U_j$. Since $\mathscr{F}$ is a sheaf, the gluability axiom gives a unique global section $s \in \Gamma(X, \mathscr{F})$ with $s|_{U_i} = \alpha_i$ for all i. Hence $\ker d = \Gamma(X, \mathscr{F})$.

We next address the lack of a δ-functor by "sheafifying" the Čech construction. Let $X, \mathcal{U}$, and $\mathscr{F}$ be as above. For each $p \geq 0$, define the *sheaf of Čech p-cochains* by:

$$\mathscr{C}^p(\mathcal{U}, \mathscr{F}) = \prod_{i_0 < \cdots < i_p} (f_{i_0, \ldots, i_p})_* \big(\mathscr{F}\big|_{U_{i_0, \ldots, i_p}}\big)$$

where $f_{i_0, \ldots, i_p} : U_{i_0, \ldots, i_p} \hookrightarrow X$ is the inclusion and $(f_{i_0, \ldots, i_p})_*$ is the pushforward. The coboundary maps $d : \mathscr{C}^p \to \mathscr{C}^{p+1}$ are defined by the same alternating-sum formula as before. The key property of this construction is:

$$\Gamma\big(X, \mathscr{C}^p(\mathcal{U}, \mathscr{F})\big) = C^p(\mathcal{U}, \mathscr{F}),$$

so the Čech cochain groups are the global sections of the Čech sheaf complex.

Lemma 6.1.15: Čech Cohomology Resolution

Let X be a topological space, $\mathcal{U}$ an open covering, and $\mathscr{F}$ a sheaf of abelian groups. Then the augmented complex

$$0 \longrightarrow \mathscr{F} \xrightarrow{\epsilon} \mathscr{C}^0(\mathcal{U}, \mathscr{F}) \xrightarrow{d} \mathscr{C}^1(\mathcal{U}, \mathscr{F}) \xrightarrow{d} \mathscr{C}^2(\mathcal{U}, \mathscr{F}) \longrightarrow \cdots$$

is exact. That is, $\mathscr{C}^\bullet(\mathcal{U}, \mathscr{F})$ is a resolution of $\mathscr{F}$.

***Proof*:**

The augmentation $\epsilon : \mathscr{F} \to \mathscr{C}^0(\mathcal{U}, \mathscr{F})$ is defined by taking the product of the natural restriction maps $\mathscr{F} \to (f_i)_*(\mathscr{F}|_{U_i})$ for each $i \in I$. Exactness at $\mathscr{C}^0$ (i.e. $\ker d^0 = \operatorname{im} \epsilon$) follows from the sheaf axioms for $\mathscr{F}$, exactly as in lemma 6.1.14.

For the exactness of the complex at $\mathscr{C}^p$ for $p \geq 1$, it suffices to check on stalks. Let $x \in X$ and choose an index $j \in I$ such that $x \in U_j$. For each $p \geq 0$, we define a *contracting homotopy* $k : \mathscr{C}^p(\mathcal{U}, \mathscr{F})_x \to \mathscr{C}^{p-1}(\mathcal{U}, \mathscr{F})_x$ as follows. A germ $\alpha_x \in \mathscr{C}^p(\mathcal{U}, \mathscr{F})_x$ is represented by a section $\alpha \in \Gamma(W, \mathscr{C}^p(\mathcal{U}, \mathscr{F}))$ over some open neighborhood W of x that we may shrink until $W \subseteq U_j$.

For any p-tuple $i_0 < \cdots < i_{p-1}$, set:

$$(k\alpha)_{i_0,\dots,i_{p-1}} = \alpha_{j,i_0,\dots,i_{p-1}}\big|_W.$$

This is well-defined because $W \subseteq U_j$ implies $W \cap U_{i_0,\dots,i_{p-1}} = W \cap U_{j,i_0,\dots,i_{p-1}}$, so the section $\alpha_{j,i_0,\dots,i_{p-1}}$ is defined on W. Passing to germs at x gives the map k on stalks. A direct computation shows:

$$(dk + kd)(\alpha) = \alpha \qquad \text{for all } \alpha \in \mathscr{C}^p(\mathcal{U}, \mathscr{F})_x,\ p \geq 1.$$

Hence k is a chain homotopy between the identity and the zero map on the stalk complex $\mathscr{C}^\bullet(\mathcal{U}, \mathscr{F})_x$ in degrees $p \geq 1$. It follows that the cohomology of this stalk complex vanishes in positive degrees, which is precisely the exactness of the resolution.

Lemma 6.1.16: Čech Cohomology and Flasque Sheaves

Let X be a topological space, $\mathcal{U}$ an open covering, and $\mathscr{F}$ a flasque sheaf of abelian groups on X. Then $\check{H}^p(\mathcal{U}, \mathscr{F}) = 0$ for all $p > 0$.

***Proof*:**
By lemma 6.1.15, the complex $0 \to \mathscr{F} \to \mathscr{C}^\bullet(\mathcal{U}, \mathscr{F})$ is a resolution. Since $\mathscr{F}$ is flasque, the sheaves $\mathscr{C}^p(\mathcal{U}, \mathscr{F})$ are flasque for each $p \geq 0$: indeed, $\mathscr{F}|_{U_{i_0,\dots,i_p}}$ is flasque (restriction of a flasque sheaf is flasque), the pushforward $(f_{i_0,\dots,i_p})_*$ preserves flasque sheaves, and a product of flasque sheaves is flasque. Thus $\mathscr{C}^\bullet(\mathcal{U}, \mathscr{F})$ is a flasque resolution of $\mathscr{F}$, and we may use it to compute derived-functor cohomology:

$$H^p(X, \mathscr{F}) = h^p\big(\Gamma(X, \mathscr{C}^\bullet(\mathcal{U}, \mathscr{F}))\big) = h^p\big(C^\bullet(\mathcal{U}, \mathscr{F})\big) = \check{H}^p(\mathcal{U}, \mathscr{F}).$$

But $\mathscr{F}$ is flasque, so $H^p(X, \mathscr{F}) = 0$ for $p > 0$ by corollary 6.1.7. Hence $\check{H}^p(\mathcal{U}, \mathscr{F}) = 0$ for $p > 0$.

Lemma 6.1.17: Čech Cohomology Comparison Map

Let X be a topological space and $\mathcal{U}$ an open covering. Then for each $p \geq 0$, there is a natural map, functorial in $\mathscr{F}$:

$$\check{H}^p(\mathcal{U}, \mathscr{F}) \longrightarrow H^p(X, \mathscr{F}).$$

***Proof*:**
Choose an injective resolution $0 \to \mathscr{F} \to \mathscr{I}^\bullet$. We also have the Čech resolution $0 \to \mathscr{F} \to \mathscr{C}^\bullet(\mathcal{U}, \mathscr{F})$ from lemma 6.1.15. By the general comparison theorem for resolutions (see [11, chapter 27]), there exists a morphism of complexes $\mathscr{C}^\bullet(\mathcal{U}, \mathscr{F}) \to \mathscr{I}^\bullet$ extending the identity on $\mathscr{F}$, unique up to homotopy. Applying $\Gamma(X, -)$ and taking cohomology gives the desired maps $\check{H}^p(\mathcal{U}, \mathscr{F}) \to H^p(X, \mathscr{F})$, and functoriality follows from the naturality of the comparison.

We are now ready for the main result. The proof uses two crucial inputs: the Čech resolution is a resolution by sheaves whose global sections compute Čech cohomology, and for a separated scheme with an affine cover, the intersections of affine opens are again affine — which is precisely where the separatedness hypothesis enters.

Theorem 6.1.18: Čech Cohomology Equivalence

Let X be a Noetherian separated scheme, $\mathcal{U}$ an open affine covering of X, and $\mathscr{F}$ a quasi-coherent sheaf on X. Then for all $p \geq 0$, the natural maps

$$\check{H}^p(\mathcal{U}, \mathscr{F}) \longrightarrow H^p(X, \mathscr{F})$$

are isomorphisms.

***Proof*:**
For $p = 0$, both sides equal $\Gamma(X, \mathscr{F})$ by lemma 6.1.14. We proceed by induction on p.

Setup. By corollary 6.2.10, we can embed $\mathscr{F}$ into a flasque, quasi-coherent sheaf $\mathscr{G}$. Let $\mathscr{Q}$ be the quotient:

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{G} \longrightarrow \mathscr{Q} \longrightarrow 0.$$

Since $\mathbf{QCoh}(X)$ is an abelian category, the quotient $\mathscr{Q}$ is quasi-coherent.

Exactness of the Čech complex. For each ordered tuple $i_0 < \cdots < i_p$, the intersection $U_{i_0,\ldots,i_p}$ is affine: it is a finite intersection of affine open subsets of a separated scheme, hence affine by proposition 3.5.8. Since $\mathscr{F}$ is quasi-coherent, the sequence:

$$0 \longrightarrow \mathscr{F}(U_{i_0,\ldots,i_p}) \longrightarrow \mathscr{G}(U_{i_0,\ldots,i_p}) \longrightarrow \mathscr{Q}(U_{i_0,\ldots,i_p}) \longrightarrow 0$$

is exact by proposition 5.1.16 (the global section functor is exact for quasi-coherent sheaves on affine schemes). Taking products over all tuples, the Čech cochain complexes form a short exact sequence:

$$0 \longrightarrow C^\bullet(\mathcal{U}, \mathscr{F}) \longrightarrow C^\bullet(\mathcal{U}, \mathscr{G}) \longrightarrow C^\bullet(\mathcal{U}, \mathscr{Q}) \longrightarrow 0.$$

Long exact sequence. This short exact sequence of complexes induces a long exact sequence in Čech cohomology. Since $\mathscr{G}$ is flasque, $\check{H}^p(\mathcal{U}, \mathscr{G}) = 0$ for $p > 0$ by lemma 6.1.16. The long exact sequence gives:

$$0 \to \check{H}^0(\mathcal{U}, \mathscr{F}) \to \check{H}^0(\mathcal{U}, \mathscr{G}) \to \check{H}^0(\mathcal{U}, \mathscr{Q}) \to \check{H}^1(\mathcal{U}, \mathscr{F}) \to 0$$

and isomorphisms $\check{H}^p(\mathcal{U}, \mathscr{Q}) \xrightarrow{\sim} \check{H}^{p+1}(\mathcal{U}, \mathscr{F})$ for each $p \geq 1$.

Comparison. Consider the corresponding long exact sequence in derived-functor cohomology:

$$0 \to H^0(X, \mathscr{F}) \to H^0(X, \mathscr{G}) \to H^0(X, \mathscr{Q}) \to H^1(X, \mathscr{F}) \to 0 \to \cdots$$

where we used $H^p(X, \mathscr{G}) = 0$ for $p > 0$ (since $\mathscr{G}$ is flasque). The comparison maps from lemma 6.1.17 give a morphism between these two long exact sequences. For $p = 0$, the comparison maps are isomorphisms (both sides equal global sections). By the five lemma applied to the four-term exact sequence above, the comparison map $\check{H}^1(\mathcal{U}, \mathscr{F}) \to H^1(X, \mathscr{F})$ is an isomorphism.

For $p \geq 2$, the long exact sequences give isomorphisms $\check{H}^p(\mathcal{U}, \mathscr{Q}) \cong \check{H}^{p+1}(\mathcal{U}, \mathscr{F})$ and $H^p(X, \mathscr{Q}) \cong H^{p+1}(X, \mathscr{F})$. Since $\mathscr{Q}$ is also quasi-coherent, we may apply the inductive hypothesis to $\mathscr{Q}$: assuming the comparison map is an isomorphism for $\mathscr{Q}$ in degree p, it follows for $\mathscr{F}$ in degree $p + 1$. This completes the induction.

6.2 Vanishing Theorems

The computation of the previous section used two key vanishing phenomena: cohomology vanishes above the dimension of the space, and cohomology of quasi-coherent sheaves vanishes on affine schemes. In this section we prove both results. These are the foundations on which virtually all further cohomological arguments rest.

The first result — the *dimension theorem* — says that if X is a Noetherian topological space of dimension n, then $H^i(X, \mathscr{F}) = 0$ for all $i > n$ and all sheaves of abelian groups $\mathscr{F}$. This is the scheme-theoretic analogue of the general principle that cohomology should not exceed the dimension of the underlying space[4].

The second result — the *affine vanishing theorem* — says that on an affine Noetherian scheme $X = \operatorname{Spec} R$, all quasi-coherent sheaves have trivial higher cohomology: $H^i(X, \mathscr{F}) = 0$ for $i > 0$. This confirms the philosophy that cohomology measures the obstruction to gluing: on an affine scheme, there is no gluing to be done. Conversely, we shall show that this vanishing property *characterizes* affine schemes among Noetherian schemes.

6.2.1 Dimension Theorem for Schemes

We begin with two lemmas about the behaviour of cohomology on Noetherian spaces. The Noetherian hypothesis enters through a single but crucial property: on Noetherian topological spaces, filtered colimits of sheaves commute with the global section functor. This fails in general.

Lemma 6.2.1: Noetherian Space and Flasque Sheaves

Let X be a Noetherian topological space. Then a filtered colimit of flasque sheaves on X is flasque.

***Proof*:**
Let $(\mathscr{F}_\alpha)$ be a filtered (directed) system of flasque sheaves on X, and let $U \subseteq V$ be open subsets. For each α, the restriction map $\mathscr{F}_\alpha(V) \to \mathscr{F}_\alpha(U)$ is surjective. Since the system is filtered, the colimit functor on abelian groups is exact, so:

$$\varinjlim_\alpha \mathscr{F}_\alpha(V) \longrightarrow \varinjlim_\alpha \mathscr{F}_\alpha(U)$$

is surjective. Now, the key point is that on a Noetherian topological space, every open subset is quasi-compact, so filtered colimits of sheaves commute with the section functor:

$$\big(\varinjlim_\alpha \mathscr{F}_\alpha\big)(U) \;\cong\; \varinjlim_\alpha \mathscr{F}_\alpha(U)$$

for every open U. This is *not* true in general: for a presheaf the colimit is computed section-wise, but for a sheaf one may need to sheafify. On a Noetherian space, however, every open cover has a finite subcover, so any section of the sheafification already comes from some $\mathscr{F}_\alpha$, and

[4]Recall from [11, chapter 27] that the corresponding statement in singular cohomology is that $H^i(M; \mathbb{Z}) = 0$ for $i > \dim M$ when M is a manifold; in de Rham cohomology, it is the vanishing of the de Rham groups above the manifold dimension.

sheafification is unnecessary. Combining the two displayed equations, the restriction map

$$\big(\varinjlim_{\alpha} \mathscr{F}_\alpha\big)(V) \longrightarrow \big(\varinjlim_{\alpha} \mathscr{F}_\alpha\big)(U)$$

is surjective, so $\varinjlim_\alpha \mathscr{F}_\alpha$ is flasque.

Lemma 6.2.2: Colimit and Cohomology Commuting

Let X be a Noetherian topological space and $(\mathscr{F}_\alpha)$ a filtered direct system of sheaves of abelian groups. Then for each $i \geq 0$ there is a natural isomorphism:

$$\varinjlim_{\alpha} H^i(X, \mathscr{F}_\alpha) \;\cong\; H^i(X, \varinjlim_{\alpha} \mathscr{F}_\alpha).$$

***Proof*:**
For each $\mathscr{F}_\alpha$, choose a flasque resolution $0 \to \mathscr{F}_\alpha \to \mathscr{I}_\alpha^\bullet$ functorially (for instance, using the Godement resolution from Example 6.1.8(3), which is canonical). The cohomology groups are then:

$$H^i(X, \mathscr{F}_\alpha) = h^i\big(\Gamma(X, \mathscr{I}_\alpha^\bullet)\big).$$

Since filtered colimits are exact in the category of abelian sheaves, taking the colimit of these resolutions yields an exact sequence:

$$0 \longrightarrow \varinjlim_{\alpha} \mathscr{F}_\alpha \longrightarrow \varinjlim_{\alpha} \mathscr{I}_\alpha^\bullet.$$

By lemma 6.2.1, since X is Noetherian, each term $\varinjlim_\alpha \mathscr{I}_\alpha^p$ is flasque. Hence $\varinjlim_\alpha \mathscr{I}_\alpha^\bullet$ is a flasque resolution of $\varinjlim_\alpha \mathscr{F}_\alpha$, and we may use it to compute cohomology:

$$H^i(X, \varinjlim_{\alpha} \mathscr{F}_\alpha) = h^i\big(\Gamma(X, \varinjlim_{\alpha} \mathscr{I}_\alpha^\bullet)\big).$$

On the other hand, filtered colimits of abelian groups commute with the cohomology functor h^i (since filtered colimits are exact):

$$\varinjlim_{\alpha} H^i(X, \mathscr{F}_\alpha) = \varinjlim_{\alpha} h^i\big(\Gamma(X, \mathscr{I}_\alpha^\bullet)\big) \cong h^i\big(\varinjlim_{\alpha} \Gamma(X, \mathscr{I}_\alpha^\bullet)\big).$$

It remains to show $\varinjlim_\alpha \Gamma(X, \mathscr{I}_\alpha^p) \cong \Gamma(X, \varinjlim_\alpha \mathscr{I}_\alpha^p)$ for each p. This holds because global sections commute with filtered colimits of sheaves on Noetherian spaces (the same property used in the proof of lemma 6.2.1). Combining these identifications completes the proof.

Corollary 6.2.3: Direct Sum and Cohomology Commuting

Let X be a Noetherian topological space and $(\mathscr{F}_\alpha)_{\alpha \in A}$ a family of sheaves of abelian groups. Then for each $i \geq 0$:

$$\bigoplus_{\alpha \in A} H^i(X, \mathscr{F}_\alpha) \;\cong\; H^i\Big(X, \bigoplus_{\alpha \in A} \mathscr{F}_\alpha\Big).$$

***Proof*:**
A direct sum is a filtered colimit of finite partial sums, and cohomology commutes with finite direct sums (since it is an additive functor). Hence the result follows from lemma 6.2.2.

The following result shows that cohomology computations on a closed subset can be performed on the ambient space by pushing forward:

Proposition 6.2.4: Cohomology of Closed Subsets

Let X be a topological space, $Y \subseteq X$ a closed subset, $j : Y \to X$ the inclusion, and $\mathscr{F}$ a sheaf of abelian groups on Y. Then for all $i \geq 0$:

$$H^i(Y, \mathscr{F}) \cong H^i(X, j_*\mathscr{F}).$$

***Proof*:**
Let $0 \to \mathscr{F} \to \mathscr{I}^\bullet$ be a flasque resolution of $\mathscr{F}$ on Y. Since the pushforward along a closed immersion preserves flasque sheaves — indeed, for $U \subseteq V$ open in X, the surjection $\mathscr{I}^p(V \cap Y) \twoheadrightarrow \mathscr{I}^p(U \cap Y)$ is precisely the restriction map for $j_*\mathscr{I}^p$ — the sequence $0 \to j_*\mathscr{F} \to j_*\mathscr{I}^\bullet$ is a flasque resolution of $j_*\mathscr{F}$ on X. Now:

$$H^i(X, j_*\mathscr{F}) = h^i\big(\Gamma(X, j_*\mathscr{I}^\bullet)\big) = h^i\big(\Gamma(Y, \mathscr{I}^\bullet)\big) = H^i(Y, \mathscr{F}),$$

where the middle equality uses $\Gamma(X, j_*\mathscr{G}) = \mathscr{G}(j^{-1}(X)) = \mathscr{G}(Y) = \Gamma(Y, \mathscr{G})$ for any sheaf $\mathscr{G}$ on Y.

By virtue of this result, we shall often write $\mathscr{F}$ in place of $j_*\mathscr{F}$ when Y is a closed subset of X, without risk of ambiguity: the cohomology groups are the same. With these tools, we can prove our first main vanishing theorem:

Theorem 6.2.5: Noetherian Space and Dimension

Let X be a Noetherian topological space of dimension n. Then for all sheaves of abelian groups $\mathscr{F}$ on X:

$$H^i(X, \mathscr{F}) = 0 \qquad \text{for all } i > n.$$

***Proof*:**
We shall need the following notation. For a closed subset $Y \subseteq X$ with inclusion $j : Y \hookrightarrow X$, write $\mathscr{F}_Y = j_*(j^{-1}\mathscr{F})$ for the "restriction of $\mathscr{F}$ to Y, pushed back to X." For an open subset $U \subseteq X$ with inclusion $i : U \hookrightarrow X$, write $\mathscr{F}_U = i_!(i^{-1}\mathscr{F})$, the extension by zero of $\mathscr{F}|_U$. If $U = X \setminus Y$, checking on stalks gives a short exact sequence:

$$0 \longrightarrow \mathscr{F}_U \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}_Y \longrightarrow 0.$$

We prove the theorem by a double induction: on the dimension of X and on the number of irreducible components.

Step 1: Reduction to X irreducible. Suppose X is reducible. Let Y be an irreducible component of X and $U = X \setminus Y$. The short exact sequence above gives a long exact sequence:

$$\cdots \to H^i(X, \mathscr{F}_U) \to H^i(X, \mathscr{F}) \to H^i(X, \mathscr{F}_Y) \to \cdots$$

It suffices to show vanishing for $\mathscr{F}_Y$ and $\mathscr{F}_U$ separately. For $\mathscr{F}_Y$: since Y is closed and irreducible with $\dim Y \leq n$, proposition 6.2.4 gives $H^i(X, \mathscr{F}_Y) = H^i(Y, j^{-1}\mathscr{F})$, which vanishes for $i > n$ by the inductive hypothesis applied to the irreducible space Y. For $\mathscr{F}_U$: the support of $\mathscr{F}_U$ is contained in $\overline{U}$, which is a closed subset with strictly fewer irreducible components than X (we have removed the component Y). By proposition 6.2.4 applied to $\overline{U}$ and induction on the number of irreducible components, we get the required vanishing. Hence we may assume X is irreducible.

Step 2: Base case $\dim X = 0$. If X is irreducible of dimension 0, its only open subsets are $\emptyset$ and X itself (a proper closed irreducible subset would give $\dim X \geq 1$). Hence the global section functor $\Gamma(X, -) : \mathbf{Ab}(X) \to \mathbf{Ab}$ is an equivalence of categories (a sheaf on X is determined by a single abelian group). In particular, $\Gamma(X, -)$ is exact, so all its derived functors vanish: $H^i(X, \mathscr{F}) = 0$ for $i > 0$.

Step 3: gluing. To verify the maps glue on the overlap $U_i \cap U_j$, we must check that $\varphi_i = \varphi_j$ when applied to the local generators $d(x_k/x_i)$. By definition, φ_i yields $\frac{1}{x_i^2}(x_i\, e_k - x_k\, e_i)$. To evaluate φ_j, we first use the quotient rule to express the generator in the U_j basis:

$$d\left(\frac{x_k}{x_i}\right) = d\left(\frac{x_k/x_j}{x_i/x_j}\right) = \frac{x_j}{x_i}\, d\left(\frac{x_k}{x_j}\right) - \frac{x_k x_j}{x_i^2}\, d\left(\frac{x_i}{x_j}\right).$$

Applying the linear map φ_j to this expression yields:

$$\begin{aligned} \varphi_j\left(d\left(\frac{x_k}{x_i}\right)\right) &= \frac{x_j}{x_i}\left[\frac{1}{x_j^2}(x_j\, e_k - x_k\, e_j)\right] - \frac{x_k x_j}{x_i^2}\left[\frac{1}{x_j^2}(x_j\, e_i - x_i\, e_j)\right] \\ &= \left(\frac{1}{x_i}\, e_k - \frac{x_k}{x_i x_j}\, e_j\right) - \left(\frac{x_k}{x_i^2}\, e_i - \frac{x_k}{x_i x_j}\, e_j\right). \end{aligned}$$

The e_j terms perfectly cancel, leaving $\frac{1}{x_i}\, e_k - \frac{x_k}{x_i^2}\, e_i$, which is exactly $\frac{1}{x_i^2}(x_i\, e_k - x_k\, e_i)$. Hence $\varphi_i|_{U_i \cap U_j} = \varphi_j|_{U_i \cap U_j}$, and the local isomorphisms glue to a global isomorphism $\Omega_{X/Y} \cong \widetilde{M}$.

Step 4: Subsheaves of $\underline{\mathbb{Z}}_U$. Let $U \subseteq X$ be open and $\mathcal{K} \subseteq \underline{\mathbb{Z}}_U$ a subsheaf. If $\mathcal{K} = 0$, there is nothing to prove. Otherwise, for each $x \in U$, the stalk $\mathcal{K}_x$ is a subgroup of $\mathbb{Z}$, hence either 0 or $d_x\mathbb{Z}$ for some positive integer d_x. Let d be the smallest positive integer occurring among the stalks. There is a non-empty open subset $V \subseteq U$ on which $\mathcal{K}|_V \cong d \cdot \underline{\mathbb{Z}}|_V \cong \underline{\mathbb{Z}}_V$ (as a subsheaf of $\underline{\mathbb{Z}}|_V$). Thus $\mathcal{K}_V \cong \underline{\mathbb{Z}}_V$, and we have:

$$0 \longrightarrow \underline{\mathbb{Z}}_V \longrightarrow \mathcal{K} \longrightarrow \mathcal{K}/\underline{\mathbb{Z}}_V \longrightarrow 0.$$

The quotient $\mathcal{K}/\underline{\mathbb{Z}}_V$ is supported on the closure $\overline{U \setminus V}$, which is a proper closed subset of X. Since X is irreducible, $\dim \overline{U \setminus V} < n$. By proposition 6.2.4 and the inductive hypothesis on dimension, we get $H^i(X, \mathcal{K}/\underline{\mathbb{Z}}_V) = 0$ for $i \geq n$. From the long exact sequence, it suffices to prove vanishing for $\underline{\mathbb{Z}}_V$.

Step 5: Completing the proof for $\underline{\mathbb{Z}}_U$. It remains to show that for any open subset $U \subseteq X$, we have $H^i(X, \underline{\mathbb{Z}}_U) = 0$ for $i > n$. Let $Y = X \setminus U$ and consider the exact sequence:

$$0 \longrightarrow \underline{\mathbb{Z}}_U \longrightarrow \underline{\mathbb{Z}} \longrightarrow \underline{\mathbb{Z}}_Y \longrightarrow 0.$$

Since X is irreducible, the constant sheaf $\underline{\mathbb{Z}}$ on X is flasque (every non-empty open subset of an irreducible space is connected, so the constant sheaf has the same sections on every non-empty open; see Example 6.1.8(2)). Hence $H^i(X, \underline{\mathbb{Z}}) = 0$ for all $i > 0$. Furthermore, Y is a proper closed subset of the irreducible space X, so $\dim Y < n$. By proposition 6.2.4 and the inductive hypothesis on dimension, $H^i(X, \underline{\mathbb{Z}}_Y) = 0$ for $i \geq n$. The long exact sequence now gives $H^i(X, \underline{\mathbb{Z}}_U) = 0$ for $i > n$, completing the proof.

6.2.2 Vanishing Theorem for Affine Schemes

We now turn to the second fundamental vanishing result: on an affine Noetherian scheme $X = \operatorname{Spec} R$, all quasi-coherent sheaves have trivial higher cohomology. The key insight is that if I is an injective R-module, then the associated sheaf $\widetilde{I}$ on $\operatorname{Spec} R$ is flasque. This is not obvious: the sections of $\widetilde{I}$ on a general open set involve a limit over distinguished opens, and there is no a priori reason for restriction maps to be surjective. The proof requires the Noetherian hypothesis in an essential way, via the Artin–Rees lemma and Noetherian induction.

Let R be a Noetherian ring, M an R-module, and $\mathfrak{a} \subseteq R$ an ideal. The $\mathfrak{a}$-*torsion submodule* of M is:

$$\Gamma_{\mathfrak{a}}(M) = \left\{ m \in M \;\middle|\; \mathfrak{a}^k m = 0 \text{ for some } k > 0 \right\}.$$

Lemma 6.2.6: $\mathfrak{a}$-Torsion of an Injective Module is Injective

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, and I an injective R-module. Then the submodule $J = \Gamma_{\mathfrak{a}}(I)$ is injective.

***Proof*:**
By Baer's criterion, it suffices to show that for any ideal $\mathfrak{b} \subseteq R$ and any homomorphism $\varphi : \mathfrak{b} \to J$, there exists a homomorphism $\psi : R \to J$ extending φ.

Since R is Noetherian, $\mathfrak{b}$ is finitely generated, and every element of J is annihilated by some power of $\mathfrak{a}$. Hence there exists $n > 0$ such that $\mathfrak{a}^n \cdot \varphi(\mathfrak{b}) = 0$, i.e. $\varphi(\mathfrak{a}^n \mathfrak{b}) = 0$. By the Artin–Rees lemma (see [**NateClassicalAlgGeo**]) applied to the inclusion $\mathfrak{b} \subseteq R$ and the $\mathfrak{a}$-adic filtration, there exists $n' \geq n$ such that $\mathfrak{a}^n \mathfrak{b} \supseteq \mathfrak{b} \cap \mathfrak{a}^{n'}$. Hence $\varphi(\mathfrak{b} \cap \mathfrak{a}^{n'}) = 0$, and φ factors through $\mathfrak{b}/(\mathfrak{b} \cap \mathfrak{a}^{n'})$.

Consider the commutative diagram:

$$\begin{tikzcd}
R \arrow[r, two heads] & R/\mathfrak{a}^{n'} \arrow[dr, dashed, "\psi'"] & & \\
\mathfrak{b} \arrow[u, hook] \arrow[r, two heads] & \mathfrak{b}/(\mathfrak{b} \cap \mathfrak{a}^{n'}) \arrow[u, hook] \arrow[r] & J \arrow[r, hook] & I
\end{tikzcd}$$

Since I is injective, the composed map $\mathfrak{b}/(\mathfrak{b} \cap \mathfrak{a}^{n'}) \to I$ extends to a map $\psi' : R/\mathfrak{a}^{n'} \to I$. The image of ψ' is annihilated by $\mathfrak{a}^{n'}$, so it is contained in $J = \Gamma_{\mathfrak{a}}(I)$. Composing with the natural projection $R \twoheadrightarrow R/\mathfrak{a}^{n'}$, we obtain the required extension $\psi : R \to J$.

Lemma 6.2.7: Localization of Injective Modules is Surjective

Let R be a Noetherian ring, I an injective R-module, and $f \in R$. Then the natural localization map $I \to I_f$ is surjective.

***Proof*:**
Let $\varphi : I \to I_f$ be the natural map and let $x \in I_f$. By definition of localization, there exist $y \in I$ and $n \geq 0$ such that $x = \varphi(y)/f^n$.

For each $i > 0$, let $\mathfrak{b}_i = \mathrm{Ann}_R(f^i)$. Since R is Noetherian, the ascending chain $\mathfrak{b}_1 \subseteq \mathfrak{b}_2 \subseteq \cdots$ stabilizes: there exists m such that $\mathfrak{b}_{m+j} = \mathfrak{b}_m$ for all $j \geq 0$. Define a homomorphism $\psi : (f^{n+m}) \to I$ by $f^{n+m} \mapsto f^m y$. This is well-defined: if $rf^{n+m} = 0$, then $r \in \mathfrak{b}_{n+m} = \mathfrak{b}_m$, so $rf^m = 0$, hence $rf^m y = 0$.

Since I is injective, Baer's criterion extends ψ to a map $\Psi : R \to I$. Let $z = \Psi(1)$. Then:

$$f^{n+m} z = f^{n+m}\Psi(1) = \Psi(f^{n+m}) = \psi(f^{n+m}) = f^m y,$$

so in I_f we have $\varphi(z) = z/1 = f^m y/f^{n+m} = y/f^n = x$. Hence φ is surjective.

Lemma 6.2.8: Associated Sheaf of Injective Module is Flasque

Let R be a Noetherian ring and I an injective R-module. Then the associated sheaf $\widetilde{I}$ on $X = \operatorname{Spec} R$ is flasque.

***Proof*:**
We proceed by Noetherian induction (cf. theorem 0.3.3) on the closed subset $Y = \mathrm{supp}(\widetilde{I})$. If Y is a single closed point, then $\widetilde{I}$ is a skyscraper sheaf, which is evidently flasque.

In the general case, it suffices to show that for any open set $U \subseteq X$, the restriction $\Gamma(X, \widetilde{I}) \to \Gamma(U, \widetilde{I})$ is surjective. If $Y \cap U = \emptyset$, then $\Gamma(U, \widetilde{I}) = 0$ and there is nothing to prove. Otherwise, choose $f \in R$ such that the distinguished open $D(f)$ satisfies $D(f) \subseteq U$ and $D(f) \cap Y \neq \emptyset$. Let $Z = X \setminus D(f)$, and for any open W let $\Gamma_Z(W, \widetilde{I})$ denote the sections of $\widetilde{I}$ over W with support in Z.

Given a section $s \in \Gamma(U, \widetilde{I})$, its restriction to $D(f)$ yields an element $s' \in \Gamma(D(f), \widetilde{I})$. By the identification $\Gamma(D(f), \widetilde{I}) \cong I_f$ (from proposition 2.1.12) and the surjectivity of $I \to I_f$ (lemma 6.2.7), there exists $t \in I = \Gamma(X, \widetilde{I})$ with $t|_{D(f)} = s'$. Let t' be the restriction of t to U. Then $s - t'$ restricts to zero on $D(f)$, so it has support in Z: $s - t' \in \Gamma_Z(U, \widetilde{I})$.

It remains to show that $\Gamma_Z(X, \widetilde{I}) \to \Gamma_Z(U, \widetilde{I})$ is surjective. Let $J = \Gamma_{\mathfrak{a}}(I)$ where $\mathfrak{a} = (f)$. Then $\Gamma_Z(X, \widetilde{I}) = J$ (the sections of $\widetilde{I}$ over all of X supported on $Z = V(f)$ are precisely the f-torsion elements of I). By lemma 6.2.6, J is an injective R-module. The support of $\widetilde{J}$ is contained in $Y \cap Z$, which is strictly contained in Y (since $D(f) \cap Y \neq \emptyset$, we have removed a non-empty open piece of Y). By the Noetherian induction hypothesis, $\widetilde{J}$ is flasque. Since $\Gamma(U, \widetilde{J}) = \Gamma_Z(U, \widetilde{I})$, the flasqueness of $\widetilde{J}$ gives the required surjectivity.

Theorem 6.2.9: Affine Noetherian Scheme: Trivial Cohomology

Let $X = \operatorname{Spec} R$ be the spectrum of a Noetherian ring R, and let $\mathscr{F}$ be a quasi-coherent sheaf on X. Then:

$$H^i(X, \mathscr{F}) = 0 \qquad \text{for all } i > 0.$$

***Proof*:**
Since $\mathscr{F}$ is quasi-coherent on the affine scheme $X = \operatorname{Spec} R$, there exists an R-module M with $\mathscr{F} \cong \widetilde{M}$ (by corollary 5.1.12). Choose an injective resolution $0 \to M \to I^\bullet$ in the category of R-modules. The associated-sheaf functor $\widetilde{(-)}$ is exact (by proposition 5.1.14), so:

$$0 \longrightarrow \widetilde{M} \longrightarrow \widetilde{I^\bullet}$$

is an exact sequence of $\mathcal{O}_X$-modules. By lemma 6.2.8, each $\widetilde{I^p}$ is flasque. Hence $\widetilde{I^\bullet}$ is a flasque resolution of $\mathscr{F} = \widetilde{M}$, and we may use it to compute cohomology:

$$H^i(X, \mathscr{F}) = h^i\big(\Gamma(X, \widetilde{I^\bullet})\big) = h^i(I^\bullet).$$

But $I^\bullet$ is an injective resolution of M, so its cohomology is $h^0(I^\bullet) = M$ and $h^i(I^\bullet) = 0$ for $i > 0$. Hence:

$$H^0(X, \mathscr{F}) = M = \Gamma(X, \mathscr{F}) \qquad \text{and} \qquad H^i(X, \mathscr{F}) = 0 \text{ for } i > 0.$$

As a useful corollary, we can embed any quasi-coherent sheaf on a Noetherian scheme into a quasi-coherent flasque sheaf. This is the tool we used in theorem 6.1.18 to compare Čech and derived-functor cohomology:

Corollary 6.2.10: Injective Embedding of Sheaves On Noetherian Schemes

Let X be a Noetherian scheme and $\mathscr{F}$ a quasi-coherent sheaf on X. Then $\mathscr{F}$ can be embedded in a quasi-coherent flasque sheaf.

***Proof*:**
Since X is Noetherian, it is quasi-compact, so it has a finite affine open cover $X = \bigcup_{j=1}^r U_j$ with $U_j = \operatorname{Spec} R_j$. For each j, let $\mathscr{F}|_{U_j} \cong \widetilde{M_j}$ for the R_j-module $M_j = \Gamma(U_j, \mathscr{F})$. Embed each M_j into an injective R_j-module I_j, and define:

$$\mathscr{G} = \bigoplus_{j=1}^{r} (\iota_j)_*(\widetilde{I}_j)$$

where $\iota_j : U_j \hookrightarrow X$ is the inclusion. For each j, the embedding $\mathscr{F}|_{U_j} \hookrightarrow \widetilde{I}_j$ induces $\mathscr{F} \to (\iota_j)_*(\widetilde{I}_j)$ (by adjunction of restriction and pushforward). Taking the direct sum gives an injective morphism $\mathscr{F} \hookrightarrow \mathscr{G}$.

Each $\widetilde{I}_j$ is flasque by lemma 6.2.8 and quasi-coherent on U_j. The pushforward $(\iota_j)_*(\widetilde{I}_j)$ is flasque (the pushforward of a flasque sheaf along any continuous map is flasque) and quasi-coherent by proposition 5.1.20[a]. A finite direct sum of flasque quasi-coherent sheaves is flasque and quasi-coherent, so $\mathscr{G}$ has the required properties.

[a] This is where the Noetherian hypothesis enters: the pushforward of a quasi-coherent sheaf along an open immersion is quasi-coherent when the target is Noetherian (or more generally, when the morphism is quasi-compact and quasi-separated), but can fail for non-Noetherian schemes.

The vanishing theorem tells us that the higher cohomology of quasi-coherent sheaves detects non-affineness. Remarkably, the converse is also true:

Theorem 6.2.11: Characterizing Affine Schemes via Cohomology

Let X be a Noetherian scheme. Then the following are equivalent:

1. X is affine.
2. $H^i(X, \mathscr{F}) = 0$ for all $i > 0$ and all quasi-coherent sheaves $\mathscr{F}$ on X.
3. $H^1(X, \mathscr{I}) = 0$ for all coherent sheaves of ideals $\mathscr{I} \subseteq \mathcal{O}_X$.

***Proof*:**
(1) ⇒ (2): This is theorem 6.2.9.

(2) ⇒ (3): Every coherent sheaf of ideals is quasi-coherent, so this is immediate.

(3) ⇒ (1): We verify the hypotheses of corollary 3.1.15: we shall find elements $f_1, \ldots, f_r \in R = \Gamma(X, \mathcal{O}_X)$ such that each X_{f_j} is affine and $(f_1, \ldots, f_r) = R$.

Step A: Finding the f_j. Let $\mathfrak{p}$ be a closed point of X and let U be an affine open neighborhood of $\mathfrak{p}$. Let $Y = X \setminus U$. Then Y and $Y \cup \{\mathfrak{p}\}$ are closed subsets, and their ideal sheaves $\mathscr{I}_Y \supseteq \mathscr{I}_{Y\cup\{\mathfrak{p}\}}$ fit in a short exact sequence:

$$0 \longrightarrow \mathscr{I}_{Y\cup\{\mathfrak{p}\}} \longrightarrow \mathscr{I}_Y \longrightarrow \kappa(\mathfrak{p}) \longrightarrow 0,$$

where $\kappa(\mathfrak{p}) = \mathcal{O}_{X,\mathfrak{p}}/\mathfrak{m}_\mathfrak{p}$ is the skyscraper sheaf at $\mathfrak{p}$. The long exact sequence gives:

$$\Gamma(X, \mathscr{I}_Y) \longrightarrow \Gamma(X, \kappa(\mathfrak{p})) \longrightarrow H^1(X, \mathscr{I}_{Y\cup\{\mathfrak{p}\}}) = 0,$$

where the last group vanishes by hypothesis (3). Hence the map $\Gamma(X, \mathscr{I}_Y) \to \kappa(\mathfrak{p})$ is surjective, so there exists $f \in \Gamma(X, \mathscr{I}_Y) \subseteq R$ with $f_\mathfrak{p} \equiv 1 \pmod{\mathfrak{m}_\mathfrak{p}}$.

By construction, f vanishes on Y, so $\mathfrak{p} \in X_f \subseteq U$. Furthermore, $X_f = U_{\bar{f}}$ where $\bar{f}$ is the image of f in $\Gamma(U, \mathcal{O}_U)$, so X_f is a distinguished open of the affine scheme U, hence affine.

Since every closed point of X has an affine neighborhood of the form X_f with $f \in R$, and X is Noetherian (hence quasi-compact) with dense closed points (by lemma 2.4.4), we can cover X by finitely many such opens: $X = X_{f_1} \cup \cdots \cup X_{f_r}$.

Step B: Showing $(f_1, \ldots, f_r) = R$. The elements $f_1, \ldots, f_r \in R$ define a morphism of sheaves $\alpha : \mathcal{O}_X^r \to \mathcal{O}_X$ by $(a_1, \ldots, a_r) \mapsto \sum_j f_j a_j$. Since the X_{f_j} cover X, the map α is surjective on stalks, hence surjective. Let $\mathscr{F} = \ker \alpha$:

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathcal{O}_X^r \xrightarrow{\alpha} \mathcal{O}_X \longrightarrow 0.$$

We need to show $H^1(X, \mathscr{F}) = 0$, for then $\Gamma(X, \mathcal{O}_X^r) \to \Gamma(X, \mathcal{O}_X)$ is surjective, giving $1 \in \operatorname{im}(\alpha)$, i.e. $\sum a_j f_j = 1$ for some $a_j \in R$, proving $(f_1, \ldots, f_r) = R$.

Filter $\mathscr{F}$ by:

$$0 = \mathscr{F} \cap \mathcal{O}_X^0 \subseteq \mathscr{F} \cap \mathcal{O}_X^1 \subseteq \cdots \subseteq \mathscr{F} \cap \mathcal{O}_X^r = \mathscr{F}$$

where $\mathcal{O}_X^j$ denotes the first j summands of $\mathcal{O}_X^r$. Each successive quotient $(\mathscr{F} \cap \mathcal{O}_X^{j+1})/(\mathscr{F} \cap \mathcal{O}_X^j)$ embeds into $\mathcal{O}_X^{j+1}/\mathcal{O}_X^j \cong \mathcal{O}_X$, so it is a coherent sheaf of ideals. By hypothesis (3), its H^1 vanishes. Climbing up the filtration via the long exact sequence, we deduce $H^1(X, \mathscr{F}) = 0$.

Hence $(f_1, \ldots, f_r) = R$, and by corollary 3.1.15, X is affine.

6.3 Application: Sheaf Cohomology of Projective Spaces

Let us now compute the cohomology of twisted sheaves on projective space — the single most important explicit computation in the cohomology of schemes. Let R be a Noetherian ring, $S_\bullet = R[x_0, \ldots, x_n]$ with the standard grading, and $X = \operatorname{Proj} S_\bullet = \mathbb{P}^n_R$. Recall from definition 5.3.4 that the Serre twisting sheaf $\mathcal{O}_X(1)$ is the invertible sheaf whose sections on the basic open $D_+(x_i)$ are the degree-1 elements of $(S_{x_i})_0$. For any sheaf $\mathscr{F}$ of $\mathcal{O}_X$-modules, we write:

$$\Gamma_*(\mathscr{F}) = \bigoplus_{d \in \mathbb{Z}} \Gamma(X, \mathscr{F}(d)),$$

which carries a natural structure of graded $S_\bullet$-module (multiplication by x_i corresponds to the natural map $\mathscr{F}(d) \to \mathscr{F}(d+1)$).

Theorem 6.3.1: Twisted Sheaf Cohomology

Let R be a Noetherian ring and let $X = \mathbb{P}^n_R$, with $n \geq 1$. Then:

1. $H^0(X, \mathcal{O}_X(d)) \cong (S_\bullet)_d$ for all $d \geq 0$, and $H^0(X, \mathcal{O}_X(d)) = 0$ for $d < 0$.
2. $H^n(X, \mathcal{O}_X(-n-1)) \cong R$.
3. The natural multiplication map

$$H^0(X, \mathcal{O}_X(d)) \times H^n(X, \mathcal{O}_X(-d-n-1)) \longrightarrow H^n(X, \mathcal{O}_X(-n-1)) \cong R$$

is a perfect pairing of finitely generated free R-modules, for each $d \geq 0$.

4. $H^i(X, \mathcal{O}_X(d)) = 0$ for $0 < i < n$ and all $d \in \mathbb{Z}$.

In summary:

$$H^i\big(\mathbb{P}^n_R,\, \mathcal{O}_{\mathbb{P}^n_R}(d)\big) = \begin{cases} (R[x_0, \ldots, x_n])_d & i = 0,\ d \geq 0, \\ \left(\dfrac{1}{x_0 x_1 \cdots x_n} R[x_0^{-1}, \ldots, x_n^{-1}]\right)_d & i = n,\ d \leq -(n+1), \\ 0 & \text{otherwise.} \end{cases}$$

6.3.2 Remark: Sheaf vs. Singular Cohomology The reader should note that this result is fundamentally different from the singular cohomology of complex projective space. When $R = \mathbb{C}$ and $d = 0$, the theorem says $H^i(\mathbb{P}^n_{\mathbb{C}}, \mathcal{O}_{\mathbb{P}^n_{\mathbb{C}}}) = 0$ for all $i > 0$, whereas the singular cohomology $H^i_{\text{sing}}(\mathbb{CP}^n; \mathbb{Z})$ is nontrivial: it equals $\mathbb{Z}$ in each even degree $0 \le i \le 2n$ and vanishes in odd degrees (see [6, section 4.1]). The discrepancy arises because sheaf cohomology with coefficients in $\mathcal{O}_X$ measures obstructions to extending *holomorphic* (or algebraic) sections, not topological ones. Singular cohomology would instead correspond to sheaf cohomology with coefficients in the constant sheaf $\underline{\mathbb{Z}}$, which we shall connect via GAGA in section 6.9.

***Proof*:**

We use Čech cohomology with respect to the standard affine cover. Let $\mathscr{F} = \bigoplus_{d\in\mathbb{Z}} \mathcal{O}_X(d)$. By corollary 6.2.3, cohomology commutes with direct sums on Noetherian spaces, so $H^i(X, \mathscr{F}) = \bigoplus_{d\in\mathbb{Z}} H^i(X, \mathcal{O}_X(d))$. Hence it suffices to compute the cohomology of $\mathscr{F}$ as a graded module and then read off each graded piece.

Let $\mathcal{U} = \{U_0, \ldots, U_n\}$ where $U_j = D_+(x_j)$ is the standard affine cover of $\operatorname{Proj} S_\bullet$. By theorem 6.1.18, the Čech cohomology $\check{H}^p(\mathcal{U}, \mathscr{F})$ computes the derived-functor cohomology $H^p(X, \mathscr{F})$. The sections of $\mathscr{F}$ on each intersection are:

$$\mathscr{F}(U_{i_0,\ldots,i_p}) = S_{x_{i_0}\cdots x_{i_p}}$$

(the localization of $S_\bullet$ at the product $x_{i_0}\cdots x_{i_p}$, retaining only degree-0 elements for each summand $\mathcal{O}_X(d)$, but over all d we obtain the full localization). The Čech complex is therefore:

$$C^\bullet(\mathcal{U}, \mathscr{F}): \quad \prod_i S_{x_i} \xrightarrow{d^0} \prod_{i<j} S_{x_i x_j} \xrightarrow{d^1} \cdots \xrightarrow{d^{n-1}} S_{x_0 x_1 \cdots x_n}.$$

Proof of (1): Computing H^0. By lemma 6.1.14, $H^0(X, \mathscr{F}) = \Gamma(X, \mathscr{F})$. The global sections of $\mathscr{F}$ are the elements lying in all localizations simultaneously: $\Gamma(X, \mathscr{F}) = \bigcap_{i=0}^n S_{x_i} \cap S = S$ by proposition 2.3.6. The graded piece of degree d is $(S_\bullet)_d$, which is a free R-module with basis $\{x_0^{m_0}\cdots x_n^{m_n} | m_i \ge 0, \sum m_i = d\}$ when $d \ge 0$, and is zero when $d < 0$ (there are no monomials of negative degree in a polynomial ring). Hence $H^0(X, \mathcal{O}_X(d)) \cong (S_\bullet)_d$ for $d \ge 0$ and $H^0(X, \mathcal{O}_X(d)) = 0$ for $d < 0$.

Proof of (2): Computing H^n. The group $H^n(X, \mathscr{F})$ is the cokernel of the last differential:

$$d^{n-1}: \prod_{j=0}^n S_{x_0\cdots\widehat{x_j}\cdots x_n} \longrightarrow S_{x_0 x_1\cdots x_n}.$$

The target $S_{x_0\cdots x_n}$ is a free R-module with basis consisting of all Laurent monomials $\{x_0^{l_0}\cdots x_n^{l_n} | l_i \in \mathbb{Z}\}$. The image of d^{n-1} is the R-submodule generated by those Laurent monomials $x_0^{l_0}\cdots x_n^{l_n}$ for which at least one $l_j \ge 0$: indeed, a monomial with $l_j \ge 0$ lies in $S_{x_0\cdots\widehat{x_j}\cdots x_n}$, so it is in the image of the j-th summand. Conversely, every element of each summand has at least one non-negative exponent.

Therefore, $H^n(X, \mathscr{F})$ is the free R-module with basis

$$\{x_0^{l_0}\cdots x_n^{l_n} | l_i < 0 \text{ for all } i\},$$

graded by total degree $\sum l_i$. In degree $-(n+1)$, there is exactly one such monomial, namely $x_0^{-1}x_1^{-1}\cdots x_n^{-1}$. Hence:

$$H^n(X, \mathcal{O}_X(-n-1)) \cong R.$$

More generally, for $d \leq -(n+1)$, the group $H^n(X, \mathcal{O}_X(d))$ is a free R-module of rank $\binom{-d-1}{n}$, and $H^n(X, \mathcal{O}_X(d)) = 0$ for $d > -(n+1)$ (since there are no "all-negative" monomials of degree greater than $-(n+1)$).

Proof of (3): The perfect pairing. For $d \geq 0$, the module $H^0(X, \mathcal{O}_X(d))$ is free over R with basis $\{x_0^{m_0}\cdots x_n^{m_n} \big| m_i \geq 0, \sum m_i = d\}$, and $H^n(X, \mathcal{O}_X(-d-n-1))$ is free with basis $\{x_0^{l_0}\cdots x_n^{l_n} \big| l_i < 0, \sum l_i = -d-n-1\}$. The multiplication map

$$H^0(X, \mathcal{O}_X(d)) \times H^n(X, \mathcal{O}_X(-d-n-1)) \longrightarrow H^n(X, \mathcal{O}_X(-n-1)) \cong R$$

sends $(x_0^{m_0}\cdots x_n^{m_n}) \cdot (x_0^{l_0}\cdots x_n^{l_n}) = x_0^{m_0+l_0}\cdots x_n^{m_n+l_n}$, and this product is nonzero in H^n if and only if all exponents remain strictly negative, i.e. $m_j + l_j < 0$ for all j. In particular, the element $x_0^{-m_0-1}\cdots x_n^{-m_n-1}$ (which has total degree $-d-n-1$) pairs with $x_0^{m_0}\cdots x_n^{m_n}$ to give $x_0^{-1}\cdots x_n^{-1}$, the generator of $H^n(X, \mathcal{O}_X(-n-1)) \cong R$. Any other basis monomial in $H^n(X, \mathcal{O}_X(-d-n-1))$ pairs with $x_0^{m_0}\cdots x_n^{m_n}$ to produce a monomial with at least one non-negative exponent, hence zero in H^n.

This shows that the monomial bases are dual to each other under the pairing, so the pairing is perfect.

Proof of (4): Vanishing of intermediate cohomology. We proceed by induction on n. For $n = 1$, the range $0 < i < 1$ is empty, so there is nothing to prove.

Let $n > 1$. We first show that every element of $H^i(X, \mathscr{F})$, for $0 < i < n$, is annihilated by some power of x_n. Localizing the Čech complex $C^\bullet(\mathcal{U}, \mathscr{F})$ at x_n yields the Čech complex of $\mathscr{F}|_{U_n}$ on the affine scheme $U_n = D_+(x_n)$ with respect to the open cover $\{U_j \cap U_n \big| 0 \leq j \leq n\}$. Each set in this cover is a distinguished affine open of U_n, and U_n itself is one of them. Since U_n is affine and $\mathscr{F}|_{U_n}$ is quasi-coherent, $H^i(U_n, \mathscr{F}|_{U_n}) = 0$ for $i > 0$ by theorem 6.2.9. Since localization is exact, it follows that $H^i(X, \mathscr{F})_{x_n} = 0$ for $i > 0$. In other words, every element of $H^i(X, \mathscr{F})$ is annihilated by some power of x_n.

Next, we show that for $0 < i < n$, multiplication by x_n induces a bijection on $H^i(X, \mathscr{F})$, forcing this module to be zero. Consider the exact sequence of graded $S_\bullet$-modules:

$$0 \longrightarrow S_\bullet(-1) \xrightarrow{\cdot x_n} S_\bullet \longrightarrow S_\bullet/(x_n) \longrightarrow 0,$$

which gives the short exact sequence of sheaves on X:

$$0 \longrightarrow \mathcal{O}_X(-1) \xrightarrow{\cdot x_n} \mathcal{O}_X \longrightarrow \mathcal{O}_H \longrightarrow 0,$$

where $H = V_+(x_n) \cong \mathbb{P}_R^{n-1}$ is the hyperplane $x_n = 0$. Twisting by all $d \in \mathbb{Z}$ and taking the direct sum:

$$0 \longrightarrow \mathscr{F}(-1) \xrightarrow{\cdot x_n} \mathscr{F} \longrightarrow \mathscr{F}_H \longrightarrow 0,$$

where $\mathscr{F}_H = \bigoplus_{d\in\mathbb{Z}} \mathcal{O}_H(d)$. The long exact sequence in cohomology gives:

$$\cdots \longrightarrow H^i(X, \mathscr{F}(-1)) \xrightarrow{\cdot x_n} H^i(X, \mathscr{F}) \longrightarrow H^i(X, \mathscr{F}_H) \longrightarrow \cdots$$

As graded $S_\bullet$-modules, $H^i(X, \mathscr{F}(-1))$ is simply $H^i(X, \mathscr{F})$ shifted by one degree, and the connecting map is multiplication by x_n.

Now $H \cong \mathbb{P}^{n-1}_R$, and $H^i(X, \mathscr{F}_H) = H^i(H, \bigoplus_d \mathcal{O}_H(d))$ by proposition 6.2.4. Applying the inductive hypothesis to $\mathbb{P}^{n-1}_R$, we get $H^i(X, \mathscr{F}_H) = 0$ for $0 < i < n-1$.

For $0 < i < n-1$*:* The long exact sequence gives $H^i(X, \mathscr{F}(-1)) \xrightarrow{\sim} H^i(X, \mathscr{F})$, i.e. multiplication by x_n is bijective. Since every element is also killed by a power of x_n, we conclude $H^i(X, \mathscr{F}) = 0$.

For $i = n-1$ *(when* $n \geq 2$*):* We must show multiplication by x_n is still bijective on $H^{n-1}(X, \mathscr{F})$. The relevant portion of the long exact sequence is:

$$H^{n-2}(X, \mathscr{F}_H) \longrightarrow H^{n-1}(X, \mathscr{F}(-1)) \xrightarrow{\cdot x_n} H^{n-1}(X, \mathscr{F}) \longrightarrow H^{n-1}(X, \mathscr{F}_H) \xrightarrow{\partial} H^n(X, \mathscr{F}(-1)).$$

First, we establish injectivity on the left. If $n \geq 3$, then $H^{n-2}(X, \mathscr{F}_H) = 0$ by the inductive hypothesis (since $0 < n-2 < \dim H = n-1$). If $n = 2$, the term on the left is $H^0(X, \mathscr{F}_H)$; in this case, the preceding map in the sequence is $H^0(X, \mathscr{F}) \to H^0(X, \mathscr{F}_H)$, which is the natural surjection $S_\bullet \to S_\bullet/(x_n)$ by part (1). Exactness then forces the map $H^0(X, \mathscr{F}_H) \to H^1(X, \mathscr{F}(-1))$ to be the zero map. In either case, multiplication by x_n is injective on $H^{n-1}(X, \mathscr{F}(-1))$.

Next, we establish surjectivity on the right. The term $H^{n-1}(X, \mathscr{F}_H)$ is the top cohomology of $H \cong \mathbb{P}^{n-1}_R$, which by part (2) is a free R-module generated by all-negative Laurent monomials in the variables $x_0, \ldots, x_{n-1}$. The connecting homomorphism ∂ maps a basis element $x_0^{l_0} \cdots x_{n-1}^{l_{n-1}}$ to the element $x_0^{l_0} \cdots x_{n-1}^{l_{n-1}} x_n^{-1}$ in $H^n(X, \mathscr{F}(-1))$. This map is visibly injective. Because ∂ is injective, exactness forces the preceding map $H^{n-1}(X, \mathscr{F}) \to H^{n-1}(X, \mathscr{F}_H)$ to be the zero map.

Therefore, multiplication by x_n is surjective on $H^{n-1}(X, \mathscr{F})$. Being both injective and surjective, multiplication by x_n is bijective on $H^{n-1}(X, \mathscr{F})$, and the annihilation argument completes the proof.

6.4 Ext Groups and Sheaf Ext

The cohomology groups $H^i(X, \mathscr{F})$ are the derived functors of $\Gamma(X, -) = \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, -)$. From this perspective, cohomology measures the failure of the global-section functor to be exact. But $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, -)$ is a special case of $\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)$ for a general $\mathcal{O}_X$-module $\mathscr{F}$; the derived functors of this more general functor are the *Ext groups*. These carry strictly more information than cohomology alone: while $H^i(X, \mathscr{G})$ measures the obstruction to extending global sections of $\mathscr{G}$, the group $\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$ measures the obstruction to extending morphisms from $\mathscr{F}$ to $\mathscr{G}$. In Serre duality, Ext groups will serve as the natural "dual" objects to cohomology groups, and computing them on projective space will be the key step.

There are two flavors of Ext: the *global Ext* groups $\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$, which are abelian groups (or modules over the global sections ring), and the *sheaf Ext* $\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$, which is itself a sheaf on X. These are related by a spectral sequence, and in the most important case — when $\mathscr{F}$ is locally free — this spectral sequence degenerates, reducing Ext computations to ordinary cohomology. This reduction is what makes the theory practical.

6.4.1 Global and Local Ext

Recall from definition 1.2.21 that for $\mathcal{O}_X$-modules $\mathscr{F}$ and $\mathscr{G}$ on a ringed space $(X, \mathcal{O}_X)$, the *sheaf Hom* is the sheaf:

$$\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})(U) = \operatorname{Hom}_{\mathcal{O}_U}(\mathscr{F}|_U, \mathscr{G}|_U),$$

and the *global Hom* is:

$$\operatorname{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = \Gamma(X, \mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})) = \mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})(X).$$

Both $\operatorname{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)$ and $\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, -)$ are left-exact functors. Since the category $\mathcal{O}_X$-**Mod** has enough injectives (by proposition 6.1.1), we may form their right derived functors:

Definition 6.4.1: Ext Groups for Sheaves

Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathscr{F}, \mathscr{G}$ be $\mathcal{O}_X$-modules.

1. The *global Ext groups* are the right derived functors of $\operatorname{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)$:

   $$\operatorname{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = R^i\operatorname{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)(\mathscr{G}).$$

   Concretely, choose an injective resolution $0 \to \mathscr{G} \to \mathscr{I}^\bullet$ and compute:

   $$\operatorname{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = h^i\big(\operatorname{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I}^\bullet)\big).$$

2. The *sheaf Ext* (or *local Ext*) functors are the right derived functors of $\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, -)$:

   $$\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = R^i\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, -)(\mathscr{G}).$$

   Concretely, choose an injective resolution $0 \to \mathscr{G} \to \mathscr{I}^\bullet$ and compute:

   $$\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = h^i\big(\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I}^\bullet)\big).$$

   This is a sheaf of $\mathcal{O}_X$-modules on X.

By convention, $\operatorname{Ext}^0_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = \operatorname{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$ and $\mathcal{E}xt^0_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = \mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$.

Note the asymmetry: we derive in the *second* variable. One can also derive in the first variable (resolving $\mathscr{F}$ by locally free sheaves), and a standard comparison argument shows this gives the same Ext groups[5]. Both Ext groups come equipped with the standard long exact sequences of derived functors:

[5] More precisely, if $\mathscr{F}$ admits a locally free resolution $\mathcal{E}^\bullet \to \mathscr{F} \to 0$ (e.g., if X is a smooth variety or if $\mathscr{F}$ is coherent on projective space), then $\operatorname{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) \cong h^i(\operatorname{Hom}_{\mathcal{O}_X}(\mathcal{E}^\bullet, \mathscr{G}))$. This is the "balancing" of Ext; see [11, chapter 27] for the proof in the module category, which carries over to sheaves.

Proposition 6.4.2: Long Exact Sequences for Ext

Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathscr{F}$ be an $\mathcal{O}_X$-module.

1. *(Covariant variable.)* A short exact sequence $0 \to \mathscr{G}' \to \mathscr{G} \to \mathscr{G}'' \to 0$ of $\mathcal{O}_X$-modules induces long exact sequences:

$$0 \to \mathrm{Hom}(\mathscr{F}, \mathscr{G}') \to \mathrm{Hom}(\mathscr{F}, \mathscr{G}) \to \mathrm{Hom}(\mathscr{F}, \mathscr{G}'') \xrightarrow{\delta} \mathrm{Ext}^1(\mathscr{F}, \mathscr{G}') \to \cdots$$

   and similarly for $\mathcal{E}xt$.

2. *(Contravariant variable.)* A short exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ of $\mathcal{O}_X$-modules induces a long exact sequence:

$$0 \to \mathrm{Hom}(\mathscr{F}'', \mathscr{G}) \to \mathrm{Hom}(\mathscr{F}, \mathscr{G}) \to \mathrm{Hom}(\mathscr{F}', \mathscr{G}) \xrightarrow{\delta} \mathrm{Ext}^1(\mathscr{F}'', \mathscr{G}) \to \cdots$$

   and similarly for $\mathcal{E}xt$.

***Proof*:**
Part (1) is the standard long exact sequence of a right derived functor applied to a short exact sequence. Part (2) follows from the corresponding result for modules (see [11, chapter 27]), adapted to the sheaf setting by computing on injective resolutions of $\mathscr{G}$ and using the left-exactness of $\mathrm{Hom}(-, \mathscr{G})$ in the first variable.

The next result shows that Ext can be computed from a locally free resolution of the first argument, which is often more convenient than resolving the second argument by injectives:

Proposition 6.4.3: Computing Ext via Locally Free Resolutions

Let X be a Noetherian scheme and let $\mathscr{F}, \mathscr{G}$ be $\mathcal{O}_X$-modules with $\mathscr{F}$ coherent. Suppose $\mathscr{F}$ admits a locally free resolution:

$$\cdots \to \mathcal{E}^{-1} \to \mathcal{E}^0 \to \mathscr{F} \to 0.$$

Then:

$$\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) \cong h^i\big(\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{E}^\bullet, \mathscr{G})\big) \qquad \text{and} \qquad \mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) \cong h^i\big(\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}^\bullet, \mathscr{G})\big).$$

In particular, if $\mathscr{F}$ is locally free, then $\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = 0$ and $\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = 0$ for all $i > 0$.

***Proof*:**
The first statement is the "balancing" of Ext. Since locally free sheaves are $\mathrm{Hom}(\mathscr{F}, -)$-acyclic (a morphism from a free module is determined by where the generators go, and this respects surjections), the locally free resolution computes the same derived functors as an injective resolution of $\mathscr{G}$. For the sheaf version, the same argument works on each open set. The vanishing for locally free $\mathscr{F}$ follows because the resolution $\mathcal{E}^\bullet = [\mathscr{F}]$ (concentrated in degree 0) is already a locally free resolution, so $h^i(\mathrm{Hom}(\mathcal{E}^\bullet, \mathscr{G})) = 0$ for $i > 0$.

The sheaf Ext is a "local" invariant, while the global Ext is obtained by "assembling" the local data via global sections. The following result makes this precise at the stalk level:

Proposition 6.4.4: Stalks of Sheaf Ext

Let $(X, \mathcal{O}_X)$ be a ringed space, $\mathscr{F}$ and $\mathscr{G}$ be $\mathcal{O}_X$-modules, and $x \in X$. Then there is a natural isomorphism:

$$\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})_x \;\cong\; \mathrm{Ext}^i_{\mathcal{O}_{X,x}}(\mathscr{F}_x, \mathscr{G}_x).$$

In particular, if $\mathscr{F}$ is locally free, then $\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = 0$ for all $i > 0$ (since the stalks $\mathscr{F}_x$ are free $\mathcal{O}_{X,x}$-modules, and Ext over a local ring vanishes for free first arguments).

***Proof*:**
Choose an injective resolution $0 \to \mathscr{G} \to \mathscr{I}^\bullet$. Taking stalks at x gives a resolution $0 \to \mathscr{G}_x \to \mathscr{I}^\bullet_x$. The stalk of the sheaf Ext is:

$$\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})_x = h^i\big(\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I}^\bullet)\big)_x = h^i\big(\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I}^\bullet)_x\big)$$

where the second equality uses the exactness of the stalk functor (by **??**). Now, $\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I}^p)_x = \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathscr{F}_x, \mathscr{I}^p_x)$ (the stalk of the sheaf Hom is the module Hom of the stalks). It remains to show that $\mathscr{I}^p_x$ is an injective $\mathcal{O}_{X,x}$-module. This follows from the next lemma.

Lemma 6.4.5: Injective Sheaves and Restriction

Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathscr{I}$ be an injective $\mathcal{O}_X$-module.

1. For any open set $U \subseteq X$, the restriction $\mathscr{I}|_U$ is an injective $\mathcal{O}_U$-module.
2. For any point $x \in X$, the stalk $\mathscr{I}_x$ is an injective $\mathcal{O}_{X,x}$-module.

***Proof*:**
(1) Let $j : U \hookrightarrow X$ denote the inclusion. The functor $j_! : \mathcal{O}_U\text{-}\mathbf{Mod} \to \mathcal{O}_X\text{-}\mathbf{Mod}$ (extension by zero) is exact and left adjoint to the restriction functor j^* (by proposition 1.3.12). Hence j^* preserves injectives: for any monomorphism $\alpha : \mathscr{F} \hookrightarrow \mathscr{G}$ of $\mathcal{O}_U$-modules, the map

$$\mathrm{Hom}_{\mathcal{O}_U}(\mathscr{G}, \mathscr{I}|_U) = \mathrm{Hom}_{\mathcal{O}_X}(j_!\mathscr{G}, \mathscr{I}) \longrightarrow \mathrm{Hom}_{\mathcal{O}_X}(j_!\mathscr{F}, \mathscr{I}) = \mathrm{Hom}_{\mathcal{O}_U}(\mathscr{F}, \mathscr{I}|_U)$$

is surjective because $j_!\alpha$ is a monomorphism (exactness of $j_!$) and $\mathscr{I}$ is injective.

(2) The stalk $\mathscr{I}_x = \varinjlim_{U \ni x} \mathscr{I}(U)$ can be computed as follows. Let $j_x : \{x\} \hookrightarrow X$ be the inclusion and consider the skyscraper functor $(j_x)_* : \mathcal{O}_{X,x}\text{-}\mathbf{Mod} \to \mathcal{O}_X\text{-}\mathbf{Mod}$. Its left adjoint is the stalk functor $\mathscr{G} \mapsto \mathscr{G}_x$. Since the stalk functor is exact, it preserves injectives by the same adjunction argument as in part (1).

As a consequence, sheaf Ext behaves well with respect to restricting to open subsets:

Proposition 6.4.6: Open Subsets and Ext Groups

Let $(X, \mathcal{O}_X)$ be a ringed space, $\mathscr{F}$ and $\mathscr{G}$ be $\mathcal{O}_X$-modules, and $U \subseteq X$ an open subset. Then there is a natural isomorphism of sheaves on U:

$$\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})\big|_U \cong \mathcal{E}xt^i_{\mathcal{O}_U}(\mathscr{F}|_U, \mathscr{G}|_U).$$

***Proof*:**
Let $0 \to \mathscr{G} \to \mathscr{I}^\bullet$ be an injective resolution on X. By lemma 6.4.5(1), the restricted sequence $0 \to \mathscr{G}|_U \to \mathscr{I}^\bullet|_U$ is an injective resolution on U. Then:

$$\mathcal{E}xt^i_{\mathcal{O}_U}(\mathscr{F}|_U, \mathscr{G}|_U) = h^i\big(\mathcal{H}om_{\mathcal{O}_U}(\mathscr{F}|_U, \mathscr{I}^\bullet|_U)\big) = h^i\big(\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I}^\bullet)|_U\big) = \mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})\big|_U,$$

where the middle equality uses the fact that $\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I}^p)|_U = \mathcal{H}om_{\mathcal{O}_U}(\mathscr{F}|_U, \mathscr{I}^p|_U)$ (the sheaf Hom is computed open-set by open-set), and the last equality uses the exactness of restriction (which commutes with taking cohomology of a complex of sheaves).

6.4.2 The Local-to-Global Spectral Sequence

The global Ext and the sheaf Ext are related by the factorization:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, -) = \Gamma(X, -) \circ \mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, -).$$

This is simply the statement that a global morphism $\mathscr{F} \to \mathscr{G}$ is a global section of the sheaf Hom. Both functors on the right are left exact, and this composition of functors is the standard setting for the Grothendieck spectral sequence[6]. For this spectral sequence to exist, we need $\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, -)$ to send injective objects to $\Gamma(X, -)$-acyclic objects. The following key lemma establishes this:

Lemma 6.4.7: Sheaf Hom of Injectives is Flasque

Let $(X, \mathcal{O}_X)$ be a ringed space, $\mathscr{F}$ an $\mathcal{O}_X$-module, and $\mathscr{I}$ an injective $\mathcal{O}_X$-module. Then $\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I})$ is a flasque sheaf.

***Proof*:**
Let $U \subseteq V$ be open subsets of X. We must show that the restriction map

$$\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I})(V) = \mathrm{Hom}_{\mathcal{O}_V}(\mathscr{F}|_V, \mathscr{I}|_V) \longrightarrow \mathrm{Hom}_{\mathcal{O}_U}(\mathscr{F}|_U, \mathscr{I}|_U) = \mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I})(U)$$

is surjective. Let $\varphi : \mathscr{F}|_U \to \mathscr{I}|_U$ be a morphism of $\mathcal{O}_U$-modules. Let $j : U \hookrightarrow V$ be the inclusion. By the adjunction $j_! \dashv j^*$, the morphism φ corresponds to a morphism $j_!(\mathscr{F}|_U) \to \mathscr{I}|_V$. The natural map $j_!(\mathscr{F}|_U) \to \mathscr{F}|_V$ gives a monomorphism in the category of $\mathcal{O}_V$-modules. Since $\mathscr{I}|_V$ is injective (by lemma 6.4.5), the morphism $j_!(\mathscr{F}|_U) \to \mathscr{I}|_V$ extends to a morphism $\psi : \mathscr{F}|_V \to \mathscr{I}|_V$. By construction, $\psi|_U = \varphi$, so the restriction map is surjective.

[6] See [11, chapter 27] for the construction of the Grothendieck spectral sequence and the conditions under which it is defined.

Since flasque sheaves are $\Gamma(X,-)$-acyclic (by corollary 6.1.7), the hypotheses of the Grothendieck spectral sequence are satisfied. We obtain:

Theorem 6.4.8: Local-to-Global Ext Spectral Sequence

Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathscr{F}, \mathscr{G}$ be $\mathcal{O}_X$-modules. Then there is a convergent spectral sequence:

$$E_2^{p,q} = H^p\big(X, \mathcal{E}xt^q_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})\big) \implies \operatorname{Ext}^{p+q}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}).$$

This is natural in both $\mathscr{F}$ and $\mathscr{G}$.

***Proof*:**
This is the Grothendieck spectral sequence for the composite functor $\operatorname{Hom}_{\mathcal{O}_X}(\mathscr{F}, -) = \Gamma(X,-) \circ \mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, -)$. By lemma 6.4.7, the functor $\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, -)$ sends injective $\mathcal{O}_X$-modules to flasque sheaves, which are $\Gamma(X,-)$-acyclic. The E_2-page of the Grothendieck spectral sequence is $R^p\Gamma \circ R^q \mathcal{H}om(\mathscr{F}, -) = H^p(X, \mathcal{E}xt^q(\mathscr{F}, -))$, and the abutment is $R^{p+q}(\Gamma \circ \mathcal{H}om(\mathscr{F}, -)) = \operatorname{Ext}^{p+q}(\mathscr{F}, -)$.

The spectral sequence simplifies dramatically in the most important case:

Corollary 6.4.9: Degeneration for Locally Free Sheaves

Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathcal{L}$ be a locally free $\mathcal{O}_X$-module of finite rank, with dual $\mathcal{L}^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$. Then for all $\mathcal{O}_X$-modules $\mathscr{G}$ and all $i \geq 0$:

$$\operatorname{Ext}^i_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G}) \;\cong\; H^i(X,\, \mathcal{L}^\vee \otimes_{\mathcal{O}_X} \mathscr{G}).$$

In particular, setting $\mathcal{L} = \mathcal{O}_X$ recovers sheaf cohomology: $\operatorname{Ext}^i_{\mathcal{O}_X}(\mathcal{O}_X, \mathscr{G}) \cong H^i(X, \mathscr{G})$.

***Proof*:**
By proposition 6.4.4, the stalk of $\mathcal{E}xt^q_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G})$ at $x \in X$ is $\operatorname{Ext}^q_{\mathcal{O}_{X,x}}(\mathcal{L}_x, \mathscr{G}_x)$. Since $\mathcal{L}$ is locally free, $\mathcal{L}_x$ is a free $\mathcal{O}_{X,x}$-module, so $\operatorname{Ext}^q_{\mathcal{O}_{X,x}}(\mathcal{L}_x, \mathscr{G}_x) = 0$ for all $q > 0$. Hence $\mathcal{E}xt^q_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G}) = 0$ for $q > 0$ (a sheaf with all zero stalks is the zero sheaf).

The spectral sequence of theorem 6.4.8 collapses at the E_2-page: only the $q = 0$ row survives, giving:

$$\operatorname{Ext}^i_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G}) \;\cong\; H^i\big(X,\, \mathcal{E}xt^0_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G})\big) = H^i\big(X,\, \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G})\big).$$

Finally, for a locally free sheaf of finite rank, the natural map $\mathcal{L}^\vee \otimes_{\mathcal{O}_X} \mathscr{G} \to \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G})$ is an isomorphism (this can be checked on stalks, where it reduces to the standard isomorphism $M^\vee \otimes N \xrightarrow{\sim} \operatorname{Hom}_R(M, N)$ for a free module M of finite rank over a ring R). Hence $\operatorname{Ext}^i(\mathcal{L}, \mathscr{G}) \cong H^i(X, \mathcal{L}^\vee \otimes \mathscr{G})$.

6.4.10 Remark: Why Locally Free is the Key Case The reader should pause to appreciate how much corollary 6.4.9 simplifies matters. For a general coherent sheaf $\mathscr{F}$, computing $\mathrm{Ext}^i(\mathscr{F}, \mathscr{G})$ requires either an injective resolution of $\mathscr{G}$ (typically enormous and non-explicit) or the full spectral sequence. But for a locally free $\mathscr{F}$, the Ext groups reduce to ordinary sheaf cohomology, which we can compute using the Čech machinery of section 6.1.1.
This is not a marginal simplification: on a smooth projective variety, *every* coherent sheaf has a finite locally free resolution (since the local rings are regular, hence have finite global dimension). So Ext between any two coherent sheaves can, in principle, be computed by resolving the first argument by locally free sheaves and applying the corollary iteratively. On projective space, the situation is even better: every coherent sheaf has a resolution by direct sums of twisted sheaves $\mathcal{O}(d)$ (by corollary 5.3.26), and the cohomology of these is known by theorem 6.3.1.

One can also give a direct proof of corollary 6.4.9 without invoking spectral sequences. Since this argument is short and illuminating, we record it:

***Proof*:**
(Alternative proof of corollary 6.4.9)
Since $\mathcal{L}$ is locally free of finite rank, it is a projective object in the category of $\mathcal{O}_X$-modules on any open set where it is free, and the functor $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, -)$ is exact (checking on stalks: if $\mathcal{L}_x \cong \mathcal{O}_{X,x}^r$, then $\mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{O}_{X,x}^r, -) = (-)^r$ is exact). Hence $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, -)$ preserves injective resolutions up to exactness. Let $0 \to \mathscr{G} \to \mathscr{I}^\bullet$ be an injective resolution. Then:

$$0 \to \mathcal{H}om(\mathcal{L}, \mathscr{G}) \to \mathcal{H}om(\mathcal{L}, \mathscr{I}^\bullet)$$

is exact, and each $\mathcal{H}om(\mathcal{L}, \mathscr{I}^p)$ is flasque by lemma 6.4.7. Thus this is a flasque resolution of $\mathcal{H}om(\mathcal{L}, \mathscr{G}) \cong \mathcal{L}^\vee \otimes \mathscr{G}$, and:

$$\mathrm{Ext}^i(\mathcal{L}, \mathscr{G}) = h^i(\mathrm{Hom}(\mathcal{L}, \mathscr{I}^\bullet)) = h^i(\Gamma(X, \mathcal{H}om(\mathcal{L}, \mathscr{I}^\bullet))) = H^i(X, \mathcal{L}^\vee \otimes \mathscr{G}).$$

6.4.3 Computations on Projective Space

We now combine the degeneration result with the computation of twisted sheaf cohomology (theorem 6.3.1) to obtain explicit Ext groups on projective space. These calculations are the foundation for Serre duality.

Proposition 6.4.11: Ext of Twisted Sheaves on Projective Space

Let R be a Noetherian ring, $X = \mathbb{P}^n_R$ with $n \geq 1$, and $a, b \in \mathbb{Z}$. Then:

$$\mathrm{Ext}^i_{\mathcal{O}_X}\big(\mathcal{O}_X(a),\, \mathcal{O}_X(b)\big) \;\cong\; H^i\big(X,\, \mathcal{O}_X(b-a)\big),$$

and hence by theorem 6.3.1:

$$\mathrm{Ext}^i_{\mathcal{O}_X}\big(\mathcal{O}_X(a),\, \mathcal{O}_X(b)\big) \;\cong\; \begin{cases} \big(R[x_0, \ldots, x_n]\big)_{b-a} & i = 0,\; b - a \geq 0, \\ R^{\binom{a-b-1}{n}} & i = n,\; b - a \leq -(n+1), \\ 0 & \text{otherwise.} \end{cases}$$

In particular:

1. $\mathrm{Ext}^i(\mathcal{O}_X, \mathcal{O}_X(d)) = H^i(X, \mathcal{O}_X(d))$ for all i, d.
2. $\mathrm{Hom}(\mathcal{O}_X(a), \mathcal{O}_X(b)) = 0$ if $b < a$, and is a free R-module of rank $\binom{b-a+n}{n}$ if $b \geq a$.
3. $\mathrm{Ext}^n(\mathcal{O}_X(a), \mathcal{O}_X(b)) \neq 0$ only when $b - a \leq -(n+1)$, i.e. $a \geq b + n + 1$.

***Proof*:**
The line bundle $\mathcal{O}_X(a)$ is locally free of rank 1, with dual $\mathcal{O}_X(a)^\vee \cong \mathcal{O}_X(-a)$ (by proposition 5.3.8). Hence by corollary 6.4.9:

$$\mathrm{Ext}^i(\mathcal{O}_X(a), \mathcal{O}_X(b)) \cong H^i(X, \mathcal{O}_X(-a) \otimes \mathcal{O}_X(b)) = H^i(X, \mathcal{O}_X(b-a)),$$

where we used $\mathcal{O}_X(c) \otimes \mathcal{O}_X(d) \cong \mathcal{O}_X(c+d)$. The explicit values follow from theorem 6.3.1. The rank in the $i = n$ case is the number of "all-negative" monomials $x_0^{l_0} \cdots x_n^{l_n}$ of total degree $b - a$ with each $l_j < 0$, which by the substitution $l_j = -m_j - 1$ (so $m_j \geq 0$ and $\sum m_j = a - b - n - 1$) equals $\binom{a-b-1}{n}$.

Now let us see how to handle sheaves that are not themselves line bundles but can be resolved by them:

Example 6.4.12: Ext Computations on Projective Space

1. *Self-Ext of the structure sheaf.* Taking $a = b = 0$ in proposition 6.4.11:

   $$\mathrm{Ext}^i(\mathcal{O}_X, \mathcal{O}_X) \cong H^i(X, \mathcal{O}_X) = \begin{cases} R & i = 0, \\ 0 & i > 0. \end{cases}$$

   This says that the structure sheaf has no non-trivial self-extensions, which is consistent with $\mathcal{O}_X$ being "rigid" as a line bundle on projective space.

2. *The dualizing line bundle.* We shall see in the next section that the *dualizing sheaf* of $\mathbb{P}^n_R$ is $\omega_X = \mathcal{O}_X(-n-1)$. Note the "dual" behaviour: by part (3) of proposition 6.4.11, the only

nonzero Ext between $\mathcal{O}_X(d)$ and ω_X is:

$$\operatorname{Ext}^n(\mathcal{O}_X(d), \omega_X) = H^n(X, \mathcal{O}_X(-n-1-d)) \cong \begin{cases} R^{\binom{d}{n}} & d \geq 0, \\ 0 & d < 0. \end{cases}$$

Meanwhile $H^0(X, \mathcal{O}_X(d)) \cong R^{\binom{d+n}{n}}$ for $d \geq 0$, and one can verify that $\binom{d+n}{n} = \binom{d}{n}$ fails in general — but the *pairing* between $H^0(X, \mathcal{O}_X(d))$ and $\operatorname{Ext}^n(\mathcal{O}_X(d), \omega_X)$ is nonetheless perfect (we will prove this via Serre duality in the next section). For $d = 0$:

$$\operatorname{Ext}^n(\mathcal{O}_X, \omega_X) = H^n(X, \omega_X) \cong R,$$

which will serve as the "trace" isomorphism $H^n(X, \omega_X) \xrightarrow{\sim} R$ in the duality theorem.

3. *Ext from the cotangent sheaf (using the Euler sequence).* Let k be a field and $X = \mathbb{P}^n_k$. The Euler sequence (by theorem 5.5.18) is:

$$0 \longrightarrow \Omega_{X/k} \longrightarrow \mathcal{O}_X(-1)^{\oplus(n+1)} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Although $\Omega_{X/k}$ is locally free (of rank n), let us use this short exact sequence to compute $\operatorname{Ext}^i(\Omega_{X/k}, \mathcal{O}_X)$ via the long exact sequence in the *contravariant* variable (proposition 6.4.2(2)):

$$\cdots \to \operatorname{Ext}^i(\mathcal{O}_X, \mathcal{O}_X) \to \operatorname{Ext}^i(\mathcal{O}_X(-1)^{n+1}, \mathcal{O}_X) \to \operatorname{Ext}^i(\Omega_{X/k}, \mathcal{O}_X) \to \operatorname{Ext}^{i+1}(\mathcal{O}_X, \mathcal{O}_X) \to \cdots$$

By proposition 6.4.11:

$$\operatorname{Ext}^i(\mathcal{O}_X(-1)^{n+1}, \mathcal{O}_X) \cong H^i(X, \mathcal{O}_X(1))^{n+1},$$

which is $(R^{n+1})^{n+1}$ for $i = 0$ and zero for $0 < i < n$. Also, $\operatorname{Ext}^i(\mathcal{O}_X, \mathcal{O}_X) = R$ for $i = 0$ and zero for $i > 0$.

For $i = 0$, the long exact sequence gives:

$$0 \to \operatorname{Hom}(\mathcal{O}_X, \mathcal{O}_X) \to \operatorname{Hom}(\mathcal{O}_X(-1)^{n+1}, \mathcal{O}_X) \to \operatorname{Hom}(\Omega, \mathcal{O}_X) \to 0$$

so $\operatorname{Hom}(\Omega_{X/k}, \mathcal{O}_X) \cong k^{(n+1)^2}/k \cong k^{n^2+2n}$. On the other hand, by corollary 6.4.9, $\operatorname{Hom}(\Omega, \mathcal{O}_X) = H^0(X, \Omega^\vee) = H^0(X, \mathcal{T}_X)$, where $\mathcal{T}_X$ is the tangent sheaf. This is the space of global vector fields on $\mathbb{P}^n_k$, which has dimension $n^2 + 2n = \dim_k \mathfrak{gl}_{n+1}(k)$: the global vector fields on projective space are precisely those induced by the action of GL_{n+1}, modulo scalars.

4. *Ext between non-locally-free sheaves (sketch).* Let $X = \mathbb{P}^2_k$ and let $C \hookrightarrow X$ be a smooth plane curve of degree d, with ideal sheaf $\mathscr{I}_C$. The short exact sequence $0 \to \mathscr{I}_C \to \mathcal{O}_X \to \mathcal{O}_C \to 0$, together with the identification $\mathscr{I}_C \cong \mathcal{O}_X(-d)$ (by corollary 5.3.16), gives:

$$0 \to \mathcal{O}_X(-d) \to \mathcal{O}_X \to \mathcal{O}_C \to 0.$$

This is a locally free resolution of $\mathcal{O}_C$ as an $\mathcal{O}_X$-module (of length 1). Hence by proposition 6.4.3:

$$\operatorname{Ext}^i_{\mathcal{O}_X}(\mathcal{O}_C, \mathscr{G}) \cong h^i\big(\operatorname{Hom}(\mathcal{O}_X, \mathscr{G}) \to \operatorname{Hom}(\mathcal{O}_X(-d), \mathscr{G})\big)$$

for any $\mathcal{O}_X$-module $\mathscr{G}$. The complex in question is $\mathscr{G}(X) \xrightarrow{\cdot f} \mathscr{G}(X)(d)$ where f is the defining equation of C, so:

$$\begin{aligned}\operatorname{Ext}^0(\mathcal{O}_C, \mathscr{G}) &= \ker\big(\mathscr{G}(X) \to \mathscr{G}(X)(d)\big),\\ \operatorname{Ext}^1(\mathcal{O}_C, \mathscr{G}) &= \operatorname{coker}\big(\mathscr{G}(X) \to \mathscr{G}(X)(d)\big),\\ \operatorname{Ext}^i(\mathcal{O}_C, \mathscr{G}) &= 0 \quad \text{for } i \geq 2.\end{aligned}$$

Taking $\mathscr{G} = \omega_X = \mathcal{O}_X(-3)$, we get $\operatorname{Ext}^1(\mathcal{O}_C, \omega_X) = \operatorname{coker}\big(H^0(X, \mathcal{O}(-3)) \to H^0(X, \mathcal{O}(d-3))\big)$. For $d \geq 3$, this is a k-vector space of dimension $\binom{d-1}{2} - 1$, which (as we will verify via Serre duality) equals $\dim_k H^0(C, \omega_C) = g$, the genus of the curve. This is not a coincidence: Serre duality will identify $\operatorname{Ext}^1(\mathcal{O}_C, \omega_X)$ with $H^0(C, \omega_C)^\vee$ via the adjunction between the closed embedding and the ambient duality.

Let us close this section by recording how the spectral sequence simplifies for coherent sheaves on projective schemes, even when the first argument is not locally free. This will be used in the proof of Serre duality:

Proposition 6.4.13: Finiteness of Ext on Projective Schemes

Let X be a projective scheme over a Noetherian ring R, and let $\mathscr{F}$ and $\mathscr{G}$ be coherent sheaves on X. Then:

1. Each $\mathcal{E}xt^q_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$ is a coherent sheaf on X.
2. Each $\operatorname{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$ is a finitely generated R-module.
3. $\operatorname{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = 0$ for $i > \dim X$ (where $\dim X$ denotes the Krull dimension).

***Proof*:**
(1) This is a local statement: on any affine open $U = \operatorname{Spec} A$ of X, the sheaves $\mathscr{F}|_U$ and $\mathscr{G}|_U$ correspond to finitely generated A-modules M and N (by corollary 5.1.12), and the sheaf Ext on U is:

$$\mathcal{E}xt^q_{\mathcal{O}_U}(\widetilde{M}, \widetilde{N}) \cong \widetilde{\operatorname{Ext}^q_A(M, N)}$$

(this follows from proposition 6.4.6 and the standard computation of Ext over a Noetherian ring via a free resolution of the finitely generated module M; each $\operatorname{Ext}^q_A(M, N)$ is finitely generated). Hence $\mathcal{E}xt^q(\mathscr{F}, \mathscr{G})$ is coherent.

(2) The local-to-global spectral sequence (theorem 6.4.8) gives $H^p(X, \mathcal{E}xt^q(\mathscr{F}, \mathscr{G})) \Rightarrow \operatorname{Ext}^{p+q}(\mathscr{F}, \mathscr{G})$. Since $\mathcal{E}xt^q(\mathscr{F}, \mathscr{G})$ is coherent and X is projective over a Noetherian ring, each cohomology group $H^p(X, \mathcal{E}xt^q(\mathscr{F}, \mathscr{G}))$ is a finitely generated R-module[a]. Since the E_2-page consists of finitely generated R-modules and the spectral sequence converges, the abutment $\operatorname{Ext}^i(\mathscr{F}, \mathscr{G})$ is finitely generated.

(3) By theorem 6.2.5, $H^p(X, -) = 0$ for $p > \dim X$. On the other hand, $\mathcal{E}xt^q(\mathscr{F}, \mathscr{G}) = 0$ for q larger than the projective dimension of $\mathscr{F}$ (which is finite since the local rings of a projective scheme over a Noetherian ring are Noetherian, hence $\mathscr{F}$ admits locally free resolutions of finite length). The spectral sequence then gives $\operatorname{Ext}^i(\mathscr{F}, \mathscr{G}) = 0$ for i sufficiently large; more precisely, for $i > \dim X$ when $\mathscr{F}$ is locally free, and in general for $i > \dim X + \operatorname{pd}(\mathscr{F})$ where pd denotes

the projective dimension. When X is smooth (hence regular), the projective dimension of any coherent sheaf is at most $\dim X$, so $\mathrm{Ext}^i(\mathscr{F}, \mathscr{G}) = 0$ for $i > 2\dim X$; but the spectral sequence gives the sharper bound $i > \dim X$ directly when the E_2-page is examined.

[a] This uses the finiteness theorem for cohomology of coherent sheaves on projective schemes, which we will prove in section 6.6 (see corollary 5.3.28). No circularity arises: the finiteness theorem depends on the twisted sheaf computation (theorem 6.3.1) and the "coherent sheaves are quotients of twisted free sheaves" result (corollary 5.3.26), both of which are already established.

6.4.14 Remark: What the Spectral Sequence Encodes The local-to-global spectral sequence should be thought of as expressing the passage from local to global information for morphisms between sheaves. On the E_2-page:

- The *bottom row* $E_2^{p,0} = H^p(X, \mathcal{H}om(\mathscr{F}, \mathscr{G}))$ captures the "global extension classes that are locally trivial." When $\mathscr{F}$ is locally free, this is the only nonzero row, and the spectral sequence degenerates.
- The *left column* $E_2^{0,q} = \Gamma(X, \mathcal{E}xt^q(\mathscr{F}, \mathscr{G}))$ captures the "local extension classes that are globally defined." These are the obstructions arising from the local algebra (e.g., the singularities of $\mathscr{F}$ or the non-regularity of the local rings).

The differentials in the spectral sequence measure the interaction between these two types of obstruction. For smooth projective varieties with locally free sheaves, all the interesting information is in the bottom row — this is why Serre duality can be stated purely in terms of cohomology groups, without mentioning Ext.

6.5 Serre Duality Theorem

In classical topology, Poincaré duality asserts that for a compact oriented manifold M of dimension n, the singular cohomology groups satisfy $H^i(M; \mathbb{R}) \cong H^{n-i}(M; \mathbb{R})^*$. The duality is mediated by integration: the cup product $H^i \times H^{n-i} \to H^n$, followed by integration over the fundamental class $H^n(M; \mathbb{R}) \xrightarrow{\sim} \mathbb{R}$, gives a perfect pairing. In Kähler geometry, this takes the refined form of Hodge duality (theorem 0.6.28): $H^{p,q}(X) \cong H^{n-p,\, n-q}(X)^*$, with the pairing mediated by the wedge product of (p, q)-forms followed by integration of the resulting top form.

Serre duality is the algebraic analogue. For a smooth projective variety X of dimension n over a field k, the role of the top differential form is played by the *canonical sheaf* $\omega_X = \det \Omega_{X/k}$, and the role of integration is played by a "trace" isomorphism $H^n(X, \omega_X) \xrightarrow{\sim} k$. The resulting duality

$$H^i(X, \mathscr{F}) \;\cong\; H^{n-i}(X,\, \mathscr{F}^\vee \otimes \omega_X)^*$$

for locally free $\mathscr{F}$ is the central structural result in the cohomology of algebraic varieties. It constrains the Hilbert polynomial, underpins the Riemann–Roch theorem, and governs the geometry of linear systems.

We first prove the duality theorem for projective space, where the "dualizing sheaf" is simply $\mathcal{O}(-n-1)$ and the computations are completely explicit. We then define the dualizing sheaf for general projective schemes and extend the theorem to smooth projective varieties.

6.5.1 Duality on Projective Space

The twisted sheaf computation (theorem 6.3.1) already contains the seed of duality: the perfect pairing of part (3) shows that $H^0(X, \mathcal{O}(d))$ and $H^n(X, \mathcal{O}(-d-n-1))$ are dual to each other. Our task is to recognize this as the $i = 0$ case of a more general duality, then extend to all i and to all coherent sheaves.

The first step is to abstract the key property of $\mathcal{O}(-n-1)$ that makes the pairing work:

Definition 6.5.1: Dualizing Sheaf

Let X be a proper scheme of dimension n over a field k. A *dualizing sheaf* for X is a coherent sheaf ω_X on X together with a k-linear map $t : H^n(X, \omega_X) \to k$, called the *trace map*, such that for every coherent sheaf $\mathscr{F}$ on X, the natural pairing:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \omega_X) \times H^n(X, \mathscr{F}) \xrightarrow{\text{apply } H^n} H^n(X, \omega_X) \xrightarrow{t} k$$

is a perfect pairing of finite-dimensional k-vector spaces. In other words, the induced map

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \omega_X) \xrightarrow{\sim} H^n(X, \mathscr{F})^*$$

is an isomorphism, where $(-)^*$ denotes the k-linear dual.
Equivalently, ω_X *represents* the functor $\mathscr{F} \mapsto H^n(X, \mathscr{F})^*$ on the category of coherent sheaves on X.

Proposition 6.5.2: Uniqueness of the Dualizing Sheaf

If a dualizing sheaf exists, it is unique up to unique isomorphism. That is, if (ω_X, t) and (ω'_X, t') are two dualizing sheaves for X, then there is a unique isomorphism $\varphi : \omega_X \xrightarrow{\sim} \omega'_X$ such that $t = t' \circ H^n(X, \varphi)$.

***Proof*:**
Both ω_X and ω'_X represent the same functor $\mathscr{F} \mapsto H^n(X, \mathscr{F})^*$. By the Yoneda lemma (theorem 0.4.22), any two representing objects are uniquely isomorphic via an isomorphism that intertwines the representing natural transformations.

Proposition 6.5.3: Dualizing Sheaf of Projective Space

Let k be a field and $X = \mathbb{P}^n_k$ with $n \geq 1$. The twisted sheaf $\omega_X = \mathcal{O}_X(-n-1)$ is a dualizing sheaf for X, with trace map $t : H^n(X, \mathcal{O}(-n-1)) \xrightarrow{\sim} k$ being the isomorphism that sends the class of $x_0^{-1} x_1^{-1} \cdots x_n^{-1}$ to $1 \in k$.

***Proof*:**
This is precisely the content of theorem 6.3.1(3) (with $R = k$). Indeed, that theorem establishes that for each $d \geq 0$, the multiplication pairing:

$$H^0(X, \mathcal{O}(d)) \times H^n(X, \mathcal{O}(-d-n-1)) \longrightarrow H^n(X, \mathcal{O}(-n-1)) \cong k$$

is a perfect pairing of finite-dimensional k-vector spaces. Since $\mathrm{Hom}(\mathcal{O}(d), \mathcal{O}(-n-1)) \cong H^0(X, \mathcal{O}(-d-n-1))$ (by corollary 6.4.9), this gives $\mathrm{Hom}(\mathcal{O}(d), \omega_X) \cong H^n(X, \mathcal{O}(d))^*$ for every d. When $d < 0$, both sides are zero (the left because there are no nonzero maps from $\mathcal{O}(d)$ to $\mathcal{O}(-n-1)$ when $d > -n-1$, the right by theorem 6.3.1(4)).

For a general coherent $\mathscr{F}$, one reduces to line bundles by choosing a presentation of $\mathscr{F}$ by direct sums of $\mathcal{O}(d)$'s (using corollary 5.3.26) and comparing the two sides via the five lemma. The argument is identical to Step 2 of the proof of theorem 6.5.5 below, so we defer the details.

The dualizing sheaf definition captures the "$i = 0$" case of duality: it says $\mathrm{Hom}(\mathscr{F}, \omega) = \mathrm{Ext}^0(\mathscr{F}, \omega) \cong H^n(X, \mathscr{F})^*$. The full Serre duality extends this to all Ext groups. To state it, we need the *Yoneda pairing*:

6.5.4 The Yoneda Pairing For $\mathcal{O}_X$-modules $\mathscr{F}, \mathscr{G}, \mathscr{H}$, the *Yoneda product* (see [11, chapter 27]) is a bilinear map:

$$\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) \times \mathrm{Ext}^j_{\mathcal{O}_X}(\mathscr{H}, \mathscr{F}) \longrightarrow \mathrm{Ext}^{i+j}_{\mathcal{O}_X}(\mathscr{H}, \mathscr{G}).$$

This is defined by splicing extensions: an element of $\mathrm{Ext}^i(\mathscr{F}, \mathscr{G})$ is (morally) an equivalence class of i-fold extensions $0 \to \mathscr{G} \to \mathcal{E}^{i-1} \to \cdots \to \mathcal{E}^0 \to \mathscr{F} \to 0$, and the Yoneda product splices two such sequences together.
Now, since $H^j(X, \mathscr{F}) = \mathrm{Ext}^j_{\mathcal{O}_X}(\mathcal{O}_X, \mathscr{F})$, the Yoneda product specializes to give a pairing:

$$\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \omega_X) \times H^{n-i}(X, \mathscr{F}) \longrightarrow H^n(X, \omega_X) \xrightarrow{t} k.$$

This defines for each i a natural map:

$$\eta^i_{\mathscr{F}} : \mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \omega_X) \longrightarrow H^{n-i}(X, \mathscr{F})^*.$$

When $\mathscr{F}$ is locally free, $\mathrm{Ext}^i(\mathscr{F}, \omega_X) \cong H^i(X, \mathscr{F}^\vee \otimes \omega_X)$ by corollary 6.4.9, and the Yoneda pairing reduces to the cup product:

$$H^i(X,\, \mathscr{F}^\vee \otimes \omega_X) \times H^{n-i}(X, \mathscr{F}) \xrightarrow{\cup} H^n(X,\, \mathscr{F}^\vee \otimes \omega_X \otimes \mathscr{F}) \xrightarrow{\mathrm{ev} \otimes \mathrm{id}} H^n(X, \omega_X) \xrightarrow{t} k,$$

where $\mathrm{ev} : \mathscr{F}^\vee \otimes \mathscr{F} \to \mathcal{O}_X$ is the evaluation map. In the Čech framework, the cup product of cochains $\alpha \in C^p(\mathcal{U}, \mathscr{F}^\vee \otimes \omega_X)$ and $\beta \in C^q(\mathcal{U}, \mathscr{F})$ is defined by:

$$(\alpha \cup \beta)_{i_0,\ldots,i_{p+q}} = \alpha_{i_0,\ldots,i_p} \otimes \beta_{i_p,\ldots,i_{p+q}}\big|_{U_{i_0,\ldots,i_{p+q}}},$$

followed by the evaluation and trace maps.

Theorem 6.5.5: Duality for Projective Space

Let k be a field, $X = \mathbb{P}^n_k$ with $n \geq 1$, and $\omega_X = \mathcal{O}_X(-n-1)$. Then for every coherent sheaf $\mathscr{F}$ on X, the Yoneda pairing induces isomorphisms:

$$\eta^i_{\mathscr{F}} : \operatorname{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \omega_X) \xrightarrow{\sim} H^{n-i}(X, \mathscr{F})^* \qquad \text{for all } 0 \leq i \leq n.$$

Both sides vanish for $i > n$.

***Proof*:**
We verify the theorem in two stages: first for line bundles, then for all coherent sheaves.

Step 1: Verification for $\mathscr{F} = \mathcal{O}_X(d)$. By corollary 6.4.9, $\operatorname{Ext}^i(\mathcal{O}(d), \omega_X) \cong H^i(X, \mathcal{O}(-d-n-1))$. The Yoneda pairing (which reduces to the cup product for locally free sheaves) gives a bilinear map:

$$H^i(X, \mathcal{O}(-d-n-1)) \times H^{n-i}(X, \mathcal{O}(d)) \longrightarrow H^n(X, \mathcal{O}(-n-1)) \xrightarrow{t} k.$$

We must show this pairing is perfect for each i. By theorem 6.3.1(4), the intermediate cohomology vanishes: $H^j(X, \mathcal{O}(m)) = 0$ for $0 < j < n$ and all $m \in \mathbb{Z}$. So for $0 < i < n$, both $H^i(X, \mathcal{O}(-d-n-1))$ and $H^{n-i}(X, \mathcal{O}(d))$ are zero and the pairing is trivially perfect (both sides vanish).

For $i = 0$: the pairing is $H^0(X, \mathcal{O}(-d-n-1)) \times H^n(X, \mathcal{O}(d)) \to k$. If $-d-n-1 < 0$ (i.e. $d > -(n+1)$), then $H^0 = 0$ and $H^n = 0$, and the pairing is perfect. If $-d-n-1 \geq 0$ (i.e. $d \leq -(n+1)$), this is the pairing of theorem 6.3.1(3) (with the degree parameter set to $-d-n-1 \geq 0$), which is perfect.

For $i = n$: by symmetry, the pairing is $H^n(X, \mathcal{O}(-d-n-1)) \times H^0(X, \mathcal{O}(d)) \to k$. If $d < 0$, both sides vanish. If $d \geq 0$, this is again the perfect pairing of theorem 6.3.1(3) with the two factors interchanged (the pairing is symmetric up to the sign conventions in the cup product, which can be checked to be trivial in this case since all sheaves involved are line bundles).

Hence $\eta^i_{\mathcal{O}(d)}$ is an isomorphism for all i and all $d \in \mathbb{Z}$. By the additivity of Ext and cohomology in direct sums, the same holds for $\mathscr{F} = \bigoplus_j \mathcal{O}(d_j)$ (any finite direct sum of line bundles).

Step 2: Extension to all coherent sheaves. Let $\mathscr{F}$ be an arbitrary coherent sheaf on $X = \mathbb{P}^n_k$. Since X has dimension n and its local rings are regular of dimension $\leq n$ (the local ring at a point of codimension c is a regular local ring of dimension c), every coherent sheaf on X admits a finite locally free resolution of length at most n. More precisely, by corollary 5.3.26 and theorem 5.3.27, we can build a resolution:

$$0 \to \mathcal{E}_m \to \mathcal{E}_{m-1} \to \cdots \to \mathcal{E}_1 \to \mathcal{E}_0 \to \mathscr{F} \to 0$$

where each $\mathcal{E}_j$ is a finite direct sum of line bundles $\mathcal{O}(d)$ and $m \leq n$. We prove $\eta^i_{\mathscr{F}}$ is an isomorphism by induction on the length m of such a resolution.

Base case ($m = 0$): Here $\mathscr{F} = \mathcal{E}_0$ is a direct sum of line bundles, handled in Step 1.

Inductive step: Suppose $m \geq 1$. Break the resolution into the short exact sequence:

$$0 \to \mathcal{K} \to \mathcal{E}_0 \to \mathscr{F} \to 0,$$

where $\mathcal{K} = \ker(\mathcal{E}_0 \to \mathscr{F})$ is the first syzygy. Since the resolution of $\mathscr{F}$ restricts to a resolution $0 \to \mathcal{E}_m \to \cdots \to \mathcal{E}_1 \to \mathcal{K} \to 0$ of $\mathcal{K}$ of length $m-1$, the inductive hypothesis gives that $\eta^i_{\mathcal{K}}$ is an isomorphism for all i.

The short exact sequence $0 \to \mathcal{K} \to \mathcal{E}_0 \to \mathscr{F} \to 0$ induces two long exact sequences: one from $\operatorname{Ext}^*(-, \omega_X)$ (contravariant in the first argument) and one from $H^{n-*}(X, -)^*$ (the dual of the covariant cohomology sequence, which is exact since dualization is exact over a field). The naturality of the Yoneda pairing ensures that the maps η^i assemble into a morphism between these long exact sequences:

$$\begin{tikzcd}
\operatorname{Ext}^{i-1}(\mathcal{E}_0, \omega) \arrow[r] \arrow[d, "\sim"', "\eta^{i-1}_{\mathcal{E}_0}"] & \operatorname{Ext}^{i-1}(\mathcal{K}, \omega) \arrow[r] \arrow[d, "\sim"', "\eta^{i-1}_{\mathcal{K}}"] & \operatorname{Ext}^{i}(\mathscr{F}, \omega) \arrow[r] \arrow[d, "\eta^{i}_{\mathscr{F}}"] & \operatorname{Ext}^{i}(\mathcal{E}_0, \omega) \arrow[r] \arrow[d, "\sim"', "\eta^{i}_{\mathcal{E}_0}"] & \operatorname{Ext}^{i}(\mathcal{K}, \omega) \arrow[d, "\sim"', "\eta^{i}_{\mathcal{K}}"] \\
H^{n-i+1}(\mathcal{E}_0)^* \arrow[r] & H^{n-i+1}(\mathcal{K})^* \arrow[r] & H^{n-i}(\mathscr{F})^* \arrow[r] & H^{n-i}(\mathcal{E}_0)^* \arrow[r] & H^{n-i}(\mathcal{K})^*
\end{tikzcd}$$

The four outer vertical maps are isomorphisms (the $\mathcal{E}_0$ maps by Step 1, the $\mathcal{K}$ maps by the inductive hypothesis). By the five lemma, the middle map $\eta^i_{\mathscr{F}}$ is also an isomorphism. This holds for every i, completing the induction.

Vanishing for $i > n$: Both sides vanish: $\operatorname{Ext}^i(\mathscr{F}, \omega_X) = 0$ for $i > n$ by proposition 6.4.13(3), and $H^{n-i}(X, \mathscr{F}) = 0$ for $n - i < 0$ trivially.

Example 6.5.6: Serre Duality on Projective Space

1. *Structure sheaf.* For $\mathscr{F} = \mathcal{O}_X$ on $X = \mathbb{P}^n_k$:
$$\operatorname{Ext}^i(\mathcal{O}_X, \mathcal{O}(-n-1)) \cong H^i(X, \mathcal{O}(-n-1)) \cong H^{n-i}(X, \mathcal{O}_X)^*.$$
For $i = n$, this gives $H^n(X, \mathcal{O}(-n-1)) \cong H^0(X, \mathcal{O}_X)^* = k$. For $0 < i < n$, both sides are zero.

2. *Cotangent sheaf on $\mathbb{P}^2$.* Let $X = \mathbb{P}^2_k$ and $\mathscr{F} = \Omega_{X/k}$. Since Ω is locally free of rank 2, duality gives:
$$H^i(X, \Omega^\vee \otimes \mathcal{O}(-3)) \cong H^{2-i}(X, \Omega)^*.$$
Now $\Omega^\vee = \mathcal{T}_X$, the tangent sheaf, and $\mathcal{T}_X \cong \mathcal{T}_{\mathbb{P}^2}$. From the dual of the Euler sequence (theorem 5.5.18):
$$0 \to \mathcal{O}_X \to \mathcal{O}_X(1)^3 \to \mathcal{T}_X \to 0,$$
we get $\mathcal{T}_X \otimes \mathcal{O}(-3) \cong \mathcal{T}_X(-3)$, whose cohomology can be computed from the twisted Euler sequence. In particular:
$$H^0(X, \Omega) \cong H^2(X, \mathcal{T}_X(-3))^* = 0$$
(no nonzero global 1-forms on $\mathbb{P}^2$), confirming a result we already knew from Example 6.1.13(1) for $\mathbb{P}^1$.

3. *Skyscraper sheaves and residues.* Let $p \in \mathbb{P}^1_k$ be a closed point and let $\mathscr{F} = \kappa(p)$ be the skyscraper sheaf at p. Then $H^0(X, \kappa(p)) = k$ and $H^i(X, \kappa(p)) = 0$ for $i > 0$ (a skyscraper sheaf has no higher cohomology). Serre duality gives:
$$\operatorname{Ext}^1(\kappa(p), \mathcal{O}(-2)) \cong H^0(X, \kappa(p))^* = k, \qquad \operatorname{Hom}(\kappa(p), \mathcal{O}(-2)) = 0.$$
The non-trivial Ext group $\operatorname{Ext}^1(\kappa(p), \omega_X) \cong k$ has a concrete interpretation: the short exact sequence $0 \to \mathcal{O}(-2) \to \mathcal{O}(-1) \to \kappa(p) \to 0$ (where the first map is multiplication by the linear form vanishing at p, suitably twisted) represents a generator. This is the algebraic avatar of the *residue map* at p: the element of $\operatorname{Ext}^1(\kappa(p), \omega)$ corresponds to extracting the residue of a meromorphic differential form at p.

6.5.2 Duality for Projective Schemes

We now extend Serre duality from projective space to projective schemes. The key new ingredient is identifying the correct dualizing sheaf. For smooth varieties, this is the canonical sheaf; for singular schemes, the dualizing sheaf exists under a mild Cohen–Macaulay hypothesis but is more subtle to construct.

Recall from definition 5.5.25 that for a smooth variety X of dimension n over a field k, the *canonical sheaf* (or *canonical bundle*) is:

$$\omega_X = \det \Omega_{X/k} = \bigwedge^n \Omega_{X/k},$$

the top exterior power of the sheaf of differentials. This is an invertible sheaf. For $X = \mathbb{P}^n_k$, the computation of theorem 5.5.18 shows that $\omega_{\mathbb{P}^n} \cong \mathcal{O}_{\mathbb{P}^n}(-n-1)$, consistent with our earlier identification.

For a smooth projective variety X of dimension n embedded in $\mathbb{P}^N_k$ as a closed subscheme, the canonical sheaf is related to $\omega_{\mathbb{P}^N}$ by the *adjunction formula*. Recall from proposition 5.5.17 that if $X \hookrightarrow \mathbb{P}^N$ is a regular embedding of codimension $c = N - n$, there is an exact sequence:

$$\mathscr{I}_X/\mathscr{I}_X^2 \longrightarrow \Omega_{\mathbb{P}^N}\big|_X \longrightarrow \Omega_X \longrightarrow 0,$$

where $\mathscr{I}_X$ is the ideal sheaf of X. Taking top exterior powers (and using that the conormal sheaf $\mathcal{N}^\vee_{X/\mathbb{P}^N} = \mathscr{I}_X/\mathscr{I}_X^2$ is locally free of rank c for a regular embedding), one obtains:

Proposition 6.5.7: Adjunction Formula

Let X be a smooth closed subvariety of dimension n in $\mathbb{P}^N_k$, with codimension $c = N - n$. Then:

$$\omega_X \;\cong\; \omega_{\mathbb{P}^N}\big|_X \otimes \det\big(\mathcal{N}^\vee_{X/\mathbb{P}^N}\big)^{-1} \;\cong\; \mathcal{O}_{\mathbb{P}^N}(-N-1)\big|_X \otimes \det(\mathcal{N}_{X/\mathbb{P}^N}).$$

If X is a complete intersection, cut out by hypersurfaces of degrees $d_1, \ldots, d_c$, then $\mathcal{N}_{X/\mathbb{P}^N} \cong \bigoplus_{j=1}^c \mathcal{O}_X(d_j)$ and hence:

$$\omega_X \cong \mathcal{O}_X\left(\sum_{j=1}^c d_j - N - 1\right).$$

Proof:
The conormal–cotangent exact sequence for the regular embedding $i : X \hookrightarrow \mathbb{P}^N$ is:

$$0 \longrightarrow \mathcal{N}^\vee_{X/\mathbb{P}^N} \longrightarrow \Omega_{\mathbb{P}^N}\big|_X \longrightarrow \Omega_X \longrightarrow 0.$$

This is exact on the left because X is smooth (equivalently, the embedding is a regular immersion; see theorem 5.5.24). Taking the top exterior power of the middle term (which has rank N) and using the multiplicativity of determinants on short exact sequences of locally free sheaves:

$$\det(\Omega_{\mathbb{P}^N}|_X) \cong \det(\mathcal{N}^\vee_{X/\mathbb{P}^N}) \otimes \det(\Omega_X),$$

so $\omega_X = \det(\Omega_X) \cong \det(\Omega_{\mathbb{P}^N}|_X) \otimes \det(\mathcal{N}^\vee_{X/\mathbb{P}^N})^{-1} = \omega_{\mathbb{P}^N}|_X \otimes \det(\mathcal{N}_{X/\mathbb{P}^N})$.

For a complete intersection, the ideal sheaf $\mathscr{I}_X$ is generated by global sections $f_1, \ldots, f_c$ of degrees $d_1, \ldots, d_c$. The conormal sheaf is $\mathscr{I}_X/\mathscr{I}_X^2 \cong \bigoplus_{j=1}^c \mathcal{O}_X(-d_j)$ (the surjection $\bigoplus \mathcal{O}(-d_j) \twoheadrightarrow \mathscr{I}_X$

induces an isomorphism on the conormal level), so $\mathcal{N}_{X/\mathbb{P}^N} = (\mathscr{I}_X/\mathscr{I}_X^2)^\vee \cong \bigoplus \mathcal{O}_X(d_j)$. Hence $\det(\mathcal{N}_{X/\mathbb{P}^N}) = \mathcal{O}_X(\sum d_j)$ and $\omega_X \cong \mathcal{O}_X(-N-1+\sum d_j)$.

Example 6.5.8: Canonical Sheaves via Adjunction

1. *Smooth plane curve of degree d.* Let $C \subset \mathbb{P}^2_k$ be a smooth curve cut out by a polynomial of degree d. Then $\omega_C \cong \mathcal{O}_C(d-3)$. For $d=1$ (a line, genus 0): $\omega_C \cong \mathcal{O}_C(-2)$, so $H^0(C,\omega_C)=0$, confirming $g=0$. For $d=3$ (an elliptic curve, genus 1): $\omega_C \cong \mathcal{O}_C$, so $H^0(C,\omega_C)=k$, confirming $g=1$. For general $d \geq 1$, the genus is $g = \dim_k H^0(C,\omega_C) = \binom{d-1}{2}$ (by proposition 7.1.13).

2. *Smooth surface of degree d in $\mathbb{P}^3$.* Let $S \subset \mathbb{P}^3_k$ be a smooth surface of degree d. Then $\omega_S \cong \mathcal{O}_S(d-4)$. For $d=4$ (a K3 surface): $\omega_S \cong \mathcal{O}_S$, so S has trivial canonical bundle. For $d<4$, ω_S has negative degree (in an appropriate sense), and for $d>4$, the canonical bundle is ample.

3. *Hypersurface of degree d in $\mathbb{P}^n$.* More generally, a smooth hypersurface $X \subset \mathbb{P}^n_k$ of degree d has $\omega_X \cong \mathcal{O}_X(d-n-1)$. The canonical class is trivial (i.e. X is Calabi–Yau in dimension $n-1$) precisely when $d=n+1$.

We can now state the full Serre duality theorem. For the proof, we shall use the dualizing sheaf of projective space together with a closed embedding to transfer duality from $\mathbb{P}^N$ to X:

Theorem 6.5.9: Serre Duality for a Projective Scheme

Let X be a projective scheme of dimension n over a field k, and suppose X admits a dualizing sheaf ω_X (with trace map $t : H^n(X,\omega_X) \to k$). Then for every coherent sheaf $\mathscr{F}$ on X, the Yoneda pairing:

$$\operatorname{Ext}^i_{\mathcal{O}_X}(\mathscr{F},\omega_X) \times H^{n-i}(X,\mathscr{F}) \longrightarrow H^n(X,\omega_X) \xrightarrow{t} k$$

induces isomorphisms:

$$\operatorname{Ext}^i_{\mathcal{O}_X}(\mathscr{F},\omega_X) \;\cong\; H^{n-i}(X,\mathscr{F})^* \qquad \text{for all } 0 \leq i \leq n.$$

Furthermore, a dualizing sheaf exists in each of the following cases:

1. X is a closed subscheme of $\mathbb{P}^N_k$ that is equidimensional and Cohen–Macaulay. In this case:
$$\omega_X \cong \mathcal{E}xt^c_{\mathcal{O}_{\mathbb{P}^N}}(\mathcal{O}_X,\, \omega_{\mathbb{P}^N}),$$
where $c = N-n$ is the codimension. In particular, if X is a smooth variety, then $\omega_X \cong \det \Omega_{X/k}$.

2. $X = \mathbb{P}^n_k$, with $\omega_X = \mathcal{O}(-n-1)$.

***Proof*:**
We give the proof for $X = \mathbb{P}^n_k$ in full (this is theorem 6.5.5, already established above) and outline the key steps for the general case.

Existence of the dualizing sheaf. For a closed subscheme $i : X \hookrightarrow \mathbb{P}^N_k$ of codimension $c = N - n$, define:

$$\omega_X = \mathcal{E}xt^c_{\mathcal{O}_{\mathbb{P}^N}}(i_*\mathcal{O}_X,\, \omega_{\mathbb{P}^N}).$$

This is a coherent sheaf on $\mathbb{P}^N$ supported on X, hence can be regarded as a coherent sheaf on X (via proposition 6.2.4). When X is Cohen–Macaulay, the local Ext vanishes in all degrees except c (since the local rings of a Cohen–Macaulay scheme of codimension c have depth exactly c, and for a module of depth c over a regular local ring, the local Ext vanishes below degree c). This ensures ω_X is concentrated in a single degree and behaves well.

The trace map $t : H^n(X, \omega_X) \to k$ is constructed from the trace map for $\mathbb{P}^N$ via the local-to-global spectral sequence (theorem 6.4.8): the isomorphism $\operatorname{Ext}^N(\mathcal{O}_X, \omega_{\mathbb{P}^N}) \cong H^N(\mathbb{P}^N, \omega_{\mathbb{P}^N})^*$ (from theorem 6.5.5 applied to $\mathbb{P}^N$) and the identification of $\operatorname{Ext}^N(\mathcal{O}_X, \omega_{\mathbb{P}^N})$ with $H^n(X, \omega_X)$ (via the spectral sequence, using the Cohen–Macaulay vanishing) combine to give the trace.

When X is smooth, the identification $\omega_X \cong \det \Omega_{X/k}$ follows from proposition 6.5.7 and the fact that $\mathcal{E}xt^c(\mathcal{O}_X, \omega_{\mathbb{P}^N}) \cong \det(\mathcal{N}_{X/\mathbb{P}^N}) \otimes \omega_{\mathbb{P}^N}|_X$ for a regular embedding of codimension c.

Duality for the general case. Given that the dualizing sheaf exists with the correct trace, the proof that $\eta^i_{\mathscr{F}}$ is an isomorphism for all coherent $\mathscr{F}$ follows the same pattern as theorem 6.5.5. One resolves $\mathscr{F}$ by sheaves for which duality is known (using corollary 5.3.26), then applies the five lemma. The key point is that on a projective scheme over k, the coherent sheaves $\mathcal{O}_X(d)$ for $d \gg 0$ satisfy duality (this can be reduced to duality on $\mathbb{P}^N$ via the closed embedding and the adjunction between i^* and i_*), and every coherent sheaf admits a finite resolution by such sheaves.

For smooth varieties, the Ext groups simplify and Serre duality takes its most elegant form:

Corollary 6.5.10: Serre Duality: Nonsingular Projective Variety

Let X be a nonsingular projective variety of dimension n over a field k, with canonical sheaf $\omega_X = \det \Omega_{X/k}$. Then for every locally free coherent sheaf $\mathscr{F}$ on X:

$$H^i(X, \mathscr{F}) \;\cong\; H^{n-i}(X,\, \mathscr{F}^\vee \otimes \omega_X)^* \qquad \text{for all } 0 \le i \le n.$$

In particular:

1. $H^n(X, \omega_X) \cong k$.
2. $H^i(X, \mathcal{O}_X) \cong H^{n-i}(X, \omega_X)^*$ for all i.
3. The *arithmetic genus* $p_a(X) = (-1)^n(\chi(\mathcal{O}_X) - 1)$ can be computed from the cohomology of the canonical sheaf via:

$$p_a(X) = (-1)^n \Big(\sum_{i=0}^{n} (-1)^i \dim_k H^i(X, \mathcal{O}_X) - 1 \Big) = \dim_k H^0(X, \omega_X) - \cdots$$

where Serre duality determines the "upper half" of the cohomology from the "lower half."

***Proof*:**
Since $\mathscr{F}$ is locally free, $\operatorname{Ext}^i(\mathscr{F}, \omega_X) \cong H^i(X, \mathscr{F}^\vee \otimes \omega_X)$ by corollary 6.4.9. Substituting into theorem 6.5.9:

$$H^i(X,\, \mathscr{F}^\vee \otimes \omega_X) \cong H^{n-i}(X, \mathscr{F})^*.$$

Replacing $\mathscr{F}$ by $\mathscr{F}^\vee \otimes \omega_X$ (which is also locally free) and using $(\mathscr{F}^\vee \otimes \omega_X)^\vee \otimes \omega_X \cong \mathscr{F}$:

$$H^i(X, \mathscr{F}) \cong H^{n-i}(X,\, \mathscr{F}^\vee \otimes \omega_X)^*.$$

Part (1) follows by setting $\mathscr{F} = \mathcal{O}_X$ and $i = 0$: $H^0(X, \mathcal{O}_X) = k$ (since X is a variety, hence integral), so $H^n(X, \omega_X) \cong H^0(X, \mathcal{O}_X)^* = k$. Part (2) is the special case $\mathscr{F} = \mathcal{O}_X$.

Example 6.5.11: Applications of Serre Duality

1. *Curves: Riemann–Roch.* Let C be a smooth projective curve of genus g over k. Serre duality gives $H^1(C, \mathcal{L}) \cong H^0(C, \mathcal{L}^\vee \otimes \omega_C)^*$ for any line bundle $\mathcal{L}$. If $\mathcal{L} = \mathcal{O}_C(D)$ for a divisor D, this becomes:

$$h^1(C, \mathcal{O}(D)) = h^0(C, \omega_C(-D)) = h^0(C, \mathcal{O}(K - D)),$$

where K is a canonical divisor (with $\mathcal{O}(K) \cong \omega_C$) and $h^i = \dim_k H^i$. Substituting into the Euler characteristic formula $\chi(\mathcal{O}(D)) = h^0(\mathcal{O}(D)) - h^1(\mathcal{O}(D))$ and using the fact that $\chi(\mathcal{O}(D)) = \deg D + 1 - g$ (which follows from the additivity of χ on short exact sequences), we recover the *Riemann–Roch theorem* for curves (cf. theorem 7.1.18):

$$h^0(C, \mathcal{O}(D)) - h^0(C, \mathcal{O}(K - D)) = \deg D + 1 - g.$$

In particular, setting $D = K$: $h^0(\omega_C) - h^0(\mathcal{O}_C) = \deg K + 1 - g$, giving $\deg K = 2g - 2$ (since $h^0(\omega_C) = g$ by definition 5.5.26 and $h^0(\mathcal{O}_C) = 1$).

2. *The Hodge numbers of a smooth surface.* Let S be a smooth projective surface over $\mathbb{C}$. Serre duality gives:

$$\begin{aligned} H^0(S, \mathcal{O}_S) &\cong H^2(S, \omega_S)^*, \\ H^1(S, \mathcal{O}_S) &\cong H^1(S, \omega_S)^*, \\ H^2(S, \mathcal{O}_S) &\cong H^0(S, \omega_S)^*. \end{aligned}$$

   Since $h^{p,q} = \dim H^q(X, \Omega^p)$ by the Dolbeault theorem (theorem 0.6.17) and GAGA, these translate to symmetries of the Hodge diamond: $h^{0,0} = h^{2,2}$, $h^{0,1} = h^{2,1}$, and $h^{0,2} = h^{2,0}$. This is the algebraic shadow of Hodge symmetry (theorem 0.6.28).

3. *Kodaira vanishing (preview).* Let X be a smooth projective variety of dimension n over $\mathbb{C}$ and let $\mathcal{L}$ be an ample line bundle. The Kodaira vanishing theorem (which requires analytic or characteristic-0 methods beyond our present scope) asserts:

$$H^i(X, \omega_X \otimes \mathcal{L}) = 0 \qquad \text{for } i > 0.$$

   By Serre duality, this is equivalent to:

$$H^{n-i}(X, \mathcal{L}^{-1}) = 0 \qquad \text{for } i > 0,$$

   i.e. $H^j(X, \mathcal{L}^{-1}) = 0$ for $j < n$. Thus Kodaira vanishing can be stated either as acyclicity of $\omega_X \otimes \mathcal{L}$ or as "low-degree vanishing" of $\mathcal{L}^{-1}$. Serre duality reduces one formulation to the other.

4. *Self-duality of abelian varieties.* Let A be an abelian variety of dimension g over k. Then $\Omega_{A/k}$ is trivial: $\Omega_{A/k} \cong \mathcal{O}_A^g$ (the tangent bundle of a group scheme is trivial, by translation). Hence $\omega_A = \det(\mathcal{O}_A^g) \cong \mathcal{O}_A$, and Serre duality gives:

$$H^i(A, \mathcal{O}_A) \cong H^{g-i}(A, \mathcal{O}_A)^*.$$

   In particular, $\dim_k H^i(A, \mathcal{O}_A) = \binom{g}{i}$ (which follows from the Künneth formula for abelian varieties, or from the degeneration of the Hodge-to-de Rham spectral sequence).

6.5.12 Remark: Beyond Smooth Varieties For singular projective schemes, Serre duality still holds under the Cohen–Macaulay hypothesis (theorem 6.5.9(1)), but the dualizing sheaf ω_X is no longer necessarily invertible. It is a coherent sheaf, and its local structure reflects the singularities of X. The theory of dualizing complexes (due to Grothendieck and Hartshorne) extends duality to arbitrary proper schemes, replacing the dualizing sheaf with a complex of sheaves $\omega_X^\bullet \in D^b(\mathbf{Coh}(X))$. In this generality, Serre duality becomes Grothendieck duality:

$$\mathbf{R}\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \omega_X^\bullet) \cong \mathbf{R}\Gamma(X, \mathscr{F})^*$$

in the derived category. For Cohen–Macaulay X, the dualizing complex is concentrated in a single degree and is the dualizing sheaf we have defined. For Gorenstein schemes (where ω_X is invertible), the theory is especially clean. We shall not pursue derived duality in these notes, but the reader should be aware that theorem 6.5.9 is a shadow of this deeper structure.

6.6 Higher Direct Images

The sheaf cohomology groups $H^i(X, \mathscr{F})$ encode global information about a single scheme X. But in practice, schemes arise in *families*: a morphism $f : X \to Y$ presents X as a family of fibers $X_y = f^{-1}(y)$ parameterized by points of Y, and we want to understand how the cohomology $H^i(X_y, \mathscr{F}|_{X_y})$ varies as y moves. The *higher direct image sheaves* $R^i f_* \mathscr{F}$ are sheaves on the base Y that package this information: they are the "relative" version of sheaf cohomology, and they reduce to ordinary cohomology when Y is a point.

The central result of this section is the *finiteness theorem*: if $f : X \to Y$ is a projective morphism of Noetherian schemes and $\mathscr{F}$ is a coherent sheaf on X, then each $R^i f_* \mathscr{F}$ is a coherent sheaf on Y. When $Y = \operatorname{Spec} k$ is a point, this says that the cohomology groups of coherent sheaves on projective varieties are finite-dimensional — a fact we have already used (via forward reference) in proposition 6.4.13 and implicitly throughout our Serre duality arguments.

6.6.1 Definition and First Properties

Recall from definition 1.3.1 that the *direct image* (or pushforward) of a sheaf $\mathscr{F}$ on X along a continuous map $f : X \to Y$ is the sheaf $f_* \mathscr{F}$ on Y defined by $f_* \mathscr{F}(V) = \mathscr{F}(f^{-1}(V))$ for open $V \subseteq Y$. When f is a morphism of ringed spaces and $\mathscr{F}$ is an $\mathcal{O}_X$-module, the pushforward $f_* \mathscr{F}$ is an $\mathcal{O}_Y$-module (via the sheaf map $f^\# : \mathcal{O}_Y \to f_* \mathcal{O}_X$). The functor $f_* : \mathcal{O}_X\textbf{-Mod} \to \mathcal{O}_Y\textbf{-Mod}$ is left exact but not right exact in general.

Definition 6.6.1: Higher Direct Image Sheaves

Let $f : X \to Y$ be a morphism of ringed spaces. The *higher direct image sheaves* are the right derived functors of f_*:

$$R^i f_* : \mathcal{O}_X\textbf{-Mod} \longrightarrow \mathcal{O}_Y\textbf{-Mod}, \qquad R^i f_*(\mathscr{F}) = h^i(f_* \mathscr{I}^\bullet),$$

where $0 \to \mathscr{F} \to \mathscr{I}^\bullet$ is an injective resolution of $\mathscr{F}$ in the category of $\mathcal{O}_X$-modules. By construction, $R^0 f_* = f_*$, and a short exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ of $\mathcal{O}_X$-modules induces a long exact sequence of $\mathcal{O}_Y$-modules:

$$0 \to f_* \mathscr{F}' \to f_* \mathscr{F} \to f_* \mathscr{F}'' \xrightarrow{\delta} R^1 f_* \mathscr{F}' \to R^1 f_* \mathscr{F} \to \cdots$$

When $Y = \{pt\}$ is a point, the morphism $f : X \to \{pt\}$ is the unique map to the terminal space, and $f_* = \Gamma(X, -)$. Hence $R^i f_* \mathscr{F} = H^i(X, \mathscr{F})$, recovering ordinary sheaf cohomology. This is the sense in which higher direct images "relativize" cohomology.

The following result is fundamental: it says that higher direct images can be computed "stalkwise" by computing cohomology on the preimages of open sets.

Proposition 6.6.2: Sheafification of Presheaf Direct Images

Let $f : X \to Y$ be a continuous map of topological spaces and let $\mathscr{F}$ be a sheaf of abelian groups on X. Then $R^i f_* \mathscr{F}$ is the sheafification of the presheaf on Y defined by:

$$V \longmapsto H^i(f^{-1}(V),\, \mathscr{F}|_{f^{-1}(V)})$$

for open $V \subseteq Y$. If this presheaf is already a sheaf (e.g., when Y has a basis of open sets V for which $f^{-1}(V)$ is "cohomologically well-behaved" in an appropriate sense), then no sheafification is needed.

***Proof*:**
Let $0 \to \mathscr{F} \to \mathscr{I}^\bullet$ be an injective resolution. For each open $V \subseteq Y$:

$$(f_* \mathscr{I}^p)(V) = \mathscr{I}^p(f^{-1}(V)).$$

The presheaf $V \mapsto h^i(\mathscr{I}^\bullet(f^{-1}(V)))$ computes $H^i(f^{-1}(V), \mathscr{F}|_{f^{-1}(V)})$ provided the restriction $\mathscr{I}^\bullet|_{f^{-1}(V)}$ is an injective (or at least acyclic) resolution of $\mathscr{F}|_{f^{-1}(V)}$. By lemma 6.4.5, the restriction of an injective sheaf to an open subset is injective, so this is indeed the case.

Now, the derived functors $R^i f_* \mathscr{F}$ are defined as the cohomology sheaves $h^i(f_* \mathscr{I}^\bullet)$. The presheaf $V \mapsto h^i((f_* \mathscr{I}^\bullet)(V))$ need not be a sheaf (taking cohomology does not commute with the sheaf condition in general), so $R^i f_* \mathscr{F}$ is the *sheafification* of this presheaf. Since $(f_* \mathscr{I}^\bullet)(V) = \mathscr{I}^\bullet(f^{-1}(V))$, the presheaf in question is $V \mapsto H^i(f^{-1}(V), \mathscr{F})$, as claimed.

Proposition 6.6.3: Stalks of Higher Direct Images

Let $f : X \to Y$ be a continuous map of topological spaces, $\mathscr{F}$ a sheaf of abelian groups on X, and $y \in Y$. Then there is a natural map:

$$(R^i f_* \mathscr{F})_y \longrightarrow H^i(X_y,\, \mathscr{F}|_{X_y}),$$

where $X_y = f^{-1}(y)$ is the (set-theoretic) fiber. This map is an isomorphism if y has a basis of open neighborhoods $\{V_\alpha\}$ such that the natural maps $\varinjlim_\alpha H^i(f^{-1}(V_\alpha), \mathscr{F}) \to H^i(X_y, \mathscr{F}|_{X_y})$ are isomorphisms.

***Proof*:**
By proposition 6.6.2, the stalk of $R^i f_* \mathscr{F}$ at y is:

$$(R^i f_* \mathscr{F})_y = \varinjlim_{V \ni y} H^i(f^{-1}(V),\, \mathscr{F}|_{f^{-1}(V)}).$$

The inclusion $X_y = f^{-1}(y) \hookrightarrow f^{-1}(V)$ for each open $V \ni y$ induces a restriction map $H^i(f^{-1}(V), \mathscr{F}) \to H^i(X_y, \mathscr{F}|_{X_y})$, and these are compatible as V varies, yielding the natural map on the colimit. The condition for this to be an isomorphism is precisely that cohomology "commutes with the limit" of shrinking V to $\{y\}$.

6.6.4 Remark: When Do Stalks Compute Fiber Cohomology? The natural map $(R^i f_* \mathscr{F})_y \to H^i(X_y, \mathscr{F}|_{X_y})$ is *not* an isomorphism in general, and the discrepancy is important. The stalk sees the cohomology of all "thickenings" of the fiber (i.e., the preimages of open neighborhoods of y), while the fiber cohomology sees only the fiber itself. These can differ when nearby fibers contribute non-trivially.
There are two important situations where the map is an isomorphism:

1. When f is proper and the topological spaces are "reasonable" (e.g., locally Noetherian schemes). In this case, the proper mapping theorem ensures that the cohomological information does not "leak" from nearby fibers.

2. When $Y = \operatorname{Spec} R$ for a local ring $(R, \mathfrak{m})$ and $y = [\mathfrak{m}]$ is the closed point. Then the only open set containing y is Y itself, so the stalk is simply $H^i(X, \mathscr{F})$ and the fiber is $X_y = X \times_Y \operatorname{Spec}(R/\mathfrak{m})$.

In the scheme-theoretic setting, the correct fiber to consider is the *scheme-theoretic fiber* $X_y = X \times_Y \operatorname{Spec} \kappa(y)$ (see definition 3.2.9), and the comparison between the stalk and the fiber cohomology involves the *formal functions theorem* (Grothendieck's theorem on formal functions), which we shall not develop in full here.

We now establish the key properties that make higher direct images useful in the scheme-theoretic setting:

Proposition 6.6.5: Quasi-Coherence of Higher Direct Images

Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism of schemes, and let $\mathscr{F}$ be a quasi-coherent sheaf on X. Then each $R^i f_* \mathscr{F}$ is quasi-coherent on Y.

***Proof*:**
The question is local on Y, so we may assume $Y = \operatorname{Spec} R$ is affine. Since f is quasi-compact and quasi-separated, the preimage $X = f^{-1}(Y)$ is a quasi-compact quasi-separated scheme. Choose a finite affine open cover $\mathcal{U} = \{U_1, \ldots, U_r\}$ of X. Each intersection $U_{i_0, \ldots, i_p} = U_{i_0} \cap \cdots \cap U_{i_p}$ is quasi-compact (by the quasi-separated hypothesis), and $\mathscr{F}$ is quasi-coherent on each.

On the affine base, the Čech-to-derived spectral sequence (which generalizes the comparison of theorem 6.1.18 to the relative setting) shows that $R^i f_* \mathscr{F}$ can be computed by a Čech-type complex. More precisely, the presheaf $V \mapsto H^i(f^{-1}(V), \mathscr{F})$ is already a sheaf when Y is affine (this uses the quasi-compact quasi-separated hypothesis), and it is computed by the Čech complex associated to the cover $\mathcal{U}$. Each term of this complex is a product of quasi-coherent sheaves on Y (since the pushforward of a quasi-coherent sheaf along a quasi-compact quasi-separated morphism is quasi-coherent, by proposition 5.1.20), and the cohomology of a complex of quasi-coherent sheaves is quasi-coherent.

6.6.2 The Finiteness Theorem

The most important structural result about higher direct images is that they preserve coherence for projective morphisms. This is the "relative" version of the statement that cohomology groups of coherent sheaves on projective varieties are finite-dimensional, and it is the technical engine behind semicontinuity theorems, base change, and the construction of Hilbert schemes.

The strategy is to reduce to the computation of cohomology of twisted sheaves on projective space (theorem 6.3.1), which we already know completely. The key tools are:

1. Every coherent sheaf on a projective scheme is a quotient of a direct sum of twisted sheaves (corollary 5.3.26).
2. Cohomology of twisted sheaves on $\mathbb{P}^n_R$ gives finitely generated R-modules (theorem 6.3.1).
3. Descending induction on i, using the dimension bound $H^i = 0$ for $i > \dim X$ (theorem 6.2.5).

Theorem 6.6.6: Finiteness of Higher Direct Images

Let $f : X \to Y$ be a projective morphism of Noetherian schemes, and let $\mathscr{F}$ be a coherent sheaf on X. Then:

1. Each $R^i f_* \mathscr{F}$ is a coherent sheaf on Y.
2. $R^i f_* \mathscr{F} = 0$ for $i > \dim X$.

***Proof*:**
Part (2) follows immediately from theorem 6.2.5: the stalks $(R^i f_* \mathscr{F})_y$ are subquotients of $H^i(f^{-1}(V), \mathscr{F})$ for open neighborhoods V of y, and $\dim f^{-1}(V) \le \dim X$.

For part (1), the question is local on Y, so we may assume $Y = \operatorname{Spec} R$ is affine. Since f is projective, there is a closed immersion $i : X \hookrightarrow \mathbb{P}^N_R$ for some N, and f factors as $f = \pi \circ i$ where $\pi : \mathbb{P}^N_R \to \operatorname{Spec} R$ is the structure morphism. By proposition 6.2.4, $R^i f_* \mathscr{F} \cong R^i \pi_*(i_* \mathscr{F})$. Since i is a closed immersion, $i_* \mathscr{F}$ is coherent on $\mathbb{P}^N_R$ (by proposition 3.1.24). Hence it suffices to prove the theorem for $\mathbb{P}^N_R$ over $\operatorname{Spec} R$, i.e., to show:

Claim: For any coherent sheaf $\mathscr{G}$ on $\mathbb{P}^N_R$, the R-module $H^i(\mathbb{P}^N_R, \mathscr{G})$ is finitely generated.

We prove the claim by descending induction on i. For $i > N$, we have $H^i = 0$ by theorem 6.2.5, and there is nothing to prove.

For the inductive step, assume that $H^j(\mathbb{P}^N_R, \mathscr{H})$ is finitely generated for all $j > i$ and all coherent $\mathscr{H}$. By corollary 5.3.26, $\mathscr{G}$ admits a surjection from a direct sum of twisted sheaves:

$$\mathcal{E} = \bigoplus_{j=1}^{s} \mathcal{O}_{\mathbb{P}^N}(d_j) \twoheadrightarrow \mathscr{G} \to 0$$

for some integers $d_1, \dots, d_s$. Let $\mathcal{K} = \ker(\mathcal{E} \twoheadrightarrow \mathscr{G})$, which is coherent (since $\mathbf{Coh}(X)$ is an abelian category). The short exact sequence $0 \to \mathcal{K} \to \mathcal{E} \to \mathscr{G} \to 0$ induces a long exact sequence:

$$H^i(\mathbb{P}^N, \mathcal{E}) \longrightarrow H^i(\mathbb{P}^N, \mathscr{G}) \longrightarrow H^{i+1}(\mathbb{P}^N, \mathcal{K}) \longrightarrow H^{i+1}(\mathbb{P}^N, \mathcal{E}).$$

By theorem 6.3.1, $H^i(\mathbb{P}^N, \mathcal{E}) = \bigoplus_j H^i(\mathbb{P}^N, \mathcal{O}(d_j))$ is a finitely generated R-module (it is even free). By the inductive hypothesis, $H^{i+1}(\mathbb{P}^N, \mathcal{K})$ is finitely generated. Hence $H^i(\mathbb{P}^N, \mathscr{G})$, which sits between two finitely generated R-modules in the exact sequence, is itself finitely generated (using the standard fact that a subquotient of a finitely generated module over a Noetherian ring is finitely generated).

The most commonly used special case is when the base is a field:

Corollary 6.6.7: Finite-Dimensionality of Cohomology

Let X be a projective scheme over a field k, and let $\mathscr{F}$ be a coherent sheaf on X. Then each cohomology group $H^i(X, \mathscr{F})$ is a finite-dimensional k-vector space.

***Proof*:**
Apply theorem 6.6.6 with $Y = \operatorname{Spec} k$. The higher direct images $R^i f_* \mathscr{F}$ are coherent sheaves on $\operatorname{Spec} k$, i.e., finite-dimensional k-vector spaces. Since $R^i f_* \mathscr{F} = H^i(X, \mathscr{F})$ when Y is a point, the result follows.

This result justifies the use of the notation $h^i(X, \mathscr{F}) = \dim_k H^i(X, \mathscr{F})$ for the *Betti numbers* of the sheaf $\mathscr{F}$, and underpins the entire theory of Hilbert polynomials and Euler characteristics. Let us also record the general version for the base that we referenced earlier:

Corollary 6.6.8: (Reminder) Pushforward of Coherent Sheaves by Projective Morphisms

Let $f : X \to Y$ be a projective morphism of Noetherian schemes. Then f_* sends coherent sheaves on X to coherent sheaves on Y, and more generally, each $R^i f_*$ sends $\mathbf{Coh}(X)$ to $\mathbf{Coh}(Y)$.

***Proof*:**
This is a restatement of theorem 6.6.6(1).

6.6.9 Remark: Properness vs. Projectivity The finiteness theorem as stated requires f to be *projective*. The natural level of generality is *proper*: Grothendieck proved that theorem 6.6.6 holds whenever f is a proper morphism of Noetherian schemes. The proof in the proper case is substantially harder: one cannot reduce to $\mathbb{P}^N$ via a closed immersion, and instead uses Chow's lemma (theorem 3.5.28) to approximate f by a projective morphism, together with a careful analysis of the resulting spectral sequence. Since every projective morphism is proper (theorem 3.5.26), our statement covers the cases most commonly encountered in practice.

6.6.3 Computation of Higher Direct Images

Higher direct images can be computed using the Čech machinery, provided the morphism is sufficiently well-behaved:

Proposition 6.6.10: Čech Computation of Higher Direct Images

Let $f : X \to Y$ be a separated morphism of Noetherian schemes, $\mathscr{F}$ a quasi-coherent sheaf on X, and $\mathcal{U} = \{U_1, \ldots, U_r\}$ a finite open affine cover of X. Then for each open affine $V = \operatorname{Spec} R \subseteq Y$, the higher direct images are computed by:

$$R^i f_*(\mathscr{F})(V) \;\cong\; \check{H}^i(\mathcal{U}|_{f^{-1}(V)}, \mathscr{F}|_{f^{-1}(V)}),$$

where $\mathcal{U}|_{f^{-1}(V)} = \{U_j \cap f^{-1}(V)\}$ is the restricted cover.

Proof:
For affine $V \subseteq Y$, the preimage $f^{-1}(V)$ is a separated Noetherian scheme (separatedness is preserved under base change), and the sets $U_j \cap f^{-1}(V)$ form an affine cover of $f^{-1}(V)$ (as U_j is affine and $f^{-1}(V)$ is separated, the intersection of two affine opens in $f^{-1}(V)$ is affine by proposition 3.5.8). By theorem 6.1.18, the Čech cohomology with respect to this affine cover computes the derived-functor cohomology:

$$\check{H}^i(\mathcal{U}|_{f^{-1}(V)}, \mathscr{F}) \cong H^i(f^{-1}(V), \mathscr{F}).$$

Since the presheaf $V \mapsto H^i(f^{-1}(V), \mathscr{F})$ is already a sheaf on affine opens of Y (as noted in proposition 6.6.5), this computes $R^i f_* \mathscr{F}$ on the basis of affine opens.

Example 6.6.11: Higher Direct Images

1. *Structure morphism of projective space.* Let $\pi : \mathbb{P}^n_R \to \operatorname{Spec} R$ be the structure morphism. Then:
$$R^i \pi_*\big(\mathcal{O}_{\mathbb{P}^n}(d)\big) \;\cong\; \widetilde{H^i(\mathbb{P}^n_R, \mathcal{O}(d))},$$
which by theorem 6.3.1 is the quasi-coherent sheaf associated to the R-module $H^i(\mathbb{P}^n_R, \mathcal{O}(d))$. In particular, $\pi_*(\mathcal{O}_{\mathbb{P}^n}) \cong \widetilde{R} = \mathcal{O}_{\operatorname{Spec} R}$ and $R^i\pi_*(\mathcal{O}_{\mathbb{P}^n}) = 0$ for $i > 0$. More generally:
$$\pi_*(\mathcal{O}(d)) \cong \begin{cases} \widetilde{(R[x_0, \ldots, x_n])_d} & d \geq 0, \\ 0 & d < 0, \end{cases}$$
and all higher direct images vanish for $0 < i < n$.

2. *Blowing up a point.* Let $Y = \mathbb{A}^2_k = \operatorname{Spec} k[x, y]$ and let $f : X = \mathrm{Bl}_0 Y \to Y$ be the blowup of the origin. This is a projective morphism with f an isomorphism over $Y \setminus \{0\}$ and fiber $f^{-1}(0) \cong \mathbb{P}^1_k$ (the exceptional divisor E). Then:
$$f_*\mathcal{O}_X \cong \mathcal{O}_Y \qquad \text{and} \qquad R^i f_* \mathcal{O}_X = 0 \text{ for } i > 0.$$
The first statement holds because f is birational and Y is normal (so $f_*\mathcal{O}_X = \mathcal{O}_Y$ by the algebraic Hartogs lemma, theorem 2.6.18). The vanishing of higher direct images reflects the fact that the fibers are either points (away from the origin) or $\mathbb{P}^1$ (at the origin), and $H^i(\mathbb{P}^1, \mathcal{O}) = 0$ for $i > 0$.

   More interesting is the behavior of the ideal sheaf of the exceptional divisor. Since $E \cong \mathbb{P}^1$ and $\mathcal{O}_X(-E)|_E \cong \mathcal{O}_{\mathbb{P}^1}(1)$ (the conormal bundle of the exceptional divisor in a blowup is the tautological bundle), we have $R^1 f_*(\mathcal{O}_X(-E)) = 0$ but $f_*(\mathcal{O}_X(-E)) \cong \mathfrak{m}_0$, the maximal ideal of the origin in $\mathcal{O}_Y$.

3. *Projection of a product.* Let C be a smooth projective curve of genus g over k, and consider the projection $f = \mathrm{pr}_2 : C \times_k \mathbb{P}^1_k \to \mathbb{P}^1_k$. Then for the structure sheaf:

$$f_* \mathcal{O}_{C \times \mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}, \qquad R^1 f_* \mathcal{O}_{C \times \mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}^g.$$

The first isomorphism follows from $H^0(C, \mathcal{O}_C) = k$. For the second: by the Künneth formula (or by proposition 6.6.10 with the product cover), the stalk of $R^1 f_* \mathcal{O}$ at a point $p \in \mathbb{P}^1$ is $H^1(C, \mathcal{O}_C) \cong k^g$, and the resulting sheaf is locally free of rank g. Since $\mathbb{P}^1$ has no non-trivial vector bundles of rank g with vanishing H^1 (by the classification of vector bundles on $\mathbb{P}^1$), we get $R^1 f_* \mathcal{O} \cong \mathcal{O}^g$.

This illustrates the general principle: the rank of $R^i f_* \mathscr{F}$ at a point y is *at least* $h^i(X_y, \mathscr{F}|_{X_y})$, with equality when the fiber dimension is constant (the precise statement is the semicontinuity theorem, which we treat in the next section).

4. *Failure for non-proper morphisms.* Let $j : \mathbb{A}^1_k \setminus \{0\} \hookrightarrow \mathbb{A}^1_k$ be the open immersion. Then $j_* \mathcal{O} = \mathcal{O}_{\mathbb{A}^1}[1/x]$ (rational functions regular away from the origin), which is *not* coherent as an $\mathcal{O}_{\mathbb{A}^1}$-module: it is quasi-coherent (corresponding to the $k[x]$-module $k[x, x^{-1}]$) but not finitely generated. Also, $R^1 j_* \mathcal{O} = 0$ (since $\mathbb{A}^1 \setminus \{0\}$ is affine, hence acyclic), but the pushforward already fails coherence. This shows that the properness (or at least projectivity) hypothesis in theorem 6.6.6 cannot be dropped.

To close this section, we record the connection between higher direct images and the Leray spectral sequence, which is the fundamental tool for computing cohomology of a space in terms of a morphism to a simpler space:

Theorem 6.6.12: Leray Spectral Sequence

Let $f : X \to Y$ be a morphism of ringed spaces and let $\mathscr{F}$ be a sheaf of $\mathcal{O}_X$-modules. Then there is a convergent spectral sequence:

$$E_2^{p,q} = H^p(Y, R^q f_* \mathscr{F}) \implies H^{p+q}(X, \mathscr{F}).$$

This is natural in $\mathscr{F}$.

***Proof*:**
This is the Grothendieck spectral sequence for the composite functor $\Gamma(X, -) = \Gamma(Y, -) \circ f_*$. We need f_* to send injective $\mathcal{O}_X$-modules to $\Gamma(Y, -)$-acyclic $\mathcal{O}_Y$-modules. Indeed, if $\mathscr{I}$ is an injective $\mathcal{O}_X$-module, then $f_* \mathscr{I}$ is flasque: for $U \subseteq V$ open in Y, the restriction $f_* \mathscr{I}(V) = \mathscr{I}(f^{-1}(V)) \to \mathscr{I}(f^{-1}(U)) = f_* \mathscr{I}(U)$ is surjective because $\mathscr{I}$ is flasque (by proposition 6.1.5) and $f^{-1}(U) \subseteq f^{-1}(V)$. Hence $f_* \mathscr{I}$ is acyclic for $\Gamma(Y, -)$, and the Grothendieck spectral sequence applies.

Example 6.6.13: Applications of the Leray Spectral Sequence

1. *Affine morphisms.* If $f : X \to Y$ is an affine morphism of Noetherian schemes and $\mathscr{F}$ is quasi-coherent, then $R^q f_* \mathscr{F} = 0$ for all $q > 0$ (since the fibers are affine). The Leray spectral sequence degenerates at E_2, giving $H^p(X, \mathscr{F}) \cong H^p(Y, f_* \mathscr{F})$. This reduces the cohomology of X to the cohomology of Y with coefficients in the pushforward.

2. *Projective bundles.* Let $\mathcal{E}$ be a locally free sheaf of rank $r+1$ on a Noetherian scheme Y and let $X = \mathbb{P}(\mathcal{E}) \to Y$ be the associated projective bundle. The higher direct images of the twisted sheaves $\mathcal{O}_X(d)$ are computed by the relative version of theorem 6.3.1: $f_* \mathcal{O}_X(d) \cong \mathrm{Sym}^d \mathcal{E}$ for $d \geq 0$, and $R^i f_* \mathcal{O}_X(d) = 0$ for $0 < i < r$. The Leray spectral sequence then computes $H^*(X, \mathcal{O}_X(d))$ in terms of $H^*(Y, \mathrm{Sym}^d \mathcal{E})$.

3. *Recovering the projection formula.* For a proper morphism $f : X \to Y$ and a locally free sheaf $\mathcal{E}$ on Y, the *projection formula* gives:

$$R^i f_*(\mathscr{F} \otimes f^* \mathcal{E}) \;\cong\; (R^i f_* \mathscr{F}) \otimes \mathcal{E}.$$

   This follows from the fact that tensoring with a locally free sheaf is exact and commutes with the formation of higher direct images (since $f^*\mathcal{E}$ is locally free on X and the tensor product with a flat module commutes with cohomology). Combined with the Leray spectral sequence, the projection formula reduces many cohomological computations to known cases.

6.6.14 Remark: The Landscape of Finiteness Results The finiteness theorem (theorem 6.6.6) is one of several finiteness results that govern the cohomology of schemes. Let us place it in context:

1. *Affine vanishing* (theorem 6.2.9): $H^i = 0$ for $i > 0$ on affine schemes with quasi-coherent coefficients. This is the strongest vanishing result, but it applies only to affine schemes.

2. *Dimension bound* (theorem 6.2.5): $H^i = 0$ for $i > \dim X$ on Noetherian spaces. This holds for *all* sheaves of abelian groups, not just quasi-coherent ones.

3. *Finiteness* (theorem 6.6.6): H^i is finitely generated for coherent sheaves on projective schemes over Noetherian rings. This does *not* say the groups vanish, only that they are "finitely complicated."

4. *Serre vanishing* (see theorem 5.3.25): for a coherent sheaf $\mathscr{F}$ on a projective scheme X, there exists d_0 such that $H^i(X, \mathscr{F}(d)) = 0$ for all $i > 0$ and all $d \geq d_0$. This says that twisting "enough" kills all higher cohomology.

Together, these results form the basic toolkit for cohomological arguments in algebraic geometry. The finiteness theorem sits at the center: it is what allows us to define numerical invariants (Euler characteristics, Hilbert polynomials, Hodge numbers) that can then be studied using the vanishing results.

6.7 Euler Characteristics and Hilbert Polynomials

With the finiteness theorem (theorem 6.6.6) in hand, we know that the cohomology groups $H^i(X, \mathscr{F})$ of a coherent sheaf on a projective scheme over a field are finite-dimensional vector spaces. This means we can *count*: we can form the alternating sum of their dimensions, obtaining the *Euler characteristic* $\chi(X, \mathscr{F})$. The Euler characteristic is a cruder invariant than the individual cohomology groups, but it has a decisive advantage: it is *additive* on short exact sequences, while the individual h^i are not. This additivity makes the Euler characteristic computable by purely algebraic means, without explicitly resolving the sheaf.

When X is a projective scheme and $\mathscr{F}$ is twisted by a line bundle $\mathcal{O}_X(d)$, the Euler characteristic $\chi(X, \mathscr{F}(d))$ is a polynomial in d: the *Hilbert polynomial*. This polynomial encodes fundamental geometric invariants of X — its dimension, degree, and arithmetic genus — and is the basis for the theory of Hilbert schemes and moduli spaces (cf. the discussion in section 3.3.6).

For curves, the interplay between the Euler characteristic and Serre duality yields the *Riemann–Roch theorem*, the single most important computational tool in the theory of algebraic curves.

6.7.1 The Euler Characteristic

Definition 6.7.1: Euler Characteristic

Let X be a projective scheme over a field k and let $\mathscr{F}$ be a coherent sheaf on X. The *Euler characteristic* of $\mathscr{F}$ is:

$$\chi(X, \mathscr{F}) = \sum_{i=0}^{\dim X} (-1)^i \dim_k H^i(X, \mathscr{F}) = \sum_{i \geq 0} (-1)^i h^i(X, \mathscr{F}),$$

where $h^i(X, \mathscr{F}) = \dim_k H^i(X, \mathscr{F})$. The sum is finite by theorem 6.2.5 and each term is finite by corollary 6.6.7.

The definition immediately connects to the one given in the curves chapter (**??**), which was stated for line bundles on curves. The key property that makes the Euler characteristic tractable is the following:

Proposition 6.7.2: Additivity of the Euler Characteristic

Let X be a projective scheme over a field k. If $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ is a short exact sequence of coherent sheaves on X, then:

$$\chi(X, \mathscr{F}) = \chi(X, \mathscr{F}') + \chi(X, \mathscr{F}'').$$

***Proof*:**
The short exact sequence induces a long exact sequence in cohomology:

$$0 \to H^0(X, \mathscr{F}') \to H^0(X, \mathscr{F}) \to H^0(X, \mathscr{F}'') \to H^1(X, \mathscr{F}') \to \cdots \to H^n(X, \mathscr{F}'') \to 0,$$

where $n = \dim X$. This is a finite exact sequence of finite-dimensional k-vector spaces. The alternating sum of dimensions in any finite exact sequence is zero (this is a standard exercise in linear algebra: if $0 \to V_0 \to V_1 \to \cdots \to V_m \to 0$ is exact, then $\sum(-1)^j \dim V_j = 0$). Grouping the terms by sheaf:

$$\sum_{i=0}^{n} \left[(-1)^i h^i(\mathscr{F}') - (-1)^i h^i(\mathscr{F}) + (-1)^i h^i(\mathscr{F}'')\right] = 0,$$

which rearranges to $\chi(\mathscr{F}) = \chi(\mathscr{F}') + \chi(\mathscr{F}'')$.

6.7.3 Remark: Euler Characteristic as a Group Homomorphism The additivity property can be rephrased as follows. The Grothendieck group $K_0(X)$ of coherent sheaves on X is the free abelian group generated by isomorphism classes $[\mathscr{F}]$ of coherent sheaves, modulo the relation $[\mathscr{F}] = [\mathscr{F}'] + [\mathscr{F}'']$ for every short exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$. The Euler characteristic defines a group homomorphism:

$$\chi : K_0(X) \longrightarrow \mathbb{Z}, \qquad [\mathscr{F}] \longmapsto \chi(X, \mathscr{F}).$$

This is the starting point for K-theory in algebraic geometry. Many invariants (the Hilbert polynomial, the Chern character, the Todd class) factor through $K_0(X)$, and their computability ultimately rests on the additivity of χ.

Example 6.7.4: Euler Characteristics

1. *Twisted sheaves on* $\mathbb{P}^n_k$. By theorem 6.3.1, the only nonzero cohomology groups of $\mathcal{O}_{\mathbb{P}^n}(d)$ are H^0 (when $d \geq 0$) and H^n (when $d \leq -(n+1)$). Hence:

$$\chi(\mathbb{P}^n_k, \mathcal{O}(d)) = \begin{cases} \binom{d+n}{n} & d \geq 0, \\ 0 & -(n+1) < d < 0, \\ (-1)^n \binom{-d-1}{n} & d \leq -(n+1). \end{cases}$$

   A remarkable fact (which we will prove in the next subsection) is that all three cases are given by a single polynomial in d:

$$\chi(\mathbb{P}^n_k, \mathcal{O}(d)) = \binom{d+n}{n} = \frac{(d+1)(d+2)\cdots(d+n)}{n!}$$

   for *all* $d \in \mathbb{Z}$. For instance, when $n = 1$: $\chi(\mathbb{P}^1, \mathcal{O}(d)) = d+1$. When $d = -1$: $\chi(\mathbb{P}^1, \mathcal{O}(-1)) = 0$, which is correct since both H^0 and H^1 vanish. When $d = -2$: $\chi(\mathbb{P}^1, \mathcal{O}(-2)) = -1$, reflecting $h^0 = 0$ and $h^1 = 1$.

2. *Structure sheaf of a curve.* Let C be a smooth projective curve of genus g over k. Then $h^0(C, \mathcal{O}_C) = 1$ (since C is integral and proper) and $h^1(C, \mathcal{O}_C) = g$ (this is one definition of the genus; see definition 7.1.12). Hence:

$$\chi(C, \mathcal{O}_C) = 1 - g.$$

By Serre duality (corollary 6.5.10), $h^1(C, \mathcal{O}_C) = h^0(C, \omega_C) = p_g$, the geometric genus (definition 5.5.26). So $\chi(C, \mathcal{O}_C) = 1 - p_g$, confirming the equivalence of the arithmetic and geometric genus for smooth curves (proposition 7.1.13).

3. *Structure sheaf of a surface.* Let S be a smooth projective surface over k. Then:

$$\chi(S, \mathcal{O}_S) = h^0(\mathcal{O}_S) - h^1(\mathcal{O}_S) + h^2(\mathcal{O}_S) = 1 - q + p_g,$$

where $q = h^1(\mathcal{O}_S)$ is the *irregularity* and $p_g = h^2(\mathcal{O}_S) = h^0(\omega_S)$ is the geometric genus (the last equality by Serre duality). The integer $\chi(\mathcal{O}_S) = 1 - q + p_g$ is a fundamental invariant in the classification of surfaces.

4. *Additivity in action.* Let $C \subset \mathbb{P}^2_k$ be a smooth plane curve of degree d. The short exact sequence $0 \to \mathcal{O}_{\mathbb{P}^2}(-d) \to \mathcal{O}_{\mathbb{P}^2} \to \mathcal{O}_C \to 0$ gives:

$$\chi(C, \mathcal{O}_C) = \chi(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}) - \chi(\mathbb{P}^2, \mathcal{O}(-d)) = 1 - \binom{-d+2}{2} = 1 - \frac{(d-1)(d-2)}{2}.$$

Hence $g = \frac{(d-1)(d-2)}{2}$, the classical genus formula for plane curves. For $d = 1$: $g = 0$ (a line). For $d = 3$: $g = 1$ (an elliptic curve). For $d = 4$: $g = 3$ (a quartic).

6.7.2 Hilbert Polynomials

The Euler characteristic of a twisted sheaf $\mathscr{F}(d)$ is a polynomial in d. This is the most important non-obvious property of the Euler characteristic, and it rests on two facts: Serre vanishing (which eventually kills all higher cohomology) and the additivity of χ (which allows induction via hyperplane sections).

Theorem 6.7.5: Hilbert Polynomial

Let X be a projective scheme over a field k, equipped with a very ample sheaf $\mathcal{O}_X(1)$, and let $\mathscr{F}$ be a coherent sheaf on X. Then there exists a polynomial $P_{\mathscr{F}} \in \mathbb{Q}[t]$, called the *Hilbert polynomial* of $\mathscr{F}$, such that:

$$P_{\mathscr{F}}(d) = \chi(X, \mathscr{F}(d)) \qquad \text{for all } d \in \mathbb{Z}.$$

The degree of $P_{\mathscr{F}}$ equals the dimension of the support of $\mathscr{F}$, and the leading coefficient is positive.

***Proof*:**
We proceed by induction on $r = \dim(\operatorname{supp}\mathscr{F})$.

Base case ($r = 0$): If $\operatorname{supp}\mathscr{F}$ is zero-dimensional, it consists of finitely many closed points (since X is Noetherian). On a zero-dimensional space, H^0 is the only nonzero cohomology group, and twisting by $\mathcal{O}_X(d)$ has no effect on a sheaf supported at finitely many points (since the stalk of $\mathcal{O}_X(d)$ at a closed point is isomorphic to $\mathcal{O}_{X,x}$, with the twist absorbed into the local isomorphism). Hence $\chi(X, \mathscr{F}(d)) = h^0(X, \mathscr{F}) = \dim_k \Gamma(X, \mathscr{F})$, a constant polynomial of degree 0.

Inductive step: Let $r \geq 1$. Choose a hyperplane section $H \in |\mathcal{O}_X(1)|$ that does not contain any associated point of $\mathscr{F}$ (this is possible because $\mathscr{F}$ has finitely many associated points and $|\mathcal{O}_X(1)|$ is base-point free by theorem 5.3.25). Then multiplication by the section $s \in H^0(X, \mathcal{O}_X(1))$ defining H gives an injective map $\mathscr{F}(-1) \xrightarrow{\cdot s} \mathscr{F}$. Let $\mathscr{G}$ be the cokernel:

$$0 \longrightarrow \mathscr{F}(-1) \xrightarrow{\cdot s} \mathscr{F} \longrightarrow \mathscr{G} \longrightarrow 0.$$

Since s vanishes on H, the sheaf $\mathscr{G}$ is supported on $\operatorname{supp}(\mathscr{F}) \cap H$, which has dimension at most $r-1$ (the hyperplane cuts the dimension by at least one, since it avoids the associated points). Twisting the whole sequence by $\mathcal{O}_X(d)$:

$$0 \longrightarrow \mathscr{F}(d-1) \longrightarrow \mathscr{F}(d) \longrightarrow \mathscr{G}(d) \longrightarrow 0.$$

By the additivity of χ:

$$\chi(X, \mathscr{F}(d)) - \chi(X, \mathscr{F}(d-1)) = \chi(X, \mathscr{G}(d)).$$

By the inductive hypothesis, $\chi(X, \mathscr{G}(d))$ is a polynomial in d of degree at most $r-1$. The function $\Delta P(d) = P(d) - P(d-1)$ is the *difference operator*; if ΔP is a polynomial of degree $\leq r-1$, then P is a polynomial of degree $\leq r$[a]. Since $\chi(X, \mathscr{F}(d))$ is an integer for all d and agrees with a polynomial in d after taking differences, it is itself a polynomial in d (with rational coefficients), of degree at most r.

To show the degree is exactly r (and the leading coefficient is positive), we use the fact that for $d \gg 0$, Serre vanishing (theorem 5.3.25) gives $\chi(X, \mathscr{F}(d)) = h^0(X, \mathscr{F}(d))$, which is a polynomial in d of degree equal to $\dim(\operatorname{supp}\mathscr{F})$ with positive leading coefficient (this follows from the theory of Hilbert functions in commutative algebra).

[a]This is the discrete analogue of the fact that if $f'(x)$ is a polynomial of degree $r-1$, then $f(x)$ is a polynomial of degree r. Concretely: the binomial coefficients $\binom{d}{j}$ form a basis for $\mathbb{Q}[d]$, and $\Delta\binom{d}{j} = \binom{d}{j} - \binom{d-1}{j} = \binom{d-1}{j-1}$, so the difference operator lowers the degree by exactly one.

Definition 6.7.6: Hilbert Polynomial and Arithmetic Genus

Let X be a projective scheme of dimension n over a field k. The *Hilbert polynomial* of X (with respect to $\mathcal{O}_X(1)$) is $P_X = P_{\mathcal{O}_X}$. The *arithmetic genus* of X is:

$$p_a(X) = (-1)^n\big(\chi(X, \mathcal{O}_X) - 1\big) = (-1)^n(P_X(0) - 1).$$

For a curve ($n = 1$): $p_a = 1 - \chi(\mathcal{O}_C) = g$. For a surface ($n = 2$): $p_a = \chi(\mathcal{O}_S) - 1 = p_g - q$.

Example 6.7.7: Hilbert Polynomials

1. *Projective space.* By Example 6.7.4(1), $P_{\mathbb{P}^n}(d) = \binom{d+n}{n}$. This is a polynomial of degree n in d with leading coefficient $1/n!$. The arithmetic genus is $p_a(\mathbb{P}^n) = (-1)^n(\binom{n}{n} - 1) = 0$.
2. *Plane curve of degree d.* Let $C \subset \mathbb{P}^2_k$ be a curve of degree d. The short exact sequence

$0 \to \mathcal{O}_{\mathbb{P}^2}(-d) \to \mathcal{O}_{\mathbb{P}^2} \to \mathcal{O}_C \to 0$ gives, after twisting by $\mathcal{O}(m)$ and taking χ:

$$P_C(m) = \binom{m+2}{2} - \binom{m-d+2}{2} = dm - \frac{d(d-3)}{2}.$$

This is a polynomial of degree 1 (as expected, since $\dim C = 1$) with leading coefficient d, which is the *degree* of C. The arithmetic genus is $p_a = 1 - P_C(0) = \frac{(d-1)(d-2)}{2}$.

3. *Hilbert polynomial of a line bundle on a curve.* Let C be a smooth projective curve of genus g and let $\mathcal{L}$ be a line bundle of degree ℓ on C. By the Riemann–Roch theorem (which we prove below):
$$P_{\mathcal{L}}(d) = \chi(C, \mathcal{L}(d)) = \ell + d \cdot \deg(\mathcal{O}_C(1)) + 1 - g.$$
In particular, $\chi(C, \mathcal{L}) = \deg \mathcal{L} + 1 - g$, confirming **??**.

4. *Twisted ideal sheaf.* Let $Z \subset \mathbb{P}^n_k$ be a closed subscheme with ideal sheaf $\mathscr{I}_Z$. Then the short exact sequence $0 \to \mathscr{I}_Z \to \mathcal{O}_{\mathbb{P}^n} \to \mathcal{O}_Z \to 0$ gives $P_{\mathscr{I}_Z}(d) = P_{\mathbb{P}^n}(d) - P_Z(d)$. The Hilbert polynomial of $\mathscr{I}_Z$ encodes the "difference" between the ambient space and the subscheme, and this is the key to defining Hilbert schemes: two subschemes are in the same Hilbert scheme component if and only if they have the same Hilbert polynomial.

The following result connects the Hilbert polynomial to the classical Hilbert function from commutative algebra. It was already stated in **??**; we can now give a cohomological proof:

Proposition 6.7.8: Hilbert Polynomial and Hilbert Function

Let $X \subseteq \mathbb{P}^N_k$ be a projective scheme and $\mathscr{F}$ a coherent sheaf on X. Then for $d \gg 0$:

$$P_{\mathscr{F}}(d) = h^0(X, \mathscr{F}(d)).$$

That is, the Hilbert polynomial agrees with the *Hilbert function* $d \mapsto h^0(X, \mathscr{F}(d))$ for all sufficiently large d.

***Proof*:**
By theorem 5.3.25 (Serre vanishing), there exists d_0 such that $H^i(X, \mathscr{F}(d)) = 0$ for all $i > 0$ and all $d \geq d_0$. For such d:

$$P_{\mathscr{F}}(d) = \chi(X, \mathscr{F}(d)) = h^0(X, \mathscr{F}(d)) - 0 + 0 - \cdots = h^0(X, \mathscr{F}(d)).$$

The degree and leading coefficient of the Hilbert polynomial have geometric meaning:

Proposition 6.7.9: Degree and Leading Coefficient

Let $X \subseteq \mathbb{P}^N_k$ be a projective scheme of dimension n and let $\mathscr{F}$ be a coherent sheaf with $\dim(\operatorname{supp}\mathscr{F}) = r$. Write:

$$P_{\mathscr{F}}(d) = \frac{a_r}{r!}\, d^r + \text{lower order terms.}$$

Then a_r is a positive integer. When $\mathscr{F} = \mathcal{O}_X$, the integer a_n is the *degree* of X in $\mathbb{P}^N$, i.e., the number of points in the intersection of X with a general linear subspace of complementary dimension (counted with multiplicity).

***Proof*:**
The positivity of a_r follows from the fact that $P_{\mathscr{F}}(d) = h^0(X, \mathscr{F}(d))$ for $d \gg 0$, and this grows like $d^r/r!$ (the Hilbert function of a module of dimension r over a polynomial ring has this growth rate; see theorem 2.6.11 for the connection between Hilbert functions and dimension). The integrality of a_r follows from the inductive proof of theorem 6.7.5: the difference $\Delta P_{\mathscr{F}}$ has leading coefficient $a_r/(r-1)!$, and by induction this integer is the degree of the support restricted to a hyperplane section.

The identification of a_n with the degree of X when $\mathscr{F} = \mathcal{O}_X$ follows from Bézout's theorem ([**NateClassicalAlgGeo**]): the intersection of X with n general hyperplanes is a zero-dimensional scheme of length a_n.

6.7.10 Remark: Invariance of the Hilbert Polynomial The Hilbert polynomial depends on the choice of embedding (or more precisely, on the choice of very ample sheaf $\mathcal{O}_X(1)$). However, the *constant term* $P_X(0) = \chi(X, \mathcal{O}_X)$ is independent of the embedding: it depends only on the abstract scheme X, since it is defined purely in terms of the cohomology of the structure sheaf. Similarly, the arithmetic genus $p_a(X) = (-1)^n(\chi(\mathcal{O}_X) - 1)$ is an intrinsic invariant. In fact, by theorem 5.5.28, the geometric genus $p_g = h^0(X, \omega_X)$ is a birational invariant for smooth varieties, and Serre duality relates it to the "top" cohomology of $\mathcal{O}_X$.
The higher coefficients of P_X do depend on the embedding, but they carry geometric information about the embedding itself (degree, sectional genus, etc.).

6.7.3 Riemann–Roch for Curves

We now bring together Serre duality and the Euler characteristic to prove the Riemann–Roch theorem for smooth projective curves. This result was stated in theorem 7.1.18; we can now give a complete proof using the cohomological tools developed in this chapter.

Recall from definition 7.1.1 that a *curve* over a field k is a separated scheme of finite type over k of pure dimension 1. A *smooth projective curve* is a smooth, projective, integral scheme of dimension 1 over k.

For a smooth projective curve C over k, the cohomology groups have the following structure: H^0 and H^1 are the only potentially nonzero groups (since $\dim C = 1$), the canonical sheaf $\omega_C = \Omega_{C/k}$ is an invertible sheaf, and Serre duality (corollary 6.5.10) gives $H^1(C, \mathcal{L}) \cong H^0(C, \mathcal{L}^\vee \otimes \omega_C)^*$ for any line bundle $\mathcal{L}$.

We need the following fundamental relationship between line bundles and divisors, which connects the degree of a divisor to the Euler characteristic:

Lemma 6.7.11: Degree and Euler Characteristic

Let C be a smooth projective curve over a field k and let $\mathcal{L}$ be a line bundle on C. Then:

$$\chi(C, \mathcal{L}) = \deg \mathcal{L} + \chi(C, \mathcal{O}_C) = \deg \mathcal{L} + 1 - g,$$

where $\deg \mathcal{L}$ is the degree of $\mathcal{L}$ (i.e., the degree of the associated Cartier divisor class; see proposition 5.4.31).

***Proof*:**
By proposition 5.4.31, every line bundle on C is of the form $\mathcal{O}_C(D)$ for some divisor D. We prove the formula by induction on the "complexity" of D, reducing to the case of effective divisors supported at a single point.

Case 1: $D = p$ a single closed point. Let $\kappa(p)$ be the residue field and let $[\kappa(p) : k] = \delta$. The short exact sequence $0 \to \mathcal{O}_C \to \mathcal{O}_C(p) \to \kappa(p) \to 0$ (where the first map is the inclusion of regular functions into functions with at most a simple pole at p, and the quotient is the skyscraper sheaf at p) gives:

$$\chi(\mathcal{O}_C(p)) = \chi(\mathcal{O}_C) + \chi(\kappa(p)) = (1 - g) + \delta = \deg(p) + 1 - g,$$

since $\deg(p) = [\kappa(p) : k] = \delta$ and $\chi(\kappa(p)) = h^0(\kappa(p)) = \delta$.

Case 2: D is a general divisor. Write $D = D' + p$ for some closed point p and divisor D' with one fewer term. The short exact sequence $0 \to \mathcal{O}_C(D') \to \mathcal{O}_C(D) \to \kappa(p) \to 0$ gives:

$$\chi(\mathcal{O}_C(D)) = \chi(\mathcal{O}_C(D')) + \delta = (\deg D' + 1 - g) + \delta = \deg D + 1 - g,$$

where we used the inductive hypothesis and $\deg D = \deg D' + \deg p = \deg D' + \delta$. Since every divisor can be written as a difference of effective divisors, and we can reduce the negative part by the same argument applied to $\mathcal{O}_C(-p)$ (with the reversed exact sequence), the formula holds for all divisors.

Theorem 6.7.12: Riemann–Roch Theorem

Let C be a smooth projective curve of genus g over a field k, and let D be a divisor on C with associated line bundle $\mathcal{L} = \mathcal{O}_C(D)$. Let K be a canonical divisor (so $\mathcal{O}_C(K) \cong \omega_C$). Then:

$$h^0(C, \mathcal{O}_C(D)) - h^0(C, \mathcal{O}_C(K - D)) = \deg D + 1 - g.$$

Equivalently, using the notation $\ell(D) = h^0(C, \mathcal{O}_C(D))$:

$$\ell(D) - \ell(K - D) = \deg D + 1 - g.$$

***Proof*:**
By Serre duality (corollary 6.5.10):

$$h^1(C, \mathcal{O}_C(D)) = \dim_k H^1(C, \mathcal{O}_C(D)) = \dim_k H^0(C, \mathcal{O}_C(D)^\vee \otimes \omega_C)^* = h^0(C, \mathcal{O}_C(K - D)),$$

where we used $\mathcal{O}_C(D)^\vee \otimes \omega_C \cong \mathcal{O}_C(-D) \otimes \mathcal{O}_C(K) \cong \mathcal{O}_C(K - D)$. Now apply lemma 6.7.11:

$$h^0(C, \mathcal{O}_C(D)) - h^1(C, \mathcal{O}_C(D)) = \chi(C, \mathcal{O}_C(D)) = \deg D + 1 - g.$$

Substituting $h^1(C, \mathcal{O}_C(D)) = h^0(C, \mathcal{O}_C(K - D))$ gives the result.

Example 6.7.13: Riemann–Roch in Practice

1. *Degree of the canonical divisor.* Setting $D = K$ in Riemann–Roch: $\ell(K) - \ell(0) = \deg K + 1 - g$. Since $\ell(K) = h^0(\omega_C) = g$ and $\ell(0) = h^0(\mathcal{O}_C) = 1$:

   $$\deg K = 2g - 2.$$

   For a smooth plane curve of degree d: $\omega_C \cong \mathcal{O}_C(d - 3)$ (by proposition 6.5.7), so $\deg K = d(d-3)$. Setting $g = (d-1)(d-2)/2$, we verify: $2g - 2 = (d-1)(d-2) - 2 = d^2 - 3d = d(d-3)$.

2. *Very ample line bundles on curves.* A line bundle $\mathcal{L}$ on a smooth curve C of genus g is very ample (hence gives an embedding into projective space) if $\deg \mathcal{L} \geq 2g + 1$. To see this: by Riemann–Roch, $\ell(D) = \deg D + 1 - g$ whenever $\deg D > 2g - 2 = \deg K$ (since then $\deg(K - D) < 0$, forcing $\ell(K - D) = 0$). Furthermore, for $\deg D \geq 2g + 1$, the linear system $|D|$ separates points and tangent vectors (the standard argument uses Riemann–Roch for $D - p$ and $D - 2p$, which have degrees $\geq 2g - 1$ and $\geq 2g - 1$ respectively, and one checks the required surjectivity).

3. *Elliptic curves.* Let C be a smooth projective curve of genus 1 with a rational point $O \in C(k)$. By Riemann–Roch with $D = 3O$:

   $$\ell(3O) = 3 + 1 - 1 = 3$$

   (since $\deg(K - 3O) = 2 \cdot 1 - 2 - 3 = -3 < 0$, so $\ell(K - 3O) = 0$). The three-dimensional space $H^0(C, \mathcal{O}(3O))$ gives a closed embedding $C \hookrightarrow \mathbb{P}^2$, and the image is a smooth plane cubic (cf. example **??**).

4. *Rational curves.* Let C have genus 0. Then $\chi(\mathcal{O}_C) = 1$ and for any line bundle $\mathcal{L}$ of degree $d \geq 0$:

   $$h^0(C, \mathcal{L}) = d + 1$$

   (since $h^1 = h^0(\omega_C \otimes \mathcal{L}^{-1})$ and $\deg(\omega_C \otimes \mathcal{L}^{-1}) = -2 - d < 0$). In particular, $h^0(\mathcal{O}_C(1)) = 2$ gives a morphism $C \to \mathbb{P}^1$ of degree 1, so $C \cong \mathbb{P}^1$. Every smooth rational curve over k with a rational point is isomorphic to $\mathbb{P}^1_k$.

6.7.14 Remark: Beyond Curves The Riemann–Roch theorem for curves is the "$n = 1$" case of a family of results that express the Euler characteristic in terms of topological and geometric invariants. In higher dimensions, the *Hirzebruch–Riemann–Roch theorem* asserts that for a smooth projective variety X of dimension n and a locally free sheaf $\mathcal{E}$:

$$\chi(X, \mathcal{E}) = \int_X \mathrm{ch}(\mathcal{E}) \cdot \mathrm{td}(T_X),$$

where $\mathrm{ch}(\mathcal{E})$ is the Chern character of $\mathcal{E}$, $\mathrm{td}(T_X)$ is the Todd class of the tangent bundle, and the integral is the degree of the component in codimension n (i.e., the intersection with the fundamental class). For curves, the Chern character of a line bundle is $1 + c_1(\mathcal{L})$ and the Todd class of T_C is $1 + \frac{1}{2}c_1(T_C) = 1 + \frac{1}{2}(2 - 2g)$, so the formula reduces to $\chi(\mathcal{L}) = \deg \mathcal{L} + 1 - g$. For surfaces, it gives the *Noether formula*:

$$\chi(\mathcal{O}_S) = \frac{1}{12}(K_S^2 + e(S)),$$

where K_S^2 is the self-intersection of the canonical class and $e(S)$ is the topological Euler characteristic. These generalizations belong to the intersection-theoretic framework developed in Part II.

6.8 Flat Morphisms and Semicontinuity

In the previous sections, we studied the cohomology of a *single* scheme. In practice, however, schemes arise in *families*: a morphism $f : X \to Y$ presents X as a family of fibers X_y parameterized by points $y \in Y$, and the higher direct images $R^i f_* \mathscr{F}$ package the cohomology of these fibers into a sheaf on the base. A fundamental question is: how does the function $y \mapsto h^i(X_y, \mathscr{F}_y)$ behave as y varies?

The answer involves two intertwined ideas. The first is *flatness*: the condition that the fibers of f vary "continuously," without sudden jumps in dimension or embedded components. Flat families are the algebraic analogue of fiber bundles, and they are the correct notion of "continuously varying family" in algebraic geometry. The second idea is *semicontinuity*: the dimensions $h^i(X_y, \mathscr{F}_y)$ can jump *up* at special points but not down, and when they are constant, the higher direct images are locally free and commute with base change.

These results are essential for moduli theory. The semicontinuity theorem ensures that the loci where cohomological invariants jump form closed subsets, and Grauert's theorem guarantees that the "generic" behavior of cohomology is well-controlled. We shall use both in the discussion of Hilbert schemes (section 3.3.6) and in relating algebraic and analytic cohomology via GAGA (section 6.9).

6.8.1 Flat Families

Definition 6.8.1: Family of Schemes

Let X, B be schemes. Then a *Family of schemes* is given by a morphism $\pi : X \to B$. The scheme B is sometimes called the *parameter space*.

The notion of a family of schemes as it stands is much too general: consider any family $\pi : X \to B$, choose $b \in B$, and create a new family by taking the disjoint union of $X \setminus \pi^{-1}(b)$ and Y for some arbitrary scheme Y where Y is sent to b. From this, it can be deduced that there is no real properties that can be deduced, and further constrictions must be given.

Recall from commutative algebra (see [11, chapter 20]) that an R-module M is *flat* if the functor $M \otimes_R -$ is exact. Geometrically, flatness of a morphism means that the fibers do not "degenerate" in an uncontrolled way:

Definition 6.8.2: Flat Morphism

Let $f : X \to Y$ be a morphism of schemes.

1. An $\mathcal{O}_X$-module $\mathscr{F}$ is *flat over* Y (or f*-flat*) if for every point $x \in X$, the stalk $\mathscr{F}_x$ is a flat $\mathcal{O}_{Y,f(x)}$-module (via the local homomorphism $f_x^\# : \mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$).
2. The morphism f itself is *flat* if $\mathcal{O}_X$ is flat over Y, i.e., if $\mathcal{O}_{X,x}$ is a flat $\mathcal{O}_{Y,f(x)}$-module for every $x \in X$.

Example 6.8.3: Flat and Non-Flat Morphisms

1. *Open immersions.* Any open immersion $j : U \hookrightarrow X$ is flat: the stalk map $\mathcal{O}_{X,x} \to \mathcal{O}_{U,x}$ is the identity for $x \in U$.
2. *Localization.* The morphism $\operatorname{Spec} R_f \to \operatorname{Spec} R$ induced by the localization $R \to R_f$ is flat, since localization is an exact functor.
3. *Finite free morphisms.* If $f : \operatorname{Spec} S \to \operatorname{Spec} R$ makes S a free R-module, then f is flat. More generally, if S is a finitely generated projective R-module, then f is flat.
4. *The family of conics.* Consider the morphism $f : V(xy - t) \to \mathbb{A}^1_k = \operatorname{Spec} k[t]$, where the fiber over $t = a \neq 0$ is the smooth hyperbola $xy = a$, and the fiber over $t = 0$ is the union of coordinate axes $xy = 0$. The coordinate ring is $k[x, y, t]/(xy - t) \cong k[x, y]$, and the map to $k[t]$ sends $t \mapsto xy$. This is flat: $k[x, y]$ is free over $k[xy] = k[t]$ (with basis given by the monomials $x^i y^j$ with either $i = 0$ or $j = 0$). The fiber over $t = 0$ is reducible, but the family is still flat because the reducibility is "expected" — the total space is smooth.
5. *A non-flat family.* The blowup morphism $f : \mathrm{Bl}_0\mathbb{A}^2 \to \mathbb{A}^2$ is *not* flat: the fiber over the origin is $\mathbb{P}^1$ (one-dimensional), while the fiber over every other point is a single point (zero-dimensional). Flatness would require the fibers to have the same dimension (at least for an equidimensional source), so this jump is an obstruction. More precisely, the structure sheaf $\mathcal{O}_{\mathrm{Bl}_0\mathbb{A}^2}$ is not flat over $\mathcal{O}_{\mathbb{A}^2,0}$.
6. *Families of subschemes.* Let $X = V(x^2 - t) \subset \mathbb{A}^1_x \times \mathbb{A}^1_t$ and let $f : X \to \mathbb{A}^1_t$ be the projection. For $t \neq 0$, the fiber is two reduced points $x = \pm\sqrt{t}$; for $t = 0$, the fiber is the non-reduced scheme $\operatorname{Spec} k[x]/(x^2)$. This family *is* flat: $k[x, t]/(x^2 - t) \cong k[x]$ is free of rank 2 over $k[t]$ (with basis $\{1, x\}$). The non-reduced fiber at $t = 0$ is the "correct" flat limit of the two-point fibers: it remembers the multiplicity. This is exactly the phenomenon we encountered with intersection multiplicities in section 2.2.2.

The key property of flat families for cohomology is that they interact well with base change. Recall from definition 3.2.3 that for a morphism $f : X \to Y$ and a base change $g : Y' \to Y$, we form the fiber product:

$$\begin{tikzcd} X' = X \times_Y Y' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "g"'] & Y \end{tikzcd}$$

For any $\mathcal{O}_X$-module $\mathscr{F}$, the *pullback* $g'^*\mathscr{F}$ is a sheaf on X', and we can form both $R^i f'_*(g'^*\mathscr{F})$ (compute cohomology on the fibers of f') and $g^*(R^i f_*\mathscr{F})$ (pull back the higher direct image). These are both sheaves on Y', and there is a natural *base change map*:

$$\beta^i : \; g^*(R^i f_*\mathscr{F}) \longrightarrow R^i f'_*(g'^*\mathscr{F}).$$

This map is not an isomorphism in general. When it is, we say that *cohomology commutes with base change in degree i.*

6.8.2 The Flat Base Change Theorem

The following theorem says that cohomology always commutes with flat base change:

Theorem 6.8.4: Flat Base Change

Let $f : X \to Y$ be a separated morphism of finite type between Noetherian schemes, and let $\mathscr{F}$ be a quasi-coherent sheaf on X that is flat over Y. Let $g : Y' \to Y$ be a flat morphism, and form the Cartesian square:

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "g"'] & Y \end{tikzcd}$$

Then for all $i \geq 0$, the base change map is an isomorphism:

$$g^*(R^i f_*\mathscr{F}) \xrightarrow{\sim} R^i f'_*(g'^*\mathscr{F}).$$

***Proof*:**
The question is local on Y and Y', so we may assume both are affine: $Y = \operatorname{Spec} R$ and $Y' = \operatorname{Spec} R'$, with the flat map g corresponding to a flat ring homomorphism $\varphi : R \to R'$.

Step 1: Reduction to Čech cohomology. Since f is separated and of finite type, X admits a finite open affine cover $\mathcal{U} = \{U_1, \ldots, U_r\}$. By proposition 6.6.10, the higher direct images are computed by the Čech complex: $R^i f_*\mathscr{F}$ is the quasi-coherent sheaf on Y associated to the R-module $h^i(C^\bullet(\mathcal{U}, \mathscr{F}))$, where:

$$C^p(\mathcal{U}, \mathscr{F}) = \prod_{i_0 < \cdots < i_p} \mathscr{F}(U_{i_0} \cap \cdots \cap U_{i_p}).$$

Each intersection $U_{i_0,\ldots,i_p}$ is affine (by separatedness), and $\mathscr{F}$ is quasi-coherent, so $\mathscr{F}(U_{i_0,\ldots,i_p})$ is an R-module.

Step 2: Base change of the Čech complex. The base change $X' = X \times_Y Y'$ has the affine cover $\mathcal{U}' = \{U'_1, \dots, U'_r\}$ where $U'_j = U_j \times_Y Y' = \operatorname{Spec}(A_j \otimes_R R')$ for $U_j = \operatorname{Spec} A_j$. The Čech complex for $g'^*\mathscr{F}$ on X' is:

$$C^p(\mathcal{U}', g'^*\mathscr{F}) = \prod_{i_0<\cdots<i_p} (g'^*\mathscr{F})(U'_{i_0,\dots,i_p}) = \prod_{i_0<\cdots<i_p} \mathscr{F}(U_{i_0,\dots,i_p}) \otimes_R R'.$$

The last equality uses the fact that for an affine base change $\operatorname{Spec}(A \otimes_R R') \to \operatorname{Spec} A$, the pullback of a quasi-coherent sheaf $\widetilde{M}$ is $\widetilde{M \otimes_R R'}$. Hence:

$$C^\bullet(\mathcal{U}', g'^*\mathscr{F}) \cong C^\bullet(\mathcal{U}, \mathscr{F}) \otimes_R R'.$$

Step 3: Flatness and cohomology. Since R' is a flat R-module (by hypothesis), the functor $- \otimes_R R'$ is exact. Exact functors commute with the formation of cohomology (i.e., with taking kernels modulo images):

$$h^i(C^\bullet(\mathcal{U}, \mathscr{F}) \otimes_R R') \cong h^i(C^\bullet(\mathcal{U}, \mathscr{F})) \otimes_R R'.$$

The left side computes $R^i f'_*(g'^*\mathscr{F})(Y')$, while the right side computes $g^*(R^i f_*\mathscr{F})(Y') = (R^i f_*\mathscr{F})(Y) \otimes_R R'$. Hence the base change map is an isomorphism on global sections over affine opens, which suffices.

The most important special case is base change to a point:

Corollary 6.8.5: Cohomology of Fibers via Base Change

Under the hypotheses of theorem 6.8.4, let $y \in Y$ be a point with residue field $\kappa(y)$, and let $X_y = f^{-1}(y)$ be the scheme-theoretic fiber (definition 3.2.9). Then:

$$(R^i f_*\mathscr{F}) \otimes_{\mathcal{O}_{Y,y}} \kappa(y) \;\cong\; H^i(X_y,\, \mathscr{F}_y),$$

where $\mathscr{F}_y = \mathscr{F}|_{X_y}$ is the restriction to the fiber, provided that $\mathscr{F}$ is flat over Y.

***Proof*:**
Apply theorem 6.8.4 to the (flat) base change $g : \operatorname{Spec}\kappa(y) \to Y$. Then $X' = X_y$, and the left side is $g^*(R^i f_*\mathscr{F}) = (R^i f_*\mathscr{F}) \otimes \kappa(y)$, while the right side is $H^i(X_y, \mathscr{F}_y)$.

Example 6.8.6: Flat Base Change in Practice

1. *Constant families.* Let $f = \mathrm{pr}_2 : X \times_k Y \to Y$ be the projection of a product (with X projective over k). Then for any coherent sheaf $\mathscr{F}$ on X, pulled back to $X \times Y$ via the first projection, the sheaf $\mathscr{F}$ is flat over Y (trivially, since it does not depend on Y at all). The base change theorem gives:

   $$R^i f_*(\mathrm{pr}_1^*\mathscr{F}) \cong H^i(X, \mathscr{F}) \otimes_k \mathcal{O}_Y,$$

   i.e., the higher direct images are the constant (free) sheaves with fiber $H^i(X, \mathscr{F})$. This is the trivial case, but it illustrates the principle: for a constant family, the cohomology does not vary.

2. *Base change to the generic point.* Let $f : X \to Y$ be a projective morphism with Y integral and $\eta \in Y$ the generic point. The base change $\operatorname{Spec} K(Y) \to Y$ (where $K(Y)$ is the function field) is flat, so:
$$(R^i f_* \mathscr{F}) \otimes \kappa(\eta) \cong H^i(X_\eta, \mathscr{F}_\eta).$$
This says the *generic rank* of $R^i f_* \mathscr{F}$ equals the cohomology of the generic fiber. Combined with semicontinuity, this tells us that the generic fiber has the "smallest" cohomology, and special fibers can only have more.

3. *Flat base change for field extensions.* Let X be a projective scheme over a field k and let K/k be a field extension. The base change $X_K = X \times_k \operatorname{Spec} K \to X$ is flat, so:
$$H^i(X_K, \mathscr{F}_K) \cong H^i(X, \mathscr{F}) \otimes_k K.$$
In particular, $h^i(X_K, \mathscr{F}_K) = h^i(X, \mathscr{F})$: the Betti numbers do not change under field extension. This is essential for reducing problems over arbitrary fields to the algebraically closed case.

Proposition 6.8.7: Associated Points and Morphisms

Let $f : X \to Y$ be a flat morphism of locally Noetherian schemes. Then:
$$\operatorname{Ass}(X) = \bigcup_{y \in \operatorname{Ass}(Y)} \operatorname{Ass}(X_y)$$
where $X_y = f^{-1}(y)$ is the fiber over y. In particular, associated points of X map to associated points of Y (among the points in the image).

***Proof*:**
This follows from the behavior of associated primes under flat base change: if $A \to B$ is flat and $\mathfrak{q} \in \operatorname{Ass}_B(B)$, then $\mathfrak{q} \cap A \in \operatorname{Ass}_A(A)$.

6.8.3 Semicontinuity

The flat base change theorem tells us that when the base change is flat, cohomology commutes with base change. But what happens for non-flat base changes, such as the inclusion of a point $y \hookrightarrow Y$? In general, the base change map $(R^i f_* \mathscr{F}) \otimes \kappa(y) \to H^i(X_y, \mathscr{F}_y)$ is only a surjection, not an isomorphism, and the fiber cohomology can be *larger* than what the stalk of the higher direct image predicts. The semicontinuity theorem makes this precise: the function $y \mapsto h^i(X_y, \mathscr{F}_y)$ is *upper semicontinuous*, meaning it can jump up on closed subsets but not on open ones.

To state the result precisely, we need a piece of commutative algebra. Let R be a Noetherian ring and $C^\bullet$ a bounded complex of finitely generated free R-modules. For any R-module M, write $h^i(C^\bullet \otimes_R M) = h^i(C^\bullet; M)$. When R is a local ring and $M = R/\mathfrak{m}$ is the residue field, the function $i \mapsto h^i(C^\bullet; R/\mathfrak{m})$ measures the fiber cohomology. The following result governs how this varies:

Lemma 6.8.8: Semicontinuity on Complexes

Let R be a Noetherian local ring with residue field $\kappa = R/\mathfrak{m}$, and let $C^\bullet$ be a bounded complex of finitely generated free R-modules. Then:

1. The function $\mathfrak{p} \mapsto \dim_{\kappa(\mathfrak{p})} h^i(C^\bullet \otimes_R \kappa(\mathfrak{p}))$ is upper semicontinuous on $\operatorname{Spec} R$: for each n, the set $\{\mathfrak{p} \in \operatorname{Spec} R \,|\, \dim h^i(C^\bullet \otimes \kappa(\mathfrak{p})) \geq n\}$ is closed.

2. The Euler characteristic $\mathfrak{p} \mapsto \chi(C^\bullet \otimes \kappa(\mathfrak{p})) = \sum(-1)^i \dim h^i(C^\bullet \otimes \kappa(\mathfrak{p}))$ is *locally constant* (it equals $\sum(-1)^i \mathrm{rk} C^i$, which is independent of $\mathfrak{p}$).

3. If $h^i(C^\bullet \otimes \kappa(\mathfrak{p}))$ is constant as a function of $\mathfrak{p}$ (say, equal to r), and R is reduced, then $h^i(C^\bullet)$ is a free R-module of rank r.

***Proof*:**

(1) The complex $C^\bullet$ is a sequence of maps between free modules: $\cdots \to R^{a_{i-1}} \xrightarrow{d^{i-1}} R^{a_i} \xrightarrow{d^i} R^{a_{i+1}} \to \cdots$. After tensoring with $\kappa(\mathfrak{p})$, these become linear maps between vector spaces: $\kappa(\mathfrak{p})^{a_{i-1}} \to \kappa(\mathfrak{p})^{a_i} \to \kappa(\mathfrak{p})^{a_{i+1}}$. Now, $\dim h^i(C^\bullet \otimes \kappa(\mathfrak{p})) = a_i - \mathrm{rk}(d^{i-1} \otimes \kappa(\mathfrak{p})) - \mathrm{rk}(d^i \otimes \kappa(\mathfrak{p}))$. The rank of a matrix can only *drop* under specialization (the vanishing of minors is a closed condition), so $\mathrm{rk}(d^j \otimes \kappa(\mathfrak{p}))$ is lower semicontinuous in $\mathfrak{p}$. Hence $\dim h^i$ is upper semicontinuous.

(2) The Euler characteristic is $\sum(-1)^i \dim h^i(C^\bullet \otimes \kappa(\mathfrak{p})) = \sum(-1)^i a_i$ (the alternating sum of the ranks of the free modules), which is independent of $\mathfrak{p}$. This uses the rank-nullity theorem: the terms involving ranks of differentials cancel in the alternating sum.

(3) If $\dim h^i(C^\bullet \otimes \kappa(\mathfrak{p}))$ is constant, then by upper semicontinuity it achieves its maximum everywhere, so the ranks of the differentials d^{i-1} and d^i are constant. By Nakayama's lemma (applied to the quotient $h^i(C^\bullet) = \ker d^i / \mathrm{im} d^{i-1}$), this forces $h^i(C^\bullet)$ to be locally free over R. If R is reduced, locally free and finitely generated implies free of constant rank.

The geometric content of this algebraic lemma is the semicontinuity theorem:

Theorem 6.8.9: Semicontinuity of Cohomology

Let $f : X \to Y$ be a projective morphism of Noetherian schemes, and let $\mathscr{F}$ be a coherent sheaf on X that is flat over Y. Then:

1. *(Upper semicontinuity.)* For each $i \geq 0$, the function

$$y \longmapsto h^i(X_y, \mathscr{F}_y)$$

is upper semicontinuous on Y: for every $n \geq 0$, the set $\{y \in Y \,|\, h^i(X_y, \mathscr{F}_y) \geq n\}$ is a closed subset of Y.

2. *(Euler characteristic is locally constant.)* The function

$$y \longmapsto \chi(X_y, \mathscr{F}_y) = \sum_{i \geq 0} (-1)^i h^i(X_y, \mathscr{F}_y)$$

is locally constant on Y.

***Proof*:**
The question is local on Y, so assume $Y = \operatorname{Spec} R$ is affine.

Constructing the representing complex. The key technical input is the existence of a *ech complex computing fiber cohomology uniformly.* Choose a finite affine open cover $\mathcal{U}$ of X. By proposition 6.6.10, the cohomology $H^i(X, \mathscr{F})$ is computed by the Čech complex $C^\bullet(\mathcal{U}, \mathscr{F})$. Since $\mathscr{F}$ is coherent and each intersection in the cover is affine, the terms $C^p(\mathcal{U}, \mathscr{F})$ are finitely generated R-modules. Since $\mathscr{F}$ is flat over $Y = \operatorname{Spec} R$, and each intersection $U_{i_0,\dots,i_p}$ is affine, the R-modules $\mathscr{F}(U_{i_0,\dots,i_p})$ are flat over R.

One can replace the Čech complex by a quasi-isomorphic bounded complex $K^\bullet$ of finitely generated *free* R-modules (by taking a free resolution of the Čech complex, truncated appropriately — this is possible because R is Noetherian)[a]. This complex has the property that:

$$h^i(K^\bullet \otimes_R \kappa(y)) \cong H^i(X_y, \mathscr{F}_y)$$

for every $y \in Y$. Indeed, tensoring with $\kappa(y)$ corresponds to restricting to the fiber, and the flatness of $\mathscr{F}$ ensures that the Čech complex commutes with base change (as in the proof of theorem 6.8.4).

Both statements now follow from lemma 6.8.8: part (1) is the upper semicontinuity of $\dim h^i(K^\bullet \otimes \kappa(y))$, and part (2) is the constancy of the Euler characteristic.

[a]More precisely, one uses the fact that a bounded above complex of flat modules over a Noetherian ring is quasi-isomorphic to a bounded above complex of free modules. Since our Čech complex is bounded, $K^\bullet$ can be chosen bounded as well.

Example 6.8.10: Semicontinuity in Families

1. *Jumping h^1 for line bundles on curves.* Let $f : \mathcal{C} \to B$ be a flat family of smooth projective curves of genus g, and let $\mathcal{L}$ be a line bundle on $\mathcal{C}$ of relative degree d (i.e., $\deg \mathcal{L}|_{C_b} = d$ for each fiber C_b). By Riemann–Roch (theorem 6.7.12):
$$h^0(C_b, \mathcal{L}_b) - h^1(C_b, \mathcal{L}_b) = d + 1 - g$$
for every $b \in B$. Since the Euler characteristic is constant, any jump in h^0 is exactly compensated by a jump in h^1. By semicontinuity, h^0 is upper semicontinuous, so the locus where h^0 is "large" (and correspondingly h^1 is large) is closed.

   Concretely, consider a pencil of plane cubics (genus 1) and a degree-0 line bundle $\mathcal{L}$. Generically, $h^0 = 0$ and $h^1 = 0$ (by Riemann–Roch, since $d + 1 - g = 0$). But at special fibers where $\mathcal{L}$ happens to be trivial, $h^0 = 1$ and $h^1 = 1$. The locus of such fibers is a closed subset of the base.

2. *The dimension of global sections of $\mathcal{O}(d)$.* Let $f : X \to \operatorname{Spec} k$ be the structure morphism of a smooth projective variety, and consider the family $\mathcal{O}_X(d)$ as d varies. Though this is not literally a flat family over a connected base, the semicontinuity principle manifests as follows: for any short exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ of coherent sheaves, the function h^0 satisfies $h^0(\mathscr{F}) \leq h^0(\mathscr{F}') + h^0(\mathscr{F}'')$, but equality need not hold. The "excess" is controlled by $H^1(\mathscr{F}')$, which is the obstruction to lifting global sections.

3. *Jumping h^0 for the structure sheaf of a degeneration.* Consider a flat family $f : X \to \mathbb{A}^1_k$ of surfaces where the generic fiber is a smooth surface with $h^0(\mathcal{O}) = 1$ and the special fiber over

0 is a reducible surface (e.g., a union of two components). If the special fiber has $h^0(\mathcal{O}) = 2$ (two connected components), then h^0 has jumped up at 0, and the locus $\{h^0 \geq 2\}$ is the closed point $\{0\}$, consistent with semicontinuity.

6.8.4 Grauert's Theorem and Cohomology and Base Change

The semicontinuity theorem says h^i can jump up. The natural follow-up question is: what happens when it does *not* jump? The answer is Grauert's theorem: constancy of h^i implies that the higher direct image is locally free and commutes with *all* base changes, not just flat ones.

Theorem 6.8.11: Grauert's Theorem (Cohomology and Base Change)

Let $f : X \to Y$ be a projective morphism of Noetherian schemes, and let $\mathscr{F}$ be a coherent sheaf on X that is flat over Y. Let $y \in Y$ be a point, and suppose the natural base change map:

$$\beta^i(y) : \; (R^i f_* \mathscr{F}) \otimes \kappa(y) \longrightarrow H^i(X_y, \mathscr{F}_y)$$

is surjective. Then:

1. $\beta^i(y)$ is an isomorphism. That is, surjectivity implies bijectivity.
2. β^i is an isomorphism in an open neighborhood of y. In particular, $R^i f_* \mathscr{F}$ is locally free near y if and only if $h^i(X_y, \mathscr{F}_y)$ is locally constant near y.
3. If $\beta^i(y)$ is an isomorphism (equivalently, surjective), then $\beta^{i-1}(y)$ is surjective if and only if it is an isomorphism.

In particular, if $h^i(X_y, \mathscr{F}_y)$ is constant on Y and Y is reduced, then $R^i f_* \mathscr{F}$ is a locally free sheaf of rank $h^i(X_y, \mathscr{F}_y)$, and the formation of $R^i f_* \mathscr{F}$ commutes with arbitrary base change $g : Y' \to Y$.

***Proof*:**
As in the proof of theorem 6.8.9, we reduce to the local algebra by replacing the higher direct images with the cohomology of a bounded complex $K^\bullet$ of finitely generated free R-modules (where $R = \mathcal{O}_{Y,y}$).

(1) The base change map $\beta^i(y)$ corresponds to the natural map $h^i(K^\bullet) \otimes_R \kappa \to h^i(K^\bullet \otimes_R \kappa)$. Write $K^\bullet$ as $\cdots \to K^{i-1} \xrightarrow{d^{i-1}} K^i \xrightarrow{d^i} K^{i+1} \to \cdots$. The surjectivity of $\beta^i(y)$ means that every cohomology class in $h^i(K^\bullet \otimes \kappa)$ lifts to $h^i(K^\bullet)$. By Nakayama's lemma (applied to the finitely generated R-module $h^i(K^\bullet)$), the map $h^i(K^\bullet) \to h^i(K^\bullet) \otimes \kappa$ is surjective if and only if it is a presentation by a minimal set of generators. The surjectivity of β^i then forces the natural map to be an isomorphism by a careful diagram chase involving the snake lemma applied to the sequences defining $\ker d^i$ and $\operatorname{im} d^{i-1}$[a].

(2) Since $\beta^i(y)$ is an isomorphism at y, the semicontinuity of h^i implies $h^i(X_{y'}, \mathscr{F}_{y'}) \leq h^i(X_y, \mathscr{F}_y)$ in a neighborhood of y. On the other hand, the rank of the locally free sheaf $R^i f_* \mathscr{F}$ can only drop on a closed subset (or rather, its fiber dimension is lower semicontinuous). The two semicontinuity conditions force constancy in a neighborhood, and the argument of part (1) applies at each nearby point.

(3) This follows from the exact sequence relating h^{i-1} and h^i in the complex $K^\bullet$: the surjectivity of β^i controls the cokernel, which in turn governs the kernel at the next level.

The final statement (for constant h^i on reduced Y) follows by combining (1) and (2) with lemma 6.8.8(3): on a reduced base, a finitely generated module with constant fiber dimension is locally free.

[a] The detailed argument proceeds as follows. By surjectivity of β^i, one shows that the map $\ker d^i \to \ker(d^i \otimes \kappa)$ is surjective. Since K^i is free, the kernel $\ker d^i$ is a submodule of a free module, hence flat (over a local ring, submodules of free modules are flat if the ring is a principal ideal domain; in general, one uses the flatness of $\mathscr{F}$ to ensure this). The surjectivity then implies that $\operatorname{im} d^{i-1} \otimes \kappa \to \operatorname{im}(d^{i-1} \otimes \kappa)$ is also well-behaved, and the result follows.

Corollary 6.8.12: Cohomology and Base Change: Clean Form

Let $f : X \to Y$ be a projective morphism of Noetherian schemes with Y integral, and let $\mathscr{F}$ be coherent and f-flat. Suppose $h^i(X_y, \mathscr{F}_y)$ is constant for all $y \in Y$. Then:

1. $R^i f_* \mathscr{F}$ is locally free of rank $h^i(X_y, \mathscr{F}_y)$.
2. For any morphism $g : Y' \to Y$, the base change map $g^*(R^i f_* \mathscr{F}) \xrightarrow{\sim} R^i f'_*(g'^* \mathscr{F})$ is an isomorphism.
3. If additionally $h^{i-1}(X_y, \mathscr{F}_y)$ is also constant, then $R^{i-1} f_* \mathscr{F}$ is locally free and commutes with base change as well.

***Proof*:**
Part (1) follows from theorem 6.8.11 together with the observation that on an integral (hence reduced) scheme, a coherent sheaf of constant fiber dimension is locally free. Part (2) follows because, once $R^i f_* \mathscr{F}$ is locally free, the base change map is an isomorphism for any g (this is immediate from the flat base change theorem, since tensoring a locally free module with anything commutes with cohomology). Part (3) follows from theorem 6.8.11(3).

Example 6.8.13: Grauert's Theorem in Practice

1. *Line bundles on a constant family of curves.* Let C be a smooth projective curve of genus g over k, and consider the family $f : C \times \operatorname{Pic}^d(C) \to \operatorname{Pic}^d(C)$ with the universal line bundle $\mathcal{L}$ of degree d. If $d > 2g - 2$, then $h^1(C, \mathcal{L}_b) = 0$ for all $b \in \operatorname{Pic}^d(C)$ (by Serre duality and degree considerations). By Grauert's theorem, $R^1 f_* \mathcal{L} = 0$ and $f_* \mathcal{L}$ is a locally free sheaf of rank $h^0 = d + 1 - g$ on $\operatorname{Pic}^d(C)$. This locally free sheaf is the *Picard bundle*, a fundamental object in the study of Jacobians.

2. *Failure of base change without flatness.* Consider the blowup $f : \mathrm{Bl}_0 \mathbb{A}^2 \to \mathbb{A}^2$ from Example 6.6.11(2). The structure sheaf $\mathcal{O}_{\mathrm{Bl}_0 \mathbb{A}^2}$ is *not* flat over $\mathbb{A}^2$ (the fiber dimension jumps). Accordingly, $R^1 f_* \mathcal{O} = 0$ even though $H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}) = 0$ for the exceptional fiber — the vanishing of R^1 does not "see" the fiber at the origin because the family is not flat there.

3. *When constancy fails: the Brill–Noether locus.* Let C be a general smooth curve of genus g and consider the family of degree-d line bundles. The *Brill–Noether locus* $W^r_d(C) = \{L \in \operatorname{Pic}^d(C) \big| h^0(C, L) \geq r + 1\}$ is a closed subvariety of $\operatorname{Pic}^d(C)$ by the semicontinuity theorem. The Brill–Noether theorem asserts that for a general curve of genus g, $W^r_d(C)$ has the

"expected dimension" $\rho = g - (r+1)(g-d+r)$ when $\rho \geq 0$, and is empty when $\rho < 0$. This is a result in the geometry of algebraic curves, and semicontinuity is the foundational tool that ensures W^r_d is an honest algebraic variety.

6.8.14 Remark: The Coherent Base Change Complex The key technical tool behind both the semicontinuity and Grauert theorems is the existence of a bounded complex $K^\bullet$ of finitely generated free R-modules representing the derived pushforward $\mathbf{R}f_*\mathscr{F}$ in the derived category of R-modules. Such a complex exists whenever f is projective and $\mathscr{F}$ is coherent and f-flat: one takes the Čech complex associated to a finite affine cover and replaces it by a free resolution.

In the language of derived categories (which we have not formally developed), the statement is: $\mathbf{R}f_*\mathscr{F} \in D^b(\mathrm{Mod}_R)$ is a *perfect complex* (quasi-isomorphic to a bounded complex of finitely generated free modules). The fiber cohomology is then $H^i(X_y, \mathscr{F}_y) = h^i(K^\bullet \otimes^{\mathbf{L}} \kappa(y))$, where $\otimes^{\mathbf{L}}$ is the derived tensor product (which equals the ordinary tensor product since $K^\bullet$ consists of free modules).

This perspective explains why the semicontinuity theorem is fundamentally a result in commutative algebra (about cohomology of complexes of free modules under base change) rather than algebraic geometry. The geometry enters only in constructing the complex $K^\bullet$ — and this construction uses the projectivity of f and the coherence of $\mathscr{F}$ in an essential way.

6.9 *GAGA

We developed two parallel cohomological machines: the algebraic cohomology of coherent sheaves on schemes (the subject of this chapter), and the analytic cohomology of coherent analytic sheaves on complex manifolds (via the Dolbeault complex, Hodge theory, and the analytic structure sheaf). Serre's GAGA theorems — *Géométrie Algébrique et Géométrie Analytique* [15] — form the bridge between these two worlds. They assert that for projective varieties over $\mathbb{C}$, the algebraic and analytic theories carry *exactly the same information*: the category of coherent algebraic sheaves is equivalent to the category of coherent analytic sheaves, and this equivalence preserves cohomology.

This is a surprising result. The algebraic world uses the Zariski topology (which is coarse: open sets are large and plentiful) and polynomial or rational functions, while the analytic world uses the classical (Euclidean) topology (which is fine: open sets can be arbitrarily small) and holomorphic functions. There are far more analytic open sets, far more analytic functions, and far more analytic sheaves than algebraic ones. Yet on projective varieties, these extra objects contribute nothing new to the cohomological theory. The rigidity of projective varieties forces the algebraic and analytic categories to coincide.

We state the GAGA theorems precisely, outline the key ideas of the proof, and develop the main consequences. This section draws heavily on the complex geometry review of chapter 1[7]; we now close the loop by proving that the algebraic cohomology developed in this chapter matches the analytic cohomology developed there.

[7]Specifically: sections 0.6 through the end of that chapter, where analytification (definition 6.9.1), analytic sheaves (**??**), and the GAGA statement (**??**) were introduced.

6.9.1 Analytification

We begin by recalling the analytification functor, which passes from algebraic geometry over $\mathbb{C}$ to complex-analytic geometry. The reader who has studied the complex geometry chapter in detail may skim this subsection; it is included for self-containedness.

Definition 6.9.1: Analytification

Let X be a scheme of finite type over $\mathbb{C}$. The *analytification* X^{an} is a complex analytic space with:

- The same underlying set as the $\mathbb{C}$-points $X(\mathbb{C})$.
- The classical (Euclidean) topology.
- The sheaf $\mathcal{O}_{X^{\mathrm{an}}}$ of holomorphic functions.

If X is smooth, then X^{an} is a complex manifold.

Let X be a scheme of finite type over $\mathbb{C}$. The set of $\mathbb{C}$-valued points $X(\mathbb{C}) = \mathrm{Hom}_{\mathbf{Sch}}(\operatorname{Spec}\mathbb{C}, X)$ carries a natural structure of complex-analytic space: locally, if $X = \operatorname{Spec}\mathbb{C}[x_1, \dots, x_n]/(f_1, \dots, f_r)$, then $X(\mathbb{C}) = V(f_1, \dots, f_r) \subseteq \mathbb{C}^n$ with the subspace topology from $\mathbb{C}^n$ and the sheaf of holomorphic functions. When X is smooth, $X(\mathbb{C})$ is a complex manifold (definition 0.6.1).

Definition 6.9.2: The Analytification Functor

Let X be a scheme of finite type over $\mathbb{C}$.

1. The *analytification* of X, denoted X^{an}, is the complex-analytic space $X(\mathbb{C})$ equipped with its natural analytic structure sheaf $\mathcal{O}_{X^{\mathrm{an}}}$ (cf. definition 6.9.1).
2. There is a natural continuous map of topological spaces $\varphi : X^{\mathrm{an}} \to X$ (the "identity on points," where X carries the Zariski topology and X^{an} carries the classical topology), and a morphism of sheaves of rings $\varphi^{\#} : \mathcal{O}_X \to \varphi_*\mathcal{O}_{X^{\mathrm{an}}}$ (sending algebraic regular functions to holomorphic functions).
3. For a coherent algebraic sheaf $\mathscr{F}$ on X, the *analytification* is the coherent analytic sheaf:
$$\mathscr{F}^{\mathrm{an}} = \varphi^*\mathscr{F} = \mathscr{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X^{\mathrm{an}}}$$
on X^{an}. This defines an exact functor:
$$(-)^{\mathrm{an}} : \mathbf{Coh}(X) \longrightarrow \mathbf{Coh}(X^{\mathrm{an}}).$$

The analytification functor has good formal properties: it is exact (since $\mathcal{O}_{X^{\mathrm{an}}}$ is flat over $\mathcal{O}_X$ — this is a non-trivial fact, reflecting the faithfully flat nature of the completion of a local ring), it preserves tensor products, duals, and exterior powers of locally free sheaves, and it sends line bundles to line bundles. Furthermore, for each $i \geq 0$, there is a natural *comparison map*:
$$\gamma^i : \; H^i(X, \mathscr{F}) \longrightarrow H^i(X^{\mathrm{an}}, \mathscr{F}^{\mathrm{an}}),$$
induced by the morphism of ringed spaces $\varphi : (X^{\mathrm{an}}, \mathcal{O}_{X^{\mathrm{an}}}) \to (X, \mathcal{O}_X)$. The GAGA theorems assert

that, for projective X, this map is an isomorphism and the functor $(-)^{\mathrm{an}}$ is an equivalence.

6.9.2 Statement of the GAGA Theorems

Theorem 6.9.3: Serre's GAGA Theorems

Let X be a projective scheme over $\mathbb{C}$. Then:

1. *(Equivalence of categories.)* The analytification functor

$$(-)^{\mathrm{an}} : \mathbf{Coh}(X) \longrightarrow \mathbf{Coh}(X^{\mathrm{an}})$$

is an equivalence of categories. In particular:

   (a) Every coherent analytic sheaf on X^{an} is the analytification of a unique (up to isomorphism) coherent algebraic sheaf on X.

   (b) Every morphism of coherent analytic sheaves on X^{an} is the analytification of a unique morphism of coherent algebraic sheaves on X.

2. *(Comparison of cohomology.)* For every coherent algebraic sheaf $\mathscr{F}$ on X and every $i \geq 0$, the comparison map

$$\gamma^i : H^i(X, \mathscr{F}) \xrightarrow{\sim} H^i(X^{\mathrm{an}}, \mathscr{F}^{\mathrm{an}})$$

is an isomorphism of finite-dimensional $\mathbb{C}$-vector spaces.

3. *(Equivalence for subschemes.)* The analytification functor induces a bijection between closed algebraic subschemes of X and closed analytic subspaces of X^{an}.

Part (2) is typically proved first and is the most important for applications; parts (1) and (3) follow from it by formal arguments. Let us outline the proof of part (2):

***Proof*:**
(Proof sketch of theorem 6.9.3(2))
The proof proceeds in three stages.

Stage 1: The case $X = \mathbb{P}^n_{\mathbb{C}}$ and $\mathscr{F} = \mathcal{O}(d)$. The algebraic cohomology $H^i(\mathbb{P}^n, \mathcal{O}(d))$ was computed in theorem 6.3.1. The analytic cohomology $H^i((\mathbb{P}^n)^{\mathrm{an}}, \mathcal{O}^{\mathrm{an}}(d))$ can be computed using the Dolbeault resolution (**??**): since $\mathbb{CP}^n$ is a compact Kähler manifold (example 0.6.22), the Hodge decomposition (theorem 0.6.28) gives $H^i(\mathbb{CP}^n, \mathcal{O}^{\mathrm{an}}) \cong H^{0,i}_{\bar{\partial}}(\mathbb{CP}^n)$, and the Hodge diamond of $\mathbb{CP}^n$ (example 0.6.30) shows this vanishes for $i > 0$. For the twisted sheaf $\mathcal{O}(d)$, the Dolbeault cohomology can similarly be computed explicitly, and one verifies that the comparison map γ^i is an isomorphism by matching dimensions on both sides.

Stage 2: Extension to all coherent sheaves on $\mathbb{P}^n_{\mathbb{C}}$. Given a coherent sheaf $\mathscr{F}$ on $\mathbb{P}^n$, choose a resolution by direct sums of twisted sheaves (corollary 5.3.26):

$$\cdots \to \mathcal{E}_1 \to \mathcal{E}_0 \to \mathscr{F} \to 0,$$

where each $\mathcal{E}_j = \bigoplus \mathcal{O}(d_{j,l})$. Since the analytification functor is exact, the analytified resolution computes the analytic cohomology. The comparison maps for the $\mathcal{E}_j$ are isomorphisms by Stage 1, and a five-lemma argument (identical to Step 2 of the proof of theorem 6.5.5) yields the isomorphism for $\mathscr{F}$.

Stage 3: Extension to projective schemes. For a general projective scheme $i : X \hookrightarrow \mathbb{P}^N_{\mathbb{C}}$, the pushforward $i_*\mathscr{F}$ is a coherent algebraic sheaf on $\mathbb{P}^N$, and $(i_*\mathscr{F})^{\mathrm{an}} = i_*^{\mathrm{an}}(\mathscr{F}^{\mathrm{an}})$ (the analytification of a pushforward along a closed embedding is the analytic pushforward of the analytification). By proposition 6.2.4, $H^i(X, \mathscr{F}) = H^i(\mathbb{P}^N, i_*\mathscr{F})$ and similarly for the analytic side. Stage 2 gives the isomorphism for $i_*\mathscr{F}$ on $\mathbb{P}^N$, hence for $\mathscr{F}$ on X.

***Proof*:**
(Proof sketch of theorem 6.9.3(1))

Full faithfulness. For coherent sheaves $\mathscr{F}, \mathscr{G}$ on X, the natural map $\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) \to \mathrm{Hom}_{\mathcal{O}_{X^{\mathrm{an}}}}(\mathscr{F}^{\mathrm{an}}, \mathscr{G}^{\mathrm{an}})$ is an isomorphism. Indeed, $\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = H^0(X, \mathcal{H}om(\mathscr{F}, \mathscr{G}))$ and $\mathrm{Hom}_{\mathcal{O}_{X^{\mathrm{an}}}}(\mathscr{F}^{\mathrm{an}}, \mathscr{G}^{\mathrm{an}}) = H^0(X^{\mathrm{an}}, \mathcal{H}om(\mathscr{F}, \mathscr{G})^{\mathrm{an}})$, and part (2) with $i = 0$ gives the isomorphism.

Essential surjectivity. Let $\mathscr{G}^{\mathrm{an}}$ be a coherent analytic sheaf on X^{an}. One must produce a coherent algebraic sheaf $\mathscr{G}$ with $\mathscr{G}^{\mathrm{an}} \cong (\mathscr{G})^{\mathrm{an}}$. The argument uses a descending induction: twisting $\mathscr{G}^{\mathrm{an}}$ by $\mathcal{O}^{\mathrm{an}}(d)$ for $d \gg 0$, the analytic Serre vanishing theorem (which holds on compact complex manifolds) gives that $\mathscr{G}^{\mathrm{an}}(d)$ is generated by its global sections. These global sections are analytic, but by part (2) applied to $\mathcal{O}_{X^{\mathrm{an}}}$ and its twists, they can be identified with algebraic sections. One then builds $\mathscr{G}$ algebraically from these generators and relations. The details require a careful interplay between the algebraic and analytic Hilbert polynomial theories; see [15] for a full account.

6.9.4 Remark: The Projectivity Hypothesis is Essential The GAGA theorems fail dramatically for non-projective (and even non-proper) varieties. The simplest example is the affine line: on $\mathbb{A}^1_{\mathbb{C}}$, the algebraic structure sheaf has $\Gamma(\mathbb{A}^1, \mathcal{O}) = \mathbb{C}[x]$ (polynomial functions), while the analytic structure sheaf has $\Gamma(\mathbb{C}, \mathcal{O}^{\mathrm{an}}) = \mathcal{O}(\mathbb{C})$ (entire holomorphic functions), which is an enormously larger ring (containing e^x, $\sin x$, etc.). Hence the global section comparison already fails.
More strikingly, there exist coherent analytic sheaves on $\mathbb{C}$ that have no algebraic counterpart: the ideal sheaf of the analytic subset $\{e^{2\pi i n} | n \in \mathbb{Z}\} = \{1\}$ in $\mathbb{C}^*$ is coherent analytic but not algebraic, since the zero set of any algebraic function on $\mathbb{C}^*$ is either finite or all of $\mathbb{C}^*$.
For *proper* but non-projective varieties, a version of GAGA still holds (due to Grothendieck): the analytification functor is an equivalence on coherent sheaves, and cohomology comparison holds. The key requirement is properness — which ensures compactness of the analytic space — not projectivity. However, our proof relied on the embedding into $\mathbb{P}^N$ and the explicit computation of twisted sheaf cohomology, so we have stated the theorem in the projective case.

6.9.3 Consequences

The GAGA theorems have far-reaching consequences that connect algebraic geometry, complex analysis, and topology. We develop the most important ones.

Corollary 6.9.5: Chow's Theorem

Every closed analytic subspace of $\mathbb{CP}^n$ is algebraic. That is, if $Z \subseteq \mathbb{CP}^n$ is a closed complex-analytic subset, then Z is the zero set of finitely many homogeneous polynomials.

***Proof*:**
A closed analytic subspace $Z \subseteq (\mathbb{P}^n_{\mathbb{C}})^{\mathrm{an}}$ is determined by its ideal sheaf $\mathscr{I}_Z^{\mathrm{an}}$, which is a coherent analytic sheaf on $(\mathbb{P}^n)^{\mathrm{an}}$. By theorem 6.9.3(1), there exists a coherent algebraic sheaf $\mathscr{I}$ on $\mathbb{P}^n_{\mathbb{C}}$ with $\mathscr{I}^{\mathrm{an}} \cong \mathscr{I}_Z^{\mathrm{an}}$. Since the analytification of an ideal sheaf is an ideal sheaf (the inclusion $\mathscr{I} \hookrightarrow \mathcal{O}_{\mathbb{P}^n}$ analytifies to $\mathscr{I}^{\mathrm{an}} \hookrightarrow \mathcal{O}_{\mathbb{P}^n}^{\mathrm{an}}$), the sheaf $\mathscr{I}$ is an ideal sheaf on $\mathbb{P}^n_{\mathbb{C}}$, defining a closed algebraic subscheme Z_{alg} with $Z_{\mathrm{alg}}^{\mathrm{an}} = Z$. By theorem 6.9.3(3), this correspondence is a bijection.

Corollary 6.9.6: Algebraic de Rham Theorem

Let X be a smooth projective variety over $\mathbb{C}$. Then the algebraic de Rham cohomology

$$H^i_{\mathrm{dR}}(X/\mathbb{C}) = \mathbb{H}^i(X,\, \Omega^\bullet_{X/\mathbb{C}})$$

(the hypercohomology of the algebraic de Rham complex $0 \to \mathcal{O}_X \to \Omega^1_{X/\mathbb{C}} \to \Omega^2_{X/\mathbb{C}} \to \cdots$) is naturally isomorphic to the singular cohomology:

$$H^i_{\mathrm{dR}}(X/\mathbb{C}) \;\cong\; H^i_{\mathrm{sing}}(X(\mathbb{C});\, \mathbb{C}).$$

***Proof*:**
By GAGA, the algebraic de Rham complex analytifies to the holomorphic de Rham complex on X^{an}, and hypercohomology is preserved:

$$\mathbb{H}^i(X,\, \Omega^\bullet_{X/\mathbb{C}}) \;\cong\; \mathbb{H}^i(X^{\mathrm{an}},\, \Omega_X^{\bullet,\mathrm{an}}).$$

Since X^{an} is a compact Kähler manifold (example 0.6.22), the holomorphic Poincaré lemma gives an exact sequence $0 \to \underline{\mathbb{C}} \to \Omega^{0,\mathrm{an}} \to \Omega^{1,\mathrm{an}} \to \cdots$, i.e., the holomorphic de Rham complex is a resolution of the constant sheaf $\underline{\mathbb{C}}$. Hence:

$$\mathbb{H}^i(X^{\mathrm{an}}, \Omega^{\bullet,\mathrm{an}}) \cong H^i(X^{\mathrm{an}}, \underline{\mathbb{C}}).$$

Finally, by the de Rham theorem (theorem 0.5.23), sheaf cohomology with coefficients in $\underline{\mathbb{C}}$ is isomorphic to singular cohomology: $H^i(X^{\mathrm{an}}, \underline{\mathbb{C}}) \cong H^i_{\mathrm{sing}}(X(\mathbb{C}); \mathbb{C})$.

Corollary 6.9.7: Comparison of Euler Characteristics

Let X be a smooth projective variety of dimension n over $\mathbb{C}$ and let $\mathscr{F}$ be a coherent algebraic sheaf on X. Then:

$$\chi_{\mathrm{alg}}(X,\mathscr{F}) = \sum_{i=0}^{n}(-1)^i \dim_{\mathbb{C}} H^i(X,\mathscr{F}) = \sum_{i=0}^{n}(-1)^i \dim_{\mathbb{C}} H^i(X^{\mathrm{an}},\mathscr{F}^{\mathrm{an}}) = \chi_{\mathrm{an}}(X^{\mathrm{an}},\mathscr{F}^{\mathrm{an}}).$$

In particular, the algebraic and analytic Hilbert polynomials coincide: the algebraic Hilbert polynomial $P_{\mathscr{F}}$ of theorem 6.7.5 equals the analytic one.

***Proof*:**
Immediate from theorem 6.9.3(2): the comparison isomorphism preserves dimensions.

Corollary 6.9.8: Hodge Decomposition via GAGA

Let X be a smooth projective variety of dimension n over $\mathbb{C}$. Then the singular cohomology of $X(\mathbb{C})$ admits a Hodge decomposition:

$$H^k_{\mathrm{sing}}(X(\mathbb{C});\, \mathbb{C}) \;\cong\; \bigoplus_{p+q=k} H^q(X,\, \Omega^p_{X/\mathbb{C}}),$$

where $H^q(X,\Omega^p_{X/\mathbb{C}})$ is the algebraic sheaf cohomology of the p-th sheaf of algebraic differentials. The *Hodge numbers* are:

$$h^{p,q}(X) = \dim_{\mathbb{C}} H^q(X,\Omega^p_{X/\mathbb{C}}),$$

and they satisfy the symmetries $h^{p,q} = h^{q,p}$ (Hodge symmetry) and $h^{p,q} = h^{n-p,\,n-q}$ (Serre duality).

***Proof*:**
By GAGA, $H^q(X,\Omega^p_{X/\mathbb{C}}) \cong H^q(X^{\mathrm{an}},(\Omega^p)^{\mathrm{an}})$. By the Dolbeault theorem (theorem 0.6.17), $H^q(X^{\mathrm{an}},(\Omega^p)^{\mathrm{an}}) \cong H^{p,q}_{\bar\partial}(X^{\mathrm{an}})$, the Dolbeault cohomology. Since X^{an} is a compact Kähler manifold, the Hodge decomposition theorem (theorem 0.6.28) gives:

$$H^k_{\mathrm{dR}}(X^{\mathrm{an}}) \cong \bigoplus_{p+q=k} H^{p,q}_{\bar\partial}(X^{\mathrm{an}}),$$

and the de Rham theorem identifies the left side with $H^k_{\mathrm{sing}}(X(\mathbb{C});\mathbb{C})$. Combining these isomorphisms gives the decomposition.

The symmetry $h^{p,q} = h^{q,p}$ is the Hodge symmetry from complex conjugation on a Kähler manifold (theorem 0.6.28). The symmetry $h^{p,q} = h^{n-p,\,n-q}$ follows from Serre duality (corollary 6.5.10): since $\omega_X = \Omega^n_{X/\mathbb{C}}$, we have $H^q(X,\Omega^p) \cong H^{n-q}(X,(\Omega^p)^\vee \otimes \Omega^n)^* \cong H^{n-q}(X,\Omega^{n-p})^*$, giving $h^{p,q} = h^{n-p,\,n-q}$.

Example 6.9.9: GAGA in Action

1. *Line bundles and the exponential sequence.* On a compact complex manifold M, the *exponential sequence* of analytic sheaves:

$$0 \longrightarrow \underline{\mathbb{Z}} \xrightarrow{2\pi i} \mathcal{O}_M^{\mathrm{an}} \xrightarrow{\exp} (\mathcal{O}_M^{\mathrm{an}})^* \longrightarrow 0$$

induces a long exact sequence in cohomology:

$$\cdots \to H^1(M, \mathcal{O}^{\mathrm{an}}) \to H^1(M, (\mathcal{O}^{\mathrm{an}})^*) \xrightarrow{c_1} H^2(M, \mathbb{Z}) \to H^2(M, \mathcal{O}^{\mathrm{an}}) \to \cdots$$

The group $H^1(M, (\mathcal{O}^{\mathrm{an}})^*)$ classifies analytic line bundles (it is the analytic Picard group), and $c_1 : \mathrm{Pic}^{\mathrm{an}}(M) \to H^2(M, \mathbb{Z})$ is the first Chern class map (definition 0.6.35).

For a smooth projective variety X over $\mathbb{C}$, GAGA identifies the analytic Picard group with the algebraic one: $\mathrm{Pic}(X) \cong \mathrm{Pic}^{\mathrm{an}}(X^{\mathrm{an}})$. The exponential sequence is *not* algebraic (the exponential map exp is transcendental), but its cohomological consequences — particularly the Chern class map and the Lefschetz theorem on $(1,1)$-classes — can be stated and applied purely algebraically via GAGA.

2. *Holomorphic vs. algebraic vector bundles.* Let X be a smooth projective variety over $\mathbb{C}$. By GAGA, the categories of algebraic and holomorphic vector bundles on X are equivalent (via **??** and its analytic counterpart, vector bundles correspond to locally free sheaves in both settings). Hence every holomorphic vector bundle on X^{an} is algebraic. This fails spectacularly for non-projective varieties: on $\mathbb{C}^*$, the flat line bundle with monodromy $e^{2\pi i\alpha}$ for irrational α is holomorphic but not algebraic.

3. *Singular cohomology from algebraic data.* Let $X = \mathbb{P}^n_{\mathbb{C}}$. The algebraic sheaf cohomology groups are: $H^i(\mathbb{P}^n, \mathcal{O}) = \mathbb{C}$ for $i = 0$ and 0 for $i > 0$ (by theorem 6.3.1). By GAGA, the same holds analytically. The Hodge decomposition gives $H^k_{\mathrm{sing}}(\mathbb{CP}^n; \mathbb{C}) = \bigoplus_{p+q=k} H^q(\mathbb{P}^n, \Omega^p)$. The Hodge diamond of $\mathbb{CP}^n$ (example 0.6.30) has $h^{p,p} = 1$ for $0 \le p \le n$ and all other $h^{p,q} = 0$, so:

$$H^k_{\mathrm{sing}}(\mathbb{CP}^n; \mathbb{C}) = \begin{cases} \mathbb{C} & k \text{ even},\ 0 \le k \le 2n, \\ 0 & k \text{ odd}. \end{cases}$$

This recovers the classical computation from algebraic topology (see [6, section 4.1]), but now derived entirely from algebraic sheaf cohomology via GAGA and Hodge theory.

4. *The Lefschetz theorem on hyperplane sections.* Let $X \subset \mathbb{P}^N_{\mathbb{C}}$ be a smooth projective variety of dimension n and let $H \subset \mathbb{P}^N$ be a hyperplane with $Y = X \cap H$ smooth. The Lefschetz hyperplane theorem asserts that the restriction map:

$$H^i_{\mathrm{sing}}(X(\mathbb{C}); \mathbb{Z}) \longrightarrow H^i_{\mathrm{sing}}(Y(\mathbb{C}); \mathbb{Z})$$

is an isomorphism for $i < n-1$ and injective for $i = n-1$. The proof uses Morse theory (or its algebraic substitute, the weak Lefschetz theorem) and is fundamentally analytic. However, via GAGA, the consequence for algebraic cohomology is immediate: the restriction maps $H^i(X, \Omega^p) \to H^i(Y, \Omega^p|_Y)$ are isomorphisms in the range dictated by Lefschetz. This has purely algebraic applications — for instance, it implies that the Picard group of a smooth hypersurface $X \subset \mathbb{P}^n$ with $n \ge 4$ is isomorphic to $\mathbb{Z}$, generated by $\mathcal{O}_X(1)$.

6.9.10 Remark: GAGA as a Principle Beyond the precise theorems stated above, GAGA functions as a *guiding principle* in algebraic geometry over $\mathbb{C}$: any "reasonable" statement about coherent sheaves and their cohomology on projective varieties that can be proved analytically also holds algebraically, and vice versa. This principle is invoked (often implicitly) throughout the subject:

- Kodaira vanishing ($H^i(X, \omega_X \otimes \mathcal{L}) = 0$ for $i > 0$ and $\mathcal{L}$ ample) is proved analytically using L^2-estimates for the $\bar{\partial}$-operator, but applies to algebraic cohomology via GAGA.
- The degeneration of the Hodge-to-de Rham spectral sequence for smooth projective varieties is proved using Kähler geometry (or, in characteristic p, by the Deligne–Illusie method), but the algebraic consequence — that the Hodge numbers $h^{p,q}$ are well-defined invariants of the algebraic variety — is what matters for classification.
- The construction of moduli spaces often proceeds by first constructing an analytic moduli space (using complex-analytic deformation theory) and then invoking GAGA to conclude that it is algebraic.

In this way, GAGA is not merely a comparison theorem but a bridge that allows free passage between the algebraic and analytic worlds, using whichever tools are most natural for the problem at hand.

6.9.4 Summary

We close with a summary of the cohomological landscape that this chapter has developed, viewed through the lens of GAGA. Let X be a smooth projective variety of dimension n over $\mathbb{C}$, and let $\mathscr{F}$ be a coherent sheaf on X. The following table summarizes the relationships between the various cohomology theories:

Algebraic	**Analytic**	**Via**
$H^i(X, \mathscr{F})$	$H^i(X^{\mathrm{an}}, \mathscr{F}^{\mathrm{an}})$	GAGA
$H^q(X, \Omega^p_{X/\mathbb{C}})$	$H^{p,q}_{\bar{\partial}}(X^{\mathrm{an}})$	GAGA + Dolbeault
$\mathbb{H}^k(X, \Omega^\bullet)$	$H^k_{\mathrm{sing}}(X(\mathbb{C}); \mathbb{C})$	Algebraic de Rham
$\chi(X, \mathscr{F})$	$\chi(X^{\mathrm{an}}, \mathscr{F}^{\mathrm{an}})$	GAGA
$\mathrm{Pic}\,(X)$	$\mathrm{Pic}^{\mathrm{an}}(X^{\mathrm{an}})$	GAGA
$\mathrm{Ext}^i(\mathscr{F}, \omega_X)$	$H^{n-i}(X, \mathscr{F})^*$	Serre duality
$h^{p,q} = h^{n-p,\,n-q}$	Hodge + Serre	GAGA + duality

The algebraic side (left column) uses only the Zariski topology, coherent sheaves, and the machinery of this chapter. The analytic side (middle column) uses the classical topology, Dolbeault cohomology, and Hodge theory from the complex geometry chapter. The right column indicates which bridge theorem connects them.

For non-projective or non-smooth varieties, the dictionary partially breaks down: GAGA requires properness, Hodge decomposition requires Kähler structure (or smoothness), and Serre duality requires Cohen — Macaulay (or at least the existence of a dualizing sheaf). The interplay between these hypotheses — determining which parts of the dictionary survive under which conditions — is a central theme of modern algebraic geometry.

7

Application: Curves, Surfaces, and a bit Beyond

It is time to reap what we have sown. Over the preceding chapters, we have built much machinery: schemes, coherent sheaves, divisors, differentials, and cohomology. The reader has persevered through the abstractions and now wish to see what benefits are attained from this generalization. This chapter is the answer.

We turn towards the concrete geometry of *curves* and *surfaces*—the one-dimensional and two-dimensional objects that have driven algebraic geometry since its inception. It is no exaggeration to say that most of the machinery of the previous chapters was *invented* to understand these objects. Riemann and his contemporaries studied algebraic curves through the lens of complex analysis; the Italians (Castelnuovo, Enriques, Severi) built an elaborate—if occasionally imprecise—theory of surfaces; and it was the desire to put these theories on firm footing that led Grothendieck to develop the scheme-theoretic framework we now command.

The philosophy of this chapter is simple: we take the abstract theorems we have proved and *use* them. Serre duality becomes the Riemann–Roch theorem. The sheaf of differentials becomes the tool that controls ramification. Euler characteristics become genus. Cohomology groups become spaces of functions and forms that we can count, embed with, and classify by. The payoff is enormous: by the end of this chapter, we will have a nearly complete understanding of the geometry of curves, and a sweeping overview of the classification of surfaces that has guided a century of research.

The chapter splits naturally into two halves. The first half (§7.1–§7.1.8) develops the theory of algebraic curves. Here the main components are the Riemann–Roch theorem, Hurwitz's formula, and the rich interplay between divisors and morphisms. The second half (§7.2–§7.2.8) turns to surfaces, where we encounter the intersection pairing, blow-ups, and the celebrated Enriques–Kodaira classification.

Throughout, we will plant signposts toward the later parts of this book. The study of families of curves leads naturally to moduli spaces and stacks (Part III). The birational geometry of surfaces—blow-ups, contractions, minimal models—is the prototype for the Minimal Model Program in arbitrary dimension (a separate volume). And the cohomological invariants that drive classification here find their core expression in étale cohomology and derived categories, each in its own volume. The reader should view this chapter not as a self-contained destination, but as the crossroads from which the highways of modern algebraic geometry radiate.

7.0.1 Conventions for this Chapter Unless otherwise stated, k denotes an algebraically closed field and all schemes are k-schemes. A *curve* will mean a complete nonsingular curve over k (see Definition 7.1.1 for the precise scheme-theoretic formulation). A *surface* will mean a smooth projective surface over k. When we say *point*, we mean *closed point*—the only other point of an integral curve is its generic point η, which we will mention explicitly when needed. We shall occasionally relax these hypotheses (particularly the algebraic closure of k) and will always say so.

7.1 Algebraic Curves

Algebraic curves are "simple" in the sense that their birational geometry is trivial—every birational map between smooth projective curves extends to an isomorphism—and yet "rich" because a single discrete birational-invariant, the *genus*, controls a vast web of geometric, arithmetic, and analytic phenomena.

There are three perspectives on algebraic curves that inform and enrich each other, and the reader should keep all three in mind as we proceed:

1. *The geometric perspective.* A curve is a one-dimensional, complete, nonsingular scheme over k. This is the language of this book, and the perspective from which we will prove our theorems. The nonsingularity assumption will be addressed in this section.

2. *The analytic perspective.* Over $\mathbb{C}$, the category of smooth projective curves (with regular maps) is equivalent to the category of compact Riemann surfaces (with holomorphic maps) by Serre's GAGA theorems (theorem 6.9.3). A Riemann surface is a complex manifold, meaning it carries a rigid *complex structure*. If we forget this complex structure, we are left with a smooth, compact, oriented 2-manifold. The topological and smooth classification of such surfaces[1] tells us they are determined entirely by their genus—the "number of holes." This gives an immediate and powerful topological intuition for many algebraic results.

3. *The algebraic perspective.* By the equivalence between smooth projective curves over k and finitely generated field extensions K/k of transcendence degree 1 (see [11, chapter 21]), studying curves is equivalent to studying function fields. This is the perspective that connects most directly to number theory: the function field of a curve over $\mathbb{F}_q$ behaves much like a number field, and many of our theorems will have arithmetic shadows. I'm thinking of actually adding this proof here, since it feels like too big a result to reference away. I'm okay with doubling the proof; being both here and in classical-alg-geo.

[1] For 2-dimensional manifolds, the topological and smooth categories are isomorphic; every topological surface admits a unique smooth structure.

These three viewpoints formalize into equivalences of categories, provided we are careful with our morphisms. When $k = \mathbb{C}$, GAGA (theorem 6.9.3) provides a direct equivalence between the category of smooth projective curves and the category of compact Riemann surfaces. Meanwhile, the functor from smooth projective curves to function fields forms an anti-equivalence[2] of categories. Throughout this chapter, we will freely draw intuition from whichever perspective is most illuminating at any given moment.

7.1.1 Definition and First Properties

Let us begin by nailing down the scheme-theoretic definition. Recall that a scheme is *integral* if it is both reduced and irreducible (proposition 2.4.19), and *regular* if all of its local rings are regular local rings (definition 2.7.3).

Definition 7.1.1: Curve

Let k be a field. A *curve over* k is a Noetherian, integral, separated scheme X of dimension 1 and of finite type over k. We say:

1. X is *complete* (or *proper*) if the structure morphism $X \to \operatorname{Spec} k$ is proper.
2. X is *nonsingular* (or *smooth*, or *regular*) if all local rings $\mathcal{O}_{X,x}$ are regular, or equivalently (since $k = \overline{k}$), if the structure morphism is smooth (theorem 2.7.10).

Let $\mathbf{Crv}_k$ be the category of all curves over k, $\mathbf{Crv}_k^{\text{prop}}$ be the subcategory of proper curves, and $\mathbf{SmProjCrv}_k$ be the category of smooth projective (i.e., complete and nonsingular) curves. As $\mathbf{SmProjCrv}_k$ is the best-behaved and most frequently used category, it will serve as our default throughout much of the text.

A few remarks are in order. First, our definition allows curves that are not complete (such as the affine line $\mathbb{A}_k^1$) and curves that are singular (such as the nodal cubic $V(y^2 - x^3 - x^2) \subseteq \mathbb{P}_k^2$). In this section, however, our standing assumption is that curves are *complete and nonsingular* unless explicitly stated otherwise.

Second, the integrality assumption means that a curve has a unique generic point η with $\overline{\{\eta\}} = X$, and the function field $\kappa(\eta) = K(X)$ is a finitely generated field extension of k of transcendence degree 1 (theorem 2.6.11). Conversely, every such field extension arises as the function field of some smooth projective curve—this is the algebraic perspective mentioned above.

Third, for a complete nonsingular curve over $k = \overline{k}$, every closed point $x \in X$ has $\mathcal{O}_{X,x}$ a discrete valuation ring (DVR)[3]. This means the closed points of X are in bijection with the DVRs of $K(X)/k$, a fact that was the classical starting point for the theory of abstract curves.

Note that the definition mentioned properness instead of projectiveness. However, all proper nonsingular curves *are* projective. We shall prove this in **??** but it is worth mentioning early to set the right intuition for the reader: curves embed in projective space.

[2] More precisely, the category of smooth projective curves over k with *dominant* morphisms is anti-equivalent to the category of finitely generated field extensions of k of transcendence degree 1. See corollary 3.6.6 and proposition 3.6.8.

[3] Indeed, $\mathcal{O}_{X,x}$ is a regular local ring of dimension 1, and a regular local ring of dimension 1 is precisely a DVR. See proposition 2.7.16 and [11, chapter 22].

Our standing assumption throughout this chapter is that curves are complete and nonsingular. But singular curves arise inevitably—as fibers in degenerating families, as images of morphisms, and as boundary points of moduli spaces—so the reader should know how they connect to the smooth theory. If X is a complete integral curve that is *not* nonsingular, the local rings $\mathcal{O}_{X,x}$ at the singular points are no longer DVRs: the clean bijection between closed points and discrete valuations of $K(X)/k$ breaks down.

While arbitrary integral curves are classified only up to birational equivalence by their function fields, smooth proper curves are classified up to exact isomorphism. Any singularity can be systematically remedied using *normalization*, which provides a birational morphism from a smooth curve. For curves, normality and nonsingularity coincide: a Noetherian local domain of dimension 1 is a DVR if and only if it is integrally closed[4].

Proposition 7.1.2: Affine Curves and Unique Proper Curve

Let $X = \operatorname{Spec} R$ be a nonsingular curve. Then there is a unique proper nonsingular curve Y up to isomorphism where $K(X) = K(Y)$.

***Proof sketch*:**
As X is finite type and affine, $X \cong \operatorname{Spec} k[x_1, ..., x_n]/I$, hence it admits a closed immersion into $\mathbb{A}^n$, which in turn naturally embeds into $\mathbb{P}^n$. Let X be re-labeled to represent the image of this sequence of embeddings. Then its projective closure $\overline{X}$ is closed and proper. It may not be nonsingular, so we take its normalization:

$$\nu \colon \widetilde{X} \longrightarrow \overline{X}$$

where $\widetilde{X}$ is the unique nonsingular curve with $K(\widetilde{X}) = K(X)$, constructed by taking the integral closure of each affine coordinate ring of X in its fraction field (definition 2.4.27). Normalization is a finite map, and finite morphisms are proper and hence preserve closed sets, giving the desired proper nonsingular curve.

One should intuitively visualize ν as pulling apart branches at a node and smoothing out a cusp:

1. For the *nodal cubic* $y^2 = x^3 + x^2$, the normalization is $\mathbb{P}^1$, with two preimages of the node (one for each branch meeting at the origin).
2. For the *cuspidal cubic* $y^2 = x^3$, the normalization is again $\mathbb{P}^1$, with a single preimage of the cusp.

These represent the two most common elementary singularities: a node (where two distinct tangent directions cross) and a cusp (where the tangent directions coincide). While a curve can have much more complicated singularities (e.g., higher-order cusps, tacnodes, or multiple crossing branches), the geometric intuition of normalization as "pulling apart" branches and "smoothing" cusps remains valid in general.

In both cases above, a singular curve of arithmetic genus 1 turns out to be geometrically *rational* ($\widetilde{X} \cong \mathbb{P}^1$). The discrepancy between the arithmetic genus of X and the geometric genus of $\widetilde{X}$ is measured by the δ-*invariant* at the singular points; see the remark after proposition 7.1.13.

[4]The "if" direction: an integrally closed (i.e. normal) Noetherian local domain of dimension 1 is a DVR; see [11, chapter 22]. The "only if" direction is proposition 2.7.17, since every DVR is a regular local ring of dimension 1.

When X is complete, $\widetilde{X}$ is precisely the "complete nonsingular model" of the function field $K(X)$ from **??**—every integral complete curve has a smooth projective model, and that model is its normalization.

Thus, we get the fundamental dictionary:

$$\left\{\begin{array}{c}\text{Smooth Projective Curves}\\ \text{over } k\end{array}\right\} \underset{\mathcal{Z}(S)}{\overset{\mathcal{I}(V)}{\rightleftarrows}} \left\{\begin{array}{c}\text{Finitely Generated Field Extensions}\\ \text{over } k\\ \text{of transcendental degree 1}\end{array}\right\} \tag{7.1}$$

7.1.3 The Analytic Perspective Over $\mathbb{C}$, smooth projective curves correspond to compact Riemann surfaces via the analytification functor (definition 6.9.1). By GAGA (theorem 6.9.3), this functor establishes an equivalence of categories between smooth projective curves over $\mathbb{C}$ and compact Riemann surfaces. In particular, every compact Riemann surface admits the structure of a smooth projective variety—a nontrivial fact whose proof requires showing that every compact Riemann surface carries a nontrivial meromorphic function (see [15]). The topological classification of compact oriented surfaces tells us that such a surface is determined up to diffeomorphism by its genus $g \geq 0$, the "number of holes." We thus inherit a topological invariant for our algebraic curves that will turn out to equal both the arithmetic and geometric genus (Proposition 7.1.13). Recall also the Hodge diamond of a curve from example 0.6.31: the single interesting Hodge number is $h^{0,1} = h^{1,0} = g$.

Next, we record a key rigidity property of maps between complete curves. This should be compared with the situation in complex analysis, where a nonconstant holomorphic map between compact Riemann surfaces is necessarily surjective (and a branched cover).

Proposition 7.1.4: Image of Complete Nonsingular Curve is Closed

Let X be a complete nonsingular curve over k, and let Y be any curve over k. Suppose $f\colon X \to Y$ is a morphism. Then either:

1. $f(X)$ is a single closed point of Y, or
2. f is surjective, Y is necessarily complete, and f is a finite morphism.

***Proof*:**
Since X is proper over k and Y is separated over k, the image $f(X) \subseteq Y$ is closed (proposition 3.5.21). As X is irreducible, so is $f(X)$. A closed irreducible subset of the curve Y is either a single closed point or all of Y (the only irreducible closed subsets of dimension 1 of a curve are the curve itself and its closed points). In the latter case, f is surjective, and Y is proper since it is the image of a proper scheme under a morphism to $\operatorname{Spec} k$.

For finiteness: since f is dominant, it induces an inclusion of function fields $K(Y) \hookrightarrow K(X)$. Both are finitely generated of transcendence degree 1 over k, so $K(X)/K(Y)$ is a finite extension; let $n = [K(X) : K(Y)]$. To show f is finite, we check the condition affine-locally. Let $V = \operatorname{Spec} S \subseteq Y$ be an affine open and let $U = f^{-1}(V)$. Since X is a complete nonsingular curve, the ring of regular functions on U is the integral closure R of S in $K(X)$. As S is a Noetherian integrally closed domain of dimension 1 (a Dedekind domain), R is a finitely generated S-module[a], hence $f^{-1}(V) = \operatorname{Spec} R$ and $f|_U$ is finite.

[a]This is the "finiteness of integral closure" for Dedekind domains; see [11, chapter 22].

This proposition has a simple but striking consequence: the *only* morphisms between complete nonsingular curves are the trivial ones (constant maps) and the finite ones (branched covers). There is no notion of an "immersion" or "partial" map—every nonconstant morphism is surjective. This rigidity is one of the key features that makes the theory of curves so clean compared to higher-dimensional geometry.

As every nonconstant morphism between complete nonsingular curves is finite and surjective, there is a basic invariants attached to such morphisms:

Definition 7.1.5: Degree of Morphism of Curves

The *degree* of f is the degree of the finite field extension:

$$\deg f \;=\; [K(X) : K(Y)].$$

One should think of $\deg f$ as the "number of sheets" of the covering: over a "generic" point $y \in Y$, the fiber $f^{-1}(y)$ consists of exactly $\deg f$ points (counted with multiplicity)[5]. In the analytic picture over $\mathbb{C}$, a degree-n map between compact Riemann surfaces is precisely an n-sheeted branched cover.

Pullback of Divisors

Weil divisors and Cartier divisors coincide on nonsingular curves (proposition 5.4.20), so we work freely with $\operatorname{WDiv} X = \bigoplus_{x \in X_{\mathrm{cl}}} \mathbb{Z} \cdot x$ and the class group $\operatorname{Cl} X$. Furthermore, as $\dim X = 1$, all (Weil) divisors are finite collections of points and hence can be thought of as "recipes" for a rational function to satisfy, namely imposing the zeros and poles required of a rational function.

We now construct the *pullback* of divisors along a finite morphism $f\colon X \to Y$ between complete nonsingular curves. For a closed point $y \in Y$, choose a uniformizer $t \in \mathcal{O}_{Y,y}$ (a generator of the maximal ideal $\mathfrak{m}_y$, so that $\nu_y(t) = 1$). For each point $x \in f^{-1}(y)$, the map $f^{\#}\colon \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ sends t to an element of the maximal ideal $\mathfrak{m}_x$. Define the map $f^*\colon \operatorname{WDiv}(Y) \to \operatorname{WDiv}(X)$ on closed points as:

$$f^*(y) \;=\; \sum_{x \in f^{-1}(y)} \nu_x(f^{\#}(t)) \cdot x \;=\; \sum_{f(x)=y} e_x \cdot x$$

where $e_x = \nu_x(f^{\#}(t))$ is the *ramification index* at x (which we shall study in earnest in §7.1.4). This is a finite sum since f is a finite morphism, and it is independent of the choice of uniformizer t: any other uniformizer has the form $t' = ut$ for a unit $u \in \mathcal{O}_{Y,y}^*$, and $\nu_x(f^{\#}(u)) = 0$ since $f^{\#}(u)$ is a unit in $\mathcal{O}_{X,x}$.

Extending by $\mathbb{Z}$-linearity, we obtain a group homomorphism $f^*\colon \operatorname{WDiv} Y \to \operatorname{WDiv} X$. One checks that f^* sends principal divisors to principal divisors[6], and hence descends to:

$$f^*\colon \operatorname{Cl} Y \to \operatorname{Cl} X.$$

[5]More precisely, $\sum_{x \in f^{-1}(y)} e_x = \deg f$ where e_x is the ramification index at x; we shall return to this in §7.1.4.
[6]If $g \in K(Y)^*$, then $f^*(\operatorname{div}_Y g) = \operatorname{div}_X(g \circ f)$: the valuation of $g \circ f$ at x equals e_x times the valuation of g at $f(x)$, which is exactly what the formula gives.

Proposition 7.1.6: Finite Map and Degrees

Let $f\colon X \to Y$ be a finite morphism of complete nonsingular curves. For any divisor D on Y:

$$\deg f^*(D) \;=\; (\deg f)\cdot(\deg D).$$

***Proof*:**
By $\mathbb{Z}$-linearity, it suffices to prove $\deg f^*(y) = (\deg f)(\deg y)$ for a single closed point $y \in Y$. Let $V = \operatorname{Spec} S \subseteq Y$ be an affine open neighbourhood of y. Since Y is a nonsingular curve, S is a Dedekind domain. Let R be the integral closure of S in $K(X)$; then $f^{-1}(V) = \operatorname{Spec} R$ and R is a finitely generated S-module (as argued in proposition 7.1.4).

Localize at the maximal ideal $\mathfrak{m}_y \subseteq S$. Set $B = S_{\mathfrak{m}_y} = \mathcal{O}_{Y,y}$, which is a DVR with uniformizer t, and let $R' = R \otimes_S B$, the semi-local ring (R' has finitely many maximal ideals) obtained by localizing R at $\mathfrak{m}_y$. Since R is a finitely generated torsion-free S-module of rank $n = [K(X) : K(Y)] = \deg f$, the localization R' is a free B-module of rank n (torsion-free and finitely generated over a PID).

The maximal ideals of R' correspond to the points $x_1, \ldots, x_r \in X$ lying over y, with $(R')_{\mathfrak{m}_i} = \mathcal{O}_{X,x_i}$ for each i. Since the maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_r \subseteq R'$ are pairwise coprime, the Chinese Remainder Theorem gives an isomorphism of k-algebras:

$$R'/tR' \;\cong\; \bigoplus_{i=1}^{r} R'_{\mathfrak{m}_i}/tR'_{\mathfrak{m}_i} \;=\; \bigoplus_{i=1}^{r} \mathcal{O}_{X,x_i}/t\,\mathcal{O}_{X,x_i}.$$

Taking dimensions over k of both sides, and noting that R'/tR' is an algebra over the residue field $\kappa(y) = B/tB$, we obtain:

$$n[\kappa(y) : k] = \dim_k R'/tR' = \sum_{i=1}^{r} \dim_k(\mathcal{O}_{X,x_i}/t\,\mathcal{O}_{X,x_i}) = \sum_{i=1}^{r} \nu_{x_i}(t)[\kappa(x_i) : k] = \sum_{i=1}^{r} e_{x_i}\deg(x_i).$$

The left-hand side is $(\deg f)(\deg y)$, and the right-hand side is $\deg f^*(y)$ by definition, yielding the claim.

This result is the scheme-theoretic incarnation of a basic topological fact: a degree-n map of compact Riemann surfaces pulls back a point (counted with multiplicity) to n points (counted with multiplicity). The next corollary is one of the most important structural results about divisors on curves.

Corollary 7.1.7: Degree Map and Integers

Let X be a complete nonsingular curve. Then principal divisors have degree 0, and the degree map induces a surjective homomorphism:

$$\deg\colon \operatorname{Cl} X \twoheadrightarrow \mathbb{Z}.$$

***Proof*:**
Let $g \in K(X)^*$ with $g \notin k^*$ (if $g \in k^*$, then $\operatorname{div} g = 0$ and the result is trivial). The element g defines a finite morphism $\varphi\colon X \to \mathbb{P}^1_k$ as follows[a]: the inclusion $k(g) \hookrightarrow K(X)$ induces a finite morphism φ to $\mathbb{P}^1_k$ (finite by proposition 7.1.4). Now, $\operatorname{div} g = \varphi^*([0]) - \varphi^*([\infty])$ where $[0]$ and $[\infty]$ are the divisors of the points 0 and ∞ on $\mathbb{P}^1_k$. By proposition 7.1.6:

$$\deg(\operatorname{div} g) = \deg \varphi^*([0]) - \deg \varphi^*([\infty]) = \deg \varphi - \deg \varphi = 0.$$

Thus the degree map $\deg\colon \operatorname{WDiv} X \to \mathbb{Z}$ kills principal divisors and descends to $\operatorname{Cl} X$. Surjectivity is clear: any closed point $x \in X$ gives a divisor of degree 1 (assuming k is algebraically closed or X has a k-rational point).

[a]Concretely, $g \in K(X)$ gives a k-algebra inclusion $k(t) \hookrightarrow K(X)$ via $t \mapsto g$, which corresponds to a dominant rational map $X \dashrightarrow \mathbb{P}^1_k$. Since X is a smooth proper curve, this extends to a morphism by the valuative criterion (theorem 3.5.25).

The kernel of the degree map, $\operatorname{Cl}^0 X = \ker(\deg\colon \operatorname{Cl} X \to \mathbb{Z})$, is an extremely important invariant of the curve. We will see in §7.1.7 that for an elliptic curve E, this group is isomorphic to $E(k)$ itself—the set of closed points equipped with the elliptic curve group law. More generally, $\operatorname{Cl}^0 X$ is the group of k-points of an abelian variety of dimension g called the *Jacobian* $\operatorname{Jac}(X)$, whose construction and properties are central to the study of abelian varieties and the moduli spaces of curves[7].

[7]The Jacobian is the identity component of the Picard scheme $\operatorname{Pic}_{X/k}$, which we will encounter in Part part III. When $k = \mathbb{C}$, it is a complex torus $\mathbb{C}^g/\Lambda$ for a lattice Λ of rank $2g$—the period lattice of the curve. For $g = 1$, the Jacobian is isomorphic to the curve itself, and this is the source of the group law on elliptic curves.

7.1.8 Arithmetic Motif: Covers of Spec $\mathbb{Z}$ Let us pause to observe the arithmetic shadow of what we have just done. Consider the ring of Gaussian integers $\mathbb{Z}[i]$ and the natural inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}[i]$. This induces a finite morphism of schemes:

$$\pi\colon \operatorname{Spec} \mathbb{Z}[i] \longrightarrow \operatorname{Spec} \mathbb{Z}$$

of degree $[\mathbb{Q}(i) : \mathbb{Q}] = 2$. It is a "two-sheeted cover" of Spec $\mathbb{Z}$, and just as for curves, the fibers over closed points exhibit exactly three behaviours familiar from algebraic number theory:

1. **Split:** Over $p \equiv 1 \pmod 4$ (e.g. $p = 5$), we have $p = (2+i)(2-i)$, so the fiber $\pi^{-1}([(p)])$ consists of two points—the primes $(2+i)$ and $(2-i)$—each with ramification index 1.
2. **Inert:** Over $p \equiv 3 \pmod 4$ (e.g. $p = 3$), the ideal (3) remains prime in $\mathbb{Z}[i]$, so the fiber is a single point with residue field $\mathbb{F}_9$. Here the "ramification index" is 1 but the residue field degree is 2.
3. **Ramified:** Over $p = 2$, we have $(1+i)^2 = 2i$, so $2 = -i(1+i)^2$. The fiber is a single point $(1+i)$ with ramification index 2.

In all cases, the "sum of ramification indices times residue field degrees" over each point equals $\deg \pi = 2$, just as in proposition 7.1.6. This is the number-theoretic incarnation of the degree formula. When we reach Hurwitz's theorem (§7.1.4), the reader should think of it as the "genus formula" for branched covers of Spec $\mathbb{Z}$—where the genus of Spec $\mathbb{Z}$ is arguably 0, and the "ramification" is measured by the discriminant. See [2] for the full analogy.

7.1.2 The Riemann–Roch Theorem

If there is one theorem that captures the essence of the theory of algebraic curves, it is Riemann–Roch. It is the theorem that converts the abstract cohomological machinery we have built into concrete geometric information: it counts sections, produces embeddings, and constrains the topology of morphisms. Virtually every result in this chapter flows from it.

The idea, in a sentence, is this: for a divisor D on a curve X of genus g, the Riemann–Roch theorem computes the coherent Euler characteristic $\chi(\mathcal{L}(D)) = \dim_k H^0(X, \mathcal{L}(D)) - \dim_k H^1(X, \mathcal{L}(D))$ in terms of the degree of D and the genus g. Geometrically, one should think of this Euler characteristic as the "expected" number of independent global sections of $\mathcal{L}(D)$, adjusted by an obstruction term (H^1) dictated by the global geometry of the curve. Combining this with Serre duality, which relates $H^1(X, \mathcal{L}(D))$ to $H^0(X, \omega_X \otimes \mathcal{L}(D)^\vee)$, we obtain a formula that balances sections of $\mathcal{L}(D)$ against sections of its "Serre dual." This balance is what makes the theorem so powerful.

Divisors on Curves: Degree, Linear Systems, and $\ell(D)$

Let X be a complete nonsingular curve over $k = \overline{k}$. We briefly consolidate the divisor theory from §5.4 in the particularly clean form it takes on curves.

Since X is a regular scheme of dimension 1, every local ring $\mathcal{O}_{X,x}$ at a closed point is a DVR, hence a UFD. In particular, X is locally factorial, and the natural map from Cartier divisors to Weil divisors is an isomorphism (proposition 5.4.20 and corollary 5.4.33):

$$\operatorname{CaCl} X \;\cong\; \operatorname{Cl} X \;\cong\; \operatorname{Pic} X.$$

Note that for a singular curve, the natural map $\operatorname{CaCl} X \to \operatorname{Cl} X$ (equivalently, $\operatorname{Pic} X \to \operatorname{Cl} X$) is *injective* (more generally it is always injective for integral schemes that are regular in codimension 1, definition 5.4.1 and proposition 5.4.20), but it need not be *surjective*. A Weil divisor supported at a singular point may fail to be locally principal, and hence may not correspond to any line bundle at all. On singular curves, *Cartier divisors* (equivalently, line bundles, elements of $\operatorname{Pic} X$) remain the correct objects for Riemann–Roch and for defining morphisms to projective space. Weil divisors are a coarser invariant that may contain extra classes not represented by line bundles. The reader should keep this distinction in mind whenever the nonsingularity hypothesis is relaxed. In this section, we continue with the assumption that all curves are nonsingular.

We write divisors additively as $D = \sum_{x \in X_{\mathrm{cl}}} n_x \cdot x$ where all but finitely many n_x are zero. The *degree* of D is $\deg D = \sum n_x$. For each divisor D, we have the associated invertible sheaf $\mathcal{L}(D)$, with the key property that for D, D' divisors:

$$\mathcal{L}(D + D') \cong \mathcal{L}(D) \otimes \mathcal{L}(D'), \qquad \mathcal{L}(-D) \cong \mathcal{L}(D)^\vee,$$
$$\mathcal{L}(D) \cong \mathcal{O}_X \iff D \sim 0 \quad \text{(linearly equivalent to zero).}$$

Recall that D is *effective*, written $D \geq 0$, if all $n_x \geq 0$. A global section $s \in H^0(X, \mathcal{L}(D))$ corresponds to a rational function $f \in K(X)^*$ with $\operatorname{div}(f) + D \geq 0$, together with the zero section.

Definition 7.1.9: Complete Linear System

The *complete linear system* of D is the set of effective divisors linearly equivalent to D:

$$|D| \;=\; \{D' \geq 0 : D' \sim D\} \;=\; \{\operatorname{div}(f) + D : f \in K(X)^*,\ \operatorname{div}(f) + D \geq 0\}.$$

There is a natural bijection $|D| \cong \mathbb{P}\big(H^0(X, \mathcal{L}(D))\big)$: the effective divisors in the class of D are parametrized by the projectivization of the space of global sections of $\mathcal{L}(D)$, since two sections $s, s' \in H^0(X, \mathcal{L}(D)) \setminus \{0\}$ define the same effective divisor if and only if $s' = \lambda s$ for some $\lambda \in k^*$.

We introduce the classical notation:

$$\ell(D) \;:=\; \dim_k H^0(X, \mathcal{L}(D)).$$

Thus $|D|$ is a projective space of dimension $\ell(D) - 1$ (and $|D| = \emptyset$ precisely when $\ell(D) = 0$). The number $\ell(D)$ is finite by theorem 5.3.27[8]. A *linear system* is any linear subspace of $|D|$, and a linear system of dimension 1 is called a *pencil*.

The following lemma is elementary but crucial; it says that divisors with sections are "non-negative" in a precise sense.

[8] One can also see this from the finiteness of cohomology on projective schemes, corollary 6.6.7.

Lemma 7.1.10: Divisors and Degree

Let D be a divisor on a complete nonsingular curve X.

1. If $\ell(D) > 0$, then $\deg D \geq 0$.
2. If $\ell(D) > 0$ and $\deg D = 0$, then $D \sim 0$ (i.e. $\mathcal{L}(D) \cong \mathcal{O}_X$).

***Proof*:**
If $\ell(D) > 0$, then $|D| \neq \emptyset$, so $D \sim D'$ for some effective $D' \geq 0$. Since $\deg D = \deg D' = \sum n_x \geq 0$ (with $n_x \geq 0$ for all x), we get (1). For (2), if additionally $\deg D = 0$, then $D' = \sum n_x \cdot x$ with all $n_x \geq 0$ and $\sum n_x = 0$, forcing $n_x = 0$ for all x, i.e. $D' = 0$, so $D \sim 0$.

The Canonical Divisor and Genus

Recall from §5.5 that the sheaf of relative differentials $\Omega_{X/k}$ on a smooth k-scheme X of dimension n is a locally free sheaf of rank n (corollary 5.5.7). When X is a smooth curve, $n = 1$, so $\Omega_{X/k}$ is an *invertible sheaf*—it is a line bundle on X.

By the definition of the canonical sheaf (definition 5.5.25), $\omega_X = \bigwedge^{\dim X} \Omega_{X/k}$. For a curve, this is simply $\omega_X = \Omega_{X/k}$. On the other hand, the dualizing sheaf in the sense of Serre duality (definition 6.5.1) is also ω_X for a smooth projective variety. These two coincide for smooth curves, which is the reason Serre duality is so immediately applicable.

Definition 7.1.11: Canonical Divisor

Let X be a complete nonsingular curve. A *canonical divisor* K (or K_X) is any divisor in the linear equivalence class corresponding to $\omega_X = \Omega_{X/k}$, i.e. $\mathcal{L}(K) \cong \omega_X$.

The canonical divisor is defined only up to linear equivalence, but its degree and the dimension $\ell(K)$ are well-defined invariants of X. We now define the fundamental discrete invariant of a curve.

Definition 7.1.12: Genus of a Curve

Let X be a complete nonsingular curve over k. The following three numbers are defined:

1. The *arithmetic genus*: $p_a(X) = 1 - P_X(0)$, where P_X is the Hilbert polynomial of X with respect to any very ample sheaf (definition 6.7.6).
2. The *geometric genus*: $p_g(X) = \dim_k H^0(X, \omega_X)$ (definition 5.5.26).
3. The *cohomological genus*: $\dim_k H^1(X, \mathcal{O}_X)$.

As we prove below, these three numbers coincide, and their common value is called simply the *genus* of X, denoted g (or $g(X)$).

Proposition 7.1.13: Equivalence of Genus

Let X be a complete nonsingular curve over $k = \overline{k}$. Then:

$$p_a(X) \;=\; p_g(X) \;=\; \dim_k H^1(X, \mathcal{O}_X).$$

***Proof*:**
Step 1: $p_a = \dim_k H^1(X, \mathcal{O}_X)$. By definition, $p_a(X) = 1 - P_X(0)$ where $P_X(n) = \chi(\mathcal{O}_X(n))$ is the Hilbert polynomial (theorem 6.7.5 and definition 6.7.6). Setting $n = 0$:

$$P_X(0) = \chi(\mathcal{O}_X) = \dim_k H^0(X, \mathcal{O}_X) - \dim_k H^1(X, \mathcal{O}_X).$$

Since X is a projective integral scheme over $k = \overline{k}$, the global sections $H^0(X, \mathcal{O}_X) = k$ (theorem 3.5.27), so $P_X(0) = 1 - \dim_k H^1(X, \mathcal{O}_X)$, and $p_a = 1 - P_X(0) = \dim_k H^1(X, \mathcal{O}_X)$.

Note also that $H^i(X, \mathcal{O}_X) = 0$ for $i \geq 2$ since $\dim X = 1$ (theorem 6.2.5), so the Euler characteristic of $\mathcal{O}_X$ is just $1 - g$ where $g = \dim_k H^1(X, \mathcal{O}_X)$.

Step 2: $\dim_k H^1(X, \mathcal{O}_X) = p_g$. Serre duality for a smooth projective variety X of dimension n over k gives a perfect pairing (theorem 6.5.9):

$$H^i(X, \mathscr{F}) \times H^{n-i}(X, \omega_X \otimes \mathscr{F}^\vee) \to H^n(X, \omega_X) \cong k.$$

Taking $n = 1$, $i = 1$, $\mathscr{F} = \mathcal{O}_X$:

$$H^1(X, \mathcal{O}_X) \;\cong\; H^0(X, \omega_X)^\vee.$$

Therefore $\dim_k H^1(X, \mathcal{O}_X) = \dim_k H^0(X, \omega_X) = p_g(X)$.

From now on, we write $g = g(X)$ for the genus. Since $g = \dim_k H^1(X, \mathcal{O}_X) \geq 0$, the genus is a non-negative integer.

The following formula is presented here for readers who have seen the equivalent in complex geometry (ex. in [**NateComplexAnalysis**]:

Proposition 7.1.14: Degree Formula for Plane Curves

Let $C \subseteq \mathbb{P}^2_k$ be a smooth projective curve of degree d. Then the genus of C is given by:

$$g(C) \;=\; \frac{(d-1)(d-2)}{2}.$$

Note that the assumption that $C \subseteq \mathbb{P}^2_k$ is restrictive but necessary: the twisted cubic will be an example of a curve that is degree 3 that naturally embeds in $\mathbb{P}^3_k$ but is rational. This result will generalize to an inequality for $\mathbb{P}^3_k$ (the universal embedding space for all curves) in theorem 7.1.61.

***Proof*:**
Since C is a closed subscheme of $\mathbb{P}^2$ defined by a single homogeneous polynomial of degree d, its ideal sheaf is $\mathcal{I}_C \cong \mathcal{O}_{\mathbb{P}^2}(-d)$. This gives the short exact sequence of sheaves on $\mathbb{P}^2$:

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^2}(-d) \longrightarrow \mathcal{O}_{\mathbb{P}^2} \longrightarrow \mathcal{O}_C \longrightarrow 0.$$

The Euler characteristic $\chi(\mathscr{F}) = \sum_i (-1)^i \dim_k H^i(X, \mathscr{F})$ is additive on short exact sequences (proposition 6.7.2), so we have:

$$\chi(\mathcal{O}_C) \;=\; \chi(\mathcal{O}_{\mathbb{P}^2}) - \chi(\mathcal{O}_{\mathbb{P}^2}(-d)).$$

From the cohomology of projective space (theorem 6.3.1), we know that the Hilbert polynomial of $\mathbb{P}^2$ is $\chi(\mathcal{O}_{\mathbb{P}^2}(n)) = \frac{(n+1)(n+2)}{2}$ for all $n \in \mathbb{Z}$. Evaluating this at $n = 0$ and $n = -d$ yields:

$$\chi(\mathcal{O}_{\mathbb{P}^2}) = 1,$$
$$\chi(\mathcal{O}_{\mathbb{P}^2}(-d)) = \frac{(-d+1)(-d+2)}{2} \;=\; \frac{(d-1)(d-2)}{2}.$$

Therefore, $\chi(\mathcal{O}_C) = 1 - \frac{(d-1)(d-2)}{2}$.

On the other hand, since C is a complete nonsingular curve, proposition 7.1.13 establishes that the genus g equals the arithmetic genus $p_a(C) = 1 - \chi(\mathcal{O}_C)$. Thus:

$$1 - g \;=\; 1 - \frac{(d-1)(d-2)}{2} \qquad \Longrightarrow \qquad g \;=\; \frac{(d-1)(d-2)}{2}.$$

Example 7.1.15: Genus Computations

1. *The projective line.* $g(\mathbb{P}^1_k) = 0$. Indeed, $H^1(\mathbb{P}^1_k, \mathcal{O}_{\mathbb{P}^1}) = 0$ by the computation of cohomology of projective space (theorem 6.3.1).

2. *Smooth plane curves.* Let $C \subseteq \mathbb{P}^2_k$ be a smooth curve of degree d. By the degree formula (proposition 7.1.14):
$$g(C) \;=\; \frac{(d-1)(d-2)}{2}.$$
This gives $g = 0$ for $d = 1, 2$ (lines and conics—both rational), $g = 1$ for $d = 3$ (elliptic curves), $g = 3$ for $d = 4$ (the first non-hyperelliptic curves), and so on. Notice the rapid growth: the genus of a plane curve of degree d is quadratic in d.

   Note: This also allows us to verify the degree of the canonical divisor for plane curves before proving it generally. By the adjunction formula (proposition 6.5.7), the canonical sheaf is $\omega_C \cong (\omega_{\mathbb{P}^2} \otimes \mathcal{O}_{\mathbb{P}^2}(C))|_C \cong \mathcal{O}_C(d-3)$. The degree of restricting $\mathcal{O}(d-3)$ to a degree d curve is $d(d-3)$. Notice that $2g - 2 = 2\left(\frac{d^2-3d+2}{2}\right) - 2 = d(d-3)$, which perfectly predicts the general formula $\deg K_X = 2g - 2$.

3. *The degree of the canonical divisor.* By Riemann–Roch (which we will prove momentarily), one can show that for *any* smooth curve of genus g:
$$\deg K_X = 2g - 2.$$
For $\mathbb{P}^1$ this gives $\deg K = -2$, consistent with $\omega_{\mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}(-2)$ (from the Euler sequence theorem 5.5.18). For a smooth plane cubic, $\deg K = 0$, consistent with $\omega_C \cong \mathcal{O}_C$—a remarkable fact!

7.1.16 Forward Reference: The Moduli Space $\mathcal{M}_g$ We have just seen that smooth projective curves over k are classified up to isomorphism by a discrete invariant $g \geq 0$—but this classification is very coarse. For a fixed genus g, there is typically a *continuous family* of non-isomorphic curves.

How large is this family? By a deformation-theoretic argument (which we will make precise in Part III), the "moduli space" parametrizing smooth curves of genus g has the following dimensions:

$$\dim \mathcal{M}_g = \begin{cases} 0 & \text{if } g = 0 \quad (\text{only } \mathbb{P}^1), \\ 1 & \text{if } g = 1 \quad (\text{the } j\text{-line}), \\ 3g - 3 & \text{if } g \geq 2. \end{cases}$$

Making this precise requires the theory of *stacks*: the space $\mathcal{M}_g$ is not a scheme (because curves have automorphisms), but a Deligne–Mumford stack. Its construction is one of the triumphs of modern algebraic geometry and will be developed in Part III of this book. For now, the reader should simply note that the genus is the "first" invariant of a curve, but far from the "last"—within each genus lies a rich geometric world.

7.1.17 Warning: The Genera Diverge for Singular Curves The equality $p_a = p_g = \dim_k H^1(X, \mathcal{O}_X)$ is a luxury of nonsingularity. For a complete integral curve X that is singular, these invariants come apart in an important and controlled way.

The *geometric genus* $g(\widetilde{X})$ is the genus of the normalization—it is a birational invariant, blind to the singularities. The *arithmetic genus* $p_a(X) = 1 - \chi(\mathcal{O}_X)$ depends on the scheme structure of X, singularities included, and is the invariant that remains constant in flat families (theorem 6.7.5). They are related by the *δ-invariant*: for each singular point $P \in X$, set

$$\delta_P \;=\; \dim_k \Big(\widetilde{\mathcal{O}}_{X,P} \,/\, \mathcal{O}_{X,P} \Big)$$

where $\widetilde{\mathcal{O}}_{X,P}$ denotes the integral closure of $\mathcal{O}_{X,P}$ in $K(X)$ (equivalently, the stalk $(\nu_* \mathcal{O}_{\widetilde{X}})_P$ of the pushforward of the structure sheaf along the normalization map ν). Then:

$$p_a(X) \;=\; g(\widetilde{X}) \;+\; \sum_{P \in X_{\mathrm{sing}}} \delta_P .$$

The δ-invariant measures the "severity" of a singularity: a node (two branches crossing transversally) has $\delta = 1$, and an ordinary cusp has $\delta = 1$ as well[a].

For example, both the nodal cubic $y^2 = x^3 + x^2$ and the cuspidal cubic $y^2 = x^3$ are degree-3 plane curves, so $p_a = \frac{(3-1)(3-2)}{2} = 1$ (example 7.1.15). Yet both have normalization $\mathbb{P}^1$, giving $g(\widetilde{X}) = 0$. The discrepancy $p_a - g(\widetilde{X}) = 1$ is accounted for by the single singular point with $\delta = 1$.

This distinction matters for moduli: when a smooth elliptic curve ($g = 1$) degenerates in a flat family to a nodal rational curve, the arithmetic genus $p_a = 1$ is preserved—as it must be, by flatness—but the geometric genus drops to 0. The "missing genus" is absorbed into the singularity. This phenomenon is the reason that the compactification $\overline{\mathcal{M}}_g$ parametrizes *stable* (i.e. nodal) curves rather than smooth ones; see box 7.1.64.

[a] More exotic singularities can have larger δ: for instance, the curve $y^2 = x^{2k+1}$ has a singularity with $\delta = k$ at the origin. The δ-invariant also equals the *colength* of $\mathcal{O}_{X,P}$ in its normalization, which in the analytic picture over $\mathbb{C}$ counts the number of "missing points" when X is compared to $\widetilde{X}$ as a topological space.

Statement and Proof of Riemann–Roch

Recall that Serre duality (theorem 6.5.9) gives, for any invertible sheaf $\mathcal{L}$ on a smooth projective curve X of genus g:

$$H^1(X, \mathcal{L}) \;\cong\; H^0(X, \omega_X \otimes \mathcal{L}^\vee)^\vee .$$

In divisor language, if $\mathcal{L} = \mathcal{L}(D)$, then $\omega_X \otimes \mathcal{L}(D)^\vee \cong \mathcal{L}(K - D)$, and hence:

$$\dim_k H^1(X, \mathcal{L}(D)) \;=\; \ell(K - D).$$

This identification transforms the Euler characteristic $\chi(\mathcal{L}(D)) = \ell(D) - \ell(K - D)$ into a quantity involving only dimensions of spaces of global sections.

Theorem 7.1.18: Riemann–Roch Theorem

Let X be a complete nonsingular curve of genus g over $k = \overline{k}$, and let K be a canonical divisor. For any divisor D on X:

$$\ell(D) - \ell(K - D) \;=\; \deg D + 1 - g.$$

Equivalently, in sheaf-cohomological language:

$$\chi(\mathcal{L}(D)) \;=\; \deg D + 1 - g.$$

Before giving the proof, let us appreciate what this theorem says. The right-hand side is *topological*: it depends only on the degree (an integer) and the genus (a topological invariant of the underlying Riemann surface, when $k = \mathbb{C}$). The left-hand side is *algebraic*: it involves the dimensions of spaces of rational functions with prescribed poles. The theorem says that these two kinds of data are linked by a single equation.

The term $\ell(K - D)$ is sometimes called the *index of speciality* of D and is denoted $i(D) := \ell(K - D)$. A divisor with $i(D) = 0$ is called *non-special*; for such divisors, Riemann–Roch simply says $\ell(D) = \deg D + 1 - g$. We shall see that "most" divisors of sufficiently large degree are non-special.

***Proof*:**
We prove the cohomological form $\chi(\mathcal{L}(D)) = \deg D + 1 - g$; the classical form follows immediately from Serre duality as explained above.

Base case: $D = 0$. We have $\chi(\mathcal{O}_X) = \dim_k H^0(X, \mathcal{O}_X) - \dim_k H^1(X, \mathcal{O}_X) = 1 - g$, since $H^0(X, \mathcal{O}_X) = k$ (theorem 3.5.27) and $g = \dim_k H^1(X, \mathcal{O}_X)$ (proposition 7.1.13). This equals $\deg 0 + 1 - g = 1 - g$.

Induction step. We show: the formula holds for D if and only if it holds for $D + P$, where P is any closed point of X.

Consider P as a closed subscheme of X, defined by the ideal sheaf $\mathcal{L}(-P)$ (a single effective Cartier divisor). The structure sheaf $\mathcal{O}_P$ is a skyscraper sheaf supported at P with stalk k, which we denote k_P. This gives the short exact sequence:

$$0 \to \mathcal{L}(-P) \to \mathcal{O}_X \to k_P \to 0.$$

Tensoring with the invertible sheaf $\mathcal{L}(D + P)$ (which is locally free of rank 1, hence flat, hence tensoring preserves exactness):

$$0 \to \mathcal{L}(-P) \otimes \mathcal{L}(D + P) \to \mathcal{L}(D + P) \to k_P \otimes \mathcal{L}(D + P) \to 0.$$

Now $\mathcal{L}(-P) \otimes \mathcal{L}(D + P) \cong \mathcal{L}(D)$, and $k_P \otimes \mathcal{L}(D + P) \cong k_P$ (tensoring a skyscraper sheaf with an invertible sheaf does not change its stalk—the invertible sheaf is locally isomorphic to $\mathcal{O}_X$ near P). So we have:

$$0 \to \mathcal{L}(D) \to \mathcal{L}(D + P) \to k_P \to 0. \tag{7.2}$$

Taking Euler characteristics and using additivity (proposition 6.7.2):

$$\chi(\mathcal{L}(D + P)) = \chi(\mathcal{L}(D)) + \chi(k_P).$$

Since k_P is a skyscraper sheaf supported at a single point, $H^0(X, k_P) = k$ and $H^i(X, k_P) = 0$ for $i > 0$, so $\chi(k_P) = 1$. Therefore:

$$\chi(\mathcal{L}(D + P)) \;=\; \chi(\mathcal{L}(D)) + 1.$$

On the other hand, $\deg(D + P) = \deg D + 1$. So the right-hand side of the Riemann–Roch formula also increases by 1 when we replace D by $D + P$. This proves the claim: Riemann–Roch holds for D if and only if it holds for $D + P$.

Thus, every divisor D can be reached from the zero divisor by finitely many steps of adding or subtracting a point[a]. By the induction step, the formula holds for D if and only if it holds for 0. Since we verified it for $D = 0$, it holds for all D, as we sought to show.

[a] Write $D = \sum n_i P_i = \sum_{n_i>0} n_i P_i - \sum_{n_j<0} |n_j| P_j$. Starting from 0, add the positive points and subtract the negative points one at a time.

The proof is strikingly simple—almost suspiciously so. The heavy lifting is hidden in two ingredients from earlier chapters: Serre duality (which required the machinery of Ext groups, the local-to-global spectral sequence, and residues) and the additivity of the Euler characteristic (which required the long exact sequence in cohomology). What the proof really shows is that the Euler characteristic $D \mapsto \chi(\mathcal{L}(D))$ is an additive function of divisors that agrees with $D \mapsto \deg D + 1 - g$ at $D = 0$ and has the same "slope." Since both sides are "discrete-linear" in D, they must agree everywhere.

7.1.19 Riemann–Roch as an Index Theorem The Riemann–Roch theorem is the simplest instance of an *index theorem*. In mathematics, an index theorem equates an analytical index (which counts the solutions to a differential equation, minus obstructions) to a purely topological invariant. In the analytic world (over $\mathbb{C}$), $\chi(\mathcal{L}) = h^0 - h^1$ computes the Fredholm index of the $\bar{\partial}$-operator (i.e. Cauchy-Riemann equations as a differential equation) on a line bundle —an analytic result that Riemann–Roch equates to the topology of the curve ($\deg D + 1 - g$). This theme is taken up in vastly greater generality in Part II:

1. The *Hirzebruch–Riemann–Roch theorem* computes $\chi(\mathscr{E})$ for any vector bundle $\mathscr{E}$ on a smooth projective variety X in terms of the Chern character $\mathrm{ch}(\mathscr{E})$ and the Todd class $\mathrm{td}(T_X)$: $\chi(\mathscr{E}) = \int_X \mathrm{ch}(\mathscr{E}) \cdot \mathrm{td}(T_X)$.
2. The *Grothendieck–Riemann–Roch theorem* (GRR) generalizes this to a relative setting: for a proper morphism $f\colon X \to Y$, it computes the Chern character of the higher direct images $R^i f_* \mathscr{E}$ in terms of $\mathrm{ch}(\mathscr{E})$ and $\mathrm{td}(T_{X/Y})$.

For a curve, the Chern character of a line bundle $\mathcal{L}(D)$ is simply $1 + \deg D$, and the Todd class of T_X is $1 + \frac{1}{2}c_1(T_X) = 1 + (1 - g)$, and HRR reduces exactly to $\chi(\mathcal{L}(D)) = \deg D + 1 - g$. The reader who is intrigued by the Riemann–Roch theorem should look forward to GRR in Part II.

Example 7.1.20: Immediate Application – Elliptic Curves

Let $E \subseteq \mathbb{P}^2_k$ be a nonsingular cubic $y^2 z = x^3 - xz^2$ with $\mathrm{char}(k) \neq 2$. This curve is not rational[a]. Let $P_0 = [0 : 1 : 0]$, the unique point at infinity, which is an inflection point: the tangent line $\{z = 0\}$ meets E with multiplicity 3, giving the divisor relation $3P_0 \sim$ (hyperplane section).

Consider the kernel $K = \ker(\deg\colon \mathrm{Cl}\, E \to \mathbb{Z})$. Define a map φ from the set of k-rational points $E(k)$ to K by $\varphi(P) = [P - P_0]$. This map is *injective*: if $[P - P_0] = [Q - P_0]$, then $P \sim Q$. But by Riemann–Roch, $\ell(P) = \deg(P) + 1 - g = 1 + 1 - 1 = 1$. The complete linear system $|P|$ has dimension $\ell(P) - 1 = 0$, meaning it consists of exactly one effective divisor, namely P itself. Thus $P = Q$.

To see *surjectivity*, take any $D \in K$, so $\deg D = 0$. By the Riemann–Roch theorem, $\ell(D + P_0) = \deg(D + P_0) + 1 - g = 1 + 1 - 1 = 1$. Thus, there exists a unique effective divisor of degree 1—which must be a single k-rational point P—such that $D + P_0 \sim P$, meaning $D \sim P - P_0$.

Geometrically, if D is supported entirely on k-rational points, one can explicitly see this by writing $D = \sum n_i(P_i - P_0)$ and using the chord-tangent process: if a line through P and Q meets E in a third point R, then $P + Q + R \sim 3P_0$, so $[P - P_0] + [Q - P_0] = [T - P_0]$ where T is the third intersection of the line P_0R with E. This iteratively reduces any such sum to $D = [P - P_0]$ for some $P \in E(k)$.

Thus φ is a bijection, and the group structure on K transfers to a group structure on $E(k)$, with identity P_0. One checks that the addition and inversion maps $+\colon E \times E \to E$ and $[-1]\colon E \to E$ are morphisms of schemes, making E a *group scheme* over k (cf. example 4.1.8). We will develop this more thoroughly in §7.1.7.

More generally, if X is any complete nonsingular curve, then $K = \mathrm{Cl}^0 X$ is the group of k-points of an abelian variety called the *Jacobian variety* $\mathrm{Jac}(X)$, and $\mathrm{Cl}\, X$ sits in a short exact sequence:

$$0 \to \mathrm{Jac}(X)(k) \to \mathrm{Cl}\, X \xrightarrow{\deg} \mathbb{Z} \to 0.$$

For surfaces and higher-dimensional varieties, the divisor class group has a richer structure: the quotient $\mathrm{Cl}\, X/\mathrm{Cl}^0 X$ becomes the *Néron–Severi group* $\mathrm{NS}(X)$, which is a finitely generated abelian group (but generally of rank > 1). This generalization will be central to the theory of surfaces in §7.2.1.

[a] We saw this in [11, chapter 21.3]. Alternatively, we will see shortly that E has genus 1, while $\mathbb{P}^1_k$ has genus 0; since the genus is a birational invariant, they cannot be isomorphic.

A quick word should be mentioned for the singular case: the Riemann–Roch theorem extends to singular curves after replacing the sheaf of Kähler differentials $\Omega_{X/k}$ with the correct substitute. On a smooth curve, $\Omega_{X/k}$ is an invertible sheaf and simultaneously serves as the dualizing sheaf for Serre duality. On a singular curve X, both of these properties fail: $\Omega_{X/k}$ acquires *torsion* at the singular points (reflecting the fact that Kähler differentials detect the non-reducedness of the diagonal at singularities), and it is no longer locally free[9]. In particular, $\Omega_{X/k}$ cannot play the role of the dualizing sheaf in Serre duality.

The correct replacement is the *dualizing sheaf* ω_X° in the sense of Serre duality (definition 6.5.1), which for a complete integral curve over $k = \overline{k}$ can be described concretely as the sheaf of *regular differentials* in the sense of Rosenlicht: meromorphic differentials η on the normalization $\widetilde{X}$ satisfying a residue condition at each preimage of a singular point[10]. On a smooth curve, $\omega_X^\circ \cong \Omega_{X/k}$, so there is nothing new in the nonsingular case. On a singular curve, ω_X° is still a rank-1 torsion-free sheaf

[9] For a concrete example, on the cuspidal cubic $X = \operatorname{Spec} k[x, y]/(y^2 - x^3)$, the module of differentials is $\Omega_{X/k} = (A\, dx \oplus A\, dy)/(3x^2\, dx - 2y\, dy)$. Consider the differential $\tau = 2x\, dy - 3y\, dx$. It is a non-zero element, but $y \cdot \tau = 2xy\, dy - 3y^2\, dx = x(3x^2\, dx) - 3x^3\, dx = 0$ in $\Omega_{X/k}$ (and similarly $x \cdot \tau = 0$). Thus τ generates a torsion submodule supported precisely at the cusp $(0, 0)$, which prevents $\Omega_{X/k}$ from being locally free.

[10] More precisely, if $\nu\colon \widetilde{X} \to X$ is the normalization and $P \in X$ is a singular point with preimages $Q_1, \ldots, Q_r$, then a meromorphic 1-form η on $\widetilde{X}$ is a *regular differential* at P if $\sum_{i=1}^r \operatorname{Res}_{Q_i}(f \cdot \eta) = 0$ for every $f \in \mathcal{O}_{X,P}$. The sheaf ω_X° is the subsheaf of $\nu_*\omega_{\widetilde{X}}(\text{poles})$ consisting of such forms. See [16, Chapter IV.2] and [18, Chapter 8] for full details.

(though not necessarily invertible), and the key miracle is that Serre duality (theorem 6.5.9) continues to hold:

$$H^1(X, \mathscr{F}) \;\cong\; \mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \omega_X^\circ)^\vee$$

for any coherent sheaf $\mathscr{F}$ on X.

With this dualizing sheaf in hand, Riemann–Roch for a *complete integral* (possibly singular) curve reads:

> *for any line bundle $\mathcal{L}$ on X,*
>
> $$\chi(\mathcal{L}) \;=\; \deg \mathcal{L} + 1 - p_a(X)$$
>
> *where $p_a(X) = 1 - \chi(\mathcal{O}_X)$ is the* arithmetic genus.

The line bundle can be More generally any rank-1 torsion-free sheaf $\mathcal{L}$; on a singular curve, these play the role that invertible sheaves play on smooth curves. The proof is identical to the smooth case: additivity of the Euler characteristic plus induction on twisting by a point. What changes is the *interpretation*: the genus that appears is p_a, not the geometric genus $g(\widetilde{X})$, and the duality term $\ell(K - D)$ must be understood via ω_X° rather than $\Omega_{X/k}$.

Due to the cohomological machinery we built up, the Euler characteristic formula $\chi(\mathcal{L}) = \deg \mathcal{L} + 1 - p_a$ works *verbatim* on any complete integral curve, smooth or not. It is only when one wants to split $\chi(\mathcal{L})$ into $h^0 - h^1$ and apply Serre duality to control each piece separately that the distinction between ω_X° and $\Omega_{X/k}$ becomes essential.

7.1.3 Applications of Riemann–Roch

Proposition 7.1.21: Degree of the Canonical Divisor

Let X be a complete nonsingular curve of genus g. Then:

$$\deg K_X \;=\; 2g - 2, \qquad \ell(K_X) = g.$$

***Proof*:**
Apply Riemann–Roch with $D = K$:

$$\ell(K) - \ell(K - K) = \deg K + 1 - g.$$

Now $\ell(K - K) = \ell(0) = \dim_k H^0(X, \mathcal{O}_X) = 1$, and $\ell(K) = p_g = g$ by proposition 7.1.13. So $g - 1 = \deg K + 1 - g$, giving $\deg K = 2g - 2$.

This formula is extremely satisfying: over $\mathbb{C}$, the degree of K_X is $2g - 2$, which is also the topological Euler characteristic of the Riemann surface with the *opposite* sign (recall $\chi_{\text{top}} = 2 - 2g$). This is no accident—it is the Gauss–Bonnet theorem in disguise.

Proposition 7.1.22: Non-Speciality for Large Degree

Let D be a divisor on a curve X of genus g. If $\deg D > 2g-2$, then $\ell(K-D) = 0$, and Riemann–Roch gives:

$$\ell(D) \;=\; \deg D + 1 - g.$$

In particular, $\ell(D) > 0$ for any divisor of degree $\geq g$.

***Proof*:**
If $\ell(K-D) > 0$, then by lemma 7.1.10, $\deg(K-D) \geq 0$, i.e. $\deg D \leq \deg K = 2g-2$. Contrapositive: if $\deg D > 2g-2$, then $\ell(K-D) = 0$.

For the "in particular": if $\deg D \geq g$, then $\ell(D) = \deg D + 1 - g \geq 1$ when D is non-special. If $\deg D \leq 2g-2$, we cannot guarantee non-speciality, but the Riemann–Roch formula still gives $\ell(D) \geq \deg D + 1 - g$, which is ≥ 1 whenever $\deg D \geq g$ (even without non-speciality, since $\ell(K-D) \geq 0$).

One of the most important applications of Riemann–Roch is the construction of embeddings of curves into projective space. Recall (definitions 5.3.21 and 5.3.23) that an invertible sheaf $\mathcal{L}$ on X is:

1. *generated by global sections* (or *base-point-free*) if the natural map $H^0(X,\mathcal{L}) \otimes_k \mathcal{O}_X \to \mathcal{L}$ is surjective,
2. *very ample* if it defines a closed immersion $\varphi_{\mathcal{L}} \colon X \hookrightarrow \mathbb{P}^N_k$ where $N+1 = \dim_k H^0(X,\mathcal{L})$.

For a line bundle $\mathcal{L}(D)$ on a curve, base-point-freeness means that for every point $P \in X$, there exists a section $s \in H^0(X,\mathcal{L}(D))$ with $s(P) \neq 0$; equivalently, $\ell(D-P) = \ell(D) - 1$ for every P. Very ampleness is a stronger condition: it requires that for every pair of (possibly equal) points $P, Q \in X$, we have $\ell(D-P-Q) = \ell(D) - 2$, i.e. sections separate points and tangent vectors[11].

Proposition 7.1.23: Base-Point-Free Criterion for Curves

Let D be a divisor on a complete nonsingular curve X of genus g. If $\deg D \geq 2g$, then $\mathcal{L}(D)$ is base-point-free.

***Proof*:**
Let $P \in X$ be any closed point. We need $\ell(D) > \ell(D-P)$, i.e. there exists a section of $\mathcal{L}(D)$ not vanishing at P. By Riemann–Roch:

$$\ell(D) = \deg D + 1 - g + \ell(K-D),$$
$$\ell(D-P) = \deg D - 1 + 1 - g + \ell(K-D+P).$$

So $\ell(D) - \ell(D-P) = 1 + \ell(K-D) - \ell(K-D+P)$. It suffices to show $\ell(K-D) = \ell(K-D+P)$. Since $\deg D \geq 2g$, we have $\deg(K-D) = 2g-2-\deg D \leq -2 < 0$, so $\ell(K-D) = 0$ by lemma 7.1.10. Also $\deg(K-D+P) = -2+1 = -1$ if $\deg D = 2g$ (or even more negative if $\deg D > 2g$), so $\ell(K-D+P) = 0$ as well. Therefore $\ell(D) - \ell(D-P) = 1 > 0$.

[11] The condition at $P = Q$ is the tangent vector condition: sections of $\mathcal{L}(D)$ must separate tangent directions at P, which means $\ell(D) - \ell(D-2P) = 2$.

Using this, completeness and projectivity coincide for nonsingular curves.

Corollary 7.1.24: Very Ample Criterion for Curves

If $\deg D \geq 2g+1$, then $\mathcal{L}(D)$ is very ample, and $\varphi_{|D|}$ embeds X as a closed subvariety of $\mathbb{P}_k^{\ell(D)-1}$.

***Proof*:**
We must show that for any two closed points $P, Q \in X$ (possibly equal), $\ell(D) - \ell(D-P-Q) = 2$. By Riemann–Roch, this reduces to showing $\ell(K-D) = \ell(K-D+P+Q)$. Since $\deg D \geq 2g+1$, we have $\deg(K - D + P + Q) \leq (2g-2) - (2g+1) + 2 = -1 < 0$, so both terms vanish by lemma 7.1.10.

The upshot is clean: *for smooth curves, "proper" and "projective" are synonymous.* We shall use these terms interchangeably from now on.

These bounds are sharp: on a hyperelliptic curve of genus g, the divisor $D = g \cdot g^1_2$ (where g^1_2 is the degree-2 linear system) has degree $2g$ and is base-point-free but not very ample. We will study this in detail in §7.1.6. Note that the equivalence of projective and proper is very much a *curve* phenomenon—it *fails* in higher dimensions[12].

With this application of Riemann–Roch, we can seamlessly link the geometric, algebraic, and scheme-theoretic models of a curve into a single equivalency.

Corollary 7.1.25: Equivalence of Curve Models

Let X be a nonsingular curve over $k = \overline{k}$, with function field $K = K(X)$. Then the following are equivalent:

1. X is projective.
2. X is proper (complete).
3. For every DVR R of K/k, the valuation ring R dominates $\mathcal{O}_{X,x}$ for some (necessarily unique) closed point $x \in X$.

***Proof*:**
(1) $\Rightarrow$ (2) is standard since projective morphisms are proper (theorem 3.5.26). The converse (2) $\Rightarrow$ (1) is exactly **??** above.

The equivalence (2) $\Leftrightarrow$ (3) is simply the valuative criterion of properness (theorem 3.5.25) applied to the morphism $X \to \operatorname{Spec} k$. Given a DVR R of K/k, properness guarantees a unique lift $\operatorname{Spec} R \to X$ sending the closed point of $\operatorname{Spec} R$ to a closed point $x \in X$, which means R dominates $\mathcal{O}_{X,x}$. Conversely, since the local rings at closed points of a nonsingular curve are exactly the DVRs of K/k, the properness criterion is satisfied. In the language of classical algebraic geometry, X is the unique *complete nonsingular model* of the function field K.

[12]There exist complete nonprojective surfaces, the simplest being Hironaka's example (example 7.3.8) of a smooth complete threefold that is not projective. The completeness-projectivity gap is what makes proper morphisms interesting as a separate notion from projective ones; see remark 6.6.9.

Proposition 7.1.26: Genus Zero Means Rational

Let X be a complete nonsingular curve of genus $g = 0$ over $k = \overline{k}$. Then $X \cong \mathbb{P}^1_k$.

***Proof*:**
Let P be any closed point of X. By Riemann–Roch:

$$\ell(P) = \deg P + 1 - g + \ell(K - P) = 1 + 1 + \ell(K - P) \geq 2.$$

By proposition 7.1.21 $\deg K = -2$, so $\deg(K - P) = -3 < 0$, forcing $\ell(K - P) = 0$ by lemma 7.1.10, giving $\ell(P) = 2$ exactly.

Since $\ell(P) = 2$, the complete linear system $|P|$ is a 1-dimensional projective space, i.e. a pencil. The corresponding morphism $\varphi_{|P|} \colon X \to \mathbb{P}^1$ has degree $\deg P = 1$, hence it is an isomorphism (a degree-1 finite morphism of smooth curves is an isomorphism of function fields, hence an isomorphism of curves by corollary 7.1.25).

Thus $\mathbb{P}^1_k$ is the *unique* smooth projective curve of genus 0 over $\overline{k}$. Over non-algebraically-closed fields, the situation is far richer: a smooth projective curve of genus 0 over k is a *conic*—a degree-2 curve in $\mathbb{P}^2_k$—and it is isomorphic to $\mathbb{P}^1_k$ if and only if it has a k-rational point (i.e. a point with residue field k). Over $\mathbb{Q}$, for instance, the conic $x^2 + y^2 + z^2 = 0$ has genus 0 but is not isomorphic to $\mathbb{P}^1_{\mathbb{Q}}$ since it has no rational points.

Example 7.1.27: Embedding Low-Genus Curves

1. *Genus 0:* As shown in proposition 7.1.26, $X \cong \mathbb{P}^1$. Taking $D = n \cdot P$ for $n \geq 1$ gives $\ell(D) = n + 1$ and the morphism $\varphi_{|D|}$ maps $\mathbb{P}^1 \to \mathbb{P}^n$ as the rational normal curve of degree n (the Veronese embedding, example 3.1.39). For $n = 2$, this is a conic in $\mathbb{P}^2$.

2. *Genus 1:* Let E be a genus one (nonsingular) curve and $O \in E$ some chosen point. We have $\deg K = 0$[a] and $\omega_E \cong \mathcal{O}_E$. Taking $D = 3O$: $\ell(3O) = 3 + 1 - 1 = 3$ (non-special since $\deg(3O) = 3 > 0 = 2g - 2$), so $|3O|$ embeds E as a cubic in $\mathbb{P}^2$. This is the standard Weierstrass embedding, which we will discuss thoroughly in §7.1.7.

3. *Genus 2:* $\deg K = 2$ and $\ell(K) = g = 2$, so $|K|$ is a pencil (a linear system of dimension 1). The canonical map $\varphi_{|K|} \colon X \to \mathbb{P}^1$ is a degree-2 map. This shall tell us that every genus-2 curve is hyperelliptic! The first non-trivial embedding uses D with $\deg D \geq 5$; taking $D = 5P$ gives $\ell(D) = 4$, so $X \hookrightarrow \mathbb{P}^3$ as a curve of degree 5.

4. *Genus 3:* $\deg K = 4$ and $\ell(K) = 3$, so $|K|$ maps $X \to \mathbb{P}^2$. If X is non-hyperelliptic, this is an embedding: X is realized as a smooth quartic in $\mathbb{P}^2$ (of degree 4, which is consistent with the classical degree formula $g = \frac{(4-1)(4-2)}{2} = 3$). If X is hyperelliptic, $|K|$ factors through the $2:1$ map to $\mathbb{P}^1$ and maps X onto a conic in $\mathbb{P}^2$.

[a] Using the Gauss-Bonnet Theorem connection mentioned earlier, this say that such curves have *flat curvature*

The canonical linear system $|K|$ is the most natural linear system on a curve: it is defined intrinsically, without any choices. Let us study it in detail.

Theorem 7.1.28: The Canonical Map and Embedding

Let X be a complete nonsingular curve of genus $g \geq 2$. Then:

1. *Existence:* The canonical linear system $|K|$ is everywhere base-point-free. Consequently, it defines a morphism
$$\varphi_K \colon X \longrightarrow \mathbb{P}^{g-1}.$$

2. *The Generic Case (Embedding):* If φ_K is injective X, then φ_K is a closed immersion. In this case, X is embedded as a curve of degree $2g-2$ in $\mathbb{P}^{g-1}$. We call this the *canonical embedding* of X.

3. *The Hyperelliptic Case (Factoring):* If φ_K is *not* injective, φ_K factors as
$$X \xrightarrow{2:1} \mathbb{P}^1 \hookrightarrow \mathbb{P}^{g-1}$$
where the first map is called the *hyperelliptic double cover*, and the second is the Veronese embedding of degree $g-1$. Thus, φ_K maps X 2-to-1 onto a rational normal curve of degree $g-1$. Curves where φ_K is not injective are called *hyperelliptic* (studied footer in section 7.1.6).

***Proof*:**
(1) Geometrically, for $|K|$ to be base-point-free means that for every point $P \in X$, there is some global differential form that does not vanish at P. Algebraically, this means imposing a vanishing condition at P must strictly reduce the dimension of the linear system, so we must show $\ell(K) - \ell(K-P) = 1$.

By Riemann–Roch,
$$\ell(K-P) = \deg(K-P) + 1 - g + \ell(P) = (2g-3) + 1 - g + \ell(P) = g - 2 + \ell(P).$$

Because $g \geq 2$, the curve is not rational. Therefore, $\ell(P) = 1$ for *every* point P (if we had $\ell(P) \geq 2$, there would exist a non-constant function with only a simple pole, giving a degree-1 isomorphism $X \cong \mathbb{P}^1$, implying $g = 0$). Substituting $\ell(P) = 1$ back yields $\ell(K-P) = g-1$. Since $\ell(K) = g$, the dimension drops by exactly 1. The linear system has no base points, and the morphism φ_K cleanly maps X into $\mathbb{P}^{g-1}$.

(2) For φ_K to be a closed immersion, it must separate both points and tangent vectors. Algebraically, this requires that imposing *two* vanishing conditions drops the dimension of the canonical system by exactly 2. That is, we need $\ell(K) - \ell(K-P-Q) = 2$ for all pairs of points P, Q (where $P = Q$ corresponds to separating tangent vectors).

Using Riemann–Roch again on the divisor $P+Q$:
$$\ell(K-P-Q) = \deg(K-P-Q) + 1 - g + \ell(P+Q) = (2g-4) + 1 - g + \ell(P+Q) = g - 3 + \ell(P+Q).$$

Subtracting this from $\ell(K) = g$, we find the dimension drop is:
$$\ell(K) - \ell(K-P-Q) = g - (g - 3 + \ell(P+Q)) = 3 - \ell(P+Q).$$

For this drop to equal 2, we must have $\ell(P+Q) = 1$. This means the divisor $P+Q$ admits no extra global sections beyond the constant functions. Conversely, the condition $\ell(P+Q) \geq 2$ for some pair P, Q means there exists a non-constant function with at most two poles. This defines a degree-2 morphism $X \to \mathbb{P}^1$ (the linear system $|P+Q|$ is a g^1_2 pencil). Thus, φ_K successfully embeds X if and only if no such g^1_2 exists; in other words, if and only if X is not hyperelliptic.

(3) Suppose X is hyperelliptic, possessing a degree-2 map to $\mathbb{P}^1$ given by a linear system $g^1_2 = |P+Q|$. We want to show that the canonical system is entirely built out of this smaller system. Let us examine the linear system $|(g-1)(P+Q)|$.

Let x, y be a basis for $H^0(X, \mathcal{O}(P+Q))$. We can form g distinct monomials of degree $g-1$: $x^{g-1}, x^{g-2}y, \ldots, y^{g-1}$. These are linearly independent sections of $\mathcal{O}((g-1)(P+Q))$, which immediately tells us $\ell((g-1)(P+Q)) \geq g$.

Now we apply Riemann–Roch to this multiple of the g^1_2:

$$\ell((g-1)(P+Q)) - \ell(K-(g-1)(P+Q)) = \deg((g-1)(P+Q)) + 1 - g = (2g-2) + 1 - g = g - 1.$$

Because we proved $\ell((g-1)(P+Q)) \geq g$, the left side of the equation forces $\ell(K-(g-1)(P+Q)) \geq 1$. This implies there is an effective divisor linearly equivalent to $K-(g-1)(P+Q)$. However, note the degree:

$$\deg(K-(g-1)(P+Q)) = (2g-2) - (g-1)(2) = 0.$$

By lemma 7.1.10, an effective divisor of degree 0 must be the trivial divisor. Therefore, we have a strict linear equivalence: $K \sim (g-1)(P+Q)$.

Geometrically the punchline is that the canonical map φ_K is given by evaluating the sections of $|K|$. Since $K \sim (g-1)(P+Q)$, these sections are exactly the degree-$g-1$ monomials in the basis x, y of our g^1_2. Evaluating a map via all monomials of degree d is precisely the definition of the d-th Veronese embedding. Thus, the canonical map evaluates the g^1_2 map $X \to \mathbb{P}^1$, and then directly applies the $(g-1)$-fold Veronese embedding $\mathbb{P}^1 \hookrightarrow \mathbb{P}^{g-1}$.

The canonical embedding is one of the most elegant constructions in algebraic geometry. It embeds the curve "using its own differential forms"—no external choices are needed. This intrinsic nature makes the canonical embedding the right tool for studying the moduli of curves: it is through the canonical embedding that one proves, for instance, that $\mathcal{M}_g$ is of general type for $g \geq 24$ (a result of Harris–Mumford and Eisenbud–Harris).

One down-side of the canonical embedding is that the embedding space $\mathbb{P}^{g-1}$ can vary as the genus varies. We can actually embed all (complete nonsingular) curves into a single projective space:

Proposition 7.1.29: Every Curve Embeds in Projective 3-Space

Let X be a complete nonsingular curve over an algebraically closed field k. Then X is isomorphic to a closed subvariety of $\mathbb{P}^3_k$.

***Proof*:**
By corollary 7.1.24, we can choose a divisor D of degree $\deg D \geq 2g+1$, which guarantees that $\mathcal{L}(D)$ is very ample. This gives a closed immersion $\varphi_{|D|} \colon X \hookrightarrow \mathbb{P}^N_k$, where $N = \ell(D) - 1$. If $N \leq 3$, we can simply compose this map with a linear inclusion $\mathbb{P}^N_k \hookrightarrow \mathbb{P}^3_k$ and we are done.

Suppose $N > 3$. We wish to simplify the embedding by projecting X into a lower-dimensional space. Choose a point $O \in \mathbb{P}^N_k \setminus X$ and consider the linear projection away from O:

$$\pi_O \colon \mathbb{P}^N_k \setminus \{O\} \longrightarrow \mathbb{P}^{N-1}_k.$$

We must ensure that the restriction $\pi_O|_X$ remains a closed immersion. For this to hold, the projection must separate points and tangent vectors:

1. *Separating points:* π_O maps two distinct points $P, Q \in X$ to the same point in $\mathbb{P}^{N-1}_k$ if and only if the line $L_{P,Q}$ connecting them passes through the center of projection O.

2. *Separating tangent vectors:* π_O fails to be an immersion at a point $P \in X$ if and only if the tangent line $T_P X$ to X at P passes through O.

To avoid these failures, we must pick O such that it avoids all secant lines and tangent lines of X. Let $\mathrm{Sec}(X)$ be the *secant variety* of X, defined as the closure of the union of all secant lines $L_{P,Q}$ for $P, Q \in X$. Since a secant line is parameterized by pairs of points $(P, Q) \in X \times X$, we can bound the dimension of the secant variety geometrically:

$$\dim \mathrm{Sec}(X) \;\leq\; \dim(X \times X) + 1 \;=\; 1 + 1 + 1 \;=\; 3.$$

(Note that tangent lines appear naturally in this locus as limits of secant lines as $Q \to P$).

Because $N \geq 4$ and our base field k is algebraically closed (and thus infinite), the ambient space $\mathbb{P}^N_k$ strictly contains the 3-dimensional variety $\mathrm{Sec}(X)$. We can therefore find a generic point $O \in \mathbb{P}^N_k \setminus \mathrm{Sec}(X)$. Projecting from this O yields a new closed immersion $X \hookrightarrow \mathbb{P}^{N-1}_k$.

We may proceed inductively. As long as the dimension of the ambient projective space is strictly greater than 3, we can always find a point outside the secant variety and project down. The process terminates when the ambient dimension reaches 3, leaving us with a closed immersion $X \hookrightarrow \mathbb{P}^3_k$.

This result confirms that $\mathbb{P}^3_k$ is large enough to house *any* smooth algebraic curve. By contrast, not every curve can be embedded in $\mathbb{P}^2_k$ (a curve in $\mathbb{P}^2_k$ of degree d has a strictly determined genus $g = \frac{(d-1)(d-2)}{2}$ (a consequence of theorem 7.1.36), and even allowing singularities, the normalization of a plane curve cannot achieve arbitrary topological types). Thus, $\mathbb{P}^3_k$ is the natural "universal" target for general curve embeddings, analogous to how the Whitney embedding theorem guarantees that any 1-dimensional smooth manifold embeds in $\mathbb{R}^3$ (and in this case we can even take $\mathbb{R}^2$ requiring $n = 2$ due to the circle)

We conclude this section with Clifford's theorem, which controls the behavior of *special* divisors.

Theorem 7.1.30: Clifford's Theorem

Let D be an effective divisor on a complete nonsingular curve X of genus $g \geq 1$ with $0 \leq \deg D \leq 2g - 2$ and $\ell(K - D) > 0$ (i.e. D is *special*). Then:

$$\ell(D) - 1 \;\leq\; \frac{\deg D}{2}.$$

Equality holds if and only if $D = 0$, $D = K$, or X is hyperelliptic and D is a multiple of the g^1_2.

***Proof*:**
The key idea is to use the tensor product $\mathcal{L}(D) \otimes \mathcal{L}(K-D) \cong \mathcal{L}(K)$ and bound $\ell(D) \cdot \ell(K-D)$ against $\ell(K)$ using the multiplication map

$$\mu\colon H^0(X, \mathcal{L}(D)) \otimes H^0(X, \mathcal{L}(K-D)) \longrightarrow H^0(X, \mathcal{L}(K)).$$

The image of μ has dimension at least $\ell(D) + \ell(K-D) - 1$: this is because given nonzero subspaces $V \subseteq K(X)$ and $W \subseteq K(X)$ of dimensions a and b respectively, the product space $V \cdot W$ has dimension at least $a + b - 1$ (fix a nonzero $w_0 \in W$; then $V \cdot w_0$ has dimension a, and each additional basis element of W contributes at least one new dimension). Since the image lies in $H^0(X, \omega_X)$, which has dimension g:

$$\ell(D) + \ell(K-D) - 1 \;\leq\; g.$$

Now apply Riemann–Roch: $\ell(D) = \deg D + 1 - g + \ell(K-D)$, so $\ell(K-D) = \ell(D) - \deg D - 1 + g$. Substituting:

$$\ell(D) + (\ell(D) - \deg D - 1 + g) - 1 \leq g \quad\Longrightarrow\quad 2\ell(D) \leq \deg D + 2 \quad\Longrightarrow\quad \ell(D) - 1 \leq \frac{\deg D}{2}.$$

The characterization of equality is more delicate and we leave it as a guided exercise[a].

[a] When equality holds, the multiplication map μ must be surjective. This forces a very rigid structure on $\mathcal{L}(D)$: either it is trivial ($D = 0$), canonical ($D = K$), or the curve is hyperelliptic and D is a multiple of the g^1_2. A clean proof can be found in [16, §IV.5].

Clifford's theorem is the "other side" of Riemann–Roch: while Riemann–Roch gives the *exact* value of $\ell(D) - \ell(K-D)$, Clifford controls $\ell(D)$ itself when the divisor is special. Together, they provide a nearly complete understanding of the function $D \mapsto \ell(D)$ on $\mathrm{Div}X$.

7.1.31 Forward Reference: Petri's Theorem and Syzygies The canonical embedding of a non-hyperelliptic curve $X \hookrightarrow \mathbb{P}^{g-1}$ opens the door to a classical line of inquiry: what are the *equations* that define the image? Petri's theorem says that for "most" curves, the ideal of X in $\mathbb{P}^{g-1}$ is generated by quadrics[a]. More precisely, the homogeneous ideal is generated in degree 2 unless X is trigonal (admits a degree-3 map to $\mathbb{P}^1$) or is a smooth plane quintic ($g = 6$).
This leads to the study of *syzygies* of canonical curves, culminating in Green's conjecture (now a theorem, proved by Voisin for general curves): the Betti numbers of the minimal free resolution of the canonical ideal detect the *Clifford index* of the curve. This is a place where the theory of *derived categories* (a separate volume) meets classical curve theory, and provides some of the most important applications of homological algebra to concrete geometry.

[a] In algebraic geometry, a *quadric* refers to the geometric hypersurface cut out by a single polynomial equation of degree 2 (a *quadratic* polynomial). To say an ideal is "generated by quadrics" means it is generated by homogeneous polynomials of degree 2.

7.1.4 Hurwitz's Theorem and Ramification

We now turn to the second central theorem of the classical theory of curves. If Riemann–Roch relates the *intrinsic* invariants of a curve (genus, divisors, sections), then Hurwitz's theorem relates the *extrinsic* geometry of a finite morphism $f: X \to Y$ to the genera of X and Y.

The intuition is topological. Over $\mathbb{C}$, a finite morphism of smooth projective curves is a branched covering of compact Riemann surfaces. Away from finitely many *branch points*, the map is a genuine covering space with $n = \deg f$ sheets. At a branch point, some of the sheets "come together"—this is ramification. The Euler characteristic (equivalently, genus) of X is determined by the Euler characteristic of Y, the degree of f, and the amount of ramification. Hurwitz's formula makes this precise in the algebraic setting, working over an arbitrary algebraically closed field.

Ramification

Throughout this section, $f: X \to Y$ is a finite morphism of complete nonsingular curves over $k = \overline{k}$, with $n = \deg f = [K(X) : K(Y)]$.

Recall that every local ring $\mathcal{O}_{X,x}$ (resp. $\mathcal{O}_{Y,y}$) at a closed point of a smooth curve is a DVR with discrete valuation ν_x (resp. ν_y). For a point $x \in X$ with $y = f(x)$, let $t \in \mathcal{O}_{Y,y}$ be a uniformizer (i.e. $\nu_y(t) = 1$). The local map $f^\#: \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ sends t to an element of $\mathcal{O}_{X,x}$, and we defined the ramification index as $e_x = \nu_x(f^\#(t))$. Let us now give this a more careful treatment.

Definition 7.1.32: Ramification Index

Let $f: X \to Y$ be a finite morphism of complete nonsingular curves, $x \in X$ a closed point, and $y = f(x)$. The *ramification index* of f at x is:

$$e_x \;=\; \nu_x(f^\#(t))$$

where t is any uniformizer of $\mathcal{O}_{Y,y}$. This is independent of the choice of t. We say:

1. f is *unramified* at x if $e_x = 1$.
2. f is *ramified* at x if $e_x > 1$. The point $y = f(x)$ is then called a *branch point*.
3. If $\operatorname{char} k = p > 0$, the ramification is *tame* if $p \nmid e_x$, and *wild* if $p \big| e_x$.

The well-definedness is immediate: if $t' = ut$ for a unit $u \in \mathcal{O}_{Y,y}^*$, then $f^\#(u)$ is a unit in $\mathcal{O}_{X,x}$ (local maps send units to units), so $\nu_x(f^\#(t')) = \nu_x(f^\#(u)) + \nu_x(f^\#(t)) = 0 + e_x$. Geometrically, e_x measures the "order of contact" between the fiber $f^{-1}(y)$ and the point x: if s is a uniformizer of $\mathcal{O}_{X,x}$, then $f^\#(t) = u \cdot s^{e_x}$ for a unit u, so the map locally looks like $s \mapsto s^{e_x}$ up to units.

Example 7.1.33: Ramification in Practice

1. *The squaring map.* Consider $f: \mathbb{P}^1 \to \mathbb{P}^1$ given by $[x : y] \mapsto [x^2 : y^2]$, which in the affine chart $y \neq 0$ is $t \mapsto t^2$. This has degree 2. At $t = 0$: the uniformizer of the target is t, and $f^\#(t) = t^2$, so $e_0 = 2$ (ramified). Similarly at $t = \infty$ (using the chart $x \neq 0$, the map is $s \mapsto s^2$ in the uniformizer $s = 1/t$), so $e_\infty = 2$. At every other point $t = a \neq 0, \infty$: the uniformizer of the target at a^2 is $t - a^2$, and $f^\#(t - a^2) = t^2 - a^2 = (t-a)(t+a)$. For the point $x = a$, $\nu_a((t-a)(t+a)) = 1$ (since $t + a$ is a unit at $t = a$ when $\operatorname{char} k \neq 2$), so $e_a = 1$ (unramified). The fiber over a^2 consists of two points $\{a, -a\}$, each with $e = 1$, confirming $\sum e_x = 2 = \deg f$.
2. *Wild ramification.* Let k have characteristic $p > 0$ and consider $f: \mathbb{A}^1 \to \mathbb{A}^1$ given by

$t \mapsto t^p - t$ (the Artin–Schreier map). At $t = 0$: $f^\#(s) = t^p - t$ where s is the uniformizer of the target at $f(0) = 0$. We compute $\nu_0(t^p - t) = 1$ (the lowest-order term is $-t$), so $e_0 = 1$—the map is unramified at the origin! In fact, the Artin–Schreier map is étale everywhere on $\mathbb{A}^1$, yet it is a nontrivial cover: the fiber over every point has p elements. This phenomenon—unramified covers that are not trivial—does not occur in the topological category (where $\pi_1(\mathbb{A}^1_{\mathbb{C}}) = 0$), and is a distinctive feature of positive characteristic.

The Ramification Divisor via Differentials

A key way to understand ramification is through the sheaf of differentials. The morphism $f \colon X \to Y$ induces, via pullback of Kähler differentials (proposition 5.5.16), a natural map of sheaves on X:

$$f^*\Omega_{Y/k} \longrightarrow \Omega_{X/k}. \tag{7.3}$$

Since both $\Omega_{Y/k}$ and $\Omega_{X/k}$ are invertible sheaves on smooth curves, this map is an injection of invertible sheaves[13] (of $\mathcal{O}_X$-modules), hence corresponds to an effective divisor: the cokernel is a torsion sheaf supported at finitely many points. Let us make this precise.

In terms of local coordinates: at a point $x \in X$ with $y = f(x)$, let s be a uniformizer of $\mathcal{O}_{X,x}$ and t a uniformizer of $\mathcal{O}_{Y,y}$. Then $f^\#(t) = u \cdot s^{e_x}$ for a unit $u \in \mathcal{O}^*_{X,x}$, and:

$$f^*(dt) \;=\; d(f^\#(t)) \;=\; d(u \cdot s^{e_x}) \;=\; (u' s^{e_x} + e_x u s^{e_x - 1})\, ds \;=\; s^{e_x - 1}(u's + e_x u)\, ds.$$

The factor s^{e_x-1} vanishes to order $e_x - 1$ at x. The factor $(u's + e_x u)$ is a unit at x *provided* $e_x \neq 0$ in k, i.e. provided the ramification at x is tame (or we are in characteristic zero). In the case of tame ramification, the map (7.3) vanishes to order exactly $e_x - 1$ at x.

Definition 7.1.34: Ramification and Branch Divisors

The *ramification divisor* of f is:

$$R \;=\; \sum_{x \in X_{\mathrm{cl}}} r_x \cdot x$$

where r_x is defined by the relation $f^*\Omega_{Y/k} \cong \Omega_{X/k} \otimes \mathcal{L}(-R)$, or equivalently, $r_x = \nu_x(f^*(dt)/ds)$ for local uniformizers. The *branch divisor* is $B = f_*(R) \in \operatorname{WDiv} Y$.
When the ramification is tame (in particular, always in characteristic 0):

$$r_x = e_x - 1.$$

When the ramification is wild ($\operatorname{char} k = p > 0$ and $p \,\big|\, e_x$), we have $r_x \geq e_x$, and the precise value depends on higher-order ramification data.

The relation $f^*\Omega_{Y/k} \cong \Omega_{X/k}(-R)$ can be rewritten in terms of canonical divisors. Since $\Omega_{Y/k} \cong \mathcal{L}(K_Y)$ and $\Omega_{X/k} \cong \mathcal{L}(K_X)$, and $f^*\mathcal{L}(K_Y) \cong \mathcal{L}(f^*K_Y)$, we obtain the fundamental *divisor relation*:

$$K_X \;\sim\; f^*K_Y + R. \tag{7.4}$$

This is the key equation from which Hurwitz's formula follows by taking degrees.

[13] It is injective because f is dominant and separable (we are assuming $\operatorname{char} k = 0$ or that the extension $K(X)/K(Y)$ is separable; see Remark 7.1.35), so the induced map $K(Y) \otimes_{K(Y)} \Omega_{K(Y)/k} \to \Omega_{K(X)/k}$ on generic fibers is injective.

7.1.35 Remark: Inseparable Maps If the field extension $K(X)/K(Y)$ is inseparable (which can only happen in characteristic $p > 0$), the map $f^*\Omega_{Y/k} \to \Omega_{X/k}$ is the *zero map*, and the preceding discussion breaks down entirely. The simplest example is the Frobenius morphism $\mathrm{Frob}\colon X \to X^{(p)}$, which is purely inseparable of degree p and for which $\mathrm{Frob}^*(\Omega_{X^{(p)}/k}) = 0$. In this book, we shall always assume our finite morphisms of curves are *separable* (i.e. $K(X)/K(Y)$ is a separable field extension) when discussing Hurwitz's formula. The inseparable case requires a more delicate analysis using Hasse–Schmidt derivations or the algebraic trace form to define the different ideal, which goes beyond our scope here; see [2] for the number field analogue.

Hurwitz's Formula

Theorem 7.1.36: Hurwitz's Formula

Let $f\colon X \to Y$ be a separable finite morphism of complete nonsingular curves of genera g_X and g_Y respectively, with $n = \deg f$. Then:

$$2g_X - 2 \;=\; n(2g_Y - 2) \;+\; \deg R$$

where $R = \sum r_x \cdot x$ is the ramification divisor. In the tamely ramified case (in particular, whenever $\mathrm{char} k = 0$):

$$2g_X - 2 \;=\; n(2g_Y - 2) \;+\; \sum_{x \in X_{\mathrm{cl}}} (e_x - 1).$$

***Proof*:**
Taking degrees of both sides of (7.4):

$$\deg K_X = \deg(f^*K_Y) + \deg R.$$

By proposition 7.1.21, $\deg K_X = 2g_X - 2$ and $\deg K_Y = 2g_Y - 2$. By the degree formula for pullbacks (proposition 7.1.6), $\deg(f^*K_Y) = n \cdot \deg K_Y = n(2g_Y - 2)$. Therefore:

$$2g_X - 2 = n(2g_Y - 2) + \deg R.$$

The second formula follows since $r_x = e_x - 1$ in the tame case.

The proof is almost embarrassingly short—the hard work was establishing the relation $K_X \sim f^*K_Y + R$ in §7.1.4, which itself rests on the theory of Kähler differentials from §5.5. This is an illustration of the scheme-theoretic philosophy: the right definitions make the proofs almost automatic.

Let us also give an alternative argument via Euler characteristics that avoids explicit mention of the ramification divisor, as it gives additional insight and generalizes more readily.

***Proof Alternative proof via Euler characteristics*:**
Consider the short exact sequence on X:

$$0 \to f^*\Omega_{Y/k} \to \Omega_{X/k} \to \Omega_{X/Y} \to 0$$

coming from the cotangent exact sequence (proposition 5.5.16) applied to the composition $X \xrightarrow{f} Y \to \operatorname{Spec} k$. Taking Euler characteristics:

$$\chi(\Omega_{X/k}) = \chi(f^*\Omega_{Y/k}) + \chi(\Omega_{X/Y}).$$

Now $\Omega_{X/k} = \omega_X \cong \mathcal{L}(K_X)$, and by Riemann–Roch (theorem 7.1.18):

$$\chi(\omega_X) = \deg K_X + 1 - g_X = (2g_X - 2) + 1 - g_X = g_X - 1.$$

For the pullback: $f^*\Omega_{Y/k} \cong f^*\omega_Y \cong \mathcal{L}(f^*K_Y)$, and

$$\chi(f^*\omega_Y) = \deg(f^*K_Y) + 1 - g_X = n(2g_Y - 2) + 1 - g_X.$$

The relative differentials $\Omega_{X/Y}$ are a torsion sheaf on X (supported at the ramification locus), hence $H^1(X, \Omega_{X/Y}) = 0$ (a torsion sheaf on a curve has no higher cohomology), so $\chi(\Omega_{X/Y}) = \dim_k H^0(X, \Omega_{X/Y}) = \deg R$. Putting it all together:

$$g_X - 1 = n(2g_Y - 2) + 1 - g_X + \deg R$$

which rearranges to $2g_X - 2 = n(2g_Y - 2) + \deg R$.

Example 7.1.37: Hurwitz in Action

1. *Double covers of* $\mathbb{P}^1$. Let $f\colon X \to \mathbb{P}^1$ be a degree-2 separable map in characteristic $\neq 2$. Then $n = 2$, $g_Y = 0$, and all ramification is tame with $e_x \in \{1, 2\}$. Let b be the number of branch points in $\mathbb{P}^1$ (since $n = 2$, each corresponds to exactly one ramification point in X where $e_x = 2$). Hurwitz gives:

$$2g_X - 2 = 2(0 - 2) + b = -4 + b, \qquad \text{so } g_X = \frac{b-2}{2}.$$

   Since g_X must be a non-negative integer, b must be even and $b \geq 2$. For $b = 2$: $g_X = 0$ (the map $[x : y] \mapsto [x^2 : y^2]$). For $b = 4$: $g_X = 1$ (elliptic curves). For $b = 6$: $g_X = 2$ (genus-2 curves, which are all hyperelliptic). More generally, a *hyperelliptic curve* of genus g is a double cover of $\mathbb{P}^1$ with $2g + 2$ branch points.

2. *Cyclic covers.* Let $f\colon X \to \mathbb{P}^1$ have degree n with total ramification ($e_x = n$) over exactly b points and no other ramification. Then:

$$2g_X - 2 = n(-2) + b(n-1), \quad \text{so} \quad g_X = \frac{(n-1)(b-2)}{2}.$$

   For $n = 3$, $b = 4$: $g_X = 2$ (a genus-2 curve). For $n = 3$, $b = 5$: $g_X = 3$ (a trigonal curve of genus 3). For $n = 3$, $b = 3$: $g_X = 1$ (an elliptic curve with an order-3 automorphism).

3. *Genus bound.* If $g_Y \geq 1$ and $n \geq 2$, then $2g_X - 2 \geq n(2g_Y - 2) + 0$, so $g_X \geq ng_Y - n + 1 \geq g_Y$. In particular, there are no nonconstant maps from a curve of genus g to a curve of genus g' where $g' > g$. This is a “topological obstruction” to the existence of maps.

7.1.5 Applications of Hurwitz's Theorem

One of the most striking applications of Hurwitz's formula is the finiteness—and in fact an explicit bound on the size—of the automorphism group of a curve of genus ≥ 2.

Theorem 7.1.38: Hurwitz's Automorphism Bound

Let X be a complete nonsingular curve of genus $g \geq 2$ over k with char$k = 0$ (or more generally, char$k \nmid |\mathrm{Aut}(X)|$). Then:

$$|\mathrm{Aut}(X)| \;\leq\; 84(g-1).$$

This bound is sharp: the Klein quartic $x^3y + y^3z + z^3x = 0$ in $\mathbb{P}^2$ has genus 3 and $|\mathrm{Aut}| = 168 = 84 \cdot 2$, achieving equality.

***Proof Sketch of proof*:**
Let $G = \mathrm{Aut}(X)$, which acts on X faithfully. The quotient $Y = X/G$ is again a smooth projective curve[a], and the quotient map $\pi\colon X \to Y$ is a finite morphism of degree $|G|$.

Apply Hurwitz's formula to π:

$$2g - 2 = |G| \cdot (2g_Y - 2) + \sum_{x \in X_{\mathrm{cl}}} (e_x - 1).$$

Since G acts transitively on each fiber, the ramification data depends only on the branch points $y_1, \ldots, y_s \in Y$ and their stabilizer orders $e_1, \ldots, e_s$ (with $e_i \geq 2$). The sum becomes $|G| \cdot \sum_{i=1}^{s}(1 - 1/e_i)$, and we get:

$$\frac{2g-2}{|G|} = 2g_Y - 2 + \sum_{i=1}^{s}\left(1 - \frac{1}{e_i}\right). \tag{7.5}$$

The left side is positive (since $g \geq 2$), so the right side must be positive. Call the right side μ. Since each $1 - 1/e_i \geq 1/2$, we have $\mu \geq 2g_Y - 2 + s/2$. For $\mu > 0$, the minimum occurs at $g_Y = 0$, $s = 3$, $e_1 = e_2 = e_3 = 2$... but this gives $\mu = -2 + 3/2 < 0$. A careful case analysis shows the minimum positive value of μ is:

$$\mu_{\min} = \frac{1}{42}, \qquad \text{achieved at } g_Y = 0,\; s = 3,\; (e_1, e_2, e_3) = (2, 3, 7).$$

Therefore $|G| = (2g-2)/\mu \leq 42(2g-2) = 84(g-1)$.

[a] Smoothness of the quotient follows from the fact that stabilizer groups at points are cyclic (being finite subgroups of $\mathrm{Aut}(\mathcal{O}_{X,x}) \cong k^*$ acting on the tangent space) and from the theory of quotients by finite groups; see proposition 4.4.2.

The proof demonstrates the power of combinatorics constrained by Hurwitz's formula. The special role of the triple $(2, 3, 7)$ reflects the fact that the $(2, 3, 7)$-triangle group is the lattice in $\mathrm{PSL}_2(\mathbb{R})$ of smallest co-area—a key connection between algebraic curves, hyperbolic geometry, and number theory.

Example 7.1.39: Automorphism Groups in Low Genus

1. *Genus* 0*:* $\mathrm{Aut}(\mathbb{P}^1_k) = \mathrm{PGL}_2(k)$, which is infinite (and 3-dimensional as an algebraic group). Hurwitz's bound does not apply since $g < 2$.

2. *Genus* 1*:* If E is an elliptic curve with origin O, then $\mathrm{Aut}(E)$ (as a pointed curve, i.e. automorphisms fixing O) is finite but can be nontrivial. Over $\mathbb{C}$: generically $\mathrm{Aut}(E, O) \cong \mathbb{Z}/2\mathbb{Z}$ (the inversion $P \mapsto -P$), but for special j-invariants $j = 0$ (resp. $j = 1728$) the automorphism group is $\mathbb{Z}/6\mathbb{Z}$ (resp. $\mathbb{Z}/4\mathbb{Z}$), corresponding to the extra symmetries of the hexagonal (resp. square) lattice.

   However, as a group scheme, E also has the translation automorphisms $\tau_P : Q \mapsto Q + P$ for every $P \in E(k)$, so the *full* automorphism group $\mathrm{Aut}(E)$ is an extension of $\mathrm{Aut}(E, O)$ by $E(k)$, which is again infinite.

3. *Genus* 2*:* Every genus-2 curve is hyperelliptic, so it comes with an involution ι (the hyperelliptic involution, definition 7.1.44), giving $|\mathrm{Aut}(X)| \geq 2$. The Hurwitz bound gives $|\mathrm{Aut}(X)| \leq 84$. The actual maximum for $g = 2$ is achieved by the Bolza curve $y^2 = x^5 - x$, which has $|\mathrm{Aut}| = 48$ over $\mathbb{C}$ (so the theoretical bound of 84 is not tight for $g = 2$).

A finite morphism $f : X \to Y$ of smooth curves is *unramified* if $e_x = 1$ for all $x \in X$, or equivalently if the ramification divisor $R = 0$, or equivalently if $\Omega_{X/Y} = 0$. Since f is automatically flat (a finite morphism between smooth curves over a field is flat[14]), an unramified finite morphism is *étale*.

When $R = 0$, Hurwitz's formula simplifies dramatically:

$$2g_X - 2 = n(2g_Y - 2).$$

Corollary 7.1.40: Étale Covers and Genus

Let $f : X \to Y$ be a finite étale morphism of complete nonsingular curves of degree n.

1. If $g_Y = 0$: then $2g_X - 2 = -2n < 0$, which is impossible for $n \geq 2$ (since $g_X \geq 0$). So $\mathbb{P}^1$ has no nontrivial finite étale covers: $\mathbb{P}^1$ is *simply connected* in the algebraic sense.

2. If $g_Y = 1$: then $g_X = 1$ as well. Étale covers of elliptic curves are elliptic curves (for instance, the multiplication-by-n map $[n] : E \to E$ is étale when $\mathrm{char}\, k \nmid n$).

3. If $g_Y \geq 2$: then $g_X = n(g_Y - 1) + 1 > g_Y$. The cover has strictly larger genus.

***Proof*:**
Immediate from Hurwitz with $\deg R = 0$.

Part (1) is a purely algebraic result that mirrors the topological fact $\pi_1(\mathbb{P}^1_{\mathbb{C}}) = 0$ (the Riemann sphere is simply connected). Part (2) reflects $\pi_1(E_{\mathbb{C}}) \cong \mathbb{Z}^2$, which has many finite quotients.

[14] Flatness follows from the fact that a finite torsion-free module over a Dedekind domain is flat; or from the "miracle flatness" theorem in dimension 1.

7.1.41 Forward Reference: The Étale Fundamental Group The étale fundamental group $\pi_1^{\text{ét}}(X, \overline{x})$ is defined as the profinite group classifying finite étale covers of X (see box 3.1.19 for an earlier preview). Over $\mathbb{C}$, GAGA (theorem 6.9.3) implies that the étale fundamental group of a smooth projective variety is the *profinite completion* of the topological fundamental group:

$$\pi_1^{\text{ét}}(X_{\mathbb{C}}) \;\cong\; \widehat{\pi_1^{\text{top}}(X(\mathbb{C}))}.$$

For a curve of genus g, π_1^{top} is the surface group with $2g$ generators and one relation, and its profinite completion is an enormously rich object. Over fields of positive characteristic or arithmetic base schemes, the étale fundamental group captures Galois-theoretic information and is one of the central objects of arithmetic geometry. This story will be developed in a separate volume.

To connect Hurwitz's theorem to number theory, this theorem has one of the most important arithmetic shadows. Let L/K be a finite extension of number fields with $[L : K] = n$, and consider the associated map of spectra of rings of integers:

$$\pi \colon \operatorname{Spec} \mathcal{O}_L \longrightarrow \operatorname{Spec} \mathcal{O}_K.$$

Although Spec $\mathcal{O}_K$ is not a "curve" in our sense (it is a scheme of finite type over Spec $\mathbb{Z}$, not over a field, and its "genus" is not defined in the usual way), the *structure* of the Hurwitz formula carries over perfectly. For each prime $\mathfrak{p}$ of $\mathcal{O}_K$, the fiber $\pi^{-1}(\mathfrak{p})$ consists of primes $\mathfrak{q}_1, \ldots, \mathfrak{q}_r$ of $\mathcal{O}_L$ with ramification indices e_i and residue field degrees f_i satisfying the fundamental identity:

$$\sum_{i=1}^{r} e_i f_i \;=\; n.$$

The *different ideal* $\mathfrak{d}_{L/K}$ is the analogue of the ramification divisor R: it is an ideal of $\mathcal{O}_L$ that measures the failure of $\Omega_{\mathcal{O}_L/\mathcal{O}_K}$ to vanish, and its norm is the *discriminant ideal* $\Delta_{L/K}$, an ideal of $\mathcal{O}_K$ corresponding to the pushforward branch divisor $B = \pi_*(R)$.

The arithmetic analogue of the branch divisor equation is the *discriminant formula*:

$$\Delta_{L/K} \;=\; \prod_{\mathfrak{p} \text{ ramified}} \mathfrak{p}^{\sum_{i=1}^{r} f_i(e_i - 1 + \text{wild terms})}.$$

In the tamely ramified case, the exponent of $\mathfrak{q}_i$ in the different is $e_i - 1$. Taking the norm down to K multiplies this exponent by the residue field degree f_i, so the exponent at $\mathfrak{p}$ is exactly $\sum f_i(e_i - 1)$, perfectly paralleling the degree of the geometric branch divisor over a point. In the wildly ramified case, the exponent is larger, just as in the geometric setting[15].

This analogy is the starting point for one of the most fruitful ideas in modern mathematics: the *arithmetic–geometric dictionary*, in which number fields play the role of function fields of curves over finite fields, primes play the role of points, and the discriminant plays the role of the branch divisor. Many of the most important theorems and conjectures in number theory—including the Riemann hypothesis for curves over finite fields (proved by Weil), the Langlands program, and the Birch–Swinnerton-Dyer conjecture—can be motivated by pushing this analogy as far as it will go.

[15]This parallel is not a coincidence: both formulas arise from the same algebraic fact about differentials in extensions of Dedekind domains. The "different" in number theory and the "ramification divisor" in geometry are the same mathematical object, viewed from different perspectives. See [2] for a comprehensive treatment.

7.1.6 Hyperelliptic Curves

In the previous section we saw that the canonical map $\varphi_K\colon X \to \mathbb{P}^{g-1}$ is either an embedding (when X is not hyperelliptic) or a 2-to-1 map onto a rational normal curve (when X is hyperelliptic). The hyperelliptic curves are thus the "exceptional case" in the canonical embedding theorem, and they deserve special attention. They are also the curves that arise most concretely in practice: every hyperelliptic curve admits an explicit equation of the form $y^2 = f(x)$, and much of the classical theory of curves was developed by studying them.

Definition and Characterization

Definition 7.1.42: Hyperelliptic Curve

A complete nonsingular curve X of genus $g \geq 2$ over $k = \overline{k}$ is *hyperelliptic* if there exists a finite morphism $\varphi\colon X \to \mathbb{P}^1$ of degree 2.

We require $g \geq 2$ in the definition to exclude $\mathbb{P}^1$ (genus 0, which trivially maps to itself by degree 2) and elliptic curves (genus 1, which always admit degree-2 maps to $\mathbb{P}^1$ but are better studied in their own right in §7.1.7).

The degree-2 map φ, when it exists, is unique up to automorphisms of $\mathbb{P}^1$:

Proposition 7.1.43: Characterizations of Hyperelliptic Curves

Let X be a complete nonsingular curve of genus $g \geq 2$ over $k = \overline{k}$. The following are equivalent:

1. X is hyperelliptic (there exists a degree-2 morphism $X \to \mathbb{P}^1$).
2. There exist two closed points $P, Q \in X$ (possibly equal) with $\ell(P + Q) \geq 2$ (i.e. X admits a g^1_2—a linear system of degree 2 and dimension 1).
3. The canonical linear system $|K_X|$ is *not* very ample.
4. The canonical map $\varphi_K\colon X \to \mathbb{P}^{g-1}$ is not a closed immersion.

Moreover, when these conditions hold, the g^1_2 is unique: any two degree-2 maps $X \to \mathbb{P}^1$ differ by an automorphism of $\mathbb{P}^1$.

***Proof*:**
(1) $\Rightarrow$ (2): A degree-2 morphism $\varphi\colon X \to \mathbb{P}^1$ corresponds to a base-point-free linear system of degree 2 and dimension 1. Taking $P + Q = \varphi^*(y)$ for some $y \in \mathbb{P}^1$, we get $\ell(P + Q) \geq 2$.

(2) $\Rightarrow$ (1): If $\ell(P + Q) \geq 2$, the linear system $|P + Q|$ is at least a pencil, giving a morphism $\varphi\colon X \to \mathbb{P}^1$ of degree $\deg(P + Q) = 2$.

(1) $\iff$ (3): This was shown in the proof of theorem 7.1.28: $|K|$ fails to be very ample precisely when there exist P, Q (possibly $P = Q$) with $\ell(P + Q) \geq 2$, which is exactly the existence of a g^1_2.

(3) $\Longleftrightarrow$ (4): A line bundle is very ample if and only if the associated morphism is a closed immersion.

Uniqueness of the g_2^1: Suppose there are two such linear systems, giving two degree-2 morphisms $f, g\colon X \to \mathbb{P}^1$. If they are not equivalent by an automorphism of $\mathbb{P}^1$, the product map $(f, g)\colon X \to \mathbb{P}^1 \times \mathbb{P}^1$ has an image C which is a curve of bidegree (a, b). The composition of the map $X \to C$ with the projections gives f and g, which both have degree 2, so $a \leq 2$ and $b \leq 2$. Since f and g are distinct, C is not a graph, forcing $a = b = 2$ and the map $X \to C$ to be birational. But the arithmetic genus of a $(2, 2)$ curve on $\mathbb{P}^1 \times \mathbb{P}^1$ is $(2-1)(2-1) = 1$. This implies $g(X) \leq 1$, contradicting $g \geq 2$. Thus the g_2^1 is unique.

The degree-2 map $\varphi\colon X \to \mathbb{P}^1$ is a separable finite morphism (assuming $\operatorname{char} k \neq 2$; we will address characteristic 2 briefly at the end). Over the generic point of $\mathbb{P}^1$, the fiber consists of two distinct points, and so $K(X)/K(\mathbb{P}^1) = K(X)/k(t)$ is a separable quadratic extension. This extension has a unique nontrivial automorphism: the *hyperelliptic involution*.

Definition 7.1.44: Hyperelliptic Involution

Let X be a hyperelliptic curve with g_2^1 given by $\varphi\colon X \to \mathbb{P}^1$. The *hyperelliptic involution* is the unique nontrivial automorphism $\iota\colon X \to X$ such that $\varphi \circ \iota = \varphi$. Equivalently, ι is the nontrivial element of $\operatorname{Gal}(K(X)/k(t)) \cong \mathbb{Z}/2\mathbb{Z}$.

For each point $y \in \mathbb{P}^1$, the fiber $\varphi^{-1}(y)$ is either two distinct points $\{x, \iota(x)\}$ (when $e_x = 1$) or a single point fixed by ι (when $e_x = 2$, a ramification point). The fixed points of ι are precisely the ramification points of φ, which we call the *Weierstrass points* of X.

By Hurwitz's formula (theorem 7.1.36):

$$2g - 2 = 2(0-2) + \sum (e_x - 1) = -4 + b$$

where b is the number of ramification points. Therefore:

$$b = 2g + 2.$$

A hyperelliptic curve of genus g has exactly $2g+2$ Weierstrass points. These are the fixed points of ι.

The hyperelliptic structure gives an explicit affine model. If $\operatorname{char} k \neq 2$, the quadratic extension $K(X)/k(t)$ is generated by an element y with $y^2 = f(t)$ for some polynomial $f \in k[t]$. Since X is nonsingular, f must have no repeated roots. The degree of f is determined by the genus:

Proposition 7.1.45: Weierstrass Models

Let X be a hyperelliptic curve of genus g over k with char$k \neq 2$. Then X is birational to the affine curve:

$$y^2 = f(x), \qquad f \in k[x], \quad \deg f \in \{2g+1, 2g+2\}, \quad f \text{ squarefree.}$$

Conversely, the smooth projective model of such a curve has genus g and is hyperelliptic. The two cases are related by a change of variable at infinity:

1. $\deg f = 2g+1$: there is a single point at infinity, which is a Weierstrass point (a branch point of φ).
2. $\deg f = 2g+2$: there are two points at infinity, neither of which is a Weierstrass point (the map φ is unramified at infinity).

***Proof Sketch*:**
The function field of X is $k(x, y)$ with $y^2 = f(x)$, and the map φ is $(x, y) \mapsto x$. The ramification points of φ correspond to the roots of f (where $y = 0$) plus possibly the point(s) at infinity, depending on the parity of $\deg f$. In both cases, the total number of branch points is $2g+2$ (either $2g+1$ finite roots plus the point at infinity, or $2g+2$ finite roots). The genus computation follows from Hurwitz.

To see that f must be squarefree: if $f(x) = (x-a)^2 g(x)$, then at the point $x = a$, $y = 0$ in the affine model, we have $y^2 = (x-a)^2 g(x)$. The Jacobian criterion (proposition 2.7.5) shows this point is singular, contradicting nonsingularity of X.

Example 7.1.46: Hyperelliptic Curves in Practice

1. **Genus 2:** Every genus-2 curve is hyperelliptic (since $\deg K = 2$ and $\ell(K) = 2$, the canonical system is the g^1_2). A typical model: $y^2 = x^5 + ax^3 + bx + c$ with $\deg f = 5$.
2. **Genus 3, hyperelliptic:** $y^2 = f(x)$ with $\deg f = 7$ or 8, squarefree. The canonical map sends $X \xrightarrow{2:1} \mathbb{P}^1 \xrightarrow{\text{Veronese}} \mathbb{P}^2$, so $\varphi_K(X)$ is a conic in $\mathbb{P}^2$. A generic genus-3 curve is *not* hyperelliptic; it is a smooth quartic in $\mathbb{P}^2$.
3. **Over $\mathbb{F}_p$:** Hyperelliptic curves over finite fields are a major topic in computational number theory and cryptography, since their Jacobians provide groups suitable for discrete-logarithm-based protocols. The Hasse–Weil bound constrains $|X(\mathbb{F}_q)|$, and counting points on hyperelliptic curves can be done efficiently.

We established in theorem 7.1.28(3) that for a hyperelliptic curve, the canonical map factors through the g^1_2. Let us make this completely explicit.

If D_0 denotes the unique g^1_2 (the divisor class of $\varphi^*(\text{point})$), then $\deg D_0 = 2$ and $\ell(D_0) = 2$. The key claim is:

Proposition 7.1.47: Canonical Class of a Hyperelliptic Curve

$K_X \sim (g-1)D_0$. In particular, the canonical linear system is $|K_X| = |(g-1)D_0|$.

***Proof*:**
We have $\deg K_X = 2g-2 = (g-1)\cdot 2 = (g-1)\deg D_0$. It suffices to show $\ell((g-1)D_0) \geq g$. To see this, consider the Weierstrass model $y^2 = f(x)$. The g_2^1 is given by $\varphi = x\colon X \to \mathbb{P}^1$, and we can set $D_0 = \varphi^*(\infty)$. The space $H^0(X, \mathcal{L}((g-1)D_0))$ consists of rational functions with poles only at the point(s) above ∞, of order $\leq 2(g-1)$. This space clearly contains the g linearly independent functions $\{1, x, x^2, \ldots, x^{g-1}\}$ (since x^j has a pole of order $2j \leq 2(g-1)$ at infinity), giving $\ell((g-1)D_0) \geq g$.

Now we apply Riemann–Roch to the divisor $(g-1)D_0$:

$$\ell((g-1)D_0) - \ell(K_X - (g-1)D_0) = \deg((g-1)D_0) + 1 - g = 2(g-1) + 1 - g = g - 1.$$

Since $\ell((g-1)D_0) \geq g$, this forces $\ell(K_X - (g-1)D_0) \geq 1$. But the degree of $K_X - (g-1)D_0$ is exactly $(2g-2) - 2(g-1) = 0$. By lemma 7.1.10, a degree-0 divisor with a global section must be principal, so $K_X - (g-1)D_0 \sim 0$, meaning $K_X \sim (g-1)D_0$.

The canonical map therefore factors as:

$$\begin{tikzcd} X \arrow[r, "\varphi", "2:1"'] \arrow[rr, bend left, "\varphi_K"] & \mathbb{P}^1 \arrow[r, hook, "\nu_{g-1}"] & \mathbb{P}^{g-1} \end{tikzcd}$$

where $\nu_{g-1}\colon \mathbb{P}^1 \hookrightarrow \mathbb{P}^{g-1}$ is the Veronese embedding of degree $g-1$ (definition 3.1.36), mapping $[s:t] \mapsto [s^{g-1} : s^{g-2}t : \cdots : t^{g-1}]$. The image $\nu_{g-1}(\mathbb{P}^1) \subseteq \mathbb{P}^{g-1}$ is the rational normal curve of degree $g-1$, and $\varphi_K(X)$ is this rational normal curve, with φ_K being 2-to-1 onto it.

7.1.7 Elliptic Curves

Elliptic curves are arguably the single most studied class of algebraic varieties. They sit at the crossroads of algebraic geometry, number theory, complex analysis, and even mathematical physics and cryptography. For us, they are the curves of genus 1—the first genus where non-trivial phenomena appear—and if there is at least one k-point, they come equipped with an inherent addition structure, i.e. a group law.

We saw in example **??** that the degree-0 Picard group of an elliptic curve is isomorphic to the curve itself, giving a group structure on its points. In example 4.1.8, we noted that this makes elliptic curves into group schemes. We now develop this theory systematically.

This section is a quick application of the theory of curves and of group-schemes. A proper in-depth coverage in the EYNTKA series can be found at [**NateEllipticCurves**].

Definition and Basic Properties

Definition 7.1.48: Elliptic Curve

An *elliptic curve* over k is a pair (E, O) where E is a complete nonsingular curve of genus 1 over k, and $O \in E(k)$ is some (nonempty) choice of k-rational point (usually called the *origin* or *identity*).

The requirement that a rational point O exist is essential: a genus-1 curve without a rational point is not an elliptic curve (over k). Over $k = \overline{k}$, every genus-1 curve has rational points (closed points with residue field k), but over non-algebraically-closed fields this can fail[16].

The first application of Riemann–Roch to elliptic curves gives the Weierstrass embedding:

Proposition 7.1.49: Elliptic Curves are Plane Cubics

Let (E, O) be an elliptic curve over k. Then the complete linear system $|3O|$ is very ample and embeds E as a smooth cubic curve in $\mathbb{P}^2_k$.

***Proof*:**
Since $g = 1$, we have $\deg K_E = 2g - 2 = 0$, so $\omega_E \cong \mathcal{O}_E$ (the canonical sheaf is trivial—a distinguishing feature of elliptic curves). By Riemann–Roch (theorem 7.1.18):

$$\begin{aligned}
\ell(O) &= 1 + 1 - 1 + \ell(K - O) = 1 + \ell(-O) = 1 \quad (\text{since } \deg(-O) = -1 < 0), \\
\ell(2O) &= 2 + 1 - 1 + \ell(K - 2O) = 2 + \ell(-2O) = 2, \\
\ell(3O) &= 3 + 1 - 1 + \ell(K - 3O) = 3 + \ell(-3O) = 3 = -3 < 0).
\end{aligned}$$

So $|3O|$ has dimension 2, defining a morphism $\varphi \colon E \to \mathbb{P}^2$. This morphism has degree $\deg(3O) = 3$. To check it is a closed immersion, we verify very ampleness: for any two points P, Q (possibly $P = Q$), we need $\ell(3O) - \ell(3O - P - Q) = 2$. Since $\ell(3O) = 3$, this means we must show $\ell(3O - P - Q) = 1$.

By Riemann–Roch applied to the divisor $D = 3O - P - Q$:

$$\ell(3O - P - Q) = \deg(3O - P - Q) + 1 - g + \ell(K - 3O + P + Q) = 1 + 0 + \ell(-3O + P + Q).$$

Because the degree of the error term is $\deg(-3O + P + Q) = -1 < 0$, it has no global sections, so $\ell(-3O + P + Q) = 0$. Therefore $\ell(3O - P - Q) = 1$ unconditionally for all pairs P, Q. This confirms $|3O|$ is very ample.

Let us make the embedding explicit. Choose a basis for $H^0(E, \mathcal{L}(nO))$ progressively:

- $H^0(E, \mathcal{L}(O)) = \langle 1 \rangle$ (the constant function, which has a pole of order ≤ 1 at O but is regular, so actually no pole).
- $H^0(E, \mathcal{L}(2O))$: contains 1 and a new function x with a pole of order exactly 2 at O (and no other poles).

[16] For example, the Selmer curve $3x^3 + 4y^3 + 5z^3 = 0$ in $\mathbb{P}^2_{\mathbb{Q}}$ has genus 1 but no $\mathbb{Q}$-rational point (this is a nontrivial arithmetic fact). It is a principal homogeneous space ("torsor") for the Jacobian elliptic curve with its natural origin.

- $H^0(E, \mathcal{L}(3O))$: contains $1, x$ and a new function y with a pole of order exactly 3 at O.

Now consider $H^0(E, \mathcal{L}(6O))$, which has dimension $\ell(6O) = 6$. The seven elements $\{1, x, y, x^2, xy, x^3, y^2\}$ all lie in this 6-dimensional space (check: each has a pole of order ≤ 6 at O), so there must be a linear relation among them. After rescaling, this relation takes the form:

$$y^2 + a_1xy + a_3y \;=\; x^3 + a_2x^2 + a_4x + a_6 \tag{7.6}$$

This is the *generalized Weierstrass equation*. It defines a cubic in $\mathbb{P}^2$ (homogenizing with a variable z: the point O maps to $[0 : 1 : 0]$).

Though this is the general form of the Weierstrass equation, in the right field by doing appropriate change of variables it can be often simplified. If $\operatorname{char}(k) \neq 2$, we can complete the square in y to eliminate the xy and y terms. We group the y terms on the left:

$$y^2 + (a_1x + a_3)y = x^3 + a_2x^2 + a_4x + a_6$$

Adding $\left(\frac{a_1x+a_3}{2}\right)^2$ to both sides, we get:

$$\left(y + \frac{a_1x + a_3}{2}\right)^2 = x^3 + a_2x^2 + a_4x + a_6 + \frac{(a_1x + a_3)^2}{4}$$

We make the substitution $y_1 = y + \frac{a_1x+a_3}{2}$. Expanding the right side and grouping by powers of x yields:

$$y_1^2 = x^3 + \left(a_2 + \frac{a_1^2}{4}\right)x^2 + \left(a_4 + \frac{a_1a_3}{2}\right)x + \left(a_6 + \frac{a_3^2}{4}\right)$$

For simplicity, let us denote these new coefficients as a_2', a_4', and a_6', so the equation becomes

$$y_1^2 = x^3 + a_2'x^2 + a_4'x + a_6'$$

Next, if $\operatorname{char}(k) \neq 3$, we can eliminate the x^2 term by completing the cube in x. We make the substitution $x = x_1 - \frac{a_2'}{3}$:

$$y_1^2 = \left(x_1 - \frac{a_2'}{3}\right)^3 + a_2'\left(x_1 - \frac{a_2'}{3}\right)^2 + a_4'\left(x_1 - \frac{a_2'}{3}\right) + a_6'$$

Expanding the cubic and quadratic terms gives:

$$y_1^2 = \left(x_1^3 - a_2'x_1^2 + \frac{a_2'^2}{3}x_1 - \frac{a_2'^3}{27}\right) + \left(a_2'x_1^2 - \frac{2a_2'^2}{3}x_1 + \frac{a_2'^3}{9}\right) + a_4'\left(x_1 - \frac{a_2'}{3}\right) + a_6'$$

Notice that the x_1^2 terms cancel out. Grouping the remaining coefficients for x_1 and the constants leaves us with an equation of the form:

$$y_1^2 = x_1^3 + Ax_1 + B$$

Dropping the subscripts, this transforms (7.6) into the *short Weierstrass form*:

$$y^2 = x^3 + Ax + B, \qquad \Delta = -16(4A^3 + 27B^2) \neq 0.$$

The condition $\Delta \neq 0$ is equivalent to the cubic having no repeated roots, which is equivalent to the curve being nonsingular. Over fields of characteristic 2 or 3, completing the square or the cube requires division by zero, so the short form is not available and one must work with (7.6).

The Group Law via Divisors

We now establish the group structure on $E(k)$ using the theory of divisors. The idea is to identify $E(k)$ with $\mathrm{Pic}^0(E)$ via $P \mapsto [P - O]$.

Theorem 7.1.50: The Group Law on an Elliptic Curve

Let (E, O) be an elliptic curve over $k = \overline{k}$. The map

$$\sigma\colon E(k) \to \mathrm{Pic}^0(E), \qquad P \mapsto \text{class of } [P] - [O],$$

is a bijection. Transporting the group structure from $\mathrm{Pic}^0(E)$ to $E(k)$, the curve E becomes an abelian group with identity O, where:

1. The addition law $P+Q = R$ is characterized by: $[P]+[Q]-[R] \sim [O]$, i.e. $P+Q+R' \sim 3O$ where R' is the "third intersection point" on the line through P and Q (and then R is the third intersection of the line through R' and O).
2. The inverse $-P$ is the third intersection point of the line connecting P and O with E.
3. The maps $+\colon E \times E \to E$ and $[-1]\colon E \to E$ are morphisms of k-schemes.

In particular, (E, O) is a group scheme over k (definition 4.1.1).

***Proof*:**

Bijectivity of σ: Injectivity: if $[P - O] = [Q - O]$ in $\mathrm{Pic}^0(E)$, then $P \sim Q$. This implies that P is an effective divisor in the complete linear system $|Q|$. But by Riemann–Roch, $\ell(Q) = 1+1-1+\ell(-Q) = 1$. Since the linear system $|Q|$ has dimension $\ell(Q)-1 = 0$, it consists of exactly one effective divisor, which is Q itself. Thus $P = Q$.

Surjectivity: let $D \in \mathrm{Pic}^0(E)$, i.e. D is a divisor of degree 0 modulo linear equivalence. Then $D + [O]$ has degree 1, and $\ell(D + O) = 1 + 1 - 1 + \ell(K - D - O) = 1 + \ell(-D - O)$. Since $\deg(-D - O) = -1 < 0$, $\ell(-D - O) = 0$, so $\ell(D + O) = 1$. This means $|D + O|$ contains a unique effective divisor, which must be a single point P (an effective divisor of degree 1 on a curve is a single point). Thus $D + [O] \sim [P]$, i.e. $D \sim [P] - [O] = \sigma(P)$.

The geometric addition law: The "chord-tangent" description in (1) follows from the fact that three collinear points P, Q, R' on a plane cubic satisfy $P+Q+R' \sim 3O$ (they are the intersection of E with a line L, and L is linearly equivalent to the line $z = 0$ that meets E in $3O$). From $P + Q + R' \sim 3O$, we get $\sigma(P) + \sigma(Q) = [P] + [Q] - 2[O] = [3O - R'] - 2[O] = [O] - [R']$, and we need $\sigma(R) = [O] - [R']$, i.e. $[R] - [O] = [O] - [R']$, giving $R + R' \sim 2O$, which means R is the third intersection of the line OR' with E.

Morphism property: One must check that the maps $(P, Q) \mapsto P+Q$ and $P \mapsto -P$ are morphisms of schemes, not merely set-theoretic maps. This can be verified by writing explicit rational formulas in the Weierstrass coordinates (a computation we omit), or more elegantly by using the functor of points (section 3.3 and definition 3.3.1): one shows that for any k-scheme T, the map $E(T) \times E(T) \to E(T)$ defined by the same divisor-theoretic construction is functorial in T, hence defines a morphism $E \times E \to E$ by Yoneda (theorem 0.4.22).

The group law on an elliptic curve is *commutative*: this follows from $\mathrm{Pic}^0(E)$ being an abelian group.

In the language of group schemes, an elliptic curve is a *commutative group scheme* of dimension 1, or equivalently an *abelian variety of dimension* 1.

A word on the singular case

What happens to the group law when an elliptic curve degenerates into a singular cubic? The group structure does not disappear—it *degenerates* from an abelian variety into a linear algebraic group. Note that elliptic curves are defined to be smooth, and hence we do not say "singular elliptic curves" but rather "singular cubic curves".

Let $C \subseteq \mathbb{P}^2_k$ be a cubic curve with a single singular point P_{sing} (and $\operatorname{char} k \neq 2, 3$). The smooth locus $C_{\text{reg}} = C \setminus \{P_{\text{sing}}\}$ is still a group variety under the chord-tangent law. The chord-tangent construction from theorem 7.1.50 still makes sense on C_{reg}: lines meeting C define effective divisors, and the same "third intersection point" recipe gives a group law on the smooth points. What fails is compactness—C_{reg} is not proper—so we lose the identification with $\operatorname{Pic}^0(C)$ and must work with the *generalized Jacobian* instead. The resulting group depends on the singularity type:

1. Node ($C\colon y^2 = x^3 + x^2$, say): the normalization is $\mathbb{P}^1$ with two preimages of the node glued together (**??**). Removing the singular point leaves $C_{\text{reg}} \cong \mathbb{G}_m = \operatorname{Spec} k[t, t^{-1}]$—the *multiplicative group* (example 4.1.4). Concretely, the group law on the smooth locus is isomorphic to $(k^*, \cdot)$.
2. Cusp ($C\colon y^2 = x^3$): the normalization is $\mathbb{P}^1$ with a single preimage of the cusp, and removing the singular point gives $C_{\text{reg}} \cong \mathbb{G}_a = \operatorname{Spec} k[t]$—the *additive group* (example 4.1.3). The group law on the smooth locus is isomorphic to $(k, +)$.

Thus the degeneration of elliptic curves mirrors a degeneration of group structures:

$$\underbrace{E}_{\text{abelian variety}} \quad \rightsquigarrow \quad \underbrace{\mathbb{G}_m}_{\text{node}} \quad \rightsquigarrow \quad \underbrace{\mathbb{G}_a}_{\text{cusp}}.$$

This trichotomy has an interesting arithmetic incarnation. If E is an elliptic curve over a local field K (say $K = \mathbb{Q}_p$), the *reduction* of E modulo the maximal ideal gives a cubic curve $\overline{E}$ over the residue field $\mathbb{F}_p$. The reduction type of E is classified by the group structure on $\overline{E}_{\text{reg}}$:

1. *Good reduction:* $\overline{E}$ is smooth; $\overline{E}_{\text{reg}} = \overline{E}$ is an elliptic curve over $\mathbb{F}_p$.
2. *Multiplicative (semistable) reduction:* $\overline{E}$ has a node; $\overline{E}_{\text{reg}} \cong \mathbb{G}_m$. This is further divided into *split* ($\mathbb{G}_m$ over $\mathbb{F}_p$) and *non-split* (a twist of $\mathbb{G}_m$).
3. *Additive (unstable) reduction:* $\overline{E}$ has a cusp; $\overline{E}_{\text{reg}} \cong \mathbb{G}_a$.

The *Néron model* of E over $\mathbb{Z}_p$ makes this classification functorial, and the *conductor* of E measures the severity of the bad reduction—directly paralleling the discriminant and ramification story from section 7.1.4. This is the starting point for the strong interplay between the geometry of singular fibers and the arithmetic of L-functions in the Birch–Swinnerton-Dyer conjecture.

The j-Invariant

Two elliptic curves over k may look different in their Weierstrass equations but be isomorphic as curves. The j-invariant is the *complete* isomorphism invariant over $\overline{k}$.

Definition 7.1.51: The j-Invariant

Let E be an elliptic curve over k with $\operatorname{char} k \neq 2, 3$, given in short Weierstrass form $y^2 = x^3 + Ax + B$. The *j-invariant* of E is:

$$j(E) = -1728 \cdot \frac{(4A)^3}{\Delta} = -1728 \cdot \frac{64A^3}{-16(4A^3 + 27B^2)} = 1728 \cdot \frac{4A^3}{4A^3 + 27B^2}.$$

The factor $1728 = 12^3$ ensures that the formula works well in all characteristics. The general definition via the a_i in (7.6) is more involved but exists in all characteristics.

Proposition 7.1.52: j-Invariant Classifies Elliptic Curves

Let $k = \overline{k}$.

1. Two elliptic curves E_1, E_2 over k are isomorphic (as curves with marked point) if and only if $j(E_1) = j(E_2)$.
2. For every $j_0 \in k$, there exists an elliptic curve E over k with $j(E) = j_0$.

In other words, the j-invariant defines a bijection between isomorphism classes of elliptic curves over $\overline{k}$ and the elements of k.

***Proof Sketch*:**
(1) An isomorphism of short Weierstrass curves $y^2 = x^3 + Ax + B$ and $y'^2 = x'^3 + A'x' + B'$ must have the form $x' = u^2x$, $y' = u^3y$ for some $u \in k^*$ (this follows from analyzing which changes of variable preserve the Weierstrass form). Under this substitution, $A' = u^4A$ and $B' = u^6B$, so $j(E') = 1728 \cdot 4(u^4A)^3/(4(u^4A)^3 + 27(u^6B)^2) = 1728 \cdot 4A^3u^{12}/(4A^3 + 27B^2)u^{12} = j(E)$.

Conversely, if $j(E) = j(E')$, the relation $4A^3/(4A^3 + 27B^2) = 4A'^3/(4A'^3 + 27B'^2)$ constrains A'/A and B'/B so that a suitable u exists.

(2) For $j_0 \neq 0, 1728$: take $A = 3j_0(1728 - j_0)$ and $B = 2j_0(1728 - j_0)^2$. For $j_0 = 0$: take $y^2 = x^3 + 1$. For $j_0 = 1728$: take $y^2 = x^3 + x$.

Over non-algebraically-closed fields, j is a necessary but not sufficient condition for isomorphism: two elliptic curves with the same j-invariant are *twists* of each other, classified by $H^1(\mathrm{Gal}(\overline{k}/k), \mathrm{Aut}(E))$. The classification of twists is an important topic in arithmetic geometry.

7.1.53 The j-Line as $\mathcal{M}_1$ The fact that elliptic curves over $\overline{k}$ are classified by a single invariant $j \in k$ means that the coarse moduli space of elliptic curves is the affine line: $\mathcal{M}_{1,1}^{\text{coarse}} \cong \mathbb{A}^1_k$. However, since elliptic curves have nontrivial automorphisms (at least the inversion $[-1]$, and more at $j = 0$ and $j = 1728$), the fine moduli space does not exist as a scheme—it is a Deligne–Mumford stack $\mathcal{M}_{1,1}$, whose coarse space is $\mathbb{A}^1$. This is the simplest nontrivial example of a moduli stack, and a perfect entry point for Part III.

The Degree-Zero Picard Group and the Jacobian

We have shown that for an elliptic curve (E, O), the map $P \mapsto [P - O]$ is an isomorphism $E(k) \xrightarrow{\sim} \operatorname{Pic}^0(E)$. Let us place this in the broader context of the Jacobian and the Picard scheme.

For a general smooth projective curve X of genus g, the group $\operatorname{Pic}^0(X)$ is far richer than X itself. It is the group of k-points of an abelian variety $\operatorname{Jac}(X)$ of dimension g, called the *Jacobian* of X:

$$\operatorname{Jac}(X)(k) \;\cong\; \operatorname{Pic}^0(X).$$

For $g = 1$ with a marked point O, the Jacobian is isomorphic to (X, O) itself—this is the content of theorem 7.1.50. For $g \geq 2$, the Jacobian is a higher-dimensional abelian variety (an abelian variety of dimension g), and the *Abel–Jacobi map*

$$\alpha \colon X \hookrightarrow \operatorname{Jac}(X), \qquad P \mapsto [P - O] \quad \text{(for a choice of base point } O\text{)},$$

embeds X into its Jacobian as a curve. The Jacobian is the "abelian envelope" of the curve—it is the universal abelian variety through which morphisms from X to abelian varieties factor (the "Albanese" property).

7.1.54 Forward Reference: The Picard Scheme and Moduli The construction $X \mapsto \operatorname{Pic}^0(X)$ is not just a group—it is functorial, and it is representable. More precisely, the functor $T \mapsto \operatorname{Pic}^0(X_T/T)$ (families of degree-0 line bundles on fibers of X over a base T) is represented by a group scheme $\operatorname{Pic}^0_{X/k}$, the *Picard scheme*. For a smooth projective curve, $\operatorname{Pic}^0_{X/k}$ is an abelian variety—the Jacobian.
The existence and properties of the Picard scheme are among the crowning achievements of Grothendieck's foundations, and they play a central role in Part III: the Picard scheme is the starting point for constructing $\mathcal{M}_g$ (via the Torelli map $X \mapsto (\operatorname{Jac}(X), \Theta)$, where Θ is the theta divisor), and it provides the link between divisor theory and moduli theory.
Also: the Néron–Severi group $\operatorname{NS}(X) = \operatorname{Pic}(X)/\operatorname{Pic}^0(X)$, which is $\cong \mathbb{Z}$ for a curve (generated by the degree map), becomes a finitely generated group of potentially large rank for surfaces and higher-dimensional varieties. This will be essential in the surface theory of §7.2.1.

Isogenies and the Endomorphism Ring

A key feature of elliptic curves is that morphisms between them (respecting the group structure) are extremely well-behaved.

Definition 7.1.55: Isogeny

An *isogeny* $\varphi\colon E_1 \to E_2$ between elliptic curves is a morphism of schemes that sends O_1 to O_2 and is nonconstant (hence surjective and finite, by proposition 7.1.4).

Any isogeny is automatically a homomorphism of group schemes[17]: $\varphi(P+Q) = \varphi(P) + \varphi(Q)$. The most important isogeny is the multiplication-by-n map:

Proposition 7.1.56: Multiplication by n

For any integer $n \neq 0$, the map $[n]\colon E \to E$ defined by $P \mapsto \underbrace{P + P + \cdots + P}_{n}$ (and $[-n] = [n] \circ [-1]$) is a finite morphism of degree n^2.

***Proof Sketch*:**
That $[n]$ is a morphism follows from the group scheme structure. It is nonconstant for $n \neq 0$ (since E has points of infinite order when $\operatorname{char} k = 0$, or by a more careful argument in positive characteristic). For the degree: the kernel $E[n] = \ker([n])$ is a finite group scheme. When $\operatorname{char} k \nmid n$, it is étale of rank n^2, so $\deg[n] = n^2$. This can be proved by computing the degree of the divisor $[n]^*(O)$ using the addition formulas, or by using the Tate module argument below.

The *n-torsion subgroup* $E[n]$ is the kernel of $[n]$: as a group scheme, it is a finite group scheme of order (rank) n^2. Its structure depends on the characteristic:

Proposition 7.1.57: Structure of the n-Torsion

Let E be an elliptic curve over $k = \overline{k}$.

1. If $\operatorname{char} k \nmid n$ (including $\operatorname{char} k = 0$): $E[n](\overline{k}) \cong (\mathbb{Z}/n\mathbb{Z})^2$ as abstract groups (and $E[n]$ is an étale group scheme over k).
2. If $\operatorname{char} k = p$ and $n = p^m$: the group scheme $E[p^m]$ has rank p^{2m}, but its group of $\overline{k}$-points is either $E[p^m](\overline{k}) \cong \mathbb{Z}/p^m\mathbb{Z}$ (the *ordinary* case) or $E[p^m](\overline{k}) \cong \{0\}$ (the *supersingular* case), depending on the Hasse invariant of E. In the supersingular case, the group scheme is connected (supported entirely at the origin).

The *endomorphism ring* $\operatorname{End}(E) = \operatorname{Hom}(E, E)$ (where Hom means isogenies plus the zero map) is a ring under addition and composition. Its structure is remarkably constrained:

[17] This is a nontrivial "rigidity" result. It follows from the fact that the map $\varphi(P+Q) - \varphi(P) - \varphi(Q)$ is a morphism $E_1 \times E_1 \to E_2$ sending (O_1, O_1) to O_2; by the rigidity lemma for abelian varieties, such a map is constant, hence identically O_2.

Proposition 7.1.58: Endomorphism Ring of an Elliptic Curve

Let E be an elliptic curve over $k = \overline{k}$. Then $\mathrm{End}(E)$ is isomorphic to one of the following:

1. $\mathbb{Z}$ (the generic case).
2. An order in an imaginary quadratic field $\mathbb{Q}(\sqrt{-d})$ (the *complex multiplication* or *CM* case). Over $\mathbb{C}$, this occurs when the lattice $\Lambda \subseteq \mathbb{C}$ with $E \cong \mathbb{C}/\Lambda$ has extra symmetry.
3. A maximal order in a quaternion algebra over $\mathbb{Q}$ (only in characteristic $p > 0$; these are the *supersingular* elliptic curves).

7.1.59 Forward Reference: The Tate Module and ℓ-adic Representations Fix a prime $\ell \neq \mathrm{char} k$. The *$\ell$-adic Tate module* of E is the inverse limit:

$$T_\ell(E) \;=\; \varprojlim_{m} E[\ell^m](\overline{k}) \;\cong\; \mathbb{Z}_\ell^2.$$

This is a free $\mathbb{Z}_\ell$-module of rank 2 carrying a continuous action of $\mathrm{Gal}(\overline{k}/k)$. The resulting representation $\rho_\ell\colon \mathrm{Gal}(\overline{k}/k) \to \mathrm{GL}_2(\mathbb{Z}_\ell)$ is the simplest example of an *ℓ-adic Galois representation*. When $k = \mathbb{F}_q$ is a finite field, the Frobenius endomorphism $\mathrm{Frob}_q \in \mathrm{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q)$ acts on $T_\ell(E)$ with characteristic polynomial $t^2 - a_q t + q$, where $a_q = q + 1 - |E(\mathbb{F}_q)|$. The *Riemann hypothesis for elliptic curves over finite fields*—proved by Hasse—states that $|a_q| \leq 2\sqrt{q}$, or equivalently that the roots of $t^2 - a_q t + q$ have absolute value $\sqrt{q}$. This is the genus-1 case of the Weil conjectures, which will be a central topic of a separate volume.
For curves of higher genus, the Tate module is replaced by the ℓ-adic cohomology group $H^1_{\text{ét}}(X, \mathbb{Z}_\ell)$, a free $\mathbb{Z}_\ell$-module of rank $2g$. The Weil conjectures (proved by Deligne for varieties of arbitrary dimension) assert that the eigenvalues of Frobenius on $H^i_{\text{ét}}$ have absolute value $q^{i/2}$—a vast generalization of Hasse's bound.

We conclude with a brief but important observation about the arithmetic of elliptic curves.

Example 7.1.60: Elliptic Curves over Number Fields
If E is an elliptic curve over a number field K (e.g. $K = \mathbb{Q}$), the group $E(K)$ of K-rational points is a finitely generated abelian group by the *Mordell–Weil theorem*:

$$E(K) \;\cong\; \mathbb{Z}^r \oplus E(K)_{\text{tors}}$$

where $r \geq 0$ is the *Mordell–Weil rank* and $E(K)_{\text{tors}}$ is finite. Computing r is an extraordinarily difficult problem, connected to the *Birch–Swinnerton-Dyer conjecture* (one of the Millennium Prize Problems), which predicts that r equals the order of vanishing of the L-function $L(E, s)$ at $s = 1$.

7.1.8 Further Topics on Curves

We collect several additional results on curves that round out the theory and connect to later chapters. This section is more survey-like in character; the reader may skip it on a first pass and return as needed.

Castelnuovo's Bound and Space Curves

Our embedding theorems (corollary 7.1.24) show that any curve can be embedded in projective space, but they say nothing about which pairs (d, g) of degree and genus can occur for a curve in a given $\mathbb{P}^r$. This is an important question, especially for "space curves" (curves in $\mathbb{P}^3$).

Theorem 7.1.61: Castelnuovo's Bound

Let $X \subseteq \mathbb{P}^r$ be a nondegenerate curve of degree d (i.e. X is not contained in any hyperplane). Write $d - 1 = m(r - 1) + \epsilon$ with $0 \leq \epsilon \leq r - 2$ (Euclidean division). Then:

$$g(X) \;\leq\; \pi(d, r) \;:=\; \binom{m}{2}(r-1) + m\epsilon.$$

The bound $\pi(d, r)$ is called the *Castelnuovo bound*, and curves achieving equality are called *Castelnuovo curves* or *curves of maximal genus*. They exist for all admissible (d, r) and have been completely classified: they lie on surfaces of minimal degree in $\mathbb{P}^r$.

Example 7.1.62: Castelnuovo's Bound in Low Dimension

1. **Plane curves ($r = 2$):** $d - 1 = m \cdot 1 + 0$, so $\pi(d, 2) = \binom{d-1}{2} = \frac{(d-1)(d-2)}{2}$. This recovers the genus formula for smooth plane curves—and shows that smooth plane curves are Castelnuovo curves.
2. **Space curves ($r = 3$):** $d - 1 = 2m + \epsilon$ with $\epsilon \in \{0, 1\}$. For example, a degree-6 curve in $\mathbb{P}^3$: $d - 1 = 5 = 2 \cdot 2 + 1$, giving $\pi(6, 3) = \binom{2}{2} \cdot 2 + 2 \cdot 1 = 4$. So a nondegenerate curve of degree 6 in $\mathbb{P}^3$ has genus ≤ 4.

We omit the proof of Castelnuovo's bound, which uses the "uniform position principle" and careful Hilbert function estimates. The interested reader can find it in [16, § IV.6] or [17, § 21.7]. The bound is a key ingredient in the study of the Hilbert scheme of curves in $\mathbb{P}^r$, which we will encounter in Part III.

Riemann–Roch in Families and Semicontinuity

One of the most powerful aspects of our cohomological machinery is that it works in *families*. Suppose $f : \mathcal{X} \to S$ is a flat proper family of curves (a flat proper morphism whose geometric fibers are smooth curves of genus g), and $\mathcal{L}$ is a line bundle on $\mathcal{X}$. For each point $s \in S$, we get a curve $X_s = f^{-1}(s)$ and a line bundle $\mathcal{L}_s = \mathcal{L}|_{X_s}$, and we may ask: how does $h^0(X_s, \mathcal{L}_s) = \ell(\mathcal{L}_s)$ vary with s?

The Euler characteristic behaves perfectly: by the flatness of f and the constancy of the Hilbert polynomial in flat families (theorem 6.7.5), $\chi(\mathcal{L}_s) = \deg \mathcal{L}_s + 1 - g$ is *constant* as s varies (since $\deg \mathcal{L}_s$ and g are locally constant in flat families). But the individual cohomology dimensions $h^i(X_s, \mathcal{L}_s)$ can jump.

The *semicontinuity theorem* (theorem 6.8.9) tells us that the jumping goes in a controlled direction:

1. The function $s \mapsto h^i(X_s, \mathcal{L}_s)$ is upper semicontinuous: it can only *jump up* on closed subsets of S.

2. Since χ is constant, if h^0 jumps up at some fiber then h^1 must also jump up to compensate.

Grauert's theorem (section 6.8.4) gives the converse: if $h^i(X_s, \mathcal{L}_s)$ is *constant* for all $s \in S$, then the higher direct image $R^i f_* \mathcal{L}$ is a locally free sheaf on S whose fiber at s is $H^i(X_s, \mathcal{L}_s)$, and moreover the base-change map is an isomorphism.

Example 7.1.63: Semicontinuity for Curves
Consider a flat family of smooth curves $\mathcal{X} \to S$ of genus $g = 3$ over a connected base S, equipped with a line bundle $\mathcal{L}$ of degree 3 on each fiber. Then $\chi(\mathcal{L}_s) = 3 + 1 - 3 = 1$ for all s. Generically, $h^0 = 1$ and $h^1 = 0$; but at special fibers where the divisor becomes "special" (i.e. $\ell(K - D) > 0$), both h^0 and h^1 jump up. The locus in S where this jump occurs is a *proper closed subset*—the "Brill–Noether locus." The study of such loci is one of the richest areas of modern curve theory.

7.1.64 Forward Reference: Deformation Theory of Curves The tangent space to the deformation functor of a smooth curve X of genus g is $H^1(X, T_X)$, where $T_X = \Omega^\vee_{X/k}$ is the tangent sheaf. By Serre duality, $H^1(X, T_X) \cong H^0(X, \omega_X^{\otimes 2})^\vee$, and by Riemann–Roch:

$$\dim H^0(X, \omega_X^{\otimes 2}) = \begin{cases} 0 & \text{if } g = 0, \\ 1 & \text{if } g = 1, \\ 3g - 3 & \text{if } g \geq 2. \end{cases}$$

This is the dimension of the moduli space $\mathcal{M}_g$! The obstructions to deformation lie in $H^2(X, T_X) = 0$ (since $\dim X = 1$), so the deformation space is always smooth. These ideas will be made precise using Schlessinger's criterion and cotangent complexes in Part III.

7.1.65 Singular Curves in Moduli: Stable Curves and $\overline{\mathcal{M}}_g$ We have seen throughout this chapter that the theory of smooth curves is remarkably clean. But the moduli space $\mathcal{M}_g$ itself is *not* clean in one crucial respect: it is not compact (not proper as an algebraic space). Geometrically, a family of smooth curves can degenerate—smooth fibers can "pinch" and acquire singularities. The semicontinuity theorem (theorem 6.8.9) tells us that cohomological invariants jump in a controlled way, and the arithmetic genus is preserved by flatness (theorem 6.7.5), but the fibers themselves leave the category of smooth curves.

To compactify $\mathcal{M}_g$, we must enlarge the class of curves we parametrize. The *Deligne–Mumford compactification* $\overline{\mathcal{M}}_g$ adds as boundary points the *stable curves*: connected, complete curves X of arithmetic genus g whose only singularities are *nodes* (ordinary double points), subject to the stability condition $|\mathrm{Aut}(X)| < \infty$.

Why nodes and not cusps or worse? Several compelling reasons converge:

1. **Flatness and p_a.** A flat family whose smooth fibers have genus g can degenerate to a nodal curve of the same arithmetic genus: nodes contribute $\delta = 1$ to the arithmetic genus (box 7.1.17), so a rational curve with g nodes has $p_a = g$, matching the smooth fibers. By contrast, a cusp also has $\delta = 1$, but cuspidal singularities are "unstable"—they can be further deformed (perturbed into nodes or smoothed entirely), so they do not represent "maximally degenerate" limits.

2. **The dualizing sheaf.** On a nodal curve X of arithmetic genus $g \geq 2$, the dualizing sheaf ω_X° (**??**) is an ample line bundle if and only if the stability condition holds. This gives the compactification a clean moduli-theoretic meaning: stable curves are precisely the curves for which ω_X° provides a canonical polarization.

3. **The group-theoretic perspective.** Nodes give the mildest possible degeneration of the group structure: on a singular fiber of a degenerating family of elliptic curves, a node yields a $\mathbb{G}_m$-fiber (multiplicative reduction), while a cusp yields a $\mathbb{G}_a$-fiber (additive reduction) (**??**). The multiplicative case is "semistable" in the sense of geometric invariant theory, while the additive case is not—this is not a coincidence.

The construction of $\overline{\mathcal{M}}_g$ as a proper Deligne–Mumford stack, and the proof that it is a projective variety (via the Knudsen–Mumford theorem), is one of the triumphs of Part III. For now, the reader should note the moral: *singular curves are not pathologies to be avoided, but essential objects that complete the moduli picture.* The theory of singular curves we have touched on in **??????** and box 7.1.17 provides exactly the foundations one needs to work at the boundary of $\overline{\mathcal{M}}_g$.

Curves over Non-Algebraically-Closed Fields

Throughout this chapter, we have assumed $k = \overline{k}$ for simplicity. Let us briefly indicate what changes over a general field k.

The main issues are:

1. **Rational points may not exist.** A smooth projective curve of genus 0 over k is isomorphic to a conic in $\mathbb{P}^2_k$, which is isomorphic to $\mathbb{P}^1_k$ if and only if it has a k-rational point (by the same Riemann–Roch argument as proposition 7.1.26, but $\ell(P) = 2$ requires a k-rational point P).

Over $\mathbb{Q}$, the existence of a rational point on a conic is determined by the Hasse–Minkowski theorem (local-global principle).

2. **Genus-1 curves without rational points.** A genus-1 curve without a k-rational point is not an elliptic curve (it has no origin for the group law). Such a curve is a *torsor* (principal homogeneous space) for its Jacobian, and its isomorphism class is an element of $H^1(k, E)$ (Galois cohomology with coefficients in the elliptic curve). The Shafarevich–Tate group $\mathrm{Sha}(E/k)$ classifies those torsors that are locally trivial at all completions of k.

3. **Regularity $\neq$ smoothness.** Over imperfect fields, a regular scheme need not be smooth (example 2.7.11). This complicates the theory of differentials and the adjunction formula.

Over finite fields $\mathbb{F}_q$, however, the theory is both tractable and extraordinarily rich:

Example 7.1.66: The Hasse–Weil Bound

Let X be a smooth projective curve of genus g over $\mathbb{F}_q$. The *Hasse–Weil bound* states:

$$\big||X(\mathbb{F}_q)| - (q+1)\big| \;\leq\; 2g\sqrt{q}.$$

For $g = 0$ ($X = \mathbb{P}^1$): $|X(\mathbb{F}_q)| = q + 1$ exactly. For $g = 1$ (elliptic curves): this is Hasse's theorem $|a_q| \leq 2\sqrt{q}$ from box 7.1.59. The general case is the *Riemann hypothesis for curves over finite fields*, proved by Weil in 1948. It is the statement that the eigenvalues of the Frobenius endomorphism on $H^1_{\text{ét}}(X, \mathbb{Q}_\ell)$ have absolute value $\sqrt{q}$, and its proof uses the theory of the Jacobian variety and intersection theory on $X \times X$. A full account will appear in a separate volume.

This concludes our treatment of curves. We have seen how the abstract cohomological machinery of the previous chapters—Riemann–Roch, Serre duality, and differentials—yields a remarkably complete picture: curves are classified by genus, their maps are controlled by Hurwitz, their embeddings by linear systems, and their families by semicontinuity. We now turn to the two-dimensional world, where the picture becomes vastly richer.

Part B: Algebraic Surfaces

7.2 Algebraic Surfaces

We now enter the world of surfaces, where the picture changes dramatically. On curves, the birational geometry was trivial: every birational map between smooth projective curves extends to an isomorphism. For surfaces, this fails spectacularly. One can *blow up* a point on a smooth surface to produce a new smooth surface that is birational but *not* isomorphic to the original. The theory of surfaces is, in large measure, the theory of understanding and controlling these blow-ups.

At the same time, surfaces are the first dimension where *intersection theory* becomes nontrivial: two curves on a surface meet in finitely many points, and the number of such points (counted with multiplicity) is a bilinear pairing on divisors. This intersection pairing is the engine that drives the entire theory, from the Riemann–Roch theorem for surfaces to the Enriques–Kodaira classification.

The reader who has absorbed the theory of curves will find that many of the same tools apply—divisors, linear systems, Serre duality, the canonical sheaf—but that the two-dimensional setting introduces genuinely new phenomena: self-intersection numbers, the Hodge index theorem, the Kodaira dimension, and the possibility of negative curves. These phenomena are not merely complications; they are the source of the extraordinary richness of surface theory.

Definition and First Examples

Definition 7.2.1: Surface

A *surface over* k is a Noetherian, integral, separated scheme X of dimension 2 and of finite type over k. We say:

1. X is *projective* if there exists a closed immersion $X \hookrightarrow \mathbb{P}^n_k$ for some n.
2. X is *nonsingular* (or *smooth*) if all local rings $\mathcal{O}_{X,x}$ are regular, or equivalently (over $k = \overline{k}$) if the structure morphism is smooth.

Unless otherwise stated, in this chapter a *surface* will mean a *smooth projective surface over* $k = \overline{k}$.

Unlike curves, where properness implies projectivity (corollary 7.1.25), there exist proper smooth surfaces that are not projective—Hironaka constructed such examples in dimension 3, and they exist in dimension 2 as well (though they are somewhat exotic). We restrict to projective surfaces to have an ample line bundle at our disposal.

Example 7.2.2: The Basic Surfaces

1. **The projective plane $\mathbb{P}^2$.** The simplest surface. It has $\operatorname{Pic}(\mathbb{P}^2) \cong \mathbb{Z}$, generated by the class of a line H. A curve of degree d on $\mathbb{P}^2$ has class dH. The canonical class is $K_{\mathbb{P}^2} = -3H$ (from the Euler sequence, theorem 5.5.18, applied to $\bigwedge^2$). Since K is anti-ample, $\mathbb{P}^2$ has "many" rational curves and is, in a sense we will make precise, as far from "general type" as possible.
2. **The product $\mathbb{P}^1 \times \mathbb{P}^1$.** This is the simplest *ruled surface*—a surface fibered over $\mathbb{P}^1$ with fibers isomorphic to $\mathbb{P}^1$. Its Picard group is $\operatorname{Pic}(\mathbb{P}^1 \times \mathbb{P}^1) \cong \mathbb{Z}^2$, generated by the classes of the two rulings $F = \{p\} \times \mathbb{P}^1$ and $G = \mathbb{P}^1 \times \{q\}$. A divisor of *bidegree* (a, b) is one of class $aF + bG$. The canonical class is $K = -2F - 2G$. This surface is not isomorphic to $\mathbb{P}^2$ (their Picard groups differ!), but it is *birational* to $\mathbb{P}^2$: an explicit birational map is given by blowing up a point of $\mathbb{P}^2$ and then contracting the strict transform of a line through it.
3. **Smooth hypersurfaces in $\mathbb{P}^3$.** A smooth surface $S \subseteq \mathbb{P}^3$ of degree d has $\operatorname{Pic}(S) \cong \mathbb{Z}$ (generated by the hyperplane class H) for $d \geq 4$ by the Noether–Lefschetz theorem[a]. The canonical class is $K_S = (d-4)H$ by the adjunction formula (proposition 6.5.7). This yields a critical trichotomy:
   (a) $d \leq 3$: K_S is anti-ample (rational surfaces).

(b) $d = 4$: $K_S = 0$ (K3 surfaces!).

(c) $d \geq 5$: K_S is ample (surfaces of general type).

4. **Hirzebruch surfaces $\mathbb{F}_n$.** For each $n \geq 0$, the nth Hirzebruch surface is $\mathbb{F}_n = \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-n))$, the projectivization of a rank-2 bundle on $\mathbb{P}^1$. It is a ruled surface with a special section C_0 satisfying $C_0^2 = -n$ and fibers F with $F^2 = 0$, $C_0 \cdot F = 1$. Notable cases: $\mathbb{F}_0 = \mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{F}_1 \cong \mathrm{Bl}_p \mathbb{P}^2$ (the blow-up of $\mathbb{P}^2$ at a point, which we will study in §7.2.4).

[a] The Noether–Lefschetz theorem says that for a *very general* smooth surface of degree ≥ 4 in $\mathbb{P}^3$, $\mathrm{Pic}\,(S) \cong \mathbb{Z}$. For *special* surfaces, the Picard group can be larger. The proof uses Hodge theory (specifically, the fact that $H^{1,1}(S) \cap H^2(S, \mathbb{Z}) = \mathbb{Z} \cdot H$ for a very general S).

7.2.3 Why Surfaces Are Harder Than Curves The central difference between curves and surfaces can be stated in one sentence: *on curves, every birational map between smooth projective models extends to an isomorphism; on surfaces, it does not.*

The reason is that on a smooth surface, one can *blow up* a closed point to produce a new smooth surface containing an *exceptional curve* $E \cong \mathbb{P}^1$ with $E^2 = -1$. This blow-up is birational (it is an isomorphism away from the point), but the new surface is not isomorphic to the old one. The theory of surfaces must therefore contend with a fundamental question that does not arise for curves: given a smooth projective surface X, is there a "simplest" or "minimal" birational model?

The answer—which is the content of the *minimal model theory* for surfaces—is yes: one can always contract (-1)-curves (by Castelnuovo's criterion, theorem 7.2.31) to arrive at a *minimal surface*. When the Kodaira dimension is ≥ 0, the minimal model is unique; when $\kappa = -\infty$, there are finitely many minimal models ($\mathbb{P}^2$ and the Hirzebruch surfaces $\mathbb{F}_n$ with $n \neq 1$).

In dimensions ≥ 3, the situation becomes far more complex: contracting curves can produce singularities, and one must allow *flips*—a story that is the subject of the Minimal Model Program (a separate volume).

Divisors and Linear Systems on Surfaces

Let X be a smooth projective surface over $k = \overline{k}$. Since X is regular, every local ring $\mathcal{O}_{X,x}$ is a regular local ring, hence a UFD (proposition 2.7.18). Therefore X is locally factorial (definition 2.4.29), and the natural map from Cartier divisors to Weil divisors is an isomorphism:

$$\mathrm{CaCl}\, X \;\cong\; \mathrm{Cl}\, X \;\cong\; \mathrm{Pic}\, X.$$

As with curves, we freely identify divisors with line bundles and use additive and multiplicative notation interchangeably.

Divisors on Surfaces

A *divisor* on X is a Weil divisor $D = \sum n_i C_i$, where the C_i are irreducible curves (codimension-1 subvarieties) on X and $n_i \in \mathbb{Z}$. Note the qualitative change from curves: on a curve, a divisor is a formal sum of *points*; on a surface, a divisor is a formal sum of *curves*. This is the shift in dimension that makes surface theory geometrically richer.

The associated invertible sheaf $\mathcal{L}(D)$ and the spaces of global sections $H^0(X, \mathcal{L}(D))$ are defined exactly as before. The complete linear system $|D|$ parametrizes effective divisors linearly equivalent to D, and is identified with $\mathbb{P}(H^0(X, \mathcal{L}(D)))$.

The Néron–Severi Group and the Picard Number

For a curve, the Picard group has a clean structure: $\operatorname{Pic}(X) \cong \operatorname{Pic}^0(X) \oplus \mathbb{Z}$, where Pic^0 is the Jacobian (connected, an abelian variety) and $\mathbb{Z}$ is the degree. For surfaces, the situation is more complex.

Definition 7.2.4: Néron–Severi Group and Picard Number

Let X be a smooth projective surface. The *Néron–Severi group* of X is:

$$\mathrm{NS}(X) \;=\; \operatorname{Pic}(X)/\operatorname{Pic}^0(X)$$

where $\operatorname{Pic}^0(X)$ is the group of line bundles that are *algebraically equivalent* to zero (equivalently, the group of k-points of the identity component of the Picard scheme $\operatorname{Pic}_{X/k}$). The *Picard number* (or *rank of the Néron–Severi group*) is $\rho(X) = \mathrm{rk}\mathrm{NS}(X)$.

By the *theorem of the base* (Néron–Severi theorem), $\mathrm{NS}(X)$ is a finitely generated abelian group. Its rank ρ is a fundamental invariant of X. Over $\mathbb{C}$, $\rho \le h^{1,1}$ (the Hodge number), with equality holding only in special cases.

Example 7.2.5: Picard Groups of Basic Surfaces

1. $\operatorname{Pic}(\mathbb{P}^2) \cong \mathbb{Z}$, generated by the hyperplane class H. Here $\operatorname{Pic}^0 = 0$, $\mathrm{NS} \cong \mathbb{Z}$, $\rho = 1$.
2. $\operatorname{Pic}(\mathbb{P}^1 \times \mathbb{P}^1) \cong \mathbb{Z}^2$, generated by the two rulings F and G. Again $\operatorname{Pic}^0 = 0$, $\mathrm{NS} \cong \mathbb{Z}^2$, $\rho = 2$.
3. For a smooth quartic surface $S \subseteq \mathbb{P}^3$ (a K3 surface): $\operatorname{Pic}^0(S) = 0$ (since $h^{0,1} = 0$), so $\operatorname{Pic}(S) = \mathrm{NS}(S)$. For a *very general* quartic, $\rho = 1$; but special quartics can have ρ as large as 20 (the maximum for K3 surfaces, achieved by the Fermat quartic $x^4 + y^4 + z^4 + w^4 = 0$ over $\overline{\mathbb{Q}}$, which has $\rho = 20$).
4. For an abelian surface A (a 2-dimensional abelian variety): $\operatorname{Pic}^0(A) \cong \widehat{A}$ (the dual abelian variety), which has dimension 2. The Néron–Severi group satisfies $1 \le \rho \le 4$. The group $\operatorname{Pic}^0(A)$ is "large"—it has the same dimension as A itself.

Ample, Nef, and Effective Divisors

On a curve, a divisor of positive degree is ample. On a surface, ampleness is more subtle. We will need several notions of "positivity" for divisors on surfaces:

1. A divisor D is *ample* if some positive multiple nD defines a closed immersion into projective space. By the Nakai–Moishezon criterion (which we will state in §7.2.1), D is ample if and only if $D^2 > 0$ and $D \cdot C > 0$ for every irreducible curve $C \subseteq X$.

2. A divisor D is *nef* ("numerically effective") if $D \cdot C \geq 0$ for every irreducible curve $C \subseteq X$.
3. A divisor D is *big* if $\ell(nD)$ grows like n^2 (quadratically, the maximum possible rate for a surface). Equivalently, the rational map $\varphi_{|nD|}$ is birational onto its image for $n \gg 0$.
4. A divisor D is *effective* if $D \geq 0$ (all coefficients non-negative).

These notions will be essential for the Enriques–Kodaira classification (§7.2.6). On curves, ample = positive degree and nef = non-negative degree, so all four notions collapse to conditions on the degree. The richer structure on surfaces is what makes their geometry so much more interesting.

7.2.6 The Cone of Curves and Duality The ampleness and nefness of divisors are best understood through the *cone of curves* $\mathrm{NE}(X) \subseteq \mathrm{NS}(X) \otimes \mathbb{R}$, defined as the convex cone generated by classes of irreducible curves. A divisor D is nef if and only if D pairs non-negatively with every element of $\mathrm{NE}(X)$, i.e. D lies in the *dual cone* $\mathrm{NE}(X)^\vee$ (the "nef cone"). The *Kleiman criterion* says that D is ample if and only if it lies in the interior of the nef cone.
In a separate volume, the *Cone Theorem* of Mori will describe the structure of $\mathrm{NE}(X)$: it is "rational polyhedral" on the K_X-negative side, generated by classes of rational curves. The extremal rays of this cone correspond to contractions of X, and the study of these contractions is the starting point of the Minimal Model Program.

7.2.1 Intersection Theory on Surfaces

We now develop the intersection pairing on a smooth projective surface—the single most important tool in the theory of surfaces. The idea is ancient and intuitive: two curves on a surface "should" meet in finitely many points, and the number of intersection points (counted with appropriate multiplicity) should depend only on the linear equivalence classes of the curves. Making this precise, and proving the resulting pairing has the properties one expects, is the content of this section.

This is the reader's first encounter with a genuine intersection product. In Part II, we will develop intersection theory in full generality (Chow rings, Chern classes, the moving lemma), but the surface case is special enough to admit a self-contained treatment using only the cohomological tools we already have. The reader should view this section as both a preview of and a motivation for the general theory.

The Intersection Pairing

Let X be a smooth projective surface over $k = \overline{k}$. We wish to define a bilinear pairing:

$$\mathrm{Pic}\,(X) \times \mathrm{Pic}\,(X) \longrightarrow \mathbb{Z}, \qquad (D_1, D_2) \mapsto D_1 \cdot D_2.$$

There are several equivalent definitions; we begin with the one that follows most naturally from our cohomological toolkit.

Definition 7.2.7: Intersection Number (Cohomological)

Let X be a smooth projective surface and D_1, D_2 divisors on X. The *intersection number* of D_1 and D_2 is:

$$D_1 \cdot D_2 \;=\; \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(-D_1)) - \chi(\mathcal{O}_X(-D_2)) + \chi(\mathcal{O}_X(-D_1 - D_2)).$$

This definition may seem opaque at first glance. The motivation comes from the inclusion-exclusion principle: if D_1 and D_2 are effective and meet transversally, then the "error" in the Euler characteristic when we subtract D_1 and D_2 separately versus subtracting their union should count exactly the points where they overlap. Let us verify that this formula gives the expected answer in simple cases.

Proposition 7.2.8: Properties of the Intersection Pairing

The intersection pairing has the following properties:

1. **Bilinearity:** $D_1 \cdot D_2$ is $\mathbb{Z}$-bilinear in D_1 and D_2.
2. **Symmetry:** $D_1 \cdot D_2 = D_2 \cdot D_1$.
3. **Dependence only on linear equivalence:** if $D_1 \sim D_1'$ and $D_2 \sim D_2'$, then $D_1 \cdot D_2 = D_1' \cdot D_2'$.
4. **Transversal effective divisors:** if C_1 and C_2 are effective divisors with no common component that meet transversally (i.e. their local equations generate the maximal ideal at each intersection point), then:
$$C_1 \cdot C_2 \;=\; |C_1 \cap C_2| \;=\; \#\{P \in X : P \in C_1 \cap C_2\}.$$
5. **General effective divisors:** if C_1 and C_2 are effective with no common component, then:
$$C_1 \cdot C_2 \;=\; \sum_{P \in C_1 \cap C_2} i_P(C_1, C_2)$$
where $i_P(C_1, C_2) = \dim_k \mathcal{O}_{X,P}/(f_1, f_2) \geq 1$ is the *local intersection multiplicity* at P (here f_1, f_2 are local equations for C_1 and C_2 at P).

***Proof*:**
(1)–(3): These follow formally from the properties of the Euler characteristic. For bilinearity, we use the additivity of χ on short exact sequences (proposition 6.7.2). The key computation is: for any three divisors A, B, C, we verify that $D_1 \cdot D_2$ as defined is linear in D_1 by replacing D_1 with $D_1 + E$ and expanding, using:

$$\chi(\mathcal{O}_X(-D_1 - E)) = \chi(\mathcal{O}_X(-D_1)) + \chi(\mathcal{O}_X(-E)) - \chi(\mathcal{O}_X) + (\text{cross terms}).$$

The cross terms assemble into $E \cdot D_2$, giving linearity. Symmetry is immediate from the definition. Dependence only on linear equivalence follows because $\chi(\mathcal{L})$ depends only on the isomorphism class of $\mathcal{L}$, hence only on the linear equivalence class of the corresponding divisor.

(4)–(5): These require a more hands-on argument. Consider the short exact sequence associated to an effective divisor C_2 on X:

$$0 \to \mathcal{O}_X(-C_2) \to \mathcal{O}_X \to \mathcal{O}_{C_2} \to 0. \tag{7.7}$$

Twisting by $\mathcal{O}_X(-C_1)$ (i.e. tensoring with $\mathcal{L}(-C_1)$, which is flat since it is locally free):

$$0 \to \mathcal{O}_X(-C_1 - C_2) \to \mathcal{O}_X(-C_1) \to \mathcal{O}_{C_2}(-C_1) \to 0$$

where $\mathcal{O}_{C_2}(-C_1) = \mathcal{O}_{C_2} \otimes \mathcal{L}(-C_1)$. Taking Euler characteristics and using (7.7) untwisted:

$$\begin{aligned} \chi(\mathcal{O}_X(-C_1 - C_2)) &= \chi(\mathcal{O}_X(-C_1)) - \chi(\mathcal{O}_{C_2}(-C_1)), \\ \chi(\mathcal{O}_X(-C_2)) &= \chi(\mathcal{O}_X) - \chi(\mathcal{O}_{C_2}). \end{aligned}$$

Substituting into the definition:

$$\begin{aligned} C_1 \cdot C_2 &= \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(-C_1)) - \chi(\mathcal{O}_X(-C_2)) + \chi(\mathcal{O}_X(-C_1 - C_2)) \\ &= \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(-C_1)) - [\chi(\mathcal{O}_X) - \chi(\mathcal{O}_{C_2})] + [\chi(\mathcal{O}_X(-C_1)) - \chi(\mathcal{O}_{C_2}(-C_1))] \\ &= \chi(\mathcal{O}_{C_2}) - \chi(\mathcal{O}_{C_2}(-C_1)). \end{aligned}$$

Now C_2 is a (possibly reducible, possibly singular) curve on X, and $\mathcal{L}(-C_1)|_{C_2}$ is a line bundle on C_2 whose degree on each irreducible component of C_2 is $-(C_1 \cdot C_{2,i})$. By the Riemann–Roch theorem on C_2 (generalized to possibly singular curves[a]):

$$\chi(\mathcal{O}_{C_2}) - \chi(\mathcal{O}_{C_2}(-C_1)) = \deg(\mathcal{L}(C_1)|_{C_2}).$$

When C_1 and C_2 have no common component, this degree equals $\sum_P i_P(C_1, C_2)$, the sum of local intersection multiplicities. When the intersection is transversal, each $i_P = 1$ and the sum is just $|C_1 \cap C_2|$.

[a]For a possibly singular or reducible curve, the quantity $\chi(\mathcal{L}) - \chi(\mathcal{O}) = \deg \mathcal{L}$ still holds by additivity and the definition of degree on each component.

The cohomological definition has the enormous advantage of being manifestly well-defined and bilinear. The price is that it is not obviously geometric—one must *prove* that it counts intersection points. Property (5) is the bridge: for effective divisors without common components, the intersection number literally counts points with multiplicity.

Example 7.2.9: Intersection Numbers on Basic Surfaces

1. **Lines in $\mathbb{P}^2$.** Let $L_1, L_2 \subseteq \mathbb{P}^2$ be two distinct lines. They are both in the class H (the hyperplane class). They meet in a single point, transversally, so $H \cdot H = 1$. More generally, curves of degrees d_1 and d_2 in $\mathbb{P}^2$ have intersection number $d_1 d_2$, recovering Bézout's theorem ([**NateClassicalAlgGeo**]).
2. **Rulings on $\mathbb{P}^1 \times \mathbb{P}^1$.** Let $F = \{p\} \times \mathbb{P}^1$ and $G = \mathbb{P}^1 \times \{q\}$ be the two rulings. Then:

$$F \cdot G = 1, \qquad F \cdot F = 0, \qquad G \cdot G = 0.$$

   The first is clear (two rulings from different families meet in one point). For $F^2 = 0$: two fibers $\{p\} \times \mathbb{P}^1$ and $\{p'\} \times \mathbb{P}^1$ with $p \neq p'$ are disjoint, so linearly equivalent effective divisors can be disjoint! This is a key difference from the curve case, where linearly equivalent effective divisors of positive degree always overlap.

3. **A curve of bidegree (a, b) on $\mathbb{P}^1 \times \mathbb{P}^1$.** If C has class $aF + bG$, then:

$$C \cdot F = b, \qquad C \cdot G = a, \qquad C^2 = 2ab.$$

For instance, the diagonal $\Delta \subseteq \mathbb{P}^1 \times \mathbb{P}^1$ has bidegree $(1, 1)$, so $\Delta^2 = 2$.

Self-Intersection and the Normal Bundle

The most novel feature of intersection theory on surfaces—something that has no analogue for curves—is the *self-intersection number* $C^2 = C \cdot C$ of a curve C on X. This number can be positive, zero, or even negative. Understanding its geometric meaning is essential for everything that follows.

The key link is the *normal bundle*:

Proposition 7.2.10: Self-Intersection via the Normal Bundle

Let C be a smooth curve on a smooth surface X. Then:

$$C^2 \;=\; \deg \mathcal{N}_{C/X}$$

where $\mathcal{N}_{C/X} = (\mathcal{I}_C/\mathcal{I}_C^2)^\vee$ is the normal bundle (an invertible sheaf on C, since C has codimension 1 in X; see definition 5.5.29).

***Proof*:**
From the proof of proposition 7.2.8, we showed $C_1 \cdot C_2 = \deg(\mathcal{L}(C_1)|_{C_2})$ for effective divisors without common components. Setting $C_1 = C_2 = C$ is problematic since C certainly shares a component with itself. But by bilinearity and invariance under linear equivalence, we can define $C^2 = C \cdot C'$ where $C' \sim C$ is any divisor linearly equivalent to C with $C' \neq C$. Alternatively, we observe that $\mathcal{L}(C)|_C$ is exactly the normal bundle: the short exact sequence

$$0 \to \mathcal{I}_C/\mathcal{I}_C^2 \to \Omega_{X/k}|_C \to \Omega_{C/k} \to 0$$

(the conormal sequence, proposition 5.5.17) dualizes to relate the normal bundle $\mathcal{N}_{C/X} = (\mathcal{I}_C/\mathcal{I}_C^2)^\vee$ to the tangent sheaves. On the other hand, the ideal sheaf of C is $\mathcal{I}_C = \mathcal{L}(-C)$, so $\mathcal{I}_C/\mathcal{I}_C^2 \cong \mathcal{L}(-C)|_C$, and $\mathcal{N}_{C/X} \cong \mathcal{L}(C)|_C$. Therefore:

$$C^2 = \deg(\mathcal{L}(C)|_C) = \deg \mathcal{N}_{C/X}.$$

Geometrically, C^2 measures how C "moves" in X:

1. $C^2 > 0$: C can be deformed within X to a nearby curve that intersects C positively. Such curves are "ample-like." Example: lines in $\mathbb{P}^2$ have $L^2 = 1$.
2. $C^2 = 0$: C can be deformed to a disjoint curve (at least numerically). Such curves typically appear as fibers of fibrations. Example: fibers of $\mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^1$.
3. $C^2 < 0$: C is "rigid"—it *cannot* be deformed to a disjoint curve. In fact, C is the unique effective divisor in its numerical class. The prototypical example is the exceptional divisor E of a blow-up, with $E^2 = -1$.

Negative self-intersection is the source of much of the richness (and difficulty) of surface theory. It is a purely two-dimensional phenomenon: on a smooth curve, a "divisor" is a sum of points, and a point obviously has non-negative "self-intersection" (which is just its degree, always ≥ 0). The existence of curves with $C^2 < 0$ on surfaces is what makes birational geometry nontrivial.

Example 7.2.11: Self-Intersection Numbers

1. **Lines in $\mathbb{P}^2$:** $L^2 = 1$. More generally, a degree-d curve has $C^2 = d^2$. This is always positive.
2. **The exceptional curve of a blow-up:** if $\pi : \mathrm{wt}\,X \to X$ is the blow-up of a point and $E = \pi^{-1}(p)$ is the exceptional divisor, then $E^2 = -1$. We will prove this in §7.2.4.
3. **The negative section of $\mathbb{F}_n$:** the section $C_0 \subseteq \mathbb{F}_n$ with $C_0^2 = -n$ is rigid when $n > 0$: there is no other effective divisor in its numerical class. For $n = 0$, $C_0^2 = 0$ and C_0 moves in a pencil (the rulings).
4. **On abelian surfaces:** the self-intersection of any curve C on an abelian surface A is $C^2 \geq 0$ by the Hodge index theorem (see §7.2.3), with $C^2 = 0$ if and only if C is a fiber of a fibration $A \to E$ over an elliptic curve.

7.2.2 The Adjunction Formula on Surfaces

The adjunction formula expresses the canonical class of a curve on a surface in terms of the canonical class of the surface and the curve's self-intersection. It is the bridge between the *intrinsic* geometry of a curve (its genus) and its *extrinsic* geometry (how it sits in the surface).

Theorem 7.2.12: The Adjunction Formula

Let C be a smooth irreducible curve on a smooth projective surface X. Then:

$$2g(C) - 2 \;=\; C \cdot (C + K_X)$$

where $g(C)$ is the genus of C and K_X is the canonical divisor of X. Equivalently, $\omega_C \cong (\omega_X \otimes \mathcal{L}(C))|_C$, i.e. $K_C \sim (K_X + C)|_C$.

***Proof*:**
The conormal exact sequence (proposition 5.5.17) for the smooth closed embedding $\iota\colon C \hookrightarrow X$ gives:

$$0 \to \mathcal{N}^\vee_{C/X} \to \Omega_{X/k}|_C \to \Omega_{C/k} \to 0 \tag{7.8}$$

where $\mathcal{N}^\vee_{C/X} = \mathcal{I}_C/\mathcal{I}_C^2 \cong \mathcal{L}(-C)|_C$ is the conormal sheaf. Taking determinants (top exterior powers) of this short exact sequence of locally free sheaves on C:

$$\bigwedge^2(\Omega_{X/k}|_C) \;\cong\; \bigwedge^1(\mathcal{N}^\vee_{C/X}) \otimes \bigwedge^1(\Omega_{C/k}).$$

Now $\bigwedge^2 \Omega_{X/k} = \omega_X$ and $\bigwedge^1 \Omega_{C/k} = \omega_C$, so:

$$\omega_X|_C \;\cong\; \mathcal{L}(-C)|_C \otimes \omega_C.$$

Rearranging: $\omega_C \cong \omega_X|_C \otimes \mathcal{L}(C)|_C \cong (\omega_X \otimes \mathcal{L}(C))|_C$. In divisor language: $K_C \sim (K_X + C)|_C$. Taking degrees on C:

$$\deg K_C = \deg(K_X|_C) + \deg(\mathcal{L}(C)|_C) = K_X \cdot C + C^2 = C \cdot (K_X + C).$$

Since $\deg K_C = 2g(C) - 2$ (proposition 7.1.21), we obtain $2g(C) - 2 = C \cdot (C + K_X)$.

Note that this is the same adjunction formula we proved earlier (propositions 5.5.30 and 6.5.7) for smooth subvarieties of smooth varieties, specialized to the case of a curve on a surface. The novelty here is that we can express the result in terms of the intersection pairing, which gives it a *numerical* flavor that is very useful in practice.

Example 7.2.13: The Adjunction Formula in Action

1. **Smooth plane curves.** Let $C \subseteq \mathbb{P}^2$ be a smooth curve of degree d. Then $C \cdot C = d^2$ (Bézout) and $C \cdot K_{\mathbb{P}^2} = C \cdot (-3H) = -3d$. Adjunction gives:
$$2g - 2 = d^2 - 3d = d(d-3), \qquad g = \frac{(d-1)(d-2)}{2}.$$
This is the genus formula we computed earlier, now recovered from the intersection pairing.

2. **Curves on $\mathbb{P}^1 \times \mathbb{P}^1$.** Let C be a smooth curve of bidegree (a, b) on $\mathbb{P}^1 \times \mathbb{P}^1$ (with $K = -2F - 2G$). Then $C^2 = 2ab$ and $C \cdot K = (aF + bG) \cdot (-2F - 2G) = -2b - 2a$, so:
$$2g - 2 = 2ab - 2a - 2b, \qquad g = (a-1)(b-1).$$
For $(a, b) = (2, 2)$: $g = 1$ (an elliptic curve as a $(2,2)$-curve). For $(a, b) = (3, 3)$: $g = 4$.

3. **Curves on K3 surfaces.** Let S be a K3 surface ($K_S = 0$) and $C \subseteq S$ a smooth curve. Then adjunction gives $2g(C) - 2 = C^2$, so $g(C) = C^2/2 + 1$. In particular, C^2 must be even, and $C^2 \geq -2$ with equality if and only if $g(C) = 0$, i.e. $C \cong \mathbb{P}^1$. Rational curves on K3 surfaces (with $C^2 = -2$) are called (-2)-*curves* or *nodal curves*, and they play a central role in the geometry of K3 surfaces.

4. **The exceptional divisor of a blow-up.** If $E \cong \mathbb{P}^1$ is the exceptional curve of a blow-up $\mathrm{Bl}_p X \to X$ of a smooth surface, then $g(E) = 0$, and adjunction gives $-2 = E^2 + K_{\mathrm{Bl}_p X} \cdot E$. We will show in §7.2.4 that $K_{\mathrm{Bl}_p X} = \pi^* K_X + E$, giving $K_{\mathrm{Bl}_p X} \cdot E = (\pi^* K_X) \cdot E + E^2 = 0 + E^2$, hence $-2 = 2E^2$, confirming $E^2 = -1$.

The adjunction formula has a useful extension to singular or reducible curves. If C is an effective divisor on X that is possibly singular or reducible (but still reduced), we define the *arithmetic genus* of C by $p_a(C) = 1 - \chi(\mathcal{O}_C)$, and the adjunction formula still holds in the form:

$$2p_a(C) - 2 \;=\; C \cdot (C + K_X). \tag{7.9}$$

This follows from the same proof: the short exact sequence $0 \to \mathcal{O}_X(-C) \to \mathcal{O}_X \to \mathcal{O}_C \to 0$ and the cohomological definition of the intersection number. When C is smooth and irreducible, $p_a(C) = g(C)$ and we recover the previous statement.

7.2.14 Forward Reference: Chow Rings and the General Intersection Product The intersection pairing $\mathrm{Pic}\,(X) \times \mathrm{Pic}\,(X) \to \mathbb{Z}$ on a smooth projective surface is the simplest instance of the *intersection product* on the Chow ring:

$$A^p(X) \otimes A^q(X) \to A^{p+q}(X)$$

where $A^p(X)$ is the group of codimension-p algebraic cycles modulo rational equivalence (definition 8.1.4). For a surface, $A^0(X) = \mathbb{Z}$ (generated by the class of X), $A^1(X) = \mathrm{Pic}\,(X)$, and $A^2(X) = \mathbb{Z}$ (generated by the class of a point). The intersection pairing is then $A^1 \times A^1 \to A^2 \cong \mathbb{Z}$. In Part II, we will define the Chow ring for smooth varieties of any dimension, prove the *moving lemma* (which allows replacing cycles by transversal ones), define Chern classes as elements of $A^*(X)$, and prove the Grothendieck–Riemann–Roch theorem. The intersection theory of surfaces, developed here "by hand," will be subsumed into this general framework.

We record one more fundamental result that constrains the intersection pairing: the Nakai–Moishezon criterion for ampleness.

Theorem 7.2.15: Nakai–Moishezon Criterion

Let D be a divisor on a smooth projective surface X. Then D is ample if and only if:

$$D^2 > 0 \qquad \text{and} \qquad D \cdot C > 0 \ \text{ for every irreducible curve } C \subseteq X.$$

***Proof Sketch*:**

The "only if" direction is easy: if D is ample, then $D \cdot C > 0$ for every curve C (since $D \cdot C = \deg(\mathcal{L}(D)|_C)$ and an ample line bundle restricted to a curve has positive degree), and $D^2 > 0$ by the same argument applied to general members of $|nD|$ for $n \gg 0$.

The "if" direction is harder and uses the Riemann–Roch theorem for surfaces (§7.2.3) and the Hodge index theorem (§7.2.3)[a]. We omit the details, referring to [16, § V.1].

[a] One shows: $D^2 > 0$ and $D \cdot H > 0$ for some ample H imply $\ell(nD) \sim n^2D^2/2$ (quadratic growth), so $|nD|$ is a large linear system for $n \gg 0$. The condition $D \cdot C > 0$ for all curves ensures that the resulting map has no contracted curves, hence is finite, hence nD is very ample for $n \gg 0$.

The Nakai–Moishezon criterion is the surface version of the simple fact that a divisor on a curve is ample if and only if it has positive degree. In higher dimensions, the criterion generalizes: D is ample on an n-dimensional projective variety if and only if $D^{\dim V} \cdot V > 0$ for every subvariety $V \subseteq X$ of every dimension $1 \leq \dim V \leq n$. This is proved in a separate volume, using the intersection theory of Part II.

7.2.3 Riemann–Roch for Surfaces

The Riemann–Roch theorem for curves (theorem 7.1.18) computed $\chi(\mathcal{L}(D))$ in terms of $\deg D$ and the genus g. We now prove the analogous result for surfaces, where the formula involves the *intersection pairing* developed in the previous section. This is the first instance of a Riemann–Roch theorem in dimension greater than one, and it illustrates how the intersection product replaces the "degree" in higher dimensions.

The Riemann–Roch Theorem for Surfaces

Theorem 7.2.16: Riemann–Roch for Surfaces

Let X be a smooth projective surface over $k = \overline{k}$, and let D be a divisor on X. Then:

$$\chi(\mathcal{L}(D)) \;=\; \frac{1}{2}\, D \cdot (D - K_X) \;+\; \chi(\mathcal{O}_X).$$

Equivalently, expanding the intersection product:

$$\chi(\mathcal{L}(D)) \;=\; \frac{D^2 - D \cdot K_X}{2} + \chi(\mathcal{O}_X).$$

Before proving this, let us compare with the curve case. For a curve of genus g, Riemann–Roch says $\chi(\mathcal{L}(D)) = \deg D + 1 - g = \deg D + \chi(\mathcal{O}_X)$. The degree on a curve is a *linear* function of D. On a surface, the Euler characteristic is a *quadratic* function of D: it involves D^2 (the self-intersection) in addition to the linear term $D \cdot K_X$. This quadratic behavior is a hallmark of dimension 2.

***Proof*:**
The strategy is the same as for curves: verify the formula at $D = 0$, then show that both sides change by the same amount when we replace D by $D + C$ for an effective divisor C.

Base case: $D = 0$. We need $\chi(\mathcal{O}_X) = \frac{1}{2} \cdot 0 \cdot (0 - K_X) + \chi(\mathcal{O}_X) = \chi(\mathcal{O}_X)$.

Induction step. Let C be a smooth irreducible curve on X. We show: the formula holds for D if and only if it holds for $D + C$. Consider the standard short exact sequence:

$$0 \to \mathcal{O}_X(-C) \to \mathcal{O}_X \to \mathcal{O}_C \to 0.$$

Tensoring with $\mathcal{L}(D + C)$:

$$0 \to \mathcal{L}(D) \to \mathcal{L}(D + C) \to \mathcal{L}(D + C)|_C \to 0.$$

Taking Euler characteristics:

$$\chi(\mathcal{L}(D + C)) - \chi(\mathcal{L}(D)) \;=\; \chi(\mathcal{L}(D + C)|_C). \tag{7.10}$$

Now $\mathcal{L}(D + C)|_C$ is a line bundle on the curve C, and by Riemann–Roch on C (theorem 7.1.18):

$$\chi(\mathcal{L}(D + C)|_C) = \deg(\mathcal{L}(D + C)|_C) + 1 - g(C) = (D + C) \cdot C + 1 - g(C).$$

By the adjunction formula (theorem 7.2.12), $2g(C) - 2 = C \cdot (C + K_X)$, so $1 - g(C) = 1 - \frac{C^2 + C \cdot K_X + 2}{2} = -\frac{C^2 + C \cdot K_X}{2}$. Therefore:

$$\begin{aligned} \chi(\mathcal{L}(D + C)|_C) &= D \cdot C + C^2 - \frac{C^2 + C \cdot K_X}{2} \\ &= D \cdot C + \frac{C^2 - C \cdot K_X}{2} \\ &= D \cdot C + \frac{1}{2} C \cdot (C - K_X). \end{aligned}$$

On the other hand, the change in the right-hand side of the Riemann–Roch formula when D is replaced by $D + C$ is:

$$\begin{aligned}
&\frac{(D+C)^2 - (D+C)\cdot K_X}{2} - \frac{D^2 - D\cdot K_X}{2} \\
&= \frac{D^2 + 2D\cdot C + C^2 - D\cdot K_X - C\cdot K_X - D^2 + D\cdot K_X}{2} \\
&= D\cdot C + \frac{C^2 - C\cdot K_X}{2}.
\end{aligned}$$

The two sides match: $\chi(\mathcal{L}(D+C)) - \chi(\mathcal{L}(D)) =$ (change in RHS). Therefore the formula holds for D if and only if it holds for $D + C$.

Conclusion. Since we can reach any divisor D from 0 by adding and subtracting smooth irreducible curves[a], and the formula holds at $D = 0$, it holds for all D.

[a]More precisely: any divisor D can be written as $D = C_1 + \cdots + C_r - C_{r+1} - \cdots - C_s$ where each C_i is a smooth irreducible curve. To see this, use the fact that any very ample linear system on a smooth surface contains smooth members (by Bertini's theorem), so $D + nH$ is represented by a smooth curve for $n \gg 0$, and H itself can be represented by a smooth curve. Thus $D = (D + nH) - nH$, a difference of smooth effective divisors.

The proof has exactly the same structure as the proof for curves: the key inputs are the additivity of χ on short exact sequences and the Riemann–Roch theorem *on the divisor C itself* (one dimension down). This is the "inductive" nature of Riemann–Roch: the formula in dimension n reduces to the formula in dimension $n - 1$ via the short exact sequence of a hypersurface section. This principle is what the Grothendieck–Riemann–Roch theorem (Part II) systematizes.

Example 7.2.17: Riemann–Roch for Surfaces in Practice

1. **$\mathbb{P}^2$:** $K_{\mathbb{P}^2} = -3H$, $\chi(\mathcal{O}_{\mathbb{P}^2}) = 1$ (since $H^i(\mathbb{P}^2, \mathcal{O}) = 0$ for $i > 0$ by theorem 6.3.1). For $D = dH$:

$$\chi(\mathcal{O}(d)) = \frac{d^2 - d(-3)}{2} + 1 = \frac{d^2 + 3d}{2} + 1 = \frac{d^2 + 3d + 2}{2} = \binom{d+2}{2}.$$

   This agrees with $\dim_k k[x, y, z]_d = \binom{d+2}{2}$ for $d \geq 0$—a reassuring check.

2. **$\mathbb{P}^1 \times \mathbb{P}^1$:** $K = -2F - 2G$, $\chi(\mathcal{O}) = 1$. For $D = aF + bG$:

$$\chi(\mathcal{O}(a,b)) = \frac{2ab - (aF + bG)\cdot(-2F - 2G)}{2} + 1 = \frac{2ab + 2a + 2b}{2} + 1 = (a+1)(b+1).$$

   Check: $H^0(\mathbb{P}^1 \times \mathbb{P}^1, \mathcal{O}(a,b)) = H^0(\mathbb{P}^1, \mathcal{O}(a)) \otimes H^0(\mathbb{P}^1, \mathcal{O}(b))$ by the Künneth formula, which has dimension $(a+1)(b+1)$ when $a, b \geq 0$, and higher cohomology vanishes.

3. **A smooth quartic in $\mathbb{P}^3$ (K3 surface):** $K_S = 0$, so:

$$\chi(\mathcal{L}(D)) = \frac{D^2}{2} + \chi(\mathcal{O}_S).$$

   We will compute $\chi(\mathcal{O}_S) = 2$ in the next subsection. Thus for an ample divisor D with $D^2 = 2d$: $\chi(\mathcal{L}(D)) = d + 2$.

The Arithmetic Genus

The *arithmetic genus* of the surface X is defined by analogy with curves:

$$p_a(X) \;=\; (-1)^{\dim X}(\chi(\mathcal{O}_X)-1) \;=\; \chi(\mathcal{O}_X)-1.$$

(The sign convention ensures $p_a \geq 0$ for surfaces of non-negative Kodaira dimension.) Thus $\chi(\mathcal{O}_X) = 1 + p_a$, and Riemann–Roch becomes:

$$\chi(\mathcal{L}(D)) = \frac{D\cdot(D-K)}{2} + 1 + p_a.$$

Setting $D = K_X$ in Riemann–Roch gives the useful identity:

$$\chi(\omega_X) = \frac{K_X^2 - K_X\cdot K_X}{2} + \chi(\mathcal{O}_X) = \chi(\mathcal{O}_X). \tag{7.11}$$

This is not surprising: by Serre duality (theorem 6.5.9), $H^i(X,\omega_X) \cong H^{2-i}(X,\mathcal{O}_X)^\vee$, so $\chi(\omega_X) = \sum(-1)^i h^i(\omega_X) = \sum(-1)^i h^{2-i}(\mathcal{O}_X) = \chi(\mathcal{O}_X)$.

Noether's Formula

The Riemann–Roch theorem for surfaces expresses $\chi(\mathcal{L}(D))$ in terms of intersection numbers and the invariant $\chi(\mathcal{O}_X)$. But what *is* $\chi(\mathcal{O}_X)$? Noether's formula gives the answer, expressing $\chi(\mathcal{O}_X)$ in terms of two fundamental numerical invariants of the surface.

Theorem 7.2.18: Noether's Formula

Let X be a smooth projective surface over $\mathbb{C}$. Then:

$$\chi(\mathcal{O}_X) \;=\; \frac{1}{12}\big(K_X^2 + \chi_{\mathrm{top}}(X)\big)$$

where $\chi_{\mathrm{top}}(X) = \sum_{i=0}^4 (-1)^i b_i(X)$ is the topological Euler characteristic (the alternating sum of Betti numbers of $X(\mathbb{C})$).

Over $\mathbb{C}$, we can use the Hodge decomposition (theorem 0.6.28) and GAGA (theorem 6.9.3) to relate algebraic and topological invariants. The Hodge numbers $h^{p,q} = \dim H^q(X,\Omega_X^p)$ satisfy $h^{p,q} = h^{q,p}$ (Hodge symmetry) and $b_k = \sum_{p+q=k} h^{p,q}$ (Hodge decomposition). For a surface:

$$\begin{aligned}
&b_0 = 1, \quad b_1 = 2h^{0,1}, \quad b_2 = h^{2,0}+h^{1,1}+h^{0,2} = 2h^{2,0}+h^{1,1},\\
&b_3 = 2h^{0,1}, \quad b_4 = 1.
\end{aligned}$$

So $\chi_{\mathrm{top}} = 2 - 4h^{0,1} + 2h^{2,0} + h^{1,1}$.

On the other hand, $\chi(\mathcal{O}_X) = h^{0,0} - h^{0,1} + h^{0,2} = 1 - q + p_g$, where we introduce the standard notation:

Definition 7.2.19: Surface Invariants

For a smooth projective surface X over $\mathbb{C}$:

1. The *irregularity*: $q = q(X) = h^{0,1}(X) = \dim H^1(X, \mathcal{O}_X)$.
2. The *geometric genus*: $p_g = p_g(X) = h^{0,2}(X) = \dim H^0(X, \omega_X) = \dim H^2(X, \mathcal{O}_X)$ (by Serre duality).
3. The *nth plurigenus*: $P_n = P_n(X) = \dim H^0(X, \omega_X^{\otimes n})$ for $n \geq 1$, with $P_1 = p_g$.

Proof Proof of Noether's formula (sketch):
The full proof of Noether's formula requires either the Hirzebruch–Riemann–Roch theorem (which computes $\chi(\mathcal{O}_X)$ via Chern numbers) or a direct computation using the Hodge index theorem and the signature formula.

Via HRR: the Hirzebruch–Riemann–Roch theorem (which we will prove in Part II) states $\chi(\mathcal{O}_X) = \int_X \mathrm{td}(T_X)$, where td is the Todd class. For a surface:

$$\mathrm{td}(T_X) = 1 + \frac{1}{2}c_1(T_X) + \frac{1}{12}(c_1(T_X)^2 + c_2(T_X)).$$

Now $c_1(T_X) = -c_1(\Omega_X) = -K_X$ (in $A^1(X)$), so $c_1(T_X)^2 = K_X^2$, and $c_2(T_X) = \chi_{\mathrm{top}}(X)$ (the topological Euler characteristic, by the Gauss–Bonnet theorem or the Poincaré–Hopf theorem). Integrating the degree-2 part over X:

$$\chi(\mathcal{O}_X) = \frac{1}{12}(K_X^2 + \chi_{\mathrm{top}}(X)).$$

Example 7.2.20: Noether's Formula in Practice

1. **$\mathbb{P}^2$:** $K = -3H$, $K^2 = 9$, $\chi_{\mathrm{top}} = 3$ (Betti numbers $1, 0, 1, 0, 1$). Noether: $\chi(\mathcal{O}) = \frac{9+3}{12} = 1$.
2. **$\mathbb{P}^1 \times \mathbb{P}^1$:** $K = -2F - 2G$, $K^2 = 8$, $\chi_{\mathrm{top}} = 4$ ($b_0 = b_4 = 1$, $b_2 = 2$, $b_1 = b_3 = 0$). Noether: $\chi(\mathcal{O}) = \frac{8+4}{12} = 1$.
3. **A smooth quartic in $\mathbb{P}^3$ (K3 surface):** $K_S = 0$, so $K_S^2 = 0$. The topological Euler characteristic of a K3 surface is $\chi_{\mathrm{top}} = 24$ (Betti numbers $1, 0, 22, 0, 1$). Noether: $\chi(\mathcal{O}_S) = \frac{0+24}{12} = 2$. So $q = 0$ and $p_g = 1$ (since $\chi = 1 - q + p_g = 2$).
4. **An abelian surface A:** $K_A = 0$ (translation-invariant 2-form), $K^2 = 0$. The Betti numbers are $1, 4, 6, 4, 1$ (since $A(\mathbb{C}) \cong \mathbb{C}^2/\Lambda$ is a 4-torus), so $\chi_{\mathrm{top}} = 0$. Noether: $\chi(\mathcal{O}_A) = 0$. Indeed $q = 2$, $p_g = 1$, $\chi = 1 - 2 + 1 = 0$.

The Hodge Diamond of a Surface

Over $\mathbb{C}$, the Hodge numbers of a smooth projective surface are conveniently arranged in a *Hodge diamond* (recall definition 0.6.29):

$$\begin{array}{ccccc} & & h^{0,0} & & \\ & h^{1,0} & & h^{0,1} & \\ h^{2,0} & & h^{1,1} & & h^{0,2} \\ & h^{2,1} & & h^{1,2} & \\ & & h^{2,2} & & \end{array} \quad = \quad \begin{array}{ccccc} & & 1 & & \\ & q & & q & \\ p_g & & h^{1,1} & & p_g \\ & q & & q & \\ & & 1 & & \end{array}$$

The symmetries are: $h^{p,q} = h^{q,p}$ (Hodge symmetry) and $h^{p,q} = h^{2-p,2-q}$ (Serre duality). The diamond is determined by the three numbers $(q, p_g, h^{1,1})$, subject to the constraint $h^{1,1} \geq \rho$ (the Picard number) and $h^{1,1} \geq 1$ (since the class of an ample divisor is a nonzero $(1,1)$-class).

Example 7.2.21: Hodge Diamonds of Basic Surfaces

1. $\mathbb{P}^2$**:** $\begin{smallmatrix} & & 1 & & \\ & 0 & & 0 & \\ 0 & & 1 & & 0 \\ & 0 & & 0 & \\ & & 1 & & \end{smallmatrix}$. Here $q = 0$, $p_g = 0$, $h^{1,1} = 1 = \rho$.

2. **K3 surface:** $\begin{smallmatrix} & & 1 & & \\ & 0 & & 0 & \\ 1 & & 20 & & 1 \\ & 0 & & 0 & \\ & & 1 & & \end{smallmatrix}$. Here $q = 0$, $p_g = 1$, $h^{1,1} = 20$, with $1 \leq \rho \leq 20$.

3. **Abelian surface:** $\begin{smallmatrix} & & 1 & & \\ & 2 & & 2 & \\ 1 & & 4 & & 1 \\ & 2 & & 2 & \\ & & 1 & & \end{smallmatrix}$. Here $q = 2$, $p_g = 1$, $h^{1,1} = 4$, with $1 \leq \rho \leq 4$.

4. **A smooth quintic surface in $\mathbb{P}^3$ (general type):** $K_S = H$, $K^2 = 5$, $\chi_{\text{top}} = 55$, $\chi(\mathcal{O}) = 5$. So $p_g = 4$ (from the exact sequence $0 \to \mathcal{O}(-5) \to \mathcal{O} \to \mathcal{O}_S \to 0$), $q = 0$, $h^{1,1} = 45$.

The Hodge Index Theorem

The intersection pairing $\text{Pic}\,(X) \times \text{Pic}\,(X) \to \mathbb{Z}$ is a symmetric bilinear form on the free part of $\text{NS}(X)$. The Hodge index theorem describes its *signature*—the number of positive and negative eigenvalues—and is one of the most important structural results in surface theory.

Theorem 7.2.22: Hodge Index Theorem

Let X be a smooth projective surface and let H be an ample divisor. Then the intersection form restricted to the orthogonal complement

$$H^{\perp} \;=\; \{D \in \text{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R} : D \cdot H = 0\}$$

is *negative definite*. Equivalently, the intersection form on $\text{NS}(X) \otimes \mathbb{R}$ has signature $(1, \rho - 1)$: exactly one positive eigenvalue and $\rho - 1$ negative eigenvalues.

The positive direction is spanned (approximately) by the ample class H itself: $H^2 > 0$ guarantees one positive eigenvalue. The theorem says that all other independent directions are negative. This is a very strong constraint on the geometry of divisors on a surface.

Proof Proof over $\mathbb{C}$**:**
Over $\mathbb{C}$, this follows from the Hodge–Riemann bilinear relations applied to the Kähler class $[\omega] \in H^{1,1}(X, \mathbb{R})$ (the class of the Kähler form associated to the ample divisor H; see definition 0.6.20 and proposition 0.6.21). The intersection pairing on $H^{1,1}(X, \mathbb{R})$ is given by the cup product:

$$(\alpha, \beta) \;=\; \int_X \alpha \wedge \beta.$$

The Hodge–Riemann relations (theorem 0.6.28) state that on the primitive cohomology $P^{1,1} = \{\alpha \in H^{1,1} : \alpha \wedge [\omega] = 0\}$ (those classes perpendicular to the Kähler class), the quadratic form $\alpha \mapsto -\int_X \alpha \wedge \alpha$ is positive definite. Restricting to algebraic classes ($\mathrm{NS}(X) \otimes \mathbb{R} \subseteq H^{1,1}(X, \mathbb{R})$) gives the result.

An algebraic proof (valid over any field) can be given using the Riemann–Roch theorem for surfaces and asymptotic estimates for $\ell(nD)$; see [16, § V.1]. The key idea is that if $D \cdot H = 0$ and $D^2 > 0$, then $\ell(nD)$ would grow quadratically, forcing $|nD|$ to be nonempty for large n. But $D \cdot H = 0$ with H ample would mean nD is trivial on the very ample system, a contradiction.

Corollary 7.2.23: Algebraic Index Inequality

Let C, D be divisors on a smooth projective surface with $C^2 > 0$. Then:

$$(C \cdot D)^2 \;\geq\; C^2 \cdot D^2$$

with equality if and only if D is numerically proportional to C (i.e. $D \equiv \frac{C \cdot D}{C^2} \cdot C$ in $\mathrm{NS}(X) \otimes \mathbb{Q}$).

***Proof*:**
This is the Cauchy–Schwarz inequality for the intersection form. Write $D = aC + D'$ where $a = (C \cdot D)/C^2 \in \mathbb{Q}$ and $D' \cdot C = 0$. Then $D^2 = a^2C^2 + D'^2$ (since $D' \cdot C = 0$). By the Hodge index theorem, $D'^2 \leq 0$ (with equality iff $D' \equiv 0$). Therefore $D^2 \leq a^2C^2 = (C \cdot D)^2/C^2$, giving $(C \cdot D)^2 \geq C^2 \cdot D^2$.

The algebraic index inequality is used constantly in surface theory. For example, it immediately implies:

Corollary 7.2.24: Negative Self-Intersection Criterion

Let C be an irreducible curve on a smooth projective surface X with H ample.

1. If $C \cdot H = 0$, then $C^2 < 0$ (assuming C is nonzero in NS).
2. If $C^2 < 0$, then C is the unique effective divisor in its numerical class.

***Proof*:**
(1) Since $H^2 > 0$ and $C \cdot H = 0$ with $C \neq 0$, the Hodge index theorem gives $C^2 < 0$.

(2) If $C' \sim_{\text{num}} C$ is another effective divisor with $C' \neq C$, then $C' - C$ is a nonzero element of NS with $C'^2 = C^2 < 0$. But $C \cdot C' = C^2 < 0$, so C and C' "anti-intersect." However, C and C' are both effective and in the same numerical class, so $C \cdot C' = C^2$. If C and C' have no common component, then $C \cdot C' \geq 0$ (a sum of local intersection multiplicities), contradicting $C^2 < 0$. Therefore $C = C'$.

This corollary explains why curves with negative self-intersection are "rigid": they cannot be deformed to other curves in the same numerical class.

7.2.25 Forward Reference: The Hodge Conjecture and Higher-Dimensional Hodge Theory The Hodge index theorem is a consequence of the Hodge–Riemann bilinear relations, which hold on all Kähler manifolds. In higher dimensions, these relations control the signature of the intersection form on $H^{p,p}$, and the *Hard Lefschetz theorem* provides the mechanism.
A famous open problem is the *Hodge conjecture*: every class in $H^{p,p}(X, \mathbb{Q})$ on a smooth projective variety X is a $\mathbb{Q}$-linear combination of algebraic cycle classes. For surfaces, this is a theorem (the Lefschetz $(1,1)$-theorem: $\mathrm{NS}(X) \otimes \mathbb{Q} = H^{1,1}(X, \mathbb{Q}) \cap H^2(X, \mathbb{Q})$), but in higher dimensions it remains one of the Clay Millennium Problems. The interplay between Hodge theory and algebraic cycles will be explored further in a separate volume.

7.2.4 Blow-ups

Blow-ups are the fundamental birational operation in the theory of surfaces. They are the mechanism by which smooth surfaces can be modified—and the source of the key difference between curves and surfaces that we flagged in box 7.2.3. Every birational morphism between smooth surfaces factors as a sequence of blow-ups (by a theorem of Zariski), so understanding blow-ups is understanding birational geometry in dimension 2.

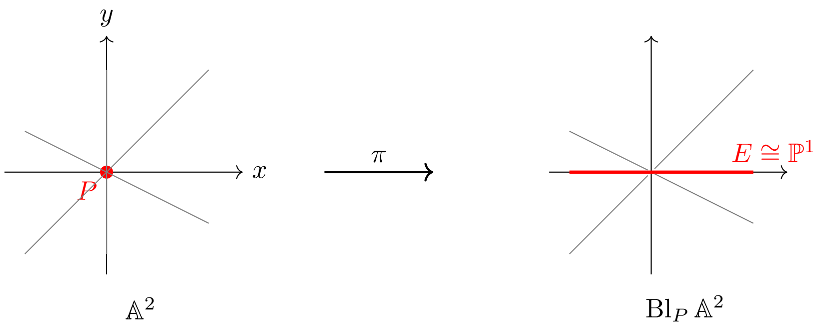
Blowing Up a Point: Geometric Motivation

Let us begin with the most geometric picture. Consider $\mathbb{A}^2_k$ with coordinates (x, y) and the origin $P = (0, 0)$. The *blow-up* of $\mathbb{A}^2$ at P replaces P by a copy of $\mathbb{P}^1$ parametrizing the *directions* through P. Every line through the origin has a well-defined slope $[x : y] \in \mathbb{P}^1$ (including the "vertical" line), and the blow-up "remembers" these directions even at the origin itself.

Concretely, define the blow-up as the closed subvariety of $\mathbb{A}^2 \times \mathbb{P}^1$:

$$\mathrm{Bl}_P \mathbb{A}^2 \;=\; \{((x, y), [s : t]) \in \mathbb{A}^2 \times \mathbb{P}^1 : xt = ys\} \;\subseteq\; \mathbb{A}^2 \times \mathbb{P}^1.$$

The projection $\pi \colon \mathrm{Bl}_P \mathbb{A}^2 \to \mathbb{A}^2$ onto the first factor is an isomorphism away from the origin: for $(x, y) \neq (0, 0)$, the ratio $[s : t] = [x : y]$ is uniquely determined. Over the origin, the fiber $\pi^{-1}(0, 0) = \{(0, 0)\} \times \mathbb{P}^1 \cong \mathbb{P}^1$: every direction is represented. This fiber is the *exceptional divisor* E.



The lines through the origin in $\mathbb{A}^2$ become curves in the blow-up that cross E at different points (corresponding to their slopes). The blow-up has "separated" the tangent directions at P. This is the reason blow-ups are used to *resolve singularities*: a node (two branches crossing at a point) can be resolved by blowing up and separating the two tangent directions (see example 3.1.30 for the formal algebraic version).

Blow-ups as Proj of Rees Algebras

We now give the scheme-theoretic definition, which works for any closed subscheme of any Noetherian scheme.

Definition 7.2.26: Blow-up

Let X be a Noetherian scheme and $\mathcal{I} \subseteq \mathcal{O}_X$ a coherent ideal sheaf defining a closed subscheme $Z \subseteq X$. The *blow-up of X along Z* (or *along $\mathcal{I}$*) is:

$$\mathrm{Bl}_Z X \;=\; \mathrm{Proj}\left(\bigoplus_{n\geq 0} \mathcal{I}^n\right)$$

where $\bigoplus_{n\geq 0}\mathcal{I}^n$ is the *Rees algebra* of $\mathcal{I}$—a graded $\mathcal{O}_X$-algebra whose degree-n piece is the nth power of the ideal sheaf (with $\mathcal{I}^0 = \mathcal{O}_X$). The natural map $\pi\colon \mathrm{Bl}_Z X \to X$ is called the *blow-up morphism*. The *exceptional divisor* is $E = \pi^{-1}(Z)_{\mathrm{red}}$ (set-theoretically, the preimage of Z).

The Proj construction here is the *relative Proj* over X (see the discussion following definition 2.3.4 and section 5.3): locally on an affine open $U = \mathrm{Spec}\,A$ of X with $\mathcal{I}|_U = \mathrm{wt}I$ for an ideal $I \subseteq A$, the blow-up restricts to $\mathrm{Proj}\bigoplus_{n\geq 0} I^n$, the classical blow-up algebra of I.

Proposition 7.2.27: Basic Properties of Blow-ups

Let $\pi\colon \mathrm{Bl}_Z X \to X$ be the blow-up of a Noetherian scheme X along a closed subscheme Z defined by $\mathcal{I}$.

1. π is a projective morphism (hence proper).
2. π is an isomorphism over $X \setminus Z$: the restriction $\pi^{-1}(X \setminus Z) \xrightarrow{\sim} X \setminus Z$ is an isomorphism. In particular, π is birational if Z has codimension ≥ 1.
3. The ideal sheaf $\mathcal{I} \cdot \mathcal{O}_{\mathrm{Bl}_Z X} = \pi^{-1}\mathcal{I} \cdot \mathcal{O}_{\mathrm{Bl}_Z X}$ is an *invertible* ideal sheaf on $\mathrm{Bl}_Z X$. It defines the exceptional divisor as a Cartier divisor: $E = V(\mathcal{I} \cdot \mathcal{O}_{\mathrm{Bl}_Z X})$.
4. **Universal property:** $\mathrm{Bl}_Z X$ is the *smallest* modification that makes $\mathcal{I}$ into a Cartier divisor. Precisely: if $f\colon Y \to X$ is any morphism such that $f^{-1}\mathcal{I} \cdot \mathcal{O}_Y$ is an invertible ideal sheaf, then f factors uniquely through π.

***Proof Sketch*:**
(1) Follows from the fact that Proj of a finitely generated graded $\mathcal{O}_X$-algebra is projective over X.

(2) Over the complement of Z, $\mathcal{I} = \mathcal{O}_X$, so $\mathcal{I}^n = \mathcal{O}_X$ for all n, and Proj of the polynomial algebra $\mathcal{O}_X[t]$ (in a single variable of degree 1) is just X itself.

(3) The tautological line bundle $\mathcal{O}_{\mathrm{Bl}}(1)$ on $\mathrm{Proj}\,(\bigoplus \mathcal{I}^n)$ satisfies $\mathcal{O}_{\mathrm{Bl}}(1) = \mathcal{I} \cdot \mathcal{O}_{\mathrm{Bl}}$ (the preimage ideal becomes principal).

(4) This is the universal property of the Proj of the Rees algebra, and can be found in [16, § IV] or [17, § 22].

Property (3) is the conceptual heart of the blow-up: the non-Cartier ideal $\mathcal{I}$ (which may be generated by more than one element) becomes Cartier on the blow-up. The exceptional divisor E is the scheme-theoretic vanishing locus of this newly-Cartier ideal.

Example 7.2.28: Blow-up of $\mathbb{A}^2$ at the Origin
Take $X = \mathbb{A}^2_k = \mathrm{Spec}\, k[x,y]$ and $Z = V(x,y)$ (the origin), with $\mathcal{I} = (x,y)$. The Rees algebra is:

$$\bigoplus_{n \geq 0} (x,y)^n \subseteq k[x,y][s,t]$$

where s,t are formal variables with $\deg s = \deg t = 1$ subject to the relation $xt = ys$ (from $x \cdot t - y \cdot s \in (x,y)^2$... more precisely, the Rees algebra is $k[x,y,s,t]/(xt - ys)$ graded by $\deg s = \deg t = 1$, $\deg x = \deg y = 0$). Then $\mathrm{Bl}_P\, \mathbb{A}^2 = \mathrm{Proj}\, k[x,y,s,t]/(xt-ys)$, which is exactly the incidence variety in $\mathbb{A}^2 \times \mathbb{P}^1$ from §7.2.4.

The exceptional divisor $E \cong \mathbb{P}^1$ is the fiber $\pi^{-1}(0,0) = \mathrm{Proj}\, k[s,t]$. On the blow-up, the ideal (x,y) becomes the ideal sheaf $\mathcal{O}(-E)$, which is invertible (generated by x on the chart $s \neq 0$ and by y on the chart $t \neq 0$).

Intersection Theory on Blow-ups

We now specialize to the case that matters most for surface theory: the blow-up $\pi\colon \mathrm{wt}X = \mathrm{Bl}_P X \to X$ of a smooth projective surface X at a closed point P.

Theorem 7.2.29: Intersection Theory on the Blow-up

Let $\pi\colon \mathrm{wt}X \to X$ be the blow-up of a smooth projective surface at a point P, with exceptional divisor $E = \pi^{-1}(P) \cong \mathbb{P}^1$. Then:

1. $\mathrm{wt}X$ is a smooth projective surface.
2. $\mathrm{Pic}\,(\mathrm{wt}X) \cong \mathrm{Pic}\,(X) \oplus \mathbb{Z}\cdot E$. Every divisor on $\mathrm{wt}X$ can be written uniquely as $\pi^*D + mE$ for $D \in \mathrm{Pic}\,(X)$ and $m \in \mathbb{Z}$.
3. The intersection numbers on $\mathrm{wt}X$ are determined by:
$$E^2 = -1, \qquad (\pi^*D_1)\cdot(\pi^*D_2) = D_1 \cdot D_2, \qquad (\pi^*D)\cdot E = 0$$
for any $D, D_1, D_2 \in \mathrm{Pic}\,(X)$.
4. The canonical class satisfies:
$$K_{\mathrm{wt}X} \;=\; \pi^*K_X + E.$$
5. For an irreducible curve $C \subseteq X$ passing through P with multiplicity $m = \mathrm{mult}_P(C)$, the *strict transform* $\mathrm{wt}C \subseteq \mathrm{wt}X$ (the closure of $\pi^{-1}(C \setminus \{P\})$) satisfies:
$$\mathrm{wt}C \;=\; \pi^*C - mE, \qquad \mathrm{wt}C \cdot E = m, \qquad \mathrm{wt}C^2 = C^2 - m^2.$$

***Proof*:**
(1) Smoothness of $\mathrm{wt}X$ follows from the explicit local description: in local coordinates (x, y) centered at P, the blow-up is covered by two affine charts $U_s = \mathrm{Spec}\,k[x, t]$ (where $y = xt$) and $U_t = \mathrm{Spec}\,k[s, y]$ (where $x = sy$), both smooth. Projectivity follows from proposition 7.2.27(1).

(2) The restriction map $\mathrm{Pic}\,(\mathrm{wt}X) \to \mathrm{Pic}\,(X \setminus \{P\})$ is surjective (since π is an isomorphism over $X \setminus \{P\}$), with kernel generated by E. Since X is smooth of dimension 2 and P is a closed point (codimension 2), the restriction $\mathrm{Pic}\,(X) \to \mathrm{Pic}\,(X \setminus \{P\})$ is an isomorphism (removing a codimension-2 subset does not change the Picard group on a smooth variety; this follows from the exact sequence for divisor class groups, proposition 5.4.12). Combining: $\mathrm{Pic}\,(\mathrm{wt}X) \cong \mathrm{Pic}\,(X) \oplus \mathbb{Z}\cdot E$.

(3) The key computation is $E^2 = -1$. We use adjunction (theorem 7.2.12): $E \cong \mathbb{P}^1$ has $g(E) = 0$, so $2 \cdot 0 - 2 = E \cdot (E + K_{\mathrm{wt}X})$. We will show $K_{\mathrm{wt}X} \cdot E = -1$ independently (from (4) below), giving $-2 = E^2 + (-1)$, hence $E^2 = -1$.

For $(\pi^*D) \cdot E = 0$: the pullback π^*D is trivial in a neighbourhood of E (since D is defined on X and π collapses E to a point, π^*D restricted to E is trivial). More precisely, $\deg(\pi^*D|_E) = 0$ since $\pi^*D|_E = \pi|_E^*(D|_P)$ and D restricted to a point is trivial.

For $(\pi^*D_1)\cdot(\pi^*D_2) = D_1 \cdot D_2$: this follows from the projection formula for the proper morphism π.

(4) The canonical class formula $K_{\mathrm{wt}X} = \pi^* K_X + E$ is proved by comparing the sheaves of 2-forms. On the chart $U_s = \operatorname{Spec} k[x,t]$ with $y = xt$:

$$dx \wedge dy = dx \wedge d(xt) = dx \wedge (x\,dt + t\,dx) = x\,dx \wedge dt.$$

So the pullback of the 2-form $dx \wedge dy$ acquires a zero of order 1 along E (which is $\{x = 0\}$ in this chart). This means $\pi^*\omega_X \cong \omega_{\mathrm{wt}X} \otimes \mathcal{L}(-E)$, i.e. $\omega_{\mathrm{wt}X} \cong \pi^*\omega_X \otimes \mathcal{L}(E)$, giving $K_{\mathrm{wt}X} = \pi^* K_X + E$.

Now we can verify: $K_{\mathrm{wt}X} \cdot E = (\pi^* K_X) \cdot E + E^2 = 0 + E^2 = E^2$, and from adjunction $-2 = E^2 + E^2 = 2E^2$, giving $E^2 = -1$ as claimed.

(5) If C passes through P with multiplicity m, then the total transform $\pi^* C$ contains E with multiplicity m (the local equation of C at P starts with terms of degree $\geq m$, and pulling back introduces a factor of x^m or y^m along E). Thus $\pi^* C = \mathrm{wt}C + mE$, giving $\mathrm{wt}C = \pi^* C - mE$. The intersection numbers follow: $\mathrm{wt}C \cdot E = (\pi^* C - mE) \cdot E = 0 - m(-1) = m$, and $\mathrm{wt}C^2 = (\pi^* C - mE)^2 = C^2 - 2m \cdot 0 + m^2(-1) = C^2 - m^2$.

Example 7.2.30: Blow-ups in Practice

1. $\mathrm{Bl}_P\,\mathbb{P}^2$**.** Let H be the class of a line in $\mathbb{P}^2$. Then $\operatorname{Pic}(\mathrm{Bl}_P\,\mathbb{P}^2) = \mathbb{Z}H \oplus \mathbb{Z}E$ (identifying $\pi^* H$ with H by abuse). A line through P has strict transform $\mathrm{wt}L = H - E$ with $\mathrm{wt}L^2 = 1 - 1 = 0$ and $\mathrm{wt}L \cdot E = 1$. A conic through P has $\mathrm{wt}C = 2H - E$ with $\mathrm{wt}C^2 = 4 - 1 = 3$.

   The surface $\mathrm{Bl}_P\,\mathbb{P}^2$ is isomorphic to the Hirzebruch surface $\mathbb{F}_1$: the pencil of lines through P becomes a ruling of $\mathbb{F}_1$, and E is the negative section with $E^2 = -1$.

   The canonical class: $K_{\mathrm{Bl}_P\,\mathbb{P}^2} = \pi^*(-3H) + E = -3H + E$. One checks $K^2 = 9 - 1 = 8$ (consistent with Noether's formula: χ_{top} drops by 1 when we blow up, giving $\chi_{\mathrm{top}} = 4$, and $\frac{8+4}{12} = 1 = \chi(\mathcal{O})$).

2. **Resolving a node.** Consider the nodal cubic $C\colon y^2 = x^3 + x^2$ in $\mathbb{A}^2$. The singular point is the origin, where $\mathrm{mult}_0(C) = 2$. Blowing up the origin in $\mathbb{A}^2$: the strict transform $\mathrm{wt}C$ satisfies $\mathrm{wt}C \cdot E = 2$ (the two branches of the node meet E at two distinct points, corresponding to the two tangent directions $y = \pm x$). The strict transform is smooth: the blow-up has *resolved* the singularity.

3. **Resolving a cusp.** The cuspidal cubic $y^2 = x^3$ has $\mathrm{mult}_0 = 2$. A single blow-up gives a strict transform with $\mathrm{wt}C \cdot E = 2$ but $\mathrm{wt}C$ is *still singular* (it has an ordinary node). A second blow-up resolves the remaining singularity. In general, resolving a cusp requires more blow-ups than resolving a node—the "worse" the singularity, the more blow-ups needed.

Castelnuovo's Contractibility Criterion

We have seen that blowing up a point on a smooth surface introduces a (-1)-curve (a smooth rational curve E with $E^2 = -1$). Castelnuovo's criterion says that the converse holds: every (-1)-curve on a smooth surface arises from a blow-up and can be "blown down."

Theorem 7.2.31: Castelnuovo's Contractibility Criterion

Let X be a smooth projective surface and $E \subseteq X$ a smooth rational curve (i.e. $E \cong \mathbb{P}^1$) with $E^2 = -1$. Then there exists a smooth projective surface Y and a point $Q \in Y$ such that $X \cong \mathrm{Bl}_Q Y$, with E as the exceptional divisor. In other words, E can be *contracted*: there exists a morphism $\pi \colon X \to Y$ that maps E to the point Q and is an isomorphism on $X \setminus E$.

***Proof Sketch*:**
The idea is to construct the contraction morphism π using linear systems. Choose an ample divisor H on X. For $n \gg 0$, the linear system $|nH + (nH \cdot E) \cdot E|$ is a subsystem that separates points and tangent vectors away from E but maps E to a single point. More precisely, one constructs a line bundle $\mathcal{L}$ on X such that $\mathcal{L} \cdot E = 0$ (so E is contracted) and $\mathcal{L}$ is ample on $X \setminus E$ (so the rest of X is embedded). The resulting morphism $\varphi_{\mathcal{L}} \colon X \to Y$ contracts E to a point, and one verifies that Y is smooth at the image point (using the fact that $E^2 = -1$ and the local structure of the contraction).

A complete proof can be found in [16, § V.5]. The key technical ingredient is Grauert's criterion for contractibility, which states that a curve C on a surface can be contracted to a (possibly singular) point if and only if the intersection matrix $(C_i \cdot C_j)$ for the components of C is negative definite. For a single smooth rational curve, this is just $E^2 < 0$, and the smoothness of the resulting point requires $E^2 = -1$.

Castelnuovo's criterion has a clean restatement in terms of the adjunction formula:

Corollary 7.2.32: Detecting (-1)-Curves

Let E be an irreducible curve on a smooth projective surface X. Then E is a (-1)-curve (i.e. $E \cong \mathbb{P}^1$ with $E^2 = -1$) if and only if:

$$E^2 = -1 \qquad \text{and} \qquad K_X \cdot E = -1.$$

Equivalently (by adjunction): $E^2 = -1$ and $g(E) = 0$.

***Proof*:**
By adjunction, $2g(E) - 2 = E^2 + K_X \cdot E = -1 + K_X \cdot E$. If $K_X \cdot E = -1$, then $2g(E) - 2 = -2$, so $g(E) = 0$ and $E \cong \mathbb{P}^1$. Conversely, $g(E) = 0$ and $E^2 = -1$ gives $K_X \cdot E = -1$.

7.2.33 Forward Reference: Extremal Contractions and the MMP Castelnuovo's contractibility criterion is the surface case of the theory of *extremal contractions* in the Minimal Model Program (MMP). In the language of the MMP:

1. The *Cone Theorem* (Mori) identifies the extremal rays of the Mori cone $\mathrm{NE}(X)$: on a surface, the extremal rays on the K_X-negative side are generated by (-1)-curves.
2. Each extremal ray corresponds to a *contraction morphism*: for a (-1)-curve, this is exactly the blow-down of Castelnuovo's criterion.
3. The process of successively contracting (-1)-curves terminates (since ρ decreases by 1 at each step) and produces the *minimal model*.

In dimension ≥ 3, contracting an extremal ray can produce a variety with *terminal singularities*, and one must perform a *flip* to continue the MMP. The existence of flips (proved by Mori in dimension 3 and by Birkar–Cascini–Hacon–McKernan in general) is one of the most important results in modern algebraic geometry. The reader should think of Castelnuovo's criterion as the "toy model" for this vast theory.

We close this section by recording the key consequence for birational geometry:

Theorem 7.2.34: Factorization of Birational Morphisms

Let $f \colon X \to Y$ be a birational morphism between smooth projective surfaces. Then f is a composition of finitely many blow-ups at closed points:

$$X = X_n \xrightarrow{\pi_n} X_{n-1} \xrightarrow{\pi_{n-1}} \cdots \xrightarrow{\pi_1} X_0 = Y$$

where each π_i is the blow-up of X_{i-1} at a closed point.

***Proof Sketch*:**
If f is not an isomorphism, then there exists a point $Q \in Y$ over which f is not defined as an isomorphism. One shows that $f^{-1}(Q)$ contains a (-1)-curve E (this uses the fact that the exceptional locus of a birational morphism of smooth surfaces consists of curves with negative definite intersection matrix, and at least one component must be a (-1)-curve). By Castelnuovo's criterion, we can contract E to get $X \to X' \to Y$, where $X \to X'$ is a blow-down. By induction on $\rho(X) - \rho(Y)$ (which decreases by 1 at each step), the process terminates.

This factorization theorem is extremely powerful: it means that the birational geometry of smooth surfaces is completely controlled by blow-ups at points. In dimension ≥ 3, no such clean factorization exists in general—birational maps may involve flips, which are not morphisms in either direction. This is another instance of the remarkable simplicity of the two-dimensional theory.

7.2.5 Ruled Surfaces and Rational Surfaces

We now study two closely related classes of surfaces: the *ruled surfaces* (fibered over a curve with $\mathbb{P}^1$-fibers) and the *rational surfaces* (birational to $\mathbb{P}^2$). These are the surfaces of Kodaira dimension $\kappa = -\infty$—the "simplest" surfaces in the classification, yet geometrically very rich.

Ruled Surfaces

Definition 7.2.35: Ruled Surface

A *ruled surface* over a smooth projective curve C is a smooth projective surface S together with a surjective morphism $p\colon S \to C$ whose fibers are all isomorphic to $\mathbb{P}^1$.
More precisely, $S = \mathbb{P}(\mathscr{E})$ for some rank-2 locally free sheaf $\mathscr{E}$ on C, where $\mathbb{P}(\mathscr{E}) = \operatorname{Proj}(\operatorname{Sym}^\bullet \mathscr{E})$ is the projectivization—the surface whose fiber over $c \in C$ is $\mathbb{P}(\mathscr{E}_c) \cong \mathbb{P}^1$.

Here the projectivization $\mathbb{P}(\mathscr{E})$ parametrizes one-dimensional *quotients* of fibers (following Grothendieck's convention), so a surjection $\mathscr{E} \twoheadrightarrow \mathcal{L}$ for a line bundle $\mathcal{L}$ on C corresponds to a section $C \hookrightarrow \mathbb{P}(\mathscr{E})$.

Two ruled surfaces $\mathbb{P}(\mathscr{E})$ and $\mathbb{P}(\mathscr{E}')$ over C are isomorphic (as surfaces over C) if and only if $\mathscr{E}' \cong \mathscr{E} \otimes \mathcal{L}$ for some line bundle $\mathcal{L}$ on C: twisting by a line bundle does not change the projectivization.

The Hirzebruch Surfaces

The most important examples are the ruled surfaces over $\mathbb{P}^1$. A rank-2 bundle on $\mathbb{P}^1$ splits as a direct sum of line bundles (by the Grothendieck–Birkhoff theorem), so $\mathscr{E} \cong \mathcal{O}(a) \oplus \mathcal{O}(b)$. Twisting by $\mathcal{O}(-b)$ gives $\mathscr{E} \otimes \mathcal{O}(-b) \cong \mathcal{O}(a-b) \oplus \mathcal{O}$, and since $a - b$ can be assumed ≥ 0 (by swapping if necessary), every ruled surface over $\mathbb{P}^1$ is isomorphic to one of the *Hirzebruch surfaces*:

Definition 7.2.36: Hirzebruch Surface

For $n \geq 0$, the nth *Hirzebruch surface* is:

$$\mathbb{F}_n \;=\; \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-n)).$$

Proposition 7.2.37: Invariants of $\mathbb{F}_n$

The Hirzebruch surface $\mathbb{F}_n$ has the following properties:

1. $\operatorname{Pic}(\mathbb{F}_n) \cong \mathbb{Z} \cdot C_0 \oplus \mathbb{Z} \cdot F$, where F is the class of a fiber of $p\colon \mathbb{F}_n \to \mathbb{P}^1$ and C_0 is the section corresponding to the surjection $\mathcal{O} \oplus \mathcal{O}(-n) \twoheadrightarrow \mathcal{O}$.
2. Intersection numbers: $F^2 = 0$, $C_0 \cdot F = 1$, $C_0^2 = -n$.
3. Canonical class: $K_{\mathbb{F}_n} = -2C_0 - (n+2)F$.
4. $\mathbb{F}_0 = \mathbb{P}^1 \times \mathbb{P}^1$ (with C_0 and F being the two rulings).
5. $\mathbb{F}_1 \cong \operatorname{Bl}_P \mathbb{P}^2$, with $C_0 = E$ (the exceptional divisor) and F the strict transform of a line through P.
6. For $n \geq 2$: C_0 is the unique irreducible curve with negative self-intersection on $\mathbb{F}_n$. In particular, $\mathbb{F}_n \not\cong \mathbb{F}_m$ for $n \neq m$.

Proof Sketch:
The Picard group and intersection numbers follow from the general theory of $\mathbb{P}(\mathscr{E})$-bundles and the tautological line bundle $\mathcal{O}_{\mathbb{P}(\mathscr{E})}(1)$; the section C_0 corresponds to $\mathcal{O}(1)$ and satisfies $C_0^2 = \deg(\mathscr{E}) \cdot (-1) + \deg(\text{surjection}) = -n$ by the self-intersection formula for sections of ruled surfaces.

For the canonical class: by the relative duality for a $\mathbb{P}^1$-bundle $p\colon S \to C$, $\omega_{S/C} \cong \mathcal{O}_S(-2) \otimes p^*(\det \mathscr{E})^\vee$, and then $\omega_S = \omega_{S/C} \otimes p^*\omega_C$. Working out for $\mathbb{F}_n$: $\det(\mathcal{O} \oplus \mathcal{O}(-n)) = \mathcal{O}(-n)$ and $\omega_{\mathbb{P}^1} = \mathcal{O}(-2)$, giving $K = -2C_0 + (n)F + (-2)F = -2C_0 - (n+2)F$.

The identification $\mathbb{F}_0 \cong \mathbb{P}^1 \times \mathbb{P}^1$ is clear. For $\mathbb{F}_1 \cong \mathrm{Bl}_P\,\mathbb{P}^2$: the pencil of lines through $P \in \mathbb{P}^2$ gives a morphism $\mathrm{Bl}_P\,\mathbb{P}^2 \to \mathbb{P}^1$ with $\mathbb{P}^1$-fibers and exceptional divisor as the (-1)-section, matching C_0 on $\mathbb{F}_1$.

For $n \geq 2$: if C is an irreducible curve on $\mathbb{F}_n$ different from C_0, write $C \sim aC_0 + bF$. Since C is effective and $C \neq C_0$, one shows $a \geq 0$ and (if $a > 0$) $b \geq an$, giving $C^2 = -a^2n + 2ab \geq a^2n \geq 0$. So C_0 is the only curve with negative self-intersection.

Example 7.2.38: Ruled Surfaces over Higher-Genus Curves
If C has genus $g \geq 1$, the ruled surfaces $\mathbb{P}(\mathscr{E}) \to C$ are much richer: rank-2 bundles on C no longer split. Their moduli form a variety of dimension $4g-3$ (for stable bundles of fixed determinant). The geometry of these ruled surfaces reflects the geometry of vector bundles on C—a topic that connects to gauge theory, the Narasimhan–Seshadri theorem, and mathematical physics.

Rational Surfaces

Definition 7.2.39: Rational Surface

A smooth projective surface X is *rational* if it is birational to $\mathbb{P}^2$.

By theorem 7.2.34, every birational map from X to $\mathbb{P}^2$ factors through blow-ups and blow-downs. So every rational surface is obtained from $\mathbb{P}^2$ (or from some $\mathbb{F}_n$) by a sequence of blow-ups. The simplest examples are:

Example 7.2.40: Rational Surfaces in Practice

1. $\mathbb{P}^2$ **itself** and all $\mathbb{F}_n$ (including $\mathbb{P}^1 \times \mathbb{P}^1 = \mathbb{F}_0$).
2. **Del Pezzo surfaces.** The blow-up of $\mathbb{P}^2$ at $r \leq 8$ points in *general position* (no three collinear, no six on a conic, no eight on a singular cubic with one of them at the singular point). These have $K^2 = 9 - r$ and are Fano surfaces (i.e. $-K$ is ample) for $r \leq 8$.
3. **The cubic surface.** A smooth cubic surface $S \subseteq \mathbb{P}^3$ is isomorphic to $\mathrm{Bl}_{P_1,\ldots,P_6}\,\mathbb{P}^2$ (the blow-up of $\mathbb{P}^2$ at 6 points in general position). It has $K_S = -H$ (the hyperplane class), $K^2 = 3$, and $\rho = 7$. The 27 lines on S correspond to: the 6 exceptional divisors E_i, the 15 strict transforms of lines $\mathrm{wt}L_{ij}$ through pairs of blown-up points, and the 6 strict transforms of conics $\mathrm{wt}Q_i$ through 5 of the 6 points. Each line ℓ satisfies $\ell^2 = -1$ and $K \cdot \ell = -1$, so every line is a (-1)-curve. The intersection pattern of these 27 lines is governed by the Weyl group $W(E_6)$—a key connection to Lie theory.

The following classical theorem gives a cohomological characterization of rationality:

Theorem 7.2.41: Castelnuovo's Rationality Criterion

A smooth projective surface X over $\mathbb{C}$ is rational if and only if $q(X) = 0$ and $P_2(X) = 0$ (irregularity zero and second plurigenus zero).

The "only if" direction is clear: q and P_2 are birational invariants, and $\mathbb{P}^2$ has $q = P_2 = 0$. The "if" direction is a theorem whose proof uses the theory of linear systems on surfaces. We omit it, referring to [16, § V.6]. Castelnuovo's criterion is remarkable because it detects rationality from just two cohomological invariants.

7.2.6 The Enriques–Kodaira Classification

We arrive at the summit of the theory of surfaces: the classification of smooth projective surfaces by their *Kodaira dimension*. This classification, developed by the Italian school (Castelnuovo, Enriques) in the early 20th century and completed by Kodaira in the 1960s over $\mathbb{C}$, organizes the zoo of surfaces into a small number of families with sharply distinct geometric behavior. It is one of the most important achievements of algebraic geometry, and it serves as the model for the classification program in all dimensions.

Kodaira Dimension

The key invariant is the growth rate of the spaces of *pluricanonical forms*.

Definition 7.2.42: Kodaira Dimension

Let X be a smooth projective variety of dimension d over k. For each $n \geq 1$, the *nth plurigenus* is $P_n(X) = \dim_k H^0(X, \omega_X^{\otimes n})$. The *Kodaira dimension* $\kappa(X)$ is defined by the growth rate of P_n:

$$\kappa(X) = \begin{cases} -\infty & \text{if } P_n = 0 \text{ for all } n \geq 1, \\ \min\left\{k \geq 0 : \limsup\limits_{n\to\infty} \dfrac{P_n}{n^k} < \infty\right\} & \text{otherwise.} \end{cases}$$

Equivalently, $\kappa(X)$ is the dimension of the image of the *pluricanonical map* $\varphi_{|nK_X|} : X \dashrightarrow \mathbb{P}^{P_n - 1}$ for n sufficiently large and divisible (or $-\infty$ if all $P_n = 0$).

For a surface, $\kappa \in \{-\infty, 0, 1, 2\}$:

1. $\kappa = -\infty$: $P_n = 0$ for all n. The canonical class K_X is "anti-effective" in a strong sense.
2. $\kappa = 0$: P_n is bounded (either 0 or 1 for each n). The canonical class is "numerically trivial."
3. $\kappa = 1$: P_n grows linearly, i.e. $P_n \sim cn$ for some $c > 0$. The pluricanonical map gives a fibration onto a curve.
4. $\kappa = 2$: P_n grows quadratically, i.e. $P_n \sim cn^2$. The pluricanonical map is birational for $n \gg 0$. These are the *surfaces of general type*.

The Kodaira dimension is a birational invariant: if X and Y are birational smooth projective surfaces, then $\kappa(X) = \kappa(Y)$. This is because the plurigenera P_n are birational invariants (a consequence of the fact that the space of regular n-canonical forms is a birational invariant for smooth varieties).

Minimal Models

Definition 7.2.43: Minimal Surface

A smooth projective surface X is *minimal* if it contains no (-1)-curves (smooth rational curves E with $E^2 = -1$). Equivalently, X cannot be obtained as a blow-up of another smooth surface.

Theorem 7.2.44: Existence and Uniqueness of Minimal Models

1. **Existence:** Every smooth projective surface admits a birational morphism to a minimal surface, obtained by successively contracting (-1)-curves (theorem 7.2.31). The process terminates since the Picard number ρ drops by 1 at each step.
2. **Uniqueness for $\kappa \geq 0$:** If $\kappa(X) \geq 0$, the minimal model is unique up to isomorphism.
3. **Non-uniqueness for $\kappa = -\infty$:** If $\kappa(X) = -\infty$, the minimal models are $\mathbb{P}^2$ and the Hirzebruch surfaces $\mathbb{F}_n$ with $n \neq 1$ (note: $\mathbb{F}_1$ is *not* minimal since C_0 is a (-1)-curve).

***Proof Sketch*:**
Existence is clear from Castelnuovo's criterion and the fact that ρ is a non-negative integer.

Uniqueness for $\kappa \geq 0$: suppose X_1 and X_2 are two minimal models in the same birational equivalence class. The birational map $X_1 \dashrightarrow X_2$ can be resolved by a smooth surface Y dominating both:

$$\begin{tikzcd} & Y \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ X_1 \arrow[rr, dashed] & & X_2 \end{tikzcd}$$

By theorem 7.2.34, π_i is a composition of blow-ups. If π_1 is not an isomorphism, then Y contains a (-1)-curve E contracted by π_1. Since $\kappa \geq 0$, one can show that $K_Y \cdot E = -1$ forces π_2 to also contract E (otherwise $\pi_{2*}(E)$ would be a curve with $K_{X_2} \cdot \pi_{2*}(E) < 0$, contradicting $\kappa \geq 0$ via the "negativity of contraction" argument). By induction, π_1 and π_2 contract the same curves, so $X_1 \cong X_2$.

For $\kappa = -\infty$: a minimal rational surface is either $\mathbb{P}^2$ (if no ruling exists) or a Hirzebruch surface $\mathbb{F}_n$ with $n \neq 1$. The non-uniqueness arises because $\mathbb{P}^2$ and $\mathbb{F}_n$ (for various $n \neq 1$) are all minimal and birational.

Overview of the Classification

We now state the Enriques–Kodaira classification. The result organizes minimal surfaces into four classes according to κ, with $\kappa = 0$ further subdivided into four types. We work over $\mathbb{C}$ throughout.

$\kappa = -\infty$: Ruled and Rational Surfaces

A minimal surface with $\kappa = -\infty$ is either $\mathbb{P}^2$ or a ruled surface $\mathbb{P}(\mathscr{E})$ over a curve C of genus $g \geq 0$. The characterization is:

$$\kappa = -\infty \quad \Longleftrightarrow \quad P_{12} = 0 \quad \Longleftrightarrow \quad X \text{ is ruled or rational.}$$

The key invariant is the *base genus*: for a ruled surface $S \to C$, the base curve C has genus $g \geq 0$. When $g = 0$, the surface is rational (birational to $\mathbb{P}^2$).

$\kappa = 0$: Four Types

Minimal surfaces with $\kappa = 0$ fall into exactly four families, distinguished by their irregularity q and geometric genus p_g. In all cases, K_X is numerically trivial (i.e. $K_X \cdot C = 0$ for all curves C), and in fact $12K_X \sim 0$ (the canonical class has finite order dividing 12 in $\mathrm{Pic}\,(X)$).

1. **Abelian surfaces** ($q = 2$, $p_g = 1$, $K_X \sim 0$): These are 2-dimensional abelian varieties—quotients $\mathbb{C}^2/\Lambda$ for a lattice Λ of rank 4. They carry a group structure (addition of points) and have trivial canonical bundle. Their Hodge diamond has $h^{1,1} = 4$ and $\chi(\mathcal{O}) = 0$.
2. **K3 surfaces** ($q = 0$, $p_g = 1$, $K_X \sim 0$): These are simply connected surfaces with trivial canonical bundle. The prototype is a smooth quartic in $\mathbb{P}^3$. Their Hodge diamond has $h^{1,1} = 20$ and $\chi(\mathcal{O}) = 2$. K3 surfaces are among the most studied objects in algebraic geometry; we discuss them further below.
3. **Enriques surfaces** ($q = 0$, $p_g = 0$, $K_X \not\sim 0$ but $2K_X \sim 0$): These can be characterized as quotients of K3 surfaces by a fixed-point-free involution. They have $\chi(\mathcal{O}) = 1$ and $h^{1,1} = 10$.
4. **Bielliptic (hyperelliptic) surfaces** ($q = 1$, $p_g = 0$): These are quotients $(E_1 \times E_2)/G$ where E_1, E_2 are elliptic curves and G is a finite group acting freely. They fiber over both elliptic curves (after appropriate quotients) and have $\chi(\mathcal{O}) = 0$.

Example 7.2.45: K3 Surfaces: A Closer Look

K3 surfaces deserve special emphasis because of their extraordinary properties:

1. **Examples:** Smooth quartics in $\mathbb{P}^3$, smooth complete intersections of type $(2, 3)$ in $\mathbb{P}^4$ or $(2, 2, 2)$ in $\mathbb{P}^5$, double covers of $\mathbb{P}^2$ branched over a smooth sextic curve, and Kummer surfaces (resolutions of $A/\{\pm 1\}$ for an abelian surface A).
2. **Deformation equivalence:** All K3 surfaces are deformation-equivalent as complex manifolds. There is a single connected 20-dimensional moduli space of complex K3 surfaces (but not all are algebraic!). Algebraic K3 surfaces form a countable union of 19-dimensional families, parametrized by their Picard lattice.
3. **The period map:** The Hodge structure on $H^2(S, \mathbb{Z})$ determines S up to isomorphism (the global Torelli theorem for K3 surfaces, proved by Piatetski-Shapiro and Shafarevich). The lattice $H^2(S, \mathbb{Z}) \cong U^3 \oplus E_8(-1)^2$ is a fixed unimodular lattice of signature $(3, 19)$, and the moduli of K3 surfaces is governed by the period domain $\mathrm{SO}(3, 19)/\mathrm{SO}(2) \times \mathrm{SO}(1, 19)$.
4. **Rational curves:** K3 surfaces contain infinitely many rational curves (proved by Bogomolov–

Hassett–Tschinkel and others). Each such curve C satisfies $C^2 = -2$ by adjunction ($K_S = 0$).

$\kappa = 1$: Elliptic Surfaces

A minimal surface with $\kappa = 1$ admits a *fibration* $f\colon X \to C$ over a curve C whose general fiber is a smooth curve of genus 1 (an elliptic curve). This fibration is the Iitaka fibration associated to the pluricanonical linear systems $|nK_X|$.

The *canonical bundle formula* describes K_X in terms of the fibration:

$$K_X \sim f^*(K_C + \mathcal{L}) + \sum_i (m_i - 1)F_i$$

where $\mathcal{L}$ is a line bundle on C of degree $\chi(\mathcal{O}_X)$, and the sum runs over the *multiple fibers* m_iF_i (fibers that are m_i times a reduced divisor). The invariants $\chi(\mathcal{O}_X)$, the genus of C, and the multiplicities m_i determine the surface.

Elliptic surfaces are ubiquitous: the "generic" surface of $\kappa = 1$ is an elliptic surface. Their study connects to the arithmetic of elliptic curves over function fields (via the Mordell–Weil group of the generic fiber), the theory of singular fibers (Kodaira's classification of singular fibers of elliptic fibrations), and string theory (where elliptic fibrations on Calabi–Yau manifolds play a central role).

$\kappa = 2$: Surfaces of General Type

A surface of *general type* is one with $\kappa = 2$, the maximal Kodaira dimension. For such surfaces, K_X is big: P_n grows like n^2, and the pluricanonical map $\varphi_{|nK_X|}$ is birational onto its image for n sufficiently large (in fact $n \geq 5$ always suffices by a theorem of Bombieri).

The numerical invariants of minimal surfaces of general type are constrained by two fundamental inequalities:

Theorem 7.2.46: Geography of Surfaces of General Type

Let X be a minimal surface of general type. Then:

1. **Bogomolov–Miyaoka–Yau inequality:** $K_X^2 \leq 3\chi(\mathcal{O}_X)$. Equality holds if and only if the universal cover of $X(\mathbb{C})$ is the unit ball in $\mathbb{C}^2$ ("ball quotients").
2. **Noether inequality:** $K_X^2 \geq 2\chi(\mathcal{O}_X) - 6$.
3. Both invariants are positive: $K_X^2 \geq 1$ and $\chi(\mathcal{O}_X) \geq 1$.

The allowed region in the (K^2, χ)-plane, bounded by the BMY inequality and the Noether line, is called the *geography* of surfaces of general type. Determining which lattice points (K^2, χ) in this region are realized by actual surfaces is a major open problem. The "boundary" surfaces (those on the Noether line or the BMY line) have very special geometry: Noether-line surfaces are fibered over curves with high-genus fibers, while BMY surfaces are ball quotients with connections to number theory.

Example 7.2.47: Surfaces of General Type

1. **Smooth hypersurfaces in $\mathbb{P}^3$ of degree $d \geq 5$:** $K_S = (d-4)H$, $K^2 = d(d-4)^2$, $\chi(\mathcal{O}) = \binom{d-1}{3}$. For $d = 5$: $K^2 = 5$, $\chi = 4$. Check: $5 \leq 3 \cdot 4 = 12$ (BMY) and $5 \geq 2 \cdot 4 - 6 = 2$ (Noether).

2. **Products of curves:** $X = C_1 \times C_2$ with $g(C_i) \geq 2$. Then $K_X = p_1^* K_{C_1} + p_2^* K_{C_2}$, $K^2 = 8(g_1 - 1)(g_2 - 1)$, $\chi(\mathcal{O}) = (g_1 - 1)(g_2 - 1)$. The BMY ratio is $K^2/\chi = 8$ (always < 9, approaching 9 for curves with many symmetries).

3. **The Godeaux surface:** the quotient of the Fermat quintic $\{x_0^5 + \cdots + x_3^5 = 0\} \subseteq \mathbb{P}^3$ by a free $\mathbb{Z}/5\mathbb{Z}$-action. It has $K^2 = 1$ and $\chi = 1$—the smallest possible invariants for a surface of general type.

7.2.7 The Classification Table

We summarize the Enriques–Kodaira classification in the following table. All surfaces are minimal and over $\mathbb{C}$.

κ	q	p_g	K^2	**Type**
$-\infty$	any	0	< 0	Ruled / Rational
0	2	1	0	Abelian surface
0	0	1	0	K3 surface
0	0	0	0	Enriques surface ($2K \sim 0$)
0	1	0	0	Bielliptic surface
1	any	any	0	Elliptic surface
2	any	≥ 1	> 0	General type

Several patterns are worth highlighting:

1. The canonical class K_X is "more positive" as κ increases: anti-ample for $\kappa = -\infty$, numerically trivial for $\kappa = 0$, semi-ample for $\kappa = 1$, and big for $\kappa = 2$. The Kodaira dimension measures "how positive K_X is."

2. The invariant K^2 is zero for $\kappa \leq 1$ (on minimal surfaces) and positive for $\kappa = 2$. This is why self-intersection of the canonical class is such a fundamental invariant.

3. The irregularity q and geometric genus p_g suffice to distinguish the four types of $\kappa = 0$ surfaces. For $\kappa = 1$ and $\kappa = 2$, more invariants are needed (the fibration structure and the geography, respectively).

7.2.48 Forward Reference: The Classification in Higher Dimensions and Positive Characteristic The Enriques–Kodaira classification of surfaces is the prototype for the *classification of algebraic varieties* in all dimensions. The organizing principle remains the same: classify by Kodaira dimension, with the Minimal Model Program providing the reduction to minimal models.

A separate volume will develop this program:

1. The *cone theorem* and *contraction theorem* generalize the theory of (-1)-curves to extremal rays of the Mori cone in any dimension.
2. *Flips* replace blow-downs when contractions produce singularities (this does not happen for surfaces, but is unavoidable in dimension ≥ 3).
3. The *abundance conjecture* (proved for surfaces, open in general) predicts that the pluricanonical map is always semi-ample for minimal varieties with $\kappa \geq 0$.

In positive characteristic, the classification has additional subtleties. For $\kappa = 0$: there exist *quasi-hyperelliptic surfaces* and *supersingular K3 surfaces* that have no analogue in characteristic zero. For general type: the BMY inequality $K^2 \leq 3\chi$ can fail in characteristic p, and Bombieri's theorem ($|5K|$ gives a birational map) requires modification. These topics remain active areas of research.

7.2.8 Further Topics on Surfaces

We conclude this chapter with several additional topics that complete the picture for surfaces and point toward future developments. As in §7.1.8, this section is more survey-like in character.

The Picard Scheme and Algebraic Equivalence

In the study of curves, we encountered the Jacobian variety $\mathrm{Jac}(X)$ representing the functor Pic^0. For surfaces, the corresponding object is the *Picard scheme* $\mathrm{Pic}_{X/k}$, a group scheme whose connected component $\mathrm{Pic}^0_{X/k}$ is an abelian variety of dimension $q = h^{0,1}(X)$ (the irregularity). The full Picard scheme has components indexed by degree (or, more precisely, by elements of the Néron–Severi group $\mathrm{NS}(X)$).

The various notions of equivalence for divisors on surfaces form a hierarchy:

1. **Linear equivalence** ($\sim$): $D_1 \sim D_2$ if $D_1 - D_2 = \mathrm{div}(f)$. This is classified by $\mathrm{Pic}(X)$.
2. **Algebraic equivalence** ($\sim_{\mathrm{alg}}$): $D_1 \sim_{\mathrm{alg}} D_2$ if there is a connected variety T and a divisor $\mathcal{D}$ on $X \times T$ such that $\mathcal{D}|_{X\times\{t_1\}} = D_1$ and $\mathcal{D}|_{X\times\{t_2\}} = D_2$ for $t_1, t_2 \in T$. The group $\mathrm{Pic}(X)/\sim_{\mathrm{alg}}$ is the Néron–Severi group $\mathrm{NS}(X)$.
3. **Numerical equivalence** ($\equiv$): $D_1 \equiv D_2$ if $D_1 \cdot C = D_2 \cdot C$ for all curves $C \subseteq X$. The group $\mathrm{Pic}(X)/\equiv$ is denoted $\mathrm{Num}(X)$.

The Néron–Severi theorem asserts that $\mathrm{NS}(X)$ is finitely generated, and a theorem of Matsusaka shows that $\mathrm{NS}(X)/\text{torsion} \cong \mathrm{Num}(X)$ (numerical and algebraic equivalence coincide modulo torsion).

For surfaces with $q = 0$ (such as $\mathbb{P}^2$, K3 surfaces, rational surfaces), $\operatorname{Pic}^0 = 0$ and all three notions coincide on $\operatorname{Pic}(X)_{\text{free}}$.

Vanishing Theorems

Vanishing theorems for cohomology of line bundles are among the most powerful tools in algebraic geometry. We state the two most important ones; their proofs require techniques beyond our current scope (Hodge theory for Kodaira vanishing, delicate positivity arguments for Kawamata–Viehweg).

Theorem 7.2.49: Kodaira Vanishing

Let X be a smooth projective variety over $\mathbb{C}$ and $\mathcal{L}$ an ample line bundle on X. Then:

$$H^i(X, \omega_X \otimes \mathcal{L}) = 0 \qquad \text{for all } i > 0.$$

Equivalently (by Serre duality): $H^i(X, \mathcal{L}^{-1}) = 0$ for $i < \dim X$.

For surfaces, Kodaira vanishing says: if $\mathcal{L}$ is ample on X, then $H^1(X, \omega_X \otimes \mathcal{L}) = H^2(X, \omega_X \otimes \mathcal{L}) = 0$. Combined with Riemann–Roch for surfaces (theorem 7.2.16), this converts χ into h^0 for ample twists of ω_X:

$$h^0(X, \omega_X \otimes \mathcal{L}) = \chi(\omega_X \otimes \mathcal{L}) = \frac{(K+D)^2 - (K+D)\cdot K}{2} + \chi(\mathcal{O}_X)$$

where D is the divisor of $\mathcal{L}$.

The following generalization is the key vanishing theorem used in the Minimal Model Program:

Theorem 7.2.50: Kawamata–Viehweg Vanishing

Let X be a smooth projective variety over $\mathbb{C}$ and D a nef and big divisor on X. Then:

$$H^i(X, \omega_X \otimes \mathcal{L}(D)) = 0 \qquad \text{for all } i > 0.$$

This extends Kodaira vanishing from ample to nef-and-big divisors—a crucial relaxation because the canonical divisor of a minimal surface of general type is nef and big but typically not ample.

7.2.51 Forward Reference: Vanishing Theorems in the MMP Kawamata–Viehweg vanishing and its refinements (Nadel vanishing, Kollár vanishing) are the technical engine of the Minimal Model Program. In a separate volume, these vanishing theorems will be used to:

1. Prove the *basepoint-free theorem*: if D is a nef divisor on a minimal variety with $aD - K$ ample for some $a > 0$, then $|mD|$ is base-point-free for $m \gg 0$.
2. Prove the *rationality theorem*: the threshold a in the cone theorem is rational.
3. Establish the *non-vanishing theorem*: if $K + D$ is nef and big, then $|m(K + D)| \neq \emptyset$ for $m \gg 0$.

These are the "big three" consequences of vanishing that drive the MMP. The reader who wishes to understand why vanishing theorems are so central should keep in mind that they convert Euler characteristic computations (which are topological) into dimension computations (which are geometric), by killing the "error terms" H^i for $i > 0$.

Arithmetic Surfaces

We close this chapter by returning to the arithmetic motif that has accompanied us throughout. An *arithmetic surface* is a 2-dimensional scheme $\mathcal{X}$ that is integral, Noetherian, and equipped with a proper flat morphism

$$f\colon \mathcal{X} \to \operatorname{Spec} \mathcal{O}_K$$

where $\mathcal{O}_K$ is the ring of integers of a number field K. The generic fiber $\mathcal{X}_\eta = \mathcal{X} \times_{\mathcal{O}_K} K$ is a curve over K, and the closed fibers $\mathcal{X}_\mathfrak{p} = \mathcal{X} \times_{\mathcal{O}_K} \mathcal{O}_K/\mathfrak{p}$ are curves over finite fields $\mathbb{F}_q$.

Thus an arithmetic surface is a "family of curves parametrized by Spec $\mathcal{O}_K$": it interpolates between a curve over a number field (the generic fiber) and curves over finite fields (the special fibers). The generic fiber is the "algebraic curve" whose arithmetic we wish to study, and the special fibers record its "reduction modulo primes."

Example 7.2.52: Arithmetic Surfaces in Practice

1. **The projective line over $\mathbb{Z}$:** $\mathcal{X} = \mathbb{P}^1_\mathbb{Z}$, with $f\colon \mathbb{P}^1_\mathbb{Z} \to \operatorname{Spec} \mathbb{Z}$. The generic fiber is $\mathbb{P}^1_\mathbb{Q}$ and the fiber over (p) is $\mathbb{P}^1_{\mathbb{F}_p}$. This is the "trivial" arithmetic surface.
2. **An elliptic curve over $\mathbb{Z}$:** Let $E\colon y^2 = x^3 - x$ over $\mathbb{Q}$. The *Néron model* $\mathcal{E} \to \operatorname{Spec} \mathbb{Z}$ is an arithmetic surface whose generic fiber is E and whose fibers over primes p are elliptic curves over $\mathbb{F}_p$ (for $p \neq 2$; the fiber over $p = 2$ is a singular curve, since E has bad reduction at 2).

The intersection theory on arithmetic surfaces is called *Arakelov theory*, named after Arakelov who developed it in the 1970s. The key idea is that Spec $\mathcal{O}_K$ is "missing" the archimedean places of K (the infinite primes), and Arakelov theory "compactifies" the arithmetic surface by adding data at infinity—specifically, a Hermitian metric on the line bundles at each archimedean place. This yields an intersection pairing

$$\mathrm{whDiv}(\mathcal{X}) \times \mathrm{whDiv}(\mathcal{X}) \to \mathbb{R}$$

(valued in $\mathbb{R}$, not $\mathbb{Z}$, because of the archimedean contributions) that extends the classical intersection pairing on surfaces. The arithmetic analogue of the adjunction formula, the Riemann–Roch theorem, and the Hodge index theorem all have Arakelov-theoretic versions, establishing a parallel between geometric and arithmetic intersection theory.

This is one of the most active frontiers of modern number theory. The Arakelov-theoretic Riemann–Roch theorem (Faltings, Gillet–Soulé) was used by Faltings in his proof of the Mordell conjecture (finiteness of rational points on curves of genus ≥ 2 over number fields), and Arakelov intersection theory continues to play a central role in the study of heights, Diophantine equations, and the arithmetic of moduli spaces.

This concludes our tour of curves and surfaces. We have traveled from the Riemann–Roch theorem for curves—the starting point of the entire theory—through Hurwitz's formula, the group law on elliptic curves, the intersection pairing on surfaces, blow-ups, and the Enriques–Kodaira classification. Along the way, we have seen the abstract machinery of schemes, cohomology, and duality transform into concrete geometric and arithmetic results.

The reader who has followed this chapter should be well-prepared for the next stages of the journey. The intersection theory of surfaces (§7.2.1) generalizes to the Chow ring and Chern classes of Part II. The minimal model theory for surfaces (§7.2.6) becomes the full Minimal Model Program, treated in a separate volume. The moduli of curves and surfaces foreshadowed throughout becomes the theory of stacks and deformation theory of Part III. And the arithmetic themes woven into every section find their expression in étale cohomology and derived categories, each treated in a separate volume. The landscape only grows richer from here.

7.3 A Glimpse Beyond: Higher-Dimensional Geometry

Throughout this chapter, we have built the theory of curves and surfaces around four interlocking themes: birational geometry (how varieties relate up to modification), intersection theory (the algebra of subvarieties meeting), classification (organizing varieties by discrete invariants), and cohomology (extracting information from sheaves). We have also noticed, repeatedly, that the passage from curves to surfaces introduced genuine complications—blow-ups, self-intersection, negative curves, the need for minimal models—that had no analogue in dimension one.

The reader may well wonder: what happens in dimension three and beyond? The short answer is that each of our four themes generalizes, but with essential new complications that required decades of effort and some of the biggest leaps mathematics of the 20th and 21st centuries to resolve. This section is a roadmap. We will state the main results and conjectures, explain the key new phenomena, and point to the parts of this book where the full theory will be developed. No proofs will be given; the goal is orientation, not rigor.

7.3.1 The Minimal Model Program

On curves, birational geometry is trivial: birational smooth projective curves are isomorphic (**??**). On surfaces, birational geometry is controlled by blow-ups and (-1)-curves: one successively contracts (-1)-curves to reach a minimal model, which is unique when $\kappa \geq 0$ (theorem 7.2.44). The entire process stays within the category of *smooth* varieties.

In dimension three and higher, this clean picture breaks. The fundamental problem is:

7.3.1 What Goes Wrong in Dimension ≥ 3 On a smooth threefold X, an *extremal contraction* (the analogue of contracting a (-1)-curve) can map a curve $C \subseteq X$ to a point. The image Y may be *singular*—unlike the surface case, where Castelnuovo's criterion guarantees smoothness of the contracted surface (theorem 7.2.31). Worse: when the contracted locus has codimension ≥ 2 (a "small contraction"), the image Y is not even $\mathbb{Q}$-factorial (Cartier divisors cannot be pulled back), and the standard intersection theory breaks down.
The solution is the *flip*: instead of accepting the singular image Y, one replaces the contracted locus by a different subvariety to obtain a new variety X^+ that is "less negative" in the K-direction. This surgery—called a *flip*—has no analogue in dimension 2.

The *Minimal Model Program* (MMP), developed by Mori, Kawamata, Kollár, Shokurov, and many others, systematizes this process. Here is the flowchart:

1. **Start** with a smooth projective variety X (or more generally, one with mild "terminal" singularities).
2. **Test:** Is K_X nef? If yes, X is a *minimal model*—stop.
3. **If not:** the Cone Theorem (Mori) guarantees the existence of a K_X*-negative extremal ray* $R \subseteq \overline{\mathrm{NE}}(X)$ and a corresponding *contraction morphism* $\varphi_R \colon X \to Y$.
4. **Three cases for the contraction:**
   (a) *Fiber type:* $\dim Y < \dim X$. Then X is a *Mori fiber space* (fibered over a lower-dimensional base with Fano fibers). This is the $\kappa = -\infty$ endpoint—stop.
   (b) *Divisorial:* The exceptional locus of φ_R has codimension 1. The image Y has terminal singularities, and $\rho(Y) = \rho(X) - 1$. Replace X by Y and go to step 2.
   (c) *Small (flipping):* The exceptional locus has codimension ≥ 2. Perform a *flip*: replace X by a new variety X^+ that agrees with X outside the exceptional locus but has "K-positive" curves where X had "K-negative" ones. Replace X by X^+ and go to step 2.
5. **Termination:** The process terminates after finitely many steps (since ρ decreases for divisorial contractions, and a separate argument handles flips).

The hard theorems, proved over the last four decades, are:

Theorem 7.3.2: Main Results of the MMP

1. **Cone Theorem** (Mori, Kawamata): The Mori cone $\overline{\mathrm{NE}}(X)$ is "rational polyhedral" on the K_X-negative side, generated by classes of rational curves.
2. **Existence of Flips** (Mori in dimension 3; Birkar–Cascini–Hacon–McKernan in general, 2010): Flips exist in all dimensions.
3. **Termination of Flips** (proved in dimension 3 by Shokurov; open in general, but the MMP terminates for varieties of general type by BCHM).
4. **Abundance Conjecture** (proved for surfaces and threefolds; open in general): If X is a minimal model (K_X nef), then K_X is *semi-ample*: some multiple $|mK_X|$ is base-point-free.

The BCHM theorem (2010) is one of the landmark results of 21st-century mathematics. It establishes the existence of minimal models for varieties of general type in all dimensions, resolving a conjecture that had been open for decades. A separate volume will develop this theory in full.

To appreciate the contrast with surfaces: on a surface, the MMP consists of at most $\rho - 1$ blow-downs, all within the smooth category, with no flips. The entire theory we developed in §7.2.4–§7.2.6 is the "trivial case" of the MMP—but it is the case that motivated the general program and provided the template.

Singularities in the MMP

The MMP forces us to work with singular varieties. The appropriate singularity classes form a hierarchy:

$$\text{smooth} \subsetneq \text{terminal} \subsetneq \text{canonical} \subsetneq \text{log terminal} \subsetneq \text{log canonical}.$$

Terminal singularities are the mildest: in dimension 2, they are smooth points (which is why the surface MMP stays smooth). In dimension 3, terminal singularities are isolated and classified (compound Du Val singularities). In higher dimensions, they are more complex but remain "mild" enough for intersection theory and vanishing theorems to work. The Kawamata–Viehweg vanishing theorem (theorem 7.2.50) extends to varieties with log terminal singularities, and this extension is what makes the MMP function.

7.3.2 Intersection Theory and Riemann–Roch in Higher Dimensions

On curves, the "intersection" of a divisor with the curve itself is just the degree. On surfaces, we developed a bilinear pairing $\operatorname{Pic}(X) \times \operatorname{Pic}(X) \to \mathbb{Z}$. In general, on a smooth projective variety X of dimension n, one needs a *multilinear* intersection product: n divisors $D_1, \ldots, D_n$ intersect in a number $D_1 \cdot D_2 \cdots D_n \in \mathbb{Z}$.

This is formalized by the *Chow ring*:

Definition 7.3.3: The Chow Ring (Preview)

For a smooth projective variety X of dimension n, the *Chow ring* is a graded commutative ring:

$$A^*(X) = \bigoplus_{p=0}^{n} A^p(X)$$

where $A^p(X)$ is the group of codimension-p algebraic cycles modulo rational equivalence (definition 8.1.4), with a ring structure given by the *intersection product*. The degree map $\deg \colon A^n(X) \to \mathbb{Z}$ (sending the class of a point to 1) completes the picture: the product $D_1 \cdots D_n \in A^n(X) \cong \mathbb{Z}$ gives the intersection number.

For surfaces, $A^0 = \mathbb{Z}$, $A^1 = \operatorname{Pic}(X)$, $A^2 = \mathbb{Z}$, and the intersection product is the pairing we defined in §7.2.1. In higher dimensions, the Chow ring is richer: A^p for $1 < p < n-1$ involves cycles of intermediate codimension (curves, surfaces, ... on X), and the intersection product is no longer just a bilinear pairing but a full ring structure.

The *moving lemma* (proved in Part II) ensures that any two cycles can be moved to intersect properly, making the product well-defined. The proof is nontrivial and uses the geometry of the ambient projective space.

Chern Classes and Riemann–Roch

In dimension 1, the Riemann–Roch theorem involves the degree of a line bundle (a single number). In dimension 2, it involves D^2 and $D \cdot K$ (two numbers, assembled from intersection theory). The pattern becomes clear in the general *Hirzebruch–Riemann–Roch theorem*:

Theorem 7.3.4: Hirzebruch–Riemann–Roch (Preview)

Let X be a smooth projective variety of dimension n and $\mathscr{E}$ a vector bundle on X. Then:

$$\chi(\mathscr{E}) \;=\; \int_X \mathrm{ch}(\mathscr{E}) \cdot \mathrm{td}(T_X)$$

where $\mathrm{ch}(\mathscr{E}) = \mathrm{rk}(\mathscr{E}) + c_1(\mathscr{E}) + \frac{1}{2}(c_1^2 - 2c_2) + \cdots$ is the *Chern character* and $\mathrm{td}(T_X) = 1 + \frac{1}{2}c_1(T_X) + \frac{1}{12}(c_1^2 + c_2) + \cdots$ is the *Todd class*. The integral $\int_X$ means "take the degree-n component and apply $\deg\colon A^n(X) \to \mathbb{Z}$."

For a line bundle $\mathcal{L}$ on a curve: $\mathrm{ch}(\mathcal{L}) = 1 + c_1(\mathcal{L})$, $\mathrm{td}(T_X) = 1 + \frac{1}{2}c_1 = 1 + (1-g)$ (on the degree-1 part), recovering $\chi(\mathcal{L}) = \deg \mathcal{L} + 1 - g$. For a line bundle on a surface: one recovers $\chi(\mathcal{L}) = \frac{1}{2}D(D-K) + \chi(\mathcal{O}_X)$ (theorem 7.2.16). For a threefold, the formula involves D^3, $D^2 \cdot K$, $D \cdot K^2$, $D \cdot c_2(X)$, and $\chi(\mathcal{O}_X)$—the complexity grows, but the structure is always the same.

The generalization is the *Grothendieck–Riemann–Roch theorem* (GRR), which computes the behavior of the Chern character under proper pushforward:

$$\mathrm{ch}(f_! \mathscr{E}) \;=\; f_*(\mathrm{ch}(\mathscr{E}) \cdot \mathrm{td}(T_{X/Y}))$$

for a proper morphism $f\colon X \to Y$. This is the "master theorem" of which HRR and the classical RR for curves are special cases (taking $Y = \operatorname{Spec} k$). Part II will develop Chern classes, prove HRR, and state GRR; Part II will also develop the intersection theory needed to make sense of the formulas.

7.3.3 The Geography of Higher-Dimensional Varieties

The Enriques–Kodaira classification organized surfaces by Kodaira dimension $\kappa \in \{-\infty, 0, 1, 2\}$. In dimension n, the Kodaira dimension takes values in $\{-\infty, 0, 1, \ldots, n\}$, and the classification by κ remains the organizing principle—but the "geography" within each class is far richer.

$\kappa = -\infty$: Fano Varieties

The higher-dimensional analogues of rational surfaces and Del Pezzo surfaces are the *Fano varieties*: smooth projective varieties X with $-K_X$ ample. These are the varieties with "the most rational curves."

A remarkable theorem constrains their complexity:

Theorem 7.3.5: Boundedness of Fano Varieties (BAB Conjecture)

Fano varieties of dimension n with at worst ϵ-log terminal singularities form a *bounded family*: they can all be parametrized by a finite number of algebraic families. In particular, there are only finitely many deformation types of smooth Fano varieties of each dimension.

This was conjectured by Alexeev and Borisov–Borisov and proved by Birkar (2019, for which he received the Fields Medal). In dimension 2, this is the elementary fact that Del Pezzo surfaces are blow-ups of $\mathbb{P}^2$ at ≤ 8 points. In dimension 3, the classification of smooth Fano threefolds (by Iskovskikh and Mori–Mukai) identifies 105 deformation families. In dimension 4 and beyond, the explicit classification is open, but boundedness is guaranteed.

$\kappa = 0$: Calabi–Yau Varieties and Holomorphic Symplectic Varieties

The $\kappa = 0$ case for surfaces gave us four types: abelian, K3, Enriques, and bielliptic. In higher dimensions, the "building blocks" of the $\kappa = 0$ world are conjectured to be three types (the *Beauville–Bogomolov decomposition*):

1. **Abelian varieties**: n-dimensional complex tori $\mathbb{C}^n/\Lambda$ with a polarization.
2. **Calabi–Yau varieties (strict sense)**: simply connected varieties with $K_X \sim 0$ and $H^i(X, \mathcal{O}_X) = 0$ for $0 < i < n$. The prototype: a smooth quintic threefold in $\mathbb{P}^4$ (the "quintic Calabi–Yau," which has $h^{1,1} = 1$ and $h^{2,1} = 101$).
3. **Irreducible holomorphic symplectic (hyperkähler) varieties**: simply connected with $H^0(X, \Omega_X^2) = \mathbb{C} \cdot \sigma$ for a nondegenerate 2-form σ. The prototype: Hilbert schemes of points on K3 surfaces.

Calabi–Yau threefolds are central to *mirror symmetry*, a profound duality (originating in string theory) that exchanges $h^{1,1}$ and $h^{n-1,1}$ between "mirror pairs" of Calabi–Yau manifolds. Mirror symmetry has led to spectacular predictions in enumerative geometry (counting rational curves), many of which have been proved rigorously. The mathematical framework for mirror symmetry involves derived categories (a separate volume) and the theory of Gromov–Witten invariants.

$\kappa = \dim X$: Varieties of General Type

The higher-dimensional analogue of surfaces of general type consists of varieties where K_X is big. The BCHM theorem guarantees the existence of a minimal model, and the *canonical model* $X_{\text{can}} = \operatorname{Proj} \bigoplus_n H^0(X, \omega_X^{\otimes n})$ is well-defined.

The moduli theory of varieties of general type is governed by the *KSBA compactification* (Kollár–Shepherd-Barron–Alexeev), which generalizes the Deligne–Mumford compactification $\overline{\mathcal{M}}_g$ of the moduli of curves. In dimension 2, the KSBA moduli space parametrizes "stable surfaces" (surfaces with semi-log-canonical singularities and ample canonical class), and its geometry is an active area of research. The boundedness of the moduli space (finiteness of components for fixed invariants) was established by Hacon–McKernan–Xu, building on the BCHM machinery.

7.3.4 Cohomological Frontiers

The cohomological tools we have used in this chapter—sheaf cohomology, Serre duality, the Euler characteristic—are just the beginning. We close with a brief survey of the cohomological theories that drive modern algebraic geometry in higher dimensions.

Étale Cohomology and the Weil Conjectures

Over $\mathbb{C}$, our cohomological invariants (Betti numbers, Hodge numbers) come from the topology of $X(\mathbb{C})$. Over finite fields $\mathbb{F}_q$, there is no underlying topological space in the classical sense, yet the Weil conjectures predict that the number of $\mathbb{F}_{q^n}$-points of a smooth projective variety satisfies remarkable properties: rationality of the zeta function, a functional equation, and the "Riemann hypothesis" (bounds on eigenvalues of Frobenius).

Grothendieck's invention of *étale cohomology* provides the ℓ-adic cohomology groups $H^i_{\text{ét}}(X, \mathbb{Q}_\ell)$ that play the role of singular cohomology over finite fields. The Weil conjectures, proved by Deligne in 1974 using étale cohomology, are:

Theorem 7.3.6: The Weil Conjectures (Deligne)

Let X be a smooth projective variety of dimension n over $\mathbb{F}_q$. Then:

1. The zeta function $Z(X,t) = \exp\left(\sum_{m=1}^{\infty} \frac{|X(\mathbb{F}_{q^m})|}{m} t^m\right)$ is a rational function of t.
2. $Z(X,t)$ satisfies a functional equation relating $Z(X,t)$ and $Z(X, q^{-n}t^{-1})$.
3. **Riemann Hypothesis:** The eigenvalues of Frobenius on $H^i_{\text{ét}}(X, \mathbb{Q}_\ell)$ have absolute value $q^{i/2}$.

For curves, this is Weil's theorem (the Hasse–Weil bound, example 7.1.66). For elliptic curves, the Riemann Hypothesis is Hasse's bound $|a_q| \leq 2\sqrt{q}$ (box 7.1.59). The full Weil conjectures in all dimensions required a vast new cohomological apparatus, and their proof by Deligne is one of the crowning achievements of 20th century mathematics. A separate volume will develop étale cohomology and present Deligne's proof.

Hodge Theory in Higher Dimensions

Over $\mathbb{C}$, the Hodge decomposition (theorem 0.6.28) expresses $H^k(X, \mathbb{C})$ as a direct sum of $H^{p,q}$ spaces. For surfaces, we used the Hodge diamond and the Hodge index theorem (theorem 7.2.22) extensively. In higher dimensions, the Hodge–Riemann bilinear relations, the Hard Lefschetz theorem, and the Lefschetz $(1,1)$-theorem all generalize, but the *Hodge conjecture* remains wide open:

Every rational (p,p)-class on a smooth projective variety over $\mathbb{C}$ is algebraic.

For $p = 1$, this is the Lefschetz $(1,1)$-theorem (true, and used implicitly throughout our surface theory: it is the reason $\mathrm{NS}(X)$ captures all algebraic classes in $H^{1,1}$). For $p \geq 2$, the conjecture is one of the Clay Millennium Prize Problems. Its resolution would profoundly clarify the relationship between topology and algebraic geometry.

Derived Categories

Perhaps the most transformative development in algebraic geometry over the past thirty years is the realization that the *derived category* $D^b(\mathbf{Coh}(X))$—the bounded derived category of coherent sheaves—is a finer invariant of X than any collection of classical invariants (Hodge numbers, Chern numbers, etc.).

Two striking results illustrate this:

1. **Mukai's theorem:** An abelian variety A and its dual wh$A$ have equivalent derived categories: $D^b(\mathbf{Coh}(A)) \simeq D^b(\mathbf{Coh}(\mathrm{wh}A))$, even though A and wh$A$ need not be isomorphic. The equivalence is given by a *Fourier–Mukai transform*—a "categorified Fourier transform" whose kernel is the Poincaré bundle on $A \times \mathrm{wh}A$.

2. **Bondal–Orlov:** If X is a smooth projective variety with K_X ample or $-K_X$ ample (Fano or anti-Fano), then X is determined up to isomorphism by its derived category. But for $\kappa = 0$ (Calabi–Yau, abelian), derived equivalences can relate non-isomorphic varieties.

The derived category is also the natural home for *homological mirror symmetry* (Kontsevich, 1994): for a mirror pair $(X, X^\vee)$ of Calabi–Yau manifolds, the conjecture predicts an equivalence $D^b(\mathbf{Coh}(X)) \simeq D^\pi(\mathrm{Fuk}(X^\vee))$ between the derived category of coherent sheaves on X and the derived Fukaya category of $X^\vee$ (a symplectic invariant). This conjecture has been proved in several cases and remains one of the most active research frontiers at the intersection of algebraic geometry, symplectic geometry, and mathematical physics. A separate volume will develop the foundations of derived categories.

7.3.7 The Road Ahead The reader who has completed this chapter has seen, in miniature, the essential phenomena of algebraic geometry: the interplay of algebra and geometry (schemes), the power of cohomology (Riemann–Roch, Serre duality), the role of positivity (ampleness, nefness, the Kodaira dimension), and the richness of classification (genus for curves, the Enriques–Kodaira table for surfaces). Parts II and III of this book develop the first two of these themes to their full generality:

Part	Theme	Generalization of...
II	Intersection Theory	§7.2.1, §7.2.3
III	Moduli and Stacks	§7.1.7, §??

Three further directions leave this book for their own volumes in the series: birational geometry and the Minimal Model Program, étale cohomology, and derived categories.
The journey from curves to the frontier of modern algebraic geometry is a long one, but the landscape is stunning. Let us continue.

7.3.5 Hironaka's Example

Shafarevich in his book outlined this example. I would like to include it here as well to actually have an example of a proper non-projective scheme.

Example 7.3.8: Hironaka's Example
HERE

Part II

Intersection Theory

7.3.9 Inspiration For now, this part will very heavily be inspired by Fulton [22] and Vakil [21]

Throughout most this chapter, schemes shall always be $\overline{k}$-schemes of finite type over an algebraically closed field. A point shall be a closed point unless stated otherwise. If needed, such a scheme shall be called an *algebraic scheme*.

8

Rational Equivalence

Recall from [**NateClassicalAlgGeo**] that we had to define *projective plane curves* to be an equivalence relation on homogeneous polynomials of $k[x, y, z]$ up to a constant ($f \sim g$ if and only if $f = \alpha g$). This was specifically done keep track of multiplicity information. By the correspondence between homogeneous and affine curves, we can without loss of generality work with $[f(x, y)], [g(x, y)] \in k[x, y]$ and choose any representative; we shall simply call these *plane curves*. Given a choice of plane curves f, g let F, G denote the resulting variety. The intersection of two (affine or projective) plane curves F, G at P was defined as:

$$i(P, f \cdot g) = \dim_K \frac{\mathcal{O}\mathbb{A}^2_k, p}{(f, g)} = \dim_K \mathcal{O}_{Z,P}$$

where $Z = \mathcal{Z}(f, g)$. It was proven that:

1. (commutative on intersection): $i(P, F \cdot G) = i(P, G \cdot F)$
2. (linear on intersection): if $F_1 + F_2$ is the curve defined by $f_1 f_2$, then $i(P, (F_1 + F_2) \cdot G) = i(P, F_1 \cdot G) = i(P, F_2 \cdot G)$
3. (invariance of representation): if F' is given by $f + gh$ for some $h \in k[x, y]$:

$$i(P, F' \cdot G) = i(P, F \cdot G)$$

4. If F, G have have a common component through P (i.e. the dimension of the irreducible variety in $F \cap G$ containing p is greater than 0), then $i(P, F \cdot G) = \infty$. Otherwise, $i(P, F \cdot G) \in \mathbb{N}_{\geq 0}$.
5. If $P \notin F \cap G$:

$$i(P, F \cdot G) = 0$$

6. if $f(x,y) = x - a$ and $g(x,y) = y - b$ (more generally $\frac{\partial(f,g)}{\partial(x,y)} \neq 0$ at p) then:

$$i(P, F \cdot G) = 1$$

7. If F has no common component with G and H and P is a simple point of F:

$$i(P, G \cdot H) \geq \min i(P, F \cdot G), i(P, F \cdot H)$$

8.1 Cycles and Rational Equivalence

The classical intersection number recalled above attaches an integer to a pair of plane curves meeting at a point. Turning that local count into a theory valid on any variety and in every dimension calls for a home for “formal combinations of subvarieties” together with an equivalence relation flexible enough to make intersection numbers well defined. That home is the Chow group, the higher-dimensional descendant of the divisor class group of definition 5.4.7: where the divisor class group records codimension-one subvarieties modulo the divisors of rational functions, the Chow group records subvarieties of *every* dimension modulo the same mechanism.

This section builds the three objects in turn, the group of cycles $Z_k(X)$, the subgroup $\mathrm{Rat}_k(X)$ of cycles rationally equivalent to zero, and the quotient $A_k(X)$. Throughout the chapter X is an algebraic scheme (a scheme of finite type over the algebraically closed field $\overline{k}$), a *subvariety* means a closed integral subscheme, and a point means a closed point unless stated otherwise.

The construction rests on the order of vanishing of a rational function along a codimension-one subvariety, introduced for divisors in section 5.4. On a variety W a prime divisor is a codimension-one subvariety V, and the local ring $\mathcal{O}_{W,V}$ is a one-dimensional local domain with fraction field the function field $k(W)$. When W is regular in codimension one this ring is a discrete valuation ring and its valuation ν_V records orders of vanishing (definition 5.4.6); the length-theoretic definition below removes the regularity hypothesis, which matters as soon as W is an arbitrary subvariety of X.

Definition 8.1.1: Order Along a Subvariety

Let W be a variety and $V \subseteq W$ a codimension-one subvariety, and set $A = \mathcal{O}_{W,V}$, a one-dimensional local domain with fraction field $k(W)$. For nonzero $a \in A$ the quotient $A/(a)$ has finite length, and the *order of* a *along* V is

$$\mathrm{ord}_V(a) = \mathrm{length}_A\left(A/(a)\right)$$

Because $\mathrm{ord}_V(ab) = \mathrm{ord}_V(a) + \mathrm{ord}_V(b)$, this extends to a homomorphism $\mathrm{ord}_V : k(W)^* \to \mathbb{Z}$ by $\mathrm{ord}_V(a/b) = \mathrm{ord}_V(a) - \mathrm{ord}_V(b)$.

When $\mathcal{O}_{W,V}$ is a discrete valuation ring, in particular whenever W is normal, this length equals the valuation $\nu_V(a)$, so definition 8.1.1 agrees with the order function of section 5.4. For a fixed $r \in k(W)^*$ the integer $\mathrm{ord}_V(r)$ vanishes for all but finitely many V (lemma 5.4.5), so each sum $\sum_V \mathrm{ord}_V(r)[V]$ below is finite.

Definition 8.1.2: Cycle

Let X be an algebraic scheme. A k-*cycle* on X is a finite formal $\mathbb{Z}$-linear combination $\sum_i n_i[V_i]$ of k-dimensional subvarieties $V_i \subseteq X$ with $n_i \in \mathbb{Z}$. The k-cycles form the free abelian group

$$Z_k(X) = \bigoplus_{\dim V = k} \mathbb{Z}\,[V]$$

on the set of k-dimensional subvarieties of X, and the group of all cycles is $Z_*(X) = \bigoplus_{k\geq 0} Z_k(X)$.

A cycle remembers only which subvarieties occur and with what integer multiplicity; it detects neither embedded nor nonreduced structure, a point made precise by the equality $Z_k(X) = Z_k(X_{\text{red}})$. Two cases sit at the extremes. A 0-cycle is a finite formal sum of closed points. At the top, if X has dimension n with n-dimensional irreducible components $X_1, \ldots, X_r$, then $Z_n(X)$ is free on $[X_1], \ldots, [X_r]$; the *fundamental cycle*

$$[X] = \sum_i \operatorname{length}_{\mathcal{O}_{X,X_i}}\left(\mathcal{O}_{X,X_i}\right)[X_i]$$

weights each component by the multiplicity with which X carries it, and reduces to $\sum_i [X_i]$ when X is reduced.

The one construction that produces relations among cycles is the divisor of a rational function. Given a $(k+1)$-dimensional subvariety $W \subseteq X$ and a nonzero $r \in k(W)^*$, its *divisor* is the k-cycle

$$[\operatorname{div}(r)] = \sum_V \operatorname{ord}_V(r)\,[V] \in Z_k(X)$$

the sum running over the codimension-one subvarieties V of W, each of which is a k-dimensional subvariety of X. The sum is finite by lemma 5.4.5, and $[\operatorname{div}(rs)] = [\operatorname{div}(r)] + [\operatorname{div}(s)]$ because ord_V is a homomorphism.

Definition 8.1.3: Rational Equivalence

A k-cycle $\alpha \in Z_k(X)$ is *rationally equivalent to zero*, written $\alpha \sim 0$, if there are finitely many $(k+1)$-dimensional subvarieties $W_i \subseteq X$ and rational functions $r_i \in k(W_i)^*$ with

$$\alpha = \sum_i [\operatorname{div}(r_i)]$$

Two k-cycles are *rationally equivalent*, written $\alpha \sim \beta$, when $\alpha - \beta \sim 0$. The cycles rationally equivalent to zero form a subgroup $\operatorname{Rat}_k(X) \leq Z_k(X)$: it is closed under addition by definition and under negation because $[\operatorname{div}(r^{-1})] = -[\operatorname{div}(r)]$.

Geometrically, a subvariety W together with a rational function r presents $[\operatorname{div}(r)]$ as the difference between the zeros and the poles of r, that is, between the fibers over 0 and ∞ of the rational map $r\colon W \dashrightarrow \mathbb{P}^1$. Rational equivalence is thus the relation generated by sweeping one cycle to another through a family parametrized by the line. In the one codimension seen before, $k = n-1$ on an n-dimensional variety, a cycle is a Weil divisor and rational equivalence is exactly the linear equivalence of definition 5.4.7; the Chow group below carries $\operatorname{Cl} X$ to all dimensions at once.

Definition 8.1.4: Chow Group

The *k-th Chow group* of an algebraic scheme X is the quotient

$$A_k(X) = Z_k(X)/\mathrm{Rat}_k(X)$$

Its elements are *k-cycle classes*, and the class of a cycle α is written $[\alpha]$, or simply α when no confusion results. The *total Chow group* is

$$A_*(X) = \bigoplus_{k\geq 0} A_k(X) = Z_*(X)/\mathrm{Rat}_*(X), \qquad \mathrm{Rat}_*(X) = \bigoplus_{k\geq 0} \mathrm{Rat}_k(X)$$

Following the cohomological convention, much of the literature grades by codimension, writing $A^p(X) = A_{n-p}(X)$ or $CH^p(X)$ when X is equidimensional of dimension n; the two indexings carry the same information, and definition 8.1.4 is the promised general form of the construction previewed for surfaces in box 7.2.14. A cycle $\sum_i n_i[V_i]$ is *positive* (or *effective*) if it is nonzero with every $n_i \geq 0$, and a cycle class is positive if it has a positive representative. Positivity is the algebraic trace of an actual geometric subvariety, and deciding which classes are positive is among the central difficulties of the subject.

Passing to the quotient collapses exactly the ambiguity that a complete variety imposes on its points.

Example 8.1.5: Zero-Cycles on the Line

On $X = \mathbb{P}^1$ any two closed points are rationally equivalent. Write t for the affine coordinate and ∞ for the point at infinity. Distinct points $p, q \in \mathbb{A}^1 \subseteq \mathbb{P}^1$ are cut out by $t-p$ and $t-q$, and the rational function $r = (t-p)/(t-q)$ has

$$\mathrm{ord}_p(r) = 1, \qquad \mathrm{ord}_q(r) = -1, \qquad \mathrm{ord}_x(r) = 0 \text{ for } x \neq p, q$$

since r takes the finite nonzero value 1 at ∞. Hence $[\mathrm{div}(r)] = [p] - [q]$ and $[p] \sim [q]$; taking $r = t - p$ instead gives $[\mathrm{div}(t-p)] = [p] - [\infty]$, so $[\infty]$ lies in the same class. Every point of $\mathbb{P}^1$ therefore represents a single class h. A rational function on $\mathbb{P}^1$ has as many zeros as poles, so $\sum_V \mathrm{ord}_V(r) = 0$ for all r, and the total-coefficient map $\sum_i n_i[p_i] \mapsto \sum_i n_i$ descends to an isomorphism

$$A_0(\mathbb{P}^1) \xrightarrow{\sim} \mathbb{Z}, \qquad h \mapsto 1$$

The affine line behaves oppositely. On $X = \mathbb{A}^1$ the function $t-p$ has the single zero p and no pole, so $[\mathrm{div}(t-p)] = [p]$ and every point satisfies $[p] \sim 0$; thus $A_0(\mathbb{A}^1) = 0$. On an affine line a point can be swept away to nothing, whereas on the complete line it cannot, and the surviving invariant of a class is its degree.

The contrast in example 8.1.5 is why completeness recurs throughout the theory: the degree of a zero-cycle is an invariant of its rational equivalence class only on a complete scheme, a point taken up in section 8.2. The remaining first properties of Chow groups are collected next; each is either immediate or follows from the localization sequence established later in the chapter.

Example 8.1.6: First Properties of Chow Groups

Let X be an algebraic scheme.

1. *Reduction.* Since cycles see only subvarieties, $A_k(X) = A_k(X_{\mathrm{red}})$ for every k.

2. *Disjoint unions.* If $X = X' \sqcup X''$ is a disjoint union of open-and-closed subschemes, then $A_k(X) = A_k(X') \oplus A_k(X'')$, as every subvariety lies in one piece.

3. *Top-dimensional cycles.* If $\dim X = n$, there is no subvariety of dimension $n+1$, so $\mathrm{Rat}_n(X) = 0$ and $A_n(X) = Z_n(X)$ is free on the n-dimensional components of X. In particular the coefficient of such a component in a class $\alpha \in A_n(X)$ is well defined.

4. *Right-exactness.* For closed subschemes $X_1, X_2 \subseteq X$ there is an exact sequence

$$A_k(X_1 \cap X_2) \to A_k(X_1) \oplus A_k(X_2) \to A_k(X_1 \cup X_2) \to 0$$

with no left-exactness in general. This is a consequence of the localization sequence developed later in the chapter.

Exercise 8.1.1

1. Let X be the reduced scheme consisting of two distinct points. Compute $Z_k(X)$ and $A_k(X)$ for all k.

2. Show that $A_0(\mathbb{A}^n) = 0$ for every $n \geq 1$.

3. Prove from the definitions that $Z_k(X) = Z_k(X_{\mathrm{red}})$ and $\mathrm{Rat}_k(X) = \mathrm{Rat}_k(X_{\mathrm{red}})$, and deduce $A_k(X) = A_k(X_{\mathrm{red}})$.

4. Let X have dimension n with n-dimensional irreducible components $X_1, \ldots, X_r$. Show that $A_n(X) = \bigoplus_{i=1}^r \mathbb{Z}\,[X_i]$.

5. Let C be a smooth projective curve over $\overline{k}$. Show that $A_0(C) \cong \mathrm{Cl}\, C = \mathrm{Pic}\, C$, and that the degree gives a short exact sequence $0 \to \mathrm{Pic}^0 C \to A_0(C) \xrightarrow{\deg} \mathbb{Z} \to 0$.

8.2 Proper Pushforward and the Degree

Cycles and their classes change under morphisms, and the first of these functorialities is covariant: a proper morphism pushes cycles forward, respects rational equivalence, and on a complete scheme produces the degree of a zero-cycle.

A proper morphism $f\colon X \to Y$ carries closed sets to closed sets, so the image of a subvariety $V \subseteq X$ is again a subvariety $W = f(V) \subseteq Y$: the image of an irreducible set is irreducible, and properness (universal closedness) makes it closed. Without properness the image can fail to be closed, as for the projection of example 3.5.24, and pushforward is not defined. The inclusion of function fields $k(W) \hookrightarrow k(V)$ is a finite extension exactly when $\dim V = \dim W$, and its degree counts the points of V over a general point of W.

Definition 8.2.1: Proper Pushforward

Let $f\colon X \to Y$ be a proper morphism and $V \subseteq X$ a subvariety with image $W = f(V)$. Set

$$\deg(V/W) = \begin{cases} [k(V) : k(W)] & \dim V = \dim W \\ 0 & \dim V > \dim W \end{cases}$$

(the extension $k(W) \hookrightarrow k(V)$ is finite in the first case), and put $f_*[V] = \deg(V/W)\,[W]$. Extending $\mathbb{Z}$-linearly gives the *pushforward* $f_*\colon Z_k(X) \to Z_k(Y)$.

Pushforward is covariant: for composable proper morphisms $(g \circ f)_* = g_* \circ f_*$, since degrees of function-field extensions multiply in towers. When $\dim V = \dim W$ the number $\deg(V/W)$ counts the sheets of the generically finite cover $V \to W$, matching the topological degree over $\mathbb{C}$. That f_* descends to Chow groups is the content of this section, and rests on how it acts on the divisor of a rational function.

Proposition 8.2.2: Pushforward of Divisor

Let $f : X \to Y$ be a proper surjective morphisms between varieties, and let $r \in k(X)^*$. Then:

1. if $\dim(Y) < \dim(X)$, then
$$f_*[\mathrm{div}(r)] = 0$$
2. if $\dim(Y) = \dim(X)$, then
$$f_*[\mathrm{div}(r)] = [\mathrm{div}(N(r))]$$
where $N(r)$ is the *norm* of r (computed as the determinant of the $R(Y)$-linear endomorphism of $R(X)$ given by multiplying by r).

***Proof*:**
The order function and the norm are multiplicative, so both sides of each identity are homomorphisms in r. Both parts reduce to two model computations; write $n = \dim X$.

Step 1: the line over a point. Let $Y = \operatorname{Spec} k$ and $X = \mathbb{P}^1$, with f the structure map. Every closed point of $\mathbb{P}^1$ has residue field $k = \overline{k}$, so $f_*[p] = [Y]$ for each p and $f_*[\mathrm{div}(r)] = \big(\sum_V \mathrm{ord}_V(r)\big)[Y]$. Writing $r = c\prod_j (t - a_j)^{m_j}$ in the affine coordinate t, the order of r at a_j is m_j and its order at ∞ is $-\sum_j m_j$ (in $s = 1/t$ one has $r = s^{-\sum_j m_j} u$ with u a unit at $s = 0$), so $\sum_V \mathrm{ord}_V(r) = 0$ and $f_*[\mathrm{div}(r)] = 0$. The same count shows that on any proper curve C over a field K a principal divisor has degree zero: normalize C and choose a finite map $\widetilde{C} \to \mathbb{P}^1_K$, then apply Step 2 to that map and this computation to $\mathbb{P}^1_K \to \operatorname{Spec} K$.

Step 2: finite morphisms of equal dimension. Suppose f is finite with $\dim Y = \dim X$, so that $K = k(Y) \subseteq L = k(X)$ is a finite extension. Fix a codimension-one subvariety $W \subseteq Y$ and let $A = \mathcal{O}_{Y,W}$, a one-dimensional local domain with fraction field K. Its integral closure B in L is a finite A-module whose maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_s$ correspond to the codimension-one subvarieties $V_i \subseteq X$ over W, with $B_{\mathfrak{m}_i} = \mathcal{O}_{X,V_i}$ and residue degree $[k(V_i) : k(W)] = [B/\mathfrak{m}_i : A/\mathfrak{m}]$. Only these V_i contribute to the coefficient of $[W]$ in $f_*[\mathrm{div}(r)]$, which is therefore $\sum_i [k(V_i) : k(W)]\,\mathrm{ord}_{V_i}(r)$, while the coefficient of $[W]$ in $[\mathrm{div}(N(r))]$ is $\mathrm{ord}_W(N(r))$. For $r \in B$ a nonzerodivisor these agree:

$$\operatorname{ord}_W(N(r)) = \operatorname{length}_A\big(A/N(r)A\big) = \operatorname{length}_A\big(B/rB\big) = \sum_i [k(V_i) : k(W)] \operatorname{ord}_{V_i}(r)$$

the middle equality is the determinant-length formula, since $N(r)$ is the determinant of multiplication by r on the finite A-module B, and the last is additivity of length across the semilocal decomposition of B, weighted by residue degrees (see [22, Appendix A] for this length bookkeeping). As $L = \operatorname{Frac}(B)$ and both sides are homomorphisms, the identity holds for all $r \in L^*$, which is part (2) for finite f.

Step 3: the general case. A proper surjective f with $\dim Y = \dim X$ is generically finite, and the coefficient of a codimension-one $W \subseteq Y$ in $f_*[\operatorname{div}(r)]$ is computed over the generic point of W, where f becomes finite; Step 2 gives (2). For (1), let $\dim Y < n$. Since $f_*[\operatorname{div}(r)] \in Z_{n-1}(Y)$ vanishes when $\dim Y < n-1$, only $\dim Y = n-1$ remains, and there $f_*[\operatorname{div}(r)] = c\,[Y]$ with c its coefficient over the generic point of Y. Base-changing to $K = k(Y)$ replaces f by a proper curve $C = X_K$ over the field K and r by a function in $k(C)^*$, and $c = \sum_{P \in C} \operatorname{ord}_P(r)\,[k(P) : K]$ is the degree of its principal divisor, which vanishes by Step 1. Hence $f_*[\operatorname{div}(r)] = 0$, completing the proof.

Theorem 8.2.3: Pushforward Preserves Rational Equivalence

Let $f\colon X \to Y$ be proper. If $\alpha \in Z_k(X)$ is rationally equivalent to zero, then so is $f_*\alpha$. Consequently f_* descends to a homomorphism

$$f_*\colon A_k(X) \to A_k(Y)$$

and $A_*(-)$ equipped with proper pushforward is a covariant functor.

***Proof*:**
It suffices to push forward a generator $\alpha = [\operatorname{div}(r)]$, where $r \in k(W)^*$ for some $(k+1)$-dimensional subvariety $W \subseteq X$. Replacing X by W and Y by $f(W)$ makes f proper and surjective, and proposition 8.2.2 identifies $f_*[\operatorname{div}(r)]$ with 0 (when $\dim f(W) < \dim W$) or with $[\operatorname{div}(N(r))]$ (when $\dim f(W) = \dim W$). Either value is a principal divisor, hence rationally equivalent to zero, so $f_*\alpha \sim 0$. Functoriality on classes then follows from $(g \circ f)_* = g_* \circ f_*$ on cycles, as we sought to show.

Definition 8.2.4: Degree of a Zero-Cycle

Let X be complete (proper over $\operatorname{Spec} k$), with structure morphism $\pi\colon X \to \operatorname{Spec} k$, and let $\alpha = \sum_P n_P[P]$ be a zero-cycle on X. The *degree* of α is

$$\deg(\alpha) = \int_X \alpha = \sum_P n_P\,[k(P) : k]$$

Since $\operatorname{Spec} k$ is a single point with no lower-dimensional subvarieties, $A_0(\operatorname{Spec} k) = Z_0(\operatorname{Spec} k) = \mathbb{Z}\,[\operatorname{Spec} k] \cong \mathbb{Z}$, and under this identification $\deg(\alpha) = \pi_*\alpha$. When the scheme is clear one writes $\int \alpha$ for $\int_X \alpha$.

By theorem 8.2.3 rationally equivalent zero-cycles have the same degree, so $\deg\colon A_0(X) \to \mathbb{Z}$ is well defined on a complete scheme. Extending by $\int_X \alpha = 0$ for $\alpha \in A_k(X)$ with $k > 0$ and using $\pi_X = \pi_Y \circ f$, any morphism $f\colon X \to Y$ of complete schemes satisfies

$$\int_X \alpha = \int_Y f_*\alpha$$

for every $\alpha \in A_*(X)$, a special case of the functoriality of pushforward.

8.2.5 Assuming Pushforward with subschemes Let $Y_1, \ldots, Y_r$ be closed subschemes of a scheme X. Let Y be a closed subscheme of X which contains all the Y_i. Given $\alpha_i \in A_*Y_i$, $i = 1, \ldots, r$, and $\beta \in A_*Y$, we will usually write "$\beta = \sum_{i=1}^r \alpha_i$ in A_*Y" in place of the precise equation $\beta = \sum_{i=1}^r \varphi_{i*}(\alpha_i)$ where φ_i is the inclusion of Y_i in Y.

Example 8.2.6: Degrees, Bézout, and the Role of Properness

1. *Bézout's theorem.* Two curves $C, D \subseteq \mathbb{P}^2$ of degrees c, d with no common component meet in a zero-cycle $C \cdot D$ of degree cd. Pushing forward to a point, $\int_{\mathbb{P}^2}[C \cdot D] = cd$: the classical intersection number from the chapter opening is the degree of a zero-cycle, and its independence of the choice of C and D in their linear systems is rational equivalence together with theorem 8.2.3.

2. *Properness is necessary.* For the non-proper projection of example 3.5.24 the image of a subvariety can fail to be closed, so f_* is not even defined; and on $\mathbb{A}^1$ every point is rationally equivalent to zero (example 8.1.5), so no degree survives. The degree is an invariant of a rational equivalence class only on a complete scheme.

3. *A finite cover of curves.* A finite morphism $\varphi\colon C \to C'$ of smooth projective curves satisfies $\varphi_*[P] = [k(P) : k(\varphi(P))]\,[\varphi(P)]$. For a point $Q \in C'$ away from the branch locus, $\varphi^{-1}(Q)$ consists of $\deg\varphi = [k(C) : k(C')]$ reduced points, so $\varphi_*\left[\varphi^{-1}(Q)\right] = (\deg\varphi)\,[Q]$; since $\deg(\varphi_*\alpha) = \deg(\alpha)$ on the complete curves, the sheet count is exactly the degree recorded by φ.

Exercise 8.2.1

1. For the finite morphism $\varphi\colon \mathbb{P}^1 \to \mathbb{P}^1$, $[x_0 : x_1] \mapsto [x_0^2 : x_1^2]$ (in characteristic $\neq 2$), compute $\varphi_*[p]$ for a closed point p and $\varphi_*[\varphi^{-1}(q)]$ for a closed point q. Verify that $\deg(\varphi_*\alpha) = \deg(\alpha)$ for every zero-cycle α, and explain where the degree $\deg\varphi = 2$ appears.

2. Let $\alpha = \sum_i n_i[P_i]$ be a zero-cycle on $\mathbb{P}^n$ over $\overline{k}$. Compute $\deg(\alpha)$ and prove it is invariant under rational equivalence.

3. Deduce from proposition 8.2.2 that a nonzero rational function on a smooth projective curve C has principal divisor of degree zero, and show by example that this fails on $\mathbb{A}^1$.

4. For the structure map $\pi\colon \mathbb{P}^1 \to \operatorname{Spec}\overline{k}$ and a rational function $r \in k(\mathbb{P}^1)^*$, verify $\int_{\mathbb{P}^1}[\operatorname{div}(r)] = \int_{\operatorname{Spec}\overline{k}} \pi_*[\operatorname{div}(r)]$, and say which hypothesis of definition 8.2.4 would fail were $\mathbb{P}^1$ replaced by $\mathbb{A}^1$.

8.3 Flat Pullback

Proper pushforward of section 8.2 is covariant: it moves cycles along a morphism in the direction of the arrow, lowering nothing and raising nothing in dimension. The second functoriality of the Chow group runs the other way. A morphism $f\colon X \to Y$ pulls a subvariety $V \subseteq Y$ back to its preimage $f^{-1}(V) \subseteq X$, and one would like to read that preimage as a cycle on X. For an arbitrary f this fails: the preimage can jump in dimension from fiber to fiber, can acquire embedded components, and carries no canonical multiplicities. The hypothesis that repairs every one of these defects at once is flatness. A flat morphism of relative dimension n has equidimensional fibers, so the preimage of a k-dimensional subvariety is pure of dimension $k+n$, and the scheme-theoretic preimage supplies the multiplicities with which its components must be counted. The resulting map $f^*\colon Z_k(Y) \to Z_{k+n}(X)$ raises dimension by n, is contravariant, and descends to Chow groups.

This section constructs f^*, records its three structural properties (it is a graded homomorphism, it composes contravariantly, and it commutes with proper pushforward across a fiber square), and proves that it respects rational equivalence. The two guiding examples are the ones every later computation rests on: restriction to an open subscheme, and the projection $X \times \mathbb{A}^n \to X$ of a trivial affine bundle.

Recall that a morphism $f\colon X \to Y$ of algebraic schemes is flat if each local ring $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,f(x)}$. Flatness is the algebraic incarnation of the geometric demand that the fibers of f vary without collapsing or jumping, and its precise dimensional consequence is what makes pullback of cycles possible.

Definition 8.3.1: Flat Morphism of Relative Dimension n

A morphism $f\colon X \to Y$ of algebraic schemes is *flat of relative dimension* n if f is flat and, for every subvariety $W \subseteq Y$ and every irreducible component Z of $f^{-1}(W)$,

$$\dim Z = \dim W + n$$

Equivalently, f is flat and every nonempty fiber of f has pure dimension n.

The equivalence stated in the definition is the mechanism the whole construction turns on, so it is worth isolating. Flatness forces the fiber dimension to be locally constant, and it forces the preimage of a subvariety to inherit that constant as an additive shift; neither holds without it.

Proposition 8.3.2: Flat Preimages Are Equidimensional

Let $f\colon X \to Y$ be a flat morphism of algebraic schemes with Y a variety, and suppose every nonempty fiber of f has pure dimension n. Then for every subvariety $W \subseteq Y$ each irreducible component of $f^{-1}(W)$ has dimension $\dim W + n$. In particular $f^{-1}(W)$ is either empty or pure of dimension $\dim W + n$.

***Proof*:**
Write $g\colon f^{-1}(W) \to W$ for the restriction of f; it is the base change of f along the closed immersion $W \hookrightarrow Y$, hence flat, and its fibers are fibers of f, so each nonempty fiber of g has pure dimension n. Thus it suffices to prove the statement for $W = Y$: that a flat morphism $g\colon X' \to W$ onto a variety W, with all nonempty fibers of pure dimension n, has $\dim Z = \dim W + n$ for

every irreducible component Z of X'.

Let $Z \subseteq X'$ be an irreducible component, with generic point η. Flatness satisfies going-down (the going-down property of flat local homomorphisms, [16, p. III.9]), so the minimal prime η of X' contracts to the minimal prime of W; that is, $g(\eta)$ is the generic point of W, and g restricts to a dominant morphism $g\big|_Z : Z \to W$ of varieties. Both are integral of finite type over $\overline{k}$, so their dimensions are transcendence degrees of function fields, and the dimension of a fiber of a dominant morphism of varieties can be read at a closed point.
The fiber-dimension theorem for the dominant morphism $g\big|_Z : Z \to W$ of varieties ([16, Exercise II.3.22]) provides a dense open $U \subseteq Z$ such that for every closed point $z \in U$, the fiber $(g\big|_Z)^{-1}(g(z))$ has dimension exactly $\dim Z - \dim W$ at z. The complement in Z of the other irreducible components of X' is also a dense open, so their intersection is nonempty and contains a closed point z; write $w = g(z) \in W$. Since z lies off the other components, X' coincides with Z near z, so the fiber $X'_w = g^{-1}(w)$ coincides with $(g\big|_Z)^{-1}(w)$ near z, and its dimension at z is $\dim Z - \dim W$. By hypothesis X'_w is nonempty of pure dimension n, so this local fiber dimension equals n. Therefore

$$n = \dim Z - \dim W$$

hence $\dim Z = \dim W + n$, as we sought to show.

The proposition is the geometric license behind the definition of flat pullback. If $V \subseteq Y$ is a k-dimensional subvariety and f is flat of relative dimension n, then $f^{-1}(V)$ is a closed subscheme of X, pure of dimension $k+n$, and one takes its associated cycle, weighting each component by the length that measures the multiplicity with which the scheme carries it. This is exactly the fundamental cycle of definition 8.1.2 applied to the scheme-theoretic preimage.

Definition 8.3.3: Flat Pullback

Let $f : X \to Y$ be a flat morphism of relative dimension n and $V \subseteq Y$ a k-dimensional subvariety. The scheme-theoretic preimage $f^{-1}(V) = X \times_Y V$ is a closed subscheme of X, pure of dimension $k+n$, with irreducible components $Z_1, \ldots, Z_r$. The *flat pullback* of $[V]$ is the $(k+n)$-cycle

$$f^*[V] = \left[f^{-1}(V)\right] = \sum_{i=1}^{r} \operatorname{length}_{\mathcal{O}_{f^{-1}(V),Z_i}} \left(\mathcal{O}_{f^{-1}(V),Z_i}\right)[Z_i]$$

the length being taken in the local ring of the preimage scheme at the generic point of Z_i. Extending $\mathbb{Z}$-linearly gives the flat pullback

$$f^* : Z_k(Y) \to Z_{k+n}(X)$$

The multiplicities are the point of the construction. The preimage as a topological set records only the components Z_i; the scheme structure records how many times f "covers" each one, and it is these lengths that make f^* compatible with divisors and hence with rational equivalence. When f is an open immersion the preimage is reduced and every length is 1, so $f^*[V] = [V \cap X]$; this is the source of the restriction map computed in the next example. The relative dimension n is the amount by which f^* raises the dimension of a cycle, so f^* is graded of degree $+n$.

Example 8.3.4: Open Restriction and the Projection $X \times \mathbb{A}^n \to X$
Two flat morphisms recur throughout the theory, and their pullbacks can be read off the definition.

1. *Open immersion.* Let $j\colon U \hookrightarrow Y$ be the inclusion of an open subscheme. Such j is flat of relative dimension 0: it is a localization at the level of local rings, and its fibers are single reduced points (over $u \in U$) or empty (over $y \notin U$). For a k-dimensional subvariety $V \subseteq Y$ the preimage $j^{-1}(V) = V \cap U$ is the open subscheme of V, which is reduced. If $V \cap U \neq \emptyset$ it is a k-dimensional subvariety of U and $j^*[V] = [V \cap U]$; if $V \subseteq Y \setminus U$ the preimage is empty and $j^*[V] = 0$. Thus flat pullback along an open immersion is the *restriction of cycles* $Z_k(Y) \to Z_k(U)$ that forgets the part of a cycle supported off U, the map at the heart of the localization sequence of section 8.4.

2. *Trivial affine bundle.* Let $p\colon X \times \mathbb{A}^n \to X$ be the projection. It is flat, being a base change of the flat morphism $\mathbb{A}^n_{\overline{k}} \to \operatorname{Spec} \overline{k}$ along $X \to \operatorname{Spec} \overline{k}$, and every fiber is a copy of $\mathbb{A}^n$, pure of dimension n; so p is flat of relative dimension n. For a k-dimensional subvariety $V \subseteq X$ the preimage is $V \times \mathbb{A}^n$, which is integral of dimension $k + n$, so
$$p^*[V] = [V \times \mathbb{A}^n]$$
the cylinder over V. No multiplicity appears because $V \times \mathbb{A}^n$ is a variety. This is the computation that will make p^* an isomorphism on Chow groups (theorem 8.4.6).

Both examples exhibit the pattern the general theory generalizes: pullback restricts a cycle to an open part in the first case and thickens it uniformly along the fiber in the second. The structural properties below show that these behaviors are functorial.

The first property collects the ways in which f^* is well behaved as a map of cycle groups. It is a graded homomorphism raising dimension by the relative dimension, and it composes contravariantly, so that a tower of flat morphisms pulls back through the composite exactly as one expects.

Proposition 8.3.5: Basic Properties of Flat Pullback

Let $f\colon X \to Y$ be flat of relative dimension n.

1. *Homomorphism.* $f^*\colon Z_k(Y) \to Z_{k+n}(X)$ is a homomorphism of abelian groups for each k, and assembles to a graded homomorphism $f^*\colon Z_*(Y) \to Z_*(X)$ of degree n.

2. *Fundamental cycle.* If Y is a variety, then $f^*[Y] = [X]$: flat pullback of the fundamental cycle of the base is the fundamental cycle of the total space.

3. *Composition.* If $g\colon Y \to Y'$ is flat of relative dimension m, then $g \circ f$ is flat of relative dimension $m + n$ and
$$(g \circ f)^* = f^* \circ g^*$$
as maps $Z_k(Y') \to Z_{k+m+n}(X)$.

4. *Identity.* $(\mathrm{id}_X)^* = \mathrm{id}$ on $Z_*(X)$.

***Proof*:**

1. *Homomorphism.* By construction f^* is defined on generators $[V]$ and extended $\mathbb{Z}$-linearly, so it is a homomorphism $Z_k(Y) \to Z_{k+n}(X)$ by fiat; proposition 8.3.2 is what guarantees the target group is $Z_{k+n}(X)$, since $f^{-1}(V)$ is pure of dimension $k+n$. Summing over k gives a degree-n graded homomorphism.

2. *Fundamental cycle.* Let Y be a variety of dimension d. Since Y is integral, its fundamental cycle is the single generator $[Y]$ with multiplicity 1, not a sum over several components. The preimage $f^{-1}(Y) = X$ is all of X, and $f^*[Y] = [X]$ is by definition the fundamental cycle of X, weighting each top-dimensional component of X by its length multiplicity. This is definition 8.1.2's fundamental cycle $[X]$, so the identity holds.

3. *Composition.* Composites of flat morphisms are flat, and relative dimensions add because fiber dimensions add along the tower: a fiber of $g \circ f$ over $y' \in Y'$ maps to the fiber $g^{-1}(y')$ of pure dimension m, with fibers of f of pure dimension n, so it has pure dimension $m+n$. Hence $g \circ f$ is flat of relative dimension $m+n$. For the cycle identity it suffices to check on a generator $[V]$ with $V \subseteq Y'$ a k-dimensional subvariety. Both sides are cycles supported on $(g \circ f)^{-1}(V) = f^{-1}(g^{-1}(V))$, so the claim is that the multiplicities match component by component. The scheme-theoretic preimage is computed by iterated fiber product,

$$(g \circ f)^{-1}(V) = X \times_{Y'} V = X \times_Y (Y \times_{Y'} V) = f^{-1}\big(g^{-1}(V)\big)$$

so the underlying scheme of $(g \circ f)^{-1}(V)$ equals the underlying scheme of $f^{-1}(g^{-1}(V))$. It remains to see that the length multiplicities agree, that is, that applying f^* to the cycle $g^*[V] = \sum_i \ell_i [W_i]$ (where W_i are the components of $g^{-1}(V)$ with lengths ℓ_i) reproduces the multiplicities of $(g \circ f)^{-1}(V)$. Fix a component Z of the common preimage, with generic point z lying over the generic point w of some W_i. Flatness of f makes $\mathcal{O}_{X,z}$ flat over $\mathcal{O}_{Y,w}$, and length is additive and multiplies through a flat local homomorphism along the fiber: the length of $\mathcal{O}_{(g \circ f)^{-1}(V),z}$ equals ℓ_i times the length of $\mathcal{O}_{f^{-1}(W_i),z}$. Summing z over the components of $f^{-1}(W_i)$ and i over the components of $g^{-1}(V)$ gives $(g \circ f)^*[V] = f^*(g^*[V])$, as required.

4. *Identity.* The identity map is flat of relative dimension 0 with $(\mathrm{id})^{-1}(V) = V$ reduced, so $(\mathrm{id}_X)^*[V] = [V]$ on generators, completing the proof.

Contravariant functoriality (part 3) is what will let a Chow-group computation be assembled from simpler flat maps, exactly as covariant functoriality of pushforward let degrees be computed in towers. The fundamental-cycle identity (part 2) is the seed of every computation in section 8.5: it identifies the generator of the top Chow group of the total space of a flat family in terms of the base.

The remaining two properties are the interaction of flat pullback with the machinery already built. The first is its compatibility with proper pushforward. Pullback and pushforward move cycles in opposite directions, and they do not commute for a single morphism; but across a fiber square, where one has a flat map and a proper map meeting at right angles, the two operations do commute. This is the cycle-level form of the projection formula, and it is the technical engine behind essentially every later compatibility.

Proposition 8.3.6: Compatibility of Flat Pullback and Proper Pushforward

Consider a fiber square of algebraic schemes

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

that is Cartesian, $X' = X \times_Y Y'$, with f proper and g flat of relative dimension n. Then f' is proper, g' is flat of relative dimension n, and

$$g^* \circ f_* = f'_* \circ g'^* \colon Z_k(X) \to Z_{k+n}(Y')$$

Consequently the identity holds on Chow groups.

***Proof*:**
Step 1: the maps in the base-changed square. Properness and flatness are stable under base change, so f' is proper and g' is flat; the fibers of g' over a point of X are fibers of g over its image in Y, so g' has the same pure relative dimension n as g. Both composites $g^* \circ f_*$ and $f'_* \circ g'^*$ therefore land in $Z_{k+n}(Y')$, and both are homomorphisms, so it suffices to check the identity on a generator $[V]$ with $V \subseteq X$ a k-dimensional subvariety.

Step 2: reduction to $X = V$. Replacing X by V, Y by $W = f(V)$, and forming the induced fiber squares, the four schemes become V, W, $V' = V \times_W W'$, $W' = W \times_Y Y'$, with $f\colon V \to W$ proper surjective and $g\colon W' \to W$ flat of relative dimension n. It suffices to prove

$$g^*(f_*[V]) = f'_*\big(g'^*[V]\big)$$

where now $f_*[V] = \deg(V/W)[W]$ by definition 8.2.1 and $g'^*[V] = [g'^{-1}(V)] = [V']$ (with its multiplicities).

Step 3: the two dimension regimes. There are two cases according to whether f drops dimension.

If $\dim V > \dim W$, then $\deg(V/W) = 0$, so the left side is $g^*(0) = 0$. On the right, $V' = V \times_W W'$ maps to W' under f' with fibers the fibers of $V \to W$, which are positive dimensional over the generic point; hence $\dim V' > \dim W'$ and every component of V' has image of strictly smaller dimension in W', so $f'_*[V'] = 0$ by the dimension clause of definition 8.2.1. Both sides vanish.

If $\dim V = \dim W$, write $d = \deg(V/W) = [k(V) : k(W)]$. The left side is $g^*(d\,[W]) = d\,[W']$, using $g^*[W] = [W'] = [g^{-1}(W)]$ (the base W is a variety and g is flat, so g^* of its fundamental cycle is the fundamental cycle of W' by proposition 8.3.5(2), which here is the total space $W' = g^{-1}(W)$). For the right side, $[V'] = g'^*[V]$ carries the pullback multiplicities flagged above, and $f'_*[V']$ must account for both those multiplicities and the residue-field degrees; flatness is what makes only the field degree survive. Fix a component W'_j of W', with generic point η_j. Localizing at η_j, the fiber of $f'\colon V' \to W'$ over η_j is Spec of $k(V) \otimes_{k(W)} k(W'_j)$: the generic fiber of $f\colon V \to W$ is $\operatorname{Spec} k(V)$, a $k(W)$-algebra of dimension d since V, W are varieties, and flat base change along $k(W) \hookrightarrow k(W'_j)$ preserves that dimension, so $\dim_{k(W'_j)}\big(k(V) \otimes_{k(W)} k(W'_j)\big) = d$. This dimension is exactly the total multiplicity that $f'_*[V']$ records over W'_j: summing the pullback multiplicity of each component of V' dominating W'_j against its residue degree over $k(W'_j)$ recovers the length d of the fiber algebra, because flatness of g (hence of g') forbids any length being created or lost in the base change. Thus $f'_*[V'] = \sum_j d\,[W'_j] = d\,[W']$, matching the left side. The two sides agree in both regimes, which establishes $g^* f_* = f'_* g'^*$ on cycles. Since each of g^*, f_*, f'_*, g'^* descends to Chow groups by theorem 8.2.3 and proposition 8.3.7 below, the identity descends as well, completing the proof.

The compatibility should be read as the statement that pushing a cycle forward and then restricting to a flat base change gives the same class as first restricting and then pushing forward within the base change. In the special case where g is an open immersion, it says that pushforward commutes with restriction to an open set, which is the reason the localization sequence of the next section is a sequence of A_*-functorial maps. In the case where f is the projection to a point, it recovers the invariance of the degree of a zero-cycle under a flat base change of the ground scheme.

Everything so far concerns cycles. To obtain a map of Chow groups, flat pullback must carry Rat_k into Rat_{k+n}. The mechanism is that flat pullback commutes with taking the divisor of a rational function, up to the multiplicities that flatness controls, so a cycle that is a sum of divisors pulls back to a sum of divisors.

Proposition 8.3.7: Flat Pullback Preserves Rational Equivalence

Let $f\colon X \to Y$ be flat of relative dimension n. If $\alpha \in Z_k(Y)$ is rationally equivalent to zero, then $f^*\alpha \in Z_{k+n}(X)$ is rationally equivalent to zero. Consequently f^* descends to a homomorphism

$$f^*\colon A_k(Y) \to A_{k+n}(X)$$

and $A_*(-)$ equipped with flat pullback is a contravariant functor on flat morphisms of algebraic schemes, raising dimension by the relative dimension.

***Proof*:**

It suffices to treat a generator of $\mathrm{Rat}_k(Y)$, namely $\alpha = [\mathrm{div}(r)]$ for a $(k+1)$-dimensional subvariety $W \subseteq Y$ and a nonzero $r \in k(W)^*$. The claim is that $f^*[\mathrm{div}(r)]$ is a sum of divisors of rational functions on subvarieties of X.

Step 1: reduction to the total space over W. The cycle $[\mathrm{div}(r)]$ is supported on W, and $f^*[\mathrm{div}(r)]$ is supported on $f^{-1}(W)$. Let $W' = f^{-1}(W) = X \times_Y W$ with its scheme-theoretic (possibly non-reduced) structure, and let $f_W\colon W' \to W$ be the restriction of f, which is flat of relative dimension n by base change. Because the components of $f^{-1}(W)$ are the components of W' and the pullback multiplicities are computed inside W', it is enough to prove that $f_W^*[\mathrm{div}(r)] \sim 0$ in $Z_{k+n}(W')$; the class then remains rationally equivalent to zero in X under the proper (closed immersion) pushforward along $W' \hookrightarrow X$, which preserves rational equivalence by theorem 8.2.3. Thus one may assume $Y = W$ is a variety and $r \in k(Y)^*$.

Step 2: the key identity on components. With Y a variety and $r \in k(Y)^*$, the divisor $[\mathrm{div}(r)] = \sum_V \mathrm{ord}_V(r)[V]$ runs over the codimension-one subvarieties $V \subseteq Y$. The assertion to prove is the flat-pullback divisor formula

$$f^*[\mathrm{div}(r)] = \left[\mathrm{div}(f^{\#}r)\right] \quad \text{on each component of } X \tag{8.1}$$

interpreted componentwise as follows. Let $X_1, \dots, X_s$ be the irreducible components of X, each of dimension $n + \dim Y$ by proposition 8.3.2, with generic points dominating Y (flatness sends generic points to the generic point of the base). On each X_j the pullback $f^{\#}r$ of r is a nonzero rational function, since $r \neq 0$ and X_j dominates Y. The claim is

$$f^*[\operatorname{div}(r)] = \sum_{j=1}^{s} m_j \,[\operatorname{div}(f^\# r\big|_{X_j})]$$

where $m_j = \operatorname{length}_{\mathcal{O}_{X,X_j}}(\mathcal{O}_{X,X_j})$ is the multiplicity of the component X_j in the fundamental cycle $[X]$. Each term on the right is a divisor of a rational function on the subvariety $(X_j)_{\mathrm{red}}$, hence rationally equivalent to zero, so (8.1) gives $f^*[\operatorname{div}(r)] \sim 0$.

Step 3: verifying the identity. Fix a codimension-one subvariety $D \subseteq X_j$, a $(k+n)$-dimensional subvariety of X, and let $V = \overline{f(D)} \subseteq Y$. Two cases arise by the relative-dimension count of proposition 8.3.2 applied to $f\big|_{X_j}$.

If $\dim V = \dim Y$, then $V = Y$ and D dominates Y; the coefficient of $[D]$ in $[\operatorname{div}(f^\# r)]$ is $\operatorname{ord}_D(f^\# r)$, computed in the discrete-valuation-style local ring $\mathcal{O}_{X_j,D}$, while D does not appear in $f^*[\operatorname{div}(r)]$ because $[\operatorname{div}(r)]$ has no component equal to Y. But $\operatorname{ord}_D(f^\# r) = 0$. The load-bearing point is that D dominates Y: its generic point ξ maps to the generic point of Y, at which r is a nonzero rational function, that is, a unit of $k(Y)$. Since $f^\# r$ is the pullback of r along the local homomorphism $\mathcal{O}_{Y,Y} = k(Y) \to \mathcal{O}_{X_j,D}$ induced on stalks at ξ, the image $f^\# r$ is a unit of $\mathcal{O}_{X_j,D}$, so $\operatorname{ord}_D(f^\# r) = 0$ and no such D contributes to either side. (A nonzero rational function need not be a unit along a codimension-one subvariety in general; here it is one precisely because D dominates the base and so sees r only at the generic point of Y.)

If $\dim V = \dim Y - 1$, then V is one of the codimension-one subvarieties appearing in $[\operatorname{div}(r)]$, and D is a component of $f^{-1}(V)$, lying on one or more of the components X_j. Two kinds of local ring enter the count, and keeping them distinct is the crux. Write

$$A = \mathcal{O}_{Y,V}, \qquad \widetilde{B} = \mathcal{O}_{X,D}$$

Here A is a one-dimensional local domain and $\operatorname{ord}_V(r) = \operatorname{length}_A(A/rA)$; the ring $\widetilde{B}$ is the local ring of the whole (possibly nonreduced) scheme X at the generic point of D, and it is flat over A because f is flat. The minimal primes of the one-dimensional ring $\widetilde{B}$ correspond to the components X_j passing through D; for such a component the quotient $B_j = \widetilde{B}/\mathfrak{p}_j = \mathcal{O}_{X_j,D}$ is the discrete-valuation-style local ring of the reduced component X_j at D, in which $\operatorname{ord}_D(f^\# r\big|_{X_j}) = \operatorname{length}_{B_j}(B_j/(f^\# r)B_j)$ is computed. When X is irreducible along D there is a single such j, but in general several components may meet along D.
The coefficient of $[D]$ in $f^*[\operatorname{div}(r)]$ is $\operatorname{ord}_V(r) \cdot e_D$, where

$$e_D = \operatorname{length}_{\mathcal{O}_{f^{-1}(V),D}}\big(\mathcal{O}_{f^{-1}(V),D}\big) = \operatorname{length}_{\widetilde{B}}\big(\widetilde{B}/\mathfrak{m}_A\widetilde{B}\big)$$

is the multiplicity of D in the pullback cycle $f^*[V]$, the second equality holding because $\mathcal{O}_{f^{-1}(V),D} = \widetilde{B} \otimes_A A/\mathfrak{m}_A = \widetilde{B}/\mathfrak{m}_A\widetilde{B}$. The coefficient of $[D]$ in $\sum_j m_j[\operatorname{div}(f^\# r\big|_{X_j})]$ is $\sum_{j\,:\,D\subseteq X_j} m_j \cdot \operatorname{ord}_D(f^\# r\big|_{X_j})$, the sum running over the components X_j that pass through D. That the two coefficients agree is two multiplicative facts chained together. First, flatness of $\widetilde{B}$ over A makes length multiply through the base change: filtering A/rA by its $\operatorname{ord}_V(r)$ composition factors $A/\mathfrak{m}_A$ and applying the exact functor $- \otimes_A \widetilde{B}$ gives

$$\operatorname{length}_{\widetilde{B}}\big(\widetilde{B}/r\widetilde{B}\big) = \operatorname{length}_A(A/rA) \cdot \operatorname{length}_{\widetilde{B}}\big(\widetilde{B}/\mathfrak{m}_A\widetilde{B}\big) = \operatorname{ord}_V(r) \cdot e_D$$

Second, order in the one-dimensional local ring $\widetilde{B}$ distributes over its minimal primes with the generic multiplicities as weights: each minimal prime $\mathfrak{p}_j$ has residue ring $B_j = \mathcal{O}_{X_j,D}$ and localization $\widetilde{B}_{\mathfrak{p}_j} = \mathcal{O}_{X,X_j}$ of length $m_j = \text{length}_{\mathcal{O}_{X,X_j}}(\mathcal{O}_{X,X_j})$, so the additivity of order over the minimal primes of a one-dimensional local ring ([22, App. A.3]) gives, for the nonzerodivisor $f^{\#}r$,

$$\text{length}_{\widetilde{B}}\big(\widetilde{B}/(f^{\#}r)\widetilde{B}\big) = \sum_{j\,:\,D\subseteq X_j} m_j \cdot \text{length}_{B_j}\big(B_j/(f^{\#}r)B_j\big) = \sum_{j\,:\,D\subseteq X_j} m_j \cdot \text{ord}_D(f^{\#}r\big|_{X_j})$$

Since $f^{\#}r$ is the image of r in $\widetilde{B}$, the left sides of the two displays coincide, whence $\text{ord}_V(r)\cdot e_D = \sum_{j\,:\,D\subseteq X_j} m_j \cdot \text{ord}_D(f^{\#}r\big|_{X_j})$: the two coefficients of $[D]$ match. When X is irreducible along D the sum has a single term and reduces to $e_D = m_j \cdot e_{D/V}$, where $e_{D/V} = \text{length}_{B_j}(B_j/\mathfrak{m}_A B_j)$ is the ramification-type multiplicity of D over V computed in the reduced component ring B_j; the pullback multiplicity e_D over the full preimage scheme then exceeds it by exactly the factor m_j. Summing over the divisors D and, on the right, over the components X_j reproduces both sides of (8.1) term by term. Hence $f^*[\text{div}(r)] = \sum_j m_j[\text{div}(f^{\#}r\big|_{X_j})]$ is a sum of divisors of rational functions, so $f^*[\text{div}(r)] \sim 0$.

Descent to Chow groups follows: $f^*(\text{Rat}_k(Y)) \subseteq \text{Rat}_{k+n}(X)$, so f^* induces $A_k(Y) \to A_{k+n}(X)$. Functoriality on classes is inherited from proposition 8.3.5(3),(4) on cycles, giving a contravariant functor as claimed, completing the proof.

The proposition completes the package: flat pullback is a well-defined operation on Chow groups, contravariant and dimension-raising, dual to the covariant dimension-preserving pushforward of section 8.2. The pair (f_*, f^*) makes A_* into a functor covariant for proper maps and contravariant for flat maps, with the compatibility of proposition 8.3.6 tying the two together across fiber squares. This is the structure the localization sequence and homotopy invariance of section 8.4 exploit, and it is why the affine-bundle projection $p\colon X \times \mathbb{A}^n \to X$, whose pullback was computed in example 8.3.4, will turn out to induce an isomorphism on Chow groups.

Example 8.3.8: Pullback to a Product and to an Open Set

The two flat morphisms of example 8.3.4 descend to Chow groups explicitly.

1. *Cylinder map.* For $p\colon X \times \mathbb{A}^1 \to X$ the induced map $p^*\colon A_k(X) \to A_{k+1}(X \times \mathbb{A}^1)$ sends the class of a subvariety V to the class of the cylinder $V \times \mathbb{A}^1$. On $X = \text{Spec}\,\overline{k}$ this reads $A_0(\text{Spec}\,\overline{k}) = \mathbb{Z} \to A_1(\mathbb{A}^1)$, sending the class of the point to $[\mathbb{A}^1]$, the fundamental class of the affine line, consistent with $p^*[Y] = [X]$ from proposition 8.3.5(2).
2. *Restriction to an open set.* For an open immersion $j\colon U \hookrightarrow Y$ the map $j^*\colon A_k(Y) \to A_k(U)$ restricts a class to U, forgetting components supported on the complement $Y \setminus U$. On $Y = \mathbb{P}^1$ with $U = \mathbb{A}^1$ this is the surjection $A_0(\mathbb{P}^1) = \mathbb{Z} \twoheadrightarrow A_0(\mathbb{A}^1) = 0$ from example 8.1.5: the class of a point on $\mathbb{P}^1$ restricts to a point of $\mathbb{A}^1$, which is rationally trivial. The kernel is generated by the class of the point at infinity, the part of $\mathbb{P}^1$ removed. This surjectivity with the removed piece generating the kernel is the prototype of the localization sequence.

Exercise 8.3.1

1. Let $f\colon X \to Y$ be flat of relative dimension 0 (for instance a finite flat morphism, or an open

immersion). Show directly from definition 8.3.3 that f^* maps $Z_k(Y) \to Z_k(X)$ for every k, preserving dimension, and that $f^*[V] = \sum_i e_i[Z_i]$ where Z_i are the components of $f^{-1}(V)$ and e_i their multiplicities in the fiber over the generic point of V.

2. Let $\pi\colon \mathbb{A}^{n+1} \setminus \{0\} \to \mathbb{P}^n$ be the tautological $\mathbb{G}_m$-bundle. It is flat of relative dimension 1. Compute $\pi^*[\mathbb{P}^k]$ for a linear subspace $\mathbb{P}^k \subseteq \mathbb{P}^n$, identifying its class as a cycle on $\mathbb{A}^{n+1} \setminus \{0\}$.

3. Give an example of a flat morphism f for which $f^*\colon A_*(Y) \to A_*(X)$ is not injective, and one for which it is not surjective. (Hint: the projection $\mathbb{P}^1 \times \mathbb{A}^1 \to \mathbb{A}^1$ and the open immersion $\mathbb{A}^1 \hookrightarrow \mathbb{P}^1$.)

4. Let $q\colon X \times \mathbb{A}^m \times \mathbb{A}^n \to X \times \mathbb{A}^m$ and $p\colon X \times \mathbb{A}^m \to X$ be the projections. Verify the composition law proposition 8.3.5(3) for $p \circ (\mathrm{id} \times \text{-})$ directly on the class of a subvariety $V \subseteq X$, confirming both routes give $[V \times \mathbb{A}^m \times \mathbb{A}^n]$.

5. Let $f\colon X \to Y$ be proper and $j\colon U \hookrightarrow Y$ an open immersion, with fiber square $X_U = f^{-1}(U) \to X$ over j. Using proposition 8.3.6, prove that $(f|_{X_U})_*(\alpha|_{X_U}) = (f_*\alpha)|_U$ in $A_*(U)$ for every $\alpha \in A_*(X)$, that is, proper pushforward commutes with restriction to an open subset.

6. Let $E \to X$ be a vector bundle of rank r over a variety X of dimension d, with projection π flat of relative dimension r. Compute $\pi^*[X]$ and identify the top Chow group $A_{d+r}(E)$, first showing that E is irreducible.

8.4 The Fundamental Sequences

The two functorialities already in hand, proper pushforward (section 8.2) and flat pullback (section 8.3), combine into the two structural results that make Chow groups computable. The first is a long-exact-sequence surrogate: cutting a scheme into a closed piece and its open complement relates the three Chow groups by a right-exact sequence, the algebraic analogue of the excision property of a homology theory. The second is homotopy invariance in its algebraic form: an affine bundle over X, and in particular the trivial bundle $X \times \mathbb{A}^n$, has the same Chow groups as X, shifted in degree by the fibre dimension. Neither result computes a single Chow group in isolation, but together they reduce the computation of $A_*(X)$ for the spaces of the next section to bookkeeping: stratify X by locally closed pieces on which the answer is known, and assemble the pieces through the sequence.

Both proofs rest on a single principle. A cycle on X is a formal combination of subvarieties; a subvariety either meets the open set U or is contained in the closed complement Z, and rational equivalence is generated by divisors of rational functions, which live on subvarieties and so obey the same dichotomy. The localization sequence is the exact translation of this observation into a statement about the three groups, and affine-bundle invariance is its first substantial payoff, extracted by Noetherian induction.

Throughout, $Z \subseteq X$ is a closed subscheme with open complement $U = X \setminus Z$, and

$$i\colon Z \hookrightarrow X, \qquad j\colon U \hookrightarrow X$$

denote the inclusions. The closed immersion i is proper, so it pushes cycles forward by definition 8.2.1: a subvariety $V \subseteq Z$ is also a subvariety of X, and $i_*[V] = [V]$ since $\deg(V/V) = 1$. The open immersion j is flat of relative dimension 0, so it pulls cycles back by definition 8.3.3: for a subvariety

$V \subseteq X$ the restriction is $j^*[V] = [V \cap U]$ when V meets U and $j^*[V] = 0$ when $V \subseteq Z$. These two operations are what the sequence is built from.

Theorem 8.4.1: Localization Sequence

Let X be an algebraic scheme, $Z \subseteq X$ a closed subscheme, and $U = X \setminus Z$ its open complement, with $i\colon Z \hookrightarrow X$ and $j\colon U \hookrightarrow X$ the inclusions. For every k the sequence

$$A_k(Z) \xrightarrow{i_*} A_k(X) \xrightarrow{j^*} A_k(U) \longrightarrow 0$$

is exact: j^* is surjective, and $\ker(j^*) = \operatorname{im}(i_*)$.

***Proof*:**
The map i_* is proper pushforward along the closed immersion (theorem 8.2.3) and j^* is flat pullback along the open immersion (proposition 8.3.7); both are defined on Chow groups, so the sequence is a sequence of homomorphisms. Exactness is checked at the two nonzero spots.

Step 1: surjectivity of j^*. A subvariety $V' \subseteq U$ is a k-dimensional closed integral subscheme of the open set U. Its closure $V = \overline{V'}$ in X is a k-dimensional subvariety of X with $V \cap U = V'$, since closure adds only points lying in the closed set Z and does not raise the dimension. Thus $j^*[V] = [V']$, and every generator of $Z_k(U)$ is hit by a cycle on X. Hence $j^*\colon Z_k(X) \to Z_k(U)$ is already surjective at the level of cycles, and a fortiori $j^*\colon A_k(X) \to A_k(U)$ is surjective.

Step 2: the inclusion $\operatorname{im}(i_*) \subseteq \ker(j^*)$. For a subvariety $V \subseteq Z$ the intersection $V \cap U$ is empty, so $j^* i_*[V] = j^*[V] = 0$. By linearity $j^* \circ i_* = 0$ on $Z_k(Z)$, hence on $A_k(Z)$, which gives $\operatorname{im}(i_*) \subseteq \ker(j^*)$.

Step 3: the inclusion $\ker(j^*) \subseteq \operatorname{im}(i_*)$. This is the substance of the theorem. Let $\alpha \in Z_k(X)$ be a cycle whose class in $A_k(X)$ lies in $\ker(j^*)$, so $j^*\alpha \sim 0$ on U; the goal is to modify α by a cycle rationally equivalent to zero on X until it is supported on Z, that is, until it lies in the image of $i_*\colon Z_k(Z) \to Z_k(X)$.

Split the cycle by whether its components meet U. Group the subvarieties occurring in α as those contained in Z and those meeting U, and write

$$\alpha = \alpha_Z + \beta, \qquad \alpha_Z \in Z_k(Z), \quad \beta = \sum_i n_i [V_i]$$

where each V_i meets U. Since $\alpha_Z = i_*\alpha_Z$ already lies in the image of i_*, it may be discarded, and it suffices to treat β. Applying j^* gives $j^*\alpha = j^*\beta = \sum_i n_i [V_i']$ with $V_i' = V_i \cap U$ the distinct k-dimensional subvarieties of U, and by hypothesis this cycle is rationally equivalent to zero on U:

$$\sum_i n_i [V_i'] = \sum_j [\operatorname{div}(r_j')] \quad \text{in } Z_k(U)$$

for finitely many $(k+1)$-dimensional subvarieties $W_j' \subseteq U$ and rational functions $r_j' \in k(W_j')^*$.

The idea is now to lift each W_j' and r_j' from U to X and compare divisors. Let $W_j = \overline{W_j'} \subseteq X$ be the closure, a $(k+1)$-dimensional subvariety of X with $W_j \cap U = W_j'$. Restriction of functions to the dense open $W_j' \subseteq W_j$ is an isomorphism of function fields $k(W_j) \xrightarrow{\sim} k(W_j')$, since a variety and a dense open subset have the same function field; let $r_j \in k(W_j)^*$ be the function corresponding to r_j'. Form the divisor $[\operatorname{div}(r_j)] \in Z_k(X)$. Its restriction to U is governed by the flat-pullback compatibility of the divisor construction: for a codimension-one subvariety $V \subseteq W_j$ meeting U, the local ring $\mathcal{O}_{W_j,V} = \mathcal{O}_{W'_j,V\cap U}$ is unchanged by passing to the open set, so $\operatorname{ord}_V(r_j) = \operatorname{ord}_{V\cap U}(r'_j)$, while a codimension-one subvariety of W_j contained in Z contributes a component of $[\operatorname{div}(r_j)]$ that lies in $Z_k(Z)$. Splitting the sum over these two kinds of V,

$$[\operatorname{div}(r_j)] = [\operatorname{div}(r'_j)] + \gamma_j, \qquad \gamma_j \in Z_k(Z) \tag{8.2}$$

where the first term on the right is read inside $Z_k(X)$ via the identification $V \leftrightarrow \overline{V}$ of subvarieties of U with their closures, and γ_j collects the codimension-one subvarieties of W_j that lie in Z.

Step 3, conclusion. Sum (8.2) over j and use $\sum_j [\operatorname{div}(r'_j)] = \beta$ as cycles on X under the closure identification (both sides are the cycle $\sum_i n_i[V_i]$, since $V_i = \overline{V'_i}$). This yields

$$\sum_j [\operatorname{div}(r_j)] = \beta + \sum_j \gamma_j$$

so that

$$\beta = \sum_j [\operatorname{div}(r_j)] - \sum_j \gamma_j$$

The first sum is a $\mathbb{Z}$-combination of divisors of rational functions on X, hence lies in $\operatorname{Rat}_k(X)$; the second sum $\gamma := \sum_j \gamma_j$ lies in $Z_k(Z)$. Therefore, in $A_k(X)$,

$$[\beta] = -[\gamma] = i_*(-[\gamma]) \in \operatorname{im}(i_*)$$

Adding back the discarded $\alpha_Z = i_*[\alpha_Z]$ gives $[\alpha] = i_*\big([\alpha_Z] - [\gamma]\big)$, exhibiting the class of α in the image of i_*. This proves $\ker(j^*) \subseteq \operatorname{im}(i_*)$, and with Steps 1 and 2 the sequence is exact, completing the proof.

The mechanism of the proof is worth isolating, because it recurs whenever Chow groups are computed by stratification. Surjectivity on the right is elementary: a subvariety of the open set extends to its closure, so no class on U is unreachable from X. The content is in the middle, and it is entirely a statement about lifting rational equivalences. A relation $j^*\alpha \sim 0$ on U is witnessed by finitely many rational functions on subvarieties of U; each such function extends across Z to a rational function on a subvariety of X (function fields do not see the boundary), and the extended divisor differs from the original only in components supported on Z. The failure of the sequence to be exact on the left, which is left as it stands, measures exactly how much rational equivalence is created by the extra room X provides over Z: a cycle on Z can become rationally equivalent to zero in X without being so in Z, because X contains $(k+1)$-dimensional subvarieties running out of Z into U. The affine line makes this concrete below.

8.4.2 Remark: No Left-Exactness The map $i_*\colon A_k(Z) \to A_k(X)$ need not be injective. Take $X = \mathbb{A}^1$, $Z = \{0\}$, and $k = 0$. Then $A_0(Z) = \mathbb{Z}$, generated by $[0]$, but $[0] \sim 0$ in $A_0(\mathbb{A}^1)$ by example 8.1.5, so $i_*[0] = 0$ and i_* is the zero map on a nonzero group. The kernel of i_* records cycles on Z that bound in X but not in Z, and there is no functorial description of it. This is why the localization sequence is only a three-term right-exact sequence, not a long exact sequence: Bloch's higher Chow groups supply the missing terms, but that theory is outside the scope of this book.

The localization sequence is stated for one closed subscheme, but its most frequent use is for a pair. Two closed subschemes X_1, X_2 of an ambient X can be compared through their union and intersection, and the resulting sequence is the Mayer-Vietoris right-exactness that was promised without proof in example 8.1.6. It is a companion to the localization sequence rather than a corollary of it: the derivation below runs directly at the level of cycles, using only that a subvariety of the union lies in one of the two pieces, so no appeal to theorem 8.4.1 is needed.

Corollary 8.4.3: Right-Exactness for Two Closed Subschemes

Let X_1 and X_2 be closed subschemes of an algebraic scheme, and set $X_1 \cup X_2$ and $X_1 \cap X_2$ (with reduced or any fixed scheme structure; Chow groups see only the reduced structure by example 8.1.6). For every k the sequence

$$A_k(X_1 \cap X_2) \xrightarrow{(\iota_{1*}, -\iota_{2*})} A_k(X_1) \oplus A_k(X_2) \xrightarrow{j_{1*}+j_{2*}} A_k(X_1 \cup X_2) \longrightarrow 0$$

is exact, where $\iota_m \colon X_1 \cap X_2 \hookrightarrow X_m$ and $j_m \colon X_m \hookrightarrow X_1 \cup X_2$ are the inclusions. There is no exactness at $A_k(X_1 \cap X_2)$ in general.

***Proof*:**
Write $Y = X_1 \cup X_2$. The argument is cleanest at the level of cycles, using the one structural fact that an irreducible closed subset of $Y = X_1 \cup X_2$ is contained in X_1 or in X_2. Consequently a subvariety $V \subseteq Y$ lies in X_1 or in X_2 (or in both, in which case $V \subseteq X_1 \cap X_2$), and because $Z_k(Y)$ is free on subvarieties, a cycle expressible both by subvarieties of X_1 and by subvarieties of X_2 has all its components in $X_1 \cap X_2$.

Step 1: surjectivity of $j_{1*} + j_{2*}$. Every subvariety of Y lies in X_1 or X_2, so at the level of cycles $Z_k(Y) = j_{1*}Z_k(X_1) + j_{2*}Z_k(X_2)$; the map $Z_k(X_1) \oplus Z_k(X_2) \to Z_k(Y)$, $(a_1, a_2) \mapsto a_1 + a_2$, is surjective. Passing to classes, $j_{1*} + j_{2*} \colon A_k(X_1) \oplus A_k(X_2) \to A_k(Y)$ is surjective.

Step 2: the composite is zero. For a subvariety $V \subseteq X_1 \cap X_2$ the two pushforwards $j_{1*}\iota_{1*}[V]$ and $j_{2*}\iota_{2*}[V]$ both equal $[V]$ in $Z_k(Y)$, since each inclusion sends $[V]$ to $[V]$. Hence $(j_{1*} + j_{2*}) \circ (\iota_{1*}, -\iota_{2*}) = j_{1*}\iota_{1*} - j_{2*}\iota_{2*} = 0$, giving $\operatorname{im}(\iota_{1*}, -\iota_{2*}) \subseteq \ker(j_{1*} + j_{2*})$.

Step 3: exactness in the middle. Let $(\eta_1, \eta_2) \in A_k(X_1) \oplus A_k(X_2)$ satisfy $j_{1*}\eta_1 + j_{2*}\eta_2 = 0$ in $A_k(Y)$. Represent η_1 and η_2 by cycles $a_1 \in Z_k(X_1)$ and $a_2 \in Z_k(X_2)$. The hypothesis says $a_1 + a_2 \sim 0$ in $Z_k(Y)$, so there are finitely many $(k+1)$-dimensional subvarieties $W_j \subseteq Y$ and $r_j \in k(W_j)^*$ with

$$a_1 + a_2 = \sum_j [\operatorname{div}(r_j)] \quad \text{in } Z_k(Y)$$

Each W_j lies in X_1 or in X_2, and the divisor $[\operatorname{div}(r_j)]$ of a rational function on W_j is a cycle supported on W_j, hence on X_1 or on X_2 accordingly. Collect the terms:

$$b_1 = \sum_{W_j \subseteq X_1} [\operatorname{div}(r_j)] \in \operatorname{Rat}_k(X_1), \qquad b_2 = \sum_{W_j \subseteq X_2,\, W_j \not\subseteq X_1} [\operatorname{div}(r_j)] \in \operatorname{Rat}_k(X_2)$$

where a W_j lying in both is assigned to the first group; then $\sum_j [\operatorname{div}(r_j)] = b_1 + b_2$ and so $a_1 + a_2 = b_1 + b_2$ in $Z_k(Y)$. Rearranging,

$$c := a_1 - b_1 = b_2 - a_2$$

The left expression $a_1 - b_1$ is a cycle supported on X_1, and the right expression $b_2 - a_2$ is a cycle supported on X_2; being the same cycle in the free group $Z_k(Y)$, it is supported on $X_1 \cap X_2$, that is $c \in Z_k(X_1 \cap X_2)$. Now compute classes. In $A_k(X_1)$, $b_1 \in \mathrm{Rat}_k(X_1)$ is zero, so $\eta_1 = [a_1] = [a_1 - b_1] = \iota_{1*}[c]$; in $A_k(X_2)$, $b_2 \in \mathrm{Rat}_k(X_2)$ is zero, so $\eta_2 = [a_2] = [a_2 - b_2] = -\iota_{2*}[c]$. Therefore

$$(\eta_1, \eta_2) = (\iota_{1*}[c], -\iota_{2*}[c]) = (\iota_{1*}, -\iota_{2*})\big([c]\big)$$

lies in the image of $(\iota_{1*}, -\iota_{2*})$. This gives $\ker(j_{1*} + j_{2*}) \subseteq \mathrm{im}(\iota_{1*}, -\iota_{2*})$, and with Step 2 exactness in the middle.

No left-exactness. The map $(\iota_{1*}, -\iota_{2*})$ need not be injective. Take X_1 and X_2 two distinct lines through the origin in $\mathbb{A}^2$, so $X_1 \cap X_2 = \{0\}$, in degree $k = 0$: then $A_0(X_1 \cap X_2) = \mathbb{Z}$ is generated by $[0]$, while $\iota_{1*}[0] = [0] \sim 0$ in $A_0(X_1) = A_0(\mathbb{A}^1)$ and likewise $\iota_{2*}[0] = 0$ in $A_0(X_2)$, so $(\iota_{1*}, -\iota_{2*})[0] = 0$ on a nonzero group. Thus there is no exactness at $A_k(X_1 \cap X_2)$, as we sought to show.

The corollary is the exact sequence recorded, without proof, as item (4) of example 8.1.6; the present derivation discharges that promissory note. Its shape is the Chow-group form of the Mayer-Vietoris sequence familiar from topology, truncated on the left for the same reason the localization sequence is: the intersection $X_1 \cap X_2$ can carry cycles that become rationally equivalent to zero once they are free to move in the larger $X_1 \cup X_2$. Where it is used most is in dévissage arguments, where a scheme is written as a union of simpler closed pieces and its Chow groups are assembled from those of the pieces and their overlaps.

A single application of the localization sequence recovers, from scratch, the computation of $A_0(\mathbb{A}^1)$ recorded in example 8.1.5, this time without touching a rational function by hand. The point is that $\mathbb{A}^1$ is $\mathbb{P}^1$ with a point removed, and the localization sequence turns the known $A_0(\mathbb{P}^1) = \mathbb{Z}$ into the answer for the open complement.

Example 8.4.4: Recomputing $A_0(\mathbb{A}^1)$ by Localization

Take $X = \mathbb{P}^1$, $Z = \{\infty\}$ the point at infinity, and $U = \mathbb{A}^1 = \mathbb{P}^1 \setminus \{\infty\}$. The localization sequence in degree 0 reads

$$A_0(\{\infty\}) \xrightarrow{i_*} A_0(\mathbb{P}^1) \xrightarrow{j^*} A_0(\mathbb{A}^1) \longrightarrow 0$$

The left group is $A_0(\{\infty\}) = \mathbb{Z}$, generated by the class of the single point ∞, and $i_*[\infty] = [\infty] \in A_0(\mathbb{P}^1)$. By example 8.1.5 the group $A_0(\mathbb{P}^1) \cong \mathbb{Z}$ is generated by the class h of any point, and $[\infty] = h$ is a generator, so i_* is surjective. By exactness $\mathrm{im}(i_*) = \ker(j^*)$, and since i_* is onto, $\ker(j^*) = A_0(\mathbb{P}^1)$; the quotient by the whole group is trivial:

$$A_0(\mathbb{A}^1) = A_0(\mathbb{P}^1)/\ker(j^*) \cong \mathrm{coker}(i_*) = \mathbb{Z}/\mathbb{Z} = 0$$

Thus $A_0(\mathbb{A}^1) = 0$, in agreement with the direct computation. The single generator of $A_0(\mathbb{P}^1)$ is killed precisely because the point removed to form $\mathbb{A}^1$ was rationally equivalent to every remaining point; removing it collapses the whole group.

With the localization sequence available, the second structural result of the section follows by Noetherian induction. Homotopy invariance in algebraic geometry takes the form of affine-bundle invariance: pulling a cycle back along the projection of an affine bundle is an isomorphism on Chow groups. The motivating case is the trivial bundle $X \times \mathbb{A}^n \to X$, which asserts that appending an affine factor does not change the Chow group beyond the expected degree shift, the algebraic counterpart of the fact that $\mathbb{A}^n$ is contractible.

The result is stated for a general affine bundle, of which the trivial bundle is the leading case.

Definition 8.4.5: Affine Bundle of Relative Dimension n

An *affine bundle of relative dimension n* over X is a scheme $p\colon E \to X$ that is locally on X a product with affine space: there is an open cover $\{U_\lambda\}$ of X and isomorphisms $p^{-1}(U_\lambda) \cong U_\lambda \times \mathbb{A}^n$ over U_λ.

A vector bundle of rank n, with its transition maps in GL_n, is such a bundle, as is any torsor under a vector bundle; the trivial example is $X \times \mathbb{A}^n$ itself. In every case p is flat of relative dimension n, so flat pullback $p^*\colon A_k(X) \to A_{k+n}(E)$ is defined by definition 8.3.3.

Theorem 8.4.6: Affine-Bundle Invariance

Let $p\colon E \to X$ be an affine bundle of relative dimension n over an algebraic scheme X. Then the flat pullback

$$p^*\colon A_k(X) \xrightarrow{\sim} A_{k+n}(E)$$

is an isomorphism for every k. In particular, for the trivial bundle, $A_k(X) \xrightarrow{\sim} A_{k+n}(X \times \mathbb{A}^n)$.

***Proof*:**
The proof runs the argument directly at relative dimension n; the fibre dimension enters only through the trivial bundle, where a global tower is available. Note that a global tower of rank-one affine bundles is *not* claimed for a general bundle: the local identification $U \times \mathbb{A}^n = (U \times \mathbb{A}^{n-1}) \times \mathbb{A}^1$ does not glue to a factorization $E \to X$ over X, since a general affine (or vector) bundle carries no global flag of subbundles. The trivial bundle $X \times \mathbb{A}^n$, being a literal product, *is* such a tower, $X \times \mathbb{A}^n = (X \times \mathbb{A}^{n-1}) \times \mathbb{A}^1$; the construction below invokes the tower only there. Fix n and prove that $p^*\colon A_k(X) \to A_{k+n}(E)$ is an isomorphism for all k, by Noetherian induction on the closed subschemes of X. For a closed subscheme $W \subseteq X$ write $E_W = p^{-1}(W)$ and $p_W\colon E_W \to W$ for the restricted bundle, again an affine bundle of relative dimension n.

Step 1: reductions. The scheme X is Noetherian, so its closed subschemes satisfy the descending chain condition and it suffices to prove: if p_W^* is an isomorphism for every proper closed subscheme $W \subsetneq X$, then p^* is an isomorphism for X. Chow groups depend only on the reduced structure (example 8.1.6), so take X reduced. If X is reducible, write $X = X_1 \cup X_2$ with X_1 an irreducible component and X_2 the union of the remaining components, both proper closed subschemes; the pullbacks $p^*_{X_1}, p^*_{X_2}, p^*_{X_1 \cap X_2}$ are isomorphisms by hypothesis. The Mayer-Vietoris sequences of corollary 8.4.3 for $X = X_1 \cup X_2$ and for $E = E_{X_1} \cup E_{X_2}$ (whose intersection is $E_{X_1 \cap X_2}$) form a commutative ladder linked by the vertical pullbacks; a diagram chase using the three outer isomorphisms shows p^* is an isomorphism for X. So assume X is a variety (reduced and irreducible), and let $\eta = \operatorname{Spec} k(X)$ be its generic point, with generic fibre $E_\eta \cong \mathbb{A}^n_{k(X)}$.

Step 2: surjectivity of p^*. It suffices to realize the class of a single $(k+n)$-dimensional subvariety $V \subseteq E$ as a pullback modulo the image of $A_{k+n}(E_Z)$ for some proper closed $Z \subsetneq X$. Let $W = \overline{p(V)} \subseteq X$. If $W \subsetneq X$ is proper, then $V \subseteq E_W$ is supported over $Z = W$, and $[V] \in \operatorname{im}\big(A_{k+n}(E_W) \to A_{k+n}(E)\big)$; by the inductive hypothesis p_W^* is onto, and compatibility of p^* with the closed immersion (the fibre square for p over $W \hookrightarrow X$ gives $p^* i_* = i_* p_W^*$ by proposition 8.3.6) writes $[V]$ as a pullback from X. If instead $W = X$, so V dominates X, restrict to the generic fibre. There $V_\eta \subseteq \mathbb{A}^n_{k(X)}$ is a closed subvariety, and its class is controlled by the trivial-bundle case of the theorem over the field $K = k(X)$: the projection $\mathbb{A}^n_K \to \operatorname{Spec} K$ has surjective flat pullback. This last fact is elementary and needs no base induction, being the trivial bundle over a point: writing the tower $\mathbb{A}^n_K = (\mathbb{A}^{n-1}_K) \times \mathbb{A}^1$ and inducting on n, the base case $A_*(\mathbb{A}^1_K)$ is generated over $A_*(\operatorname{Spec} K)$ by the pullback (every closed point of $\mathbb{A}^1_K$ is the divisor of its minimal polynomial, so $A_0(\mathbb{A}^1_K) = 0$, exactly as in example 8.1.5 with K in place of $\bar{k}$), and the tower propagates this up the coordinates. Thus $[V_\eta]$ is rationally equivalent on $\mathbb{A}^n_{k(X)}$ to a pullback $p^*_\eta(a_\eta)$ of a class a_η on $\operatorname{Spec} k(X)$. Spreading out the finitely many rational functions that realize this equivalence, together with a_η, to an open $U \subseteq X$ shows $[V]|_{E_U} = p^*_U(\alpha_U)$ for a class $\alpha_U \in A_k(U)$ and hence $[V] - p^*(\alpha_V)$ (with $\alpha_V \in A_k(X)$ lifting α_U via surjectivity of j^*, theorem 8.4.1) restricts to 0 on E_U, so its support lies over the proper closed $Z = X \setminus U$. In every case

$$[V] = p^*(\alpha_V) + \gamma_V \quad \text{in } A_{k+n}(E), \qquad \gamma_V \in \operatorname{im}\big(A_{k+n}(E_Z) \to A_{k+n}(E)\big)$$

for some proper closed $Z \subsetneq X$ and $\alpha_V \in A_k(X)$. By the inductive hypothesis p^*_Z is onto $A_*(E_Z)$, so $\gamma_V = i_* p^*_Z(\delta_V) = p^*(i_*\delta_V)$ for a class δ_V on Z, again by proposition 8.3.6. Hence $[V] = p^*(\alpha_V + i_*\delta_V)$ and p^* is surjective.

Surjectivity of p^* is now in hand for every X; injectivity is the remaining point, and it is not a formal consequence of the localization ladder, since the two rows are only right-exact and provide no left term to chase against. It is proved by exhibiting a left inverse to p^* for the trivial bundle and reducing the general bundle to that case.

Step 3: a left inverse for the trivial bundle. The construction is carried out for a single affine coordinate over an arbitrary base Y; the trivial bundle $X \times \mathbb{A}^n$ is a tower of n such single coordinates, and the rank-n left inverse is the composite of the n single-coordinate left inverses (Step 3d). Take $E = Y \times \mathbb{A}^1$, with coordinate t on the second factor and projection $p\colon Y \times \mathbb{A}^1 \to Y$, and define a homomorphism $\sigma\colon Z_{k+1}(Y \times \mathbb{A}^1) \to Z_k(Y)$ on a generator $[V]$, $V \subseteq Y \times \mathbb{A}^1$ a $(k+1)$-dimensional subvariety, by

$$\sigma[V] = \begin{cases} p_*\big[\operatorname{div}_V(t|_V)\big] & t|_V \in k(V)^* \text{ nonconstant} \\ 0 & t|_V \text{ constant} \end{cases}$$

extended additively; here "$t|_V$ constant" is read to include the degenerate case $t|_V \equiv 0$ (when $V \subseteq Y \times \{0\}$), which likewise contributes 0, since then $t|_V \notin k(V)^*$ and no divisor $\operatorname{div}_V(t|_V)$ is formed. When $t|_V$ is nonconstant, $[\operatorname{div}_V(t|_V)]$ is a k-cycle on V; the pushforward p_* of this particular cycle is well-defined even though $p|_V\colon V \to \overline{p(V)} \subseteq Y$ need not be proper. The point is that t is a regular function on $Y \times \mathbb{A}^1$, so its restriction $t|_V$ is regular on V and $\operatorname{div}_V(t|_V)$ is effective, with support contained in the zero locus $V \cap (Y \times \{0\}) \subseteq Y \times \{0\}$. On $Y \times \{0\}$ the projection p is the closed immersion identifying $Y \times \{0\}$ with Y, in particular proper, so $p_*[\operatorname{div}_V(t|_V)]$ is a well-defined k-cycle on Y by definition 8.2.1, computed component by component over the locus where $t|_V$ vanishes. Geometrically $\sigma[V]$ is the fibre of V over $t = 0$, projected to Y: the specialization of V to the zero section.

Step 3a: $\sigma \circ p^* = \mathrm{id}$. For a k-dimensional subvariety $W \subseteq Y$ the pullback is $p^*[W] = [W \times \mathbb{A}^1]$, and t restricts to the coordinate on $W \times \mathbb{A}^1$, which is nonconstant with $[\operatorname{div}_{W\times\mathbb{A}^1}(t)] = [W \times \{0\}]$ (the coordinate t has its single zero along $W \times \{0\}$ and no pole on $W \times \mathbb{A}^1$). Pushing forward, $p_*[W \times \{0\}] = [W]$, so $\sigma(p^*[W]) = [W]$. By additivity $\sigma \circ p^* = \mathrm{id}$ on $Z_k(Y)$, hence on $A_k(Y)$ once σ is known to descend.

Step 3b: σ respects rational equivalence. A generator of $\mathrm{Rat}_{k+1}(Y \times \mathbb{A}^1)$ is $[\mathrm{div}_S(g)]$ for a $(k+2)$-dimensional subvariety $S \subseteq Y \times \mathbb{A}^1$ and $g \in k(S)^*$, and unwinding the definitions gives $\sigma[\mathrm{div}_S(g)] = p_*\big(\sum_V \mathrm{ord}_V(g)\,[\mathrm{div}_V(t|_V)]\big)$, the outer sum over the codimension-one subvarieties $V \subseteq S$ on which $t|_V$ is nonconstant (a V with $t|_V$ constant, including $V \subseteq Y \times \{0\}$ where $t|_V \equiv 0$, drops out, contributing 0). This iterated divisor is symmetric in the two functions g and $t|_S$ modulo a principal divisor on S:

$$\sum_V \mathrm{ord}_V(g)\,[\mathrm{div}_V(t)] - \sum_V \mathrm{ord}_V(t)\,[\mathrm{div}_V(g)] \in \mathrm{Rat}_k(S) \tag{8.3}$$

The relation (8.3) is the commutativity of two divisor operations on the $(k+2)$-dimensional S, equivalently the well-definedness of intersecting a cycle with a Cartier divisor; its proof requires the divisor-theoretic machinery developed in the next chapter of this Part and is deferred there, since the tools are not yet available. Granting (8.3), $\sigma[\mathrm{div}_S(g)]$ differs from $p_*\big(\sum_V \mathrm{ord}_V(t)\,[\mathrm{div}_V(g)]\big)$ by the pushforward of a principal divisor, and $\sum_V \mathrm{ord}_V(t)\,[\mathrm{div}_V(g)]$ is itself a $\mathbb{Z}$-combination of principal divisors on the subvarieties V, so its pushforward lies in $\mathrm{Rat}_k(Y)$ by theorem 8.2.3. Either way $\sigma[\mathrm{div}_S(g)] \in \mathrm{Rat}_k(Y)$, so σ descends to

$$\sigma\colon A_{k+1}(Y \times \mathbb{A}^1) \to A_k(Y), \qquad \sigma \circ p^* = \mathrm{id}$$

In particular p^* is injective for the trivial rank-one bundle over any base Y, and being surjective by Step 2, it is an isomorphism there. The left inverse σ commutes with proper pushforward along closed immersions and with flat restriction to opens, since each of p_* and $\mathrm{div}(t)$ does; this compatibility is what carries the trivial-bundle result across the localization sequence in Step 3c.

Step 3d: the trivial rank-n bundle. For $E = X \times \mathbb{A}^n$ with coordinates $t_1, \dots, t_n$, the tower

$$X \times \mathbb{A}^n \xrightarrow{q_n} X \times \mathbb{A}^{n-1} \xrightarrow{q_{n-1}} \cdots \xrightarrow{q_1} X$$

factors $p = q_1 \circ \cdots \circ q_n$ into single-coordinate trivial rank-one bundles $q_m\colon (X \times \mathbb{A}^{m-1}) \times \mathbb{A}^1 \to X \times \mathbb{A}^{m-1}$, so $p^* = q_n^* \circ \cdots \circ q_1^*$ by functoriality of flat pullback (proposition 8.3.5). Applying Step 3b with $Y = X \times \mathbb{A}^{m-1}$ gives a left inverse σ_m to each q_m^*; the composite $\sigma := \sigma_1 \circ \cdots \circ \sigma_n\colon A_{k+n}(X \times \mathbb{A}^n) \to A_k(X)$ satisfies $\sigma \circ p^* = \mathrm{id}$, so p^* is injective, and with Step 2 it is an isomorphism for the trivial bundle. The compatibility of each σ_m with proper pushforward along closed immersions and with flat restriction to opens passes to the composite, so σ has the same compatibility used in the next step.

Step 3c: injectivity for a general bundle. Let $p\colon E \to X$ be an affine bundle of relative dimension n over the variety X and $\alpha \in A_k(X)$ with $p^*\alpha = 0$. Choose a dense open $U \subseteq X$ over which E trivializes, $E_U \cong U \times \mathbb{A}^n$, and set $Z = X \setminus U$, a proper closed subscheme. Flat base change (proposition 8.3.5) gives $p_U^*(j^*\alpha) = j^*(p^*\alpha) = 0$, and p_U^* is injective by Step 3d applied to the trivial bundle over U; hence $j^*\alpha = 0$. By the localization sequence for $Z \subseteq X$ (theorem 8.4.1), $\alpha = i_*\beta$ for some $\beta \in A_k(Z)$. Pushing β into E through the fibre square over $Z \hookrightarrow X$ (proposition 8.3.6) gives $p^*\alpha = p^*i_*\beta = i_*(p_Z^*\beta) = 0$, so $p_Z^*\beta$ lies in the kernel of $i_*\colon A_{k+n}(E_Z) \to A_{k+n}(E)$. Over the total space E the left inverse of Step 3d, restricted to the trivializing locus and extended by $\sigma_Z := (p_Z^*)^{-1}$ over E_Z (the inductive isomorphism), assembles into a retraction $r_E\colon A_{k+n}(E) \to A_k(X)$ of p^*; the compatibility noted at the end of Step 3d makes the square with i_* commute, so that $r_E \circ i_* = i_* \circ \sigma_Z$ on $A_{k+n}(E_Z)$. Evaluating at $p_Z^*\beta$,

$$i_*\beta = i_*\big(\sigma_Z p_Z^*\beta\big) = r_E\big(i_* p_Z^*\beta\big) = r_E(0) = 0$$

so $\alpha = i_*\beta = 0$. Hence p^* is injective.

> Steps 2 and 3 show p^* is an isomorphism for every X and every relative dimension n, completing the Noetherian induction; the tower entered only for the trivial bundle (Step 3d), where it is an actual global factorization, so no global tower was claimed for a general bundle, completing the proof.

The theorem is the algebraic-geometric shadow of a topological triviality: the projection $X \times \mathbb{A}^n \to X$ is a homotopy equivalence when $\mathbb{A}^n = \mathbb{C}^n$ is given the classical topology, so it induces isomorphisms on singular homology, and Chow groups behave the same way. The degree shift by n is the algebraic bookkeeping for the real dimension shift $2n$ on the topological side, since a k-cycle (complex dimension k) has real dimension $2k$ and A_{k+n} matches $H_{2(k+n)} = H_{2k+2n}$. The proof uses none of this: surjectivity of p^* needs only that every zero-cycle on an affine line over a field is rationally trivial ($A_0(\mathbb{A}^1_K) = 0$), propagated up the base by localization, while injectivity needs a left inverse, the specialization to the zero section. The one step left open, the divisor symmetry (8.3), is exactly the input that makes intersecting with a Cartier divisor well defined on rational equivalence, the payload of the next chapter; the isomorphism here is the first place that well-definedness is needed, and the forward reference is deliberate rather than a gap. The reason the argument needs Noetherian induction rather than a direct computation is the same reason the localization sequence is not left-exact: one cannot control what rational equivalences a cycle acquires on E without dismantling the base into pieces small enough that the bundle trivializes and the fibrewise vanishing takes over.

Exercise 8.4.1

1. Let $Z \subseteq X$ be closed with complement U. Show directly from theorem 8.4.1 that $A_k(U) \cong A_k(X)/\operatorname{im}\big(A_k(Z) \to A_k(X)\big)$, and use this to prove that if $A_k(Z) = 0$ then $j^*\colon A_k(X) \to A_k(U)$ is an isomorphism.
2. Let $X = \mathbb{A}^2$ and let $Z = \{0\}$ be the origin, so $U = \mathbb{A}^2 \setminus \{0\}$ is the punctured plane. Using the localization sequence together with the values $A_k(\mathbb{A}^2)$ (namely $A_2(\mathbb{A}^2) = \mathbb{Z}$ and $A_k(\mathbb{A}^2) = 0$ for $k < 2$, which you may take from section 8.5), compute $A_k(U)$ for all k.
3. Let X be a variety of dimension n and $U \subseteq X$ a nonempty open subset. Show that $A_n(X) \to A_n(U)$ is an isomorphism, and that both are $\mathbb{Z}$, generated by the fundamental class.
4. Compute $A_*(\mathbb{P}^n \times \mathbb{A}^m)$ in terms of $A_*(\mathbb{P}^n)$, assuming the value $A_k(\mathbb{P}^n) = \mathbb{Z}$ for $0 \le k \le n$ from section 8.5. State the answer as a graded group.
5. Let $p\colon E \to X$ be a vector bundle of rank r over an algebraic scheme X, with zero section $s\colon X \hookrightarrow E$. Affine-bundle invariance makes $p^*\colon A_k(X) \to A_{k+r}(E)$ an isomorphism. Explain why one cannot conclude $A_*(E) \cong A_*(X)$ by inverting p^* with a proper pushforward p_*, identifying the precise obstruction, and describe the degree by which p^* and the closed-immersion pushforward s_* each shift a cycle class.
6. Take X_1 and X_2 to be two distinct lines through the origin in $\mathbb{A}^2$, meeting only at $\{0\}$. Write out the sequence of corollary 8.4.3 in degree 0 and verify its exactness by hand, given that A_0 of a line is 0 and A_0 of a point is $\mathbb{Z}$.

8.5 First Computations

The machinery of the preceding sections is now assembled: cycles, rational equivalence, proper pushforward, flat pullback, and the two fundamental sequences. This section puts it to work on the spaces one meets first. We compute the Chow groups of affine space, of projective space, and of a smooth projective curve, and record the shape the answer takes for a projective bundle. These are the building blocks from which nearly every later computation is assembled, and they already display the two extremes of the theory: affine space, whose cycles almost all sweep away to nothing, and projective space, whose cycles are pinned down by a single integer in each dimension.

The pattern to watch is how the localization sequence and affine-bundle invariance from section 8.4 do the actual work. Affine space is an affine bundle over a point, so its Chow groups collapse; projective space is built from affine space by adding a hyperplane at infinity, so an induction along the localization sequence peels off one $\mathbb{Z}$ per dimension. Every computation below is an instance of one of these two moves.

Affine space is the first computation, and $\mathbb{A}^n$ retracts, in the sense of rational equivalence, all the way down to a point: every subvariety of dimension less than n can be swept off to zero, and only the fundamental class in top dimension survives. This is the higher-dimensional form of the observation $A_0(\mathbb{A}^1) = 0$ from example 8.1.5, and it follows in one line from affine-bundle invariance once one notices that $\mathbb{A}^n$ is an affine bundle over $\operatorname{Spec} \overline{k}$.

Proposition 8.5.1: Chow Groups of Affine Space

For every $n \geq 0$ the Chow groups of affine space are

$$A_k(\mathbb{A}^n) = \begin{cases} \mathbb{Z}\,[\mathbb{A}^n] & k = n \\ 0 & k \neq n \end{cases}$$

the group $A_n(\mathbb{A}^n)$ being free of rank one on the fundamental class $[\mathbb{A}^n]$.

***Proof*:**
The structure morphism $\pi\colon \mathbb{A}^n \to \operatorname{Spec} \overline{k}$ exhibits $\mathbb{A}^n$ as the trivial affine bundle of relative dimension n over a point. Affine-bundle invariance (theorem 8.4.6) states that flat pullback

$$\pi^*\colon A_k(\operatorname{Spec} \overline{k}) \xrightarrow{\sim} A_{k+n}(\mathbb{A}^n)$$

is an isomorphism for every k. The point $\operatorname{Spec} \overline{k}$ has a single subvariety, itself, of dimension 0, so $A_0(\operatorname{Spec} \overline{k}) = Z_0(\operatorname{Spec} \overline{k}) = \mathbb{Z}\,[\operatorname{Spec} \overline{k}]$ and $A_k(\operatorname{Spec} \overline{k}) = 0$ for $k \neq 0$. Transporting these through π^* gives $A_{k+n}(\mathbb{A}^n) = 0$ for $k \neq 0$, that is $A_j(\mathbb{A}^n) = 0$ for $j \neq n$, and $A_n(\mathbb{A}^n) = \pi^*\big(\mathbb{Z}\,[\operatorname{Spec} \overline{k}]\big)$. Flat pullback of the fundamental cycle of the base is the fundamental cycle of the total space (definition 8.3.3), so $\pi^*[\operatorname{Spec} \overline{k}] = [\mathbb{A}^n]$, and $A_n(\mathbb{A}^n) = \mathbb{Z}\,[\mathbb{A}^n]$ is free of rank one, completing the proof.

Affine space thus has no cycles in low dimension worth remembering, a fact that reads two ways. From below it says that any k-dimensional subvariety $V \subseteq \mathbb{A}^n$ with $k < n$ bounds: there is a family of cycles, parametrized by the line, sweeping $[V]$ to zero. From above it isolates the one invariant that does survive, the coefficient of the fundamental class, which for a top cycle records the multiplicity with which $\mathbb{A}^n$ appears in it. The vanishing in low dimension is what makes affine space so poor a stage for enumerative geometry, and simultaneously what makes it the natural chart in which to localize a computation on a larger space. This second role is exactly how projective space is handled next.

The contrast to keep in mind is the affine line. A single point of $\mathbb{A}^1$ is a 0-cycle in dimension $0 < 1$, so proposition 8.5.1 predicts $A_0(\mathbb{A}^1) = 0$, recovering the computation of example 8.1.5: the coordinate function $t - p$ has $[\mathrm{div}(t-p)] = [p]$, presenting the point as a principal divisor and hence as zero in the Chow group. The proposition is the assertion that this collapse happens in every dimension below the top, uniformly and for the same reason.

Projective space is the opposite. Nothing sweeps to zero for free, because a rational function on a subvariety of $\mathbb{P}^n$ has as many zeros as poles, and the surviving invariant in each dimension is an integer. The computation is an induction that peels $\mathbb{P}^n$ apart along a hyperplane: the closed subscheme $\mathbb{P}^{n-1} \subseteq \mathbb{P}^n$ at infinity, with open complement the affine chart $\mathbb{A}^n$. Affine space contributes nothing below the top by proposition 8.5.1, so the localization sequence forces every class of $\mathbb{P}^n$ in dimension $k < n$ to come from $\mathbb{P}^{n-1}$, and the induction runs.

The motivating principle is that projective space is *cellular*: it is stratified by the chain

$$\mathbb{P}^0 \subset \mathbb{P}^1 \subset \cdots \subset \mathbb{P}^{n-1} \subset \mathbb{P}^n$$

of linear subspaces, each obtained from the previous by adding an affine cell $\mathbb{A}^k = \mathbb{P}^k \setminus \mathbb{P}^{k-1}$. The Chow groups turn out to be free on the classes of these cells, one generator in each dimension. This is the algebraic shadow of the cell decomposition that computes the singular homology of $\mathbb{C}\mathrm{P}^n$ in [6], where the same chain of linear subspaces yields $H_{2k}(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) = \mathbb{Z}$ and $H_{\mathrm{odd}} = 0$.

Theorem 8.5.2: Chow Groups of Projective Space

For every $n \geq 0$ and every $0 \leq k \leq n$, the Chow group $A_k(\mathbb{P}^n)$ is free abelian of rank one, generated by the class $[\mathbb{P}^k]$ of any linear subspace $\mathbb{P}^k \subseteq \mathbb{P}^n$ of dimension k:

$$A_k(\mathbb{P}^n) = \mathbb{Z}\,[\mathbb{P}^k], \qquad 0 \leq k \leq n$$

and $A_k(\mathbb{P}^n) = 0$ for $k > n$. Any two linear subspaces of the same dimension are rationally equivalent, so the generator does not depend on the choice.

***Proof*:**
The argument is an induction on n, run through projection from a point, which realizes the complement of a point in $\mathbb{P}^n$ as a line bundle over $\mathbb{P}^{n-1}$. Write h_k for the class $[\mathbb{P}^k]$ of a linear k-plane; the claim is that $A_k(\mathbb{P}^n) = \mathbb{Z}\,h_k$ is free of rank one for $0 \leq k \leq n$, and 0 otherwise. Groups $A_k(\mathbb{P}^n)$ with $k > n = \dim \mathbb{P}^n$ vanish because $\mathbb{P}^n$ has no subvariety of that dimension, so only $0 \leq k \leq n$ is at issue.

Step 1: base cases $n = 0$ and $k = 0$. For $n = 0$ the space $\mathbb{P}^0 = \operatorname{Spec} \overline{k}$ is a point, with $A_0(\mathbb{P}^0) = \mathbb{Z}\,[\mathbb{P}^0]$ and $A_k(\mathbb{P}^0) = 0$ for $k \neq 0$ by proposition 8.5.1, which is the claim. For general n and $k = 0$, every closed point of $\mathbb{P}^n$ is rationally equivalent to every other: two points p, q both lie on a line $\ell \cong \mathbb{P}^1 \subseteq \mathbb{P}^n$, and on ℓ they are rationally equivalent by example 8.1.5, so $[p] \sim [q]$ in $\mathbb{P}^n$. Thus $A_0(\mathbb{P}^n)$ is cyclic, generated by $h_0 = [p]$. Since $\mathbb{P}^n$ is complete, the degree map $\deg\colon A_0(\mathbb{P}^n) \to \mathbb{Z}$ of definition 8.2.4 is well defined and sends $h_0 = [p] \mapsto [k(p) : \overline{k}] = 1$, so h_0 has infinite order and $A_0(\mathbb{P}^n) = \mathbb{Z}\,h_0$ is free of rank one.

Step 2: projection from a point as a line bundle. Fix $n \geq 1$ and a closed point $P \in \mathbb{P}^n$. Choosing coordinates with $P = [0 : \cdots : 0 : 1]$, linear projection away from P is a morphism

$$q \colon \mathbb{P}^n \setminus \{P\} \to \mathbb{P}^{n-1}, \qquad [x_0 : \cdots : x_n] \mapsto [x_0 : \cdots : x_{n-1}]$$

whose fiber over a point $[a] \in \mathbb{P}^{n-1}$ is the line through P and $[a]$ with P removed, an affine line $\mathbb{A}^1$. This exhibits $\mathbb{P}^n \setminus \{P\}$ as a line bundle over $\mathbb{P}^{n-1}$, the total space of $\mathcal{O}_{\mathbb{P}^{n-1}}(1)$; in particular it is an affine bundle of relative dimension 1. Only the last clause is used below. Affine-bundle invariance (theorem 8.4.6) makes flat pullback

$$q^* \colon A_{k-1}(\mathbb{P}^{n-1}) \xrightarrow{\sim} A_k(\mathbb{P}^n \setminus \{P\}) \tag{8.4}$$

an isomorphism for every k.

Step 3: removing the point via localization. The point $\{P\}$ is a closed subscheme of $\mathbb{P}^n$ with open complement $\mathbb{P}^n \setminus \{P\}$. The localization sequence (theorem 8.4.1) for this pair reads

$$A_k(\{P\}) \xrightarrow{i_*} A_k(\mathbb{P}^n) \xrightarrow{j^*} A_k(\mathbb{P}^n \setminus \{P\}) \to 0 \tag{8.5}$$

exact at $A_k(\mathbb{P}^n)$ and $A_k(\mathbb{P}^n \setminus \{P\})$. The point contributes $A_k(\{P\}) = 0$ for every $k \geq 1$ and $A_0(\{P\}) = \mathbb{Z}\,[P]$. Hence for $k \geq 1$ the left term vanishes and j^* is an isomorphism, which combined with (8.4) gives, for $k \geq 1$,

$$A_k(\mathbb{P}^n) \xrightarrow[j^*]{\sim} A_k(\mathbb{P}^n \setminus \{P\}) \xleftarrow[q^*]{\sim} A_{k-1}(\mathbb{P}^{n-1}) \tag{8.6}$$

so $A_k(\mathbb{P}^n) \cong A_{k-1}(\mathbb{P}^{n-1})$ as abelian groups.

Step 4: the induction. Fix k with $1 \leq k \leq n$. Iterating (8.6) k times descends both indices by k:

$$A_k(\mathbb{P}^n) \cong A_{k-1}(\mathbb{P}^{n-1}) \cong \cdots \cong A_0(\mathbb{P}^{n-k})$$

and $A_0(\mathbb{P}^{n-k}) = \mathbb{Z}$ is free of rank one by Step 1. Together with the case $k = 0$ of Step 1, this proves $A_k(\mathbb{P}^n) \cong \mathbb{Z}$ is free of rank one for every $0 \leq k \leq n$. For $k > n$ the group vanishes as noted at the outset.

Step 5: the generator is the class of a linear subspace. It remains to identify the free generator with $h_k = [\mathbb{P}^k]$. Trace a linear $\mathbb{P}^k \subseteq \mathbb{P}^n$ through (8.6). Choose the point $P \in \mathbb{P}^k$; projection of the linear subspace $\mathbb{P}^k$ from the point $P \in \mathbb{P}^k$ is again linear, of one lower dimension, so $q(\mathbb{P}^k \setminus \{P\})$ is a linear $\mathbb{P}^{k-1} \subseteq \mathbb{P}^{n-1}$. Restricting the bundle $q \colon \mathbb{P}^n \setminus \{P\} \to \mathbb{P}^{n-1}$ of Step 2 over this $\mathbb{P}^{k-1}$ realizes $\mathbb{P}^k \setminus \{P\}$ as the affine bundle over $\mathbb{P}^{k-1}$; concretely $q^{-1}(\mathbb{P}^{k-1}) = \mathbb{P}^k \setminus \{P\}$ as a reduced irreducible subvariety, so flat pullback gives $q^*[\mathbb{P}^{k-1}] = [q^{-1}(\mathbb{P}^{k-1})] = [\mathbb{P}^k \setminus \{P\}]$ with multiplicity 1. Hence by construction of (8.6) the class $h_k = [\mathbb{P}^k]$ satisfies $j^* h_k = [\mathbb{P}^k \setminus \{P\}] = q^*[\mathbb{P}^{k-1}]$, so h_k corresponds to h_{k-1} under the isomorphism $A_k(\mathbb{P}^n) \cong A_{k-1}(\mathbb{P}^{n-1})$. Descending to $A_0(\mathbb{P}^{n-k})$, the class h_k maps to $h_0 = [\mathrm{pt}]$, which is the free generator by Step 1. Thus h_k is a generator of $A_k(\mathbb{P}^n) = \mathbb{Z}$. The top case $k = n$ is consistent with the fundamental-class computation $A_n(\mathbb{P}^n) = Z_n(\mathbb{P}^n) = \mathbb{Z}\,[\mathbb{P}^n]$ of example 8.1.6, since $\mathbb{P}^n$ is irreducible and $h_n = [\mathbb{P}^n]$.

Step 6: independence of the linear subspace. Any two linear k-planes $\Lambda, \Lambda' \subseteq \mathbb{P}^n$ have classes differing by an integer multiple of the generator; the previous step assigns each of them the generator h_0 under the descent to $A_0(\mathbb{P}^{n-k})$, so $[\Lambda] = [\Lambda']$ in $A_k(\mathbb{P}^n)$. Concretely a projective linear transformation carrying Λ to Λ' sweeps out a rational equivalence between them, so the generator h_k is independent of the chosen k-plane, completing the induction and the proof.

The theorem says that the Chow groups of $\mathbb{P}^n$ take the simplest possible form: one copy of $\mathbb{Z}$ in each dimension from 0 to n, with a canonical generator, the class of a linear subspace. Writing the total group additively,

$$A_*(\mathbb{P}^n) = \bigoplus_{k=0}^{n} \mathbb{Z}\, h_k, \qquad h_k = [\mathbb{P}^k]$$

so $A_*(\mathbb{P}^n)$ is free of rank $n+1$. In codimension indexing the generator $h_k = [\mathbb{P}^k]$ has codimension $n-k$, and it is common to write $h = [\mathbb{P}^{n-1}]$ for the hyperplane class and record h_k as the intersection h^{n-k} of $n-k$ hyperplanes; the multiplicative structure that makes this literal is the intersection product developed in a later chapter. For now the additive statement is what matters: on projective space, a k-cycle carries exactly one integer of information, its degree, the coefficient of h_k.

This is the precise sense in which projective space is rigid where affine space is soft. A subvariety $V \subseteq \mathbb{P}^n$ of dimension k and degree d (its degree as a projective variety, the number of points in which it meets a general linear $\mathbb{P}^{n-k}$) has class $[V] = d\, h_k$, and no amount of rational equivalence can change d. The degree is a complete invariant of the class, and rational equivalence on $\mathbb{P}^n$ is, for irreducible cycles, exactly the relation "same dimension and same degree" (and for arbitrary cycles, "same degree in each dimension"). The comparison with singular homology is exact: under the cycle class map to $H_{2k}(\mathbb{C}\mathrm{P}^n)$ studied later, h_k maps to the generator, and the freeness of $A_k(\mathbb{P}^n)$ mirrors the freeness of $H_{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$ ([6]).

The one-dimensional case deserves its own statement, both because curves are where the theory began and because on a curve the Chow group is an object the reader already knows under another name. On a smooth projective curve C the only cycles are 0-cycles and the fundamental class, and the 0-cycles modulo rational equivalence are precisely the divisor classes. This ties the general construction back to the divisor class group $\mathrm{Cl}\, C$ and the Picard group $\mathrm{Pic}\, C$ of section 5.4.

Proposition 8.5.3: Chow Groups of a Smooth Projective Curve

Let C be a smooth projective curve over $\overline{k}$. Then

$$A_1(C) = \mathbb{Z}\,[C], \qquad A_0(C) \cong \mathrm{Cl}\, C = \mathrm{Pic}\, C$$

and $A_k(C) = 0$ for $k \neq 0, 1$. Under the second isomorphism the degree of a zero-cycle corresponds to the degree of a divisor, giving a short exact sequence

$$0 \to \mathrm{Pic}^0 C \to A_0(C) \xrightarrow{\deg} \mathbb{Z} \to 0$$

where $\mathrm{Pic}^0 C$ is the group of divisor classes of degree zero.

***Proof*:**
Since $\dim C = 1$, the only nonzero groups are $A_0(C)$ and $A_1(C)$, and $A_k(C) = 0$ for $k \neq 0, 1$. The curve C is irreducible of dimension 1, so by example 8.1.6 its top group is $A_1(C) = Z_1(C) = \mathbb{Z}\,[C]$, free on the fundamental class.

The zero-cycles. A 0-cycle on C is a finite $\mathbb{Z}$-combination of closed points, so $Z_0(C)$ is the group $\mathrm{WDiv}\, C$ of Weil divisors on C. The subgroup $\mathrm{Rat}_0(C)$ of cycles rationally equivalent to zero is generated by the cycles $[\mathrm{div}(r)]$ for $r \in k(W)^*$ ranging over 1-dimensional subvarieties $W \subseteq C$; the only such W is C itself, so $\mathrm{Rat}_0(C)$ is generated by the principal divisors $[\mathrm{div}(r)]$ with $r \in k(C)^*$. Thus $\mathrm{Rat}_0(C)$ is exactly the group $\mathrm{Prin}\, C$ of principal divisors in the sense of definition 5.4.6, and

$$A_0(C) = Z_0(C)/\mathrm{Rat}_0(C) = \mathrm{WDiv}\, C/\,\mathrm{Prin}\, C = \mathrm{Cl}\, C$$

is the divisor class group of definition 5.4.7. Because C is smooth, the divisor-to-line-bundle correspondence of section 5.4 identifies $\mathrm{Cl}\, C$ with $\mathrm{Pic}\, C$.

The degree sequence. On a smooth projective curve linearly equivalent divisors have equal degree: if $D \sim D'$ then $\deg D = \deg D'$, since the Riemann-Roch theorem (theorem 7.1.18) gives $\chi(\mathcal{L}(D)) = \deg D + 1 - g$, a quantity depending only on the class of D, so $\deg D$ is determined by it. Hence deg descends to a homomorphism $\deg\colon A_0(C) \to \mathbb{Z}$, agreeing under the identification $A_0(C) = \mathrm{Cl}\, C$ with the degree of a divisor class. It is surjective because a single closed point has degree $[k(p) : \overline{k}] = 1$, and its kernel is the group $\mathrm{Pic}^{\,0} C$ of degree-zero classes by definition. This gives the stated short exact sequence, completing the proof.

The proposition recovers, in the language of Chow groups, the entire divisor theory of a curve. The degree map splits the class group into a discrete part, the copy of $\mathbb{Z}$ recording degree, and a connected part $\mathrm{Pic}^{\,0} C$, the Jacobian, carrying the continuous moduli of divisor classes of a fixed degree. For $C = \mathbb{P}^1$ the Jacobian is trivial and $A_0(\mathbb{P}^1) = \mathbb{Z}$, in agreement with example 8.1.5 and with the case $n = 1$ of theorem 8.5.2; for a curve of genus $g \geq 1$ the group $\mathrm{Pic}^{\,0} C$ is a g-dimensional abelian variety and $A_0(C)$ is strictly larger than $\mathbb{Z}$, so the degree is no longer a complete invariant of a class. The Chow group of a curve is therefore the first place in the theory where an interesting continuous invariant appears, and it is the model on which the higher-dimensional theory is built.

The Chow group $A_0(C)$ also illustrates the failure of the total-degree map to see everything. On $\mathbb{P}^n$ the degree was a complete invariant of a k-cycle class; on a curve of positive genus it is not, because two points $p, q \in C$ need not be rationally equivalent even though $[p]$ and $[q]$ have the same degree. Their difference $[p] - [q]$ is a nonzero class in $\mathrm{Pic}^{\,0} C$ precisely when p and q are not linearly equivalent, which for $g \geq 1$ is the generic situation. This is the geometric content of the statement that a curve of positive genus admits no nonconstant rational function of degree one.

The last computation of the section is stated but not proved, because its proof needs Segre classes, which are developed later. It generalizes the projective-space computation from the trivial bundle to an arbitrary vector bundle: the Chow groups of a projectivized vector bundle are a free module over the Chow groups of the base, on the powers of the tautological class. This is the projective-bundle formula, and it is the engine that will later define Chern classes.

Given a vector bundle E of rank $r + 1$ on a scheme X, write $\mathbb{P}(E) \to X$ for the associated projective bundle, whose fiber over a point x is the projective space $\mathbb{P}(E_x) \cong \mathbb{P}^r$ of one-dimensional subspaces of the fiber E_x. Let $\xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1)) \in A^1(\mathbb{P}(E))$ be the first Chern class of the Serre-twist line bundle $\mathcal{O}_{\mathbb{P}(E)}(1)$ (the dual of the tautological subbundle, with the "one-dimensional subspaces" convention above), viewed as an operator on cycles. The projective-bundle formula states that the flat pullback π^* together with capping against powers of ξ makes $A_*(\mathbb{P}(E))$ a free module over $A_*(X)$ on the classes $1, \xi, \ldots, \xi^r$:

$$A_*(\mathbb{P}(E)) = \bigoplus_{i=0}^{r} \xi^i \cap \pi^* A_*(X)$$

Concretely, every class on $\mathbb{P}(E)$ is uniquely $\sum_{i=0}^{r} \xi^i \cap \pi^* \alpha_i$ with $\alpha_i \in A_*(X)$. Taking $X = \mathrm{Spec}\, \overline{k}$ and $E = \overline{k}^{\,n+1}$ recovers $\mathbb{P}(E) = \mathbb{P}^n$ and the free-module statement collapses to $A_*(\mathbb{P}^n) = \bigoplus_{i=0}^{n} \mathbb{Z}\, \xi^i$ of theorem 8.5.2, with ξ^i the class of a linear subspace of codimension i. The full statement, and the definition of the Chern class operators it rests on, come in the chapter on cones and Segre classes; the formula is recorded here because it is the natural common generalization of the two computations above, and because it is the shape every later Chow-group computation of a bundle takes.

Before the exercises, the three worked instances of the theory are collected side by side, so the common mechanism is visible in one place.

Example 8.5.4: The Chow Groups of $\mathbb{P}^1$, $\mathbb{P}^2$, and a Plane Curve

1. *The projective line.* For $n = 1$, theorem 8.5.2 gives $A_0(\mathbb{P}^1) = \mathbb{Z}\,[\mathrm{pt}]$ and $A_1(\mathbb{P}^1) = \mathbb{Z}\,[\mathbb{P}^1]$, so

$$A_*(\mathbb{P}^1) = \mathbb{Z}\,[\mathbb{P}^1] \oplus \mathbb{Z}\,[\mathrm{pt}]$$

   The degree-0 part $A_0(\mathbb{P}^1) = \mathbb{Z}$ agrees with example 8.1.5, and with proposition 8.5.3 for the genus-zero curve $\mathbb{P}^1$, whose Jacobian $\mathrm{Pic}^0\mathbb{P}^1$ is trivial.

2. *The projective plane.* For $n = 2$,

$$A_*(\mathbb{P}^2) = \mathbb{Z}\,[\mathbb{P}^2] \oplus \mathbb{Z}\,[\mathbb{P}^1] \oplus \mathbb{Z}\,[\mathrm{pt}]$$

   with $A_2(\mathbb{P}^2) = \mathbb{Z}\,[\mathbb{P}^2]$, $A_1(\mathbb{P}^2) = \mathbb{Z}\,[\mathbb{P}^1]$ generated by the class of a line, and $A_0(\mathbb{P}^2) = \mathbb{Z}\,[\mathrm{pt}]$. A plane curve $D \subseteq \mathbb{P}^2$ of degree d has class $[D] = d\,[\mathbb{P}^1] = d\,h_1$ in $A_1(\mathbb{P}^2)$, since D is cut out by a form of degree d and hence is rationally equivalent to d lines (the pencil spanned by the form and ℓ^d for a linear form ℓ sweeps one to the other). Two plane curves of degrees c and d therefore have classes meeting, after the intersection product is available, in cd points, which is Bézout's theorem ([**NateClassicalAlgGeo**]) read off from $h_1 \cdot h_1 = h_0$.

3. *A plane curve as a curve in its own right.* Let $C \subseteq \mathbb{P}^2$ be a smooth plane curve of degree d, so C is a smooth projective curve of genus $g = \binom{d-1}{2}$. Two computations of its Chow groups coexist. Viewed inside $\mathbb{P}^2$, the class $[C] = d\,h_1 \in A_1(\mathbb{P}^2)$ records only its degree. Viewed intrinsically, proposition 8.5.3 gives $A_0(C) \cong \mathrm{Cl}\,C = \mathrm{Pic}\,C$ with its degree sequence $0 \to \mathrm{Pic}^0 C \to A_0(C) \to \mathbb{Z} \to 0$, and for $d \geq 3$ the genus is positive, so $\mathrm{Pic}^0 C$ is nontrivial and $A_0(C)$ is strictly larger than $\mathbb{Z}$. The pushforward $A_0(C) \to A_0(\mathbb{P}^2) = \mathbb{Z}$ along the inclusion is the degree map, collapsing all of $\mathrm{Pic}^0 C$ to zero: the intrinsic Chow group of the curve sees the moduli that the ambient projective plane cannot.

Exercise 8.5.1

1. Compute $A_*(\mathbb{A}^2 \setminus \{p\})$, the Chow groups of the affine plane with one closed point removed. (Use the localization sequence for the closed point inside $\mathbb{A}^2$ together with proposition 8.5.1.)

2. Let $H \cong \mathbb{P}^{n-1}$ be a hyperplane in $\mathbb{P}^n$ and $U = \mathbb{P}^n \setminus H \cong \mathbb{A}^n$ its complement. Using the localization sequence, show directly that $j^*\colon A_k(\mathbb{P}^n) \to A_k(\mathbb{A}^n)$ is an isomorphism for $k = n$ and the zero map for $k < n$, and reconcile this with theorem 8.5.2.

3. Let $Q \subseteq \mathbb{P}^3$ be a smooth quadric surface. Assuming the isomorphism $Q \cong \mathbb{P}^1 \times \mathbb{P}^1$ and that $A_*(\mathbb{P}^1 \times \mathbb{P}^1)$ is free on the classes of a point, the two rulings, and the fundamental class, determine the rank of each $A_k(Q)$ and describe generators. Compare with $A_*(\mathbb{P}^2)$ and explain why $A_1(Q)$ has rank 2 rather than 1.

4. For $\mathbb{P}^m$ and $\mathbb{P}^n$, use theorem 8.5.2 and the projective-bundle formula (with $\mathbb{P}^m \times \mathbb{P}^n = \mathbb{P}(\mathcal{O}^{\oplus(m+1)})$ over $\mathbb{P}^n$) to show that $A_k(\mathbb{P}^m \times \mathbb{P}^n)$ is free of rank equal to the number of pairs

(i, j) with $0 \leq i \leq m$, $0 \leq j \leq n$, and $i + j = k$.

5. Let C be a smooth projective curve of genus 1 over $\overline{k}$. Show that $A_0(C)$ is not finitely generated, and identify the classes $[p] - [q]$ of differences of points inside $\operatorname{Pic}^0 C$. Contrast with $A_0(\mathbb{P}^1)$.

6. Let $X \subseteq \mathbb{A}^3$ be the affine cone $x^2 + y^2 = z^2$ over a conic, a surface singular at the origin. Using $A_k(X) = A_k(X_{\mathrm{red}})$ and the vanishing of $A_k(\mathbb{A}^n)$ in low degree as a guide, determine $A_2(X)$ and explain why the singular point does not contribute a class in $A_0(X)$ that survives rational equivalence.

Part III

Moduli, Deformation Theory, and Stacks

(I feel like this is a natural next part)

Index

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